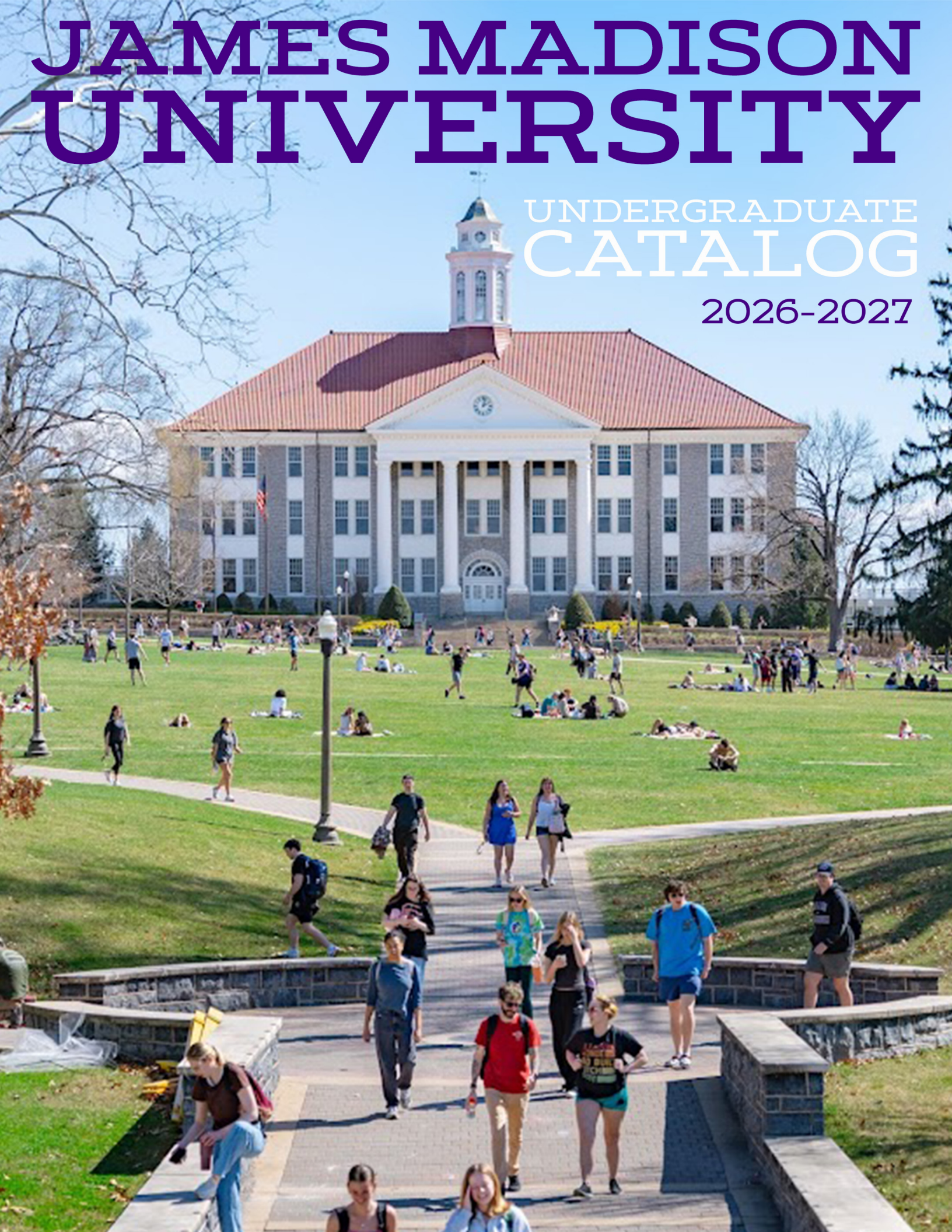




JAMES MADISON UNIVERSITY

UNDERGRADUATE
CATALOG
2026-2027

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PUBLISHING STATEMENT

This catalog is not a contract; it presents the requirements, offerings, information and regulations in effect at the time of publication. The university reserves the right to change any requirement or other provision at any time through university-approved curriculum processes. The university endeavors to provide notice of any change through the advising process or through other means of communications; however, it is the responsibility of each individual student to become and remain aware of all applicable requirements and provisions that may apply to the student.

The university operator can be reached at (540) 568-6211.

James Madison University
800 South Main Street
Harrisonburg, VA 22807

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MESSAGE FROM THE PRESIDENT

Dear JMU Student,

Welcome to possibilities!

This James Madison University 2026-2027 Catalog encompasses the range and scope of learning, discovery and exploration that await you as a JMU student. These classes, degree programs and hands-on experiences encompass our long commitment to providing students with the knowledge, skills and opportunities that will help you chart a successful future.

Regardless of the degree you pursue, I encourage you to consult with advisors who can help you navigate your journey. Talk with fellow students and faculty members to find the classes that will benefit you best. And, above all, explore the unexpected. Challenge yourself with a class outside your comfort zone or in a discipline you've never studied. Find opportunities for research, community engagement or international study to create a learning experience uniquely right for you.

Ahead are a wealth of JMU possibilities. Make them your own.

Go Dukes!

James C. Schmidt, President

ABOUT THE UNIVERSITY

James Madison University is a public, research university and is the only university in America named for James Madison. The university provides a high-quality experience for students enrolled at the undergraduate, graduate and doctoral levels. Innovative academic programs, founded in a broad range of the liberal arts, are informed and refined by advances in technology and societal needs. Academic offerings are augmented by co-curricular activities that help our future graduates develop the knowledge, skills and character they need to thrive.

JMU faculty are known for blending inspired teaching with rigorous scholarship, with an emphasis on involving undergraduate and graduate students in their research. Through this distinctive teacher-scholar model, JMU has transformed from a respected regional comprehensive university into a nationally recognized R2 doctoral research institution.

Mission Statement

We are a community committed to preparing students to be educated and enlightened citizens who lead productive and meaningful lives.

History

Since its establishment in 1908, James Madison University has grown from a small state normal and industrial school for women to a coeducational comprehensive university and evolved into a nationally recognized R2 doctoral research institution with a Fall 2025 enrollment of 23,157 students.

The university was founded in 1908 as the State Normal and Industrial School for Women at Harrisonburg, with Julian Ashby Burruss as its first president. The school opened its doors in 1909 with an enrollment of 209 students and a faculty of 15. Its first 20 graduates received diplomas in 1911. In 1914, the name of the school was changed to the State Normal School for Women at Harrisonburg. The school received authorization to award bachelor's degrees in 1916. During this initial period of development, Burruss' administration established the campus plan and constructed six buildings.

After Burruss resigned in 1919, Dr. Samuel Page Duke became the second president. Duke's administration erected nine major buildings. In 1924, the university became the State Teachers College at Harrisonburg and continued under that name until 1938, when it was named Madison College in honor of James Madison, the fourth president of the United States. In 1946, the Duke administration admitted men as regular day students.

Following the retirement of Duke, Dr. G. Tyler Miller became the third president of the university in 1949 and remained until 1970. Miller's administration enlarged the campus by 240 acres and constructed 19 buildings. The administration also revamped the curriculum. In 1954, the expanding school received authority to grant master's degrees. The university became a coeducational institution and first enrolled Black students in 1966.

Dr. Ronald E. Carrier became JMU's fourth president in 1971. His administration transformed Madison College into a university. In 1977, the institution adopted its current name, James Madison University. The Carrier administration nearly tripled the number of students and university faculty members and constructed some 30 major campus buildings. Doctoral degrees were authorized in 1994.

Dr. Linwood H. Rose was named JMU's fifth president in September 1998. Under his leadership, JMU was continually recognized in national publications as one of the nation's finest institutions of its type. More than 20 new academic programs were implemented, 25 major buildings were constructed and a Phi Beta Kappa chapter was installed. The university successfully completed its first capital campaign.

Jonathan R. Alger, JD, took office in July 2012 as the sixth president of James Madison University. Under his leadership, JMU developed a vision to be "the national model of the engaged university" with a strategic plan focused on engaged learning and community and civic engagement. In 2022, JMU's Carnegie classification was elevated to R2, establishing the university as a Doctoral University with High Research Activity, and the highly successful Unleashed comprehensive campaign raised over \$250M. Academic programs in education, health care and information technology expanded under his purview.

Mr. Charles King was appointed interim president July 1, 2024. He previously served as the university's senior vice president for administration and finance and chief financial officer for 25 years, retiring in 2021. During this academic year, JMU was recognized by the Carnegie Commission as a 2025 Research 2: High Spending and Doctorate Production institution, progress on the Carrier Library renovation progressed substantially, and the JMU Dukes football team won their first bowl title. The university also established its first Doctor of Education degree.

In March 2024, Dr. James C. Schmidt was named the seventh president of James Madison University. His presidency began with an extensive listening tour involving internal and external JMU faculty, staff, students, alumni, university partners, and civic, legislative and business leaders across the commonwealth. By the end of his first year, President Jim – informed with feedback from the entire campus community – drafted a new comprehensive strategic plan focused on identifying and building on JMU's distinctive strengths that will take JMU to the next level.

JMU has become a national model for developing work-based learning experiences, advancing a teacher-scholar model and promoting access. Ranked as the "top college in Virginia for getting a job," 95% of JMU undergraduates in the class of 2024 achieved a career outcome within a year; 96.7% of graduates with advanced degrees achieved the same.

Administration

The general responsibility for the administration of the university has been assigned to the president, who is appointed by the JMU Board of Visitors. When the board is in recess, its executive committee may exercise the power of the board.

Assisting the president in the administration of the university are the provost and vice president for academic affairs; the vice president for access and enrollment; the vice president for administration and finance; the vice president for marketing and strategic communications; the vice president for philanthropy and engagement; the vice president for research, economic development and innovation; the vice president for student affairs; the vice president and chief of staff; and the director of intercollegiate athletics.

Appointment to these positions and to the university's instructional and administrative faculty is made by the JMU Board of Visitors upon the recommendation of the president.

Board of Visitors

David Rexrode, Fredericksburg, VA (Rector)

Nicole Palya Wood, Doswell, VA (Vice Rector)

Lieutenant Colonel Jeff Bolander, USMC, Retired, McGaheysville, VA

Larry Caudle, Jr., White Stone, VA

Carly Fiorina, Lorton, VA

Thomas Galati, Penn Laird, VA

Heather Hedrick, Richmond, VA

The Honorable Kay Coles James, Richmond, VA

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John Lynch, Virginia Beach, VA

Angela Reddix, Virginia Beach, VA

Steven Smith, Bristol, TN

Nikki Thacker, Richmond, VA

Jack White, Fairfax Station, VA

Lisa Winn Bryan, Chesterfield, VA

Gram Sher, Virginia Beach, VA (Student Representative)

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David C. Kirkpatrick, Ph.D.

Vice President and Chief of Staff

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G. Fletcher Linder, Ph.D.

Interim Provost and Vice President for Academic Affairs

Timothy M. Miller, Ed.D.

Vice President for Student Affairs

Towana H. Moore, M.B.A.

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Vice President for Research, Economic Development and Innovation

Melinda J. Wood, M.S.

Vice President for Access and Enrollment

Deans

Bethany Blackstone, Ph.D.

Dean, Honors College

Michael E. Busing, Ph.D.

Dean, College of Business

Rubén G. Graciani, M.F.A.

Dean, College of Visual and Performing Arts

P. Grady Dixon, Ph.D.

Dean, College of Integrated Science and Engineering

Mark E. L'Esperance, Ph.D.

Dean, College of Education

G. Fletcher Linder, Ph.D.

Dean, University Studies

Sharon E. Lovell, Ph.D.

Dean, College of Health and Behavioral Studies

Bethany P. Nowviskie, Ph.D.

Dean, Libraries

Samantha Bates Prins, Ph.D.

Dean, College of Science and Mathematics

Nicholas J. Swartz, Ph.D.

Dean, School of Professional and Continuing Education

Linda M. Thomas, J.D., Ph.D.

Dean, The Graduate School

Traci A. Zimmerman, Ph.D.

Dean, College of Arts and Letters

Accreditation

James Madison University is accredited by the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC) to award baccalaureate, masters, educational specialist and doctorate degrees. Questions about the accreditation of James Madison University may be directed in writing to the Southern Association of Colleges and Schools Commission on Colleges at 1866 Southern Lane, Decatur, GA 30033-4097, by calling (404) 679-4500, or by using information available on SACSCOC's website (<https://www.sacscoc.org/>).

Additional Accreditation

ABET, Incorporated

AACSB International – The Association to Advance Collegiate Schools of Business

American Chemical Society

Accreditation Council for Education in Nutrition and Dietetics

Accreditation Council for Occupational Therapy Education

American Psychological Association

Association of University Health Programs in Health Administration

Accreditation Review Commission on Education for the Physician Assistant, Inc.

Commission for Collegiate Nursing Education

Commission on Accreditation of Athletic Training Education

Computing Accreditation Commission of ABET, Inc.

Council on Academic Accreditation in Audiology and Speech Language

Pathology of the American Speech-Language and Hearing Association

Council for Accreditation of Counseling and Related Educational Programs

Council for the Accreditation of Academic Programs for Speech
Pathology and Clinical Audiology
Council for the Accreditation of Educator Preparation
Council on Social Work Education (CSWE), Commission on Accreditation
(COA)
National Association of School Psychologists
National Association of Schools of Art and Design
National Association of Schools of Dance
National Association of Schools of Music
National Association of Schools of Theatre
Network of Schools of Public Policy, Affairs and Administration
Review Commission on Education for the Physician Assistant

JMU Alumni

Office of Alumni Relations

Phone: (540) 568-6234

JMU Alumni Association Website (<https://www.alumni.jmu.edu>)

The JMU Alumni Association (JMUAA) believes in the motto "Dukes from Day One, Alumni for Life." Representing more than 170,000 Dukes, the JMUAA endeavors to connect and serve all Alumni and to be the primary resource for Alumni engagement with James Madison University. The JMUAA also works to champion the achievements of our Alumni, and to build a culture of sustained investment in JMU, its students, and its community through Alumni time, talent, treasure, and testimony. Together, JMU Alumni across the globe give back to the university in various ways with the desire to continue growing the reach, impact, and reputation of our alma mater. The JMUAA has developed its strategic priorities with the intent to continue building on our previous successes, as well as to find new ways to connect and better serve our ever-growing membership.

The JMUAA is a non-dues-based organization that operates in a unique interdependent model, led by a network of Alumni volunteers that work collaboratively with the university's Office of Alumni Relations (OAR). Together, we give back to the university in ways that further the reach and reputation of the JMU we know and love.

Find out more about the JMU Alumni Association and upcoming events at our website (<https://www.alumni.jmu.edu>).

JMU Foundation

The James Madison University Foundation, Inc. (<http://www.jmu.edu/foundation/>), a 501(c)(3) organization, was formed in 1969 to support the university — its students, faculty, staff, programs and facilities. The foundation receives, manages and stewards all charitable contributions for JMU. The foundation accepts gifts of cash, securities, real estate, personal property and more. The foundation's **mission** is to support and serve the James Madison University community. As we carry out that obligation, our **vision** is to provide an ever-growing and sustainable source of private support for the university community by promoting a prudent investment policy, stewardship and earning the trust of our constituents. As we seek to fulfill our ambitions, we are guided by an abiding commitment to the university, **valuing** trust, collaboration, innovation, accountability and excellence in all we do.

College of Arts and Letters

Traci A. Zimmerman, **Dean**

Melinda J. Adams, **Associate Dean**

Rebecca "Becky" Childs, **Associate Dean**

Siân E. White, **Associate Dean**

Shannon L. Wilson, **Director of Professional Development and Engagement**

Phone: (540) 568-6334

MSC: 2105

Location: Harrison Hall, Suite 1109

College of Arts and Letters Website (<http://www.jmu.edu/cal>)

Mission Statement

The College of Arts and Letters empowers students to become reflective lifelong learners who can address today's social, cultural, and political challenges through inclusive awareness, creative thinking, problem solving, communication, and responsive leadership.

Vision

To be a champion for liberal arts education, empowered locally and engaged globally.

Values

- **Stewardship:** The College of Arts and Letters is committed to the responsible, ethical, equitable, and transparent use of our time and our financial, scholastic, environmental, and human resources.
- **Collaboration:** The College of Arts and Letters embraces the power and benefits of working together to achieve results and recognizes that only through collaborative governance and shared decision-making among leadership, faculty, and staff will it realize its full potential.
- **Faculty and Staff Development:** The College of Arts and Letters invests in our employees' wellbeing and life-long learning. The College prioritizes professional development and cultivates work environments in which faculty and staff flourish both professionally and personally.
- **Equity:** The College of Arts and Letters commits to maintaining a fair and just environment where all are welcome, respected, and able to contribute to the academic experience.
- **Integrity:** The College of Arts and Letters holds its leadership, its faculty, and its staff to the highest standards of honesty, ethical behavior, and mutual respect and to full accountability in decision making.
- **Shared Governance:** The College of Arts and Letters upholds its commitment to shared governance through inclusive participation in planning and decision-making and meaningful accountability that fosters a climate of mutual trust.
- **Academic Excellence:** The College of Arts and Letters promotes high standards in teaching and research, upholds academic freedom and academic integrity for faculty and students, and challenges students to think and create in original and complex ways.

Academic Programs and Cross-Disciplinary Activities

Students may select from a broad spectrum of majors, minors, and graduate programs in the seven departments and three schools. In

addition, the college offers a wide array of cross-disciplinary minors, pre-professional programs, general education courses, annual events and supporting services, some of which reach out to the regional community. Information about these programs may be found on the "Majors and Programs" tab (<https://www.jmu.edu/cal/academics/index.shtml>) of the College of Arts and Letters website (<https://www.jmu.edu/cal/index.shtml>).

Academic Units

School of Communication Studies (p. 49)

Lori Britt, **Director**

Department of English (p. 58)

Sarah Hagelin, **Academic Unit Head**

Department of History (p. 68)

P. David Dillard, **Academic Unit Head**

Department of Justice Studies (p. 75)

Peggy Plass, **Academic Unit Head**

School of Media Arts and Design (p. 84)

Joe Hinshaw, **Director**

Department of Philosophy and Religion (p. 92)

Kenneth Pearce, **Academic Unit Head**

Department of Political Science (p. 95)

Kerry F. Crawford, **Academic Unit Head**

Department of Sociology and Anthropology (p. 100)

Abigail Wightman, **Academic Unit Head**

Department of World Languages and Cultures (p. 104)

Cynthia Chalupa, **Academic Unit Head**

School of Writing, Rhetoric and Technical Communication (p. 107)

Michael Smith, **Director**

Centers and Institutes

African, African American, and Diaspora (AAAD) Studies Center

Delores Phillips, Director

Phone: (540) 568-7772

Email: phillidp@jmu.edu

The African, African American, and Diaspora (AAAD) Studies Center provides interdisciplinary knowledges, programming and initiatives related to the peoples, cultures and institutions of Africa and the African Diaspora. Each year, the center hosts the Annual African, African American, and Diaspora Studies Interdisciplinary Conference, which welcomes scholars across the world.

The American Political Reform & Innovation Lab (APRILab)

Tim LaPira, Director

Phone: (540) 568-5309

Email: lapiratm@jmu.edu

The American Political Reform & Innovation Lab (APRILab) is a nonpartisan, civic- and community-engaged political science and data science research lab dedicated to institutional political reform.

Furious Flower Poetry Center

Lauren Alleyne, Executive Director

Phone: (540) 568-8883

Email: alleynlx@jmu.edu

The nation's first academic center for Black poetry, the Furious Flower Poetry Center is committed to ensuring the visibility, inclusion and critical consideration of Black poets in American letters, as well as in the whole range of educational curricula. Our programming seeks to cultivate an appreciation for poetry among students of all levels—from elementary to graduate school and beyond. In addition, Furious Flower seeks to support and promote Black poets at all stages of their careers and to preserve the history of Black poets for future generations.

Innovation Center for Youth Justice

Rita Poteyeva, Director

Phone: (540) 568-7124

Email: poteyemx@jmu.edu

The ICYJ is a partnership between James Madison University and the RFK National Resource Center for Juvenile Justice. Our shared mission and goals include national and international leadership for positive innovation in policy and practice in youth justice systems.

Institute for Constructive Advocacy and Dialogue

Lori Britt, Co-Director

Phone: (540) 568-5028

Email: brittll@jmu.edu

Rob Alexander, Co-Director

Phone: (540) 568-3771

Email: alexanrw@jmu.edu

ICAD designs and guides conversations that help communities think together productively and use dialogue and deliberation across differences to address the challenges and opportunities of civic life.

The Mahatma Gandhi Center for Global Nonviolence

Daniel Beers, Director

Phone: (540) 568-8180

Email: beersdj@jmu.edu

The Mahatma Gandhi Center for Global Nonviolence (p. 169) serves as a local and global hub to promote the understanding of justice through education, scholarship and engagement.

Center for Public Broadcasting/WMRA-WEMC

Matt Bingay, Director

Phone: (540) 568-6221

Email: wmra@jmu.edu

The Center for Public Broadcasting's mission is to inform, connect and engage communities through journalism, broadcasting and outreach. It serves over 50,000 listeners in the Shenandoah Valley, Charlottesville and Farmville areas.

The center comprises a network of non-commercial public radio stations, serving Harrisonburg at 90.7 FM, Charlottesville at 103.5 FM, Lexington at 89.9 FM and Winchester at 94.5 FM, and one standalone station, WEMC, serving Harrisonburg at 91.7 FM. The WMRA (<https://www.wmra.org/>) network stations are licensed to the James Madison University Board of Visitors. WEMC is licensed to Eastern Mennonite University but is operated entirely by the center. Much of WMRA's programming

is rebroadcast by WMLU 91.3 FM, the station owned by Longwood University in Farmville, Virginia. WMRA and WEMC are members of National Public Radio, and are affiliated with Public Radio International and American Public Media. The stations broadcast 24 hours per day year-round, offering extensive in-depth news coverage and classical, folk, blues and jazz music. The center also operates Valley Voice Radio Reading Service for the print-impaired. Listeners and local businesses contribute nearly three-quarters of the center's annual budget.

Annual Events

CAL Career Readiness Conference

Shannon Wilson, Director

Phone: (540) 568-8904

Email: wilso4sl@jmu.edu

The College of Arts and Letters Career Readiness Conference is a multi-day event open to CAL students and features JMU alumni from all CAL majors discussing the wide range of career opportunities available to graduates with a liberal arts degree. The program of alumni panels, career-readiness workshops and networking opportunities provides an opportunity to learn more about the transition from college to post-graduate life. The event cultivates and strengthens relationships among alumni, current students, and CAL faculty and staff.

Madison Writing Awards

Madison Writing Awards (MWA) is a university-wide biennial competition that celebrates the importance of writing and sophisticated rhetorical practice across all undergraduate academic programs. Winners are recognized at a spring awards reception, and the top submissions receive generous cash prizes. These awards reflect the commitment of the School of Writing, Rhetoric and Technical Communication and the College of Arts and Letters to promote cross-disciplinary dialogue and engagement as well as to acknowledge the power of writing in all its forms and contexts.

JMU Arts & Letters Undergraduate Research Conference

Melinda Adams

Phone: (540) 568-3377

Email: adams2mj@jmu.edu, CAL_UGRC@jmu.edu

The JMU Arts & Letters Undergraduate Research Conference features outstanding research and/or creative works by undergraduates working in any humanities, social science, or communication discipline. Held every spring, it attracts students from across the eastern United States and has become one of the largest humanities and social science undergraduate research conferences in the region.

College of Business

Michael E. Busing, **Dean**

Hui H. Sono, **Associate Dean, Academic Affairs**

Cheryl Yoder Mast, **Assistant Dean, Finance and Administration**

Phone: (540) 568-2785

MSC: 0207

Location: Hartman Hall, 3009-3020

College of Business Website (<https://www.jmu.edu/cob/index.shtml>)

Mission Statement

The JMU College of Business is a learning community committed to excellence in:

- Preparing students to be engaged, principled business professionals and leaders;
- Advancing scholarship in business disciplines; and
- Enhancing organizational performance through our outreach activities.

Vision Statement

To be regarded as a leader in preparing collaborative business partners engaged with ideas and the world.

Values

- **Integrity:** We are a community dedicated to honesty, mutual respect, ethical reasoning, and responsible behavior.
- **Intellectual Growth:** We value academic excellence achieved through the intellectual curiosity and growth of both faculty and students, and through the creation and maintenance of a challenging and rigorous learning environment that encourages critical thinking and life-long learning.
- **Community:** We value a supportive and inclusive culture that provides equal opportunity to succeed, where diverse ideas, backgrounds, and experiences are celebrated and given voice to strengthen our community, creating a sense of belonging.
- **Engagement:** We value an engaged, active learning environment inside and outside the classroom. We enrich the student experience through mutually beneficial internal and external relationships.
- **Innovation/Collaboration:** We value initiative, creativity, collaboration, and entrepreneurial spirit. We promote new ideas and solutions that advance intellectual growth and have a positive impact.

Overview

The College of Business offers baccalaureate degree programs leading to a Bachelor of Business Administration (B.B.A.), a Bachelor of Arts (B.A.), and a Bachelor of Science (B.S.). All degree programs offered by the College of Business are accredited by AACSB International – The Association to Advance Collegiate Schools of Business. The following academic majors are offered as a B.B.A.: accounting, computer information systems, economics, finance, international business, business management, and marketing. Students may also earn a B.A. degree or B.S. degree in economics. Quantitative finance is offered as a B.S. degree. The B.B.A. degree is comprised of essentially the same general structure, consisting of the following four components:

- General Education component
- B.B.A. core component

- Major component
- Electives

More information can be found on the B.B.A. Requirements (p. 16) tab.

Declaration of a Business Major or Minor

JMU students seeking to change/declare any major or minor in the College of Business (CoB) must be in "good standing," meaning they must have a cumulative grade point average of 2.0 or higher at the time when they submit a change/declaration of major/minor request. Students will follow the Office of the Registrar's procedure for making such requests. Students are expected to closely follow explicit departmental guidelines and deadlines for completing the process.

Declaring a major does not supersede the formal admission/progression standards of the College of Business for enrollment in COB 300 or junior-level courses in the CoB major programs. Declaring a major does not supersede admission requirements for specific academic programs in the CoB.

Minors are administered by their sponsoring department and may have additional requirements.

Business as a Second Major

A student enrolled in any JMU degree program other than the B.B.A. program who wishes to select a B.B.A. major field (i.e., accounting, computer information systems, economics, finance, international business, business management or marketing) as a second major must meet the following courses:

- Successful completion of all B.B.A. core courses (refer to the B.B.A. core component section); and
- Successful fulfillment of all requirements specific to the major selected.
- Note that since the B.B.A. is a different degree than most B.A. and B.S. degrees at JMU, a student essentially has a double degree rather than just a double major.

Transfer Credit Policy

Transfer of Lower-level Courses that are offered as Upper-level at JMU

The College of Business (CoB) normally does not award transfer credit for courses that were taken at the 100-200 level if those courses are offered at the 300-400 level at JMU.

Transfer of Courses Toward the Major

Each program in the CoB will accept no more than two courses of transfer credit toward the major except in extraordinary circumstances. The major is defined as the course work required by a major field of study. For a course to be considered for transfer credit in the student's major, it must have been completed in an AACSB-accredited business program, or a similarly accredited program, at a four-year university. Certain majors within the CoB may have more restrictive policies on transfer credit.

Limit on Business Course Transfers

In order to earn a B.B.A. degree from the JMU CoB, a student must complete at least fifty percent of the business credit hours required for the degree at JMU. Required business credit hours include the B.B.A. core and the courses required for the major. The following B.B.A. core courses

do not count toward meeting this requirement: COB 191, ECON 200, and ECON 201.

Academic Units

School of Accounting (p. 39)
John Lauck, **Director**

Department of Computer Information Systems and Business Analytics (p. 51)
Jeremy Ezell, **Academic Unit Head**

Department of Economics (p. 55)
Nevin Cavusoglu, **Academic Unit Head**

Department of Finance and Business Law (p. 60)
Elias J. Semaan, **Academic Unit Head**

Department of International Business (p. 74)
Irvine "Bud" Clarke, III, **Academic Unit Head**

Department of Management (p. 79)
Laura M. Leduc, **Interim Academic Unit Head**

Department of Marketing (p. 80)
Theresa B. Clarke, **Academic Unit Head**

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		
COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	
MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28
B.B.A. Upper-Level Core Component		
The following courses comprise the B.B.A. Upper-Level Core Component:		
COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3
COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.
² COB 241 must be taken before COB 242.

³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.
⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.
⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.
⁶ All four COB 300 courses must be taken during the same semester.
⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.
⁸ COB 487 must be taken during the senior year.

Major Component

The seven B.B.A. majors offered by the College of Business consist of 24-33 major credit hours. Students pursuing a B.B.A. degree may take major courses after progressing to formal admission (see below) and after they have enrolled/completed COB 300. The major program component of the B.B.A. degree is designed to be completed in three traditional semesters after COB 300 in person. Therefore, students should plan their course work with that timetable in mind.

Admission into the College of Business permits a student to enroll in COB 300 and to be formally admitted as a business B.B.A. major. However, some majors in the college may have standards and prerequisites that exceed those of the college as a whole. Hence, a student must meet both the College of Business requirements and the requirements of the major in which the student seeks to enroll. Refer to the appropriate sections of the catalog for specific degree requirements for the individual majors in the College of Business.

Recommended Course Sequence

A potential course sequence showing a four-year graduation track is shown below.

Course	Title	Credits
First Year		
COB 191	Business Analytics I ¹	3
COB 204	Computer Information Systems	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
MATH 205 or MATH 235	Applied Calculus [C3QR] or Calculus I [C3QR]	3
Credits		15
Second Year		
any courses listed above not completed in the first year, and:		
COB 202	Interpersonal Skills	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting	3
COB 291	Business Analytics II	3
Credits		12
Total Credits		27

¹ MATH 220 may be substituted for COB 191

Progression Standards for the College of Business ¹

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory

progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, are described below.

Students are formally admitted into the College of Business as a B.B.A. major and permitted to register for COB 300 when the following requirements are met:

- Submission of a completed application for admission into the College of Business is submitted to the COB Advising and Support Center according to the schedule shown below.
 - December 1 for the upcoming spring semester
 - February 15 for the upcoming summer session (if offered)
 - May 1 for the upcoming fall semester
- Successful completion with a GPA of 2.7 (or higher) in the following business core courses: COB 191, COB 202, COB 204, COB 241, COB 291, ECON 200, and ECON 201. Only grades earned at JMU will be used to calculate the B.B.A. core GPA.
- Successful completion of calculus and COB 242.

No core course needed for COB 300 admission may be attempted more than two times. An "attempted course" is one in which a final letter grade, including "W", "WP" and "WF", is recorded on the student's official JMU transcript. Furthermore, during or after a first attempt a student may face delayed registration permission for a second enrollment.

The university policy for calculating repeat-credit and repeat-forgive will be followed.

Students who fail to meet the requirements described above will be denied admission into the College of Business and will be referred to JMU University Exploratory Advising for assistance in choosing another major.

Students should consult with the major department for further progression requirements in the specific major.

¹ Requirements to be admitted into the College of Business are effective based on when a student intends to enroll in COB 300. Students are advised to consult the College of Business Advising and Support Center for current requirements.

College of Education

Mark L'Esperance, **Dean**

Oris Griffin, **Associate Dean for Student and Faculty Affairs**

Amy D. Thelk, **Associate Dean of Internal Supports and Accountability**

Katie Dredger, **Interim Associate Dean for Research, Scholarship and Curriculum**

Dara Hall, **Executive Director of Teacher Education and Student Success**

Phone: (540) 568-6572

MSC: 6907

Location: Memorial Hall, Suite 3175

College of Education Website (<https://www.jmu.edu/coe/index.shtml>)

Mission Statement

The mission of the James Madison University College of Education is to prepare educators and leaders to thrive in a complex global environment. The college engages in collaborative, transformative and meaningful work through evidence-based practice to empower all learners, foster continuous growth, and advocate for equal protection for all. The personal and professional development of students is accomplished by emphasizing excellence and continuous innovation in quality undergraduate, graduate and professional programs. The College of Education is distinguished through faculty and student achievements, academic rigor, excellence in teaching, student and faculty interactions and relationships, technological innovations and national recognitions. The college maintains relevance through active and growing interactions with other colleges within the university and with local, state, regional, national and international communities. The college is committed to providing:

- Undergraduate programs that support initial teacher licensure, as well as programs that provide in-depth specialty studies and opportunities for the development of professional knowledge and skills.
- Graduate programs that support initial teacher licensure, advanced preparation and studies that emphasize advanced knowledge in a specialty area and the development of effective leadership and professional skills for addressing the needs of a changing society.
- Continuing professional development and in-service programs in cooperation with public and private schools and agencies, other colleges, institutions and businesses.

The undergraduate and graduate educator preparation programs are accredited by the Council for the Accreditation of Educator Preparation and approved by the Virginia State Board of Education.

The basic philosophy of the college is reflected in these goals:

- To educate individuals for the multiple professions included in the college at both the undergraduate and graduate levels, not merely by transmitting skills and knowledge but by stimulating creativity, developing cognitive abilities and encouraging the testing of hypotheses and reinterpretation of the human experience.
- To encourage a balanced faculty orientation toward teaching, research, scholarship, community service and professionalism that recognizes individual strengths and preferences of the college's faculty.
- To create an environment that fosters an atmosphere of open communication among students, faculty members and community.
- To anticipate societal needs and provide necessary resources for implementing effective off-campus programs now and in the future.

The college offers undergraduate and graduate programs designed for initial teacher licensure, advanced programs for teachers, educational leadership, educational technology, adult education, human resource development, teacher leadership and military science.

Programs and Licensure

The College of Education offers undergraduate majors, minors and pre-professional education programs across a range of concentrations in both teaching and non-teaching areas. Majors in education lead to a Bachelor of Science that has been specially designed to meet a vital nationwide need for trained and committed teachers. To ensure that you gain the skills needed for this challenging profession, education majors may require additional content, such as a second major, a minor or an area of emphasis. Students must apply after meeting program admission requirements. Please see the College of Education website (<https://www.jmu.edu/coe/index.shtml>) for details. Your academic adviser will provide you with more information.

Advisement

Students who wish to pursue a teacher education program of study are required to declare the appropriate major and complimentary areas as listed below. Students should apply to the specific teacher licensure program their first semester at JMU or as soon as possible. Consult your adviser and the College of Education website (<https://www.jmu.edu/coe/index.shtml>) for details.

Licensure Area	Education Major	Additional Requirements
Art Education (PreK-12)	Studio Art, Design and Art History	N/A
Dance Education (PreK-12)	Dance	N/A
Elementary Education (PreK-6)	Elementary Education	N/A
Foreign Language Education (PreK-12)	Modern Foreign Languages - French, Spanish or German	N/A
Health and Physical Education (PreK-12)	Kinesiology	N/A
Inclusive Early Childhood Education (PreK-3 and Special Education birth-age 5)	Inclusive Early Childhood Education	N/A
Middle Education (6-8)	Middle Grades Education	One or two concentration area minors required: Language Arts, Mathematics, Science, Social Studies
Music Education (PreK-12)	Music (concentration in Instrumental or Vocal)	N/A

Secondary Education (6-12)	Secondary Education and Content Major (double major) or Content Major and Minor in Pre-professional Secondary Education	Content Major options: Biology, Chemistry, Earth Science, English, History, Mathematics, Physics or Political Science
Special Education (K-12)	Special Education (concentration in general and adapted)	N/A
Theatre Education (PreK-12)	Theatre	N/A

The College of Education offers the following undergraduate minors:

- Educational Media (p. 482)
- Military Leadership (p. 532)
- Teaching English to Speakers of Other Languages (p. 573)

The following endorsements are also available:

- Algebra I

Undergraduate students pursuing licensure to teach by completion of graduate M.A.T. programs should:

- Meet requirements indicated by the respective program prior to submitting an application to The Graduate School.
- Apply for admission to The Graduate School according to program deadlines.
- Complete all pre-professional studies requirements before enrolling in graduate courses in education.
- See the relevant program pages for the M.A.T. and the M.Ed. degrees and for teacher licensure in the identified areas.

Professional Education Unit

The mission of the James Madison University professional education unit is to prepare caring, knowledgeable, skilled and reflective educators who believe that all students can learn and succeed. Our candidates and faculty are committed to lifelong learning and aspire to meet educational needs in a changing, pluralistic and democratic society. The personal and professional development of candidates is accomplished by emphasizing excellence and continuous innovation in quality undergraduate, graduate and professional programs.

The Professional Education Unit is comprised of all programs across the university designed to lead to licensure or advanced study in education.

These following initial licensure programs are offered at the undergraduate level:

- Art Education
- Dance Education
- Early Childhood Education
- Elementary Education
- Foreign Language Education
- Inclusive Early Childhood Education
- Middle School Grades Education
- Music Education

- Secondary Education¹
- Special Education K-12: General and Adapted Curriculum
- Theatre Education

These initial licensure programs and concentrations are offered at the graduate level for those having baccalaureate degrees:

- Early Childhood Special Education
- Middle Grades Education
- Physical and Health Education
- Secondary Education¹
- Special Education K-12: Adapted Curriculum
- Special Education K-12: General Curriculum

¹ Secondary education licensure programs include biology, chemistry, earth science, physics, English, mathematics, history and social sciences.

Advanced programs and concentrations are offered at the graduate level for licensed teachers, other school personnel and those in the education sector:

- Adult Education and Human Resource Development
- Curriculum, Instruction, and Educational Equity
- Educational Leadership
- Leadership, Equity, Innovation & Learning
- Learning Design and Technology
- Literacy Education
- Spanish Language and Culture for Educators
- Teacher Leadership
- Teaching English to Speakers of Other Languages

Professional Education Coordinating Council

The Professional Education Coordinating Council (PECC) is the official university governing body responsible for the preparation of teachers and other school personnel. PECC membership includes coordinators or representatives of all initial licensure and advanced study programs in education, school representatives, the director of assessment and the directors of the Education Support Center and the Educational Technology and Media Center. The Dean of the College of Education serves ex officio as head of the Professional Education Unit and will designate a member of the Dean's Office to serve as the chair of PECC. The Professional Education Handbook outlines all PECC policies and is found on the Education Support Center homepage at <http://www.jmu.edu/coe/esc>.

Teacher Education Conceptual Framework

The JMU conceptual framework is a guiding set of principles, beliefs and concepts that provide a basis for designing, implementing, monitoring, assessing and changing programs that prepare teachers and other educators who work closely with children and others in school settings. We use the following standards of our profession as our guide:

- CAEP (Council for the Accreditation of Teacher Preparation) Standards
- InTASC (Interstate Teacher Assessment and Support Consortium) Standards

- Professional Association and Professional Accreditation Standards
- Virginia Performance Standards
- Educational Justice

Field Experiences

Field experiences (including practica and student teaching internships) are required for candidates in all programs of the professional education unit. The number and nature of these experiences may differ based on program structure and candidates' individual needs and/or goals. Transportation and other arrangements for the practicum and internship courses/experiences will be the candidate's responsibility.

Student Teaching

Student teaching is required as an integral part of the sequence of professional experiences in all teacher education programs.

Its purpose is to enable pre-service teachers to apply acquired skills, understandings and attitudes in PreK-12 classrooms or comprehensive child development programs. Each individual licensure program determines the length of its particular student teaching experience.

The Education Support Center coordinates the student teaching program with participating school divisions, assigning all candidates to their student teaching sites and facilitating the requirements of the experience. Experienced teachers serve as cooperating teachers who coach and mentor the student teachers in their classrooms. University supervisors have the primary responsibility for the supervision and evaluation of student teachers. Student teaching is graded on a credit/no-credit (undergraduates) or satisfactory/unsatisfactory (graduates) basis.

Candidates must student teach in the area for which they are seeking licensure or endorsement. A candidate enrolled in multiple teacher licensure programs or in programs with multiple endorsement areas must complete a student teaching experience in each area.

Student teaching placements are made in accredited Virginia public schools, but may be made within accredited Virginia private schools, programs and agencies with program approval.

Most placements are made within approximately one hour's driving distance from campus. Some programs also place students in divisions across Virginia, e.g. in northern Virginia, Richmond, Tidewater areas and Roanoke. Other local and non-local placement sites may be assigned in accordance with individual program and/or student needs. All placements are based on availability and efficiency of appropriate supervision. Student teaching is a full-time experience. Permission to take additional course work other than that required by the program will be made only in exceptional cases. Student teachers should not expect to work or participate in excessive extracurricular activities during student teaching. Students with problems and/or special needs must contact the director of the ESC.

Opportunities exist for qualified candidates to complete a portion of their student teaching at international locations. Those interested in pursuing this option must meet additional requirements and have permission of their programs. Refer to the ESC website (<https://www.jmu.edu/coe/esc/index.shtml>) for additional information.

Student Teaching Criteria

To be approved for student teaching, all candidates must:

- Complete all stated requirements for admission to teacher education.
- Attend a mandatory student teaching information session and apply for student teaching.
- Complete all course work, assessments, certificates, and stated requirements prior to student teaching.
- Be recommended for student teaching by their licensure program.
- Meet any additional admission and retention standards of their academic department or school.
- Submit paperwork to show that the student is free from exposure to communicable tuberculosis.

Professional behaviors are expected of all teacher candidates throughout the program. School divisions routinely require a background check prior to student teaching. Convictions for criminal offenses or charges pending may result in schools refusing to permit a placement for practicum or student teaching. There may also be serious consequences when being considered for a license to teach in Virginia and in other states. Education programs are aligned with Virginia's licensure requirements; therefore, we require that you disclose in a written email any pending misdemeanor, felony charges, or convictions within one week to either your advisor/area coordinator or Academic Unit Head, and/or the Director of the Education Support Center (ESC). The Director of the ESC will guide the disclosure process which is in place to work with the candidate on a case-by-case basis to discuss if and how this might impact the candidate's program progression, field experience, and pathway to licensure.

Application Deadlines

Student teaching applications are accepted during the fall semester for the next academic year. It is the student's responsibility to be aware of all application requirements and deadlines.

Teacher Education Licensure

Candidates should apply for a Virginia teaching license upon completion of a teacher education program; this license is not issued automatically. Virginia licensure requirements include the following assessments:

- Praxis Subject Assessments for most licensure areas.
- Reading for Virginia Educators (RVE) for selected programs.

In addition, all candidates shall provide evidence of certification of training in emergency first aid, CPR (including hands-on practice) and the use of AED, as well as Cultural Competency, Dyslexia and Restraint and Seclusion training.

Licensure Criteria

A statement indicating completion of an approved teacher education program will be entered on the candidate's transcript after:

- education requirements have been met, including a cumulative 2.5 GPA for undergraduates and a 3.0 for graduates;
- degree is awarded; and
- scores (sufficient for subject area) on all state mandated assessments are achieved.

Out-of-State Licensure

It is recommended that candidates applying for out-of-state licenses first obtain the Virginia license. Out-of-state licensure requirements and application forms must be obtained directly from the desired state agencies. Other states may have additional testing and GPA requirements

the applicant must meet. See the ESC website (<https://www.jmu.edu/coe/esc/index.shtml>) for more information.

Academic Units

Department of Early, Elementary and Reading Education (p. 54)
Joi Merritt, **Academic Unit Head**

Department of Educational Foundations and Exceptionalities (p. 56)
Dannette Murray, **Academic Unit Head**

Department of Learning, Technology and Leadership Education (p. 78)
Michele Estes, **Academic Unit Head**

Department of Middle, Secondary and Mathematics Education (p. 86)
Angela Webb, **Interim Academic Unit Head**

Department of Military Science (p. 87)
Andrew Bryant, **Academic Unit Head**

Education Support Center

Executive Director: Dara Hall

The Education Support Center (ESC) (<http://www.jmu.edu/coe/esc>) assists students with their pursuit of a teaching license:

- monitors admission to, and retention in, the professional education program;
- provides advising support;
- coordinates field experiences for all programs;
- approves applications for Virginia Licensure; and
- serves as the center for information about professional education programs.

Information and application materials for admission to teacher education, appeals, registration and preparation for entry assessments, student teaching and licensure are available on the ESC website. Also provided is information regarding costs associated with teacher education programs.

Educational Technology and Media Center

Phone: (540) 568-6302

The Educational Technology and Media Center (ETMC) (<http://www.jmu.edu/coe/etmc>) supports students, faculty, and staff in the effective and innovative use of educational technologies for teaching and learning. Through access, instruction, and guided exploration, the ETMC promotes meaningful integration of technology in educational contexts.

The ETMC houses James Madison University's children's and young adult literature collection, comprising more than 15,000 volumes that support coursework, instructional planning, and licensure preparation. The center also maintains a K–12 textbook collection and Praxis® testing study guides to assist students preparing for teacher licensure.

The facility provides access to a range of maker- and media-focused technologies, including 3D printing, laser cutting, and precision 2D cutting tools; laminating and button-making equipment; entry-level and programmable educational robotics; and a dedicated podcasting studio for audio recording and production. These resources support design thinking, computational thinking, creativity, and instructional design across disciplines.

The ETMC offers technology equipment for checkout, including laptops and iPads, to support academic work, field experiences, and instructional

practice. For students pursuing licensure in Virginia's schools, the ETMC provides opportunities to explore, evaluate, and apply instructional technologies aligned with contemporary classroom practices.

Admission to Teacher Education

Candidates who want to pursue a course of study leading to the initial Virginia teaching license must be admitted to the teacher education program. Admission is a prerequisite to most education courses; candidates not admitted to teacher education will be blocked from registering for those courses.

Admission Criteria

- Declare education program and submit application
- Two satisfactory professional references and a pre-professional self-assessment
- Grades of "C" or better in
 - WRTC 103 or equivalent (Writing)
 - ENG course or equivalent (Literature)
 - EDUC 175, PSYC 160 or equivalent (Human Development)
 - EDUC 200 or equivalent (Foundations of Education)
- Math Entry Assessment: a sufficient ALEKS math placement score or Praxis Core Mathematics Academic Skills for Educators Tests (or SAT or ACT exemption for the math assessment)
- Child Abuse Prevention (CAP) certificate
- Cultural Competency certificate
- Cumulative GPA of 2.5 or higher
- First Aid/CPR/AED certificate
- No record of felony conviction or misdemeanors involving children or drugs or founded complaint of child abuse or neglect
- Specific program requirements (Consult the program area coordinator for specific information related to any additional requirements for acceptance into their programs)

Deadline

First year students generally apply to Teacher Education by April 15th of their first year and complete all admission requirements by the following semester.

Transfer, post-baccalaureate and graduate candidates should apply during their first term of enrollment at JMU. Students must complete all requirements for admission into teacher education programs in order to enroll in required education courses. A full schedule of application and admission deadlines can be found on the ESC website. (<https://www.jmu.edu/coe/esc/index.shtml>)

College of Health and Behavioral Studies

Sharon E. Lovell, **Dean**

Kevin J. Apple, **Associate Dean**

Kirk Armstrong, **Interim Associate Dean**

Linda Plitt Donaldson, **Associate Dean**

Douglas R. Hochstetler, **Associate Dean**

Phone: (540) 568-2705

MSC: 4101

Location: Health and Behavioral Studies Building, Room 5051

College of Health and Behavioral Studies Website (<https://www.jmu.edu/chbs>)

Mission, Vision and Values

Mission

We engage students, faculty and communities in learning, scholarship and service in health and behavioral studies to inspire responsible contributions to our world.

Vision

CHBS will be a leader in collaborative education, scholarship and practice that drive real world advancements.

Values

We promote and support departments, programs and initiatives that advance a culture of excellence and uphold the following values:

- **Scholarship & Professional Development:** Participate actively in research and continual professional development. Research refers to systematic examination of phenomena using scientific methods and includes: Basic Research—leads to understanding of fundamental questions; Applied Research—relates basic research to find solutions to everyday problems; Translational Research—relates basic research to professional applications that enhance human health and well-being in a multi-disciplinary environment. Professional Development includes activities designed to enhance professional knowledge and skills.
- **Innovation:** Pursue creative approaches/trends that strive toward new and effective solutions, while respecting successful traditions and established best practices.
- **Diversity:** Appreciate that salient differences, including intersectionality, among and between constituencies create opportunities for all individuals to have a voice in CHBS endeavors.
- **Engaged Learning:** Employ teaching and learning strategies within various curricula to foster effective thinking, discernment, discovery and lifelong learning skills.
- **Integrity:** Act in a manner consistent with stated personal, professional, college and university values.
- **Collaboration:** Communicate and recognize commonalities, respect differences, and search cooperatively for possibilities to engage in interprofessional and interdisciplinary work.
- **Collegiality:** Respect that everyone has a role to play in achieving the CHBS vision. This shared purpose can be achieved while valuing each other's contributions.

- **Community:** In and beyond our college, impact-driven interactions feature trust, respect, mutuality, service and associations inclusive of multiple viewpoints.

Cross Disciplinary Programs, Outreach Programs, Partnerships, Academic Centers and Institutes

The College of Health and Behavioral Studies places a high value on partnerships with the community. These partnerships and our outreach programs are integral to our academic programs and assist us in meeting our responsibility to participate in efforts to enhance the well-being of our community. We value the impact of experiential activities on the enrichment of student learning. Many of the programs within the college are cross disciplinary in nature, reflecting our commitment and supporting the mission of the college. Further details about these cross disciplinary programs are provided in the minors and Academic Institutes and Centers sections of the catalog and on the CHBS website.

Academic Units

Department of Communication Sciences and Disorders (p. 47)

Ayasakanta Rout, **Academic Unit Head**

Department of Health Professions (p. 66)

Jamie Frye, **Interim Academic Unit Head**

Department of Health Sciences (p. 67)

Jeffrey Herrick, **Academic Unit Head**

Department of Kinesiology (p. 76)

Elizabeth Edwards, **Academic Unit Head**

School of Nursing (p. 90)

Melody Eaton, **Director**

Department of Psychology (p. 97)

Monica Reis-Bergan, **Interim Academic Unit Head**

Department of Graduate Psychology (p. 62)

Robin Anderson, **Academic Unit Head**

Department of Social Work (p. 98)

Simon Funge, **Academic Unit Head**

Majors and Minors

Students may select from a variety of majors, minors, programs and concentrations that are available through the departments in the College of Health and Behavioral Studies. Programs offered include the following:

Majors

- Communication Sciences and Disorders, B.A. (p. 223)
- Communication Sciences and Disorders, B.S. (p. 225)
- Dietetics, B.S. (p. 245)
- Health Sciences, B.S. (p. 287)
- Health Services Administration, B.S. (p. 291)
- Kinesiology, B.S. (p. 348)
- Nursing, B.S.N. (p. 388)
- Nursing, R.N. to B.S.N. (p. 391)
- Psychology, B.A. (p. 407)

- Psychology, B.S. (p. 410)
- Social Work, B.S.W. (p. 423)

Minors

- Chronic Illness Minor (p. 462)
- Coaching Education Minor (p. 468)
- Family Studies Minor (p. 495)
- Substance Use Disorder Education Minor (p. 572)
- Youth Justice Minor (p. 581)

Graduate Programs

Five of our eight departments provide graduate programs. JMU's graduate programs in the College of Health and Behavioral Studies offer the following degrees:

- Assessment and Measurement, Ph.D. (<https://catalog.jmu.edu/programs/assessment-measurement-phd/>)
- Athletic Training, M.S.A.T. (<https://catalog.jmu.edu/programs/athletic-training-msat/>)
- Audiology, Au.D. (<https://catalog.jmu.edu/programs/audiology-aud/>)
- Clinical and School Psychology, Psy.D. (<https://catalog.jmu.edu/programs/clinical-school-psychology-psyd/>)
- Communication Sciences and Disorders, Ph.D. (<https://catalog.jmu.edu/programs/counseling-supervision-phd/>)
- Counselor Education, College Student Personnel Administration Concentration, M.Ed. (<https://catalog.jmu.edu/programs/counselor-education-college-student-personnel-administration-concentration-med/>)
- Counseling and Supervision, Ph.D. (<https://catalog.jmu.edu/programs/counseling-supervision-phd/>)
- Education, PHETE (Physical & Health Education Teacher Education) Concentration, M.A.T. (<https://catalog.jmu.edu/programs/education-phete-physical-health-education-teacher-education-concentration-mat/>)
- Kinesiology, Exercise Physiology Concentration, M.S. (<https://catalog.jmu.edu/programs/kinesiology-exercise-physiology-concentration-ms/>)
- Nursing Practice, D.N.P. (<https://catalog.jmu.edu/programs/nursing-practice-dnp/>)
- Nursing, M.S.N. (<https://catalog.jmu.edu/programs/nursing-msn/>)
- Occupational Therapy, M.O.T. (<https://catalog.jmu.edu/programs/occupational-therapy-mot/>)
- Physician Assistant Studies, M.P.A.S. (<https://catalog.jmu.edu/programs/physician-assistant-studies-mpas/>)
- Psychological Sciences, Applied Research Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-applied-research-concentration-ma/>)
- Psychological Sciences, Behavior Analysis Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-behavior-analysis-concentration-ma/>)
- Psychological Sciences, Experimental Psychology Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-experimental-psychology-concentration-ma/>)
- Psychological Sciences, Quantitative Psychology Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-quantitative-psychology-concentration-ma/>)
- School Psychology and Counselor Education, Clinical Mental Health Counseling Concentration, M.A./Ed.S. (<https://catalog.jmu.edu/programs/school-psychology-counselor-education-clinical-mental-health-counseling-concentration-ma-eds/>)

programs/school-psychology-counselor-education-clinical-mental-health-counseling-concentration-ma-eds/)

- School Psychology and Counselor Education, School Counseling Concentration, M.Ed./Ed.S. (<https://catalog.jmu.edu/programs/school-psychology-counselor-education-school-counseling-concentration-med-eds/>)
- School Psychology and Counselor Education, School Psychology Concentration, M.A./Ed.S. (<https://catalog.jmu.edu/programs/school-psychology-counselor-education-school-psychology-ma-eds/>)
- Speech Language Pathology, M.S. (<https://catalog.jmu.edu/programs/speech-language-pathology-ms/>)

College of Integrated Science and Engineering

P. Grady Dixon, **Dean**

M. Afzal Upal, **Associate Dean**

Michael O. Lam, **Interim Associate Dean**

Phone: (540) 568-8841

MSC: 4116

Location: 701 Carrier Drive, Harrisonburg, VA 22807

College of Integrated Science and Engineering Website (<https://www.jmu.edu/cise/index.shtml>)

Mission

The College of Integrated Science and Engineering encompasses programs whose common focus is solving problems that impact the world at large through both the application of science and technology and the consideration of the social context in which the problem sits.

Over the past several decades, remarkable developments in science and technology have altered our lives and our society, bringing both great opportunities and challenges. Continued prosperity and societal harmony depend on the integration of scientific knowledge, the creative use of engineering design, technical capabilities in computing and other fields, the application of ethical principles and an understanding and appreciation of cultural commonalities and differences. Consequently, there is a need for individuals who understand the importance of specialization, as well as cross disciplinary connections, and also the integration of knowledge for practical application. These individuals must have the flexibility to be able to operate in an environment of uncertainty and complexity, the drive to seize such opportunities as they arise, and the vision and creativity to create new opportunities as needed. Our faculty is dedicated to producing graduates with a scientific and technical knowledge base and a matching set of interpersonal, organizational and technical skills. To this end, our faculty members not only educate our students, but they also inspire and serve as role models.

The college places importance on carrying out its role within the community of Academic Affairs, working collaboratively with other colleges and supporting division-wide programs and priorities.

Mission Statement

The College of Integrated Science and Engineering, by supporting its constituent academic programs, seeks to build a community of students, faculty and staff who share a common purpose of improving our world through the cultivation of integrated sciences and engineering.

Goals

The goals of the College of Integrated Science and Engineering are:

- To promote a student-centered focus in teaching, scholarship and service that encompasses excellence, collegiality and professionalism.
- To foster, among both faculty and students, life-long professional development, personal growth and commitment to ethical behavior.
- To contribute to the betterment of society at local, regional, national and global levels with a focus on sustainable solutions.
- To prepare students to enter professions of value to our community, through the study of applied science and engineering within a social context, or to undertake advanced study.

- To support a community of faculty that pursues high-quality, innovative and cross disciplinary instruction, scholarship, and service.
- To emphasize innovation.
- To encourage partnerships both within and outside the university.

Academic Units

Department of Computer Science (p. 53)

Vetria L. Byrd, **Academic Unit Head**

Department of Engineering (p. 57)

Scott Paulson, **Interim Academic Unit Head**

School of Integrated Sciences (p. 70)

Stephanie B. Stockwell, **Co-Director (Integrated Science and Technology)**

Henry Way, **Co-Director (Geography and Intelligence Analysis)**

Academic Programs

Students may select from a variety of majors, minors, programs and concentrations that are available through the departments in the College of Integrated Science and Engineering. Programs offered include the following:

Majors

- Biotechnology, B.S. (p. 211)
- Computer Science, B.S. (p. 242)
- Engineering, B.S. (p. 261)
- Geography, B.A. (p. 271) / Geography, B.S. (p. 275)
- Information Technology, B.S. (p. 319)
- Integrated Science and Technology, B.S. (p. 321)
- Intelligence Analysis, B.S. (p. 325)

Minors

- Computer Science Minor (p. 472)
- Environmental Humanities Minor (p. 485)
- Environmental Information Systems Minor (p. 487)
- Environmental Management Minor (p. 488)
- Environmental Science Minor (p. 490)
- Environmental Studies Minor (p. 492)
- Geography Minor (p. 500)
- Humanitarian Affairs Minor (p. 514)
- Integrated Science and Technology Minor (p. 516)
- Materials Science Minor (p. 524)
- Robotics Minor (p. 560)
- Science, Technology and Society Minor (p. 562)
- Urban and Regional Studies Minor (p. 575)

Graduate Programs

The College of Integrated Science and Engineering offers the following graduate degrees:

- Computer Science, Cybersecurity Concentration, M.S. (<https://catalog.jmu.edu/programs/computer-science-cybersecurity-concentration-ms/>)
- Integrated Science and Technology, M.S. (<https://catalog.jmu.edu/programs/integrated-science-technology-ms/>)

- Environmental Management and Sustainability concentration – joint degree program with the University of Malta

Cross Disciplinary Programs, Outreach Programs, Partnerships and Academic Centers

The College of Integrated Science and Engineering highly values partnerships with the community. These partnerships and our outreach programs are integral to our academic programs and assist us in fulfilling the responsibility we feel to serve our community. We value the impact of experiential activities on the enrichment of student learning. A listing of CISE centers, outreach programs and partnerships may be found on the CISE website (<https://www.jmu.edu/cise/>).

College of Science and Mathematics

Samantha C. Prins, **Dean**

Steven J. Whitmeyer, **Associate Dean for Research and Scholarship**

Mindy Capaldi, **Associate Dean for Faculty Support and Development**

Brycelyn M. Boardman, **Assistant Dean**

Rocky Parker, **Assistant Dean**

Cody Fouts, **Student Success Director**

Phone: (540) 568-3508

MSC: 4114

Location: EnGeo Building Suite 3122-3132, 801 Carrier Drive

College of Science and Mathematics Website (<http://www.jmu.edu/csm>)

Mission Statement

We are a diverse community of faculty, students, and staff. Our faculty are both teachers and scholars who are dedicated to advancing knowledge through excellence in teaching, inclusive high-impact learning experiences, and collaborative and original research. We foster critical thinking, student learning, and career readiness, while serving the broader community to promote a deeper understanding of the role and value of science and mathematics in society.

Vision Statement

To be a fully supported, inclusive and interdisciplinary scholarly community that drives discovery, prepares the next generation of professionals, and advances society.

Values

- **Integrity:** We are a community dedicated to honesty, mutual respect, and ethical reasoning.
- **Community:** We value a supportive and inclusive culture that allows every student, staff, and faculty member to accomplish science and mathematics together. Diverse perspectives, ideas, backgrounds, and experiences are celebrated and given voice to strengthen our sense of belonging and help us contribute meaningfully to our community.
- **Engagement:** We value an engaged, active learning environment in the classroom, lab, and field settings. We invest in outreach and communication as a means to enhance and contextualize the role of science and mathematics in society.
- **Exploration/Discovery:** Curiosity drives what we do—intellectually and scientifically. Our faculty conduct collaborative research that is impact driven and/or advances their disciplines. We actively mentor student research and provide all students access to state-of-the-art equipment and facilities.
- **Investment:** Our faculty and administrators support our activities with opportunities for professional development and mentoring. Faculty seek and gain external funding for our activities through grants and other resources.

Academic Units

Department of Biology (p. 43)

Kyle Seifert, **Academic Unit Head**

Department of Chemistry and Biochemistry (p. 45)

Linette M. Watkins, **Academic Unit Head**

Department of Geology and Environmental Science (p. 61)

Kristen St. John, **Interim Academic Unit Head**

Department of Mathematics and Statistics (p. 82)

Krishnan "Ravi" Shankar, **Academic Unit Head**

Department of Physics and Astronomy (p. 94)

Brett Taylor, **Academic Unit Head**

Programs

The college offers a variety of academic programs, majors, minors, concentrations, cross disciplinary programs and tracks. Most of these are listed below. For an explanation and contact point of each, visit the college website (<https://www.jmu.edu/csm/index.shtml>).

- American Chemical Society accredited degree (p. 217)
- Anatomy and Physiology
- Applied Physics (p. 399)
- Astronomy minor (p. 453)
- Biochemistry concentration (p. 217)
- Biochemistry and molecular biology minor (p. 455) for students not majoring in biotechnology
- Biology, B.A. (p. 198), B.S. (p. 204), M.S. (<https://catalog.jmu.edu/programs/biology-ms/>), and minor (p. 454)
- Biophysical chemistry major (p. 220)
- Biotechnology major (p. 211)
- Chemical education concentration (p. 217)
- Chemistry/business concentration (p. 217)
- Climate science minor (p. 466)
- Computational analytics minor (p. 470)
- Computational and applied mathematics
- Computational science concentration
- Data analytics minor (p. 477)
- Data science concentration
- Environmental earth science major (p. 285)
- Ecology and environmental biology concentration (p. 204)
- Environmental and engineering geology (p. 281)
- Environmental management minor (p. 488)
- Environmental science minor (p. 490)
- Environmental studies minor (p. 492)
- Forestry Biology Dual degree (B.S.+ M.F.) program (p. 204)
- Fundamental physics (p. 399)
- Geology major (p. 281) and minor (p. 501)
- Geophysics minor (p. 502)
- Materials chemistry concentration (p. 217)
- Materials science minor (p. 524)
- Mathematics, B.A. (p. 355), B.S. (p. 357), M.Ed. (<https://catalog.jmu.edu/programs/mathematics-med/>), and minor (p. 526)
- Microbiology concentration (p. 204)
- Neuroscience concentration (p. 204)
- Physics – individual option
- Physics, B.A. (p. 397), B.S. (p. 399), and minor (p. 541)
- Physics minor for engineers (p. 540)
- Pre-Dentistry (p. 546)
- Pre-Medicine (p. 547)
- Pre-Occupational Therapy (p. 549)
- Pre-Optometry (p. 550)

- Pre-Pharmacy (p. 551)
- Pre-Physical Therapy (p. 553)
- Pre-Physician Assistant (p. 554)
- Pre-Veterinary medicine (p. 557)
- Statistics, B.S. (p. 433)

Teaching Licensure Areas:

- Algebra I Add-on endorsement
- Earth Science Teaching Licensure
- Mathematics for Middle Grades Teachers Minor
- Secondary Biology
- Secondary Mathematics
- Secondary Chemistry
- Secondary Physics

Resource and Service Centers

The college also supports the following resource and service centers, collections, events and outreach programs that enhance teaching, scholarly activity and community relations.

Astronomy Park

Contact: Sean T. Scully

Email: scullyst@jmu.edu

Phone: (540) 568-4511

Website: (<https://www.jmu.edu/planetarium/star-parties.shtml>)

Located on the east side of campus near the Physics and Chemistry building is a permanent area for sky observing on campus. There are permanent mounts for two portable 11-inch and two 14-inch, computer-controlled telescopes. This site provides a convenient area for sky observing for introductory astronomy students. Students can easily see the moon, planets, nebulae, galaxies, star clusters as well as the sun using the appropriate solar filters. The department is also equipped with CCD cameras, spectrometers and multiple solar filters that provide more advanced students with experience in astrophotography and data collection techniques. The public is invited to attend public star gazes which are held several times each semester

Center for Computational Mathematics and Modeling

Contact: Roger Thelwell

Email: thelwerj@jmu.edu

Phone: (540) 568-5103

This cross-disciplinary institute for scientific computing houses state-of-the-art graphics workstations and a 16 PII node Beowulf computer system. The Beowulf computer system is a parallel computing environment that can be used on large-scale problems. Faculty and students will have access to this "supercomputer" from the center and from their offices. The center also operates an Immersive 360o Visualization System. The center uses mathematics both to simulate real-world phenomena and to generate visual data.

Faculty members from the sciences, economics and business disciplines interact with mathematicians to model problems that they are researching with undergraduate students.

Center for Engineering Geoscience and Geohazards (CEGGA)

Contact: Yonathan Admassu

Email: admassyx@jmu.edu

Phone: (540) 568-5016

The Center for Engineering Geoscience and Geohazards in Appalachia (CEGGA) is established to provide a collaborative platform to exchange ideas and best practices through its annual forum. CEGGA will also solicit research funding for geohazard related research affecting the Appalachian region.

Center for Materials Science

Contact: Chris Hughes

Email: hugheswc@jmu.edu

The educational mission of the Center for Materials Science (<https://www.jmu.edu/chemistry/instrumentation/materials-characterization.shtml>) is to develop and maintain an innovative interdisciplinary and multidisciplinary undergraduate program in materials science that will increase the maturation of students, their research experience and their employment opportunities. The mission includes the integration of undergraduate education with basic and applied research in materials science.

Faculty in the Center for Materials Science have expertise in a wide variety of areas including inorganic and organic synthesis, microfabrication, nanotechnology, thin film growth and surface modification, materials characterization, and modeling and simulation of complex systems. The facilities include a class 10000 clean room, electron beam lithography, and many types of microscopy and other analytical techniques.

Collaborative work is welcome and can include consultation with faculty, assignment of student projects or simply access to facilities.

Department of Chemistry & Biochemistry LC/MS Facility

Contact: Christine A. Hughey

Email: hugheyca@jmu.edu

Phone: (540) 568-6633

The JMU Mass Spectrometry Research & Teaching facility (<https://www.jmu.edu/chemistry/instrumentation/lcms.shtml>), housed within the Department of Chemistry and Biochemistry, was established in 2010 with two Major Research Instrumentation (MRI) grants from the National Science Foundation.

The mass spectrometry facility houses four LC/MS instruments, one DART TOF and one GC/MS. The LC/MS instruments include: an Agilent 6460 triple quadrupole (QQQ) mass spectrometer coupled to two Rapid Resolution LC pumps and a diode array detector, an Agilent 6224 time of flight (TOF) mass spectrometer coupled to an Infinity UHPLC pump, and Agilent 6530 and 6546 quadrupole time-of-flight (q-TOF) mass spectrometers each coupled to Infinity UHPLC pumps with DAD detectors. All four LC/MS instruments are equipped with an electrospray source. The time of flight instruments afford the high mass accuracy and high mass resolving power necessary for identification of unknowns in complex mixtures. The MS/MS capability of the q-TOFs afford additional structural information. The sensitivity of the QQQ makes this instrument ideal for targeted small molecule quantitation. The DART TOF is used for the direct analysis of analytes from solid or liquid samples. The quadrupole GC/MS is used for separation, characterization and quantitation of volatile compounds. Collectively, these instruments provide a robust platform for the qualitative and quantitative analysis of both large and small molecules.

Electron Microscopy Center

Contact: Xiaofeng "Harry" Hu

Email: huxx@jmu.edu

Phone: (540) 568-8115

The Electron Microscopy Center is a multi-user shared facility serving the entire university community for teaching and research. The center also offers demonstrations, trainings and imaging services to other schools, specialized educational programs and local industries. It houses two scanning electron microscopes, Zeiss Sigma 300 VP and LEO 1430 VP, and sample preparation equipment as metal sputter coater, carbon coater and critical point dryer.

JMU Aeroacoustics Lab

Contact: Caroline Lubert

Email: lubertcp@jmu.edu

Phone: (540) 568-2922

JMU's Aeroacoustics Lab (<https://www.jmu.edu/mathstat/undergrad-research/aeroacoustics-lab.shtml>) is a research lab centered around a purpose-built 2m x 5m x 5m Anechoic Chamber. This soundproof room is used to conduct simultaneous acoustical (4 B&K microphones at different angles to the test section) and optical (a Schlieren system of mirrors and lenses) measurements. The lab offers research experiences to many students in the field of aeroacoustics – jet noise modeling. The specific application of current interest within the lab is rocket launch noise. This project is carried out in conjunction with two Virginia partners, NASA Wallops Flight Facility (WFF) and Northrop Grumman Innovation Systems (NGIS).

JMU Center for Genome and Metagenome Studies

Contact: Karen Barnard-Kubow

Email: barnarkb@jmu.edu

Phone: (540) 568-6225

The mission of JMU Center for Genome and Metagenome Studies (<http://www.jmu.edu/genomics>) (CGEMS) is to support innovative research and training in the methods and principles of genomics, metagenomics and bioinformatics to provide an exemplary learning experience for undergraduate and graduate students. CGEMS houses Next Generation DNA sequencers (NGS) and the computing infrastructure for analysis and storage of the sequence data that they generate, including the Illumina MiniSeq short read and the Oxford Nanopore MinION long read sequencers. The center supports and fosters collaboration among researchers and students in a variety of departments and colleges. CGEMS also sponsors seminars and provides outreach to other higher education institutions in Virginia as well as area schools.

JMU Greenhouse

Contact: Heather P. Griscom

Email: griscohp@jmu.edu

Phone: (540) 568-5525

The biology department's greenhouse is organized into a head house section and several compartments that can be controlled individually for experimental studies. Each compartment has independent temperature and lighting controls and an automatic watering system. The greenhouse currently contains a collection of plant species used for teaching classes and conducting research, including a population of African violets from an endangered species native to Kenya.

JMU Herbarium

Contact: Conley K. McMullen

Email: mcmullck@jmu.edu

Phone: (540) 568-3805

The JMU Herbarium (<https://www.jmu.edu/biology/about/facilities.shtml>), located in Bioscience 1003, comprises 20 full-size and 14 half-size herbarium cabinets that house approximately 18,000 pressed and dried plant specimens. The herbarium is recognized in Index Herbariorum, an international listing of the world's significant herbaria and carries the acronym JMUH. JMUH presently houses a teaching and research collection of bryophytes, ferns and fern allies, gymnosperms and angiosperms, as well as a small collection of algae. These collections are predominantly from Virginia (particularly the Shenandoah Valley) and West Virginia, although specimens from Hawaii, Poland and the Galápagos Islands are also included. The earliest specimens date from 1825. Also present in the herbarium are a full-size drying oven, upright specimen freezer, three swing-arm dissecting microscopes, 20+ complete plant presses and collecting tools. In addition, JMUH houses a small library that contains a variety of Floras (including Flora of North America), botany textbooks, identification keys and journals. The latter include the American Fern Journal, Botanical Journal of the Linnean Society, Castanea, Rhodora and Systematic Botany).

JMU JEM Lab

Contact: Laura Taalman

Email: taalmala@jmu.edu

Phone: (540) 568-3355

Originally known as the JMU Math MakerLab, the JEM Lab (<https://www.jmu.edu/mathstat/jemlab/index.shtml>) offers opportunities for experimental and exploratory research in mathematics to students who are early in their undergraduate careers. Students who work in the JEM Lab leverage a wide variety of computational tools and resources to explore mathematics. JEM Lab students are usually first-year or second-year undergraduates, with no previous experience with any mathematical or design software packages, and that is how we like it! In 2021 the JEM Lab became a member of Geometry Labs United, a consortium of technology research labs at national and international research universities. The JEM Lab is proud to be the first and only Geometry Lab at a primarily undergraduate institution.

JMU Math & Stat: Learning Suite

Contact: Sanduni Palliyage

Email: palliysp@jmu.edu

Phone: (540) 568-7213

The JMU Math & Stat Learning Suite (<https://www.jmu.edu/mathstat/learningsuite.shtml>) is housed in the basement of Roop Hall, Room G25, and offers tutoring support in a variety of first year undergraduate mathematics and statistics courses. Students can get free tutoring help in Math 155, Math 205, Math 220, Math 231/232, Math 235 from department faculty. The Learning Suite offers a welcoming environment and tutoring of over 20 hours a week.

JMU Meteorite Collection

Contact: Sean Scully

Email: scullyst@jmu.edu

Phone: (540) 540-568-2659

The James Madison University Meteorite Collection (<https://www.jmu.edu/physics/meteorite-collection.shtml>) is a growing collection

of the many sorts of meteorites to strike the Earth, and is located in Miller Hall in the lobby of the John C. Wells Planetarium. The display is open to the public year-round during university business hours, and after hours by special arrangement.

JMU Regional Undergraduate Laser Facility

Contact: Oleksandr Kokhan

Email: kokhanox@jmu.edu

Phone: (540) 568-1656

The JMU Regional Undergraduate Laser Facility (<https://www.jmu.edu/chemistry/instrumentation/lasers.shtml>) supports research and education in advanced optical spectroscopy. The facility houses a large number of continuous-wave and pulsed lasers together with a rich collection of detectors, digitizers and other equipment for characterization and handling of laser beams.

John C. Wells Planetarium

Contact: Sean Scully

Email: scullyst@jmu.edu

Phone: (540) 568-2659

Located in Miller Hall, the planetarium (<https://www.jmu.edu/planetarium/index.shtml>) serves as a teaching laboratory for both the undergraduates and the local community alike. The facility is used as a resource for introductory astronomy classes and well as welcoming school groups from the region. Several public planetarium shows are offered every month that vary with the seasons. The planetarium is equipped with a GOT0- Chronos/Digistar-7 hybrid planetarium system that offers full dome video as well as exceptionally clear and accurate simulations of the night sky.

Madison Accelerator Laboratory (MAL)

Contact: Adriana Banu, Scientific Program Coordinator

Email: banula@jmu.edu

Phone: (540) 568-8940

Contact: Tilda Pendleton, Laboratory Manager

Email: pendlesj@jmu.edu

Phone: (540) 568-2650

The Madison Accelerator Laboratory (<https://sites.lib.jmu.edu/mal>) (MAL) is a multidisciplinary research user facility available for JMU faculty and students as well as for higher education institutions in Virginia and beyond. The facility features a converted medical electron linear accelerator (linac) with electron beam energies from 5-14 MeV and X-ray beam with endpoint energies from 6-15 MeV. The linac can be used for a variety of applications from basic nuclear physics and radiation effects testing to probing materials via Photon Activation Analysis, a non-destructive and versatile tool that provides high sensitivity for elemental chemical identification, particularly useful for irreplaceable or rare samples. In addition, MAL features a 140 kV X-ray imager with 3-D imaging capabilities allowing non-destructive internal imaging of materials, biological samples and archaeological/historical artifacts.

Microscopy Facility

Contact: Kristopher E. Kubow

Email: kubowke@jmu.edu

Phone: (540) 568-4521

The Biology Department's Microscopy Facility (<http://www.jmu.edu/microscopy>) is equipped with several light and fluorescence microscopes, including a Nikon C1 Confocal Laser Scanning Microscope, enabling time

lapse imaging, 3-D image reconstruction and fluorescence imaging. The facility has a dedicated staff member who can provide training on the equipment and help faculty and students with any microscopy aspects of their research projects.

Mineral Museum

Contact: Elizabeth Johnson and Eric Pyle

Email: johns2ea@jmu.edu; pyleej@jmu.edu

Phone: (540) 568-6130

Housed with the Department of Geology, the JMU Mineral Museum (<https://www.jmu.edu/mineralmuseum/>) contains more than 700 exceptionally beautiful display specimens that provide mineralogy students with outstanding visual examples of some of the finest crystals from around the world. Each year, numerous educational groups, mineralogical societies and individual collectors visit the collection.

Science and Mathematics Learning Center

Contact: Laura Miller, Director of Science and Mathematics Learning Center

Email: mille4lk@jmu.edu

Phone: (540) 568-6203

The College of Science and Mathematics has established a Learning Center for Science and Mathematics (<http://www.jmu.edu/smlc>). The center, which is a part of the JMU Student Success Center, provides extra help with math and science for students in general education and beginning science courses. The center is staffed by five full-time coordinators and carefully selected upper-level science and mathematics majors.

Shenandoah Valley Regional NMR Facility

Contact: Jun Yin

Email: yinjax@jmu.edu

Phone: (540) 568-3683

The Shenandoah Valley Regional NMR Facility (<https://www.jmu.edu/chemistry/instrumentation/nmr/index.shtml>) was established with grants from The National Science Foundation (9650132) and The Merck Foundation with matching funds provided by James Madison University, Eastern Mennonite University and Bridgewater College.

The NMR Facility is comprised of three NMR spectrometers: 300, 400 and 600. The 600 features Bruker's state of art AVANCE NEO Console. Both 300 and 400 are AVANCE III HD system, the 300 is equipped with a Sample Case that supports 24 sample for sequential or batch automation.

These instruments are housed at JMU and can be accessed remotely by the Regional NMR Consortium. The group is composed of chemists from Bridgewater College, Eastern Mennonite University, James Madison University, Mary Baldwin University and the University of Virginia.

Shenandoah Valley Regional Science Fair

Contact: Corey Cleland

Email: clelandc@jmu.edu

Phone: (434) 284-0543

The Shenandoah Valley Regional Science Fair has been administered by the JMU science faculty for over four decades. The science fair is a competition open to all students in grades 6-12 who live in Virginia's Shenandoah Valley.

Statistical Consulting Center

Contact: Hasan Hamdan

Email: statconsult@jmu.edu

Through this office, statistics faculty members and students provide JMU and the local community with assistance in the design and analysis of statistical surveys and experiments. Students obtain practical experience and an appreciation for the impact of statistical methods on today's society.

Wriggling Organisms and Modeling (WORM) Lab

Contact: Eva Strawbridge

Email: strawbem@jmu.edu

Phone: (540) 568-6938

The WORM Lab (<https://www.jmu.edu/mathstat/undergrad-research/worm-lab.shtml>) is an experimental space for research and undergraduate education at the interface of mathematics and biology. The primary focus of our research group is the numerical, analytical, and experimental study of the locomotion of the 1mm long worm *C. elegans*. The lab currently consists of a wet lab and a microscope dark room which houses an in-house built worm-tracking microscope as well as a Brookfield DV-III Rheometer. Research students in the WORM Lab conduct independent theoretical and experimental investigations into the question of the biomechanics of locomotion.

College of Visual and Performing Arts

Rubén Graciani, **Dean**
 Eric Guinivan, **Associate Dean**
 Karin Tollefson-Hall, **Associate Dean**

Phone: (540) 568-7131

MSC: 2107

Location: Shirley Hanson Roberts Center
 College of Visual and Performing Arts Website (<http://www.jmu.edu/arts/>)

Mission Statement

The College of Visual and Performing Arts empowers our communities to collaborate, envision and achieve their creative and professional aspirations. We explore the arts and design as a vehicle for cultural advancement, transformation and engagement.

Goals

The programs in the college are committed to achieving the following common objectives:

- To prepare students to be articulate, effective and inspiring performers, educators, creators, scholars and professionals in the arts.
- To attain recognition and leadership in the arts at the regional, national and global levels.
- To enhance, develop and sustain undergraduate and graduate programs of distinction.
- To support cultural, aesthetic and intellectual diversity, and to foster interdisciplinary exchange.
- To offer students instruction and learning experiences that incorporate the latest technology, research and practices.
- To engage the surrounding community as an active partner in promoting and experiencing the arts.

Academic Units

School of Art, Design and Art History (p. 41)
 MiKyoung Lee, **Director**

School of Music (p. 88)
 John Allemeier, **Director**

School of Theatre and Dance (p. 103)
 Ilana Morgan, **Director**

Forbes Center for the Performing Arts

Phone: (540) 568-7000
 Forbes Center Website (<https://www.jmu.edu/forbescenter/>)

Comprised of the Dorothy Thomasson Estes Center for Theatre and Dance and the Shirley Hanson Roberts Center for Music Performance, the Forbes Center houses five state-of-the-art performance venues: the Mainstage Theatre (450 seats), the Concert Hall (600 seats), the Recital Hall (196 seats), the Studio Theatre (200 seats) and the Earlynn J. Miller Dance Theatre (200 seats).

The center is home to the Masterpiece Season, a variety of cultural events for JMU and the entire community. The schools of Art, Design and Art History; Music; and Theatre and Dance all take an active role in this

series. The season also includes the Encore Series and Family Series featuring visiting artists such as Brittney Griffin, DanceWorks Chicago, Orpheus Chamber Orchestra, Chris Thile, Nekisha Durrett and Pablo Delano.

Galleries

Interim Director: Kevin Kwon
Phone: (540) 568-6407
 Galleries Website

ArtWorks Gallery

ArtWorks (<https://artworksgallery.wixsite.com/artworksgallery/>) is JMU's student-run gallery, exhibiting work by undergraduate and graduate students. The gallery features moveable walls; hardwood floors; large windows; and a high, open ceiling. Exhibits change, on average, every four weeks. The gallery is located a short walk from Duke Hall on the second floor of 131 Grace Street. The artWorks Gallery is a curricular component of the School of Art, Design and Art History.

Duke Hall Gallery of Fine Art

James Madison University's Duke Hall Gallery (<https://www.jmu.edu/dukehallgallery/index.shtml>) is a dynamic exhibition space that reframes the historical arc of art and design while inviting bold new directions in theory and experimentation. More than a professional gallery, it is a crossroads where students, scholars and creative practitioners from around the world come into conversation. A nexus for contemporary creativity and forward-looking change, the gallery champions diverse perspectives and expands opportunities for meaningful cultural exchange. As a teaching space, the gallery offers internships each semester and is a vital part of many School of Art, Design and Art History classes, including museum studies.

New Image Gallery

Director: Corrine Diop
Email: diopcj@jmu.edu
Phone: (540) 568-3318/6485

New Image Gallery (<https://www.jmu.edu/sadah/galleries/new-image/index.shtml>) is a university gallery featuring contemporary photography and new media by regional, national and international artists. New Image Gallery is located on the second floor of 131 Grace Street, JMU, Harrisonburg, VA. New Image Gallery is a curricular component of the School of Art, Design and Art History.

Office of Creative Propulsion

Director: Kathryn Nusa Logan
Email: logankn@jmu.edu
Phone: (540) 568-5656

The Office of Creative Propulsion (<https://www.jmu.edu/arts/ocp/index.shtml>) is an entity within JMU's College of Visual and Performing Arts. CVPA's ongoing work is animated by a belief that the arts - by themselves and as contributions to multidisciplinary efforts - generate personal and social meaning, inspiration and change. Our communities are ripe with possibilities for artists, designers, educators and scholars who can leverage their experiences and education to reimagine and disrupt traditional practices; stretch and break artistic boundaries; collaborate on multidisciplinary teams; synthesize data from the world around them to identify problems and opportunities; and act boldly, justly and inclusively to create their own unique path. Through facilitating integrative courses, multidisciplinary collaborations, community engagement and participatory arts experiences, OCP supports students

and faculty in ideating, launching, and refining arts-based efforts that promote more equitable and inclusive opportunities for meaningful arts engagement.

The Office of Creative Propulsion includes:

Madison Art Collection

Director: Virginia Soenksen

Email: soenksvm@jmu.edu

Phone: (540) 568-2661

The Madison Art Collection (MAC) (<http://www.jmu.edu/madisonart/>) is an encyclopedic collection of over 35,000 works of art and artifacts from a wide range of cultures. As the only art museum in the Shenandoah Valley, MAC provides unique opportunities for students, faculty and the surrounding community to access art of the world. The Madison Art Collection's public gallery space, showcases two thematic exhibitions a year featuring objects from the permanent collection and is open and free to all. Internships offer undergraduate students the opportunity to develop practical skills in a museum environment, including curation, collections management and exhibition design. Students and scholars interested in accessing objects in the Madison Art Collection should contact the director. MAC facilities are located in 131 W. Grace Street.

Honors College

Bethany Blackstone, **Dean**

Allison Fagan, **Associate Dean**

Félix Wang, **Associate Dean**

Phone: (540) 568-6953

Location: Hillcrest House

Honors College Website (<http://www.jmu.edu/honors>)

Mission

The Honors College seeks to develop a diverse community of inquisitive and aspirational learners from all parts of campus through distinctive and challenging educational experiences. Our core values include independent, student-centered learning; interdisciplinary training; collaboration; and active and informed service and leadership.

Programs

- Honors Interdisciplinary Studies Minor (p. 509)

School of Professional & Continuing Education

Nicholas Swartz, **Dean**
Sarah MacDonald, **Associate Dean**

Email: swartznj@jmu.edu

Phone: (540) 568-4253

Location: Ice House, 127 W. Bruce Street, MSC 6906

School of Professional & Continuing Education Website (<https://www.jmu.edu/pce/index.shtml>)

School of Professional & Continuing Education (SPCE), a unit within Academic Affairs, confers three degrees through the Adult Degree Program (B.A., B.S., B.I.S.) and partners with academic units across campus to facilitate a range of certificates, degrees and courses, youth programs, a lifelong learning institute for seniors, and professional development programs. SPCE's mission is to enrich learning across lifespans through transformative education and enrichment.

Senior Citizen Tuition Waiver

Legal residents of the State of Virginia who have reached 60 years of age before the beginning of an academic term and who have a taxable income that did not exceed \$23,850 for the year proceeding the term may register for and enroll in courses as full-time or part-time students and pay no tuition but will incur a \$20 application fee as well as fees established for the purpose of paying for course materials, such as laboratory fees. Senior citizens who have a taxable income higher than \$23,850 may choose to register to audit courses under the same policy, subject to instructor and department approval. Senior citizens shall be subject to the admissions requirements of the institution and a determination by the institution of its ability to offer the course or courses for which the senior citizen registers. A senior citizen shall only be admitted to a course in which enrollment is sought after all tuition-paying students have been accommodated. A senior citizen tuition waiver form must be submitted to Professional & Continuing Education to determine eligibility according to section 23-38:54 of the Code of Virginia.

Forms can be found on the SPCE website (<https://www.jmu.edu/pce/index.shtml>).

Non-degree Seeking Student Admission

The non-degree seeking student classification includes adult non-degree students, high school non-degree students and teacher licensure students. Individuals seeking enrollment as a non-degree seeking student must complete the "Non-degree Seeking Student Application." Virginia residents must also complete the "Checklist and Application for Virginia In-state Tuition Rates." A non-refundable \$20 application fee must accompany the application. Non-degree seeking students must submit the application and processing fee each year they wish to enroll in courses, and are discontinued at the end of each summer.

Non-degree seeking students can enroll and register for up to 11 hours of credit per semester. If additional credits are needed, students should contact SPCE. Course prerequisites may apply. Courses at the 500 or 600 level and above require approval by the appropriate department head. Non-degree seeking students can complete the non-degree student application and the in-state form by going to the SPCE website (<https://www.jmu.edu/pce/>) and clicking "Apply."

The courses taken in the non-degree seeking student category carry university credit, and they may be transferred into a degree program, once admitted, at the discretion of the program. JMU reserves the right to deny admission to non-degree seeking students; admission as a non-degree seeking student does not guarantee admission to a degree program at the university. If a student has been denied admission as an undergraduate student, they must wait one full year from the date of the denial to register as a non-degree seeking student.

Admission of Veterans

School of Professional & Continuing Education (SPCE) encourages veterans to apply for admission as full- or part-time students.

JMU does not determine eligibility for federal veterans' education benefits. Apply through the U.S. Department of Veterans Affairs (VA). The U.S. Department of Veterans Affairs website will guide students through the process, and the VA will notify JMU of veterans' eligibility. The Registrar's Office processes the claims and other paperwork required so students receive payments on time.

For information regarding veterans' education benefits contact:

Veterans Coordinator
Office of the Registrar, MSC 3528
James Madison University
Harrisonburg, VA 22807
(540) 568-4769
www.jmu.edu/registrar/veterans (<http://www.jmu.edu/registrar/veterans>)

Adult Degree Program

Email: adp@jmu.edu

Phone: (540) 568- 4253

Location: Ice House, 127 W. Bruce Street, Room 332

Adult Degree Program Website (<https://www.jmu.edu/adultdegree>)

Mission

The Adult Degree Program delivers a flexible, dynamic and rigorous degree completion curriculum to meet the individual education and career goals of our students. ADP's non-residential program offers a framework for students with significant transfer credit and/or prior learning experience to design their own academic plan.

Admission

To be eligible for the ADP, applicants must be at least 22 years of age, have at least 30 transferable college credits and have a minimum cumulative GPA of 2.0 or higher for all prior coursework.

With very limited exceptions, students in the Individualized Study major (the ADP's academic major) cannot transfer to another major at JMU.

Experiential Learning

The Adult Degree Program offers several pathways for students to participate in experiential learning opportunities. Students can choose to participate in workforce related experiences, intensive research or international programs.

Degrees Offered

Students in the ADP have the opportunity to earn a Bachelor of Individualized Study (B.I.S.) in individualized study (p. 304), a Bachelor of Arts (B.A.) in individualized study (p. 302) or a Bachelor of

Science (B.S.) in individualized study (p. 313). The university degree requirements differ by degree type.

The Graduate School

Linda Thomas, **Dean**

Drew Danzell, **Interim Associate Dean**

Laura Ryman, **Assistant Dean**

Phone: (540) 568-6131

The Graduate School Website (<http://www.jmu.edu/grad>)

Holland Yates Hall, Suite 4050

100 E. Grace Street

MSC 6702

Harrisonburg, VA 22807

Overview

The Graduate School coordinates graduate and post-graduate education throughout the university. There have been over 22,779 graduate degrees awarded through 2022, and enrollment growth and ongoing development of nationally and internationally recognized graduate programs are key strategic initiatives of the university.

Mission

Graduate Education at James Madison University serves the public good by providing graduate students with high-quality academic experiences and professional development through its diverse, collaborative, and engaged community.

History

Founded in 1908 and located in the center of Virginia's Shenandoah Valley, James Madison University is a public comprehensive university. The university offers degree programs at the bachelor's, master's, educational specialist and doctoral levels. The total enrollment exceeds 22,000 with over 1,900 graduate students. JMU annually receives more than 2,600 graduate applications.

The campus consists of more than 720 acres and features nearly 150 major buildings. In the more than 100 years since its founding, JMU has grown from a state normal and industrial school for women to today's comprehensive university. In 1938 the institution was named Madison College in honor of the fourth president of the United States and in 1977, the name was changed to James Madison University. The JMU Graduate School was established in 1954, when the State Board of Education authorized the university to offer programs leading to the Master of Science in Education degree.

Currently, seven academic colleges offer more than 70 graduate programs. JMU has the best graduation rate in the U.S. for large, master's level public universities, the best college for employment in Virginia and the most recommended public university in the United States.

Authorization

James Madison University is authorized to offer graduate programs leading to master's, Educational Specialist, Doctor of Audiology, Doctor of Education, Doctor of Philosophy, Doctor of Psychology, Doctor of Musical Arts and Doctor of Nursing Practice degrees.

Many graduate programs also offer concentration areas.

JMU Libraries

Phone: (540) 568-6150

JMU Libraries Website (<http://www.lib.jmu.edu>)

Monica Rogers, B. Grover N. Saunders, Ari E. Short, Greg C. Spitzer, James R. West

At JMU, the libraries are here to support the academic and creative success of all students. We offer comfortable and safe places to learn, explore and collaborate in four locations on campus.

Carrier Library, near the Quad, features 195,000 square feet of newly renovated space. Its features include study spaces with stunning views, a media production studio, and a student kitchenette. Carrier Library also houses the Book Lab (a space for preservation, conservation, and book arts) and Distinctive Collections, home to the Libraries' rare and unique materials.

Rose Library, on East campus, provides five floors of bright, welcoming spaces for both individual and collaborative study.

Both **Carrier** and **Rose** Library offer a makerspace, café, computer lab, printers and a secure 24-hour study area.

The Music Library, located in the basement of the Music Building, primarily serves the School of Music, but its specialized resources, computer lab, printers and study spaces are open to all students.

The Educational Technology & Media Center (ETMC) in Memorial Hall primarily serves the College of Education, but its study spaces and printers are available to everyone.

We provide access to millions of resources—books, journals, music recordings, videos and other resources. We also offer a liaison librarian for each major at JMU who can help students navigate these resources to find the best sources for their topic. Students can receive support from the Libraries' knowledgeable staff online, at a service desk, or by appointment.

Faculty

Professors

Andrea H. Adams, Jamie M. Calcagno-Roach, Katherine "Kathy" E. Clarke, Brian C. Flota, Rebecca B. French, Kelly D. Giles, Steven W. Holloway, Jennifer "Jenne" M. Klotz, Juhong Christie Liu, Elizabeth L. Price, Lara E. Sapp, Carolyn F. Schubert, Yasmeen L. Shorish, Kristen Shuyler, Brian A. Sullivan, Galen Talis, Liz Thompson, Stefanie E. Warlick, Malia E. Willey

Associate Professors

Howard S. Carrier, Elizabeth "Liz" T. Chenevey, Elaine R. Kaye, Mark G. Lane, Jessica L. Lantz, Kate N. Morris, David S. Vess, R. Nicole Wilson, Nathaniel Taeho Yu

Assistant Professors

Liana C. Bayne-Lin, J. Caitlin Birch, Samantha Harmon, Valerie L. Linsinbigler, Caitlin MacKenzie Mannion, Kathryn Shenk, Zhenhuan "Henry" Yang

Instructor

Abiodun Stephen Ijeluola

Administrative Faculty

David M. Black, Gregory "Greg" H. Brown, Layne Carpenter, Victoria "Tori" Groene-Zeigler, Eric J. Hansen, Charles "Bill" W. Hartman, Kevin J. Hegg,

University Studies

Fletcher Linder, **Dean**

Email: lindergf@jmu.edu

Phone: (540) 568-5270

Audrey Burnett, **Associate Dean**

Email: burnetaj@jmu.edu

Phone: (540) 568-8950

Scott Paulson, **Associate Dean**

Email: paulsosa@jmu.edu

Phone: (540) 568-4567

University Studies (<https://www.jmu.edu/universitystudies/index.shtml>) is the Dean's office for Exploratory students, Interdisciplinary Liberal Studies (IdLS), Independent Scholars, and General Studies. It is also the academic unit supporting the Center for Assessment and Research Studies, Ethical Reasoning in Action, the Assessment and Testing Center, Online Strategy and Compliance, and some elective courses. University Studies is located in Gabbin Hall.

School of Accounting

Director: John R. Lauck

Location: Zane Showker Hall

School of Accounting Website (<http://www.jmu.edu/cob/accounting>)

Mission Statement

The JMU School of Accounting is a learning community that is assertively inclusive and committed to preparing students to serve the public interest as principled and engaged professionals who are able to apply accounting and business knowledge for the betterment of individuals, organizations and society.

The school also offers a Master of Science in accounting. Its mission is to prepare students for success in the accounting profession by strengthening the students' technical expertise, enhancing their understanding of professional responsibility and improving their business skills necessary to compete in today's complex and ever-changing business environment.

Goals

The accounting faculty has identified strategic and tactical goals in three major areas: learning objectives for students, intellectual contributions for faculty and service to our profession and the community.

Learning Environment Objectives

We engage in the scholarship of teaching to prepare students for professional accounting careers in public accounting, industry or the public sector. We provide a learning environment that encourages a commitment to lifelong learning and develops a diverse set of skills in students, including technical competence, information technology proficiency, critical thinking, teamwork and communication.

Intellectual Contributions Objectives

We engage in the scholarships of discovery, application of knowledge and instructional development to advance knowledge in the field of accounting, to improve business practice, to encourage and support innovative teaching methods and curricula and to foster the intellectual and professional growth of our faculty.

Service Objectives

We serve our school, college, university, accounting profession and business, and academic communities through active participation and leadership in academic, professional and business organizations.

Career Opportunities

The programs of study offered by the accounting program provide an educational experience intended to prepare students for a variety of careers in the accounting, financial, auditing, consulting and information systems fields. Some of the job titles held by graduates of our program include the following:

- Audit Partner
- Business Analyst
- Business Consultant
- Chief Financial Officer
- Comptroller
- Computer Systems Consultant
- Controller

- Cost Accountant
- Cost Analyst
- Division Controller
- Forensic Accountant
- Fraud Examiner
- Information Systems Coordinator
- Internal Auditor
- Systems Consultant
- Tax Adviser
- Tax Partner

Our graduates find employment with all of the "Big Four" international public accounting firms, dozens of national, regional and local public accounting firms, government agencies and major international companies. Many of our graduates work for business consulting firms. The field of accounting has one of the highest demands for new graduates of any area in today's market.

Co-curricular Activities and Organizations

The Financial Professionals Club (FPC) is an organization open to all aspiring or current JMU business majors. Through professional development, community service, and social events, the FPC provides opportunities for students to

1. learn about careers in the financial information professions,
2. interact with prospective employers, and
3. serve through leadership positions. The FPC hosts many joint events with the Beta Alpha Psi honor society.

Beta Alpha Psi is the honors organization for accounting, finance, quantitative finance and computer information systems majors.

Programs of Study

The School of Accounting offers both an undergraduate and graduate program. The undergraduate degree consists of the four-year major in accounting, leading to the Bachelor of Business Administration (B.B.A.) degree. During their junior year, students may choose to meet with their advisor to discuss applying for the Master of Science in Accounting program.

Students who choose to complete the four-year undergraduate program will have the requisite accounting, business and general education to pursue a variety of career opportunities in accounting and business including assurance, consulting, corporate accounting/finance, financial analysis/venture capital, financial planning, non-for-profit/government and tax consulting services.

Students are advised that 150 hours of college education are required to be licensed as a Certified Public Accountant in most states. The Accounting, M.S. is therefore recommended for those electing a career in public accounting. A student must be admitted to the Accounting, M.S. program in order to enroll in graduate courses. Admission to the program is competitive. Admission requirements are included in the graduate catalog. Students may apply following their junior year and early application for admission is encouraged.

Faculty

Professors

Luis Betancourt, Paul A. Copley, Alexander L. Gabbin, David C. Hayes,
John R. Lauck, Irana J. Scott

Associate Professors

John W. Briggs, Andrew P. Duxbury, James H. Irving, Rebekah D. Moore,
Matthew R. Reidenbach, Robert C. Richardson, Nicole Wright

Assistant Professor

Chaoping Li

Senior Lecturers

Espy "Ted" Browning, Heidi Derstine, Eileen M. Shifflett

Lecturers

Leah Kratz, Melissa Mayhew, Marilla Melcher

Programs and Courses

Major

- Accounting, B.B.A. (p. 171)

Graduate Program

- Accounting, M.S. (<https://catalog.jmu.edu/programs/accounting-ms/>)

Courses

- Accounting (ACTG) (<https://catalog.jmu.edu/courses/actg/>)

School of Art, Design and Art History

Director: MiKyoung Lee

Email: lee28mx@jmu.edu

Phone: (540) 568-6216/6661

Location: Duke Hall, Room 1011

School of Art, Design and Art History Website (<https://www.jmu.edu/sadah/index.shtml>)

Mission Statement

We collaborate as creators and scholars to advance diversity and independent thought through rigorous practice and research.

Goals

The School of Art, Design and Art History discerns the following five distinct teaching goals:

- To prepare future professionals with a global perspective.
- To develop and license future leaders in the field of education.
- To enrich the general education of the JMU community.
- To offer professional instruction for students who wish to enrich their education with a second major or minor.
- To present outstanding contemporary exhibition programs.

To meet these goals, the School of Art, Design and Art History offers three degrees, a choice of five majors and select areas of emphasis. PK-12 Licensure is available to every School of Art, Design and Art History major, regardless of their degree track. Minors are offered in select areas. All degrees require a minimum of 120 credit hours. A full description of degree requirements is listed under degree requirements.

Career Opportunities and Marketable Skills

Art students at JMU have the opportunity to prepare for a wide variety of art and art-related career fields. Because the School of Art, Design and Art History educates art students as critical thinkers and creative problem solvers, graduates have an enhanced ability to think independently, respond flexibly, work productively and compete successfully for employment in career fields that value original and thoughtful creativity. Some possible careers include:

- Advertising Designer
- Architectural Designer
- Artist
- Art Critic
- Art Educator
- Art Historian
- Ceramicist
- Conservator
- Gallery Owner
- Graphic Designer
- Illustrator
- Industrial Designer
- Jeweler
- Multimedia Designer
- Museum Curator/Educator
- Painter

- Photographer
- Printmaker
- Sculptor
- Textile Designer

To discuss specific career options, students should contact the school and make an appointment with a member of the faculty.

Accreditation

Supplementing JMU's general accreditation, all degrees offered by the School of Art, Design and Art History are professionally accredited by the National Association of Schools of Art and Design. In addition, the Art Education program is also accredited by the National Council for Accreditation of Teacher Education (now CAEP).

SADAH Hardware and Software Requirements

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department's website (<https://www.jmu.edu/sadah/index.shtml>).

Faculty

Professors

Aderonke A. Adesanya, Sarah T. Brooks, Corinne J. Diop, David H. Ehrenpreis, Jori Erdman, David Hardy, Richard D. Hilliard, Beth Hinderliter, Laura R. Katzman, MiKyoung Lee, Dawn M. McCusker, Robert J. Mertens, John W. Ott, Mark D. Rooker, Rebecca Silberman, Greg Stewart, William L. Tate II, Karin L. Tollefson-Hall, Lisa M. Tubach, William H. K. Wightman, Susan D. Zurbrigg

Associate Professors

Dymphna G. DeWild, Kevin B. Phaup, Hannah K. Sions, Allyson Mellberg Taylor

Assistant Professors

Zenab Bastawala, Nicholas Brinen, Dylan Krueger, Nieko McDaniel, Jonathan McMillan, Feixue Mei, Joon Park, Anna Marie Smith, Brett Taylor, Nicholas E. Wickersham, Traci Wile, Sangjun Yoo

Programs and Courses

Majors

- Art, B.A. (p. 184)
- Art, B.F.A. (p. 187)
- Art, Architectural Design Major, B.F.A. (p. 182)
- Art, Graphic Design Major, B.F.A. (p. 192)
- Art, B.S. (p. 190)
- Art, Industrial Design Major, B.S. (p. 195)
- Art History, B.A. (p. 179)

Minors

- Art Minor (p. 450)
- Art History Minor (p. 451)
- Illustration Minor (p. 515)

Cross-Disciplinary Minor

- Book Arts Minor (p. 456)

Licensure Programs

- Art Education Professional PK-12 Teaching Licensure (p. 582)

Graduate Programs

- Art, M.F.A. (<https://catalog.jmu.edu/programs/art-mfa/>)

Courses

- Architectural Design (ARCD) (p. 610)
- Art (ART) (<https://catalog.jmu.edu/courses/art/>)
- Art and Design Foundations (FDS) (p. 617)
- Art Education (ARED) (<https://catalog.jmu.edu/courses/ared/>)
- Art History (ARTH) (<https://catalog.jmu.edu/courses/arth/>)
- Arts Entrepreneurship (ARTE) (p. 621)

Department of Biology

Academic Unit Head: Kyle Seifert

Email: seiferkx@jmu.edu

Phone: (540) 568-6733

Location: Bioscience Building, Room 2001C

Department of Biology Website (<http://www.jmu.edu/biology>)

Associate Academic Unit Head: Heather Griscom

Email: griscoh@jmu.edu

Location: Bioscience Building, Room 1028H

Assistant Academic Unit Head and Graduate Program Director: Janet Daniel

Email: danie2jc@jmu.edu

Location: Bioscience Building, Room 2028C

Mission Statement

The Department of Biology holds as its primary core value a commitment to providing superlative teaching for students. To accomplish this mission, we will create an environment for learning that will include opportunities for undergraduate research, a broadly-based academic program, a supportive, diverse and collaborative faculty, an understanding of the process of science and a recognition of the importance of community outreach and involvement.

Career Opportunities and Advanced Fields of Study

- Anthropology
- Aquatic Science
- Biodiversity
- Bioinformatics
- Biotechnology
- Botany
- Clinical Laboratory Sciences
- Dentistry
- Ecology and Environmental Science
- Epidemiology
- Forensic Science
- Forestry
- Genetic Counseling
- Graduate School in the Biological Sciences
- Immunology
- Landscape Architecture
- Medicine
- Microbiology
- Microscopy
- Neurobiology
- Nursing
- Occupational Therapy
- Optometry
- Pharmacology
- Physical Therapy
- Physician Assistant
- Physiology

- Research Assistant
- Scientific Writing
- Secondary Education
- U.S. Fish and Wildlife Service
- Veterinary Medicine
- Virology
- Zoology

Students interested in pursuing any of these career opportunities should contact the biology office. An appropriate advisor will be assigned for mentoring and course selection.

Co-curricular Activities and Organizations

Biology majors participate in activities such as:

- Weekly departmental seminars
- Tri-Beta, a national biology society
- Pre-professional health clubs and honor society
- Summer and academic-year research opportunities
- Summer courses at biological field stations
- Internships with various organizations
- Aiding in teaching as student assistants
- Presenting papers at meetings
- Volunteering at Sentara RMH and with the rescue squad

Departmental Research

Currently, the biology department has research strengths in the following areas:

- Anatomy
- Cell Biology and Genetics
- Comparative and Functional Morphology
- Developmental Biology
- Ecology, Evolution, Behavior and Systematics
- Genomics
- Microbiology
- Neurobiology
- Physiology
- Plant Biology

More information regarding faculty research can be found on the Department of Biology website (<https://www.jmu.edu/biology/research/index.shtml>).

Faculty

Professors

Corey L. Cleland, Steven G. Cresawn, Raymond A. Enke, Mark L. Gabriele, Heather P. Griscom, Christine L. May, Conley K. McMullen, Kyle Seifert, Morgan M. Steffen, Bruce A. Wiggins, Roshna E. Wunderlich

Associate Professors

Sharon K. Babcock, Timothy A. Bloss, Justin W. Brown, Janet Chen Daniel, Katrina E. Gobetz, Susan R. Halsell, Joseph A. Harsh, Kristopher E. Kubow, Pradeep V. Menon, Dana L. Moseley, Michael R. Parker, Terrie K. Rife, Kimberly H. Slekar, Bisi T. Velayudhan, George S. Vidal, Marquis Walker, Louie L. Wurch

Assistant Professors

Karen B. Barnard-Kubow, Andrea Berardi, Leone M. Brown, Bryan A. Cage, Jaira Ferreira de Vasconcellos, David Hembry, Tolulope Kayode, Rosario Marroquin-Flores, Geisianny Moreira, Emily Naylor, Kelsey Reider, Lauren Sarringhaus, Laura Tipton, Kristin Yoshimura, Ning Zhang

Senior Lecturers

Elizabeth A. Doyle, Oliver Hyman, Thomas Hynd, Jessica L. Marquardt, Caylin Murray

Lecturers

Kimberly J. Bolyard, Mary-Susan Burnett, Tracy Deem, Mark Ellet Jr., Patrick J. Kilkenny, Frank May

Instructors

William Flint, Joey Powers, Julia Stutzman, Robert Walters

Staff

Luther Brefo, Christal Curry, Virginia Fulginiti, Lisa Miller

Annual Events

New Graduate Student Orientation

Contact: Janet C. Daniel

Email: daniel2jc@jmu.edu

Phone: (540) 568-2322

Every August, the department welcomes incoming first year graduate students with a one-day orientation to the program, faculty, fellow students and departmental procedures. Students will receive guidance in the transition into graduate student life, expectations and responsibilities.

Biology Department Seminar Series

Contact: Lee Brown

Email: brown8lm@jmu.edu

Phone: (540) 568-6098

Contact: David Hembry

Email: hembrydh@jmu.edu

The departmental seminar series features speakers addressing a variety of research methodologies and programs. They are given by JMU scientists, visiting scientists and graduate students. Every March, first year graduate students give a presentation of their thesis proposals to the department. Seminars are held in the Biosciences Building Room 1007 from 12:40-1:20pm every Friday during the academic year.

Programs and Courses

Majors

- Biology, B.A. (p. 198)
- Biology, B.S. (p. 204)

Cross-Disciplinary Major

- Biotechnology, B.S. (p. 211)

Minor

- Biology Minor (p. 454)

Cross-Disciplinary Minor

- Biochemistry and Molecular Biology Minor (p. 455)

Graduate Program

- Biology, M.S. (<https://catalog.jmu.edu/programs/biology-ms/>)

Courses

- Biology (BIO) (<https://catalog.jmu.edu/courses/bio/>)
- Biotechnology (BIOT) (p. 632)

Department of Chemistry and Biochemistry

Academic Unit Head: Linette M. Watkins

Email: watkinlm@jmu.edu

Phone: (540) 568-6246

Location: Physics and Chemistry Building, Room 1186

Department of Chemistry and Biochemistry Website (<http://www.jmu.edu/chemistry>)

Mission Statement

The Department of Chemistry and Biochemistry offers the B.S. degree for a major in chemistry, with concentrations that meet the American Chemical Society Accredited Programs requirements for programs in chemistry, biochemistry and materials chemistry, and concentrations in chemistry general program, chemistry secondary education, and chemistry/business. In addition, the department offers a B.S. degree in chemistry with a major in biophysical chemistry. It also offers minors in chemistry, biochemistry and molecular biology, and materials science. The programs are designed to provide the theoretical and practical instruction in chemistry and related areas to prepare students for careers in chemistry, biochemistry, medicine, dentistry, paramedical areas, forensic sciences, chemical engineering and other technology-based areas. The department also recognizes its responsibility to provide courses for non-chemistry majors who need a basic understanding of the principles of chemistry either for their chosen major or their general education.

Career Opportunities

- Graduate school in chemistry, biochemistry or related areas (ACS Certified Degrees preferred)
- Professional employment as a chemist or biochemist (ACS Certified Degrees preferred)
- Professional school (medical, dental, veterinary, pharmacy, business and law)
- Chemical engineering
- Environmental science
- Food or fragrance chemistry
- Forensic science
- Immunology
- Industrial hygiene
- Pharmaceutical chemistry
- Pharmacology
- Production supervision
- Quality control
- Research assistant
- Science educator
- Scientific writing
- Some forms of development work
- Technical library science
- Toxicology
- Other fields where critical problem solving and technical communication skills are valued

Co-curricular Activities and Organizations

- American Chemical Society Student Affiliate Chapter
- Alpha Chi Sigma Professional Fraternity (Coed)
- Iota Sigma Pi
- National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE)
- Society for the Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
- Pre-Professional Health clubs and societies
- Summer and academic year research
- Attend and present at local, regional, or national professional conferences
- Employment as a learning assistant
- Science outreach in the local schools and community
- Advocacy for diversity, equity and inclusion in STEM

Teaching Licensure

There are two paths to licensure as a high school teacher. One is a 4-year path with a double major in Chemistry Education and Secondary Education. A more common option is a 5-year path with the major in Chemistry with a concentration in Chemistry Secondary Education, and adding a 5th year Master of Arts in Teaching (M.A.T.).

It is critical that students seeking licensure consult regularly with both their education advisor and their major advisor to support their progression through the programs. For a full description of the program in secondary teaching, refer to the Department of Middle, Secondary and Mathematics Education (p. 86), in addition to the College of Education (p. 18)/Professional Education Unit section of the catalog.

Faculty

Professors

Donna S. Amenta, Ashleigh E. Baber, Christopher E. Berndsen, Brycelyn M. Boardman, Kevin L. Caran, Thomas C. DeVore, Christine A. Hughey, Oleksandr Kokhan, Gina MacDonald, Debra L. Mohler, Barbara A. Reisner, Linette M. Watkins, Nathan T. Wright, Yanjie Zhang

Associate Professors

Scott B. Lewis, Gretchen M. Peters

Assistant Professors

Lindsay Caesar, Hui Chen, James Gilhula, Travis Marshall-Roth

Senior Lecturers

Daniel E. Blumling, Qingsheng Liu, David L. Wilson Jr.

Lecturers

Caitlin Coplan, Kristen E. Funck, W. Christopher Hollinsed, Oscar Judd, Brittany Long

Programs and Courses

Majors

- Chemistry, B.S. (p. 217)
- Chemistry, Biophysical Chemistry Major, B.S. (p. 220)

Cross-Disciplinary Major

- Biotechnology, B.S. (p. 211)

Minor

- Chemistry Minor (p. 460)

Courses

- Chemistry (CHEM) (p. 634)

Department of Communication Sciences and Disorders

Academic Unit Head: Ayasakanta Rout

Email: routax@jmu.edu

Phone: (540) 568-6440

Location: Health and Behavioral Studies Building, Room 1024

Department of Communication Sciences and Disorders Website (<https://www.jmu.edu/chbs/csd/index.shtml>)

Mission

The Department of Communication Sciences and Disorders engages students, faculty and members of the community to advance the state of knowledge through basic and applied research, innovative professional and pre-professional education, and enhanced clinical practices to better serve those with communication disorders.

The department is committed to promoting comprehensive, state-of-the-art undergraduate pre-professional education that includes discipline-specific course work and observation. In keeping with university requirements, this includes a broad-based General Education component. The department also delivers graduate-level course work and practicum experiences for those interested in entering professional practice in either speech-language pathology or audiology, a university teaching and research position, or a management/administrative position in service delivery settings.

The department is committed to advancing the state of knowledge in both basic and applied aspects of communication sciences and disorders through its doctoral research degrees and the research activities of its faculty and students and to providing service to the profession, university and client communities at local, state, national and international levels. Through the Audiology and Speech-Language Clinics, the department provides services as part of the clinical teaching component of its mission and serves as a clinical research resource for students and faculty.

Goals

- Providing course work and observation at the undergraduate level, including study of the underlying science and development of human communication, and an introduction to disorders that may occur in human communication.
- Providing course work at the undergraduate level that will prepare students for:
 - Graduate study in the areas of audiology or speech-language pathology
 - Graduate school in a related discipline

Programs of Study

The department offers a B.A. (p. 223) and B.S. degree in communication sciences and disorders (p. 225). The department also offers the M.S. in speech language pathology (<https://catalog.jmu.edu/programs/speech-language-pathology-ms/>) (professional preparation), the Au.D. (Doctor of Audiology) (<https://catalog.jmu.edu/programs/audiology-aud/>) and the Ph.D. in communication sciences and disorders (<https://catalog.jmu.edu/programs/communication-sciences-disorders-phd/>) with emphases in audiology, speech-language pathology and/or speech and hearing science. Inquiries concerning these graduate programs should be directed to the dean of The Graduate School or the appropriate department graduate program director. A master's degree

is the minimum requirement for competency/certification endorsed by the American Speech-Language-Hearing Association and for Virginia licensure in speech-language pathology. In audiology, a doctoral degree is the minimum requirement for competency/certification endorsed by the American Speech-Language-Hearing Association and for new Virginia licensure applicants.

The Audiology and Speech-Language Clinics

The JMU Audiology clinic and the JMU Speech-Language clinic provide the following services for individuals with communication disorders across the life span to the local community. Appointments for services may be made by any member of the community.

Areas of Service Delivery

- Speech, language, hearing and/or balance assessments
- Intervention programs in speech, language and hearing disorders
- Preventative and educational consultation
- Referrals for other professional services when indicated

Career Opportunities

Individuals with a graduate degree in audiology or speech-language pathology can pursue a variety of career options:

- Audiologist
- Clinical Educator
- Director of Speech and Hearing Services
- Early Intervention Specialist
- Hearing Scientist
- Private Practitioner
- Researcher
- Speech-Language Pathologist
- Speech Scientist
- University Professor

Individuals with a bachelor's degree may be qualified to pursue other career options:

- Audiology assistant
- Speech-language pathology assistant

Keep in mind that graduate study is required for most positions in this field. Some individuals who major in communication sciences and disorders go on for further study in special education, medicine and psychology.

Co-curricular Organizations

- National Student Speech-Language-Hearing Association
- Student Academy of Audiology

Graduate Admission Criteria

Specific admission requirements and prerequisite course work for individual programs are listed with each program description. Applications will be reviewed in accordance with the department's holistic admission philosophy. We commit to the mission and vision of JMU's Department of Communication Sciences and Disorders, by adopting a holistic review process that fosters inclusive excellence

and values the experiences and attributes that students from diverse backgrounds bring to our university. We welcome students who can think critically and who embody compassion, collaboration, adaptability, resilience and openness to feedback.

Students applying for admission to either the M.S. program in Speech-Language Pathology (<https://catalog.jmu.edu/programs/speech-language-pathology-ms/>) or the Au.D. program (<https://catalog.jmu.edu/programs/audiology-aud/>) should be aware that some of the clinical placement sites in which students are required to complete clinical practica in order to graduate will require the student to produce a current criminal records check. It is the student's responsibility to arrange for the check, to keep it current in order to comply with the requirements of the various clinical sites, and to advise the university and department if the status of the student's criminal record changes at any time during the student's program.

Vision

We will be recognized for bridging science and clinical practice.

The Department of Communication Sciences and Disorders provides graduate-level course work and practicum experiences for those interested in entering professional practice in either speech-language pathology or audiology, university teaching and research positions, or management/administrative positions in service delivery settings. The department is committed to advancing the state of knowledge in both basic and applied aspects of communication sciences and disorders through its master's and doctoral research degrees and the research activities of its faculty and students, and to providing service to the profession, university and client communities at local, state, national and international levels. The Department's Speech-Language-Hearing Clinic provides outreach services to the region as part of the clinical teaching component of its mission and provides a clinical research resource for students and faculty.

The audiology and speech-language pathology clinical training graduate programs in the Department of Communication Sciences and Disorders are accredited by the Council on Academic Accreditation in Audiology and Speech-Language Pathology of the American Speech-Language-Hearing Association.

Audiology and Speech-Language-Hearing Clinics

The JMU Speech-Language Pathology and Audiology Clinics, both operated by the Department of Communication Sciences and Disorders, provide evaluation and clinical intervention services for individuals with speech, language and hearing problems. The priority for services offered through these clinics is determined by the needs of practicum students. Appointments for a consultation or evaluation may be made by any member of the general public or university community. Professional services are provided by expert certified speech-language pathologists and audiologists who supervise the practicum of students.

Financial Aid

Graduate assistantships are available on a competitive basis to graduate students. In addition, fellowships funded by the Scottish Rite Foundation of Virginia are available to graduate students admitted to the graduate program in speech-language pathology. Scottish Rite Fellowships are restricted to Virginia residents who intend to pursue positions working with children who have speech and language impairments in the state

of Virginia. Contact the appropriate graduate program director in the department for further information.

Faculty

Professors

Rory A. DePaolis, Carol C. Dudding, Christina Kuo, Erin G. Piker, Ayasakanta Rout, Geralyn R. Timler

Associate Professors

Erin E. Kamarunas, Yingjiu Nie, Emily R. Zane

Assistant Professors

Erin S. Clinard, Nikki Go, Devon P. Major, Surani G. Nakkawita, Lars Rogenmoser

Senior Lecturer

Melissa M. Garber

Lecturers

Shiree C. Harbick, Ingrid D. Hinkley, Beth I. McCall, Sarah D. Teter

Programs and Courses

Majors

- Communication Sciences and Disorders, B.A. (p. 223)
- Communication Sciences and Disorders, B.S. (p. 225)

Graduate Programs

- Audiology, Au.D. (<https://catalog.jmu.edu/programs/audiology-aud/>)
- Communication Sciences and Disorders, Ph.D. (<https://catalog.jmu.edu/programs/communication-sciences-disorders-phd/>)
- Speech Language Pathology, M.S. (<https://catalog.jmu.edu/programs/speech-language-pathology-ms/>)

Courses

- Communication Sciences and Disorders (CSD) (<https://catalog.jmu.edu/courses/csd/>)

School of Communication Studies

Director: Lori L. Britt

Email: brittll@jmu.edu

Phone: (540) 568-6228

Assistant Director: Lee Mayfield

Location: Harrison Hall, Room 1276

School of Communication Studies Website (<http://www.jmu.edu/commstudies>)

Mission

We are a community engaged in teaching, researching and practicing communication as central to cultivating ethical, productive and meaningful lives and relationships.

Goals

As the school offers courses and programs in communication studies, the school seeks to fulfill the following goals:

- To prepare all JMU students who take a communication studies course to use oral communication skills effectively.
- To prepare students with a major or minor in communication studies for graduate and professional study as well as for careers in communication and communication-related professions.
- To provide co-curricular opportunities which enhance and reinforce communication competencies for all JMU students.

Career Opportunities and Marketable Skills

Students of communication studies develop skill competencies required for effective interaction and leadership in communities, workplaces and groups. Some of these skills include:

- Presentational Speaking
- Empathic listening
- Interacting in small groups
- Using nonverbal communication skills such as use of space, voice, gaze and facial expressions
- Constructing persuasive messages and campaigns
- Conducting research that leads to insights about human communication patterns
- Presenting information digitally
- Analyzing communication at all levels including interpersonal, small group, organizational and public
- Advocating successfully
- Using language effectively
- Writing effectively
- Understanding and applying communications research and theory
- Mediating and resolving conflicts
- Collaborative problem solving

These skills support careers in such fields as:

- Public Relations
- Strategic Communication
- Consulting
- Entertainment

- Branding and Digital/Social Media
- Hospitality
- Management
- Human Resources
- Politics
- Sales
- Teaching
- Corporate Communication and Public Affairs
- Grassroots and Non-Profit Advocacy

Study in communication studies also serves as valuable pre-professional preparation for graduate and professional studies in the following fields:

- Communication
- Counseling
- Law
- Management
- Ministry

Co-curricular Activities and Organizations

To enhance courses and programs in communication studies, the school offers a variety of co-curricular activities and organizations. Involvement in some of these activities can earn students practicum credit.

Activities and Organizations

- Debate Team: Affords all JMU students interested in debating intercollegiate tournament competition and local audience experiences.
- JMU Speech Team: Intercollegiate tournament competition and local audience experiences for all JMU students interested in public speaking and the oral interpretation of literature can be acquired through individual events team participation.
- Lambda Pi Eta: A chapter of a national honorary organization for students interested in communication.
- Public Relations Student Society Association: An award-winning student organization for students pursuing careers in public relations.
- Bluestone Communications: An award-winning student run Public Relations firm that works with real clients in strategic PR planning, primary market research, social media counseling, creative work, media relations and special event planning.
- Institute for Constructive Advocacy and Dialogue: Students taking coursework in facilitation and group communication can work with the institute in designing and facilitating group processes and gain practicum credit.

Faculty

Professors

Carlos G. Alemán, Melissa W. Alemán, Timothy C. Ball, Matthew P. Brigham, Lori L. Britt, Peter K. Bsumek, Eric M. Fife, Lindsey A. Harvell-Bowman, Corey A. Hickerson, C. Leigh Nelson, Sharlene T. Richards, Jennifer G. Rosier, Daniel K. Schill, Toni S. Whitfield, Chang Wan "Isaac" Woo

Associate Professors

Alison A. Bodkin, Michael L. Broderick, Kathryn D. Hobson, Lars J. Kristiansen, Andrea Martinez Gonzalez, Paul E. Mabrey III, Mallory L. Marsh, Shana L. Meganck, Tobias J. Reynolds-Tylus

Assistant Professors

Ebenezer Aidoo, Victoria Bertram, Sarah Caban, Reslie Cortés, Kristen E. Okamoto

Senior Lecturers

Annick H. Dupal, C. Lee Mayfield

Lecturers/Instructors

Andrew Barnes, Nichole Barnes, Neil S. Butt, Julie A. Gochenour, Kenny Hopkinson, Taylor Johnson, Jenda K. Krauklis, Roberta Mitchell, Jason Mollica, Nathan Selove, Ashlyn Johns Serrels, Grace Smith

Programs and Courses

Majors

- Communication Studies, B.A. (p. 227)
- Communication Studies, B.S. (p. 233)

Minors

- Communication Studies Minor (p. 469)
- Cultural Communication Minor (p. 475)
- Health Communication Minor (p. 506)

Cross-Disciplinary Minors

- Political Communication Minor (p. 542)
- Sport Communication Minor (p. 569)

Graduate Program

- Communication and Advocacy, M.A. (<https://catalog.jmu.edu/programs/communication-advocacy-ma/>)

Department of Computer Information Systems and Business Analytics

Academic Unit Head: Jeremy D. Ezell

Email: ezelljd@jmu.edu

Phone: (540) 568-3072

Location: Zane Showker Hall, Room 5036

The Department of Computer Information Systems and Business Analytics Website (<http://www.jmu.edu/cob/cis>)

Mission Statement

The Department of Computer Information Systems and Business Analytics is committed to:

- Educating students by creating an active, experiential learning environment that prepares them to apply knowledge of information systems, operations, business analytics and information and cybersecurity management for the betterment of organizations and society; and
- Serving the academic community and business communities through appropriate research and service.

Objectives

Computer information systems (CIS) is offered as a major through the Department of Computer Information Systems and Business Analytics. This program prepares business students for careers as information systems, business analytics, and information and cybersecurity management professionals. The program of study focuses on the development and management of information systems in a business environment. Students develop the technical skills and organizational insights required to analyze, design, implement and administer information systems in a data driven world. The CIS curriculum includes hands-on projects, laboratory exercises, case analysis and business simulations to build strong technical and analytical skills, effective oral and written communication skills, and the ability to work independently and in team-oriented environments. Students are offered the opportunity to gain practical experience through internships. The department faculty endorses the program educational objectives listed below.

The CIS B.B.A. program will produce graduates with the ability to:

- Exhibit leadership in a role as subject matter expert or consultant.
- Exhibit confidence in combining technical, business, and interpersonal skills to innovate and solve problems in a team environment.
- Effectively adapt and apply changing technology to business opportunities and challenges.
- Work effectively in the IS field to make a positive contribution to society.

Career Opportunities

Computer information systems professionals analyze business opportunities and problems, then design and build solutions using the power of information technologies. Students in the CIS program gain the business and technical skills that prepare them to move quickly from technical to leadership roles within the organization.

- Consultant
- Business Analyst

- Business Analytics Specialist
- Business Intelligence Specialist
- Computer Forensics Specialist
- IT/IS Auditor
- IT/IS Consultant
- Risk Analyst
- Security Consultant
- Systems Analyst
- Telecommunications Analyst
- IT/IS Manager
- Application Development Manager
- Business Owner (IT/IS Industry)
- Chief Information Officer
- Chief Security Officer
- Data Warehousing Manager
- Information Systems Manager
- Program Manager
- Project Manager
- IT Operations Professional
- Database Administrator
- Information Technology Trainer
- Network and Systems Administrator
- Security Specialist
- Software Engineer
- Web Developer

Co-curricular Activities and Organizations

The award-winning JMU Chapter of the Association for Information Systems (AIS), a national professional organization, was founded in 2014 and serves to advance knowledge and to promote excellence in the practice and study of information systems. Along with providing premier networking opportunities with top consulting firms, AIS provides student members opportunities to engage in national student competitions, in-house tutoring activities, professional development workshops, and various fundraising and social events.

Accreditation

The B.B.A. in computer information systems is accredited by the Computing Accreditation Commission of ABET (<http://www.abet.org>).

Faculty

Professors

Michael E. Busing, Thomas W. Dillon, Chenqgi Guo, Jeffrey L. May, Raktim Pal, Ping Wang

Associate Professors

Dmytro Babik, Amy J. Connolly, Jeremy D. Ezell, Leigh A. Mutchler, Luis J. Novoa, Rhonda A. Syler, Elham Torabi, Baback Vaziri

Assistant Professors

Puzant Balozian, Kiarash M. Hassani, Kamal Lamsal, Dan Liu, Jiajia Qu, E. Teymourian, Hoda Atef Yekta, Yuyun Zhong

Senior Lecturers

Laura Atkins, Shawn R. Lough

Lecturers

Jeffrey Driscoll, J. Greg Gabriel, James Jewett, Faker Zouaoui

Programs and Courses

Major

- Computer Information Systems, B.B.A. (p. 239)

Minors

- Business Analytics Minor (p. 458)
- Computer Information Systems Minor (p. 471)

Courses

- Business Analytics (BSAN) (p. 633)
- Computer Information Systems (CIS) (p. 651)

Department of Computer Science

Academic Unit Head: Vetricia L. Byrd

Email: byrdvl@jmu.edu

Phone: (540) 568-3335

Location: King Hall, Room 222

Department of Computer Science Website (<http://www.jmu.edu/cise/cs>)

Mission

The Department of Computer Science strives to be an intellectual community that explores the broad field of computing, and provides an accessible, rigorous and continually refined curriculum to our students. With small class sizes and emphasis on state-of-the-art pedagogy, we educate graduates to develop technology to solve problems, communicate clearly, practice their craft responsibly and achieve professional success. We foster a sense of shared community with our students, giving them opportunities for leadership, professional development and scholarly pursuits.

Career Opportunities and Marketable Skills

Computing technology pervades modern society and demand for computing professionals is strong and projected to remain strong for the foreseeable future. Careers in computing range from technical positions specifying, designing, building and maintaining networks and systems of all kinds, through project leadership and technical management. The Computer Science major prepares students for entry-level technical positions as programmers, software developers, requirements analysts, software designers, testers, software quality assurance professionals, system architects, network engineers, information security specialists and computing consultants. The Information Technology major equips students with the necessary skills to work on projects that have the potential to make a significant impact on society. From developing life-saving medical technologies to addressing environmental challenges, Information Technology can contribute to a better world. Additionally, this major allows for various career options such as network administration, cybersecurity analysis, data analytics, cloud computing administration and development, and more.

Admission and Retention Requirements

Information on admission to programs within the Department of Computer Science, retention and minimum grade requirements may be found on the individual program pages.

Faculty

Professors

Florian P. Buchholz, Samy S. El-Tawab, M. Hossain Heydari, Christopher Johnson, Michael S. Kirkpatrick, Michael O. Lam, Christopher S. Mayfield, Emil H. Salib, Nathan R. Sprague, Brett C. Tjaden, Xunhua "Steve" Wang, Dee A. B. Weikle

Associate Professors

Mohamed S. Aboutabl, John C. Bowers, Vetricia L. Byrd, Zhuojun Duan, Suk Jin Lee, Kevin P. Molloy, Mona Rizvi, Ahmad Salman, Michael C. Stewart

Assistant Professors

Adeen Ayub, Prajakta Belsare, Moh Khalid Hansan, Bradley McCoy, Chandani Shrestha, Isaac Wang

Senior Lecturer

Philip Riley

Lecturer

Alvin Chao, Nareman Hamdan

Programs and Courses

Majors

- Computer Science, B.S. (p. 242)
- Information Technology, B.S. (p. 319)

Minor

- Computer Science Minor (p. 472)

Cross-Disciplinary Minor

- Robotics Minor (p. 560)

Graduate Program

- Computer Science, Cybersecurity Concentration, M.S. (<https://catalog.jmu.edu/programs/computer-science-cybersecurity-concentration-ms/>)

Courses

- Computer Science (CS) (<https://catalog.jmu.edu/courses/cs/>)
- Information Technology (IT) (p. 736)

Department of Early, Elementary and Reading Education

Academic Unit Head: Joi Merritt

Email: merritjd@jmu.edu

Phone: (540) 568-3066

Location: Memorial Hall, Room 3100

Department of Early, Elementary and Reading Education Website (<http://www.jmu.edu/coe/eere>)

Assistant Academic Unit Head: Christopher Hass

Email: hasscl@jmu.edu

Phone: (540) 568-3632

Literacy Graduate Program Director: Mark Lewis

Email: lewis6ma@jmu.edu

Phone: (540) 568-6255

Mission

The Department of Early, Elementary and Reading Education is committed to preparing caring, knowledgeable, skilled and reflective professionals who are committed to teaching all children. The department emphasizes supervised clinical placements that provide candidates with experiences in diverse settings that have a wide range of ages and abilities. Graduates are eligible for additional endorsements and/or licensure in Virginia.

The Department of Early, Elementary and Reading Education offers undergraduate and graduate programs in literacy, early and elementary education. All programs include extensive fieldwork to ensure varied opportunities for experiences working with children and families.

Faculty

Professors

John T. Almarode, Aaron T. Bodle, Shin Ji Kang, Smita Mathur, Joi D. Merritt

Associate Professors

Chelsey M. Bahlmann Bollinger, Abigail A. Amoaka Kayser, Maryam Sadat Sharifian, Emma S. Thacker

Assistant Professors

Christopher Hass, Alexa M. Quinn, Alexis Rutt

Senior Lecturer

Eric A. Imbrescia

Instructors

Alicia R. Berry, Jennifer Dameron, Elizabeth Ochoa, Daniel L. Sawyer, Bonnie Yoder

Advisors

Annette Fornadel, Hannah Smith

Programs and Courses

Major

- Elementary Education, B.S. (p. 258)

Graduate Programs

- Education, Literacy Education Concentration, M.Ed (<https://catalog.jmu.edu/programs/education-literacy-education-concentration-med/>)
- Literacy Education Certificate (<https://catalog.jmu.edu/programs/literacy-education-certificate/>)

Courses

- Early Childhood Education (ECED) (<https://catalog.jmu.edu/courses/eced/>)
- Literacy Education (LED) (<https://catalog.jmu.edu/courses/led/>)
- Elementary Education (ELED) (<https://catalog.jmu.edu/courses/eled/>)

Department of Economics

Academic Unit Head: Nevin Cavusoglu

Email: cavusonx@jmu.edu

Phone: (540) 568-8752

Location: Zane Showker Hall, Room 6019

Department of Economics Website (<http://www.jmu.edu/cob/economics>)

Mission

The Department of Economics is committed to students' intellectual development by fostering an understanding and appreciation of economic forms of explanation and their relationships to other social sciences. Economics faculty members are dedicated to sound and effective pedagogy, to scholarship of the highest quality and to outreach to the local and business community.

The department provides an intellectual foundation for the appreciation and understanding of economic theory and policy. This foundation is developed within a broader educational perspective that stresses the importance of imaginative thinking, free inquiry and the pursuit of life-long learning. In this way, the program prepares students with the economic literacy necessary to cope with the challenges inherent in a world of accelerating change.

Goals

- Help students develop analytical and critical thinking skills.
- Promote cross disciplinary forms of instruction.
- Seek continuous improvement in the quality of classroom instruction.
- Serve the community through outreach services.
- Help students pursue careers and additional education.

Marketable Skills

- Analytical thinking capabilities highly valued by business, government and the nonprofit sector.
- Writing and research skills applicable to a wide variety of careers.
- Statistical and econometric skills used in business and finance.
- Preparation in critical thinking valued by graduate schools, including law, business, and arts and sciences.
- Analytical skills valued by employers for internships in business, government and consulting.

Co-curricular Activities and Organizations

- Economics Club (open membership)
- Omicron Delta Epsilon (national honor society in economics)

Credit by Examination

Credit in ECON 200 will be granted to students who achieve a grade of 4 or 5 on the Advanced Placement Test in Macroeconomics administered by the Educational Testing Service. Credit in ECON 201 will be granted to students who achieve a grade of 4 or 5 on the Advanced Placement Test in Microeconomics.

Faculty

Professors

Nevin Cavusoglu, Joanne M. Doyle, William C. Grant, Scott R. Milliman, Angela M. Smith

Associate Professors

Benjamin N. Hamilton, Syed Muhammed Hussain, Piyusha Mutreja, John Robert Subrick

Assistant Professors

Sayahnika Basu, Huiyi Chen, Tingting Ding, Daniel Harper, Mrithyurijayan Nilayamgode

Senior Lecturer

Zachary J. Gochenour

Lecturers

Zafer Akin, Linghui Han, Philip S. Heap, John Kruggel, Michael Mamo, Azat Nurmukhametov, Jingya Sun

Programs and Courses

Majors

- Economics, B.A. (p. 248)
- Economics, B.B.A. (p. 251)
- Economics, B.S. (p. 255)

Minor

- Economics Minor (p. 481)

Courses

- Economics (ECON) (p. 662)

Department of Educational Foundations and Exceptionalities

Academic Unit Head: Dannette Murray

Email: murraydy@jmu.edu

Phone: (540) 568-4952

Location: Memorial Hall, Room 3126A

The Department of Educational Foundations and Exceptionalities Website (<http://www.jmu.edu/coe/efex/index.shtml>)

Assistant Academic Unit Head: Katrina Maynard

Email: maynarkl@jmu.edu

Phone: (540) 568-6508

Location: Memorial Hall, Room 3126B

Graduate Program Director: Joshua Pulos

Email: pulosjm@jmu.edu

The Department of Educational Foundations and Exceptionalities offers programs in special education, inclusive early childhood education, and teaching English to speakers of other languages. All programs include extensive fieldwork to ensure varied experience working with children and families.

Faculty

Professors

Katya Koubek, Dannette Y. Murray, Timothy G. Thomas

Associate Professors

Sharon L. Blatz, Carmen Ruth Bosch, Tiara S. Brown, Andrew Markelz, Joshua M. Pulos, Benjamin S. Riden, Sara M. Snyder, Kristofor R. Wiley

Assistant Professors

Sara Germansky, Vivian Lee, Diana Meza

Instructors

Katie Cunningham, Lauren Heberling, Katrina Maynard, Christy McKee

Advisors

Annette Fornadel, Hannah Smith

Programs and Courses

Majors

- Inclusive Early Childhood Education, B.S. (p. 300)
- Special Education, B.S. (p. 421)

Minor

- Teaching English to Speakers of Other Languages Minor (p. 573)

Graduate Programs

- Education, Spanish Language and Culture for Educators Concentration, M.Ed. (<https://catalog.jmu.edu/programs/education-spanish-language-culture-educators-concentration-med/>)
- Education, Teaching English to Speakers of Other Languages (TESOL) Concentration, M.Ed. (<https://catalog.jmu.edu/programs/education-teaching-english-speakers-languages-tesol-concentration-med/>)
- Special Education, Adapted Curriculum K-12 Concentration, M.A.T. (<https://catalog.jmu.edu/programs/special-education-adapted-curriculum-k12-concentration-mat/>)

- Special Education, Behavior Specialist Concentration, M.Ed. (<https://catalog.jmu.edu/programs/special-education-behavior-specialist-concentration-med/>)
- Special Education, Early Childhood Special Education Concentration (Birth-Age 5), M.A.T. (<https://catalog.jmu.edu/programs/special-education-early-childhood-special-education-concentration-birthage-5-mat/>)
- Special Education, General Curriculum K-12 Concentration, M.A.T. (<https://catalog.jmu.edu/programs/special-education-general-curriculum-k12-concentration-mat/>)
- Special Education, M.A.T. (<https://catalog.jmu.edu/programs/special-education-mat/>)

Courses

- Education (EDUC) (<https://catalog.jmu.edu/courses/educ/>)
- Exceptional Education (EXED) (<https://catalog.jmu.edu/courses/exed/>)
- Inclusive Early Childhood Education (IECE) (<https://catalog.jmu.edu/courses/iece/>)
- Spanish Language and Culture Program (SLCP) (<https://catalog.jmu.edu/courses/slcp/>)
- Teaching English to Speakers of Other Languages (TESL) (<https://catalog.jmu.edu/courses/tesl/>)

Department of Engineering

Interim Academic Unit Head: Scott Paulson

Email: paulsosa@jmu.edu

Phone: (540) 568-8869

Location: Engineering/Geosciences 1122

Department of Engineering Website (<http://www.jmu.edu/engineering>)

Mission

Through an engineering curriculum grounded in effective instructional practices, we nurture an engaged educational community of conscientious and adaptable learners who develop solutions for the betterment of society.

Objectives

The JMU Engineering program provides an eight-semester project-driven platform for students to contribute engineering work for sustainable societies. We seek engineers who are eager to:

- understand the needs of people and communities, effectively communicate with project partners, and provide leadership for diverse teams,
- make beneficial solutions for society and the planet using innovation, design thinking, engineering analyses, and an understanding of complexities involved in real projects.
- function effectively in the work environment with integrity and professionalism, and
- demonstrate a commitment to professional development through self-motivated learning, career advancement, or advanced studies.

The Bachelor of Science in Engineering is a single cross disciplinary engineering degree that integrates many traditional engineering disciplines with course work in business, project management, engineering, design and liberal arts. The foci of the program are sustainability, engineering design and project delivery.

Engineering for a sustainable world is, in short, a body of knowledge and set of holistic analytical design skills that contribute to the development of products, processes, services and infrastructure that simultaneously protect the environment, conserve resources and meet human needs at an acceptable financial cost. By reframing traditional engineering practice, sustainability provides a way of moving toward the development of sustainable societies, where human quality of life is advanced with a minimum impact on finite resources and the environment.

This modern, project-based, engineering curriculum spans many areas of engineering to create adaptable engineers with practical know-how.

The curriculum is ABET accredited and prepares students for the Fundamentals of Engineering (FE) pre-licensure examination. Graduates will be prepared to succeed in the engineering workforce or in advanced engineering degree programs by accumulating a professional portfolio of engineering project experiences throughout the curriculum.

Career Opportunities

Upon graduation, alumni will be prepared for a wide range of opportunities in the engineering workforce or in graduate school. Typical fields of engineering that students will be prepared to enter include sustainable design, process design, product design, process engineering, project management and systems engineering. Other industry options

include product service system design, technical sales, management training and technical marketing.

A wide range of graduate school options include master's and doctoral programs in civil engineering, environmental engineering, industrial engineering, materials engineering, mechanical engineering and systems engineering. Other postgraduation options include business school, law school, AmeriCorps, Peace Corps, military service, entrepreneurship (starting a small business), applied science fields, international experiences, medical school and careers in politics/public policy.

Some examples of the industries that hire engineers include, among others, design and build companies, aeronautic firms, automobile manufacturers, colleges and universities, computer service and software firms, consulting firms, energy systems firms, federal contractors, federal, state and local governments (e.g., NASA, EPA, NIST, DOD, DOE), non-profit agencies, manufacturing firms, inspection agencies, mining and petroleum firms, pharmaceutical and medical research companies, research and development laboratories, telecommunication companies, and waste management and recycling firms.

Admission and Retention Requirements

Information on admission to programs within the Department of Engineering, retention and minimum grade requirements may be found on the individual program pages.

Faculty

Professor

Kyle G. Gipson, Steven R. Harper, S. Keith Holland, Samuel A. Morton III, Robert J. Prins, Bradley A. Striebig

Associate Professors

Daniel I. Castaneda, Jason B. Forsyth, Shraddha Joshi, Heather A. Kirkvold

Assistant Professors

John Barbish, Pedro Puente, Christine Toh, Steven R. Woodruff

Lecturers

Brent D. Cunningham

Programs and Courses

Major

- Engineering, B.S. (p. 261)

Courses

- Engineering (ENGR) (p. 670)

Department of English

Academic Unit Head: Sarah Hagelin

Location: Keezell Hall, Room 215

Department of English Website (<https://www.jmu.edu/english/index.shtml>)

Mission

Our mission is to prepare our students to be appreciative yet critical readers and creators of language, texts, and ideas, attending to the powerful link between the literary arts and the human condition. We develop in students the ability to write with clarity, creativity, and rhetorical power, and to recognize and explore the complex interaction between texts of various genres and diverse contexts of production and reception, that they may emerge with superior dexterity of reasoning, a heightened empathy, and an expanded worldview.

Goals

To build upon the skills introduced in the General Education program, the Department of English strives to:

- Provide students advanced instruction in writing styles ranging from expository and creative writing to literary criticism.
- Help students master advanced skills in analytical and critical thinking.
- Develop skills in research and information access.

Career Opportunities

Careers that involve critical thinking, document analysis, or oral and written communication such as:

- Career counseling
- Corporate recruiting
- Consulting
- Editing
- Educational administration
- Film critic
- Graduate school in English, creative writing, law and library science
- Human resources
- Insurance underwriting
- Intelligence analyst
- Law
- Library science (librarian, archivist, preservationist)
- Literary agent
- Literacy instructor
- Lobbyist
- Magazine writing and editing (print and online)
- Marketing
- Newspaper reporting/foreign correspondent
- Non-profit advocacy and administration
- Public relations and public affairs officer
- Publishing
- Research
- Speechwriter

- Stockbroker
- Teaching

Co-curricular Activities and Organizations

- English Student Ambassadors
- Iris (a literary and arts magazine)
- Sigma Tau Delta (the National English Honor Society)
- Sister Speak (JMU's feminist journal)
- Word is Born (Creative Writing organization)

Faculty

Professors

Lauren K. Alleyne, Katey M. Castellano, Erica Bleeg Cavanagh, Becky Childs, Allison E. Fagan, Annette R. Federico, Sarah Hagelin, Brooks E. Hefner, Mollie A. Godfrey, Debali Mookerjee-Leonard, Besi Muhonja, Bethany P. Nowviskie, Mark C. Rankin, Matthew E. Rebhorn, Mary Thompson, Siân E. White

Associate Professors

David Babcock, Johannes Burgers, Amanda J. Gerber, Dawn M. Goode, Laura Henigman, Dennis H. Lo, Heidi L. Pennington, Delores B. Phillips, Sofia Samatar, Greg Wrenn

Assistant Professors

Jason Baltazar, Maria José Delgadillo, Rebecca Klaver, Courtney Murray Ross, Breanne Weber

Programs and Courses

Major

- English, B.A. (p. 265)

Minor

- English Minor (p. 483)

Licensure Program

- English Teaching Licensure (p. 586)

Graduate Program

- English, M.A. (<https://catalog.jmu.edu/programs/english-ma/>)

Courses

- English (ENG) (<https://catalog.jmu.edu/courses/eng/>)

Cross-Disciplinary Minors

English participates in the following cross disciplinary minors:

- African, African American and Diaspora Studies Minor (p. 446)
- American Studies Minor (p. 448)
- Book Arts Minor (p. 456)
- British Communication and Media Minor (p. 457)
- Classical Studies Minor (p. 465)
- Creative Writing Minor (p. 473)
- Environmental Humanities Minor (p. 485)
- Environmental Studies Minor (p. 492)
- Film Studies Minor (p. 497)
- Latin American, Latinx and Caribbean Studies Minor (p. 518)
- Medical Humanities Minor (p. 527)

- Medieval and Renaissance Studies Minor (p. 529)
- Middle Eastern Communities and Migrations Minor (p. 530)
- Modern European Studies Minor (p. 533)
- Queer Studies Minor (p. 559)
- Women's, Gender and Sexuality Studies Minor (p. 576)
- World Literature Minor (p. 578)

Department of Finance and Business Law

Academic Unit Head: Elias J. Semaan

Email: semaanej@jmu.edu

Phone: (540) 568-7323

Location: Zane Showker Hall, Room 3137

Department of Finance and Business Law Website (<https://www.jmu.edu/cob/finance/index.shtml>)

Mission

The Department of Finance and Business Law, together with our partners in the business community, is committed to preparing students to be principled, productive contributors to business and society.

Values

To support its mission, the Department of Finance and Business Law is committed to the following values:

- **Ethics:** Develop decision makers who are respectful of stakeholders and the rule of law.
- **Engagement:** Connect students and faculty with stakeholders in the local and global communities.
- **Critical Thinking:** Foster intellectual curiosity and analytical skills to enable an understanding of the role of finance and business law in the global economy.
- **Innovation:** Challenge students to develop their own solutions to financial issues in a dynamic economy.

Career Opportunities

The finance major is designed to prepare students for careers in the financial management of industrial and commercial enterprises; commercial, retail and mortgage banking; investment analysis and portfolio management; real estate; insurance; finance positions in federal, state and local governments; and graduate study. The quantitative finance major prepares students for careers in risk analytics, investment banking, trading, risk management, financial engineering, and many other finance positions which require applied mathematics and computational skills.

Job titles held by graduates of the programs include:

- Business Systems Analyst
- Capital Budgeting Manager
- Chief Investment Officer
- Consultant
- Consultant/Financial Advisory
- Corporate Model Analyst
- Currency Trader
- Data Analyst
- Financial Advisor
- Financial Analyst
- Financial Engineer
- Internal Auditor
- Investment Banker
- Lending Officer
- Loan Analyst

- Market Analyst
- Operations Analyst
- Pricing Analyst
- Private Wealth Manager
- Project Finance Manager
- Portfolio Manager
- REIT Analyst
- Research Analyst
- Risk Analyst/manager
- Treasurer/CFO

Co-curricular Activities and Organizations

- Financial Management Association Student Chapter
- Madison Investment Fund
- Madison Venture Group
- Women In Business

Degree and Major Requirements

The Department of Finance and Business Law offers programs leading to the Bachelor of Business Administration degree in finance and the Bachelor of Science degree in quantitative finance. As part of the JMU assessment program, graduating seniors are required to participate in assessment activities. Assessment information is used to assist the College of Business and department faculty in curricula.

Faculty

Professors

Jaideep Chowdhury, Jason D. Fink, Kristin E. Fink, Qingfeng "Wilson" Liu, Elias J. Semaan, Hui He Sono, Carl J. Ullrich

Associate Professors

Carl F. Larsson, Adam Usman, Weiwei Zhang

Assistant Professors

Danmo Lin, Yu He, Xu Niu, Hanzhi (Frank) Xu

Senior Lecturers

William "Brack" B. Collier, David L. Parker, Ronald J. Rubin

Lecturer

E. Ozcan, Robert Richardson

Programs and Courses

Majors

- Finance, B.B.A. (p. 268)
- Quantitative Finance Major with a Minor in Economics and Mathematics, B.S. (p. 415)
- Quantitative Finance/Mathematics Double Major, B.S. (p. 417)

Courses

- Business Law (BLAW) (p. 633)
- Finance (FIN) (p. 688)

Department of Geology and Environmental Science

Interim Academic Unit Head: Kristen St. John

Email: stjohnke@jmu.edu

Phone: (540) 568-6675

Location: Engineering/Geosciences, Room 3232

Department of Geology & Environmental Science Website (<https://www.jmu.edu/geology/>)

Mission Statement

Our mission is to serve two vital needs of the JMU students. First, the majors in geology present high-quality programs of specialized study focusing on Earth materials, internal and external Earth processes, analysis of Earth history and application of geology to environmental and engineering issues. In support of this mission is a commitment to foster the ability to think analytically and to communicate both within the discipline and with non-scientists. Course work and research experiences prepare the student for postgraduate study or professional careers that are subject to rapidly changing societal needs. Second, our department strives to enhance the university's general education program by offering timely and challenging courses that provide insight into Earth processes and human-environment interactions. These courses promote life-long liberal learning by fostering critical thinking and an awareness of natural science.

Goals

Provide a stimulating, intellectual environment for students in geology and environmental science that will generate interest and enthusiasm for learning and will provide a solid foundation for graduate work and careers in geology and environmental science.

Teach science as science is practiced. Since the advancement of scientific knowledge often occurs within a social context – collaboration among scientists, conferences, seminars – the goal is to develop a similar mode of operation for the geology program.

Provide high-quality, relevant general studies courses within the discipline that focus on the fundamental science in the societal context (e.g., environmental change, climate change, hazards). These courses will incorporate critical thinking and an appreciation of human-environment interactions.

Career Opportunities

- Earth Science Teacher
- Engineering Geologist
- Environmental Geologist
- Environmental Scientist/Specialist
- Geochemist
- Government or Industry Geologist
- Geological Oceanographer
- Geomorphologist
- Geophysicist
- Hydrogeologist
- Meteorologist
- Science Museum Curator
- Paleoclimatologist/Paleoceanographer

- Paleontologist
- Petroleum Geologist
- Soil Scientist

Co-curricular Activities and Organization

The department encourages majors and minors to participate in the student Geology Club, which sponsors field trips, camping excursions, and hosts educational activities for elementary school students. Geology majors are encouraged to apply to be laboratory teaching assistants and research assistants for faculty in the department. Majors and minors are also strongly encouraged to become members of one or more of the following geoscience organizations and to present their research at the affiliated regional or national meetings:

- Geological Society of America (GSA)
- American Association of Petroleum Geologists (AAPG)
- American Geophysical Union (AGU)
- National Association of Geoscience Teachers (NAGT)

Faculty

Professors

Steven J. Baedke, Louis Scott Eaton, Elizabeth A. Johnson, Stephen A. Leslie, Eric J. Pyle, Kristen St. John, Steven Whitmeyer

Associate Professors

Yonathan Admassu, Chiara Elmi, Angel A. Garcia, William E. Lukens, R. Shane McGary, Shelley Whitmeyer

Assistant Professors

Kirsty McKenzie, Dhanuska B. Wijesinghe

Instructors

Matthew Heller, Chris Kaznosky, Susie Shomo

Lecturer

Natalie Bursztyn

Programs and Courses

Majors

- Geology, B.A. (p. 279)
- Geology, B.S. (p. 281)
- Geology, Environmental Earth Science Major, B.A. (p. 283)
- Geology, Environmental Earth Science Major, B.S. (p. 285).

Minors

- Geology Minor (p. 501)
- Geophysics Minor (p. 502)

Cross-Disciplinary Minor

- Climate Science Minor (p. 466)

Licensure Program

- Earth Science Teaching Licensure (p. 585)

Courses

- Geology (GEOL) (<https://catalog.jmu.edu/courses/geol/>)

Department of Graduate Psychology

Academic Unit Head: Robin Anderson

Phone: 540-568-6439

Email: ander2rd@jmu.edu

Department of Graduate Psychology Website (<https://www.jmu.edu/chbs/gradpsyc/index.shtml>)

Mission

Our mission is to transform students into outstanding practitioners and scholars of psychology and counseling. We work to create a community that celebrates excellence, and that values learning, scholarship and service to others.

Overview

James Madison University offers eight graduate programs designed for individuals who want to pursue advanced training in psychology and counseling. These programs share the goals of academic enrichment, refinement of research skills, development of applied skills, and personal and professional growth.

Faculty

Professors

Robin D. Anderson, Stephanie Crockett, Christine E. DeMars, Sara J. Finney, Keston H. Fulcher, Tammy D. Gilligan, Robert J. Harmison, John D. Hathcoat, Gregg Henriques, Viki Kelchner, Michele L. Kielty, Deborah Kipps-Vaughan, Cara Meixner, Dena A. Pastor, Elena A. Savina, Angela Renee Staton, Trevor F. Stokes, Deborah C. Sturm, David E. Szwedo

Associate Professors

Kelly C. Atwood, Yu Bao, Megan R. Good, Steve Grande, Tiffany C. Hornsby, Joseph J. LeBlanc, Brian C. Leventhal

Assistant Professors

Kenson G. Hiatt, Chendong Li, Caitlyn R. Owens, Philip J. Schulte

Instructors

Hugh Brown, Ginger Griffin, Tonya Lambert Delp, Michelle Richards-Witt

Clinician in Residence

Timothy Schulte

Programs and Courses

Graduate Degree Programs

- Assessment and Measurement, Ph.D. (<https://catalog.jmu.edu/programs/assessment-measurement-phd/>)
- Clinical and School Psychology, Psy.D. (<https://catalog.jmu.edu/programs/clinical-school-psychology-psyd/>)
- Counseling and Supervision, Ph.D. (<https://catalog.jmu.edu/programs/counseling-supervision-phd/>)
- Counselor Education, College Student Personnel Administration Concentration, M.Ed. (<https://catalog.jmu.edu/programs/counselor-education-college-student-personnel-administration-concentration-med/>)
- Psychological Sciences, Applied Research Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-applied-research-concentration-ma/>)

- Psychological Sciences, Behavior Analysis Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-behavior-analysis-concentration-ma/>)
- Psychological Sciences, Experimental Psychology Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-experimental-psychology-concentration-ma/>)
- Psychological Sciences, Quantitative Psychology Concentration, M.A. (<https://catalog.jmu.edu/programs/psychological-sciences-quantitative-psychology-concentration-ma/>)
- School Psychology and Counselor Education, Clinical Mental Health Counseling Concentration, M.A./Ed.S. (<https://catalog.jmu.edu/programs/school-psychology-counselor-education-clinical-mental-health-counseling-concentration-ma-eds/>)
- School Psychology and Counselor Education, School Counseling Concentration, M.Ed./Ed.S. (<https://catalog.jmu.edu/programs/school-psychology-counselor-education-school-counseling-concentration-med-eds/>)
- School Psychology and Counselor Education, School Psychology, M.A./Ed.S. (<https://catalog.jmu.edu/programs/school-psychology-counselor-education-school-psychology-ma-eds/>)

Graduate Certificate Programs

- College Student Personnel Administration Certificate (<https://catalog.jmu.edu/programs/college-student-personnel-administration-certificate/>)

Courses

- Psychology (PSYC) (<https://catalog.jmu.edu/courses/psyc/>)

Application Dates

Refer to individual programs for application requirements. Programs may require a criminal history check as part of the final admissions process.

Assessment and Measurement Doctoral Program

The application opens October 15 and the target date is January 10. All application materials must be received by January 10 to ensure review by the program.

Clinical and School Psychology Doctoral Program

The application date for the fall semester is December 1. All application forms and supporting materials are due at this time.

Counseling and Supervision Doctoral Program

The application date for the summer semester is January 7. The committee conducts screening interviews in late January and early February.

Counseling: Clinical Mental Health and School Counseling Programs

January 2 is the deadline to be guaranteed consideration. Reviews start December 1. Selected students will be asked to interview in February

Counseling: College Student Personnel Administration Program

The application date for the fall semester is January 15. The program will give preference to applications received by this date. The committee conducts screening interviews in February.

Counseling: School Counseling Programs

January 2 is the deadline to be guaranteed consideration. Reviews start December 1. Selected students will be asked to interview in February.

Psychological Sciences Program

The application date for the fall semester is January 10. All application forms and supporting materials are due at this time. The program conducts screening interviews in February.

School Psychology Program

The application date for the fall semester is February 1. The program will give preference to applications received by that date. The committee conducts required on-campus screening interviews in late February and early March.

Hart School of Hospitality, Sport and Recreation Management

Director: David Shonk

Email: shonkdj@jmu.edu

Phone: (540) 568-7885

Location: Godwin Hall, Room 361

Hart School Website (<http://www.jmu.edu/hartschool>)

Mission

We engage in transformational education through innovative teaching, scholarship and service.

Vision

We strive to be a leader in impacting the future of global hospitality, sport and recreation.

Core Values

- Excellence
- Knowledge
- Student Focus
- Relationships
- Diversity
- Innovation
- Ethical Decision-Making
- Optimism

Hospitality Management

Career Opportunities: Readiness for the 21st Century and Beyond

For almost 50 years, the hallmark of the hospitality program at JMU's Hart School is its belief in creating a LEARN BY DOING environment that provides the foundational understanding of business principles and puts a services spin on it. This formula has led to thousands of successful leaders in the hospitality profession and related leadership positions globally.

Technology and artificial intelligence continue to evolve, and the Hart School Difference provides students with the necessary skills to compete and excel in this new world. The spirit of innovation, leadership, collaboration, ethical reasoning and communication are interwoven in all coursework.

Each Fall and Spring, over 50 employers come to campus for the Hart School Internship and Career Fair. These employers' goal is simple, to hire as many JMU HM majors as possible for internships and full-time job employment. More details at <https://www.jmu.edu/hartschool/caps/career-fairs/spring/index.shtml>

Dukes helping Dukes is our philosophy of networking. Our alumni stay connected, want to hire fellow JMU Dukes, and together are making a difference around the world.

Finally, there are no guarantees, but rest assured that by being a hospitality management major in the Hart School, students will be taught by exceptional educators who care about their success both in and out of the classroom. Upon graduation, students will have a set of **transferable** skills that readies them for the challenges of the 21st century and

beyond. Students will not only be prepared to join the workforce but to be leaders in the changing world of hospitality leadership and related services business.

Co-curricular Activities and Organizations

- Professional Convention Management Association (PCMA) is a student organization that provides both educational and social programs to the hospitality management major. This group plans and coordinates an annual trip to the PCMA Annual Convention.
- The National Society of Minorities in Hospitality (NSMH) is a student organization that explores the issues, challenges and opportunities for minorities in mega-industry. It participates in both regional and national conventions in association with industry leaders.
- The James Madison University Student Chapter of Club Management Association of America (CMAA) is instrumental in exposing students to the profession of club management and its many career opportunities.
- Eta Sigma Delta (ESD) is an honor society recognizing hospitality and tourism students for outstanding academic achievement, meritorious service and demonstrated professionalism.

Sport and Recreation Management

Career Opportunities

Graduates with this degree will be employed in professional sport organizations, semi-professional sport organizations, collegiate athletics, sport marketing agencies, sport broadcasting venues, facility management in both sport and recreation, community recreation agencies, commercial recreation agencies, theme parks, military recreation as well as hotels, resorts and other hospitality and entertainment venues.

Careers in sport and recreation management are plentiful. The listing below offers examples of possible career paths and is not meant to be comprehensive.

- Fitness/Health Club Manager
- Sportscaster
- Aerobics Instructor
- Sports Agent
- General Manager
- Athletic Coach
- Athletic Director
- Sports Event Coordinator
- Media Relations Specialist
- Sports Information Specialist
- YMCA Youth Leader
- Recreation Professional
- Sports Marketing Specialist
- Community Center Director
- Corporate Fitness Leader
- Director of Stadium Operations
- Director of Ticket Operations
- Professional Sports Scout
- Promotion Director
- Youth Programs Director
- Campus Recreation Director

- Athletic Contract Manager
- Sports Camp Director

Co-Curricular Activities

In order to enhance the educational, experiential, networking and professional opportunities for our students, sport and recreation management (SRM) offers and encourages involvement in a wide variety of co-curricular activities and SRM-related conferences and organizations. Co-curricular activities also include required practicum and internship experiences for which academic credit is given. Below is a list of SRMs most prominent activities:

Conferences and Events

- Sports Events Marketing Experience (SEME)
- Sports Industry Networking and Career Conference (SINC)
- MIT Sloan Analytics Conference
- National Intramural Recreational Sports Association (NIRSA) Annual Conference and Recreational Sports Expo
- National Recreation and Parks Association (NRPA) National Conference
- Association of Outdoor Recreation Education Annual Conference

Professional Organizations

- National Intramural Recreational Sports Association (NIRSA)
- National Recreation and Parks Association (NRPA)
- North American Society for Sport Management (NASSM)
- North American Society for the Sociology of Sport (NASSS)
- Association of Outdoor Recreation and Education
- Society of Health and Physical Educators (SHAPE; formerly American Alliance for Health, Physical Education, Recreation and Dance)
- Sport Marketing Association (SMA)

JMU Clubs and Organizations

- Phi Epsilon Kappa
- Sigma Rho Mu
- SRM Major's Club (SLAM)
- Women in Sport and Recreation Management
- Sports Clubs

Faculty

Professors

Reginald F. Foucar-Szocki, David J. Shonk, Julia Wallace Carr, Stephanie T. West

Associate Professors

Benjamin H. Carr Jr., Augustus Hallmon, Gilpatrick Hornsby, Yuan (Pam) Li, Tony Johoo Kim

Assistant Professors

Sungjai "Jay" Hong, Yiran (Kevin) Liu, Sandhya Manjunath, Jonathan Oliveira

Senior Lecturers

Miguel B. Baltazar, Alyssa T. Bosley, Theresa L. Lind

Lecturers

Steven R. Bobbitt, Guy deBrun, Donna Fickes, Dominic Palumbo, Ben Swartz

Programs and Courses

Majors

- Hospitality Management, B.S. (p. 298)
- Sport and Recreation Management, B.S. (p. 431)

Graduate Program

- Sport and Recreation Leadership, M.S. (<https://catalog.jmu.edu/programs/sport-recreation-leadership-ms/>)

Courses

- Hospitality Management (HM) (p. 726)
- Sport and Recreation Leadership (SRL) (<https://catalog.jmu.edu/courses/srl/>)
- Sports & Recreation Management (SRM) (p. 855)

Department of Health Professions

Interim Academic Unit Head: Jamie Frye

Email: healthprofessions@jmu.edu

Phone: (540) 568-7658

Location: Health and Behavioral Studies, Room 3092
Health Professions Website (<http://healthprof.jmu.edu>)

Mission

The mission of the Department of Health Professions is to produce leaders in the dynamic healthcare industry through innovative practices in education, leadership, scholarship, and service.

At the undergraduate level, the department offers a Bachelor of Science degree in Dietetics and a Bachelor of Science degree in Health Services Administration.

The graduate programs in health professions are dedicated to preparing students to become evidence-based critical thinkers in the health and healthcare fields. Specifically, these programs build upon the undergraduate health sciences programs by providing a more detailed knowledge base that is fortified by self-directed learning experiences and the development of practical, clinical and/or research skills.

Vision

To be recognized as a leader in professional healthcare education and practice.

Career Opportunities

- Athletic Trainer (Graduate Program)
- Registered Dietitian/Nutritionist (Graduate Program)
- Health Administrator
- Occupational Therapist (Graduate Program)
- Physician Assistant (Graduate Program)

Co-curricular Organizations

- Madison Athletic Training Student Association
- Health Administration Student Association
- JMU Dietetic Association
- JMU Physician Assistant Student Society
- JMU Student Occupational Therapy Association
- Pre-Occupational Therapy Association
- Pre-Physical Therapy Society
- Pre-Physician Assistant Club
- Coalition of Occupational Therapy Advocates for Diversity

Degree Offerings

The Department of Health Professions offers the following degrees:

- Bachelor of Science in Dietetics
- Bachelor of Science in Health Services Administration
- Master of Science in Applied Nutrition
- Master of Science in Athletic Training
- Master of Occupational Therapy
- Master of Physician Assistant Studies

Special Admission Requirements

Admission to JMU does not guarantee admission to all academic majors and minors. Special applications are required for admission to the upper-level courses of the dietetics program and the health services administration program. Please check program websites for application timelines and deadlines. More information about admission requirements can be found on the program pages.

Admission requirements for programs in the Department of Health Professions vary by program. Refer to the specific program for admission criteria and deadlines.

Faculty

Professors

Jeremy D. Akers, Kirk J. Armstrong, Jamie L. Frye, Paula J. Maxwell, April L. Temple, Danielle M. Torisky, Gerald R. Weniger

Associate Professors

Ana L. Dengo-Flores, John W. Goetschius, Michelle Hesse, Kristina C. Liskey, Connie L. Peterson, Sharon F. Maiewski, Abby J. Massey, Whitney R. Simmons, Ashley Woolfolk Skelly, Sarah R. Spranger, Jennifer R. Walsh, France M. Weaver

Assistant Professors

Zainab A. Ahmed, Julie Patrick Bonham, Sarah Collins, Heidi Hull, Thomas M. Newman, Molly Parker, Amy Russell Yun, Laura Tice

Lecturer

Katherine A. Harrison

Programs and Courses

Majors

- Dietetics, B.S. (p. 245)
- Health Services Administration, B.S. (p. 291)

Graduate Programs

- Applied Nutrition, M.S. (<https://catalog.jmu.edu/programs/applied-nutrition-ms/>)
- Athletic Training, M.S.A.T. (<https://catalog.jmu.edu/programs/athletic-training-msat/>)
- Occupational Therapy, M.O.T. (<https://catalog.jmu.edu/programs/occupational-therapy-mot/>)
- Physician Assistant Studies, M.P.A.S. (<https://catalog.jmu.edu/programs/physician-assistant-studies-mpas/>)

Courses

- Athletic Training Education Program (ATEP) (<https://catalog.jmu.edu/courses/atep/>)
- Health Services Administration (HSA) (p. 711)
- Nutrition (NUTR) (<https://catalog.jmu.edu/courses/nutr/>)
- Occupational Therapy (OT) (<https://catalog.jmu.edu/courses/ot/>)
- Physician Assistant Studies (PA) (<https://catalog.jmu.edu/courses/pa/>)

Department of Health Sciences

Academic Unit Head: Jeffrey E. Herrick

Assistant Academic Unit Head: Theresa M. Enyeart Smith

Email: healthsciences@jmu.edu

Phone: (540) 568-6510

Location: Health and Behavioral Studies, Room 3091

Department of Health Sciences Website (<http://www.healthsci.jmu.edu>)

Mission Statement

Through education, scholarship and service we enhance the health and well-being of individuals and populations.

Goals

The goals of the Department of Health Sciences are to:

- Educate future health professionals through inclusive and engaging learning environments.
- Provide professional service infused with inclusive excellence to the department, college, university and global community.
- Conduct, disseminate and publish cutting-edge research/scholarships in health sciences.
- Contribute to the health and well-being of the JMU community.
- Support the General Education program at JMU.

Co-curricular Organizations

- Eta Sigma Gamma (Health Sciences Honor Society)
- Phi Epsilon Kappa
- Pre-Med Association
- Pre-Occupational Therapy Association
- Pre-Physical Therapy Society
- Pre-Physician Assistant Club

Faculty

Professors

Stephanie L. Baller, Audrey J. Burnett, Theresa M. Enyeart Smith, Dayna Henry, Jeffrey E. Herrick, Catherine Zeman

Associate Professors

Duke Biber, Jamie Gilbert, Raihan Kahn, Jill Lassiter, M. Katherine Ott Walter, Debra L. Sutton

Assistant Professors

Towfiqul Alam, Eugenia Camacho Fernandez, Biswadeep Dhar, Precious Patrick Edet, Matthew Holman, Lauren Kauffman, Matthew Lovesky, Brandon Mendenhall, Gazi Sakir Mohammond Pritom, Kayla Tillet, Brooks Youngmark, Sojib Bin Zaman

Senior Lecturer

Timothy M. Howley

Lecturers

William Boucher, Jordan McCann, Christopher Smith, Sherri Wilson

Academic Advisors

Laura W. Blosser, Alisabeth Gheen

Programs and Courses

Major

- Health Sciences, B.S. (p. 287)

Courses

- Health Sciences (HTH) (<https://catalog.jmu.edu/courses/htb/>)

Department of History

Academic Unit Head: P. David Dillard

Email: history@jmu.edu

Phone: (540) 568-6132

Location: Wilson Hall 3034

Department of History Website (<https://www.jmu.edu/history/index.shtml>)

Mission

We regard history as an ever-evolving discipline based on the rigorous collection and critical analysis of evidence to explore the diversity of human experiences over time. As historians, we narrate the past by interpreting our sources, advancing informed perspectives on past social, political and cultural realities, while also providing essential contextual knowledge for the present. By teaching students how to interrogate the historical record, evaluate arguments based on the strength of the evidence, and communicate complex ideas effectively, we prepare students to become discerning, productive, and knowledgeable participants in a global society.

Inside the Classroom

Students of history will become well-rounded and engaged citizens by:

- Acquiring knowledge of the human condition through the study of political, cultural, and social experiences of people around the world.
- Learning the historical contexts of major political, intellectual, social, economic, religious, and environmental events and transformations.
- Understanding the major cultural developments in fine and popular arts in societies around the world in their historical contexts.
- Discerning how people from a variety of societies hold differing values, ethics, and legal concepts and comprehending how these ideas relate to each other and to Western ideas and values.

Students of history will gain the following skills

- Reading, writing, and speaking effectively and critically.
- Making informed judgments based on existing evidence.
- Locating printed and online information to research a topic exhaustively.
- Critically evaluating textual evidence by identifying its thesis, sources, conclusions, perspective, bias, and reliability.

Beyond the Classroom

While at JMU, students of history will bring the insights and skills they learn in the classroom to their relationships and experiences with their broader communities.

Possibilities for such interactions include:

- Leading efforts to promote open dialogue, civic engagement, and public service.
- Fostering interdisciplinary relationships by engaging with the larger scholarly community at the university.
- Participating in partnerships and events that expand and challenge their own worldview.
- Studying abroad to deepen their understanding of cultural differences and non-US historical perspectives and experiences.
- Becoming involved in professional work through internships.

Career Opportunities

Students graduating with a major or minor in history leave JMU with skills that are sought after by a wide range of employers. History graduates are adaptive learners and able project managers. As a result, they are particularly marketable in a world where professions change rapidly. Some of the career paths that history majors follow include:

- Advertising
- Archives and Libraries
- Business
- Communication
- Digital humanities
- Education
- Government
- Information Management
- Journalism
- Law
- Non-Profit and Non-Governmental Organizations
- Project Management
- Public History and Museums
- Public Policy
- Research
- Teaching
- Writing and Editing

Undergraduate work in history is also a successful foundation for advanced degrees in the humanities, law, the social sciences, medicine, and other disciplines.

Co-curricular Activities and Organizations

- Phi Alpha Theta (National Honor Society for History)
- The Madison Historians

Faculty

Professors

Rebecca N. Brannon, Christian S. Davis, Philip D. Dillard, Evan J. Friss, Michael D. Gubser, Maura E. Hametz, Shah Mahmoud Hanifi, Kevin R. Hardwick, Yongguang Hu, Gabrielle M. Lanier, Kristen L. McCleary, Fawn-Amber Montoya, Margaret M. Mulrooney, David Owusu-Ansah, Steven A. Reich, Michael J. Seth, Emily M. Westkaemper, Andrew D. Witmer

Associate Professors

Jessica B. Davidson, Timothy J. Fitzgerald, Mary K. Gayne, Philip M. Herrington, Colleen M. Moore, Alison D. Sandman

Assistant Professors

Neel Amin, Eva Carrara, Akil A. Cornelius, Jonathan Jones, Sarah R. Meiners

Programs and Courses

Major

- History, B.A. (p. 293)

Minor

- History Minor (p. 508)

Licensure Program

- History Teaching Licensure (p. 587)

Graduate Program

- History, M.A. (<https://catalog.jmu.edu/programs/history-ma/>)

Courses

- History (HIST) (<https://catalog.jmu.edu/courses/hist/>)

Cross-Disciplinary Minors

Students majoring or minoring in history might consider adding a cross-disciplinary minor. The programs below are supported by history faculty and include courses that can be double-counted in the history major or minor.

- African, African American and Diaspora Studies Minor (p. 446)
- American Studies Minor (p. 448)
- Asian Studies Minor (p. 452)
- Classical Studies Minor (p. 465)
- Disability Studies Minor (p. 480)
- Environmental Humanities Minor (p. 485)
- Family Studies Minor (p. 495)
- Historical Archaeology Minor (p. 507)
- Latin American, Latinx and Caribbean Studies Minor (p. 518)
- Medieval and Renaissance Studies Minor (p. 529)
- Middle Eastern Communities and Migrations Minor (p. 530)
- Modern European Studies Minor (p. 533)
- Russian, Eurasian and Slavic Studies Minor (p. 561)
- Science, Technology and Society Minor (p. 562)
- Women's, Gender and Sexuality Studies Minor (p. 576)

School of Integrated Sciences

Co-Director: Stephanie Stockwell
Phone: (540) 568-8980
Email: stockswb@jmu.edu

Co-Director: Henry Way
Phone: (540) 568-2740
Email: wayha@jmu.edu

Location: EnGeo 2122
 School of Integrated Sciences Website (<https://www.jmu.edu/sis/>)

Geography

Co-Director: Dudley Bonsal
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Location: EnGeo 2113

Co-Director: Kayla Yurco
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Location: EnGeo 2109

Kimberly Stewart, **Advisor**
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Phone: (540) 568-6199
Location: EnGeo 2108

Geography Website (<https://www.jmu.edu/cise/geography/index.shtml>)

Mission

The Geography Program empowers, inspires and motivates students to become critical thinkers. Students will be prepared for lifelong learning and be able to respond to intellectual challenges with interest, excitement and competence. The Program trains students to be geographers and global citizens who are actively involved in the world around them. Graduates have the skills to make a professional contribution to the field, and to compete in graduate programs as well as in the job market. They are highly qualified for careers in both research and development, and are equipped for long-term success as professionals in the field of geography.

Through the study of geography, students will:

- Understand and properly use the terminology and concepts that are central to the discipline of geography, and explain how these concepts evolved over time.
- Effectively use appropriate geospatial technologies to address questions about human interactions within the built and natural environments.
- Be productive participants in research efforts aimed at measuring, describing, analyzing and explaining the underlying processes giving rise to geographic phenomena.
- Work effectively in multidisciplinary teams.
- Evaluate human-environment interactions from a holistic point of view that addresses geographic, as well as political, social, economic and ethical factors affecting those interactions.
- Demonstrate civic responsibility and appreciation for cultural and physical diversity from local to global scales.

Career Opportunities

Geography majors achieve career success in many areas. Recent graduates have become environmental scientists and specialists, geographic information scientists and technologists, urban and regional planners, international development specialists, geospatial intelligence analysts, forest and wildlife managers and conservationists, and climatologists and meteorologists.

Co-curricular Activities and Organizations

- American Society for Photogrammetry and Remote Sensing
- Geography Club
- Gamma Theta Upsilon (International Geographical Honor Society)
- Geospatial Intelligence Certificate Program (accredited by the U.S. Geospatial Intelligence Foundation)

Integrated Science and Technology

Director: Christopher Bachmann
Email: bachmacg@jmu.edu
Phone: (540) 568-2735
Location: King 316

Coordinator for Students: Paul W. Henriksen
Email: henrikpw@jmu.edu
Phone: (540) 568-2755
Location: King 321

Integrated Science and Technology Website (<https://www.jmu.edu/cise/isat/index.shtml>)

Mission Statement

The Integrated Science and Technology Program prepares graduates to excel in a complex technological world. Our curriculum empowers students to provide globally responsible multi-disciplinary solutions to important problems with sensitivity to social and ethical complexities.

The foundational concepts of the ISAT program, which distinguishes it from other science- and technology-based programs, is its integration of multiple disciplines by focusing on problem-solving and systems thinking. The unique integrative character of the program is carried by the curriculum content, pedagogy and departmental culture.

Goals

We measure our success by achieving the following goals.

ISAT graduates will be able to do the following:

- Apply and integrate mathematics, physical science, biological science and technology.
- Apply sound experimental methodology.
- Understand the professional requirements for the acquisition and use of information and data.
- Work effectively in multidisciplinary teams.
- Solve technological problems and understand their societal implications.
- Understand and apply the principles of professional ethics.
- Communicate effectively on social, scientific and technical matters.
- Analyze science and technology within broader global, political, economic and social contexts.
- Become autonomous, self-directed learners who recognize the need for lifelong learning.

- Use the computer as an effective problem-solving tool.
- Examine a problem and assemble the tools and knowledge needed to solve it.

Career Opportunities

The ISAT program prepares students for an agile professional career. ISAT graduates are technological problem solvers, clear and confident communicators, and experienced collaborators. This combination of hard and soft skills provides a framework for success in a wide variety of rapidly changing fields. Examples include careers in data science and machine learning, renewable energy, remote sensing, technical and operational consulting, research and development, regulation and policy, advanced manufacturing, and environmental management.

Most ISAT students are employed in the public or private sectors following graduation. Some graduates pursue advanced degrees in areas such as nanotechnology, materials science, system dynamics, engineering, business, sustainability, computer science and public policy.

The flexibility of the ISAT program allows students to select courses that inspire and strategically prepare them for the careers or graduate programs of their choice.

Co-curricular Activities and Organizations

- ISAT Honor Society (IHS)
- Environmental Management, JMU Student Chapter of Air and Waste Management Association
- Association for Facilities Engineering, JMU Student Chapter
- Virginia Biotechnology Association (VaBio), JMU Student Chapter
- Society of Automotive Engineers International, JMU Student Chapter
- IEEE Computer Society, JMU Student Chapter
- Armed Forces Communications and Electronics Association (AFCEA), JMU Student Chapter
- American Wind Energy Association, JMU Student Chapter
- Collaboration for Environment, Health and Safety (CEHS)
- Theta Tau, JMU Student Chapter
- Alpha Epsilon Omega, JMU Student Chapter

Program Accreditation

The ISAT Bachelor's degree program is accredited by the Applied and Natural Science Accreditation Commission of the Accreditation Board for Engineering and Technology (ABET).

Intelligence Analysis

Director: Stephen Marrin

Email: marrinsp@jmu.edu

Phone: (540) 568-2746

Advisor: Kimberly Stewart

Email: stewa3ka@jmu.edu

Phone: (540) 568-6199

Location: EnGeo 2108

Intelligence Analysis Website (<https://www.jmu.edu/ia>)

Mission Statement

The Bachelor of Science degree in Intelligence Analysis provides a multi-disciplinary education for students who seek careers as analysts, with a specialization in intelligence analysis. The degree integrates knowledge from a variety of academic disciplines (philosophy, history, political

science, technology, business) and combines that with professionally oriented knowledge and skills. Students learn innovative ways to structure their thinking to assess complex real-world problems, along with how technology can be employed to acquire data, evaluate that data, and communicate it effectively to others.

Career Opportunities

IA students can find employment in an array of government agencies (both civilian and military), federal contracting and consulting firms, select U.S. and multinational corporations, and state and local law enforcement.

Admission to Intelligence Analysis

Information on admission to the major may be found on the individual program page: Intelligence Analysis, B.S. (p. 325)

Co-curricular Activities and Organizations

- JMU Intelligence Analysis Club
- Women in Intelligence and National Security Student Organization

Faculty

Professors

Christopher G. Bachmann, Mace L. Bentley, Zachary J. Bortolot, Robert N. Brent, Christie-Joy Brodrick Hartman, Jennifer E. Coffman, Steven P. Frysinger, Noel Hendrickson, Cynthia A. Klevickis, Robert A. Kolvoord, Stephen P. Marrin, David K. McGraw, Jonathan J. Miles, Carole L. Nash, Amanda Glenn Sanson, Jonathan H. Spindel, Stephanie Stockwell, Jeffrey D. Tang, Anthony A. Teate, Linda M. Thomas, M. Afzal Upal, Henry Way

Associate Professors

Dudley B. Bonsal, Shannon N. Conley, Orlandrew Danzell, Jennifer Davis, Tobias Gerken, Roderick H. MacDonald, Galen B. Murton, John A. Robinson, Rebecca A. Simmons, Xuebin Wei, Emily York, Kayla Yurco, Hao Zhang

Assistant Professors

Phillip M. Baxter, Mesude Bayrakci Boz, Qingqing Chen, Stefano Colafranceschi, Adam Dohrenwend, Zachary Hallberg, Tyler Mitchell, Ehren Reid Von Moler, Rachel Palkovitz, Giangiuseppe Pili, Timothy J. Prestby, Brooke Rose, Samar Saleh, Jared Stoltzfus, Kevin Varga, Raafat M. Zaini

Senior Lecturer

Paul W. Henriksen

Programs and Courses

Majors

- Geography, B.A. (p. 271)
- Geography, B.S. (p. 275)
- Integrated Science and Technology, B.S. (p. 321)
- Intelligence Analysis, B.S. (p. 325)

Minors

- Geography Minor (p. 500)
- Integrated Science and Technology Minor (p. 516)

Graduate Programs

- Cyber Intelligence Certificate (<https://catalog.jmu.edu/programs/cyber-intelligence-certificate/>)

- Integrated Science and Technology, M.S. (<https://catalog.jmu.edu/programs/integrated-science-technology-ms/>)

Courses

- Geography (GEOG) (<https://catalog.jmu.edu/courses/geog/>)
- Intelligence Analysis (IA) (<https://catalog.jmu.edu/courses/ia/>)
- Integrated Science and Technology (ISAT) (<https://catalog.jmu.edu/courses/isat/>)

Cross-Disciplinary Programs

The School of Integrated Sciences participates in several cross disciplinary programs. These include:

- Biotechnology, B.S. (p. 211)
- Environmental Information Systems Minor (p. 487)
- Environmental Management Minor (p. 488)
- Environmental Science Minor (p. 490)
- Environmental Studies Minor (p. 492)
- Global Supply Chain Management Minor (p. 504)
- Materials Science Minor (p. 524)
- Science, Technology and Society Minor (p. 562)
- Urban and Regional Studies Minor (p. 575)

Department of Interdisciplinary Liberal Studies

Academic Unit Head: Phil Frana

Email: idl@jmu.edu

Phone: (540) 568-5260

Location: Gabbin Hall, Room 118

Interdisciplinary Liberal Studies Website (<http://www.jmu.edu/idls>)

Interdisciplinary Liberal Studies (IDLS) offers programs in the tradition of a liberal education, focusing on transferable skills and the methods and knowledge from multiple disciplines. Our programs allow students flexibility to pursue their interests and combine them to create unique learning opportunities.

Faculty

Professors

Matthew A. Chamberlin, Jared Featherstone, Philip L. Frana, Connie Frigo, Michael J. Klein, G. Fletcher Linder, Scott A. Paulson, Mikaela Schmitt-Harsh

Associate Professors

H. Gelfand, Jennifer M. Mangan

Assistant Professors

Johnathan Walker, Felix Wang

Programs and Courses

Majors

- Individualized Study, B.A. with a Major in General Studies (p. 315)
- Individualized Study, B.A. with a major in Independent Scholars (p. 317)
- Individualized Study, B.S. with a major in General Studies (p. 316)
- Individualized Study, B.S. with a major in Independent Scholars (p. 318)

Courses

- Independent Scholars (IND) (p. 733)
- Individualized Study (IS) (p. 733)

Department of International Business

Academic Unit Head: Irvine "Bud" Clarke III

Email: clarkeix@jmu.edu

Phone: (540) 568-3049

Location: Hartman Hall, Room 1135

Department of International Business Website (<http://www.jmu.edu/cob/international-business/>)

Mission

The mission of the Department of International Business is to provide nationally competitive undergraduate education in the study of international business. The program is cross-disciplinary and designed to prepare students for participation in organizations that require a broad business perspective, applicable skills in global business and an understanding of the complexity of the globalized world. The program also prepares students for graduate programs in business and related fields.

Goals

- To provide a strong foundation in the international implications for the business disciplines.
- To develop students' cultural understanding through course work and a required international experience.
- To enable students to gain proficiency in a second language.
- To enable students to develop extensive knowledge of a specific region of the world related to the second language.
- To develop students' competencies in addressing specific managerial issues related to international business, such as ethical considerations, managing foreign exchange risk, managing a multicultural work force, etc.
- To facilitate students' continuing development of written and oral communication skills.

Career Opportunities

- Consumer marketing and business-to-business marketing, including sales, consumer relations and market analysis.
- Consulting, including management analysis, strategic planning, expansion overseas and market entry strategies.
- Marketing research, including digital marketing and account analysis.
- Corporate finance, financial analyst, project management, junior analyst-finance, planning and administration.
- Capital management, credit manager, assistant treasurer – funds flow and risk management.
- Banking and financial institutions, lending officer and marketing officer.
- Foreign commercial service officer in the U.S. Department of Commerce.
- Careers that require analytic acumen and the ability to adapt quickly to diverse and changing environments.

Co-curricular Activities and Organizations

- Global Commerce Club: This student club is open to all JMU students. It sponsors speakers, informational sessions and social events.
- American Marketing Association: This student club is affiliated with the American Marketing Association, a national marketing

- association for marketing professionals, faculty and students. MMA is open to all JMU students. MMA is comprehensive with its marketing programming and offers students information and activities in direct marketing, retailing and marketing management.
- Pi Sigma Epsilon: This is a professional fraternity, which focuses on programming and extracurricular experiences in marketing, personal selling and sales management.
- Financial Management Association: FMA is an organization designed for those interested in pursuing a career in the financial arena.

Faculty

Professors

Nevin Cavusoglu, Irvine "Bud" Clarke III, Laura M. Leduc, Qingfeng "Wilson" Liu, Hui He Sono, Mert Tokman

Associate Professors

Piyusha Mutreja

Assistant Professors

Kira Haensel, Mrithyunjayan Nilayamgode

Lecturers

Philip S. Heap, Fadi Majdoub, David L. Parker

Programs and Courses

Major

- International Business, B.B.A. (p. 336)

Courses

- International Business (IB) (p. 750)

Department of Justice Studies

Academic Unit Head: Peggy S. Plass

Email: plassms@jmu.edu

Phone: (540) 568-7124

Location: Darcus Johnson Hall 2013

Department of Justice Studies Website (<http://www.jmu.edu/justicestudies>)

Mission

Justice Studies is committed to offering a cross-disciplinary, intellectually-challenging and vocationally-relevant course of study for persons interested in academic or applied careers in justice studies at the community, national or global level.

Goals

To carry out this mission, the department seeks to:

- Help students develop a comprehensive understanding of justice studies.
- Empower students to analyze and explain justice and injustice and their impact on individuals, communities, institutions and nations.
- Sharpen students' abilities to think and reason critically, to practice sound methodological skills, to communicate effectively, and to develop solutions.
- Prepare students to utilize and produce scholarship in the field of justice studies.
- Equip students with essential competencies, knowledge and skills for a broad spectrum of pre-professional, career, and community and civic pathways.

Career and Professional Opportunities

Justice Studies offers students opportunities to develop a wide range of skills, knowledge, competencies, and habits of mind that can be applied within many career, professional, community, and educational settings in the future. Examples of career fields which our graduates have pursued include the following:

- Peace Corps
- AmeriCorps
- Governmental and non-governmental organizations
- Law enforcement
- Corrections
- Judicial services
- Law and legal careers
- Advocacy/Lobbying
- Policy analysis
- Nonprofit organizations (national and international)
- Human and Social services
- Domestic and international security
- Intelligence analysis
- Diplomatic Corps
- Researcher/data analyst
- Program evaluation
- Academia/Education
- Graduate and professional study

Faculty

Professors

Taimi L. Castle, Benjamin D. Meade, Peggy S. Plass, Christine M. Robinson, Heather L. Scheuerman

Associate Professors

Daniel Beers, Gianluca De Fazio, Tara N. Parsons Kristiansen, Graciela Perez, Margarita Poteyeva, Case Watkins

Assistant Professors

Krista Billingsley, Melissa M. Švigelj

Lecturer

Daisy Breneman

Programs and Courses

Majors

- Justice Studies, B.A. (p. 338)
- Justice Studies, B.S. (p. 343)

Cross-Disciplinary Minors

- Criminal Justice Minor (p. 474)
- Humanitarian Affairs Minor (p. 514)
- Social Justice Studies Minor (p. 563)
- Youth Justice Minor (p. 581)

Courses

- Justice Studies (JUST) (p. 756)

Department of Kinesiology

Academic Unit Head: Liz Edwards

Email: jmukinesiology@jmu.edu

Phone: (540) 568-6145

Location: Godwin Hall, Room 307

Department of Kinesiology Website (<https://www.jmu.edu/chbs/kinesiology/>)

Mission

The Department of Kinesiology is dedicated to the development of future leaders in professions that maximize the potential of individuals and society through physical activity. Programs include exercise science and teacher education in physical education and health.

The department is committed to providing:

- Outstanding undergraduate programs based on the criteria of relevant professional associations which will enable graduates to achieve success in their professional endeavors.
- Programs that build upon the strong liberal studies background provided through General Education.
- Opportunities that challenge students to think critically
- Use of technological advances.
- An appreciation of the global community.
- Quality graduate programs that complement the undergraduate programs and provide qualified students with an opportunity for advanced study in the disciplines of kinesiology.
- Contributions to the university's general education curriculum through programs designed to promote lifelong fitness and wellness.
- Service to JMU, the professions and local community through our unique knowledge and expertise.
- Research and development projects that push back the boundaries of knowledge and promote effective practice in the kinesiology disciplines.

Our aim is to develop professionals committed to engaging with diverse communities in ways that help all individuals reach their full potential. We are dedicated to appreciating differences in perspective, identity, and experience through cultivating class environments, curriculum, and pre-professional opportunities that align with the spirit of inclusion. This mission ultimately gets us closer to our vision of high-quality education.

Career Opportunities and Marketable Skills

The Exercise Science concentration provides students with the opportunity to gain knowledge and applied experiences to enter fields such as:

- Fitness and Wellness Industry
 - Personal Trainers
 - Corporate Wellness/Fitness Directors
 - Certified Exercise Physiologists
 - Strength and Conditioning Coaches
- Health Care Industry (when combined with a pre-professional health minor and graduate studies)
 - Physical Therapy
 - Occupational Therapy

- Physician Assistant
- Medicine
- Athletic Training
- Cardiac Rehabilitation
- Medical Technician

The Physical and Health Education concentration provide students with the opportunity to gain knowledge and applied experiences to enter the field of teaching Health and Physical Education pre-K-12.

The Coaching Education Minor provide students with the opportunity to gain knowledge and applied experiences to enter fields such as:

- Coaching youth and interscholastic teams
- College/University coaching
- Strength and Conditioning coaching

Faculty

Professors

Sarah Carson Sackett, Elizabeth S. Edwards, Trent A. Hargens, Douglas R. Hochstetler, Nicholas D. Luden, Catherine A. McKay, Thomas E. Moran, Susan B. Nye, Michael J. Saunders, Christopher J. Womack

Associate Professors

Lori Gano-Overway, Stephanie P. Kurti-Luden, Eric D. Magrum

Assistant Professors

Andrew D'Lugos

Senior Lecturer

Chelsea Duncan

Lecturers

Robert J. Lifka, Jana R. Walters

Programs and Courses

Major

- Kinesiology, B.S. (p. 348)
 - The exercise science concentration prepares students for direct entry into the fitness and wellness industries, as well as prepares students for graduate studies in exercise physiology and many allied health related fields. Contact: Dr. Chris Womack (womackcx@jmu.edu).
 - The physical and health education teacher education (PHETE) concentration prepares students to develop and administer high quality physical and health education classes in PreK-12. Students must complete the 1-year MAT in Physical and Health Education Teacher Education to meet initial licensure requirements. Contact: Dr. Susan Nye (nyesb@jmu.edu).

Minor

- Coaching Education Minor (p. 468)
 - This minor provides students with the opportunity to gain knowledge and applied experiences to enter fields such as coaching youth, interscholastic and college teams. Contact: Dr. Lori Gano-Overway (ganoovla@jmu.edu).

Graduate Programs

- Kinesiology, Exercise Physiology Concentration, M.S. (<https://catalog.jmu.edu/programs/kinesiology-exercise-physiology-concentration-ms/>)

- This program prepares students to become practicing exercise physiologists in fields such as cardiac rehabilitation, to assume leadership positions in fitness and wellness industries, or for continuing studies in physiology or health-related fields. This program offers thesis and non-thesis options. Program Director: Dr. Stephanie Kurti-Luden (kurtisp@jmu.edu).
- Education, PHETE (Physical and Health Education Teacher Education) Concentration, M.A.T. (<https://catalog.jmu.edu/programs/education-phete-physical-health-education-teacher-education-concentration-mat/>)
 - This 1-year program is a teacher-licensure program that must be combined with specific undergraduate coursework (see PHETE concentration above). Students who did not complete JMU's Kinesiology, PHETE Concentration may have to complete additional undergraduate coursework to be eligible for admission to the program and teacher licensure upon completion. Program Director: Dr. Susan Nye (nyesb@jmu.edu).
- Physical and Health Education Teacher Education, M.A. (<https://catalog.jmu.edu/programs/physical-health-education-teacher-education-ma/>)
 - This 12-month graduate program is for teachers who already possess a teaching license and want to seek training in the areas of health and physical education. Program Director: Dr. Susan Nye (nyesb@jmu.edu).
- Sport Coaching Graduate Certificate (<https://catalog.jmu.edu/programs/sport-coaching-graduate-certificate/>)
 - Program Director: Lori Gano-Overway (ganoovla@jmu.edu)

Courses

- Kinesiology (KIN) (<https://catalog.jmu.edu/courses/kin/>)

Department of Learning, Technology and Leadership Education

Academic Unit Head: Michele Estes

Email: estesmd@jmu.edu

Phone: (540) 568-8012

Location: Memorial Hall, Room 3310B

Department of Learning, Technology and Leadership Education Website
(<http://www.jmu.edu/coe/ltle>)

Graduate Programs Coordinator: Danielle Maxham

Email: gallagda@jmu.edu

Phone: 540-568-3298

The Department of Learning, Technology and Leadership Education houses undergraduate courses and minors, graduate certificates, and graduate degrees that prepare students for advanced learning in adult education and human resource development, educational leadership, learning design and technology, and teacher leadership. Students are prepared for careers in education in a wide range of professional settings.

The LTLE department mission is to address complex problems through authentic inquiry into human resources, leadership, and the learning sciences. Our vision is to effect equitable, sustainable change in education, business, industry, and community.

Faculty

Professors

Edward J. Brantmeier, Michele D. Estes, Oris T. Griffin, Dana L. Haraway, William L. White

Associate Professors

James P. Bywater, Tabitha K. Coates, Ying Jin, Monica Smith-Woofter

Assistant Professors

Parama Chaudhuri, Eileen Oliver-Eggert, Amelia Underwood, Katherine Walters

Clinical Associate Instructors

Cecil Snead

Programs and Courses

Minor

- Educational Media Minor (p. 482)

Graduate Programs

- Adult Education/Human Resource Development, M.S.Ed. (<https://catalog.jmu.edu/programs/education-adult-education-human-resource-development-msed/>)
- Education, Ed.D. (<https://catalog.jmu.edu/programs/education-edd/>)
- Education, Educational Leadership Concentration, M.Ed. (<https://catalog.jmu.edu/programs/education-educational-leadership-concentration-med/>)
- Education, Learning Design and Technology Concentration, M.Ed. (<https://catalog.jmu.edu/programs/education-learning-design-technology-concentration-med/>)
- Education, Teacher Leadership Concentration, M.Ed. (<https://catalog.jmu.edu/programs/education-teacher-leadership-concentration-med/>)

- Educational Leadership Certificate (<https://catalog.jmu.edu/programs/educational-leadership-certificate/>)
- Learning Design and Technology Graduate Certificate (<https://catalog.jmu.edu/programs/learning-design-technology-graduate-certificate/>)
- Teacher Leadership Certificate (<https://catalog.jmu.edu/programs/teacher-leadership-certificate/>)

Courses

- School Administration (ADSU) (<https://catalog.jmu.edu/courses/adsu/>)
- Adult Education Human Resource Development (AHRD) (<https://catalog.jmu.edu/courses/ahrd/>)
- Curriculum, Instruction and Educational Equity (CIEE) (<https://catalog.jmu.edu/courses/ciee/>)
- Human Resource Development (HRD) (p. 728)
- Learning, Design and Technology (LDT) (<https://catalog.jmu.edu/courses/ldt/>)
- Leadership, Equity, Innovation and Learning (LEIL) (<https://catalog.jmu.edu/courses/leil/>)
- Learning, Technology and Leadership Education (LTLE) (<https://catalog.jmu.edu/courses/ltle/>)
- Teacher Leadership (TCLD) (<https://catalog.jmu.edu/courses/tcld/>)

Department of Management

Interim Academic Unit Head: Laura M. Leduc

Email: leducm@jmu.edu

Phone: (540) 568-5171

Location: Zane Showker Hall, Room 4035

Department of Management Website (<http://www.jmu.edu/cob/management>)

Mission

The mission of the Department of Management is to provide students with a nationally competitive undergraduate education in the field of management studies. The Department of Management major prepares future organizational leaders to manage effectively in a dynamic global business environment. The Department of Management faculty strive to develop students' theoretical and practical managerial expertise, focusing on interpersonal and team skills, situational and data analysis, innovation and creativity, problem solving, evidence-based decision making, and ethical reasoning.

Goals

The overall goals of the Department of Management (MGT) program are:

- To prepare graduates of the MGT major for a wide variety of management careers and leadership roles.
- To provide instruction relating to the study and practice of management to undergraduate and graduate students in the College of Business.

Core Management Objectives

After studying management, students will possess an advanced understanding of:

- The functions of management: Management majors will understand the basic managerial functions of planning, organizing, leading and controlling and how these functions are used to achieve organizational objectives in dynamic, culturally diverse and competitive global business environments.
- The legal and ethical environment of management: Management majors will recognize and understand the behavioral implications of the ethical and legal boundaries of managerial decision making and subsequent actions.
- Effective decision making and problem solving: Management majors will develop and enhance critical thinking, analytical and decision-making skills. These skills will be demonstrated in students' ability to engage successfully in situational analysis, problem recognition, integration of information from diverse sources, use of logic and reasoning to develop solutions and alternative courses of action, and in the implementation of action plans.
- Organizational strategy and design: Management majors will be able to identify and explain elements of the internal and external environments that influence the formulation, implementation and evaluation of organizational strategy, structure and design.

Career Opportunities

- Project Manager
- Human Resources Generalist or Specialist
- Management Consultant
- Business Analyst

- Operations or Logistics Manager
- Supply Chain Analyst
- Labor Relations Specialist
- Small Business Owner/Manager
- Customer Service/Relations Manager
- Sales Manager
- Retail Manager
- Facility/Events Manager

Co-curricular Activities and Organizations

- Sigma Iota Epsilon (SIE): National honors fraternity for management majors.
- Society for Human Resource Management (SHRM): Student chapter of a national organization that provides students with opportunities to learn more about the field of human resource management through speakers, facility tours, networking and other activities.
- Association for Supply Chain Management (APICS): JMU chapter of APICS, a national professional organization that provides research, education and certification programs that foster supply chain excellence, innovation and resilience.
- The JMU Society of Real Estate (SRE): A professional business organization that allows students to explore different facets of the real estate industry through speaker events, workshops, company site visits and other activities.

Faculty

Professors

Laura M. Leduc, Fariss-Terry I. Mousa, Marshall W. Pattie, William J. Ritchie III, Matthew A. Rutherford, Ali Shahzad

Associate Professors

Sergiy D. Dmytryiev, C.K. Lee, David R. Peterson, Eric M. Stark

Assistant Professors

Sara J. Bianco, Cody Bradley, Kira Haensel, Jinhee Moon

Senior Lecturers

Robert G. Eliason, Carol W. Hamilton, Fernando A. Pargas, Christina C. Roeder, Ruairi R. Walsh

Lecturers

Sarah Dodson

Programs and Courses

Major

- Business Management, B.B.A. (p. 214)

Courses

- Management (MGT) (p. 770)

Department of Marketing

Academic Unit Head: Theresa B. Clarke

Email: clarketb@jmu.edu

Phone: (540) 568-3238

Location: Showker Hall, Room 4024

Department of Marketing Website (<http://www.jmu.edu/cob/marketing>)

Mission

The mission of the Department of Marketing is to develop highly successful marketing professionals who deliver exceptional value to their customers, employers and communities and to advance the practice of marketing through academic and professional engagement.

The marketing curriculum focuses on traditional and emerging marketing skills needed in various business settings and industries. The curriculum seeks to develop competence in five areas: factual and conceptual knowledge, problem-solving skills, communication skills, experiential learning and the use of information technology. Students will be prepared to enter corporate, small business or not-for-profit environments with highly valued skills and an understanding of the need for continuous learning.

Goals

- To deliver a solid foundation of the concepts and theories of the marketing discipline, including market environmental issues; strategic market planning, implementation and evaluation; integrated marketing communications; buyer behavior; professional selling; market analytics; and development of marketing programs.
- To engage students in critical thinking processes, requiring an in-depth analysis of qualitative and quantitative market data and the development of subsequent marketing strategies based on this analysis.
- To enable students to evaluate marketing alternatives and commit to a course of action, using financial, organizational, environmental and ethical criteria as bases for decision making.
- To equip students with a diverse array of information technology tools and techniques, empowering them to improve marketing and overall business performance and deliver greater value to customers.
- To foster students' ongoing growth in soft skill success including active listening, writing, and oral communication, while also nurturing essential professional qualities such as initiative, curiosity, problem-solving, resilience, accountability, and flexibility.
- To develop students' marketing-based interpersonal skills, such as group-based negotiation, consensus building, delegation and performance evaluation.
- To integrate classroom instruction with real-world industry practices across the marketing curriculum, facilitating marketing students' access to hands-on experiential learning opportunities relevant to various marketing career paths.
- To introduce students to the fundamentals of professional selling and develop students' interpersonal communication skills through face-to-face role-play scenarios.

Career Opportunities

- Professional sales: product and service sales positions across a broad range of companies and industries as sales development representatives, business development representatives, account

managers, inside sales representatives and outside sales representatives.

- Consumer marketing and business-to-business marketing: market analysis, product development, services marketing, global marketing and customer relationship management.
- Digital marketing: search engine marketing, search engine optimization, mobile marketing, digital account management, content marketing, social media marketing and marketing analytics.
- Not-for-profit marketing: fund-raising, direct marketing and customer service.
- Product/Brand management: product specialist, channel development and promotional marketing.
- Retailing: store management, merchandise buying, trend analysis, omnichannel retail management and visual merchandising.

Co-curricular Activities and Organizations

- AMA (<https://www.amajmu.com/>): This student club is affiliated with the American Marketing Association, a national marketing organization for marketing professionals, faculty, and students. AMA is open to all JMU students. AMA is comprehensive with its marketing programming and offers students information and activities in marketing management.
- DECA (https://beinvolved.jmu.edu/DECA/club_signup): Collegiate DECA is a student-driven business organization that values competence, innovation, integrity, and teamwork. DECA prepares emerging leaders and entrepreneurs with interests in marketing, finance, hospitality, and management.
- Delta Mu Sigma (<https://www.jmu.edu/cob/students/student-organizations.shtml>): DMS is a local honor society dedicated to the commitment of growth, innovation, and integrity of digital marketing scholarship and practice. Admission is by application only.
- Madison Sales Club (<https://www.jmu.edu/cob/centers/center-for-professional-sales/students/index.shtml>): This organization is open to all majors and strives to help all students interested in sales and give them the knowledge and skills they need to be successful in a professional sales role.
- Mu Kappa Tau (<https://pse.org/mkt/>): M.K.T. is a national scholastic honor society for marketing majors. Admission is by invitation only.
- Scratch Pad (Ad Club) (<https://www.scratchpadjmu.com/>): Scratch Pad is JMU's only student advertising club and a student chapter of the American Advertising Federation (AAF). This organization is open to all majors and competes in the National Student Advertising Competition each year.

Faculty

Professors

Irvine "Bud" Clarke III, Theresa B. Clarke, William T. Faranda, Mert Tokman

Associate Professors

Gizem Atav, Kelly A. Naletelich, Timucin Ozcan, Janna M. Parker

Assistant Professors

Ali Anwar, Joshua Groves, Carissa Kim

Senior Lecturers

Canessa N. Collins, Lindsay M. King, Robert M. McMillen, Cathleen S. Snyder, Richard Tate

Lecturers

Steve W. Kozak, Fadi Majdoub, Brooke Rhodes, Amy Strunk

Programs and Courses

Major

- Marketing, B.B.A. (p. 353)

Courses

- Marketing (MKTG) (p. 773)

Department of Mathematics and Statistics

Academic Unit Head: Krishnan "Ravi" Shankar

Email: mathstat@jmu.edu

Phone: (540) 568-6184

Location: Roop Hall, Room 305

Department of Mathematics and Statistics Website (<http://www.jmu.edu/mathstat>)

Mission

The Department of Mathematics and Statistics provides a program of study in the mathematical sciences that meets the needs of a wide variety of students and makes a continuing contribution to the advancement of mathematical and statistical knowledge and its dissemination. The program provides opportunities for in-depth study that can lead to careers as analysts, mathematicians and statisticians in private and public sectors, teachers of mathematics, and further study in graduate school. The program provides support for the mathematical and statistical needs of students in the natural sciences, integrated sciences, social sciences, and professional and pre-professional programs. The program meets the general education needs of all students, providing an understanding of mathematical and statistical thinking and approaches to problem solving.

We are committed to promoting mathematics and statistics as an art of human endeavor as well as a fundamental method of inquiry into the sciences and a vast array of other disciplines. We are committed to encouraging an attitude of appreciation and support for mathematics and statistics in current university students and, through them, the next generation of citizens. We are also committed to fostering an appreciation for the effective use of applied mathematics and statistics in connection with and support of other disciplines for those students majoring in other subjects.

Goals

As a major in mathematics or statistics, a student can expect to use and build on skills such as:

- Thinking critically
- Formulating and solving problems
- Communicating solutions clearly, both orally and in writing

These skills have been gained in previous courses in mathematics, statistics and other areas. As the breadth of knowledge of the subject grows, students gain an increased understanding and appreciation of the fact that mathematics is truly a universal language whose creation and applications cut across all boundaries of race, class, culture and time.

There also will be opportunities for students to experience the challenges and rewards of faculty-mentored research, individually or as a member of a team, as they investigate mathematical and statistical problems that extend beyond those normally encountered in the classroom. Students increase their abilities to prove theorems, understand complex structures and apply mathematics and statistics in many real-world settings. The program students choose will make it possible for them to acquire strong preparation for professional applications in mathematics and statistics, teaching, natural and social sciences or other technical areas, and for graduate work.

Programs

The Department of Mathematics and Statistics offers the B.A. and B.S. degrees with a major in mathematics and the B.S. degree with a major in statistics. There is a program for a major in mathematics that qualifies a student for initial teaching licensure. Minors are offered in mathematics, statistics, computational analytics and data analytics. The department also offers the M.Ed. degree in Mathematics.

The department also recognizes the importance of providing courses for non-specialists who need to make effective use of mathematics or statistics in their chosen careers.

The university is an institutional/educational member of the American Mathematical Society, the American Statistical Association, the Mathematical Association of America and the Society for Industrial and Applied Mathematics.

Students are strongly encouraged to participate in the numerous undergraduate research opportunities as well as individual and small group projects available in the department. Opportunities exist through the Center for Computational Mathematics and Modeling, the JMU Experimental Mathematics Lab, the Statistical Consulting Center, honors theses and independent studies with individual faculty mentors.

Majors in the department are expected to participate in assessment activities. Assessment information is used to assist faculty members in modifying curricula.

Faculty

Professors

Elizabeth A. Arnold, Elizabeth T. Brown, Alex Capaldi, Mindy Capaldi, Lihua Chen, Joshua E. Ducey, Steven T. Garren, Hasan Hamdan, Nusrat Jahan, Brant C. Jones, Caroline P. Lubert, Stephen K. Lucas, Kane Nashimoto, Hala Y. Nelson, Minah Oh, Samantha C. Bates Prins, Jason D. Rosenhouse, Alexis L. Stevens, Eva M. Strawbridge, Laura Taalman, Roger J. Thelwell, Anthony L. Tongen, Leonard A. Van Wyk, D. Brian Walton, Debra Polignone Warne, Paul G. Warne, Ling Xu

Associate Professors

Richard P. Busi, David L. Duncan, Rebecca E. Field, Katie S. Quertermous, Dinesh R. Sharma, John (Zig) M. Siegfried, John J. B. Webb, James C. Willingham, Prabhathi Withana Gamage, Celestine A. Woodruff

Assistant Professors

Meric Augat, Johnathan Bush, Beth S. Cochran, Vladislav Kokushkin, Behnaz Moradi-Jamei, Ali Tavasoli, Zev Woodstock

Senior Lecturers

Arlene R. Casiple, Jennifer S. Gibson, Colleen Watson

Lecturers

Stephanie Berry, Rachel Koch, Jie Shi Liew, Sanduni Palliyage, Elizabeth Ben Ward

Programs and Courses

Majors

- Mathematics, B.A. (p. 355)
- Mathematics, B.S. (p. 357)
- Statistics, B.S. (p. 433)

Minors

- Computational Analytics Minor (p. 470)
- Data Analytics Minor (p. 477)
- Mathematics for Elementary and Middle Grades Teachers Minor (p. 525)
- Mathematics Minor (p. 526)
- Statistics Minor (p. 570)

Graduate Programs

- Data Science for Society Graduate Certificate (<https://catalog.jmu.edu/programs/data-science-society-graduate-certificate/>)
- Mathematics, M.Ed. (<https://catalog.jmu.edu/programs/mathematics-med/>)

Courses

- Data Sciences (DATA) (<https://catalog.jmu.edu/courses/data/>)
- Mathematics (MATH) (<https://catalog.jmu.edu/courses/math/>)

School of Media Arts and Design

Director: Joe Hinshaw

Email: smad@jmu.edu

Phone: (540) 568-7007

Location: Harrison Hall, Room 0283

School of Media Arts and Design Website (<http://www.jmu.edu/smad>)

Mission Statement

The School of Media Arts and Design prepares students to serve as creative, reflective and highly skilled media producers and storytellers, sensitive to the personal and social impact of the messages they create and to the important role media can play in advancing the public interest.

Vision

To be a national leader in media education, providing innovative programs that embrace and integrate traditional media concepts, values and skills with new and evolving technologies.

We value:

- Inquiry-based learning, both independent and collaborative, that emphasizes the creative process and is distinctly innovative, challenging, cooperative and collegial.
- Historical, legal and ethical literacy and critical thinking skills that enable students to reach thoughtful and principled decisions throughout their careers.
- Diverse communities and global perspectives acquired through study and experience.
- Strong co-curricular, extracurricular and internship activities that enrich education and enhance an understanding of the role of media in society.

Goals

Graduating SMAD students will:

- communicate a message via a well-researched, targeted story appropriate for the audience and medium.
- compose written products characteristic of entry-level media arts and design professionals.
- apply ethical decision-making to the media production process.
- integrate prior learning in order to analyze, plan, create, evaluate, revise and collaborate on a capstone-level creative product.
- apply critical thinking and media literacy skills when creating and consuming media, including appropriate and ethical use of AI.
- produce and disseminate media while exercising current legal rights and responsibilities.

As faculty and staff in James Madison University's School of Media Arts and Design, we acknowledge the impact we and our students have on representation and narrative, as well as the role media can play in advancing public interest. The power of media and our responsibility as producers and storytellers are at the forefront of our experiential learning and critical analysis. We are dedicated to inclusive excellence, and we encourage activities that enrich our shared experience. We are here to uplift underrepresented communities in our academic pursuits, and we strive to build a community free of prejudice, intimidation and discrimination through our learning and teaching, scholarship and service.

Career Opportunities

The school integrates the fields and study of mass communication with an emphasis on effective visual storytelling across four areas of concentration: creative advertising, digital video and cinema, interactive design, and journalism. It offers a program of study that prepares students for careers that incorporate mass media related areas of expertise and for graduate study in mass communication and related areas.

Co-curricular Activities

Majors and interested non-majors are encouraged to participate in co-curricular activities and organizations associated with the school. Co-curricular activities entail practical media arts and design experiences for which credit is available through practica or other applied courses; however, students may participate in most co-curricular opportunities without receiving course credit.

- The Breeze is the student newspaper.
- BreezeTV is the student broadcasting activity.
- Curio is one of several student-produced magazines in SMAD.
- WMRA-FM is the campus-based public radio station serving the Shenandoah Valley.
- WXJM-FM is the student radio station.
- ESPN+ productions of JMU Athletics

Special Admission and Retention Requirements, Minimum Grades, and Concentration Changes

Information on admission to the School of Media Arts and Design, retention and minimum grade requirements, and concentration changes may be found on the individual program pages.

Faculty

Professors

Dolores Flamiano, M. Joseph Hinshaw, Shelly L. Hokanson, Dietrich Maune, Imelda C. O'Reilly, Kevin J. Reynolds, Shaun Wright, Alexandra M. Vilela, Nan Zheng

Associate Professors

Ryan T. Alessi, Nefin Dinç, Chen Guo, Adrienne Hooker, Elisabeth M. Kvernen, A. Talé Mitchell, Morgan W. Smalls, David Wang

Assistant Professors

Olivia Booker, Michelle Herman, Angela Hayslett, Elim Hernandez, Ryan C. Parkhurst, Natascha Roelsgaard

Lecturers

Damon Dillman, Matthew Leech, Joseph R. Loyacano III

Programs and Courses

Majors

- Media Arts and Design, B.A. (p. 359)
- Media Arts and Design, B.S. (p. 364)

Teaching Endorsement

- Media Arts and Design Journalism Teaching Endorsement (p. 588)

Courses

- Media Arts and Design (SMAD) (p. 782)

Department of Middle, Secondary and Mathematics Education

Interim Academic Unit Head: Angela W. Webb

Email: webbwaw@jmu.edu

Phone: (540) 568-5834

Location: Memorial Hall, Suite 3200

Assistant Academic Unit Head: Kristina J. Doubet

Email: doubetkj@jmu.edu

Graduate Program Director: Melanie Shoffner

Email: shoffnme@jmu.edu

Department of Middle, Secondary and Mathematics Education Website
(<http://www.jmu.edu/coe/msme>)

The Department of Middle, Secondary and Mathematics Education houses undergraduate and graduate programs that prepare students for teaching careers in K-12 schools. All programs include extensive fieldwork to ensure varied experiences working with children and families in school settings.

Faculty

Professors

Mary Elizabeth Cancienne, Eric M. Carbaugh, Michelle D. Cude, Kristina J. Doubet, Mary "Katie" Dredger, Mark E. L'Esperance, Mark A. Lewis, Amanda G. Sawyer, Melanie L. Shoffner, Ann H. Wallace, Angela W. Webb

Associate Professors

Zareen G. Aga, Robbie L. Higdon, Sarah M. Lupo

Assistant Professors

Leonard Richards, Jr., Marcus Wolfe

Advisor

Amber Loyacano

Programs and Courses

Majors

- Middle Grades Education, B.S. (p. 369)
- Secondary Education, B.S. (p. 419)

Minor

- Pre-Professional Secondary Education (SEED) Minor (p. 555)

Graduate Programs

- Education, Initial Licensure in Middle Education Concentration, M.A.T. (<https://catalog.jmu.edu/programs/education-initial-licensure-middle-education-concentration-mat/>)
- Education, Initial Licensure in Secondary Education Concentration, M.A.T. (<https://catalog.jmu.edu/programs/education-initial-licensure-secondary-education-concentration-mat/>)
- Education, Secondary Education (Grades 6-12) Concentration, M.A.T. (<https://catalog.jmu.edu/programs/education-secondary-education-grades-612-concentration-mat/>)

Courses

- Math Education (MAED) (<https://catalog.jmu.edu/courses/maed/>)
- Middle Education (MIED) (<https://catalog.jmu.edu/courses/mied/>)
- Middle Education/Secondary Education (MSSE) (<https://catalog.jmu.edu/courses/msse/>)
- Vocational Education (VOED) (p. 864)

Department of Military Science

Academic Unit Head: LTC Andrew Bryant

Email: bryantav@jmu.edu

Phone: (540) 568-6264

Location: Memorial Hall, Suite 1123J

Department of Military Science Website (<https://www.jmu.edu/rotc/index.shtml>)

Mission

The James Madison University Army ROTC Duke Battalion mission is to recruit, inspire, develop and commission competent and confident warrior leaders of character as officers in the United States Army, Army Reserve or Army National Guard.

Goals

The program offers two-, three- and four-year options, allowing students to complete the professional military education requirements to earn a commission as an Army officer upon graduation. The three- and four-year programs consist of a basic course and an advanced course. A two-year option allows students with at least two academic years remaining in either undergraduate or graduate studies to complete a compressed basic course during the summer prior to enrolling in the advanced course in order to complete all requirements for commissioning as a second lieutenant in the active Army, Army National Guard or Army Reserves. Additionally, students not intending to pursue a career in the military can gain valuable leadership, teambuilding and communication skills by taking the basic military science courses without any obligation, which will transfer into marketable civilian job skills.

Career Opportunities and Marketable Skills

Army ROTC provides students with highly marketable leadership skills. The curriculum imparts leadership principles, concepts of human development, and aspects of health and fitness. Practical application of classroom instruction in lab develops one's leadership style, communication and organizational skills, and strengthens personal character. Development of these skills implicitly builds one's self-confidence, discipline and professional attributes.

Army Reservists or Army National Guardsmen who are continuing their education full time may be eligible for the Simultaneous Membership Program, which combines Reserve Forces duty with Army ROTC on campus and enables the student to earn tuition assistance and additional monetary benefits. Graduates of the program earn an Army commission and may serve four years in career areas as diverse as medical service, signal, infantry, law enforcement, aviation or nursing.

There are also opportunities for students seeking advanced degrees to delay going on active duty in order to pursue a graduate study program in law, medicine or other professional certification areas. Further, 2 and 3 year Army ROTC scholarships are possible through a competitive process for undergraduate and graduate students with no prior Army ROTC experience.

Co-curricular Activities and Organizations

- Color Guard
- JMU Cannon Crew
- JMU Ranger Group

- Army Ten Miler/Half Marathon Team
- Ranger Challenge

Special Admission and Retention Requirements

Advanced military science courses are normally taken during the junior and senior years, or during graduate school. Qualified students pursuing a commission as a second lieutenant are contracted and paid a subsistence allowance of \$420 per month for up to 10 months during the school year. Prior to commissioning, each cadet must successfully complete the four-week ROTC Cadet Summer Training (CST) at Ft. Knox, Kentucky. Cadets must maintain at least a 2.8 GPA, meet Department of War medical fitness standards, as well as pass the Army Fitness Test and height/weight control standards.

Faculty

Professor

Andrew Bryant

Assistant Professors

Garrett Adelizzio, Andrew Eshelman, Jesse Krantz, Kyle Lynch

Instructors

Craig Austin, Bobby Delaporte, Craig Gibson, Chance Platz, Mark Turner, Chris Walton, Ricky Williams

Programs and Courses

Minors

- Leadership in Military Affairs Minor (Air Force) (p. 520)
- Military Leadership Minor (p. 532)

Programs

- Air Force Reserve Officers Training Corps (p. 597)
- Military Science (p. 599)

Courses

- Military Science (MSCI) (p. 791)

School of Music

Director: John Allemeier

Email: allemejm@jmu.edu

Phone: (540) 568-3614

Location: Music Building, Room 130

School of Music Website (<http://www.jmu.edu/music>)

Associate Director: John Peterson

Email: peter2jr@jmu.edu

Phone: (540) 568-5368

Graduate Program Director and Admissions: William Dabback

Email: dabbacwm@jmu.edu

Phone: (540) 568-6719

Mission

The JMU School of Music is a community dedicated to changing lives through music by cultivating creativity, artistry, and curiosity. The School of Music provides the highest level of musical training in a comprehensive program that prepares students in conducting, music education, music industry, music performance, theory, composition, music history and music studies and to prepare them to be thoughtful and productive citizens. To that end, the School of Music is dedicated to:

- Selecting undergraduate and graduate majors and minors who have demonstrated a commitment to developing their musical skills and talents.
- Motivating music enthusiasts to explore musical concepts by exposing them to and including them in music performance, composition and education.
- Fostering a sense of community that encourages intellectual curiosity, creative endeavor and respect for all.
- Encouraging excellence from faculty members as educators, researchers, performers, clinicians and supporters so that they develop students into motivated, competent professionals and outstanding world citizens.
- Providing music majors and non-music majors with knowledge of music and developing appropriate skill levels and musicianship.
- Offering curricula that prepare students to be professionals in music performance, composition, education or industry.
- Broadening students' understanding of music through innovative teaching, creative experiences and scholarly research.
- Providing a wide variety of cultural events for the JMU and Shenandoah Valley communities.
- Exposing students to current technology employed in the music field, such as computers, music instruction software, electronic devices, and advanced audio and visual equipment.
- Preparing D.M.A. students to teach at the college level, not just in their principal areas but also in many of the core curriculum classes, such as theory, music history and music appreciation.

The School of Music is an accredited institutional member of the National Association of Schools of Music.

Career Opportunities

The various programs offered by the school can lead students to a wide range of careers. Programs in performance or composition are intended for students who desire to continue their musical training in graduate

programs that will prepare them for professional careers in performance, composition and/or teaching at the college level.

Those who elect the emphasis in music industry will be prepared for positions in a broad area of music-business occupations and for admission to graduate professional schools of business.

The music education program prepares students to teach vocal and/or instrumental music in public schools.

The music theatre program is designed to prepare vocal music majors for the field of popular Broadway musical theatre performance as well as for careers in opera and operetta.

The jazz studies program prepares students for the rigors of graduate study in jazz performance, equips students with the jazz vocabulary necessary for professional performance, and provides knowledge and skills necessary for employment in private instruction or as a jazz specialist in public schools.

The following list of careers is only a small sample of possibilities:

- Artists' manager
- Composer
- Conductor
- Entertainment lawyer
- Music educator
- Music journalist
- Music librarian
- Music software developer
- Orchestra manager
- Performing artist
- Professional accompanist
- Professor/teacher
- Record producer

Co-curricular Activities and Organizations

These activities are open to all JMU students without audition:

- Concert Band
- Marching Royal Dukes
- Tenor/Bass and Soprano/Alto Choruses

These activities are open to all JMU students with audition or instructor permission:

- Brass Ensembles
- Chamber Orchestra
- Chorale
- Clarinet Choir
- Flute Choir
- Guitar Ensemble
- Horn Choir
- Jazz Band
- Jazz Chamber Ensembles
- Jazz Ensemble
- Madison Baroque
- Madison Singers
- Monticello Strings

- Opera Theatre
- Opera Theatre Orchestra
- Percussion Ensemble
- Piano Accompanying
- Steel Drum Band
- String Ensembles
- Symphonic Band
- Symphony Orchestra
- Treble Chamber Choir
- Trombone Choir
- Trumpet Ensemble
- Tuba and Euphonium Ensemble
- Wind Symphony
- Woodwind Ensembles

Class Fees

There is a once per semester fee for enrollment in MUAP 200, MUAP 214 and MUAP 300. See MyMadison for details.

Music Scholarships

The entrance audition also serves as the scholarship audition. All music scholarships are awarded on the basis of merit. Recorded auditions will not qualify a prospective student for consideration to receive a music scholarship.

A transfer student on scholarship at another school can only be considered for a scholarship at JMU if the music executive of the current school sends a written release to the director of the School of Music.

Faculty

Professors

John M. Allemeier, Sonya G. Baker, Susan N. Barber, Christopher J. Carrillo, Beth E. Chandler Cahill, Christopher Cicconi, Andrew M. Connell, David H. Cottrell, William M. Dabback, Gabriel T. Dobner, Carl P. Donakowski, Charles J. Dotas, Jonathan B. Gibson, Eric R. Guinivan, Wanchi Huang, Sarunas Jankauskas, Andrew B. Lankford, John R. Peterson, Lori E. Piitz, David J. Pope, Kevin J. Stees, Paulo R. Steinberg, Jo-Anne van der Vat-Chromy, Ian R. Zook

Associate Professors

Pedro R. Aponte, Casey Cangelosi, Kira Omelchenko, Diane K. Phoenix-Neal, Samuel S. Suggs, Jamison Walker, Jeanette M. Zyko

Assistant Professors

Joshua Borths, Rachel Grimsby, Masayoshi Ishikawa, Gui Hwan Lee, Judith Ofcarcik, Andria Savage, Jingyi Zhang

Lecturers

Amy Birdsong, Kirsten Blair, Chris DeVona, Michael Hadary, Scott D. Ridders

Adjunct Faculty

Katherine Axtell, Adam Larrabee, David Newman, Scott Zane Smith, Rhonda Stees, Aaron Trumbore, Brenda Witmer

Programs and Courses

Majors

- Music, B.A. (p. 375)
- Music, B.M. (p. 378)

Minors

- General Music Minor (p. 499)
- Jazz Studies Minor (p. 517)
- Music Industry Minor (p. 537)

Cross-Disciplinary Minor

- Music and Human Services Minor (p. 536)

Teaching Licensure Program

- Music Teaching Licensure (p. 590)

Graduate Programs

- Music Performance, Pedagogy and Literature, D.M.A. (<https://catalog.jmu.edu/programs/music-performance-pedagogy-literature-dma/>)
- Music, Composition Concentration, M.M. (<https://catalog.jmu.edu/programs/music-composition-concentration-mm/>)
- Music, Conducting Concentration, M.M. (<https://catalog.jmu.edu/programs/music-conducting-concentration-mm/>)
- Music, Music Education Concentration, M.M. (<https://catalog.jmu.edu/programs/music-music-education-concentration-mm/>)
- Music, Performance Concentration, M.M. (<https://catalog.jmu.edu/programs/music-performance-concentration-mm/>)

Courses

- Applied Music (MUAP) (<https://catalog.jmu.edu/courses/muap/>)
- Music (MUS) (<https://catalog.jmu.edu/courses/mus/>)
- Music Education (MUED) (<https://catalog.jmu.edu/courses/mued/>)
- Music Industry (MUI) (p. 803)
- Music Pedagogy (MUPED) (<https://catalog.jmu.edu/courses/muped/>)

School of Nursing

Director: Melody Eaton

Email: eatonmk@jmu.edu

Phone: (540) 568-6314

Location: Health and Behavioral Studies Building, Room 4094

School of Nursing Website (<http://www.nursing.jmu.edu>)

Associate Director, Undergraduate Programs: Jamie Robinson

Associate Director, Graduate Programs: Andrea Knopp

Mission

We engage students, faculty and communities through dynamic and innovative nursing education, practice and scholarship to influence health in our world.

Vision

To be a SON nationally recognized for academic excellence and innovative contributions to health care.

Values

We are committed to:

- Integrity: being honest, sincere and just in all of our endeavors
- Caring: sharing compassion, kindness and authenticity with those we encounter
- Excellence: pursuing the highest quality in teaching, learning, scholarship and service
- Collaboration: cultivating partnerships built on respect, trust and commitment
- Advocacy: providing a voice for those we serve and promoting policies that improve healthcare for all
- Inclusivity: honoring the richness that diverse perspectives bring to our world
- Resilience: achieving inner strength and wisdom by adapting to life's challenges

Career Opportunities and Marketable Skills

- Preparation as a professional nurse leading to a Bachelor of Science in Nursing (B.S.N.) degree.
- Eligibility to take the National Certification and Licensure (NCLEX) Exam to become a registered nurse (R.N.).
- Preparation for entry-level positions in a wide range of health care settings.
- Upon completion of the programs of study, students may be eligible for leadership positions in a wide range of health care settings.
- Eligibility for APRN and CNL certification exams with excellent certification pass rate
- Preparation for evidence-based advanced practice clinical or leadership roles in nursing and healthcare
- Student engagement in scholarship
- Global opportunities for practicums
- High ranking SON, US News and World Report

Co-curricular Activities and Organizations

- JMU Chapter of the Virginia Nursing Student Association
- Omega Zeta of Sigma Theta Tau International Nursing Honor Society
- B.S.N. Student Advisory Council
- Nursing Students without Borders
- Virginia Student Nurses Association

Accreditation

The Bachelor of Science in Nursing program is approved by the Virginia Board of Nursing. The baccalaureate, master's, and doctorate of nursing practice programs at James Madison University are accredited by the:

Commission on Collegiate Nursing Education (<http://www.ccnaccreditation.org>)
655 K Street, NW, Suite 750
Washington, DC 20001
(202) 463-6930

Program Options

The School of Nursing offers three baccalaureate program options. The Traditional B.S.N. and Fast Flex B.S.N. prepares traditional undergraduate students to practice as an entry-level professional nurse. The R.N. to B.S.N. is a degree completion pathway that provides flexible learning opportunities for licensed R.N.s who have an associate degree or diploma to complete the baccalaureate degree.

Graduate Programs offered at JMU School of Nursing include Master of Science of Nursing (M.S.N.) or Doctor of Nursing Practice (D.N.P.). The M.S.N. concentrations are advanced practice RN (nurse practitioner) as a Family Nurse Practitioner (FNP), Adult Gerontology Primary Care Nurse Practitioner, Nurse Midwifery or Psychiatric Mental Nurse Practitioner. Leadership MSN concentrations are Clinical Nurse leader or Nurse Administrator. DNP offerings include post masters DNP or post BSN-DNP in either APRN FNP, AGPCNP or Healthcare Systems Leadership.

Admission and Retention Requirements

Information on admission to programs within the School of Nursing, retention and minimum grade requirements may be found on the individual program pages.

Special Expenses

A differential tuition of \$104 per credit hour is added to the standard charge for courses carrying the NSG prefix. This charge applies to all students, both in-state and out-of-state, at the undergraduate and graduate levels, with exception of students enrolled in the R.N. to B.S.N. program.

Faculty

Professors

Sandra L. Annan, Christine A. Argenbright, Holly Buchanan, Melody K. Eaton, Amy Graham, Andrea F. Knopp, Erica J. Lewis, Erika Metzler Sawin

Associate Professors

Modjadj Choshi, Elizabeth K. Herron, Bridget Houlahan, Christina K. Lam, Jamie Robinson, Donna Schminkey, Karen Weeks

Assistant Professors

Stephanie Armstrong, Christa Berthiaume, Callie Bradley, Lauren Childers, Teona Eriogou, Jessica Fuller, Lisa Hesser, Bethany L. Landes, Holly

Mason, Lauren K. Mullen, Kimberly Priode, Nancy L. Puffenbarger,
Julianne Secrist, Julie Sutherland, Tina Switzer, Hui Zhao

Instructors

Kymber Jennings Beers, Anna Cho, Melody King, Gina Parish-Gibson,
Emily Stevens, Brandi Walton

Senior Lecturer

Lisa J. Carmines

Lecturers

Scott Beckler, Catherine Brown, Therese Collins, Tracy E. Smith, Ronnicia
Thomas, Tim Whelden

Programs and Courses

Majors

- Nursing, B.S.N. (p. 388)
- Nursing, R.N. to B.S.N. (p. 391)

Graduate Programs

- Nursing, M.S.N. (<https://catalog.jmu.edu/programs/nursing-msn/>)
- Nursing Practice, D.N.P. (<https://catalog.jmu.edu/programs/nursing-practice-dnp/>)

Certificates

- Health Policy Certificate (<https://catalog.jmu.edu/programs/health-policy-certificate/>)

Courses

- Nursing (NSG) (<https://catalog.jmu.edu/courses/nsg/>)

Department of Philosophy and Religion

Academic Unit Head: Kenneth Pearce

Email: pearcekl@jmu.edu

Main Office Phone: (540) 568-6394

Department of Philosophy and Religion Website (<http://www.jmu.edu/philrel>)

Mission

The academic study of philosophy and religion prepares students to meet people with diverse beliefs in a rapidly changing world. Students completing a major in Philosophy or Global Religions and Cultures will be able to think critically, rigorously analyze arguments, listen carefully to the viewpoints of others, construct views that are based on evidence and reasoning, and communicate clearly and effectively in writing and in speaking. They will grapple with the insights of some of the greatest intellectual figures in history as they learn to contextualize ideas and develop nuanced understandings of the world around them.

Emphases of the Philosophy Major

Philosophy is a discipline that explores fundamental questions about the meaning of life, the nature of knowledge and the mind, what actions are moral, and what exists. Students completing a Philosophy major with a concentration in General Philosophy or Interdisciplinary Philosophy will become versed in the major movements, problems, writings and concepts in the history of Western philosophy. They will also have the opportunity to study other philosophical traditions and to learn about ongoing philosophical debates today. The program concentrates on contemporary movements such as analytic philosophy, existentialism, American philosophy, and feminist philosophy, as well as on the major subdivisions of philosophy, including logic, ethics, metaphysics, epistemology, aesthetics, philosophy and law, philosophy of science, and philosophy of religion.

Students completing a major in philosophy will gain skills in critical thinking, problem solving, analytical writing, and rationally compelling speaking and writing.

Emphases of the Global Religions and Cultures Major

Global Religions and Cultures is an interdisciplinary field of inquiry into the belief systems, ritual practices, identities, and values that shape communities and societies. The Global Religions and Cultures major ensures that students gain breadth of exposure to diverse religious and cultural traditions from across the globe as well as the opportunity to pursue specialized knowledge in a particular tradition or contemporary issue. Students completing a major in Global Religions and Cultures will train in a range of disciplinary approaches and methods employed across the humanities as they learn to:

- Analyze historical and contemporary expressions of religions in their relevant social, political and cultural contexts.
- Construct informed interpretations of sacred texts by integrating skills in critical reading, critical thinking and contextualization.
- Immerse themselves in the perspective of worldviews different from their own and engage critically yet generously across those differences.

- Pose increasingly sophisticated questions regarding both "solvable" problems as well as "unanswerable" questions about the construction of ultimate reality.
- Apply problem solving and critical thinking skills by conducting research, exhibiting rigorous inquiry, and weighing evidence.
- Demonstrate proficient written and oral communication in a variety of settings for different audiences, formulating reasonable arguments based on evidence.
- Practice sound information literacy through verifying the reliability and relevance of sources, including detecting bias.

Career Opportunities and Marketable Skills

Majors in Philosophy and Global Religions and Cultures greatly enhance students' ability to navigate a dynamic and changing world by fostering open-mindedness, religious and philosophical literacy, a grounding in ethics, a habit of logical thinking, structured and disciplined argumentation, and practices in rigorous analysis. Our majors also equip students with solid skills in analysis, research, writing, speaking and presentations. As a result, our graduates find rewarding and meaningful work in a wide variety of career fields. Many of our alumni pursue careers in intelligence analysis, communications, government contracting, public health, business, public service, the non-profit sector, the technology sector, education, counseling, or university administration. Many attend graduate school, law school, medical school, or divinity school. Throughout their time in our department, majors are guided along their career paths with the help of dedicated faculty advisors as well as the University Career Center. Our students often combine our majors with second majors from other departments, which may also enhance their employment opportunities.

Student Activities and Organizations in Philosophy and Religion

The department has excellent opportunities for building student community, engaging in student social activities and fostering intellectual development outside of the classroom, including a student-led religion society, a philosophy and film club, a philosophy honor society (Phi Sigma Tau), and a religion honor society (Theta Alpha Kappa).

Preparation for Law School

Coordinator: Mark Piper

Email: pipermc@jmu.edu

Phone: (540) 568-3531

Students who plan to attend law school should seriously consider Philosophy as an undergraduate major. Philosophy majors have historically scored among the highest on the LSAT, GRE and GMAT. Philosophy courses emphasize the kinds of skills that prepare students for the LSAT and the law school curriculum: reading, comprehending and analyzing complex texts; organizing and synthesizing information and drawing reasonable inferences from it; analyzing and evaluating the reasoning and arguments of others; and researching and writing essays and papers.

Law schools recommend that students choose an undergraduate major that challenges them and provides them with an understanding of what shapes human experience. Philosophy does an outstanding job on

both counts. The requirements of the major leave students plenty of opportunity to acquire a broad education by exploring other areas.

Preparation for Theological Studies

Coordinator: Jennifer Aycock

Email: aycockjl@jmu.edu

Phone: (540) 568-6394

Students join the pre-theology program to prepare for careers in the service of religious communities. Usually, these careers—such as ministry, religious education, counseling and chaplaincy—require specialized training and graduate degrees from theological seminaries, rabbinical schools or university divinity schools. The pre-theology advisor guides students toward courses and experiential learning at JMU that will provide an academic foundation for graduate studies in theology as well as an opportunity to develop their practical skills or vocational gifts. Students in this program may major in any field, although the Association for Theological Schools recommends substantial pre-professional training in philosophy and religion.

In the Department of Philosophy and Religion, pre-theology students will find a robust offering of courses ranging from the interpretation of religious texts to theology and ethics, and from the histories of particular religious traditions to cross-cultural topics in global religion. Class assignments develop students' skills in critical thinking, ethical reasoning, scriptural hermeneutics, and written and oral communication, all of which theological schools highly value. The pre-theology advisor may direct students to relevant courses in foreign languages, social work, non-profit studies, humanitarian affairs, education or justice studies to complement their academic training in religion. Qualified students are encouraged to undertake independent studies or write honors theses in order to explore in greater depth the theological topics most important to them.

The department also offers students the opportunity to receive academic credit for internships with non-profit organizations and religious organizations in order to help students explore the particular forms of religious service they would like to pursue. Making connections with faith leaders within their own traditions helps students access larger networks of tradition-specific resources and contacts as they reflect on their goals and decide where to apply for further study in theology. We encourage students to visit various theological schools during the application season and to meet with representatives that the department invites to campus to discuss theology programs with students.

Faculty

Professors

Charles Bolyard, Robert Brown, Christian E. Early, Patrick Fleming, Emily O. Gravett, Steven H. Hoeltzel, Alan Kirk, William M. Knorpp Jr., Tracy A. Luper, Sushil Mittal, Kenneth L. Pearce, Mark C. Piper, Anne van Leeuwen, Andrea L. Veltman

Associate Professors

Christina A. Kilby, David C. Kirkpatrick, Cyril V. Uy II

Assistant Professors

Pia Antolic-Piper, Jennifer L. Aycock, Helen Buckwalter, Ian Hensley, Alberto G. Urquidez

Programs and Courses

Majors

- Philosophy and Religion, B.A.; Major in Global Religions and Cultures (p. 393)
- Philosophy and Religion, B.A.; Major in Philosophy (p. 395)

Minors

- Christian Studies Minor (p. 461)
- Ethics Minor (p. 494)
- Global Religions and Cultures Minor (p. 503)
- Philosophy Minor (p. 539)

Courses

- Philosophy (PHIL) (p. 813)
- Religion (REL) (p. 835)

Department of Physics and Astronomy

Academic Unit Head: Brett Taylor

Email: physics@jmu.edu

Phone: (540) 568-6109

Location: Physics/Chemistry Building, Suite 2186

Department of Physics Website (<https://www.jmu.edu/physics/index.shtml>)

Mission

The mission of the Department of Physics and Astronomy is the preparation of scientifically enlightened citizens. Science literacy is promoted by the production of teachers, researchers, technical professionals and knowledgeable individuals through the integration of classroom and experiential learning.

Vision

We strive to be a leading undergraduate physics and astronomy department by building a research-active, student-centered community.

Values

Excellence, integrity and mutual respect are the core values that define our department. The vigorous pursuit of research with undergraduates is central to extending our understanding of nature and the engagement of students directly in the practice of physics and astronomy.

Service to the university, the public and the profession is essential for continued vitality of science education and research.

Goals

To help students:

- Appreciate the role of science in society and the historical development of physics in the ongoing quest to discover the structure of the universe.
- Gain an understanding of the basic principles and the experimental basis of the various fields of physics and the logical relationships of the various fields.
- Develop competence in using computers for computation, data acquisition, numerical control, device development, and information acquisition and processing.
- Become capable problem solvers using techniques that require mathematical skills, conceptual and mathematical models, order-of-magnitude estimates, and an understanding of limiting cases.
- Develop competence in designing, constructing, and using laboratory instruments, and to draw valid conclusions from experimental data.
- Improve written and oral technical communication skills.

Co-Curricular and Organizations

- Society of Physics Students

Faculty

Professors

L. Adriana Banu, Harold M. Butner, Anca Constantin, Klebert B. Feitosa, Keigo Fukumura, Kevin L. Giovanetti, W. Christopher Hughes, Gabriel Niculescu, M. Ioana Niculescu, Sean T. Scully, Brett Taylor

Associate Professors

Masoud Kaveh Baghbadorani, Costel Constantin, Mark E. Mattson, Ilarion V. Melnikov, Giovanna Scarel, Kendra L. Letchworth Weaver

Assistant Professors

Sachith Dissanayake, Tehani K. Finch

Senior Lecturer

Jason Sterlace

Lecturer

Jason Czak

Programs and Courses

Majors

- Physics, B.A. (p. 397)
- Physics, B.S. (p. 399)

Minors

- Astronomy Minor (p. 453)
- Physics Minor (p. 541)
- Physics Minor for Engineers (p. 540)

Courses

- Astronomy (ASTR) (p. 622)
- Interscience Research (ISCI) (p. 751)
- Physics (PHYS) (p. 817)

Department of Political Science

Academic Unit Head: Kerry Crawford

Email: crawfokf@jmu.edu

Phone: (540) 568-6952

Location: Miller Hall, Room 2123

Department of Political Science Website (<http://www.jmu.edu/polisci>)

Mission Statement

The Department of Political Science offers strong major and minor programs sharing a focus on public concerns and public service. We are committed to providing our students with the tools and competence to succeed in their lives, their graduate education and their careers by instilling academic rigor, information access and research skills, dedication to life-long learning and respect for diversity in cultures, nations and institutions of democracy.

Goals

To carry out the above mission, the Department of Political Science seeks to:

- Offer strong major programs: the B.A. in political science (p. 405), the B.S. in public administration (p. 413) and the B.A. in international affairs (p. 327).
- Offer strong minor programs: political science (p. 543), public policy & administration (p. 558); non-profit studies (p. 538) and political communication (p. 542) (cross-disciplinary).
- Offer strong graduate programs: the M.A. in political science with a concentration in European Union policy studies (<https://catalog.jmu.edu/programs/political-science-european-union-policy-studies-concentration-ma/>) and the Master of Public Administration (<https://catalog.jmu.edu/programs/public-administration-mpa/>).
- Enhance critical thinking, communication and information gathering skills.
- Foster active learning through research, simulations and internships.

Career Opportunities

The political science department offers programs that lead students to careers in the following fields:

- Administration and management
- Federal, state and local government
- International organizations
- Law
- Non-profit organizations
- Politics
- Private sector

To enhance the marketability of its students, the political science department supports career-related internships. For information contact the political science office, Miller Hall, Room 2120, (540) 568-6149 or contact the internship coordinator, Fred Mayhew (mayhewfd@jmu.edu).

Co-curricular Activities and Organizations

- Pi Alpha Alpha: public administration honor society
- Pi Sigma Alpha: political science honor society
- Pre-Law Society

- Sigma Iota Rho: international affairs honor society
- PASO: JMU Public Affairs Student Organization

Programs in the Department of Political Science

The Department of Political Science offers the B. A. degree with a major in political science, the B.S. degree with a major in public administration and the B.A. degree with a major in international affairs. The Department also offers the M.A. in political science with concentration in European Union policy studies (a 10-month program based in Florence, Italy) and the M.P.A. in public administration.

Washington Semester

Program Director: Edward Yang

Phone: (202) 422-6161

Email: yangyx@jmu.edu

Location: 1400 16th St. NW, Suite 320, Washington, D.C., 20036

Website: (<https://www.jmu.edu/polisci/washington.shtml>)

The Washington Semester program enables students to explore the wealth of political and cultural opportunities available in the Washington, D.C. area. The program includes semester-length internships, courses taught on-site, panel discussions by people in various professions, cultural activities, and networking opportunities with JMU alumni working in the DC area.

The Washington Semester program is open to all majors and is offered every semester. Students not majoring in political science, international affairs, or public administration should consult with their departments about how the internship and coursework fits their academic plans; some departments may be willing to arrange course substitutions.

Financial Assistance

A limited number of graduate assistantships are available on a competitive basis. Assistantships are limited to nine paid graduate hours of tuition each fall and spring semester. Students must pay for any additional hours each semester at the tuition rate based on residency status.

All relevant regulations in the undergraduate and graduate catalogs are applicable.

Faculty

Professors

Melinda J. Adams, Jessica R. Adolino, Robert W. Alexander, Bethany Blackstone, Charles H. Blake II, Andreas Broscheid, Jennifer Byrne, Martin Cohen, Kerry F. Crawford, Kathleen M. Ferraiolo, Abe Goldberg, Scott J. Hammond, John W. Hulsey, Manal Jamal, David A. Jones, Bernd Kaussler, Jonathan W. Keller, Timothy M. LaPira, Hak-Seon Lee, Howard L. Lubert, Elizabeth Oldmixon, Liliokanaio Peaslee, Robert N. Roberts, Kenneth R. Rutherford, John A. Scherpereel, Valerie A. Sulfaro, Amanda Cleveland Teye, Kristin N. Wylie, Yi Edward Yang

Associate Professors

Silvia D'Amato, Keith A. Grant, Frederick D. Mayhew, Nicholas J. Swartz, Jennifer A. Taylor

Assistant Professors

Bryan Green, Sombo Muzata, Christopher Price, Jaimee Swift

Lecturers

DeAnne Chenoweth, Robin Leiter-White

Instructors

Joel Hensley

Adjunct Instructors

Katherine Allen, Angela Adami, Frederigo Argentieri, Kadir Ayhan, Christine Bakker, Michelle Bixler, John Cencich, Daniela Chironi, Gianni Del Panta, Jeremy Kleidosty, Evangelia Koundouraki, Anna Lavizzari, Jessica Minnis-McClain, Nicole Neitzey, John Newman, Cindy Noftsinger, Nathaniel Riddle, Patrick Shea, Craig Short, Tammi Slovinsky, Lynne Weikart

Programs and Courses

Majors

- International Affairs, B.A. (p. 327)
- Political Science, B.A. (p. 405)
- Public Administration, B.S. (p. 413)

Minors

- Political Science Minor (p. 543)
- Public Policy and Administration Minor (p. 558)

Cross-Disciplinary Minors

- Nonprofit Studies Minor (p. 538)
- Political Communication Minor (p. 542)

Graduate Programs

- Political Science, European Union Policy Studies Concentration, M.A. (<https://catalog.jmu.edu/programs/political-science-european-union-policy-studies-concentration-ma/>)
- Public Administration, M.P.A. (<https://catalog.jmu.edu/programs/public-administration-mpa/>)

Courses

- International Affairs (INTA) (p. 750)
- Political Science (POSC) (<https://catalog.jmu.edu/courses/posc/>)
- Public Administration (PUAD) (<https://catalog.jmu.edu/courses/puad/>)
- Public Policy and Administration (PPA) (p. 832)

Department of Psychology

Interim Academic Unit Head: Monica Reis-Bergan

Email: ugpsyc@jmu.edu

Phone: (540) 568-6114

Location: Miller Hall, Room 1120

Department of Psychology Website (<https://www.jmu.edu/chbs/psyc/index.shtml>)

Mission Statement

The mission of the Department of Psychology is to educate students in, and contribute to, the science of psychology.

Goals

The vision of the Department of Psychology is to sustain and advance a nationally recognized department focused on promoting scientific proficiency within the cross-disciplinary field of psychology. To that end, the curriculum is designed to provide students with a strong foundation in psychological science that reflects multiple perspectives within the discipline. The hallmarks of our program are learning opportunities that promote the skills necessary to conduct and evaluate research and to interpret findings. The department trains our students to be responsive to emerging trends and apply their knowledge of psychological science as responsible global citizens.

The Department of Psychology is a dynamic contributor to the General Education program and the university community through the delivery of high-quality courses and other educational experiences emphasizing psychological science. We also contribute to the M.A. program in psychological science. Our faculty strive to create and implement creative and effective pedagogy, embracing the dual role of teacher/scholar. The department fosters a collegial and collaborative environment within which divergent opinions as well as cultural diversity are respected, valued and promoted.

Faculty

Professors

Jeffrey T. Andre, Kevin J. Apple, Suzanne C. Baker, Kenneth E. Barron, Jeffrey S. Dyche, Michael D. Hall, Daniel D. Holt, Jessica G. Irons, Krisztina V. Jakobsen, Natalie A. Kerr, Jaime L. Kurtz, Sharon E. Lovell, Claire W. Lyons, Kethera A. J. Moore, Monica J. Reis-Bergan, Jessica Salvatore, Bryan K. Saville, Melanie L. Shoup-Knox, Tracy E. Zinn

Associate Professors

Benjamin T. Blankenship, Robyn L. Kondrad, Kala J. Melchiori, Sri Siddhi N. Upadhyay

Assistant Professors

Curtis A. Bradley, Haeyoon Chung, Yanbin Li, Janna Taft Young, Heather A. Yarger

Senior Lecturers

Kimberly D. R. DuVall

Lecturers

Daniel George, Taj Rollins

Programs and Courses

Majors

- Psychology, B.A. (p. 407)
- Psychology, B.S. (p. 410)

Courses

- Psychology (PSYC) (<https://catalog.jmu.edu/courses/psyc/>)

Department of Social Work

Academic Unit Head: Simon P. Funge

Email: SOWK@jmu.edu

Phone: (540) 568-6980

Location: Health and Behavioral Studies Building, Room 2084
Department of Social Work Website (<http://www.jmu.edu/socwork>)

Mission Statement

The Department of Social Work prepares generalist social workers committed to addressing a broad range of social challenges and strengthening the well-being of diverse individuals, families, groups, organizations and communities. Grounded in ethical practice, a generalist perspective and cultural humility, the program prepares students to support and enhance the welfare of individuals, families, groups and communities while challenging systemic barriers to equity, social progress and human rights..

The department offers a B.S.W. degree program accredited by the Council on Social Work Education. In addition, it offers a minor in family studies (with family ecology or gerontology foci).

The Department of Social Work is committed to:

- Preparing students to work effectively in a broad spectrum of social service agencies to address poverty, and challenge multiple forms of oppression, social injustice, and other human rights violations.
- Preparing students for life-long professional learning and advanced academic study by providing an environment geared toward achieving academic excellence.
- Advancing excellence and integrity through the development of high-quality academic programs through high-impact teaching and engaged learning.
- Making contributions to professional knowledge through varied forms of scholarship.
- Responding to the profession and communities by providing service and continuing education opportunities.

Career Opportunities and Marketable Skills

Career Opportunities

- Aging services
- Child and adult day care centers
- Children and youth services
- Community action agencies
- Criminal justice agencies
- Domestic violence programs
- Family service agencies
- Homeless shelters
- Hospitals/home health programs/hospices
- Income maintenance programs
- Legal services agencies
- Behavioral and mental health services
- Services to people with developmental disabilities
- Nursing homes and residential communities
- Public child welfare/child protection services
- Residential treatment facilities

- School programs
- Substance use programs
- Vocational rehabilitation services

Marketable Skills

- Advocacy
- Assessment
- Case management/brokering
- Communication
- Community outreach
- Crisis intervention
- Group facilitation
- Intake/referral
- Intervention/service planning
- Interviewing
- Networking
- Policy analysis
- Problem solving
- Program development/evaluation
- Recording/writing
- Relationship-building/interpersonal
- Research
- Service provision
- Team/group/collaborative

Co-curricular Activities and Organizations

The Social Work Organization (SWO) offers the opportunity to socialize, meet professionals and volunteer in the community. Membership in SWO is open to any student interested in a career in the helping professions.

Phi Alpha Honor Society promotes academic excellence among social work students. Membership in Phi Alpha is by invitation to students with a minimum of nine credit hours in required social work course holding a major grade point average (GPA) of 3.25 and a cumulative GPA of 3.0.

Admission and Retention Requirements

Information on admission to programs within the Department of Social Work, retention and minimum grade requirements may be found on the individual program pages.

Faculty

Professors

Simon P. Funge, Linda Plitt Donaldson, Hyong S. Yeom

Associate Professor

Nancy T. Poe

Assistant Professors

John Gyorko, Shawn McNally, Natasha Miller

Instructors

Hyuntae Kim, Carmel Murphy Norris

Programs and Courses

Major

- Social Work, B.S.W. (p. 423)

Cross-Disciplinary Minor

- Family Studies Minor (p. 495)

Courses

- Family (FAM) (p. 687)
- Social Work (SOWK) (p. 842)

Department of Sociology and Anthropology

Academic Unit Head: Abigail Wightman

Phone: (540) 568-6213

Location: Sheldon Hall, Room 123

Department of Sociology and Anthropology Website (<http://www.jmu.edu/socanth>)

Sociology Program Coordinator: Benjamin Brewer

Email: brewerbd@jmu.edu

Phone: (540) 568-7391

Location: Sheldon Hall, Room 212

Anthropology Program Coordinator: Julie Solometo

Email: solomejp@jmu.edu

Phone: (540) 568-6981

Location: Sheldon Hall, Room 207

Mission

Our quest is to understand why humans behave as they do and the real world is our laboratory. By examining human activity in all of its diverse and complex forms, the work of sociologists and anthropologists forms the basis for confronting many of our world's most challenging issues.

Sociology

Sociologists study the causes and consequences of social inequality as well as the social aspects of the environment, technology, politics, health, globalization, social movements, social justice, markets, mass media/popular culture, and more.

As a sociology major, you can apply the skills you learn to real-world needs in volunteer projects or internship experiences that will enhance your resumé. Through course work and projects, you can gain hands-on experience doing sociology, which might include: immersion in the daily life of groups, interview methodology, examining recorded interaction, interpreting historical documents, analyzing large national surveys, and producing sophisticated social critique.

Why are these skills important? By examining how our everyday actions are connected to larger social, economic, and political affairs, you can address social possibilities and problems that are important to non-profit organizations, government agencies, and a range of other settings such as management, health care, and marketing. You can be the change!

Anthropology

Anthropology's goal is to document the diversity of ways humans live and have lived throughout the world. Anthropologists go "off road," working directly with people and their artifacts, to examine their beliefs and values, how they make a living, how they express themselves, how they interact, and how they affect and are affected by the natural world.

Why is this important? An increasingly globalized and multi-cultural world requires anthropological knowledge and insight. Anthropology is essential to students interested in working in both the US and overseas on contemporary issues in areas such as social policy work, international development, public health, community-based advocacy, environmental and social justice, and cultural preservation.

Students in the JMU Anthropology program get to develop their interests by taking a wide range of classes in cultural, linguistic and biological anthropology and archaeology. At JMU, Anthropology majors can also

do individually-driven practical and hands-on research in internships, anthropology labs, excavations and field-schools, and on study-abroad programs. Join us: learn what is going on in the world and help to make it a better place.

Student Research Opportunities

Students in the Sociology and Anthropology majors have a wide range of opportunities to develop the rigorous research skills necessary for a strong entry into their professional lives or graduate school.

In the Classroom: Students learn research skills in courses such as SOCI 300, SOCI 480, ANTH 435, ANTH 445, and ANTH 455. These are all small, upper-level seminar-style classes that offer direct hands-on learning.

Outside of the Classroom: Students in Sociology and Anthropology are highly engaged in issues that impact people's daily lives. So it is not surprising that our majors also practice their research skills outside of the classroom. These opportunities include:

- Field Schools and Study Abroad Programs
- Department Summer Study Abroad Programs
- Archaeology Field Schools
- The JMU Archaeology Research Lab
- The Gender and Science Ethnography Lab
- Internships
- Service-Learning Experiences
- Research and Teaching Assistantships with Professors

Career Opportunities

By majoring in Sociology or Anthropology you will join graduates who have gone on to careers in non-profit organizations, medicine, public policy, counseling, communications, urban planning, historical preservation, environmental justice, teaching, journalism, international development and more.

Faculty

Professors

Benjamin D. Brewer, Bethany P. Bryson, Liam M. Buckley, Chris R. Colocousis, Kerry M. Dobransky, Matthew B. Ezzell, Rebecca E. Howes-Mischel, Richard R. Lawler, Amy L. Paugh, Stephen C. Poulson, Joseph H. Spear, Kimiko Tanaka, Megan E. Tracy, David A. Trouille

Associate Professors

Di Hu, Lisa L. Porter, Julie P. Solometo, Abigail Wightman

Assistant Professors

Taylor Callaway, Eduardo Duran, Jalia Joseph, Brian Walter, Samantha L. Yausy

Visiting Assistant Professor

Hannah Smith Brennan

Senior Lecturer

Yamilette D. Chacon

Programs and Courses Majors

- Anthropology, B.A. (p. 173)
- Anthropology, B.S. (p. 176)

- Sociology, B.A. (p. 425)
- Sociology, B.S. (p. 428)

Minors

- Anthropology Minor (p. 449)
- Sociology Minor (p. 567)

Cross-Disciplinary Minor

- Historical Archaeology Minor (p. 507)

Courses

- Anthropology (ANTH) (p. 603)
- Sociology (SOCI) (p. 845)

School of Strategic Leadership Studies

Academic Unit Head and Graduate Program Director: Margaret F. Sloan

Phone: (540) 568-7020

Email: ssls@jmu.edu

School of Strategic Leadership Studies Website (<https://www.jmu.edu/leadership/index.shtml>)

Mission

James Madison University offers an innovative doctoral program in Strategic Leadership Studies.

This program emphasizes business principles, accountability, and leadership theory and applications, all areas important for new organizational, nonprofit and higher education leaders. Students are instructed in the broader visionary perspectives necessary for effective leadership. The program emphasizes practical applications grounded in sophisticated research skills needed for data-based, innovative decision-making in current and future practice.

The leadership course work encompasses models, theories and processes which are tested through application to situations faced by practicing leaders. In addition to leadership course work, students enroll in clusters of courses centered on research methodology including: measurement; statistics and accountability; business administration; managerial finance and accounting for decision-making and control; and electives in organizational science and leadership, postsecondary analysis and leadership, or nonprofit and community leadership. All students study strategic management and advanced leadership dynamics.

Faculty

Professors

Margaret F. Sloan

Associate Professors

Benjamin S. Selznick, Adam J. Vanhove, Nara Yoon

Assistant Professor

Minjong Jun

Programs and Courses

Graduate Program

- Strategic Leadership Studies, Ph.D. (<https://catalog.jmu.edu/programs/strategic-leadership-studies-phd/>)

Courses

- Leadership Studies (LEAD) (<https://catalog.jmu.edu/courses/lead/>)

School of Theatre and Dance

Director: Ilana Morgan

Email: theatredance@jmu.edu

Phone: (540) 568-6342

Location: Forbes Center for the Performing Arts

School of Theatre and Dance Website (<http://www.jmu.edu/theatredance>)

Mission

The School of Theatre and Dance is rooted in the belief that the relevant artist is the thinking artist who couples the mastery and embodiment of concrete skills with knowledge of and sensitivity to the cultural environments of ideas, artistic forms and other persons. The school trains and educates artists, scholars and teachers in the rich traditions and current practices of dance, theatre and musical theatre. The school's programs blend liberal arts education and critical thinking with intensive pre-professional training and practice. Faculty foster an environment that values and cultivates creative, passionate, disciplined, curious, innovative, engaged, articulate, collaborative and independent-thinking artists and scholars. Committed to a teaching approach that emphasizes mentorship and individual attention to students, faculty members of the school empower students in the development of their own personal strengths, provide them with tools and opportunities to realize their potential, and equip them for successful work in professional environments, graduate programs and to be lifelong learners.

Goals

To realize this mission, the School of Theatre and Dance strives to:

- Develop in students the ability to work productively and sensitively in creative and collaborative processes.
- Offer multiple opportunities for students to self-initiate and self-produce work in a context supportive of experimentation.
- Produce performances and creative experiences of high quality.
- Present performances by and learning experiences with visiting artists.
- Motivate students to take active responsibility for their work, processes and careers.
- Promote, support and enable faculty professional development so that teachers continue to provide the education for students that only growing, current and active creative artists and scholars can provide.
- Inspire and prepare students to be advocates for the arts.
- Foster an understanding of the roles of the artist and the arts in society.
- Advance dynamic partnerships with diverse communities.

Career Opportunities

For information about career opportunities in theatre, dance and musical theatre, contact the appropriate coordinator:

- Theatre Coordinator: Emily Becher-McKeever, becheree@jmu.edu
- Dance Coordinator: Suzanne Miller-Corso, mille2sm@jmu.edu
- Musical Theatre Coordinator: Kate Arecchi, arecchk@jmu.edu

Co-curricular Activities and Organizations

The School of Theatre and Dance offers a number of co-curricular activities and organizations. They allow students to apply and experiment

with the theories they learn in the classroom and gain practical experience in their field.

- Alpha Psi Omega
- Associate Dance Ensemble
- BIPOC of STAD
- Contemporary Dance Ensemble
- Dance Studio Productions
- Dance Theatre
- Dukes Players
- IMPACT
- Madison New Works Lab
- Mainstage Productions
- National Honor Society for Dance Arts
- Stratford Players
- Studio Theatre Productions
- USITT Student Chapter
- Virginia Repertory Dance Company
- Weird Cousins Improv Group
- Workshop Productions

Faculty

Professors

Dennis Beck, Ryan Corrison, Ingrid De Sanctis, Rubén Graciani, Suzanne Miller-Corso, Ilana Morgan, Matthew Pardo, Peter Zazzali

Associate Professors

Kate Arecchi, Emily Becher-McKeever, John Burgess, Rachel Rhoades, Brian Smallwood

Assistant Professors

Christian Barillas, Tara Burns, Jillian Courtney, Kathryn Lumpkin, Simon Marland, James Ogden, Javier Padilla, Suwatana Rockland

Lecturer

Mona Merhi

Programs and Courses

Majors

- Theatre, B.A. (p. 435)
- Theatre, Dance Major; B.A. (p. 438)
- Theatre, Musical Theatre Major, B.A. (p. 440)

Minors

- Dance Minor (p. 476)
- Theatre Minor (p. 574)

Licensure Programs

- Dance Teaching Licensure (p. 584)
- Theatre Teaching Licensure (p. 591)

Courses

- Dance (DANC) (p. 658)
- School of Theatre and Dance (STAD) (p. 842)
- Theatre (THEA) (p. 858)

Department of World Languages and Cultures

Academic Unit Head: Cynthia Chalupa

Email: chalupcs@jmu.edu

Phone: (540) 568-6128

Location: Keezell Hall, Room 301

Department of World Languages and Cultures Website (<https://www.jmu.edu/languages/index.shtml>)

Mission

The Department of World Languages and Cultures (WLC) is a central site on campus for the implementation of James Madison University's internationalization efforts. Through its course offerings, study abroad programs, organizations and clubs, and faculty with research areas in diverse languages and cultures, the department serves as a gateway to global learning.

The department offers students and the community a broad range of courses and programs that help develop language and intercultural competence, important skills for the job market, as well as a global perspective and a nuanced understanding of worldwide cultures, both ancient and modern.

The department's mission is to ensure that all graduates who have taken languages have sufficient proficiency to communicate in the target culture and use the language in the workplace. WLC prepares language majors and minors for success as professionals in a wide variety of career fields and enhances the education of students by providing them with basic to advanced language proficiency and intercultural skills that complement their chosen major. Language study also helps develop written, spoken, and interpersonal skills in English, that supports students' learning in other fields and in their first language. By teaching an understanding and appreciation of global cultures and broadening students' perspectives to global communities both in and outside of the United States, the department plays a leading role in internationalizing JMU's curriculum.

The department also plays an important role in International Business (p. 336) and International Affairs (p. 327) majors and in the university cross-disciplinary minors by offering courses for the Asian Studies Minor (p. 452), Medieval and Renaissance Studies Minor (p. 529), Middle Eastern Communities and Migrations Minor (p. 530), Modern European Studies Minor (p. 533), Russian, Eurasian and Slavic Studies Minor (p. 561), Latin American, Latinx and Caribbean Studies Minor (p. 518), World Literature Minor (p. 578), Classical Studies Minor (p. 465) and Linguistics Minor (p. 522).

Goals and Objectives

The department strives to fulfill this mission by pursuing the following goals. Within each area, students will be able to:

Knowledge

- Demonstrate a familiarity with a variety of linguistic principles in both the target language and their own;
- Describe the creative works of the target culture and be familiar with the historical and cultural contexts in which they were created;
- Discuss humanistic ideas, both within the target cultures and beyond;

- Describe the cultural foundations and history of the languages studied and the target cultures in which they are spoken;
- Identify similarities and differences between the target culture and their own.

Skills

- Demonstrate interpretive (reading and listening), interpersonal (speaking and writing spontaneously), and presentational (speaking and writing in planned speech) skills in both English and the target language;
- Link their language skills to other disciplines they are studying and their use at the workplace;
- Think critically, analytically, and objectively;
- Research a topic thoroughly, identify the pros and cons of AI use, and incorporate a wide variety of resources when presenting and writing in the target language;
- Analyze the pros and cons of translation tools and explain their shortcomings in language learning.

Experiences

- Describe the importance of language to human experience and individual and cultural identities;
- Engage in meaningful conversations about cultures other than their own and the products created by those cultures;
- Describe their own cultural and linguistic competencies and develop realistic goals regarding their progress toward language proficiency;
- See the world through the perspectives of cultures beyond their own.

Attitudes

- Embrace life-long learning;
- Approach issues from cross-disciplinary, global and historical perspectives;
- Recognize and acknowledge diversity among cultures;
- Respect the opinions and beliefs of other people(s) while defining and advancing their own;
- Recognize the contingent nature of knowledge.

Program

The department offerings address student needs and interests. Our program:

- Offers majors and minors in a variety of languages;
- Links language learning to other disciplines to help students develop job-ready skills and improve their opportunities at home and globally;
- Encourages and supports the pursuit of a second major and/or minor in a language and another discipline;
- Offers advanced courses for cross-disciplinary programs;
- Offers courses that fulfill the B.A. language requirement;
- Contributes to the Honors program and the success of its students;
- Provides General Education courses and reinforces Madison Foundations goals.

Faculty

The faculty members of the department strive to:

- Meet the ever-evolving needs of the students;
- Develop innovative teaching techniques, courses, and programs that incorporate student inquiry and job-readiness;
- Engage consistently in scholarship;
- Provide dutiful service to the college, the university and the profession;
- Be a role model for students as educators and as human beings.

Student Objectives

Language majors will be able to:

- Apply critical thinking skills in a variety of settings;
- Demonstrate digital and media literacy and apply AI responsibly to their study of languages and cultures;
- Demonstrate an understanding and appreciation of global cultures and their products, practices and perspectives.

Department Objectives

The Department of World Languages and Cultures offers programs designed to:

- Promote the development of linguistic and cultural proficiency among students;
- Develop written, spoken, and interpersonal skills in English, that support their learning in other fields and in their first language;
- Develop an appreciation of global cultures and languages;
- Prepare students for the workplace through the development of spoken and written presentational and research skills both in the target language and their first language;
- Cultivate job candidates for positions in a wide variety of professional fields including technology, business and government, education, medicine and nursing, tourism, and the arts.

Career Opportunities

- Banking
- Criminal Justice
- Education
- Diplomacy
- Foreign Service
- Imports/Exports
- Interpretations
- Law
- Management
- Marketing
- Medicine (including Nursing, Occupational and Physical Therapy, and Public Health)
- Ministry
- Museum work
- Non-governmental Organizations
- Public Relations
- Social work
- Translation

Co-curricular Organizations

The department supports the following organizations:

- Le Cercle Francophone
- Chinese Tea Club
- Dobro Slovo
- German Club
- Il Circolo Culturale Italiano
- Phi Sigma Iota
- Russian Club
- El Club de Español
- Sigma Delta Pi

International Education and Study Abroad

The department plays an important role in the university's study abroad mission. Majors are strongly encouraged to spend time abroad in the target culture. Teaching licensure candidates must demonstrate an advanced-low level of language proficiency; study abroad can help candidates achieve this goal. Students are urged to check with their language advisors to see which courses taken abroad count toward the language major. Language majors/double majors must take 33 credit hours of language and culture courses specified by the department. Students are required to take 18 credits on JMU's main campus.

Language minors must take 18 credits of language courses specified by the department. Students are required to take 9 credits on JMU's main campus.

Placement in the Language Curriculum

The score a student receives from the online placement test will be valid for registering in a language class that takes place up to and including the second semester of the sophomore year at JMU. A student who waits until their junior year to begin language study will be required to repeat the online placement test. Students will be placed in a class according to the new score, even if this means starting in a lower-level class than was indicated by the first placement test.

Students who receive the Seal of Biliteracy from their high school, will automatically place into 232 in the language. If they have a high score on the test taken for the Seal, they should contact the WLC Academic Unit Head for additional placement options.

Students who receive AP/IB credit should contact the Academic Unit Head for accurate placement based on their test scores.

The WLC faculty strongly recommend that students begin their required language classes no later than the second semester of the sophomore year and ideally as soon as possible after taking the online placement exam.

Semester Registering	Accepted Placement Score
Fall, first year	First year test
Spring, first year	First year test
Fall, sophomore year	First year test
Spring, sophomore year	First year test
Fall, junior year (and all later semesters)	Must retake test in spring of sophomore year or later in order to register for language classes for this semester and all later semesters

Heritage speakers who would like to pursue a language major or minor may be placed through the language placement test or through other

methods demonstrating education in the target culture. Check the department website for procedures relating to placement.

- Swahili (SWA) (p. 857)
- World Languages and Cultures (WLC) (p. 867)

For placement in languages not offered by the department, contact the WLC Academic Unit Head.

Faculty

Professors

José Ignacio Barrio Olano, Cynthia Chalupa, Peter Eubanks, Jennifer R. Lang-Rigal, Dorothee Polanz, Tomás Regalado-López

Associate Professors

Giulia Cardillo, Diana Galarreta-Aima, Marianne Mason, Yenisei Montes de Oca, Amina Saidou, Aram Shahin, Holly A. Yanacek

Assistant Professors

Verónica Dávila Ellis

Senior Lecturers

Laura Cangiano, Veronica Haun, Karina Kline-Gabel, John Tkac

Instructors

Eduardo Javier Calleja, David Campora, Mariagrazia Fiorello, Stephen Gerome, Iris C. Haseloff, Samuel Hernández, Hugo Moreira Blanco, Mélanie Munier, C. Valentina Riso, Nada Salim

Programs and Courses

Major

- Modern Foreign Languages, B.A. (p. 371)

Minors

- Business Spanish Minor (p. 459)
- Medical Spanish Minor (p. 528)
- Modern Foreign Language Minor (p. 534)
- Spanish-English Translation and Interpretation Minor (p. 568)

Language Teaching Licensure Program

- Modern Foreign Languages Teaching Licensure (p. 589)

Courses

- Arabic (ARAB) (p. 607)
- American Sign Language (ASL) (p. 602)
- Chinese (CHIN) (p. 638)
- Classics (CLAS) (p. 640)
- Foreign Language (FL) (<https://catalog.jmu.edu/courses/fl/>)
- Foreign Language Education (FLED) (p. 691)
- French (FR) (p. 691)
- German (GER) (p. 703)
- Greek (GRK) (p. 708)
- Italian (ITAL) (p. 752)
- Japanese (JAPN) (p. 755)
- Korean (KOR) (p. 766)
- Latin (LAT) (p. 766)
- Linguistics (LING) (p. 770)
- Persian (PERS) (p. 813)
- Portuguese (PORT) (p. 828)
- Russian (RUS) (p. 840)
- Spanish (SPAN) (p. 849)

School of Writing, Rhetoric and Technical Communication

Director: Michael Smith

Email: smithmx@jmu.edu

Phone: (540) 568-6004

Location: Harrison Hall, Room 2252

School of Writing, Rhetoric and Technical Communication Website
(<http://www.jmu.edu/wrtc/index.shtml>)

Mission Statement

We guide and inspire writing professionals.

Vision

Changing how the world values writers.

Career Opportunities

In the WRTC major, students learn the kinds of research, analytical and reasoning skills that will allow them to become successful professionals in a wide range of fields. WRTC graduates can expect career opportunities in writing, editing or production positions with a variety of business, educational or industry employers, including publishing houses, marketing companies, law firms, journalism outlets, computer hardware and software companies, health care providers, engineering firms, environmental organizations, nonprofits, educational institutions and writing consultant groups. WRTC graduates also choose to further their studies in MA, MS, MBA, MFA, PhD and law school programs.

Co-Curricular and Professional Opportunities

Work-Based Learning Courses

All students in the WRTC major are required to enroll in at least one course designated as a work-based learning course. This requirement allows students to apply classroom concepts to real-world communication tasks, build professional skills, and gain experience working with organizations in fields like business, government, or nonprofits. It also helps students develop the confidence, workplace readiness, and practical experience needed to successfully secure and thrive in future internships. Courses fulfilling the work-based learning requirement include WRTC 480, WRTC 484, WRTC 486, and WRTC 488. In addition, several other courses include a work-based learning requirement.

Internships

The WRTC internship is a requirement for all WRTC majors. The internship allows students to apply content knowledge and skills acquired in their WRTC coursework in a professional context to design, write, edit and produce documents for internship providers in academia, business, industry and government. Students are supported throughout the process, from finding an internship placement to reflecting on the experience. Information about internships may be obtained through the WRTC website.

RhetTech Journal

RhetTech is WRTC's student-run, online journal. It publishes high-quality compositions in a variety of genres from undergraduate students working in the disciplines of writing, rhetoric and technical communication across

the country. *RhetTech* is created and managed by students enrolled in WRTC 328. These practicum students develop evaluation criteria, read and discuss each submission, and work individually with writers to prepare their work for publication.

JMURJ

The *James Madison Undergraduate Research Journal* is a peer-reviewed academic journal dedicated to promoting, publishing and sharing the text- and media-based research and scholarship created by undergraduate students in all JMU disciplines. *JMURJ* editorial board members collaborate with students, faculty and administrators in a university-wide effort to advance JMU undergraduate scholarship, research and intellectual work.

The Bluestone yearbook

The Bluestone is James Madison University's award-winning yearbook. *The Bluestone* was first published in 1910 and has been issued every year since, making it the university's oldest publication. *The Bluestone* is published in the spring, covering all events and activities from approximately March of one year to March of the following year. The book is created and produced by a student staff led by the Editor(s)-in-Chief. WRTC students frequently serve in roles on the editorial board and in staff positions such as writers and designers.

Semester in London

WRTC students can participate in the Semester in London program through the School of Media Arts and Design and can enroll in the cross-listed courses WRTC 351 and WRTC 360. Both courses fulfill requirements in the British Communication and Media minor (p. 457), and WRTC 360 also fulfills a requirement for the Modern European Studies minor (p. 533).

Pre-Law

Pre-law at JMU is not a major, but a number of WRTC majors elect to pursue law school after graduation. WRTC prepares these students by equipping them with strong analytical, research, and persuasive writing skills that are essential for success in law school. Through coursework focused on argumentation, critical thinking, and rhetorical awareness, students learn to craft clear, compelling legal-style arguments. This foundation helps graduates transition confidently into the rigorous reading, writing, and reasoning demands of law school. Relevant coursework includes WRTC 301, WRTC 314, WRTC 330, WRTC 340, WRTC 460 and WRTC 462.

Scholarship and Award Opportunities

JB Hancock Scholarship for Writing and New Media

The JB Hancock Scholarship awards at least \$1,000 annually to one or more WRTC majors who have created a compelling piece of writing or new media communication that demonstrates both the importance of and excellence in connecting with audiences and communicating clearly and responsibly in a manner that is factual and persuasive.

Whitten Maher Memorial Scholarship for Writing and Design

The Whitten Maher Memorial Scholarship for Writing and Design was established to acknowledge and support undergraduate work that conveys the compassion, intensity and well-informed reason of its namesake. The scholarship is administered by WRTC but is open to all returning full- or part-time James Madison University undergraduate

students in all academic disciplines. Each scholarship recipient is awarded at least \$1,000 toward the following year's tuition.

Madison Writing Awards

The Madison Writing Awards (MWA), housed in WRTC, is a university-wide competition that celebrates writing across the curriculum in all undergraduate academic programs, regardless of medium and genre, from traditional papers to multimodal productions. These awards reflect the commitment of James Madison University, the College of Arts and Letters, and the School of Writing, Rhetoric and Technical Communication to prepare students for educated and enlightened global citizenship through the outlets of writing and rhetoric. The MWA biennial awards ceremony features a showcase of winning pieces as well as the presentation of cash prizes.

Faculty

Professors

Jennifer M. Almjeld, Angela Crow, Steven S. Lunsford, Michael Smith, Traci A. Zimmerman

Associate Professors

Lori Beth De Hertogh, Susan R. Ghiaciuc, Elisabeth C. Gumnior, Laura K. Miller

Assistant Professors

Justiss W. Burry, Meghalee Das, Jeremy R. Merritt, Mollie Stambler, Tiffany Wilfgar

Senior Lecturers

Danielle R. DeRise, Michael Hickman, Kevin Jefferson, Cynthia S. Martin, Karen M. McDonnell, Jay Varner

Lecturers

Nancy El Gendy, Tiajuana LaTanya Tillman

Programs and Courses

Majors

- Writing, Rhetoric and Technical Communication, B.A. (p. 442)
- Writing, Rhetoric and Technical Communication, B.S. (p. 444)

Minors

- Writing, Rhetoric and Technical Communication Minor (p. 580)

Graduate Program

- Writing, Rhetoric and Technical Communication, M.A. (<https://catalog.jmu.edu/programs/writing-rhetoric-technical-communication-ma/>)
- Writing, Rhetoric, and Technical Communication, M.S. (<https://catalog.jmu.edu/programs/writing-rhetoric-technical-communication-ms/>)

Courses

- Writing, Rhetoric, and Technical Communication (WRTC) (<https://catalog.jmu.edu/courses/wrtc/>)

Cross-Disciplinary Minors

WRTC contributes courses to the following cross-disciplinary minors:

- Digital Studies Minor (p. 478)
- Disability Studies Minor (p. 480)

- Entrepreneurship Minor (p. 484)
- Environmental Humanities Minor (p. 485)
- Environmental Science Minor (p. 490)
- Environmental Studies Minor (p. 492)
- Humanitarian Affairs Minor (p. 514)
- Legal Studies Minor (p. 521)
- Medical Humanities Minor (p. 527)
- Science, Technology and Society Minor (p. 562)
- Substance Use Disorder Education Minor (p. 572)
- Women's, Gender and Sexuality Studies Minor (p. 576)

GENERAL EDUCATION: THE HUMAN COMMUNITY

Sarah Brooks and Elizabeth Brown, **Interim Co-Directors of General Education**

Email: gened@jmu.edu

Mission Statement

In the liberal arts tradition, General Education: The Human Community aspires to create informed global citizens of the 21st century.

James Madison University's General Education program is committed to creating an academic environment that supports inclusive excellence, and we affirm that inclusive excellence advances understanding and is integral to a liberal arts education. We acknowledge the history of exclusion nationally and at James Madison University. We embrace the process of making JMU's General Education program accessible, affirming, and action-oriented.

By placing inclusive excellence at the center of our mission, we seek to implement strategies that reimagine the relevance of a liberal arts education. We challenge our community of students, faculty, and staff to engage in personal and collective reflection, development, and action.

Philosophy

General Education: The Human Community is the core academic program of James Madison University in which students come to understand how distinct disciplines look at the world from different vantage points. Courses in The Human Community are organized into five areas, each emphasizing unique tools, rationales and methodologies. Taken together, courses in a student's chosen major and The Human Community complement and complete each other. Both are integral and essential components of a student's full and proper education.

Goals

Students understand the historical and contemporary distinctions and interconnections among people, institutions, and communities that create, preserve and transmit culture and knowledge in the arts, sciences, mathematics, social sciences and humanities.

Students become skilled in questioning, investigating, analyzing, evaluating and communicating.

Students participate in a variety of aesthetic and civic experiences reflecting human concerns and values that transcend the limits of specialization.

Organization and Learning Outcomes

The Human Community is a coherent, learning outcomes-based academic program. These outcomes are met by completing 14 requirements organized in five areas.

Area Credit Hours

- Madison Foundations (p. 111): 9 Credit Hours
- Arts and Humanities (p. 110): 9 Credit Hours
- The Natural World (p. 114): 10 Credit Hours

- American and Global Perspectives (p. 109): 7 Credit Hours
- Sociocultural and Wellness Area (p. 113): 6 Credit Hours

American and Global Perspectives

John A. Scherpereel, **American and Global Perspectives Coordinator**

Courses in American and Global Perspectives require students to think critically about their own society and its relationship to the larger global community. These courses develop responsible and enlightened global citizenship by examining a wide variety of the processes that shape the human experience. Students will complete one course that focuses on the American experience and one course that examines the global experience.

American and Global Perspectives Learning Outcomes

The American Experience

After completing an **American Experience** course students will be able to identify, conceptualize and evaluate:

- Social and political processes and structures using quantitative and qualitative data.
- Primary sources from diverse perspectives relating to American history, political institutions and society.
- The evolution of intellectual concepts shaping American democratic institutions, including issues involving power, inequity and justice.
- The complexity and diversity of American politics, society and culture.
- Intentions and consequences of America's engagement in global affairs.
- How the historical exclusion of various social identities influences political, social, cultural and economic development.

The Global Experience

After completing a **Global Experience** course students will be able to identify, conceptualize and evaluate:

- Basic global problems.
- Global political, social, cultural and economic systems that shape societies.
- Issues involved in analyzing societies different from one's own.
- Strategies to achieve diversity, equity, inclusion, justice and access both locally and globally.
- Diverse theoretical models to analyze global problems.
- The value and complexity of global diversity in all its forms.

American and Global Perspectives Requirements

American and Global Perspectives consists of seven credits distributed across two program requirements: **The American Experience** and **The Global Experience**. Students will choose one course to satisfy each of these two requirements. **American and Global Perspectives** requirements may be completed concurrently or individually, in any order. Students may not take both POSC 200 and POSC 225 to complete the **American and Global Perspectives** requirements.

The American Experience [C4AE]

Each American Experience course provides students with an understanding of the major themes and concepts that structure American life today. HIST 225 does so through a contextual and document-based study of the American historical experience that emphasizes the interaction of people, ideas and social movements. JUST 225 frames questions regarding historic and contemporary events in terms of issues of justice, highlighting how societal structures interact with individual lives and vice versa. POSC 225 focuses on the evolution and contemporary operation of the American political system by examining its fundamental principles and current dynamics.

Code	Title	Credits
HIST 225	U.S. History [C4AE]	4
JUST 225	Justice and U.S. Society [C4AE]	4
POSC 225	U.S. Government [C4AE]	4

The Global Experience [C4GE]

Each of the courses in the Global Experience is an investigation into a series of global issues that are of great importance to the human community. Topics discussed will vary from course to course. Issues are examined in a systemic context that allows students to see connections between disciplines. The unifying theme is an analysis of overarching structures at the global level that condition people's behavior and which are shaped by that behavior. From this perspective the study of global issues requires more than studying current events; it involves placing these global issues in a systemic context.

Code	Title	Credits
AAAD 200	Introduction to African, African American and Diaspora Studies [C4GE]	3
ANTH 195	Cultural Anthropology [C4GE]	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
GEOG 200	Geography: The Global Dimension [C4GE]	3
POSC 200	Global Politics [C4GE]	3
SOCI 110	Social Issues in a Global Context [C4GE]	3

Arts and Humanities

Dennis C. Beck, **Arts and Humanities Coordinator**

Arts and Humanities courses show students what it means to live lives enriched by reflection, imagination and creativity. They do so by offering each individual a cross-disciplinary experience within the arts and humanities, those areas of endeavor that humans have long valued for their intrinsic worth and that invite a deeper appreciation of the human experience. The broadly stated goals for **Arts and Humanities** are:

- To introduce students to cultural, historical, aesthetic and theoretical expressions of and questions about human experience.
- To expose students to multiple academic disciplines in the arts and humanities and their methods and unique perspectives.
- To inspire a deeper awareness of how the interplay between culture and expression affects both collective and individual identities.
- To foster appreciation of the aesthetic and formal qualities of literary, visual and performing arts.
- To engage students in thinking critically and communicating clearly about enduring questions concerning human life, culture and history.

Arts and Humanities Learning Outcomes

Human Questions and Contexts

After completing a **Human Questions and Contexts** course students will be able to:

- Critically examine their own and others' viewpoints on and responses to the world.
- Apply the methods of the discipline(s) studied to material from the humanities.
- Explain and reflect on complex ideas about the human world.
- Demonstrate an understanding of broader cultural, historical, or conceptual contexts of particular issues, ideas, objects, or events—past and present.
- Experience and appreciate the value, meaning, and context of humanities events (such as exhibits, films, performances or public lectures).

Visual and Performing Arts

After completing a **Visual and Performing Arts** course students will be able to:

- Experience and understand the arts as integral to human life.
- Explain how art and society are interrelated.
- Demonstrate disciplinary literacy (vocabulary, concepts, creative processes) in an art form.
- Produce an informed response to the formal, contextual and aesthetic qualities of artistic works.

Literature

After completing a **Literature** course students will be able to:

- Generate increasingly nuanced questions about literature and explain why those questions matter.
- Use appropriate vocabulary and methods to analyze specific literary expressions of culture, modes of interpretation, and the relationship between the reader, the author, and the text.
- Define ways that texts serve as arguments and identify rhetorical and formal elements that inform these arguments.
- Recognize some of the cultural, social, political, economic, and creative contexts in which texts are produced and understand that readers may interpret and evaluate literature from a variety of standpoints.
- Articulate ways in which literature gives us access to the diversity of human experience and helps us understand our lives through storytelling.

Arts and Humanities Requirements

Arts and Humanities consist of nine credits distributed across three program requirements: **Human Questions and Contexts**, **Visual and Performing Arts**, and **Literature**. Students will choose one course to satisfy each of these three requirements. **Arts and Humanities** requirements may be completed concurrently or individually, in any order.

Human Questions and Contexts [C2HQC]

AMST 200 takes a cross-disciplinary approach to questions about American identity and shows how they reflect a complex interplay of cultural, historical, religious and ideological perspectives. The ANTH and HIST courses introduce students to the great cultures of the world by surveying the common patterns of experience that characterized

Western, Middle Eastern, Asian, African, Meso- and South American societies in the past. The HUM and LAXC courses are cross-disciplinary, in-depth explorations of specific topics, cultures, periods or themes. The PHIL and REL courses explore the great inquiries into human existence and the ways different cultures across different time periods constructed their responses to questions concerning humans' existence and their relationship to nature, ultimate reality and the universe. Thus, all of the courses in **Human Questions and Contexts** emphasize central questions about the human condition and ways of studying values and beliefs as they are shaped by class, gender, race, historical events, philosophy and religion.

Code	Title	Credits
AMST 200	Introduction to American Studies [C2HQC]	3
ANTH 205	Buried Cities, Ancestral Places: The Rise and Fall of Early Human Societies [C2HQC]	3
HIST 101	World History to 1500 [C2HQC]	3
HIST 102	World History since 1500 [C2HQC]	3
HUM 250	Foundations of Western Culture [C2HQC] (Topics vary by section. Examples include: Ancient Greece, Ancient Mediterranean)	3
HUM 251	Modern Perspectives [C2HQC] (Topics vary by section. Examples include: The 'Catalan Question': a Socio-cultural and Historical Approach, Creating a Culture of Tolerance in Islamic Spain, and Italian Society and Culture after WWII)	3
HUM 252	Global Cultures [C2HQC] (Topics vary by section. Examples include: Global South Asia, West Africa, Arab Streets: Meeting of East and West, Japanese Culture: Past and Present)	3
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]	3
PHIL 101	Introduction to Philosophy [C2HQC]	3
REL 101	Religions of the World [C2HQC]	3
REL 102	Religion, Spirituality and the Meaning of Life [C2HQC]	3

Visual and Performing Arts [C2VPA]

ARTH 205 and ARTH 206 are global art history surveys that introduce students to the visual arts, whose history often has been interconnected with developments in music, dance and theatre/film. These surveys are organized chronologically, but focus distinctly on artistic perception and experience. ARTH 204 asks pressing questions of select, major monuments, examining how images and works of architecture most reflect current values as well as those from the past. MUS 206 explores history and the arts through the study of music: its development, aesthetics, forms and styles; and its context within the cultural communities that produced it. ART 200 and MUS 200 are introductions to art or music in general culture; DANC 215 considers the historic and cultural significance of dance as well as the components of dance performance; THEA 210 studies theatre as an art form in its cultural contexts including acting, directing, design, costuming and lighting; MUS 203 explores America's music landscape and examines the interconnections among music, art and literature in historical periods.

Code	Title	Credits
ART 200	Art Today: Themes in Contemporary Art [C2VPA]	3
ARTH 204	Art, Power and Society [C2VPA]	3

ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
DANC 215	The Dance Experience [C2VPA]	3
MUS 200	Music and Culture [C2VPA]	3
MUS 203	Music in America [C2VPA]	3
MUS 206	Introduction to Global Music [C2VPA]	3
THEA 210	Introduction to Theatre [C2VPA]	3

Literature [C2L]

The literature surveys and studies provide students with extensive reading experiences of representative genres and authors and various critical approaches to literary texts, as well as opportunities to explore the complex ways that literature both reflects and helps change or create the cultural and intellectual contexts of the times in which texts are written. Students are expected to learn strategies for reading and interpreting any literary text so that they come to deepen their appreciation of the aesthetics, rhetorical strategies and meaning of a range of literary texts. Through the humanistic study of literature, students will also obtain a better understanding of themselves and their own culture as well as those of others.

Literature and Writing Infusion

The courses that satisfy the **Literature** requirement are designated as writing-infused. Students will write a minimum of 5,000 words (approximately 15 pages double-spaced in a standard font) in assignments that may include both informal and formal, ungraded and graded forms. The extensive opportunity to produce and receive feedback on various genres of academic writing will help students sharpen their responses to interesting and thought-provoking texts and promote more engaged and sophisticated reading strategies.

Code	Title	Credits
ENG 221	Literature and Ideas [C2L]	3
ENG 222	Literature and Genres [C2L]	3
ENG 235	Survey of English Literature: From Beowulf to the Eighteenth Century [C2L]	3
ENG 236	Survey of English Literature: Eighteenth Century to Modern [C2L]	3
ENG 239	Studies in World Literature [C2L]	3
ENG 247	Survey of American Literature: From the Beginning to the Civil War [C2L]	3
ENG 248	Survey of American Literature: From the Civil War to the Modern Period [C2L]	3
ENG 260	Survey of African American Literature [C2L]	3
HUM 200	Great Works [C2L] (Topics vary by section. Examples include: Narratives of Belonging and Identity; Survey of Traditional and Modern Chinese Literature; Modern Arabic Short Stories; Utopia, Dystopias)	3

Madison Foundations

Sarah T. Brooks, **Madison Foundations Coordinator**

Madison Foundations is the cornerstone of General Education: The Human Community at JMU, with course work requiring students to demonstrate:

- Critical thinking skills
- Effective oral presentation skills
- Effective writing skills
- Competency in information literacy

Students will be able to use reading, writing, human communication, critical thinking, and information literacy skills for inquiring, learning, thinking and communicating in their personal, academic and civic lives. Competence in these areas is fundamental to subsequent study in major and professional programs. Therefore, all students complete Madison Foundations requirements during their first academic year, including their first summer, at JMU.

Madison Foundations Learning Outcomes

After completing Madison Foundations, students should be able to use reading, writing, human communication, critical thinking and information literacy skills for inquiring, learning, thinking and communicating in their personal, academic and civic lives.

Critical Thinking

After completing a Critical Thinking course students will be able to:

- Identify the basic components of arguments, including premises, supporting evidences, assumptions, conclusions and implications;
- Evaluate claims and sources for clarity, credibility, reliability, accuracy and relevance;
- Evaluate arguments for soundness, strength and completeness;
- Demonstrate an intellectual disposition to be fair-minded in considering evidence, arguments and alternative points of view.

Human Communication

After completing a Human Communication course students will be able to:

- Explain the fundamental processes that significantly influence communication.
- Construct messages consistent with the diversity of communication purpose, audience, context and ethics.
- Respond to messages consistent with the diversity of communication purpose, audience, context and ethics.
- Utilize information literacy skills expected of ethical communicators.

Writing

After completing a Writing course students will be able to:

- Demonstrate an awareness of rhetorical knowledge, which includes the ability to analyze and act on understandings of audiences, purposes and contexts in creating and comprehending texts.
- Employ critical thinking, which includes the ability through reading, research and writing, to analyze a situation or text and make thoughtful decisions based on that analysis.
- Employ writing processes, which include prewriting, drafting, engaging with feedback, and revising.
- Recognize the many characteristics of effective messages and how these may vary according to genres, situations, and audiences.

- Show flexibility in composition strategies by applying information literacy and other modern literacies.

Information Literacy

Information Literacy is developed in all courses within Madison Foundations. Additionally, the Human Communication courses incorporate targeted learning modules with graded quizzes. After this broad, holistic curricular development of research skills and information literacy, students will be able to:

- Recognize the components of scholarly work and that scholarship can take many forms.
- Demonstrate persistence and employ multiple strategies in research and discovery processes.
- Identify gaps in their own knowledge and formulate appropriate questions for investigations in academic settings.
- Evaluate the quality of information and acknowledge expertise.
- Use information effectively in their own work and make contextually appropriate choices for sharing their scholarship.
- Use information ethically and legally.

Madison Foundations Requirements

Madison Foundations consists of nine credits distributed across four program requirements: **Critical Thinking, Human Communication, Writing and Information Literacy**. These requirements are satisfied with the completion of three courses: one each specifically for **Critical Thinking, Human Communication, and Writing**. These three courses collectively satisfy the **Information Literacy** requirement as well.

Completion of all requirements in **Madison Foundations** is required of students in their first academic year at JMU, and the courses may be taken in any order. **Madison Foundations** courses are not repeatable without permission, which is given based on course availability. There are no overrides available in **Madison Foundations** courses during the Fall and Spring semesters. Summer term is an additional opportunity to complete **Madison Foundations** coursework.

Critical Thinking [C1CT]

Each of the courses satisfying the **Critical Thinking** requirement presents students with various techniques and approaches to critical thinking such as analyzing and evaluating information, arguments, premises and concepts. Critical thinking fosters inquiry and problem-solving abilities. Depending upon the course, the content focuses on the function of language, basic business principles, issues in recent history, mediated communication, informal logical reasoning or problem-solving in science and technology.

Code	Title	Credits
BUS 160	Applied Critical Thinking in Business [C1CT]	3
EDUC 102	Critical Questions in Education [C1CT]	3
HIST 150	Critical Issues in Recent Global History [C1CT]	3
ISAT 160	Problem Solving Approaches in Science and Technology [C1CT]	3
PHIL 120	Critical Thinking [C1CT]	3
PHIL 150	Ethical Reasoning [C1CT]	3
SMAD 150	Mediated Communication: Issues and Skills [C1CT]	3

Additional Information

Students pursuing a Bachelor of Arts (B.A.) degree may not use either PHIL 120 or PHIL 150 to fulfill the B.A. philosophy course requirement.

Students who have received credit for one **Critical Thinking** class are not eligible to receive credit for a second **Critical Thinking** class without permission.

Human Communication [C1HC]

Each of the courses satisfying the **Human Communication** requirement introduces students to the study of human communication as a process. Emphasis is on examining the role of self-concept, perception, culture, verbal and nonverbal dimensions in the communication process; applying critical listening skills; and developing skills in oral presentations. Depending upon the course, the content focuses on an overview of the principles and practices of interpersonal, small group and public communication, or constructing informative and persuasive speeches with an emphasis on individual public speaking contexts, or constructing informative and persuasive group presentations.

Code	Title	Credits
SCOM 121	Fundamental Human Communication: Presentations [C1HC]	3
SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC]	3
SCOM 123	Fundamental Human Communication: Group Presentations [C1HC]	3

Additional Information

Students who have received credit for one **Human Communication** class are not eligible to receive credit for a second **Human Communication** class.

Writing [C1W]

The course that satisfies the **Writing** requirement provides students with the tools to study the role of the writer, the purpose of documents, and the contexts and audience expectations within which documents are produced. Writing is taught as a process involving invention, collaboration, research and revision. Students read, analyze and create texts in multiple genres using traditional and digital communication tools. Content areas may include public and professional discourse, culture, humanities, and technology.

Code	Title	Credits
WRTC 103	Rhetorical Reading and Writing [C1W]	3

Sociocultural and Wellness Area

Kethera A.J. Moore, **Sociocultural and Wellness Area Coordinator**

Through studying the many variables that influence human behavior in contemporary society, students gain an understanding of the relationship between the individual and a diverse community and develop a sense of responsibility for self and community. Students explore how individuals develop and function in the social, psychological, emotional, physical and spiritual dimensions.

Sociocultural and Wellness Area Learning Outcomes

Wellness Domain

- After completing a **Wellness Domain** course students will be able to:
- Understand the dimensions of wellness, the various factors affecting each dimension and how dimensions are interrelated.
 - Understand the relationship between personal behaviors and lifelong health and wellness.
 - Assess their own levels of health and wellness and understand how these levels impact their quality of life.
 - Identify and implement strategies to improve their wellness.

Sociocultural Domain

- After completing a **Sociocultural Domain** course students will be able to:
- Understand how individual and sociocultural factors interact in the development of the beliefs, behaviors and experiences of oneself and others.
 - Discern the extent to which sources of information about the sociocultural domain are reputable and unbiased.
 - Evaluate the extent to which the approach to and uses of psychosocial research are ethical and appropriate.

Sociocultural and Wellness Area Requirements

The **Sociocultural and Wellness Area** consists of six credits distributed across two program requirements: the **Wellness Domain** and the **Sociocultural Domain**. Students will choose one course to satisfy each of these two requirements. **Sociocultural and Wellness Area** requirements may be completed concurrently or individually, in any order.

Students are expected to complete **Sociocultural and Wellness Area** course work during their first two years at the university and are strongly encouraged to complete their **Wellness Domain** coursework in their first year.

Wellness Domain [C5W]

- Each of the courses satisfying the **Wellness Domain** requirement examines the individual behaviors that influence the dimensions of health and wellness.
- Students will gain knowledge and conduct self-assessments relating to a variety of health topics. In addition, students will learn strategies that improve lifetime health and wellness. Courses include a physical wellness component as a part of the course requirements.

Code	Title	Credits
HTH 100	Personal Wellness [C5W]	3
KIN 100	Lifetime Fitness and Wellness (2,2) [C5W]	3

Sociocultural Domain [C5SD]

Each of the courses satisfying the **Sociocultural Domain** requirement focuses on sociocultural and psychological aspects of individuals interacting within societal contexts. These courses enable students to develop ethical and scientifically-based critical thinking about human behavior and social interaction.

Students study the formation and functions of social relationships and reflect on personal responsibilities to diverse communities within which people function throughout life. Students explore sociocultural and psychological aspects of personal belief systems, self-identity and assumptions about others.

Code	Title	Credits
EDUC 175	Learning, Behavior and Engagement in Context [C5SD]	3
PSYC 101	General Psychology [C5SD]	3
PSYC 160	Life Span Human Development [C5SD]	3
SOCI 140	Microsociology: The Individual in Society [C5SD]	3
WGSS 200	Introduction to Women's, Gender and Sexuality Studies [C5SD]	3

The Natural World

Elizabeth T. Brown, **The Natural World Coordinator**

Scientific investigations into the natural world use analytical methods to evaluate evidence, build and test models based on that evidence, and develop theories. Quantitative reasoning, broadly construed, is humanity's most powerful language for prediction and analysis of physical and human phenomena. Courses in **The Natural World** provide students with the opportunity to develop methodological awareness and problem-solving skills in the sciences and quantitative disciplines at the college level. Students will be introduced to the scientific method along with a substantial body of scientific facts, concepts, models and theories. They will also gain experience in using quantitative methods to obtain knowledge about the natural, mathematical, and social worlds. This area is cross-disciplinary, thereby demonstrating boundaries and connections among mathematics, statistics, the sciences and other aspects of culture.

The General Education program recommends that students begin **The Natural World** during their first year and complete it by the end of their sophomore year if possible. Individual courses satisfy requirements in a number of major and professional programs. Students are encouraged to select appropriate courses in **The Natural World** on the basis of their backgrounds, interests and educational objectives.

The Natural World Learning Outcomes

After completing courses in **The Natural World** students will be able to meet the following objectives:

- Describe the methods of inquiry that lead to mathematical truth and scientific knowledge and be able to distinguish science from pseudoscience.
- Use theories and models as unifying principles that help us understand natural phenomena and make predictions.
- Recognize the interdependence of applied research, basic research and technology, and how they affect society.
- Identify the interdependence of natural systems with the social, economic and ethical aspects of global and local issues.
- Use graphical, symbolic and numerical methods to analyze, organize and interpret natural phenomena.
- Discriminate between association and causation and identify the types of evidence used to establish causation.
- Formulate hypotheses, identify relevant variables and design experiments to test hypotheses.

- Evaluate the credibility, use and misuse of scientific and mathematical information in scientific developments and public-policy issues.

The Natural World Requirements

The **Natural World** consists of 10 credits distributed across four program requirements representing four different aspects of scientific knowledge: **Quantitative Reasoning, Physical Principles, Natural Systems and Lab Experience**. Students will choose one course to satisfy each of these four requirements. **Quantitative Reasoning** consists of mathematics or statistics courses. **Physical Principles and Natural Systems** consist of science courses. The requirements may be completed in any order, except in the case of courses denoted by an asterisk (*), which have a mathematics and/or science prerequisite or corequisite. In addition, students are required to have at least one lab experience. Certain courses are designed for future teachers, and enrollment in these courses may be limited to Education majors; these courses are indicated with a double asterisk (**).

Quantitative Reasoning [C3QR]

Students build mathematical models of systems and learn to understand, interpret and analyze data that is numerical in nature.

Code	Title	Credits
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	4
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	3
MATH 103	The Nature of Mathematics [C3QR]	3
MATH 105	Statistical Literacy [C3QR]	3
MATH 110	Mathematics for Elementary and Middle School Teachers I [C3QR] (Students who have taken MATH 107 prior to Fall 2021 may count that course in place of MATH 110 towards General Education)	4
MATH 205	Applied Calculus [C3QR]	3
MATH 220	Elementary Statistics [C3QR]	3
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 231	Calculus with Functions I [C3QR]	3
MATH 233	Calculus with Modeling I [C3QR]	3
MATH 235	Calculus I [C3QR]	4

Physical Principles [C3PP]

Each of the courses satisfying the Physical Principles requirement investigates underlying principles of nature. These principles are applied to build models, often quantitative in nature, that explore and explain a variety of natural phenomena.

Code	Title	Credits
ASTR 120	The Solar System [C3PP]	3
ASTR 121	Stars, Galaxies, and Cosmology [C3PP]	3
CHEM 120	Concepts of Chemistry [C3PP]	3
CHEM 131	General Chemistry I [C3PP] ^{*,1}	3
ENVT 101	Perspectives in Environmental Science [C3PP]	3
ISAT 100	Environmental and Energy Sustainability [C3PP]	3
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP] (includes lab))	4
ISCI 101	Physics, Chemistry and the Human Experience [C3PP]	3
ISCI 172	Physical Science for Teachers [C3PP] ^{**}	3

PHYS 121	The Physical Nature of Light and Sound (3, 1) [C3PP] (includes lab))	4
PHYS 125	Physics with Biological Applications [C3PP]	3
PHYS 140	College Physics I [C3PP] ²	3
PHYS 215	Energy and the Environment [C3PP]	3
PHYS 240	University Physics I [C3PP]	3

¹ CHEM 131L required lab corequisite

² PHYS 140L required lab corequisite

Natural Systems [C3NS]

Each of the courses satisfying the **Natural Systems** requirement explores the behavior of earth and life systems. Students will investigate interactions within these systems, between the systems and their environment, and with society.

Code	Title	Credits
ANTH 196	Biological Anthropology [C3NS]	3
BIO 103	Contemporary Biology [C3NS]	3
BIO 140	Foundations of Biology I [C3NS]	3
BIO 270	Human Physiology [C3NS]	3
GEOL 102	Environment: Earth [C3NS]	3
GEOL 110	Physical Geology [C3NS] ^{*,1}	3
GEOL 115	Earth Systems and Climate Change [C3NS]	3
GEOL 210	Applied Physical Geology [C3NS] [*]	3
GEOL 211	Introduction to Oceanography [C3NS]	3
ISAT 113	Biotechnology Issues in Science and Technology [C3NS] ^{*,2}	3
ISCI 171	Earth and Planetary Science for Teachers [C3NS] ^{**}	3
PSYC 122	The Science of Vision and Audition [C3NS]	3

¹ GEOL 110L required lab corequisite

² ISAT 113L required lab corequisite

Lab Experience [C3L]

Each of the courses satisfying the **Lab Experience** emphasizes the observational and experimental nature of science. Through hands-on experiential learning, students will make observations and use them to test predictions and hypotheses.

Students must take **Physical Principles** or **Natural Systems** course with a lab included, or:

Code	Title	Credits
ANTH 196L	Scientific Perspectives in Biological Anthropology [C3L] ^{*,1}	1
BIO 140L	Foundations of Biology I Laboratory [C3L] ^{*,2}	1
BIO 270L	Human Physiology Laboratory [C3L] ^{*,3}	1
CHEM 131L	General Chemistry Laboratory [C3L] ^{*,4}	1
GEOL 110L	Physical Geology Laboratory [C3L] ^{*,5}	1
GEOL 115L	Earth Systems and Climate Change Lab [C3L] ^{*,6}	1
ISAT 113L	Biotechnology Issues in Science and Technology Laboratory [C3L] ^{*,7}	1
ISCI 104	Scientific Perspectives [C3L]	1
ISCI 173	Life and Environmental Science for Teachers [C3L] ^{**}	3

PHYS 125L	Physics with Biological Applications Laboratory [C3L] ^{*,8}	1
PHYS 140L	General Physics Laboratory I [C3L] ^{*,9}	1
PHYS 240L	University Physics Laboratory I [C3L] ^{*,10}	1

¹ ANTH 196 required prerequisite or corequisite

² BIO 140 required prerequisite or corequisite

³ BIO 270 required prerequisite or corequisite

⁴ CHEM 131 required corequisite

⁵ GEOL 110 required corequisite

⁶ GEOL 115 required prerequisite or corequisite

⁷ ISAT 113 required corequisite

⁸ PHYS 125 required corequisite

⁹ PHYS 140 required corequisite

¹⁰ PHYS 240 required prerequisite or corequisite

UNDERGRADUATE DEGREE REQUIREMENTS

All undergraduate degrees at James Madison University are designed to be completed in four years.

Bachelor of Arts (B.A.)

To meet requirements for the Bachelor of Arts Degree, students must meet both a philosophy and foreign language requirement. If a student uses a course to satisfy a General Education requirement, they may not also use that same course to satisfy a Bachelor of Arts Degree requirement. See the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml) for more information.

Code	Title	Credits
	General Education requirement ¹	41
	Foreign language requirement ²	0-12
	Philosophy requirement ³	3
	Major requirements and university electives	62-76
Total Credits		120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures placement test.

³ Any PHIL course except PHIL 120 or PHIL 150, provided that course has not been applied towards General Education requirements. Consult the list of courses satisfying Bachelor of Arts degree requirements on the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml).

Bachelor of Business Administration (B.B.A.)

Code	Title	Credits
	General Education requirement ¹	41
	B.B.A. core courses	45-46
	Major requirements and university electives	24-46
Total Credits		120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

Bachelor of Fine Arts (B.F.A.)

Code	Title	Credits
	General Education requirement ¹	41
	Core program – art courses	24
	Art emphasis courses	54

Electives	1
Total Credits	120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

Bachelor of Individualized Study (B.I.S.)

Code	Title	Credits
	General Education requirement ¹	41
	Major requirements	30
	Electives	49
Total Credits		120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

Bachelor of Music (B.M.)

Code	Title	Credits
	General Education requirement ¹	41
	Core music program courses	30
	Major concentration courses and electives	43-59
Total Credits		120-128

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

Bachelor of Science (B.S.)

To meet requirements for the Bachelor of Science Degree, students must meet both a scientific literacy and quantitative requirement. If a student uses a course to satisfy a General Education requirement, they may not also use that same course to satisfy a Bachelor of Science Degree requirement. See the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml) for more information.

Code	Title	Credits
	General Education requirement ¹	41
	Quantitative requirement ²	3
	Scientific literacy requirement ²	3-4
	Major requirements and electives	73-77
Total Credits		120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

² In addition to General Education. Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml).

Bachelor of Science in Nursing (B.S.N.)

Code	Title	Credits
	General Education requirement ¹	41
	Prerequisites, nursing major requirements and electives	79
Total Credits		120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

Bachelor of Social Work (B.S.W.)

Code	Title	Credits
	General Education requirement ¹	41
	Core program – social work courses	44
	Social work electives	6
	Electives	29
Total Credits		120

¹ The General Education program contains a set of requirements each student must fulfill. Some General Education requirements may be fulfilled by requirements within a student's major.

UNDERGRADUATE ADMISSIONS

Office of Admissions (<http://www.jmu.edu/admissions>)

Phone: (540) 568-5681

Email: admissions@jmu.edu

Fax: (540) 568-3332

Location: Holland Yates Hall, MSC 0101

The enrollment for the university is authorized by the State Council of Higher Education for Virginia. The number of students admitted each year is limited by the resources available to the university.

Visiting the University

We encourage prospective students to attend a group information session and a walking tour of campus. To check our visitation schedule and make a reservation, visit our website (<http://www.jmu.edu/admissions>).

First-Year Student Admission

JMU's admission process is competitive. The admissions committee works to select the strongest candidates from a high-quality applicant pool. Our primary focus is on the applicant's academic history, which includes program of study, rigor of program and academic achievements. Optional items that contribute to the applicant's profile include standardized test scores, a letter of recommendation, extracurricular activities and a personal statement. Each applicant is rated in the following four areas, listed in order of importance:

Academic Program

Students who challenge themselves with the upper-level courses offered in their high school (i.e., Advanced Placement (AP) classes, International Baccalaureate (IB), dual enrollment (DE), in addition to Honors-level courses) increase their competitiveness. Although schools provide different opportunities, applicants should pursue the most demanding college preparatory programs available. Strong students come from different types of schools, therefore we evaluate applicants within the context of their high schools.

Competitive applicants should minimally have the following:

- Four years of math with at least one full year beyond algebra II (i.e. pre-calculus, trigonometry, discrete math, statistics, math analysis, etc.). Consumer math, personal finance, and algebra functions and data analysis will not be evaluated as a full year beyond algebra II;
- Three years of laboratory science (preferably biology, chemistry and physics or an AP, IB or DE science). General science or earth science does not count as a lab science);
- Three or more years of the same foreign language or two years of two different foreign languages;
- Four years of English; and
- Four years of social studies.

Academic Achievement

To evaluate achievement in high school, we evaluate grades in the core subject areas: mathematics, English, foreign language, social studies and lab sciences. A competitive candidate will have mostly A's and B's in

core courses. We look at performance throughout the entire high school career.

Standardized Tests

Submitting SAT I or ACT scores is optional and not required to complete the application.

Extracurricular Activities

We are looking for quality rather than quantity and consider the variety and depth of involvement in clubs, organizations and athletics. Community service and part-time jobs fall into the extracurricular category, also.

Application Deadlines

Early Action Admission

Early Action is non-binding early notification admission to qualified first-year student applicants. Applications must be submitted by November 1 to be considered for Early Action admission. Successful candidates will be notified in mid-January and must submit a tuition deposit by May 1 to enroll at the university. Most students who are not selected for early notification are considered under the university's regular admission process.

Students who apply Early Action do not have an advantage over students who apply Regular Decision.

Regular Decision Admission

Regular Decision is the final application deadline for first-year admission consideration. Applications must be submitted by January 15 to be considered for regular decision admission. Regular Decision is non-binding. All applicants will receive notification of their admission status by the first of April. Admitted students must submit a tuition deposit by May 1 to enroll at the university.

Application Procedure for First-Year Student Admission

To apply for First-Year admission to the university, applicants must:

- Submit the online application for undergraduate admission along with the electronic application fee. This fee is not refundable or transferable and will not be credited to the student's account.
- Request that an official transcript from the high school be sent through a secure provider such as Naviance/Parchment, e-Scrip Safe, etc. or through the U.S. Postal mail:
 - James Madison University Office of Admissions
MSC 0101
James Madison University
100 E. Grace Street
Harrisonburg, VA 22807
- (Optional) Applicants may request SAT (code: 5392) or ACT (code: 005392) scores be sent to James Madison University directly from the College Board or American College Testing, respectively.

Information furnished on applications and all other university documents and records must be accurate and complete without evasion or misrepresentation. Submitting inaccurate or incomplete documents is cause for rejection or dismissal from the university.

Transfer Student Admission

To transfer to JMU, a student should:

- Have completed or be in the process of completing at least 24 credits at the college or university level, after graduating from high school or earning a GED, at the time of application.
- Successfully complete college coursework in the following areas: English, math, lab science and social science to be competitive. The more college-level coursework a student completes, the less emphasis the admission committee places on the high school transcript.
- Be in good standing and eligible to continue or graduate from their previous institution(s). It is the student's responsibility to provide the JMU Office of Admissions with official transcripts of work completed from all colleges attended. Concealment of previous attendance at a college or university is cause for cancellation of admission and registration. (Students with holds on their academic records will not be considered for admission until holds are released).

The university recommends students have a "B" cumulative grade point average (3.0 on a 4.0 scale) to be competitive for admission.

Application Procedure for Transfer Admission

To apply for transfer admission to the university, applicants must:

- Submit the online application for undergraduate admission along with the electronic application fee. This fee is not refundable or transferable and will not be credited to the student's account.
- Request official transcripts be sent from all colleges attended prior to the application deadline (including dual enrollment).
- Submit an official high school transcript or a copy of a GED certificate in addition to college transcripts, regardless of the number of college credits completed or the number of years out of high school.
- Indicate a major.
- (Optional) Complete a 650-word personal statement for review by the Admissions Committee.

All materials are due by October 15 for spring admission, February 1 for summer admission and March 1 for fall admission.

Evaluation of Transfer Credits

Credit will be awarded for those courses equivalent to courses offered at JMU in which the student has earned a grade of "C" or better.

After the student has been approved for admission, the Office of the Registrar will evaluate the transcript(s) of each transfer student to show the credits accepted by the university. The academic unit head of the program in which the student is majoring will determine the credits required for graduation.

With the exception of some community college degrees in General Studies, JMU General Education requirements will be waived for those students who have completed either an Associate of Arts, Associate of Science, or Associate of Arts and Sciences degree at a Virginia community college. For the complete listing of degrees associated with a Virginia Community College that waive all General Education requirements, go to Transfer Advising. Students who complete one of these associate degrees and are offered admission will receive junior-level status; however, due to varying major and degree requirements, junior-level status does not guarantee graduation in two years. For more details, consult the Office of Admissions or the JMU Virginia Community College Transfer Guide.

International Admission

The Office of Admissions is responsible for the admission and enrollment of undergraduate international students. It also evaluates "A" Level examinations for academic credit.

Credit will be awarded for those courses equivalent to courses offered at JMU in which the student has earned a grade of "C" or better.

All non-U.S. citizens and nonpermanent residents of the U.S. must complete the online application for undergraduate admission.

In addition to regular first-year student and transfer admission requirements, international students must present evidence of English proficiency and documentation of sufficient financial resources.

The university welcomes international applications and is authorized by federal law to enroll non-immigrant alien students. Requests for information concerning the admission of undergraduate international students should be directed to:

Office of International Admissions (<http://www.jmu.edu/admissions/apply/international.shtml>), MSC 0101
James Madison University 100 E. Grace Street
Harrisonburg, VA 22807
Phone: (540) 568-7865

Credit for Military Service

JMU encourages veterans to apply for admission as full-time or part-time students.

Information regarding VA Educational Benefits is available from the veterans' coordinator, Student Success Center, 5th floor.

Veterans and service members may submit their Joint Services Transcript (JST) or Community College of the Air Force (CCAF) transcript for credit review. This credit will be determined based on recommendations in A Guide to the Evaluation of Educational Experiences in the Armed Services. Consult the Office of the Registrar.

If a veteran or service member does not have a JST or CCAF, they may submit their DD-214 for review. Less than one year of active duty service provides three credits of health or kinesiology elective credit. One year or greater provides six credits of health and kinesiology elective credit. Students may not receive credit from both the JST/CCAF and DD-214.

Credit is allowed for courses taken at the Defense Language Institute. The amount of credit varies with the type of courses successfully completed. JMU must receive an official transcript. Credit allowed is based on their recommendation in A Guide to the Evaluation of Educational Experiences in the Armed Services. For those languages not offered at JMU, a maximum of 12 hours is accepted.

Advanced Placement

Applicants for admission who have completed advanced work in secondary school may use AP test scores to apply for course credit at JMU. Information about AP exams is available at the College Board website (<https://apstudent.collegeboard.org/takingtheexam/about-exams>). Once AP credit has been posted to the transcript, it will not be removed unless it was posted in error, or the student subsequently repeats the course at JMU and earns a passing grade (D- or better). For a full list of available AP credit opportunities at JMU, refer to the Advanced Placement table.

2026-2027 Academic Year Advanced Placement Courses

The score necessary to earn college credit at JMU, the corresponding course title at JMU and the credit hours which can be earned appear below. The grading scale is from one to five with five being the highest score. Credit hour equivalencies are reviewed annually by academic units and therefore cannot be appealed.

Placement Course	Minimum Required Score	JMU Equivalent	Credit Hours Earned	Meets General Education Requirement
2-D Art and Design	4	FDS 102 or ART 292	3	No
3-D Art and Design	4	ART 000 or ART 292	3	No
African American Studies	4	GNED 142	3	Yes
African American Studies	3	AAAD 000	2	No
Art History	4	ARTH 206	3	Yes
Biology	4	BIO 140/BIO 140L & BIO 150/BIO 150L or ISAT 113, ISCI 104 & BIO 000 ¹	8	Yes
Biology	3	BIO 140/BIO 140L	4	Yes
Calculus AB	4 or 5	MATH 235	4	Yes
Calculus AB	3	MATH 205	3	Yes
Calculus BC	4 or 5	MATH 235 & MATH 236	8	Yes
Calculus BC	3	MATH 206 plus MATH 205 or MATH 235 depending on the subscore earned for CALC AB content	6 or 7	Yes
Chemistry	4	Majors: CHEM 131 & CHEM 132	6	Yes
Chemistry	4	Nonmajors: CHEM 131 & CHEM 131L and CHEM 132 & CHEM 131L-CHEM 132L or CHEM 120 & CHEM 000 ²	8	Yes
Chemistry	3	CHEM 120 and CHEM 000 ² or ISCI 101 and ISCI 104	4	Yes

Chemistry	4	Non majors: GEOL 115, ISCI 104 & ISAT 000 ³	8	Yes
Chemistry	4	ISAT majors: ISAT 112 & ISAT 000 ³ However, to earn credit for ISAT 112, students must earn a 4 or higher on both the Environmental Science exam and the Chemistry exam.		Yes
Chinese Language and Culture	5	CHIN 300-level	3	No
Chinese Language and Culture	4	CHIN 232	3	No
Chinese Language and Culture	3	CHIN 231	3	No
Comparative Government	4	POSC 240	3	No
Computer Science A	5	CS 149	3	No
Computer Science A	3	CS 000	3	No
Computer Science Principles	3	CS 000	3	No
Drawing	4	ART 000 or ART 292	3	No
Economics (Micro)	4	ECON 201	3	No
Economics (Macro)	4	ECON 200	3	Yes
English Language and Composition	4	WRTC 103	3	Yes
English Literature and Composition	5	GNED 123 ⁴	3	Yes

Environmental Science	4	ISAT majors: ISAT 112 & ISAT 000 ³	8	Yes
		However, to earn credit for ISAT 112, students must earn a 4 or higher on both the Environmental Science exam and the Chemistry exam.		
		Non majors: GEOL 115, ISCI 104 & ISAT 000 ³		
Environmental Science	3	ENVT 101	4	Yes
French Language or French Literature	5	FR 300-level	3	No
French Language or French Literature	4	FR 232	3	No
French Language or French Literature	3	FR 231	3	No
Geography, Human	3	GEOG 200	3	Yes
German Language	5	GER 300-level	3	No
German Language	4	GER 232	3	No
German Language	3	GER 231	3	No
Government: U.S.	4	POSC 225	4	Yes
History: European	3	HIST 000	3	No
History: U.S.	4	HIST 225	4	Yes
History: World	4	HIST 102	3	Yes
Italian Language and Culture	5	ITAL 300-level	3	No
Italian Language and Culture	4	ITAL 232	3	No
Italian Language and Culture	3	ITAL 231	3	No
Japanese Language and Culture	5	JAPN 300-level	3	No

Japanese Language and Culture	4	JAPN 232	3	No
Japanese Language and Culture	3	JAPN 231	3	No
Latin	4 or 5	LAT 232	3	No
Latin	3	LAT 231	3	No
Music Theory – Aural Subscore	5	MUS 143	1	No
Physics 1	3 or 4	PHYS 140 & PHYS 140L	4	Yes
Physics 2	3	PHYS 150 & PHYS 150L	4	No
Physics C: Mechanics	4	PHYS 240 & PHYS 240L	4	No
Physics C: Mechanics	3	PHYS 140 & PHYS 140L	4	Yes
Physics C: Electricity and Magnetism	4	PHYS 250 & PHYS 250L	4	No
Physics C: Electricity and Magnetism	3	PHYS 150 & PHYS 150L	4	Yes
Precalculus	3	MATH 155	3	No
Psychology	4	PSYC 101	3	Yes
Spanish Language	5	SPAN 300-level	3	No
Spanish Language	4	SPAN 232	3	No
Spanish Language	3	SPAN 231	3	No
Spanish Literature	5	SPAN 300-level	3	No
Spanish Literature	4	SPAN 232	3	No
Spanish Literature	3	SPAN 231	3	No
Statistics	3	MATH 220 or ISAT 251	3	Yes

¹ BIO 000 does not count toward major or minor requirements in biology or toward General Education program requirements but is elective credit toward a degree.

² Awards vary by major. CHEM 000 does not count toward any major or minor requirements in chemistry or toward General Education program requirements but is only elective credit toward a degree.

³ ISAT 000 does not count toward major or minor requirements in integrated science and technology or toward General Education program requirements but is elective credit toward a degree.

⁴ GNED 123 fulfills the General Education program Literature requirement.

College Level Examination Program (CLEP) Awards

College-Level Examination Program (CLEP) is a national program that offers students credit by computer-based examination. This allows students to have the opportunity to obtain recognition for achievement in specific college courses. Students who participated in the College-Level Examination Program may be awarded course credit that satisfies a General Education requirement or elective. Prospective students who complete CLEP examinations and achieve an appropriate score (usually a 50 or above) are eligible to earn college credit. Evaluation of credit will be performed in coordination with the appropriate departmental office and cannot be appealed.

2026-2027 CLEP Exams

Subject	C-level (50)	B-level (subject-dependent)
American Government	3 credits: POSC 000	N/A
American Literature	6 credits: ENG 247 & ENG 248	N/A
Analyzing & Interpreting Literature	6 credits: ENG 000	N/A
Biology	6 credits: BIO 000	N/A
Calculus	3 credits: MATH 205	N/A
Chemistry	3 credits: CHEM 000	N/A
College Algebra	3 credits: MATH 155	N/A
College Composition	3 credits: WRTC 000	N/A
College Composition Modular	3 credits WRTC 000	N/A
College Mathematics	3 credits: MATH 000	N/A
English Literature	6 credits: ENG 247 & ENG 248	N/A
Financial Accounting	None	N/A
French Language	3 credits: FR 101	6 credits: FR 101 & FR 102
German Language	3 credits: GER 101	6 credits: GER 101 & GER 102
History of the US I	3 credits: HIST 000	N/A
History of the US II	3 credits: HIST 000	N/A
Human Growth & Development	3 credits: PSYC 160	N/A
Humanities	3 credits: HUM 000	N/A
Information Systems	3 credits: COB 204	N/A
Introduction to Business Law	3 credits: COB 218	N/A
Introduction to Educational Psychology	3 credits: PSYC 001	N/A
Introductory Psychology	3 credits: PSYC 001	N/A
Introductory Sociology	3 credits: SOCI 101	N/A
Natural Sciences	6 credits: ISCI 001	N/A
Precalculus	3 credits: MATH 155	4 credits: MATH 135
Principles of Macroeconomics	3 credits: ECON 200	N/A

Principles of Management	3 credits: MGT 000	N/A
Principles of Marketing	3 credits: MKTG 380	N/A
Principles of Microeconomics	3 credits: ECON 201	N/A
Social Sciences and History	6 credits: HIST 001	N/A
Spanish Language	3 credits: SPAN 101	6 credits: SPAN 101 & SPAN 102
Spanish with Writing	3 credits: SPAN 101	6 credits: SPAN 101 & SPAN 102
Western Civilization I	6 credits: HIST 000	N/A
Western Civilization II	6 credits: HIST 000	N/A

International Examinations Credit

JMU will follow the recommendations of the National Council on the Evaluation of Foreign Educational Credentials when awarding academic credit based on international examinations. Questions regarding the evaluation of international examinations should be directed to the Office of Admissions.

Cambridge International Examinations and United Kingdom "A" Level Examinations

Students who participated in the Cambridge International Examination program or the United Kingdom "A" Level examinations may be awarded general education or other credit. Credit is evaluated by the appropriate academic unit. Once CIE credit has been posted to the transcript, it will not be removed unless it was posted in error, or the student subsequently repeats the course at JMU and earns a passing grade (D- or better).

For a full listing of available Cambridge International Examination courses and credit opportunities, refer to the table below.

2026-2027 Cambridge International Examination Courses

The grade necessary to earn college credit at JMU, the corresponding course title at JMU and the credit hours that may be earned appear in the Cambridge International Examination table. The acceptable grading scale is A, B, or C, although the acceptable grade for credit may vary between courses and the level of the exam taken. Credit hour equivalencies are reviewed annually by academic units.

A-Level Exams

Cambridge International Exam	Accepted Grades for Credit	JMU Equivalent Course	Credit Hours Earned
Arabic	A,A*	ARAB 232	3
Arabic	B,C,D,E	ARAB 231	3
Biology	A	BIO 103, ISCI 104, 8 BIO 000	
Biology	B,C,D,E	BIO 103, ISCI 104	4
Chemistry	A	CHEM 131, CHEM 131L, CHEM 132, CHEM 132L	8
Chemistry	B	CHEM 131, CHEM 131L	4
Chemistry	C,D,E	ISCI 101, ISCI 104	4

Chinese	A,A*	CHIN 232	3
Chinese	B,C,D,E	CHIN 231	3
Computer Science	A,B	CS 000, CS 000	6
Computer Science	C,D,E	CS 000	3
English Language	A,B,C	WRTC 103	3
English (Literature in)	A,B	ENG 221, ENG 222	6
English (Literature in)	C,D,E	ENG 221	3
French – Language	A,A*	FR 232	3
French – Language	B,C,D,E	FR 231	3
German – Language	A,A*	GER 232	3
German – Language	B,C,D,E	GER 231	3
Mathematics	A*,A,B	MATH 235, MATH 000	7
Mathematics	C,D,E	MATH 000	3
Mathematics further	A*,A,B,C,D,E	MATH 000	3
Physical Education	A,B,C,D,E	KIN 000	3
Physics	A,B	PHYS 140, PHYS 140L, PHYS 150, PHYS 150L	8
Physics	C,D,E	ISCI 101, ISCI 104	4
Portuguese – Language	A,B,C,D,E	PORT 231	3
Psychology	A,B,C	PSYC 101, PSYC 000	6
Psychology	D,E	PSYC 002	3
Sociology	A,B,C	SOCI 101	3
Sociology	D,E	SOCI 000	3
Spanish – Language	A*,A	SPAN 232	3
Spanish – Language	B,C,D,E	SPAN 231	3

AS-Level Exams

Cambridge International Exam	Accepted Grades for Credit	JMU Equivalent Course	Credit Hours Earned
Arabic – Language	A	ARAB 232	3
Arabic – Language	B,C,D,E	ARAB 231	3
Biology	A,B,C,D,E	BIO 103, ISCI 104	4
Chemistry	A,B,C,D,E	ISCI 101, ISCI 104	4
Chinese – Language	A,B,C,D,E	CHIN 231	3

Computer Science	A,B,C,D,E	CS 000	3
French – Language	A	FR 232	3
French – Language	B,C,D,E	FR 231	3
German – Language	A	GER 232	3
German – Language	B,C,D,E	GER 231	3
Hinduism	A,B	REL 310	3
Japanese – Language	A,B,C,D,E	JAPN 231	3
Mathematics	A,B	MATH 000	3
Mathematics further	A,B,C,D,E	MATH 000	3
Physics	A,B,C,D,E	ISCI 101, ISCI 104	4
Portuguese – Language	A,B,C,D,E	PORT 231	3
Psychology	A,B,C,D,E	PSYC 002	3
Spanish – Language	A,B,C,D,E	SPAN 232	3
Spanish – First Language	A,B,C,D,E	SPAN 232	3

United Kingdom "A" Level Examinations

Credit will be awarded to those students who receive a "C" or higher on an "A" level exam. Credit will be comparable to completing the two-course introductory sequence (six credit hours) in the subject.

French Baccalaureate

Students with scores of 12/20 or higher on examinations in subjects studied at a coefficient of 4 or 5 may be awarded general education or other credit. Evaluation of credit will be directed by University Programs in coordination with the appropriate departmental undergraduate office. The "Total Score" of the French Baccalaureate is not used to determine credit eligibility.

German Abitur

Students who participated in the Thirteenth Class (Dreizehnte Klasse) German Abitur program may be awarded general education and other credit for examination scores of 10 or higher (15-point scale). Evaluation of credit will be directed by University Programs in coordination with the appropriate departmental undergraduate office.

International Baccalaureate

JMU recognizes the International Baccalaureate diploma and individual IB courses by awarding credit on IB higher-level examinations in essentially the same manner in which credit is allowed for Advanced Placement courses.

Typically, credit hours will be awarded for each higher-level examination on which a score of five, six or seven has been earned. Credit hour equivalencies are reviewed annually by academic unit.

Some standard-level examinations will also be considered for credit. The university encourages the completion of the IB diploma and will give special consideration for admission to students who have completed the

IB program. The grading scale is from one to seven, with seven being the highest score.

For a full listing of available IB courses and credit opportunities, refer to the International Baccalaureate Courses table.

2026-2027 International Baccalaureate Courses

This information is subject to change at the discretion of James Madison University. For the 2026-2027 academic year, the scores displayed in the online catalog supersede the scores in the printed undergraduate catalog.

Once IB credit has been posted to the transcript, it will not be removed unless it was posted in error, or the student subsequently repeats the course at JMU and earns a passing grade (D- or better).

Higher-Level IB Courses

International Baccalaureate Course	Minimum Required Score	JMU Equivalent	Credit Hours Earned
Biology	7, 6, 5	BIO 140, BIO 140L, BIO 150, BIO 150L	8
Business Management	6	BUS 160	3
Chemistry	5	Majors: CHEM 131, CHEM 132	6
Chemistry	5	Nonmajors: CHEM 131, CHEM 131L, CHEM 132, CHEM 132L	8
Chemistry	4	ISCI 101	3
Computer Science	5	CS 000	3
Design Technology	6, 7	ART 102 & ART 292	6
Design Technology	5	ART 102	3
Design Technology	4	ART 292	3
Economics	6	ECON 201 & ECON 200	6
Economics	4,5	ECON 201	3
Language A, Language and Literature (INTBA English)	5	WRTC 103	3
Language A: Literature	5	ENG 221	3
Foreign Languages (Same course numbers in each language)	7	300 + elective	6
Foreign Languages (Same course numbers in each language)	6	231-232	6

Foreign Languages (Same course numbers in each language)	5	231	3
Foreign Languages (Same course numbers in each language)	4	102	4
Geography	6	GEOG 200	3
Geography	4,5	GEOG 000	3
Global Politics	4, 5, 6, 7	POSC 200	3
History: Africa	5,6,7	HIST 263 + HIST 000	6
History: Africa	4	HIST 263	3
History: Americas	5,6,7	HIST 225	4
History: Americas	4	HIST 000	3
History: All Other Regions	4-7	HIST 000	3-6
Mathematics	5	MATH 235	4
Mathematics: Analysis and Approaches	5, 6, 7	MATH 235	4
Mathematics: Analysis and Approaches	4	MATH 000	3
Mathematics: Applications and Interpretation	4, 5, 6, 7	MATH 220	3
Physics	5	PHYS 140 & PHYS 150	6
Physics	4	ISCI 101	3
Psychology	4	PSYC 101	3
Social and Cultural Anthropology	5,6,7	ANTH 195 + ANTH 000	6
Social and Cultural Anthropology	4	ANTH 195	3
Visual Arts	4	ART 000	3

Standard-Level IB Courses

International Baccalaureate Course	Minimum Required Score	JMU Equivalent	Credit Hours Earned
Chemistry	6	Majors: CHEM 131 only	3
Chemistry	6	Nonmajors: CHEM 131 - CHEM 131L	4
Chemistry	5	ISCI 101	3
Computer Science	7, 6, 5	CS 000	3
Design Technology	7, 6, 5	ART 292	3
Economics	6	ECON 201	3
Economics	5	ECON 000	3

Foreign Languages (same course numbers in each language)	7	232	3
Foreign Languages (same course numbers in each language)	6	231	3
Foreign Languages (same course numbers in each language)	5	102	4
Geography	5	GEOG elective	3
Global Politics	5, 6, 7	POSC 200	3
History	5	HIST 000	3
Math Analysis & Approaches	5, 6, 7	MATH 205	3
Math Applications & Interpretations	5, 6, 7	MATH 000	3
Philosophy	5	PHIL 101	3
Physics	6	PHYS 140 & PHYS 150	6
Physics	5	ISCI 101	3
Psychology	6	PSYC 101	3
Social and Cultural Anthropology	5	ANTH 195	3
Visual Arts	6	ART 000	3
World Religions	5	REL 101	3

Special Admissions Requirements: College of Visual and Performing Arts

In addition to regular first year student and transfer admission requirements, prospective and current students who want to major in music, musical theatre or dance must also complete an on-campus audition. Incoming freshmen, transfers and change-of-major students intending to major in architectural design, graphic design, industrial design or the B.F.A. in art are required to submit an online portfolio. Visit CVPA's admissions website (<https://www.jmu.edu/arts/admissions/index.shtml>) for information regarding the artistic admissions process for all applicable College of Visual and Performing Arts majors.

Art

Phone: (540) 568-6216/6661

Email: art-arthistory@jmu.edu

Visit [jmu.edu/sadah](https://www.jmu.edu/sadah) (<https://www.jmu.edu/sadah/index.shtml>) for information for prospective students (<https://www.jmu.edu/artandarthistory/admissions/undergraduate.shtml>).

Art History Majors

Students intending to major in art history are not required to submit a portfolio, but should follow the regular JMU admissions process. To enroll in the museum studies concentration in art history, students must have a 3.3 GPA in the major and a minimum of nine credits in Art History and General Education courses.

Architectural Design, Art, Graphic Design and Industrial Design Majors

All prospective freshmen, transfer and change of major students intending to major in architectural design, graphic design, industrial design or studio art must meet JMU admission requirements as stated in the Undergraduate Catalog. In addition to meeting JMU admissions requirements, all students seeking the B.F.A. degree with majors in architectural design, art or graphic design, or the B.S. degree with a major in industrial design are required to submit a portfolio to the school's SlideRoom account (James Madison University College of Visual and Performing Arts - SlideRoom - Login). The submission of a portfolio is optional for students seeking the B.A. or B.S. with a major in art. Additionally, all transfer and change of major students, except those seeking the B.A. or B.S. with a major in art, will need to upload unofficial transcripts and a statement of intent to the school's SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study. The submission of a portfolio is seen as evidence of a student's interest and potential for future success in art.

All students, regardless of degree type, who wish to be considered for a scholarship must submit a portfolio to SlideRoom and attend a SADAH Portfolio Feedback day. SADAH hosts in-person and virtual Portfolio Feedback days. Prospective students may register for the events on the school's portfolio page. All scholarship awards are based on merit and vary in amount.

The School of Art, Design and Art History offers students the opportunity to meet professors who will look over their portfolio and offer critiques prior to the digital submission, through several on-campus and off-campus portfolio feedback days. This portfolio feedback event will provide an opportunity for feedback on actual artwork, as well as an opportunity to meet with faculty from the school. Tours of facilities and program info sessions will take place at on-campus events. All prospective first-year, transfer and change of major students are strongly encouraged to attend a portfolio feedback day prior to submitting their portfolio. For these in-person feedback sessions, it is highly recommended that the student show actual works of art; however, a portfolio that consists of digital or printed images is acceptable. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website for the dates of the upcoming portfolio feedback days and for additional portfolio requirements.

Students artistically admitted to the School of Art, Design and Art History participate in a secondary review process after completing foundations coursework to be accepted to their prospective major. Students seeking official acceptance into the graphic design major must enroll in GRPH 208 and submit a portfolio representing work completed in GRPH 101; GRPH 200 and GRPH 206. Students not admitted may reapply the following semester.

Students seeking official acceptance into the architectural design major must enroll in ARCD 208 and submit a portfolio representing work completed in ARCD 200 and ARCD 202. Students not admitted may reapply the following year.

Students seeking official acceptance into the advanced industrial design courses must enroll in INDU 208 and submit a portfolio representing work completed in INDU 200, INDU 202, GRPH 101, INDU 220, INDU 210 and independent design related work. Students not admitted may reapply one time.

Music

Phone: (540) 568-3851

Email: music_admit@jmu.edu

School of Music Website (<https://www.jmu.edu/music/index.shtml>)

Admissions to the Bachelor of Music concentrations in the School of Music requires an audition on a primary instrument and an additional SlideRoom application. Students should plan to audition during the academic year preceding their first semester of enrollment. Auditions for music are given on specific dates between November and February. It is important that students indicate their intention to participate in the audition process by January 15.

Admission to the Bachelor of Arts major requires an additional SlideRoom application and an online interview. Students that are seeking faculty lessons will also need to audition.

Application processes and requirements are listed on the School of Music website.

Theatre and Dance

Phone: (540) 568-6342

Email: theatredance@jmu.edu

School of Theatre and Dance Website (<https://www.jmu.edu/theatredance/index.shtml>)

Admissions to the dance, musical theatre and theatre majors are by audition for performers or portfolio review for design/technical theatre, theatre studies (directing, dramaturgy, theatre scholarship, etc.) or theatre education students.

Theatre and musical theatre auditions/portfolio reviews are held between November and February and by video/virtually for first-year and transfer student applicants. Refer to the school website for dates and more information on requirements for admission.

Dance major auditions are held between November and February and by video/virtually for first-year and transfer student applicants. Refer to the school website for dates and more information on requirements for admission.

Special Admissions Requirements: School of Nursing

Phone: (540) 568-6314

School of Nursing Website (<https://www.jmu.edu/chbs/nursing/index.shtml>)

In addition to the regular first-year student and transfer admission requirements, students who intend to major in nursing must submit a secondary B.S.N. Admission Application. Applications, admission criteria and admission deadlines may be found on the School of Nursing website.

Special Expenses

A differential tuition of \$104 per credit hour is added to the standard charge for courses carrying the NSG prefix. This charge applies to all students, both in-state and out-of-state, at the undergraduate and graduate levels, with the exception of students enrolled in the R.N. to B.S.N. program.

TUITION AND FEES

Email: ubo@jmu.edu

Phone: (540) 568-6505

Student Success Center, Room 5100

University Business Office Website (<https://www.jmu.edu/ubo>)

Tuition and Fees

Tuition and fee charges are available online at the University Business Office website (<https://www.jmu.edu/ubo/index.shtml>). Students can utilize the Cost of Attendance Calculator to see the estimated cost of attendance for their program of study. The tuition and fees listed contribute to general maintenance and operation, instruction, and other university service costs, including recreational and health-service facility costs. The fees also support student activities such as the Student Government Association, University Program Board and student publications.

The amounts listed for tuition do not include additional tuition for certain academic programs, academic fees, course fees or the cost of books or supplies. Certain courses offered through the College of Business and the School of Nursing require additional tuition charges. Some graduate programs have specific tuition rates. Specialized tuition rates for graduate programs can be found at the UBO website (<https://www.jmu.edu/ubo/tuition-and-fees/graduate.shtml>).

The university reserves the right to provide some or all instruction and related academic activities through alternative methods of delivery, including remote delivery. It also reserves the right to change the method of delivery before or during an academic term in the event of a health or safety emergency or other circumstance when it determines that such change is necessary or in the best interests of the campus community. Tuition and mandatory fees will not be reduced or refunded if the university changes the delivery method for any or all of an academic session.

In the event that circumstances beyond the control of the university render the continued provision of certain auxiliary services or programs, including housing and dining services, pursuant to the normal course of operations impossible, unsafe or otherwise impracticable, the university reserves the right to cease or alter the means of delivery of such services or programs. In such an event, payments of the housing fee, dining fee and comprehensive fee, which funds the provision of certain services or programs, previously made by the student may be refunded on a pro rata basis for the period for which such services or programs were altered or made unavailable to the student. The student understands that any refund of the housing, dining or comprehensive fee issued by the university pursuant to this paragraph may be subject to modification or reduction, to the extent the university reasonably deems necessary, in order to address costs already incurred by the university in preparing and providing all such services and/or programs.

Audit and internship registrations are billed at the same rate as for-credit courses. Departmental examination for credit incurs a nonrefundable fee billed to the student account.

The university reserves the right to adjust tuition and fee charges because of rising costs or other conditions upon approval of the JMU Board of Visitors.

For a full description of the tuition and fee rates, refer to the University Business Office website (<https://www.jmu.edu/ubo>).

Billing Policies

Students and their Authorized Users will be notified by email when their initial electronic billing statement is available in Madison Money Manager (M3).

Students are responsible for reviewing their account in MyMadison/M3 after any registration, course, or meal-plan change and for paying the full balance by the due date to avoid late fees and holds. Students are strongly encouraged to set their parents up as "Authorized Users" in the M3 system to assure the timely payment of their student account and to avoid unnecessary delays in communicating information.

Students are expected to access financial information through M3 in MyMadison. Failure to receive email notification of billing statements will not grant immunity in financial matters. Failure to attend or to pay does not cancel charges. Failure to access available data through self-service access via MyMadison or to read and comply with university regulations will not exempt students from financial penalties. To qualify for a refund or release of charges a student must officially drop a course by the drop/add deadline posted by the Office of the Registrar.

Payment of Tuition and Fees

Payments may be made in person at the University Business Office, through the mail, or by check or credit card online. More information about payment options is available online at the University Business Office website (<https://www.jmu.edu/ubo/student-accounts/pay.shtml>).

Payment arrangements must be finalized prior to the due date in order to prevent penalties on the student account. Payment may be made by any combination of personal payments, installment payment plan and/or financial aid that covers the entire balance due. Personal payments may be made by a combination of personal check, cash, money order, cashier's check, electronic check and credit cards. A list of payment options may be found at the UBO website (<https://www.jmu.edu/ubo/student-accounts/pay.shtml>).

Debts owed to the university are governed by the following policies:

- Until a student's account is paid in full, they will be ineligible for readmission or registration for a future semester.
- Upon recommendation of the Director of the University Business Office and with the approval of the Associate Vice President for Finance, students in debt to the university may be suspended from their classes or may be withdrawn.
- A \$50 per check fee is assessed for electronic or paper checks returned unpaid to the university. If a check is returned, the University Business Office staff will notify the student by email. If the student does not make appropriate alternative payment as specified, the student's class registration will be canceled and the student may be withdrawn from the university. When a check is returned, any future payment by personal check will require 15 business days to clear before a hold is released.
- There is a late fee applied to all delinquent accounts. If a student has a balance due that is not covered by financial aid or the installment payment plan by the payment due date, a late payment fee will be applied to the account which cannot be waived. Financial aid must be accepted and the installment plan must be set up by the due date to

avoid the late payment fee. The late payment fee is 3% of the balance due.

Collection of Past Due Accounts Outstanding Debts/Delinquent Accounts

Students with outstanding debts are denied any registration activity and receipt of a diploma until all debts are paid in full, except as required by federal or state laws and regulations. Student accounts are subject to the financial policies of James Madison University, as specified in the undergraduate catalog. Unless students resolve the debt, the university will advance the matter to the next step in the collection process. Specific collection activities and university collection agents may be found at the UBO website (<https://www.jmu.edu/ubo/student-accounts/collections.shtml>).

Collections in the Commonwealth of Virginia

The university pursues debt in accordance with the requirements set forth by the Commonwealth of Virginia in the Virginia Debt Collection Act. Virginia state law requires that the university make every attempt to collect past due amounts owed to state agencies. If, after 60 days, full payment of a debt has not been received, the student account will be placed with a collection agency or the Virginia Office of the Attorney General. Accounts turned over to the Office of the Attorney General are subject to additional fees and penalties pursuant to Virginia law.

Students also risk tarnishing their credit rating and will be subject to further enforcement proceedings. Collection efforts are costly to the student. Avoid them by paying on time. Collection fees cannot be appealed. If a student finds that his/her account has been referred to a collection company, they should contact the company immediately to make payment arrangements.

Additionally, the account can be listed by the Credit Bureau as a bad debt, a delinquent account can be collected in full from income tax refunds, lottery winnings, or other refunds due from the state, and the Attorney General's Office may pursue a court judgement on the account. Timely payment is strongly encouraged so that collection efforts can be avoided.

Setoff Debt Collection Act

Under the provisions of this act, an individual's Virginia income tax refund or lottery winnings will be subject to the university's claim for any unpaid balance of tuition and fees. Any communication disputing an amount owed must be submitted in writing to the Director of the University Business Office.

Eligibility for In-State Tuition

Eligibility for in-state tuition charges is based on the provisions of Section 23.1-500 of the Code of Virginia in effect on the first day of classes for each term. A link to the Code of Virginia guidelines can be found on the University Business Office website. This statute limits in-state tuition to those with Virginia domiciliary status. Domicile is defined as the "present, fixed home to which you return following temporary absences and at which you intend to stay indefinitely." If there is any question of the right to classification as a domicile of Virginia, it is the student's obligation, prior to the first day of classes for the semester, to raise the question with JMU administrative officials. Students who exceed the minimum number of credit hours required for graduation by 25 percent should refer to the UBO website for additional tuition charges.

Dependent Students

To qualify for in-state tuition, a dependent student or an un-emancipated minor shall establish by clear and convincing evidence that, for a period of at least one year prior to the date of the alleged entitlement, the person through whom they claim eligibility was domiciled in Virginia and had abandoned any previous domicile, if another existed. A link to the Code of Virginia, as well as Residency Guidelines, can be found on the University Business Office website under the section "Residency Requirements." Certain exceptions apply to Active Duty Military and veteran dependents.

Independent Students

The statute defines an independent student as one whose parents have surrendered the right to his/her care, custody and earnings; have ceased to support him/her and have not claimed him/her as a dependent on federal and state income tax returns for at least 12 months prior to the alleged eligibility. Certain exceptions apply to Active Duty Military and veteran students.

To qualify for in-state tuition, an independent student shall establish by clear and convincing evidence that for a period of at least one year immediately prior to the date of the alleged entitlement, they were domiciled in Virginia and had abandoned any previous domicile, if such existed.

Domiciliary status shall not ordinarily be conferred by the performance of acts which are auxiliary to fulfilling educational objectives or are required or routinely performed by temporary residents of the commonwealth. Mere physical presence or residence primarily for educational purposes shall not confer domiciliary status. A matriculating student who has entered an institution and is classified as an out-of-state student shall be required to rebut by clear and convincing evidence the presumption that they are in the commonwealth for the purpose of attending school and not as a bona fide domiciliary.

In-state Tuition Application

Initial determinations of eligibility are made by the Office of Admissions (incoming first year students and transfers), The Graduate School (graduate students) and the School of Professional and Continuing Education (for continuing education and non-degree students). Decisions on re-entry students are also made by these respective offices. Reclassification decisions on currently enrolled degree-seeking students are made by the Director of the University Business Office.

To establish eligibility after entering JMU as an out-of-state student, an applicant must complete the reclassification form posted on the University Business Office website. For information on special provisions of Section 23.1-500, contact the Office of Admissions, The Graduate School, Professional and Continuing Education or the University Business Office.

Once a student receives an initial determination of eligibility, they may appeal for a review of the application by contacting the office which made the decision within 30 days. Appeal for a final administrative review of the decision to deny in-state tuition may be made to the university's Residency Appeals Committee. If there is any question of the right to classification as a domicile of Virginia, it is the student's obligation prior to or at the time of registration to raise the question with the administrative officials of James Madison University. Any party aggrieved by a final administrative decision shall have the right to petition within 30 days for a review by the Circuit Court of Rockingham County.

Returning students may apply for reclassification from out-of-state to in-state status through the University Business Office. The Reclassification

Application Form can be obtained at the UBO website under "Residency." Denial of a request for such reclassification may be appealed within 30 days to the Associate Vice President for Finance with a final appeal within 30 days to the Residency Appeals Committee. Petition for review of this final appeal must be made within 30 days to the Circuit Court of Rockingham County.

A change to in-state status may be made only when the completed application for reclassification form is received by the University Business Office prior to the first day of classes for that semester. Students are responsible for paying out-of-state tuition rates until in-state status has been approved. Refer to the University Business Office website and specifically the section "Residency" for the reclassification form and links to the Code of Virginia.

Housing and Meal Plans

Part-time students cannot purchase a housing contract without prior approval of the Office of Residence Life. Students residing in university housing are required to have a Meal Plan with their housing contract.

Students residing in off-campus housing have a variety of meal plan options available through the JMU Dining website or by visiting Dining HQ, located on the first floor of D-Hall. All meal plans are billed directly to student accounts as year-long contracts, with charges applied each semester.

Refunds

An overpayment of a student account from personal funds can be either refunded or posted to a future semester as a payment. An overpayment created by a credit card payment must be refunded back to the credit card. Financial Aid refunds are processed automatically in accordance with applicable Title IV regulations. All overpayments are refunded to the student except for Parent PLUS loans, which are refunded as specified by the parent in the PLUS application. All student refunds are processed through the university's refund vendor. A refund preference must be selected by using the "My Refunds" link in MyMadison. Questions concerning refund procedures can be directed to the University Business Office .

Parents who are eligible to receive a refund on their federal Parent PLUS loan will receive a paper check at the home address of record.

Withdrawal Refunds

Students who officially withdraw from the university by the semester deadline will be refunded all tuition and mandatory fees pursuant to the official refund schedule.

Withdrawing after the end of this period will not reduce tuition charges. Meal plan fees will be prorated from the dining hall opening date. After the deadline dates, refunds will be for only a pro rata share of the meal plan fee. Room refunds will be made in accordance with the Residence Life refund schedule as stated in the current Residential Contract.

For further information on withdrawal from the university, see the section Withdrawal from the University or this link (<https://www.jmu.edu/ubo/student-accounts/refunds.shtml>).

Room and Tuition Deposits

New Students

For new students, a deposit is required to confirm their acceptance of the offer of admission. When a student pays the deposit, the money is

applied to the student's account. The deadline for this deposit is May 1. Refunds after the stated deadlines will be made only for personal illness certified by a physician or other extenuating circumstances approved by the Director of Admissions.

Returning Students

For returning students who want to live in university residential facilities, a signed Residential Contract must be submitted online to the Office of Residence Life (<http://www.jmu.edu/orl>). A contract fee will be assessed when the contract is submitted.

FINANCIAL AID, SCHOLARSHIPS AND STUDENT EMPLOYMENT

Financial Aid

Phone: (540) 568-7820

Email: fin_aid@jmu.edu

Location: Student Success Center, Room 5200

Financial Aid Website (<http://www.jmu.edu/financialaid/>)

The Office of Financial Aid and Scholarships helps qualified students secure a financial aid package designed to meet their financial needs. An aid package may consist of grants, scholarships, loans and work-study. Students interested in information on financial assistance programs should visit the financial aid website (<https://www.jmu.edu/financialaid/>), contact the Office of Financial Aid and Scholarships at the above address or send an email to fin_aid@jmu.edu. The website includes a link to JMU Terms and Conditions for Financial Aid – Consumer Information, which provides information regarding general financial aid rules and required disclosures.

Application Procedures and Deadlines

All financial aid applicants must undergo a standardized federal "needs analysis" by completing the Free Application for Federal Student Aid (FAFSA). To receive priority consideration, it is essential that applicants ensure their FAFSA has reached the federal government by March 1, prior to the academic year for which they are seeking financial assistance. Failure to apply by the priority filing date may cause delays in receiving aid and can result in less attractive aid packages.

A student must complete a FAFSA before financial aid eligibility can be determined for the following sources of aid:

- Federal Pell Grant
- Federal Supplemental Educational Opportunity Grant
- Commonwealth Award
- Virginia Guaranteed Assistance Program
- Federal Subsidized Direct Loan
- Federal Unsubsidized Direct Loan
- Federal Parent PLUS Direct Loan
- Federal Work-Study Program
- Need-based JMU Foundation Scholarships
- University Grants

When students file the FAFSA, the federal processor calculates their Student Aid Index (SAI). The SAI is an estimate of the family's ability to contribute to the student's overall educational expenses for one year. JMU calculates the student's financial "need" by subtracting the SAI from the Cost of Attendance (described later). Due to limited funding, grants are awarded to students who have the highest financial need, and in many cases to those who met the priority FAFSA filing date.

Financial aid recipients must complete the FAFSA each school year. Amounts and types of assistance may vary from year to year. If funds are available, the Office of Financial Aid and Scholarships continues to assist students who meet the following conditions:

- Complete the FAFSA, with precedence given to those who apply by the priority filing date of March 1, prior to the academic year for which they are seeking financial assistance.
- Meet general eligibility requirements for aid as defined by the FAFSA.
- Maintain Satisfactory Academic Progress.

The financial aid office sends an electronic notification to students offered financial assistance by JMU. The financial aid offer has important information, so the recipient should follow all instructions to ensure the completion of required forms. Students can find consumer information regarding the financial aid process, including pertinent rules and regulations, through the financial aid website. If any of the information included in the financial aid offer is incorrect, the student should immediately notify the Office of Financial Aid and Scholarships. Financial aid is offered based on FAFSA information, as well as the student's status at JMU (e.g., academic level, enrollment status and residency).

Parents of dependent undergraduate students may qualify for a Federal Parent PLUS Direct Loan. For those eligible to apply for the Parent PLUS, the financial aid office sends aid notifications to parents with instructions for applying. The parent aid notice does not include any information about student aid offers.

Federal and state regulations also require the Office of Financial Aid and Scholarships to consider any outside sources of assistance when offering financial aid. These outside sources can be JMU scholarships, private scholarships, veteran's benefits, tuition waivers, etc. The student's financial aid package may fluctuate throughout the year based on changes in FAFSA information, JMU status or the receipt of additional aid. Financial aid offers are usually sent to returning students in early summer. New students may receive financial aid offers in the spring prior to enrollment.

Cost of Attendance

An important part of determining a student's eligibility for financial aid is calculating a Cost of Attendance. In accordance with federal regulations, JMU has developed a Cost of Attendance (i.e., budget) for anticipated expenses a student may incur during the current school year. These expenses include tuition, housing, food, books and supplies, travel and personal. Housing and food can refer to either residence hall or off-campus living expenses, depending upon a student's response on the FAFSA. Expenses are also considered for students who live at home with parents or relatives, but the Cost of Attendance is lower than for those living elsewhere. Travel expenses include items such as gasoline, vehicle maintenance and insurance. Personal expenses include laundry, clothing and entertainment. Many of the elements in the Cost of Attendance are estimates, so it is possible for a student to spend more or less than anticipated during any given year.

Repeating Coursework

For information on how repeat courses may affect a student's financial aid, see the terms and conditions (<https://www.jmu.edu/financialaid/terms-and-conditions.shtml>) on the financial aid website (<https://www.jmu.edu/financialaid/terms-and-conditions.shtml>).

Classes and Program of Study Applicability

Schools can only consider classes that count toward degree requirements when offering financial aid. For example, if a student takes a full-time course load (i.e., 12 undergraduate credits) and nine of the 12 hours are financial aid eligible, then that student could only receive financial aid for nine hours. For information on determining what classes can be counted for financial aid purposes, see the terms and conditions

(<https://www.jmu.edu/financialaid/terms-and-conditions.shtml>) on the financial aid website (<https://www.jmu.edu/financialaid/terms-and-conditions.shtml>).

Satisfactory Academic Progress

In order to qualify for financial assistance, federal regulations indicate that a student must meet certain academic requirements as determined by the Office of Financial Aid and Scholarships. State, federal and some institutional aid programs are subject to the Satisfactory Academic Progress (SAP) policy. For a complete description of the policy, refer to the SAP website (<https://www.jmu.edu/financialaid/learn/sap/index.shtml>).

Understanding Satisfactory Academic Progress (SAP)

In accordance with federal regulations, the Office of Financial Aid and Scholarships must ensure that students meet both qualitative (grade-based) and quantitative (time-related) requirements before certifying eligibility for financial aid. The term qualitative refers to grade point average (GPA). Quantitative components include both pace and maximum time. Refer to the SAP website for a detailed description of each SAP component.

Students who have not completed the required number of hours or achieved the required cumulative GPA are not eligible to receive financial aid until such time that they meet the requirements. If extenuating circumstances contributed to students' inability to meet SAP requirements, they may request reconsideration by submitting an Appeal Form with appropriate supporting documentation. This form is available on the SAP website (<https://www.jmu.edu/financialaid/learn/sap/index.shtml>).

Refer to the JMU Terms and Conditions for Financial Aid (<https://www.jmu.edu/financialaid/terms-and-conditions.shtml>) – Consumer Information for more information on how repeating courses may affect financial aid.

Evaluation Process

All undergraduate students who file a Free Application for Federal Student Aid (FAFSA) are subject to the SAP policy. Upon receipt of the results of a student's FAFSA each year, the Office of Financial Aid and Scholarships will evaluate the student's SAP status before offering financial assistance. If students were unable to meet SAP requirements during their previous enrollment, they will receive notification regarding their area(s) of deficiency. Students who do not meet SAP standards cannot receive financial aid.

Appeals Process

If extenuating circumstances contributed to students' inability to meet SAP requirements, they may request reconsideration by submitting an Appeal Form with appropriate supporting documentation. This form is available on the SAP website (<https://www.jmu.edu/financialaid/learn/sap/index.shtml>), along with current appeal deadlines.

When students resolve their academic deficiency midyear, an appeal is still required and must include all the normal components of that process, including an explanation of extenuating circumstances and supporting documentation.

Students choosing not to appeal may become eligible for assistance in the subsequent award year by attending the current year without financial aid and resolving their academic deficiency by the end of it.

Grants

Federal Pell Grant

Pell grants are generally awarded only to undergraduate students who are seeking their first bachelor's degree and whose Student Aid Index (SAI) falls within the federally prescribed range.

Federal Supplemental Educational Opportunity Grant

SEOG is awarded to Pell-eligible students with the highest financial need as long as funds continue to be available.

Commonwealth Award

The Commonwealth Award is a need-based grant for Virginia residents who are seeking an undergraduate degree. The maximum award will not exceed the cost of tuition and fees. Awards will be made as long as funds are available, with precedence given to those who met the FAFSA priority filing date.

Virginia Guaranteed Assistance Program

VGAP is a need-based grant for undergraduate, full-time Virginia residents who graduated from a Virginia high school with at least a 2.5 GPA. The maximum award will not exceed the cost of tuition, fees and an allowance for books. Awards will be made as long as funds are available, with precedence given to those who met the FAFSA priority filing date.

University Grants

These institutional grant programs are primarily available to undergraduate students. The FAFSA is used to determine each student's need level, and grants are awarded accordingly. Awards are made as long as funds continue to be available.

Student Loans

Federal Direct Loan Program

The Direct Loan (subsidized and unsubsidized) is a long-term, low-interest loan for which undergraduate students may apply. Interest rates for Direct Loans disbursed on or after July 1, 2026 – June 30, 2027 are as follows:

- Undergraduate Subsidized Direct Loans = 6.52%
- Undergraduate Unsubsidized Direct Loans = 6.52%

Once disbursed this interest rate is fixed over the life of the loan. Interest rates on new loans will change on an annual basis each July 1st. Visit the financial aid website for up-to-date information and interest rates.

For a subsidized loan, the government will pay the interest while the student is in school. For an unsubsidized loan, the student can either pay the interest while in school or have it capitalized (i.e., added to the principal). Payments on the principal amount do not begin until six months after the student graduates or drops below half-time status.

Students must be making satisfactory academic progress and be enrolled at least half time for the period covered by the loan. Each year, dependent undergraduate students may borrow up to \$5,500 at the first-year level (no more than \$3,500 subsidized), up to \$6,500 at the sophomore level (no more than \$4,500 subsidized) and up to \$7,500 at the junior and senior levels (no more than \$5,500 subsidized). Independent undergraduate students may borrow no more than \$9,500 at the first-year level, up to \$10,500 at the sophomore level and no more than \$12,500 at the junior and senior levels.

Dependent undergraduate students may not borrow more than \$31,000 in Direct Loan funds during their undergraduate career (no more than

\$23,000 subsidized). Independent students may borrow no more than \$57,500 during their undergraduate career (no more than \$23,000 subsidized). For students who received prior Federal Stafford Loans at JMU or another institution, the career total is the sum of all Direct and Stafford Loans.

Federal Parent PLUS Direct Loan

Biological parents, adoptive parents or any stepparent of a dependent undergraduate student listed on the FAFSA may apply for a loan through the Parent PLUS Direct Loan program. The borrower must be a citizen or permanent resident of the United States. In addition, the student must be making satisfactory academic progress and be enrolled at least half time for the period covered by the loan. A student must complete a FAFSA before Parent PLUS Direct Loan eligibility can be determined.

The interest rate on the Parent PLUS disbursed after July 1, 2026 – June 30, 2027 is 9.07%. Once disbursed, this interest rate is fixed over the life of the loan. Interest rates on new loans will change on an annual basis each July 1st. Visit the financial aid website for up-to-date information and interest rates.

Interest begins to accrue on the date of the first loan disbursement. The first payment is due after the last disbursement for the loan period. Parents who wish to delay repayment on the PLUS loan should contact the Direct Loan Servicing Center.

Alternative/Private Loans

Some banks offer credit-based alternative loans to students and parents who either do not qualify for the Direct or PLUS Direct loans or cannot receive enough money through these loan programs to cover their educational expenses. Terms of these private loans vary. Undergraduate borrowers are typically required to have a credit-worthy co-signer. The financial aid office strongly encourages students and parents to exhaust other sources of aid before pursuing an alternative loan. Interested individuals may obtain more information about alternative loan options from the financial aid website.

Scholarships

JMU Scholarships

Many scholarships for students are established through the JMU Foundation and individual university departments. Scholarships are awarded either through the Office of Financial Aid and Scholarships or by the appropriate college or division according to established criteria. Awards are based upon merit and/or need. To be considered for need-based scholarships, students must complete the FAFSA. For information on specific scholarships, students should visit the scholarships website (<https://www.jmu.edu/scholarships/>).

Private Off-Campus Scholarships

Private off-campus scholarships include those awarded to students by outside (non-JMU) organizations. These scholarships are credited to the student's account upon receipt of the funds. If this type of scholarship is to be used to pay tuition and fees, the funds must be received prior to the payment due date for that semester.

Mail all off-campus scholarship checks to:

University Business Office
MSC 3516
738 South Mason Street
Harrisonburg, VA 22807

The student is responsible for compliance with the provisions of the scholarship (e.g., grade reporting and verification of attendance).

Virginia Military Survivors and Dependents Education Program (VMSDEP)

This program provides eligible students, as confirmed by the Virginia Department of Veterans Services (DVS), with waiver of all tuition and mandatory fees at a Virginia public college or university. In addition, as funds are available, eligible students may receive a stipend to offset other educational expenses, such as room and board.

Virginia Line of Duty Waiver

Students whose parent or spouse was disabled or killed in the line of duty while employed or serving as a public safety officer or firefighter with the Commonwealth of Virginia or one of its political subdivisions shall be entitled to free undergraduate tuition and the payment of required fees under certain conditions.

University Withdrawal

If students withdraw from the university, the University Business Office may adjust their charges based upon their withdrawal date and the JMU Refund Policy. For the university refund policy, refer to the University Business Office.

Student Employment

Information can be found on the Student Employment website (<https://www.jmu.edu/student-employment>). JMU employs both graduate and undergraduate students in academic, administrative or service-oriented areas. Students must be degree-seeking and enrolled to be employed in these positions. They receive payment for their services via direct deposit twice a month. Wages earned in student positions are not applied directly toward the cost of tuition; however, they serve as a source of income for weekly living expenses. There are three work programs at JMU.

Federal Work-Study Program

Students establish eligibility for Federal Work-Study employment by demonstrating financial need through their FAFSA. Those who are offered Federal Work-Study will need to apply and interview with employers to secure a position; however, employment is not guaranteed. These jobs provide a student with the opportunity to earn a paycheck throughout the year. If the student reports FWS earnings as need-based employment on the FAFSA, then the money earned through this program is not counted as income when determining financial aid eligibility for that year.

Institutional Employment

Institutional employment positions are on-campus positions available to degree-seeking JMU students regardless of financial need. To obtain additional information concerning available on-campus positions, refer to PageUp (<https://jobs.jmu.edu/jobs/search>). There are approximately 3,000 Institutional Employment positions available on campus each year. Students may not work more than 20 hours per week in any on-campus position during the fall and spring semesters.

Off-Campus Part-Time Jobs

The Off-Campus Part-Time Job Program is designed to assist students in securing off-campus, part-time employment regardless of their financial aid eligibility. The program's coordinator works with local employers

to promote hiring JMU students and to assist with advertising their opportunities. Additionally, the program is centered on creating real-world experiences for students that will not only increase self-knowledge, but also develop marketable skills that will provide a solid foundation for securing career options beyond graduation.

ACADEMIC POLICIES AND PROCEDURES

Academic Standing and Continued Enrollment

Continued enrollment at JMU depends upon an undergraduate student's ability to maintain satisfactory academic progress toward attaining a degree. The university measures this ability by the student's cumulative grade point average.

To assist students in maintaining satisfactory progress, JMU has adopted academic standards designed to provide early identification of students who are experiencing academic difficulty and to provide timely intervention through academic support programs.

Academic Good Standing

An undergraduate student who maintains a cumulative grade point average of at least 2.0 is considered to be in academic good standing and is eligible for continued enrollment at JMU.

Academic Notice

A student with a cumulative grade point average of 2.0 or above who earns a semester/term GPA below 2.0 is given an "academic notice." Academic notice alerts a student to the consequences of continued poor academic performance and is communicated to the student and the student's advisor. Academic notice does not appear on the student's permanent academic record and the transcript notation for the semester/term will reflect "Academic Good Standing."

Academic Probation

Academic probation is an indication of serious academic difficulty and applies whenever a student's cumulative grade point average falls below 2.0. A student who is placed on academic probation at the end of a semester/term may enroll for the following semester/term; however, a student on academic probation will be restricted to a course load of no more than 13 credit hours each semester until their cumulative grade point average has improved sufficiently enough to remove them from academic probation. The university might also require students on academic probation to confer regularly with their academic advisors, participate in academic skills development programs and/or utilize specific JMU academic resources.

Academic Suspension Threshold

A student will be placed on academic suspension alert or academic suspension if that student's cumulative grade point average falls below the academic suspension threshold for a student with their total number of hours for standing.

Total Number of Hours for Standing ¹	Cumulative GPA for Academic Suspension Threshold
1-27	less than 1.500
28-44	less than 1.650
45-59	less than 1.750
60-74	less than 1.850
75-89	less than 1.900
90-104	less than 1.994

105-119	less than 1.999
120 or above	less than 2.000

¹ Total Number of Hours for Standing includes all hours attempted at JMU plus:

- Credit hours transferred to JMU.
- Credit hours earned by departmental exam and external examination (AP, etc.) credit.
- Credit hours for courses taken on a credit/no credit basis (whether the final grade was CR or NC). For instance, in the example below a student has attempted 60 credit hours at JMU and received 30 credits for transfer work and two hours of no credit work. The hours attempted plus the no credit and transfer hours place this student in the 92 credit hour category, even though the total number of hours earned is equal to 82. With a cumulative GPA of 1.991, the student is subject to suspension.

JMU Hrs. Attempte	Trans Hrs.	CR/NC Earned	JMU Hrs. Earned	Cum. Hrs. Earned	Hrs. for Standing	Quality Points Earned	Cum. GPA
60	30	2	52	82	92	119.5	1.991

Academic Suspension Alert

A student will be placed on academic suspension alert if:

- their cumulative GPA is below the academic suspension threshold at the end of their first semester of enrollment at JMU,
- their cumulative GPA is below the academic suspension threshold at the end of a summer session, but their academic status was better than academic suspension at the end of the immediately preceding spring semester,
- their cumulative GPA is below the academic suspension threshold at the end of a fall semester, and they have no prior academic suspensions.

A student who is placed on academic suspension alert may enroll for the following semester/term; however, a student on academic suspension alert will be restricted to a course load of no more than 13 credit hours each semester until they reach academic good standing. The university might also require students on academic suspension alert to confer regularly with their academic advisors, participate in academic skills development programs and/or utilize specific JMU academic resources.

Academic Suspension

A student will be placed on academic suspension if:

- their current term of enrollment is not better than the first term of enrollment and their cumulative GPA is below the academic suspension threshold at the end of a spring semester.
- their cumulative GPA is below the academic suspension threshold at the end of a summer session, and their academic status was academic suspension at the end of the immediately preceding spring semester

As a rule, academic suspension will be invoked at the end of spring semester (and summer session for students who attend summer session); however, in exceptional cases where academic performance is persistently unsatisfactory, or in cases where students fail to meet

continued enrollment or conditions, the university may choose to suspend students at the end of fall semester.

The period of academic suspension is a minimum of twelve months following a first suspension, and a minimum of twenty-four months following a second suspension. A student who is academically suspended for a second time is eligible to apply to return to JMU by agreeing to apply for the Transfer Equivalent Option (TEO) upon their approved return to JMU. Re-entry is not guaranteed.

Academic Dismissal

A student who is suspended a third time will be permanently dismissed from the university. No appeal of this dismissal will be considered, nor will a student dismissed for academic deficiency be allowed to exercise the Transfer Equivalent Option (TEO).

Academic Standing If Enrolled in Fewer than Nine Credit Hours During First Term of Enrollment

A student who takes fewer than nine credit hours in their first term/semester at JMU will not be reviewed for academic standing that term/semester.

Adding a Course, Changing Sections or Changing Credit Options

Students may add a course and change sections or credit options according to deadlines and instructions published on the Registrar's website. Students should note that adding a course may result in a tuition increase.

Dropping and withdrawing both result in the termination of a student's enrollment in that course.

Dropping a Course

Dropping a course must be completed before the drop deadline. There is no fee for dropping a course, and the dropped course will not appear on the student's transcript. Depending on the student's course load, dropping a course might result in a tuition reduction.

Withdrawing from a Course

A student may terminate enrollment in a course by withdrawing from the course after the drop deadline but before the course adjustment deadline. A student who withdraws from a course will receive a grade of "W" for the course, and this grade will be recorded (and remain) on the student's transcript regardless of the status of the student in the course at the time of the withdrawal. Additionally, tuition charges will apply for a withdrawn course.

In extraordinary situations, a student unable to complete some course requirements after the course adjustment deadline (approximately one week after midterm grades are due for a regular semester) may request that the instructor consider awarding a grade of "WP" (Withdrawal While Passing) or "WF" (Withdrawal While Failing). A student should not assume that a late withdrawal will be provided by the instructor. There is no obligation for the instructor to assign a "WP" or "WF" grade. The instructor determines the form (e.g., verbal, written) and timing of requests for a "WP" or "WF" grade. The student must ensure that the request is made in an appropriate manner and at an appropriate time. In response to such a request, the instructor may choose to record a grade of "WP" or "WF," but is not obligated to do so and may record any grade other than "W." The course instructor may also suggest that the student

contact the Office of the Registrar about withdrawing from the university. Withdrawing from a course will not result in a tuition reduction.

Students considering withdrawing from a course should be aware that graduate and professional schools and future employers might hold differing opinions on a student's withdrawal from a course. For this reason, a student should withdraw from a course only after serious consideration.

Course Load

The university considers full-time enrollment in a term to be a minimum of 12 credit hours. In all programs, the normal load per semester is 15 or 16 credit hours. A student with a cumulative grade point average of 3.25 or better may register for as many as 21 credit hours per semester. Any student in academic good standing may take a maximum of 19 credit hours without securing special permission.

Students in academic good standing who wish to exceed these credits per semester limitations at JMU and/or through concurrent enrollment elsewhere, must secure permission from the head of the academic unit in which they are majoring.

A student on academic suspension alert or probation may not take more than 13 credit hours per semester without appropriate approval from the dean of their major college.

Students should note that an undergraduate course load of at least 12 credit hours a semester is required for a student to live in a residence hall.

Academic Probation and Course Load

Students on academic probation must get the approval of the office of the dean of their major college if they wish to take more than 13 credit hours per semester.

Summer Session Course Load

During summer session, undergraduate students are advised to carefully consider their academic course load. Students are reminded that summer course work is intensive because of the condensed instructional time-frame and are encouraged to plan their summer schedules with the demanding workload in mind.

Attendance

A student's participation in the work of a course is clearly a precondition to receiving credit in that course. The university recognizes that the nature of a student's participation in the work of a course cannot be prescribed on a campus-wide basis. For this reason, classroom attendance is not a matter subject to regulation by the university. Policies related to class absences and attendance are determined by the instructor for each course taught. As the faculty member is most knowledgeable of the course content, and presence is tied so closely to their teaching methods and the discipline, the instructor is best positioned to create the requirements for their individual courses. All faculty are required to develop their own policies for their courses and share those policies with their students in the course syllabus.

Federal Financial Aid regulations require that all students be reported as having initially attended or participated in academic activity by the instructor of record after census date. Students being reported as having not attended nor participated in academic activity may be withdrawn administratively from the class.

Do not assume that your lack of attendance or participation will result in your administrative removal from the class. If you wish to drop or withdraw from a class, you need to do so via the enrollment process and by prescribed deadlines.

Absence Policy

Instructors' policies govern how many excused absences will be allowed and how these excused absences will be handled in their classes. Certain absences are often considered legitimate; some of which are required excused absences as written in Academic Affairs Policy #16: Attendance and Excused Absences:

- Absences required by applicable laws, such as court-imposed or active military service obligations.
- Religious observance where the nature of the observance prevents the student from attending class.
- Scheduled necessary medical procedures.
- Participation in intercollegiate (NCAA) athletic competitions.
- Official university events that have been confirmed by the Office of Student Academic Success and Enrollment Management.
- University recognized academic functions that the student is required to attend (e.g., music, debate, workshop, academic conferences, etc.).

Faculty are responsible for providing an attendance policy for each course. The attendance policy must state any mandatory and/or unrepeatable components of the course and the procedures for requesting and obtaining approval for excused absences.

Students are responsible for reviewing and understanding the attendance policy. Students who plan to have multiple absences should notify the faculty prior to the Drop-Add period and determine with the faculty if mutually acceptable alternative methods exist for completing the missed classroom, lab, clinical/field or other required activities. This allows students to drop the course if it is determined that missed academic activities cannot be rescheduled in a reasonable fashion or that the absences would prevent adequate mastery of the material. Students are expected to submit verification of scheduled absences to the faculty by no later than the first class period of the second week of the semester, if possible. Examples of unexpectedly rescheduled absences (initially scheduled for one time, then changed with limited notice) and unforeseeable unscheduled absences (compelling, verifiable circumstances beyond the student's control) include:

- Activity season was extended due to achieving berth in playoffs (verified by the athletic director).
- Death in the immediate family (verified by the funeral director and/or copy of obituary).
- Disaster services or armed services activation for duty (verified by copy of official notification).
- Inclement weather postponement of the event or postponement and rescheduling of the event by external parties, e.g., illness of featured speaker (verified by the event coordinator).
- Jury duty or court-ordered appearance (verified by the clerk of the court).
- Student illness or injury (verified by health care provider).
- Other unavoidable, compelling, verifiable absence.

Students are to notify the faculty of each unexpected absence and determine with the faculty if mutually acceptable alternative methods

exist for completing or demonstrating mastery of missed learning activities within one week of becoming aware of the projected absence.

If required by the faculty, students must provide verification for any absence before the absence when possible, or upon returning to class when advance notice is not feasible. Faculty may consider the absence as unexcused if the student fails to comply with published notification and verification timeframes or procedures (See Academic Affairs Policy #16: Attendance and Excused Absences).

Auditing

In order to audit a class, a student must have the permission of the instructor and the academic unit head of the academic unit offering the class. Audited courses will not affect a student's attempted and earned credit hours or grade point average, but tuition will be billed for the course.

Catalog of Record

For General Education and university degree requirements, students are generally subject to the curricular and graduation requirements contained in the undergraduate catalog in effect upon their initial enrollment at JMU. Students who leave the university and return after a minimum two calendar years absence will have their requirements updated to reflect the curricular and graduation requirements contained in the undergraduate catalog in effect for the term in which they return to JMU.

A student who declares or changes a major, minor, preprofessional program or concentration will complete the requirements in effect for the catalog year in which the declaration or change was made. With approval from the academic unit head overseeing the academic program, the catalog requirement term can be adjusted forward or backward.

Aging Credit

With the exception of undergraduates enrolled in adult degree programs, students are expected to complete all baccalaureate major and degree requirements within seven years of their original entry date to JMU or previous higher educational institution(s). If required by the academic unit, academic work completed more than seven years prior to the students' anticipated graduation date might be subject to review by the major and minor academic units and the General Education Program for applicability to the undergraduate degree program. Additional standards may apply in programs leading to licensure or certification.

Changes in Requirements

James Madison University is a progressive educational institution, and its offerings and requirements in the undergraduate catalog are under continual examination and revision. Revisions often affect currently enrolled students. Although the university attempts to alert students to these revisions through the academic advising process, individual reports of academic progress and various campus publications, responsibility for meeting all curricular and other graduation requirements rests with the students, who are encouraged to consult regularly with their program advisors.

Classification

The classification of students depends upon the number of credit hours they have earned.

Classification	Credit Hours
First Year	Fewer than 28
Sophomore	28-59
Junior	60-89
Senior	More than 89

Course Adjustment

A course adjustment is any change to a student’s registered course schedule. A course adjustment can include any of the following: changing a credit option, changing a section, adding a course, dropping a course or withdrawing from a course. Deadlines for processing specific course adjustments are stated on the Registrar’s website. The end of the course adjustment period (approximately one week after midterm grades are due for a regular semester) is the deadline for adding a course with department permission, withdrawing from a course, changing credit options and changing the repeat option for a semester course. Each student is responsible for knowing their registered course schedule and for making any desired course adjustments prior to the published registration deadlines.

Confidentiality of Educational Records

The Family Educational Rights and Privacy Act of 1974

James Madison University adheres to and annually informs students of the Family Educational Rights and Privacy Act of 1974, as amended. This act, with which the institution intends to fully comply, was designated to protect the privacy of educational records. Under the Family Educational Rights and Privacy Act (FERPA), students have certain rights with respect to their education records. These rights include:

- The right to inspect and review the student’s education records within 45 days of the day the university receives a request for access.
 - The student should submit to the registrar, dean, head of the academic unit or other appropriate official written requests that identify the record(s) they wish to inspect. The university official will make arrangements for access and notify the student of the time and place where the records may be inspected. If the records are not maintained by the university official to whom the request was submitted, that official shall advise the student of the correct official to whom the request should be addressed.
- The right to request the amendment of the student’s education records that the student believes are inaccurate or misleading.
 - The student may ask the university to amend a record that they believe is inaccurate or misleading. The student should write the university official responsible for the record, clearly identify the part of the record they want changed, and specify why it is inaccurate or misleading. If the university decides not to amend the record as requested by the student, the university will notify the student of the decision and advise the student of their right to a hearing regarding the request for amendment. Additional information regarding the hearing procedures will be provided to the student when notified of the right to a hearing.
- The right to consent to disclosures of personally identifiable information contained in the student’s education records, except to the extent that FERPA authorizes disclosure without consent.

One exception, which permits disclosure without consent, is disclosure to school officials with legitimate educational interests. A school official is a person employed by the university in an administrative, supervisory, academic or research, or support staff position (including law enforcement unit personnel and health staff); a person or company

with whom the university has contracted or outsourced institutional services or functions (such as an attorney, auditor, food service or bookstore provider, or collection agent); a person serving on the Board of Visitors; a volunteer performing services for the institution; or a student serving on an official committee, such as a disciplinary or grievance committee, or assisting another school official in performing their tasks. A school official has a legitimate educational interest if the official needs to review an education record in order to fulfill their professional responsibility. Upon request, the university may disclose education records without consent to officials of another school in which a student seeks or intends to enroll.

The following is considered "Directory Information" at James Madison University and may be made available to the general public unless the student notifies the Office of the Registrar in person or in writing within five days after the first day of class registration: Student’s name, major and minor fields of study, college of major and year (first year, sophomore, etc.), enrollment status (full-time/part-time) including credit hours, dates of attendance, degree sought and time, degrees conferred, awards and honors conferred, participation in officially recognized activities and sports, weight and height of members of athletic teams, the most recent previous educational agency or institution attended by the student, and fraternity and/or sorority and educational societies.

Enrolled students may "opt in" to three additional data elements of directory information: address, email address and telephone number. By "opting in" to these additional data elements, the student is authorizing JMU to release this information to any third-party entities who request any directory information on JMU students. If a student wishes to "opt in" to these additional three elements, they can log in to MyMadison, Select Student Center, Personal Information, and then Student Privacy.

Students have the right to file a complaint with the U.S. Department of Education concerning alleged failures by James Madison University to comply with the requirements of FERPA.

The name and address of the office that administers FERPA is:

Family Policy Compliance Office (<https://www2.ed.gov/policy/gen/reg/ferpa/>)
U.S. Department of Education
400 Maryland Avenue, SW
Washington, D.C. 20202-4605

More detailed information concerning JMU’s records policy is available from the James Madison University Policies Manual Policy 2112 (<https://www.jmu.edu/JMUpolicy/>), The Family Educational Rights and Privacy Act.

Credit Hours

The JMU academic calendar is based on the semester system. The unit of credit is the credit hour. Each credit hour represents the combination of in-class and out-of-class work:

1. A minimum of one 50-minute class period of faculty instruction or its equivalent each week over a 15-week semester for all forms of instruction (or equivalent total time for blocks of instruction different than 15-week semesters). For purposes of this definition, instruction consists of regular and substantive interaction or guided interaction, which may include in-person lectures, synchronous or asynchronous online lectures, seminar, discussions, studios, supervised group work, and examinations,
or

A minimum of two 50-minute class periods for each week in the semester (or equivalent total time for blocks of instruction different than 15-week semesters) for laboratory, studio or fieldwork, and

2. A minimum of two hours of out-of-class student work or academic engagement each week to meet appropriate course objectives (or equivalent total time for blocks of instruction different than 15-week semesters). For purposes of this definition, out-of-class student work consists of the time a student spends outside of direct faculty instruction to fulfill course objectives, which may include reading assignments, problem assignments, quizzes and examinations, online and face-to-face group work, writing papers(s), and preparation/follow up for labs or field experience.

Credit/No-credit Course Registration

The credit/no-credit option has been established to encourage students to explore academic areas with which they are unfamiliar. Such academic exploration allows students to fully integrate field-based learning activities into appropriate programs of study. The design of the credit/no-credit option allows students to participate in courses outside of their major and minor fields of concentration without jeopardizing their academic records. In some cases, the credit/no-credit option might also help to reduce academic pressures and competition for grades.

Students electing to take courses under this option should be selective in choosing the courses that they take as credit/no-credit. Graduate and professional schools and future employers, however, might hold differing opinions of such a nontraditional grading system. For this reason, students should consult with their academic advisors for information concerning the inclusion of credit/no-credit course grades within their programs.

Students are eligible to take a course on a credit/no-credit basis if they have completed at least 28 credit hours at JMU and have attained a 2.25 cumulative grade point average or higher. Transfer students may take courses on the credit/no-credit option only if they have completed 28 credit hours with at least 14 hours at JMU.

JMU allows students to register for kinesiology activity courses on a credit/no-credit basis at any time without regard to minimum hours completed or grade point average. Also, grades for student teaching are assigned on a credit/no-credit basis. Except for field-based courses (which may be defined by an academic unit as credit/no-credit), the following courses may not be taken as credit/no-credit:

- Courses used to meet General Education requirements
- Courses used to meet declared major requirements
- Courses used to meet declared minor requirements
- Any course listed by subject name in the major or minor program, even if the course does not specifically satisfy a requirement for the major or minor
- Courses being repeated that were previously taken for a letter grade
- Degree requirements

Students should also keep in mind the following factors when considering the credit/no-credit option:

- Students can take up to 15 credit hours on a credit/no-credit basis.
- The university limits the maximum number of credit/no-credit courses a student can take to four credit hours per semester or one course of more than four credit hours per semester.

- Student teaching is not counted as a part of these credit/no-credit totals.
- Students registering for classes should complete the required paperwork with the Office of the Registrar for courses that they plan to take on a credit/no-credit basis, and all changes to and from the credit/no-credit option must be completed by the end of the course adjustment period.
- The credit/no-credit option will only apply to final grades. All course work and quizzes will be graded as if the student were taking the course for graded credit.
- Students taking a course on a credit/no-credit basis will only be identified to the instructor after final grades have been submitted.
- Instructors will submit letter grades on the grade roster and the student administration system will convert all grades of "A" through "C" to "CR" (credit) for those students enrolled under the credit/no-credit option. (The university makes exceptions to this process for field-based courses designated by the academic unit.) Students will receive credit hours, but they will not receive quality points for the work completed. Therefore, the grade of "CR" will not affect a student's cumulative grade point averages.
- A grade of "NC" (no-credit) will be recorded for all grades of "C-" and below, and the student will not receive credit hours or quality points for the grade of "NC." The fact that the student took the course will remain on that student's transcript, but the grade assigned in that class will not affect that student's cumulative grade point average.

Credit Opportunities

The university offers options for students to earn credit toward their degree outside the traditional classroom setting:

- Admission of High School Students as Special Students pursuing dual enrollment credit
- Admission of first year students with Advanced Placement, or other approved test scores that result in course credit
- Independent Study while enrolled as a student

Credit by Examination

College degrees represent growth and maturity in certain fundamental knowledge and skills rather than a mere accumulation of credit hours. In recognition of the fact that some persons may achieve academic competence through nontraditional means, such as private study, technical employment or prior instruction, JMU endorses the concept of credit by examination.

By permitting a student to earn credit by academic unit examination for knowledge already gained, highly motivated or academically advanced persons are able to accelerate their program. It is the student's responsibility to ascertain what preparation and background are necessary for taking advantage of this means of acceleration, before attempting an examination.

Any enrolled undergraduate student may apply to take an examination for credit in selected courses in the undergraduate curriculum. Permission to take an examination for credit must be obtained from the head of the academic unit in which credit is sought. A student will not be permitted to take the examination for credit for a course which satisfies any of the following:

- The student is presently enrolled in the course.
- The student has previously completed the course.

- The course is a prerequisite for a course in which the student is currently enrolled or has completed.
- The course is numbered at a lower level than other courses in which the student is currently enrolled or has previously completed, unless the express approval of the academic unit head is secured.
- The course appears previously on the student's record with a grade assigned due to a violation of the Honor System.

Each academic unit will use its own discretion in developing the form of the examination and in determining the procedure to be followed. Students may earn as many as 30 credit hours through credit by academic unit examination, with no more than 12 credit hours in any one discipline. To receive credit, a student must make a grade of "C" or better on the examination. Each academic unit shall determine what constitutes a "C" for that academic unit. No academic unit shall require a grade higher than a "C" for passing the examination. The examination for a course can be taken only once in a given semester. A nonrefundable fee will be charged to the student's account. Each academic unit will determine its time schedule for giving examinations.

Independent Study

Every academic unit at JMU offers a course designed to give capable students an opportunity to do faculty-supervised independent studies. Such courses often carry more than the normal three credit hours for a semester's work. In addition, these independent study courses allow especially capable students to work at their own, often accelerated pace. Arrangements for independent study should be made with individual faculty members.

Final Examinations

Students are expected to attend final examinations during the times scheduled for those examinations. Extenuating circumstances, however, might prompt faculty members to approve a student's request for an exception to attending the final examination. Students whose requests for exceptions are disapproved by the relevant faculty members have the right to appeal to the relevant academic unit head or academic dean. No appeal will be favorably considered without prior consultation with the faculty member.

Final Examinations Missed Due to Inclement Weather or Emergency

In response to inclement weather and other emergencies, the university may be forced to cancel final exams. When the university closes due to weather or other types of emergencies, faculty will administer regularly scheduled final examinations at an official make-up time designated by the university. The official make-up time will be designated as part of the closing announcement. Unless otherwise notified, make-up examination locations will be the same as locations for regularly scheduled exams.

If it is determined that exams cannot be given because of inclement weather or other emergency, faculty will assign final grades to students based on the course requirements completed prior to the regularly scheduled exam dates.

Grade Review Procedure

Maintaining standards of excellence and the integrity of the teaching/learning process are important values to JMU faculty. The university and its faculty members also recognize that grades may sometimes be inappropriately assigned. If such disagreements occur, students have a right to voice their opinion concerning a particular course grade. Evaluation of student work and assignment of grades on the basis of

academic criteria are the responsibilities and prerogative exercised by the faculty member. A faculty member has the responsibility to evaluate student performance on a fair and scholarly basis in accordance with university policy.

It is the student's responsibility to maintain all documentation for their classes, including copies of assignments and grades earned.

Grade Change Procedure

If a student (graduate, undergraduate or post-baccalaureate) believes that a grade was assigned in error, because of a mistake in calculation or an error in recording a grade, the student should consult the faculty member (or faculty members, in the case of a jointly taught course) to resolve the discrepancy.

It is the student's responsibility to maintain all documentation for their classes, including copies of assignments and grades earned. If the faculty member agrees that a change should be made, the faculty member should initiate a Grade Change request through the MyMadison Faculty Tab.

The only basis for this type of change is an error in grade assignment or calculation.

If the faculty member does not agree to change a grade based on an error in recording the grade, the student may activate the grade review process listed below.

Grade Review Process

If a student (graduate, undergraduate or post-baccalaureate) believes that a final course grade was unfairly awarded, that student may initiate the grade review process. Students should be aware that, as a result of review, a grade may be raised, lowered or left the same. Evaluation of student work and assignment of grades on the basis of academic criteria are the responsibilities of and prerogative exercised by the faculty member teaching that particular course. Grades should be assigned on a fair and scholarly basis. Grounds for grade review are limited to two categories.

1. The grade was assigned in a manner other than that listed in the course syllabus or as amended by the faculty member with appropriate notice.
2. The grade was assigned in a manner other than that used for other students in the class.

This policy applies to student grade review requests that do not concern discrimination, harassment, sexual misconduct, or Title IX sexual harassment. Policies for complaints concerning these matters are governed by Policies 1302, 1324, 1340 and 1346 and students are referred to the Office of Equal Opportunity and Title IX Office.

Activating the Grade Review Process

To activate the grade review process, the student should follow these steps.

1. The student submits a Grade Review Form (available from the Registrar's website) to the appropriate faculty member by Monday of the third week of classes in the regular semester that follows the semester for which the contested grade was given. The student should attach a written explanation of reasons for the dispute, including any documentation relating to the disputed grade.
2. The student communicates with the faculty member by Friday of the third week of classes to attempt to resolve the concern.

- If the student and the faculty member reach an agreement that the grade should be changed, the faculty member changes the grade by initiating a Grade Change request through the MyMadison Faculty Tab.
 - If no resolution is reached, the faculty member signs the Grade Review Form and records a written response on the reverse side of the form. The faculty member returns the original copy of this form to the student, retains a copy of the form for their personal records and forwards a copy to the relevant academic unit head by Friday of the fourth week of classes.
3. The student must contact the relevant academic unit head by the Friday of the fifth week of classes to request review of statement and response.
 4. The academic unit head then meets with the student and confers with the relevant faculty member.
 - The academic unit head signs the Grade Review Form and records a written response on the reverse side of the form by Friday of the seventh week of classes. The student receives the original copy of this form, the relevant faculty member receives a copy of the form and the sender retains a copy of the form.
 - If all involved parties agree that the grade should be changed, the faculty member initiates a Grade Change request through MyMadison Faculty Tab.

After the review process outlined has been completed, if the academic unit head and faculty member agree that a grade should not be changed, a student can also request that the form, documentation and responses be reviewed by the dean of the college in which the class was taught. The college dean's responsibility is to ascertain whether all parties have had an opportunity to present all relevant facts and have received a fair and impartial hearing at each level, and to review whether the faculty member has acted appropriately in assigning the grade. To enter this phase of the process, a student should follow these procedures.

1. The student contacts the dean by Friday of the eighth week of classes and requests that the dean review the overall process.
2. The college dean reviews the process to be sure the student and the faculty member have had a fair hearing, and whether the faculty member has acted appropriately in assigning the grade. If the relevant college dean believes that due process was not followed during the review process, or that the faculty member has not acted appropriately in assigning the grade, the dean consults with the relevant faculty member and academic unit head or General Education area coordinator in an attempt to resolve the dispute.
3. The relevant college dean sends a written response to all involved parties by Friday of the tenth week of classes. This written response is appended to the Grade Review Form. The dean returns the original copy to the student, retains a copy for themselves and sends a copy to the relevant academic unit head and the relevant faculty member. If it is agreed that the student's grade should be changed, the relevant faculty member initiates a Grade Change request through the MyMadison Faculty Tab.

If the dean determines that the grade will not be changed, there is no further review available to the student. The entire process will not extend past the end of the semester following the contested grade with the exception of grades given for summer session courses.

Special Circumstances

Students who spend a semester abroad should submit the Grade Review Form to the relevant faculty member by the Monday of the third week of

classes in the semester following the semester during which the grade was given. These students must also notify the faculty member of their semester absence from campus.

Students can complete a written explanation of the reasons for the dispute and can submit relevant documentation upon their return to campus, but that student cannot submit the request after the end of that semester.

If a faculty member is not on campus when a student wants to initiate a grade review, the student should contact the academic unit head to begin the process. The academic unit head will contact the faculty member for their decision on whether the grade should be changed.

Grades for block courses are considered to be officially recorded at the end of the semester; therefore, a grade review process for a block grade should follow the same procedure as for a semester course.

Graduation

Graduation Requirements

The academic unit head, or designee, verifies completion of major and minor course requirements for graduation. The Office of the Registrar verifies total credits earned, the General Education program, degree requirements and the cumulative GPA earned at the university, as well as other university-wide requirements.

To receive a degree from JMU, students must:

- Meet the General Education requirements.
- Have a minimum of 120 earned credit hours accepted by JMU.
- Have a cumulative grade point average of 2.0 or better at JMU.
- Have a cumulative grade point average of 2.0 or better in the major and minor academic plans at JMU.
- Meet the major and degree requirements of one of the curricula leading to the degree for which they are candidates.
- Have been enrolled at JMU a minimum of two semesters.
- Have earned a minimum of 25% of credit hours at JMU (30 credits for degree programs of 120 credits).
- Have earned at least 50% of credit hours accepted by JMU from accredited senior (four-year) institutions of higher education, including JMU (60 credits for degree programs of 120 credits).

A student who has applied to graduate may participate in commencement exercises only if the student has fulfilled or is reasonably expected to fulfill all applicable graduation requirements prior to the date of the commencement exercises. A student who is reasonably expected to fulfill all applicable graduation requirements no later than the end of summer session may participate in the spring commencement exercises immediately preceding the summer term. A student who is reasonably expected to fulfill all applicable graduation requirements no later than the end of winter session may participate in the fall commencement exercises immediately preceding the winter session.

Attendance at commencement exercises is expected.

Permission to participate in (or actual participation in) commencement exercises does not mean or imply that a student has fulfilled all applicable graduation requirements.

Graduation with Honors

Before becoming eligible for graduation with honors, a student must successfully complete the following:

- Enrollment at JMU for a minimum of four semesters (fall and spring).
- Completion of a minimum of 60 attempted and earned credit hours at JMU.
- A minimum cumulative grade point average of 3.50 on all course work completed at James Madison University, including any work completed at JMU as a dual enrollment student prior to undergraduate matriculation and/or any work attempted and earned beyond four semesters or 60 credit hours.

Graduation honors will be determined as specified based on all course work attempted and earned at JMU.

Cumulative Averages Required for Graduation Honors

Honors	Average
Cum laude	3.50-3.699
Magna cum laude	3.70-3.899
Summa cum laude	3.9 and above

These standards apply only to students entering JMU for the first time in fall 2015 and thereafter. Requirements for graduation with honors for students who entered JMU prior to 2015 are regulated by the catalog in effect at the time they enrolled.

All grades received in all courses attempted at JMU will be used to calculate the grade point average in consideration for graduation with honors. Students who have utilized the Transfer Equivalent Option upon their return to JMU do not qualify for graduation with honors.

Graduation honors will be noted in the commencement program if the student has completed four semesters and 60 credits at JMU. In-progress semesters and credits will not qualify a student for commencement program listing, nor for commencement ceremony Honors cords. The graduation honors listed in the program will be based on the grade point average at the end of the semester preceding the semester in which final graduation requirements are met. However, for students who participate in the May ceremony but complete requirements in summer session, the honors noted in the commencement program will be based on the grade point average earned at the end of the preceding fall semester. Final graduation honors recorded on the diploma and transcript will be determined by the grade point average at the end of the semester in which all graduation requirements are met.

Graduation Awards

The title of Valedictorian is an honor bestowed by the Faculty Senate on behalf of the JMU faculty. Criteria were established by the Faculty Senate. It is given to the May graduate with the highest grade point average who has also earned at least 100 credit hours at James Madison University as of the fall semester prior to the student's May graduation. The student must have completed an average of at least 14 credit hours per fall and spring semester during their career at JMU. The grade point average will be calculated using grades recorded on the official transcript through the fall semester of the student's last year of undergraduate enrollment.

The Academic Excellence Award is an honor bestowed by the JMU Faculty Senate on behalf of the JMU faculty. Criteria were established by the Faculty Senate. It is for the student graduating in August or December who holds a grade point average that equals or exceeds that of the

valedictorian and who has earned at least 100 credit hours at James Madison University. The student must have completed an average of at least 14 credit hours per fall and spring semester during his/ her career at JMU. The grade point average will be calculated using grades recorded on the official transcript through the spring semester of the student's last year of undergraduate enrollment. When more than one student qualifies for the Valedictorian or Academic Excellence Awards, each qualified student will be recognized.

Grievance Procedure for Students

This policy applies to student grievances related to the instructional process that do not concern grades, discrimination or harassment. Policies for grievances concerning these matters are outlined elsewhere.

To initiate the grievance procedure, the student should submit to the academic unit head a written statement explaining the reason for the grievance. Supportive documentation should also be included. The academic unit head meets with the student and confers with the relevant faculty member. Following these meetings, the unit head initiates the process as follows.

1. Each academic unit head shall appoint an advisory committee to hear student grievances. The committee shall consist of three faculty members from the academic unit, appointed by the academic unit head. All committee members must be free of any actual or perceived conflict of interest with the student. The advisory committee may take any of the following actions:
 - Examine materials submitted by the student and the party grieved against ("respondent").
 - Interview the student and the respondent.
 - Interview any witnesses requested by the student, the respondent or the committee.
 - Request additional materials from any person or entity relevant to the charges.
 - Make a recommendation on the grievance to the academic unit head.
2. The academic unit head may accept the recommendation of the committee, reject the recommendation, or partially accept and partially reject the recommendation. The academic unit head will take any action they deem appropriate on the grievance.
3. If either the student or the respondent is dissatisfied with the action taken by the academic unit head, the action may be appealed to the dean. The decision of the dean is final.
4. If the academic unit head is the party against whom the grievance is filed, the dean will receive the report of the committee and stand in the place of the head of the academic unit for the purpose of making the decision on the grievance. If the dean is the party against whom the grievance is filed, the Provost and Senior Vice President for Academic Affairs will handle any appeal.
5. Following the final disposition of the grievance, a brief written summary of the complaint and outcome is filed with the dean as per University Policy 3110: Student Complaint Log.

Following exhaustion of campus-based procedures, students may direct complaints to the State Council of Higher Education for Virginia. Additional information is available from their website (<https://www.schev.edu/students/resources/student-complaints>).

Online students should also refer to the Complaint Procedure for Online Students, #3 under Procedures for Academic Matters.

Harassment

It is the established policy of JMU to provide a work and study environment for faculty and staff members and students free from all forms of harassment, intimidation and exploitation. Prohibited harassment may include offensive verbal or physical conduct. JMU prohibits discrimination and harassment that is based on protected characteristics as defined within JMU Policies 1302 and 1324. Questions, assistance or violations related to discrimination and/or harassment on the basis of a protected characteristic should be directed to the university's Office of Equal Opportunity via the OEO website (<https://www.jmu.edu/access-and-enrollment/o eo/index.shtml>), email (oeo@jmu.edu), or phone (540-568-6991).

Students may deal with harassment in any of the following ways. If a student believes that they have been harassed by a staff member, faculty member or a student employee, the student should take one or more of the following actions:

- Discuss the matter with the faculty or staff member involved, explaining why a particular comment or action was offensive.
- Discuss the matter with the immediate supervisor of the faculty or staff member, giving an account of the comment or action in question.

If the student believes that they have been harassed by a student, the student should take one or more of the following actions:

- Discuss the matter with the accused, explaining why a particular comment or action was offensive.
- Bring a charge of harassment to the Office of Student Accountability and Restorative Practices.

If a student employee believes that they have been a victim of harassment, they should take one or more of the following actions:

- Discuss the matter with the accused, explaining why a particular comment or action was offensive.
- Discuss the matter with the immediate supervisor of the accused, giving an account of the comment or action in question.
- Contact the student employment office at (540) 568-3269.

Regardless of who the alleged harasser is, a student may discuss the matter with the Associate Vice President for Student Affairs, the director of the Counseling Center or the Office of Equal Opportunity. The student will be advised of proper university procedures that can be pursued. If requested, complaints will be held in confidence and counseling will be provided. No investigation or action against the accused person will be taken on a student's behalf unless the student consents to be identified, if necessary, to the individual accused in connection with the investigation.

A student also has the option of filing a formal charge of harassment with the U.S. Department of Education. Contact information for the U.S. Department of Education can be obtained by contacting the Office of Equal Opportunity via the OEO website (<https://www.jmu.edu/access-and-enrollment/o eo/index.shtml>), email (oeo@jmu.edu), or phone (540-568-6991).

Honor System

The academic program at JMU operates under an Honor System that dates back to the 1909-10 academic session. Students adopted the present Honor System in order to uphold individual and community integrity. Each student is expected to observe complete honesty in all

academic matters and to report instances where another student has violated the Honor System.

A student Honor Council administers the Honor System, and every student who matriculates at the university, whether graduate or undergraduate, becomes a member of the Honor System. The university expects the cooperation of faculty members and administrators in upholding this Honor System. The Student Handbook provides full information on the Honor System, and the Honor Council office provides students with assistance in understanding Honor System policy. All incoming JMU students are required to complete an online Honor System Tutorial and test during their first semester.

The Honor Council encourages all members of the JMU community to familiarize themselves with the Honor Code and Honor System procedures.

Inclement Weather or Emergency

When the university campus in Harrisonburg is closed due to weather or other emergency, all Harrisonburg-based in-person classes are cancelled on that day. In situations where the university closes mid-way through the day (e.g., 1:00 pm), all classes must cease at that time even if there is class time remaining. When the university is delayed in opening, classes can convene when the university opens, but must end at the regularly scheduled class time. During closures, faculty members may not require students to attend events, classes, laboratories, or any other functions on campus.

Makeup Days for Classes

When it is necessary to cancel classes due to weather or other emergencies, faculty have several options for making up the missed instructional time (See Policy 1309: Closings, Class Cancellations and Exam Postponements Due to Inclement Weather or Emergencies for more information).

- Incorporate the course content missed on that day into other classes throughout the semester.
- Hold class at a time acceptable to all class members other than the regularly scheduled time. Time and location will be arranged by the academic unit.
- Prepare the lecture asynchronously and allow the students to review the lecture in a reasonable amount of time.
- Provide students with alternate assignments that can be completed through electronic means.
- Hold class through electronic means, with the understanding that there may be no academic support services from IT or the Libraries. Students who are not able to attend class virtually, due to lack of internet, living situation, etc. must not be penalized for missing class.

Faculty are encouraged to provide a written statement, in their syllabus/ syllabi, of their method for addressing class cancellations due to inclement weather conditions or emergency situations.

Major Information

Students entering JMU should confer with their advisors to determine a major program of study. If entering students have not decided on a specific major, they may declare as exploratory. JMU encourages exploratory students to discuss their interests with representatives from University Advising, professors, academic unit heads and fellow students to find a major program best suited to each student's goals and interests.

Failure to do so could extend the time that students will need to fulfill graduation requirements. Students who would like assistance in identifying career options related to their specific majors can utilize the University Career Center, which helps students decide on career direction by assessing their career interests, skills and abilities as well as providing information about career options.

Declaration of Major

All students must declare their major by the beginning of their sophomore year. To declare a major, students will make their request through the Major Minor Change link on the student center in MyMadison. The request will be forwarded to the academic unit they wish to enter. Academic units accept students on the basis of their academic records and on the satisfactory completion of other criteria the academic units might establish. Academic units assign advisors for students who are beyond the level of a first-year student.

Transfer students who enter JMU with 30 or more credit hours must declare a major upon acceptance to the university.

Change of Major

Students who would like to change their major will make their request through the Major Minor Change link on the student center in MyMadison. The request will be forwarded to the head of the academic unit they wish to enter. Academic units accept students on the basis of their academic records and on the satisfactory completion of other criteria the academic units might establish. Academic units assign advisors for students who are accepted as majors.

Misconduct in Research and Other Scholarly Work

Policy 2205: Misconduct in Research and Scholarship, applies to all individuals involved in the performance of scholarly and creative activity and research conducted at JMU, whether performed under external or internal funding. It applies to faculty, staff, students, and others at the university engaged in scholarship, research, research training, and activities related to scholarship, research or research training, whether funded or not, and regardless of the funding source. It applies to all researchers and scholars, including sub-recipients of all grants and awards.

Misconduct, as defined under this policy, means fabrication, falsification, plagiarism or other practices that seriously deviate from those that are commonly accepted within the scientific and academic community for proposing, conducting or reporting research. Misconduct by a student under this policy may result in disciplinary action up to and including expulsion from the university, loss of fellowship or scholarship and potential criminal prosecution.

Non-Returning Students and Leave of Absence

Students who plan to leave JMU, either temporarily or permanently, must officially notify the university. To initiate a proper separation from JMU, students should visit the Registrar's website for more information.

Prerequisite and Eligibility Requirements

Students should consult the appropriate catalog or class schedule details to determine prerequisite or eligibility requirements for course selections. Without special permission, students cannot take for credit a course for which the prerequisite has not been met or for which they are not eligible.

In addition, students cannot take for credit a course that is a prerequisite to a course they have already taken. Many courses require other courses as prerequisites. Students should be aware that a course in which they receive a grade below "C" probably does not adequately prepare them to take a course that requires the first as a prerequisite. Before continuing to a succeeding course, students should discuss their level of preparation with the instructor of the second course and decide whether or not they need to retake the first course to improve their understanding as well as their grade.

Re-Entry to the University

Degree-seeking students who have not attended JMU for one or more semesters and who wish to return to their studies at JMU are classified as re-entry students. The re-entry process is contingent upon the student's length of absence and academic standing at the time of departure.

To avoid delays in registration, the Intent to Enroll Form (for students who have been absent from JMU for less than two years) or the Undergraduate Re-entry Form (for students who have been absent for two or more years) must be submitted by the indicated deadline for the anticipated semester of return. The Intent to Enroll Form and the Re-entry Form are available online. All registration holds must be cleared before the student will be eligible to register.

Semester	Deadline
Fall Semester	July 1
Spring Semester	November 1
Summer Session	April 1

Re-entry after Absence of Less Than Two Calendar Years

Students in Good Standing

Students who left the university in good standing and who have been absent for one semester or more, but less than two calendar years, must submit an Intent to Enroll Form to the Office of the Registrar.

Students on Academic Probation

Students who left the university on academic probation, and who have been absent for one semester or more, must submit an Intent to Enroll Form directly to the Assistant Vice Provost for Academic Student Services, MSC 7506. A personal statement citing the reason for departure and interim activities must accompany the form. Re-entry and continued enrollment are not guaranteed, but are contingent upon review by an academic review committee chaired by the Assistant Vice Provost for Academic Student Services. Such review may result in denial or conditional re-entry.

Students on Academic Suspension

Students who are placed on a first academic suspension may follow the appeal process stated in their suspension notification or apply for reentry after the suspension period. A personal statement citing the reason for departure and interim activities must accompany all Intent to Enroll Forms. Students must submit all documents to the Assistant Vice Provost for Academic Student Services, MSC 7506. Re-entry and continued enrollment are not guaranteed but contingent upon review by an academic review committee chaired by the Assistant Vice Provost for Academic Student Services. Such review may result in denial or conditional re-entry.

With prior approval from the dean of the major college and the Office of the Registrar, a suspended student may choose to take courses

at another institution. Course work completed during the period of suspension may be considered as part of the criteria for re-entry, but courses taken at another institution cannot be used to raise the grade point average at JMU. Students should review the Academic Suspension Threshold chart and consult with their advisors and/or deans to understand fully the impact that transfer credit may have on their JMU academic standing once they return.

Re-entry after Second Suspension or Absence Exceeding Two Calendar Years

Students who are eligible to apply for re-entry after an absence of two or more calendar years, regardless of their academic standing at the time of departure, must apply for readmission through the Office of Academic Student Services. Readmission is not automatic but subject to committee review of the application, the personal statement citing the reason for absence and evaluation of work completed at JMU as well as work completed at other institutions during the student's absence from JMU.

If a degree-seeking student leaves James Madison University after earning a portion of General Education credits and earns an approved Virginia Community College System (VCCS) Associate Degree, JMU will accept it in fulfillment of General Education requirements. Students who earn the approved degree while on academic suspension status from James Madison University are only eligible to re-enter James Madison University by agreeing to apply for the Transfer Equivalent Option upon their approved return to James Madison University.

Registration

Semester course listings are available on the university's internet accessible information system (<https://mymadison.jmu.edu/>). The university expects all students to register on the dates indicated in the registration calendar.

Credit is not allowed in any course for which the student is not duly registered, and registration is not complete until all fees for the semester have been paid. Students must complete course work during the semester in which they are registered for a course.

Student Assessment

James Madison University requires students to take a series of student outcomes assessments prior to their graduation. These assessments are held at three stages of students' academic careers, including:

- as entering students
- at the mid-undergraduate point, typically their sophomore year
- as graduating seniors, both university-wide and in their academic major(s)

Testing occurs at the first stage before the fall semester, at the second stage on a scheduled Assessment Day in the spring semester and at the third stage during the graduation semester as part of the Senior Compass requirements. Testing of seniors in their academic major occurs on the spring Assessment Day or is embedded in courses or requirements of the program. The university encourages students to review program requirements for details.

Assessment enables the collection of data on student learning outcomes across academic majors, the general education program, student affairs programs, and university-wide initiatives. Information gathered through assessment helps JMU understand how well the institution is supporting

student learning and to make improvements to the academic experience. In addition, assessment within degree programs allows academic units to determine whether students are achieving established learning outcomes. Because these assessments inform improvement efforts at JMU, students are required to participate.

Entering students who do not complete assessments during the fall assessment interval will receive a hold on their records and will be unable to add fall semester courses or to register for spring semester courses until the assessments are completed. Students who do not complete the mid-undergraduate assessments (spring Assessment Day) will receive a hold on their records and will be unable to add spring semester courses or register for fall semester courses until the assessments are completed. Graduating students who do not complete Senior Compass requirements prior to commencement will not be able to receive their diploma.

JMU does not report individual student scores; the university does, however, report aggregated test results to measure how programs and services contribute to student learning and development. Assessment results are reported within JMU and to external audiences. Internally, group-level assessment results are shared with faculty committees and administrators across the campus to improve university programs. Externally, JMU releases findings to the state and institutional accrediting bodies. The Center for Assessment and Research Studies (<https://www.jmu.edu/assessment/index.shtml>) provides support to JMU programs by assisting with the development of learning outcomes and the administration of assessments.

Semester Honors List

To qualify for the honor of being placed on the President's List, a student must earn a grade point average of 3.900 or above and be considered by the registrar to be carrying a full-time course load of graded credit hours (i.e., exclusive of credit hours taken on a credit/no credit or audit basis).

To qualify for the Dean's list, a student must meet the President's List requirement for course load and earn a grade point average of 3.500-3.899.

Students Seeking Second Baccalaureate Degrees

Students Currently Enrolled

A currently enrolled student may earn two different baccalaureate degrees through concurrent or consecutive enrollment at James Madison University. To earn a second degree, students must:

- Earn a minimum of 150 credit hours, including a minimum of 60 hours at JMU.
- Meet all degree and university graduation requirements for both degree programs.
- Meet all prerequisite and course requirements in two different majors.

A student who has met graduation requirements for one degree may participate in the commencement activities for that degree and continue with the second degree provided the following occur:

- The student has officially declared the second degree and major prior to the submission of the graduation application for the first degree, and both degrees and majors appear on the student's academic transcript.

- At the time of completion of the first degree, the student must have completed at least 12 credit hours in the major required for the second degree. A student who has earned fewer credits may not continue with the second degree if they decide to complete the first degree and participate in commencement exercises.
- No lapse in enrollment (not including summer) occurs between the completion of the first degree and the continued pursuit of the second degree.

Students Not Currently Enrolled

Students who have earned a baccalaureate degree from JMU or from another accredited college or university and want to pursue an additional, different, bachelor's degree and major program have limited options. Students may earn a second, different, baccalaureate degree at JMU in the following programs:

- Adult Degree (Individualized Study, B.A.; Individualized Study, B.I.S.; Individualized Study, B.S.)
- R.N. to B.S.N.

Student Teaching

Student teaching should be a full-time experience. Only in exceptional cases will additional course work be approved during the student teaching period.

Moreover, student teachers cannot expect to work or participate in excessive extracurricular activities during their student teaching period as any interference with student teaching might lower the quality of the individual's performance. Students with problems or special needs must contact the Education Support Center. All student teaching grades are assigned on a credit/no-credit basis.

Transcript

The permanent academic record, or transcript, is the official record of a student's classes, credits and grades earned to date, and it includes the date of graduation, credential received and date of withdrawal or dismissal, if applicable. Conferral information displayed on the transcript will include degree type, major, concentration, minor and certificate names. Posted grades or academic credits of any type will not be removed unless a verified error was made or the student repeats, through JMU enrollment, any posted test or transfer credit and earns a passing grade (D- or better). No requested additions, deletions or changes to the transcript will be permitted after a student graduates.

The Office of the Registrar may release a student's transcript upon receipt of an approved request from the student or former student.

Transfer Equivalent Option for Readmitted Students

A student who returns to JMU after a separation of two years and who maintains a minimum 2.0 GPA for 12 credit hours attempted at JMU in consecutive terms (summer may be included but is not required) after their return to JMU is eligible to apply for the transfer equivalent option. This option allows a student a quality point status equivalent to that of a transfer student admitted to the university.

The student's new GPA will initially be that attempted in the 12 credit hours taken upon readmission. To exercise this option, the student must submit a Transfer Equivalent Option (TEO) Form to the Assistant Vice Provost for Academic Student Services.

The following regulations govern this option:

- The option must be exercised within 30 days of receiving written correspondence from the Assistant Vice Provost for Academic Student Services indicating that TEO requirements have been met.
- A student with outstanding "I" grades is not eligible to apply.
- A student who leaves the university while in good standing is not eligible to apply.
- Credit/no-credit courses do not count toward the 12 attempted credit hours needed for application.
- If a student's TEO requirements are met over multiple consecutive terms/semesters, and/or more than 12 credit hours are attempted while working toward application of TEO, eligibility for TEO will be determined by the average earned on all credit hours attempted after their return to JMU.
- All grades will remain a part of the transcript.
- An eligible student will receive degree credit for only those courses for which grades of "C" or better were earned prior to re-entry.
- Quality points earned for any course completed prior to readmission will not count in determining the new cumulative GPA. Therefore, a student exercising the TEO option will not qualify for graduation with honors.
- A student whose application for the transfer equivalent policy has been approved may request that credit hours for pre-approved courses taken at another institution be transferred to JMU. The request will be reviewed by the office of the major college dean and the Office of the Registrar.
- The option will be extended only once during a student's enrollment at JMU.

A student interested in this option must apply for re-entry to the university through the Office of Academic Student Services according to established deadlines.

Semester	Deadline
Fall Semester	July 1
Spring Semester	November 1
Summer Session	April 1

Transfer of Credit from Other Institutions

A student wishing to earn credits at another institution should declare their intent to do so by completing the process outlined on the Office of the Registrar's website. In certain circumstances, official prior approvals may be required via the transfer credit approval form. Transfer requests for courses not listed in the transfer equivalency guide should be accompanied by course descriptions and/or syllabi as part of the process.

The General Education program allows students to transfer credit for a particular course based on a comparison of course objectives and content at JMU with those at the other institution. Once students have initially enrolled for classes at JMU they are expected to complete the General Education program through JMU enrollment. Any exceptions to this policy must be approved by the Office of General Education.

After students have enrolled for classes, they may use transfer courses to satisfy up to three General Education program requirements.

Special exceptions to this policy will be reviewed on an individual basis. Such exceptions may include those students who have been

discontinued from JMU, or those who wish to participate in study abroad programs or other special circumstances.

The student is responsible for having an official transcript sent to the Office of the Registrar when the work has been completed and final grades have been posted. Credit hours will be awarded for approved courses carrying a grade of "C" or better (2.0 quality points).

Once transfer credit has been posted to the transcript, it will not be removed unless it was posted in error, or the student subsequently repeats the course at JMU and earns a passing grade (D- or better).

Undergraduate Grading System

Maintaining standards of excellence and the integrity of the teaching/learning process are important values to JMU faculty. Evaluation of student work and assignment of grades on the basis of academic criteria are the responsibilities and prerogative exercised by the faculty member. A faculty member has the responsibility to evaluate student performance on a fair and scholarly basis in accordance with university policy.

The university makes grades available to students through the online student information system (MyMadison (<https://mymadison.jmu.edu/>)) at the end of each semester. Mid-semester grades, if assigned, in all courses are also made available through the online student information system (MyMadison (<https://mymadison.jmu.edu/>)) to new first year and transfer students in their first year at JMU. Grades are not mailed to students.

The academic achievement of a student in a specific course is expressed by letters as in the following table.

Grade	Description
A	Superior
B	Good
C	Average
D	Passing
F	Failure
I	Incomplete
CR	Credit for average or better work
NC	No credit awarded
WP	Withdrawal while passing
WF	Withdrawal while failing
W	Withdrawal

A course in which a grade of "F" is received does not result in earned credit hours, but does count as credit hours attempted in computing the grade point average. A course in which a grade of "NC", "WP", "WF" or "W" is received neither results in earned credit hours nor counts as credit hours attempted in computing the grade point average.

In certain circumstances, a grade of "I" (recorded by faculty and denoting incomplete work in a given course) will be given for a course. An "I" grade not resolved prior to the applicable incomplete grade deadline will automatically be converted to an "F" grade at the end of the next regular semester. Any student receiving an "I" grade should follow the university policy on incomplete grades.

Grade Point Average

The university computes a student's grade point average for any period by dividing the number of quality points earned during that period by the number of credit hours attempted during the same period. For instance,

during a period in which a student attempts 16 credit hours and earns 40 quality points, the student's GPA is 2.5.

A student's cumulative GPA is computed by dividing the total number of quality points earned at JMU by the total number of credit hours attempted at JMU.

Grades Not Reported

The "NR" grade is used to indicate that a faculty member did not report a grade, or grades, for one or more section(s).

Courses in which a student receives a grade of "NR" will need to be changed to the grade the student earned by the end of the next regular semester, or the grade is recorded as a "WP". Faculty may change the "NR" grade to any grade other than "W" or "I" at any time before the end of the next regular semester. Or, if the grade has converted to "WP", the faculty member may change that grade to any grade other than "W" or "I", provided the student has not graduated.

Incomplete Grades

The "I" grade is used to indicate incomplete work in a given course and is awarded only when a student is unable to complete course work because of illness or another equally compelling reason.

Courses in which a student receives a grade of "I" must be completed by the end of the next regular semester, or the grade is recorded as an "F". Faculty may require a shorter time period in which the student must complete all required work. (See the University Calendar for the dates by which grade changes must be reported to the Office of the Registrar.) A student seeking a grade of "I" must make that request to the relevant faculty member before the end of the semester in which that course is taking place.

Quality Points

Letter grades and quality points express the academic achievement of a student in a specific course. Quality point values are earned for each semester credit hour. The grade of "B" in a three credit hour course earns nine quality points; the grade of "C" in the same course would earn six quality points.

Grade	Per Credit Hour
A	4.00
A-	3.70
B+	3.30
B	3.00
B-	2.70
C+	2.30
C	2.00
C-	1.70
D+	1.30
D	1.00
D-	0.7
F	0
I (Incomplete)	0
CR (Credit for work which is at the 2.000 level or above)	0
NC (No credit awarded)	0

WP (Withdrawal while passing)	0
WF (Withdrawal while failing)	0
W (Withdrawal)	0

The following do not affect quality points earned, credit hours attempted or GPA:

- Grades received at other institutions.
- Audited courses.
- "I," "CR," "NC," "W," "WP" or "WF" grades.

Repeating Courses

A student may repeat any of the graded courses that they have taken during an undergraduate career at JMU. All grades will be included when calculating the student's grade point average. There are, however, certain exceptions to this rule.

Students may elect to repeat up to two courses during their enrollment as an undergraduate at JMU with a "repeat/forgiveness" option. As a result of the "repeat/forgiveness" option, the university will exclude the previous grade and credit hours earned for the repeated course when it calculates the student's cumulative GPA and earned credit hours total, regardless of whether the previous grade was higher or lower than the repeat attempt.

Both grades will appear on the transcript, and the recalculation of the cumulative GPA will occur after the "repeat/forgiveness" option attempt. The student must either declare the "repeat/forgiveness" option at registration or "edit" the course repeat option via MyMadison prior to the end of the course adjustment period. A student may not exercise the "repeat/forgiveness" option for courses in which that student was assigned a grade as a result of an Honor Code violation. Courses taught on a topic basis are generally repeatable; however, these courses may only be designated with a "repeat credit" option or "repeat/forgiveness" option if both course attempts have the same topic (i.e. a second attempt of the course HUM 200: Western Classics may only be assigned as a repeat credit or repeat/forgiveness credit if the original attempt of HUM 200 was taken with the topic "Western Classics").

All grades will appear on the student's transcript, but a course that has been repeated will only be counted once toward satisfying graduation requirements. Courses taken at other institutions do not qualify to be taken as "repeat forgiveness." Courses taken under the Credit/No Credit grading basis may not be repeated if credit was awarded for the first attempt. Courses taken in a different JMU career (i.e. Continuing Education) and subsequently transferred to an undergraduate JMU career are not repeatable. Students may request approval to retake these courses utilizing the audit grading basis only.

Withdrawal from the University

Students withdraw from the university when their enrollment is terminated before students have completed the semester for which they registered. Students who decide to withdraw must complete the Non-Returning/Leave of Absence Notice process (https://www.jmu.edu/registrar/students/leaving_jmu.shtml).

Students may contact the Office of the Registrar by email at withdrawal@jmu.edu, or by phone at 540-568-6281 with questions regarding the withdrawal process.

Academic Terms and Definitions

Degree Requirements

A degree is an academic title conferred on students who complete a unified program of study. Degrees vary according to the major program. JMU offers eight undergraduate degrees that incorporate course requirements unique to the degree and major discipline. Majors culminating in Bachelor of Arts and Bachelor of Science degrees share common sets of course requirements appropriate to the degree with specific distinctions.

Bachelor of Arts degree: The B.A. is distinguished by its humanistic emphasis. Students who complete a B.A. may satisfy the degree requirements by taking courses that advance their understanding of human culture through analysis of ideas; perception of differences; appreciation of art and creative products through understanding art forms, beauty, and symmetry; knowledge of theories and principles of form, substance, argument and philosophy; understanding of the interaction between language and culture; and achievement of linguistic competency in a second language.

Bachelor of Science degree: The B.S. is distinguished by its scientific emphasis. Students who complete a B.S. may satisfy the degree requirements by taking courses that advance their understanding of the use of scientific analysis, experimentation and the application of scientific principles and facts in solving problems; understanding of the crucial role of mathematical reasoning; and understanding analysis and techniques in comprehending problems in the natural or social sciences.

A list of courses that satisfy either Bachelor of Arts or Bachelor of Science degree requirements is available on the Registrar's website.

Program of Study Components

A program of study, or course of study, refers to any set of courses and experiences identified by the university as satisfying the requirements of a program or program component for a student or students. The following terms describe programs of study at James Madison University.

- **Major:** A major is a prescribed set of required and elective courses in a specialized area of study within an undergraduate degree program. The major must consist of a minimum of 30 credits hours. The student must formally declare the major for it to appear on the transcript of record.
- **Minor:** A minor (undergraduate) is a cohesive set of required and elective courses that connotes knowledge and skills in a discipline outside, but not at the depth of, a major. A minor must be a minimum of 18 credit hours, with the typical range being 18-24 credit hours. The minor is designed for students who are not majoring in the same area (students may not earn a major and minor in the same discipline). The student must formally declare the major for it to appear on the transcript of record. A minor is not required for graduation except when a student's declared major requires it.
- **Concentration:** A concentration is a prescribed set of courses within a major/program that provides specialization according to the student's interest and/or career goals. A concentration may be required or optional, depending on the parent program. Beginning Fall 2025, newly developed concentrations require a minimum of nine credit hours. The student must formally declare the concentration for it to appear on the transcript of record.
- **Cognate:** A set of courses outside the major that is designed to complement other components of the student's course of study and

to support selected professional goals. A cognate is not a required component of a program, unless specifically incorporated as such by the major. A student does not have to formally declare a cognate and it will not appear on the transcript of record.

- **Core:** A defined group of courses within a particular major or minor that is required of all students completing that major or minor.
- **Track:** A prescribed set of courses within a concentration. A track is not a required component of all concentrations. A student does not have to formally declare a track and it will not appear on the transcript of record.
- **Certificate Program:** A certificate is a formal award certifying the completion of a specialized post-secondary education program that is approved by SCHEV. A certificate consists of 9-24 credit hours and can be completed in conjunction with, or independent of, a degree program. A student must formally declare a certificate program and it will appear on the transcript of record.
- **Preparation Program:** A defined set of courses or experiences that serves as prerequisites to upper or graduate-level professional program courses; or, that prepares students for eligibility to obtain a license issued by an agency, group, or professional organization. A preparation program may coincide with a major, minor or concentration, or it may be comprised of courses from various disciplines and concentrations. A student must formally declare a preparation program and/or the associated required majors, minors, or concentrations in accordance with the program of study required for preparation. A preparation program will not appear on the transcript of record.

These definitions apply to all programs of study except in cases where program nomenclature specified by the accrediting agency differs from the nomenclature stated by James Madison University.

Military Service

Class Registration for Active Duty Students

James Madison University supports active duty students in the armed forces by providing assistance with class registration when necessary and by request. Active duty students needing assistance should provide a copy of their active duty orders to the Office of the Registrar prior to the first day of class to qualify for assistance under this policy. The Office of the Registrar will serve as an additional resource for the student and the academic unit(s) to assist with the creation of an appropriate class schedule to ensure the service member remains on track to degree completion in a timely manner.

Section 103 Policy

In accordance with Section 103 of the Veterans Benefits and Transition Act of 2018, James Madison University has adopted the following policy for compliance, effective August 1, 2019:

James Madison University will allow an individual to attend and participate in a course of education if the individual provides a Certificate of Eligibility under Chapter 31 or Chapter 33. The individual is permitted to attend the course beginning the date the student provides the Certificate of Eligibility until the earlier date VA provides payment to the school or 90 days after the school certifies tuition and fees. James Madison University will not impose any penalty, including late fees, denial or access to classes, libraries or facilities, or require the student to borrow additional funds due to the inability to meet his or her financial obligations to the institution as a result of delayed payments for education assistance under Chapter 31 or Chapter 33. This policy only pertains to the VA payment of the tuition, comprehensive fee and individual course fees

that are paid directly to the university. All other charges on a student's account, including but not limited to out-of-state tuition difference, remaining percentage of in-state tuition if not 100% eligible for Post 9/11, room rent, internet fee, meal plan, parking fines and/or health center charges are required to be paid upfront by the student by the University Business Office deadline each semester. Late fees and holds may be assessed for these charges.

Section 103 allows the university to require additional actions in order to qualify for this policy. James Madison University requires Chapter 31 and Chapter 33 eligible students to submit their JMU Request for Veterans Educational Benefits form by the first Friday of the semester to the School Certifying Official to avoid a hold or late fee on the account for the VA issued tuition/fee payment. The Certificate of Eligibility is also required for first time benefit users at James Madison University.

Veterans Access, Choice and Accountability Act of 2014

James Madison University complies with veteran student regulations regarding tuition rates. The following individuals shall be charged a rate of tuition not to exceed the in-state rate for tuition and fees purposes:

- A Veteran using educational assistance under either chapter 30 (Montgomery GI Bill[®] – Active Duty Program) or chapter 33 (Post-9/11 GI Bill[®]), of title 38, United States Code, who lives in the Commonwealth of Virginia while attending a school located in the Commonwealth of Virginia (regardless of their formal State of residence).
- Anyone using transferred Post-9/11 GI Bill[®] benefits (38 U.S.C. § 3319) who lives in the Commonwealth of Virginia while attending a school located in the Commonwealth of Virginia (regardless of their formal State of residence).
- A spouse or child using benefits under the Marine Gunnery Sergeant John David Fry Scholarship (38 U.S.C. § 3311(b)(9)) who lives in the Commonwealth of Virginia while attending a school located in the Commonwealth of Virginia (regardless of their formal State of residence).
- An individual using educational assistance under chapter 31, Veteran Readiness and Employment (VR&E) who lives in the Commonwealth of Virginia while attending a school located in the Commonwealth of Virginia (regardless of their formal State of residence) effective for courses, semesters, or terms beginning after March 1, 2019.
- A spouse or child using Chapter 35 Dependents Educational Assistance (DEA) benefits who lives in the Commonwealth of Virginia while attending a school located in the Commonwealth of Virginia (regardless of their formal State of residence).
- An individual using MGIB Chapter 1606 Selected Reserves benefits who lives in the Commonwealth of Virginia while attending a school located in the Commonwealth of Virginia (regardless of their formal State of residence) effective after August 1, 2026.
- Anyone described above while he or she remains continuously enrolled (other than during regular scheduled breaks between courses, semesters, or terms) at the same institution. The person so described must be using educational benefits under either chapter 30, chapter 31, chapter 33, chapter 35 or chapter 1606 of title 38, United States Code.

The in-state tuition provisions do not apply to those individuals on active duty using benefits under the Post-9/11 GI Bill[®] and Montgomery GI Bill[®] - Active Duty.

To be considered for in-state tuition through the Veterans Access, Choice and Accountability Act of 2014, the student must submit their Chapter

33 Post 9/11, Chapter 30 MGIB Active Duty, Chapter 35 DEA, or Chapter 1606 MGIB SR Certificate of Eligibility or Chapter 31 VRE authorization to the Veteran Benefit's Coordinator in the JMU Office of the Registrar no later than the first day of classes for the semester. If the student claims eligibility but the COE is not available, the Veteran's Coordinator can verify eligibility with Veteran's Affairs. In this case, student's VA benefit application must be approved with VA by the first day of classes.

Please visit the GI Bill (<https://www.va.gov/resources/in-state-tuition-rates-under-the-veterans-choice-act/>)[®] website for more information regarding the Choice Act.

Veterans Education Benefits Eligibility

Students who want to request Veterans Education Benefits for a term must complete and submit The JMU Request for Veterans Education Benefits form to the School Certifying Official. Students who are ineligible to enroll in classes in a given term do not qualify for Veterans Education Benefits for that term.

Veterans Educational Benefits Grievance Procedure

The Virginia State Approving Agency (SAA), is the approving authority of education and training programs in Virginia. SAA investigates complaints of GI Bill[®] beneficiaries. While most complaints should initially follow the school grievance policy, if the situation cannot be resolved at the school, the beneficiary should contact the SAA office via email saa@dvs.virginia.gov.

GI Bill[®] is a registered trademark of the U.S Department of Veterans Affairs (VA). More information about education benefits offered by VA is available at the official U.S. government website (<http://www.benefits.va.gov/gibill>).

GI Bill[®] Advertising Statement

This institution is approved to offer GI Bill[®] educational benefits by the Virginia State Approving Agency.

GI Bill[®] is a registered trademark of the U.S Department of Veterans Affairs (VA). More information about education benefits offered by VA is available at the official U.S. government website (<http://www.benefits.va.gov/gibill>).

Short Term Military Leave

Faculty members are expected to make reasonable academic accommodations or opportunities for students to complete course assignments and/or exams without penalty to the course grade for class absence(s) or missed deadlines due to short term mandatory military training or obligations. Students will provide faculty members with official military documentation (paper, electronic orders or a unit's memorandum) with as much advance notification as possible for absences that will result from temporal responsibilities of their military obligations. For time-sensitive state or federal emergencies/activations where written documentation may not be available until the end of the obligation, the student is responsible for securing those orders to provide to faculty members upon return to the university.

Support for Armed Services Active Duty

Relief, Refund and Reinstatement Tuition Guidelines

James Madison University supports students called to active duty in the armed services by providing for tuition relief and refunds, and for reinstatement of students whose documented service in the uniformed services has required their sudden withdrawal or prolonged absence from their enrollment in the institution. Included is service in the uniformed

services whether voluntary or involuntary on active or inactive duty in the Armed Forces, including such service by a member of the National Guard or Reserve. When a JMU student is under call or ordered to active duty resulting in a full university withdrawal or prolonged absence, the following provisions will apply.

Tuition Charges and Student Account Balances

If an enrolled student is deployed to active duty military service during the semester, the student can:

- Drop all classes and all tuition charges will be waived with:
 - Personal payments refunded
 - Financial aid adjusted as required
 - Financial aid refunds repaid by student if appropriate
- Maintain all or part of enrollment with:
 - Tuition adjusted accordingly
 - Financial aid adjusted as required
 - Financial aid refunds repaid by student if appropriate

If there are unpaid student account balances at time of deployment, the university will work with individual students on payment arrangements. No collection actions will occur during deployment; however, students must resolve any unpaid balances prior to subsequent enrollment.

Room and Board

If an enrolled student is deployed to active duty military service during the semester, the student's dining and residence hall contracts will be adjusted as follows:

- Board fees will be prorated from the dining hall opening date.
- A refund of the room rent will be issued.

Textbooks

When a student is called to active duty, a full refund for textbooks purchased for the semester in progress is available through the university bookstore by presenting the textbooks and a copy of the applicable military orders.

Deposits

For new students, an enrollment deposit is required to confirm their acceptance of the offer of admission. If a student is deployed to active duty military service before beginning the planned semester of enrollment, the deposit will be refunded.

For returning students, deposits made with the intent of securing facilities or services in a future session will be refunded in full.

Academic Credit

If an enrolled student is deployed to active duty military service during the semester, the student will have three options concerning grades assigned for the semester in which the call to active duty occurs.

- If the student leaves at any time during the semester and elects to receive a full tuition refund, no notation of courses or grades will be recorded on the student's transcript.
- If the student elects to receive an "I" (Incomplete), the regulation regarding conversion of an "I" to "F" grade will be suspended until the student returns to campus. Should the student not return to JMU, the "I" will revert to a "W" (Withdrawal).
- If the student leaves at a point in the semester after which a significant amount of work has been completed, the student may

request the assignment of a grade for work completed. This option requires joint agreement of the student and faculty member(s).

Reinstatement: Re-entry to the University (military)

Formerly enrolled undergraduate degree-seeking students who have not attended JMU for one or more semesters and who wish to return to their studies at JMU are classified as re-entry students. The re-entry process is contingent upon the student's length of absence and academic standing at the time of departure. To avoid delays in registration, the Intent to Enroll Form (for undergraduate students who have been absent from JMU for less than two years) or the Office of Admissions Re-entry Form (for students who have been absent for two or more years) must be submitted by the indicated deadline for the anticipated semester of return. All registration holds must be cleared before the student will be eligible to register.

Re-entry after Absence of Less Than Two Calendar Years

Undergraduate Students in Good Standing

Students who left the university in good standing and who have been absent for one semester or more must submit an Intent to Enroll Form located on the Registrar's website (<https://www.jmu.edu/registrar/index.shtml>).

Undergraduate Students on Academic Probation or Suspension

Students who were on academic probation or suspension when they left the university for active military duty, and who have been absent for one semester or more, must submit an Intent to Enroll Form directly to the Assistant Vice Provost for Academic Student Services. A personal statement describing the student's intended academic and study plans must accompany the form. Re-entry and continued enrollment are not guaranteed, but are contingent upon review by an academic review committee chaired by the Assistant Vice Provost for Academic Student Services. Such review may result in denial or conditional re-entry.

Re-entry after Absence of Two Calendar Years or More

Students who have been absent for two or more calendar years, are eligible for re-entry if (1) they return to the University after a cumulative absence of not more than five years, and (2) they provide notice of intent to return to the University not later than three years after the completion of the period of service. (Exceptions to these time periods may be found in the Higher Education Opportunity Act (HEOA) of 2008.) Students on academic probation or suspension when they left the university may be assigned conditional re-entry. Students must submit the Undergraduate Re-Entry form for readmission via the Office of Academic Student Services as well as proof of military service such as the DD-214.

Students returning after an absence of two calendar years and who do not meet the two conditions above must submit the Undergraduate Re-Entry form for readmission via the Office of Academic Student Services. Readmission is not automatic but subject to committee review of the application, the personal statement and evaluation of work completed at JMU, as well as work completed at other institutions during the student's absence from JMU.

If a degree-seeking student leaves James Madison University after earning a portion of General Education credits and earns an approved Virginia Community College System (VCCS) Associate Degree, JMU will accept it in fulfillment of General Education requirements. Students who earn the approved degree while on academic suspension status from James Madison University are only eligible to re-enter James Madison

University by agreeing to apply for the Transfer Equivalent Option upon their approved return to James Madison University.

Reinstatement into Specific Undergraduate Program of Study

When a student with a declared major wishes to return to the same program at the university after an absence of more than two years, the course work taken prior to withdrawal must be reviewed by an advisor/academic unit head prior to the student's re-enrollment. This counseling is required to assure that the previously completed course work is current and applicable to the major. The advisor will assist the student to develop an appropriate academic plan for degree completion.

Students without declared majors wishing to return to study should consult with an advisor in University Advising to receive assistance in developing an academic plan.

Deferral of Enrollment

If an undergraduate student has been admitted to James Madison University and is called to active military duty or enlists in the National Guard before enrolling, the student may request a deferral of admission using the process below.

The student must submit a letter to the director of admissions requesting a deferment. In the letter the student should provide the reason for the request (call to active duty or National Guard boot camp will not end until after the start of the semester that the student intended to enroll) and indicate the term they wish to enroll (the term can be changed if needed).

The request is reviewed by the director of admissions who sends a letter notifying the student that their request has been granted and indicates when JMU expects the student to enroll. The director of admissions will also inform the following offices of the deferment: Registrar's Office; University Business Office; Financial Aid and Scholarships; Orientation Office; Residence Life, Career and Academic Planning; and University Advising (for transfer students).

During the semester prior to scheduled enrollment, JMU will send the student a letter explaining what needs to be done prior to the start of their enrollment at JMU.

If the student's discharge from the service is delayed, the student contacts the director of admissions and requests an extension of their deferment. The director will approve the delay and inform the following offices: University Business Office; Financial Aid and Scholarships; Orientation Office; Residence Life, Career and Academic Planning; and University Advising (for transfer students).

Documentation

Students should complete Withdrawal or Leave of Absence forms and include a copy of the applicable military orders to qualify for the considerations detailed in this policy.

The Withdrawal Process is appropriate for students who are enrolled in a semester, are deployed to active duty military service and must begin that duty before the current semester has ended. To initiate a proper separation from JMU, students should visit the Registrar's website for more information.

A Leave of Absence is appropriate for students who are deployed to active duty military service, but do not need to begin duty during a current semester; for example, if duty will begin during a future semester before

classes begin. The Leave of Absence Form is available online from the Registrar's website.

Evaluation of Undergraduate Transfer Credits for Military Students

Credit will be awarded for those courses equivalent to courses offered at JMU in which the student has earned a grade of "C" or better.

The Office of the Registrar and the academic unit head of the program in which the student is majoring will determine the credits required for graduation.

With the exception of some Virginia community college degrees in General Studies, JMU General Education requirements will be waived for those students who have completed either the Associate of Arts, Associate of Science, or the Associate of Arts and Sciences degree at a Virginia community college. For the complete listing of degrees associated with a Virginia Community College that waive all General Education requirements, go to the Transfer Advising website at <http://www.jmu.edu/transfer>. Students who complete one of these associate degrees and are offered admission will receive junior-level status; however, due to varying major and degree requirements, junior-level status does not guarantee graduation in two years. For more details, consult the Office of Admissions or the JMU Virginia Community College Transfer Guide.

Types of Credit Awarded

Type of Credit	Applies To	Reviewed By
Credits from other institutions of higher learning	All undergraduate degree seeking students	Office of the Registrar and the academic unit head of the major program of study
Military transcript (i.e. JST, CCAF)	All undergraduate degree seeking students	Office of the Registrar using the ACE guide recommendation on the military transcript
CLEP	All undergraduate degree seeking students	Office of the Registrar and the academic unit head of the major program of study
DSST exams	Students in the Adult Degree Program	Adult degree program and the academic unit heads
Excelsior College examination	Students in the RN to BSN program	Nursing department and the Office of the Registrar
Individualized portfolio evaluation	Students in the Adult Degree program	Adult Degree Program
Learning acquired in military service (DD- 214)	All undergraduate degree seeking students	Office of the Registrar when student does not have military transcript (JST, CCAF)

For questions pertaining to transfer credit for military education and training, contact the Office of the Registrar at (540) 568-6281 or transfer_credit@jmu.edu.

STUDENT SERVICES AND RESOURCES

Academic Affairs

The Division of Academic Affairs (p. 152) offers the following resources for students:

- Center for Faculty Innovation (p. 154)
- Center for Global Engagement (p. 154)
- The Learning Centers (p. 159)
- Pre-Professional Health Advising (p. 158)
- Office of the Registrar (p. 157)
- University Advising (p. 160)

Access and Enrollment

The Division of Access and Enrollment (<https://www.jmu.edu/access-and-enrollment/index.shtml>) offers the following resources for students:

- Office of Equal Opportunity (p. 157)
- Title IX Office (p. 160)

Administration and Finance

The Division of Administration and Finance (p. 153) offers the following resources for students:

- Business Services (p. 153)
- JMU Police Department (p. 156)

Student Affairs

The Division of Student Affairs (p. 158) offers the following resources for students:

- Center for Multicultural Student Services (p. 154)
- Community Engagement & Volunteer Center (p. 154)
- Counseling Center (p. 155)
- Dean of Students (p. 156)
- Office of Disability Services (p. 156)
- Office of Orientation and Transition (p. 157)
- Office of Residence Life (p. 157)
- Office of Student Accountability and Restorative Practices (OSARP) (p. 157)
- Student Government Association (p. 158)
- Student Handbook (p. 158)
- Student Life (p. 158)
- Student Success Center (p. 159)
- University Career Center (p. 161)
- University Health Center (p. 162)
- University Recreation (p. 162)
- Victim Advocacy Services (p. 163)

Academic Affairs

Phone: (540) 568-3429

Academic Affairs Website (<http://www.jmu.edu/academic-affairs>)

Mission Statement

We are an inclusive academic community committed to excellence in the pursuit and dissemination of knowledge. We advance intellectual and creative discovery through transformative learning experiences that positively impact our lives and communities.

R2/National University Status Statement

As an R2 institution, JMU is committed to creating and sustaining an environment where faculty can invest deeply in their identities as teachers, scholars and creatives. As teacher-scholars, we embrace discovery and exploration across disciplines and borders in support of real-world impact.

JMU offers our faculty and staff the flexibility to grow and evolve across a career, and we understand that student-centered teaching, research and service are core to our educational mission. They can be integrated and should be celebrated. As we grow into the R2 space, our mission and vision remain the same, but opportunities expand as we reimagine the ways our work supports teaching, scholarship, creative activity and service.

- **Teaching:** We value many forms of teaching that intersect with—and are enhanced by—our research, scholarship and creative activity. Inclusive high-impact pedagogies bolster student learning and critical thinking with respect to disciplinary content, as well as supplement skills development and workforce readiness.
- **Scholarship/Creative Activity:** We involve students at all levels in scholarship and creative activity in collaboration with JMU faculty and staff. We value all forms of research and creative activity, including community-driven and interdisciplinary scholarship that responds to the needs of our community, commonwealth and beyond.
- **Service:** Our service contributes to the operation of our institution and the well-being of its people. We also share academic expertise with organizations and community partners beyond the university and contribute to the betterment of our disciplines and fields.

Shared Governance Statement

Shared governance is a form of governance in higher education through which core institutional stakeholders—primarily administrative leadership, faculty, and governing boards—work together to make decisions that impact the institution. Facilitated by structures and processes, shared governance ensures that those with the most relevant expertise have a primary decision-making role in their respective areas.

Shared governance and academic freedom are inextricably linked and essential for fulfilling JMU's academic mission. Bound by the norms of honesty, integrity, and responsibility, academic freedom is the principle that scholars and educators can teach, research, and express ideas related to their scholarly expertise without censorship or fear of retribution. Academic freedom also protects the right of a faculty member to speak freely when participating in institutional governance, as well as to speak freely as a citizen. Shared governance at JMU rests on a commitment to transparent and meaningful collaboration in a framework of academic freedom.

In the context of shared governance, faculty, administrators, and board members have the following primary responsibilities:

- **Administrative Leadership:** setting and carrying out institutional vision, overseeing financial health and viability, and leading strategic planning.

- Faculty: establishing and maintaining academic quality and responsibility, including teaching, curriculum, scholarship and creative inquiry, and setting and upholding standards in their disciplines.
- Governing Board: hiring/supervising the president, fiduciary oversight, and accountability for fulfilling institutional mission.

Administration and Finance

Phone: (540) 568-6434

The Division of Administration and Finance (<http://www.jmu.edu/adminfinance>) is committed to the preparation of students to be educated and enlightened citizens who will lead productive and meaningful lives. The division supports the university's commitment to excellence by empowering our staff to communicate effectively with the university community and provide proactive approaches to satisfy customer expectations.

Business Services

Bookstore

Location: 211 Bluestone Drive, MSC 2902

Phone: (540) 568-6121

The JMU Bookstore (<https://www.jmu.edu/bookstore>) stocks all textbooks: new and used condition, digital and rental formats, required by the academic units, as well as a large selection of general and technical books for faculty and students. The store also sells school supplies, office computer supplies, Apple and Dell computers, software, clothing, gifts, and greeting cards. For students' convenience, the bookstore provides services including special orders for books, textbook buy back, gift cards, and repairs for Apple and Dell computers.

Card and Print Services

Location: Foundation Hall, MSC 5722

Phone: (540) 568-6446 and (540) 568-7300

Location: Engineering/Geosciences (EnGeo; formerly HHS), Room 1002, MSC 4311

Phone: (540) 568-8731

Card (<https://www.jmu.edu/cardctr>) and Print Services (<http://www.jmu.edu/printservices>) supports the university community by providing identification, access and printing solutions across campus. The office issues the JMU access card (JACard), the official identification card for all students, faculty and staff. The JACard provides access to residence halls, campus facilities, meal plan access, Dining Dollars and FLEX declining balance accounts, which are administered in partnership with Dining Services and the University Business Office. Dining Dollars allow students to eat at on-campus dining locations and are exempt from sales tax. FLEX accounts may be used for purchases and services across campus, including the bookstore, copy machines, computer labs, library fees, parking, mail services, UREC, UPB movies, the University Health Center, the Forbes Center, all dining facilities and on-campus vending machines. In addition, the office supports a Mobile ID app available on Apple and Android devices, which offers access to many of the same campus services as a convenient supplement to the physical card.

Card and Print Services also operates two campus print locations that provide a full range of photocopying and digital printing services. These services include academic coursepack production, copyright permissions, full-color and black-and-white digital printing, binding,

laminating, wide-format posters and other print-related products. Both locations offer computer labs where students, faculty, staff and community members can work on projects and access production equipment to complete their printing needs.

Dining Services

Location: D-Hall 1st floor

Phone: (540) 568-6751

Email: dining@jmu.edu

We know that dining (<http://www.jmu.edu/dining>) is an integral part of campus life. Each day is an opportunity to serve delicious food, host great events and provide an overall memorable dining experience. Convinced that nutritious food is essential to a healthy lifestyle and strong academics, we build our menus with health in mind and make eating a well-balanced meal easy.

Meal Plan Basics

Explore our meal plan options, available to residents and commuters, at our website (<https://jmu.mydininghub.com/>) or visit us at our D-Hall 1st floor office. Students living on campus are required to purchase a meal plan, and all plans include Meal Punches and Dining Dollars. Not sure which meal plan to choose? Consider personal eating habits, class schedule and leisure activities each semester before choosing a meal plan that fits. Meal plan "punches" are not active during university breaks.

Dining at a Glance

Meals or Meal Punches are included with a meal plan and reload daily or weekly. Use them at our all-you-care-to-eat dining halls, D-Hall and E-Hall, or for a "Duke Deal" meal exchange available at select retail locations. Visit our website for complete details.

Dining Dollars can be used at all on-campus locations, including dining halls, retail brands, concessions and vending machines. Any unused Dining Dollars will only roll over from fall to spring, not year to year. Need to add more funds to your meal plan? No problem, add Dining Dollars GOLD anytime through the University Business Office (<https://www.jmu.edu/ubo/index.shtml>) (UBO). JMU Flex and credit/debit card payments are also accepted.

Dining Dollars Gold can be added at any time through UBO (<https://www.jmu.edu/ubo/index.shtml>). Why go for the gold? You don't need to be on a meal plan to purchase them and any unused Dining Dollars Gold will roll over each semester as long as you're a student at JMU.

Whether you choose Dining Dollars or Dining Dollars Gold, enjoy these benefits:

- They're tax-free, so guaranteed savings over cash, credit, or Flex Dollars
- More spending power than meal "punches"
- They can be used for guest purchases

Allergies or Special Dietary Needs At JMU Dining

We recognize and accommodate food allergies preferences, and special dietary needs in a way that's personalized, sensitive and supportive. In fact, all employees undergo certified allergy training, and we pride ourselves on working with you to make sure you can make informed food choices. For more information reach out to our registered dietitian at (540) 568-5731.

Sustainability

At JMU Dining, we consistently strive to become more sustainable by embedding environmental stewardship in our everyday operations. Our programs and policies increase sustainable food purchasing, educate and empower students to lead a sustainable lifestyle, prioritize circularity, and reduce food waste. Together we can make a positive impact on both people and the planet.

Parking and Transit Services

Location: Champions Drive Parking Deck, 280 Champions Drive, MSC 1301

Phone: (540) 568-3300

Email: parkingservices@jmu.edu

Parking and Transit Services (<http://www.jmu.edu/parking>) offers annual, semester and daily parking permits to eligible members of the university community, provides parking arrangements for special events on campus, collaborates with the Harrisonburg Department of Public Transportation to provide safe and reliable bus service, and manages the university's alternative transportation programs including bicycle, carpool, and rideshare options. The department also maintains multi-space pay stations in eight locations across the campus that are intended to provide convenient, short-term metered parking. In addition, permit holders have access to electric vehicle charging stations at eight locations for an hourly fee. Parking permits are available to campus visitors, free of charge, upon presentation of a valid driver's license and vehicle registration. Please remember that any vehicle parked on university-owned or leased property must be registered with JMU or pay for metered parking in designated areas.

Center for Faculty Innovation

Location: Rose Library, Fifth Floor

Phone: (540) 568-4846

The Center for Faculty Innovation (CFI) (<http://www.jmu.edu/cfi/>) provides comprehensive professional development opportunities for JMU instructional faculty at all levels and stages of their careers. CFI initiatives are designed to enhance academic culture and support innovations in teaching, scholarship, and well-being. The CFI is staffed by educational developers who, as teacher scholars in their academic disciplines, are committed to creating a sense of shared community grounded in academic excellence.

The CFI pursues its vision of comprehensive professional development for all faculty through collaborative partnerships with the JMU community by implementing and assessing a range of scholarly-based programs. Further, the CFI embraces issues that transcend the individual faculty role, striving to make meaningful contributions to local dialogues and global dynamics. Finally, the CFI strives to enhance academic culture through scholarly initiatives that reflect on the quality and changing nature of higher education.

JMU graduate students are welcome to participate in most CFI activities.

Center for Global Engagement

Location: 100 East Grace Street, MSC 5731

Phone: (540) 568-5209

Fax: (540) 568-3310

The Center for Global Engagement (<http://www.jmu.edu/global>) provides high-quality global education opportunities for our campus community, supports our international students and scholars, and assists in the internationalization efforts at JMU. From our robust study abroad programs (p. 594) to programming that connects our international community, to global virtual exchange, the CGE offers a full range of global experiences.

Center for Multicultural Student Services

Location: Student Success Center, Suite 1312, MSC 3504

Phone: (540) 568-6636

The Center for Multicultural Student Services (CMSS) (<http://www.jmu.edu/multicultural>) is an integral part of the student experience at James Madison University.

In addition to providing a home away from home to the multicultural community, the Center for Multicultural Student Services supports the journey towards degree completion by focusing on student success, holistic student development (academically, socially, culturally, professionally and personally) and intergroup community engagement by cultivating and strengthening networking opportunities.

CMSS is home to 50+ student affinity-based groups (including fraternities and sororities). We create space and opportunities for students to develop a sense of belonging through staff and peer-to-peer connections. CMSS offers HRD 123, a 3-credit hour course that develops leadership, facilitation and presentation skills to prepare students for college, the workforce and to lead peers through campus dialogues on critical topics. CMSS is open to all students and hosts a variety of opportunities to engage in a Pre-Weeks of Welcome program, experiential learning trips, leadership conferences and educational programs that center an array of students' affinities and interests. "There's a Place for YOU at JMU!"

The office works diligently to provide the following:

- Student leadership development
- Student engagement opportunities
- Student development (academically, professionally, socially, culturally, personally)
- Opportunities to educate, celebrate, value and uplift various cultural heritages, traditions and experiences
- Peer and staff mentorship, advising and overall student support services to undergird the following but not limited to:
 - Student success
 - Sense of belonging
 - Community engagement and workforce preparedness
 - Retention to degree completion

Community Engagement & Volunteer Center

Location: Student Success Center, 2nd Floor Room 2200, MSC 1011

Phone: (540) 568-6366

The Community Engagement & Volunteer Center (CEVC) (<https://www.jmu.edu/cevc>) connects students and faculty with meaningful, community-based experiences to co-create just and sustainable communities. CEVC cultivates mutually beneficial partnerships between

students, faculty, and community organizations that address community-identified priorities and contribute to student learning.

Explore: If you're new to community engagement, short on time, or don't know where to start, drop by CEVC for guidance or sign up for our monthly newsletter of opportunities. To explore ways to make positive change in the greater Harrisonburg community:

- Participate in DMAD, a pre-orientation service experience,
- Create a Volunteer Now profile and browse opportunities,
- Volunteer for a one-time opportunity in the Shenandoah Valley, or
- Attend a community or educational event with CEVC.

Immerse: You can build relationships, deepen your civic understanding, and address pressing social issues through sustained engagement with a community initiative. To immerse yourself in community change:

- Join a weekly Dukes Volunteer opportunity,
- Participate in an Alternative Break,
- Take a Community Engaged Learning course, or
- Connect your club or organization to service.

Lead: Inspire your peers, develop career-readiness, and increase your ability to create a more just world. To lead peers on campus and in the community:

- Work as a CEVC Program Coordinator,
- Lead an Alternative Break,
- Facilitate DMAD for incoming Dukes, or
- Become Madison Community Scholar.

Whether you choose to engage in a one-time event, a semester-long project, or a leadership role, you can contribute to positive change in any of these priority areas:

- Ending hunger and homelessness
- Promoting health and well-being
- Protecting the environment
- Supporting youth and education
- Welcoming immigrants and refugees

Counseling Center

Location: Student Success Center, Room 3100, MSC 0801

Phone: (540) 568-6552

The Counseling Center (<http://www.jmu.edu/counselingctr>) provides free, confidential personal counseling services to all JMU students. Initial visits do not require an appointment as the Counseling Center offers walk-in hours for initial assessments Monday-Friday 10am-3pm.

JMU students experiencing a mental health emergency (e.g., thoughts or plans to kill yourself, thoughts of seriously harming others, recent sexual assault) can call or come into the Counseling Center during business hours to speak with a clinician. After regular business hours or on weekends, call the Counseling Center and select "Option 1" to connect to the after-hours crisis line.

Counseling services typically provided at the Counseling Center involve issues such as relationship problems, self-esteem, depression, anxiety/stress, eating and body image concerns, and difficulty adjusting to college life.

Treatment options provided by the Counseling Center include:

- **Specialized Workshops:** There are several specialized workshops to assist JMU students with a variety of goals, including: reducing anxiety symptoms, improving mood, increasing coping strategies, regulating emotions, managing distress, fostering healthy relationships and improving self-care. These programs include numerous evidence-based strategies and interventions. These free drop-in workshops are one of the quickest ways to reduce symptoms.
- **Group Counseling:** Group counseling is a great way to address a variety of issues within a clinician facilitated small group of participants who meet weekly to discuss their concerns. Groups typically focus on a specific experience, identity, or symptoms (e.g., DBT, Grief, Nature Nurture Ecotherapy, ReSET:Perfectionism, Understanding Self and Others).
- **Individual Counseling:** Individual counseling provides students with the opportunity to freely explore any personal problems or concerns which have a negative impact on the quality of their life. Due to the high demand for clinical services, individual counseling services at the Counseling center are typically brief, student generally meet with their clinician for approximately 3-5 sessions, depending upon treatment needs and availability. Students who request or require longer term treatment are provided with a referral to a clinician or other services in the local community.
- **Case Management Services:** The Case Managers provide services to students seeking care from the Counseling Center who are assessed as needing additional support due to symptom presentation, lack of resources/means, and elevated risk to themselves or others.
- **Sexual Trauma Empowerment Program (STEP):** The Counseling Center provides a variety of clinical, educational, consultation, and referral services for survivors of sexual trauma.
- **Psychiatric Services:** The Counseling Center Psychiatric staff are trained medical professionals who specialize in prescribing medications that may be helpful to students dealing with mental health concerns. Psychiatric services are only available to students who are currently engaged in ongoing treatment at the Counseling Center. Once students are stabilized on medication, they will be transferred to a community provider in Harrisonburg or back home. Students who are interested in only medication, need a refill, or medication management will be referred to a provider in the community.
- **Consultation:** The Counseling Center staff provide consultation, in person or over the phone, to students, faculty, staff, and others who are concerned about the distressed, unusual, problematic, or potentially harmful behavior of others.
- **Outreach Services:** Educational programming on a wide variety of mental health topics may be requested through the Counseling Center website.
- **Workshops:** The Counseling Center offers a variety of free workshops on topics such as assertiveness, self-care, and sleep hygiene to JMU students throughout the semester.
- **Equine Assisted Psychotherapy:** This non-traditional talk therapy approach is a hands-on, experiential process that helps empower students to find solutions to problems.
- **Wilderness Therapy:** These programs are a collaborative effort between the CC and UREC and use adventure and the outdoors as the platform to improve relational and health coping skills as well as to build self-confidence and self-efficacy.
- **Self-Care Spaces:** The Counseling Center offers two unique spaces for self-care, these are available to all JMU students; you do not need

to be a client of the Counseling Center to use these spaces. The Oasis is a great relaxation and stress management resource. The Studio is a great space to engage in expressive arts.

Dean of Students

Location: Student Success Center, Room 3010, MSC 3534

Phone: (540) 568-6468

The Dean of Students (<http://www.jmu.edu/deanofstudents>) team promotes well-being, interpersonal growth, and learning through advocacy and collaborative partnerships with students, employees, and families.

The office responds to the informational and personal concerns of students, parents, faculty and staff. Staff in the Dean of Students are committed to helping students and others in the areas of:

- Supporting and advising students and families during challenging situations through the university care program, Madison Cares.
- Normalizing setback and obstacles by sharing the stories of our community and getting students connected to a mentor-based Rebound program to create plans for success and to meet wellness goals.
- Notifying faculty of extended absences from class when a student is unable to attend due to extenuating circumstances and will be out for more than two days.
 - Such notifications are sent as a means to officially notify a student's professors of their situation. It is the responsibility of the student to contact each professor upon their return regarding the professor's policy of missed classes, assignments, etc.
- Referring students to resources to increase the likelihood of their success.
- Provide individualized student-centered life skills and learning strategies coaching through the Learning Success Strategies program.
- Consulting with faculty, staff and students regarding distressed and disruptive students; referrals to appropriate resources.

JMU Police Department

Location: Anthony-Seeger Hall, MSC 6810

Phone: (540) 568-6913

The James Madison University Police Department (<http://www.jmu.edu/police/>) consists of law enforcement and cadets. The office supports and advances the educational purposes of the university through the provision of a safe and secure environment for learning, working and personal development.

Police

The university police are commissioned Virginia officers with comprehensive law enforcement powers on JMU property and the City of Harrisonburg. University police continually patrol JMU's campus and facilities, providing full-service protection 24 hours a day, 7 days a week to the JMU campus community and surrounding areas.

The Campus Police Cadet Program carefully screens and trains students to aid the university police. Cadets are on duty from 7 p.m. until 2 a.m. during most weeknights and until 3 a.m. on weekends. Cadets are responsible for patrolling the campus and securing academic and

administrative buildings each evening. In addition, they can provide escort services to students walking across campus.

Office of Disability Services

Location: Student Success Center, Suite 1202

Email: disability-svcs@jmu.edu

Phone: (540) 568-6705

Fax: (540) 568-7099

ODS Website (<http://www.jmu.edu/ods>)

ODS collaborates with the JMU community providing programs and support services that promote and advocate for accessible learning environments that value disability. Services include:

- Equal access to university programs and services
- Provision and coordination of reasonable accommodations
- Disability-related support services

Disability services is comprised of the following areas:

Accessible Media & Technology

Email: alt-media@jmu.edu

Accessible Media and Technology (<http://www.jmu.edu/ods/accommodations/accessible-media>) provides course materials in alternative formats for students with qualifying disabilities, manages the accessible technology labs and supports university faculty and staff with designing accessible course materials. Services include:

- Textbooks and articles in alternative formats such as audio, large print, accessible PDFs and Braille
- Captioning for videos and recorded audio
- Educational programs on creating accessible course materials

Accessible Technology Computer Labs with specialized software and furniture are located in Rose Library (Room 1204) and the Student Success Center (Suite 1202).

Exam Accommodation Services

Email: odsexam@jmu.edu

Exam Accommodation Services (<https://www.jmu.edu/ods/accommodations/academic/testing/index.shtml>) offers exams in accessible media formats, proctoring services, a reduced distraction environment and extended exam time through the Office of Disability Services. Accommodations are provided within parameters agreed on between the professor, the student and ODS.

Housing Accommodation Request Committee

Email: specialhousingrequest@jmu.edu

The Housing Accommodation Request Committee (<https://www.jmu.edu/orl/housing/accommodations.shtml>) reviews requests for housing accommodations such as air conditioning and elevator access due to functional limitations stemming from disability or medical condition. Please check the website listed above for deadlines.

Screening & Referral Service

This office (<http://www.jmu.edu/ods/accommodations/screening-and-referral.shtml>) provides a two-hour clinical interview exploring a broad range of historical, academic, study habits/skills, academic frustrations and emotional components. A screening does not yield a diagnosis; rather, it assists in the decision-making process for recommending the next steps for the student.

Office of Equal Opportunity

Location: 1017 Harrison St., MSC 5802

Phone: (540) 568-6991

Email: oeo@jmu.edu

The Office of Equal Opportunity (<http://www.jmu.edu/oeo>) administers JMU's commitment to promoting equal opportunity, compliance, and adherence to relevant federal and state laws for the entire JMU community. JMU adheres to a strict policy that prohibits discrimination of any kind in JMU's hiring practices, admissions, academics or programs involving JMU employees or students. The Office of Equal Opportunity is a resource available to all university community members seeking guidance regarding the protections and rights provided in JMU Policies 1302 and 1324 and under state and federal law. JMU policy 1302 is the university's nondiscrimination policy and policy 1324 outlines the procedures for adjudication of allegations of protected characteristic discrimination, harassment and retaliation.

The responsibility for overall coordination, monitoring and information dissemination about JMU's program of equal opportunity, non-discrimination and work force analysis is assigned to the Office of Equal Opportunity. The OEO also monitors JMU's compliance with applicable federal and state laws assigned to the department.

Any university community member who believes they have been subjected to discrimination, harassment or retaliation due to age, color, disability, gender expression, gender identity, genetic information (including family medical history), marital status, military status (including veteran status), national origin (including ethnicity), parental status, political affiliation, pregnancy (including childbirth or related medical conditions), race, religion, sex, sexual orientation, or on any basis protected by law, unless otherwise permitted or required by law may file a complaint with the OEO. The OEO investigates complaints of discrimination, harassment and retaliation on the basis of these protected characteristics. The OEO provides consultation to assist individuals as they explore available options and resources related to non-discrimination policies at JMU.

Office of Orientation and Transition

Location: Student Success Center, Suite 2100, MSC 1010

Phone: (540) 568-1787

The Office of Orientation and Transition (<http://www.jmu.edu/orientation>) is committed to preparing undergraduate students for the transition to becoming a Duke, no matter if students are experiencing college for the first time or joining us from another institution. It all starts with the One Book Experience, followed by attending Orientation and then Weeks of Welcome to kick-off the semester. New students will be able to interact with their upper-class peers for support along the way including the Orientation Peer Advisers (OPAs), Orientation Operations Assistants

(OOAs) and First year Orientation Guides (FROGs). Our programs and services are designed to introduce new students to all that JMU offers.

Office of Residence Life

Location: Huffman Hall, MSC 2401

Phone: (540) 568-4663

The Office of Residence Life (<http://www.jmu.edu/orl>) oversees multiple functions related to living and learning at James Madison University. Offices and programs within the department focus on student learning, student development and successful transitions throughout the university experience.

Learning Initiatives focuses on the development of resident students by offering many services and programs specifically targeted toward this group. Community Development oversees the training and supervision of residence hall staff, develops policies and procedures for hall operations, and administers the delivery of programs and services designed to promote student success. Housing Operations manages all university-sponsored housing facilities, including the administration of contracts and room assignments. Housing Operations also oversees Conference Services. Business Operations coordinates all maintenance and housekeeping services and manages residence hall summer projects.

- Business Operations: (540) 568-7576
- Community Development: (540) 568-3501
- Director's Office: (540) 568-6275
- Learning Initiatives: (540) 568-3598
- Housing Operations: (540) 568-4663

Office of Student Accountability and Restorative Practices (OSARP)

Location: Student Success Center, Room 2122

Phone: (540) 568-6218

Email: osarp@jmu.edu

OSARP (<http://www.jmu.edu/osarp>) facilitates student development and restoration through a fair process that holds students accountable to community and university standards in order to support student and community success. Our office CARES about student, organizational, and community success through: Collaboration, Accountability, Restoration, Education & Support.

Office of the Registrar

Location: Student Success Center, Room 5300, MSC 3528

Phone: (540) 568-6281

The Office of the Registrar (<http://www.jmu.edu/registrar>) is responsible for:

- Athletic certification
- Class enrollment
- Class schedule preparation
- Academic Advisement Reports
- Diploma issuance
- Enrollment verifications
- Graduation
- Student records

- Transfer credit evaluation
- Transcript issuance
- Veterans Affairs

Pre-Professional Health Advising

Director: Sharon Babcock

Assistant Director: Katie Fisher

Phone: (540) 568-6652

Email: pph@jmu.edu

Pre-Professional Health Advising Website (<http://www.jmu.edu/pph>)

Pre-professional health (PPH) advising offers programs and services for undergraduate students who are preparing for professional health careers that require masters or doctoral education. Grounded in our mission to support the next generation of health professionals by committing to student centered development, PPH offers structured programs, academic guidance, and developmental advising to help students build competitive and well-informed pathways to graduate professional schools.

PPH minors outline the courses, activities, experiences and competencies associated with successful admission to graduate professional programs. PPH advisors work collaboratively with students to support academic planning, facilitate career-related exploration and informed decision-making, and guide development of competitive applications to graduate professional programs. Application advising and evaluation services are offered to current students and alumni.

Health profession programs use a holistic review, evaluating the breadth and depth of undergraduate preparation. PPH students are encouraged to choose a major aligned with their interests and strengths. Competitive applicants must demonstrate excellence in academic accomplishments, prerequisite completion, meaningful clinical and service experiences, career exploration, and preparation for standardized exams. Quality letters of recommendation are also a critical component of the application process, therefore PPH students are encouraged to establish mentoring relationships with faculty, clinicians and supervisors.

Incoming first-year and transfer students can declare a PPH program without departmental review. Requests from current students to declare a PPH minor are reviewed prior to assigning an advisor. For minors with specific admission standards, those standards must be met before a declaration request is approved.

PPH Minors

- Pre-Athletic Training (p. 545)
- Pre-Dentistry (p. 546)
- Pre-Medicine (p. 547)
- Pre-Occupational Therapy (p. 549)
- Pre-Optometry (p. 550)
- Pre-Pharmacy (p. 551)
- Pre-Physical Therapy (p. 553)
- Pre-Physician Assistant (p. 554)
- Pre-Veterinary Medicine (p. 557)

Student Affairs

Phone: (540) 568-3685

Student Affairs Website (<http://www.jmu.edu/studentaffairs>)

Mission

To engage all students in accessible and inclusive learning, growth, and personal development opportunities and experiences.

Student Government Association

Location: Madison Union, Room 331, JMU Box 8022

Phone: (540) 568-6376

The Student Government Association (<https://www.jmu.edu/sga/>) of James Madison University is an organization dedicated to collaborating with all members of its community to advocate for student opinion, while fostering a proactive, inclusive environment.

SGA promotes the welfare of students by providing the medium through which students can actively voice their concerns and by serving as a liaison between the students, faculty and administration at JMU.

Student Handbook

The Student Handbook (<http://www.jmu.edu/handbook>) contains all standards of conduct and policies for individual students, recognized student organizations and student groups; explains the accountability processes facilitated by the Office of Student Accountability & Restorative Practices; and outlines the processes to address allegations of hazing, sexual misconduct and Title IX sexual harassment. It is important to know that processes and policies can change; the online Student Handbook (<https://www.jmu.edu/osarp/handbook/index.shtml>) is the current version used by the Office of Student Accountability & Restorative Practices. It is your responsibility to know and follow federal, state and local laws as well as JMU policies.

Student Life

Fraternity & Sorority Life

Location: The Union 207, MSC 3508

Phone: (540) 568-4195

Fraternity & Sorority Life (<https://info.jmu.edu/fsl/>) works with social fraternities and sororities to foster cooperation and communication amongst the chapters, the university and the community. Staff members advise the InterFraternity and Panhellenic Councils. Staff members empower students and organizations to strive for qualities of belonging, responsibility, relationship and partnership building, leadership, learning and development, and civic engagement that define exceptional fraternity and sorority experiences.

Off-Campus Life

Location: The Union 205, MSC 4214

Phone: (540) 568-6071

Off-Campus Life (OCL) (<http://www.jmu.edu/ocl>) is here to assist students in gaining necessary knowledge and skills to make informed decisions within the off-campus community, to provide support and resources to off-campus students allowing them to maintain an academic focus and to inspire and empower students to be responsible citizens who create safe, values-oriented communities. Additionally, OCL operates The Pantries, which are free resources for students offering food, hygiene products, limited school supplies and more. The Pantries

are located in Taylor Down Under, Room 112, ENGO 0302, and Memorial Hall 2105. They are all open to all students, no questions asked.

OCL also administers the Utility Deposit Assistance Program (UDAP). UDAP is a JMU sponsored program that allows students to obtain utility connections without paying the up-front deposit costs. JMU becomes your guarantor for payments up to specified limits for water, electric and gas. This \$40 contract can be purchased through the student tab in MyMadison.

Student Leadership and Involvement

Location: The Union 320, MSC 3502
Phone: (540) 568-8157

Student Leadership and Involvement (<http://www.jmu.edu/osl/sli/index.shtml>) is home to student organizations, student leadership development, campus programming and the University Program Board. Student Leadership and Involvement provides a wide range of educational, social and wellness events, as well as support services for all JMU student organizations. The Belnvolved website is a platform for all student organizations, students and departments. Student Organization Night at the beginning of each semester offers an opportunity for all students to meet members of student organizations and discover how to become involved with them. Student leadership development offers programs like MYMOM, Pathfinders, Explore Madison, Reading and Leading, and Madison Leads. Students can take advantage of customized workshops to help people learn more about themselves, how to connect with others and how to build teams more effectively.

Facilities and Event Services

The Union, MSC 3506
Phone: (540) 568-5555

Student Success Center, MSC 1027
Phone: (540) 568-7747

Festival Conference and Student Center, MSC 4201
Phone: (540) 568-1715

Three buildings house the programs and services which Student Life provides for the JMU community. They are the Union and Grafton-Stovall Theatre, Student Success Center, and Festival Conference & Student Center. Facilities and Event Services address physical building, operations and information concerns in these spaces.

Festival Conference and Student Center

Location: Festival, MSC 4201
Phone: (540) 568-8932

Festival Conference and Student Center (<http://www.jmu.edu/festival>) provides gathering places for the campus, with meeting rooms, assembly spaces, lounges, support services and dining available. It provides a comprehensive approach to the coordination of services necessary for campus events and coordination of scheduled meetings and conferences throughout the year.

The Union and Grafton-Stovall Theatre

Location: The Union 245, MSC 3506
Phone: (540) 568-6330

The Union (<http://www.jmu.edu/theunion>) and Grafton-Stovall Theatre provide gathering places for the campus, with meeting rooms,

assembly spaces, lounges and support services available. It provides a comprehensive approach to the coordination of services necessary for campus events and coordination of scheduled meetings and conferences throughout the year.

Taylor Down Under

Location: The Union, 1st Floor
Phone: (540) 568-7853 or (540) 568-5555

Located on the ground floor of The Union, this huge space is comprised of a lounge, billiard tables, foosball, darts, board games, Merge Coffee Bar and programming space. The TDU lounge area is a popular hangout for all students. There is a computer lab for student use, wireless printing, evening entertainment on the TDU Stage, comfortable seating and an information desk.

University Program Board

Location: The Union 324, MSC 3505
Phone: (540) 568-8157

Many great ways to engage on campus are hosted through the University Program Board (UPB) (<https://info.jmu.edu/upb/>), a student organization advised through Student Leadership and Involvement. Concerts, films, speakers and numerous other activities designed to complement the educational mission of JMU are arranged by the UPB. The board also hosts collaborative programs with other student organizations and JMU departments. Students can join the UPB organization or attend events!

Student Success Center

Location: Student Success Center, Room 1010, MSC 1027
Phone: (540) 705-6081
Student Success Center Website (<http://www.jmu.edu/successcenter>)

At JMU, all students can be successful if they rise to the challenges they face as students, take full advantage of the many academic and student support services available to them on the campus, and participate deeply in engagement opportunities on campus and beyond.

Student success succinctly describes JMU's pervasive philosophy of collaborative, campus-wide efforts to coordinate and support services that: help students become more efficient, effective and engaged learners; address students' physical and emotional health; and provide advice and assistance for a variety of financial, curricular, technical and employment concerns.

The James Madison University Student Success Center houses several university departments, resources and functions that support student learning, student health and student services.

The Learning Centers

Location: Student Success Center, Suite 1100
Phone: (540) 568-1759

JMU's Learning Centers (<http://www.jmu.edu/learning>) support students, faculty and staff through the following programs and free services:

Communication Center

Location: Student Success Center, Room 1155
Phone: (540) 568-4818

The Communication Center (<http://www.jmu.edu/commcenter>) provides resources and assistance with digital and oral communication projects and promotes students' communication excellence through attention to process, innovation and audience-centered design. The center offers consultations, class workshops, and online resources for faculty and students across campus. The center provides the following services:

- Speech preparation assistance
- Assistance with speech outlines and research
- Speech anxiety reduction strategies
- Developing audience-centered presentations and visual aids
- Enhancement of speech delivery and style

English Language Consultations

Location: Student Success Center, Room 1155

Phone: (540) 568-2881

Website: (<http://www.jmu.edu/learning/multilingual-services>)

In addition to opportunities to work on academic skills in a cooperative environment, multilingual learners can seek consultation on such topics as:

- Reading, writing, listening and speaking
- Written or oral academic assignments from all disciplines, including theses and dissertations, as well as publications
- American academic culture

Peer Academic Coaching

Location: Student Success Center, Room 1144

Phone: (540) 568-4370

Academic success coaching (<https://www.jmu.edu/learning/aca.shtml>) is for students who want comprehensive guidance and resources to address all aspects of their academic success. Refer to the coaching website (<https://www.jmu.edu/learning/aca.shtml>) for more information and to schedule an appointment with a faculty member or peer coach.

Peer Assisted Study Sessions

Location: Student Success Center, Room 1141

Phone: (540) 568-4370

Peer Assisted Study Sessions (PASS) (<http://www.jmu.edu/pass>) help students successfully complete historically challenging courses. Students work together in regularly scheduled out-of-class study sessions that are facilitated by trained peer educators. These sessions are designed to help students master course content and develop their organizational, study and learning skills. Refer to the PASS website for a current list of supported courses.

Science & Math Learning Center

Location: Student Success Center, Room 1107

Phone: (540) 568-3379

The Science and Math Learning Center (SMLC) (<http://www.jmu.edu/smlc>) provides support to JMU students enrolled in first- and second-year science and mathematics courses by providing a secure, supportive learning environment that fosters independent thinking. The center provides a walk-in tutoring service by both faculty and trained peer tutors in the following subject areas: chemistry, mathematics, philosophy, physics and statistics. Refer to the SMLC website (<http://www.jmu.edu/>

[smlc](http://www.jmu.edu/smlc)) for a current list of supported courses. The SMLC is equipped with an educational computing lab.

University Writing Center

Location: Student Success Center, Room 1121

Phone: (540) 568-1759 or (540) 568-2986

The University Writing Center (<http://www.jmu.edu/uwc>) works directly with student and faculty writers, provides resources on writing strategies and supports writing across campus. The center provides the following services:

- Individualized writing consultations
- Writing space for Writing Center users
- Faculty consultations for designing assignments and responding to student writing
- In-class workshops on writing-related issues for any academic course or department
- Supports students working on digital assignments (formerly "DigiComm")

Title IX Office

Location: 4035 Holland Yates Hall, MSC 7806

100 E. Grace Street

Phone: (540) 568-5219

Email: titleix@jmu.edu

The Title IX Office (<https://www.jmu.edu/access-and-enrollment/title-ix/index.shtml>) participates in the coordination of JMU's compliance with Title IX of the Education Amendments of 1972 (20 U.S.C. § 1681 et seq.), university policies 1340, 1346, and 1324, and other applicable federal and state laws prohibiting discrimination on the basis of sex. JMU's prohibition of discrimination on the basis of sex includes dating and domestic violence, sexual assault, sexual exploitation, sexual harassment, stalking, and non-consensual relationships as defined in Policy 1340 and Policy 1346.

The Title IX Office is responsible for receiving and responding to reports of misconduct as defined in Policy 1340 and Policy 1346; offering supportive measures and information about confidential on- and off-campus resources for individuals involved in a report of misconduct; investigating formal complaints in a fair and impartial manner; and monitoring reports for patterns or systemic problems. The Title IX Office also provides support for students experiencing pregnancy and related conditions.

The responsibility for overall coordination, monitoring and information dissemination about JMU's Title IX program is assigned to the Title IX Coordinator. Inquiries or complaints may be directed to the Title IX Coordinator, Laura Sider Jost, via the Title IX Office website (<https://www.jmu.edu/access-and-enrollment/title-ix/index.shtml>), email (titleix@jmu.edu) or phone (540-568-5219).

University Advising

Location: Student Success Center 4045 MSC 1024

Phone: (540) 568-7350

University Advising Website: (<http://www.jmu.edu/advising>)

JMU students are responsible for the educational choices they make; however, they make these choices in a supportive environment. Effective

academic advising is a relationship between the student and their advisor. Students must be aware of their own interests, values, and goals; knowledgeable about relevant policies and requirements; and willing to take initiative in seeking assistance when needed. Our academic advisors respond to their students as individuals, are aware of their progress, and help them make connections between academic choices and career possibilities.

University Advising supports students from matriculation to graduation, assisting with their academic planning and contributing to their academic success. University Advising hosts the Advising Hub on the 4th floor of the Student Success Center for general advising questions, maintains the academic advising website for the university, and develops advising resources for students and faculty. University Advising hosts and supports the following programs:

First-Year Academic Advising

Academic advising is vital to student success. First-year advisors assist students in shaping their educational experience to meet specific intellectual, personal, and career goals. Advisors do not decide their students' goals or educational program but instead help them learn how to develop appropriate goals, make good choices that enhance learning and personal growth, and succeed in and out of the classroom.

During the summer of their first year, all new students are assigned to first-year advisors who discuss with them our programs of study, academic policies and procedures, incoming credit, and class enrollment procedures. These advisors continue to work with first-year students until midway through the spring semester, at which time those students with a declared major are assigned to advisors within their academic unit. Students who are still exploring majors are assigned to an exploratory academic advisor in University Advising. Students must declare their major by the beginning of their sophomore year. Please refer to Academic Policies and Procedures for information on declaring a major.

Academic Advising for New Transfer Students

University Advising coordinates advising and course registration for new transfer students matriculating to JMU in the fall and spring semesters. All new transfer students who have earned at least 30 transfer credits must declare a major upon entering the university and are advised by an academic advisor in their major. Advisors support new transfer students in reviewing transfer credit, enrolling in classes, developing academic goals, exploring career opportunities, connecting to campus resources, and understanding academic policies and procedures.

Exploratory Advising

University Advising assists students who are not sure which major they want to pursue or who wish to explore changing their program of study. Additionally, our academic advisors help students who are exploring the possibility of adding a second major/minor. Need help? Visit our website to learn more or make an appointment through MyMadison Connect (<https://jamesmadisonuniversity1.my.site.com/s/>), selecting "Other Advising Support".

Major Academic Advising

University Advising provides full-time academic advising support for select academic programs with large student enrollments, helping students stay on track with major requirements. Full-time academic

advisors hold joint appointments between the academic unit and University Advising.

International Study Center Advising

The international advisor helps international students make progress from the International Study Center to JMU by assisting them with General Education requirements, selection of a major, graduation requirements, and course registration.

Madison Advising Peers

Madison Advising Peers (MAPs) are JMU students who care about their fellow students and are committed to helping them achieve their college goals. MAPs work in conjunction with faculty advisors by providing supplemental academic advising information and assisting other JMU undergraduate students who may have general advising questions. A MAP can be a supportive and friendly resource for students who feel overwhelmed and confused. Visit the MAPs in SSC 4045 or email madisonadvisingpeers@jmu.edu with questions.

JMU Next Transfer Advising

The Transfer Initiatives office maintains online resources and provides academic advising to help prospective students understand transfer equivalencies and requirements for guaranteed admission. The three JMU Next Transfer Advisors can help any prospective transfer student from any college or university through virtual advising or in-person visits to Virginia community college campuses. They can also meet with current JMU students who have questions about transfer credit and coursework.

University Career Center

Location: Student Success Center, Room 3250, MSC 1016
Phone: (540) 568-6555

The University Career Center's (UCC) (<https://www.jmu.edu/career/>) mission is to Empower Students. Cultivate opportunities. Forge partnerships. Launch careers. This includes *all* class years and *every* major. Building upon your career knowledge and networking confidence far before it's time for you to begin job/internship searching will give you a competitive edge against other applicants, impress employers, and help to increase overall job satisfaction.

Whether you have your heart set on a career path, or have no idea what you want, UCC's trusted team of career advisors and staff help Dukes become comfortable and confident in their ability to shape their own future. UCC assists students with this through:

Career Exploration

- UNST 102: An elective 1-credit hour course which helps students investigate career options
- Career Guide to JMU Majors: An online resource that provides detailed information about majors and how they link to potential career options
- Decision-Making Assistance: Students can meet individually with career advisors to explore potential careers that match their strengths and interests.
- Career Exploration Assessments: UCC offers 2 assessments, FOCUS and the Strong Interest Inventory, to assist students in identifying

majors and careers of interest as they relate to the student's interests and values.

- Nurse phone line
- Athletic training

Finding a Job or Internship

- **Handshake:** Students can upload their resumes and apply for entry-level jobs and internships specifically curated for college students using this online resource.
- **Resume Writing:** The office provides resume workshops and a variety of resume review opportunities (including specialized 1:1 appointments).
- **Interview Practice:** The office offers practice interview events (PIE) in person or through our online platform, BigInterview. Employers also regularly visit campus and use virtual platforms, like Handshake and Zoom, to conduct interviews with JMU students for jobs and internships.
- **Networking Opportunities:** UCC hosts many networking events throughout the year to connect students with JMU alumni and employers in various fields through panels, socials, and more.
- **Career Fairs:** The JMU Career and Internship Fair (formerly the University Career Fair), and other career events, bring employers to campus each semester to provide information to students about job and internship opportunities. Several colleges and majors also partner with UCC to offer specialized career fairs for certain industries, including education, nursing, hospitality, sport and recreation management, integrated science and engineering, and business.

Preparing for Graduate School

- **Graduate School Resources:** Specialized 1:1 appointments. Available online; topics include financing graduate school, choosing a graduate school, applying to graduate school, etc.

University Health Center

Location: Student Success Center, MSC 7901

Phone: (540) 568-6178

Fax: (540) 568-6176

The University Health Center (<http://www.jmu.edu/healthcenter>) provides students with confidential medical care and learning opportunities so they can develop the knowledge and abilities to manage their health. The health center is open weekdays, 8am-4:30pm. Appointments can be made only through MyJMUChart, or by calling. Visits to the health center are partially covered by student fees. Some provider visits are \$5 and there are minimal charges for specialty services, such as lab tests and procedures. Students are responsible for payment at the time of service; the health center does not bill insurance. Services include:

- Allergy shots
- General medical appointments
- International travel consultations
- Lab services
- Care coordination services
- Pharmacy - prescriptions and over-the-counter items
- Nutrition counseling
- X-ray services
- Walk-in care for certain medical conditions that require prompt attention
- GYN care

New students at JMU are required by state law to submit specific information about immunizations. JMU does **not** require a physical exam, but you must comply with the immunization requirement and TB screening form. The Health Center has an interface with the Virginia Immunization System, so some immunizations may automatically come across into your record.

For students starting Fall semester health requirements are due by July 6. For students starting Spring semester, these are due by January 5. It is mandatory that you complete the following steps before attending classes:

1. Log in to your MyJMUChart Patient Portal (<https://www.jmu.edu/healthcenter/medicalservices/my-jmu-health.shtml>) and review the set of instructions for new students, including what tabs to click on and what to upload.
 - a. Check the status of your immunizations.
 - b. Complete the TB Assessment.
 - c. Upload your insurance information.
2. Check your JMU email and secure messages on MyJMUChart Patient Portal for confirmation that your record has been verified and is complete. Your pre-entry health requirements are not complete until you receive that confirmation. An enrollment hold impacting your ability to enroll in future semester classes and a fine will be placed on your account if your immunizations and TB Assessment survey are not deemed complete by Health Center staff.

University Recreation

Location: University Recreation Center, MSC 3901

Phone: (540) 568-8732

The University Recreation (UREC) (<http://www.jmu.edu/recreation>) team is here to motivate and support students to live well-balanced and active lives. UREC is located centrally on campus and JMU's outdoor recreation facility, University Park (UPARK), is located adjacent to the JMU Arboretum. Students only need their JACard or UREC App to enter these facilities, as membership is included with student fees.

UREC is a vast recreation center with gyms, courts, tracks, aquatic areas and fitness spaces, as well as an equipment center that provides sporting equipment and locker and towel services. UREC also features climbing and bouldering walls, a demonstration kitchen, a café, massage therapy rooms, a meditation studio, personal training, fitness assessment, nutrition counseling, small group training, bike repair and athletic training. UPARK has an open event lawn, tennis, sand volleyball and basketball courts, a sports turf, pavilion, disc golf course, and TEAM Challenge Course.

UREC provides programs and services that promote healthy lifestyles while providing opportunities to develop fitness and wellness habits in the areas of adventure, aquatics, fitness, group exercise, intramural sports, sports clubs, nutrition, safety and wellness.

Online registration (<https://URECregister.jmu.edu>) is available and programs with fees can also be registered for in-person at UREC using FLEX. Students can host immediate family members of all ages for free during family hours, which are between 6pm Friday and 6pm Sunday.

Students can also purchase up to two passes for guests ages 16 and up during all operational hours.

Victim Advocacy Services

Location: Student Success Center, Suite 3200, MSC 0801

Phone: (540) 568-6251

Email: victimadvocacy@jmu.edu

JMU Victim Advocacy (<http://www.jmu.edu/victimadvocacy>) services act as a safe, confidential, welcoming and inclusive resource for JMU students who have experienced sexual violence, relationship abuse, or stalking, whether recently or in the past. We also serve family, friends, and other support persons to student survivors. We welcome all who may have questions, who need support and guidance, and who are seeking resources and connection. Whatever you decide is right for you, Victim Advocates are here to help.

To connect with an advocate, visit, call, or email us (victimadvocacy@jmu.edu).

Advocates are responsive to email and phone during our regular business hours, Monday-Friday, 8am–4:30pm. Our advocates respond as soon as possible during these business hours and no later than one business day from the time you initiate contact.

Victim Advocacy services are free, trauma-informed and guided by and individualized to your needs. Our advocates collaborate with multiple offices across campus and in the community and can explain the role of a variety of options for support. Commonly, advocacy conversations may include information about or accompaniment to:

- Time-sensitive health and safety resources
- Criminal reporting options
- Campus reporting options
- Protective orders
- Medical care
- Counseling services
- Academic, housing and other accommodations through Title IX
- Other resources specific to your needs

You are welcome to bring a friend or support person along for your visit. Victim Advocates are also available to meet with students or faculty who are seeking help supporting a survivor.

Environmental Programs

At JMU, environmental study (<http://www.jmu.edu/environment>) is embedded in a variety of disciplines. Students from throughout the university have the opportunity to study environmental issues by majoring in an environmental program or choosing a minor that complements any major.

Majors and Concentrations

Students wishing to major in a field that addresses the environment can choose from the following:

- Biology, B.S. (p. 204): The biology major offers a concentration in ecology and environmental biology.
- Geology, Environmental Earth Science Major, B.A. (p. 283) and B.S. (p. 285): This major is designed to integrate all the Earth sciences in a systems approach to understanding the Earth.
- Economics, B.A. (p. 248), Economics, B.B.A. (p. 251), and Economics, B.S. (p. 255): Students can pursue a concentration in environmental and natural resource economics within the economics major.
- Engineering, B.S. (p. 261): The JMU engineering program empowers and motivates students to engineer systems for sustainable societies.
- Geography, B.S.: (p. 275) The geography science major, housed in the School of Integrated Sciences (p. 70), offers a concentration in environmental conservation, sustainability and development.
- Geology, B.S. (p. 281): Within the geology major, students can select a concentration in environmental and engineering geology.
- Integrated Science and Technology, B.S. (p. 321): The ISAT major offers a concentration in environment.

Minors

Any of the environment minors may be taken in conjunction with any STEM (Science, Technology, Engineering and Mathematics) area major. They also may be taken in conjunction with a major from any other JMU academic area in order to explore environmental issues from differing perspectives. The minor coordinators can assist a student in selecting a combination of major and minor study that best responds to a student's particular environmental interests and career goals.

Students should be aware that most minors have prerequisites, meaning that certain courses must be completed before a student can enroll in other courses. Consult with the minor advisor for additional information and recommendations for scheduling.

- Climate Science (p. 466): The Climate Science Minor introduces students to the domain knowledge and analytical skills necessary for understanding and investigating Earth's climate system. The program serves as an organizing framework for students interested in climate science and will broaden disciplinary expertise for advanced study, graduate programs and/or employment.
- Environmental Humanities (p. 485): This cross disciplinary minor allows students to articulate broad historical and cultural views of the complex interactions among humans and the natural and material environment. Students will analyze and produce expressive, creative work that illustrates the complexity of human interaction with the environment.
- Environmental Information Systems (p. 487): The cross disciplinary minor in environmental information systems is designed for

undergraduates interested in using computer and information management technology to solve environmental problems and improve environmental stewardship.

- Environmental Management (p. 488): The cross disciplinary environmental management minor prepares students to apply the principles of environmental science and engineering to contemporary environmental problems in natural resource, industrial and public policy contexts. The minor is particularly suitable for students interested in professional careers in business, natural resources management, and environmental policy and planning.
- Environmental Science (p. 490): The cross disciplinary environmental science minor draws from courses that focus on the application of scientific concepts and principles to the understanding of environmental problems and their solutions. This is a multidisciplinary program that can be elected by any student. For example, students pursuing programs ranging from the physical, natural or social sciences, to education, journalism or business, all benefit from this broadly based environmental curriculum.
- Environmental Studies (p. 492): The cross disciplinary environmental studies minor provides a cross disciplinary education engaging socio-cultural, scientific and technical issues raised by the oft-conflicting needs and desires of globally interacting societies.

Centers

Center for Materials Science

Director: Chris Hughes

Email: hugheswc@jmu.edu

Location: 1109 EnGEO, 801 Carrier Drive, MSC 4113

The Center for Materials Science at James Madison University was established in 1996 as a resource to integrate undergraduate education with basic and applied research in materials science. The center is a cross disciplinary initiative of the College of Integrated Science and Engineering (p. 24), the College of Science and Mathematics (p. 26), and the Department of Engineering (p. 57). Faculty in five different departments participate in the center. The center provides students the opportunity to develop broad cross disciplinary skills and knowledge in the science of materials.

Center for Health and Environmental Communication

Director: Peter Bsumek

Email: bsumekpk@jmu.edu

Director: Sharlene Thompson

Email: thompssr@jmu.edu

The Center for Healthmental Communication (CHEC) (<http://www.jmu.edu/chec/>) is housed in the School of Communication Studies. The CHEC utilizes the expertise of the school's faculty and students to facilitate original academic research, coordinate community outreach and advocacy projects, and serve as a community resource on matters related to understanding and improving public and private communication in the health and environmental contexts.

The CHEC recognizes that how we communicate is as important as what we communicate. Central to its mission is facilitating constructive and productive communication between experts and non-experts, between clients and providers, between communities, and between community members and decision makers.

JMU Farm

Faculty Sponsor: Carole Nash

Phone: (540) 568-6805

Email: nashcl@jmu.edu

The JMU Farm, located in Port Republic, consists of 12 hectares (30 acres) of second growth forest, 300 meters (1000 feet) of frontage on the North River (just upstream of the formation of the Shenandoah River) and an historic brick house. This facility is intended to:

provide an ideal space in which to disseminate environmental information to school students and community groups.

provide an inviting off-campus venue for workshops, meetings and symposia.

The facility can be reserved for JMU-related events by contacting the Office of Academic Resources at (540) 568-3744.

Research and Outreach Programs

Collaboration for Environment, Health and Safety

Website (<http://www.jmu.edu/cehs>)

Office of Environmental Stewardship and Sustainability

Website (<http://www.jmu.edu/stewardship>)

Center for the Advancement of Sustainable Energy

Director: Jonathan Miles

Phone: (540) 568-3044

Email: milesjj@jmu.edu

The Center for the Advancement of Sustainable Energy (CASE) focuses on research, education, and outreach to advance sustainable energy deployment by providing energy-related services to a wide variety of stakeholders including local governments, state agencies, landowners, non-governmental organizations, businesses, and academic communities.

Virginia Clean Cities

Director: Alleyn Harned

Phone: (540) 568-8896

Email: harnedas@jmu.edu

Virginia Clean Cities advances air quality improvement, economic opportunity, and energy security through deployment of alternative fuel vehicles and infrastructure, education programs, and other petroleum reduction activities.

Institute for Innovation in Health and Human Services

Director: Linda Plitt Donaldson

Phone: (540) 568-2642

Email: plittldx@jmu.edu

Location: 2031 Deyerle Ave.

Mission

IIHHS (<https://www.jmu.edu/iihhs/index.shtml>) is an interdisciplinary community-engaged hub that brings together students, staff, faculty, and the community to advance a culture of health and equity through education, clinical practice, research, innovative programs and advocacy.

The following programs, clinics, and activities connect our campus with communities while advancing scholarship and achieving exceptional learning experiences.

Alvin V. Baird Attention and Learning Disabilities Center

Director: Trevor Stokes, stokestf@jmu.edu

The mission of the Alvin V. Baird Attention and Learning Disabilities Center is to develop and promote evidence-based interventions for children and adolescents with attention and learning challenges, while educating families, teachers, students and professionals about best practices for their assessment and treatment. Attention and learning factors are present across a range of developmental and psychological disorders, which are the focus of activities at the Baird Center. The Baird Center operates two clinics:

- The Applied Behavior Analysis Parent-Child Interaction Clinic (ABAP) provides assessment and treatment intervention services as well as parent consultation, training and support for children with behavioral and neurodevelopmental disabilities such as autism, intellectual disability, attention deficit hyperactivity, oppositional defiant behavior, brain injury and cerebral palsy. The clinic is staffed by JMU faculty licensed as Behavior Analysts and Clinical Psychologists who work directly with children and families and also supervise students working in the clinic programs.
- The Interprofessional Autism Clinic (IPAC) provides in-depth assessment and multi-disciplinary intervention for children ages 2-10 years old with Autism Spectrum Disorder or suspected Autism. The clinic is staffed with a licensed occupational therapist, a licensed speech and language pathologist, a licensed behavior analyst and clinical psychologist, in addition to graduate and undergraduate students enrolled in each of these academic programs. The clinic provides treatment as a simultaneous merge of methods from these disciplines while approaches are individualized to meet each child's unique needs.

Audiology Clinic

Director: Melissa Garber, garbermm@jmu.edu

The James Madison University Audiology Clinic (JMUAC) is part of the university's teaching, research and community service programs.

The JMUAC, within the Department of Communication Sciences and Disorders and supported by the Institute for Innovation in Health and Human Services, provides quality hearing healthcare and education to children and adults from the JMU community and surrounding areas.

Services are provided by graduate students under the direction of licensed and certified Clinical Doctors of Audiology.

Campus Suicide Prevention Center of Virginia

Director: Danette Gibbs, gibbsdc@jmu.edu (gibbsdc@jmu.edu)

The goal of the Campus Suicide Prevention Center of Virginia is to reduce risk for suicide on Virginia's college and university campuses. Specifically, this program supports the individuals and teams on each campus as they work to build the infrastructure necessary to promote mental health for all students, identify and support those with mental health concerns, and effectively respond to individuals who are at risk for suicide.

Caregivers Community Network

Director: Kathy Guisewite, guisewkf@jmu.edu

In partnership with the Valley Program for Aging Services (VPAS), Community Network (CCN) provides services, companionship and support for those who care for frail older family members. CCN also provides services for those with memory loss or Alzheimer's disease. CCN can help to give caregivers a break and provide valuable time for attention to family concerns. Services are evidence-based and customized for each individual.

Counseling and Psychological Services

Director: Kelly Atwood, atwoodkc@jmu.edu

Counseling and Psychological Services (CAPS) is a teaching, research and service mental health clinic. CAPS offers affordable outpatient mental health services to the Harrisonburg and Rockingham County community while providing students in the Department of Graduate Psychology experience in assessment and treatment of psychological problems. CAPS is equipped to provide individual, couple and family therapy as well as to conduct intellectual and psychological assessments across the lifespan.

Brain Injury Connections

Director: Anne Taetzsch Fitzgerald, taetzsch@jmu.edu

Brain Injury Connections, a partnership program, implements a multi-year region-wide grant to provide case management and supportive services to families and individuals recovering from brain injuries. It assists with access to services for re-learning daily living skills and with strategies for returning to school or work.

Community Health Worker Training and Technical Assistance Program

Associate Director, Community Health Education: Kati Derrick, derrickck@jmu.edu

The CHW Training and TA Center empowers individuals to promote community health through comprehensive training, technical support, and mentorship, equipping them to address health disparities and connect community members to resources.

Healthcare for the Homeless Suitcase Clinic

Faculty Contact: Erika Metzler Sawin, sawinem@jmu.edu and Kymber Beers, beers2kn@jmu.edu

The Healthcare for the Homeless Suitcase Clinic provides health care at the point of contact in local shelters and human service agencies. Specifically, it provides on-site primary care services to homeless clients enrolled in local shelters while simultaneously providing case management services as needed. Staffed by a nurse practitioner and nurse case manager, this new model of healthcare delivery to homeless populations exemplifies a model of service that aims to break the cycle of homelessness by providing healthcare delivery with positive long term results.

Health Education Design Group

Director: Suzanne Grossman, grossmsx@jmu.edu

The Health Education Design Group (HEDG) produces educational media in partnership with Virginia agencies and JMU programs. HEDG brings together expert research and a skilled media production team to develop creative video and online educational and training materials for health-related projects. It also works with programs and agencies to plan and coordinate conferences and other live training sessions.

Healthy Families Page County (HFPC) & Healthy Families Shenandoah County (HFSC)

Program Director: JoLynne Bartley, bartl2jx@jmu.edu

Program Manager, Page County: Mandy Duley, duleyad@jmu.edu

Program Manager, Shenandoah County: Brittany Silvestri, silvesba@jmu.edu

Healthy Families programs are based on a national program model and the two that operate in IIHHS provide education, resources and support for the most vulnerable first-time parents in Page and Shenandoah Counties through voluntary home visiting services.

Gus Bus Early Learning Program

Program Director: Jolynne Bartley, bartl2jx@jmu.edu

Program Coordinator: Rachel Gagliardi, gagliarl@jmu.edu

This initiative is a mobile literacy program serving low-income neighborhoods and day care centers in Harrisonburg, Rockingham County and Page County with a customized vehicle that provides a book exchange program, story time, nutritional support and resource referral information for families in need.

Interprofessional Services for Learning Assessment

Director: Kelly Atwood, atwoodkc@jmu.edu

The ISLA program offers diagnostic evaluation, consultation and support services for adults enrolled in a college or university. Evaluation teams consisting of professionals from clinical psychology, communication sciences, clinical neuropsychology, special education and nursing identify

areas of need and design positive ways to promote meaningful learning and educational experiences.

Interprofessional and Interdisciplinary Education

Director: Vesna Hart, hartvx@jmu.edu

IIHHS serves a vital role in interprofessional and interdisciplinary education through its workshops and IPE courses that provide critical interprofessional and interdisciplinary learning opportunities for our students.

Occupational Therapy Clinical Education Services

Director: Heidi Hull, hullhr@jmu.edu

A pediatric occupational therapy practice established to address a significant gap in locally available occupational therapy services, this clinic offers teaching, research, practice and service opportunities, and expands the interprofessional education and practice capacities of the university.

Page County Integrated PCBH Program

Director: Kelly Atwood, atwoodkc@jmu.edu

The Page county Integrated Primary Care and Behavioral Health Program is a partnership with IIHHS and Page Memorial Hospital (PMH) to integrate behavioral health and primary care services in a Rural Health clinic in Page County. This program provides interprofessional student education and training, research opportunities, and direct patient care using the integrated model. Behavioral health services include psychological testing and assessment, neuropsychological screening and testing, individual counseling and consultation with medical providers.

Precious Time Program

Faculty Director: Julianne Secrist, secrisjj@jmu.edu

Program Director: Ciana Graham-Shelton, graha2cx@jmu.edu

The Claude Moore Foundation awarded this program grant funds to provide respite care to families who have special needs children. This respite program relies on students from nursing, social work, psychology, and other health and human service majors to provide caregivers with supportive assistance in meeting the demands of their family responsibilities.

Rural Engagement and Capacity Building Hub

Program Director: JoLynne Bartley, bartl2jx@jmu.edu

The REACH vision is to facilitate rural-led partnerships to promote community well-being. The purpose of REACH is to collaborate with rural communities, addressing their identified barriers and inequities in order to improve health and overall quality of life. REACH efforts will build on established projects and partnerships involving JMU faculty and staff, and over time, REACH will serve as a connection point and resource for community organizations, universities and others across the state to address the unique barriers to improving health and overall quality of life in rural communities.

Shenandoah Valley Child Development Clinic

Clinic Director: Ginger Griffin, griffigl@jmu.edu

The Shenandoah Valley Child Development Clinic (CDC) provides individualized, cross disciplinary evaluations that may include medical, social work, nursing, educational, psychological, speech/language and audiology components depending upon the specialized needs of the child/adolescent.

Children/adolescents evaluated may have developmental, educational, emotional or behavioral concerns. The CDC provides care coordination services and assists families in decisions that address their developmental, educational, emotional or behavioral concerns. Services are provided on a sliding fee scale and Medicaid/FAMIS are accepted. Training opportunities are available in the CDC for students from a variety of disciplines.

Shenandoah Valley Migrant Education Program

Program Director: Jolynne Bartley, bartl2jx@jmu.edu

Program Coordinator: Nancy Resendiz, resendnx@jmu.edu

The Migrant Education Program provides free, supplemental education services to children and youth aged 3-21 of migrant and highly mobile agricultural workers. Services include tutoring/mentoring, school readiness initiatives, dropout prevention activities, educational interpretations (Spanish/English) and facilitation of families' stabilization in the community. The SVMEP serves as a point of contact for the Hispanic Services Council, a networking organization of agencies interested in the Latino population.

Speech-Language-Hearing Applied Laboratory

Director: Ingrid Hinkley, kmklh4@jmu.edu

The JMU Speech-Language-Hearing Applied Laboratory, formerly referred to as the JMU Speech and Hearing Center, provides communication evaluation and treatment services to individuals with known or suspected speech and/or hearing impairments.

This center assists residents of the Shenandoah Valley ranging in age from infants to senior citizens. Hearing testing and aid advising is available for those with concerns regarding hearing. Evaluation and treatment of communication impairments, including speech, sound disorders, language impairments, voice disorders and stuttering problems, are additional services offered in the applied laboratory.

Graduate students supervised by faculty who are licensed audiologists or speech-language pathologists serve as clinicians in this lab.

Sexual Health Education Virginia (SexEdVA)

Director: Kati Derrick, derrickc@jmu.edu

The SexEdVA initiative is a leader in supporting adolescent sexual health through training and technical assistance, curriculum development,

research and advocacy across the Commonwealth of Virginia. Following are SexEdVA programs housed at the Institute:

- **Personal Responsibility Education Program (PREP)** is a broad-based and collaborative effort that emphasizes both abstinence and contraception and addresses the adulthood preparation subjects of healthy relationships, parent-child communication, educational and career success, healthy life skills, and adolescent development in order to significantly reduce the number of teen pregnancies in Harrisonburg, Rockingham and Page counties of Virginia.
- **Sexual Risk Avoidance Education Program (SRAE)** implements strategies to build protective factors that promote healthy behaviors among young people ages 10-19 living in communities with disproportionately high rates of teen pregnancy.
- **Vision of You (VOY)** is an interactive, self-paced online sexuality education program that uses engaging video, animation, interactive components and gamification principles to provide information, skills and change sexual health behavior.

The Health Place

Director: JoLynne Bartley, bartl2jx@jmu.edu

The Health Place, a satellite of the Institute for Innovation in Health and Human Services, promotes collaborative and interprofessional health and human services that are affordable, accessible, responsive to and advance the health needs of Page County residents.

Waynesboro Outreach Program (Promotores de Salud)

Director: Kati Derrick, derrickc@jmu.edu

The Waynesboro Outreach Program provides health promotion and education materials to raise awareness among Spanish speakers living in the Shenandoah River area about the impact of mercury found in the South River and South Fork Shenandoah Rivers and to provide education on how to reduce the risks associated with fish consumption.

Youth Data Survey

Director: Suzanne Grossman, grossmx@jmu.edu (grossmsx@jmu.edu)

IIHHS has been working with the community since 1996 to provide key data about local youth risk behaviors and produce information that enables the community to develop priority areas around youth issues. Every two years in the fall the survey is administered to half of all 8th, 10th and 12th graders in Harrisonburg City and Rockingham County and a report is published in the summer following survey administration.

Mahatma Gandhi Center for Global Nonviolence

Director: Dr. Taimi Castle (castletl@jmu.edu)

Phone: (540) 568-5929

Location: Darcus Johnson Hall 1005, MSC 1205

Email: GandhiCenter@jmu.edu

Website (<https://www.jmu.edu/gandhicenter/index.shtml>)

Mission

Our mission is to promote justice through education, scholarship and service.

Leadership

The Gandhi Center is led by a director who reports to the academic unit head of the Department of Justice Studies (p. 75). The center is under the College of Arts and Letters (p. 13) and is supported by an advisory committee. The advisory committee is composed of faculty and community members who assist with fundraising efforts.

Teaching

The Gandhi Center brings students and faculty together for discussion, learning and outreach across disciplines, engaging faculty and students in the justice projects of the center. Faculty and community members support the educational and engagement missions of the center by providing regular workshops and seminars designed to advance the study and understanding of justice. The center works closely with groups across campus with similar goals and assists in sponsoring lectures during the annual Beitzel Lecture Series and Symposium. JUSTPEACE is a week of educational programming that began in 2022 and features presentations and workshops on a variety of justice topics.

Research

In 2021, the permanent site for the Justice Research Lab was established in the Gandhi Center suite (DJH 1005A). The lab is a collaborative research space where interdisciplinary scholars research justice into action. The lab is home to the peer-reviewed, scholarly journal, the *International Journal on Responsibility*. The Gandhi Center also supports an annual research fellowship.

Student Interns

The Gandhi Center facilitates and supports a variety of initiatives. Over the years, student interns have implemented and organized events and programs such as gandhiFEST, Global Nonviolence Peace Camp for Children and Puppies for Peace. Student interns also manage marketing and social media for the center. Service with the Gandhi Center includes a diverse and varied set of activities focusing on both local and global community needs.

Volunteers and interns can join with existing programs or design their own program/project.

The Gandhi Award/Service Award

The Mahatma Gandhi Global Nonviolence Award is bestowed upon individuals with global recognition who believe humans everywhere are

to be peacemakers, support nonviolence, love their enemies, seek justice, share their possessions with those in need, and express and demonstrate these beliefs in their words, life and actions. The inaugural award was given in 2007 to the Most Reverend Desmond Tutu, Archbishop Emeritus of Cape Town, South Africa and 1984 Nobel Peace Laureate. The second award was given in 2009 jointly to former U.S. President and 2002 Nobel Peace Laureate Jimmy Carter and former first lady Rosalynn Carter.

The Service Award is presented to a local community member who advances nonviolence and social justice. In 2013 Dr. Vida Huber and Dr. Howard Zehr (Restorative Justice) were the inaugural recipients. Dr. Calvin Redekop and PAX received the award in 2015.

The inaugural Mahatma Gandhi Center for National Nonviolence Award was bestowed on the Honorable John Lewis in 2021. David Hogg and March for Our Lives received the award in 2022.

The Gandhi Statue

The government of India presented a larger-than-life-size bronze statue of Mahatma Gandhi as a gift to the university in recognition of the work of the Gandhi Center. The statue was dedicated and unveiled on October 2, 2008, the International Day of Nonviolence and the birth anniversary of the Mahatma, by His Excellency Ronen Sen, Ambassador of India to the United States of America. The statue, which is located on the ground floor of JMU's Rose Library, was the first of Mahatma Gandhi in the Commonwealth of Virginia.

Nelson Institute for International and Public Affairs

Director: Tara Parsons

Phone: (540) 568-7645

Location: Darcus Johnson Hall, Room 2009, MSC 1205

Email: parsonstn@jmu.edu

Mission

Nelson Institute for International and Public Affairs (<http://www.jmu.edu/nelsoninstitute>) is a concrete manifestation of James Madison University's commitment to meet both the educational requirements of its students in a changing world and its own need to respond to the "real world" challenges faced by society today.

Goals

The institute advances several university priorities:

- Offering quality academic programs.
- Being innovative in developing programs.
- Developing optimal student competencies in written and oral communication, critical thinking, information systems, and quantitative literacy.

Activities

The Nelson Institute Seminars provide justice studies majors and other interested JMU students the opportunity to engage in directed, practical, problem-solving exercises in the field of justice studies in small group ("task force") settings under the supervision of a faculty member.

The seminars provide a space where students are charged with finding a solution to a contemporary policy problem. They offer a unique opportunity for students to engage in policy-oriented research that integrates classroom instruction, out-of-class group learning activities, and civic engagement and service learning opportunities. The institute also provides students with the opportunity to present their findings to a regional or national audience.

As a complement to the seminars, the Nelson Institute brings guest speakers to JMU who are experts in the subject of the seminar to enhance the learning experience. Funding is made available to help students present their research at regional and national conferences as well as to travel to talk with government, NGO and private sector officials about the policy problem they are working on. Students present their results at a variety of venues including the College of Arts and Letters' (p. 13) student conference, Mad Rush, the conference organized by the Department of World Languages and Cultures (p. 104), and regional conferences such as the Virginia Social Science Association meeting.

Accounting, B.B.A.

Program Description

The Bachelor of Business Administration degree in accounting requires a minimum of 120 credit hours of undergraduate course work.

Certifications

Many graduates desire to gain certifications in their selected area of specialization. The Certified Public Accountant (CPA) is the best known of these certifications. To obtain a CPA license in Virginia, candidates must meet a 150-hour education requirement, including 24 credits in accounting and 24 in business. The masters in accounting program is highly recommended for those electing a career in public accounting.

In addition to the CPA exam, graduates of our program sit for the Certified Management Accountant (CMA) exam, the Certified Internal Auditor (CIA) exam, the Certified Information Systems Auditor (CISA) exam and others.

Transfer Credit

All upper-division accounting course work (<https://catalog.jmu.edu/courses/actg/>) (300-499) must be completed at JMU.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business (p. 15) catalog page.

Accounting Major Progression Standards

Accounting majors are required to earn a minimum grade of "B-" in both COB 241 and COB 242.

Students are required to earn a minimum grade of "C-" in ACTG 343 before continuing to ACTG 344.

No required 300- or 400-level accounting major course may be attempted more than two times. An "attempted course" is one in which a final letter grade, including "W", "WP" and "WF" is recorded on the student's official JMU transcript.

Degree Requirements

Degree Requirements	Credit Hours
General Education requirement	41
B.B.A. core requirement	45-46
Major requirements	24
University electives	9-10
Total	120

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		

COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	
MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28
B.B.A. Upper-Level Core Component		
The following courses comprise the B.B.A. Upper-Level Core Component:		
COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3
COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.
² COB 241 must be taken before COB 242.
³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.
⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.
⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.
⁶ All four COB 300 courses must be taken during the same semester.
⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.
⁸ COB 487 must be taken during the senior year.

Major Requirements

To receive a B.B.A. degree in accounting, students must have a minimum cumulative grade point average of 2.0 in the major. Major courses are those with an ACTG prefix.

Code	Title	Credits
Required Courses		
ACTG 305	Spreadsheet Skills and Data Analysis for Accounting Information	3
ACTG 310	Data Analytics for Accountants	3
ACTG 313	Accounting Information Systems	3
ACTG 343	Corporate Financial Reporting I	3
ACTG 344	Corporate Financial Reporting II	3
ACTG 377	Federal Income Tax Accounting	3
ACTG 410	Auditing and Assurance	3
Subtotal		21

Electives		
Choose one of the following:		3
ACTG 413	Cybersecurity and Accounting Information Protection Strategies	
ACTG 475	Strategic Cost Accounting and Analytics	
ACTG 491	Advanced Data Analytics and Emerging Technologies in Accounting and Forensics	
Subtotal		3
Total Credits		24

ACTG 344	Corporate Financial Reporting II	3
Credits		6
Second Semester		
ACTG 410	Auditing and Assurance	3
ACTG 475	Strategic Cost Accounting and Analytics	3
COB 487	Strategic Management	3
Any remaining General Education or electives required for graduation; consult your advisor.		
Credits		9
Total Credits		15

Recommended Schedule for Majors

First Two Years

Students planning to major in accounting must be accepted into the College of Business. Admission requirements to the College of Business require completion of 9 lower-level B.B.A. core courses. It is expected that the lower-level B.B.A. core curriculum will be completed during the first two years of study along with General Education and elective courses. Students will work with an advisor to develop their schedule of courses for the first two years. It is recommended that accounting majors complete ACTG 305 in the spring of their sophomore year.

Code	Title	Credits
ACTG 305	Spreadsheet Skills and Data Analysis for Accounting Information	3

Third and Fourth Years

Students accepted into the College of Business will begin major courses in their junior year.

Third Year

Course	Title	Credits
First Semester		
ACTG 310	Data Analytics for Accountants	3
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
Credits		15
Second Semester		
ACTG 343	Corporate Financial Reporting I	3
ACTG 313 or ACTG 377	Accounting Information Systems or Federal Income Tax Accounting	3
COB 318	Legal and Ethical Environment of Business	3
General Education or electives		9
Credits		18
Total Credits		33

Fourth Year

Up to seven hours of core requirements in economics and calculus may also be taken for General Education credit. Students who take the General Education packages and courses recommended by the College of Business will have only 39 credit hours of additional B.B.A. core requirements.

Course	Title	Credits
First Semester		
Any 300-level accounting course not completed in the third year.		
ACTG 313 or ACTG 377	Accounting Information Systems or Federal Income Tax Accounting	3

Anthropology, B.A.

Program Description

Anthropology's goal is to document the diversity of ways humans live and have lived throughout the world. Anthropologists go "off road," working directly with people and their artifacts, to examine their beliefs and values, how they make a living, how they express themselves, how they interact, and how they affect and are affected by the natural world.

Why is this important? An increasingly globalized and multicultural world requires anthropological knowledge and insight. Anthropology is essential to students interested in working in both the U.S. and overseas on contemporary issues in areas such as social policy work, international development, public health, community-based advocacy, environmental and social justice, and cultural preservation.

Students develop their interests by taking a wide range of classes in cultural, linguistic and biological anthropology and archaeology. Anthropology majors can also do individually-driven, practical and hands-on research in internships, anthropology labs, excavations and field-schools, and study-abroad programs.

Degree Requirements

Degree Requirements	Credit Hours
General Education	41
Foreign language requirement (intermediate level required)	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	41
University electives	29-41
Total	120

Major Requirements

To earn a B.A. or B.S. degree in anthropology, students complete 41 credit hours in the major. Given the diverse opportunities the discipline provides, the major is designed to allow students the opportunity to work closely with their advisors to develop a curriculum appropriate to their personal and professional interests. Those students wishing to do so may elect to pursue a concentration in one of the three sub-disciplines of cultural, biological or archaeological anthropology.

Up to two elective courses from a discipline outside of anthropology may be applied to the major. Elective courses from outside of the program must be approved by the student's advisor and must be at the 300- or 400-level.

Anthropology Concentrations

General Concentration

The general program provides students with a holistic introduction to the breadth of anthropology highlighting experience in the subdisciplines of cultural, archaeological and biological anthropology, as well as introductory experiences in linguistics. The program is designed to provide students with a well-rounded understanding of the discipline in preparation for advanced graduate training or as an adjunct to their personal and professional aspirations.

Code	Title	Credits
Required General Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
Subtotal		13
Electives		
One elective from each of the following: cultural anthropology, biological anthropology and archaeology		9
Other elective(s) ^{2,3}		15
Choose one of the following:		4
ANTH 435	Ethnographic Genres and Methods	
ANTH 445	Methods and Research Perspectives in Biological Anthropology	
ANTH 455	Archaeology: Methods of Analysis and Interpretation	
Subtotal		28
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside the program.

³ Students should consider including ANTH 305 as one of their electives to learn more about the subdiscipline of linguistic anthropology.

Archaeology Concentration

Archaeology is the study of the development and change of human societies from the prehistoric past to the present through the recovery, analysis and interpretation of material remains. Our program emphasizes an anthropological approach to archaeology and has close ties to anthropology's other subdisciplines. Archaeology has developed its own body of theories and methods for addressing the human story, many of which are focused on understanding what human material culture reveals about peoples' lives.

The archaeology concentration builds student expertise in a variety of areas of archaeological practice, including field methods, the analysis and interpretation of archaeological data, and an understanding of the culture histories of many world regions. Elective courses and independent projects allow students to explore a variety of facets of archaeological theory and practice, with an emphasis on hands-on experience and engagement with real world data and artifacts. Students work with faculty to determine what electives within and outside the program best serve their scholarly and professional interests. Archaeology concentrators can benefit from expertise gained in upper-level electives in history, the geographic sciences, geology, art history, and biology. Intensive training in field methods, though not required for the concentration, is strongly encouraged and is provided by a summer archaeological field school. Those interested in historical archaeology should consider the cross-disciplinary Historical Archaeology Minor (p. 507).

Code	Title	Credits
Required Archaeology Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3

ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
ANTH 455	Archaeology: Methods of Analysis and Interpretation	4
Subtotal		17
Electives		
Upper division courses in cultural and biological anthropology		6
Electives ²		11-12
Choose one of the following:		3-4
ANTH 410	Spatial Analysis for Social Research ³	
ANTH 490	Special Studies in Anthropology ³	
Choose one of the following regional archaeology courses:		3
ANTH 205	Buried Cities, Ancestral Places: The Rise and Fall of Early Human Societies [C2HQC]	
ANTH 250	Anthropology of the American Southwest	
ANTH 303	Andean Archaeology	
ANTH 311	Archaeology of Virginia	
ANTH 327	Ancient North American Civilizations	
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside of the program. GEOL 442, ARTH 394/HIST 394 and ARTH 396/HIST 396 are particularly recommended.

³ Students electing to take ANTH 490 to satisfy this requirement will complete a project that requires the analysis and interpretation of archaeological data. This course requires permission of instructor and must be taken for 3 credits. Total credits for the archaeology concentration must still add to at least 41.

Biological Anthropology Concentration

The focus of biological anthropology is the study of human biology from an evolutionary perspective. Biological anthropology is interested in understanding how and why the human species became what it is today. Thus, it involves the study of human evolution, human biology and its variation, human ecology (how humans interrelate with their environment) and primate behavior and biology (to place humans in the proper comparative context). Biological anthropologists also recognize that human culture, and learned behavior in general, are fundamentally important to understanding the human condition which leads them to emphasize a bio-cultural approach in which both biology and culture are integrated into a holistic understanding of humanity.

Students work closely with biological anthropology faculty to choose electives from both within and outside of the department to refine their own research and scholarly interests. Upper-level electives in biology, psychology and/or geographic sciences are recommended depending on the student's particular goals. Students might consider taking a minor or second major in these disciplines. Students are strongly encouraged to gain practical experience in biological anthropology through study abroad, internships or independent study with faculty.

Code	Title	Credits
Required Biological Anthropology Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
ANTH 445	Methods and Research Perspectives in Biological Anthropology	4
Subtotal		17
Electives		
Upper-division courses in archaeology and cultural anthropology		6
Electives ^{2,3}		12
Choose two of the following:		6
ANTH 308	Fundamentals of Forensic Anthropology	
ANTH 315	Human Evolution	
ANTH 316	Human Evolutionary Psychology	
ANTH 317	Primate Evolutionary Ecology	
ANTH 319	Human Osteology	
ANTH 320	Plagues and Pandemics	
Subtotal		24
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside of the program. BIO 270/BIO 270L and BIO 290/BIO 290L are accepted electives.

³ In addition to biological anthropology courses, students are encouraged to take electives from across the breadth of cultural and linguistic anthropology and archaeology.

Cultural Anthropology Concentration

Cultural anthropology is at the core of anthropology. It provides students with in-depth experience in the interpretation and comparison of cultures. It is closely linked to the humanities and to other social sciences. Students learn what culture is, how different cultural systems and forms of social organization work, how language both reflects and constitutes culture, and methodological and theoretical frameworks for interpreting cultural differences and similarities. Students work closely with cultural anthropology faculty to choose a series of electives from both within and outside of the department to refine their own research interests.

Students are encouraged (but not required) to become proficient in a foreign language and to develop a regional area of specialization through course work or a minor (e.g., Latin American studies, Africana studies, Middle Eastern studies, Asian studies). Outside upper-level electives are recommended in history, sociology, economics, religion, modern foreign languages and political science. Students are encouraged to pursue study abroad, ethnographic field school and internship opportunities.

Code	Title	Credits
Required Cultural Anthropology Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1

ANTH 375	Anthropological Theory ¹	3
ANTH 435	Ethnographic Genres and Methods	4
Subtotal		17
Electives		
Upper-division courses in archaeology and biological anthropology		6
Electives ^{2,3}		15
Choose one of the following Regional Cultural Anthropology courses:		3
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	
ANTH 280	Peoples and Cultures of Sub-Saharan Africa	
ANTH 295	People and Cultures of East Asia	
ANTH 306	Japanese Society and Culture	
ANTH 364	U.S./Latin American Borders	
ANTH 368	Contemporary American Culture	
ANTH 436	Afro-Latin America	
Subtotal		24
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside the program.

³ Suggested electives include ANTH 305, additional area studies courses and upper-level topical courses.

Recommended Schedule for Majors

Transfer students on a two-year course of study should change "Year" in this sequence to "Semester."

The following is an example of a four-year course of study for a student seeking a degree in anthropology:

Course	Title	Credits
First Year		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
General Education, B.A. degree requirements, and/or university electives		21
Credits		30
Second Year		
ANTH 201	The Discipline of Anthropology	1
Anthropology electives and/or concentration courses		9
General Education, B.A. degree requirements, and/or university electives		20
Credits		30
Third Year		
ANTH 375	Anthropological Theory	3
Anthropology electives and/or concentration courses		6
General Education, B.A. degree requirements, and/or university electives		21
Credits		30
Fourth Year		
Choose one of the following:		4
ANTH 435	Ethnographic Genres and Methods	
ANTH 445	Methods and Research Perspectives in Biological Anthropology	
ANTH 455	Archaeology: Methods of Analysis and Interpretation	
Anthropology electives and/or concentration courses		9

General Education, B.A. degree requirements, and/or university electives	17
Credits	30
Total Credits	120

Anthropology, B.S.

Program Description

Anthropology’s goal is to document the diversity of ways humans live and have lived throughout the world. Anthropologists go “off road,” working directly with people and their artifacts, to examine their beliefs and values, how they make a living, how they express themselves, how they interact, and how they affect and are affected by the natural world.

Why is this important? An increasingly globalized and multicultural world requires anthropological knowledge and insight. Anthropology is essential to students interested in working in both the U.S. and overseas on contemporary issues in areas such as social policy work, international development, public health, community-based advocacy, environmental and social justice, and cultural preservation.

Students develop their interests by taking a wide range of classes in cultural, linguistic and biological anthropology and archaeology. Anthropology majors can also do individually-driven, practical and hands-on research in internships, anthropology labs, excavations and field-schools, and study-abroad programs.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement	3-4
Major requirements	41
University electives	35-36
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary.

² MATH 220 is strongly recommended for those students who have not taken that course or an equivalent as part of their General Education.

Major Requirements

To earn a B.A. or B.S. degree in anthropology, students complete 41 credit hours in the major. Given the diverse opportunities the discipline provides, the major is designed to allow students the opportunity to work closely with their advisors to develop a curriculum appropriate to their personal and professional interests. Those students wishing to do so may elect to pursue a concentration in one of the three sub-disciplines of cultural, biological or archaeological anthropology. Up to two elective courses from a discipline outside of anthropology may be applied to the major. Elective courses from outside of the program must be approved by the student’s advisor and must be at the 300- or 400-level.

Anthropology Concentrations

General Concentration

The general program provides students with a holistic introduction to the breadth of anthropology highlighting experience in the subdisciplines of cultural, archaeological and biological anthropology, as well as introductory experiences in linguistics. The program is designed to provide students with a well-rounded understanding of the discipline

in preparation for advanced graduate training or as an adjunct to their personal and professional aspirations.

Code	Title	Credits
Required General Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
Subtotal		13
Electives		
One elective from each of the following: cultural anthropology, biological anthropology and archaeology		9
Other elective(s) ^{2,3}		15
Choose one of the following:		4
ANTH 435	Ethnographic Genres and Methods	
ANTH 445	Methods and Research Perspectives in Biological Anthropology	
ANTH 455	Archaeology: Methods of Analysis and Interpretation	
Subtotal		28
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside the program.

³ Students should consider including ANTH 305 as one of their electives to learn more about the subdiscipline of linguistic anthropology.

Archaeology Concentration

Archaeology is the study of the development and change of human societies from the prehistoric past to the present through the recovery, analysis and interpretation of material remains. Our program emphasizes an anthropological approach to archaeology and has close ties to anthropology’s other subdisciplines. Archaeology has developed its own body of theories and methods for addressing the human story, many of which are focused on understanding what human material culture reveals about peoples’ lives.

The archaeology concentration builds student expertise in a variety of areas of archaeological practice, including field methods, the analysis and interpretation of archaeological data, and an understanding of the culture histories of many world regions. Elective courses and independent projects allow students to explore a variety of facets of archaeological theory and practice, with an emphasis on hands-on experience and engagement with real world data and artifacts. Students work with faculty to determine what electives within and outside the program best serve their scholarly and professional interests. Archaeology concentrators can benefit from expertise gained in upper-level electives in history, the geographic sciences, geology, art history, and biology. Intensive training in field methods, though not required for the concentration, is strongly encouraged and is provided by a summer archaeological field school. Those interested in historical archaeology should consider the cross-disciplinary Historical Archaeology Minor (p. 507).

Code	Title	Credits
Required Archaeology Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
ANTH 455	Archaeology: Methods of Analysis and Interpretation	4
Subtotal		17
Electives		
Upper division courses in cultural and biological anthropology		6
Electives ²		11-12
Choose one of the following:		3-4
ANTH 410	Spatial Analysis for Social Research ³	
ANTH 490	Special Studies in Anthropology ³	
Choose one of the following regional archaeology courses:		3
ANTH 205	Buried Cities, Ancestral Places: The Rise and Fall of Early Human Societies [C2HQC]	
ANTH 250	Anthropology of the American Southwest	
ANTH 303	Andean Archaeology	
ANTH 311	Archaeology of Virginia	
ANTH 327	Ancient North American Civilizations	
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside of the program. GEOL 442, ARTH 394/HIST 394 and ARTH 396/HIST 396 are particularly recommended.

³ Students electing to take ANTH 490 to satisfy this requirement will complete a project that requires the analysis and interpretation of archaeological data. This course requires permission of instructor and must be taken for 3 credits. Total credits for the archaeology concentration must still add to at least 41.

Biological Anthropology Concentration

The focus of biological anthropology is the study of human biology from an evolutionary perspective. Biological anthropology is interested in understanding how and why the human species became what it is today. Thus, it involves the study of human evolution, human biology and its variation, human ecology (how humans interrelate with their environment) and primate behavior and biology (to place humans in the proper comparative context). Biological anthropologists also recognize that human culture, and learned behavior in general, are fundamentally important to understanding the human condition which leads them to emphasize a bio-cultural approach in which both biology and culture are integrated into a holistic understanding of humanity.

Students work closely with biological anthropology faculty to choose electives from both within and outside of the department to refine their own research and scholarly interests. Upper-level electives in biology, psychology and/or geographic sciences are recommended depending on the student's particular goals. Students might consider taking a minor or second major in these disciplines. Students are strongly encouraged

to gain practical experience in biological anthropology through study abroad, internships or independent study with faculty.

Code	Title	Credits
Required Biological Anthropology Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
ANTH 445	Methods and Research Perspectives in Biological Anthropology	4
Subtotal		17
Electives		
Upper-division courses in archaeology and cultural anthropology		6
Electives ^{2,3}		12
Choose two of the following:		6
ANTH 308	Fundamentals of Forensic Anthropology	
ANTH 315	Human Evolution	
ANTH 316	Human Evolutionary Psychology	
ANTH 317	Primate Evolutionary Ecology	
ANTH 319	Human Osteology	
ANTH 320	Plagues and Pandemics	
Subtotal		24
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside of the program. BIO 270/BIO 270L and BIO 290/BIO 290L are accepted electives.

³ In addition to biological anthropology courses, students are encouraged to take electives from across the breadth of cultural and linguistic anthropology and archaeology.

Cultural Anthropology Concentration

Cultural anthropology is at the core of anthropology. It provides students with in-depth experience in the interpretation and comparison of cultures. It is closely linked to the humanities and to other social sciences. Students learn what culture is, how different cultural systems and forms of social organization work, how language both reflects and constitutes culture, and methodological and theoretical frameworks for interpreting cultural differences and similarities. Students work closely with cultural anthropology faculty to choose a series of electives from both within and outside of the department to refine their own research interests.

Students are encouraged (but not required) to become proficient in a foreign language and to develop a regional area of specialization through course work or a minor (e.g., Latin American studies, Africana studies, Middle Eastern studies, Asian studies). Outside upper-level electives are recommended in history, sociology, economics, religion, modern foreign languages and political science. Students are encouraged to pursue study abroad, ethnographic field school and internship opportunities.

Code	Title	Credits
Required Cultural Anthropology Concentration Courses		
ANTH 195	Cultural Anthropology [C4GE]	3

ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
ANTH 201	The Discipline of Anthropology	1
ANTH 375	Anthropological Theory ¹	3
ANTH 435	Ethnographic Genres and Methods	4
Subtotal		17
Electives		
Upper-division courses in archaeology and biological anthropology		6
Electives ^{2,3}		15
Choose one of the following Regional Cultural Anthropology courses:		3
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	
ANTH 280	Peoples and Cultures of Sub-Saharan Africa	
ANTH 295	People and Cultures of East Asia	
ANTH 306	Japanese Society and Culture	
ANTH 364	U.S./Latin American Borders	
ANTH 368	Contemporary American Culture	
ANTH 436	Afro-Latin America	
Subtotal		24
Total Credits		41

¹ Students should take two of ANTH 195, ANTH 196 or ANTH 197 and at least one anthropology elective before taking ANTH 375.

² Students may take up to two advisor-approved electives at the 300 or 400 level from courses outside the program.

³ Suggested electives include ANTH 305, additional area studies courses and upper-level topical courses.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
General Education, B.S. degree requirements, and/or university electives		21
Credits		30
Second Year		
ANTH 201	The Discipline of Anthropology	1
Anthropology electives and/or concentration courses		9
General Education, B.S. degree requirements, and/or university electives		20
Credits		30
Third Year		
ANTH 375	Anthropological Theory	3
Anthropology electives and/or concentration courses		6
General Education, B.S. degree requirements, and/or university electives		21
Credits		30
Fourth Year		
Choose one of the following:		4
ANTH 435	Ethnographic Genres and Methods	
ANTH 445	Methods and Research Perspectives in Biological Anthropology	
ANTH 455	Archaeology: Methods of Analysis and Interpretation	
Anthropology electives and/or concentration courses		9
General Education, B.S. degree requirements, and/or university electives		17
Credits		30
Total Credits		120

Art History, B.A.

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Program Description

The Bachelor of Arts in art history is intended for students who wish to study the history of the visual arts, including the cultural and social context in which they were created. It is the mission of the art history program to educate students in global and cross-cultural perspectives on the arts.

Students majoring in art history will develop a general knowledge of exemplary monuments, artists, art media, art periods and artistic styles. Students will also:

- Describe, analyze and interpret the form and content of individual works of art in relation to their cultures and historical contexts.
- Become familiar with art history theory and methods of analysis and criticism.
- Research and write about significant artists, artistic events, periods and artistic concepts.

The art history program encourages majors and minors to participate in internship opportunities at museums, galleries and other art related settings, and it provides up-to-date information about jobs and graduate programs in art history and related fields. Visiting scholars, campus art exhibitions, trips to major museums and the JMU Study Abroad program provide students with opportunities to study and apply their art history knowledge outside the classroom. Upon completion of the art history major, students will have been exposed to a broad background and knowledge of the opportunities for graduate school and employment. The art history program also advocates cross disciplinary education and actively supports students who double major with other disciplines. Throughout the student's undergraduate years, faculty provide individualized curricular advising and career mentorship.

Internship/Independent Study Credit

A maximum of three credits of art history internship, independent study or honors course work may be applied toward the major in art history. A student may petition the art history coordinator to apply six credit hours toward the art history major if the internship or independent study course work is directly relevant to the student's interests and career goals.

Requirements for Art History as a Second Major

Students in any degree program may study art history as a second major by completing the 39 credit art history requirement. Students studying art history as a second major do not need to complete the requirements for the B.A. degree if their first major will complete the requirements for a different baccalaureate degree. However, non-B.A. degree students completing art history as a second major are strongly encouraged to complete the B.A. foreign language requirement.

Admission Requirements

Students intending to major in art history are not required to submit a portfolio, but should follow the regular JMU admissions process.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	39
University electives ³	25-37
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures placement test.

³ Depends on number of hours needed to fulfill the B.A. foreign language requirement. If they are completed in six hours, students should use the larger number of general electives to complete a second major or minor that complements art or art history or, for students intending to pursue graduate degrees, to gain reading knowledge of a second foreign language.

Major Requirements

The major in art history requires 39 credit hours in art history and studio courses, as the following chart shows. At least six of these credit hours must be 400-level art history courses.

Core courses at the 200 level are broad interpretive overviews of art history that combine lecture and discussion to permit students to learn the content and chronology of world art history; to learn how art historians collect, analyze and synthesize evidence; and engage in introductory exercises in the discipline. They generally include short formal writing assignments and exams with essay sections.

Distributives at the 300 level also combine lecture and discussion, more narrowly survey specific periods or cultures and address aspects of art historical methodology. These courses require longer formal writing assignments and include student research. Courses at the 400 level are seminars on specialized topics that center on advanced student research. A single course may not fulfill more than one distribution requirement.

Code	Title	Credits
Required Courses		
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
ARTH 300	Art History Seminar	3
ARTH 400	Art History Research Portfolio	0
Subtotal		9
Non-Western Elective		
Choose one of the following:		3
ARTH 312	African Art: Sub-Saharan	

ARTH 325	Topics in Non-Western Art and Architecture	
ARTH 418	Modern and Contemporary African Art	
ARTH 419	Topics in African Art	
ARTH 424	Arts of Ancient Egypt	
ARTH 425	Advanced Topics in Non-Western Art	
ARTH 484	Art of the Americas	
Subtotal		3
Cross-Cultural Elective		
Choose one of the following:		3
ARTH 326	Topics in Cross-Cultural Art	
ARTH 332	Islamic Art and Architecture	
ARTH 426	Advanced Topics in Cross-Cultural Art	
ARTH 440	City of Jerusalem	
ARTH 446	Renaissance Art and the East	
ARTH 488	African American Art	
Subtotal		3
Pre-Modern Elective		
Choose one of the following:		3
ARTH 312	African Art: Sub-Saharan	
ARTH 327	Topics in Pre-Modern Art	
ARTH 332	Islamic Art and Architecture	
ARTH 424	Arts of Ancient Egypt	
ARTH 427	Advanced Topics in Pre-Modern Art	
ARTH 440	City of Jerusalem	
ARTH 446	Renaissance Art and the East	
ARTH 484	Art of the Americas	
Subtotal		3
Modern and Contemporary Elective		
Choose one of the following:		3
ARTH 304	History of Photography	
ARTH 328	Topics in Modern and Contemporary Art	
ARTH 360	Nineteenth-Century Art	
ARTH 372	Modern Art from 1900-1945	
ARTH 380	American Art to 1870	
ARTH 382	American Art from 1870	
ARTH 404	Topics in the History of Photography	
ARTH 406	Monticello	
ARTH 418	Modern and Contemporary African Art	
ARTH 419	Topics in African Art	
ARTH 428	Advanced Topics in Modern and Contemporary Art	
ARTH 471	Commemoration and Controversy: Public Art in America	
ARTH 472	Modern Art Since 1945	
ARTH 488	African American Art	
Subtotal		3
Additional Courses		
Art history electives (300-400 level) ¹		12
Studio art and design electives (any level) ²		6
Subtotal		18
Internship/Independent Study Credit		

A maximum of three credits of art history internship, independent study or honors course work may be applied toward the major in art history. A student may petition the art history coordinator to apply six credit hours toward the art history major if the internship or independent study course work is directly relevant to the student's interests and career goals.

Total Credits	39
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¹ A maximum of three credits from ARTH 490, ARTH 495 or ARTH 499 may count toward the major.

² ART 200 may not count as a studio art and design elective.

Museum Studies Concentration

The museum studies concentration enriches the art history curriculum by offering course work that examines the critical role that museums have played in constructing the discipline and pedagogy of art history. History and theory-oriented classes will introduce students to the role and function of museums in society and the ways in which museums both reflect and perpetuate the values of the cultures that create them. Experiential practica or internship courses will expose students to the wide range of work conducted in museums, including curation, collections management, conservation, education, design and installation, media and public relations, publications, development and administration.

While the concentration is academic and not vocational, it offers students valuable hands-on experience beneficial for admission into graduate school and entry into the competitive market of art-related professions. Virginia is a state that boasts a plethora of art museums, house museums, history museums, and historic and archaeological sites. Art history majors who complete the concentration will be well positioned for seeking employment in regional and state museums.

The museum studies concentration consists of five courses (15 credit hours). Students are required to complete three core courses and two elective courses.

Code	Title	Credits
Required Courses		
ARTH/HIST 394	Introduction to Museum Work	3
ARTH/HIST 408	The Museum: History and Controversies	3
ARTH 495	Internship in Art History	3
Subtotal		9
Electives		
Choose two of the following:		6
ARTH 301	Institutions and Professions in Art History	
ARTH 309	Topics in Museum Studies	
ARTH/HIST 396	History Beyond the Classroom: Public History	
ARTH/HIST 406	Monticello	
ARTH 409	Topics in Museum Studies	
ARTH 491	Exhibition Seminar	
ARTH/ANTH/HIST 492	History from Things: Material Culture	
ARTH/HIST 493	Historic Preservation	
ANTH 494	Field Techniques in Archaeology	

HIST 440	The History Museum	
Subtotal		6
Total Credits		15

Recommended Schedule For Majors

Course	Title	Credits
First Year		
ARTH 205 or ARTH 206	Survey of World Art I: Prehistoric to Renaissance [C2VPA] or Survey of World Art II: Renaissance to Modern [C2VPA]	3
Foreign language courses		6
Madison Foundations		9
General Education courses and/or university electives		12
Credits		30
Second Year		
ARTH 205 or ARTH 206	Survey of World Art I: Prehistoric to Renaissance [C2VPA] or Survey of World Art II: Renaissance to Modern [C2VPA]	3
Choose foreign language (if needed) or general electives		6
General Education requirement		18
Studio Art elective		3
Credits		30
Third Year		
ARTH 300	Art History Seminar	3
Non-Western art history elective		3
Western art history electives		6
University electives		3
Studio Art elective		3
General Education requirement		12
Credits		30
Fourth Year		
Art History electives		12
University electives		18
Credits		30
Total Credits		120

Art, Architectural Design Major, B.F.A.

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Program Description

The Bachelor of Fine Arts degree in Art with a major in architectural design educates future design leaders. It is an intensive studio-based program focused on rigorous design processes. Architectural design is approached holistically—emphasizing investigations into the nature of materials and objects, space, history and theory, programmatic invention, functional poetics, and collaboration across disciplines. The curriculum instills purpose, craft, technical competencies, and versatile thinking. We are an inclusive program that develops skilled and empathetic architectural designers.

The Architectural Design program is a pre-professional major that prepares students for graduate studies as well as entry-level positions in a variety of design fields including architecture, interior design and landscape architecture. Students should be aware that educational requirements for architectural licensing vary from state to state.

The program is enhanced by frequent field trips as well as guest critics and speakers. Vibrant student organizations including a chapter of the National Organization for Minority Architecture students and the American Institute of Architecture Students provide leadership opportunities for students and connections to the profession.

Admission Requirements

Admission to the architectural design major is selective and competitive for a limited number of reserved seats in upper-division (300-400 level) architectural design courses. Declaration of architectural design as a major and completion of lower-division (100-200) prerequisite art courses does not guarantee admission into the program. Admission to upper-division ARCD courses is based on completion of all prerequisite art courses and merit as determined by faculty review of portfolios submitted in satisfaction of the course ARCD 208.

ARCD 208 is a 0 credit, pass/fail course that functions as a prerequisite to enrollment in 300-400 level architectural design courses. ARCD 208 portfolios are reviewed in the sophomore year prior to pre-registration. Students should contact the area coordinator of architectural design to determine the exact time and location for the ARCD 208 portfolio review.

ARCD 200 and ARCD 202 are restricted to Architectural Design students who have passed FDS 102, FDS 104 and FDS 106 only. Students are restricted from taking any 300-400 level architectural design studio courses until ARCD 208 has been taken and passed. Accepted students who receive a passing grade for ARCD 208 will be able to register for ARCD upper-division courses for the following semester.

Accepted students who receive a passing grade for ARCD 208 are required to furnish a computer that meets or exceeds the recommendations of the program listed on the Bookstore's website (<https://www.jmu.edu/bookstore/computers/recommendations.shtml>).

Students not accepted will have one opportunity to reapply or may choose to continue in another major within the SADAH.

Special Admission Requirements Architectural Design, Art Studio, Graphic Design and Industrial Design Majors

All prospective freshmen, transfer and change of major students intending to major in architectural design, graphic design or studio art must meet JMU admission requirements as stated in the Undergraduate Catalog. In addition to meeting JMU admission requirements, all students seeking the BFA with a major in architectural design, art studio or graphic design, or the BS with a major in industrial design are required to submit a portfolio to the school's SlideRoom account (<https://jmuart.slideroom.com>). The submission of a portfolio is optional for students seeking the BA and BS with a major in art studio. All students regardless of degree type, who wish to be considered for a scholarship must submit a portfolio to SlideRoom. The submission of a portfolio is seen as evidence of a student's interest and potential for future success in art. The deadline for submitting portfolios to SADAH's SlideRoom site can be found on the School's Portfolio (<https://www.jmu.edu/sadah/admissions/portfolios.shtml>) page. There is a \$10 charge for this submission.

Transfer and Change of Major Students

Transfer and change of major students seeking the BFA with a major in architectural design, art studio, or graphic design or the BS with a major in industrial design are required to submit a portfolio and also need to upload unofficial transcripts and a statement of intent to the school's SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study.

On-Campus Portfolio Feedback Days and Scholarships

All students who would like to be considered for scholarships are required to attend one of the on-campus or virtual portfolio feedback days. Students will show their original art and participate in a faculty interview. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website (<https://www.jmu.edu/sadah/index.shtml>) for the dates of the upcoming portfolio feedback days and for additional portfolio requirements. All scholarship awards are based on merit and vary in amount. Tours of facilities and program information sessions will take place at on-campus events.

Off-Campus Portfolio Feedback Days

The SADAH also offers students the opportunity for faculty critiques on their portfolio, prior to their digital submission and before an on-campus or virtual portfolio scholarship interview. The off-campus portfolio feedback events provide an opportunity for students to meet SADAH's faculty. All prospective first-year, transfer and change of major students are strongly encouraged to attend a portfolio feedback day, on or off campus, prior to submitting their portfolio.

Architectural Design

Students seeking official acceptance into the architectural design major must enroll in ARCD 208 and submit a portfolio representing work completed in ARCD 200 and ARCD 202. Students not admitted may reapply the following year.

SADAH Computer Requirements for Architectural Design

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH

considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department’s website (<https://www.jmu.edu/sadah/index.shtml>).

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Major requirements	78
University electives	1
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.
It is recommended that students interested in pursuing graduate studies in architecture take PHYS 140 and MATH 205 in fulfillment of General Education credits in The Natural World.

Major Requirements

This major requires 78 credit hours in art, art history and art-related courses; 45 credits must focus on architectural design.

Code	Title	Credits
Foundation Courses		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
FDS 108	Expressive and Expanded Drawing	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Subtotal		18
Required Architectural Design Courses		
ARCD 200	Architect Design Studio I	6
ARCD 202	Architect Design Studio II	6
ARCD 208	Portfolio Review	0
ARCD 220	Digital Processes	3
ARCD 300	Architect Design Studio III	6
ARCD 302	Architect Design Studio IV	6
ARCD 330	Materials and Methods	3
ARCD 370	History of Architecture	3
ARCD 376	Modern Architecture	3
ARCD 400	Architect Design Studio V	6
ARCD 402	Architect Design Studio VI	6
ARCD 440	Professional Design Practices	3
ARCD 496	Internship	0
Subtotal		51
Art, Design and Art History Electives		

Art, design and art history electives may include courses from general fine arts, graphic design, industrial design, art education, architectural design, internships and art history. ¹

Subtotal	9
Total Credits	78

¹ Excluding courses taken in the required major, as well as ARTH 490, ARTH 495 and ARTH 499.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
General Education		12
Art Foundations		18
Credits		30
Second Year		
ARCD 200	Architect Design Studio I	6
ARCD 202	Architect Design Studio II	6
ARCD 220	Digital Processes	3
ARCD 208	Portfolio Review	0
ARCD 376	Modern Architecture	3
General Education		12
Credits		30
Third Year		
ARCD 300	Architect Design Studio III	6
ARCD 302	Architect Design Studio IV	6
ARCD 330	Materials and Methods	3
ARCD 370	History of Architecture	3
ARCD 440	Professional Design Practices	3
ARCD 496	Internship ¹	0
General Education and/or Art, Design and Art History Elective Courses		12
Credits		33
Fourth Year		
ARCD 400	Architect Design Studio V	6
ARCD 402	Architect Design Studio VI	6
Art, Design and Art History Elective Courses		6
General Education		9
Credits		27
Total Credits		120

¹ ARCD 496 is typically completed during the summer.

Art, B.A.

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Program Description

The Bachelor of Arts degree in Art is designed for students who seek a breadth of knowledge in the visual arts and who have an interest or background in the humanities and foreign language. The art faculty is made up of practicing artists who work closely with the students, and students also work closely with each other as they develop their studio practice. Visiting artists and scholars, workshops, field trips and frequent on-campus exhibitions of professional artwork enhance student engagement with contemporary art. Additionally, students have opportunities to exhibit and present their artwork, assist in running on-campus galleries and art studio labs, and partake in other hands-on art experiences.

Students graduating with a major in art will:

- Display technical command of at least one medium and display fluency in a range of media
- Effectively analyze and apply two- and three-dimensional design principles and devices
- Develop an independent line of creative inquiry that embraces risk-taking and inventive solutions
- Knowledgeably critique and contextualize their artwork to cultivate expressive ideas and meaningful content
- Demonstrate awareness of contemporary art and art history within a global context

Special Admission Requirements

Architectural Design, Art Studio, Graphic Design and Industrial Design Majors

All prospective first year, transfer and change of major students intending to major in architectural design, graphic design or studio art must meet JMU admission requirements as stated in the Undergraduate Catalog. In addition to meeting JMU admission requirements, all students seeking the BFA with a major in architectural design, art studio or graphic design, or the BS with a major in industrial design are required to submit a portfolio to the school's SlideRoom account (<https://jmuart.slideroom.com>). The submission of a portfolio is optional for students seeking the BA and BS with a major in art studio. All students regardless of degree type, who wish to be considered for a scholarship must submit a portfolio to SlideRoom. The submission of a portfolio is seen as evidence of a student's interest and potential for future success in art. The deadline for submitting portfolios to SADAH's SlideRoom site can be found on the School's Portfolio (<https://www.jmu.edu/sadah/admissions/portfolios.shtml>) page. There is a \$10 charge for this submission.

Transfer and Change of Major Students

Transfer and change of major students seeking the BFA with a major in architectural design, art studio, or graphic design or the BS with a major in industrial design are required to submit a portfolio and also need to upload unofficial transcripts and a statement of intent to the school's SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study.

On-Campus Portfolio Feedback Days and Scholarships

All students who would like to be considered for scholarships are required to attend one of the on-campus or virtual portfolio feedback days. Students will show their original art and participate in a faculty interview. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website (<https://www.jmu.edu/sadah/index.shtml>) for the dates of the upcoming portfolio feedback days and for additional portfolio requirements. All scholarship awards are based on merit and vary in amount. Tours of facilities and program information sessions will take place at on-campus events.

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SADAH Computer Requirements for Art, B.A.

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department's website (<https://www.jmu.edu/sadah/index.shtml>).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language courses (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	45
University electives	19-31
Total ³	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Language and Cultures' placement test.

³ The optional art education concentration requires more than 120 credits.

Major Requirements

The art major requires a minimum of 45 credit hours in art and art history.

Code	Title	Credits
Foundation Courses		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Subtotal		15
3-D Studio Art Courses		
Choose one of the following:		3
ART 222	Introductory Ceramics	
ART 230	Introduction to Fibers Processes	
ART 240	Introductory Metals	
ART 280	Introductory Sculpture: Idea Material Process	
Subtotal		3
Studio Art Electives		
18 credit hours of Studio Art electives must consist of studio-based courses with an ART or FDS prefix. ¹		18
Subtotal		18
Studio Art Emphasis		
In addition to the 18 credit hours of studio art electives above, 6 additional credit hours of upper-level Studio Art electives (300-400 level) must be taken in a single studio area (i.e. Ceramics, Fibers, Metals, Sculpture, Printmaking and Immersive Media, Photography or Painting). Only courses with an ART prefix count toward this requirement.		6
Subtotal		6
Art History Elective		
Students must take 3 credit hours of art history at the 300-400 level. ²		3
Subtotal		3
Total Credits		45

¹ Excluding ART 200, ART 105, ART 304*, ART 305, ART 405, ART 491, ART 493 and ART 496.

* Students who elect the concentration in art education may count ART 304 towards this requirement.

² Students may not count ARTH 397, ARTH 490, ARTH 495, ARTH 497, or ARTH 499 towards this requirement.

Art Education Concentration

Coordinator: William Wightman

Phone: (540) 568-5144

Email: wightmwh@jmu.edu

Students pursuing the B.A. in Art who desire to obtain K-12 Licensure should declare a concentration in art education. The concentration requires 42 credit hours, 3 of which may double-count for major requirements. An additional 3 credits may double count towards general education.

It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For a complete description of

admission and retention policies and procedures for teacher education, refer to the College of Education (p. 18).

Students seeking the concentration in art education professional PK-12 teaching licensure are encouraged to consult regularly with the program coordinator of art education. The undergraduate degree leading to licensure must include the following minimum requirements in art:

- Nine credits must be earned in art history and art appreciation.
- Six credits must be earned in ceramics and crafts, with a minimum of one course in ceramics.
- Six credits must be earned in three-dimensional media, with one course in sculpture.
- Twelve credits must be earned in four different two-dimensional media.
- Twenty-seven studio credits, with a minimum of six credits in each of three separate studio areas.

Code	Title	Credits
Art Education Concentration Requirements		
ARED 300	Elementary Art Education Methods	3
ARED 302	Secondary Art Education Methods	3
ARED 400	Visual Arts Across the Curriculum	3
ARED 480	Student Teaching ¹	16
ARED 490	Field Experiences in Art Education ²	6
ART 304	Methods of Art Criticism	3
EDUC 200	Foundations of Education in the United States	3
LED 420	Content Area Literacy, K-12	2
PSYC 160	Life Span Human Development [C5SD]	3
Total Credits		42

¹ Taken twice for a total of 16 credits.

² Taken concurrently with ARED 300, ARED 302 and ARED 400 for 2 credits three times for a total of 6 credits.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
Madison Foundations		9
General Education requirement		9
Credits		30
Second Year		
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Required 3-D studio art course		3
Foreign language courses (if needed)		6
General Education requirement		12
Studio Art electives (any level)		6
Credits		30
Third Year		
Art history elective (300-400 level)		3
Studio art electives (any level)		9
General electives		6
B.A. philosophy requirement		3

General Education requirement	9
Credits	30
Fourth Year	
Studio art emphasis (300-400 level)	6
Studio art or general electives	24
Credits	30
Total Credits	120

Art, B.F.A.

Coordinator: Mark Rooker
Phone: (540) 568-6410
Email: rookermd@jmu.edu

Program Description

The Bachelor of Fine Arts degree in Art offers concentrations in ceramics, fibers, metals, painting, photography, printmaking, and sculpture. It is designed for students who aspire to develop a professional art practice as expansively defined in today's world. Students are prepared to produce and exhibit art and to pursue further study at the graduate level.

The art faculty is made up of practicing artists who work closely with the students, and students also work closely with each other as they develop their studio practice. Visiting artists and scholars, workshops, field trips and frequent on-campus exhibitions of professional artwork enhance student engagement with contemporary art. Additionally, students have opportunities to exhibit and present their artwork, to assist in running on-campus galleries and studio art labs and to partake in other hands-on art experiences.

Students graduating with a major in art will:

- Display technical command of at least one medium and display fluency in a range of media
- Effectively analyze and apply two- and three-dimensional design principles and devices
- Develop an independent line of creative inquiry that embraces risk-taking and inventive solutions
- Knowledgeably critique and contextualize their artwork to cultivate expressive ideas and meaningful content
- Demonstrate awareness of contemporary art and art history within a global context

Special Admission Requirements

Architectural Design, Art, Graphic Design and Industrial Design Majors

All prospective first year, transfer and change of major students intending to major in architectural design, graphic design or studio art must meet JMU admission requirements as stated in the Undergraduate Catalog. In addition to meeting JMU admission requirements, all students seeking the B.F.A. with a major in architectural design, art studio or graphic design, or the B.S. with a major in industrial design are required to submit a portfolio to the school's SlideRoom account (<https://jmuart.slideroom.com>). The submission of a portfolio is optional for students seeking the B.A. and B.S. with a major in art studio. All students regardless of degree type, who wish to be considered for a scholarship must submit a portfolio to SlideRoom. The submission of a portfolio is seen as evidence of a student's interest and potential for future success in art. The deadline for submitting portfolios to SADAH's SlideRoom site can be found on the School's Portfolio (<https://www.jmu.edu/sadah/admissions/portfolios.shtml>) page. There is a \$10 charge for this submission.

Transfer and Change of Major Students

Transfer and change of major students seeking the B.F.A. with a major in architectural design, art studio, or graphic design or the B.S. with a major in industrial design are required to submit a portfolio and also need to upload unofficial transcripts and a statement of intent to the school's

SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study.

On-Campus Portfolio Feedback Days and Scholarships

All students who would like to be considered for scholarships are required to attend one of the on-campus or virtual portfolio feedback days. Students will show their original art and participate in a faculty interview. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website (<https://www.jmu.edu/sadah/index.shtml>) for the dates of the upcoming portfolio feedback days and for additional portfolio requirements. All scholarship awards are based on merit and vary in amount. Tours of facilities and program information sessions will take place at on-campus events.

Off-Campus Portfolio Feedback Days

The SADAH also offers students the opportunity for faculty critiques on their portfolio, prior to their digital submission and before an on-campus or virtual portfolio scholarship interview. The off-campus portfolio feedback events provide an opportunity for students to meet SADAH's faculty. All prospective first-year, transfer and change of major students are strongly encouraged to attend a portfolio feedback day, on or off campus, prior to submitting their portfolio.

SADAH Computer Hardware and Software Requirements for Art, B.F.A.

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department's website (<https://www.jmu.edu/sadah/index.shtml>).

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Major requirements	81
University elective	1
Total ²	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² The optional art education concentration requires more than 120 credits.

Major Requirements

The major requires 81 credit hours in art, art history and art related courses.

Code	Title	Credits
Foundation Courses		
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3

ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
FDS 108	Expressive and Expanded Drawing	3
Subtotal		18
Core Art Courses		
ART 105	Introduction to Studio Art	3
ART 305	Professional Practices in Studio Art	3
ART 405	Studio Art Capstone Thesis Seminar	3
Subtotal		9
Introductory Studio Courses		
ART 222	Introductory Ceramics	3
ART 230	Introduction to Fibers Processes	3
ART 240	Introductory Metals	3
ART 252	Introductory Painting	3
ART 260	Introductory Black and White Darkroom Photography	3
ART 271	Introductory Printmaking: A Survey of Analog Printmaking Methods	3
ART 280	Introductory Sculpture: Idea Material Process	3
Subtotal		21
Studio Concentrations		
Students must take 12 credits at the 300- and 400-level in a single medium from the following concentration options. See below for a list of courses in each concentration.		12
Ceramics		
Fibers		
Metals		
Painting		
Photography		
Printmaking		
Sculpture		
Subtotal		12
Art History Electives (300-400 Level)		
Student must take 6 credits of art history electives at the 300-400 level. ¹		6
Subtotal		6
Art and Art-Related Electives		
Art and art-related electives may include courses from art, graphic design, architectural design, art education independent study, industrial design, internships and study abroad studio courses but excludes courses taken to fulfill the studio concentration.		15
Subtotal		15
Total Credits		81

¹ Excludes ARTH 397, ARTH 490, ARTH 495, ARTH 497 and ARTH 499.

Ceramics Concentration

Code	Title	Credits
ART 320	Molds and Casting: Ceramics	3
ART 322	Surface Development: Ceramics	3

ART 324	Potter's Wheel	3
ART 420	Advanced Ceramics (repeatable to 9 credits)	1-3
ART 429	Portfolio Development: Ceramics	3

Fibers Concentration

Code	Title	Credits
ART 330	Intermediate Fiber Processes	3
ART 332	Woven Structures	3
ART 334	Iterative Structures	3
ART 430	Advanced Fiber Processes	1-3
ART 432	Fiber Installation Art	3
ART 434	Collaboration in Fiber Arts	3
ART 439	Portfolio Development: Fibers	3

Metals Concentration

Code	Title	Credits
ART 340	Molds and Casting: Metals	3
ART 342	Knifemaking	3
ART 344	Welding and Metal Fabrication	3
ART 346	Color on Metal	3
ART 440	Advanced Metals	1-3
ART 449	Portfolio Development: Metals	3

Painting Concentration

Code	Title	Credits
ART 352	Intermediate Painting (repeatable for credit)	3
ART 452	Advanced Painting (repeatable for credit)	1-3
ART 459	Portfolio Development: Painting	3

Photography Concentration

Code	Title	Credits
ART 360	Digital Photography	3
Choose three of the following:		3-9
ART 362	Experimental Black and White Photography	
ART 364	Large Format Photography	
ART 460	Alternative Photographic Processes	
ART 462	Wonder and Illusion: Origins of Photography	
ART 464	The Photograph/Video as Document	
ART 466	Performance for the Lens	
ART 468	Screen-Based Photography/Video	
ART 469	Portfolio Development: Photography	

Printmaking Concentration

Code	Title	Credits
ART 372	Mechanical Reproduction(s): Relief, Intaglio and Technology	3
ART 373	Graphic Edition(s): Lithography, Silkscreen and Digital Processes	3
ART 470	Advanced Printmaking (repeatable to 6 credits)	1-3
ART 474	Expanded Print Media: Contemporary Printmaking and Interdisciplinary Practices	3
ART 479	Portfolio Development: Printmaking and Expanded Media	3

Sculpture Concentration

Code	Title	Credits
ART 380	Conceptual Practice: Sculpture	3
ART 480	Expanded Field: Sculpture (repeatable to 9 credits)	1-3
ART 489	Portfolio Development: Sculpture	3

Art Education Concentration

Coordinator: William Wightman

Phone: (540) 568-5144

Email: wightmwh@jmu.edu

Students pursuing art education as part of the B.F.A. in Art complete two concentrations, one in their chosen studio art area and one in art education. The concentration in art education requires 42 credit hours, 15 of which apply towards the art and art-related electives requirement in the major. Students complete an additional 27 credit hours for the art education concentration. The total credit hours for the degree, major and both concentrations are 143.

It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For a complete description of admission and retention policies and procedures for teacher education, refer to the College of Education (p. 18).

Students seeking the concentration in art education professional PK-12 teaching licensure are encouraged to consult regularly with the program coordinator of art education. The undergraduate degree leading to licensure must include the following minimum requirements in art:

- Nine credits must be earned in art history and art appreciation.
- Six credits must be earned in ceramics and crafts, with a minimum of one course in ceramics.
- Six credits must be earned in three-dimensional media, with one course in sculpture.
- Twelve credits must be earned in four different two-dimensional media.
- Twenty-seven studio credits, with a minimum of six credits in each of three separate studio areas.

Code	Title	Credits
Art Education Concentration Requirements		
ARED 300	Elementary Art Education Methods	3
ARED 302	Secondary Art Education Methods	3
ARED 400	Visual Arts Across the Curriculum	3
ARED 480	Student Teaching ¹	16
ARED 490	Field Experiences in Art Education ²	6
ART 304	Methods of Art Criticism	3
EDUC 200	Foundations of Education in the United States	3
LED 420	Content Area Literacy, K-12	2
PSYC 160	Life Span Human Development [C5SD]	3
Total Credits		42

¹ Taken twice for a total of 16 credits.

² Taken concurrently with ARED 300, ARED 302 and ARED 400 for 2 credits three times for a total of 6 credits.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
FDS 108	Expressive and Expanded Drawing	3
ART 105	Introduction to Studio Art	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Madison Foundations		9
Credits		30
Second Year		
ART 222	Introductory Ceramics	3
ART 230	Introduction to Fibers Processes	3
ART 240	Introductory Metals	3
ART 252	Introductory Painting	3
ART 260	Introductory Black and White Darkroom Photography	3
ART 271	Introductory Printmaking: A Survey of Analog Printmaking Methods	3
ART 280	Introductory Sculpture: Idea Material Process	3
General Education requirement		10
Credits		31
Third Year		
Art history elective (300-400 level)		6
Introductory studio course		3
Art and art-related electives		9
Studio concentration		6
General Education requirement		3
ART 305	Professional Practices in Studio Art	3
Credits		30
Fourth Year		
ART 405	Studio Art Capstone Thesis Seminar	3
Art and art-related electives		6
Studio concentration		9
General Education requirement		10
University Elective(s)		1
Credits		29
Total Credits		120

Art, B.S.

Coordinator: Mark Rooker

Phone: (540) 568-6410

Email: rookerm@jmu.edu

Program Description

The Bachelor of Science degree in Art is designed for students who seek a breadth of knowledge in the visual arts and who have an interest or background in math, the natural sciences or the social sciences. The art faculty is made up of practicing artists who work closely with the students, and students also work closely with each other as they develop their studio practice. Visiting artists and scholars, workshops, field trips and frequent on-campus exhibitions of professional artwork enhance student engagement with contemporary art. Additionally, students have opportunities to exhibit and present their artwork, to assist in running on-campus galleries and studio art labs and to partake in other hands-on art experiences.

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Transfer and change of major students seeking the BFA with a major in architectural design, art studio, or graphic design or the BS with a major in industrial design are required to submit a portfolio and also need to upload unofficial transcripts and a statement of intent to the school's SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study.

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All students who would like to be considered for scholarships are required to attend one of the on-campus or virtual portfolio feedback days. Students will show their original art and participate in a faculty interview. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website (<https://www.jmu.edu/sadah/index.shtml>) for the dates of the upcoming portfolio feedback days and for additional portfolio requirements. All scholarship awards are based on merit and vary in amount. Tours of facilities and program information sessions will take place at on-campus events.

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Degree Requirements

Degree Requirements	Credit Hours
General Education requirement ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement ²	3
Major requirements	45
University electives	28
Total ³	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirement.

³ The optional art education concentration requires more than 120 credits.

Major Requirements

The art major requires 45 credit hours in art and art history courses.

Code	Title	Credits
Foundation Courses		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3

FDS 106	Form and Space	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Subtotal		15
3-D Studio Art Courses		
Choose one of the following:		
ART 222	Introductory Ceramics	
ART 230	Introduction to Fibers Processes	
ART 240	Introductory Metals	
ART 280	Introductory Sculpture: Idea Material Process	
Subtotal		3
Studio Art Electives		
18 credit hours of studio art electives must consist of studio-based courses with an ART or FDS prefix. ¹		18
Subtotal		18
Studio Art Emphasis		
In addition to the 18 credit hours of studio art electives above, 6 additional credit hours of upper-level studio art electives (300-400 level) must be taken in a single studio area (i.e. Ceramics, Fibers, Metals, Sculpture, Printmaking and Immersive Media, Photography or Painting). Only courses with an ART prefix count toward this requirement.		6
Subtotal		6
Art History Elective		
Students must take 3 credit hours of art history at the 300-400 level. ²		3
Subtotal		3
Total Credits		45

¹ Excluding ART 200, ART 304*, ART 305, ART 491, ART 493 and ART 496.

* Students who elect the concentration in art education may count ART 304 towards this requirement.

² Students may not count ARTH 397, ARTH 490, ARTH 495, ARTH 497, or ARTH 499 towards this requirement.

Concentration in Art Education

Coordinator: William Wightman

Phone: (540) 568-5144

Email: wightmwh@jmu.edu

Students pursuing the B.S. in Art who desire to obtain K-12 Licensure may declare a concentration in art education. The concentration requires 42 credit hours, 3 of which may double-count for major requirements. An additional 3 credits may double count towards general education.

It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For a complete description of admission and retention policies and procedures for teacher education, refer to the College of Education.

Students seeking the concentration in art education professional PK-12 teaching licensure are encouraged to consult regularly with the program coordinator of art education. The undergraduate degree leading to licensure must include the following minimum requirements in art:

- Nine credits must be earned in art history and art appreciation.
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- Six credits must be earned in three-dimensional media, with one course in sculpture.
- Twelve credits must be earned in four different two-dimensional media.
- Twenty-seven studio credits, with a minimum of six credits in each of three separate studio areas.

Code	Title	Credits
Art Education Concentration Requirements		
ARED 300	Elementary Art Education Methods	3
ARED 302	Secondary Art Education Methods	3
ARED 400	Visual Arts Across the Curriculum	3
ARED 480	Student Teaching ¹	16
ARED 490	Field Experiences in Art Education ²	6
ART 304	Methods of Art Criticism	3
EDUC 200	Foundations of Education in the United States	3
LED 420	Content Area Literacy, K-12	2
PSYC 160	Life Span Human Development [C5SD]	3
Total Credits		42

¹ Taken twice for a total of 16 credits.

² Taken concurrently with ARED 300, ARED 302 and ARED 400 for 2 credits three times for a total of 6 credits.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
General Education courses		18
Credits		30
Second Year		
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Required studio art elective		3
Scientific Literacy Requirement (in addition to General Education)		3
General Education requirement		9
Studio Art electives (any level)		6
General elective		3
Credits		27
Third Year		
Art history elective (300-400 level)		3
Studio Art electives (any level)		9
General elective		3
Quantitative Requirement (in addition to General Education)		3
General Education requirement		12
Credits		30
Fourth Year		
Studio art emphasis (300-400 level)		6
General Education courses and university electives		27
Credits		33
Total Credits		120

Art, Graphic Design Major, B.F.A.

Coordinator: David Hardy

Phone: (540) 568-6833

Email: hardydl@jmu.edu

Program Description

The Bachelor of Fine Arts degree in Art with a major in graphic design is rooted in fundamental design principles, research and applications, while addressing current and emerging technologies through digital practice. The study of the historical and cultural context of the field broadens students' knowledge and nurtures their creativity.

Program objectives for the graphic design student are:

- Articulate ideas and understand the fundamental elements and principles of graphic design while using correct terminology pertinent to the field.
- Demonstrate a working knowledge of artistic methods, craft and formal structure as vehicles to enhance a solutions communicative value.
- Demonstrate proficiency in both traditional and digital ways of making.
- Develop diverse problem-solving methodologies that combine theory, research, analytical skills and conceptualization to create effective graphic design solutions.
- Develop a professional graphic design portfolio reflective of one's knowledge and abilities that encompasses the contemporary design market.

The graphic design program objectives are enhanced by frequent field trips to design-related events, studio visits, guest speakers and vetted internship opportunities which provide students with exposure to top-notch professionals in the field. On campus, students can get involved in student clubs and organizations, such as student chapters of AIGA: The Professional Association for Design. There are opportunities to engage with visiting designers and exhibitions through the Duke Hall Gallery of Fine Art as well as collaborate with faculty and peers for shows at JMU galleries. Graphic design majors also have many opportunities to gain professional experience working for a variety of JMU offices and organizations.

Admission Requirements

Architectural Design, Art Studio, Graphic Design and Industrial Design Majors

All prospective freshmen, transfer and change of major students intending to major in architectural design, graphic design or studio art must meet JMU admission requirements as stated in the Undergraduate Catalog. In addition to meeting JMU admission requirements, all students seeking the BFA with a major in architectural design, art studio or graphic design, or the BS with a major in industrial design are required to submit a portfolio to the school's SlideRoom account (<https://jmuart.slideroom.com/>). The submission of a portfolio is optional for students seeking the BA and BS with a major in art studio. All students, regardless of degree type, who wish to be considered for a scholarship must submit a portfolio to SlideRoom. The submission of a portfolio is seen as evidence of a student's interest and potential for future success in art. The deadline for submitting portfolios to SADAH's SlideRoom

site can be found on the School's Portfolio page (<https://www.jmu.edu/sadah/admissions/portfolios.shtml>).

Transfer and Change of Major Students

Transfer and change of major students seeking the BFA with a major in architectural design, art studio, or graphic design or the BS with a major in industrial design are required to submit a portfolio and also need to upload unofficial transcripts and a statement of intent to the school's SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study.

On-Campus Portfolio Feedback Days and Scholarships

All students who would like to be considered for scholarships are required to attend one of the on-campus or virtual portfolio feedback days. Students will show their original art and participate in a faculty interview. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website (<https://www.jmu.edu/sadah/index.shtml>) for the dates of the upcoming portfolio feedback days and for additional portfolio requirements. All scholarship awards are based on merit and vary in amount. Tours of facilities and program information sessions will take place at on-campus events.

Off-Campus Portfolio Feedback Days

The SADAH also offers students the opportunity for faculty critiques on their portfolio, prior to their digital submission and before an on-campus or virtual portfolio scholarship interview. The off-campus portfolio feedback events provide an opportunity for students to meet SADAH's faculty. All prospective first-year, transfer and change of major students are strongly encouraged to attend a portfolio feedback day, on or off campus, prior to submitting their portfolio.

Graphic Design

Students seeking official acceptance into the graphic design major must enroll in GRPH 208 and submit a portfolio representing work completed in GRPH 101, GRPH 200 and GRPH 206. Students not admitted may reapply the following semester.

Admission to the graphic design major is selective and competitive for a limited number of reserved seats in upper-division (300-400 level) graphic design courses. Declaration of graphic design as a major and completion of lower-division (100-200) prerequisite ART and GRPH courses does not guarantee admission into the program. Admission to upper-division GRPH courses is based on completion of all prerequisite GRPH courses and merit as determined by a faculty review of portfolios submitted in satisfaction of GRPH 208.

GRPH 208 is a 0 credit, pass/fail course that functions as a prerequisite to enrollment in all 300-400 level graphic design courses. Students should enroll in GRPH 208 during the semester following completion of lower-level GRPH courses, or concurrently with GRPH 206. GRPH 208 portfolios are reviewed each semester. Students enrolled in GRPH 208 will be emailed at the beginning of the semester and will be informed of the exact time and location of the GRPH 208 portfolio review.

Any art major may take lower-level GRPH courses, but is restricted from taking any 300-400 level graphic design course until GRPH 208 has been taken and passed.

Accepted students who receive a passing grade for GRPH 208 will be able to register for upper-division courses GRPH for the following semester and are required to furnish a Macintosh laptop computer and acquire

an Adobe Creative Cloud subscription. Purchase recommendations can be found at the Bookstore's website (<https://www.jmu.edu/bookstore/computers/recommendations.shtml>).

Students not accepted will have one opportunity to reapply in the following semester or may choose to continue in another major within the School of Art, Design and Art History.

SADAH Computer Requirements for Graphic Design

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department's website (<https://www.jmu.edu/sadah/index.shtml>).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Major requirements	78
University electives	1
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

Major Requirements

This major requires 78 credit hours in art, art history and art-related courses; 36 are graphic design required and elective courses.

Code	Title	Credits
Foundation Courses		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
FDS 108	Expressive and Expanded Drawing	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Subtotal		18

Required Art Studio Courses

Choose three of the following:		9
ART 220 or ART 222	Introductory Ceramics: Potter's Wheel Introductory Ceramics	
ART 232	Introduction to Fiber Arts: Structures	
ART 240	Introductory Metals	
ART 252	Introductory Painting	
ART 260	Introductory Black and White Darkroom Photography	
ART 270	Printmaking: Screenprint	

or ART 272	Printmaking: Relief and Intaglio	
or ART 274	Printmaking: Lithography	
ART 276	Introductory Book Arts: Materials and Structures	
ART 280	Introductory Sculpture: Idea Material Process	
Subtotal		9
Major Requirements		
ARTH 303	History of Design	3
GRPH 101	Digital Media Design I	3
GRPH 200	Image Making	3
GRPH 201	Digital Media Design II	3
GRPH 206	Typography I	3
GRPH 208	Portfolio Review	0
GRPH 306	Typography II	3
GRPH 312	Web Design	3
GRPH 313	Interactive Media	3
GRPH 408	Brand Identity	3
GRPH 410	Graphic Design Portfolio	3
GRPH 492	Topics in Graphic Design	3
Subtotal		33
Major Electives		
Choose six credit hours from the following:		6
GRPH 300	Illustration	
GRPH 301	Digital Media Design III	
GRPH 304	Package Design	
GRPH 375	Letterpress	
GRPH 376	Intermediate Book Arts: Concept, Content, Form	
GRPH 380	Wayfinding, Signage, Exhibit Design	
GRPH 390	Independent Study in Graphic Design	
GRPH 392	Topics in Graphic Design	
GRPH 496	Internship in Graphic Design ¹	
Subtotal		6
Additional Courses		
Art and design electives ²		9
Art history electives (300-400 level) ³		3
Subtotal		12
Total Credits		78

¹ A maximum of six credit hours of practicum or internship may count toward the degree.
² Art and design electives may include courses from architectural design, art education, graphic design, industrial design, studio, studio assistant and internships. Excludes all art history courses and ART 200.
³ Excludes ARTH 303, ARTH 490, ARTH 495 and ARTH 499.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
First Semester		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
Madison Foundations		3

General Education courses	3
Credits	15
Second Semester	
FDS 106 Form and Space	3
ARTH 206 Survey of World Art II: Renaissance to Modern [C2VPA]	3
GRPH 101 Digital Media Design I	3
General Education courses	6
Credits	15
Second Year	
First Semester	
FDS 108 Expressive and Expanded Drawing	3
GRPH 200 Image Making	3
Two-D or Three-D Studio Art Courses	3
General Education courses	6
Credits	15
Second Semester	
GRPH 201 Digital Media Design II	3
GRPH 206 Typography I	3
GRPH 208 Portfolio Review	0
Two-D or Three-D Studio Art Courses	3
General Education courses	6
Credits	15
Third Year	
First Semester	
ARTH 303 History of Design	3
GRPH 306 Typography II	3
GRPH 312 Web Design	3
General Education courses	6
Credits	15
Second Semester	
GRPH 313 Interactive Media	3
Two-D or Three-D Studio Art Courses	3
Art history elective (300-400 level)	3
General Education courses	6
Credits	15
Fourth Year	
First Semester	
GRPH 408 Brand Identity	3
GRPH 492 Topics in Graphic Design	3
Two-D or Three-D Studio Art Courses	3
GRPH Concentration Elective	6
General Education courses	3
Credits	18
Second Semester	
GRPH 410 Graphic Design Portfolio	3
Art and art-related electives	6
General Education courses	2
University elective	1
Credits	12
Total Credits	120

Art, Industrial Design Major, B.S.

Coordinator: Anna Smith
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Program Description

The Bachelor of Science degree in Art with a major in industrial design is an accredited professional program with an interdisciplinary, user-centered approach. Program objectives for industrial design students are to:

- Employ diverse creative problem-finding skills.
- Utilize design theory and research to understand user culture, context, and requirements.
- Demonstrate a working knowledge of industry practices, conceptualization skills, and problem-solving.
- Design and articulate complex solutions that are aesthetically and culturally appropriate, effective, and efficient.
- Exhibit proficiency in both traditional and digital ways of making.
- Critique and contextualize designs to cultivate ideas and meaningful solutions.
- Create a professional industrial design portfolio reflective of one's knowledge and abilities that encompass current expectations in the professional practice of industrial design.

The industrial design program objectives are enhanced by frequent field trips to design events, guest speakers, workshops, and internship opportunities. Design students have an opportunity to engage with design leaders in the field through field trips, events and participation in a professional design organization.

Admissions Requirements

Architectural Design, Art Studio, Graphic Design and Industrial Design Majors

All prospective freshmen, transfer and change of major students intending to major in architectural design, graphic design or studio art must meet JMU admission requirements as stated in the Undergraduate Catalog. In addition to meeting JMU admission requirements, all students seeking the BFA with a major in architectural design, art studio or graphic design, or the BS with a major in industrial design are required to submit a portfolio to the school's SlideRoom account (<https://jmuart.slideroom.com/>). The submission of a portfolio is optional for students seeking the BA and BS with a major in art studio. All students, regardless of degree type, who wish to be considered for a scholarship must submit a portfolio to SlideRoom. The submission of a portfolio is seen as evidence of a student's interest and potential for future success in art. The deadline for submitting portfolios to SADAH's SlideRoom site can be found on the School's Portfolio page.

Transfer and Change of Major Students

Transfer and change of major students seeking the BFA with a major in architectural design, art studio, or graphic design or the BS with a major in industrial design are required to submit a portfolio and also need to upload unofficial transcripts and a statement of intent to the school's SlideRoom account that explains their educational and artistic goals, articulating the reasons for choosing this area of study.

On-Campus Portfolio Feedback Days and Scholarships

All students who would like to be considered for scholarships are required to attend one of the on-campus or virtual portfolio feedback days. Students will show their original art and participate in a faculty interview. If time-based media examples (video and animation) are included, students will need to bring their own digital device for presentation purposes. Refer to the SADAH website for the dates of the upcoming portfolio feedback days and for additional portfolio requirements. All scholarship awards are based on merit and vary in amount. Tours of facilities and program information sessions will take place at on-campus events.

Off-Campus Portfolio Feedback Days

The SADAH also offers students the opportunity for faculty critiques on their portfolio, prior to their digital submission and before an on-campus or virtual portfolio scholarship interview. The off-campus portfolio feedback events provide an opportunity for students to meet SADAH's faculty. All prospective first-year, transfer and change of major students are strongly encouraged to attend a portfolio feedback day, on or off campus, prior to submitting their portfolio.

Industrial Design

Admission to the industrial design major is selective and competitive for a limited number of reserved seats in upper-division (300-400 level) industrial design courses. Declaration of industrial design as a major and completion of lower-division (100-200) prerequisite art and design courses does not guarantee admission into the program. Admission to upper-division INDU courses is based on completion of all prerequisite courses and merit as determined by faculty review of portfolios submitted in satisfaction of the course INDU 208 Portfolio Review.

After completing INDU 200, INDU 210 and INDU 220 (offered fall semester only), students enroll in INDU 202, INDU 212 and INDU 222 (all offered spring semester only); students must also enroll in INDU 208, a 0 credit, pass/fail course that functions as a prerequisite to enrollment in 300-400 level industrial design advanced studio courses.

The INDU 208 portfolios are reviewed during the spring semester, prior to registration for the following fall semester. The exact time and location for the INDU 208 portfolio review will be announced by INDU faculty to all INDU sophomore level (200) courses in the spring semester, well in advance of the portfolio review date.

INDU 200, INDU 210, and INDU 220 are restricted to BS of ID majors who have passed FDS 102, FDS 104 and FDS 106 only, and students are restricted from taking any 300-400 level industrial design advanced studio courses until INDU 202, INDU 212, INDU 222, and INDU 208 have been taken and passed.

Accepted students who receive a passing grade for INDU 208 will be able to register for INDU upper division advanced studio courses for the following semester. Students not accepted will have one opportunity to reapply within the following academic year or may choose to continue in another major within the SADAH.

SADAH Computer Requirements for Industrial Design

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department's website (<https://www.jmu.edu/sadah/index.shtml>).

Degree Requirements

Degree Requirements	Credit Hours
General Education requirements ¹	41
Quantitative requirement (in addition to General Education courses)	3
Scientific literacy requirement (in addition to General Education courses) ¹	3
Major requirements	78
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major. Three credits of General Education and three credits of the B.S. scientific literacy requirement are fulfilled the major requirements, making the total 120 credits.

Major Requirements

The industrial design major requires 78 credit hours in art, art history and art-related courses. Of these hours, 42 credits must be in INDU courses.

Code	Title	Credits
Foundation Courses		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
Subtotal		9
Required Major Courses		
GRPH 101	Digital Media Design I	3
INDU 200	Design Matters	3
INDU 202	Industrial Design Studio	3
INDU 208	Portfolio Review	0
INDU 210	Making Is Thinking	3
INDU 212	Materials and Processes for Manufacturing	3
INDU 220	Visual Communication for Industrial Design I	3
INDU 222	Visual Communication for Industrial Design II	3
INDU 360	Professional Practice in Industrial Design	3
Subtotal		24
Advanced Studios		
Choose 24 credit hours from the following:		24
INDU 300	Product Design	
INDU 302	Transportation Studio	
INDU 400	Systems and Services	
INDU 402	Social Innovation and Social Entrepreneurship	
INDU 404	Industrial Design Advanced Special Topics Studio	
INDU 406	Senior Thesis Studio (Instructor Permission Required)	
INDU 408	Swing Studio Option	
Subtotal		24

Art and Design History Courses

ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
ARTH 303	History of Design	3
Subtotal		9

Art and Design Electives

Art and design electives may include studio-based courses from industrial design, graphic design, architectural design, studio, studio assistantships and internships. Excludes courses taken to fulfill the concentration courses, advanced studios, and art and design history. ¹		6
Subtotal		6

Interdisciplinary Electives

Society and Culture. Choose one of the following:		3
ANTH 195	Cultural Anthropology [C4GE]	
ANTH 205	Buried Cities, Ancestral Places: The Rise and Fall of Early Human Societies [C2HQC]	
ANTH 323	Anthropology and Photography	
SOCI 311	Sociology of the Environment	
SOCI 315	Technology and Society	
SOCI 337	Sociology of Gender	

Business, Marketing and Entrepreneurship. Choose one of the following:		3
COB 318	Legal and Ethical Environment of Business	
MKTG 380	Principles of Marketing	
MGT 305	Organizational Behavior and Human Resource Management	
MGT 310	Business Fundamentals for Entrepreneurs	
MGT 472	New Venture Creation	
Subtotal		6

Total Credits		78
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¹ Also excludes ART 200, ART 105, ART 304 and ART 305.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
FDS 106	Form and Space	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
General Education		12
Quantitative Reasoning		3
Credits		30
Second Year		
INDU 200	Design Matters	3
INDU 202	Industrial Design Studio	3
INDU 208	Portfolio Review	0
INDU 210	Making Is Thinking	3
INDU 212	Materials and Processes for Manufacturing	3
INDU 220	Visual Communication for Industrial Design I	3
INDU 222	Visual Communication for Industrial Design II	3
GRPH 101	Digital Media Design I	3

General Education and Interdisciplinary Elective	9
Credits	30
Third Year	
ARTH 303 History of Design	3
INDU 360 Professional Practice in Industrial Design	3
General Education courses and university elective	9
Advanced Studios	12
Art and Design Elective	3
Credits	30
Fourth Year	
Advanced Studios	12
Art and Design Electives	3
General Education	9
Interdisciplinary Elective	6
Credits	30
Total Credits	120

Biology, B.A.

Program Description

The department offers a four-year B.A. degree program for a major in biology and for a major in biology qualifying for the Secondary Collegiate Professional License and is distinguished from the four-year B.S. by two key university requirements (the completion of an intermediate level foreign language and three credit hours in philosophy) and several key program requirements (a capstone experience, writing courses, and required Integrated Science and Technology and Anthropology courses). Students may not major in both biology and biotechnology.

Credit by Examination

When evidence of sufficient background or preparation is presented, the Department of Biology offers credit by examination in many of its non-lab courses at the discretion of the course instructor or coordinator. Students seeking such credit should make arrangements with the course instructor or coordinator and obtain approval of the department head.

Teaching Licensure

Biology majors need courses in physics, geology and organic chemistry for many science education positions.

In addition to the general education and academic major requirements, biology majors desiring secondary teacher licensure must be admitted to teacher education, complete the pre-professional program in secondary education at the undergraduate level and complete the graduate level Master of Arts in Teaching degree.

It is critical that students seeking licensure consult regularly with both their education advisor and their major advisor to support their progression through the programs. For a full description of the program in secondary teaching, refer to the Department of Middle, Secondary and Mathematics Education in addition to the College of Education.

Special Requirements

In order for BIO 270/BIO 270L and BIO 290/BIO 290L credit to be transferred, both semesters of an Anatomy and Physiology course (A&P I and A&P II) must be completed with a minimum grade of "C" from the same institution. A single semester of either of these sequences transfers as BIO 000. Only BIO 290/BIO 290L can be applied toward biology major electives; BIO 270/BIO 270L does not apply toward biology electives.

Practical hands-on experience in the field and/or laboratory is important content of laboratory-based courses. Transfer credit for courses including laboratories will only be awarded for those having skill and application content comparable to that of JMU courses. Elective biology credits applicable toward the major or minor are transferred as specific courses or as BIO 200, BIO 426 or BIO 427. A maximum of eight credit hours of first year (100) level BIO courses may be applied to the biology major or minor requirements.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language requirements ²	0-12
Philosophy requirement (in addition to General Education courses)	3

Biology Major and Capstone ³ requirements	36-39
Cognate requirements	26
University Electives	9-24
Total	120

- ¹ The General Education program contains a set of requirements each student must fulfill. Some requirements are fulfilled by courses within the major. It is strongly recommended that students work with their advisors to identify strategic selections for their general education requirements. The Natural World (10 credit hours) is entirely covered by the Biology B.A. major requirements, and therefore those credits are accounted for only once in the 120 credit hour total.
- ² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures' placement test.
- ³ If the capstone experience is comprised of BIO credit, this can also count toward the 20 credits of BIO electives, reducing the total number of required courses.

Major Requirements

Biology majors seeking a B.A. must complete 36 credit hours of biology courses including 20 credit hours of biology electives, plus a capstone course (3 cr. hr., which may or may not be BIO credit). Specific requirements include 16 credit hours of four core courses (BIO 140/BIO 140L, BIO 150/BIO 150L, BIO 240/BIO 240L and BIO 250/BIO 250L), at least two upper-division laboratory courses, and one course from a list of courses with an emphasis on organismal diversity. In addition, Biology B.A. majors must complete a set of required cognate courses in chemistry, mathematics/statistics, writing, ISAT and anthropology. Students are encouraged to participate in independent research with a faculty mentor and are required to participate in a capstone experience. Credits earned doing research in biology will count toward the biology major, but some restrictions apply.

When requested, senior biology majors are expected to participate in program assessment test activities as a graduation requirement. Assessment information helps the department modify the curriculum to meet student needs.

Students completing the biology major must earn a minimum grade of "C-" in BIO 140, BIO 140L, BIO 150, BIO 150L, CHEM 131, CHEM 131L and CHEM 132.

Code	Title	Credits
Core Requirements		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	4
Subtotal		16
Additional Biology Requirements		

Students pursuing the B.A. in biology must complete at least 20 credit hours of approved biology courses beyond the core; at least 14 of these must be at the 300 and 400 levels. Up to 6 credits of 200-level BIO courses may count towards the 20 credit requirement, excluding BIO 270/BIO 270L.

At least one BIO course must be from the list of 300/400-level organismal diversity courses (see list below).

Two BIO 300/400-level courses must have a laboratory component (see list below).

Courses in both organismal diversity and laboratory lists may count for both requirements.

Three credits of independent research (BIO 496, up to a maximum of 2 credits, BIO 497 and/or BIO 499A/BIO 499B/BIO 499C) may be used for one, but only one, of the laboratory courses.

Subtotal 20

Capstone Requirement

Students must complete a Capstone experience, which will be decided upon with the input of the student's faculty advisor. In addition to the courses listed below, capstone experiences may include a study abroad experience, a JMU X-Labs class, or other class as approved by the advisor.¹

BIO 493 Pre-Veterinary Student Internship

BIO 494 Internship in Biology

BIO 497 Advanced Biological Research

BIO 499A Senior Project

BIO 499B Senior Project

BIO 499C Senior Project

ENVT 400 Capstone Seminar in Environmental Problem Solving

HON 499A Honors Senior Project

Subtotal 0-3

Cognate Requirements

Chemistry Cognate Requirements:²

CHEM 131 General Chemistry I [C3PP] 3

CHEM 131L General Chemistry Laboratory [C3L] 1

CHEM 132 General Chemistry II 3

CHEM 132L General Chemistry Laboratory 1

Subtotal 8

Mathematics and Statistics Cognate Requirements

Choose one in each of the following sets of courses. Mathematics and Statistics beyond the minimum requirement is desirable in many areas of biology. Consult your advisor about which courses are appropriate.

MATH 220 Elementary Statistics [C3QR] 3

or MATH 229 Introduction to Applied Statistics Using R [C3QR]

Choose one of the following: 3

BIO 454 Introduction to Biometrics

BIO 456 Ecological Models and Data

DATA 200 Introduction to Data

MATH 231 Calculus with Functions I [C3QR]

MATH 235 Calculus I [C3QR]

MATH 268 Data Analysis and Visualization

MATH 309 SAS Programming and Data Management

MATH 321 Analysis of Variance and Experimental Design

MATH 322 Applied Linear Regression

MATH 324 Applied Nonparametric Statistics

MATH 325 Survey Sampling Methods

MATH 327 Categorical Data Analysis

MATH 354 Introduction to Biometrics

MATH 368 Statistical Computing

Subtotal 6

Transdisciplinary Cognate Requirements

Additional recommended courses will depend on the career goals of the student and should be discussed with the advisor.

ANTH 195 Cultural Anthropology [C4GE] 3

ISAT 171 Governance of Sociotechnical Systems 3

Choose two of the following: 6

ANTH 360 Medical Anthropology

ANTH 373 Anthropology and the Environment

ENVT 303 Topics in Environment

ENVT/REL 366 Religion, Animals and the Environment

GEOG 323 The Geography of Human Genetics, Infectious Diseases and Diet

GEOG 325 Environmental Ethics

GEOG 330 Weather, Climate and Society

GEOG 342 Management and Protection of Natural Resources

GEOG/ISAT 429 Sustainability: An Ecological Perspective

ISAT 112 Environmental Issues in Science and Technology (2,3) [C3PP]

ISAT 271 Technology, Science and Society

ISAT 320 Fundamentals of Environmental Science and Technology I

ISAT/WGSS 485 Gender Issues in Science

PHIL 410 Philosophy of Science

REL 372 Religion and Medicine, East and West

REL 380 Religion and Science

SCOM 354/ WRTC 326 Environmental Communication and Advocacy

SCOM 370 Introduction to Health Communication

SCOM 465/ WRTC 416 Rhetoric of Environmental Science and Technology

SOCI 311 Sociology of the Environment

SOCI 315 Technology and Society

SOCI 359 The Sociology of Climate Denial

SOCI 375 Medical Sociology

UNST 389 Campus Health Communication

WRTC 358 Writing About Science and Technology

WRTC 458 Scientific and Medical Communication

Subtotal 12

Total Credits

62-65

¹ If the capstone course is a BIO course, it may be double-counted toward the 20 hours of biology electives, which accounts for the range of 36-39 biology major credits.

² Additional courses in chemistry (i.e. CHEM 241, CHEM 242, CHEM 353, CHEM 354, CHEM 355 and/or BIO 361) are all valuable courses toward

careers in biology, and should be discussed with an advisor as elective options based on a student's career interests.

Organismal diversity and laboratory courses

Code	Title	Credits
Choose at least one organismal diversity course from the following list:		3-4
BIO 310	Marine Biology	
BIO 320	Comparative Anatomy of Vertebrates	
BIO 325	Medical Parasitology	
BIO 327	Biological Illustration	
BIO 340	Morphology and Anatomy of Vascular Plants	
BIO 357	Galápagos Islands: Biodiversity and Conservation	
BIO 360	Plant Biology	
BIO 364	Human Uses of Plants	
BIO 386	Field Botany	
BIO 402	Forest Ecology	
BIO/GEOL 405	Vertebrate Paleontology	
BIO 412	Mammalogy	
BIO 419	Animals and their Environments	
BIO 430	Ornithology: Research and Current Advances	
BIO 449	Insect Ecology	
BIO 474	Reproductive Physiology	
BIO 484	Aquatic Microbiology	
BIO 486	Systematics of Vascular Plants	
Choose at least two laboratory courses from the following list:		2-8
BIO 318	Experimental Vertebrate Development	
BIO 320	Comparative Anatomy of Vertebrates	
BIO 340	Morphology and Anatomy of Vascular Plants	
BIO 346	Laboratory in Bacterial Pathogenomics	
BIO 348L	Medical Microbiology Laboratory	
BIO 350	Paleobiology	
BIO 354	Global Climate Change and Life: Ecological and Biological Impacts of Climate Variability	
BIO 360L	Plant Biology Laboratory	
BIO 364L	Laboratory in Human Uses of Plants	
BIO 370	Human and Animal Physiology	
BIO 380	Advanced Microbiology Laboratory Techniques	
BIO 386	Field Botany	
BIO 387L	Environmental Microbiology Laboratory	
BIO/GEOL 400	Geology and Ecology of the Bahamas	
BIO 402	Forest Ecology	
BIO 410	Advanced Human Anatomy	
BIO 412	Mammalogy	
BIO 419	Animals and their Environments	
BIO 425L	Immunology Laboratory	
BIO 427	Topics in Biology with Laboratory	
BIO 430	Ornithology: Research and Current Advances	
BIO 432	Light Microscopy	
BIO 445	Neurobiology	
BIO 446	Experimental Neurobiology	
BIO 449	Insect Ecology	

BIO 452	Population Ecology
BIO 454	Introduction to Biometrics
BIO 455	Plant Physiology
BIO 456	Ecological Models and Data
BIO 457	Biological Applications of Geographic Information Systems
BIO 459	Freshwater Ecology
BIO 460	Plant Biotechnology
BIO 480L	Advanced Molecular Biology Lab
BIO 481	Genomics
BIO 482	Human Histology
BIO 483	Bioinformatics
BIO 484	Aquatic Microbiology
BIO 486	Systematics of Vascular Plants
BIO 490	Mechanics of Animal Movement

Additional Information

One Topics in Biology (BIO 426/BIO 427) may be used to fulfill the organismal diversity requirement, with departmental permission. One BIO 427 can be used to satisfy an upper-level laboratory requirement, with departmental permission. BIO 426 cannot be used to satisfy a laboratory requirement. These course numbers can be repeated with a change in topic, but only 12 credits from BIO 426/BIO 427 can be applied toward the 36-hour biology course requirement.

When choosing additional biology courses to complete the 36 credit hour requirement, students are strongly encouraged to discuss their career interests with an advisor who can help select courses best suited to their needs. Students are encouraged to participate in independent research and teaching courses with a faculty mentor, though a maximum of eight credits of BIO 492, BIO 493, BIO 494, BIO 496, BIO 497, BIO 499A/BIO 499B/BIO 499C and ISCI 450 can be counted toward the biology major.

Biology Concentrations

Anatomy and Physiology Concentration

Concentration Coordinator: Jess Marquardt

Email: marquajl@jmu.edu

The Biology Department offers a concentration in anatomy and physiology for all Biology majors with an interest in exploring the relationship between structure and function of humans and/or animals. Further, the concentration will provide a strong foundation for students interested in clinical health care, veterinary medicine, and/or research in related fields.

Students choosing a concentration in anatomy and physiology must meet all the requirements for the biology major as well as the following additional requirements. Courses for this concentration can also satisfy biology major requirements.

Complete a total of at least 17 credit hours from the following courses.

Code	Title	Credits
Anatomy Core Requirements		
Choose one of the following:		4
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	1,2,3
BIO 320	Comparative Anatomy of Vertebrates	

Subtotal	4
Physiology Core Requirements	
BIO 370 Human and Animal Physiology ²	4
Subtotal	4
Anatomy and Physiology Electives	
Complete 9 hours from the following:	9
Choose at least one course from the following primarily anatomical or physiological courses: ⁴	
BIO 318 Experimental Vertebrate Development	
BIO 320 Comparative Anatomy of Vertebrates ³	
BIO 410 Advanced Human Anatomy ²	
BIO 412 Mammalogy ^{1,2}	
BIO 467 Human Cardiovascular Physiology	
BIO 468 Endocrinology	
BIO 472 Human Metabolism	
BIO 474 Reproductive Physiology ¹	
BIO 482 Human Histology ²	
BIO 490 Mechanics of Animal Movement ²	
Choose remaining credit hours from the list of primarily anatomical or physiological courses above, or from the following courses:	
BIO 301 Introductory Neuroscience	
BIO 304 Cell and Molecular Biology	
BIO 425 Immunology	
BIO 425L Immunology Laboratory	
BIO 445 Neurobiology ²	
BIO 446 Experimental Neurobiology ²	
BIO 451 Current Topics in Human Development and Evolution	
Subtotal	9
Total Credits	17

¹ Meets the biology major requirement for an organismal diversity course.

² Meets one of the biology major requirements for laboratory/field courses.

³ May be used to fulfill either the Anatomy Core Requirement or as an Anatomy and Physiology Elective, but not both.

⁴ Note: BIO 270 and BIO 270L cannot be used to fulfill this requirement.

Ecology and Environmental Biology Concentration

Concentration Coordinator: Bruce Wiggins

Email: wigginba@jmu.edu

The biology department offers a concentration within the biology major for students with interest in ecology, field biology, natural resources, environmental biology, conservation biology, evolution, animal behavior and organismal biology. Students choosing a concentration in ecology and environmental biology must meet all of the requirements for the biology major as well as the following additional requirements. Courses for this concentration can also satisfy biology major requirements.

Students must complete at least 20 credit hours from the following list.

With prior approval from the concentration coordinator, BIO 426 and/or BIO 427 may be substituted. Students are strongly encouraged to discuss their career interests with an advisor who can help select courses best

suited to their needs. In addition to the courses listed below, a maximum of eight credits of BIO 496, BIO 497 and BIO 499A/BIO 499B/BIO 499C can be counted toward the EEB concentration.

Code	Title	Credits
Ecology and Environmental Biology Concentration Courses		
Select at least 20 credit hours from the following list:		20
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory ¹	
BIO 310	Marine Biology ¹	
BIO 320	Comparative Anatomy of Vertebrates ^{1,2}	
BIO 327	Biological Illustration	
BIO 340	Morphology and Anatomy of Vascular Plants ^{1,2}	
BIO/MATH 342	Mathematical Models in Biology	
BIO 350	Paleobiology ²	
BIO 354	Global Climate Change and Life: Ecological and Biological Impacts of Climate Variability	
BIO 357	Galápagos Islands: Biodiversity and Conservation ¹	
BIO 360	Plant Biology ¹	
BIO 370	Human and Animal Physiology ²	
BIO 386	Field Botany ^{1,2}	
BIO 387	Environmental Microbiology	
BIO 387L	Environmental Microbiology Laboratory ²	
BIO/PSYC 395	Comparative Animal Behavior	
BIO/GEOL 400	Geology and Ecology of the Bahamas ²	
BIO 402	Forest Ecology ²	
BIO 404	Evolutionary Analysis	
BIO/GEOL 405	Vertebrate Paleontology ¹	
BIO 412	Mammalogy ^{1,2}	
BIO 418	Conservation Biology	
BIO 419	Animals and their Environments ^{1,2}	
BIO 430	Ornithology: Research and Current Advances ^{1,2}	
BIO 447	Evolution and Ecology of Infectious Disease	
BIO 449	Insect Ecology ^{1,2}	
BIO 451	Current Topics in Human Development and Evolution	
BIO 452	Population Ecology ²	
BIO 453	Microbial Ecology and Evolution	
BIO 454	Introduction to Biometrics ²	
BIO 455	Plant Physiology ²	
BIO 456	Ecological Models and Data ²	
BIO 457	Biological Applications of Geographic Information Systems ²	
BIO 459	Freshwater Ecology ²	
BIO 484	Aquatic Microbiology ^{1,2}	
BIO 486	Systematics of Vascular Plants ^{1,2}	
Subtotal		20
Cognate Courses		
Students must take two additional cognate courses: ³		
Choose one Statistics course:		3-4
MATH 321	Analysis of Variance and Experimental Design	
MATH 322	Applied Linear Regression	
MATH 324	Applied Nonparametric Statistics	

MATH 325	Survey Sampling Methods	
BIO 454	Introduction to Biometrics ²	
BIO 456	Ecological Models and Data ²	
Choose one Geographic Information Systems course:		3-4
GEOG 215	Introduction to Geographic Information Science and Cartography	
BIO 457	Biological Applications of Geographic Information Systems ²	
Subtotal		6-8
Total Credits		26-28

¹ Meets the biology major requirement for an organismal diversity course.

² Meets one of the biology major requirements for laboratory/field courses.

³ Note: Biology courses that fulfill these additional cognates (BIO 454, BIO 456, and BIO 457) can also be used to fulfill other Biology major requirements.

Ecology and Environmental Biology Concentration Additional Information

It is highly recommended that students take additional upper-level degree elective courses in geography/GIS (such as GEOG 466 or GEOG 467) and in statistics (such as MATH 322 or MATH 324). Students should consult with their advisor about which courses are appropriate.

Microbiology Concentration

Concentration Coordinator: Steve Cresawn

Email: cresawsg@jmu.edu

The biology department offers a concentration within the biology major for students with interest in bacteriology, virology, immunology, parasitology and infectious disease. Students choosing a concentration in microbiology must meet all the requirements for the biology major, including the following specific requirements.

Complete a total of at least 18 credit hours from the following courses.

Code	Title	Credits
General Microbiology Requirements		
BIO 245	General Microbiology	3
BIO 245L	General Microbiology Laboratory	1
BIO 246	Microbiology Seminar	1
Subtotal		5
Laboratory Requirement		
Choose at least one of the following laboratory courses:		1-3
BIO 203	Viral Discovery	
ISAT 203	Viral Discovery	
BIO 204	Viral Genome and Bioinformatics	
ISAT 204	Viral Genome and Bioinformatics	
BIO 346	Laboratory in Bacterial Pathogenomics ¹	
BIO 348L	Medical Microbiology Laboratory ¹	
BIO 380	Advanced Microbiology Laboratory Techniques ¹	
BIO 387L	Environmental Microbiology Laboratory ¹	
BIO 425L	Immunology Laboratory ¹	
BIO 484	Aquatic Microbiology ¹	
Subtotal		1-3

Microbial Requirements

If a student takes an elective course that is listed in both Microbial Pathogenesis and Microbial Ecology categories below, the course may only satisfy one category requirement, but not both.

Microbial Pathogenesis - Choose at least one of the following: 3

BIO 325	Medical Parasitology ²
BIO 348	Medical Microbiology
BIO 425	Immunology
BIO 444	Virology ³
BIO 447	Evolution and Ecology of Infectious Disease ³

Microbial Ecology - Choose at least one of the following: 3

BIO 387	Environmental Microbiology
BIO 444	Virology ³
BIO 447	Evolution and Ecology of Infectious Disease ³
BIO 453	Microbial Ecology and Evolution
BIO 484	Aquatic Microbiology ²

Subtotal 6

Microbiology Electives

Microbiology courses from any of the above listed courses to reach 18 credit hours. 4-6

Subtotal 4-6

Total Credits 18

¹ Meets one of the biology major requirements for an upper-level BIO 300-400 level laboratory course.

² Meets one of the biology major requirements for an organismal diversity course.

³ May be used to fulfill either the Microbial Ecology or Microbial Pathogenesis requirement, but not both.

Neuroscience Concentration

Concentration Coordinator: George Vidal

Email: vidalgx@jmu.edu

The biology department offers a concentration within the biology major for students with interests in cell and molecular, systems, and behavioral aspects of the nervous system and its functionality. Students choosing a concentration in neuroscience must meet all of the requirements for the biology major in addition to the following requirements. Courses for this concentration can also satisfy biology major requirements.

Code	Title	Credits
Neuroscience Concentration Courses		
Choose one of the following:		3
BIO 301	Introductory Neuroscience	
BIO/PSYC 385	Biopsychology	
Choose one of the following:		3-4
BIO/CSD 415	Neuroanatomy and Neurogenic Communication Disorders ¹	
BIO 445	Neurobiology ²	
Subtotal		6-7
Neuroscience Concentration Electives		
Complete 9 credits of the following:		9
Choose at least one course from the following principally neurobiology courses:		
BIO/PSYC 375	Sensation and Perception	

BIO/CSD 415	Neuroanatomy and Neurogenic Communication Disorders ¹	
BIO 443	Understanding Techniques in Neuroscience	
BIO 445	Neurobiology ^{1,2}	
BIO 446	Experimental Neurobiology ²	
BIO 448	Developmental Neurobiology	
BIO 478	Aging and Neurodegeneration	
Choose remaining credit hours from the list of principally neurobiology courses above, or from the following courses:		
BIO 316	Animal Development ²	
BIO 480	Advanced Molecular Biology	
BIO 480L	Advanced Molecular Biology Lab ²	
BIO 481	Genomics ²	
BIO 482	Human Histology ²	
Subtotal		9
Total Credits		15-16

¹ BIO 445 and BIO 415/CSD 415: The course not taken as part of required neuroscience concentration core may be used to fulfill elective requirements.

² Meets one of the biology major requirements for a laboratory/field course.

Neuroscience Concentration Additional Information

Students should choose additional BIO courses at the 300-400 level, including a course that fulfills the organismal diversity major requirement, and are strongly encouraged to discuss their career interests with an advisor who can help select courses best suited to their needs. With prior approval from the concentration coordinator, BIO 426 and/or BIO 427 may be substituted. It is recommended that students participate in independent research (BIO 496/BIO 497/BIO 499A/BIO 499B/BIO 499C) with a faculty mentor.

Recommended Schedule for Majors

First semester first year biology majors are encouraged to start with a 14-15 hour course load. This will generally include BIO 140/BIO 140L, CHEM 131 and CHEM 131L, and/or a math course plus General Education. The work load will then be increased in the second semester based on the level of first semester success.

Course	Title	Credits
First Year		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
Mathematics course (General Education: The Natural World)		3
Madison Foundations ¹		9
Credits		28
Second Year		
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	4
Mathematics Course (fulfills General Education: The Natural World)		3

B.A. degree requirements (foreign language, philosophy) and/or university electives	12
General Education ²	9
Credits	32
Third Year	
BIO electives, to include a BIO 300-400 level lab and an organismal diversity course	10
Transdisciplinary Cognates	6
B.A. degree requirements (foreign language) and/or university electives	7
General Education	7
Credits	30
Fourth Year	
BIO electives, to include a BIO 300-400 level laboratory (7.00 if Capstone is a BIO course)	7
Capstone experience	3
Transdisciplinary Cognates	3
General Education and university electives	17
Credits	30
Total Credits	120

¹ including WRTC 103

² including ANTH 195

Biology, B.S.

Program Description

The department offers a four-year B.S. degree program for a major in biology and for a major in biology qualifying for the Secondary Collegiate Professional License. Students may not major in both biology and biotechnology.

When requested, senior biology majors are expected to participate in program assessment test activities as a graduation requirement. Assessment information helps the department modify the curriculum to meet student needs.

Credit by Examination

When evidence of sufficient background or preparation is presented, the Department of Biology offers credit by examination in many of its non-lab courses at the discretion of the course instructor or coordinator. Students seeking such credit should make arrangements with the course instructor or coordinator and obtain approval of the department head.

Teaching Licensure

Biology majors need courses in physics, geology and organic chemistry for many science education positions.

In addition to the general education and academic major requirements, biology majors desiring secondary teacher licensure must be admitted to teacher education, complete the pre-professional program in secondary education at the undergraduate level and complete the graduate level Master of Arts in Teaching degree.

It is critical that students seeking licensure consult regularly with both their education advisor and their major advisor to support their progression through the programs. For a full description of the program in secondary teaching, refer to the Department of Middle, Secondary and Mathematics Education, in addition to the College of Education.

Special Requirements

In order for BIO 270/BIO 270L and BIO 290/BIO 290L credit to be transferred, both semesters of an Anatomy and Physiology course (A&P I and A&P II) must be completed with a minimum grade of "C" from the same institution. A single semester of either of these sequences transfers as BIO 000. Only BIO 290/BIO 290L can be applied towards biology major electives; BIO 270/BIO 270L does not apply towards biology electives.

Practical hands-on experience in the field and/or laboratory is important content of laboratory-based courses. Transfer credit for courses including laboratories will only be awarded for those having skill and application content comparable to that of JMU courses. Elective biology credits applicable toward the major or minor are transferred as specific courses or as BIO 200, BIO 426 or BIO 427. A maximum of eight credit hours of first year (100) level BIO courses may be applied to the biology major or minor requirements.

Biology, Forest Biology, Dual Degree

Students are able to enroll in a dual degree program that makes it possible for the student to earn a Bachelor of Science (B.S) degree in Biology from James Madison University and a Master of Forestry (M.F.)

degree from Virginia Tech in five years. See the dual degree program tab (p. 209) for more information.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education) ²	3-4
Biology requirements	40
Cognate requirements	24-30
Total	120

- ¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.
- ² The quantitative and scientific literacy requirements may be met by courses required in the major

Major Requirements

Biology majors must complete 40 credit hours of biology courses including 16 credit hours at the 300 and 400 level. Specific requirements include four core courses (BIO 140/BIO 140L, BIO 150/BIO 150L, BIO 240/BIO 240L and BIO 250/BIO 250L), at least two upper-division laboratory courses and one course from a list of courses with an emphasis on organismal diversity. In addition, B.S. biology majors must complete a set of cognate courses in chemistry, mathematics, statistics and physics that are listed below. Students are encouraged to participate in independent research with a faculty mentor. Credits earned doing research will count toward the biology major but some restrictions apply.

Students completing the biology major must earn a minimum grade of "C-" in BIO 140,BIO 140L, BIO 150, BIO 150L, CHEM 131, CHEM 131L and CHEM 132.

Code	Title	Credits
Core Requirements		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	4
Subtotal		16
Additional Biology Requirements		
Students in the B.S. biology major must complete at least 24 credit hours of approved biology courses and at least 16 of these must be at the 300 and 400 levels. Up to 8 credit hours of 200 level BIO courses may count towards the 24 credit hour requirement, excluding BIO 270/BIO 270L.		24

At least one course must be from the list of 300/400-level organismal diversity courses (see list below).

Two 300/400-level courses must have a laboratory component (see list below).		
Courses in both lists may count for both requirements.		
Three credits of independent research (BIO 496 (up to a maximum of 2 credit hours), BIO 497 and/or BIO 499A/BIO 499B/BIO 499C) may be used for one, but only one, of the laboratory courses.		
Subtotal		24
Cognate Requirements		
<i>Chemistry Cognate Requirement</i>		
Chemistry beyond the minimum requirement is recommended in many areas of biology. Consult with your advisor about which courses are appropriate. Students who take BIO 361/CHEM 361 in addition to CHEM 242, CHEM 353, CHEM 354 or GEOL 355/CHEM 355 will have BIO 361/CHEM 361 applied toward their BIO 300-400-level elective requirements. BIO 361/CHEM 361 may not be used for both a chemistry cognate and a biology major elective.		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
CHEM 241	Organic Chemistry I	3
Choose one of the following:		3
BIO/CHEM 361	Biochemistry I	
CHEM 242	Organic Chemistry II	
CHEM 353	Environmental Chemistry	
CHEM 354	Environmental Chemistry Field Camp (summer only)	
CHEM/GEOL 355	Geochemistry of Natural Waters	
Subtotal		14
<i>Statistics Cognate Requirement</i>		
Statistics beyond the minimum requirement is desirable in many areas of biology. Consult your advisor about which courses are appropriate. Choose one of the following:		3-4
MATH 220	Elementary Statistics [C3QR]	
MATH 229	Introduction to Applied Statistics Using R [C3QR]	
Subtotal		3-4
<i>Mathematics Cognate Requirement</i>		
Mathematics beyond the minimum requirement is desirable in many areas of biology. Consult your advisor about which courses are appropriate. Choose one of the following:		3-4
BIO 454	Introduction to Biometrics	
BIO 456	Ecological Models and Data	
DATA 200	Introduction to Data (This course does not fulfill the B.S. quantitative requirement.)	
MATH 231	Calculus with Functions I [C3QR]	
MATH 235	Calculus I [C3QR]	
MATH 268	Data Analysis and Visualization	
MATH 309	SAS Programming and Data Management	
MATH 321	Analysis of Variance and Experimental Design	
MATH 322	Applied Linear Regression	
MATH 324	Applied Nonparametric Statistics	
MATH 325	Survey Sampling Methods	
MATH 327	Categorical Data Analysis	
MATH 354	Introduction to Biometrics	

MATH 368	Statistical Computing	
Subtotal		3-4
<i>Physics Cognate Requirement</i>		
Choose one of the following options:		4-8
PHYS 125 & 125L	Physics with Biological Applications [C3PP] and Physics with Biological Applications Laboratory [C3L]	
PHYS 140 & 140L & PHYS 150 & PHYS 150L	College Physics I [C3PP] and General Physics Laboratory I [C3L] and College Physics II and General Physics Laboratory II	
PHYS 240 & 240L & PHYS 250 & PHYS 250L	University Physics I [C3PP] and University Physics Laboratory I [C3L] and University Physics II and University Physics Laboratory II	
Subtotal		4-8
Total Credits		64-70

¹ Chemistry beyond the minimum requirement is recommended in many areas of biology. Consult with your advisor about which courses are appropriate. Students who take BIO 361/CHEM 361 in addition to CHEM 242, CHEM 353, CHEM 354 or GEOL 355/CHEM 355 will have BIO 361/CHEM 361 applied toward their BIO 300-400-level elective requirements. BIO 361/CHEM 361 may not be used for both a chemistry cognate and a biology major elective.

Organismal diversity and laboratory courses

Code	Title	Credits
Choose at least one organismal diversity course from the following list:		
BIO 310	Marine Biology	
BIO 320	Comparative Anatomy of Vertebrates	
BIO 325	Medical Parasitology	
BIO 327	Biological Illustration	
BIO 340	Morphology and Anatomy of Vascular Plants	
BIO 357	Galápagos Islands: Biodiversity and Conservation	
BIO 360	Plant Biology	
BIO 364	Human Uses of Plants	
BIO 386	Field Botany	
BIO 402	Forest Ecology	
BIO 405	Vertebrate Paleontology	
GEOL 405	Vertebrate Paleontology	
BIO 412	Mammalogy	
BIO 419	Animals and their Environments	
BIO 430	Ornithology: Research and Current Advances	
BIO 449	Insect Ecology	
BIO 474	Reproductive Physiology	
BIO 484	Aquatic Microbiology	
BIO 486	Systematics of Vascular Plants	
Choose at least two laboratory courses from the following list:		2-8
BIO 318	Experimental Vertebrate Development	
BIO 320	Comparative Anatomy of Vertebrates	
BIO 340	Morphology and Anatomy of Vascular Plants	
BIO 346	Laboratory in Bacterial Pathogenomics	

BIO 348L	Medical Microbiology Laboratory
BIO 350	Paleobiology
BIO 354	Global Climate Change and Life: Ecological and Biological Impacts of Climate Variability
BIO 360L	Plant Biology Laboratory
BIO 364L	Laboratory in Human Uses of Plants
BIO 370	Human and Animal Physiology
BIO 380	Advanced Microbiology Laboratory Techniques
BIO 386	Field Botany
BIO 387L	Environmental Microbiology Laboratory
BIO/GEOL 400	Geology and Ecology of the Bahamas
BIO 402	Forest Ecology
BIO 410	Advanced Human Anatomy
BIO 412	Mammalogy
BIO 419	Animals and their Environments
BIO 425L	Immunology Laboratory
BIO 427	Topics in Biology with Laboratory
BIO 430	Ornithology: Research and Current Advances
BIO 432	Light Microscopy
BIO 445	Neurobiology
BIO 446	Experimental Neurobiology
BIO 449	Insect Ecology
BIO 452	Population Ecology
BIO 454	Introduction to Biometrics
BIO 455	Plant Physiology
BIO 456	Ecological Models and Data
BIO 457	Biological Applications of Geographic Information Systems
BIO 459	Freshwater Ecology
BIO 460	Plant Biotechnology
BIO 480L	Advanced Molecular Biology Lab
BIO 481	Genomics
BIO 482	Human Histology
BIO 483	Bioinformatics
BIO 484	Aquatic Microbiology
BIO 486	Systematics of Vascular Plants
BIO 490	Mechanics of Animal Movement

Additional Information

One Topics in Biology (BIO 426/BIO 427) may be used to fulfill the organismal diversity requirement, with departmental permission. One BIO 427 can be used to satisfy an upper-level laboratory requirement, with departmental permission. BIO 426 cannot be used to satisfy a laboratory requirement. These course numbers can be repeated with a change in topic, but only 12 credits from BIO 426/BIO 427 can be applied toward the 40-hour biology course requirement.

When choosing additional biology courses to complete the 40 credit hour requirement, students are strongly encouraged to discuss their career interests with an advisor who can help select courses best suited to their needs. Students are encouraged to participate in independent research and teaching courses with a faculty mentor, though a maximum of eight credits of BIO 492, BIO 493, BIO 494, BIO 496, BIO 497, BIO 499A/BIO 499B/BIO 499C and ISCI 450 can be counted toward the biology major.

Biology Concentrations

Anatomy and Physiology Concentration

Concentration Coordinator: Jess Marquardt

Email: marquajl@jmu.edu

The Biology Department offers a concentration in anatomy and physiology for all Biology majors with an interest in exploring the relationship between structure and function of humans and/or animals. Further, the concentration will provide a strong foundation for students interested in clinical health care, veterinary medicine, and/or research in related fields.

Students choosing a concentration in anatomy and physiology must meet all the requirements for the biology major as well as the following additional requirements. Courses for this concentration can also satisfy biology major requirements.

Complete a total of at least 17 credit hours from the following courses.

Code	Title	Credits
Anatomy Core Requirements		
Choose one of the following:		4
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	
BIO 320	Comparative Anatomy of Vertebrates ^{1,2,3}	
Subtotal		4
Physiology Core Requirements		
BIO 370	Human and Animal Physiology ²	4
Subtotal		4
Anatomy and Physiology Electives		
Complete 9 hours from the following:		9
Choose at least one course from the following primarily anatomical or physiological courses: ⁴		
BIO 318	Experimental Vertebrate Development	
BIO 320	Comparative Anatomy of Vertebrates ³	
BIO 410	Advanced Human Anatomy ²	
BIO 412	Mammalogy ^{1,2}	
BIO 467	Human Cardiovascular Physiology	
BIO 468	Endocrinology	
BIO 472	Human Metabolism	
BIO 474	Reproductive Physiology ¹	
BIO 482	Human Histology ²	
BIO 490	Mechanics of Animal Movement ²	
Choose remaining credit hours from the list of primarily anatomical or physiological courses above, or from the following courses:		
BIO 301	Introductory Neuroscience	
BIO 304	Cell and Molecular Biology	
BIO 425	Immunology	
BIO 425L	Immunology Laboratory	
BIO 445	Neurobiology ²	
BIO 446	Experimental Neurobiology ²	
BIO 451	Current Topics in Human Development and Evolution	

Subtotal	9
Total Credits	17

¹ Meets the biology major requirement for an organismal diversity course.

² Meets one of the biology major requirements for laboratory/field courses.

³ May be used to fulfill either the Anatomy Core Requirement or as an Anatomy and Physiology Elective, but not both.

⁴ Note: BIO 270 and BIO 270L cannot be used to fulfill this requirement.

Ecology and Environmental Biology Concentration

Concentration Coordinator: Bruce Wiggins

Email: wigginsba@jmu.edu

The biology department offers a concentration within the biology major for students with interest in ecology, field biology, natural resources, environmental biology, conservation biology, evolution, animal behavior and organismal biology. Students choosing a concentration in ecology and environmental biology must meet all of the requirements for the biology major as well as the following additional requirements. Courses for this concentration can also satisfy biology major requirements.

Students must complete at least 20 credit hours from the following list.

With prior approval from the concentration coordinator, BIO 426 and/or BIO 427 may be substituted. Students are strongly encouraged to discuss their career interests with an advisor who can help select courses best suited to their needs. In addition to the courses listed below, a maximum of eight credits of BIO 496, BIO 497 and BIO 499A/BIO 499B/BIO 499C can be counted toward the EEB concentration.

Code	Title	Credits
Ecology and Environmental Biology Concentration Courses		
Select at least 20 credit hours from the following list:		20
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory ¹	
BIO 310	Marine Biology ¹	
BIO 320	Comparative Anatomy of Vertebrates ^{1,2}	
BIO 327	Biological Illustration	
BIO 340	Morphology and Anatomy of Vascular Plants ^{1,2}	
BIO/MATH 342	Mathematical Models in Biology	
BIO 350	Paleobiology ²	
BIO 354	Global Climate Change and Life: Ecological and Biological Impacts of Climate Variability	
BIO 357	Galápagos Islands: Biodiversity and Conservation ¹	
BIO 360	Plant Biology ¹	
BIO 370	Human and Animal Physiology ²	
BIO 386	Field Botany ^{1,2}	
BIO 387	Environmental Microbiology	
BIO 387L	Environmental Microbiology Laboratory ²	
BIO/PSYC 395	Comparative Animal Behavior	
BIO/GEOL 400	Geology and Ecology of the Bahamas ²	
BIO 402	Forest Ecology ²	
BIO 404	Evolutionary Analysis	
BIO/GEOL 405	Vertebrate Paleontology ¹	

BIO 412	Mammalogy ^{1,2}
BIO 418	Conservation Biology
BIO 419	Animals and their Environments ^{1,2}
BIO 430	Ornithology: Research and Current Advances ^{1,2}
BIO 447	Evolution and Ecology of Infectious Disease
BIO 449	Insect Ecology ^{1,2}
BIO 451	Current Topics in Human Development and Evolution
BIO 452	Population Ecology ²
BIO 453	Microbial Ecology and Evolution
BIO 454	Introduction to Biometrics ²
BIO 455	Plant Physiology ²
BIO 456	Ecological Models and Data ²
BIO 457	Biological Applications of Geographic Information Systems ²
BIO 459	Freshwater Ecology ²
BIO 484	Aquatic Microbiology ^{1,2}
BIO 486	Systematics of Vascular Plants ^{1,2}
Subtotal	20

Cognate Courses

Students must take two additional cognate courses: ³

Choose one Statistics course:		3-4
MATH 321	Analysis of Variance and Experimental Design	
MATH 322	Applied Linear Regression	
MATH 324	Applied Nonparametric Statistics	
MATH 325	Survey Sampling Methods	
BIO 454	Introduction to Biometrics ²	
BIO 456	Ecological Models and Data ²	
Choose one Geographic Information Systems course:		3-4
GEOG 215	Introduction to Geographic Information Science and Cartography	
BIO 457	Biological Applications of Geographic Information Systems ²	
Subtotal		6-8
Total Credits		26-28

¹ Meets the biology major requirement for an organismal diversity course.

² Meets one of the biology major requirements for laboratory/field courses.

³ Note: Biology courses that fulfill these additional cognates (BIO 454, BIO 456, and BIO 457) can also be used to fulfill other Biology major requirements.

Ecology and Environmental Biology Concentration Additional Information

It is highly recommended that students take additional upper-level degree elective courses in geography/GIS (such as GEOG 466 or GEOG 467) and in statistics (such as MATH 322 or MATH 324). Students should consult with their advisor about which courses are appropriate.

Microbiology Concentration

Concentration Coordinator: Steve Cresawn

Email: cresawsg@jmu.edu

The biology department offers a concentration within the biology major for students with interest in bacteriology, virology, immunology, parasitology and infectious disease. Students choosing a concentration in microbiology must meet all the requirements for the biology major, including the following specific requirements.

Complete a total of at least 18 credit hours from the following courses.

Code	Title	Credits
General Microbiology Requirements		
BIO 245	General Microbiology	3
BIO 245L	General Microbiology Laboratory	1
BIO 246	Microbiology Seminar	1
Subtotal		5
Laboratory Requirement		
Choose at least one of the following laboratory courses:		1-3
BIO 203	Viral Discovery	
ISAT 203	Viral Discovery	
BIO 204	Viral Genome and Bioinformatics	
ISAT 204	Viral Genome and Bioinformatics	
BIO 346	Laboratory in Bacterial Pathogenomics ¹	
BIO 348L	Medical Microbiology Laboratory ¹	
BIO 380	Advanced Microbiology Laboratory Techniques ¹	
BIO 387L	Environmental Microbiology Laboratory ¹	
BIO 425L	Immunology Laboratory ¹	
BIO 484	Aquatic Microbiology ¹	
Subtotal		1-3
Microbial Requirements		
If a student takes an elective course that is listed in both Microbial Pathogenesis and Microbial Ecology categories below, the course may only satisfy one category requirement, but not both.		
Microbial Pathogenesis - Choose at least one of the following:		3
BIO 325	Medical Parasitology ²	
BIO 348	Medical Microbiology	
BIO 425	Immunology	
BIO 444	Virology ³	
BIO 447	Evolution and Ecology of Infectious Disease ³	
Microbial Ecology - Choose at least one of the following:		3
BIO 387	Environmental Microbiology	
BIO 444	Virology ³	
BIO 447	Evolution and Ecology of Infectious Disease ³	
BIO 453	Microbial Ecology and Evolution	
BIO 484	Aquatic Microbiology ²	
Subtotal		6
Microbiology Electives		
Microbiology courses from any of the above listed courses to reach 18 credit hours.		4-6
Subtotal		4-6
Total Credits		18

¹ Meets one of the biology major requirements for an upper-level BIO 300-400 level laboratory course.

² Meets one of the biology major requirements for an organismal diversity course.

³ May be used to fulfill either the Microbial Ecology or Microbial Pathogenesis requirement, but not both.

Neuroscience Concentration

Concentration Coordinator: George Vidal

Email: vidalgx@jmu.edu

The biology department offers a concentration within the biology major for students with interests in cell and molecular, systems, and behavioral aspects of the nervous system and its functionality. Students choosing a concentration in neuroscience must meet all of the requirements for the biology major in addition to the following requirements. Courses for this concentration can also satisfy biology major requirements.

Code	Title	Credits
Neuroscience Concentration Courses		
Choose one of the following:		3
BIO 301	Introductory Neuroscience	
BIO/PSYC 385	Biopsychology	
Choose one of the following:		3-4
BIO/CSD 415	Neuroanatomy and Neurogenic Communication Disorders ¹	
BIO 445	Neurobiology ²	
Subtotal		6-7
Neuroscience Concentration Electives		
Complete 9 credits of the following:		9
Choose at least one course from the following principally neurobiology courses:		
BIO/PSYC 375	Sensation and Perception	
BIO/CSD 415	Neuroanatomy and Neurogenic Communication Disorders ¹	
BIO 443	Understanding Techniques in Neuroscience	
BIO 445	Neurobiology ^{1,2}	
BIO 446	Experimental Neurobiology ²	
BIO 448	Developmental Neurobiology	
BIO 478	Aging and Neurodegeneration	
Choose remaining credit hours from the list of principally neurobiology courses above, or from the following courses:		
BIO 316	Animal Development ²	
BIO 480	Advanced Molecular Biology	
BIO 480L	Advanced Molecular Biology Lab ²	
BIO 481	Genomics ²	
BIO 482	Human Histology ²	
Subtotal		9
Total Credits		15-16

¹ BIO 445 and BIO 415/CSD 415: The course not taken as part of required neuroscience concentration core may be used to fulfill elective requirements.

² Meets one of the biology major requirements for a laboratory/field course.

Neuroscience Concentration Additional Information

Students should choose additional BIO courses at the 300-400 level, including a course that fulfills the organismal diversity major requirement, and are strongly encouraged to discuss their career interests with an

advisor who can help select courses best suited to their needs. With prior approval from the concentration coordinator, BIO 426 and/or BIO 427 may be substituted. It is recommended that students participate in independent research (BIO 496/BIO 497/BIO 499A/BIO 499B/BIO 499C) with a faculty mentor.

Recommended Schedule for Majors

First semester first year biology majors are encouraged to start with a 14-15 hour course load. This will generally include BIO 140/BIO 140L, CHEM 131/CHEM 131L, and/or a math course plus General Education. The work load will then be increased in the second semester based on the level of first semester success.

Course	Title	Credits
First Year		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L] ¹	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
CHEM 131	General Chemistry I [C3PP] ¹	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II ¹	3
CHEM 132L	General Chemistry Laboratory	1
Mathematics courses ¹		7
General Education		9
Credits		32
Second Year		
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	4
CHEM 241	Organic Chemistry I	3
CHEM elective		3
Mathematics course		3
General Education		12
Credits		29
Third Year		
Upper-level biology laboratory courses		8
Biology elective		3
General Education		7
Electives		11
Credits		29
Fourth Year		
Upper-level biology laboratory course		4
Biology electives		9
Physics courses		4
General Education		3
Electives		10
Credits		30
Total Credits		120

¹ Fulfills General Education: The Natural World.

Program Description

James Madison University Liaison: Heather Griscom
Phone: (540) 568-5525
Email: griscohph@jmu.edu

This dual degree program makes it possible for the student to earn a Bachelor of Science (B.S.) degree in biology (p. 204) from James Madison University and a Master of Forestry (M.F.) degree from Virginia

Tech (<https://catalog.vt.edu/graduate/degree-programs/forestry-forest-products/>) in five years.

During the first three years at JMU, the student must complete 89 credit hours, including all JMU general education requirements, the cognate courses in chemistry and mathematics, a course in geology and at least 24 semester hours in biology courses, as shown below.

During the fourth year of study, when the student is in residence at Virginia Tech as an undergraduate student in an accelerated advanced degree program, the student will take further courses (31 credits) approved by the JMU Department of Biology for credit toward the B.S. in biology from JMU.

A total of 38 semester hours of biology and biology-related courses (taken at JMU and Virginia Tech) will be required for the JMU B.S. in biology, which will be conferred after the fourth year of study. If the student's academic record is satisfactory, they will be admitted into the graduate program of the Department of Forestry at Virginia Tech, where they will spend an additional year taking forest biology and natural resources courses to obtain the degree of Master of Forestry. To apply for the dual degree program, the student must have the permission of the dean of the College of Science and Mathematics. Information about the program can be obtained from the faculty liaison. Students are encouraged to inquire as early as possible in their undergraduate careers.

Recommended Schedule

Course	Title	Credits
First Year		
Fall Semester		
First Year (JMU)		
BIO 110	First Year Seminar in Biology	1
BIO 140	Foundations of Biology I [C3NS]	3
BIO 140L	Foundations of Biology I Laboratory [C3L]	1
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
Madison Foundations: Critical Thinking		3
Madison Foundations: Communication		3
Credits		15
Spring Semester		
BIO 150	Foundations of Biology II	3
BIO 150L	Foundations of Biology II Laboratory	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
MATH 220	Elementary Statistics [C3QR]	3
Madison Foundations: Writing Credits		3
Credits		14
Second Year		
Fall Semester		
Second Year (JMU)		
BIO 250	Quantitative Ecology and Evolution	3
BIO 250L	Quantitative Ecology and Evolution Laboratory	1
CHEM 241	Organic Chemistry I	3
MATH 321	Analysis of Variance and Experimental Design	3
Arts and Humanities: Visual and Performing Arts		3
Credits		13
Spring Semester		
BIO 240	Genetics	3
BIO 240L	Genetics Laboratory	1
CHEM 355	Geochemistry of Natural Waters	3
GEOL 110	Physical Geology [C3NS]	3

American and Global Perspectives: The American Experience	4
Credits	14
Third Year	
Fall Semester	
Third Year (JMU)	
BIO 402 Forest Ecology	4
American and Global Perspectives: The Global Experience	3
Arts and Humanities: Human Questions and Contexts	3
Sociocultural and Wellness Area: Wellness Domain	3
Elective or BIO	4
Credits	17
Spring Semester	
BIO 457 Biological Applications of Geographic Information Systems	4
ECON 201 Introduction to Microeconomics	3
Sociocultural and Wellness Area: Sociocultural Domain	3
Arts and Humanities: Literature	3
Elective or BIO	4
Credits	17
Fourth Year	
Fall Semester	
Fourth Year (Virginia Tech)	
FOR courses at Virginia Tech	15
Credits	15
Spring Semester	
FOR courses at Virginia Tech	15
Credits	15
Total Credits	120

Biotechnology, B.S.

Academic Unit Head, Biology/Biotechnology: Dr. Kyle Seifert

Phone: (540) 568-6733

Email: seiferkx@jmu.edu

Location: Biosciences Building, Room 2001

Program Description

JMU Biotechnology is a 4-year interdisciplinary major leading to a bachelor of science degree. The program is shared among the Departments of Biology, Chemistry and Biochemistry and Integrated Science and Technology (ISAT). Students undertake a rigorous curriculum rich with hands-on laboratory experiences, critical analyses of both the "how" and the "why" of biotechnological solutions, and the development of transferable skills needed to keep up in a rapidly changing field. The combination of both scientific/technical depth and cross-disciplinary breadth allow graduates to pursue diverse career paths in industry, government and advanced studies.

Biotechnology majors complete 47-50 credit hours of science foundation courses, 17 credit hours of biotechnology transition and core courses, and 15 credit hours of electives. Advanced courses are selected from an extensive list of Biology, Chemistry and Biochemistry, ISAT, Engineering and Mathematics and Statistics offerings. Students work with assigned academic advisors to develop individualized portfolios of upper level courses that best suit their unique interests and goals. Up to six credits of biotechnology undergraduate research or internship experiences can be applied to the upper level electives requirement.

Students are eligible to apply for the General Business Minor. Students may not receive dual credit toward the Biotechnology major for 300- and 400-level Biology courses that are also applied toward the Biology major. Biotechnology students pursuing a minor in ISAT may not substitute 100-level Biology or Chemistry courses for ISAT foundation courses. Biotechnology majors are not eligible for the Biochemistry and Molecular Biology Minor.

Please note that Biotechnology majors are not permitted to double major in Biology, Chemistry or ISAT or declare Biology, Chemistry and ISAT concentrations.

Degree Requirements

Degree Requirements	Credit Hours
General Education ^{1,2}	41
Quantitative requirement ^{3,4}	3
Scientific Literacy requirement ^{3,4}	3-4
Major requirements	79-82
Total	120

¹ The General Education program contains a set of requirements that each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The Natural World required courses can be fulfilled within the major requirements and do not require additional credit hours.

³ In addition to General Education.

⁴ The quantitative and scientific literacy requirements are fulfilled within the major requirements and do not require additional credit hours.

Major Requirements

Code	Title	Credits
Science Foundation Courses		
CHEM 131 & CHEM 132	General Chemistry I [C3PP] and General Chemistry II	6
CHEM 131L & CHEM 132L	General Chemistry Laboratory [C3L] and General Chemistry Laboratory	2
CHEM 241	Organic Chemistry I	3
CHEM 242 & 242L	Organic Chemistry II and Organic Chemistry Laboratory	5
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory ¹	4
PHYS 140 & 140L & PHYS 150 & PHYS 150L	College Physics I [C3PP] and General Physics Laboratory I [C3L] and College Physics II and General Physics Laboratory II	8
or PHYS 240 & 240L & PHYS 250 & PHYS 250L	University Physics I [C3PP] and University Physics Laboratory I [C3L] and University Physics II and University Physics Laboratory II	
MATH 231 & MATH 232	Calculus with Functions I [C3QR] and Calculus with Functions II	4-6
or MATH 233 & MATH 234	Calculus with Modeling I [C3QR] and Calculus with Modeling II	
or MATH 235	Calculus I [C3QR]	
MATH 220	Elementary Statistics [C3QR]	3-4
or MATH 285	Data Analysis	
Subtotal		47-50
Biotechnology Transition and Core Courses		
BIOT 260	Biotechnology Seminar	1
ISAT 305	Instrumentation and Measurement in Biotechnology (0, 2)	1
ISAT 451	Biotechnology in Industry and Agriculture	3
ISAT 456	Ethical, Legal and Social Implications of Biotechnology	3
CHEM 361	Biochemistry I	3
CHEM 366L	Biochemistry Laboratory	2
BIO 480 & 480L	Advanced Molecular Biology and Advanced Molecular Biology Lab	4
Subtotal		17
Biotechnology Elective Courses		
Up to 6 credits of undergraduate research or internship experiences can be applied to the upper level electives requirement.		
Choose 15 credit hours of the following:		15
BIO/ISAT 204	Viral Genome and Bioinformatics	
BIO 301	Introductory Neuroscience	
BIO 304	Cell and Molecular Biology	
BIO 310	Marine Biology	
BIO 316	Animal Development	

BIO 324	Human Genetics
BIO 325	Medical Parasitology
BIO/MATH 342	Mathematical Models in Biology
BIO 348	Medical Microbiology
BIO 348L	Medical Microbiology Laboratory
BIO 360	Plant Biology
BIO 364	Human Uses of Plants
BIO 364L	Laboratory in Human Uses of Plants
BIO 370	Human and Animal Physiology
BIO 387	Environmental Microbiology
BIO 387L	Environmental Microbiology Laboratory
BIO 425	Immunology
BIO 425L	Immunology Laboratory
BIO 432	Light Microscopy
BIO 443	Understanding Techniques in Neuroscience
BIO 444	Virology
BIO 445	Neurobiology
BIO 446	Experimental Neurobiology
BIO 447	Evolution and Ecology of Infectious Disease
BIO 448	Developmental Neurobiology
BIO 451	Current Topics in Human Development and Evolution
BIO 454	Introduction to Biometrics
BIO 455	Plant Physiology
BIO 460	Plant Biotechnology
BIO 472	Human Metabolism
BIO 477	The Genetics of Cancer
BIO 478	Aging and Neurodegeneration
BIO 481	Genomics
BIO 482	Human Histology
BIO 483	Bioinformatics
BIO 484	Aquatic Microbiology
BIO 490	Mechanics of Animal Movement
BIO 494	Internship in Biology
BIOT 494	Internship in Biotechnology
BIOT 495	Techniques in Biotechnology
BIOT 496	Biotechnology Literature Research
BIOT 497	Research in Biotechnology
BIOT 499A	Honors Capstone Project
BIOT 499B	Honors Capstone Project
BIOT 499C	Honors Capstone Project
CHEM 270	Inorganic Chemistry I
CHEM 331	Physical Chemistry I
CHEM 336L	Applied Physical Chemistry Laboratory
CHEM 351	Analytical Chemistry
CHEM 352	Instrumental Analysis
CHEM 352L	Instrumental Analysis Laboratory
CHEM 362	Biochemistry II
CHEM 440	Intermediate Organic Chemistry
CHEM 445	Polymer Chemistry
CHEM 450	Nuclear and Radiation Chemistry

ENGR 472	Environmental Engineering
ENGR 498	Advanced Topics in Engineering (select topics, such as bioprocess engineering)
ISAT 351	Biotechnology for the New Millennium II
ISAT 452	Medical Biotechnology
ISAT 454	Computer Applications in Biotechnology
ISAT 455	Regulatory Issues Biotechnology
ISAT 459	Awareness and Understanding of Chemical, Biological and Radiological Weapons of Mass Destruction
ISAT 485	Gender Issues in Science
ISAT 487	Science, Technology and Society Futures Laboratory
MATH 318	Introduction to Probability and Statistics
MATH 321	Analysis of Variance and Experimental Design
MATH 322	Applied Linear Regression
MATH 421	Applied Multivariate Statistical Analysis
MATH 485	Selected Topics
Subtotal	15
Total Credits	79-82

¹ Students who wish to take BIO 250/BIO 250L in lieu of BIO 245/BIO 245L are encouraged to seek approval for this substitution from the biotechnology program director.

Additional Information

When choosing additional 300-400 level courses to complete the credit hour requirement, students are strongly encouraged to discuss their career interests with an advisor who can help select courses best suited to their needs. Students are encouraged to participate in independent biotechnology research courses with a faculty mentor, though a maximum of six credits of BIOT 495, BIOT 496, BIOT 497, BIOT 499A, BIOT 499B and BIOT 499C can be counted toward the biotechnology major.

Recommended Schedule for Majors

First semester, first year biotechnology majors are encouraged to start with a 14-15 hour course load. This will generally include a biology course (four credit hours), CHEM 131 and CHEM 131L, and/or a math course, plus General Education. In some cases, the general chemistry sequence may be postponed when math placement scores suggest that the development of quantitative reasoning skills should come first. The work load may then be adjusted in the second semester based on the level of success in the first semester.

Note: Other recommended first-year courses include BIO 203 or ISAT 203 and BIO 204 or ISAT 204.

Course	Title	Credits
First Year		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
CHEM 131	General Chemistry I [C3PP] ¹	3
CHEM 131L	General Chemistry Laboratory [C3L] ¹	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1

MATH 220	Elementary Statistics [C3QR]	3
General Education courses (including Madison Foundations) and/or degree requirements		12
Credits		31
Second Year		
BIOT 260	Biotechnology Seminar	1
ISAT 305	Instrumentation and Measurement in Biotechnology (0, 2)	1
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 242L	Organic Chemistry Laboratory	2
General Education (from Arts and Humanities, American and Global Perspectives and Sociocultural Wellness Area), university electives and degree requirements		11
Credits		29
Third Year		
CHEM 361 or BIO 361	Biochemistry I or Biochemistry I	3
CHEM 366L	Biochemistry Laboratory	2
BIO 480	Advanced Molecular Biology	3
Biotechnology Electives		4
Physics courses		8
General Education: from Arts and Humanities, American and Global Perspectives and Sociocultural and Wellness Area		7
Electives		4
Credits		31
Fourth Year		
ISAT 451	Biotechnology in Industry and Agriculture	3
ISAT 456	Ethical, Legal and Social Implications of Biotechnology	3
Biotechnology Electives		11
General Education: from Arts and Humanities, American and Global Perspectives and Sociocultural and Wellness Area		6
Electives		6
Credits		29
Total Credits		120

¹ Fulfills The Natural World. Some students may be advised to postpone their general chemistry courses if their math placement scores are not sufficiently high.

Business Management, B.B.A.

Program Description

The major in business management offers three concentrations that prepare students for careers in a wide range of industries, including government organizations, manufacturing, consulting, retail, banking, pharmaceuticals, consumer products, insurance and more.

Students will develop skills in leadership, teamwork, critical thinking and analytical decision making as they apply key managerial concepts and theories learned in the classroom to practical experiences.

Major coursework topics include organizational leadership, evidence-based decision making, innovation and technology management, project management, human resource management, entrepreneurship and consulting.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business (p. 15) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education requirements ¹	41
B.B.A. lower-level core courses	27
B.B.A. upper-level core courses	18
Management major requirements	24
Electives	8-11
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		
COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	

MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28
B.B.A. Upper-Level Core Component		
The following courses comprise the B.B.A. Upper-Level Core Component:		
COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3
COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.
² COB 241 must be taken before COB 242.
³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.
⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.
⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.
⁶ All four COB 300 courses must be taken during the same semester.
⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.
⁸ COB 487 must be taken during the senior year.

Major Requirements

Students planning to major in business management must complete the 27-28 hours of lower-division B.B.A. core curriculum prior to enrolling in upper-division core courses normally taken in the first semester of their junior year. It is expected that the lower-division core curriculum will be completed during the first two years of study along with most of the university general education curriculum. Failing to complete all the lower-division core requirements on time will delay enrollment in upper-division core and major courses. Because of the demands of group, team and outside of classroom applied assignments, students will not be allowed to enroll in more than four courses with a MGT prefix in any given semester.

Business management majors must choose one of the following options to complete the required eight courses in the MGT major:

- Option 1: Five MGT core courses, one concentration and one elective
- Option 2: Five MGT core courses and three MGT electives.

Code	Title	Credits
Core Courses		
MGT 320	Management of Innovation and Technology	3
MGT 325	Project Management	3
MGT 357	Evidence-Based Decision Making	3
MGT 365	Fundamentals of Human Resource Management	3
MGT 390	Organizational Leadership	3
Subtotal		15
Concentration and Electives		

Students choose one concentration from the following options and one additional major elective; or three additional MGT electives.

Concentration options (see below for details):	
Human Resource Management	
Innovation and Entrepreneurship	
Management Consulting	
Subtotal	9
Total Credits	24

Concentrations

Human Resource Management Concentration

The human resource management (HRM) concentration is designed to help students broaden and deepen their understanding of the acquisition, development, management and retention of human capital in organizations. This focus area is suggested for students who intend to work in human resource (HR) generalist or specialist positions, students planning on a career in labor relations or employment law, students who plan to start or manage a business, and students who want to attend graduate school in an HR-related field. MGT 365 is considered the first course in this concentration.

Required Courses

Code	Title	Credits
MGT 470	Acquiring and Developing Human Capital	3
MGT 475	Managing and Retaining Human Capital	3
Total Credits		6

Innovation and Entrepreneurship Concentration

The innovation and entrepreneurship (ENT1) concentration is designed to provide students with an introduction to the role of innovation in business development, a solid foundation in the principle elements of entrepreneurship and small business management, and the opportunity to gain hands-on experience working with a team to create a new business venture. This focus area is suggested for students interested in creating and managing a new business venture, students intending to work for larger firms that encourage innovation and intrapreneurship, and students planning to work in small or family run businesses. MGT 320 is considered the first course in this concentration.

Required Courses

Code	Title	Credits
MGT 471	Entrepreneurship	3
MGT 472	New Venture Creation	3
Total Credits		6

Management Consulting Concentration

The management consulting concentration is designed to develop and enhance students' critical thinking, problem solving and decision-making skills. In this concentration students engage in research and analysis to build a strong foundation for learning and implementing the business analysis and consulting process. This focus area is suggested for students interested in working as a business analyst, consultant or solutions manager, or in any position that requires strong analytical and problem-solving skills. MGT 357 is considered the first course in this concentration.

Required Courses

Code	Title	Credits
MGT 457	Management Consulting Fundamentals	3
MGT 467	Advanced Management Consulting	3
Total Credits		6

Recommended Schedule for Majors

Course	Title	Credits
Third Year		
First Semester		
For all Business Management, B.B.A students.		
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
General Education or elective		3
Credits		15
Total Credits		15

Recommended Schedule for Business Management Majors

Course	Title	Credits
Third Year		
Second Semester		
COB 318	Legal and Ethical Environment of Business	3
General Education or electives		6
Choose three of the following MGT core requirements:		9
MGT 320	Management of Innovation and Technology	
MGT 325	Project Management	
MGT 357	Evidence-Based Decision Making	
MGT 365	Fundamentals of Human Resource Management	
MGT 390	Organizational Leadership	
Credits		18
Fourth Year		
COB 487	Strategic Management	3
Remaining two of five MGT core course requirements		6
General Education or electives as needed.		12
Choose one of the following options:		9
Option 1: One concentration (two courses) plus one MGT elective		
Option 2: Three MGT electives.		
Credits		30
Total Credits		48

Recommended Schedule for Human Resource Management Concentration

Course	Title	Credits
Third Year		
Second Semester		
COB 318	Legal and Ethical Environment of Business	3
MGT 365	Fundamentals of Human Resource Management	3
General Education or electives		6
Choose two of the following core requirements:		6
MGT 320	Management of Innovation and Technology	
MGT 325	Project Management	
MGT 357	Evidence-Based Decision Making	
MGT 390	Organizational Leadership	
Credits		18
Fourth Year		
COB 487	Strategic Management	3

MGT 470	Acquiring and Developing Human Capital	3
MGT 475	Managing and Retaining Human Capital	3
Remaining two of five MGT core course requirements		6
One MGT elective course		3
General Education or electives as needed.		12
Credits		30
Total Credits		48

Recommended Schedule for Innovation and Entrepreneurship Concentration

Course	Title	Credits
Third Year		
Second Semester		
MGT 320	Management of Innovation and Technology	3
General Education or electives		6
Choose two of the following MGT core requirements:		6
MGT 325	Project Management	
MGT 357	Evidence-Based Decision Making	
MGT 365	Fundamentals of Human Resource Management	
MGT 390	Organizational Leadership	
Credits		15
Fourth Year		
COB 487	Strategic Management	3
MGT 471	Entrepreneurship	3
MGT 472	New Venture Creation	3
Remaining two of five MGT core course requirements		6
One MGT elective		3
General Education or electives as needed		12
Credits		30
Total Credits		45

Recommended Schedule for Management Consulting Concentration

Course	Title	Credits
Third Year		
Second Semester		
COB 318	Legal and Ethical Environment of Business	3
MGT 357	Evidence-Based Decision Making	3
General Education or electives		6
Choose two of the following MGT core requirements:		6
MGT 320	Management of Innovation and Technology	
MGT 325	Project Management	
MGT 365	Fundamentals of Human Resource Management	
MGT 390	Organizational Leadership	
Credits		18
Fourth Year		
COB 487	Strategic Management	3
MGT 457	Management Consulting Fundamentals	3
MGT 467	Advanced Management Consulting	3
Remaining two of five MGT core course requirements		6
One MGT elective		3
Credits		18
Total Credits		36

Chemistry, B.S.

Program Description

The Department of Chemistry and Biochemistry offers the B.S. degree in chemistry, with concentrations that meet the American Chemical Society Accredited Programs requirements. The programs are designed to provide the theoretical and practical instruction in chemistry and related areas to prepare students for careers in chemistry, biochemistry, medicine, dentistry, paramedical areas, forensic sciences, chemical engineering and other technology-based areas.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education) ²	3-4
Major and cognate requirements	50-52
Concentration	10-31
Electives	3-26
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The quantitative and scientific literacy requirements are met by courses required in the major.

Major Requirements

Students completing a B.S. degree in Chemistry must select either a major in Chemistry or Biophysical Chemistry. They cannot earn a major in both areas.

Students majoring in Chemistry must also select one of six concentrations and complete all course work in that concentration. The credit hours for major requirements will vary based on the chosen concentration.

Code	Title	Credits
Core Requirements for all Concentrations		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 132	General Chemistry II	3
CHEM 135L	Special General Chemistry Laboratory I ¹	1
CHEM 136L	Special General Chemistry Laboratory II ¹	2
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 270	Inorganic Chemistry I	3
CHEM 287L	Integrated Inorganic/Organic Laboratory I	2
CHEM 288L	Integrated Inorganic/Organic Laboratory II	2
CHEM 331	Physical Chemistry I	3
CHEM 351	Analytical Chemistry	4
CHEM 361	Biochemistry I	3
CHEM 481	Literature and Seminar I	1
CHEM 482	Literature and Seminar II	1

Subtotal		34
Required Cognate Courses for all Concentrations		
MATH 235 or MATH 231 & MATH 232	Calculus I [C3QR] Calculus with Functions I [C3QR] and Calculus with Functions II	4-6
MATH 236	Calculus II	4
PHYS 240	University Physics I [C3PP]	3
PHYS 240L	University Physics Laboratory I [C3L]	1
PHYS 250	University Physics II	3
PHYS 250L	University Physics Laboratory II	1
Subtotal		16-18
Concentrations		
Students, in consultation with their faculty advisor, must choose one of the following concentrations (see below):		
Concentration I: American Chemical Society Certified Curriculum in Chemistry Concentration		
Concentration II: American Chemical Society Certified Curriculum in Biochemistry Concentration		
Concentration III: American Chemical Society Certified Curriculum in Materials Chemistry Concentration		
Concentration IV: Chemistry Secondary Education Concentration		
Concentration V: General Program in Chemistry Concentration		
Concentration VI: Chemistry/Business Program		
Subtotal		10-31
Total Credits		60-83

¹ CHEM 131L-CHEM 132L may be substituted.

Electives

The well-prepared student is encouraged to take as many of the additional departmental offerings as possible as electives with attention being given to junior and/or senior research projects.

Three credits of undergraduate research (CHEM 390/CHEM 497/CHEM 499A) may be used to satisfy chemistry elective requirements if the research project is completed with the same faculty mentor over multiple semesters.

Chemistry Concentrations

All ACS Certified Programs require a minimum of 400 hours of laboratory and research.

Concentration I: American Chemical Society Certified Curriculum in Chemistry Concentration

Students completing concentration I must complete the courses below in addition to core courses.

Code	Title	Credits
CHEM 352	Instrumental Analysis	3
CHEM 352L	Instrumental Analysis Laboratory	2
CHEM 432	Physical Chemistry II	3
CHEM 438L	Physical Chemistry Laboratory	2
CHEM 470	Inorganic Chemistry II	3
MATH 237	Calculus III	4

MATH 238	Linear Algebra with Differential Equations	4
Total Credits		21

Concentration II: American Chemical Society Certified Curriculum in Biochemistry Concentration

This program meets the recommended undergraduate degree requirements of the American Society for Biochemistry and Molecular Biology for a major in biochemistry. Students completing concentration II must complete the courses below in addition to core courses.

Code	Title	Credits
CHEM 352	Instrumental Analysis	3
CHEM 352L	Instrumental Analysis Laboratory	2
CHEM 362	Biochemistry II	3
CHEM 366L	Biochemistry Laboratory	2
CHEM 432	Physical Chemistry II	3
CHEM 438L	Physical Chemistry Laboratory	2
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4
BIO 480 & 480L	Advanced Molecular Biology and Advanced Molecular Biology Lab	4
Total Credits		31

Concentration III: American Chemical Society Certified Curriculum in Materials Chemistry Concentration

Students completing concentration III must complete the courses below in addition to core courses.

Code	Title	Credits
CHEM 336L	Applied Physical Chemistry Laboratory	2
CHEM 352	Instrumental Analysis	3
CHEM 352L	Instrumental Analysis Laboratory	2
CHEM 375	An Introduction to Materials Science	3
CHEM 432	Physical Chemistry II	3
CHEM 445	Polymer Chemistry	3
CHEM 445L	Polymer Chem Lab	1
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations	4
Choose one of the following:		3-4
CHEM 485 & 485L	Science of the Small and Science of the Small Laboratory	
ENGR 314	Materials and Intermediate Solid Mechanics	
ENGR 498	Advanced Topics in Engineering (when topic is appropriate)	
GEOL 390	Laboratory Techniques in Geology (2,2)	
Total Credits		28-29

Concentration IV: Chemistry Secondary Education Concentration

This concentration is designed for students wishing to pursue secondary education teacher licensure in chemistry. Students in this concentration should also declare the Pre-Professional Secondary Education (SEED)

minor. The SEED minor, in conjunction with the bachelor of science degree in chemistry with a concentration in chemistry secondary education, prepares students to enter the 5th year master's degree program in education that leads to teacher licensure.

Code	Title	Credits
BIO 140	Foundations of Biology I [C3NS]	3
BIO 140L	Foundations of Biology I Laboratory [C3L]	1
CHEM 325	Chemical Hazards and Laboratory Safety	1
CHEM 336L	Applied Physical Chemistry Laboratory	2
CHEM 352	Instrumental Analysis	3
CHEM 352L	Instrumental Analysis Laboratory	2
GEOL 355	Geochemistry of Natural Waters	3
Total Credits		15

Concentration V: General Program in Chemistry Concentration

Students completing concentration V must complete the Degree and Major courses and the courses below in addition to core courses.

Code	Title	Credits
CHEM 336L	Applied Physical Chemistry Laboratory	2
CHEM 352	Instrumental Analysis	3
CHEM 352L	Instrumental Analysis Laboratory	2
Upper division chemistry elective		3
Total Credits		10

Concentration VI: Chemistry/Business Program

This program is designed for business-oriented chemistry students preparing for careers in patent law, technical sales, technical service and related areas. Chemistry Business students should declare a General Business Minor for ease in enrolling in College of Business courses. Students completing concentration VI must complete the courses below in addition to core courses.

Code	Title	Credits
CHEM 336L	Applied Physical Chemistry Laboratory	2
ACTG 244	Accounting for Non-Business Majors	3
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
FIN 345	Finance Fundamentals for Non-Business Majors	3
MATH 220	Elementary Statistics [C3QR]	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
Approved chemistry, science and/or COB courses ¹		5
Total Credits		31

¹ Although business electives may be taken by students in this concentration, the total number of business credit hours may not exceed 27.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 132	General Chemistry II	3
CHEM 135L or CHEM 131L	Special General Chemistry Laboratory I or General Chemistry Laboratory [C3L]	1
CHEM 136L or CHEM 132L	Special General Chemistry Laboratory II or General Chemistry Laboratory	2
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
General Education requirement or Electives		10
Credits		31
Second Year		
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 270	Inorganic Chemistry I	3
CHEM 287L	Integrated Inorganic/Organic Laboratory I	2
CHEM 288L	Integrated Inorganic/Organic Laboratory II	2
PHYS 240	University Physics I [C3PP]	3
PHYS 240L	University Physics Laboratory I [C3L]	1
PHYS 250	University Physics II	3
PHYS 250L	University Physics Laboratory II	1
General Education requirement or Electives		9
Credits		30
Third Year		
CHEM 331	Physical Chemistry I	3
CHEM 351	Analytical Chemistry	4
CHEM 352	Instrumental Analysis	3
CHEM 352L	Instrumental Analysis Laboratory	2
CHEM 361	Biochemistry I	3
CHEM 481	Literature and Seminar I	1
CHEM 482	Literature and Seminar II	1
Courses in the concentration, General Education or Electives		13
Credits		30
Fourth Year		
Courses in the concentration, General Education or Electives		29
Credits		29
Total Credits		120

Chemistry, Biophysical Chemistry Major, B.S.

Program Description

This versatile major allows the student to focus on obtaining a solid foundation in chemistry and applying physical science concepts and techniques to solving a broad range of biological problems. The major emphasizes problem-solving, hands-on learning and creative, interdisciplinary thinking. The major interfaces well with neuroscience, molecular biology, biochemistry, artificial intelligence, data science, nanoscience, materials science and engineering. This interdisciplinary major provides a solid foundation for success in many types of graduate and professional schools as well as a broad range of teaching, industrial, and government jobs.

Students cannot major in both Chemistry and Biophysical Chemistry.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education) ²	3-4
Major requirements and electives	86-90
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The quantitative and scientific literacy requirements for this degree are met by courses required for the major.

Major Requirements

Code	Title	Credits
Core Courses		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 132	General Chemistry II	3
CHEM 135L	Special General Chemistry Laboratory I ¹	1
CHEM 136L	Special General Chemistry Laboratory II ¹	2
CHEM 270	Inorganic Chemistry I	3
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 287L	Integrated Inorganic/Organic Laboratory I ²	2
CHEM 288L	Integrated Inorganic/Organic Laboratory II	2
CHEM 351	Analytical Chemistry	4
CHEM 361	Biochemistry I	3
CHEM 363	Biophysical Chemistry (Literature & Seminar included)	3
CHEM 367L	Biochemistry Laboratory (fall)	2
CHEM 368L	Biophysical Chemistry Laboratory (spring)	2
CHEM 331	Physical Chemistry I	3
CHEM 336L	Applied Physical Chemistry Laboratory	2
CHEM 432	Physical Chemistry II	3

BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 480 & 480L	Advanced Molecular Biology and Advanced Molecular Biology Lab	4
Subtotal		56

Physics and Mathematics Background Courses

MATH 231 & MATH 232 or MATH 235	Calculus with Functions I [C3QR] and Calculus with Functions II or Calculus I [C3QR]	4-6
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations	4
PHYS 240 & 240L	University Physics I [C3PP] and University Physics Laboratory I [C3L]	4
PHYS 250 & 250L	University Physics II and University Physics Laboratory II	4
Subtotal		24-26

Electives

Choose at least two of the following:		6-8
CHEM/PHYS/ MATS 375	An Introduction to Materials Science	
CHEM 362	Biochemistry II	
CHEM 440	Intermediate Organic Chemistry	
CHEM 445	Polymer Chemistry	
CHEM 470	Inorganic Chemistry II	
BIO 324	Human Genetics	
BIO 445	Neurobiology	
PHYS 260	University Physics III	
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	
PHYS 326	Biophysics	
PHYS 381	Materials Characterization (Lecture/Lab) (Lecture/ Lab)	
BIO/MATH 342	Mathematical Models in Biology	
CHEM 390	Problems in Chemistry	
CHEM 497 or CHEM 499	Undergraduate Chemical Research or Honors	
BIO 494 or BIO 497	Internship in Biology or Advanced Biological Research	
PHYS 398 or PHYS 498 or PHYS 499	Independent Study in Physics or Astronomy or Undergraduate Research in Physics or Astronomy or Honors	
MATH 497 or MATH 499	Undergraduate Research or Honors Thesis	
MATH 498 or MATH 499	Honors Thesis	
MATH 499	Honors Thesis	
Any 300-400 level CHEM, PHYS or MATH course pre-approved by advisor		
Subtotal		6-8

Total Credits **86-90**

¹ CHEM 131L-CHEM 132L may be substituted.

² CHEM 242L may be substituted.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 132	General Chemistry II	3
CHEM 135L	Special General Chemistry Laboratory I	1
CHEM 136L	Special General Chemistry Laboratory II	2
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
General Education requirement or Electives		10
Credits		31
Second Year		
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 270	Inorganic Chemistry I	3
CHEM 287L	Integrated Inorganic/Organic Laboratory I	2
CHEM 288L	Integrated Inorganic/Organic Laboratory II	2
BIO 240 & 240L	Genetics and Genetics Laboratory	4
MATH 237	Calculus III	4
PHYS 240	University Physics I [C3PP]	3
PHYS 250	University Physics II	3
PHYS 240L	University Physics Laboratory I [C3L]	1
PHYS 250L	University Physics Laboratory II	1
General Education requirement or Electives		3
Credits		32
Third Year		
CHEM 331	Physical Chemistry I	3
CHEM 336L	Applied Physical Chemistry Laboratory	2
CHEM 351	Analytical Chemistry	4
CHEM 361	Biochemistry I	3
MATH 238	Linear Algebra with Differential Equations	4
General Education requirement or Electives		14
Credits		30
Fourth Year		
CHEM 363	Biophysical Chemistry	3
CHEM 367L	Biochemistry Laboratory	2
CHEM 368L	Biophysical Chemistry Laboratory	2
CHEM 432	Physical Chemistry II	3
BIO 480	Advanced Molecular Biology	3
General Education requirement or Electives		14
Credits		27
Total Credits		120

College of Business, European Business Concentration

Program Description

The European business concentration is composed of COB 300A, COB 300B, COB 300C and COB 300D (Integrative Business: Management, Finance, Operations, Marketing—12 credit hours) and COB 301 (European Integration, Culture and History—3 credit hours) both taught in Belgium as part of the Semester in Antwerp (p. 594) program.

These courses will be taught in the context of the European Union and, as a part of the curriculum, students will visit and study European businesses (e.g., European high-tech startup companies, businesses in the Port of Antwerp) and the institutions that comprise the European Union (e.g., the Council of Europe, the European Parliament). Students will visit business, governmental and cultural institutions in such countries as France, England, Germany, Luxembourg, Norway, the Netherlands, Austria and others, depending on which semester the student goes abroad.

Students in this concentration will:

- Acquire an in-depth understanding of the European business environment.
- Learn about the institutions of the European Union.
- Gain an understanding of the issues surrounding economic and monetary union.
- Apply the integrated functional systems to the European market.
- Understand the role of culture in conducting business in Europe.

This concentration is available to B.B.A. students within the College of Business and only for students who complete COB 300A, COB 300B, COB 300C and COB 300D and COB 301 with the Semester in Antwerp program (p. 594).

Concentration Requirements

Code	Title	Credits
Integrative Business		
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
Subtotal		12
European Integration, Culture and History		
COB 301	European Integration, Culture and History	3
Subtotal		3
Total Credits		15

Communication Sciences and Disorders, B.A.

Program Director: Yingjiu Nie
Email: nieyx@jmu.edu

Program Description

The B.A. and B.S. degree program in Communication Sciences and Disorders prepares students to work in healthcare and related fields and to apply foundational knowledge and skills to entry-level graduate education and training for audiology or speech-language pathology or a relevant area of study (e.g., psychology, education). The degree program provides students with the knowledge and skills to describe components of communication in functional contexts and recognize barriers to and/or solutions for enhanced communication. The program focuses on foundational scientific knowledge in communication, swallowing and hearing processes, while providing an understanding of symptomology of disorders, clinical management and standards of ethical conduct in communication sciences and disorders.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course(s) (in addition to General Education courses)	3
Major requirements	54-56
University electives	10-22
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

Code	Title	Credits
Required Courses		
CSD 200	Introduction to Communication Sciences and Disorders	3
CSD 207	Phonetics	3
CSD 208	Anatomy and Physiology of the Speech and Hearing Mechanism	3
CSD 209	Speech and Hearing Science	3
CSD 300	Language Development	3
CSD 301	Audiology	3
CSD 314	Pediatric Speech and Language Disorders	3
CSD 318	Aural Rehabilitation	3
CSD 412	Professional Issues and Person-Centered Care in Communication Sciences and Disorders	3

CSD 415	Neuroanatomy and Neurogenic Communication Disorders ¹	3
CSD 416	Adult Neurogenic Communication Disorders	3
PSYC 101	General Psychology [C5SD]	3
PSYC 160	Life Span Human Development [C5SD]	3
EXED 200	Foundations of Exceptional Education	3
MATH 220	Elementary Statistics [C3QR]	3
CSD 470	Methods and Observation in Speech Language Pathology	3
or CSD 471	Methods and Observation in Audiology	
Subtotal		48

Basic Science Requirements		
Requirement to meet basic science requirements for ASHA certification standards. Some graduate programs may have more specific requirements. Students are advised to check their programs of interest early to ensure they are taking the appropriate courses.		
One course with BIO prefix ²		3-4
One course with either PHYS or CHEM prefix ²		3-4
Subtotal		6-8
Total Credits		54-56

¹ CSD students must take CSD 415; BIO 415 may not be substituted.

² BIO, PHYS and CHEM courses may also be used to fulfill General Education requirements.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
CSD 200	Introduction to Communication Sciences and Disorders	3
PSYC 101	General Psychology [C5SD]	3
Biological sciences course		3
General Education requirement and/or B.A. degree requirements		21
Credits		30
Second Year		
CSD 207	Phonetics	3
CSD 208	Anatomy and Physiology of the Speech and Hearing Mechanism	3
CSD 209	Speech and Hearing Science	3
PSYC 160	Life Span Human Development [C5SD]	3
EXED 200	Foundations of Exceptional Education	3
MATH 220	Elementary Statistics [C3QR]	3
Physics or chemistry course		3
General Education requirement and/or B.A. degree requirements		9
Credits		30
Third Year		
CSD 300	Language Development	3
CSD 301	Audiology	3
CSD 314	Pediatric Speech and Language Disorders	3
CSD 318	Aural Rehabilitation	3
General Education, electives and/or B.A. degree requirements		18
Credits		30
Fourth Year		
CSD 412	Professional Issues and Person-Centered Care in Communication Sciences and Disorders	3
CSD 415	Neuroanatomy and Neurogenic Communication Disorders	3
CSD 416	Adult Neurogenic Communication Disorders	3

CSD 470 or CSD 471	Methods and Observation in Speech Language Pathology or Methods and Observation in Audiology	3
General Education, electives and/or B.A. degree requirements		18
Credits		30
Total Credits		120

Communication Sciences and Disorders, B.S.

Program Director: Yingjiu Nie
Email: nieyx@jmu.edu

Program Description

The B.A. and B.S. degree program in Communication Sciences and Disorders prepares students to work in healthcare and related fields and to apply foundational knowledge and skills to entry-level graduate education and training for audiology or speech-language pathology or a relevant area of study (e.g., psychology, education). The degree program provides students with the knowledge and skills to describe components of communication in functional contexts and recognize barriers to and/or solutions for enhanced communication. The program focuses on foundational scientific knowledge in communication, swallowing and hearing processes, while providing an understanding of symptomology of disorders, clinical management and standards of ethical conduct in communication sciences and disorders.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education)	3-4
Major requirements	60-62
University electives	10-13
Total	120

¹ The General Education programs contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses within the major.
² The quantitative literacy requirement may be fulfilled by courses required for the major.

Major Requirements

Code	Title	Credits
Required Courses		
CSD 200	Introduction to Communication Sciences and Disorders	3
CSD 207	Phonetics	3
CSD 208	Anatomy and Physiology of the Speech and Hearing Mechanism	3
CSD 209	Speech and Hearing Science	3
CSD 300	Language Development	3
CSD 301	Audiology	3
CSD 314	Pediatric Speech and Language Disorders	3
CSD 318	Aural Rehabilitation	3
CSD 412	Professional Issues and Person-Centered Care in Communication Sciences and Disorders	3
CSD 415	Neuroanatomy and Neurogenic Communication Disorders ¹	3
CSD 416	Adult Neurogenic Communication Disorders	3

PSYC 101	General Psychology [C5SD]	3
PSYC 160	Life Span Human Development [C5SD]	3
EXED 200	Foundations of Exceptional Education	3
MATH 220	Elementary Statistics [C3QR]	3
CSD 470	Methods and Observation in Speech Language Pathology	3
or CSD 471	Methods and Observation in Audiology	
Subtotal		48

Basic Science Requirements		
Requirement to meet basic science requirements for ASHA certification standards. Some graduate programs may have more specific requirements. Students are advised to check their programs of interest early to ensure they are taking the appropriate courses.		
One course with BIO prefix ²		3-4
One course with either PHYS or CHEM prefix ²		3-4
Subtotal		6-8
Restrictive Electives		
Select 6 credits from the following options:		
One course with BIO prefix (in addition to the CSD basic science requirement) ²		
One course with MATH prefix ^{2,3}		
One course with SOCI prefix ²		
CSD 420	Introduction to Sign Language	
Subtotal		6
Total Credits		60-62

¹ CSD students must take CSD 415; BIO 415 may not be substituted.
² BIO, PHYS, CHEM, MATH and SOCI courses may also be used to fulfill General Education or B.S. degree requirements.
³ In addition to MATH 220.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
CSD 200	Introduction to Communication Sciences and Disorders	3
PSYC 101	General Psychology [C5SD]	3
Biological sciences course		3
General Education requirement and/or B.S. degree requirements		21
Credits		30
Second Year		
CSD 207	Phonetics	3
CSD 208	Anatomy and Physiology of the Speech and Hearing Mechanism	3
CSD 209	Speech and Hearing Science	3
PSYC 160	Life Span Human Development [C5SD]	3
EXED 200	Foundations of Exceptional Education	3
MATH 220	Elementary Statistics [C3QR]	3
Physics or chemistry course		3
General Education requirement and/or B.S. degree requirements		9
Credits		30
Third Year		
CSD 300	Language Development	3
CSD 301	Audiology	3
CSD 314	Pediatric Speech and Language Disorders	3
CSD 318	Aural Rehabilitation	3

General Education, elective/restrictive elective and/or B.S. degree requirements		18
Credits		30
Fourth Year		
CSD 412	Professional Issues and Person-Centered Care in Communication Sciences and Disorders	3
CSD 415	Neuroanatomy and Neurogenic Communication Disorders	3
CSD 416	Adult Neurogenic Communication Disorders	3
CSD 470 or CSD 471	Methods and Observation in Speech Language Pathology or Methods and Observation in Audiology	3
General Education courses and/or electives		18
Credits		30
Total Credits		120

Communication Studies, B.A.

Program Description

The School of Communication Studies offers the Bachelor of Science and Bachelor of Arts degrees with a major in communication studies. Students must take at least 42 hours of work in communication studies beyond the General Education requirement.

All programs must include at least four elements:

- Fifteen hours of required core communication studies courses.
- Three hours of an advanced research methods course.
- Eighteen hours of communication concentration courses.
- Six hours of elective courses in communication studies.

Admission to the Major

Admission to JMU does not guarantee admission to the School of Communication Studies. All students interested in majoring in the program must apply for a limited number of spaces by enrolling in and successfully completing SCOM 240 and SCOM 240L. In order to register for these classes, students must change their major to "SCOM declared" to register for SCOM 240 and SCOM 240L by no later than March 1 for fall registration and by October 1 for spring registration. Students who miss the deadline are still encouraged to declare the major as SCOM will continue to process applications in the order in which they were received if there are still seats available in SCOM 240/SCOM 240L. Declaring the major by this date does not guarantee availability of courses. The School of Communication Studies reviews performance in the SCOM 240/SCOM 240L class to make decisions about acceptance into the major each semester.

Admission to the major is based on availability to the most qualified students as determined by performance in SCOM 240 and SCOM 240L.

Students who are not admitted are permitted one more attempt (an attempt is any time the student receives a grade for the course on the transcript including W, WP, or WF) to successfully meet performance requirements in SCOM 240/SCOM 240L by enrolling the next regular semester or in the summer if the student desires. For students retaking SCOM 240/SCOM 240L (whether repeat or repeat/forgive), the school will only consider the grades earned in the second attempt.

Successful completion of a major in the School of Communication Studies most often requires a minimum of four semesters after a student is fully admitted to the School. Depending upon the student's particular circumstances and degree progress, more than four semesters may be required for completing the major. Once admitted to the School of Communication Studies, a student cannot retroactively apply more than nine hours of SCOM courses, including SCOM 240 and SCOM 240L, to their major (SCOM 121/SCOM 122/SCOM 123 does not count toward those nine hours).

Policy for Students Transferring from Another Institution

Students applying from other institutions are held to the same policies and guidelines as other applicants. However, once admitted to the School of Communication Studies, transfer students may petition for SCOM credit for courses taken at previous institutions. As with other applicants, they may only use up to nine hours of SCOM courses taken before

full admittance to the major to satisfy major requirements, including SCOM 240 and SCOM 240L.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	42
University electives ³	22-34
Total	120

- ¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.
- ² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.
- ³ Students are encouraged to explore academic areas outside the Communication Studies major by pursuing a second major or minor.

Major Requirements

Outside the Major Credit Requirement

SCOM majors are required to complete a minimum of 42 credit hours in the major and 72 credits outside of the major, which includes General Education, the Bachelor of Arts or Bachelor of Science requirements, and any minor or second major credits. To support interdisciplinary learning, students are encouraged to pursue a second major or minor.

Limitations in Applied Courses

No more than six hours combined credit in SCOM 316, SCOM 317, SCOM 318, SCOM 319, SCOM 390 and SCOM 495 may be counted toward a major in communication studies unless approved by the academic unit head.

Double Counting

A student with a Communication Studies major or minor may use courses that satisfy their second major or minor requirements to satisfy requirements for their Communication Studies program. There is no limit on the number of classes a student may double count towards their Communication Studies programs.

Code	Title	Credits
Communication Studies Major Core		
SCOM 240	Introduction to Communication Theory	2
SCOM 240L	Communication Theory Lab	1
SCOM 242	Presentational Speaking	3
SCOM 280	Introduction to Communication Research ¹	3
SCOM 341	Persuasion	3
SCOM 358	Business and Professional Communication Studies	3
Subtotal		15
Advanced Research Methods		

Choose one of the following:	3
SCOM 381 Rhetorical Research Methods	
SCOM 383 Quantitative Experimental Research Methods	
SCOM 385 Qualitative Research Methods	
SCOM 386 Survey Research Methods	
SCOM 388 Campaign Research Methods	
Subtotal	3
Communication Studies Concentration	
Complete one of the following Communication Studies Concentrations. See below for more information.	18
Advocacy Studies	
Cultural Communication	
Generalist	
Health Communication Studies	
Interpersonal Communication Studies	
Organizational Communication Studies	
Public Relations Studies	
Subtotal	18
Communication Studies Electives	
Complete 6 credits of Communication Studies Electives	6
Subtotal	6
Total Credits	42

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

Communication Studies Concentrations

Advocacy Studies Concentration

This concentration prepares students to understand, critically evaluate and engage the communication theories, processes, media institutions and communication technologies that citizens, political leaders, government officials, public administrators, interest groups and community service organizations use to campaign, deliberate, adjudicate, govern and advocate for social change. Skilled advocates adopt, develop and implement diverse sets of communication strategies that help clients articulate interests and goals, determine social systems and audiences most likely to achieve goals, identify effective media and appropriate goal oriented messages, and develop plans for implementing change.

Code	Title	Credits
Advocacy Studies Concentration Core		
SCOM 342	Argument and Advocacy	3
SCOM 352	Communication and Social Movements	3
SCOM 353	American Political Culture and Communication	3
SCOM 442	Advanced Topics in Advocacy Studies	3
Subtotal		12
Advocacy Studies Concentration Electives		
Choose two of the following:		6
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM/WGSS 302	Third Wave Ecofeminism	
SCOM 331	Communication and Conflict	
SCOM 343	Contemporary Rhetorical Theory and Practice	

SCOM 346	Free Speech in America	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	
SCOM 356	Immigrant Advocacy	
SCOM/SMAD 357	Youth, Communication and Culture	
SCOM 371	Disability and Advocacy	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
SCOM 431	Legal Communication	
SCOM 453	Political Campaign Communication	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 470	Health Communication Campaigns	
SCOM/POSC/ SMAD 472	Media and Politics	
Subtotal		6
Total Credits		18

Cultural Communication Concentration

This concentration empowers students to communicate reflectively in a diverse, multicultural and global society. With a distinct focus on critical and cultural communication theories and social justice frameworks, students are equipped with opportunities and tools necessary for understanding culturally complex circumstances, issues and needs that arise in personal relationships, civic settings, and other social and mediated contexts. Skills developed in this concentration prepare students with confidence to more effectively listen, analyze, adapt and respond to communication contexts in ways that cultivate inclusive environments, cultures and just relations.

Code	Title	Credits
Cultural Communication Concentration Core		
SCOM 248	Intercultural Communication	3
SCOM 349	Ethnographic Approaches to Communication Studies	3
SCOM 448	Communication, Culture and Identity	3
Subtotal		9
Communication Skills		
Choose one of the following:		3
SCOM 261	Public Relations Techniques I: Written	
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 344	Oral Interpretation	
SCOM 447	Communication Tools for Organizational Challenges	
Subtotal		3
Cultural Communication Concentration Electives		
Choose two of the following:		6
SCOM/ANTH 305	Language and Culture	
SCOM 347	Communication, Diversity and Popular Culture	

SCOM 348	Communication and Gender	
SCOM 352	Communication and Social Movements	
SCOM 355	Food as Communication	
SCOM 356	Immigrant Advocacy	
SCOM 357	Youth, Communication and Culture	
SCOM 371	Disability and Advocacy	
SCOM 395	Study Abroad Seminar ¹	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
Subtotal		6
Total Credits		18

¹ Only 3 credits of SCOM 395 may be applied to this requirement.

Generalist Concentration

This concentration allows students to combine interests in several areas of communication study. The generalist focus allows students to choose options that create intersections that would support specific career outcomes, such as focusing on health communication and organizational communication to prepare for a career in health organizations, or a focus on advocacy and culture to prepare for work in humanitarian and social justice fields.

Not all listed courses may be available to generalist concentrators as some are reserved for particular concentrations. However, if students have already taken these courses when they were in a different concentration, they will count toward the generalist concentration requirements. Please also note that some of these courses have prerequisites.

Code	Title	Credits
Communication Skills		
Choose one of the following:		3
SCOM 261	Public Relations Techniques I: Written	
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 344	Oral Interpretation	
SCOM 361	Public Relations Techniques II: Visual	
SCOM 367	Advanced Public Relations Writing	
SCOM 447	Communication Tools for Organizational Challenges	
Communication Theory and Context		
Choose two of the following:		6
SCOM 245	Signs, Symbols and Social Interaction	
SCOM 248	Intercultural Communication	
SCOM 260	Introduction to Public Relations	
SCOM/WGSS 302	Third Wave Ecofeminism	
SCOM/ANTH 305	Language and Culture	
SCOM 306	Special Topics in Organizational Communication	
SCOM 313	Topics in Communication Studies	
SCOM 314	Communication in Romantic Relationships	

SCOM 320	Introduction to Interpersonal Communication	
SCOM 330	Special Topics in Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 338	The Art of Public Debate	
SCOM 345	Nonverbal Communication	
SCOM 346	Free Speech in America	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM/WGSS 348	Communication and Gender	
SCOM 349	Ethnographic Approaches to Communication Studies	
SCOM 350	Organizational Communication	
SCOM/WRTC 351	Visual Rhetoric	
SCOM 352	Communication and Social Movements	
SCOM 353	American Political Culture and Communication	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	
SCOM 355	Food as Communication	
SCOM 356	Immigrant Advocacy	
SCOM 357	Youth, Communication and Culture	
SCOM 362	Public Relations Theory and Concepts	
SCOM 363	International Public Relations	
SCOM 365	Sports Public Relations	
SCOM 369	Topics in Engaged Public Relations	
SCOM 370	Introduction to Health Communication	
SCOM 395	Study Abroad Seminar	
SCOM 413	Advanced Topics in Communication Studies	
SCOM 414	Attachment Communication	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
SCOM 425	Leadership Communication	
SCOM 431	Legal Communication	
SCOM 440	Family Communication	
SCOM/HIST 441	Oral History	
SCOM 442	Advanced Topics in Advocacy Studies	
SCOM 448	Communication, Culture and Identity	
SCOM 450	Advanced Studies in Organizational Communication	
SCOM 453	Political Campaign Communication	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 460	Public Relations Management	
SCOM 461	Public Relations Campaigns	
SCOM 462	Student-Run Public Relations Firm	
SCOM 468	Interpersonal Communication in Health Contexts	
SCOM 470	Health Communication Campaigns	
SCOM 471	Culture and Health Communication	
SCOM/POSC/ SMAD 472	Media and Politics	
Subtotal		9

Advanced Communication Theory and Context

Choose one of the following:	3
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SCOM 413	Advanced Topics in Communication Studies	
SCOM 414	Attachment Communication	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
SCOM 425	Leadership Communication	
SCOM 431	Legal Communication	
SCOM 440	Family Communication	
SCOM/HIST 441	Oral History	
SCOM 442	Advanced Topics in Advocacy Studies	
SCOM 448	Communication, Culture and Identity	
SCOM 450	Advanced Studies in Organizational Communication	
SCOM 453	Political Campaign Communication	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 460	Public Relations Management	
SCOM 461	Public Relations Campaigns	
SCOM 462	Student-Run Public Relations Firm	
SCOM 468	Interpersonal Communication in Health Contexts	
SCOM 470	Health Communication Campaigns	
SCOM 471	Culture and Health Communication	
SCOM/POSC/ SMAD 472	Media and Politics	
Subtotal		3
Generalist Concentration 300/400-Level Electives		
Two SCOM courses at the 300 or 400 level		6
Subtotal		6
Total Credits		18

Health Communication Studies Concentration

This concentration prepares students for careers in local, state or federal health agencies, hospitals/medical practices, non-profits, health-related industry, private research groups/foundations, and health education/promotion consulting firms. Students will gain strong writing and research skills and gain knowledge in the areas of: health communication, provider-patient communication and health campaigns.

Code	Title	Credits
Health Communication Concentration Core		
SCOM 370	Introduction to Health Communication	3
SCOM 468	Interpersonal Communication in Health Contexts	3
SCOM 470	Health Communication Campaigns	3
Subtotal		9
Communication Skills		
Choose one of the following:		3
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 447	Communication Tools for Organizational Challenges	
Subtotal		3
Health Communication Concentration Electives		
Choose two of the following:		6
SCOM 303	Special Topics in Health Communication (may be taken twice when topic changes)	

SCOM 320	Introduction to Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 345	Nonverbal Communication	
SCOM 350	Organizational Communication	
SCOM 371	Disability and Advocacy	
SCOM 440	Family Communication	
SCOM 448	Communication, Culture and Identity	
SCOM 471	Culture and Health Communication	
Subtotal		6
Total Credits		18

Interpersonal Communication Studies Concentration

Students analyze how messages are used to manage personal relationships in social and professional contexts. By examining research, theory and processes of interpersonal communication, students learn important principles for managing impressions and building rapport, identifying needs and pursuing influence goals, reducing relational uncertainty, and adapting to change. Students completing the concentration learn to communicate the value of healthy relationships, are well prepared for graduate school, and develop relational skills for excelling in careers such as peer counseling, education, family and human services, sales and management, aging services, ministry, and community leadership.

Code	Title	Credits
Interpersonal Communication Concentration Core		
SCOM 320	Introduction to Interpersonal Communication	3
SCOM 440	Family Communication	3
Subtotal		6
Communication Skills		
Choose one of the following:		3
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 447	Communication Tools for Organizational Challenges	
Subtotal		3
Communication Depth Elective		
Choose one of the following:		3
SCOM 248	Intercultural Communication	
SCOM 260	Introduction to Public Relations	
SCOM 350	Organizational Communication	
SCOM 370	Introduction to Health Communication	
Subtotal		3
Interpersonal Communication Concentration Electives		
Choose two of the following:		6
SCOM 314	Communication in Romantic Relationships	
SCOM 330	Special Topics in Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 345	Nonverbal Communication	
SCOM 348	Communication and Gender	
SCOM 414	Attachment Communication	
SCOM 468	Interpersonal Communication in Health Contexts	

Subtotal	6
Total Credits	18

Organizational Communication Studies Concentration

Organizational communication involves the study of the ways people interact within organizations, such as business, government, education and nonprofit groups. The Organizational Communication concentration offers students the knowledge and skills for organizing people, information and ideas, and provides opportunities for hands on practice. Coursework prepares students to organize productive talk, facilitate training and skill development, lead through communication, and apply theories of organizing to improve practice in the world. This concentration prepares students for careers in such things as business, consulting, training and event planning and for the study of law.

Code	Title	Credits
Organizational Communication Core		
SCOM 350	Organizational Communication	3
SCOM 450	Advanced Studies in Organizational Communication	3
SCOM 447	Communication Tools for Organizational Challenges	3
Subtotal		9
Organization Communication Concentration Electives		
Choose two of the following:		6
SCOM 306	Special Topics in Organizational Communication	
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 425	Leadership Communication	
SCOM 495	Internship in Communication Studies ¹	
Subtotal		6
Organization Communication Concentration Breadth Electives		
Choose one of the following:		3
SCOM 320	Introduction to Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 345	Nonverbal Communication	
SCOM/WGSS 348	Communication and Gender	
SCOM 370	Introduction to Health Communication	
Subtotal		3
Total Credits		18

¹ Only 3 credits of SCOM 495 may be applied to this requirement.

Public Relations Studies Concentration

This concentration focuses on the management of communication between organizations and their internal and external publics with the goal of mutual understanding and influence. Public relations practitioners work with specific audiences relative to focused goals. Preparation for a career in public relations should include a broad educational base and a variety of communication skills.

Code	Title	Credits
Public Relations Concentration Core		
SCOM 260	Introduction to Public Relations	3
SCOM 261	Public Relations Techniques I: Written	3

SCOM 362	Public Relations Theory and Concepts	3
Subtotal		9
Advanced Public Relations Skills		
SCOM 361	Public Relations Techniques II: Visual	3
or SCOM 367	Advanced Public Relations Writing	
Subtotal		3
Public Relations in Context		
Choose one of the following:		3
SCOM 363	International Public Relations	
SCOM 365	Sports Public Relations	
SCOM 369	Topics in Engaged Public Relations	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 460	Public Relations Management	
Subtotal		3
Public Relations Capstone		
SCOM 461	Public Relations Campaigns	3
or SCOM 462	Student-Run Public Relations Firm	
Subtotal		
Total Credits		18

Recommended Schedule for Communication Studies Majors

Generally, students will start their program in Communication Studies no earlier than their second semester and often at the beginning of their sophomore year. This is a suggested progression of major coursework. We recommend students take 30 credit hours per year to stay on track to meet all requirements in four years.

Course	Title	Credits
First Year		
Declared Students (End of First-Year or Beginning of Second Year)		
Complete the following:		30
Declare the major on MyMadison ¹		
Choose one of the following:		
SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC] ²	
SCOM 123	Fundamental Human Communication: Group Presentations [C1HC] ²	
Fulfill as many General Education requirements as possible		
Research and select a minor or second major and begin coursework		
Take requirements for B.A. Degree (Language at 232 level, Philosophy class)		
Credits		30
Second Year		
Complete the following:		30
SCOM 240	Introduction to Communication Theory	
SCOM 240L	Communication Theory Lab	
SCOM 242	Presentational Speaking	
SCOM 341	Persuasion	
Begin concentration specific coursework		
Continue General Education coursework		
B.A. language and philosophy requirements		
Continue minor, second major or other university coursework		
Credits		30
Third Year		
Complete the following:		30
SCOM 280	Introduction to Communication Research	
SCOM 358	Business and Professional Communication Studies	
Continue concentration requirements		

Complete advanced research methods course ³	
Complete general SCOM Electives	
Explore internship opportunities to count for SCOM elective credit	
Complete General Education coursework	
Continue minor, second major or other university coursework	
Credits	30
Fourth Year	
Complete the following:	30
Complete concentration requirements	
Complete minor, second major or other university coursework	
Credits	30
Total Credits	120

¹ Until fully-admitted, SCOM declared students may take only 9 hours of SCOM major credit.

² Either SCOM 122 or SCOM 123 are required to enroll in SCOM major course, SCOM 240/ SCOM 240L.

³ SCOM 240 is a pre-requisite.

Communication Studies, B.S.

Program Description

The School of Communication Studies offers the Bachelor of Science and Bachelor of Arts degrees with a major in communication studies. Students must take at least 42 hours of work in communication studies beyond the General Education requirement.

All programs must include at least four elements:

- Fifteen hours of required core communication studies courses.
- Three hours of an advanced research methods course.
- Eighteen hours of communication concentration courses.
- Six hours of elective courses in communication studies.

Admission to the Major

Admission to JMU does not guarantee admission to the School of Communication Studies. All students interested in majoring in the program must apply for a limited number of spaces by enrolling in and successfully completing SCOM 240 and SCOM 240L. In order to register for these classes, students must change their major to "SCOM declared" to register for SCOM 240 and SCOM 240L by no later than March 1 for fall registration and by October 1 for spring registration. Students who miss the deadline are still encouraged to declare the major as SCOM will continue to process applications in the order in which they were received if there are still seats available in SCOM 240/SCOM 240L. Declaring the major by this date does not guarantee availability of courses. The School of Communication Studies reviews performance in the SCOM 240/SCOM 240L class to make decisions about acceptance into the major each semester.

Admission to the major is based on availability to the most qualified students as determined by performance in SCOM 240 and SCOM 240L.

Students who are not admitted are permitted one more attempt (an attempt is any time the student receives a grade for the course on the transcript including W, WP, or WF) to successfully meet performance requirements in SCOM 240/SCOM 240L by enrolling the next regular semester or in the summer if the student desires. For students retaking SCOM 240/SCOM 240L (whether repeat or repeat/forgive), the school will only consider the grades earned in the second attempt.

Successful completion of a major in the School of Communication Studies most often requires a minimum of four semesters after a student is fully admitted to the School. Depending upon the student's particular circumstances and degree progress, more than four semesters may be required for completing the major. Once admitted to the School of Communication Studies, a student cannot retroactively apply more than nine hours of SCOM courses, including SCOM 240 and SCOM 240L, to his/her major (SCOM 121/SCOM 122/SCOM 123 does not count toward those nine hours).

Policy for Students Transferring from Another Institution

Students applying from other institutions are held to the same policies and guidelines as other applicants. However, once admitted to the School of Communication Studies, transfer students may petition for SCOM credit for courses taken at previous institutions. As with other applicants, they may only use up to nine hours of SCOM courses taken before

full admittance to the major to satisfy major requirements, including SCOM 240 and SCOM 240L.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement ²	3
Scientific Literacy Requirement ²	3
Major Requirements	42
University Elective(s) ³	31
Total	120

- ¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.
- ² In addition to course work taken to fulfill General Education requirement.
- ³ Students are encouraged to explore academic areas outside the Communication Studies major by pursuing a second major or minor.

Major Requirements

Outside the Major Credit Requirement

SCOM majors are required to complete a minimum of 42 credit hours in the major and 72 credits outside of the major, which includes General Education, the Bachelor of Arts or Bachelor of Science requirements, and any minor or second major credits. To support interdisciplinary learning, students are encouraged to pursue a second major or minor.

Limitations in Applied Courses

No more than six hours combined credit in SCOM 316, SCOM 317, SCOM 318, SCOM 319, SCOM 390 and SCOM 495 may be counted toward a major in communication studies unless approved by the academic unit head.

Double Counting

A student with a Communication Studies major or minor may use courses that satisfy their second major or minor requirements to satisfy requirements for their Communication Studies program. There is no limit on the number of classes a student may double count towards their Communication Studies programs.

Code	Title	Credits
Communication Studies Major Core		
SCOM 240	Introduction to Communication Theory	2
SCOM 240L	Communication Theory Lab	1
SCOM 242	Presentational Speaking	3
SCOM 280	Introduction to Communication Research ¹	3
SCOM 341	Persuasion	3
SCOM 358	Business and Professional Communication Studies	3
Subtotal		15
Advanced Research Methods		
Choose one of the following:		3
SCOM 381	Rhetorical Research Methods	
SCOM 383	Quantitative Experimental Research Methods	
SCOM 385	Qualitative Research Methods	

SCOM 386	Survey Research Methods	
SCOM 388	Campaign Research Methods	
Subtotal		3
Communication Studies Concentration		
Complete one of the following Communication Studies Concentrations. See below for more information.		18
Advocacy Studies		
Cultural Communication		
Generalist		
Health Communication Studies		
Interpersonal Communication Studies		
Organizational Communication Studies		
Public Relations Studies		
Subtotal		18
Communication Studies Electives		
Complete 6 credits of Communication Studies Electives		6
Subtotal		6
Total Credits		42

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

Communication Studies Concentrations

Advocacy Studies Concentration

This concentration prepares students to understand, critically evaluate and engage the communication theories, processes, media institutions and communication technologies that citizens, political leaders, government officials, public administrators, interest groups and community service organizations use to campaign, deliberate, adjudicate, govern and advocate for social change. Skilled advocates adopt, develop and implement diverse sets of communication strategies that help clients articulate interests and goals, determine social systems and audiences most likely to achieve goals, identify effective media and appropriate goal oriented messages, and develop plans for implementing change.

Code	Title	Credits
Advocacy Studies Concentration Core		
SCOM 342	Argument and Advocacy	3
SCOM 352	Communication and Social Movements	3
SCOM 353	American Political Culture and Communication	3
SCOM 442	Advanced Topics in Advocacy Studies	3
Subtotal		12
Advocacy Studies Concentration Electives		
Choose two of the following:		6
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM/WGSS 302	Third Wave Ecofeminism	
SCOM 331	Communication and Conflict	
SCOM 343	Contemporary Rhetorical Theory and Practice	
SCOM 346	Free Speech in America	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	

SCOM 356	Immigrant Advocacy	
SCOM/SMAD 357	Youth, Communication and Culture	
SCOM 371	Disability and Advocacy	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
SCOM 431	Legal Communication	
SCOM 453	Political Campaign Communication	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 470	Health Communication Campaigns	
SCOM/POSC/ SMAD 472	Media and Politics	
Subtotal		6
Total Credits		18

Cultural Communication Concentration

This concentration empowers students to communicate reflectively in a diverse, multicultural and global society. With a distinct focus on critical and cultural communication theories and social justice frameworks, students are equipped with opportunities and tools necessary for understanding culturally complex circumstances, issues and needs that arise in personal relationships, civic settings, and other social and mediated contexts. Skills developed in this concentration prepare students with confidence to more effectively listen, analyze, adapt and respond to communication contexts in ways that cultivate inclusive environments, cultures and just relations.

Code	Title	Credits
Cultural Communication Concentration Core		
SCOM 248	Intercultural Communication	3
SCOM 349	Ethnographic Approaches to Communication Studies	3
SCOM 448	Communication, Culture and Identity	3
Subtotal		9
Communication Skills		
Choose one of the following:		3
SCOM 261	Public Relations Techniques I: Written	
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 344	Oral Interpretation	
SCOM 447	Communication Tools for Organizational Challenges	
Subtotal		3
Cultural Communication Concentration Electives		
Choose two of the following:		6
SCOM/ANTH 305	Language and Culture	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM 348	Communication and Gender	
SCOM 352	Communication and Social Movements	
SCOM 355	Food as Communication	
SCOM 356	Immigrant Advocacy	

SCOM 357	Youth, Communication and Culture	
SCOM 371	Disability and Advocacy	
SCOM 395	Study Abroad Seminar ¹	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
Subtotal		6
Total Credits		18

¹ Only 3 credits of SCOM 395 may be applied to this requirement.

Generalist Concentration

This concentration allows students to combine interests in several areas of communication study. The generalist focus allows students to choose options that create intersections that would support specific career outcomes, such as focusing on health communication and organizational communication to prepare for a career in health organizations, or a focus on advocacy and culture to prepare for work in humanitarian and social justice fields.

Not all listed courses may be available to generalist concentrators as some are reserved for particular concentrations. However, if students have already taken these courses when they were in a different concentration, they will count toward the generalist concentration requirements. Please also note that some of these courses have prerequisites.

Code	Title	Credits
Communication Skills		
Choose one of the following:		3
SCOM 261	Public Relations Techniques I: Written	
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 344	Oral Interpretation	
SCOM 361	Public Relations Techniques II: Visual	
SCOM 367	Advanced Public Relations Writing	
SCOM 447	Communication Tools for Organizational Challenges	
Communication Theory and Context		
Choose two of the following:		6
SCOM 245	Signs, Symbols and Social Interaction	
SCOM 248	Intercultural Communication	
SCOM 260	Introduction to Public Relations	
SCOM/WGSS 302	Third Wave Ecofeminism	
SCOM/ANTH 305	Language and Culture	
SCOM 306	Special Topics in Organizational Communication	
SCOM 313	Topics in Communication Studies	
SCOM 314	Communication in Romantic Relationships	
SCOM 320	Introduction to Interpersonal Communication	
SCOM 330	Special Topics in Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 338	The Art of Public Debate	

SCOM 345	Nonverbal Communication	
SCOM 346	Free Speech in America	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM/WGSS 348	Communication and Gender	
SCOM 349	Ethnographic Approaches to Communication Studies	
SCOM 350	Organizational Communication	
SCOM/WRTC 351	Visual Rhetoric	
SCOM 352	Communication and Social Movements	
SCOM 353	American Political Culture and Communication	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	
SCOM 355	Food as Communication	
SCOM 356	Immigrant Advocacy	
SCOM 357	Youth, Communication and Culture	
SCOM 362	Public Relations Theory and Concepts	
SCOM 363	International Public Relations	
SCOM 365	Sports Public Relations	
SCOM 369	Topics in Engaged Public Relations	
SCOM 370	Introduction to Health Communication	
SCOM 395	Study Abroad Seminar	
SCOM 413	Advanced Topics in Communication Studies	
SCOM 414	Attachment Communication	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
SCOM 425	Leadership Communication	
SCOM 431	Legal Communication	
SCOM 440	Family Communication	
SCOM/HIST 441	Oral History	
SCOM 442	Advanced Topics in Advocacy Studies	
SCOM 448	Communication, Culture and Identity	
SCOM 450	Advanced Studies in Organizational Communication	
SCOM 453	Political Campaign Communication	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 460	Public Relations Management	
SCOM 461	Public Relations Campaigns	
SCOM 462	Student-Run Public Relations Firm	
SCOM 468	Interpersonal Communication in Health Contexts	
SCOM 470	Health Communication Campaigns	
SCOM 471	Culture and Health Communication	
SCOM/POSC/ SMAD 472	Media and Politics	
Subtotal		9

Advanced Communication Theory and Context

Choose one of the following:		3
SCOM 413	Advanced Topics in Communication Studies	
SCOM 414	Attachment Communication	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
SCOM 425	Leadership Communication	

SCOM 431	Legal Communication	
SCOM 440	Family Communication	
SCOM/HIST 441	Oral History	
SCOM 442	Advanced Topics in Advocacy Studies	
SCOM 448	Communication, Culture and Identity	
SCOM 450	Advanced Studies in Organizational Communication	
SCOM 453	Political Campaign Communication	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 460	Public Relations Management	
SCOM 461	Public Relations Campaigns	
SCOM 462	Student-Run Public Relations Firm	
SCOM 468	Interpersonal Communication in Health Contexts	
SCOM 470	Health Communication Campaigns	
SCOM 471	Culture and Health Communication	
SCOM/POSC/ SMAD 472	Media and Politics	
Subtotal		3
Generalist Concentration 300/400-Level Electives		
Two SCOM courses at the 300 or 400 level		6
Subtotal		6
Total Credits		18

Health Communication Studies Concentration

This concentration prepares students for careers in local, state or federal health agencies, hospitals/medical practices, non-profits, health-related industry, private research groups/foundations, and health education/promotion consulting firms. Students will gain strong writing and research skills and gain knowledge in the areas of: health communication, provider-patient communication and health campaigns.

Code	Title	Credits
Health Communication Concentration Core		
SCOM 370	Introduction to Health Communication	3
SCOM 468	Interpersonal Communication in Health Contexts	3
SCOM 470	Health Communication Campaigns	3
Subtotal		9
Communication Skills		
Choose one of the following:		3
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 447	Communication Tools for Organizational Challenges	
Subtotal		3
Health Communication Concentration Electives		
Choose two of the following:		6
SCOM 303	Special Topics in Health Communication (may be taken twice when topic changes)	
SCOM 320	Introduction to Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 345	Nonverbal Communication	
SCOM 350	Organizational Communication	
SCOM 371	Disability and Advocacy	

SCOM 440	Family Communication	
SCOM 448	Communication, Culture and Identity	
SCOM 471	Culture and Health Communication	
Subtotal		6
Total Credits		18

Interpersonal Communication Studies Concentration

Students analyze how messages are used to manage personal relationships in social and professional contexts. By examining research, theory and processes of interpersonal communication, students learn important principles for managing impressions and building rapport, identifying needs and pursuing influence goals, reducing relational uncertainty, and adapting to change. Students completing the concentration learn to communicate the value of healthy relationships, are well prepared for graduate school, and develop relational skills for excelling in careers such as peer counseling, education, family and human services, sales and management, aging services, ministry, and community leadership.

Code	Title	Credits
Interpersonal Communication Concentration Core		
SCOM 320	Introduction to Interpersonal Communication	3
SCOM 440	Family Communication	3
Subtotal		6
Communication Skills		
Choose one of the following:		3
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 342	Argument and Advocacy	
SCOM 447	Communication Tools for Organizational Challenges	
Subtotal		3
Communication Depth Elective		
Choose one of the following:		3
SCOM 248	Intercultural Communication	
SCOM 260	Introduction to Public Relations	
SCOM 350	Organizational Communication	
SCOM 370	Introduction to Health Communication	
Subtotal		3
Interpersonal Communication Concentration Electives		
Choose two of the following:		6
SCOM 314	Communication in Romantic Relationships	
SCOM 330	Special Topics in Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 345	Nonverbal Communication	
SCOM 348	Communication and Gender	
SCOM 414	Attachment Communication	
SCOM 468	Interpersonal Communication in Health Contexts	
Subtotal		6
Total Credits		18

Organizational Communication Studies Concentration

Organizational communication involves the study of the ways people interact within organizations, such as business, government, education and nonprofit groups. The Organizational Communication concentration

offers students the knowledge and skills for organizing people, information and ideas, and provides opportunities for hands on practice. Coursework prepares students to organize productive talk, facilitate training and skill development, lead through communication, and apply theories of organizing to improve practice in the world. This concentration prepares students for careers in such things as business, consulting, training and event planning and for the study of law.

Code	Title	Credits
Organizational Communication Core		
SCOM 350	Organizational Communication	3
SCOM 450	Advanced Studies in Organizational Communication	3
SCOM 447	Communication Tools for Organizational Challenges	3
Subtotal		9
Organization Communication Concentration Electives		
Choose two of the following:		6
SCOM 306	Special Topics in Organizational Communication	
SCOM 325	Small Group Communication	
SCOM 340	Principles and Processes of Interviewing	
SCOM 425	Leadership Communication	
SCOM 495	Internship in Communication Studies ¹	
Subtotal		6
Organization Communication Concentration Breadth Electives		
Choose one of the following:		3
SCOM 320	Introduction to Interpersonal Communication	
SCOM 331	Communication and Conflict	
SCOM 345	Nonverbal Communication	
SCOM/WGSS 348	Communication and Gender	
SCOM 370	Introduction to Health Communication	
Subtotal		3
Total Credits		18

¹ Only 3 credits of SCOM 495 may be applied to this requirement.

Public Relations Studies Concentration

This concentration focuses on the management of communication between organizations and their internal and external publics with the goal of mutual understanding and influence. Public relations practitioners work with specific audiences relative to focused goals. Preparation for a career in public relations should include a broad educational base and a variety of communication skills.

Code	Title	Credits
Public Relations Concentration Core		
SCOM 260	Introduction to Public Relations	3
SCOM 261	Public Relations Techniques I: Written	3
SCOM 362	Public Relations Theory and Concepts	3
Subtotal		9
Advanced Public Relations Skills		
SCOM 361	Public Relations Techniques II: Visual	3
or SCOM 367	Advanced Public Relations Writing	
Subtotal		3

Public Relations in Context

Choose one of the following:		3
SCOM 363	International Public Relations	
SCOM 365	Sports Public Relations	
SCOM 369	Topics in Engaged Public Relations	
SCOM 456	Fundraising Campaigns and Donor Motivation	
SCOM 460	Public Relations Management	
Subtotal		3
Public Relations Capstone		
SCOM 461	Public Relations Campaigns	3
or SCOM 462	Student-Run Public Relations Firm	
Subtotal		
Total Credits		18

Recommended Schedule for Communication Studies Majors

Generally, students will start their program in Communication Studies no earlier than their second semester and often at the beginning of their sophomore year. This is a suggested progression of major coursework. We recommend students take 30 credit hours per year to stay on track to meet all requirements in four years.

Course	Title	Credits
First Year		
Declared Students (End of First-Year or Beginning of Second Year)		
Complete the following:		30
Declare the major on MyMadison ¹		
Choose one of the following:		
SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC] ²	
SCOM 123	Fundamental Human Communication: Group Presentations [C1HC] ²	
Fulfill as many General Education requirements as possible		
Research and select a minor or second major and begin coursework		
Take requirements for B.S. Degree		
Credits		30
Second Year		
Complete the following:		30
SCOM 240	Introduction to Communication Theory	
SCOM 240L	Communication Theory Lab	
SCOM 242	Presentational Speaking	
SCOM 341	Persuasion	
Begin concentration coursework		
Continue General Education coursework		
Continue B.S. degree requirements		
Continue minor, second major or other university coursework		
Credits		30
Third Year		
Complete the following:		30
SCOM 280	Introduction to Communication Research	
SCOM 358	Business and Professional Communication Studies	
Continue concentration requirements		
Complete advanced research methods course ³		
Complete general SCOM electives		
Explore internship opportunities to count for SCOM elective credit		
Complete General Education coursework		
Continue minor, second major or other university coursework		
Credits		30

Fourth Year

Complete the following: 30

Complete concentration requirements

Complete minor, second major or other university coursework

Credits	30
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Total Credits	120
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¹ Until fully-admitted, SCOM declared students may take only 9 hours of SCOM major credit.

² Either SCOM 122 or SCOM 123 are required to enroll in SCOM major course, SCOM 240/SCOM 240L.

³ SCOM 280 is a pre-requisite.

Computer Information Systems, B.B.A.

Program Description

The B.B.A. in computer information systems prepares students to leverage technology strategically in business environments. Students develop technical proficiency, business acumen and analytical skills to design, implement and manage information systems that drive organizational success. The curriculum emphasizes problem-solving, systems development, data management, cybersecurity and emerging technologies such as artificial intelligence and cloud computing. Graduates are well-equipped for careers in consulting, systems analysis, project management, cybersecurity and business analytics, among other fields.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
B.B.A. core courses ²	39
CIS major requirements	27-28
Electives ³	12
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Assumes that MATH 205 and ECON 200 are taken as General Education courses.

³ Credit hours may vary. See advisor for details.

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		
COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	
MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28

B.B.A. Upper-Level Core Component

The following courses comprise the B.B.A. Upper-Level Core Component:

COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3

COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.

² COB 241 must be taken before COB 242.

³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.

⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.

⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.

⁶ All four COB 300 courses must be taken during the same semester.

⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.

⁸ COB 487 must be taken during the senior year.

Major Requirements

Students planning to major in Computer Information System (CIS) must complete the 27-28 hours of lower-division B.B.A. core curriculum prior to enrolling in upper-division core courses normally taken in the first semester of their junior year. It is expected that the lower-division core curriculum will be completed during the first two years of study along with most of the university general education curriculum. Failing to complete all the lower-division core requirements on time will delay enrollment in upper-division core and major courses.

Code	Title	Credits
Core Courses		
CIS 221	Principles of Programming ¹	3
CIS 304	Enterprise Architecture	3
CIS 330	Database Design and Application	3
CIS 454	Systems Analysis and Design	3
CIS 484	Information Systems Development and Implementation	3
Subtotal		
Concentration		
Students must choose one of the following concentrations. See below for details. ²		12-13
Business Analytics Concentration		
Information Systems Concentration		
Information and Cybersecurity Management Concentration		
Subtotal		27-28
Total Credits		27-28

¹ CS 149 or ISAT 252 or IA 241 with a grade of "C" or better or enrollment in CS 159 may be substituted for CIS 221.

² Students must declare a concentration by the deadline stated under concentrations on the CIS & BSAN department webpage (<https://www.jmu.edu/cob/cis/academics/index.shtml#concentrations>).

Concentrations

Business Analytics Concentration

Business analytics is a technical approach to analyzing problems and making business-related decisions. It uses statistical methods, management science techniques, machine learning and mathematical modeling to forecast the implications of various choices and identify the best alternatives. Business analytics focuses on the effective use of data and information to provide fact-based insights and drive positive business actions. Combined with the ability to analyze, design and implement information system, including applications and databases, the computer information systems major with a concentration in business analytics prepares students for careers in consulting, business analysis and data analysis. Students enrolled in the business analytics concentration of the computer information systems major are not permitted to earn a minor in business analytics. However, students enrolled in the business analytics concentration may switch to the information systems or information and cybersecurity management concentrations of the computer information systems major. After the switch, such students may pursue the business analytics minor provided they meet prerequisite requirements of the 300-level BSAN courses. See the business analytics minor for admission requirements.

Code	Title	Credits
Required Courses		
BSAN 390	Statistical Modeling for Business Analytics	3
BSAN 391	Decision Modeling for Business Analytics	3
BSAN 393	Data Mining for Business Analytics	3
or CIS 393	Data Mining for Business Analytics	
CIS 463	Business Intelligence	3
Total Credits		12

Information Systems Concentration

The information systems concentration prepares students to analyze, design and implement information system, including applications, databases, networks and architectures. Elective courses allow students to acquire more detailed knowledge in security, project management, web development and consulting. The computer information systems major with a concentration in information systems prepares students for careers in consulting, business analysis, data analysis and systems analysis.

Code	Title	Credits
Required Courses		
CIS 301	Cloud Server Management	1
CIS 320	Principles of Computing Networks	3
CIS 331	Intermediate Computer Programming	3
CIS Electives		6

Information Systems Concentration Electives

Students majoring in CIS are highly encouraged to complete an internship in computer information systems between their junior and senior years for non-academic credit.¹

CIS 354	Advanced Visual Basic Programming
CIS 366	Web Design and Development
CIS 393	Data Mining for Business Analytics
or BSAN 393	Data Mining for Business Analytics
CIS 420	Advanced Topics in Computing Networks
CIS 424	Computer Security Management

CIS 425	Defensive Cybersecurity
CIS 428	Mobile Computing and Security
CIS 434	Information Technology Consulting
CIS 463	Business Intelligence
CIS 464	Information Systems Project Management
CIS 490	Special Studies in Computer Information Systems or Business Analytics
CIS 498	Special Topics in Computer Information Systems
Total Credits	13

¹ CIS 361

Information and Cybersecurity Management Concentration

The Information and Cybersecurity Management concentration prepares students to help organizations protect data/information assets and information systems. Students will learn to detect and respond to threats and assess risks and vulnerabilities while ensuring confidentiality, integrity, and availability of organizational data and systems. Students will learn techniques and industry best practices of risk assessment, incident response, and continuity planning. Program objectives also include developing an understanding of behavioral, legal, privacy, and ethical considerations of data and information systems protection. The Computer Information Systems major with a concentration in Information and Cybersecurity Management prepares students for careers in information security analysis, incident response, risk and vulnerability analysis, continuity planning, computer forensics, cybersecurity governance, and cybersecurity consulting.

Code	Title	Credits
Required Courses		
CIS 301	Cloud Server Management	1
CIS 320	Principles of Computing Networks	3
CIS 424	Computer Security Management	3
CIS 425	Defensive Cybersecurity	3
CIS Electives from approved elective list below		3
Subtotal		13

Information and Cybersecurity Management Concentration Electives

Students majoring in CIS are highly encouraged to complete an internship in computer information systems between their junior and senior years for non-academic credit (CIS 361).

CIS 420	Advanced Topics in Computing Networks
CIS 428	Mobile Computing and Security
CIS 434	Information Technology Consulting
CIS 464	Information Systems Project Management
CIS 498	Special Topics in Computer Information Systems ¹

Total Credits **13**

¹ CIS 498 topic must be related to cybersecurity and approved by CIS-BSAN department head.

Cooperative Education Concentration

Coordinator: Tom Dillon

The optional cooperative education concentration in CIS major offers highly qualified undergraduate majors the opportunity to participate

in a six- to eight-month professional experience with well-recognized industry leaders in information technology and operations. Students will be awarded 12 hours of academic credit that will substitute for two required courses in the CIS curriculum and for six credits of special topics (CIS 498). Substitution for the two specific courses will be made based on the structure and context of the co-op experience and in cooperation with the co-op firm.

Students must demonstrate competency via examination in the two required courses selected in order to receive credit in those courses. Students who have at least a 3.0 GPA, are majors in CIS and have fulfilled all of their COB core requirements (COB 487) are eligible to apply on a competitive basis through the CIS & BSAN department.

Students who wish to participate in a co-op program must apply both to the CIS & BSAN department and the participating firm at least three months in advance of the start of the co-op. Co-ops typically begin in January or May and last six to eight months. Course substitutions must be approved in conjunction with the co-op coordinator in the CIS & BSAN department and the co-op coordinator in the firm.

A program of study must be placed on file for each student who is accepted for a co-op prior to beginning the co-op experience. Students may participate in a co-op during their junior or senior years, but they are limited to one co-op. Students who want to participate in a co-op as postgraduates may do so as special students. These students will receive a certificate on successful completion of the co-op experience. *Prerequisite: CIS majors with junior standing and a minimum 3.0 grade point average.*

Recommended Schedule for Majors

Computer information systems majors should follow the course schedule described here to complete the final two years of their program. It is possible to deviate from this program, but care must be taken to ensure that all course prerequisites are met.

First Two Years

Students normally take the lower-division B.B.A. core curriculum along with many of the General Education curriculum. All lower-division core requirements must be completed before enrolling in the upper-division core courses. It is recommended that CIS 221 be completed in the second semester of the second year.

Course	Title	Credits
First Year		
B.B.A. lower division core and general education courses		30
Credits		30
Second Year		
B.B.A. lower division core and general education courses		30
Credits		30
Third Year		
First Semester		
Students who are able to complete B.B.A. core requirements early enough to take COB 300 in the second semester of the second year should take additional CIS courses in the first semester of the third year and subsequent semesters. Consult your advisor.		
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
CIS 304	Enterprise Architecture	3
Credits		15

Second Semester		
BSAN 390	Statistical Modeling for Business Analytics	3
CIS 330	Database Design and Application	3
COB 318	Legal and Ethical Environment of Business	3
Other major, general education and/or university elective courses		6
Credits		15
Fourth Year		
First Semester		
BSAN 391	Decision Modeling for Business Analytics	3
CIS 331	Intermediate Computer Programming	3
CIS 361	Computer Information Systems Internship	0
CIS 454	Systems Analysis and Design	3
Other major, general education and/or university elective courses		6
Credits		15
Second Semester		
Students majoring in CIS are highly encouraged to complete an internship in computer information systems for non-academic credit (CIS 361).		
CIS 393 or BSAN 393	Data Mining for Business Analytics or Data Mining for Business Analytics	3
CIS 463	Business Intelligence	3
COB 487	Strategic Management	3
Other major, general education and/or university elective courses		6
Credits		15
Total Credits		120

Computer Science, B.S.

Program Description

The Bachelor of Science degree in Computer Science prepares graduates to excel in the field of computing. All students complete required coursework in programming, mathematics, data structures, algorithms, software engineering, computer systems, and programming languages. Students also choose elective courses from a variety of computing subfields including robotics, artificial intelligence, human-computer interaction, cyber defense, database systems, and web applications. Throughout the curriculum, students apply their knowledge by completing many software development projects both individually and on a team, using a variety of languages and systems.

Admission Requirements

Admission to JMU does not guarantee admission to the Computer Science major. Admission is limited, but the department will strive to accommodate all interested students who have demonstrated the potential to succeed in CS. Students may enroll in CS courses beyond CS 149, CS 159 and CS 227/MATH 227 only if they are fully admitted to the CS major or minor.

Students interested in the major must complete the application process described below to request full admission.

Students who wish to apply for full admission to the CS major must:

- Be enrolled at JMU
- Have declared the CS major
- Have completed a minimum of 15 credit hours
- Have completed or be currently enrolled in CS 159 and CS 227/MATH 227/MATH 245
- Achieve grades of "B-" or better in CS 159 and CS 227/MATH 227/MATH 245
- Submit an official application that includes an essay of no more than 250 words to address a prompt that demonstrates students' commitment to the values of the Department of Computer Science.

Additional application and acceptance information:

- Students will be granted full admission to the CS major as space permits based on their grades in CS 159 and CS 227/MATH 227/MATH 245 and faculty evaluation of their potential to succeed in the CS major.
- There are two application cycles, once in the fall for spring admission, and once in the spring for fall admission.
- Students will be notified of their CS major admission status no later than January 1st for admission for the spring semester, and no later than May 31st for admission for the fall semester.
- Students may apply to the CS major and/or minor two times, in total.
- Students who have attempted CS 149, CS 159 or CS 227/MATH 227/MATH 245 at JMU may not attempt them elsewhere (i.e. they must make any second attempt(s) at JMU).
- Non-transfer students must take CS 159 and CS 227/MATH 227/MATH 245 at JMU.
- MATH 245 may be substituted for CS 227.
- An applicant's CS and calculus grades will be considered during the admission process, with a strong emphasis on CS grades. Final

admission decisions will be made by a committee of CS faculty members.

Incoming transfer students will be placed into appropriate CS courses for their first semester after consultation with a transfer advisor based on placement test performance and other considerations. Students who are placed into CS 149 will be subject to the same eligibility requirements and application process as non-transfer students. Transfer students who are placed into courses other than CS 149 must apply for full admission at the end of their first semester. They will be evaluated by a committee of CS faculty members based on their performance in the first semester as well as the other elements of the admissions application. Transfer students must have credit for or be taking CS 159 and CS 227/MATH 227/MATH 245 the semester they apply for full admission to the major.

Transfer students who are denied admission on their first attempt may continue to take appropriate CS courses during a subsequent semester as determined through consultation with a transfer advisor. Admission decisions for the second attempt will be based on the full record of coursework at JMU.

Progressing in the Major and Minimum Grades

Students may enroll in each of CS 149, CS 159, CS 240 and CS 261 at most twice. Most CS courses require a minimum grade of "C-" (minimum of "B-" in CS 149 and CS 227) in prerequisite courses, including those taken in the Department of Mathematics and Statistics. Students must achieve a minimum cumulative grade point average of 2.0 in all courses used to satisfy CS major degree requirements.

Students must earn a minimum grade of "C-" in the following courses to count for the CS major: CS 240, CS 261, CS 327, CS 361, Calculus (MATH 231 or MATH 235), and Statistics (MATH 220, MATH 229, or MATH 318).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3
Major requirements (listed below)	49-52
University electives	24-27
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

Major Requirements

Code	Title	Credits
CS 149	Introduction to Programming	3
CS 159	Advanced Programming	3
CS 240	Algorithms and Data Structures	3
CS 261	Computer Systems I	3
CS 327	Discrete Structures II	3
CS 345	Software Engineering	3

CS 361	Computer Systems II	3
CS 430	Programming Languages	3
Computer Science electives at the 300-level or above, with at least two above CS 332		9
Choose one introductory discrete math course:		3
CS 227	Discrete Structures I	
MATH 227	Discrete Structures I	
MATH 245	Discrete Mathematics	
Choose one calculus sequence:		4-6
MATH 235	Calculus I [C3QR]	
MATH 231 & MATH 232	Calculus with Functions I [C3QR] and Calculus with Functions II	
Choose one statistics course:		3-4
MATH 220	Elementary Statistics [C3QR]	
MATH 229	Introduction to Applied Statistics Using R [C3QR]	
MATH 318	Introduction to Probability and Statistics	
Choose one algorithms course:		3
CS 412	Applied Algorithms	
CS 452	Design and Analysis of Algorithms	
Choose one systems elective:		3
CS 432	Compilers	
CS 450	Operating Systems	
CS 455	Advanced Computer Networking	
CS 456	Computer Architecture	
CS 470	Parallel and Distributed Systems	
Total Credits		49-52

Additional Information

The credit/no-credit option may not be applied to any courses specifically listed above, nor may that option be applied to computer science electives.

Certificate of Completion

Periodically, the department may offer a collection of two or more advanced courses in a particular area of study.

Students successfully completing those courses will obtain a certificate of completion in that area of study. Examples of possible certificate programs include networking, software engineering and information security.

U.S. Government Requirements for Computer Scientists

The U.S. government standard for occupational category GS- 1550: Computer Science Series includes a requirement of 15 hours in statistics and mathematics including differential and integral calculus. This means that students considering a career as a computer scientist with the U.S. government (including DoD, NASA, etc.) must complete more math courses than the minimum requirement for a B.S. degree. Recommended calculus sequences for these students are MATH 235-MATH 236 or MATH 231- MATH 232-MATH 236. However, only the U.S. Office of Personnel Management can give final approval of individual qualifications.

Sample Plan of Study for Majors

This is just an illustration of a path through our curriculum.

Course	Title	Credits
First Year		
Fall Semester		
CS 149	Introduction to Programming	3
MATH 231	Calculus with Functions I [C3QR] ¹	3
General Education requirement		9
Credits		15
Spring Semester		
CS 159	Advanced Programming	3
CS 227	Discrete Structures I	3
MATH 232	Calculus with Functions II ¹	3
General Education requirement		6
Credits		15
Second Year		
Fall Semester		
CS 240	Algorithms and Data Structures	3
CS 345	Software Engineering	3
General Education requirement		6
Electives		3
Credits		15
Spring Semester		
CS 261	Computer Systems I	3
MATH 220	Elementary Statistics [C3QR] ²	3
General Education requirement		6
Electives		3
Credits		15
Third Year		
Fall Semester		
CS 327	Discrete Structures II	3
CS 361	Computer Systems II	3
General Education requirement		6
Electives		3
Credits		15
Spring Semester		
CS 430	Programming Languages	3
CS Systems Elective		3
General Education requirement		3
Electives		6
Credits		15
Fourth Year		
Fall Semester		
CS Algorithm Elective		3
CS Elective		3
Electives		9
Credits		15
Spring Semester		
CS Electives		6
Electives		9
Credits		15
Total Credits		120

¹ Many students opt to take the two-course calculus sequence (MATH 231 and MATH 232), but if you have the appropriate ALEKS score, you could take MATH 235 to fulfill your calculus requirement.

² Many students opt to take Elementary Statistics (MATH 220) to fulfill our statistics requirement, but we have two other course options: MATH 229 or MATH 318.

Dietetics, B.S.

Program Description

Students completing the Bachelor of Science in Dietetics are prepared for careers in or relevant to nutrition and dietetics, including areas of clinical, public health, food service and research. The Dietetics Program Professional Nutrition and Dietetics Concentration at James Madison University is a Didactic Program in Dietetics (DPD) that is accredited by the Accreditation Council for Education in Nutrition and Dietetics (ACEND) of the Academy of Nutrition and Dietetics:

120 S. Riverside Plaza, Suite 2190
Chicago, IL 60606-6995
800-877-1600, ext. 5400

Those seeking to earn the Registered Dietitian Nutritionist (RDN) credential upon graduation must complete an ACEND-accredited graduate program, coordinated program or dietetic internship paired with a graduate degree to be eligible for the national registration exam.

Special Admission Requirements and Program Prerequisite Courses

Any student admitted to JMU can declare dietetics as their major and will be permitted to enroll in NUTR 280 and NUTR 295. However, to make progress in the major beyond the first few courses, students need to apply and be admitted to the program.

Students wishing to be admitted to the dietetics program at JMU must apply by February 15 in the spring semester during which admission requirements will be met, usually the sophomore year.

To be admitted, students must have a cumulative GPA of at least 2.7 in the prerequisite courses listed under the requirements section and a minimum grade of "C-" in each course.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3
Scientific Literacy requirement (in addition to General Education) ²	3-4
Prerequisite requirements for admission to the major ¹ (7 credits are counted in General Education and 4 credits are counted in Scientific Literacy)	20 (11)
Foundational courses (16 credits fulfill General Education requirements)	(16)
Dietetics major requirements including concentration	32-60
University electives ³	3-31
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Many of the General Education requirements may be fulfilled through courses that are listed as prerequisite and foundational courses for the major.

² The scientific literacy requirement is met by courses required in the major.

³ These electives may include additional NUTR courses.

Prerequisite Courses

Code	Title	Credits
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L] ¹	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L] ¹	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
HTH 210	Medical Terminology	3
NUTR 280	Nutrition for Wellness	3
NUTR 295	Foundations of Nutrition Practice	2
Total Credits		20

¹ May also fulfill General Education or B.S. degree requirements.

Foundational Courses

The following courses also fulfill General Education requirements:

Code	Title	Credits
MATH 220	Elementary Statistics [C3QR]	3
PSYC 101 or PSYC 160	General Psychology [C5SD] Life Span Human Development [C5SD]	3
HIST 225 or POSC 225	U.S. History [C4AE] U.S. Government [C4AE]	4
SCOM 121 or SCOM 122 or SCOM 123	Fundamental Human Communication: Presentations [C1HC] Fundamental Human Communication: Individual Presentations [C1HC] Fundamental Human Communication: Group Presentations [C1HC]	3
WRTC 103	Rhetorical Reading and Writing [C1W] ¹	3
Total Credits		16

Retention and Receiving Didactic Program in Dietetics Verification Statement

To remain in the major, students must maintain a minimum GPA of 2.8 in the major and earn a minimum grade of "C-" in all NUTR and HTH courses. In order to earn the Accreditation Council for Education in Nutrition and Dietetics verification statement, students in the Professional Concentration must also pass program core knowledge assessments. Transfer credit will not be accepted for the following courses: NUTR 295, NUTR 363, NUTR 482 and NUTR 484.

Major Requirements

Code	Title	Credits
Dietetics Core Courses		
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
HTH 354	U.S. and Global Health Care Systems	3
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	1
NUTR 340	The Science and Culture of Food	3

NUTR 350	Emerging Research in Nutrition and Dietetics	3
NUTR 384	Clinical Nutrition I	3
NUTR 385	Nutrition Throughout the Life Cycle	3
NUTR 482	Nutrition and Metabolism	3
Subtotal		23
Concentrations		
Complete one of the following concentrations. See below for more information.		9-37
Professional Nutrition and Dietetics		
Pre-Professional Nutrition and Health		
Subtotal		9-37
Total Credits		32-60

Concentrations

Professional Nutrition and Dietetics Concentration

This concentration is accredited by the Accreditation Council on Education for Nutrition and Dietetics as a Didactic Program in Dietetics for students pursuing an accredited graduate program, coordinated program or dietetic internship after graduation.

Code	Title	Credits
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4
CHEM 241 & 241L	Organic Chemistry I and Concepts of Organic Chemistry Laboratory	4
CHEM 260 & 260L	Concepts of Biochemistry and Concepts of Biochemistry Laboratory	4
NUTR 363	Quantity Food Production	4
NUTR 380	Global Nutrition	3
NUTR 394	Culinary Nutrition Therapy	3
NUTR 465	Management in Dietetics	3
NUTR 484	Clinical Nutrition II	3
NUTR 485	Community Nutrition	3
NUTR 486	Counseling Skills for Dietetics	3
NUTR 495	Senior Seminar in Dietetics	3
Total Credits		37

Pre-Professional Nutrition and Health Concentration

This concentration is for students preparing for a professional health program at the graduate level and especially students concurrently enrolled in a university minor. Students in this concentration must complete 9-10 credits of NUTR electives from the following courses.

Code	Title	Credits
Nutrition Electives		
Choose 9-10 credit hours from the following:		9-10
NUTR 363	Quantity Food Production	
NUTR 380	Global Nutrition	
NUTR 382	Sports Nutrition	
NUTR 394	Culinary Nutrition Therapy	
NUTR 455	Exercise and Nutrition in Chronic Disease	
NUTR 465	Management in Dietetics	
NUTR 484	Clinical Nutrition II	
NUTR 485	Community Nutrition	

NUTR 486	Counseling Skills for Dietetics
NUTR 495	Senior Seminar in Dietetics
Total Credits	9-10

Recommended Schedule for Majors - Professional Nutrition and Dietetics Concentration

Course	Title	Credits
First Year		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
NUTR 295	Foundations of Nutrition Practice	2
Choose one of the following:		3
PSYC 101	General Psychology [C5SD] ¹	
PSYC 160	Life Span Human Development [C5SD] ¹	
Quantitative Requirement (B.S. degree requirement)		3
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Madison Foundations - Critical Thinking (any course in Madison Foundations Critical Thinking area)		3
Madison Foundations - Communication (any course in Madison Foundations Communication area)		3
General Education requirement		6
Credits		27
Second Year		
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L] ¹	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory ¹	4
HTH 210	Medical Terminology	3
POSC 225	U.S. Government [C4AE] ¹	4
MATH 220	Elementary Statistics [C3QR] ¹	3
NUTR 280	Nutrition for Wellness	3
General Education requirement		9
Credits		30
Third Year		
CHEM 241	Organic Chemistry I	3
CHEM 241L	Concepts of Organic Chemistry Laboratory	1
CHEM 260	Concepts of Biochemistry	3
CHEM 260L	Concepts of Biochemistry Laboratory	1
HTH 354	U.S. and Global Health Care Systems	3
NUTR 340	The Science and Culture of Food	3
NUTR 350	Emerging Research in Nutrition and Dietetics	3
NUTR 380	Global Nutrition	3
NUTR 384	Clinical Nutrition I	3
NUTR 385	Nutrition Throughout the Life Cycle	3
NUTR 394	Culinary Nutrition Therapy	3
Credits		29
Fourth Year		
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	1
NUTR 363	Quantity Food Production	4
NUTR 465	Management in Dietetics	3
NUTR 482	Nutrition and Metabolism	3
NUTR 484	Clinical Nutrition II	3
NUTR 485	Community Nutrition	3
NUTR 486	Counseling Skills for Dietetics	3
NUTR 495	Senior Seminar in Dietetics	3

Electives	3
Credits	34
Total Credits	120

¹ These courses may count for both General Education and the major depending on General Education course choice.

Note: NUTR courses in Years 3 and 4 are offered only one semester per year.

Recommended Schedule for Majors - Pre-Professional Nutrition and Health Concentration

Course	Title	Credits
First Year		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
MATH 220	Elementary Statistics [C3QR]	3
PSYC 101 or PSYC 160	General Psychology [C5SD] or Life Span Human Development [C5SD]	3
Quantitative Requirement		3
General Education requirement		15
Credits		32
Second Year		
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	4
BIO 290L & BIO 290	Human Anatomy Laboratory and Human Anatomy	4
HTH 210	Medical Terminology	3
NUTR 280	Nutrition for Wellness	3
NUTR 295	Foundations of Nutrition Practice	2
General Education requirement		13
Credits		29
Third Year		
HTH 354	U.S. and Global Health Care Systems	3
NUTR 340	The Science and Culture of Food	3
NUTR 350	Emerging Research in Nutrition and Dietetics	3
NUTR 384	Clinical Nutrition I	3
NUTR 385	Nutrition Throughout the Life Cycle	3
NUTR or General Electives		15
Credits		30
Fourth Year		
NUTR 482	Nutrition and Metabolism	3
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	1
NUTR or General Electives		25
Credits		29
Total Credits		120

Economics, B.A.

Program Description

Economics majors choose from a B.A., B.S. or B.B.A. degree. The B.A. and B.S. degrees are traditional liberal arts degrees that lead to a variety of career and graduate school options; the B.B.A. is designed to prepare students for careers in business.

The minimum requirement for a B.A. degree in economics is 36 credit hours of economics, including 21 credit hours of core courses and 15 credit hours of electives. It is also necessary for the student to complete the foreign language and philosophy requirements for a B.A. degree as well as complete the General Education: The Human Community program.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business (p. 15) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course(s) (in addition to General Education courses)	3
Major requirements (listed below) and electives	64-76
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

Code	Title	Credits
Required Courses		
ECON 200	Introduction to Macroeconomics [C4GE] ¹	3
ECON 201	Introduction to Microeconomics ¹	3
ECON 284	Introduction to Quantitative Analysis in Economics ^{1,2}	3
ECON 331	Intermediate Microeconomic Theory ³	3
ECON 332	Intermediate Macroeconomic Theory ³	3
ECON 385	Econometrics ^{3,4}	3
ECON 488	Senior Capstone Seminar in Economics	3
Economics electives (at least six credits must be at the 400 level) ⁵		15
Subtotal		36

Additional Required Courses

Choose one of the following calculus courses:		3
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 232	Calculus with Functions II	
MATH 233	Calculus with Modeling I [C3QR]	
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	
Choose one of the following statistics courses:		
MATH 220	Elementary Statistics [C3QR]	3
or COB 191	Business Analytics I	
Subtotal		6
Total Credits		42

¹ Students need a minimum grade of "C" in ECON 200, ECON 201 and ECON 284 as a prerequisite for advanced economics courses.

² Students may substitute COB 291 for ECON 284.

³ Students need to complete ECON 331, ECON 332 and ECON 385 with a minimum grade of "C" to graduate with an Economics degree.

⁴ Students must complete ECON 385 with a minimum grade of "B-" in order to choose ECON 483 or ECON 485 as an elective option.

⁵ Not including ECON 488, ECON 490 or ECON 499A

Economics Concentrations

Concentration in Environmental and Natural Resource Economics

Students who elect a major in Economics with a concentration in Environmental and Natural Resource Economics are required to meet degree requirements and complete the following Economics courses (which can also be used to satisfy degree requirements) *as well as* the classes specified *in either* Option A or Option B.

Code	Title	Credits
Environmental and Natural Resource Economics Concentration Courses		
Choose two of the following:		6
ECON 305	Environmental Economics	
ECON 340	Economics of Natural Resources	
ECON 382	Urban Economics	
Choose one of the following:		1-3
ECON 394	Economics Internship (topic must be related to Environment and Natural Resources)	
ECON 490	Special Studies in Economics (topic must be related to Environment and Natural Resources)	
Subtotal		7-9
Electives		
One 200- or 300-level economics elective		3
Two 400-level economics electives		6
Subtotal		9
Options		
Choose one of the following options:		9-12
Option A (12 Credits)		
For students with an interest in environmental sustainability as it applies to forests, fisheries and wildlife.		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	

BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory
Choose one of the following:	
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory
GEOG 210	Physical Geography (2, 2)
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]

Option B (9-11 Credits)

For students interested in global climate change and environmental sustainability (including traditional approaches to pollution prevention and control).

Choose one of the following:	
GEOG 210	Physical Geography (2, 2)
GEOL 115	Earth Systems and Climate Change [C3NS]
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]
Choose two of the following:	
GEOG 325	Environmental Ethics
GEOG 327	Climatology
GEOG 329	Global Climate Change
GEOG 427	Water Resources of the World
GEOG/ISAT 429	Sustainability: An Ecological Perspective
HIST 427	U.S. Environmental History
ISAT 320	Fundamentals of Environmental Science and Technology I
ISAT 321	Fundamentals of Environmental Science and Technology II
ISAT 422	Industrial Environmental Management
ISAT 427	Industrial Hygiene
ISAT 428	Industrial Ecology
SOCI 311	Sociology of the Environment
Subtotal	9-12
Total Credits	25-30

Concentration in Financial Economics

Students who elect a major in Economics with a concentration in Financial Economics are required to meet Economics degree requirements and complete the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Financial Economics Concentration Courses		
ECON/FIN 372	International Finance and Payments	3
ECON 430	Monetary Theory	3
or ECON 483	Applied Time Series Analysis	
It is also required that the following 12 credits of Finance courses be used to satisfy six credits of the 300-level Economics elective credits required for the degree. ¹		
FIN 301	Principles of Finance	3
or COB 300B	Integrative Business: Finance	

FIN 360	Analytical Methods in Finance	3
FIN 371	Principles of Investments	3
FIN 380	Elemental and Derivative Securities Analysis	3
Subtotal		18
Elective		
One 400-level ECON elective (not including ECON 490, ECON 499A)		3
Subtotal		3
Total Credits		21

¹ Note that this makes it possible to receive a degree in Economics with a concentration in Financial Economics with no more than 30 credits in Economics.

Concentration in International and Development Economics

Students who elect a major in Economics with a concentration in International and Development Economics are required to meet degree requirements and complete the courses listed below (which can also be used to satisfy degree requirements).

Code	Title	Credits
International and Development Economics Concentration Courses		
Choose four of the following:		12
ECON 312	Comparative Economic Systems	
ECON 365	Economic Development	
ECON 370	International Trade and Trade Policies	
ECON/FIN 372	International Finance and Payments	
ECON 437	Economic Growth	
Subtotal		12
Electives		
Two 400-level economics electives		6
Subtotal		6
Total Credits		18

Concentration in Political Economy

Students who elect a major in Economics with a concentration in political economy are required to meet degree requirements and complete four of the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Political Economy Concentration Courses		
Choose four of the following:		12
ECON 312	Comparative Economic Systems	
ECON 326	Public Finance	
ECON 327	Game Theory	
ECON 405	Political Economy	
ECON 426	Theory of Public Choice	
ECON 455	Economics of Regulated Industries	
Subtotal		12
Electives		
Two 400-level economics electives ¹		0-6
Subtotal		0-6
Total Credits		12-18

¹ ECON 405, ECON 426 and/or ECON 455 also fulfill this 400-level elective requirement if taken to satisfy the 12 credit hour concentration requirement.

Concentration in Quantitative Economics

Students who elect a major in Economics with a concentration in Quantitative Economics are required to meet degree requirements and complete the following courses (which can also be used to satisfy degree requirements). Note that along with the required 12 credit hours in the Quantitative Economics concentration, students are required to complete additional elective courses to satisfy the major requirements.

Code	Title	Credits
Quantitative Economics Concentration Courses		
ECON 483	Applied Time Series Analysis	3
or ECON 485	Advanced Econometrics	
Choose two of the following:		6
ECON 384	Mathematical Economics	
ECON 431	Advanced Microeconomic Theory	
ECON 432	Advanced Macroeconomics	
ECON 437	Economic Growth	
ECON 483	Applied Time Series Analysis ¹	
ECON 485	Advanced Econometrics ¹	
Subtotal		9
Electives		
One 300- or 400-level economics elective in addition to the courses taken to satisfy the credit hour concentration requirement		3
Subtotal		3
Total Credits		12

¹ If taken as a required course above, students cannot also apply the course in this category.

Concentration in Socioeconomics

Students who elect a major in Economics with a concentration in Socioeconomics are required to meet Economics degree requirements and complete the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Socioeconomics Concentration Courses		
Choose four of the following:		12
ECON 306	The Economics of Women and the Family	
ECON 328	Behavioral Economics	
ECON 329	Health Economics	
ECON 340	Economics of Natural Resources	
ECON 360	Labor Economics	
ECON 365	Economic Development	
ECON 382	Urban Economics	
ECON 427	Experimental Economics	
Subtotal		12
Electives		
Two 400-level economics electives		6

Subtotal	
Total Credits	18

Recommended Schedule for B.A. in Economics Majors
First Two Years

During the first two years, students should complete:

Code	Title	Credits
Most of the General Education program		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
ECON 284	Introduction to Quantitative Analysis in Economics	3
COB 191	Business Analytics I	3
Choose one of the following:		3-4
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 233	Calculus with Modeling I [C3QR]	
MATH 235	Calculus I [C3QR]	

Additional Information

Students are encouraged to take ECON 331 and/or ECON 332 in their sophomore year (provided that the course prerequisites have been met).

Third and Fourth Years

B.A. economics majors should complete ECON 331, ECON 332, and preferably ECON 385 by the end of their junior year. While most majors will complete the 400-level requirements in economics during their senior year, students may take a 400-level course during their junior year if the prerequisite for the course has been met. ECON 488 should be taken during the senior year.

Economics, B.B.A.

Program Description

Economics majors choose from a B.A., B.S. or B.B.A. degree. The B.A. and B.S. degrees are traditional liberal arts degrees that lead to a variety of career and graduate school options; the B.B.A. is designed to prepare students for careers in business.

The B.B.A. degree in economics requires a minimum of 120 credit hours of undergraduate work.

The minimum requirement for a B.B.A. degree in economics is 33 credit hours of economics including 18 credit hours of core courses and 15 credit hours of electives. B.B.A. students also complete the core business requirements as well as complete the General Education program.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business catalog page (p. 15).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
B.B.A. core courses ²	45-46
Major Requirements	33 ³
Electives	6-7
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Up to seven credit hours of core requirements in economics and calculus may also be taken for General Education credit. Students who take the General Education packages and courses recommended by the College of Business will have only 39 credit hours of additional B.B.A. core requirements.

³ 6 credits in the Economics, B.B.A. major requirements overlap with the B.B.A. core component.

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		
COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3

ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	
MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28
B.B.A. Upper-Level Core Component		
The following courses comprise the B.B.A. Upper-Level Core Component:		
COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3
COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.

² COB 241 must be taken before COB 242.

³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.

⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.

⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.

⁶ All four COB 300 courses must be taken during the same semester.

⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.

⁸ COB 487 must be taken during the senior year.

Major Requirements

Code	Title	Credits
ECON 200	Introduction to Macroeconomics [C4GE] ¹	3
ECON 201	Introduction to Microeconomics ¹	3
ECON 331	Intermediate Microeconomic Theory ²	3
ECON 332	Intermediate Macroeconomic Theory ²	3
ECON 385	Econometrics ^{2,3}	3
ECON 488	Senior Capstone Seminar in Economics	3
Economics electives ^{4,5}		15
Total Credits		33

¹ Students need a minimum grade of "C" in ECON 200 and ECON 201 as a prerequisite for advanced economics courses.

² Students need to complete ECON 331, ECON 332 and ECON 385 with a minimum grade of "C" to graduate with an Economics degree.

³ Students must complete ECON 385 with a minimum grade of "B-" in order to choose ECON 483 or ECON 485 as an elective option.

⁴ At least six credits must be at the 400 level, not including ECON 488, ECON 490 or ECON 499A

⁵ Students declaring an economics concentration will fulfill these electives with concentration requirements.

Economics Concentrations

Concentration in Environmental and Natural Resource Economics

Students who elect a major in Economics with a concentration in Environmental and Natural Resource Economics are required to meet degree requirements and complete the following Economics courses (which can also be used to satisfy degree requirements) *as well as* the classes specified *in either* Option A or Option B.

Code	Title	Credits
Environmental and Natural Resource Economics Concentration Courses		
Choose two of the following:		6
ECON 305	Environmental Economics	
ECON 340	Economics of Natural Resources	
ECON 382	Urban Economics	
Choose one of the following:		1-3
ECON 394	Economics Internship (topic must be related to Environment and Natural Resources)	
ECON 490	Special Studies in Economics (topic must be related to Environment and Natural Resources)	
Subtotal		7-9
Electives		
One 200- or 300-level economics elective		3
Two 400-level economics electives		6
Subtotal		9
Options		
Choose one of the following options:		9-12
<i>Option A (12 Credits)</i>		
For students with an interest in environmental sustainability as it applies to forests, fisheries and wildlife.		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	
Choose one of the following:		
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	
GEOG 210	Physical Geography (2, 2)	
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	
<i>Option B (9-11 Credits)</i>		
For students interested in global climate change and environmental sustainability (including traditional approaches to pollution prevention and control).		
Choose one of the following:		
GEOG 210	Physical Geography (2, 2)	
GEOL 115	Earth Systems and Climate Change [C3NS]	
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	
Choose two of the following:		
GEOG 325	Environmental Ethics	

GEOG 327	Climatology	
GEOG 329	Global Climate Change	
GEOG 427	Water Resources of the World	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
HIST 427	U.S. Environmental History	
ISAT 320	Fundamentals of Environmental Science and Technology I	
ISAT 321	Fundamentals of Environmental Science and Technology II	
ISAT 422	Industrial Environmental Management	
ISAT 427	Industrial Hygiene	
ISAT 428	Industrial Ecology	
SOCI 311	Sociology of the Environment	
Subtotal		9-12
Total Credits		25-30

Concentration in Financial Economics

Students who elect a major in Economics with a concentration in Financial Economics are required to meet Economics degree requirements and complete the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Financial Economics Concentration Courses		
ECON/FIN 372	International Finance and Payments	3
ECON 430	Monetary Theory	3
or ECON 483	Applied Time Series Analysis	
It is also required that the following 12 credits of Finance courses be used to satisfy six credits of the 300-level Economics elective credits required for the degree. ¹		
FIN 301	Principles of Finance	3
or COB 300B	Integrative Business: Finance	
FIN 360	Analytical Methods in Finance	3
FIN 371	Principles of Investments	3
FIN 380	Elemental and Derivative Securities Analysis	3
Subtotal		18
Elective		
One 400-level ECON elective (not including ECON 490, ECON 499A)		3
Subtotal		3
Total Credits		21

¹ Note that this makes it possible to receive a degree in Economics with a concentration in Financial Economics with no more than 30 credits in Economics.

Concentration in International and Development Economics

Students who elect a major in Economics with a concentration in International and Development Economics are required to meet degree requirements and complete the courses listed below (which can also be used to satisfy degree requirements).

Code	Title	Credits
International and Development Economics Concentration Courses		
Choose four of the following:		12
ECON 312	Comparative Economic Systems	
ECON 365	Economic Development	
ECON 370	International Trade and Trade Policies	
ECON/FIN 372	International Finance and Payments	
ECON 437	Economic Growth	
Subtotal		12
Electives		
Two 400-level economics electives		6
Subtotal		6
Total Credits		18

Concentration in Political Economy

Students who elect a major in Economics with a concentration in political economy are required to meet degree requirements and complete four of the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Political Economy Concentration Courses		
Choose four of the following:		12
ECON 312	Comparative Economic Systems	
ECON 326	Public Finance	
ECON 327	Game Theory	
ECON 405	Political Economy	
ECON 426	Theory of Public Choice	
ECON 455	Economics of Regulated Industries	
Subtotal		12
Electives		
Two 400-level economics electives ¹		0-6
Subtotal		0-6
Total Credits		12-18

¹ ECON 405, ECON 426 and/or ECON 455 also fulfill this 400-level elective requirement if taken to satisfy the 12 credit hour concentration requirement.

Concentration in Quantitative Economics

Students who elect a major in Economics with a concentration in Quantitative Economics are required to meet degree requirements and complete the following courses (which can also be used to satisfy degree requirements). Note that along with the required 12 credit hours in the Quantitative Economics concentration, students are required to complete additional elective courses to satisfy the major requirements.

Code	Title	Credits
Quantitative Economics Concentration Courses		
ECON 483	Applied Time Series Analysis	3
or ECON 485	Advanced Econometrics	
Choose two of the following:		6
ECON 384	Mathematical Economics	
ECON 431	Advanced Microeconomic Theory	
ECON 432	Advanced Macroeconomics	

ECON 437	Economic Growth	
ECON 483	Applied Time Series Analysis ¹	
ECON 485	Advanced Econometrics ¹	
Subtotal		9
Electives		
One 300- or 400-level economics elective in addition to the courses taken to satisfy the credit hour concentration requirement		3
Subtotal		3
Total Credits		12

¹ If taken as a required course above, students cannot also apply the course in this category.

Concentration in Socioeconomics

Students who elect a major in Economics with a concentration in Socioeconomics are required to meet Economics degree requirements and complete the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Socioeconomics Concentration Courses		
Choose four of the following:		12
ECON 306	The Economics of Women and the Family	
ECON 328	Behavioral Economics	
ECON 329	Health Economics	
ECON 340	Economics of Natural Resources	
ECON 360	Labor Economics	
ECON 365	Economic Development	
ECON 382	Urban Economics	
ECON 427	Experimental Economics	
Subtotal		12
Electives		
Two 400-level economics electives		6
Subtotal		
Total Credits		18

Recommended Schedule for Majors
First Two Years

During the first two years, students should complete:

The 30-31 hour, lower-division B.B.A. core curriculum (failing to complete these courses before the first semester of the junior year could delay admission to the degree program and enrollment in COB 300A/COB 300B/COB 300C/COB 300D.

Code	Title	Credits
Most of the General Education curriculum		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
COB 191	Business Analytics I	3
Choose one of the following:		3-4
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	
MATH 205	Applied Calculus [C3QR]	

MATH 231	Calculus with Functions I [C3QR]
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MATH 235	Calculus I [C3QR]
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Third and Fourth Years

B.B.A. economics majors should take COB 300A, COB 300B, COB 300C, and COB 300D before the spring semester of their junior year and complete ECON 331, ECON 332, and preferably ECON 385 by the end of their junior year.

While most majors will complete the 400-level requirements in economics during their senior year, students may take a 400-level course during their junior year if the prerequisite for the course has been met. ECON 488 should be taken during the senior year.

Economics, B.S.

Program Description

Economics majors choose from a B.A., B.S. or B.B.A. degree. The B.A. and B.S. degrees are traditional liberal arts degrees that lead to a variety of career and graduate school options; the B.B.A. is designed to prepare students for careers in business.

The minimum requirement for a B.S. degree in economics is 36 credit hours of economics including 21 credit hours of core courses and 15 credit hours of electives. It is also necessary for the student to complete the quantitative and scientific literacy requirements for a B.S. degree as well as complete the General Education program.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business catalog page (p. 15).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement ²	3-4
Major requirements (listed below) and electives	72-73
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirements.

Major Requirements

Code	Title	Credits
Required Economics, B.S. Courses		
ECON 200	Introduction to Macroeconomics [C4GE] ¹	3
ECON 201	Introduction to Microeconomics ¹	3
ECON 284	Introduction to Quantitative Analysis in Economics ^{1,2}	3
ECON 331	Intermediate Microeconomic Theory ³	3
ECON 332	Intermediate Macroeconomic Theory ³	3
ECON 385	Econometrics ^{3,4}	3
ECON 488	Senior Capstone Seminar in Economics	3
Economics electives (at least six credits must be at the 400 level) ⁵		15
Subtotal		36
Additional Required Courses		
Choose one of the following calculus courses:		3-4
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 233	Calculus with Modeling I [C3QR]	

MATH 235	Calculus I [C3QR]	
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	
Choose one of the following statistics courses:		
MATH 220	Elementary Statistics [C3QR]	3
or COB 191	Business Analytics I	
Subtotal		6-7
Total Credits		42-43

¹ Students need a grade of "C" or better in ECON 200, ECON 201, and ECON 284 as a prerequisite for advanced economics courses.

² Students may substitute COB 291 for ECON 284.

³ Students need to complete ECON 331, ECON 332, and ECON 385 with a grade of "C" or better to graduate with an Economics degree.

⁴ Students must complete ECON 385 with a grade of "B-" or better in order to choose ECON 483 or ECON 485 as an elective option.

⁵ Not including ECON 488, ECON 490 or ECON 499A

Economics Concentrations

Concentration in Environmental and Natural Resource Economics

Students who elect a major in Economics with a concentration in Environmental and Natural Resource Economics are required to meet degree requirements and complete the following Economics courses (which can also be used to satisfy degree requirements) *as well as* the classes specified *in either* Option A or Option B.

Code	Title	Credits
Environmental and Natural Resource Economics Concentration Courses		
Choose two of the following:		6
ECON 305	Environmental Economics	
ECON 340	Economics of Natural Resources	
ECON 382	Urban Economics	
Choose one of the following:		1-3
ECON 394	Economics Internship (topic must be related to Environment and Natural Resources)	
ECON 490	Special Studies in Economics (topic must be related to Environment and Natural Resources)	
Subtotal		7-9
Electives		
One 200- or 300-level economics elective		3
Two 400-level economics electives		6
Subtotal		9
Options		
Choose one of the following options:		9-12
<i>Option A (12 Credits)</i>		
For students with an interest in environmental sustainability as it applies to forests, fisheries and wildlife.		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	
Choose one of the following:		

BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory
GEOG 210	Physical Geography (2, 2)
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]

Option B (9-11 Credits)

For students interested in global climate change and environmental sustainability (including traditional approaches to pollution prevention and control).

Choose one of the following:

GEOG 210	Physical Geography (2, 2)
GEOL 115	Earth Systems and Climate Change [C3NS]
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]

Choose two of the following:

GEOG 325	Environmental Ethics
GEOG 327	Climatology
GEOG 329	Global Climate Change
GEOG 427	Water Resources of the World
GEOG/ISAT 429	Sustainability: An Ecological Perspective
HIST 427	U.S. Environmental History
ISAT 320	Fundamentals of Environmental Science and Technology I
ISAT 321	Fundamentals of Environmental Science and Technology II
ISAT 422	Industrial Environmental Management
ISAT 427	Industrial Hygiene
ISAT 428	Industrial Ecology
SOCI 311	Sociology of the Environment

Subtotal 9-12

Total Credits 25-30

Concentration in Financial Economics

Students who elect a major in Economics with a concentration in Financial Economics are required to meet Economics degree requirements and complete the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Financial Economics Concentration Courses		

ECON/FIN 372	International Finance and Payments	3
ECON 430	Monetary Theory	3
or ECON 483	Applied Time Series Analysis	

It is also required that the following 12 credits of Finance courses be used to satisfy six credits of the 300-level Economics elective credits required for the degree.¹

FIN 301	Principles of Finance	3
or COB 300B	Integrative Business: Finance	
FIN 360	Analytical Methods in Finance	3
FIN 371	Principles of Investments	3
FIN 380	Elemental and Derivative Securities Analysis	3
Subtotal		18

Elective		
One 400-level ECON elective (not including ECON 490, ECON 499A)		3
Subtotal		3
Total Credits		21

¹ Note that this makes it possible to receive a degree in Economics with a concentration in Financial Economics with no more than 30 credits in Economics.

Concentration in International and Development Economics

Students who elect a major in Economics with a concentration in International and Development Economics are required to meet degree requirements and complete the courses listed below (which can also be used to satisfy degree requirements).

Code	Title	Credits
International and Development Economics Concentration Courses		
Choose four of the following:		12
ECON 312	Comparative Economic Systems	
ECON 365	Economic Development	
ECON 370	International Trade and Trade Policies	
ECON/FIN 372	International Finance and Payments	
ECON 437	Economic Growth	
Subtotal		12
Electives		
Two 400-level economics electives		6
Subtotal		6
Total Credits		18

Concentration in Political Economy

Students who elect a major in Economics with a concentration in political economy are required to meet degree requirements and complete four of the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Political Economy Concentration Courses		
Choose four of the following:		12
ECON 312	Comparative Economic Systems	
ECON 326	Public Finance	
ECON 327	Game Theory	
ECON 405	Political Economy	
ECON 426	Theory of Public Choice	
ECON 455	Economics of Regulated Industries	
Subtotal		12
Electives		
Two 400-level economics electives ¹		0-6
Subtotal		0-6
Total Credits		12-18

¹ ECON 405, ECON 426 and/or ECON 455 also fulfill this 400-level elective requirement if taken to satisfy the 12 credit hour concentration requirement.

Concentration in Quantitative Economics

Students who elect a major in Economics with a concentration in Quantitative Economics are required to meet degree requirements and complete the following courses (which can also be used to satisfy degree requirements). Note that along with the required 12 credit hours in the Quantitative Economics concentration, students are required to complete additional elective courses to satisfy the major requirements.

Code	Title	Credits
Quantitative Economics Concentration Courses		
ECON 483	Applied Time Series Analysis	3
or ECON 485	Advanced Econometrics	
Choose two of the following:		6
ECON 384	Mathematical Economics	
ECON 431	Advanced Microeconomic Theory	
ECON 432	Advanced Macroeconomics	
ECON 437	Economic Growth	
ECON 483	Applied Time Series Analysis ¹	
ECON 485	Advanced Econometrics ¹	
Subtotal		9
Electives		
One 300- or 400-level economics elective in addition to the courses taken to satisfy the credit hour concentration requirement		3
Subtotal		3
Total Credits		12

¹ If taken as a required course above, students cannot also apply the course in this category.

Concentration in Socioeconomics

Students who elect a major in Economics with a concentration in Socioeconomics are required to meet Economics degree requirements and complete the following courses (which can also be used to satisfy degree requirements).

Code	Title	Credits
Socioeconomics Concentration Courses		
Choose four of the following:		12
ECON 306	The Economics of Women and the Family	
ECON 328	Behavioral Economics	
ECON 329	Health Economics	
ECON 340	Economics of Natural Resources	
ECON 360	Labor Economics	
ECON 365	Economic Development	
ECON 382	Urban Economics	
ECON 427	Experimental Economics	
Subtotal		12
Electives		
Two 400-level economics electives		6
Subtotal		
Total Credits		18

Recommended Schedule for Majors

First Two Years

During the first two years, students should complete:

Code	Title	Credits
Most of the General Education program		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
ECON 284	Introduction to Quantitative Analysis in Economics	3
COB 191	Business Analytics I	3
Choose one of the following:		3-4
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 233	Calculus with Modeling I [C3QR]	
MATH 235	Calculus I [C3QR]	
Total Credits		15-16

Additional Information

Students are encouraged to take ECON 331 and/or ECON 332 in their sophomore year (provided that the course prerequisites have been met).

Third and Fourth Years

B.S. economics majors should complete ECON 331, ECON 332, and preferably ECON 385 by the end of their junior year. While most majors will complete the 400-level requirements in economics during their senior year, students may take a 400-level course during their junior year if the prerequisite for the course has been met. ECON 488 should be taken during the senior year.

Elementary Education, B.S.

Program Description

The Elementary Education Program prepares candidates to teach students in grades PreK-6. Drawn from research and theories in child development, teaching and learning, the course work and field experiences prepare teacher candidates to employ a cross-disciplinary approach to instruction that reflects an understanding of the diverse nature of learners and their families.

The JMU Elementary Education Program seeks to foster in its candidates an empathic understanding of the ways that children are affected by social contexts and by the children's own strengths and needs; the knowledge and pedagogical skills to support each child's right to success; and belief in the value of each child. Our candidates are guided in:

- Critically challenging conventional wisdom and common practices to identify hidden assumptions and activities that constrain or privilege some at the expense of others.
- Learning to ask questions and developing an inquiring approach motivated by the desire to understand the world in its myriad complexities.
- Reflecting deeply on and constructing positive relationships with others.
- Expressing knowledge, skills and attitudes in ways that communicate with others and provide a forum for the creative and academic expression of the profession and the self.
- Developing an appreciation for the global connection of all humanity and our interdependence on the finite natural resources of the earth.
- Experiencing life among people whose social contexts are unlike the candidates' own to broaden and deepen respect for and sensitivity to various cultures and social contexts.
- Knowing and appreciating the process of human unfolding throughout the cycles of life from conception onward, particularly throughout the period of childhood and preadolescence.
- Valuing the literacies and literate backgrounds of all learners to develop equitable, inclusive instruction that honors diverse learners at all developmental phases.
- Reflecting critically on the ways we embody and enact a sustainable sense of self.
- Building on interdisciplinary connections among content areas to create inclusive, responsive and effective instruction for all learners.

The courses in the elementary education program are organized to integrate academic content with pedagogical practice throughout the major. Field experiences are provided along with course work to enable candidates to apply their knowledge in a variety of settings. Candidates must be fully accepted into teacher education to begin the ELED course work in the third year.

Assessment occurs during each semester, and candidate performance is reviewed at the end of each semester. Candidates must demonstrate satisfactory performance before moving on to the next semester. Satisfactory performance includes a minimum grade of "C" in all education course work and an overall 2.5 GPA, demonstration of professional behaviors, and acceptable performance in practica and on key assessments.

Candidates should note that they must be fully admitted to teacher education prior to registering for the courses offered during Year 3 of the elementary education program. Candidates should also note that, on occasion, Virginia Department of Education or other accrediting agencies may change requirements after the catalog copy is approved. Therefore, it is especially important for candidates to confer with their education advisors on a regular basis for the most current requirements. Completion of the Elementary Education major leads to a Virginia teaching license with an endorsement in PK-6 Elementary Education. To be recommended for licensure, all candidates must meet the following requirements:

- Complete General Education requirements.
- Complete the Elementary Education pre-professional program.
- Complete the disciplinary courses required for the elementary education degree.
- Meet all admission and retention criteria for teacher education including satisfactory scores on the Praxis Core Math, RVE and Praxis Elementary Multiple Subjects tests.
- Meet admission requirements for the Elementary Education program.
- Meet performance and behavior standards as indicated by ratings on the program Professional Dispositions Checklist.

Admission Requirements

Students must meet specific requirements to be admitted to programs within the College of Education (p. 18). These criteria can be found on the College of Education (p. 18) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Scientific Literacy ²	3-4
Quantitative Reasoning ³	3
Major Requirements	65-70
University Electives	2-8
Total	120

¹ Many of the pre-professional requirements may also count toward general education requirements.

² In addition to General Education. Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml). May be fulfilled through the pre-professional requirements.

³ MATH 210 fulfills this requirement.

Pre-professional Requirements

These courses may fulfill General Education requirements and/or university electives for this program.

Students majoring in Elementary Education will be required to complete JMU General Education with additional courses related to Bachelor of Science requirements (Quantitative and Scientific Literacy). Given required VDOE competencies associated with PK-6 teacher licensure, students will be required to enroll in the courses below.

All students are required to take both MATH 110 and MATH 210 to meet VDOE competencies.

Code	Title	Credits
MATH 110 & MATH 210	Mathematics for Elementary and Middle School Teachers I [C3QR] and Mathematics for Elementary and Middle School Teachers II	8
Subtotal		8
Arts and Humanities (C2HQC)		
HIST 101 or HIST 102	World History to 1500 [C2HQC] World History since 1500 [C2HQC]	3
Subtotal		3
The Natural World (C3T1G1)		
ISCI 171	Earth and Planetary Science for Teachers [C3NS]	3
ISCI 172	Physical Science for Teachers [C3PP]	3
ISCI 173 or BIO 140	Life and Environmental Science for Teachers [C3L] Foundations of Biology I [C3NS]	3
Subtotal		9
American and Global Perspectives (C4AE)		
ANTH 195 or GEOG 200	Cultural Anthropology [C4GE] Geography: The Global Dimension [C4GE]	3
HIST 225 or POSC 225	U.S. History [C4AE] U.S. Government [C4AE]	4
Subtotal		7
Sociocultural and Wellness Area (C5SD)		
PSYC 160 or EDUC 175	Life Span Human Development [C5SD] Learning, Behavior and Engagement in Context [C5SD]	3
Subtotal		3
Total Credits		30

Major Requirements

Code	Title	Credits
Elementary Education Core Requirements		
EDUC 200	Foundations of Education in the United States	3
ELED 272	Introduction to Early and Elementary Education	3
ELED 343	Children and Mathematics II	3
ELED 450	Planning Instruction and Assessment	3
ELED 455	Home, School, Communities	3
ELED 485	Guiding Student Behavior: Responsive Classroom Management	3
EXED 200	Foundations of Exceptional Education	3
LED 300	Reading Instruction and Assessment for Diverse Learners PK-6	3
Subtotal		24
Elementary Education Practicum/Student Teaching Requirements		
ELED 321	Practicum with a Focus on Learners and Learning	1
ELED 322	Practicum with a Focus on Curriculum Integration and Guiding Behavior	1
ELED 400	Elementary Education Final Practicum (taken twice for a total of 2 credits)	2
ELED 480	Student Teaching in Elementary Education (taken twice for a total of 16 credits)	16
Subtotal		20
Disciplinary Requirements or Concentration		
Select one of the following:		21-26

Elementary Education Disciplinary Requirements (see below)	
Concentration in Global Education (see below)	
Subtotal	21-26
Total Credits	65-70

Elementary Education Disciplinary Requirements

These courses are required of ELED majors who do not elect the concentration in Global Education.

Code	Title	Credits
ELED 208	Child Development: Birth Through Adolescence	3
ELED 310	Diversity in Elementary Education	3
ELED 332	Children and Science	3
ELED 333	Children and Mathematics I	3
ELED 334	Children and Social Studies	3
LED 350	Writing Instruction, Assessment and Technology for Diverse Learners PK-6	3
LED 462	Disciplinary Literacy for Diverse Classrooms PK-6	3
Total Credits		21

Concentration in Global Education

Students in the Concentration in Global Education take the courses listed below in place of the “Elementary Education Disciplinary Requirements” above.

Code	Title	Credits
ELED 200	Introduction to Global Education	3
ELED 270	Global Diversity in Elementary Education	3
ELED 278	Child Development in a Global Context: Birth Through Adolescence	3
ELED 372	Children and Science in a Global Context	3
ELED 373	Children and Mathematics I in a Global Context	3
ELED 374	Children and Social Studies in a Global Context	3
ELED 375	Seminar in Global Education Praxis I	1
ELED 475	Seminar in Global Education Praxis II	1
LED 370	Writing Instruction, Assessment and Technology for Diverse Global Learners (PK-6)	3
LED 473	Disciplinary Literacy for Diverse Classrooms in PK-6 Global Contexts	3
Total Credits		26

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
EDUC 200	Foundations of Education in the United States	3
EXED 200	Foundations of Exceptional Education	3
General Education courses, minor courses and/or university electives		9
Credits		15
Spring Semester		
EDUC 175 or PSYC 160	Learning, Behavior and Engagement in Context [C5SD] or Life Span Human Development [C5SD]	3
General Education courses, minor courses and/or university electives		12
Credits		15
Second Year		
Fall Semester		
ELED 208	Child Development: Birth Through Adolescence	3

ELED 272	Introduction to Early and Elementary Education	3
ELED 310	Diversity in Elementary Education	3
General Education courses, minor courses and/or university electives		6
Credits		15
Spring Semester		
General Education courses, minor courses and/or university electives		15
Credits		15
Third Year		
Fall Semester		
ELED 321	Practicum with a Focus on Learners and Learning	1
ELED 332	Children and Science	3
ELED 333	Children and Mathematics I	3
LED 300	Reading Instruction and Assessment for Diverse Learners PK-6	3
General Education courses, minor courses and/or university electives		6
Credits		16
Spring Semester		
ELED 322	Practicum with a Focus on Curriculum Integration and Guiding Behavior	1
General Education courses, minor courses and/or university electives		4
ELED 334	Children and Social Studies	3
ELED 343	Children and Mathematics II	3
LED 350	Writing Instruction, Assessment and Technology for Diverse Learners PK-6	3
Credits		14
Fourth Year		
Fall Semester		
ELED 400	Elementary Education Final Practicum	1
ELED 450	Planning Instruction and Assessment	3
ELED 480	Student Teaching in Elementary Education	8
ELED 485	Guiding Student Behavior: Responsive Classroom Management	3
Credits		15
Spring Semester		
ELED 400	Elementary Education Final Practicum	1
ELED 455	Home, School, Communities	3
ELED 480	Student Teaching in Elementary Education	8
LED 462	Disciplinary Literacy for Diverse Classrooms PK-6	3
Credits		15
Total Credits		120

Recommended Schedule for Global Education Concentration

Course	Title	Credits
First Year		
Fall Semester		
EDUC 200	Foundations of Education in the United States	3
EXED 200	Foundations of Exceptional Education	3
General Education courses, minor courses and/or university electives		9
Credits		15
Spring Semester		
EDUC 175 or PSYC 160	Learning, Behavior and Engagement in Context [C5SD] or Life Span Human Development [C5SD]	3
ELED 200	Introduction to Global Education	3
General Education courses, minor courses and/or university electives		9
Credits		15
Second Year		
Fall Semester		
General Education courses, minor courses and/or university electives		15
Credits		15

Spring Semester		
ELED 278	Child Development in a Global Context: Birth Through Adolescence	3
ELED 272	Introduction to Early and Elementary Education	3
ELED 270	Global Diversity in Elementary Education	3
General Education courses, minor courses and/or university electives		4
Credits		13
Third Year		
Fall Semester		
ELED 321	Practicum with a Focus on Learners and Learning	1
ELED 372	Children and Science in a Global Context	3
ELED 373	Children and Mathematics I in a Global Context	3
LED 300	Reading Instruction and Assessment for Diverse Learners PK-6	3
General Education courses, minor courses and/or university electives		6
Credits		16
Spring Semester		
ELED 322	Practicum with a Focus on Curriculum Integration and Guiding Behavior	1
ELED 374	Children and Social Studies in a Global Context	3
ELED 343	Children and Mathematics II	3
LED 370	Writing Instruction, Assessment and Technology for Diverse Global Learners (PK-6)	3
ELED 375	Seminar in Global Education Praxis I	1
General Education courses, minor courses and/or university electives		4
Credits		15
Fourth Year		
Fall Semester		
ELED 400	Elementary Education Final Practicum	1
ELED 450	Planning Instruction and Assessment	3
ELED 480	Student Teaching in Elementary Education	8
ELED 485	Guiding Student Behavior: Responsive Classroom Management	3
Credits		15
Spring Semester		
ELED 400	Elementary Education Final Practicum	1
ELED 455	Home, School, Communities	3
ELED 475	Seminar in Global Education Praxis II	1
ELED 480	Student Teaching in Elementary Education	8
LED 472	Literacy Assessment and Instruction in the Content Areas for the Middle Grades	3
Credits		16
Total Credits		120

Engineering, B.S.

Program Description

JMU Engineering supports an ABET accredited, 4-year interdisciplinary Bachelor of Science degree program. The degree program affords a world-class undergraduate engineering experience based on a philosophy of learning by doing. It integrates many traditional engineering disciplines with course work in project management, engineering, design and liberal arts with foci on sustainable design and project delivery.

Admission to the Major

Admission to JMU does not guarantee admission to the engineering major. Full admission to the engineering major is restricted due to the project-intensive nature of the undergraduate engineering curriculum. The Engineering Academic Unit will strive to accommodate all interested students who have demonstrated the potential to succeed in the engineering major, typically at the sophomore-level. If more students meet the standards for admission than there are available seats, then enrollment will be limited.

Any student can declare the engineering major in order to take the prerequisite courses with an ENGR prefix. To progress beyond the prerequisite courses with an ENGR prefix, students must be fully admitted to the engineering major. For full admission to the engineering major, students must meet the standards for admission.

The standards for admission for on-campus students include:

- Being enrolled at JMU.
- Having declared the engineering major.
- Having completed a minimum of 15 credit hours at JMU.
- Having completed the following courses (or approved equivalents): ENGR 112, ENGR 231, MATH 232 or MATH 235, and PHYS 240.
- Having completed the following courses (or approved equivalents) with a grade of "C-" or better: ENGR 112 and ENGR 231.
- Submitting an official application that includes a written response of no more than 250 words that demonstrates the students' motivation to pursue an engineering academic career.

The standards for admission for transfer students include:

- Being enrolled at JMU.
- Having declared the engineering major.
- Having completed a minimum of 15 credit hours at JMU.
- Having completed the following courses (or approved equivalents): ENGR 301, ENGR 311 or ENGR 312 or ENGR 313 or ENGR 314 or ENGR 315 or ENGR 317 or ENGR 331, and ENGR 321.
- Having completed the following courses (or approved equivalents) with a grade of "C-" or better: ENGR 301 and ENGR 321.
- Submitting an official application that includes a written response of no more than 250 words that demonstrates the students' motivation to pursue an engineering academic career.

If more students meet the standards for admission than there are seats available, then enrollment will be limited. When enrollment is limited, a committee of engineering faculty members will evaluate students' potential to succeed in the engineering major by considering the following:

- The average weighted GPA of the highest letter grades earned in ENGR 112 and ENGR 231 (or approved equivalents) for on-campus students; or the average weighted GPA of the highest letter grades earned in ENGR 301 and ENGR 321 (or approved equivalents) for transfer students.
- The average weighted GPA of the highest letter grades earned across MATH 232 or MATH 235, and PHYS 240 (or approved equivalents) for on-campus students; or the average weighted GPA of the highest letter grades earned across ENGR 311 or ENGR 312 or ENGR 313 or ENGR 314 or ENGR 315 or ENGR 317 or ENGR 331 (or approved equivalents) for transfer students.
- A written response within the official application that will be evaluated by a committee of engineering faculty members.

Any of the preceding ENGR courses may be taken for a final letter grade no more than twice to be considered for full admission to the engineering major, excluding final letter grades of W, WP, and WF. The Engineering Academic Unit Head may permit an exception due to extraordinary circumstances, such as illness or other extenuating circumstances.

On-campus students' ENGR grades will be considered with a stronger emphasis than students' MATH and PHYS grades. Transfer students' overall academic performance while at JMU will be considered. Final admission decisions will be made by a committee of engineering faculty members.

Admission for on-campus students occurs at two points during the academic year: at the end of the spring semester and at the start of the fall semester. Admission for transfer students occurs at two points during the academic year: at the end of the fall semester and at the end of the spring semester. Students will be notified of their engineering major admission status no later than May 31st for admission for the upcoming fall semester and no later than January 1st for admission for the upcoming spring semester.

Students who are not admitted during the admission point may elect to be considered at the next admission point, if they so choose, though the same admission criterion applies. Students may apply to the engineering major no more than two times in total. The Engineering Academic Unit Head may permit an exception due to extraordinary circumstances, such as illness or other extenuating circumstances.

Progressing in the Major

Students who are fully admitted to the Engineering major and who fail to meet minimum grade requirements in any course required for the Engineering major across two attempts (excluding final letter grades of W, WP and WF) are subject to losing their status as being fully admitted to the Engineering major. Students who lose their status as being fully admitted to the Engineering major may elect to be considered for full admission at the next admission point, if they so choose, though the same admission criterion applies. Students may apply to the Engineering major no more than two times in total. The Engineering Academic Unit Head may permit an exception due to extraordinary circumstances, such as illness or other extenuating circumstances.

Transfer Information

For students transferring from a program with an existing articulation agreement, the student coordinator will work with the Engineering Academic Unit Head to assess the relative merit of transfer students with qualifying transcripts.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education) ²	3
Major requirements and approved electives	95
Total	129

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirement. Some credits may be fulfilled by courses required within the major.

Major Requirements

Code	Title	Credits
ENGR Prerequisites		
Students must earn a minimum grade of "C-" in the following courses (or approved equivalents) to be fully admitted to the engineering major.		
ENGR 112	Engineering Skills and Decisions	3
ENGR 231	Engineering Design I: Product Design and Development	3
Subtotal		6
MATH and PHYS Prerequisites		
Students must complete the following courses (or approved equivalents) to be fully admitted to the engineering major.		
MATH 235	Calculus I [C3QR] ¹	4
PHYS 240	University Physics I [C3PP]	3
Subtotal		7
Required Courses		
The following courses (or approved equivalents) are required to complete the engineering major.		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations ²	4
PHYS 240L	University Physics Laboratory I [C3L]	1
PHYS 250 & 250L	University Physics II and University Physics Laboratory II	4
ENGR 212 & 212L	Statics and Introductory Solid Mechanics and Statics and Introductory Solid Mechanics Laboratory ³	4
ENGR 226	Engineering Project Management ³	3
ENGR 232	Engineering Design II: Human-Centered Design and Systems Analysis ³	3
ENGR 311 & 311L	Fluid Mechanics and Fluid Mechanics Laboratory	4

ENGR 312 & 312L	Thermodynamics and Heat Transfer and Thermodynamics and Heat Transfer Laboratory	4
ENGR 313 & 313L	Circuits and Devices and Circuits and Devices Laboratory	4
ENGR 314 & 314L	Materials and Intermediate Solid Mechanics and Materials and Intermediate Solid Mechanics Laboratory	4
ENGR 315	Computational Methods in Engineering	3
ENGR 317	Principles of Sustainable Engineering	3
ENGR 321	Engineering Practicum I ³	1
ENGR 323	Engineering Practicum II ³	1
ENGR 324	Engineering Project I ³	1
ENGR 331	Engineering Design III: Design Thinking and Fabrication Methods	3
ENGR 421	Engineering Practicum III ³	1
ENGR 422	Engineering Project II ³	1
ENGR 424	Engineering Capstone	2
Subtotal		67

ENGR Electives

A total of fifteen credit hours of approved electives are required to complete the engineering major, including at least twelve credit hours of engineering technical electives. Students may not use courses required for the engineering major to fulfill approved elective requirements.

ENGR Technical Electives

12 credits of ENGR Technical Electives may include (but are not limited to): ⁴

ENGR 345	Foundations of Computer Engineering
ENGR 360	Structural Analysis
ENGR 387	Foundations of Brewery Engineering and Systems
ENGR 426	Product Development and Entrepreneurial Engineering
ENGR 440	Mechatronics
ENGR 445	Embedded System Design
ENGR 452	Mechanisms in Motion: Applied Kinematics
ENGR 453	Machine Design
ENGR 457	Electromechanical Systems Analysis
ENGR 461	Structural Systems Analysis
ENGR 462	Foundations of Structural Engineering Design
ENGR 463	Concrete Infrastructure
ENGR 466	Mechanical Vibrations and Structural Dynamics
ENGR 471	Water in Africa
ENGR 472	Environmental Engineering
ENGR 477	Sustainable Systems Analysis
ENGR 478	Water Resources Engineering
ENGR 481	Principles of Process Engineering
ENGR 487	Brewery Engineering and Systems Practicum

Other Approved Electives

At most three credit hours of any other non-required 300- or 400-level courses at JMU (both ENGR and non-ENGR courses are eligible), excluding ENGR 301 and ENGR 401. This may include ENGR 490, ENGR 496, ENGR 497, ENGR 498, ENGR 499A, ENGR 499B and ENGR 499C. This may also include courses that qualify as engineering technical electives and that are not used to meet the engineering technical elective requirement.

Total Credits **95**

¹ It is allowable to substitute MATH 235 with both MATH 231 and MATH 232.

² It is allowable to substitute MATH 238 with both MATH 300 and MATH 336.

³ A minimum grade of "C-" is required to apply this course towards degree requirements.

⁴ Technical electives must be non-required, 300- or 400-level ENGR courses, excluding ENGR 301, ENGR 401, ENGR 490, ENGR 496, ENGR 497, ENGR 498, ENGR 499A, ENGR 499B and ENGR 499C.

Concentrations

Each student, in consultation with their engineering academic advisor, may choose to pursue one of the following program concentrations.

Civil and Environmental Systems Concentration

The civil and environmental systems concentration prepares engineering students to analyze structural systems as well as to design, model, and build infrastructure that abide by codes, specifications, and responsible stewardship of natural resources. Through projects and coursework, this concentration provides opportunities to learn about computational techniques in analyzing and modeling structures, building codes and construction standards, design of steel and concrete infrastructure, and sustainable practices that protect the natural environment.

Students must take 3 required credit hours and 12 elective credit hours to complete the civil and environmental systems concentration.

Code	Title	Credits
Required Civil and Environmental Systems Concentration Course		
ENGR 360	Structural Analysis	3
Subtotal		3
Civil and Environmental Systems Concentration Electives		
Select 12 credits from the following:		12
ENGR 461	Structural Systems Analysis	
ENGR 462	Foundations of Structural Engineering Design	
ENGR 463	Concrete Infrastructure	
ENGR 466	Mechanical Vibrations and Structural Dynamics	
ENGR 472	Environmental Engineering	
ENGR 477	Sustainable Systems Analysis	
ENGR 478	Water Resources Engineering	
ENGR 498	Advanced Topics in Engineering (Civil and Environmental - must be taken for 3.00 credit hours)	
Subtotal		12
Total Credits		15

3 Geo-Environmental Systems Concentration

The geo-environmental systems concentration prepares engineering students to analyze environmental systems as well as to design and build engineered processes that control pollution or restore environmental functions to natural ecosystems. Through projects and coursework, this concentration provides opportunities to learn about developing sustainable agricultural practices, designing water treatment systems for sub-Saharan African communities, developing carbon credits for carbon markets, stream and shore restoration projects, and life cycle analysis to engineer more sustainable products and processes.

Students must take 3 required credit hours and 12 elective credit hours to complete the geo-environmental systems concentration.

The 12 elective credit hours may be fulfilled from the geo-environmental systems concentration ENGR electives. Or students may fulfill the 12 elective credit hours with 9 elective credit hours from the geo-environmental systems concentration ENGR electives and an additional 3 elective credit hours from the geo-environmental systems concentration GEOL electives.

Code	Title	Credits
Required Geo-Environmental Systems Concentration Course		
ENGR 472	Environmental Engineering	3
Subtotal		3
Geo-Environmental Systems Concentration ENGR Electives		
Select 9-12 credit hours of the following:		9-12
ENGR 471	Water in Africa	
ENGR 477	Sustainable Systems Analysis	
ENGR 478	Water Resources Engineering	
ENGR 481	Principles of Process Engineering	
ENGR 498	Advanced Topics in Engineering (Geo-Environmental - must be taken for 3.00 credit hours)	
Subtotal		9-12
Geo-Environmental Systems Concentration GEOL Electives		
Select 0-3 credit hours of the following:		0-3
GEOL 360	GIS for the Geoscientist ¹	
GEOL 410	Engineering Geology (2, 2) ¹	
GEOL 460	Hydrogeology (2, 2) ¹	
Subtotal		0-3
Total Credits		15

¹ GEOL 210 is a prerequisite.

Electro-Mechanical Systems Concentration

The electro-mechanical systems concentration prepares engineering students to design, analyze, and build engineered systems that exist at the intersections of mechanical, electrical, and computer engineering. Through projects and coursework, this concentration provides opportunities to learn about various electromechanical systems such as autonomous/electric vehicles, robotics, industrial internet of things, process control design, or any other system that senses, responds physically or electrically, and is under computational control.

Students must take 15 elective credit hours to complete the electro-mechanical systems concentration.

Code	Title	Credits
Electro-Mechanical Systems Concentration Electives		
Select 15 credit hours of the following:		15
ENGR 345	Foundations of Computer Engineering	
ENGR 440	Mechatronics	
ENGR 445	Embedded System Design	
ENGR 452	Mechanisms in Motion: Applied Kinematics	
ENGR 453	Machine Design	
ENGR 457	Electromechanical Systems Analysis	
ENGR 466	Mechanical Vibrations and Structural Dynamics	
ENGR 498	Advanced Topics in Engineering (Electro-Mechanical Systems - must be taken for 3.00 credit hours)	
Total Credits		15

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
ENGR 112	Engineering Skills and Decisions	3
MATH 235	Calculus I [C3QR]	4
PHYS 240 & 240L	University Physics I [C3PP] and University Physics Laboratory I [C3L]	4
General Education		3
Credits		14
Spring Semester		
ENGR 231	Engineering Design I: Product Design and Development	3
MATH 236	Calculus II	4
PHYS 250 & 250L	University Physics II and University Physics Laboratory II	4
General Education		3
Credits		14
Second Year		
Fall Semester		
ENGR 232	Engineering Design II: Human-Centered Design and Systems Analysis	3
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
MATH 237	Calculus III	4
General Education		6
Credits		17
Spring Semester		
ENGR 212 & 212L	Statics and Introductory Solid Mechanics and Statics and Introductory Solid Mechanics Laboratory	4
ENGR 226	Engineering Project Management	3
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
MATH 238	Linear Algebra with Differential Equations	4
General Education		3
Credits		18
Third Year		
Fall Semester		
ENGR 311 & 311L	Fluid Mechanics and Fluid Mechanics Laboratory	4
ENGR 313 & 313L	Circuits and Devices and Circuits and Devices Laboratory	4
ENGR 315	Computational Methods in Engineering	3
ENGR 321	Engineering Practicum I	1
ENGR 331	Engineering Design III: Design Thinking and Fabrication Methods	3

General Education		3
Credits		18
Spring Semester		
ENGR 312 & 312L	Thermodynamics and Heat Transfer and Thermodynamics and Heat Transfer Laboratory	4
ENGR 314 & 314L	Materials and Intermediate Solid Mechanics and Materials and Intermediate Solid Mechanics Laboratory	4
ENGR 323	Engineering Practicum II	1
ENGR 324	Engineering Project I	1
Approved engineering elective		3
General Education		3
Credits		16
Fourth Year		
Fall Semester		
ENGR 317	Principles of Sustainable Engineering	3
ENGR 421	Engineering Practicum III	1
ENGR 422	Engineering Project II	1
Approved engineering elective		3
General Education		7
Credits		15
Spring Semester		
ENGR 424	Engineering Capstone	2
Approved engineering elective		3
Approved engineering elective		3
Approved engineering elective		3
General Education		6
Credits		17
Total Credits		129

English, B.A.

Program Description

Students majoring in English earn the B.A. degree unless they are completing an additional major in a degree program other than the B.A. The minimum requirement for a major in English is 36 credit hours of coursework in the major.

The department recommends that students choose a variety of courses from among different periods and topics. Students progress through the major by taking ENG 221, ENG 222, ENG 223, ENG 299, and ENG 399. Students select among 300-level courses to meet intermediate degree requirements and complete at least two 400-level Advanced Studies courses. They also fulfill three distributional overlay requirements at the 300 and 400 level.

Courses taken to fulfill Arts and Humanities requirements may also fulfill requirements in the English major.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	36
University electives	28-40
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement exam.

Major Requirements

Code	Title	Credits
Core Courses		
ENG 221	Literature and Ideas [C2L]	3
ENG 222	Literature and Genres [C2L]	3
ENG 223	Literature and Cultures	3
ENG 299	Introduction to Literary Studies ¹	3
ENG 399	Advanced Writing in Literary Studies	3
Subtotal		15
Restricted Electives		
In addition to the core courses above, students must:		
Choose one course at the 200 or 300 level		3
Choose four courses from the 300 level		12
Choose two approved 400-level courses from the "Advanced Studies in..." courses		6
See list of courses below to meet intermediate and advanced studies overlay requirements:		

One course at the 300 or 400 level, pre-1700	
One course at the 300 or 400 level, pre-1900	
One course from the Identity, Diversity and Power selections	
Subtotal	21
Total Credits	36

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

300 or 400 Level, Pre-1700

Code	Title	Credits
ENG 301	Old English Language and Literature	3
ENG 306	The Bible as Literature	3
ENG 311	Medieval Literature and Culture	3
ENG 313	Studies in Early Modern Literature and Culture	3
ENG 316	Early Modern Drama	3
ENG 317	Studies in Shakespeare	3
ENG 319	Teaching Shakespeare	3
ENG 320	Shakespeare on the Page and Stage in London	3
ENG 401	Advanced Studies in Medieval Literature	3
ENG 402	Advanced Studies in British Literature Before 1700	3

300 or 400 Level, Pre-1900

Code	Title	Credits
ENG 301	Old English Language and Literature	3
ENG 306	The Bible as Literature	3
ENG 311	Medieval Literature and Culture	3
ENG 313	Studies in Early Modern Literature and Culture	3
ENG 316	Early Modern Drama	3
ENG 317	Studies in Shakespeare	3
ENG 319	Teaching Shakespeare	3
ENG 320	Shakespeare on the Page and Stage in London	3
ENG 321	Restoration and Eighteenth-Century British Prose and Poetry	3
ENG 322	Restoration and Eighteenth-Century British Drama	3
ENG 325	Romantic Literature	3
ENG 327	Gothic Literature	3
ENG 329	Victorian Literature	3
ENG 330	Nineteenth-Century British Novel	3
ENG 342	Early American Literature	3
ENG 343	Antebellum American Literature	3
ENG 344	Late Nineteenth-Century American Literature	3
ENG 352	The American Novel to 1914	3
ENG 363	Native American Literature	3
ENG 401	Advanced Studies in Medieval Literature	3
ENG 402	Advanced Studies in British Literature Before 1700	3

Identity, Diversity and Power

Code	Title	Credits
ENG 358	Oral Literature	3
ENG 359	Studies in African American Literature	3
ENG 360	Introduction to Ethnic American Literature	3
ENG 361	African American Fiction	3

ENG 362	African American Poetry	3
ENG 363	Native American Literature	3
ENG 364	Introduction to Asian American Literature	3
ENG 367	Latinx Literature	3
ENG 368	Women's Literature	3
ENG 369	Feminist Literary Theory	3
ENG 370	Queer Literature	3
ENG 374	Contemporary Global Anglophone Literature	3
ENG 377	Introduction to African Literature	3
ENG 378	Studies in South Asian Literature	3
ENG 379	Literature and Empire	3
ENG 408	Advanced Studies in African American Literature	3
ENG 423	Advanced Studies in Gender and Sexuality in Literature	3
ENG 432	Advanced Studies in African Literature	3
ENG 434	Advanced Studies in Latin American Literature in Translation	3
ENG 466	Advanced Studies in Women's Literature	3

400 Level

Code	Title	Credits
ENG 401	Advanced Studies in Medieval Literature	3
ENG 402	Advanced Studies in British Literature Before 1700	3
ENG 403	Advanced Studies in British Literature After 1700	3
ENG 405	Advanced Studies in Anglophone Literature	3
ENG 407	Advanced Studies in American Literature	3
ENG 408	Advanced Studies in African American Literature	3
ENG 410	Advanced Studies in Author	3
ENG 413	Advanced Studies in Literature and Ideas	3
ENG 414	Advanced Studies in Genre	3
ENG 415	Advanced Studies in Textuality and the History of the Book	3
ENG 417	Advanced Studies in Linguistics and the English Language	3
ENG 420	Advanced Studies in Theory and Cultural Studies	3
ENG 423	Advanced Studies in Gender and Sexuality in Literature	3
ENG 430	Advanced Studies in Comparative Literature	3
ENG 432	Advanced Studies in African Literature	3
ENG 433	Advanced Studies in Arabic Literature	3
ENG 434	Advanced Studies in Latin American Literature in Translation	3
ENG 439	Advanced Studies in Major Authors of Literature in Spanish in Translation	3
ENG 466	Advanced Studies in Women's Literature	3

Concentrations

In consultation with their advisors, students may choose nine credits of courses at the 300- or 400-level that acquaint them with the scope of one of the fields below. Students may consult with their advisors and individual professors about additional courses that may, because of content covered in a particular semester, be eligible for a concentration by special permission.

British Literature

Select appropriate courses in consultation with advisor.

American Literature

Select appropriate courses in consultation with advisor.

World Literature

Code	Title	Credits
ENG 305	Mythology	3
ENG 306	The Bible as Literature	3
ENG 358	Oral Literature	3
ENG 360	Introduction to Ethnic American Literature	3
ENG 361	African American Fiction	3
ENG 362	African American Poetry	3
ENG 363	Native American Literature	3
ENG 364	Introduction to Asian American Literature	3
ENG 368	Women's Literature	3
ENG 377	Introduction to African Literature	3
ENG 378	Studies in South Asian Literature	3
ENG 379	Literature and Empire	3
ENG 405	Advanced Studies in Anglophone Literature	3
ENG 408	Advanced Studies in African American Literature	3
ENG 423	Advanced Studies in Gender and Sexuality in Literature	3
ENG 430	Advanced Studies in Comparative Literature	3
ENG 432	Advanced Studies in African Literature	3
ENG 433	Advanced Studies in Arabic Literature	3
ENG 434	Advanced Studies in Latin American Literature in Translation	3
ENG 436	Studies in German Literature	3
or GER 436	Studies in German Literature	
ENG 437	Studies in Italian Literature	3
or ITAL 437	Studies in Italian Literature	
ENG 438	Studies in Russian Literature	3
or RUS 438	Studies in Russian Literature	
ENG 439	Advanced Studies in Major Authors of Literature in Spanish in Translation	3
ENG 466	Advanced Studies in Women's Literature	3

Environmental Literature and Writing

Code	Title	Credits
ENG 363	Native American Literature	3
ENG 371	Literature and the Environment	3
ENG 390	The Environmental Imagination	3

Language and Theory

Code	Title	Credits
ENG 301	Old English Language and Literature	3
ENG 303	History of the English Language	3
ENG 308	Introduction to Linguistics	3
ENG 309	Traditional English Grammar	3
ENG 310	Structure of the English Language	3
ENG 348	Studies in Linguistics and the English Language	3
ENG 358	Oral Literature	3

ENG 366	Contemporary Critical Theory	3
ENG 369	Feminist Literary Theory	3
ENG 417	Advanced Studies in Linguistics and the English Language	3
ENG 420	Advanced Studies in Theory and Cultural Studies	3

Print Culture

Code	Title	Credits
ENG 313	Studies in Early Modern Literature and Culture	3
ENG 358	Oral Literature	3
ENG 415	Advanced Studies in Textuality and the History of the Book	3

Gender and Sexuality Studies

Code	Title	Credits
ENG 327	Gothic Literature	3
ENG 368	Women's Literature	3
ENG 369	Feminist Literary Theory	3
ENG 370	Queer Literature	3
ENG 423	Advanced Studies in Gender and Sexuality in Literature	3
ENG 466	Advanced Studies in Women's Literature	3

Race and Ethnicity Studies

Code	Title	Credits
ENG 358	Oral Literature	3
ENG 360	Introduction to Ethnic American Literature	3
ENG 361	African American Fiction	3
ENG 362	African American Poetry	3
ENG 363	Native American Literature	3
ENG 364	Introduction to Asian American Literature	3
ENG 377	Introduction to African Literature	3
ENG 378	Studies in South Asian Literature	3
ENG 379	Literature and Empire	3
ENG 408	Advanced Studies in African American Literature	3
ENG 432	Advanced Studies in African Literature	3
ENG 433	Advanced Studies in Arabic Literature	3

Creative Writing

Students with a strong interest in writing should include in their program nine credit hours of electives from the following courses.

Code	Title	Credits
Choose three of the following:		9
ENG 347	Playwriting	
or THEA 347 Playwriting		
ENG 390	The Environmental Imagination	
ENG 391	Introduction to Creative Writing - Nonfiction	
ENG 392	Introduction to Creative Writing - Poetry	
ENG 393	Introduction to Creative Writing - Fiction	
ENG 483	Narrative Form	
ENG 484	Poetic Craft and Creativity	
ENG 493	Advanced Creative Nonfiction	
ENG 494	Advanced Poetry Writing	

ENG 495	Advanced Fiction Writing
ENG 496	Advanced Topics in Creative Writing

Additional Information for English Teaching Licensure
For students interested in pursuing English teacher licensure, refer to the English Teaching Licensure program (p. 586) for specific requirements.

Additional Information
Other courses may be applied to these requirements via course directive. Students should consult advisors and the departmental semester course offerings about other options.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Foreign language requirement		6
General Education requirement		24
Credits		30
Second Year		
Two of three required courses ¹		6
ENG 299	Introduction to Literary Studies	3
General Education requirement		15
Foreign language requirement		6
Credits		30
Third Year		
Remaining requirement from General Education		3
English 300-level courses		12
Other elective(s)		15
Credits		30
Fourth Year		
English 400 level advanced studies courses		6
English 300-400 level courses		9
Other elective(s)		15
Credits		30
Total Credits		120

¹ From among ENG 221-ENG 260

Finance, B.B.A.

Program Description

The B.B.A. degree with a major in finance requires a minimum of 120 credit hours of undergraduate course work. Students enrolled at James Madison University who wish to change their major to finance should go to the Student Center in MyMadison to submit a request. Students must be in good academic standing to change their major to finance and, if they have taken any FIN prefix courses at the time of the change request, must meet the prerequisites for the required courses in the finance major.

Students who plan to major in finance and earn a score on the Math Placement Exam sufficient for placement into MATH 235 are strongly encouraged to enroll in MATH 235.

Required major courses provide finance majors with a foundation in financial management, investments and institutions. Electives within the major permit students to obtain an additional emphasis and explore other areas within the field of finance.

Transfer Credit

Each program in the CoB will accept no more than two courses of transfer credit toward the major except in extraordinary circumstances. The major is defined as the course work required by a major field of study in addition to the lower and upper-level BBA core courses. For a course to be considered for acceptance in the student's major, it must have been completed in an AACSB International-accredited business program, or a similarly accredited program, at a four-year university.

In general, all finance course work must be completed at JMU. Transfer credit for finance courses is awarded only in certain circumstances. In no case will transfer credit be awarded for FIN 488. Contact the department head for more information on transfer credit.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business (p. 15) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
B.B.A. core courses ²	45-46
Finance major requirements	24
Electives	9-10
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major

² Up to seven hours of core requirements in economics and calculus may also be taken for General Education credit.

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		
COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	
MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28
B.B.A. Upper-Level Core Component		
The following courses comprise the B.B.A. Upper-Level Core Component:		
COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3
COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.

² COB 241 must be taken before COB 242.

³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.

⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.

⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.

⁶ All four COB 300 courses must be taken during the same semester.

⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.

⁸ COB 487 must be taken during the senior year.

Major Requirements

Code	Title	Credits
Finance Major Core Courses		
FIN 360	Analytical Methods in Finance	3
FIN 365	Intermediate Financial Management	3
FIN 371	Principles of Investments	3
FIN 488	Advanced Financial Policy	3
Subtotal		12
Electives or Concentration		

In addition to these core courses, the student majoring in finance must successfully complete 12 credit hours of finance elective courses. Finance electives include any 300- level or 400-level finance course. Students may use these 12 credits to complete a concentration in financial analysis, financial technology and analytics, or risk management (see below). ¹

A student may choose up to one of the following courses to satisfy the finance elective credit hours: ²

ACTG 313	Accounting Information Systems	12
ACTG 343	Corporate Financial Reporting I	
ACTG 344	Corporate Financial Reporting II	
ACTG 377	Federal Income Tax Accounting	
BLAW 360	Legal Analysis and Business Persuasion	
BLAW 470	Securities Regulation	
BLAW 480	Estate Planning	
BLAW 494	White Collar Crime	
BLAW 495	Contract Law, Sales and Secured Transactions	
BLAW 496	The Law of Business Organizations, Negotiable Property Instruments and Property	
BLAW 497	Legal Aspects of International Business	
BLAW 498	Special Topics in Business Law	
CIS 304	Enterprise Architecture	
CIS 330	Database Design and Application	
CIS 393	Data Mining for Business Analytics	
CIS 463	Business Intelligence	
BSAN 393	Data Mining for Business Analytics	
Subtotal		12
Total Credits		24

¹ FIN 301, FIN 345 and the finance major core courses do not count as finance electives.

² A finance major cannot take more than 10 credit hours of FIN prefix courses in any one semester. In addition, three credits of FIN 499 (FIN 499A/FIN 499B/FIN 499C) will count toward the finance electives.

Concentrations

Though not required, finance majors may elect a concentration in financial analysis, financial technology and analytics, or risk management. Students electing these concentrations take specific electives in place of the required four electives for the major. Finance majors who wish to declare a concentration may do so once accepted into COB 300 or during any semester following. A student must earn a 2.0 in the courses designated for a concentration in order to have successfully completed the concentration.

Financial Analysis Concentration

Many James Madison University finance graduates go on to careers in financial analysis, and the need for financial analysts remains strong. The financial analysis concentration helps prepare students for these careers and also Levels I and II of the Chartered Financial Analyst exam.

Students wishing to declare and complete the financial analysis concentration must have a minimum grade of "C" in COB 241 and COB 242.

Code	Title	Credits
Required Courses		
FIN 362	Financial Analysis	3
FIN 378	Fixed Income Analysis	3
FIN 380	Elemental and Derivative Securities Analysis	3
Subtotal		9
Electives		
Choose one of the following:		3
ACTG 343	Corporate Financial Reporting I	
FIN 420	Financial Planning and Wealth Management	
FIN 434	Fundamentals of Financial Data Science	
FIN 455	Global Finance	
FIN 475	Financial Modeling and Risk Analysis	
Subtotal		3
Total Credits		12

Financial Technology and Analytics Concentration

The changing landscape of finance requires increased knowledge of technology. The financial technology and analytics concentration will prepare the students to be agile in financial information systems and expose them to various analytical and visualization tools.

Code	Title	Credits
Required Courses		
FIN 434	Fundamentals of Financial Data Science	3
FIN 471	Portfolio Management	3
FIN 475	Financial Modeling and Risk Analysis	3
Subtotal		9
Electives		
Choose one of the following:		3
BSAN 393	Data Mining for Business Analytics	
CS 149	Introduction to Programming	
CIS 304	Enterprise Architecture	
CIS 330	Database Design and Application	
CIS 393	Data Mining for Business Analytics	
Subtotal		3
Total Credits		12

Risk Management Concentration

A focus in risk management is designed for finance majors pursuing a more in-depth review of the issues facing organizations and the tools needed to address those uncertainties. In the risk management concentration, students focus on the theory of risk management, risk identification, risk measurement and applications in the form of risk modeling techniques such as Value-at-Risk and Monte Carlo simulations.

Code	Title	Credits
Required Courses		
FIN 380	Elemental and Derivative Securities Analysis	3
FIN 450	Financial Risk Management	3
FIN 451	Risk Management II	3
FIN 475	Financial Modeling and Risk Analysis	3
Total Credits		12

Recommended Schedule for Majors

First Two Years

Students planning to major in finance must complete the 27-28 credit hour, lower-division B.B.A. core curriculum prior to enrolling in upper-division core courses, normally taken in the first semester of the junior year. It is expected that the lower-division core curriculum will be completed during the first two years of study along with most of the university General Education curriculum. Failing to complete all lower-division B.B.A. core requirements on time will delay enrollment in upper-division core and major courses.

Third and Fourth Years

Finance majors will follow the course schedule below to complete the final two years of their program. It is possible to deviate from this program, but care must be taken to ensure that all course prerequisites are met. Finance majors are encouraged to enroll in FIN 360 during the same semester as COB 300 or during the semester following completion of COB 300. Note that students taking FIN 360 concurrently with COB 300 will have more choices in finance electives in subsequent semesters because FIN 360 is a prerequisite for most finance courses. It is anticipated that students will complete the finance requirements in three semesters following COB 300.

Course	Title	Credits
First Year		
General Education courses, B.B.A. core requirements and/or university electives		30
Credits		30
Second Year		
General Education courses, B.B.A. core requirements and/or university electives		30
Credits		30
Third Year		
First Semester		
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
FIN 360	Analytical Methods in Finance	3
Credits		15
Second Semester		
FIN 365	Intermediate Financial Management	3
FIN 371	Principles of Investments	3
Finance elective		3
General Education courses and/or university electives		6
Credits		15
Fourth Year		
First Semester		
Finance elective		6
General Education courses and/or university electives		9
Credits		15
Second Semester		
COB 487	Strategic Management ¹	3
FIN 488	Advanced Financial Policy	3
Finance elective		3
General Education courses and/or university electives		6
Credits		15
Total Credits		120

¹ COB 487 is required during a student's senior year but may be taken either semester.

Geography, B.A.

Program Description

Students complete a common core that gives a firm foundation in essential geographical knowledge which includes human geography, physical geography, geospatial techniques, statistics, and human/land relations.

Beyond the core, concentrations encourage a deeper understanding of the discipline and the relevant skills to address critical problems faced by humanity. These concentrations can be customized to the student's interests (Custom) or focus on a particular area within Geography (Applied Geographic Information Science; Environment, Conservation, Sustainability and Development).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course(s) (in addition to General Education courses)	3
Major requirements	50
Electives	14-26
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major. ISAT 251 for the math requirement in The Natural World is strongly recommended, as is GEOG 200 in American and Global Perspectives.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures placement test.

Major Requirements

In addition to the geography core courses, students must choose a concentration, listed in the "Concentrations" section. All courses for the major must be taken on a graded basis. Students must earn a minimum grade of "C" in each of the core courses as they are prerequisites to most concentration courses.

Code	Title	Credits
Core Courses		
GEOG 202	Introduction to Your Geographical Future	1
GEOG 210	Physical Geography (2, 2)	4
GEOG 215	Introduction to Geographic Information Science and Cartography	3
GEOG 216	Earth Observation and Satellite Navigation Systems	3
GEOG 280	Human Geography: The Cultural Landscape	3
GEOG 308	Environment, Nature and Society	3
Advanced Sequence GEOG 400-level (selected in consultation with the advisor) ^{1,2}		6
Subtotal		23

Statistics Requirement		
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	3
or MATH 220	Elementary Statistics [C3QR]	
Subtotal		3
Concentration		
Students choose a concentration from the options listed below.		24
Total Credits		50

¹ For Honors students, the advanced sequence is GEOG 499A, GEOG 499B, GEOG 499C.

² Students may opt for a capstone experience that entails three or six hours of independent research (GEOG 490), with the approval of and in close consultation with a project advisor.

Geography Concentrations

Applied Geographic Information Science (AGIS) Concentration

JMU geography graduates with an AGIS concentration are prepared to gain professional employment in government and industry, and to go on to graduate programs. Public agencies where they find employment include local and regional planning agencies, mapping organizations, intelligence agencies, and agencies focused on the environment and public land management. JMU AGIS graduates find opportunities in industry with many companies, including federal and local government contractors, geospatial software companies, companies that collect and process geospatial data such as satellite imagery and street map data, and environmental consulting firms. Multiple top graduate programs have accepted AGIS graduates to continue their studies in geography and other fields that relate to geospatial technologies.

In addition to the geography core requirements, students in the AGIS concentration must complete the following course work.

Code	Title	Credits
Applied Geographic Information Science (AGIS) Concentration Courses		
GEOG 365	Cartography and Geospatial Visualization	3
GEOG 366	Intermediate Geographic Information Science	3
GEOG 367	Programming for Geographic Information Science	3
GEOG 385	Digital Image Processing in Remote Sensing	3
Subtotal		12
Electives		
Up to one concentration-related course from another program may be used as an elective with the approval of the student's advisor.		
Choose 9 credits of the following:		9
GEOG 355	Global Navigation Satellite Systems	
GEOG 406	Forest Inventory: A Geospatial Approach	
GEOG 464	Advanced Cartography	
GEOG 465	Topics in GIS	
GEOG 466	GIS and Geographic Databases	
GEOG 467	GIS Project Management	
GEOG 469	Applications of Geographic Information Systems	
GEOG 485	Advanced Topics in Remote Sensing	
GEOG 487	Advanced Cloud-Based Remote Sensing	
GEOG 490	Senior Research or Field Practicum	

GEOG 491	International Studies	
GEOG 495	Internship in Geography	
GEOG 497	Independent Study	
Subtotal		9
AGIS Cognate Course		
Choose 3 credits of the following:		3
GEOG 200	Geography: The Global Dimension [C4GE]	
GEOG/HUMN 301	Introduction to Natural Disasters Response	
GEOG 310	Environmental Issues	
GEOG 315	Field Studies in Geography	
GEOG 320	Land Use Geography	
GEOG 322	Agricultural Systems	
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet	
GEOG 325	Environmental Ethics	
GEOG 327	Climatology	
GEOG 329	Global Climate Change	
GEOG 330	Weather, Climate and Society	
GEOG 331	Geography of Virginia	
GEOG 332	Geography of Europe	
GEOG 334	Geography of East and Southeast Asia	
GEOG 335	Geography of Africa	
GEOG 336	Environmental Hazards: A Focus on Southeast Asia	
GEOG 337	Geography of Latin America	
GEOG 338	Geography of the Philippine Islands: Problems and Possibilities	
GEOG 339	Geography of the Caribbean	
GEOG 340	Biogeography	
GEOG 341	Wilderness Studies	
GEOG 342	Management and Protection of Natural Resources	
GEOG 343	Wildlife Management	
GEOG 344	Globalization and Development	
GEOG 345	Geography of Poverty	
GEOG 348	Indigenous Geographies	
GEOG 350	Topics in Geography	
GEOG 355	Global Navigation Satellite Systems	
GEOG/HUMN 360	GIS for Humanitarian Assistance	
GEOG 374	Population Geography	
GEOG 375	Political Geography	
GEOG 376	Urban Geography	
GEOG 377	Feminist Geography	
GEOG 380	Cultural Geography	
GEOG 410	Geography and Film	
GEOG 415	Environment, Landscape and Culture	
GEOG 427	Water Resources of the World	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
GEOG 430	Dynamics of the Atmosphere	
GEOG 440	Advanced Biogeography	
GEOG 447	Climates Through Time	

GEOG 470	Senior Seminar in Environmental Conservation, Sustainability and Development	
GEOG 476	Sustainable Cities Seminar	
GEOG 477	Gendered Geographies	
BIO 402	Forest Ecology	
ISAT 425	Environmental Hydrology	
ISAT 473	Local Agriculture and Farm Internships	
ISAT 474	Political Ecology	
Subtotal		3
Total Credits		24

Geography Custom Concentration

In addition to the geography core requirements, students in the customized concentration must complete one the following requirements:

Code	Title	Credits
Methods Course		
Choose one of the following:		3
GEOG 315	Field Studies in Geography	
GEOG 341	Wilderness Studies	
GEOG 355	Global Navigation Satellite Systems	
GEOG 365	Cartography and Geospatial Visualization	
GEOG 366	Intermediate Geographic Information Science	
GEOG 367	Programming for Geographic Information Science	
GEOG 385	Digital Image Processing in Remote Sensing	
Concentration Electives		
Students will choose from a set of courses, approved by a program advisor, on the basis of how they fit together thematically and support the educational objectives of the geography program. Students must take a minimum of 21 credit hours at the 300 level or above (in addition to the methods course). At least 6 of these 21 credit hours must be at the 400 level or above.		21
Total Credits		24

Environmental Conservation, Sustainability and Development Concentration

The Environmental Conservation, Sustainability and Development (ECSD) concentration focuses on the geographical contexts within which people and places interact. Required and elective course work allows students to explore spatial and temporal patterning between human communities and the natural environment at multiple scales. The curriculum addresses global issues such as global climate change and globalization; environment and human interactions including political, economic, physical and ethical factors; human and ecological aspects of sustainable development; natural resource management including energy, forests, wildlife and biodiversity; cultural ecology; regional geography; and population issues. JMU geography graduates with an ECSD concentration are prepared to gain professional employment and to go on to graduate programs.

In addition to the geography core requirements, students in the Environmental Conservation, Sustainability and Development concentration must complete the following course work.

Code	Title	Credits
Senior Seminar Course		
Choose one of the following:		3
GEOG 410	Geography and Film	
GEOG 415	Environment, Landscape and Culture	
GEOG 427	Water Resources of the World	
GEOG 429	Sustainability: An Ecological Perspective	
GEOG 430	Dynamics of the Atmosphere	
GEOG 440	Advanced Biogeography	
GEOG 447	Climates Through Time	
GEOG 470	Senior Seminar in Environmental Conservation, Sustainability and Development	
GEOG 476	Sustainable Cities Seminar	
GEOG 477	Gendered Geographies	
GEOG 491	International Studies	
Subtotal		3
ECSD Electives		
These 300-level courses are identified on the concentration form, which will be reviewed and approved by the student's academic advisor.		
In consultation with the academic advisor, a student may select one non-geography course as an elective.		
Choose 18 credit of the following:		18
GEOG 200	Geography: The Global Dimension [C4GE]	
GEOG/HUMN 301	Introduction to Natural Disasters Response	
GEOG 310	Environmental Issues	
GEOG 315	Field Studies in Geography	
GEOG 320	Land Use Geography	
GEOG 322	Agricultural Systems	
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet	
GEOG 325	Environmental Ethics	
GEOG 327	Climatology	
GEOG 329	Global Climate Change	
GEOG 330	Weather, Climate and Society	
GEOG 331	Geography of Virginia	
GEOG 332	Geography of Europe	
GEOG 334	Geography of East and Southeast Asia	
GEOG 335	Geography of Africa	
GEOG 336	Environmental Hazards: A Focus on Southeast Asia	
GEOG 337	Geography of Latin America	
GEOG 338	Geography of the Philippine Islands: Problems and Possibilities	
GEOG 339	Geography of the Caribbean	
GEOG 340	Biogeography	
GEOG 341	Wilderness Studies	
GEOG 342	Management and Protection of Natural Resources	
GEOG 343	Wildlife Management	
GEOG 344	Globalization and Development	
GEOG 345	Geography of Poverty	
GEOG 348	Indigenous Geographies	

GEOG 350	Topics in Geography	
GEOG 355	Global Navigation Satellite Systems	
GEOG/HUMN 360	GIS for Humanitarian Assistance	
GEOG 374	Population Geography	
GEOG 375	Political Geography	
GEOG 376	Urban Geography	
GEOG 377	Feminist Geography	
GEOG 380	Cultural Geography	
GEOG 406	Forest Inventory: A Geospatial Approach	
GEOG 410	Geography and Film	
GEOG 415	Environment, Landscape and Culture	
GEOG 427	Water Resources of the World	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
GEOG 430	Dynamics of the Atmosphere	
GEOG 440	Advanced Biogeography	
GEOG 447	Climates Through Time	
GEOG 470	Senior Seminar in Environmental Conservation, Sustainability and Development	
GEOG 476	Sustainable Cities Seminar	
GEOG 477	Gendered Geographies	
GEOG 490	Senior Research or Field Practicum	
GEOG 491	International Studies	
GEOG 495	Internship in Geography	
GEOG 497	Independent Study	
BIO 402	Forest Ecology	
ISAT 425	Environmental Hydrology	
ISAT 473	Local Agriculture and Farm Internships	
ISAT 474	Political Ecology	
Subtotal		18
Cognate Course		
Choose one of the following:		3
GEOG 365	Cartography and Geospatial Visualization	
GEOG 366	Intermediate Geographic Information Science	
GEOG 385	Digital Image Processing in Remote Sensing	
Subtotal		3
Total Credits		24

Recommended Schedule for Majors

Course	Title	Credits
First Year		
GEOG 202	Introduction to Your Geographical Future	1
GEOG 210	Physical Geography (2, 2) ¹	4
GEOG 215	Introduction to Geographic Information Science and Cartography	3
GEOG 216	Earth Observation and Satellite Navigation Systems ¹	3
GEOG 280	Human Geography: The Cultural Landscape	3
MATH 105	Statistical Literacy [C3QR] ²	3
General Education, BA degree requirements and electives		13
Credits		30
Second Year		
GEOG 308	Environment, Nature and Society (If ECSD concentration)	3

ISAT 251 or MATH 220	Topics in Applied Statistics in ISAT [C3QR] or Elementary Statistics [C3QR]	3
Minor and concentration requirements		9
General Education, BA degree requirements and electives		12
Credits		27
Third Year		
GEOG 308	Environment, Nature and Society (If AGIS or custom concentration) ³	3
Concentration and minor requirements		18
General Education, BA degree requirements and electives		12
Credits		33
Fourth Year		
Advanced sequence GEOG 400-level courses		6
Concentration and minor requirements		21
Electives		3
Credits		30
Total Credits		120

¹ It is recommended that GEOG 210 and GEOG 216 not be taken during a student's first semester.

² Enroll in MATH 105 if the ALEKS score is <46. If score placed the student into MATH 220, MATH 220 may be taken as early as the spring of the first year. Alternatively, ISAT 251 can be taken during the sophomore year.

³ It is recommended that GEOG 365 and GEOG 366 not be taken in the same semester.

Geography, B.S.

Program Description

Students complete a common core that gives a firm foundation in essential geographical knowledge which includes human geography, physical geography, geospatial techniques, statistics, and human/land relations.

Beyond the core, concentrations encourage a deeper understanding of the discipline and the relevant skills to address critical problems faced by humanity. These concentrations can be customized to the student's interests (Custom) or focus on a particular area within Geography (Applied Geographic Information Science; Environment, Conservation, Sustainability and Development).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3
Scientific Literacy requirement (in addition to General Education)	3-4
Major requirements	50
Electives	22-23
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major. ISAT 251 for the math requirement in The Natural World is strongly recommended, as is GEOG 200 in American and Global Perspectives.

Major Requirements

In addition to the geography core courses, students must choose a concentration, listed in the "Concentrations" section. All courses for the major must be taken on a graded basis. Students must earn a minimum grade of "C" in each of the core courses as they are prerequisites to most concentration courses.

Code	Title	Credits
Core Courses		
GEOG 202	Introduction to Your Geographical Future	1
GEOG 210	Physical Geography (2, 2)	4
GEOG 215	Introduction to Geographic Information Science and Cartography	3
GEOG 216	Earth Observation and Satellite Navigation Systems	3
GEOG 280	Human Geography: The Cultural Landscape	3
GEOG 308	Environment, Nature and Society	3
Advanced Sequence GEOG 400-level (selected in consultation with the advisor) ^{1,2}		6
Subtotal		23
Statistics Requirement		
ISAT 251 or MATH 220	Topics in Applied Statistics in ISAT [C3QR] / Elementary Statistics [C3QR]	3
Subtotal		3

Concentration	
Students choose a concentration from the options listed below.	24
Total Credits	50

¹ For Honors students, the advanced sequence is GEOG 499A, GEOG 499B, GEOG 499C.

² Students may opt for a capstone experience that entails three or six hours of independent research (GEOG 490), with the approval of and in close consultation with a project advisor.

Geography Concentrations

Applied Geographic Information Science (AGIS) Concentration

JMU geography graduates with an AGIS concentration are prepared to gain professional employment in government and industry, and to go on to graduate programs. Public agencies where they find employment include local and regional planning agencies, mapping organizations, intelligence agencies, and agencies focused on the environment and public land management. JMU AGIS graduates find opportunities in industry with many companies, including federal and local government contractors, geospatial software companies, companies that collect and process geospatial data such as satellite imagery and street map data, and environmental consulting firms. Multiple top graduate programs have accepted AGIS graduates to continue their studies in geography and other fields that relate to geospatial technologies.

In addition to the geography core requirements, students in the AGIS concentration must complete the following course work.

Code	Title	Credits
Applied Geographic Information Science (AGIS) Concentration Courses		
GEOG 365	Cartography and Geospatial Visualization	3
GEOG 366	Intermediate Geographic Information Science	3
GEOG 367	Programming for Geographic Information Science	3
GEOG 385	Digital Image Processing in Remote Sensing	3
Subtotal		12
Electives		
Up to one concentration-related course from another program may be used as an elective with the approval of the student's advisor.		
Choose 9 credits of the following:		9
GEOG 355	Global Navigation Satellite Systems	
GEOG 406	Forest Inventory: A Geospatial Approach	
GEOG 464	Advanced Cartography	
GEOG 465	Topics in GIS	
GEOG 466	GIS and Geographic Databases	
GEOG 467	GIS Project Management	
GEOG 469	Applications of Geographic Information Systems	
GEOG 485	Advanced Topics in Remote Sensing	
GEOG 487	Advanced Cloud-Based Remote Sensing	
GEOG 490	Senior Research or Field Practicum	
GEOG 491	International Studies	
GEOG 495	Internship in Geography	
GEOG 497	Independent Study	
Subtotal		9

AGIS Cognate Course

Choose 3 credits of the following: 3

GEOG 200	Geography: The Global Dimension [C4GE]
GEOG/HUMN 301	Introduction to Natural Disasters Response
GEOG 310	Environmental Issues
GEOG 315	Field Studies in Geography
GEOG 320	Land Use Geography
GEOG 322	Agricultural Systems
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet
GEOG 325	Environmental Ethics
GEOG 327	Climatology
GEOG 329	Global Climate Change
GEOG 330	Weather, Climate and Society
GEOG 331	Geography of Virginia
GEOG 332	Geography of Europe
GEOG 334	Geography of East and Southeast Asia
GEOG 335	Geography of Africa
GEOG 336	Environmental Hazards: A Focus on Southeast Asia
GEOG 337	Geography of Latin America
GEOG 338	Geography of the Philippine Islands: Problems and Possibilities
GEOG 339	Geography of the Caribbean
GEOG 340	Biogeography
GEOG 341	Wilderness Studies
GEOG 342	Management and Protection of Natural Resources
GEOG 343	Wildlife Management
GEOG 344	Globalization and Development
GEOG 345	Geography of Poverty
GEOG 348	Indigenous Geographies
GEOG 350	Topics in Geography
GEOG 355	Global Navigation Satellite Systems
GEOG/HUMN 360	GIS for Humanitarian Assistance
GEOG 374	Population Geography
GEOG 375	Political Geography
GEOG 376	Urban Geography
GEOG 377	Feminist Geography
GEOG 380	Cultural Geography
GEOG 410	Geography and Film
GEOG 415	Environment, Landscape and Culture
GEOG 427	Water Resources of the World
GEOG/ISAT 429	Sustainability: An Ecological Perspective
GEOG 430	Dynamics of the Atmosphere
GEOG 440	Advanced Biogeography
GEOG 447	Climates Through Time
GEOG 470	Senior Seminar in Environmental Conservation, Sustainability and Development
GEOG 476	Sustainable Cities Seminar
GEOG 477	Gendered Geographies

BIO 402	Forest Ecology
ISAT 425	Environmental Hydrology
ISAT 473	Local Agriculture and Farm Internships
ISAT 474	Political Ecology
Subtotal	3
Total Credits	24

Geography Custom Concentration

In addition to the geography core requirements, students in the customized concentration must complete one the following requirements:

Code	Title	Credits
Methods Course		
Choose one of the following:		3
GEOG 315	Field Studies in Geography	
GEOG 341	Wilderness Studies	
GEOG 355	Global Navigation Satellite Systems	
GEOG 365	Cartography and Geospatial Visualization	
GEOG 366	Intermediate Geographic Information Science	
GEOG 367	Programming for Geographic Information Science	
GEOG 385	Digital Image Processing in Remote Sensing	
Concentration Electives		
Students will choose from a set of courses, approved by a program advisor, on the basis of how they fit together thematically and support the educational objectives of the geography program. Students must take a minimum of 21 credit hours at the 300 level or above (in addition to the methods course). At least 6 of these 21 credit hours must be at the 400 level or above.		21
Total Credits		24

Environmental Conservation, Sustainability and Development Concentration

The Environmental Conservation, Sustainability and Development (ECSD) concentration focuses on the geographical contexts within which people and places interact. Required and elective course work allows students to explore spatial and temporal patterning between human communities and the natural environment at multiple scales. The curriculum addresses global issues such as global climate change and globalization; environment and human interactions including political, economic, physical and ethical factors; human and ecological aspects of sustainable development; natural resource management including energy, forests, wildlife and biodiversity; cultural ecology; regional geography; and population issues. JMU geography graduates with an ECSD concentration are prepared to gain professional employment and to go on to graduate programs.

In addition to the geography core requirements, students in the Environmental Conservation, Sustainability and Development concentration must complete the following course work.

Code	Title	Credits
Senior Seminar Course		
Choose one of the following:		3
GEOG 410	Geography and Film	
GEOG 415	Environment, Landscape and Culture	
GEOG 427	Water Resources of the World	

GEOG 429	Sustainability: An Ecological Perspective
GEOG 430	Dynamics of the Atmosphere
GEOG 440	Advanced Biogeography
GEOG 447	Climates Through Time
GEOG 470	Senior Seminar in Environmental Conservation, Sustainability and Development
GEOG 476	Sustainable Cities Seminar
GEOG 477	Gendered Geographies
GEOG 491	International Studies
Subtotal	3
ECSD Electives	
These 300-level courses are identified on the concentration form, which will be reviewed and approved by the student's academic advisor.	
In consultation with the academic advisor, a student may select one non-geography course as an elective.	
Choose 18 credit of the following:	18
GEOG 200	Geography: The Global Dimension [C4GE]
GEOG/HUMN 301	Introduction to Natural Disasters Response
GEOG 310	Environmental Issues
GEOG 315	Field Studies in Geography
GEOG 320	Land Use Geography
GEOG 322	Agricultural Systems
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet
GEOG 325	Environmental Ethics
GEOG 327	Climatology
GEOG 329	Global Climate Change
GEOG 330	Weather, Climate and Society
GEOG 331	Geography of Virginia
GEOG 332	Geography of Europe
GEOG 334	Geography of East and Southeast Asia
GEOG 335	Geography of Africa
GEOG 336	Environmental Hazards: A Focus on Southeast Asia
GEOG 337	Geography of Latin America
GEOG 338	Geography of the Philippine Islands: Problems and Possibilities
GEOG 339	Geography of the Caribbean
GEOG 340	Biogeography
GEOG 341	Wilderness Studies
GEOG 342	Management and Protection of Natural Resources
GEOG 343	Wildlife Management
GEOG 344	Globalization and Development
GEOG 345	Geography of Poverty
GEOG 348	Indigenous Geographies
GEOG 350	Topics in Geography
GEOG 355	Global Navigation Satellite Systems
GEOG/HUMN 360	GIS for Humanitarian Assistance
GEOG 374	Population Geography
GEOG 375	Political Geography
GEOG 376	Urban Geography

GEOG 377	Feminist Geography
GEOG 380	Cultural Geography
GEOG 406	Forest Inventory: A Geospatial Approach
GEOG 410	Geography and Film
GEOG 415	Environment, Landscape and Culture
GEOG 427	Water Resources of the World
GEOG/ISAT 429	Sustainability: An Ecological Perspective
GEOG 430	Dynamics of the Atmosphere
GEOG 440	Advanced Biogeography
GEOG 447	Climates Through Time
GEOG 470	Senior Seminar in Environmental Conservation, Sustainability and Development
GEOG 476	Sustainable Cities Seminar
GEOG 477	Gendered Geographies
GEOG 490	Senior Research or Field Practicum
GEOG 491	International Studies
GEOG 495	Internship in Geography
GEOG 497	Independent Study
BIO 402	Forest Ecology
ISAT 425	Environmental Hydrology
ISAT 473	Local Agriculture and Farm Internships
ISAT 474	Political Ecology
Subtotal	18
Cognate Course	
Choose one of the following:	3
GEOG 365	Cartography and Geospatial Visualization
GEOG 366	Intermediate Geographic Information Science
GEOG 385	Digital Image Processing in Remote Sensing
Subtotal	3
Total Credits	24

Recommended Schedule for Majors

Course	Title	Credits
First Year		
GEOG 202	Introduction to Your Geographical Future	1
GEOG 210	Physical Geography (2, 2) ¹	4
GEOG 215	Introduction to Geographic Information Science and Cartography	3
GEOG 216	Earth Observation and Satellite Navigation Systems ¹	3
GEOG 280	Human Geography: The Cultural Landscape	3
MATH 105	Statistical Literacy [C3QR] ²	3
General Education, B.S. degree requirements and electives		13
Credits		30
Second Year		
GEOG 308	Environment, Nature and Society (If ECSD concentration)	3
MATH 220	Elementary Statistics [C3QR]	3
Minor and/or concentration requirements		9
General Education, B.S. degree requirements and electives		12
Credits		27
Third Year		
GEOG 308	Environment, Nature and Society (If AGIS or custom concentration) ³	3
Concentration and minor requirements		18

General Education, B.S. degree requirements and electives	12
Credits	33
Fourth Year	
Advanced sequence GEOG 400-level courses	6
Concentration and minor requirements	21
Electives	3
Credits	30
Total Credits	120

¹ It is recommended that GEOG 210 and GEOG 216 not be taken during a student's first semester.

² Enroll in MATH 105 if the ALEKS score is <46. If score placed the student into MATH 220, MATH 220 may be taken as early as the spring of the first year. Alternatively, ISAT 251 can be taken during the sophomore year.

³ It is recommended that GEOG 365 and GEOG 366 not be taken in the same semester.

Geology, B.A.

Program Description

The B.A. degree in Geology provides students a broad foundation for understanding the structure of the Earth, Earth materials and how the physical environment is formed and altered.

Students may not major in both Geology and Environmental Earth Science.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language Requirement ²	0-12
Philosophy Requirement (in addition to General Education courses)	3
Major requirements	64-68
University Electives	6-22
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student’s chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures placement test.

Major Requirements

Code	Title	Credits
Core Requirements		
GEOL 230	Evolution of Earth (3, 2)	4
GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 377	Earth Surface Processes (2, 2)	3
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	2
Subtotal		10
Major Requirements		
GEOL 102	Environment: Earth [C3NS]	3
or GEOL 110	Physical Geology [C3NS]	
or GEOL 115	Earth Systems and Climate Change [C3NS]	
or GEOL 210	Applied Physical Geology [C3NS]	
GEOL 110L	Physical Geology Laboratory [C3L]	1
or GEOL 115L	Earth Systems and Climate Change Lab [C3L]	
GEOL 280	Mineralogy (3, 2)	4
GEOL 300	Introduction to Petrology (3, 3)	4
GEOL 330	Sedimentology, Stratigraphy and Paleoclimatology	4
GEOL 365	Structural Geology, Tectonics and Climate	4
Subtotal		20
Major Electives		

Select 16 credit hours of GEOL courses at the 200-level or above. Students are encouraged to work with their advisor to select courses appropriate to their goals. GEOL 210, GEOL 301, GEOL 320, GEOL 367, GEOL 401 and GEOL 477 do not count as upper level electives for the B.A. in Geology.		16
Subtotal		16
Cognate Science and Mathematics Courses		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
MATH 205 or MATH 235 or MATH 231 & MATH 232	Applied Calculus [C3QR] Calculus I [C3QR] Calculus with Functions I [C3QR] and Calculus with Functions II	3-6
MATH 220 or MATH 229 or MATH 236	Elementary Statistics [C3QR] Introduction to Applied Statistics Using R [C3QR] Calculus II	3-4
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
CHEM 132 & 132L or PHYS 150 & 150L	General Chemistry II and General Chemistry Laboratory College Physics II and General Physics Laboratory II	4
Subtotal		18-22
Total Credits		64-68

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	4
General Education		7
Credits		15
Spring Semester		
GEOL 230	Evolution of Earth (3, 2)	4
MATH 205	Applied Calculus [C3QR]	3
General Education		7
Credits		14
Second Year		
Fall Semester		
GEOL 280	Mineralogy (3, 2)	4
MATH 220	Elementary Statistics [C3QR]	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
General Education		3
GEOL electives		3
Credits		17
Spring Semester		
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
GEOL 300	Introduction to Petrology (3, 3)	4
General Education		3
Philosophy BA Requirement		3
Credits		14
Third Year		
Fall Semester		
GEOL 330	Sedimentology, Stratigraphy and Paleoclimatology	4

GEOL electives and/or General Education courses	7
Foreign language requirement	3
Credits	14
Spring Semester	
GEOL 291 Writing and Communicating in the Geosciences	1
GEOL 365 Structural Geology, Tectonics and Climate	4
GEOL 377 Earth Surface Processes (2, 2)	3
Foreign language requirement	3
General Education	3
Credits	14
Fourth Year	
Fall Semester	
GEOL 497 Work-Based Learning Experiences in Geology and Environmental Science	1
GEOL electives, university electives, or General Education courses	13
Foreign language requirement	3
Credits	17
Spring Semester	
GEOL 497 Work-Based Learning Experiences in Geology and Environmental Science	1
GEOL electives, university electives, or General Education courses	11
Foreign language requirement	3
Credits	15
Total Credits	120

Geology, B.S.

Program Description

The B.S. degree in Geology prepares students for professional employment in the geosciences, graduate school and professional licensure in geology. Concentrations are offered for Solid Earth & Tectonics, Life, Ocean, & Climate, and Environmental & Engineering Geology. This degree path also requires a capstone field experience.

Students may not major in both Geology and Environmental Earth Science.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education) ²	3-4
Major requirements	69-72
University Electives	17-20
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² The quantitative and scientific literacy requirements are met by courses required by the major.

Major Requirements

Code	Title	Credits
Core Requirements		
GEOL 230	Evolution of Earth (3, 2)	4
GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 377	Earth Surface Processes (2, 2)	3
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	2
Subtotal		10
Major Requirements		
GEOL 102	Environment: Earth [C3NS]	3
or GEOL 110	Physical Geology [C3NS]	
or GEOL 115	Earth Systems and Climate Change [C3NS]	
or GEOL 210	Applied Physical Geology [C3NS]	
GEOL 110L	Physical Geology Laboratory [C3L]	1
or GEOL 115L	Earth Systems and Climate Change Lab [C3L]	
GEOL 280	Mineralogy (3, 2)	4
GEOL 300	Introduction to Petrology (3, 3)	4
GEOL 330	Sedimentology, Stratigraphy and Paleoclimatology	4
GEOL 365	Structural Geology, Tectonics and Climate	4
GEOL 399	Field Geology	4
Subtotal		24
Concentrations		
Students must select one of the following concentrations. The requirements for each concentration are listed below.		12

Environmental and Engineering Geology Concentration		
Life, Ocean and Climate Concentration		
Solid Earth and Tectonics Concentration		
General Geology Concentration		
Subtotal		12
Cognate Science and Mathematics Courses		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
MATH 231 & MATH 232 or MATH 235	Calculus with Functions I [C3QR] and Calculus with Functions II or Calculus I [C3QR]	4-6
MATH 220 or MATH 229 or MATH 236	Elementary Statistics [C3QR] or Introduction to Applied Statistics Using R [C3QR] or Calculus II	3-4
PHYS 140 & 140L or PHYS 240 & 240L	College Physics I [C3PP] and General Physics Laboratory I [C3L] or University Physics I [C3PP] and University Physics Laboratory I [C3L]	4
PHYS 150 & 150L or PHYS 250 & 250L	College Physics II and General Physics Laboratory II or University Physics II and University Physics Laboratory II	4
Subtotal		23-26
Total Credits		69-72

Concentrations

Students must take 12 credits from one of the following concentrations.

Environmental and Engineering Geology Concentration		
The environmental and engineering geology concentration provides students with the necessary skills to enter the environmental and/or geotechnical engineering industry. Students will have the opportunity to take courses that focus on soil classification, ground water flow mechanics, water chemistry, soil/rock mechanics, geomorphology and geophysics.		
Code	Title	Credits
Choose at least 2 of the following courses:		6
GEOL 311	Mineral Resources and the Environment (2,2)	
GEOL 355	Geochemistry of Natural Waters	
GEOL 410	Engineering Geology (2, 2)	
GEOL 430	Environmental Soil Science (2, 2)	
GEOL 442	Field Geophysics	
GEOL 460	Hydrogeology (2, 2)	
Choose at least 1 of the following courses:		3
GEOL 360	GIS for the Geoscientist	
GEOL 390	Laboratory Techniques in Geology (2,2)	
GEOL 489	Quantitative Methods in Geology	

Life, Ocean and Climate Concentration

This concentration is meant for students interested in academic or professional careers that requires a background in the evolution of the Earth system, life and climate. The recommended course work is designed to provide students with content knowledge and research skills

to consider questions of evolution, extinction and global climate change from a geologic perspective.

Code	Title	Credits
Choose at least 2 of the following courses:		6-7
GEOL 350	Paleobiology	
GEOL 400	Geology and Ecology of the Bahamas	
GEOL 405	Vertebrate Paleontology	
GEOL 406	Paleoclimatology and Paleoceanography	
Choose at least 1 of the following courses:		3
GEOL 360	GIS for the Geoscientist	
GEOL 390	Laboratory Techniques in Geology (2,2)	
GEOL 489	Quantitative Methods in Geology	

Solid Earth and Tectonics Concentration

The solid Earth and tectonics track is designed for students who are interested in studying the evolution of the Earth and the methods to study it. The recommended course work is designed to provide students with content knowledge and skills to investigate the structure of the Earth. Possible professional opportunities include, but are not limited to, mining, seismic investigations and geologic mapping.

Code	Title	Credits
Choose at least 2 of the following courses:		4-7
GEOL 302	Sedimentary Petrology	
GEOL 307	Planetary Geology (2, 2)	
GEOL 311	Mineral Resources and the Environment (2,2)	
GEOL 440	Geophysics (3, 2)	
Choose at least 1 of the following courses:		3
GEOL 360	GIS for the Geoscientist	
GEOL 390	Laboratory Techniques in Geology (2,2)	
GEOL 489	Quantitative Methods in Geology	

General Geology Concentration

Students select 12 credit hours of 300- or 400-level GEOL courses. Students are encouraged to work with their advisor to select courses appropriate to their goals. GEOL 301, GEOL 320, GEOL 367, GEOL 401 and GEOL 477 do not count as upper level electives for the B.S. in Geology.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	4
General Education		7
Credits		15
Spring Semester		
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
GEOL 230	Evolution of Earth (3, 2)	4
MATH 235	Calculus I [C3QR]	4
General Education		3
Credits		15

Second Year		
Fall Semester		
GEOL 280	Mineralogy (3, 2)	4
GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 330	Sedimentology, Stratigraphy and Paleoclimatology	4
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
General Education		3
Credits		16
Spring Semester		
GEOL 300	Introduction to Petrology (3, 3)	4
PHYS 150 & 150L	College Physics II and General Physics Laboratory II	4
General Education		6
Credits		14
Third Year		
Fall Semester		
Upper-level GEOL courses and/or General Education courses		15
Credits		15
Spring Semester		
GEOL 365	Structural Geology, Tectonics and Climate	4
GEOL 377	Earth Surface Processes (2, 2)	3
GEOL 399	Field Geology	4
Upper-level GEOL concentration courses		4
Credits		15
Fourth Year		
Fall Semester		
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	1
GEOL electives, university electives, or General Education courses		14
Credits		15
Spring Semester		
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	1
GEOL electives, university electives, or General Education courses		14
Credits		15
Total Credits		120

Geology, Environmental Earth Science Major, B.A.

Program Description

The B.A. degree in Geology with the Environmental Earth Science major gives students a broad foundation for understanding Earth's environments (i.e., geosphere, biosphere, cryosphere, hydrosphere and atmosphere).

Students may not major in both Geology and Environmental Earth Science.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language Requirement ²	0-12
Philosophy Requirement (in addition to General Education courses)	3
Major Requirements (including cognate sciences and math)	72
University Electives	2-14
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures' placement exam.

Major Requirements

Code	Title	Credits
Core Requirements		
GEOL 230	Evolution of Earth (3, 2)	4
GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 377	Earth Surface Processes (2, 2)	3
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	2
Subtotal		10
Major Required Courses		
GEOL 102 or GEOL 110 or GEOL 115 or GEOL 210	Environment: Earth [C3NS] Physical Geology [C3NS] Earth Systems and Climate Change [C3NS] Applied Physical Geology [C3NS]	3
GEOL 110L or GEOL 115L	Physical Geology Laboratory [C3L] Earth Systems and Climate Change Lab [C3L]	1
GEOL 167	History and Philosophy of the Geosciences	3
GEOL 211	Introduction to Oceanography [C3NS]	3
GEOL 320	Meteorology	3
GEOL 330 or GEOL 365	Sedimentology, Stratigraphy and Paleoclimatology Structural Geology, Tectonics and Climate	4
GEOL 367	Genesis of Solid Earth Materials (3, 2)	4

GEOL 477	Contemporary Issues in the Geosciences	4
Subtotal		25
Major Upper-Level Electives		
Students must complete 18 credit hours of 300- or 400-level BIO, CHEM, GEOL or ENVT courses chosen in consultation with their advisor.		18
Subtotal		18
Cognate Science and Mathematics Area		
BIO/GEOL 350	Paleobiology	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
MATH 220 or MATH 229	Elementary Statistics [C3QR] Introduction to Applied Statistics Using R [C3QR]	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
CHEM 132 & 132L or PHYS 150 & 150L	General Chemistry II and General Chemistry Laboratory College Physics II and General Physics Laboratory II	4
Subtotal		19
Total Credits		72

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	4
GEOL 167	History and Philosophy of the Geosciences	3
General Education		4
Credits		15
Spring Semester		
GEOL 230	Evolution of Earth (3, 2)	4
MATH 220	Elementary Statistics [C3QR]	3
General Education		9
Credits		16
Second Year		
Fall Semester		
GEOL 211	Introduction to Oceanography [C3NS]	3
GEOL 367	Genesis of Solid Earth Materials (3, 2)	4
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
General Education courses or language requirement		3
Credits		14
Spring Semester		
GEOL 320	Meteorology	3
GEOL 377	Earth Surface Processes (2, 2)	3
PHYS 150 & 150L	College Physics II and General Physics Laboratory II	4
General Education courses and/or language requirement		6
Credits		16
Third Year		
Fall Semester		
GEOL 350	Paleobiology	4
GEOL major electives, General Education courses and/or language requirement		12
Credits		16

Spring Semester

GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 365	Structural Geology, Tectonics and Climate	4
GEOL major electives, General Education courses and/or language requirement		10
Credits		15

Fourth Year**Fall Semester**

GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	1
GEOL major electives or university electives		12
Credits		13

Spring Semester

GEOL 477	Contemporary Issues in the Geosciences	4
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	1
GEOL major electives or university electives		10
Credits		15
Total Credits		120

Geology, Environmental Earth Science Major, B.S.

Program Description

The B.S. degree in Geology with the Environmental Earth Science major prepares students for environmental Earth Science careers in government and the private sector by providing knowledge and skills to perform surveys, understand regulations, identify and assess environmental impacts, and develop mitigation plans. The major is the recommended pathway for Earth science teacher licensure in Virginia.

Students may not major in both Geology and Environmental Earth Science.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education courses) ²	3
Scientific Literacy requirement (in addition to General Education courses) ²	3-4
Major Requirements	75-85
University Electives	4-14
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar’s website. (http://www.jmu.edu/registrar/BA_and_BS_Degree_Courses.shtml)

Major Requirements

Code	Title	Credits
Core Courses		
GEOL 230	Evolution of Earth (3, 2)	4
GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 377	Earth Surface Processes (2, 2)	3
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	2
Subtotal		10
Required Courses		
GEOL 102	Environment: Earth [C3NS]	3
or GEOL 110	Physical Geology [C3NS]	
or GEOL 115	Earth Systems and Climate Change [C3NS]	
or GEOL 210	Applied Physical Geology [C3NS]	
GEOL 110L	Physical Geology Laboratory [C3L]	1
or GEOL 115L	Earth Systems and Climate Change Lab [C3L]	
GEOL 167	History and Philosophy of the Geosciences	3
GEOL 211	Introduction to Oceanography [C3NS]	3
GEOL 320	Meteorology	3
GEOL 330	Sedimentology, Stratigraphy and Paleoclimatology	4
or GEOL 365	Structural Geology, Tectonics and Climate	

GEOL 367	Genesis of Solid Earth Materials (3, 2)	4
GEOL 477	Contemporary Issues in the Geosciences	4
Subtotal		25
Major Upper-Level Electives or Related Minor		
Students must select one of the following options to complete this requirement:		
Climate Science Minor		
Environmental Science Minor		
Pre-Professional Secondary Education (SEED) Minor. SEED minors must select GEOL 365 and either GEOL 307 or ASTR 220 for teacher licensure requirements.		
18 credit hours of 300- or 400-level BIO, CHEM, GEOL or ENVT courses chosen in consultation with their advisor.		
Subtotal		18-25
Cognate Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
or BIO/GEOL 350	Paleobiology	
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
or PHYS 150 & 150L	College Physics II and General Physics Laboratory II	
MATH 205	Applied Calculus [C3QR]	3-6
or MATH 235	Calculus I [C3QR]	
or MATH 231 & MATH 232	Calculus with Functions I [C3QR] and Calculus with Functions II	
MATH 220	Elementary Statistics [C3QR]	3
or MATH 229	Introduction to Applied Statistics Using R [C3QR]	
Subtotal		22-25
Total Credits		75-85

¹ SEED minors must complete GEOL 365.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	4
GEOL 167	History and Philosophy of the Geosciences	3
General Education		3
		Credits
		14
Spring Semester		
GEOL 230	Evolution of Earth (3, 2)	4
MATH 205	Applied Calculus [C3QR]	3
General Education		9
		Credits
		16

Second Year**Fall Semester**

GEOL 211	Introduction to Oceanography [C3NS]	3
GEOL 367	Genesis of Solid Earth Materials (3, 2)	4
MATH 220	Elementary Statistics [C3QR]	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
General Education		3
Credits		17

Spring Semester

GEOL 320	Meteorology	3
GEOL 377	Earth Surface Processes (2, 2)	3
General Education		4
Required minor courses		3
Credits		13

Third Year**Fall Semester**

BIO 350	Paleobiology	4
Required minor courses		10
Credits		14

Spring Semester

GEOL 291	Writing and Communicating in the Geosciences	1
GEOL 365	Structural Geology, Tectonics and Climate	4
General Education		8
Required minor courses or university electives		3
Credits		16

Fourth Year**Fall Semester**

GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	1
Required minor courses or university electives		14
Credits		15

Spring Semester

GEOL 477	Contemporary Issues in the Geosciences	4
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science	1
Required minor courses or university electives		10
Credits		15
Total Credits		120

Health Sciences, B.S.

Program Description

The B.S. degree in health sciences consists of health science course requirements in addition to General Education requirements and B.S. requirements.

The health sciences degree is designed for students interested in understanding the causes of disease, strategies for promoting wellness, and the scientific basis and methodologies for analysis of health concerns. The degree combines a broad foundation of health-related course work with a choice of preparatory courses suitable for entry into graduate programs in medicine, dentistry, optometry, occupational therapy, physical therapy, pharmacy, public health, physician assistant studies, athletic training and veterinary medicine.

Special Admission Requirements

Admission to JMU does not guarantee admission to all academic majors and minors.

Declaration of Major Deadlines

Deadlines for change or declaration of major forms are:

Semester	Deadline
Summer	March 1
Fall	March 1
Spring	October 1 of the previous year

Change of major requests received after the deadline will be processed the following semester. Students changing their major to health sciences after March 1 of the sophomore year should expect an additional semester(s) to complete the program. The number of additional semesters required to complete the program will depend on the timing of the change to health sciences and the number of summer courses completed.

Assessment

Graduating majors must participate in Health Sciences senior assessment activities. Assessment information helps the faculty modify the curriculum to meet student needs.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement (in addition to General Education) ²	3
Scientific Literacy Requirement (in addition to General Education) ²	3-4
Major Requirements	65-66
University Electives	13-14
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary.

² Quantitative and scientific literacy requirements may be fulfilled with major required courses.

Major Requirements

Code	Title	Credits
Core Courses		
CHEM 120	Concepts of Chemistry [C3PP]	3-4
or CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	
or CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	
MATH 220	Elementary Statistics [C3QR]	3
or MATH 229	Introduction to Applied Statistics Using R [C3QR]	
HTH 100	Personal Wellness [C5W]	3
HTH 150	Introduction to Health Sciences	2
HTH 210	Medical Terminology	3
HTH 231	Population Health Determinants	3
HTH 245	Foundations of Infectious Disease	3
HTH 255	Chronic Disease and Disabilities	3
HTH 320	Statistical Methods for Health Science Research	3
HTH 351	Health Behavior Change	3
HTH 354	U.S. and Global Health Care Systems	3
HTH 408	Health Research Methods	3
HTH 423	Ethics and Critical Thinking in Health	3
HTH 450	Epidemiology	3
Subtotal		41-42
Select 16		
Choose 16 credits from the following:		16
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	
BIO 240 & 240L	Genetics and Genetics Laboratory	
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory ¹	
BIO 324	Human Genetics	
BIO 370	Human and Animal Physiology	
CHEM 241 & 241L	Organic Chemistry I and Concepts of Organic Chemistry Laboratory	
CHEM 242 & 242L	Organic Chemistry II and Organic Chemistry Laboratory	
CHEM 260	Concepts of Biochemistry	
HTH 352	Environmental Health	
HTH 380	Health Communication	
HTH 422	Public Health and Healthcare Policy	
HTH 441	Biomechanics for the Healthcare Professional	
HTH 458	Health Program Planning and Evaluation	
KIN 302 & 302L	Exercise Physiology and Exercise Physiology Laboratory ²	
NUTR 280	Nutrition for Wellness	
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	

PHYS 150 & 150L	College Physics II and General Physics Laboratory II	
PSYC 101	General Psychology [C5SD]	
PSYC 160	Life Span Human Development [C5SD]	
PSYC 250	Introduction to Abnormal Psychology ³	
Subtotal		16
Select 8		
Choose 8 credit hours from the following:		8
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	
Subtotal		8
Total Credits		65-66

¹ Will not count for the 16 credits in this category unless also taking BIO 270/BIO 270L and BIO 290/BIO 290L.

² Pre-Athletic Training minors only.

³ Pre-Occupational Therapy, Pre-Physical Therapy or Pre-Physician Assistant minors only.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
HTH 100	Personal Wellness [C5W]	3
MATH 220 or MATH 229	Elementary Statistics [C3QR] or Introduction to Applied Statistics Using R [C3QR]	3
HTH 150	Introduction to Health Sciences	2
Courses from Select 16		6
General Education requirement		18
Credits		32
Second Year		
Choose one of the following options:		4
Option 1:		
CHEM 120	Concepts of Chemistry [C3PP]	
Option 2:		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	
Option 3:		
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	
HTH 210	Medical Terminology	3
HTH 231	Population Health Determinants	3
HTH 245	Foundations of Infectious Disease	3
HTH 255	Chronic Disease and Disabilities	3
General Education requirement and Electives		8
Choose one of the following options:		4
Option 1:		
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	
Option 2:		
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	
Option 3:		
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	
Credits		28

Third Year		
HTH 320	Statistical Methods for Health Science Research	3
HTH 351	Health Behavior Change	3
HTH 354	U.S. and Global Health Care Systems	3
General Education, select 16 courses and electives		17
Choose one of the following options:		4
Option 1:		
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	
Option 2:		
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	
Option 3:		
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	
Credits		30
Fourth Year		
HTH 408	Health Research Methods	3
HTH 423	Ethics and Critical Thinking in Health	3
HTH 450	Epidemiology	3
General Education, select 16 courses, pre-professional courses and/or electives		21
Credits		30
Total Credits		120

Recommended Schedule for Accelerated Pathway to Athletic Training Students

After completing the 3 years listed below, students segue into the MSAT program. See the graduate catalog for a required schedule for the fourth and fifth years.

Course	Title	Credits
First Year		
Choose one of the following options:		3-4
Option 1:		
CHEM 120	Concepts of Chemistry [C3PP]	
Option 2:		
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	
HTH 100	Personal Wellness [C5W]	3
HTH 150	Introduction to Health Sciences	2
MATH 220	Elementary Statistics [C3QR]	3
PSYC 101	General Psychology [C5SD]	3
General Education requirement		18
Credits		32-33
Second Year		
ATEP 205	Introduction to Athletic Training ¹	3
BIO 270	Human Physiology [C3NS]	3
BIO 270L	Human Physiology Laboratory [C3L]	1
BIO 290	Human Anatomy	3
BIO 290L	Human Anatomy Laboratory	1
HTH 210	Medical Terminology	3
HTH 231	Population Health Determinants	3
HTH 354	U.S. and Global Health Care Systems	3
NUTR 280	Nutrition for Wellness	3
PSYC 160	Life Span Human Development [C5SD]	3
General Education requirement		7
Credits		33
Third Year		
ATEP 306	Athletic Injury Recognition ¹	3
HTH 245	Foundations of Infectious Disease	3
HTH 255	Chronic Disease and Disabilities	3

HTH 320	Statistical Methods for Health Science Research	3
HTH 351	Health Behavior Change	3
HTH 408	Health Research Methods	3
HTH 423	Ethics and Critical Thinking in Health	3
HTH 450	Epidemiology	3
KIN 302	Exercise Physiology	3
KIN 302L	Exercise Physiology Laboratory	1
Choose one of the following options:		4
Option 1:		
KIN 306 & 306L	Human Biomechanics and Human Biomechanics Laboratory	
Option 2:		
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	
General Education requirement		3
Credits		35
Total Credits		100-101

¹ This course is recommended as an elective credit for accelerated pathway students.

Accelerated Pathway to the Master of Science in Athletic Training

Health Science majors interested in an athletic training career are eligible to apply to an accelerated pathway (3+2) leading to the Master of Science in Athletic Training (MSAT) degree. This pathway provides a unique opportunity for highly motivated and academically talented students to fast-track their education. The purpose of this pathway is to prepare students to become certified athletic trainers with the clinical knowledge, skills and professional attitudes that will enable them to deliver health care effectively in various clinical settings. Highly motivated and academically talented students will complete both the undergraduate degree in Health Sciences and graduate degree in Athletic Training programs in 5 years, compared to the traditional 6 years for completing both degrees. Students interested in this pathway must complete a very specific set of course requirements as part of the health sciences degree. See those details below.

Admission and Application Requirements

Students interested in the accelerated pathway will begin their undergraduate studies with a focused curriculum designed to meet the prerequisites for their graduate program while meeting their major requirements. Upon successful completion of the required coursework at a high academic standard, students will transition into the graduate program.

Interested students will need to:

1. Declare an undergraduate major in health sciences.
2. Discuss their intent to pursue the 3+2 accelerated pathway to the Master of Science in Athletic Training with their academic advisor and pre-professional health advisor (if applicable).
3. Submit their declaration of intent for pursuing the 3+2 accelerated pathway to the Master of Science in Athletic Training. Ideally, this declaration is made as a first-year student at JMU. Submitting the declaration of intent early allows adequate time to plan and schedule year two and year three courses.

Students pursuing the 3+2 accelerated pathway to the Master of Science in Athletic Training are required to meet the following criteria for admission:

- Preferred GPA of 2.7 or higher
- Successful completion of the following MSAT prerequisite courses with a minimum grade of "C-" by the end of the third year. The JMU courses in parentheses are part of the health sciences major and fulfill the prerequisite requirements:
 - General Psychology (PSYC 101)
 - Nutrition (NUTR 280)
 - Statistics or Research Methods (MATH 220, HTH 320 or HTH 408)
 - Human or Mammalian Anatomy & Physiology (BIO 270/BIO 270L and BIO 290/BIO 290L)
 - Physics or Biomechanics (PHYS 140/PHYS 140L)
 - General Chemistry (CHEM 120 or CHEM 131/CHEM 131L)
 - Exercise Physiology (KIN 302/KIN 302L)
- Students who are uncertain whether a course meets prerequisites should email athletictraining@jmu.edu and include a copy of the syllabus from the semester they took the course.
- The following courses are not prerequisites but are highly recommended: ATEP 205, ATEP 306
- Submission of a formal application to the Master of Science in Athletic Training program through ATCAS and the Graduate School of JMU by September 1 of the third year.
- Successful completion of an admission interview during September of the third year.
- Successful completion of major courses by the end of the third year.

Program Structure and Timeline

The 3+2 accelerated pathway to the Master of Science in Athletic Training allows undergraduate students to enroll in and apply 21-22 graduate credit hours toward the completion of the B.S. in Health Sciences. For a list of courses, please see the MSAT page in the graduate catalog.

In their first year, students meet with academic major advisors and indicate interest in the accelerated pathway, declaring their intent to pursue this pathway, while beginning their major coursework. In the fall of their third year, students apply to the MSAT program. Students must complete all undergraduate major and prerequisite coursework by the end of the spring semester of their third year at JMU. Students admitted to the accelerated MSAT program will begin their graduate coursework in May after their third year at JMU. Admitted students take 21-22 credit hours at the graduate level during the summer after their third year and the fall of their fourth year. These 21-22 graduate credits will be applied to their undergraduate transcript to fulfill the remaining credits needed for the undergraduate degree. Students enrolled in graduate coursework are required to adhere to The Graduate School Academic Requirements for progression. The undergraduate degree will be conferred at the end of the fall semester of year 4, allowing students to participate in December commencement. Students transition to graduate student status for the spring of the fourth year and complete the remaining courses in the graduate program over the following semesters. Upon successful completion of all courses in the MSAT program, students will earn their master's degree at the end of their fifth year at JMU. Students will participate in the May commencement.

Students pursuing the accelerated pathway should meet with their academic advisor each semester to discuss their progress toward

undergraduate degree completion and staying on track with MSAT prerequisites to ensure prerequisite and graduate requirements are being met.

Health Services Administration, B.S.

Program Description

The program in health services administration prepares students for entry-level managerial, administrative and staff positions in various health organizations including hospitals, health systems, managed care organizations, health insurance companies, retirement and long-term care facilities, ambulatory care organizations, physician practices, public health facilities, and governmental offices. In addition, the program provides the foundation for graduate studies in health administration and related fields. The health services administration program is a Full Certified undergraduate program by the Association of University Programs in Health Administration (AUPHA).

Admission Requirements and Minimum Grades

Any student admitted to JMU can declare Health Services Administration as their major and enroll in prerequisite courses. However, to progress in the major beyond the prerequisite courses, students must apply and be admitted to the program.

Students wishing to be admitted to the health services administration program at JMU must successfully complete the prerequisite courses listed below. To be eligible to apply, students must

1. have completed or be enrolled in the prerequisite courses listed below at the time of application and receive a minimum grade of "C-".
2. submit a personal statement addressing
 - the student's understanding of the profession of health administration
 - volunteer or other experience with health care services
 - personal goals for a career in health services administration.

Applications are due September 15 for spring semester admission or February 15 for fall semester admission. Specific application requirements may be found on the health services administration program website (<http://healthprof.jmu.edu/hsa/>).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3
Scientific Literacy requirement (in addition to General Education)	3
Prerequisite courses	9
Major requirements ²	58
University Electives	12
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Students automatically meet the B.S. quantitative and scientific literacy requirements by completing the major requirements. No more than 30 hours may be taken in the College of Business.

Prerequisites

Students must have completed or be enrolled in the following prerequisite courses and receive a minimum grade of "C-" to apply for and continue in the major.

Code	Title	Credits
Prerequisite Courses		
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors	3
or COB 204	Computer Information Systems	
ECON 201	Introduction to Microeconomics	3
HTH 354	U.S. and Global Health Care Systems	3
Total Credits		9

Required Courses

Code	Title	Credits
ACTG 244	Accounting for Non-Business Majors	3
FIN 345	Finance Fundamentals for Non-Business Majors	3
HSA 358	Health Services Administration Cornerstone	3
HSA 363	Health Economics ¹	3
HSA 365	Healthcare Ethics and the Law	3
HSA 366	Health Politics and Policy ²	3
HSA 385	Professional Development Seminar	1
HSA 452	Hospital Organization and Administration	3
HSA 454	Internship in Health Organizations	3
HSA 455	Long Term Care Organization and Administration ¹	3
HSA 456	Principles of Ambulatory Care Operations (B.S. degree requirement) ¹	3
HSA 463	Healthcare Quality Management and Analytics ²	3
HSA 464	Healthcare Finance	3
HSA 480	Health Services Administration Capstone	3
HTH 210	Medical Terminology	3
HTH 231	Population Health Determinants	3
HTH 320	Statistical Methods for Health Science Research (B.S. degree requirement)	3
HTH 450	Epidemiology	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
Total Credits		58

¹ Offered only in fall semester.

² Offered only in spring semester.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
General Education requirement		31
Credits		31
Second Year		
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors	3
ECON 201	Introduction to Microeconomics	3
HTH 354	U.S. and Global Health Care Systems	3

MATH 220	Elementary Statistics [C3QR]	3
General Education requirement		9
Electives		7
Credits		28
Third Year		
ACTG 244	Accounting for Non-Business Majors	3
FIN 345	Finance Fundamentals for Non-Business Majors	3
HTH 320	Statistical Methods for Health Science Research (B.S. degree requirement)	3
HSA 358	Health Services Administration Cornerstone	3
HSA 365	Healthcare Ethics and the Law	3
HSA 385	Professional Development Seminar	1
HTH 210	Medical Terminology	3
HTH 231	Population Health Determinants	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
General electives		3
Credits		31
Fourth Year		
HTH 450	Epidemiology	3
HSA 363	Health Economics ¹	3
HSA 366	Health Politics and Policy ²	3
HSA 452	Hospital Organization and Administration	3
HSA 454	Internship in Health Organizations	3
HSA 455	Long Term Care Organization and Administration ¹	3
HSA 456	Principles of Ambulatory Care Operations (B.S. degree requirement) ¹	3
HSA 463	Healthcare Quality Management and Analytics ²	3
HSA 464	Healthcare Finance	3
HSA 480	Health Services Administration Capstone	3
Credits		30
Total Credits		120

¹ Offered only in fall semester.

² Offered only in spring semester.

History, B.A.

Program Description

History courses introduce students to the way historians think and use evidence. The 100- and 200-level courses serve both general education and history majors with an overview of how to approach history. 300-level courses allow majors and interested non-majors to dive into historical thinking and argumentation through courses highlighting different themes and areas of the world. 400-level courses offer capstone-guided research experience intended for majors and experienced history students.

The requirements for a major in history consist of introductory and upper-level courses. All courses introduce students to the nature of history. In addition to involving reading, writing and critical thinking, these courses help students develop skills in using historical thinking and conducting research and interpretation for the 21st-century world. Majors in history are strongly encouraged to continue study in foreign languages beyond the minimum university requirement and, when appropriate, to integrate their foreign language studies into their history classes.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses) (requirement for B.A. degree)	3
Major requirements	35-47
University electives	17-41
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

The major requires five core courses. The first two introductory courses include HIST 101 and HIST 102. HIST 225 is the final introductory course. HIST 300 is a required course in research methods and thinking about how to do history. Entering majors should take HIST 300 as soon as possible. HIST 299 helps students to gain knowledge and confidence to translate academic learning to the post-college workforce.

All majors take a 300-level course that emphasizes the "big picture" of global history and the way ideas spread across space and time. All majors also take a 300-level course that applies historical thinking to today's world and to skills outside the classroom. This can include internships, courses that use the dedicated History Studio and courses that delve into the archives. Students may choose from the list of courses that meet these requirements.

Majors also complete two capstone courses at the 400-level that offer opportunities to develop and share their own historical interpretations

with a wider audience. For students writing a senior honors thesis, only three hours of HIST 499C may be counted among the two 400-level courses required for the major.

Majors must also take three elective courses at the 300- or 400-levels. Students may elect a major concentration of either public history or teacher education. A concentration is not required. Students who elect a concentration must fulfill all the specific requirements for those programs.

Code	Title	Credits
Core Requirements		
HIST 101	World History to 1500 [C2HQC]	3
HIST 102	World History since 1500 [C2HQC]	3
HIST 225	U.S. History [C4AE]	4
HIST 299	Putting History Skills to Work	1
HIST 300	Historical Thinking	3
Subtotal		14

History Majors (No Concentration Option)

These are requirements outside the core for upper-level courses.

Connected History:

The Connected history requirement offers classes that think historically about big-picture issues in a global context. Classes might consider regions or the globe. These classes guide students in how to use historical inquiry to understand deep connections and cause and effect relationships across time and place. We intend this to be world history in its broadest sense.

Choose one of the following:		3
HIST 307	The Trans-Atlantic Slave Trade	
HIST 314	Plagues, Saints and Empires: The World before 1500	
HIST 315	Global History of Sport	
HIST 317	Plagues and Pandemics in Modern World History	
HIST 318	Our Better Angels?: The World History of Violence	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 328	Towers, Time and Technology on the Eve of World War I	
HIST 333	Maps, Money, Manufacture and Trade	
HIST 334	A Global History of Nationalism	
HIST 353	The Global Middle East	
HIST 365	Topics in Connected History	
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	
HIST 373	History of the Ottoman Empire, 1453-1922	

Applying History:

Applying history classes move beyond the confines of the traditional classroom. These classes apply historical thinking and historical source-based evidence in real world settings. These can take different forms. Internships are an option here. So are classes that focus on acquiring and applying digital and technological skills, or on using history to think about public policy. Others guide students in archival research either online or in an in-person archive. Many study abroad classes also fulfill these goals.

Choose one of the following:		3
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	
HIST 324	Terror and Survival in American Lynching	

HIST 335	Topics in Applying History	
HIST 340	Internship in History	
HIST 343	History of Today	
HIST 345	Pills and Potions: Histories of U.S. Health Care and Medicine	
HIST 351	War in Afghanistan: Local and Global Perspectives	
HIST 357	Fates of Empires	
HIST 358	Modern Africa, ca. 1800-1990s	
HIST 359	History and Public Policy	
HIST 360	Research Apprenticeship in History	
HIST 362	American Material Religion	
HIST 374	The Southern Plantation	
HIST 396	History Beyond the Classroom: Public History	
Research Capstone Requirement:		
Majors need to take two courses at the 400-level. These courses are designed to offer all students guided research experiences. Students may choose any two courses they wish at this level.		6
Elective Courses:		
Students need to take three additional elective courses chosen from 300- and 400-level courses. Students have free choice of these elective courses. Students may not double-count courses that meet other specific requirements in history. In other words, students could choose to take two different courses that meet the connected history requirement. One would count toward the connected history requirement and the other would count as an elective course for the major. ¹		9
Subtotal		21
Total Credits		35

¹ HIST 300 and HIST 391 do not count toward this requirement.

Concentrations

History Teacher Education Concentration

Do you want to teach history in a middle or high school setting? A history major is good preparation for a professional path in social studies education which allows students to combine a passion for history with a clear professional path. Students may choose between a four-year program and a five-year program that lead to preparation for teacher licensure. The five-year program (including a pre-professional secondary education minor and social studies (SS) minor, and then application and acceptance to an M.A.T. program in the fifth year) allows students to extend and deepen their historical knowledge and skill at historical thinking, while learning to apply it in field experiences. The four-year program (a double major in secondary education and history, with a minor in SS) provides an accelerated program for motivated students to get into the classroom sooner.

Students interested in teaching licensure must also complete the SS minor and either the Secondary Education Major or the Secondary Education Minor.

Students who take POSC 457 cannot also take or receive credit for HIST 357.

Pre-professional secondary education minors (5-year program) should ordinarily take POSC 457 for the SS minor and therefore not receive credit

for HIST 357. If they do take HIST 357 instead (with prior permission), it cannot double-count for the major and SS minor.

Secondary Education majors (four-year program): HIST 357 can double-count with the SS minor.

Code	Title	Credits
Thinking with Economics		
Choose one of the following:		3
HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies	
HIST 333	Maps, Money, Manufacture and Trade	
HIST 469	A History of International Development in the Twentieth Century	
HIST 470	Modern Africa	
or another economic history class approved by advisor		
Subtotal		3
Virginia History Requirement		
HIST 303	Virginia's Place in #VastEarlyAmerica	3
or HIST 350	Virginia	
Subtotal		3
Connected History		
The Connected history requirement offers classes that think historically about big-picture issues in a global context. Classes might consider regions or the globe. These classes guide students in how to use historical inquiry to understand deep connections and cause and effect relationships across time and place. We intend this to be world history in its broadest sense.		
Choose one of the following:		3
HIST 307	The Trans-Atlantic Slave Trade	
HIST 314	Plagues, Saints and Empires: The World before 1500	
HIST 315	Global History of Sport	
HIST 317	Plagues and Pandemics in Modern World History	
HIST 318	Our Better Angels?: The World History of Violence	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 328	Towers, Time and Technology on the Eve of World War I	
HIST 333	Maps, Money, Manufacture and Trade	
HIST 334	A Global History of Nationalism	
HIST 353	The Global Middle East	
HIST 365	Topics in Connected History	
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	
HIST 373	History of the Ottoman Empire, 1453-1922	
Subtotal		3
Applying History Requirement for Teacher Education		
Applying history classes move beyond the confines of the traditional classroom. These classes apply historical thinking and historical source-based evidence in real world settings. Students in the history teacher education program should consider choosing courses that apply history in K-12 classrooms. ^{1,2}		
Choose one of the following:		3
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	
HIST 324	Terror and Survival in American Lynching	

HIST 335	Topics in Applying History	
HIST 340	Internship in History	
HIST 343	History of Today	
HIST 345	Pills and Potions: Histories of U.S. Health Care and Medicine	
HIST 351	War in Afghanistan: Local and Global Perspectives	
HIST 357	Fates of Empires	
HIST 358	Modern Africa, ca. 1800-1990s	
HIST 359	History and Public Policy	
HIST 360	Research Apprenticeship in History	
HIST 362	American Material Religion	
HIST 374	The Southern Plantation	
HIST 396	History Beyond the Classroom: Public History	
Subtotal		3
Research Capstone Requirement		
Majors need to take two courses at the 400-level. These courses are designed to offer all students guided research experiences. Students may choose any two courses they wish at this level.		6
Subtotal		6
Elective Courses		
Students need to take one elective course chosen from 300- and 400-level courses. Students have free choice of that elective course. Students may generally not double-count courses that meet other specific requirements in history. ³		3
Subtotal		3

¹ HIST 357 concentrates on thinking through K-12 pedagogies.

² For students in the 4-year history teacher education program, HIST 357 double-counts for a similar requirement in the SS minor.

³ Students may double-count a course toward the Thinking with Economics requirement and another requirement if the course is approved for both requirements. In that instance, students will then need to complete one additional elective history course at the 300- or 400-level to complete the required credit hours in the major. HIST 300 and HIST 391 do not count toward this requirement.

History Teacher Education Concentration Total: 35 Credit Hours

Public History Concentration

Historians today practice their discipline in a variety of careers as well as in more traditional academic settings. Those historians who work in museums, archives, government agencies, libraries, historic preservation organizations, businesses, contract history firms, cultural resource management firms and historic sites are known as public historians because they use their skills as historians to serve a public audience. The concentration in Public History trains students in the broad range of skills and issues associated with public history while providing them with a solid general background in history.

Students pursuing the public history concentration augment their foundation of traditional history courses by taking introductory and specialized public history courses and completing a semester-long internship. History majors opting to pursue the public history concentration will complete all history requirements and additional courses designed to train students in public history.

Students are required to complete two public history core courses as well as meeting all history department requirements. Two of the three

elective courses should be chosen from the list of primary electives; the remaining elective course may be chosen from either the primary or secondary list of elective courses.

Code	Title	Credits
Centering Public History Requirement		
HIST 396	History Beyond the Classroom: Public History	3
Subtotal		3
Connected History		
The Connected history requirement offers classes that think historically about big-picture issues in a global context. Classes might consider regions or the globe. These classes guide students in how to use historical inquiry to understand deep connections and cause and effect relationships across time and place. We intend this to be world history in its broadest sense.		
Choose one of the following:		3
HIST 307	The Trans-Atlantic Slave Trade	
HIST 314	Plagues, Saints and Empires: The World before 1500	
HIST 315	Global History of Sport	
HIST 317	Plagues and Pandemics in Modern World History	
HIST 318	Our Better Angels?: The World History of Violence	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 328	Towers, Time and Technology on the Eve of World War I	
HIST 333	Maps, Money, Manufacture and Trade	
HIST 334	A Global History of Nationalism	
HIST 353	The Global Middle East	
HIST 365	Topics in Connected History	
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	
HIST 373	History of the Ottoman Empire, 1453-1922	
Subtotal		3
Applying History Requirement for Public History		
All students planning to graduate with a concentration in public history need to complete an approved internship. This internship counts as the applying history requirement.		
HIST 340	Internship in History	3
Subtotal		3
Public History Primary Courses		
Students need to take two or three courses from this list of approved public history courses. Students who take two of these courses also need to complete one course from the secondary public history list. Students who complete three of these courses do not need to take a course from the public history secondary list.		
Choose 6-9 credit hours of the following:		6-9
HIST 331	Historical Archaeology	
HIST 349	History, Site, Memory	
HIST 366	Topics in Public History	
HIST 394	Introduction to Museum Work	
HIST 407	Digital History	
HIST 438	Workshop in Public and Local History	
HIST 440	The History Museum	
HIST 441	Oral History	
HIST 492	History from Things: Material Culture	

HIST 493	Historic Preservation	
Subtotal		6-9
Public History Secondary Courses		
Students either take three courses from the Public History primary courses list or two from that list and one from this Public History secondary list. Between the two requirements, students must take 9 credit hours without double-counting a course.		
Select 0-3 credit hours of the following:		0-3
HIST 326	How Cars Changed Everything	
HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies	
HIST 338	United States Urban History	
HIST 360	Research Apprenticeship in History	
HIST 391	Travel Studies Seminar (Summer in Ghana Program or Travel Studies Seminar: Paris and Normandy: Civilization, Culture, and Memory: North of England Programme)	
HIST 402	Workshop in Colonial American Life	
HIST 403	Workshop in Civil War Virginia	
HIST 406	Monticello	
HIST 408	The Museum: History and Controversies	
HIST 452	Parks and Recreation: History of National Parks	
ANTH 494	Field Techniques in Archaeology	
Subtotal		0-3
Research Capstone Requirement		
Majors need to take two courses at the 400 level. These courses are designed to offer all students guided research experiences. Students may choose any two courses they wish at this level.		6
Subtotal		6
Elective Courses for Public History Students		
Students need to take one elective course at the 300 or 400 level in addition to the other requirements for this program. Students have free selection of that course from any 300 or 400 level course in the major. ¹		3-9
Subtotal		3-9

¹ In the event that a student takes a course that meets both the public history primary or secondary course requirement and the research capstone requirement, the student will take an additional elective history course at the 300 or 400 level to complete the required credit hours. Any 300 or 400 level course in history meets this requirement. HIST 300 and HIST 391 do not count toward this requirement.

Public History Concentration Total 41-47 Credit Hours

Major Requirements Total: 35-47 Credit Hours

Recommended Schedule for Majors
Recommended Schedule for Majors without a Specific Concentration

(Non- History Teacher Education Concentration)

Course	Title	Credits
First Year		
HIST 101	World History to 1500 [C2HQC]	3
HIST 102	World History since 1500 [C2HQC]	3
HIST 225	U.S. History [C4AE]	4

Foreign language requirement		6
General Education requirement		12
Credits		28
Second Year		
We encourage majors to begin taking 300 level courses in the major in their second year after completing General Education courses, but waiting until the third year to begin taking 400 level courses in the major.		
HIST 300	Historical Thinking ¹	3
Foreign language requirement (if needed) or electives		6
General Education requirement		12
University Elective(s)		9
Credits		30
Third Year		
Philosophy course		3
History electives (mix 300 and 400 levels)		9
General Education courses and/or university electives		18
Credits		30
Fourth Year		
HIST 299	Putting History Skills to Work	1
Electives		22
History electives (mix 300 and 400 levels)		9
Credits		32
Total Credits		120

¹ Students are encouraged to take HIST 300 by the end of the second year.

Recommended Schedule for the Major with History
Teacher Education Concentration

A more detailed recommendation for combining the majors and minors leading to licensure can be found on the History department website.

Course	Title	Credits
First Year		
Students who have a double-major in which the other major is a BS degree may be subject to the BS degree requirements instead, which do not include a foreign language. Be sure to talk to your advisor.		
HIST 101	World History to 1500 [C2HQC]	3
HIST 102	World History since 1500 [C2HQC]	3
HIST 225	U.S. History [C4AE]	4
Foreign language requirement are required if doing the B.A. degree.		6
General Education requirement		12
Credits		28
Second Year		
We encourage majors to begin taking 300 level courses in the major in their second year after completing General Education courses, but waiting until the third year to begin taking 400 level courses in the major.		
HIST 303 or HIST 350	Virginia's Place in #VastEarlyAmerica or Virginia	3
HIST 333	Maps, Money, Manufacture and Trade	3
HIST 300	Historical Thinking ¹	3
Foreign language requirement (if needed) or electives		6
General Education courses and/or university electives		15
Credits		30
Third Year		
HIST 357	Fates of Empires ²	3
Philosophy course		3
History electives (mix 300 and 400 levels)		9
General Education courses and/or university electives		17
Credits		32
Fourth Year		
History electives (mix 300 and 400 levels)		9

Electives	20
Student teaching for students in the 4-year program	
HIST 299	Putting History Skills to Work
	1
Credits	
	30
Total Credits	
	120

¹ Students are encouraged to take HIST 300 by the end of their second year.

² HIST 357 students in the 4-year program [students in 5-year program should take POSC 457 fall of senior year]

Hospitality Management, B.S.

Program Information

Hospitality Management at James Madison University prepares students to be world class business leaders in the fast-paced, ever-changing and diverse hospitality industry. Students engage in transformational education as part of an extensive curriculum focused on high quality, student-centered educational experiences by participating in practical, applied, and experiential learning opportunities.

Goals

- To expose students to a rigorous academic and experiential learning program, including national certifications, two internships and over 1,000 hours of professional work experience prior to graduation.
- To provide an intimate educational setting of small classes, team learning experiences and meaningful personal contacts with faculty and industry professionals.
- To promote the use and development of technology, critical thinking, innovation and communication skills in hospitality leadership.
- To ensure that all HM students have a meaningful choice of job opportunities upon graduation.
- To build a set of leadership skills while providing students exceptional financial acumen.
- To maintain a faculty of knowledgeable and respected industry professionals, dedicated to continuous improvement via internships, authorship, industry contacts and participation at national and international industry conferences.

Declaring or Changing Major into Hospitality Management

If you are a current JMU student wishing to declare or change a major you must complete a request through MyMadison. After you complete the request for changing majors into Hospitality Management, you are required to contact a member of the Hart School Advisory Team (Scott Arbogast, arbogastw@jmu.edu; Jason Gochenour, gochenjl@jmu.edu) to complete the declaration process. Visit the Hart School's website (<https://www.jmu.edu/hartschool/caps/advising/index.shtml>) to schedule an appointment with either Scott or Jason.

The HM major within the Hart School has cutoff dates. Students are only admitted twice a year, October 1 and March 1. Any student wishing to take Hart School courses in the spring semester or any summer term session would be required to contact the Hart School advisory team and declare the change of major on or before October 1. Any student wishing to take Hart School courses in the Fall semester would be required to contact the Hart School advisory team and declare the change of major on or before March 1.

Degree Requirements

The B.S. degree in Hospitality Management (HM) requires a minimum of 120 credit hours of undergraduate course work. In addition to all HM core and elective courses, all HM majors must complete either the general business minor or the entrepreneurship minor.

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3

Scientific literacy requirement (in addition to General Education) 3-4

Major requirements 52

General Business Minor or Entrepreneurship Minor 18
2,3

University Electives 2-3

Total 120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Successful completion of these courses with a 2.0 GPA will qualify the student for a general business minor; however, it is the responsibility of the student to complete the necessary paperwork in the College of Business to apply for the minor.

³ The 18 credit hour minors contain 6 credit hours which are fulfilled in the major core requirements (ACTG 244, MKTG 380).

Major Requirements

Code	Title	Credits
Core Courses		
HM 100	First-Year Seminar	1
HM 111	Overview of Hospitality and Tourism Management	3
HM 112	Foundations of Leadership of Hospitality Management	3
HM 222	Hospitality Professional Work Experience	3
HM 251	Cost Control and Budgeting	3
HM 252	Culinary Arts and Menu Management	3
HM 321	Introduction to Event Management	3
HM 322	Lodging, Clubs and Resort Operations and Technology	3
HM 333	Internship in Hospitality Management	3
HM 371	Profit Optimization and Revenue Management	3
HM 422	Hospitality Human Resources Management	3
HM 442	Hospitality Seminar	3
HM 444	Supervisory Hospitality Internship	3
ACTG 244	Accounting for Non-Business Majors ^{1, 2}	3
MKTG 380	Principles of Marketing ¹	3
Subtotal		43
Hospitality Management Electives		
Choose 9 credit hours from the following:		9
HM 101	Lecture Series in Hospitality Leadership and Innovation	
HM 201	Foundations of Hospitality, Sport and Recreation Management	
HM 210	Tourism in a Global Context	
HM 221	Special Event Management: Le Gourmet	
HM 244	Disney Internship	
HM 261	Hospitality Data Analytics	
HM 298	Special Studies in Hospitality Management	
HM 316	Country Club Management	
HM 361	Italian Culture and Wine	
HM 362	Italian Gastronomy	
HM 414	Beverage Management and Marketing	

HM 417	Advanced Event Management	
HM 419	Napa and Sonoma Wine and Culture	
HM 490	Special Studies in Hospitality and Tourism Management	
HM 498	Special Topics in Hospitality and Tourism Management	
HM 499	Honors	
Subtotal		9
Total Credits		52

Hospitality Management Elective Courses	6
Credits	30
Total Credits	120

¹ ACTG 244 and MKTG 380 require junior standing.

² COB 242 may substitute for ACTG 244.

Required Minor: 18 Credit Hours

All majors must declare either the general business minor (p. 498) or the entrepreneurship minor (p. 484) through the College of Business. Two courses required in the hospitality management major, ACTG 244 and MKTG 380, are required courses for the general business minor and elective options for the entrepreneurship minor.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
HM 100	First-Year Seminar	1
HM 111	Overview of Hospitality and Tourism Management	3
HM 112	Foundations of Leadership of Hospitality Management	3
Madison Foundations		9
General Education		14
Credits		30
Second Year		
HM 222	Hospitality Professional Work Experience	3
HM 251	Cost Control and Budgeting	3
HM 252	Culinary Arts and Menu Management	3
ACTG 244	Accounting for Non-Business Majors	3
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors	3
ECON 201	Introduction to Microeconomics	3
Quantitative Requirement		3
General Education		9
Credits		30
Third Year		
HM 321	Introduction to Event Management	3
HM 322	Lodging, Clubs and Resort Operations and Technology	3
HM 333	Internship in Hospitality Management	3
HM 371	Profit Optimization and Revenue Management	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
General Education and University Elective(s)		9
Hospitality Management Elective Courses		3
Credits		30
Fourth Year		
HM 422	Hospitality Human Resources Management	3
HM 442	Hospitality Seminar	3
HM 444	Supervisory Hospitality Internship	3
FIN 345	Finance Fundamentals for Non-Business Majors	3
University Elective(s) / scientific literacy		12

Inclusive Early Childhood Education, B.S.

Program Description

The Bachelor of Science pre-professional program in Inclusive Early Childhood Education enables students to gain knowledge and understanding of early childhood inclusion (Birth-3rd grade) and practices that promote access, participation and support for young children and their families. Students completing the four-year degree program, when combined with the teacher dual licensure option, are prepared to serve as teachers of students with and without disabilities in a variety of inclusive educational placements.

The licensure program is designed to prepare resilient, culturally responsive educators who are advocates for young children and youth with and without disabilities that are not only qualified for the complexity of their professional roles but also are reflective problem-solvers. This program offers a dual licensure in PreK-3rd grade and Early Childhood Special Education (birth-5 years).

The program includes extensive field experiences which occur each semester beginning in the student's sophomore year. Assessment of candidate performance includes evaluation of performance in individual courses and practicum, as well as other criteria. Students are required to demonstrate satisfactory performance, which includes a minimum grade of "C" in all major and concentration coursework and a "satisfactory" score on all practicum evaluations (i.e. all indicators on the evaluation are within the satisfactory range). Key assessments will be used to identify those candidates who are not making satisfactory progress towards advancement to the next sequences of courses and experiences. At various points throughout the program, there are formative and summative assessments where faculty committees review the overall performance of each candidate. Formative assessments will be used to advise candidates and develop a plan of action for addressing any concerns that have been identified by the faculty. Summative assessments will be used to identify those candidates who are not making satisfactory progress toward advancement to the next sequences of courses and experiences. If progress is unsatisfactory, the candidate will not be allowed to continue until any identified issues are corrected. In some cases, a candidate will be allowed to continue in the next semester, but there will be a plan of action for addressing any concerns that have been identified by the faculty. Students need to take MATH 110 and MATH 210 in order to be prepared for the Math Praxis.

Every student is assigned an advisor. Students should plan on regularly consulting their advisor.

It is important for students to understand that the undergraduate major in Inclusive Early Childhood Education does not lead to teacher-education licensure unless licensure requirements are completed. To qualify for a recommendation for licensure in Inclusive Early Childhood Education [Early Childhood Special Education (Birth-5 years) and general education curriculum (PreK-3rd Grade)], students must complete licensure requirements in addition to requirements for the Inclusive Early Childhood Education undergraduate major. These requirements include 22 credit hours of practicum and student teaching experiences. Please note that these credit hour requirements are included in the 120 credit hour requirements for the Bachelor of Science in Inclusive Early Childhood Education degree.

Admission Requirements

Students must meet specific requirements to be admitted to programs within the College of Education (p. 18). These criteria can be found on the College of Education (p. 18) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement ²	3
Scientific Literacy Requirement ²	3-4
IECE Major	51
Licensure Requirements or Electives	24
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credit hours may be fulfilled by requirements within a student's major. General Education courses must include EDUC 175 or PSYC 160. While required for the major professional studies core, credits for EDUC 175 or PSYC 160 are counted in the General Education total of 41 credit hours.

² In addition to General Education. Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml).

Major Requirements

Code	Title	Credits
IECE Professional Studies Core		
EDUC 175	Learning, Behavior and Engagement in Context [C5SD]	3
or PSYC 160	Life Span Human Development [C5SD]	
EDUC 200	Foundations of Education in the United States	3
EDUC 310	Teaching in a Diverse Society	3
EXED 200	Foundations of Exceptional Education	3
Subtotal		12
Required IECE Courses		
IECE 300	Issues and Trends in Inclusive Early Childhood Education	3
IECE 303	Development of Young Children Birth Through Age 8	3
IECE 322	Curriculum and Instruction in Inclusive Early Childhood Education	3
IECE 324	Assessment of the Young Child	3
IECE 355	Foundations of Inclusive Emergent and Early Literacy for Inclusive Early Childhood Education (IECE)	3
IECE 429	Teaching Early Childhood Learners with Complex Care Needs (IECE)	3
IECE 450	Contemporary Family Issues in Inclusive Education	3
IECE 456	Foundations of Literacy (PK-3) for Inclusive Early Childhood Education (IECE)	3
IECE 460	Instructional Practices in Numeracy	3
IECE 462	Instructional Practices in Natural Sciences for Young Children	3

IECE 464	Instructional Practices in Social Studies for Young Children	3
IECE 466	Managing Classrooms and Guiding Behavior	3
IECE 470	Individualized Behavior Intervention for Young Children	3
Subtotal		39
Licensure Requirements or Electives		
Students who do not choose the licensure option will choose 24 credits of electives.		
IECE 321	Initial Field Experience in IECE Part 1	1
IECE 331	Initial Field Experience in IECE Part 2	1
IECE 421	Intermediate Field Experience in IECE Part 1	2
IECE 431	Intermediate Field Experience in IECE Part 2	2
IECE 461	Advanced Field Experience in IECE Part 1	2
IECE 471	Advanced Field Experience in IECE Part 2	2
IECE 480	Student Teaching in Inclusive Early Childhood Education (taken twice for 10 credits)	10
IECE 481	Seminar in Inclusive Early Childhood Education (taken twice for 4 credits)	4
Subtotal		24
Total Credits		75

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
General Education		9
EDUC 200	Foundations of Education in the United States	3
EXED 200	Foundations of Exceptional Education	3
Credits		15
Spring Semester		
General Education		9
EDUC 310	Teaching in a Diverse Society	3
PSYC 160 or EDUC 175	Life Span Human Development [C5SD] or Learning, Behavior and Engagement in Context [C5SD]	3
Credits		15
Second Year		
Fall Semester		
General Education		7
IECE 300	Issues and Trends in Inclusive Early Childhood Education	3
IECE 303	Development of Young Children Birth Through Age 8	3
IECE 321	Initial Field Experience in IECE Part 1	1
Credits		14
Spring Semester		
General Education		7
IECE 322	Curriculum and Instruction in Inclusive Early Childhood Education	3
IECE 331	Initial Field Experience in IECE Part 2	1
IECE 355	Foundations of Inclusive Emergent and Early Literacy for Inclusive Early Childhood Education (IECE)	3
IECE 466	Managing Classrooms and Guiding Behavior	3
Credits		17
Third Year		
Fall Semester		
General Education		3
B.S. Degree Requirements		3
IECE 421	Intermediate Field Experience in IECE Part 1	2

IECE 450	Contemporary Family Issues in Inclusive Education	3
IECE 464	Instructional Practices in Social Studies for Young Children	3
IECE 456	Foundations of Literacy (PK-3) for Inclusive Early Childhood Education (IECE)	3
Credits		17
Spring Semester		
IECE 324	Assessment of the Young Child	3
IECE 431	Intermediate Field Experience in IECE Part 2	2
IECE 460	Instructional Practices in Numeracy	3
IECE 462	Instructional Practices in Natural Sciences for Young Children	3
General Education		3
B.S. Degree Requirements		3
Credits		17
Fourth Year		
Fall Semester		
IECE 461	Advanced Field Experience in IECE Part 1	3
IECE 470	Individualized Behavior Intervention for Young Children	3
IECE 480	Student Teaching in Inclusive Early Childhood Education	5
IECE 481	Seminar in Inclusive Early Childhood Education	2
Credits		13
Spring Semester		
IECE 429	Teaching Early Childhood Learners with Complex Care Needs (IECE)	3
IECE 471	Advanced Field Experience in IECE Part 2	2
IECE 480	Student Teaching in Inclusive Early Childhood Education	5
IECE 481	Seminar in Inclusive Early Childhood Education	2
Credits		12
Total Credits		120

Individualized Study, B.A.

Phone: (540) 568- 4253

Email: adp@jmu.edu

Location: Ice House, 127 W. Bruce Street, Room 332

Adult Degree Program Website (<http://www.jmu.edu/adultdegree>)

Program Description

The Adult Degree Program/Bachelor of Individualized Study delivers a flexible, dynamic and rigorous degree completion curriculum to meet the individual education and career goals of our students. ADP/BIS's non-residential program offers a framework for modern learners with transfer credit and/or prior learning experience to design their own academic plan.

Experiential Learning

The Adult Degree Program offers several pathways for students to participate in experiential learning opportunities. Students can choose to participate in workforce related experiences, intensive research or international programs.

Degrees Offered

Students in the ADP have the opportunity to earn a Bachelor of Individualized Study (B.I.S.) in individualized study, a Bachelor of Arts (B.A.) in individualized study or a Bachelor of Science (B.S.) in individualized study. The university degree requirements differ by degree type.

Admission

To be eligible for the ADP, applicants must be at least 22 years of age, have at least 30 transferable college credits and have a minimum cumulative GPA of 2.0 for all prior coursework.

With very limited exceptions, students in the individualized study major (the ADP's academic major) cannot transfer to another major at JMU.

Degree Requirements

The Bachelor of Arts (B.A.) in individualized study requires 41 general education credits, an additional philosophy course (3.00 credit hours) and foreign language at the intermediate level (6.00 credit hours).

Degree Requirements	Credit Hours
General Education courses ¹	41
Foreign language courses (intermediate level required) ²	0-14
Philosophy course (in addition to General Education courses) ³	3
Major concentration courses and electives	62-76
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures placement test.

³ Any PHIL course except PHIL 120 or PHIL 150. Consult the list of courses satisfying Bachelor of Arts degree requirements on the Registrar's website.

Major Requirements

Students must complete all of the following requirements:

- IS 200
- A minimum of 120 credit hours, including 41 General Education credits
- A minimum of 30 credit hours earned from JMU
- At least 60 credit hours earned from a four-year institution
- IS 498, an online interdisciplinary research course with a capstone project coordinated by the instructor and includes a final written and oral presentation; minimum grade of "C" required

Students must also complete an experiential component of at least three credits from the options listed below:

- IS 206 (required for students pursuing credit for prior learning experience only)
- IS 300
- A study abroad course or another approved experiential course
- A practicum or student teaching course
- An additional three credits of IS 498¹

¹ If taking additional three credits of IS 498, they must be taken concurrently in the final semester, and will be an intensive research experience focusing on a single research topic.

Additional Requirements

- A program of study (POS) for meeting the requirements of degree completion must be submitted to the ADP office for approval while enrolled in the IS 200 course.
- A student's concentration, or field of study, consists of a minimum of 30 credits, at least 24 of which must be upper-division credit (300- and 400-level courses). The concentration includes classes from two or more academic disciplines and represent a coherent body of knowledge, which may require course work beyond the 30 minimum credit hours.
- No more than 30 credit hours of course work can be selected from courses taught by the College of Business. This 30 credit hour limit includes COB courses transferred from other colleges and nontraditional credit earned in the COB.
- The concentration must be completed with a GPA of 2.0 or better.
- A minimum of 15 credit hours in the concentration must be JMU credits.
- Students must participate in assessment activities used to measure student learning outcomes and modify curriculum.
- There is no residency requirement; however, program agreements that extend beyond six years may need to be amended if university policies change that affect the original program of study.

Individualized Study Concentration

This is the focus area of study in the individualized study major. The concentration is interdisciplinary and consists of classes from two or

more academic program areas to meet each student's educational and/or professional goals.

- Minimum of 30 credits.
- 24 credits must be completed in 300- or 400-level courses.
- IS 498 – online interdisciplinary research course with a capstone project, included in the 24 upper-level course requirements.

Mechanisms for Earning Credit

The number of credits, which may be accepted as transfer credit or earned through the following methods, is determined by the student's approved POS. It is reasonable to expect that the unique educational needs, background and personal circumstances of each student will determine to a great extent the manner in which credit has been and will be earned.

- Transfer credit awarded by the registrar's office.
- Credit by departmental examination.
- Credit for approved sponsored learning or independent studies.
- Regular course work.
- Credit for experiential learning through portfolio of prior learning assessment (PLA).¹
- Credit for educational programs of the Armed Forces (JST).
- College Level Examination Program (CLEP).¹
- American Council on Education College Credit Recommendation Service (ACE).¹

¹ No more than 30 non-traditional credits can be counted towards the degree requirements of 120 credit hours which includes PLA, JST, CLEP and ACE.

Prior Learning Assessment (PLA)

Prior learning assessment (PLA) is an assessment of knowledge and skills acquired prior to or outside of enrollment in higher education for the purpose of earning college-level credit. The term portfolio is used to describe a collection of evidence to support the PLA process. Through the portfolio, a student makes a case by succinctly identifying, articulating and documenting mastery of college level learning or its equivalence.

PLA is an option for a student who has gained significant learning through experiences including, but not limited to: workplace training, volunteer service, civic activities, conferences, workshops, vocational interests, travel and independent reading.

Prior learning credit depends on each student's individual experiences and may not be a viable option for every student. The process entails extensive preparation, excellent academic writing skills and the ability to meet deadlines. Before a portfolio is submitted for approval, students must complete IS 206. There is no guarantee that a portfolio submitted for credit will be approved.

College Credit Recommendation Service

ADP accepts some but not all credits as recommended in the National Guide to Educational Credit for Training Programs and the Guide to Educational Credit by Examination prepared by the American Council on Education. Additional information on program policies and procedures specific to JST is available through the ADP website.

College-Level Examination Program

Effective Fall 2024, students who participated in the College-Level Examination Program may be awarded course credit that satisfies a General Education requirement or elective. Prospective students who complete CLEP examinations and achieve an appropriate score (usually a 50 or above) are eligible to earn college credit. Evaluation of credit will be performed in coordination with the appropriate departmental office and cannot be appealed.

JMU will accept these scores electronically from the testing agency. Please contact College Board for information on providing your exam results to JMU. More information can be found at <https://clep.collegeboard.org/scores>.

Credit/No-Credit

Students enrolled in the ADP may complete courses on a credit/no-credit basis if they have completed 28 credit hours of college level course work and have a 2.25 cumulative GPA.

Courses in the area of concentration, including the senior capstone research course, may not be taken on a credit/no-credit basis. Other provisions of the credit/no-credit option, as described in this catalog, will also be in effect.

Online Interdisciplinary Tracks

Online interdisciplinary tracks were developed in cooperation with various academic units at the university. Students can use these tracks when developing their concentration.

- Business and Information Technology
- Small Business Management and Entrepreneurship
- Human Resource Development and Management
- Professional Workplace Communication
- Healthcare Leadership
- Child Development
- Family and Consumer Sciences
- Inclusive Early Childhood Education with teacher licensure¹
- Early Childhood Education Development with teacher licensure¹
- Elementary Education with teacher licensure¹
- Special Education with teacher licensure¹

¹ Only available to students in teacher licensure programs

Individualized Study, B.I.S.

Phone: (540) 568-4253

Email: adp@jmu.edu

Location: Ice House, 127 W. Bruce Street, Room 332

Adult Degree Program Website (<http://www.jmu.edu/adultdegree>)

Mission

The Adult Degree Program/Bachelor of Individualized Study delivers a flexible, dynamic and rigorous degree completion curriculum to meet the individual education and career goals of our students. ADP/BIS's non-residential program offers a framework for modern learners with transfer credit and/or prior learning experience to design their own academic plan.

Degrees Offered

Students in the ADP have the opportunity to earn a Bachelor of Individualized Study (B.I.S.) in individualized study, a Bachelor of Arts (B.A.) in individualized study or a Bachelor of Science (B.S.) in individualized study. The university degree requirements differ by degree type.

Admission

To be eligible for the ADP, applicants must be at least 22 years of age, have at least 30 transferable college credits and have a minimum cumulative GPA of 2.0 for all prior coursework.

With very limited exceptions, students in the individualized study major (the ADP's academic major) cannot transfer to another major at JMU.

Concentration Admissions

Students must meet specific requirements to be admitted to programs within the College of Education (p. 18). These criteria can be found on the College of Education (p. 18) catalog page.

Degree Requirements

The Bachelor of Individualized Study (B.I.S.) requires each student works with the ADP academic adviser to create an individual program of study that includes a minimum of 41 General Education credits.

Degree Requirements	Credit Hours
General Education courses ¹	41
Major requirements ²	30-73
Electives	Up to 49
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Includes education courses leading to licensure in teacher education.

Major Requirements

Students must complete all of the following requirements:

- IS 200
- A minimum of 120 credit hours, including 41 General Education credits
- A minimum of 30 credit hours earned from JMU

- At least 60 credit hours earned from a four-year institution
- IS 498, an online interdisciplinary research course with a capstone project coordinated by the instructor and includes a final written and oral presentation; a minimum grade of "C" required

Students must also complete an experiential component of at least three credits from the options listed below:

- IS 206 (required for students pursuing credit for prior learning experience only)
- IS 300
- A study abroad course or another approved experiential course
- A practicum or student teaching course
- An additional three credits of IS 498¹

¹ If taking additional three credits of IS 498, they must be taken concurrently in the final semester, and will be an intensive research experience focusing on a single research topic.

Additional Requirements

- A program of study (POS) for meeting the requirements of degree completion must be submitted to the ADP office for approval while enrolled in the IS 200 course.
- A student's concentration, or field of study, consists of a minimum of 30 credits, at least 24 of which must be upper-division credit (300- and 400-level courses). The concentration includes classes from two or more academic disciplines and represent a coherent body of knowledge, which may require course work beyond the 30 minimum credit hours.
- No more than 30 credit hours of course work can be selected from courses taught by the College of Business. This 30 credit hour limit includes COB courses transferred from other colleges and nontraditional credit earned in the COB.
- The concentration must be completed with a GPA of 2.0 or better.
- A minimum of 15 credit hours in the concentration must be JMU credits.
- Students must participate in assessment activities used to measure student learning outcomes and modify curriculum.
- There is no residency requirement; however, program agreements that extend beyond six years may need to be amended if university policies change that affect the original program of study.

Individualized Study Concentration

This is the focus area of study in the individualized study major. The concentration is interdisciplinary and consists of classes from two or more academic program areas to meet each student's educational and/or professional goals.

- Minimum of 30 credits.
- 24 credits must be completed in 300- or 400-level courses.
- IS 498 – online interdisciplinary research course with a capstone project, included in the 24 upper-level course requirements.

Mechanisms for Earning Credit

The number of credits, which may be accepted as transfer credit or earned through the following methods, is determined by the student's approved POS. It is reasonable to expect that the unique educational needs, background and personal circumstances of each student will

determine to a great extent the manner in which credit has been and will be earned.

- Transfer credit awarded by the registrar's office.
- Credit by departmental examination.
- Credit for approved sponsored learning or independent studies.
- Regular course work.
- Credit for experiential learning through portfolio of prior learning assessment (PLA).¹
- Credit for educational programs of the Armed Forces (JST).
- College Level Examination Program (CLEP).¹
- American Council on Education College Credit Recommendation Service (ACE).¹

¹ No more than 30 non-traditional credits can be counted towards the degree requirements of 120 credit hours which includes PLA, JST, CLEP and ACE.

Prior Learning Assessment (PLA)

Prior learning assessment (PLA) is an assessment of knowledge and skills acquired prior to or outside of enrollment in higher education for the purpose of earning college-level credit. The term portfolio is used to describe a collection of evidence to support the PLA process. Through the portfolio, a student makes a case by succinctly identifying, articulating and documenting mastery of college level learning or its equivalence.

PLA is an option for a student who has gained significant learning through experiences including, but not limited to: workplace training, volunteer service, civic activities, conferences, workshops, vocational interests, travel and independent reading.

Prior learning credit depends on each student's individual experiences and may not be a viable option for every student. The process entails extensive preparation, excellent academic writing skills and the ability to meet deadlines. Before a portfolio is submitted for approval, students must complete IS 206. There is no guarantee that a portfolio submitted for credit will be approved.

College Credit Recommendation Service

ADP accepts some but not all credits as recommended in the National Guide to Educational Credit for Training Programs and the Guide to Educational Credit by Examination prepared by the American Council on Education. Additional information on program policies and procedures specific to JST is available through the ADP website.

College-Level Examination Program

Effective Fall 2024, students who participated in the College-Level Examination Program may be awarded course credit that satisfies a General Education requirement or elective. Prospective students who complete CLEP examinations and achieve an appropriate score (usually a 50 or above) are eligible to earn college credit. Evaluation of credit will be performed in coordination with the appropriate departmental office and cannot be appealed.

JMU will accept these scores electronically from the testing agency. Please contact College Board for information on providing your exam results to JMU. More information can be found at <https://clep.collegeboard.org/scores>.

Credit/No-Credit

Students enrolled in the ADP may complete courses on a credit/no-credit basis if they have completed 28 credit hours of college level course work and have a 2.25 cumulative GPA.

Courses in the area of concentration, including the senior capstone research course, may not be taken on a credit/no-credit basis. Other provisions of the credit/no-credit option, as described in this catalog, will also be in effect.

Online Interdisciplinary Tracks

Online interdisciplinary tracks were developed in cooperation with various academic units at the university. Students can use these tracks when developing their concentration.

- Business and Information Technology
- Small Business Management and Entrepreneurship
- Human Resource Development and Management
- Professional Workplace Communication
- Healthcare Leadership
- Child Development
- Family and Consumer Sciences
- Inclusive Early Childhood Education with teacher licensure
- Early Childhood Education Development with teacher licensure
- Elementary Education with teacher licensure
- Special Education with teacher licensure

Education, Early Childhood Education Development (ECED) Concentration, A.A.S. to B.I.S.

This fully online program is for students who have completed an Associate of Applied Science (A.A.S.) degree in Early Childhood from a Virginia Community College. It leads to licensure in grades PreK-3 and a Bachelor of Individualized Study degree with a concentration in Early Childhood Education Development from JMU.

There are 60 credits that must be earned at a four-year institution and 30 of those must be earned at JMU.

All students must attend an orientation meeting at JMU the semester or summer prior to starting the program. This orientation will be offered online. Program offers free tutoring for Math Assessment, Praxis Elementary Subject Assessments (Reading & L.A Test Code 5002, Mathematics Test Code 5003, Science Test Code 5005, and Social Studies Test Code 5004), and Praxis Teaching Reading: Elementary Test (5205) (new version of RVE).

Field experience includes practica and student teaching. Practicum courses are to extend through the full semester. Student teaching must take place in public school or a licensed/accredited preschool. It is approximately 12 weeks with the possibility of teaching in your own classroom.

Candidates must be fully accepted into teacher education to begin the education program course work in the Summer Semester Year 1.

Minimum Grade Requirement

Assessment occurs during each semester, and candidate performance is reviewed at the end of each semester. Candidates must demonstrate

satisfactory performance before moving on to the next semester. Satisfactory performance includes a minimum grade of "C" in all education course work and an overall 2.5 GPA, demonstration of professional behaviors, and acceptable performance in practicum and on key assessments.

Candidates should note that they must be fully admitted to teacher education prior to registering for the courses offered during Summer Semester Year 1 of the Early Childhood Education program. Candidates should also note that, on occasion, the Virginia Department of Education (VDOE) or other accrediting agencies may change requirements after the catalog copy is approved. Therefore, it is especially important for candidates to confer with their advisors on a regular basis for the most current requirements. Completion of the Early Childhood Education program leads to a Virginia teaching license with an endorsement in PK-3 Early Childhood Education. To be recommended for licensure, all candidates must meet the following requirements:

- Complete General Education requirements.
- Complete the Early Childhood Education pre-professional program.
- Complete the disciplinary courses required for the Early Childhood Education degree.
- Meet all admission and retention criteria for teacher education including satisfactory scores on the Math Assessment, Praxis Elementary Subject Assessments (Reading & L.A. Test Code 5002, Mathematics Test Code 5003, Science Test Code 5005, and Social Studies Test Code 5004), and Praxis Teaching Reading: Elementary Test (5205) (new version of RVE).
- Meet admission requirements for the Early Childhood Education program.
- Meet performance and behavior standards as indicated by ratings on the program Professional Dispositions Checklist.

Professional Studies Core Requirements: 9 Credit Hours

The following courses are required for licensure. These courses may be taken prior to or while enrolled in the program:

Code	Title	Credits
EDUC 200	Foundations of Education in the United States	3
PSYC 160	Life Span Human Development [C5SD]	3
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Total Credits		9

ECED Concentration Requirements

Early Childhood Education Content Requirements

Code	Title	Credits
ECED 304	Diversity, Equity and Justice in Early Childhood Education	3
ECED 372	Introduction to Early Childhood Education	3
ECED 412	Natural and Social Sciences for Young Children	3
ECED 442	The Young Child	3
ECED 443	Practicum in Primary Grades	1
ECED 444	Mathematics Content and Pedagogy for Early Childhood Education I	3
ECED 445	Mathematics Content and Pedagogy for Early Childhood Education II	3
ECED 461	Integrated Day Practicum	1
ECED 480	Student Teaching ECED	8

EXED 325	Inclusive Practices in Early Childhood	3
EXED 402	Positive Behavior Supports and Individualized Interventions for Young Children (must be taken for three credit hours)	3
IS 498	Individualized Study Project (must be taken for three credit hours)	3
LED 355	Inclusive Early Literacy Development and Acquisition	3
LED 456	Inclusive Literacy Learning in the Primary Grades	3
Total Credits		43

General Education Requirements: 21 Credit Hours

Code	Title	Credits
	General Education courses	21
Total Credits		21

Education, Elementary Education (ELED) Concentration, B.I.S.

This fully online program is for students who are continuously employed as instructional assistants in schools and have completed 60 credits and/or have an associate's degree. Other relevant employment considered on a case-by-case basis. A transferable associate degree from any VCCS institution will provide the smoothest transition to the JMU BIS course work. Other associate degrees from accredited institutions are considered on a case-by-case basis. It leads to licensure in grades PreK-6 and a Bachelor of Individualized Study degree with a concentration in Elementary Education Development from JMU.

There are 60 credits that must be earned at a four-year institution and 30 of those must be earned at JMU.

Candidates should note that they must be fully admitted to teacher education prior to registering for the courses offered during Summer Semester Year 1 of the Elementary Education Program. Candidates should also note that, on occasion, Virginia Department of Education (VDOE) or other accrediting agencies may change requirements after the catalog copy is approved. Therefore, it is especially important for candidates to confer with their education advisor on a regular basis for the most current requirements. Completion of the Elementary Education concentration leads to a Virginia teaching license with an endorsement in PreK-6 Elementary Education. To be recommended for licensure, all candidates must meet the following requirements:

- Complete General Education requirements or have the VCCS waiver.
- Complete the Elementary Education pre-professional program.
- Complete the disciplinary courses required for the Elementary Education concentration.
- Meet all admission and retention criteria for teacher education including satisfactory scores on the Math Assessment, Praxis Elementary Subject Assessments (Reading & L.A. Test Code 5002, Mathematics Test Code 5003, Science Test Code 5005, and Social Studies Test Code 5004), and Praxis Teaching Reading: Elementary Test (5205) (new version of RVE).
- Meet admission requirements for the Elementary Education Program.
- Meet performance and behavior standards as indicated by ratings on the program Professional Dispositions Checklist.

All students must attend an orientation meeting at JMU the semester or summer prior to starting the program. This orientation is offered online.

Please note, the program offers free tutoring for the assessments listed above.

Candidates must be fully accepted into teacher education to begin their education program course work in the Summer Semester Year 1.

Field experience includes practicums and student teaching. Practicum courses are to extend through the full semester. Student teaching must take place in public school and is 12 weeks with the possibility of teaching in your own classroom.

Minimum Grade Requirement

Assessment occurs during each semester, and candidate performance is reviewed at the end of each semester. Candidates must demonstrate satisfactory performance before moving on to the next semester. Satisfactory performance includes a minimum grade of "C" in all education course work and an overall 2.5 GPA, demonstration of professional behaviors, and acceptable performance in practicum and on key assessments.

The Elementary Education Program prepares candidates to teach students in grades PreK-6. Drawn from research and theories in child development, teaching, and learning, the course work and field experiences prepare teacher candidates to employ a cross-disciplinary approach to instruction that reflects an understanding of the diverse nature of learners and their families.

The JMU Elementary Education Program seeks to foster in its candidates an empathic understanding of the ways that children are affected by social contexts and by the children's own strengths and needs; the knowledge and pedagogical skills to support each child's right to success; and belief in the value of each child. Our candidates are guided in:

- Critically challenging conventional wisdom and common practices to identify hidden assumptions and activities that constrain or privilege some at the expense of others.
- Learning to ask questions and developing an inquiring approach motivated by the desire to understand the world in its myriad complexities.
- Reflecting deeply on and constructing positive relationships with others.
- Expressing knowledge, skills and attitudes in ways that communicate with others and provide a forum for the creative and academic expression of the profession and the self.
- Developing an appreciation for the global connection of all humanity and our interdependence on the finite natural resources of the earth.
- Experiencing life among people whose social contexts are unlike the candidates' own to broaden and deepen respect for and sensitivity to various cultures and social contexts.
- Knowing and appreciating the process of human unfolding throughout the cycles of life from conception onward, particularly throughout the period of childhood and preadolescence.
- Valuing the literacies and literate backgrounds of all learners to develop equitable, inclusive instruction that honors diverse learners at all developmental phases.
- Reflecting critically on the ways we embody and enact a sustainable sense of self.
- Building on interdisciplinary connections among content areas to create inclusive, responsive and effective instruction for all learners.

The courses in the Elementary Education Program are organized to integrate academic content with pedagogical practice throughout the concentration. Field experiences are provided along with course work to enable candidates to apply their knowledge in a variety of settings.

Pre-professional Requirements

These courses may fulfill General Education requirements and/or university electives for this program.

Students concentrating in Elementary Education will be required to complete JMU General Education with additional courses related to Bachelor of Science requirements (Quantitative and Scientific Literacy). Given required VDOE competencies associated with PreK-6 teacher licensure, students will be required to enroll in the courses below:

All students are required to take 6 credits of math to meet VDOE competencies.

All students are required to take 6 credits of history to meet VDOE competencies.

All students are required to take 10 credits from the following list of courses:

Code	Title	Credits
6 credits of math to meet VDOE competencies		6
6 credits of history to meet VDOE competencies		6
10 credits from the following courses:		10
ISCI 171	Earth and Planetary Science for Teachers [C3NS]	
	or ANTH 196 Biological Anthropology [C3NS]	
ISCI 101	Physics, Chemistry and the Human Experience [C3PP]	
	or ISCI 172 Physical Science for Teachers [C3PP]	
ISCI 173	Life and Environmental Science for Teachers [C3L]	
ANTH 195	Cultural Anthropology [C4GE]	
	or GEOG 200 Geography: The Global Dimension [C4GE]	
HIST 225	U.S. History [C4AE]	
	or POSC 225 U.S. Government [C4AE]	
ANTH 196L	Scientific Perspectives in Biological Anthropology [C3L]	
ASTR 120	The Solar System [C3PP]	
ASTR 121	Stars, Galaxies, and Cosmology [C3PP]	
ISCI 104	Scientific Perspectives [C3L]	
PHYS 121	The Physical Nature of Light and Sound (3, 1) [C3PP]	
PHYS 140	College Physics I [C3PP]	
Total Credits		22

Professional Studies Core Requirements

The following courses are required for licensure. These courses may be taken prior to or while enrolled in the program:

Code	Title	Credits
EDUC 200	Foundations of Education in the United States	3
ELED 310	Diversity in Elementary Education	3
EXED 200	Foundations of Exceptional Education	3
PSYC 160	Life Span Human Development [C5SD]	3

WRTC 103	Rhetorical Reading and Writing [C1W]	3
Total Credits		15

ELED Concentration Requirements

Elementary Education Content Requirements: 50 Credit Hours

Code	Title	Credits
ELED 272	Introduction to Early and Elementary Education	3
ELED 321	Practicum with a Focus on Learners and Learning (must be taken for one credit hour)	1
ELED 322	Practicum with a Focus on Curriculum Integration and Guiding Behavior (must be taken for one credit hour)	1
ELED 332	Children and Science	3
ELED 334	Children and Social Studies	3
ELED 400	Elementary Education Final Practicum	1
ELED 444	Mathematics Content and Pedagogy for Elementary Education I	3
ELED 445	Mathematics Content and Pedagogy for Elementary Education II	3
ELED 450	Planning Instruction and Assessment	3
ELED 455	Home, School, Communities	3
ELED 480	Student Teaching in Elementary Education	8
ELED 485	Guiding Student Behavior: Responsive Classroom Management	3
EXED 325	Inclusive Practices in Early Childhood	3
IS 498	Individualized Study Project (must be taken for three credit hours)	3
LED 300	Reading Instruction and Assessment for Diverse Learners PK-6	3
LED 350	Writing Instruction, Assessment and Technology for Diverse Learners PK-6	3
LED 462	Disciplinary Literacy for Diverse Classrooms PK-6	3
Total Credits		50

Electives: 0-15 Credit Hours

Number of elective credit hours required is contingent on the four-year institution 60 credit hour requirement. Elective options include:

Code	Title	Credits
ELED 490	Special Studies in Elementary Education	1-3
EXED 310	Survey of Emotional Disturbance	3

Additional courses may fulfill elective credits with approval from advisor.

Education, Inclusive Early Childhood Education (IECE) Concentration, B.I.S.

This fully online program is for students who are continuously employed as instructional assistants in schools and have completed 60 credits and/or have an associate's degree. Other relevant employment considered on a case-by-case basis. A transferable associate degree from any VCCS institution will provide the smoothest transition to the JMU BIS course work. Other associate degrees from accredited institutions are considered on a case-by-case basis.

This program is a Bachelor's of Individualized Study with a concentration in Inclusive Early Childhood Education. Students completing this program are prepared to serve as teachers of students with and without

disabilities in a variety of educational placements and service delivery models including early intervention, hospital education, ECSE and public elementary settings

Completers of this program earn a degree and are eligible for a Virginia teaching license in areas birth-age 5 special education and PreK-3rd grade general education.

There are 60 credits that must be earned at a four-year institution and 30 of those must be earned at JMU.

Candidates should note that they must be fully admitted to teacher education prior to registering for the courses offered during Summer Semester Year 1 of their education program. Candidates should also note that, on occasion, Virginia Department of Education (VDOE) or other accrediting agencies may change requirements after the catalog copy is approved. Therefore, it is especially important for candidates to confer with their education advisor on a regular basis for the most current requirements. Completers of this program earn a degree and are eligible for a Virginia teaching license in areas birth-age 5 special education and PreK-3rd grade general education. To be recommended for licensure, all candidates must meet the following requirements:

- Complete General Education requirements or have the VCCS waiver.
- Complete the Inclusive Early Childhood Education pre-professional program.
- Complete the disciplinary courses required for the Inclusive Early Childhood Education concentration
- Meet all admission and retention criteria for teacher education including satisfactory scores on the Math Assessment, Praxis Elementary Subject Assessments (Reading & L.A Test Code 5002, Mathematics Test Code 5003, Science Test Code 5005, and Social Studies Test Code 5004), and Praxis Teaching Reading: Elementary Test (5205) (new version of RVE).
- Meet admission requirements for the Inclusive Early Childhood Education program.
- Meet performance and behavior standards as indicated by ratings on the program Professional Dispositions Checklist.

The Bachelor of Science In Individualized Study with a concentration in Inclusive Early Childhood Education enables students to gain knowledge and understanding of early childhood inclusion (Birth-3rd grade) and practices that promote access, participation, and support for young children and their families. Students completing the degree program, when combined with the teacher dual licensure option, are prepared to serve as teachers of students with and without disabilities in a variety of inclusive educational placements.

The licensure program is designed to prepare resilient, culturally responsive educators who are advocates for young children and youth with and without disabilities that are not only qualified for the complexity of their professional roles but also are reflective problem-solvers. This program offers a dual licensure in PreK-3rd grade and Early Childhood Special Education (birth-5 years).

Minimum Grade Requirement

Assessment occurs during each semester, and candidate performance is reviewed at the end of each semester. Candidates must demonstrate satisfactory performance before moving on to the next semester. Satisfactory performance includes a minimum grade of "C" in all education course work and an overall 2.5 GPA, demonstration of

professional behaviors, and acceptable performance in practicum and on key assessments.

The program includes extensive field experiences. Assessment of candidate performance includes evaluation of performance in individual courses and practicum, as well as other criteria. At various points throughout the program, there are formative and summative assessments where faculty committees review the overall performance of each candidate. Formative assessments will be used to advise candidates and develop a plan of action for addressing any concerns that have been identified by the faculty. Summative assessments will be used to identify those candidates who are not making satisfactory progress toward advancement to the next sequences of courses and experiences. If progress is unsatisfactory, the candidate will not be allowed to continue until any identified issues are corrected. In some cases, a candidate will be allowed to continue in the next semester, but there will be a plan of action for addressing any concerns that have been identified by the faculty.

Every student is assigned an advisor. Students should plan on regularly consulting their advisor.

It is important for students to understand that the undergraduate concentration in Inclusive Early Childhood Education does not lead to Teacher-Education licensure unless licensure requirements are completed. To qualify for a recommendation for licensure in Inclusive Early Childhood Education [Early Childhood Special Education (Birth-5 years) and general education curriculum (PreK-3rd Grade)], students must complete licensure requirements in addition to requirements for the Inclusive Early Childhood Education undergraduate concentration.

Professional Studies Core Requirements: 15 Credit Hours

The following courses are required for licensure. These courses may be taken prior to or while enrolled in the program:

Code	Title	Credits
EDUC 200	Foundations of Education in the United States	3
EDUC 310	Teaching in a Diverse Society	3
EXED 200	Foundations of Exceptional Education	3
PSYC 160	Life Span Human Development [C5SD]	3
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Total Credits		15

Inclusive Early Childhood Education Content Requirements

Code	Title	Credits
IECE 300	Issues and Trends in Inclusive Early Childhood Education	3
IECE 303	Development of Young Children Birth Through Age 8	3
IECE 321	Initial Field Experience in IECE Part 1	1
IECE 322	Curriculum and Instruction in Inclusive Early Childhood Education	3
IECE 324	Assessment of the Young Child	3
IECE 355	Foundations of Inclusive Emergent and Early Literacy for Inclusive Early Childhood Education (IECE)	3
IECE 421	Intermediate Field Experience in IECE Part 1	2

IECE 429	Teaching Early Childhood Learners with Complex Care Needs (IECE)	3
IECE 450	Contemporary Family Issues in Inclusive Education	3
IECE 456	Foundations of Literacy (PK-3) for Inclusive Early Childhood Education (IECE)	3
IECE 460	Instructional Practices in Numeracy	3
IECE 461	Advanced Field Experience in IECE Part 1	3
IECE 462	Instructional Practices in Natural Sciences for Young Children	3
IECE 464	Instructional Practices in Social Studies for Young Children	3
IECE 466	Managing Classrooms and Guiding Behavior	3
IECE 470	Individualized Behavior Intervention for Young Children	3
IECE 480	Student Teaching in Inclusive Early Childhood Education (must be taken twice for a total of 12 credits)	12
IECE 481	Seminar in Inclusive Early Childhood Education	1
IS 498	Individualized Study Project	3
Total Credits		61

Education, Special Education (SPED) Concentration, B.I.S.

This fully online program is for students who are continuously employed as instructional assistants in schools and have completed 60 credits and/or have an associate's degree. Other relevant employment considered on a case-by-case basis. A transferable associate degree from any VCCS institution will provide the smoothest transition to the JMU BIS course work. Other associate degrees from accredited institutions are considered on a case-by-case basis.

This program leads to a bachelor's degree and a teaching license in both Special Education K-12: Accessing the General Curriculum and Special Education K-12: Adapted Curriculum. Students must have a transferable associate's degree to complete the program as described. The program focuses on developing teachers who are knowledgeable about the characteristics, diagnosis and remediation of children with disabilities accessing the general education curriculum and in adapted curriculum settings. They will learn from dedicated and engaged faculty members in synchronous online courses.

Completers of this program earn a degree and are eligible for a Virginia teaching license in areas in both Special Education K-12: Accessing the General Curriculum and Special Education K-12: Adapted Curriculum.

There are 60 credits that must be earned at a four-year institution and 30 of those must be earned at JMU.

Candidates should note that they must be fully admitted to teacher education prior to registering for the courses offered during Summer Semester Year 1 of their education program. Candidates should also note that, on occasion, Virginia Department of Education (VDOE) or other accrediting agencies may change requirements after the catalog copy is approved. Therefore, it is especially important for candidates to confer with their education advisor on a regular basis for the most current requirements. Completers of this program earn a degree and are eligible for a Virginia teaching license in areas in both Special Education K-12: Accessing the General Curriculum and Special Education K-12: Adapted

Curriculum. To be recommended for licensure, all candidates must meet the following requirements:

- Complete General Education requirements or have the VCCS waiver.
- Complete the Special Education Pre-professional Program.
- Complete the disciplinary courses required for the Special Education concentration.
- Meet all admission and retention criteria for Teacher Education including satisfactory scores on the Math Assessment, and Praxis Teaching Reading: Elementary Test (5205) (new version of RVE).
- Meet admission requirements for the Special Education program.
- Meet performance and behavior standards as indicated by ratings on the program Professional Dispositions Checklist.

All students must attend an orientation meeting at JMU the semester or summer prior to starting the program. This orientation is offered online. Please note, the program offers free tutoring for the assessments listed above.

Candidates must be fully accepted into teacher education to begin their program course work in the Summer Semester Year 1.

Field experience includes practicum and student teaching. Students teach for twelve weeks in a general curriculum education classroom and twelve weeks in an adapted curriculum education classroom. One of the above placements will be in an elementary setting and the other will be in a middle/high school setting. This includes the possibility of teaching in your own classroom with the last placement.

Minimum Grade Requirement

Assessment occurs during each semester, and candidate performance is reviewed at the end of each semester. Candidates must demonstrate satisfactory performance before moving on to the next semester. Satisfactory performance includes a minimum grade of "C" in all education course work and an overall 2.5 GPA, demonstration of professional behaviors, and acceptable performance in practicum and on key assessments.

The licensure program is designed to prepare resilient, culturally-responsive educators who are advocates for children and youth with disabilities and who are not only qualified for the complexity of their professional roles but also are reflective problem solvers.

Assessment of candidate performance includes evaluation of performance in individual courses. Students are required to demonstrate satisfactory performance, which includes a minimum grade of "C" in all major and concentration coursework and a "satisfactory" score on all key assessments. If a student does not pass a key assessment, they will be given one additional attempt to retake the key assessment after remediation sessions with a designated faculty member. Key assessments and course grades will be used to identify those candidates who are not making satisfactory progress toward advancement to the next sequence of courses. The assessments will be used to advise candidates and develop a plan of action to address and support any concerns that have been identified by the faculty. In some cases, a candidate will be allowed to continue in the next semester, but there will be a plan of action for addressing any concerns that have been identified by the faculty. Every student is assigned an advisor. Students should plan on consulting their advisor regularly.

Professional Studies Core Requirements

The following courses are required for licensure. These courses may be taken prior to or while enrolled in the program:

Code	Title	Credits
EDUC 200	Foundations of Education in the United States	3
EDUC 310	Teaching in a Diverse Society	3
EXED 200	Foundations of Exceptional Education	3
PSYC 160	Life Span Human Development [C5SD]	3
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Total Credits		15

SPED Concentration Requirements

Code	Title	Credits
Required Courses		
EDUC 303	Foundations of Curriculum and Instruction	3
EXED 303	Foundations of Classroom and Behavior Management	3
EXED 341	Characteristics of Learners with Disabilities Accessing the General Curriculum	3
EXED 342	Characteristics of Learners with Low Incidence Disabilities Accessing the Adapted Curriculum	3
EXED 407	Transitions of Learners with Disabilities (must be taken for three credit hours)	3
EXED 419	Foundations and Theories of Literacy for K-12	3
EXED 474	Assessment and Evaluation in Special Education	3
EXED 475	Program Planning in Special Education	3
EXED 478	Supervised Practice in Special Education Methods (must be taken for two credit hours)	2
EXED 480	Student Teaching in Special Education (must be taken twice for a total of eight credit hours)	8
EXED 484	Inclusive Practices in Special Education (must be taken for three credit hours)	3
EXED 485	Systematic Behavioral Support and Interventions	3
IS 498	Individualized Study Project (must be taken for three credit hours)	3
Subtotal		43
Accessing the General Curriculum Courses		
EXED 405	Alternative Math Instructional Methods and Supports for Students with Disabilities	3
EXED 450	Principles of Specialized Reading Instruction	3
EXED 477	Remedial Methods in Special Education	3
Subtotal		9
Adapted Curriculum Courses		
EXED 429	Teaching Learners with Complex Care Needs	3
EXED 436	Communication Language and Literacy for Learners on the Adapted Curriculum	3
EXED 473	Methods of Instruction for Learners Accessing the Adapted Curriculum	3
Subtotal		9
Total Credits		61

Education, Early Childhood Education Development (ECED) Concentration, A.A.S. to B.I.S.

Recommended Schedule for ECED Concentration

Additional courses may be needed each semester, depending on previous courses taken/transferred.

Course	Title	Credits
First Year		
Fall Semester		
No practicum, but must have access to children for assignments in the following courses. Grades PK-1 are preferable.		
ECED 372	Introduction to Early Childhood Education	3
ECED 444	Mathematics Content and Pedagogy for Early Childhood Education I	3
LED 355	Inclusive Early Literacy Development and Acquisition	3
Credits		9
Spring Semester		
ECED 442	The Young Child	3
ECED 443	Practicum in Primary Grades	1
ECED 445	Mathematics Content and Pedagogy for Early Childhood Education II	3
LED 456	Inclusive Literacy Learning in the Primary Grades	3
Credits		10
Summer Semester		
HIST 101 or HIST 102	World History to 1500 [C2HQC] or World History since 1500 [C2HQC]	3
GEOG 200	Geography: The Global Dimension [C4GE]	3
ISCI 101 or ISCI 172	Physics, Chemistry and the Human Experience [C3PP] or Physical Science for Teachers [C3PP]	3
Credits		9
Second Year		
Fall Semester		
ECED 304	Diversity, Equity and Justice in Early Childhood Education	3
ECED 412	Natural and Social Sciences for Young Children	3
ECED 461	Integrated Day Practicum	1
EXED 325	Inclusive Practices in Early Childhood	3
Credits		10
Spring Semester		
EXED 402	Positive Behavior Supports and Individualized Interventions for Young Children (must be taken for three credit hours)	3
Choose one of the following:		3
ART 200	Art Today: Themes in Contemporary Art [C2VPA]	
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	
MUS 200	Music and Culture [C2VPA]	
Choose one of the following Critical Thinking courses:		3
BUS 160	Applied Critical Thinking in Business [C1CT]	
EDUC 102	Critical Questions in Education [C1CT]	
HIST 150	Critical Issues in Recent Global History [C1CT]	
PHIL 120	Critical Thinking [C1CT]	
PHIL 150	Ethical Reasoning [C1CT]	
SMAD 150	Mediated Communication: Issues and Skills [C1CT]	
Credits		9
Summer Semester		
HTH 100 or KIN 100	Personal Wellness [C5W] or Lifetime Fitness and Wellness (2,2) [C5W]	3
Choose one of the following:		3
SCOM 121	Fundamental Human Communication: Presentations [C1HC]	

SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC]	
SCOM 123	Fundamental Human Communication: Group Presentations [C1HC]	
Credits		6
Third Year		
Fall Semester		
ECED 480	Student Teaching ECED	8
IS 498	Individualized Study Project (must be taken for three credit hours)	3
Credits		11
Total Credits		64

Education, Elementary Education (ELED) Concentration, B.I.S.

Recommended Schedule for ELED Concentration

Additional courses may be needed each semester, depending on previous courses taken/transferred.

Course	Title	Credits
First Year		
Fall Semester		
ELED 272	Introduction to Early and Elementary Education	3
ELED 310	Diversity in Elementary Education	3
EXED 200	Foundations of Exceptional Education	3
ISCI 171	Earth and Planetary Science for Teachers [C3NS]	3
Credits		12
Spring Semester		
ELED 321	Practicum with a Focus on Learners and Learning (must be taken once for a total of one credit hour)	1
ELED 444	Mathematics Content and Pedagogy for Elementary Education I	3
ISCI 101 or ISCI 172	Physics, Chemistry and the Human Experience [C3PP] or Physical Science for Teachers [C3PP]	3
LED 300	Reading Instruction and Assessment for Diverse Learners PK-6	3
Credits		10
Summer Semester		
ELED 332	Children and Science	3
ELED 334	Children and Social Studies	3
EXED 310	Survey of Emotional Disturbance	3
Credits		9
Second Year		
Fall Semester		
ELED 322	Practicum with a Focus on Curriculum Integration and Guiding Behavior (must be taken once for a total of one credit hour)	1
ELED 445	Mathematics Content and Pedagogy for Elementary Education II	3
LED 350	Writing Instruction, Assessment and Technology for Diverse Learners PK-6	3
Credits		7
Spring Semester		
ELED 400	Elementary Education Final Practicum	1
ELED 450	Planning Instruction and Assessment	3
ELED 485	Guiding Student Behavior: Responsive Classroom Management	3
ISCI 173	Life and Environmental Science for Teachers [C3L]	3
Credits		10
Summer Semester		
EXED 325	Inclusive Practices in Early Childhood	3

IS 498	Individualized Study Project (must be taken for three credit hours)	3
LED 462	Disciplinary Literacy for Diverse Classrooms PK-6	3
Credits		9
Third Year		
Fall Semester		
ELED 455	Home, School, Communities	3
ELED 480	Student Teaching in Elementary Education	8
Credits		11
Total Credits		68

Education, Inclusive Early Childhood Education (IECE) Concentration, B.I.S.

Recommended Schedule for IECE Concentration

Additional courses may be needed each semester, depending on previous courses taken/transferred.

Course	Title	Credits
First Year		
Fall Semester		
IECE 300	Issues and Trends in Inclusive Early Childhood Education	3
IECE 303	Development of Young Children Birth Through Age 8	3
IECE 321	Initial Field Experience in IECE Part 1	1
IECE 322	Curriculum and Instruction in Inclusive Early Childhood Education	3
Credits		10
Spring Semester		
IECE 355	Foundations of Inclusive Emergent and Early Literacy for Inclusive Early Childhood Education (IECE)	3
IECE 421	Intermediate Field Experience in IECE Part 1	2
IECE 466	Managing Classrooms and Guiding Behavior	3
IECE 464	Instructional Practices in Social Studies for Young Children	3
Credits		11
Summer Semester		
IECE 429	Teaching Early Childhood Learners with Complex Care Needs (IECE)	3
IECE 462	Instructional Practices in Natural Sciences for Young Children	3
Credits		6
Second Year		
Fall Semester		
IECE 324	Assessment of the Young Child	3
IECE 456	Foundations of Literacy (PK-3) for Inclusive Early Childhood Education (IECE)	3
IECE 460	Instructional Practices in Numeracy	3
IECE 461	Advanced Field Experience in IECE Part 1	3
Credits		12
Spring Semester		
IECE 480	Student Teaching in Inclusive Early Childhood Education	6
IECE 481	Seminar in Inclusive Early Childhood Education	1
Credits		7
Summer Semester		
IECE 450	Contemporary Family Issues in Inclusive Education	3
IECE 470	Individualized Behavior Intervention for Young Children	3
Credits		6
Third Year		
Fall Semester		
IECE 480	Student Teaching in Inclusive Early Childhood Education	6

IS 498	Individualized Study Project (must be taken for three credit hours)	3
Credits		9
Total Credits		61

Education, Special Education (SPED) Concentration, B.I.S.

Recommended Schedule for SPED Concentration

Additional courses may be needed each semester, depending on previous courses taken/transferred.

Course	Title	Credits
First Year		
Fall Semester		
EDUC 303	Foundations of Curriculum and Instruction	3
EXED 341	Characteristics of Learners with Disabilities Accessing the General Curriculum	3
EXED 342	Characteristics of Learners with Low Incidence Disabilities Accessing the Adapted Curriculum	3
EXED 474	Assessment and Evaluation in Special Education	3
Credits		12
Spring Semester		
EXED 419	Foundations and Theories of Literacy for K-12	3
EXED 429	Teaching Learners with Complex Care Needs	3
EXED 484	Inclusive Practices in Special Education (must be taken for three credit hours)	3
Credits		9
Summer Semester		
EXED 405	Alternative Math Instructional Methods and Supports for Students with Disabilities	3
EXED 475	Program Planning in Special Education	3
EXED 303	Foundations of Classroom and Behavior Management	3
Credits		9
Second Year		
Fall Semester		
EXED 450	Principles of Specialized Reading Instruction	3
EXED 473	Methods of Instruction for Learners Accessing the Adapted Curriculum	3
EXED 477	Remedial Methods in Special Education	3
EXED 478	Supervised Practice in Special Education Methods (must be taken for two credit hours)	2
Credits		11
Spring Semester		
EXED 480	Student Teaching in Special Education (must be taken for four credit hours)	4
EXED 485	Systematic Behavioral Support and Interventions	3
Credits		7
Summer Semester		
EXED 407	Transitions of Learners with Disabilities (must be taken for three credit hours)	3
EXED 436	Communication Language and Literacy for Learners on the Adapted Curriculum	3
Credits		6
Third Year		
Fall Semester		
EXED 480	Student Teaching in Special Education (must be taken for four credit hours)	4
IS 498	Individualized Study Project (must be taken for three credit hours)	3
Credits		7
Total Credits		61

Individualized Study, B.S.

Phone: (540) 568- 4253

Email: adp@jmu.edu

Location: Ice House, 127 W. Bruce Street, Room 332

Adult Degree Program Website (<http://www.jmu.edu/adultdegree>)

Program Description

The Adult Degree Program/Bachelor of Individualized Study delivers a flexible, dynamic, and rigorous degree completion curriculum to meet the individual education and career goals of our students. ADP/BIS's non-residential program offers a framework for modern learners with transfer credit and/or prior learning experience to design their own academic plan.

Experiential Learning

The Adult Degree Program offers several pathways for students to participate in experiential learning opportunities. Students can choose to participate in workforce related experiences, intensive research or international programs.

Degrees Offered

Students in the ADP have the opportunity to earn a Bachelor of Individualized Study (B.I.S.) in individualized study, a Bachelor of Arts (B.A.) in individualized study or a Bachelor of Science (B.S.) in individualized study. The university degree requirements differ by degree type.

Admission

To be eligible for the ADP, applicants must be at least 22 years of age, have at least 30 transferable college credits and have a minimum cumulative GPA of 2.0 for all prior coursework.

With very limited exceptions, students in the individualized study major (the ADP's academic major) cannot transfer to another major at JMU.

Degree Requirements

The Bachelor of Science (B.S.) in individualized study requires 41 general education credits, an additional scientific literacy course (3.00 credit hours) and an additional quantitative course (3.00 credit hours).

Degree Requirements	Credit Hours
General Education courses ¹	41
Quantitative requirement (in addition to General Education courses) ²	3
Scientific Literacy requirement (in addition to General Education courses) ²	3-4
Major concentration courses and electives	73-77
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar's website.

Major Requirements

Students must complete all of the following requirements:

- IS 200
- A minimum of 120 credit hours, including 41 General Education credits
- A minimum of 30 credit hours earned from JMU
- At least 60 credit hours earned from a four-year institution
- IS 498, an online interdisciplinary research course with a capstone project coordinated by the instructor and includes a final written and oral presentation; minimum grade of "C" required

Students must also complete an experiential component of at least three credits from the options listed below:

- IS 206 (required for students pursuing credit for prior learning experience only)
- IS 300
- A study abroad course or another approved experiential course
- A practicum or student teaching course
- An additional three credits of IS 498¹

¹ If taking additional three credits of IS 498, they must be taken concurrently in the final semester, and will be an intensive research experience focusing on a single research topic.

Additional Requirements

- A program of study (POS) for meeting the requirements of degree completion must be submitted to the ADP office for approval while enrolled in the IS 200 course.
- A student's concentration, or field of study, consists of a minimum of 30 credits, at least 24 of which must be upper-division credit (300- and 400-level courses). The concentration includes classes from two or more academic disciplines and represent a coherent body of knowledge, which may require course work beyond the 30 minimum credit hours.
- No more than 30 credit hours of course work can be selected from courses taught by the College of Business. This 30 credit hour limit includes COB courses transferred from other colleges and nontraditional credit earned in the COB.
- The concentration must be completed with a GPA of 2.0 or better.
- A minimum of 15 credit hours in the concentration must be JMU credits.
- Students must participate in assessment activities used to measure student learning outcomes and modify curriculum.
- There is no residency requirement; however, program agreements that extend beyond six years may need to be amended if university policies change that affect the original program of study.

Individualized Study Concentration

This is the focus area of study in the individualized study major. The concentration is interdisciplinary and comprised of classes from two or more academic program areas to meet each student's educational and/or professional goals.

- Minimum of 30 credits.
- 24 credits must be completed in 300- or 400-level courses.
- IS 498. Individualized Study Project – online interdisciplinary research course with a capstone project, included in the 24 upper-level course requirements.

Mechanisms for Earning Credit

The number of credits, which may be accepted as transfer credit or earned through the following methods, is determined by the student's approved POS. It is reasonable to expect that the unique educational needs, background and personal circumstances of each student will determine to a great extent the manner in which credit has been and will be earned.

- Transfer credit awarded by the registrar's office.
- Credit by departmental examination.
- Credit for approved sponsored learning or independent studies.
- Regular course work.
- Credit for experiential learning through portfolio of prior learning assessment (PLA).¹
- Credit for educational programs of the Armed Forces (JST).
- College Level Examination Program (CLEP).¹
- American Council on Education College Credit Recommendation Service (ACE).¹

¹ No more than 30 non-traditional credits can be counted towards the degree requirements of 120 credit hours which includes PLA, JST, CLEP and ACE.

Prior Learning Assessment (PLA)

Prior learning assessment (PLA) is an assessment of knowledge and skills acquired prior to or outside of enrollment in higher education for the purpose of earning college-level credit. The term portfolio is used to describe a collection of evidence to support the PLA process. Through the portfolio, a student makes a case by succinctly identifying, articulating and documenting mastery of college level learning or its equivalence.

PLA is an option for a student who has gained significant learning through experiences including, but not limited to: workplace training, volunteer service, civic activities, conferences, workshops, vocational interests, travel and independent reading.

Prior learning credit depends on each student's individual experiences and may not be a viable option for every student. The process entails extensive preparation, excellent academic writing skills and the ability to meet deadlines. Before a portfolio is submitted for approval, students must complete IS 206. There is no guarantee that a portfolio submitted for credit will be approved.

College Credit Recommendation Service

ADP accepts some but not all credits as recommended in the National Guide to Educational Credit for Training Programs and the Guide to Educational Credit by Examination prepared by the American Council on Education. Additional information on program policies and procedures specific to JST is available through the ADP website.

College-Level Examination Program

Effective Fall 2024, students who participated in the College-Level Examination Program may be awarded course credit that satisfies a General Education requirement or elective. Prospective students who complete CLEP examinations and achieve an appropriate score (usually a 50 or above) are eligible to earn college credit. Evaluation of credit will be

performed in coordination with the appropriate departmental office and cannot be appealed.

JMU will accept these scores electronically from the testing agency. Please contact College Board for information on providing your exam results to JMU. More information can be found at <https://clep.collegeboard.org/scores>.

Credit/No-Credit

Students enrolled in the ADP may complete courses on a credit/no-credit basis if they have completed 28 credit hours of college level course work and have a 2.25 cumulative GPA.

Courses in the area of concentration, including the senior capstone research course, may not be taken on a credit/no-credit basis. Other provisions of the credit/no-credit option, as described in this catalog, will also be in effect.

Online Interdisciplinary Tracks

Online interdisciplinary tracks were developed in cooperation with various academic units at the university. Students can use these tracks when developing their concentration.

- Business and Information Technology
- Small Business Management and Entrepreneurship
- Human Resource Development and Management
- Professional Workplace Communication
- Healthcare Leadership
- Child Development
- Family and Consumer Sciences
- Inclusive Early Childhood Education with teacher licensure¹
- Early Childhood Education Development with teacher licensure¹
- Elementary Education with teacher licensure¹
- Special Education with teacher licensure¹

¹ Only available to students in teacher licensure programs

Individualized Study, General Studies Major, B.A.

Program Description

The Individualized Study, B.A. with major in General Studies is a self-designed major where students combine coursework from at least two thematic areas to tailor their course of study to their unique interests.

General studies provides students with a flexible plan of study that meets the student's personal, academic and professional goals. It is intended for students whose interests differ from established majors, and who want to focus their studies across courses and departments to better meet their career goals or preparation for further education.

The core curriculum provides students with common experiences where students:

- identify a combination of subjects that allow them to pursue an academic plan tailored to their individual interests.
- become familiar with methods of inquiry that inform ways of knowing in more than one traditional academic discipline.
- demonstrate growth in written and oral communication, critical thinking, and intellectual independence.
- complete a senior project that synthesizes information and skills from the student's coursework into a scholarly product.

Students must complete a minimum of 15 credit hours of major requirements at the 300-level or higher. Courses may be double-counted between the major and degree requirements or general education requirements. In this case, the student will take additional university electives to reach a total of 120 credits.

Admissions

Admission requirements for the general studies major include:

- Students must have successfully completed a minimum of 75 credit hours
- Students must meet with a General Studies advisor to discuss (and agree upon) a plan of study that aligns with the student's goals
- Students can have no other declared majors
- Approval by the IdLS academic unit head must be granted

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Foreign language courses (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses) ³	3
Major requirements	42-48
University Electives	16-34
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the

student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures' placement test.

³ Any PHIL course except PHIL 120 or PHIL 150. Consult the list of courses satisfying Bachelor of Arts degree requirements on the Registrar's website.

Major Requirements

Code	Title	Credits
Core Requirements		
IDLS 200	Interdisciplinary Scholarship	3
IDLS 300	Individualized Major Workshop	3
Senior Project		
Students must take 6 credits from the following options:		6
IDLS 486	Internship and Field Experience	
IDLS 498	Interdisciplinary Senior Project	
Interest Area Requirements		
Students must select two or three interest areas with approval from the advisor. See below for details.		
Total Credits		42-48

Interest Area Requirements

- Students must select either two or three interest areas (each interest area requires approval from the advisor).
- Students must complete a minimum of 12 credits in each interest area, 3 credits of which must be at the 300-level or above.
- Students must complete a minimum of 30 total credits in the interest areas if they select two interest areas; they must complete a minimum of 36 credits in the interest areas if they select three interest areas.
- Students selecting two interest areas must complete 18 credits in one area and 12 credits in the other or 15 credits in each of the two interest areas.
- Students selecting three interest areas must complete a minimum of 12 credits in each of the interest areas.
- Within the 30-36 credits in the interest areas, a minimum of 15 credits must be at the 300-level or above.

Individualized Study, General Studies Major, B.S.

Program Description

The Individualized Study, B.S. with major in General Studies is a self-designed major where students combine coursework from at least two thematic areas to tailor their course of study to their unique interests.

General studies provides students with a flexible plan of study that meets the student's personal, academic and professional goals. It is intended for students whose interests differ from established majors, and who want to focus their studies across courses and departments to better meet their career goals or preparation for further education.

The core curriculum provides students with common experiences where students:

- identify a combination of subjects that allow them to pursue an academic plan tailored to their individual interests.
- become familiar with methods of inquiry that inform ways of knowing in more than one traditional academic discipline.
- demonstrate growth in written and oral communication, critical thinking, and intellectual independence.
- complete a senior project that synthesizes information and skills from the student's coursework into a scholarly product.

Students must complete a minimum of 15 credit hours of major requirements at the 300-level or higher. Courses may be double-counted between the major and degree requirements or general education requirements. In this case, the student will take additional university electives to reach a total of 120 credits.

Admissions

Admission requirements for the general studies major include:

- Students must have successfully completed of a minimum of 75 credit hours
- Students must meet with a General Studies advisor to discuss (and agree upon) a plan of study that aligns with the student's goals
- Students can have no other declared majors
- Approval by the IdLS academic unit head must be granted

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Quantitative requirement (in addition to General Education courses) ²	3
Scientific Literacy requirement (in addition to General Education courses) ²	3-4
Major requirements	42-48
University electives	24-31
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² Some requirements may be fulfilled by courses within the major.

Major Requirements

Code	Title	Credits
Core Requirements		
IDLS 200	Interdisciplinary Scholarship	3
IDLS 300	Individualized Major Workshop	3
Senior Project		
Students must take 6 credits from the following options:		6
IDLS 486	Internship and Field Experience	
IDLS 498	Interdisciplinary Senior Project	
Interest Area Requirements		
Students must select two or three interest areas with approval from the advisor. See below for details.		30-36
Total Credits		42-48

Interest Area Requirements

- Students must select either two or three interest areas (each interest area requires approval from the advisor).
- Students must complete a minimum of 12 credits in each interest area, 3 credits of which must be at the 300-level or above.
- Students must complete a minimum of 30 total credits in the interest areas if they select two interest areas; they must complete a minimum of 36 credits in the interest areas if they select three interest areas.
- Students selecting two interest areas must complete 18 credits in one area and 12 credits in the other or 15 credits in each of the two interest areas.
- Students selecting three interest areas must complete a minimum of 12 credits in each of the interest areas.
- Within the 30-36 credits in the interest areas, a minimum of 15 credits must be at the 300-level or above.

Individualized Study, Independent Scholars Major, B.A.

Program Description

Individualized curricula using existing courses across participating JMU units will be constructed with faculty oversight to help students pursue their own inquiry goals. Each student’s curriculum will include:

- 12 credits of required coursework, which includes a cross-disciplinary exploration course; skill-enhancing workshops; and 6 credit senior project.
- 36 credits of course work aligned with the student’s academic goals.
- the plans for relevant field experiences, internships and/or study abroad programs

For students in the honors program, the IS senior project will fulfill the senior honors project requirement.

Admission to the Major

- Acceptance into the IS major is based on successful completion of the IDLS 200 introductory course.
- Interested students will meet with IS faculty to discuss enrolling in the IS introductory course (IDLS 200), which is taught each semester.
- Decisions about acceptance into IS will be made by the IS faculty. Successful applicants will develop a convincing inquiry proposal and feasible curriculum, that clearly outlines a topic of inquiry, and demonstrates exceptional intellectual curiosity, communication ability and independent research potential.
- Students not accepted into IS at the end of IDLS 200 will be carefully advised into other majors and can use the three IDLS credits as electives.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Philosophy course ²	3
Foreign Language	0-12
Major requirements	48
Electives	16-28
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirement.

Major Requirements

Code	Title	Credits
Core Requirements		
IDLS 200	Interdisciplinary Scholarship	3
IDLS 300	Individualized Major Workshop	3
Subtotal		6
Senior Project		
Students must take 6 credits from the following options:		6

IDLS 486	Internship and Field Experience	
IDLS 498	Interdisciplinary Senior Project	
IDLS 499A	Honors	
IDLS 499B	Honors	
IDLS 499C	Honors	
Subtotal		6
Major Electives		
Students must select 36 credits of coursework beyond general education and B.A. degree requirements. Students must work with their advisor to select a cohesive set of courses that aligns with their plan of study and academic goals.		36
Total Credits		48

Individualized Study, Independent Scholars Major, B.S.

Program Description

Individualized curricula using existing courses across participating JMU units will be constructed with faculty oversight to help students pursue their own inquiry goals. Each student’s curriculum will include:

- 12 credits of required coursework, which includes a cross-disciplinary exploration course; skill-enhancing workshops; and 6 credit senior project.
- 36 credits of course work aligned with the student’s academic goals.
- the plans for relevant field experiences, internships and/or study abroad programs

For students in the honors program, the IS senior project will fulfill the senior honors project requirement.

Admission to the Major

- Acceptance into the IS major is based on successful completion of the IDLS 200 introductory course.
- Interested students will meet with IS faculty to discuss enrolling in the IS introductory course (IDLS 200), which is taught each semester.
- Decisions about acceptance into IS will be made by the IS faculty. Successful applicants will develop a convincing inquiry proposal and feasible curriculum, that clearly outlines a topic of inquiry, and demonstrates exceptional intellectual curiosity, communication ability, and independent research potential.
- Students not accepted into IS at the end of IDLS 200 will be carefully advised into other majors and can use the three IDLS credits as electives.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement ²	3-4
Major requirements	48
Electives	23-24
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirement.

Major Requirements

Code	Title	Credits
Core Requirements		
IDLS 200	Interdisciplinary Scholarship	3
IDLS 300	Individualized Major Workshop	3
Subtotal		6
Senior Project		
Students must take 6 credits from the following options:		6

IDLS 486	Internship and Field Experience
IDLS 498	Interdisciplinary Senior Project
IDLS 499A	Honors
IDLS 499B	Honors
IDLS 499C	Honors
Subtotal	
Major Electives	
Students must select 36 credits of coursework beyond general education and B.S. degree requirements. Students must work with their advisor to select a cohesive set of courses that aligns with their plan of study and academic goals.	
Total Credits	48

Information Technology, B.S.

Program Description

The Bachelor of Science degree in Information Technology focuses on highly relevant skills in cybersecurity, computer networking, and end-user design and development. The information technology degree goes beyond the science behind computers, teaching students how to design, develop, test and maintain solutions in a wide range of computing and networking application areas. Information technology studies are integrated to provide students with the knowledge and skills to meet the computer technology needs of business, government, healthcare, education and other organizations. In addition to core competencies in programming and networking, security and privacy, and other application-focused computing fields, the information technology major features a junior-level project to address a community need. Also, students in the information technology major are required to complete a two-semester senior capstone project, allowing them to apply the range of their abilities in a real-world context. Students in the information technology major are also required to select nine credit hours of IT senior class electives from various fields such as cyber-security, web/mobile development and computer networking.

Progressing in the Major and Minimum Grades

To be admitted to and progress in the major, students must meet the following criteria:

- Achieve a minimum grade of "B-" in CS 149 to advance to IT 160 and IT 201.
 - For students transferring in CS 149, complete an assessment to determine their level of preparedness. Students may be encouraged to retake the course at JMU.
 - Students may attempt CS 149 twice.
 - Students who take CS 149 at JMU and wish to repeat the course must take their second attempt at JMU.
- Performance in IT 160 and IT 201 determines if a student is eligible to continue in the major. 50-68 total students are admitted per year (48 to start in the fall; 20 to start in the spring). The selection criteria are based on the grades a student earns in IT 160 and IT 201:
 - Students must earn a minimum grade of "C" in IT 160 and IT 201.
 - Students can repeat IT 160 and IT 201 twice, at most. The grade achieved in the repeat attempt of IT 160 and/or IT 201 will be the grade used for admission.
 - If more than 68 students receive a "C" or better in IT 160 and IT 201, then the top 68 students based upon their average grades in those courses will be able to continue. Grades in MATH 235 or MATH 231/MATH 232 will be used to break ties.
 - Students not admitted on the first attempt may reapply the next semester.
- Additional Requirements:
 - Achieve a minimum grade of "C" in IT 215 (prerequisite for an upper-level course).
 - Achieve a minimum grade of "C-" in IT 333 (prerequisite for several upper-level courses).

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement ²	3
Major Requirements	62-65
University Electives	8-11
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credit hours may be fulfilled by courses required in the major.

² In addition to General Education. Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml). The scientific literacy requirement may be met by courses required in the major.

Major Requirements

Code	Title	Credits
Core Courses		
CS 149	Introduction to Programming ¹	3
IT 160	Data Structures and Advanced Programming	3
IT 201	Computational Structures and Logic	3
IT 203	Information Security and Privacy	3
IT 212	Digital Electronics	3
IT 215	Introduction to Telecommunications and Computer Networking	3
IT 240	Database Administration	3
IT 301	Web Technologies	3
IT 302	Societal and Ethical Issues in Information Technology	3
IT 311	Operating Systems Administration	3
IT 313	Community Projects	3
IT 333	Advanced Networking for Information Technology	3
IT 340	Data Science and Machine Learning	3
IT 347	Interactive Computing Systems	3
Subtotal		42
Choose one of the following options:		4-6
Option 1:		
MATH 235	Calculus I [C3QR]	
Option 2:		
MATH 231 & MATH 232	Calculus with Functions I [C3QR] and Calculus with Functions II	
Choose one of the following:		3-4
MATH 220	Elementary Statistics [C3QR]	
MATH 229	Introduction to Applied Statistics Using R [C3QR]	
MATH 318	Introduction to Probability and Statistics	
Subtotal		7-10
Capstone		
IT 444	Capstone Project Design	1
IT 445	Capstone Project Implementation	3
Subtotal		4

Major Electives

Choose nine credit hours from the following:		9
IT 334	Computer Cyber Crime, Forensics and Auditing	
IT 341	Mobile Applications	
IT 342	Server Administration	
IT 415	Data Visualization Fundamentals and Techniques	
IT 433	Internship in Information Technology	
IT 435	Applied Cryptography	
IT 443	Cloud Computing and IoT Networks	
IT 460	Advanced Cybersecurity	
IT 461	Internetworking	
IT 462	Network Applications Development	
IT 465	Wireless Networking and Security	
Subtotal		9
Total Credits		62-65

¹ Students who cannot enroll immediately in CS 149 due to low MATH placement score can take IT 101 while they prepare for CS 149. IT 101 will count as an elective.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
CS 149	Introduction to Programming	3
MATH 231	Calculus with Functions I [C3QR] ¹	3
General Education, degree requirements or elective courses		9
Credits		15
Spring Semester		
IT 160	Data Structures and Advanced Programming	3
IT 201	Computational Structures and Logic	3
MATH 232	Calculus with Functions II ¹	3
General Education, degree requirements or elective courses		6
Credits		15
Second Year		
Fall Semester		
IT 215	Introduction to Telecommunications and Computer Networking	3
IT 203	Information Security and Privacy	3
General Education, degree requirements or elective courses		9
Credits		15
Spring Semester		
IT 212	Digital Electronics	3
IT 240	Database Administration	3
MATH 220	Elementary Statistics [C3QR] ²	3
General Education, degree requirements or elective courses		6
Credits		15
Third Year		
Fall Semester		
IT 301	Web Technologies	3
IT 302	Societal and Ethical Issues in Information Technology	3
IT 333	Advanced Networking for Information Technology	3
General Education, degree requirements or elective courses		6
Credits		15
Spring Semester		
IT 311	Operating Systems Administration	3
IT 313	Community Projects	3
IT 340	Data Science and Machine Learning	3

IT 347	Interactive Computing Systems	3
General Education, degree requirements or elective courses		3
Credits		15
Fourth Year		
Fall Semester		
IT 444	Capstone Project Design	1
General Education degree requirements or elective courses		14
Credits		15
Spring Semester		
IT 445	Capstone Project Implementation	3
General Education, degree requirements or elective courses		12
Credits		15
Total Credits		120

¹ Many students opt to take the two-course calculus sequence (MATH 231 and MATH 232), but if you have the appropriate ALEKS score, you could take MATH 235 to fulfill your calculus requirement.

² Many students opt to take Elementary Statistics (MATH 220) to fulfill our statistics requirement, but we have two other course options: MATH 229 or MATH 318.

Integrated Science and Technology, B.S.

Program Description

The Bachelor of Science degree in Integrated Science and Technology produces graduates broadly acquainted with basic science, technology and social science. All students pursue a common program through their sophomore year that provides a foundation of science and its technological applications. Studies are integrated to include mathematics, statistics, physics, chemistry, biology, computation, environmental science, modern production, energy, and the role of science and technology in society. During their junior and senior years, students pursue advanced studies in critical sectors of industry and government, including applied biotechnology, energy, environment and sustainability, industrial and manufacturing systems, public interest technology and science, and applied computing. Each student selects a concentration in these areas and pursues additional study that culminates in a capstone project. Students rely heavily upon systems thinking, analytical methods, and computation as problem-solving tools throughout the curriculum, work in teams extensively, and engage in laboratory experiences.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement ²	2
Scientific Literacy Requirement ²	3-4
Major and Concentration Requirements ²	78-79
University Electives	0-1
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² The quantitative and scientific literacy requirements of the B.S. degree are fulfilled by courses in the ISAT major.

Major Requirements

Code	Title	Credits
Issues in Science and Technology		
This sequence of four courses engages students in the practice of science, both to motivate and to provide understanding of science and technology in the context of important current social issues. Current areas from which issues are selected are living systems, the environment, modern production and energy.		
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	4
ISAT 113 & 113L	Biotechnology Issues in Science and Technology [C3NS] and Biotechnology Issues in Science and Technology Laboratory [C3L]	4
ISAT 211	Issues in Modern Production Systems	3
ISAT 212	Energy Issues in Science and Technology	3
Subtotal		14
Analytical Methods		

This sequence of five courses provides students with basic methods and tools for understanding and analyzing problems in science and technology. Subjects are taught in an integrated manner with applications as the unifying factor. Topics include calculus, elements of the physical sciences, statistics, project management, the computer, knowledge-based systems, and instrumentation and measurement.

ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	4
ISAT 152	Topics in Applied Physics in ISAT	4
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	3
ISAT 252	Programming and Problem Solving	3
ISAT 300	Applied Computing, Instrumentation and Measurement (usually taken in the third year)	3
Subtotal		17

Social Context of Technology and Science

This two-course sequence introduces the student to the broader issues encountered in science and technology problem-solving, particularly social, ethical, economic and legal issues.

ISAT 171	Governance of Sociotechnical Systems	3
ISAT 271	Technology, Science and Society	3
Subtotal		6

Holistic Problem Solving

ISAT 190	Introduction to Holistic Problem Solving in the ISAT Major	1
ISAT 290	Introduction to Systems Thinking for Complex Problems	3
ISAT 390	Holistic Problem-Solving Workshop I	3
ISAT 391	Holistic Problem-Solving Workshop II	3
Subtotal		10

Strategic Sectors in Science and Technology

Students complete instruction in strategic sectors during their junior year. Students must choose two of the sectors and complete the full sector sequence for each of the two. The strategic sectors, developed from lists of national critical technologies, represent areas of current strategic importance in the world economy. The current areas for the sectors are:

Applied Biotechnology

Applied Computing

Energy

Environment and Sustainability

Industrial and Manufacturing Systems

These are the same topic areas as for the concentrations in ISAT. Sector level classes are 300 level and concentration classes are 400 level.

Choose two of the following sector options: 13-14

Applied Biotechnology

ISAT 305 Instrumentation and Measurement in Biotechnology (0, 2)

ISAT 350 Biotechnology for the New Millennium I

ISAT 351 Biotechnology for the New Millennium II

Applied Computing

ISAT 340 Database Design, Implementation and Management

ISAT 341 Machine Learning and Data Science

Energy

ISAT 301	Instrumentation and Measurement in Energy (0,2)	
ISAT 310	Energy Fundamentals I	
ISAT 311	Role of Energy in Modern Society	
<i>Environment and Sustainability</i>		
ISAT 320	Fundamentals of Environmental Science and Technology I	
ISAT 321	Fundamentals of Environmental Science and Technology II	
<i>Industrial and Manufacturing Systems</i>		
ISAT 303	Production Systems Automation Lab	
ISAT 330	Sustainable Manufacturing Systems	
ISAT 331	Cyber-Physical Industry	
Subtotal		13-14
Senior Capstone Project		
ISAT 490	Capstone Experience I	1
ISAT 491	Senior Capstone Project I	1
ISAT 492	Senior Capstone Project II	2
ISAT 493	Senior Capstone Project III	2
Subtotal		6
Concentration Requirements		
Students may focus their program of study by taking four additional courses in a particular area of concentration (see below for requirements). This area should align with one of the strategic sectors completed in the junior year.		12
Applied Biotechnology		
Applied Computing		
Energy		
Environment and Sustainability		
Industrial and Manufacturing Systems		
Public Interest Technology and Science		
Tailored Concentration ¹		
Subtotal		12
Total Credits		78-79

¹ Students also have the option to tailor their area of concentration with the help of their advisor and the approval of the ISAT program director.

Concentrations

The courses listed below may be taken to fulfill concentration requirements. If you wish to substitute other courses, you must obtain concentration advisor approval. Four courses (twelve credit hours) are required in addition to your senior project for a single concentration.

Applied Biotechnology Concentration

Code	Title	Credits
Choose 12 credit hours from the following:		
ISAT 451	Biotechnology in Industry and Agriculture	
ISAT 452	Medical Biotechnology	
ISAT 454	Computer Applications in Biotechnology	
ISAT 455	Regulatory Issues Biotechnology	
ISAT 456	Ethical, Legal and Social Implications of Biotechnology (Fall and Spring)	

ISAT 459	Awareness and Understanding of Chemical, Biological and Radiological Weapons of Mass Destruction	
ISAT 485	Gender Issues in Science	
Total Credits		12

Applied Computing Concentration

Code	Title	Credits
Choose 12 credit hours from the following:		
ISAT 440	Cultural, Ethical, Legal and Social Issues in Computing	
ISAT 441	Simulation and Modeling	
ISAT 445	Software Engineering and Agile Development	
ISAT 447	Interaction Design	
ISAT 449	Emerging Topics in Applied Data Science	
Total Credits		12

Energy Concentration

Code	Title	Credits
Required Energy Concentration Courses		
ISAT 411	Energy Economics and Policy (Fall)	3
ISAT 413	Options for Energy Efficiency (Fall)	3
Subtotal		6
Energy Concentration Electives		
Choose two of the following:		
ISAT 410	Sustainable Energy Development (Spring)	
ISAT 414	Energy Fundamentals II (Spring)	
ISAT 416	International Energy Studies (summer study abroad only)	
Subtotal		6
Total Credits		12

Environment and Sustainability Concentration

Code	Title	Credits
Choose 12 credit hours from the following:		
ISAT 420	Environmental Analysis and Modeling	
ISAT 421	Environmental Policy and Regulation	
ISAT 422	Industrial Environmental Management	
ISAT 423	Environmental Remediation	
ISAT 424	Natural Resource Management	
ISAT 425	Environmental Hydrology	
ISAT 426	Environmental Information Systems ¹	
ISAT 427	Industrial Hygiene	
ISAT 428	Industrial Ecology	
ISAT 429	Sustainability: An Ecological Perspective	
ISAT 473	Local Agriculture and Farm Internships	
ISAT 474	Political Ecology	
Total Credits		12

¹ EIS minors only

Industrial and Manufacturing Systems Concentration

Code	Title	Credits
Choose 12 credit hours from the following:		12
ISAT 430	Materials and Manufacturing Processes	
ISAT 431	Advanced Manufacturing Processes and Systems	
ISAT 432	Materials Selection in Product Design	
ISAT 433	Foundations of Lean and Six Sigma	
ISAT 434	Industrial Informatics	
ISAT 435	Lean Six Sigma Applications	
ISAT 437	Technological Innovation and Sustainable Design	
Total Credits		12

Public Interest Technology and Science Concentration

Code	Title	Credits
Choose 12 credit hours from the following:		12
ISAT 411	Energy Economics and Policy	
ISAT 421	Environmental Policy and Regulation	
ISAT 440	Cultural, Ethical, Legal and Social Issues in Computing	
ISAT 455	Regulatory Issues Biotechnology	
ISAT 456	Ethical, Legal and Social Implications of Biotechnology	
ISAT 485	Gender Issues in Science	
ISAT 483	Topics in Science, Technology and Society	
ISAT 487	Science, Technology and Society Futures Laboratory	
Total Credits		12

Tailored Concentration

Code	Title	Credits
Students have the option to tailor their area of concentration with the help of their advisor and the approval of the ISAT program director.		12
Total Credits		12

ISAT Major Potential Course Substitutions

Under certain circumstances, students may substitute some required ISAT courses:

- Students who have come to the ISAT major from other majors or institutions may substitute some required ISAT courses if they have already taken equivalent courses. Once a student declares the ISAT major, they should complete the ISAT courses as required in the program above.
- Students who are double-majoring in a major that requires specific courses that overlap in content with required ISAT courses may request course substitutions.
- Students majoring in ISAT preparing for higher education in health careers (dentistry, optometry, medicine, pharmacy, physical therapy, veterinary) may substitute some required ISAT courses if they take equivalent courses required by the pre-professional health programs.

Contact Paul Henriksen, Coordinator for Students, for more information.

ISAT Courses	Potential Substitutions
ISAT 113/ISAT 113L	BIO 140/BIO 140L or BIO 240/BIO 240L
ISAT 112	CHEM 131

ISAT 152	(PHYS 140, PHYS 140L and PHYS 150) or (PHYS 240, (PHYS 140L or PHYS 240L) and PHYS 250)
ISAT 151	MATH 205, (MATH 231 and MATH 232) or MATH 235
ISAT 251	COB 191, MATH 220 or MATH 318

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
ISAT 113 & 113L	Biotechnology Issues in Science and Technology [C3NS] and Biotechnology Issues in Science and Technology Laboratory [C3L]	4
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	4
ISAT 190	Introduction to Holistic Problem Solving in the ISAT Major	1
General Education / University Elective(s)		6
Credits		15
Spring Semester		
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	4
ISAT 152	Topics in Applied Physics in ISAT	4
ISAT 171	Governance of Sociotechnical Systems	3
General Education / University Elective(s)		4
Credits		15
Second Year		
Fall Semester		
ISAT 212	Energy Issues in Science and Technology	3
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	3
ISAT 271	Technology, Science and Society	3
General Education / University Elective(s)		6
Credits		15
Spring Semester		
ISAT 211	Issues in Modern Production Systems	3
ISAT 252	Programming and Problem Solving	3
ISAT 290	Introduction to Systems Thinking for Complex Problems	3
General Education / University Elective(s)		6
Credits		15
Third Year		
Fall Semester		
ISAT Strategic Sector I		3
ISAT Strategic Sector I Lab		1
ISAT Strategic Sector II		3
ISAT 300	Applied Computing, Instrumentation and Measurement	3
ISAT 390	Holistic Problem-Solving Workshop I	3
ISAT 490	Capstone Experience I	1
Credits		14
Spring Semester		
ISAT Strategic Sector I		3
ISAT Strategic Sector II		3
ISAT Strategic Sector II Lab		1
ISAT 391	Holistic Problem-Solving Workshop II	3
ISAT 491	Senior Capstone Project I	1
General Education / University Elective(s)		4
Credits		15
Fourth Year		
Fall Semester		
Students take the third senior project class and on the average two concentration classes.		

ISAT 492	Senior Capstone Project II	2
ISAT Concentration I		3
ISAT Concentration II		3
General Education / University Elective(s)		8
Credits		16

Spring Semester

Students take the last senior project class and the rest of their four concentration classes.

ISAT 493	Senior Capstone Project III	2
ISAT Concentration III		3
ISAT Concentration IV		3
General Education / University Elective(s)		7
Credits		15
Total Credits		120

Intelligence Analysis, B.S.

Program Description

The Bachelor of Science degree in Intelligence Analysis provides a multi-disciplinary education for students who seek careers as analysts, with a specialization in intelligence analysis.

The degree integrates knowledge from a variety of academic disciplines (philosophy, history, political science, technology, business) and combines that with professionally oriented knowledge and skills.

Students learn innovative ways to structure their thinking to assess complex real-world problems, along with how technology can be employed to acquire data, evaluate that data, and communicate it effectively to others.

Admission Requirements

Currently Enrolled Students

The IA program will strive to accommodate all interested students who adopt IA upon admission to JMU. The size of the cohort could be limited beyond fall semester of sophomore year.

To be eligible to continue to progress in the major, students must complete the following courses with a minimum grade of "C":

Code	Title	Credits
IA 150	Introduction to Intelligence Analysis	3
IA 200	Introduction to National Security Intelligence	3
IA 240	Technology Applications in a Networked World	3
IA 261	Hypothesis Testing	3

If more than 58 students receive a "C" or better in IA 200, IA 240 and IA 261, then the top 58 students based upon their average grades in those courses will be able to continue.

Transfer Students

Transfer students must also take the four required courses during their first fall semester, as these courses are not offered in the spring. It is recommended that transfer students talk with an IA advisor regarding the selection of a minor that can be useful for the major if they begin during spring semester.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement ²	3
Major requirements	60
Subject Matter Specialty	18
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Credits may be fulfilled with courses taken in the major.

Major Requirements

Code	Title	Credits
IA Foundations Courses		
IA 150	Introduction to Intelligence Analysis	3
IA 175	Introduction to Intelligence Analysis Domains	3
IA 200	Introduction to National Security Intelligence	3
IA 250	Research Design and Writing for Intelligence Analysis	3
IA 300	Seminar on Issues in Intelligence Analysis	3
IA 405	Ethics, Law and Intelligence Analysis	3
IA 440	Intelligence Analysis Capstone Project Preparation	3
IA 450	Capstone Project in Intelligence Analysis	3
Subtotal		24
Technology Methods Courses		
IA 240	Technology Applications in a Networked World	3
IA 241	Introduction to Programming and Data Science	3
IA 340	Data Mining, Modeling and Knowledge Discovery	3
IA 342	Visualization Methods, Technologies and Tools for Intelligence Analysis	3
Subtotal		12
Reasoning Methods Courses		
IA 261	Hypothesis Testing	3
IA 312	Causal Analysis	3
IA 313	Counterfactual Reasoning	3
IA 314	Strategy Assessment (Rational Decision Theory)	3
Subtotal		12
Statistics Course		
Choose one of the following: ¹		3
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	
MATH 220	Elementary Statistics [C3QR]	
Subtotal		3
IA Elective Courses		
Students must take three of the following IA elective courses:		9
IA 341	System Dynamics Modeling, Simulation and Analysis	
IA 343	Introduction to Artificial Intelligence and Machine Learning for Intelligence Analysis	
IA 360	Terrorism and Counterterrorism	
IA 363	Apocalypticism, Religious Terrorism and Peace	
IA 365	Cyber Intelligence	
IA 370	Advanced Competitive Intelligence	
IA 371	Foundations of Anti-Money Laundering	
IA 375	Political Geography	
IA 459	Awareness and Understanding of Chemical, Biological and Radiological Weapons of Mass Destruction	
IA 460	All Hazards Response and Management Systems	
IA 480	Selected Topics in Intelligence Analysis ²	
IA 495	Internship in Intelligence Analysis	
Subtotal		9
Subject Matter Specialty		

Students must complete a minor or a second major in another field (at least 18 credit hours). With the approval of an IA advisor, students may, in special cases, design their own custom concentration of six courses that fit together thematically and support the educational objectives of the IA program.

Total Credits **78**

¹ If a student has an ALEKS score <46, they should enroll in MATH 105 in their first year. Students who place into MATH 220 may take it in their sophomore year at JMU. Alternatively, they may take ISAT 251 during their sophomore year.

² IA 480 may be taken three times.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
IA 150	Introduction to Intelligence Analysis	3
IA 175	Introduction to Intelligence Analysis Domains	3
IA 200	Introduction to National Security Intelligence	3
General Education requirement		18
Courses for minor or second major		3
Credits		30
Second Year		
IA 240	Technology Applications in a Networked World	3
IA 241	Introduction to Programming and Data Science	3
IA 250	Research Design and Writing for Intelligence Analysis	3
IA 261	Hypothesis Testing	3
IA 312	Causal Analysis	3
Courses for minor or second major		6
General Education courses, statistics requirement, and/or B.S. degree requirements		9
Credits		30
Third Year		
IA 300	Seminar on Issues in Intelligence Analysis	3
IA 313	Counterfactual Reasoning	3
IA 314	Strategy Assessment (Rational Decision Theory)	3
IA 340	Data Mining, Modeling and Knowledge Discovery	3
IA 342	Visualization Methods, Technologies and Tools for Intelligence Analysis	3
IA 405	Ethics, Law and Intelligence Analysis	3
IA electives		3
General Education		6
Courses for minor or second major		3
Credits		30
Fourth Year		
IA 440	Intelligence Analysis Capstone Project Preparation	3
IA 450	Capstone Project in Intelligence Analysis	3
IA electives		6
Courses for minor or second major		6
Courses for second major or electives		12
Credits		30
Total Credits		120

International Affairs, B.A.

Coordinator: Keith Grant
Location: Miller Hall, Room 2167
Email: grantka@jmu.edu
Phone: (540) 568-4336

Program Description

The major in international affairs provides an interdisciplinary understanding of foreign cultures and societies, the dynamics of world politics, and other nations' world views and their consequent actions. A liberal arts program (B.A. degree), the international affairs major combines interdisciplinary, intercultural and multilingual education. The courses for these concentrations stem from a broad variety of traditional liberal studies disciplines. The major also allows students to incorporate a minor in African, African American, and Diaspora (AAAD) Studies (p. 446); Asian Studies (p. 452); Latin American, Latinx, and Caribbean Studies (p. 518); Middle Eastern Communities and Migrations (p. 530); or Russian, Eurasian and Slavic Studies (p. 561). The international affairs major facilitates (but does not require) participation in Study Abroad programs. Because of limited course offerings, certain specializations may require language training from another university.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses) ³	3
Major requirements	49-50
University electives	14-27
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary.

² The international affairs major requires the equivalent of three years of language study. Students should complete two 300 or 400 level foreign language courses with instruction in that language. Courses counting towards the Language Requirement may not be used elsewhere within the major. Students majoring in international affairs who use proficiency in a native language to meet the INTA language requirement may have this requirement waived through assessment by the Department of World Languages and Cultures. Students should contact the Department of World Languages and Cultures to arrange the assessment and request the assessment be sent to Dr. Grant, the INTA coordinator. Each student's foreign language must be appropriate for their track. French is acceptable for the Africa and Middle East tracks.

³ If students have taken a PHIL course to meet their General Education requirements, they must take an additional course to meet the B.A. degree philosophy requirement. PHIL 120 and PHIL 150 cannot be used to fulfill the B.A. philosophy requirement.

Major Requirements

International affairs is a 49-credit hour major, with a 31-credit common core and 18 credits of concentration study. Students choose one of four concentrations: foreign policy and global governance; global human development; international security and conflict; or comparative study, with tracks on Africa, Asia, Europe, Latin America or the Middle East. The international affairs major requires that all students complete a common core. Students should also be aware that required courses may not be offered every semester; therefore, careful planning needs to occur. Key requirements should not be left until just before graduation. Any course substitutions must be approved by the international affairs coordinator in consultation with other faculty members.

Students may use three credits of INTA 301 for international/cross-area or area credit. INTA 301 will always count as a political science course. The precise way INTA 301 will count toward the major will be determined in consultation with the Washington Semester advisor and the international affairs coordinator.

Code	Title	Credits
Core Courses		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
ECON 270	International Economics ¹	3
INTA 295	CrossNational Research Skills ²	4
INTA 489	Senior Seminar in International Affairs ³	3-4
POSC 230	International Relations	3
POSC 240	Comparative Politics	3
POSC 370	U.S. Foreign Policy	3
Two 300 or 400 level foreign language courses with instruction in that language ⁴		6
Subtotal		31-32
Concentrations		
Complete one of the following concentrations:		18
Foreign Policy and Global Governance		
Global Human Development		
International Conflict and Security		
Comparative Study (Africa, Asia, Europe, Latin America or the Middle East)		
Subtotal		18
Total Credits		49-50

¹ Students should complete these core economics requirements before enrolling in upper-level economics courses. Students double-majoring in economics and international affairs may substitute ECON 370 for ECON 270 to meet their core requirement.

² Completion of the General Education: The Natural World: Quantitative Reasoning and either POSC 230 or POSC 240 are prerequisites for INTA 295. POSC 295 may substitute for INTA 295 with approval by the INTA coordinator.

³ Students must complete all of the INTA Core Courses before registering for INTA 489.

⁴ The international affairs major requires the equivalent of three years of language study. Students should complete two 300 or 400 level foreign language courses with instruction in that language. Courses counting towards the Language Requirement may not be used elsewhere within the major. Students majoring in international affairs who use proficiency in a native language to meet the INTA language

requirement may have this requirement waived through assessment by the Department of World Languages and Cultures. Students should contact the Department of World Languages and Cultures to arrange the assessment and request the assessment be sent to Dr. Grant, the INTA coordinator. Each student's foreign language must be appropriate for their track. French is acceptable for the Africa and Middle East tracks.

Concentrations

Foreign Policy and Global Governance Concentration

Code	Title	Credits
Political Dynamics of Foreign Policy and Global Governance		
Choose one of the following:		3
JUST 372	Ethics and International Politics	
POSC 372	Ethics and International Politics	
POSC 395	International Law	
POSC 396	International Organizations	
POSC 458	International Political Analysis	
Subtotal		3
Historical Patterns		
Choose one of the following:		3
HIST 263	Africa	
HIST 274	Modern East Asia	
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	
HIST 309	French History since 1648	
HIST 312	Japan since 1600	
HIST 318	Our Better Angels?: The World History of Violence	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 332	History of Twentieth-Century Spain	
HIST 334	A Global History of Nationalism	
HIST 344	Magic Mountains, Man-eating Beasts and Unquiet Woods: An Environmental History of British India	
HIST 348	Maps and Mobility in Islamic World History	
HIST 351	War in Afghanistan: Local and Global Perspectives	
HIST 353	The Global Middle East	
HIST 358	Modern Africa, ca. 1800-1990s	
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	
HIST 373	History of the Ottoman Empire, 1453-1922	
HIST 377	History of Korea	
HIST 378	China in the Modern World	
HIST 385	The Russian Empire	
HIST 386	The Soviet Union: A Socialist Experiment	
HIST 388	Germany since 1871	
HIST 398	European Military History	
HIST 436	Afro-Latin America	
HIST 437	Latin America and Latin Americans through Film: Focus on the Twentieth Century	
HIST 444	Revolution and Social Change in Latin America	
HIST 447	South America	
HIST 448	Gender in Latin America and the Iberian World	
HIST 449	Women and Fascism	

HIST 462	Human Trafficking in African History
HIST 463	Pandemics and Epidemics in African History
HIST 469	A History of International Development in the Twentieth Century
HIST 470	Modern Africa
HIST 472	Military, Militarism and Warfare in African History
HIST 473	Early Modern Islamic Empires
HIST 474	Stalinism: Approaches, Experiences and Legacies
HIST 478	Eastern Europe
HIST 485	Colonial Knowledge Production in the Middle East and South Asia
HIST 486	Europe Since 1914
HUM 252	Global Cultures [C2HQC]
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]
Subtotal	3
Regional and Cross Regional Sociopolitical Dynamics	
Choose one of the following:	
ANTH 195	Cultural Anthropology [C4GE]
ECON 312	Comparative Economic Systems
GEOG 280	Human Geography: The Cultural Landscape
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet
GEOG 325	Environmental Ethics
GEOG 380	Cultural Geography
HIST 334	A Global History of Nationalism
POSC 200	Global Politics [C4GE]
POSC 340	Politics of Development
POSC 341	Social Movements in the U.S. and Abroad
POSC 344	Politics of the European Union
POSC 345	Politics of Western Europe
POSC 346	Politics of Central and Eastern Europe
POSC 350	Latin American Politics
POSC 353	African Politics
POSC 354	Politics of the Middle East
POSC 355	East Asian Politics
POSC 356	Politics of China
POSC 376	Introduction to Black Politics
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation
POSC 383	Women and Politics in Comparative Perspective
REL 315	Women and Religion
REL 323	Comparative Ethics East and West
REL 373	Race and Religion
REL 383	The Global Refugee Crisis and Religion
SOCI 336	Race and Ethnicity
WGSS 383	Women and Politics in Comparative Perspective
Subtotal	3

Depth Component

Choose three courses from the following, from at least two disciplines. At least one course must be in a discipline other than history or political science. Cross-listed courses may count as either field of study, but only count once within the major. 9

Communication:	
SCOM 248	
Economics:	
ECON 365	Economic Development
ECON 370	International Trade and Trade Policies
ECON 372	International Finance and Payments
or FIN 372	International Finance and Payments
Geography:	
GEOG 325	Environmental Ethics
GEOG 344	Globalization and Development
GEOG 374	Population Geography
GEOG 375	Political Geography
History:	
HIST 469	A History of International Development in the Twentieth Century
Justice Studies:	
JUST 331	Human Rights in Theory and Practice
JUST 353	Justice and Development
JUST 356	Refugees and Humanitarian Response
JUST 357	Environmental Justice
JUST 372	Ethics and International Politics
JUST 375	Genocide in the 20th Century
JUST 376	Global Social Movements
Political Science:	
POSC 331	Human Rights in Theory and Practice
POSC 340	Politics of Development
POSC 363	Hacking for Diplomacy
POSC 372	Ethics and International Politics
POSC 379	International Laws of War
POSC 383	Women and Politics in Comparative Perspective
POSC 389	Hacking for National Security and Defense
POSC 393	Human Security
POSC 394	International Conflict
POSC 395	International Law
POSC 396	International Organizations
POSC 397	The Politics of International Economic Relations
POSC 430	International Security and Conflict Management
POSC 435	International Terrorism
POSC 440	Post-Conflict Societies
POSC 458	International Political Analysis
Religion:	
HUM 252	Global Cultures [C2HQC] (When topic is Gandhi, Nonviolence and Global Transformation)
REL 385	Religion and the Law of War
Women's, Gender and Sexuality Studies:	
WGSS 383	Women and Politics in Comparative Perspective
Subtotal	9
Total Credits	18

Global Human Development Concentration

Code	Title	Credits
Political Dynamics of Global Human Development		
Choose one of the following:		3

POSC 340	Politics of Development
POSC 393	Human Security
POSC 397	The Politics of International Economic Relations
Subtotal	3
Historical Patterns	
Choose one of the following:	3
HIST 263	Africa
HIST 274	Modern East Asia
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas
HIST 309	French History since 1648
HIST 312	Japan since 1600
HIST 318	Our Better Angels?: The World History of Violence
HIST 321	Gender and Sexuality in Modern Europe
HIST 332	History of Twentieth-Century Spain
HIST 334	A Global History of Nationalism
HIST 344	Magic Mountains, Man-eating Beasts and Unquiet Woods: An Environmental History of British India
HIST 348	Maps and Mobility in Islamic World History
HIST 351	War in Afghanistan: Local and Global Perspectives
HIST 353	The Global Middle East
HIST 358	Modern Africa, ca. 1800-1990s
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States
HIST 373	History of the Ottoman Empire, 1453-1922
HIST 377	History of Korea
HIST 378	China in the Modern World
HIST 385	The Russian Empire
HIST 386	The Soviet Union: A Socialist Experiment
HIST 388	Germany since 1871
HIST 398	European Military History
HIST 436	Afro-Latin America
HIST 437	Latin America and Latin Americans through Film: Focus on the Twentieth Century
HIST 444	Revolution and Social Change in Latin America
HIST 447	South America
HIST 448	Gender in Latin America and the Iberian World
HIST 449	Women and Fascism
HIST 462	Human Trafficking in African History
HIST 463	Pandemics and Epidemics in African History
HIST 469	A History of International Development in the Twentieth Century
HIST 470	Modern Africa
HIST 472	Military, Militarism and Warfare in African History
HIST 473	Early Modern Islamic Empires
HIST 474	Stalinism: Approaches, Experiences and Legacies
HIST 478	Eastern Europe
HIST 485	Colonial Knowledge Production in the Middle East and South Asia
HIST 486	Europe Since 1914
HUM 252	Global Cultures [C2HQC]
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]

Subtotal	3
Regional and Cross Regional Sociopolitical Dynamics	
Choose one of the following:	3
ANTH 195	Cultural Anthropology [C4GE]
ECON 312	Comparative Economic Systems
GEOG 280	Human Geography: The Cultural Landscape
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet
GEOG 325	Environmental Ethics
GEOG 380	Cultural Geography
POSC 200	Global Politics [C4GE]
HIST 334	A Global History of Nationalism
POSC 340	Politics of Development
POSC 341	Social Movements in the U.S. and Abroad
POSC 344	Politics of the European Union
POSC 345	Politics of Western Europe
POSC 346	Politics of Central and Eastern Europe
POSC 350	Latin American Politics
POSC 353	African Politics
POSC 354	Politics of the Middle East
POSC 355	East Asian Politics
POSC 356	Politics of China
POSC 371	Topics in Comparative Politics
POSC 376	Introduction to Black Politics
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation
POSC 383	Women and Politics in Comparative Perspective
REL 315	Women and Religion
REL 323	Comparative Ethics East and West
REL 373	Race and Religion
REL 383	The Global Refugee Crisis and Religion
SOCI 336	Race and Ethnicity
WGSS 383	Women and Politics in Comparative Perspective
Subtotal	3
Depth Component	
Choose three courses from the following, from at least two disciplines. At least one course must be in a discipline other than history or political science.	9
Communication:	
SCOM 248	Intercultural Communication
Economics:	
ECON 365	Economic Development
ECON 370	International Trade and Trade Policies
ECON 372	International Finance and Payments
FIN 372	International Finance and Payments
Geography:	
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet
GEOG 325	Environmental Ethics
GEOG 330	Weather, Climate and Society
GEOG 344	Globalization and Development
GEOG 374	Population Geography
History:	

HIST 318	Our Better Angels?: The World History of Violence
HIST 321	Gender and Sexuality in Modern Europe
HIST 462	Human Trafficking in African History
HIST 469	A History of International Development in the Twentieth Century
Justice Studies:	
JUST 331	Human Rights in Theory and Practice
JUST 354	Native Justice and Resistance
JUST 355	Solutions to Global Poverty
JUST 356	Refugees and Humanitarian Response
JUST 357	Environmental Justice
JUST 372	Ethics and International Politics
JUST 376	Global Social Movements
Political Science:	
POSC 331	Human Rights in Theory and Practice
POSC 340	Politics of Development
POSC 341	Social Movements in the U.S. and Abroad
POSC 363	Hacking for Diplomacy
POSC 372	Ethics and International Politics
POSC 376	Introduction to Black Politics
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation
POSC 383	Women and Politics in Comparative Perspective
POSC 389	Hacking for National Security and Defense
POSC 393	Human Security
POSC 395	International Law
POSC 396	International Organizations
POSC 397	The Politics of International Economic Relations
POSC 440	Post-Conflict Societies
Religion:	
HUM 252	Global Cultures [C2HQC] (When topic is Gandhi, Nonviolence and Global Transformation)
REL 314	Hope or Terror? Gandhi and the Other 9/11
REL 315	Women and Religion
REL 373	Race and Religion
REL 383	The Global Refugee Crisis and Religion
REL 385	Religion and the Law of War
Theatre:	
THEA 303	Topics in Theatre (when topic is appropriate)
Subtotal	9
Total Credits	18

International Conflict and Security Concentration

Code	Title	Credits
Political Dynamics of International Conflict and Security		
Choose one of the following:		3
JUST 372	Ethics and International Politics	
POSC 372	Ethics and International Politics	
POSC 379	International Laws of War	
POSC 394	International Conflict	
POSC 430	International Security and Conflict Management	
Subtotal		3
Historical Patterns		

Choose one of the following:	3	GEOG 280	Human Geography: The Cultural Landscape
HIST 263	Africa	GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet
HIST 274	Modern East Asia	GEOG 325	Environmental Ethics
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	GEOG 380	Cultural Geography
HIST 309	French History since 1648	HIST 334	A Global History of Nationalism
HIST 312	Japan since 1600	POSC 200	Global Politics [C4GE]
HIST 318	Our Better Angels?: The World History of Violence	POSC 340	Politics of Development
HIST 321	Gender and Sexuality in Modern Europe	POSC 341	Social Movements in the U.S. and Abroad
HIST 332	History of Twentieth-Century Spain	POSC 344	Politics of the European Union
HIST 334	A Global History of Nationalism	POSC 345	Politics of Western Europe
HIST 344	Magic Mountains, Man-eating Beasts and Unquiet Woods: An Environmental History of British India	POSC 346	Politics of Central and Eastern Europe
HIST 348	Maps and Mobility in Islamic World History	POSC 350	Latin American Politics
HIST 351	War in Afghanistan: Local and Global Perspectives	POSC 353	African Politics
HIST 353	The Global Middle East	POSC 354	Politics of the Middle East
HIST 358	Modern Africa, ca. 1800-1990s	POSC 355	East Asian Politics
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	POSC 356	Politics of China
HIST 373	History of the Ottoman Empire, 1453-1922	POSC 371	Topics in Comparative Politics
HIST 377	History of Korea	POSC 376	Introduction to Black Politics
HIST 378	China in the Modern World	POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation
HIST 385	The Russian Empire	POSC 383	Women and Politics in Comparative Perspective
HIST 386	The Soviet Union: A Socialist Experiment	REL 315	Women and Religion
HIST 388	Germany since 1871	REL 323	Comparative Ethics East and West
HIST 398	European Military History	REL 373	Race and Religion
HIST 436	Afro-Latin America	REL 383	The Global Refugee Crisis and Religion
HIST 437	Latin America and Latin Americans through Film: Focus on the Twentieth Century	SOCI 336	Race and Ethnicity
HIST 444	Revolution and Social Change in Latin America	WGSS 383	Women and Politics in Comparative Perspective
HIST 447	South America	Subtotal	3
HIST 448	Gender in Latin America and the Iberian World	Depth Component	
HIST 449	Women and Fascism	Choose three courses from the following, from at least two disciplines. At least one course must be in a discipline other than history or political science.	9
HIST 462	Human Trafficking in African History	Anthropology:	
HIST 463	Pandemics and Epidemics in African History	ANTH 366	Anthropology of War
HIST 469	A History of International Development in the Twentieth Century	Communication:	
HIST 470	Modern Africa	SCOM 248	Intercultural Communication
HIST 472	Military, Militarism and Warfare in African History	SCOM 331	Communication and Conflict
HIST 473	Early Modern Islamic Empires	Economics:	
HIST 474	Stalinism: Approaches, Experiences and Legacies	ECON 327	Game Theory
HIST 478	Eastern Europe	History:	
HIST 485	Colonial Knowledge Production in the Middle East and South Asia	HIST 318	Our Better Angels?: The World History of Violence
HIST 486	Europe Since 1914	HIST 351	War in Afghanistan: Local and Global Perspectives
HUM 252	Global Cultures [C2HQC]	HIST 397	War and Society in U.S. History
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]	HIST 398	European Military History
Subtotal	3	Justice Studies:	
Regional and Cross Regional Sociopolitical Dynamics		JUST 331	Human Rights in Theory and Practice
Choose one of the following:	3	JUST 356	Refugees and Humanitarian Response
ANTH 195	Cultural Anthropology [C4GE]	JUST 372	Ethics and International Politics
ECON 312	Comparative Economic Systems	JUST 375	Genocide in the 20th Century
		Political Science:	
		POSC 363	Hacking for Diplomacy
		POSC 372	Ethics and International Politics

POSC 379	International Laws of War	
POSC 389	Hacking for National Security and Defense	
POSC 393	Human Security	
POSC 394	International Conflict	
POSC 395	International Law	
POSC 396	International Organizations	
POSC 430	International Security and Conflict Management	
POSC 435	International Terrorism	
POSC 440	Post-Conflict Societies	
POSC 458	International Political Analysis	
Religion:		
REL 363	Apocalypticism, Religious Terrorism and Peace	
REL 385	Religion and the Law of War	
Subtotal		9
Total Credits		18

Comparative Study Concentration

The following international/cross-area courses and area courses are required in addition to the core courses listed under Major Requirements. Students in the comparative study concentration must complete the requirements for all three areas.

International and Cross-Area Courses

Each course must come from a different field of study. The potential fields are listed below.

Complete two courses: 6 Credit Hours.

No course taken for this requirement may be used to fulfill a requirement elsewhere in the major.

Code	Title	Credits
Cross-Cultural		
ANTH 195	Cultural Anthropology [C4GE]	3
SCOM 248	Intercultural Communication	3
SOCI 336	Race and Ethnicity	3
Economics		
ECON 312	Comparative Economic Systems	3
ECON 365	Economic Development	3
ECON 372	International Finance and Payments	3
Geography		
GEOG 280	Human Geography: The Cultural Landscape	3
GEOG 325	Environmental Ethics	3
GEOG 344	Globalization and Development	3
GEOG 375	Political Geography	3
GEOG 380	Cultural Geography	3
History		
HIST 318	Our Better Angels?: The World History of Violence	3
HIST 334	A Global History of Nationalism	3
HIST 469	A History of International Development in the Twentieth Century	3
Humanities		
HUM 252	Global Cultures [C2HQC] (When topic is Gandhi, Nonviolence and Global Transformation)	3
Justice Studies		
JUST 331	Human Rights in Theory and Practice	3

JUST 354	Native Justice and Resistance	3
JUST 355	Solutions to Global Poverty	3
JUST 356	Refugees and Humanitarian Response	3
JUST 357	Environmental Justice	3
JUST 372	Ethics and International Politics	3
JUST 375	Genocide in the 20th Century	3
JUST 376	Global Social Movements	3
Political Science		
POSC 331	Human Rights in Theory and Practice	3
POSC 340	Politics of Development	3
POSC 341	Social Movements in the U.S. and Abroad	3
POSC 363	Hacking for Diplomacy	3
POSC 372	Ethics and International Politics	3
POSC 376	Introduction to Black Politics	3
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation	3
POSC 379	International Laws of War	3
POSC 383	Women and Politics in Comparative Perspective	3
POSC 389	Hacking for National Security and Defense	3
POSC 393	Human Security	3
POSC 394	International Conflict	3
POSC 395	International Law	3
POSC 396	International Organizations	3
POSC 397	The Politics of International Economic Relations	3
POSC 430	International Security and Conflict Management	3
POSC 435	International Terrorism	3
POSC 458	International Political Analysis	3
Religion		
REL 315	Women and Religion	3
REL 323	Comparative Ethics East and West	3
REL 373	Race and Religion	3
REL 383	The Global Refugee Crisis and Religion	3
REL 385	Religion and the Law of War	3
Women's, Gender and Sexuality Studies		
WGSS 383	Women and Politics in Comparative Perspective	3

Area Requirements

Students may choose from five tracks to fulfill the area requirement: Africa, Asia, Europe, Latin America or the Middle East. Students must complete four courses for their specific area. These courses must come from at least three fields of study. The potential fields are listed below. At least one of these courses must be a history course. Students with advanced language skills may also enroll in a 400-level literature course to meet the culture requirement. The following lists the potential courses for each of these areas.

Africa Track

These courses must come from at least three fields of study. The potential fields are listed below.

Complete four courses: 12 Credit Hours

No course taken for this requirement may be used to fulfill a requirement elsewhere in the major.

Code	Title	Credits
Africana Studies		
AAAD 200	Introduction to African, African American and Diaspora Studies [C4GE]	3
AAAD 400	Advanced Topics in African, African American and Diaspora Studies	3
Culture		
ANTH 280	Peoples and Cultures of Sub-Saharan Africa	3
ARTH 210	African Art and Culture in the Humanities	3
ARTH 312	African Art: Sub-Saharan	3
ARTH 418	Modern and Contemporary African Art	3
ARTH 424	Arts of Ancient Egypt	3
ENG 377	Introduction to African Literature	3
ENG 432	Advanced Studies in African Literature	3
REL 250	Introduction to Islam	3
REL 305	Black Skin, Divine Realities? Exploring Islam in Africa	3
REL 330	Religions of Africa and the African Diaspora	3
Economics		
ECON 365	Economic Development	3
Geography		
GEOG 335	Geography of Africa	3
History		
HIST 263	Africa	3
HIST 358	Modern Africa, ca. 1800-1990s	3
HIST 436	Afro-Latin America	3
HIST 462	Human Trafficking in African History	3
HIST 463	Pandemics and Epidemics in African History	3
HIST 470	Modern Africa	3
HIST 472	Military, Militarism and Warfare in African History	3
Political Science		
POSC 340	Politics of Development	3
POSC 353	African Politics	3
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation	3

Asia Track

These courses must come from at least three fields of study. At least one course must be History. The potential fields are listed below.

Complete four courses: 12 Credit Hours

No course taken for this requirement may be used to fulfill a requirement elsewhere in the major.

Code	Title	Credits
Culture		
ANTH 295	People and Cultures of East Asia	3
ANTH 306	Japanese Society and Culture	3
CHIN 446	Special Topics in Chinese Literature	3
CHIN 447	Special Topics in Chinese Culture	3
CHIN 490	Special Studies in Chinese	1-4
JAPN 446	Special Topics in Japanese Literature	3
JAPN 447	Special Topics in Japanese Culture	3
JAPN 490	Independent Study in Japanese	1-4

KOR 446	Special Topics in Korean Literature	3
KOR 447	Special Topics in Korean Culture	3
KOR 490	Independent Study in Korean	1-4
REL 213	Introduction to Ancient Chinese Wisdom Traditions	3
REL 216	(Re)thinking Order and Chaos: An Introduction to Hinduism	3
REL 217	The Mystery of the True Path: An Introduction to Daoism	3
REL 288	Introduction to Buddhism	3
REL 314	Hope or Terror? Gandhi and the Other 9/11	3
REL 388	Tibetan Buddhism	3
REL 488	Buddhism in the Modern World	3
Geography		
GEOG 334	Geography of East and Southeast Asia	3
GEOG 336	Environmental Hazards: A Focus on Southeast Asia	3
GEOG 338	Geography of the Philippine Islands: Problems and Possibilities	3
History		
HIST 274	Modern East Asia	3
HIST 311	China to 1600	3
HIST 312	Japan since 1600	3
HIST 344	Magic Mountains, Man-eating Beasts and Unquiet Woods: An Environmental History of British India	3
HIST 351	War in Afghanistan: Local and Global Perspectives	3
HIST 377	History of Korea	3
HIST 378	China in the Modern World	3
HIST 473	Early Modern Islamic Empires	3
HIST 485	Colonial Knowledge Production in the Middle East and South Asia	3
HUM 252	Global Cultures [C2HQC]	3
Political Science		
POSC 355	East Asian Politics	3
POSC 356	Politics of China	3

Europe Track

These courses must come from at least three fields of study. The potential fields are listed below.

Complete four courses: 12 Credit Hours

No course taken for this requirement may be used to fulfill a requirement elsewhere in the major.

All students in this track must complete POSC 344.

Code	Title	Credits
Culture		
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
ARTH 313	Masterpieces of Italian Renaissance Art [C2VPA] (Semester in Florence only)	3
ARTH 314	Masterpieces of Spanish Art [C2VPA] (Semester in Spain only)	3
ARTH 316	Experiencing Art in London [C2VPA] (Semester in London only)	3

ENG 321	Restoration and Eighteenth-Century British Prose and Poetry	3
ENG 322	Restoration and Eighteenth-Century British Drama	3
ENG 340	Modern British and Irish Literature	3
ENG 341	Contemporary British and Irish Literature	3
ENG 375	Irish and Anglo-Irish Literature	3
ENG 402	Advanced Studies in British Literature Before 1700	3
FR 307	History of French Culture and Society	3
FR 335	A Journey through Literature	3
FR 425	Transformative Texts	3
FR 375	Business and Society in the Francophone World	3
FR 386	French Short Stories	3
FR 420	Stories and Histories in French Culture	3
FR 446	Special Topics in Francophone Literature	3
FR 447	Special Topics in Francophone Culture	3
GER 307	History of Germanic Cultures and Societies	3
GER 308	Modern German Culture and Society	3
GER 386	German Short Stories	3
GER 464	German Theater	3
GER 446	Special Topics in German Literature	3
GER 447	Special Topics in German Culture	3
GER 465	German Cinema	3
HIST 373	History of the Ottoman Empire, 1453-1922	3
HIST 476	Revolutionary Russia	3
ITAL 308	Contemporary Italian Culture and Society	3
ITAL 375	Business and Society in Italy	3
ITAL 386	Italian Short Stories	3
ITAL 421	Stories and Histories: Italian Theatre	3
ITAL 422	Voices of Today	3
ITAL 447	Special Topics in Italian Culture	3
ITAL 465	Italian Cinema	3
REL 325	Catholicism in the Modern World	3
REL 375	Nineteenth-Century Philosophy and Theology	3
RUS 308	Introduction to Russian Culture	3
RUS 446	Special Topics in Russian Literature	3
RUS 447	Special Topics in Russian-Speaking Cultures	3
SPAN 307	History of Spanish Culture and Society	3
SPAN 386	Spanish Short Stories	3
SPAN 407	Aspects of Contemporary Spanish Culture and Society	3
SPAN 425	Transformative Texts	3
SPAN 426	Voices of the Past: Works of the Golden Age	3
SPAN 446	Special Topics in Spanish Literature	3
SPAN 447	Special Topics in Spanish-Speaking Cultures	3
SPAN 422	Voices of Today: Wounds and Words in Post-War Spain	3
SPAN 464	Spanish Theater	3
SPAN 465	Spanish-Language Cinema	3
History		
HIST 306	A History of the Body in the West	3
HIST 307	The Trans-Atlantic Slave Trade	3
HIST 308	Contemporary Italian History and Culture	3
HIST 309	French History since 1648	3

HIST 321	Gender and Sexuality in Modern Europe	3
HIST 332	History of Twentieth-Century Spain	3
HIST 382	Europe in the 20th Century	3
HIST 383	Early England	3
HIST 385	The Russian Empire	3
HIST 386	The Soviet Union: A Socialist Experiment	3
HIST 388	Germany since 1871	3
HIST 389	Revolution and Radical Change in Modern Europe	3
HIST 398	European Military History	3
HIST 449	Women and Fascism	3
HIST 474	Stalinism: Approaches, Experiences and Legacies	3
HIST 476	Revolutionary Russia	3
HIST 478	Eastern Europe	3
HIST 482	French History Seminar	3
HIST 486	Europe Since 1914	3

Political Science

POSC 344	Politics of the European Union	3
POSC 345	Politics of Western Europe	3
POSC 346	Politics of Central and Eastern Europe	3

Latin America Track

These courses must come from at least three fields of study. At least one course must be History. The potential fields are listed below.

Complete four courses: 12 Credit Hours

No course taken for this requirement maybe used to fulfill a requirement elsewhere in the major.

Code	Title	Credits
Culture		
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	3
ANTH 303	Andean Archaeology	3
REL 324	Latinx and Latin American Christianity	3
SPAN 308	Latin American Cultures	3
SPAN 385	Latin American Short Story	3
SPAN 423	Voices of Today: Latin American Poetry	3
SPAN 421	Stories and Histories: Latin American Novel	3
SPAN 446	Special Topics in Spanish Literature	3
SPAN 447	Special Topics in Spanish-Speaking Cultures	3
SPAN 485	Business and Society in Latin America	3
Geography		
GEOG 339	Geography of the Caribbean	3
History		
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	3
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	3
HIST 436	Afro-Latin America	3
HIST 437	Latin America and Latin Americans through Film: Focus on the Twentieth Century	3
HIST 444	Revolution and Social Change in Latin America	3
HIST 447	South America	3
HIST 448	Gender in Latin America and the Iberian World	3

LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]	3
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Political Science

POSC 350	Latin American Politics	3
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation	3

Middle East Track

These courses must come from at least three fields of study. The potential fields are listed below.

Complete four courses: 12 Credit Hours

No course taken for this requirement maybe used to fulfill a requirement elsewhere in the major.

Code	Title	Credits
Culture		
ARAB 308	Contemporary Islamicate Civilization	3
ARAB 335	A Journey through Literature	3
ARAB 375	Business and Society in the Arab World	3
ARAB 385	The Arabian Nights	3
ARAB 386	Arabic Fables and Short Stories	3
ARAB 405	The Arabic Novel	3
ARAB 428	Muhammad and the Qur'an	3
ARAB 446	Special Topics in Arabic Literature	3
ARAB 447	Special Topics in Arabic Civilization and Culture	3
ARAB 465	Arabic Cinema	3
ARTH 424	Arts of Ancient Egypt	3
ENG 433	Advanced Studies in Arabic Literature	3
REL 220	Introduction to Judaism	3
REL 250	Introduction to Islam	3
REL 306	Gender and Sexuality in Islam	3
REL 307	Knowledge, Identity and Power in Islamic Societies	3
Economics		
ECON 365	Economic Development	3
History		
HIST 348	Maps and Mobility in Islamic World History	3
HIST 351	War in Afghanistan: Local and Global Perspectives	3
HIST 353	The Global Middle East	3
HIST 373	History of the Ottoman Empire, 1453-1922	3
HIST 473	Early Modern Islamic Empires	3
HIST 485	Colonial Knowledge Production in the Middle East and South Asia	3
Political Science		
POSC 340	Politics of Development	3
POSC 354	Politics of the Middle East	3

Comparative Study Concentration Total: 18 Credit Hours

Internships

A maximum of three credits in the major may be earned through an internship. Students must consult with the international affairs coordinator **prior** to doing an internship to check on its applicability to the major.

Study Abroad and Washington Semester

Many of the courses available to students through JMU's Office of International Programs and the Washington Semester program can be used to fulfill requirements in the INTA major. Students planning to go abroad should keep this in mind in working out their schedules and should consult the Washington Semester advisor and the INTA coordinator for course approvals and substitutions.

If you have questions about the Washington Semester, review the program website (<http://www.jmu.edu/polisci/washington.shtml>) and contact the program director, David Jones.

Recommended Schedule for Majors

Students entering the major having already completed their General Education requirements should contact their advisor for guidance.

The following is an example of a four-year course of study for a student seeking a degree:

Course	Title	Credits
First Year		
POSC 230	International Relations	3
POSC 240	Comparative Politics	3
World language		6
Madison Foundations		9
General Education requirements, including Quantitative Reasoning requirement		9
Credits		30
Second Year		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
POSC 370	U.S. Foreign Policy	3
World language		6
General Education requirements		15
Credits		30
Third Year		
ECON 270	International Economics	3
INTA 295	CrossNational Research Skills	4
World language courses at the 300-level or above		6
Complete General Education requirements		5
Coursework for an INTA concentration		9
Coursework for university electives, second major or minor		3
Credits		30
Fourth Year		
INTA 489	Senior Seminar in International Affairs	4
B.A. Philosophy requirement		3
Complete INTA concentration		9
Coursework for university electives, second major or minor		14
Credits		30
Total Credits		120

International Business, B.B.A.

Program Description

The B.B.A. degree in international business requires a minimum of 120 credit hours of undergraduate work. This consists of general education courses, B.B.A. core courses and major requirements. The major requirements include international business courses, advanced language courses and courses in one of three concentrations.

Each student in the major is required to complete an "international experience" (study, intern or work abroad). The international experience requirement is for a minimum of six weeks outside of the United States, unless express permission is granted by the program director for another option. The primary language spoken in the country chosen by each student for the study/work abroad requirement is not to be English and would optimally be the language chosen by the student to meet the international business language requirement. Approval of the international experience requirement must be obtained from the program director (approval forms are available online). International students should see the program director to determine appropriate requirements for this aspect of the curriculum.

Students majoring in international business must carefully plan their individual course work because not all required courses are offered each semester.

As part of the JMU assessment program, graduating seniors are required to participate in assessment activities. Assessment information is used to assist the COB faculty in modifying curricula and co-curricular events.

Admission and Retention Requirements

Students planning to major in international business must complete the 27 to 28 hour, lower-level B.B.A. curriculum prior to enrolling in upper-level B.B.A. courses, normally taken in the first semester of the junior year. It is expected that the lower-level B.B.A. core curriculum along with most of the university General Education curriculum will be completed during the first two years of study. Failing to complete all lower-level core requirements on time will delay enrollment in upper-level core and major courses until at least the second semester of the junior year.

The foreign language requirements of the major begin at the advanced level. Consequently, students with little or no foreign language training must use elective credits to obtain proficiency through the intermediate level.

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business (p. 15) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ^{1,2}	41
B.B.A. core courses	45-46
Major requirements (including language and concentration) ³	36

University Electives	0-5
Total	120

- ¹ Students are encouraged to choose general education courses having a foreign or international content. The program office has a list of these courses. The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary. ECON 200 and MATH 205 or MATH 235 double-count for general education and B.B.A. core requirements.
- ² Some language and General Education courses may be completed during the semester abroad.
- ³ Students are required to spend the equivalent of at least six weeks abroad. See below for details.

B.B.A. Core Component

Code	Title	Credits
B.B.A. Lower-Level Core Component		
B.B.A. majors must complete all of the B.B.A. core components as part of their degree program. The following courses comprise the B.B.A. Lower-Level Core Component:		
COB 191	Business Analytics I ¹	3
COB 202	Interpersonal Skills	3
COB 204	Computer Information Systems	3
COB 241	Financial Accounting	3
COB 242	Managerial Accounting ^{2,3}	3
COB 291	Business Analytics II ⁴	3
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR] ⁵	
MATH 235	Calculus I [C3QR] ⁵	
Subtotal		27-28
B.B.A. Upper-Level Core Component		
The following courses comprise the B.B.A. Upper-Level Core Component:		
COB 300A	Integrative Business: Management ^{3,6}	3
COB 300B	Integrative Business: Finance ^{3,6}	3
COB 300C	Integrative Business: Operations ^{3,6}	3
COB 300D	Integrative Business: Marketing ^{3,6}	3
COB 318	Legal and Ethical Environment of Business ⁷	3
COB 487	Strategic Management ⁸	3
Subtotal		18
Total Credits		45-46

¹ MATH 220 may be substituted for COB 191.

² COB 241 must be taken before COB 242.

³ COB 242 is a prerequisite for COB 300 but is not used in calculating the B.B.A. core GPA.

⁴ COB 191 or equivalent and MATH 205 or equivalent are prerequisites for COB 291.

⁵ Calculus is required but not used in calculating the B.B.A. core GPA. Students may substitute MATH 231, MATH 233 or ISAT 151 to fulfill the calculus requirement. MATH 235 is preferred for Economics and Finance majors.

⁶ All four COB 300 courses must be taken during the same semester.

⁷ COB 300 is a prerequisite or corequisite for COB 318. COB 318 is a prerequisite for COB 487 and BLAW 497.

⁸ COB 487 must be taken during the senior year.

Major Requirements

Code	Title	Credits
Required International Business Courses		
BLAW 497	Legal Aspects of International Business	3
IB 200	Introduction to International Business	3
IB 480	International Business Theory and Policy	3
MGT 340	International Management	3
Subtotal		12
Required Language Courses		
Students are required to take 12 upper-level credit hours in a language of their choosing.		
Language Foundations: (language-prefix) 300 or 301 or 320 or 322		3
Culture and Civilization: (language-prefix) 307 or 308		3
Business: (language-prefix) 330		3
Business and Society: (language-prefix) 375 ¹		3
Subtotal		12
Concentration		
Students must complete one of the following three concentrations. See below for details.		
General Business		
Marketing		
Finance		
Subtotal		12
Total Credits		36

¹ Students choosing Spanish should take eitherSPAN 485 or SPAN 486 to fulfill this requirement.

Concentrations

General Business

Code	Title	Credits
ECON 270	International Economics	3
FIN 355	International Financial Management	3
MKTG 460	Global Marketing	3
International Business approved elective		3
Total Credits		12

Marketing

Code	Title	Credits
ECON 270	International Economics	3
MKTG 460	Global Marketing	3
Choose two of the following:		6
MKTG 384	Integrated Marketing Communications	
MKTG 385	Consumer Behavior	
MKTG 430	Professional Selling	
MKTG 470	Strategic Internet Marketing	

MKTG 482	Marketing Analytics
Total Credits	12

Finance

Code	Title	Credits
ECON 370	International Trade and Trade Policies	3
ECON 372	International Finance and Payments	3
FIN 355	International Financial Management	3
FIN 360	Analytical Methods in Finance	3
Total Credits		12

Recommended Schedule for Majors

First Two Years

Students planning to major in international business must complete the lower-division B.B.A. core curriculum prior to enrolling in upper-division core courses, which are normally taken in the first semester of the junior year. It is expected that the lower-division B.B.A. core curriculum will be completed during the first two years of study along with most of the university General Education curriculum. Failing to complete all lower-division B.B.A. core requirements on time will delay enrollment in upper-division core and major courses.

Course	Title	Credits
Third Year		
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
COB 318	Legal and Ethical Environment of Business	3
ECON 270	International Economics	3
General Education		6
Language major requirements		6
Credits		30
Fourth Year		
BLAW 497	Legal Aspects of International Business	3
COB 487	Strategic Management	3
IB 480	International Business Theory and Policy	3
MGT 340	International Management	3
Language major requirements		6
Concentration courses		12
Credits		30
Total Credits		60

Justice Studies, B.A.

Program Description

The Bachelor of Arts degree in justice studies provides an expansive and holistic education in the broad, interdisciplinary field of justice studies, including and beyond the subfields represented in the concentration areas (criminal justice and criminology, global justice and policy, and social justice engagement). The major offers a rigorous and vocationally-relevant course of study for persons interested in academic or applied careers at the community, national or global level.

Students can complete the B.A. degree in justice studies with or without a concentration area (refer to requirements for each option).

Transfer Credit

A maximum of two courses may be transferred into the justice studies major from other institutions. No transfer credit will be given for any course offered in the major at the 300 or 400 level.

Declaration and Progression Information

Because of the way students progress through the major, it takes at least five semesters to complete the requirements. Only declared Justice Studies majors can enroll in JUST 200. Students must take JUST 200 first; it is the only course students are allowed to take during their first semester in the major. The semester following successful completion of JUST 200, students may begin to take the 200-level foundation courses. During the third semester in the major, students may begin to take 300-level Justice Studies courses. Also be aware that MATH 220 is a prerequisite for JUST 399 Justice Research Methods, which is in turn a prerequisite for JUST 400 Senior Seminar in Justice Studies.

New first-year and transfer students should declare the Justice Studies major when they register for orientation.

For currently enrolled students, the process for declaring majors is online; see the Registrar's website (<https://www.jmu.edu/registrar/index.shtml>) for additional information. Students must declare the major by March 1 (for summer and fall enrollment) or October 1 (for spring enrollment).

Please contact the Justice Studies academic advisor, Daisy Breneman, with any questions about declaring the major.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ^{1,2}	41
Philosophy course (in addition to any taken for General Education)	3
Foreign language courses (intermediate level required) ³	0-12
Major requirements	41
University Electives	20-38
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² MATH 220 (prerequisite for JUST 399) will fulfill a General Education requirement.

³ The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

Code	Title	Credits
Core Courses		
JUST 100	Justice Studies Proseminar	1
JUST 200	Introduction to Justice Studies	3
JUST 399	Justice Research Methods ¹	4
JUST 400	Senior Seminar in Justice Studies	3
Subtotal		11
Fairness and Responsibility Component Requirement		
Choose one of the following: ²		3
JUST 328	Race, Class and Justice	
JUST 339	Mapping Justice	
JUST 341	Gender and Justice	
JUST 354	Native Justice and Resistance	
JUST 356	Refugees and Humanitarian Response	
JUST 374	War and Justice	
JUST 385	Disability and Justice	
JUST 387	Gender Identity, Law and Justice	
Subtotal		3
Foundation Course Requirements for General Justice Studies Majors		
The following requirements are only for majors without a concentration.		
JUST 210 or JUST 212	Crime and Criminal Justice Theories of Crime and Criminal Justice	3
JUST 221 or JUST 223	Social Justice Theories Social Justice Interventions and Policies	3
JUST 235 or JUST 237	Justice in the Global Community Dilemmas in Global Justice	3
Subtotal		9
Electives for General Justice Studies Majors		
Students select six additional courses from the following list. At least four of the six electives must have a JUST identifier. ³		
HIST 338	United States Urban History	
JUST 300	Perspectives on Comparative Justice	
JUST 301	Advanced Level Special Topics in Justice Studies	
JUST 307	Justice and Crime	
JUST 308	Social Psychology of Justice	
JUST 309	Global Crime	
JUST 311	Women, Crime and Criminal Justice	
JUST 312	Mass and Serial Murder	
JUST 316	Human Development and Crime	
JUST 317	Victimization of Children	
JUST 318	Sex Offenders	
JUST 320	Organized Crime	
JUST 323	Comparative Criminal Justice	
JUST 324	Death Penalty	
JUST 326	Victimology	

JUST 327	Criminal Law
JUST 328	Race, Class and Justice
JUST 329	Perspectives on Law
JUST 330	Corrections
JUST/POSC 331	Human Rights in Theory and Practice
JUST 334	Media and Justice
JUST 339	Mapping Justice
JUST 341	Gender and Justice
JUST 343	Justice and Society
JUST 345	Restorative Justice
JUST 350	Justice and Globalization
JUST 351	Building Democracy
JUST 352	Global Migration
JUST 353	Justice and Development
JUST 354	Native Justice and Resistance
JUST 355	Solutions to Global Poverty
JUST 356	Refugees and Humanitarian Response
JUST 357	Environmental Justice
JUST 365	Justice in Literature, Film and Art
JUST 372	Ethics and International Politics
JUST 373	Rebuilding Post Conflict Societies
JUST 374	War and Justice
JUST 375	Genocide in the 20th Century
JUST 376	Global Social Movements
JUST 377	Global Futures
JUST 385	Disability and Justice
JUST 386	Sexual Orientation, Law and Justice
JUST 387	Gender Identity, Law and Justice
JUST/POSC 392	Peace Studies
POSC 395	International Law
JUST 397	Legal Research and Information Literacy
JUST 401	Internship in Justice Studies
JUST 402	Advanced Research in Justice Studies
JUST 403	Nelson Institute Seminar
JUST 404	Practicum in Community-Based Research
JUST 405	Research Assistantship in Justice Studies
JUST 499	Honors
PHIL 335	The Individual, The State and Justice
POSC 326	Civil Rights
POSC 348	The Politics of Cultural Pluralism
POSC 372	Ethics and International Politics
POSC 383	Women and Politics in Comparative Perspective
POSC 396	International Organizations
PSYC 310	The Psychology of Women and Gender
PSYC 415	Forensic Psychology
REL 322	Sociology of Religion
REL 450	Religion and Society
SCOM 342	Argument and Advocacy
SOCI 322	Sociology of Religion
SOCI 336	Race and Ethnicity

SOCI 354	Social Inequality
SOCI 367	Sociology of Sexuality
Subtotal	18
Total Credits	41

¹ MATH 220 is a prerequisite for JUST 399.

² Any course not used to fulfill this requirement may be used as elective credit.

³ Students may select only one of the following courses for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.

Justice Studies Concentrations

Concentration A: Criminal Justice and Criminology

This concentration focuses on the nature, causes and solutions for crime, primarily, but not exclusively, in the United States.

Code	Title	Credits
Concentration A: Required Foundation Courses		
JUST 210	Crime and Criminal Justice	3
JUST 212	Theories of Crime and Criminal Justice	3
Subtotal		6
Concentration A: Foundation Course Electives		
Choose one of the following:		3
JUST 221	Social Justice Theories	
JUST 223	Social Justice Interventions and Policies	
JUST 235	Justice in the Global Community	
JUST 237	Dilemmas in Global Justice	
Subtotal		3

Concentration A: Electives

Choose six additional courses from the following list. At least four of the six electives must have a JUST identifier. ¹

JUST 301	Advanced Level Special Topics in Justice Studies ²
JUST 307	Justice and Crime
JUST 308	Social Psychology of Justice
JUST 309	Global Crime
JUST 311	Women, Crime and Criminal Justice
JUST 312	Mass and Serial Murder
JUST 316	Human Development and Crime
JUST 317	Victimization of Children
JUST 318	Sex Offenders
JUST 320	Organized Crime
JUST 324	Death Penalty
JUST 327	Criminal Law
JUST 328	Race, Class and Justice
JUST 329	Perspectives on Law
JUST 330	Corrections
JUST 334	Media and Justice
JUST 339	Mapping Justice
JUST 341	Gender and Justice
JUST 343	Justice and Society
JUST 345	Restorative Justice

JUST 354	Native Justice and Resistance	
JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST 361	Terrorism	
JUST 365	Justice in Literature, Film and Art ²	
JUST 374	War and Justice	
JUST 385	Disability and Justice	
JUST 386	Sexual Orientation, Law and Justice	
JUST/WGSS 387	Gender Identity, Law and Justice	
JUST 397	Legal Research and Information Literacy	
JUST 401	Internship in Justice Studies	
JUST 404	Practicum in Community-Based Research ²	
JUST 402	Advanced Research in Justice Studies ²	
JUST 403	Nelson Institute Seminar ²	
JUST 405	Research Assistantship in Justice Studies	
PSYC 415	Forensic Psychology ³	
Subtotal		18
Total Credits		27

¹ Note: Other courses may be substituted with your advisor's and the department head's consent.

² May be taken for elective credit in Concentration A, Concentration B and Concentration C when the topic is appropriate. Students may select only one of the following for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.

³ Course is only open to students who are double majoring in Psychology.

Concentration B: Global Justice and Policy

This concentration explores issues of justice in global context. Most prominent among them are questions of security, equity and equality. Courses address the individual, group and state dimensions of these and related issues in a diverse set of policy areas including democratization, cultural identity, developmental, environmental protection, conflict resolution and human rights.

Code	Title	Credits
Concentration B: Required Foundation Courses		
JUST 235	Justice in the Global Community	3
JUST 237	Dilemmas in Global Justice	3
Subtotal		6
Concentration B: Foundation Course Electives		
Choose one of the following:		3
JUST 210	Crime and Criminal Justice	
JUST 212	Theories of Crime and Criminal Justice	
JUST 221	Social Justice Theories	
JUST 223	Social Justice Interventions and Policies	
Subtotal		3
Concentration B: Electives		
Students in Concentration B select six additional elective courses from the following list. At least four must have a JUST identifier: ¹		18
JUST 301	Advanced Level Special Topics in Justice Studies ²	
JUST 309	Global Crime	

JUST 320	Organized Crime	
JUST 324	Death Penalty	
JUST/POSC 331	Human Rights in Theory and Practice	
JUST 334	Media and Justice ²	
JUST 339	Mapping Justice	
JUST 341	Gender and Justice	
JUST 345	Restorative Justice	
JUST 350	Justice and Globalization	
JUST 351	Building Democracy	
JUST 352	Global Migration	
JUST 353	Justice and Development	
JUST 354	Native Justice and Resistance	
JUST 355	Solutions to Global Poverty	
JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST 361	Terrorism	
JUST 365	Justice in Literature, Film and Art ²	
JUST/POSC 372	Ethics and International Politics	
JUST 373	Rebuilding Post Conflict Societies	
JUST 374	War and Justice	
JUST 375	Genocide in the 20th Century	
JUST 376	Global Social Movements	
JUST 377	Global Futures	
JUST 385	Disability and Justice	
JUST 386	Sexual Orientation, Law and Justice	
JUST/WGSS 387	Gender Identity, Law and Justice	
JUST/POSC 392	Peace Studies	
JUST 397	Legal Research and Information Literacy	
JUST 401	Internship in Justice Studies ²	
JUST 402	Advanced Research in Justice Studies ²	
JUST 403	Nelson Institute Seminar ²	
JUST 404	Practicum in Community-Based Research ²	
JUST 405	Research Assistantship in Justice Studies	
PHIL 335	The Individual, The State and Justice	
POSC 395	International Law	
POSC 396	International Organizations	
SCOM 342	Argument and Advocacy	
Any 300-level anthropology course that is centered on a world region (other than North America) ³		
Any 300-level history course that is centered on a world region (other than North America) and that includes coverage of the 20th century ³		
Any 300-level political science course that is regionally focused ³		
Any 300-level religion course with a contemporary and international focus ³		
Subtotal		18
Total Credits		27

¹ Note: Other courses may be substituted with your advisor's and the department head's consent.

- ² May be taken for elective credit in Concentration A, Concentration B and Concentration C when the topic is appropriate. Students may select only one of the following for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.
- ³ Only one course from this option may be counted toward the major; approval for course selection from Justice Studies and the academic unit offering the course is also required.

Concentration C: Social Justice Engagement

This concentration is designed to investigate what is fair, equitable and just for society. Emphasizing the oppression and liberation of vulnerable, exploited and marginalized populations, this curriculum promotes sustainable and just solutions to social, political and economic problems.

Code	Title	Credits
Concentration C: Required Foundation Courses		
JUST 221	Social Justice Theories	3
JUST 223	Social Justice Interventions and Policies	3
Subtotal		6
Concentration C: Foundation Course Electives		
Choose one of the following:		3
JUST 210	Crime and Criminal Justice	
JUST 212	Theories of Crime and Criminal Justice	
JUST 235	Justice in the Global Community	
JUST 237	Dilemmas in Global Justice	
Subtotal		3
Concentration C: Electives		
Students in Concentration C select six additional elective courses from the following list. At least four must have a JUST identifier: ¹		18
HIST 338	United States Urban History	
JUST 301	Advanced Level Special Topics in Justice Studies ²	
JUST 308	Social Psychology of Justice	
JUST 309	Global Crime	
JUST 311	Women, Crime and Criminal Justice	
JUST 324	Death Penalty	
JUST 334	Media and Justice ²	
JUST 339	Mapping Justice	
JUST 341	Gender and Justice	
JUST 343	Justice and Society	
JUST 352	Global Migration	
JUST 354	Native Justice and Resistance	
JUST 355	Solutions to Global Poverty	
JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST 361	Terrorism	
JUST 365	Justice in Literature, Film and Art	
JUST 375	Genocide in the 20th Century	
JUST 376	Global Social Movements	
JUST 385	Disability and Justice	
JUST 386	Sexual Orientation, Law and Justice	
JUST/WGSS 387	Gender Identity, Law and Justice	
JUST/POSC 392	Peace Studies	

JUST 397	Legal Research and Information Literacy
JUST 401	Internship in Justice Studies ²
JUST 402	Advanced Research in Justice Studies ²
JUST 403	Nelson Institute Seminar ²
JUST 404	Practicum in Community-Based Research ²
JUST 405	Research Assistantship in Justice Studies
POSC 326	Civil Rights
POSC 348	The Politics of Cultural Pluralism
POSC 383	Women and Politics in Comparative Perspective
PSYC 310	The Psychology of Women and Gender ³
REL/SOCI 322	Sociology of Religion
REL 371	Religion and Disability
REL 372	Religion and Medicine, East and West
REL 450	Religion and Society
SOCI 336	Race and Ethnicity
SOCI 354	Social Inequality
SOCI 367	Sociology of Sexuality
Subtotal	18
Total Credits	27

- ¹ Note: Other courses may be substituted with your advisor's and the department head's consent.
- ² May be taken for elective credit in Concentration A, Concentration B and Concentration C when the topic is appropriate. Students may select only one of the following for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.
- ³ Course is only open to students who are double majoring in Psychology.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
JUST 200	Introduction to Justice Studies	3
MATH 220	Elementary Statistics [C3QR] (Can be counted for General Education, The Natural World) ¹	3
Foundation course		3
General education, degree requirements and/or university electives		21
Credits		30
Second Year		
Foundation courses		6
Justice Studies electives		6
General education, degree requirements and/or university electives		18
Credits		30
Third Year		
JUST 100	Justice Studies Proseminar	1
JUST 399	Justice Research Methods	4
Fairness and Responsibility course		3
Justice Studies electives		9
General education, degree requirements and/or university electives		13
Credits		30
Fourth Year		
JUST 400	Senior Seminar in Justice Studies	3
Justice Studies electives		3

General education, degree requirements and/or university electives	24
Credits	30
Total Credits	120

¹ prerequisite for JUST 399

Justice Studies, B.S.

Program Description

The Bachelor of Science degree in justice studies provides an expansive and holistic education in the broad, interdisciplinary field of justice studies, including and beyond the subfields represented in the concentration areas (criminal justice and criminology, global justice and policy, and social justice engagement). The major offers a rigorous and vocationally-relevant course of study for persons interested in academic or applied careers at the community, national or global level.

Students can complete the B.S. degree in justice studies with or without a concentration area (refer to requirements for each option below).

Transfer Credit

A maximum of two courses may be transferred into the justice studies major from other institutions. No transfer credit will be given for any course offered in the major at the 300 or 400 level.

Declaration and Progression Information

Because of the way students progress through the major, it takes at least five semesters to complete the requirements. Only declared Justice Studies majors can enroll in JUST 200. Students must take JUST 200 first; it is the only course students are allowed to take during their first semester in the major. The semester following successful completion of JUST 200, students may begin to take the 200-level foundation courses. During the third semester in the major, students may begin to take 300-level Justice Studies courses. Also be aware that MATH 220 is a prerequisite for JUST 399 Justice Research Methods, which is in turn a prerequisite for JUST 400 Senior Seminar in Justice Studies.

New first-year and transfer students should declare the Justice Studies major when they register for orientation.

For currently enrolled students, the process for declaring majors is online; see the Registrar's website (<https://www.jmu.edu/registrar/index.shtml>) for additional information. Students must declare the major by March 1 (for summer and fall enrollment) or October 1 (for spring enrollment).

Please contact the Justice Studies academic advisor, Daisy Breneman, with any questions about declaring the major.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ^{1,2}	41
Quantitative requirement (in addition to General Education) ²	3
Scientific literacy requirement (in addition to General Education)	3-4
Major requirements	41
University electives	31-32
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² MATH 220 (prerequisite for JUST 399) may count as either General Education or Quantitative requirement.

Major Requirements

Code	Title	Credits
Core Courses		
JUST 100	Justice Studies Proseminar	1
JUST 200	Introduction to Justice Studies	3
JUST 399	Justice Research Methods ¹	4
JUST 400	Senior Seminar in Justice Studies	3
Subtotal		11
Fairness and Responsibility Component Requirement		
Choose one of the following: ²		3
JUST 328	Race, Class and Justice	
JUST 339	Mapping Justice	
JUST 341	Gender and Justice	
JUST 354	Native Justice and Resistance	
JUST 356	Refugees and Humanitarian Response	
JUST 374	War and Justice	
JUST 385	Disability and Justice	
JUST 387	Gender Identity, Law and Justice	
Subtotal		3
Foundation Course Requirements for General Justice Studies Majors		
The following requirements are only for majors without a concentration.		
JUST 210	Crime and Criminal Justice	3
or JUST 212	Theories of Crime and Criminal Justice	
JUST 221	Social Justice Theories	3
or JUST 223	Social Justice Interventions and Policies	
JUST 235	Justice in the Global Community	3
or JUST 237	Dilemmas in Global Justice	
Subtotal		9
Electives for General Justice Studies Majors		
Students select six additional courses from the following list. At least 18 four of the six electives must have a JUST identifier. ³		
HIST 338	United States Urban History	
JUST 300	Perspectives on Comparative Justice	
JUST 301	Advanced Level Special Topics in Justice Studies	
JUST 307	Justice and Crime	
JUST 308	Social Psychology of Justice	
JUST 309	Global Crime	
JUST 311	Women, Crime and Criminal Justice	
JUST 312	Mass and Serial Murder	
JUST 316	Human Development and Crime	
JUST 317	Victimization of Children	
JUST 318	Sex Offenders	
JUST 320	Organized Crime	
JUST 323	Comparative Criminal Justice	
JUST 324	Death Penalty	
JUST 326	Victimology	
JUST 327	Criminal Law	
JUST 328	Race, Class and Justice	

JUST 329	Perspectives on Law
JUST 330	Corrections
JUST/POSC 331	Human Rights in Theory and Practice
JUST 334	Media and Justice
JUST 339	Mapping Justice
JUST 341	Gender and Justice
JUST 343	Justice and Society
JUST 345	Restorative Justice
JUST 350	Justice and Globalization
JUST 351	Building Democracy
JUST 352	Global Migration
JUST 353	Justice and Development
JUST 354	Native Justice and Resistance
JUST 355	Solutions to Global Poverty
JUST 356	Refugees and Humanitarian Response
JUST 357	Environmental Justice
JUST 361	Terrorism
JUST 365	Justice in Literature, Film and Art
JUST 372	Ethics and International Politics
JUST 373	Rebuilding Post Conflict Societies
JUST 374	War and Justice
JUST 375	Genocide in the 20th Century
JUST 376	Global Social Movements
JUST 377	Global Futures
JUST 385	Disability and Justice
JUST 386	Sexual Orientation, Law and Justice
JUST 387	Gender Identity, Law and Justice
JUST/POSC 392	Peace Studies
JUST 397	Legal Research and Information Literacy
JUST 401	Internship in Justice Studies
JUST 402	Advanced Research in Justice Studies
JUST 403	Nelson Institute Seminar
JUST 404	Practicum in Community-Based Research
JUST 405	Research Assistantship in Justice Studies
JUST 499	Honors
PHIL 335	The Individual, The State and Justice
POSC 326	Civil Rights
POSC 348	The Politics of Cultural Pluralism
POSC 372	Ethics and International Politics
POSC 383	Women and Politics in Comparative Perspective
POSC 395	International Law
POSC 396	International Organizations
PSYC 310	The Psychology of Women and Gender
PSYC 415	Forensic Psychology
REL 322	Sociology of Religion
REL 450	Religion and Society
SCOM 342	Argument and Advocacy
SOCI 322	Sociology of Religion
SOCI 336	Race and Ethnicity
SOCI 354	Social Inequality

SOCI 367	Sociology of Sexuality
Subtotal	18
Total Credits	41

¹ MATH 220 is a prerequisite for JUST 399.

² Any course not used to fulfill this requirement may be used as elective credit.

³ Students may select only one of the following courses for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.

Justice Studies Concentrations

Concentration A: Criminal Justice and Criminology

This concentration focuses on the nature, causes and solutions for crime, primarily, but not exclusively, in the United States.

Code	Title	Credits
Concentration A: Required Foundation Courses		
JUST 210	Crime and Criminal Justice	3
JUST 212	Theories of Crime and Criminal Justice	3
Subtotal		6
Concentration A: Foundation Course Electives		
Choose one of the following:		3
JUST 221	Social Justice Theories	
JUST 223	Social Justice Interventions and Policies	
JUST 235	Justice in the Global Community	
JUST 237	Dilemmas in Global Justice	
Subtotal		3

Concentration A: Electives

Choose six additional courses from the following list. At least four of the six electives must have a JUST identifier. ¹

JUST 301	Advanced Level Special Topics in Justice Studies ²
JUST 307	Justice and Crime
JUST 308	Social Psychology of Justice
JUST 309	Global Crime
JUST 311	Women, Crime and Criminal Justice
JUST 312	Mass and Serial Murder
JUST 316	Human Development and Crime
JUST 317	Victimization of Children
JUST 318	Sex Offenders
JUST 320	Organized Crime
JUST 324	Death Penalty
JUST 327	Criminal Law
JUST 328	Race, Class and Justice
JUST 329	Perspectives on Law
JUST 330	Corrections
JUST 334	Media and Justice
JUST 339	Mapping Justice
JUST 341	Gender and Justice
JUST 343	Justice and Society
JUST 345	Restorative Justice
JUST 354	Native Justice and Resistance

JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST 361	Terrorism	
JUST 365	Justice in Literature, Film and Art ²	
JUST 374	War and Justice	
JUST 385	Disability and Justice	
JUST 386	Sexual Orientation, Law and Justice	
JUST/WGSS 387	Gender Identity, Law and Justice	
JUST 397	Legal Research and Information Literacy	
JUST 401	Internship in Justice Studies	
JUST 404	Practicum in Community-Based Research ²	
JUST 402	Advanced Research in Justice Studies ²	
JUST 403	Nelson Institute Seminar ²	
JUST 405	Research Assistantship in Justice Studies	
PSYC 415	Forensic Psychology ³	
Subtotal		18
Total Credits		27

¹ Note: Other courses may be substituted with your advisor's and the department head's consent.

² May be taken for elective credit in Concentration A, Concentration B and Concentration C when the topic is appropriate. Students may select only one of the following for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.

³ Course is only open to students who are double majoring in Psychology.

Concentration B: Global Justice and Policy

This concentration explores issues of justice in global context. Most prominent among them are questions of security, equity and equality. Courses address the individual, group and state dimensions of these and related issues in a diverse set of policy areas including democratization, cultural identity, developmental, environmental protection, conflict resolution and human rights.

Code	Title	Credits
Concentration B: Required Foundation Courses		
JUST 235	Justice in the Global Community	3
JUST 237	Dilemmas in Global Justice	3
Subtotal		6
Concentration B: Foundation Course Electives		
Choose one of the following:		3
JUST 210	Crime and Criminal Justice	
JUST 212	Theories of Crime and Criminal Justice	
JUST 221	Social Justice Theories	
JUST 223	Social Justice Interventions and Policies	
Subtotal		3
Concentration B: Electives		
Students in Concentration B select six additional elective courses from the following list. At least four must have a JUST identifier. ¹		18
JUST 301	Advanced Level Special Topics in Justice Studies ²	
JUST 309	Global Crime	
JUST 320	Organized Crime	

JUST 324	Death Penalty	
JUST/POSC 331	Human Rights in Theory and Practice	
JUST 334	Media and Justice ²	
JUST 339	Mapping Justice	
JUST 341	Gender and Justice	
JUST 345	Restorative Justice	
JUST 350	Justice and Globalization	
JUST 351	Building Democracy	
JUST 352	Global Migration	
JUST 353	Justice and Development	
JUST 354	Native Justice and Resistance	
JUST 355	Solutions to Global Poverty	
JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST 361	Terrorism	
JUST 365	Justice in Literature, Film and Art ²	
JUST/POSC 372	Ethics and International Politics	
JUST 373	Rebuilding Post Conflict Societies	
JUST 374	War and Justice	
JUST 375	Genocide in the 20th Century	
JUST 376	Global Social Movements	
JUST 377	Global Futures	
JUST 385	Disability and Justice	
JUST 386	Sexual Orientation, Law and Justice	
JUST/WGSS 387	Gender Identity, Law and Justice	
JUST/POSC 392	Peace Studies	
JUST 397	Legal Research and Information Literacy	
JUST 401	Internship in Justice Studies ²	
JUST 402	Advanced Research in Justice Studies ²	
JUST 403	Nelson Institute Seminar ²	
JUST 404	Practicum in Community-Based Research ²	
JUST 405	Research Assistantship in Justice Studies	
PHIL 335	The Individual, The State and Justice	
POSC 395	International Law	
POSC 396	International Organizations	
SCOM 342	Argument and Advocacy	
Any 300-level anthropology course that is centered on a world region (other than North America) ³		
Any 300-level history course that is centered on a world region (other than North America) and that includes coverage of the 20th century ³		
Any 300-level political science course that is regionally focused ³		
Any 300-level religion course with a contemporary and international focus ³		
Subtotal		18
Total Credits		27

¹ Note: Other courses may be substituted with your advisor's and the department head's consent.

- ² May be taken for elective credit in Concentration A, Concentration B and Concentration C when the topic is appropriate. Students may select only one of the following for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.
- ³ Only one course from this option may be counted toward the major; approval for course selection from Justice Studies and the academic unit offering the course is also required.

Concentration C: Social Justice Engagement

This concentration is designed to investigate what is fair, equitable and just for society. Emphasizing the oppression and liberation of vulnerable, exploited and marginalized populations, this curriculum promotes sustainable and just solutions to social, political and economic problems.

Code	Title	Credits
Concentration C: Required Foundation Courses		
JUST 221	Social Justice Theories	3
JUST 223	Social Justice Interventions and Policies	3
Subtotal		6
Concentration C: Foundation Course Electives		
Choose one of the following:		3
JUST 210	Crime and Criminal Justice	
JUST 212	Theories of Crime and Criminal Justice	
JUST 235	Justice in the Global Community	
JUST 237	Dilemmas in Global Justice	
Subtotal		3
Concentration C: Electives		
Students in Concentration C select six additional elective courses from the following list. At least four must have a JUST identifier. ¹		18
HIST 338	United States Urban History	
JUST 301	Advanced Level Special Topics in Justice Studies ²	
JUST 308	Social Psychology of Justice	
JUST 309	Global Crime	
JUST 311	Women, Crime and Criminal Justice	
JUST 324	Death Penalty	
JUST 334	Media and Justice ²	
JUST 339	Mapping Justice	
JUST 341	Gender and Justice	
JUST 343	Justice and Society	
JUST 352	Global Migration	
JUST 354	Native Justice and Resistance	
JUST 355	Solutions to Global Poverty	
JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST 361	Terrorism	
JUST 365	Justice in Literature, Film and Art	
JUST 375	Genocide in the 20th Century	
JUST 376	Global Social Movements	
JUST 385	Disability and Justice	
JUST 386	Sexual Orientation, Law and Justice	
JUST/WGSS 387	Gender Identity, Law and Justice	
JUST/POSC 392	Peace Studies	

JUST 397	Legal Research and Information Literacy
JUST 401	Internship in Justice Studies ²
JUST 402	Advanced Research in Justice Studies ²
JUST 403	Nelson Institute Seminar ²
JUST 404	Practicum in Community-Based Research ²
JUST 405	Research Assistantship in Justice Studies
POSC 326	Civil Rights
POSC 348	The Politics of Cultural Pluralism
POSC 383	Women and Politics in Comparative Perspective
PSYC 310	The Psychology of Women and Gender ³
REL/SOCI 322	Sociology of Religion
REL 371	Religion and Disability
REL 372	Religion and Medicine, East and West
REL 450	Religion and Society
SOCI 336	Race and Ethnicity
SOCI 354	Social Inequality
SOCI 367	Sociology of Sexuality
Subtotal	18
Total Credits	27

- ¹ Note: Other courses may be substituted with your advisor's and the department head's consent.
- ² May be taken for elective credit in Concentration A, Concentration B and Concentration C when the topic is appropriate. Students may select only one of the following for elective credit: JUST 401, JUST 402, JUST 403, JUST 404 or JUST 405. Students should seek approval from their advisor or the department head.
- ³ Course is only open to students who are double majoring in Psychology.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
JUST 200	Introduction to Justice Studies	3
MATH 220	Elementary Statistics [C3QR] ^{1,2}	3
Foundation course		3
General education, degree requirements and/or university electives		21
Credits		30
Second Year		
Foundation courses		6
Justice Studies electives		6
General education, degree requirements and/or university electives		18
Credits		30
Third Year		
JUST 100	Justice Studies Proseminar	1
JUST 399	Justice Research Methods	4
Fairness and Responsibility course		3
Justice Studies electives		9
General education, degree requirements and/or university electives		13
Credits		30
Fourth Year		
JUST 400	Senior Seminar in Justice Studies	3
Justice Studies electives		3
General education, degree requirements and/or university electives		24
Credits		30
Total Credits		120

¹ Prerequisite for JUST 399

² Can be counted for General Education, The Natural World.

Kinesiology, B.S.

Program Description

The kinesiology major program leads to the Bachelor of Science degree. Students are advised to familiarize themselves with the B.S. degree requirements.

Students are advised to be certain they complete all General Education requirements before applying for graduation.

Students who major in kinesiology work toward a B.S. degree by selecting and completing one of the following concentrations:

- Exercise Science
- Physical and Health Education Teacher Education

Exercise Science Concentration Admission Policy

Any student can declare the exercise science concentration; however, students must apply to be fully admitted to the major for a limited number of spaces. Declaring the exercise science concentration does not guarantee full admission to the major.

Applications to begin upper-level Exercise Science are December 15th (fall semester start) and July 31st (spring semester start). Prior to submitting their application, students must have completed the following coursework:

Code	Title	Credits
KIN 100	Lifetime Fitness and Wellness (2,2) [C5W]	3
KIN 202	Biological Foundations of Kinesiology	3

Two of the following:

Code	Title	Credits
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	4
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4

Students will need to complete the final pre-requisite course (the remaining Biology or Chemistry course) with a "C-" or better prior to beginning upper-level Exercise Science coursework. Students may utilize one repeat-forgive course as part of their application; the repeated attempt must have been completed at the time of application.

Students denied admission may reapply during the next admission cycle for a total of two applications.

Physical and Health Education Teacher Education Concentration Admission Policy

The admission and retention standards are designed to instill a professional attitude and commitment to the profession of teaching. The admission and retention standards complement the JMU Teacher

Education application process. The following outlines the admission and retention standards and procedures for PHETE.

PHETE Admission Standards

1. Full acceptance to the JMU Teacher Education Program
2. Submission of:
 - Work Experience Verification Form
 - Professional Resume
 - Personal Statement
3. Demonstration and appreciation of professional dispositions
4. Successful completion of the following course work:

Code	Title	Credits
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
EDUC 200	Foundations of Education in the United States	3
PSYC 160	Life Span Human Development [C5SD] ¹	3
KIN 224	Skill Laboratory: Court and Field Games	3
KIN 303	Motor Development and Learning ¹	3
KIN 312	The Profession of Teaching Health and Physical Education ¹	3

¹ Minimum grade of "C" required in PSYC 160, KIN 303 and KIN 312.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement (in addition to General Education) ²	3
Scientific Literacy Requirement (in addition to General Education) ²	3-4
Kinesiology Core Courses	21
Major, Concentration, and Kinesiology Elective Requirements ³	46-55
University Electives	0-16
Total	120

- ¹ The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary.
- ² Quantitative and scientific literacy requirements may be fulfilled with major required courses.
- ³ The number of credits taken to meet major and concentration requirements may vary if courses are double-counted for General Education.

Major Requirements

Code	Title	Credits
Required Courses		
BIO 290	Human Anatomy	3
BIO 290L	Human Anatomy Laboratory	1
KIN 100	Lifetime Fitness and Wellness (2,2) [C5W]	3
KIN 201	Introduction to Kinesiology as a Profession	2
KIN 202	Biological Foundations of Kinesiology	3
KIN 303	Motor Development and Learning	3

NUTR 280	Nutrition for Wellness	3
MATH 220	Elementary Statistics [C3QR]	3
Subtotal		21
Kinesiology Concentrations		
Choose one of the following:		46-54
Exercise Science		
Physical and Health Education Teacher Education		
Subtotal		46-54
Total Credits		67-75

Exercise Science Concentration

The concentration prepares students to pursue careers in the fitness industry as well as graduate education in exercise science, pre-professional or an allied health field.

Students interested in athletic training, occupational therapy, physical therapy, medicine or other pre-professional health programs should review the Pre-Professional Health Advising section for prerequisite courses and recommendations for entrance to graduate professional health programs. Pre-professional health program coordinators are available to assist students with career planning and preparation.

Code	Title	Credits
Exercise Science Concentration Requirements		
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L] ¹	4
KIN 302 & 302L	Exercise Physiology and Exercise Physiology Laboratory	4
KIN 306 & 306L	Human Biomechanics and Human Biomechanics Laboratory	4
KIN 321 & 321L	Exercise Testing and Prescription and Exercise Testing and Prescription Laboratory	4
KIN 381	Practicum in Exercise Science	3
KIN 481	Internship in Exercise Science	4-12
MATH 205	Applied Calculus [C3QR] ¹	3
PHYS 140	College Physics I [C3PP] ¹	3
PHYS 140L	General Physics Laboratory I [C3L] ¹	1
Subtotal		34-42
Psychological/Behavioral Course		
Choose one of the following:		3
KIN 305	Psychological Foundations of Sport and Performance	
KIN 353	Developing Mental and Life Skills through Sport and Physical Education	
KIN 426	Psychological Foundations of Exercise and Health Behavior	
Subtotal		3
Departmental Electives		
Choose nine credit hours of the following:		9
KIN 304	Historical and Philosophical Foundations of Sport and Physical Activity	
KIN 329	Social Aspects of Sport	
KIN 424	Exercise and Nutrition in Chronic Disease	

KIN 425	Concepts of Strength and Conditioning	
KIN 428	Advanced Topics in Exercise Science and Leadership	
KIN 429	Special Topics in Adapted Physical Education	
KIN 430	Exercise Across the Lifespan	
KIN 431	Environmental Exercise Physiology	
KIN 432	Physiology of Endurance Performance	
KIN 433	Exercise, Cardiovascular Disease and ECG Assessment	
KIN 440	Exercise Immunology	
KIN 441	Strength and Conditioning II: Advanced Training Methods and Concepts	
KIN 442	Strength and Conditioning III: Philosophy, Leadership and Managerial Responsibilities	
KIN 443	Advanced Respiratory Physiology in Exercise Science	
KIN 490	Special Studies in Kinesiology and Recreation	
Subtotal		9
Total Credits		46-54

¹ This course may be counted for both General Education and major degree requirements.

Physical and Health Education Teacher Education Concentration

The Physical and Health Education Teacher Education (PHETE) concentration is committed to educating future leaders in the profession of teaching physical and health education. The program encourages the teacher candidate to advocate for effective educational changes, work collaboratively and collectively for social change, and to continue to grow professionally.

The PHETE concentration is intended for students who wish to pursue a Master of Arts in Teaching (M.A.T.) degree. Upon successful completion of the first four years, students earn a Bachelor of Science (B.S.) degree in kinesiology. It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For more information contact the PHETE coordinator, Dr. Susan Nye, Godwin Hall, Room 315, (540) 568-4865, nyesb@jmu.edu.

Concentration Requirements

Code	Title	Credits
EDUC 200	Foundations of Education in the United States	3
KIN 221	Skill Laboratory: Rhythmic and Dance Activities	3
KIN 222	Skill Laboratory: Teaching Fitness and Wellness in Schools	3
KIN 223	Skill Laboratory: Individual and Lifetime Activities	3
KIN 224	Skill Laboratory: Court and Field Games	3
KIN 225	Skill Laboratory: Wilderness and Adventure Education	3
KIN 309	School Health Content	3
KIN 312	The Profession of Teaching Health and Physical Education	3
KIN 353	Developing Mental and Life Skills through Sport and Physical Education	3
KIN 410	School Health Content for PHETE	3

KIN 417	Adapted Physical Education	3
KIN 418	Assessment in Elementary Physical Education	3
KIN 420	Instructional Methods in Elementary Physical Education	3
KIN 421	Elementary Curriculum in Physical Education	3
KIN 480	Student Teaching in Physical Education	8
LED 420	Content Area Literacy, K-12	2
PSYC 160	Life Span Human Development [C5SD]	3
Total Credits		55

PHETE Graduate Requirements

Information on the graduate requirements can be found on the page for the 5th-year M.A.T. in Education with a Concentration in Physical and Health Education Teacher Education (<https://catalog.jmu.edu/programs/education-phete-physical-health-education-teacher-education-concentration-mat/>).

Recommended Schedule for Exercise Science Concentration

Course	Title	Credits
First Year		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
KIN 100	Lifetime Fitness and Wellness (2,2) [C5W]	3
KIN 201	Introduction to Kinesiology as a Profession	2
MATH 220	Elementary Statistics [C3QR]	3
General Education		18
Credits		30
Second Year		
BIO 270	Human Physiology [C3NS]	3
BIO 270L	Human Physiology Laboratory [C3L]	1
BIO 290	Human Anatomy	3
BIO 290L	Human Anatomy Laboratory	1
KIN 202	Biological Foundations of Kinesiology	3
MATH 205	Applied Calculus [C3QR]	3
NUTR 280	Nutrition for Wellness	3
General Education requirement		10
Elective or minor courses		3
Credits		30
Third Year		
KIN 302	Exercise Physiology	3
KIN 302L	Exercise Physiology Laboratory	1
KIN 303	Motor Development and Learning	3
KIN 306	Human Biomechanics	3
KIN 306L	Human Biomechanics Laboratory	1
KIN 321 & 321L	Exercise Testing and Prescription and Exercise Testing and Prescription Laboratory	4
KIN 381	Practicum in Exercise Science	3
PHYS 140	College Physics I [C3PP]	3
PHYS 140L	General Physics Laboratory I [C3L]	1
Minor or general elective courses		8
Credits		30
Fourth Year		
One psychological/behavioral course from list above		3
KIN 481	Internship in Exercise Science	4
KIN electives		9

Minor or general elective courses	14
Credits	30
Total Credits	120

Recommended Schedule for PHETE Concentration

Course	Title	Credits
First Year		
KIN 100	Lifetime Fitness and Wellness (2,2) [C5W]	3
KIN 201	Introduction to Kinesiology as a Profession	2
General Education		25
Credits		30
Second Year		
BIO 290	Human Anatomy	3
BIO 290L	Human Anatomy Laboratory	1
KIN 202	Biological Foundations of Kinesiology	3
KIN 353	Developing Mental and Life Skills through Sport and Physical Education	3
General Education		8
Electives		7
PSYC 160	Life Span Human Development [C5SD]	3
Credits		28
Third Year		
Fall Semester		
EDUC 200	Foundations of Education in the United States	3
KIN 224	Skill Laboratory: Court and Field Games	3
KIN 303	Motor Development and Learning	3
KIN 312	The Profession of Teaching Health and Physical Education	3
NUTR 280	Nutrition for Wellness	3
Credits		15
Spring Semester		
KIN 222	Skill Laboratory: Teaching Fitness and Wellness in Schools	3
KIN 225	Skill Laboratory: Wilderness and Adventure Education	3
KIN 309	School Health Content	3
LED 420	Content Area Literacy, K-12	2
MATH 220	Elementary Statistics [C3QR]	3
Electives		1
Credits		15
Fourth Year		
Fall Semester		
KIN 223	Skill Laboratory: Individual and Lifetime Activities	3
KIN 421	Elementary Curriculum in Physical Education	3
KIN 417	Adapted Physical Education	3
KIN 420	Instructional Methods in Elementary Physical Education	3
Electives		3
Credits		15
Spring Semester		
KIN 221	Skill Laboratory: Rhythmic and Dance Activities	3
KIN 410	School Health Content for PHETE	3
KIN 418	Assessment in Elementary Physical Education	3
KIN 480	Student Teaching in Physical Education	8
Credits		17
Total Credits		120

Recommended Schedule for Accelerated Pathway to Athletic Training Students

After completing the 3 years below, students segue into the MSAT program. See the Athletic Training, M.S.A.T. page (<https://catalog.jmu.edu/programs/athletic-training-msat/>) for more information.

Course	Title	Credits
First Year		
Fall Semester		
KIN 100	Lifetime Fitness and Wellness (2,2) [C5W]	3
WRTC 103	Rhetorical Reading and Writing [C1W]	3
General Education courses, including American Experience		10
Credits		16
Spring Semester		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
HTH 210	Medical Terminology	3
KIN 201	Introduction to Kinesiology as a Profession	2
MATH 220	Elementary Statistics [C3QR]	3
PSYC 101	General Psychology [C5SD]	3
General Education		3
Credits		18
Second Year		
Fall Semester		
BIO 290	Human Anatomy	3
BIO 290L	Human Anatomy Laboratory	1
KIN 202	Biological Foundations of Kinesiology	3
MATH 205	Applied Calculus [C3QR]	3
General Education		6
Credits		16
Spring Semester		
ATEP 205	Introduction to Athletic Training	3
BIO 270	Human Physiology [C3NS]	3
BIO 270L	Human Physiology Laboratory [C3L]	1
NUTR 280	Nutrition for Wellness	3
PHYS 140	College Physics I [C3PP]	3
PHYS 140L	General Physics Laboratory I [C3L]	1
Credits		14
Third Year		
Fall Semester		
KIN 302	Exercise Physiology	3
KIN 302L	Exercise Physiology Laboratory	1
KIN 303	Motor Development and Learning	3
KIN 321	Exercise Testing and Prescription	3
KIN 321L	Exercise Testing and Prescription Laboratory	1
KIN 381	Practicum in Exercise Science	3
KIN electives		3
Credits		17
Spring Semester		
KIN 306	Human Biomechanics	3
KIN 306L	Human Biomechanics Laboratory	1
KIN 481	Internship in Exercise Science	4-12
KIN psychology requirement (see above)		3
KIN electives		3
General Education		3
Credits		17-25
Total Credits		98-106

Accelerated Pathway to the Master of Science in Athletic Training

Kinesiology majors who choose a concentration in exercise science and are interested in an athletic training career are eligible to apply to an accelerated pathway (3+2) leading to the Master of Science in Athletic Training (MSAT) degree. The purpose of this pathway is to prepare students to become certified athletic trainers with the clinical knowledge, skills and professional attitudes that will enable them to deliver health care effectively in various clinical settings. Highly motivated and academically talented students will complete both the undergraduate degree in kinesiology and graduate degree in athletic training programs in 5 years, compared to the traditional 6 years for completing both degrees. Students interested in this pathway must complete a very specific set of course requirements as part of the kinesiology degree. See those details below.

Admission and Application Requirements

Students interested in the accelerated pathway will begin their undergraduate studies with a focused curriculum designed to meet the prerequisites for their graduate program while meeting their major requirements. Upon successful completion of the required coursework at a high academic standard, students will transition into the graduate program.

Interested students will need to:

1. Declare an undergraduate major in kinesiology with a concentration in exercise science.
2. Attain admission into the exercise science concentration (see admission policy above).
3. Discuss their intent to pursue the 3+2 accelerated pathway to the Master of Science in Athletic Training with their academic advisor and pre-professional health advisor (if applicable).
4. Submit their declaration of intent for pursuing the 3+2 accelerated pathway to the Master of Science in Athletic Training. Ideally, this declaration is made as a first-year student at JMU. Submitting the declaration of intent in the first year allows adequate time to plan and schedule year two and year three courses.

Students pursuing the 3+2 accelerated pathway to the Master of Science in Athletic Training are required to meet the following criteria for admission:

- Preferred GPA of 2.7 or higher
- Successful completion of the following MSAT prerequisite courses with a minimum grade of "C-" by the end of the third year. The JMU courses in parentheses are part of the exercise science concentration in kinesiology and fulfil the prerequisite requirements:
 - General Psychology (PSYC 101)
 - Nutrition (NUTR 280)
 - Statistics or Research Methods (MATH 220)
 - Human or Mammalian Anatomy & Physiology BIO 270/BIO 270L;and BIO 290/BIO 290L)
 - Physics or Biomechanics (PHYS 140/PHYS 140L or KIN 306/KIN 306L)
 - General Chemistry (CHEM 131/CHEM 131L)
 - Exercise Physiology (KIN 302/KIN 302L)

- Students who are uncertain whether a course meets prerequisites should email athletictraining@jmu.edu and include a copy of the syllabus from the semester they took the course.
- The following courses are not prerequisites but are highly recommended: ATEP 205, ATEP 306
- Submission of formal application to the Master of Science in Athletic Training program through ATCAS and the Graduate School of JMU by September 1 of the third year.
- Successful completion of an admission interview during September of the third year.
- Successful completion of major courses by the end of the third year.

Program Structure

The 3+2 accelerated pathway to the Master of Science in Athletic Training allows undergraduate students to enroll in and apply 21-22 graduate credit hours toward the completion of the B.S. in Kinesiology. For a list of courses, please see the MSAT page (<https://catalog.jmu.edu/programs/athletic-training-msat/>).

In their first year, students meet with academic major advisors and indicate interest in the accelerated pathway, declaring their intent to pursue this pathway, while beginning their major coursework. In the fall of their third year, students apply to the MSAT program. Students must complete all undergraduate major and prerequisite coursework by the end of the spring semester of their third year at JMU. Students admitted to the accelerated MSAT program will begin their graduate coursework in May after their third year at JMU. Admitted students take 21-22 credit hours at the graduate level during the summer after their third year and the fall of their fourth year. These 21-22 graduate credits will be applied to their undergraduate transcript to fulfill the remaining credits needed for the undergraduate degree. Students enrolled in graduate coursework are required to adhere to The Graduate School Academic Requirements for progression. The undergraduate degree will be conferred at the end of the fall semester of year 4, allowing students to participate in December commencement. Students transition to graduate student status for the spring of the fourth year and complete the remaining courses in the graduate program over the following semesters. Upon successful completion of all courses in the MSAT program, students will earn their master's degree at the end of their fifth year at JMU. Students will participate in the May commencement.

Students pursuing the accelerated pathway should meet with their academic advisor each semester to discuss their progress toward undergraduate degree completion and staying on track with MSAT prerequisites to ensure prerequisite and graduate requirements are being met.

Marketing, B.B.A.

Program Description

Marketing majors conform to the general structure of the B.B.A. degree program. The B.B.A. degree in marketing requires a minimum of 120 credit hours of undergraduate course work.

Students planning to major in marketing must complete the 27-28 hour, lower-level B.B.A. core requirements (see the College of Business section for the list of courses) prior to enrolling in upper-level B.B.A. core and marketing courses. Upper-level courses normally begin in the first semester of the junior year. The lower-level core curriculum should be completed during the first two years of study along with most of the university General Education curriculum. Failing to complete all lower-level B.B.A. core requirements on time will delay enrollment in upper-level core and major courses.

Admission and Retention Requirements

College of Business Progression Standards

Any student admitted to JMU can declare any major offered by the College of Business. However, students must make satisfactory progress in order to be formally admitted into the College of Business. Requirements to continue pursuing a B.B.A. degree, often referred to as progression standards or formal admission, can be found on the College of Business catalog page (p. 15).

Optional Marketing Concentration Declaration Deadline

For students who plan to pursue an optional marketing concentration in either Professional Sales or Digital Marketing, students must declare it in MyMadison by October 1 (to register for spring classes) and March 1 (to register for fall classes).

Transfer Credit Policy

The marketing program will accept no more than two marketing courses for transfer credit toward the major. These courses must be from AACSB institutions. In addition to this general College of Business policy, there are restrictions on which courses will be accepted for transfer credit.

Students must take the following courses at JMU:

Code	Title	Credits
MKTG 482	Marketing Analytics	3
MKTG 485	Marketing Management	3

Degree Requirements

Degree Requirements	Credit Hours
General Education requirements ¹	41
B.B.A. lower-level core courses	27-28
B.B.A. upper-level core courses	18
Marketing major requirements	27
Electives	7-13
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary.

Major Requirements

The marketing curriculum consists of 27 credit hours in marketing. The required courses equip students with the knowledge and skills all marketers should have. Elective courses allow students to pursue areas of special interest and prepare for specific career tracks in marketing. Students may choose to declare and complete an optional concentration in Professional Sales or Digital Marketing based on their selection of marketing electives.

Code	Title	Credits
Marketing Curriculum Required Core Courses		
MKTG 384	Integrated Marketing Communications	3
MKTG 385	Consumer Behavior	3
MKTG 430	Professional Selling	3
MKTG 482	Marketing Analytics	3
MKTG 485	Marketing Management	3
Subtotal		15
Concentration or Elective Courses		
Beyond the 15 credit hours of core courses, students must take an additional 12 credits of MKTG electives. Concentration-focused MKTG electives may count toward both the Marketing Elective requirement and an optional concentration.		12
Subtotal		12
Total Credits		27

Marketing Electives

Code	Title	Credits
Choose a total of 12 credits from the following options:		12
MKTG 331	Creative Marketing Design: Digital Vector Techniques ¹	
MKTG 332	Creative Marketing Design: Visual Editing and Document Styling ¹	
MKTG 386	Services Marketing	
MKTG 388	Omnichannel Retailing	
MKTG 410	Social Media Marketing ¹	
MKTG 450	Business-to-Business (B2B) Marketing ²	
MKTG 455	Sales Leadership ²	
MKTG 460	Global Marketing	
MKTG 462	International Logistics	
MKTG 465	Customer Relationship Management ²	
MKTG 466	Advanced Professional Selling ²	
MKTG 470	Strategic Internet Marketing ¹	
MKTG 477	Digital Marketing Practicum ¹	
MKTG 480	Product Development and Management	
MKTG 490	Special Studies in Marketing	
MKTG 494	Marketing Internship	
MKTG 498	Special Topics in Marketing	
MKTG 499C	Senior Honors Project	
Total Credits		12

¹ These digital marketing-focused electives may be applied toward the digital marketing concentration.

² These sales-focused marketing electives may be applied toward the professional sales concentration.

Concentrations

The Department of Marketing offers two optional concentrations. Students are not required to select a concentration but may declare a concentration in MyMadison if desired by October 1 (to register for spring classes) and March 1 (to register for fall classes).

Digital Marketing Concentration

The digital marketing sector is for students interested in paid search, display advertising, search engine optimization, content marketing, email marketing and social media. Students will be prepared to complete industry-recognized certifications and will learn to use appropriate industry tools such as website builders, email service providers, content creation tools, social media planning dashboards and keyword planning tools. Students will learn the development process for digital marketing platforms and to connect the creation of materials to marketing strategies and measurable objectives. Courses in this optional concentration build on the content of MKTG 384 so students should complete this course as soon as possible after completing COB 300.

Students must complete at least six credits of digital marketing-focused electives to complete the digital marketing concentration.

Code	Title	Credits
Digital Marketing-Focused Electives		
Choose six credits from the following:		6
MKTG 331	Creative Marketing Design: Digital Vector Techniques	
MKTG 332	Creative Marketing Design: Visual Editing and Document Styling	
MKTG 410	Social Media Marketing	
MKTG 470	Strategic Internet Marketing	
MKTG 477	Digital Marketing Practicum	

Professional Sales Concentration

This concentration prepares students for careers in professional sales that focus on business customers and, in most cases, on the selling of relatively complex and technical products and services.

Courses in this concentration build on the content of MKTG 430. Students should complete this course as soon as possible after completing COB 300.

Students must complete at least six credits of sales-focused marketing electives to complete the professional sales concentration.

Code	Title	Credits
Sales-Focused Marketing Electives		
Choose six credits from the following:		6
MKTG 450	Business-to-Business (B2B) Marketing	
MKTG 455	Sales Leadership	
MKTG 465	Customer Relationship Management	
MKTG 466	Advanced Professional Selling (a required sales-focused marketing elective for students pursuing the professional sales concentration)	
Total Credits		6

schedule, but care must be taken to ensure that all course prerequisites are met.

Course	Title	Credits
First Year		
General Education courses and B.B.A. Core requirements		30
Credits		30
Second Year		
General Education courses and B.B.A. Core requirements		30
Credits		30
Third Year		
First Semester		
COB 300A	Integrative Business: Management	3
COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
Electives		3
Credits		15
Second Semester		
MKTG 384	Integrated Marketing Communications	3
MKTG 430	Professional Selling	3
COB 318	Legal and Ethical Environment of Business	3
Electives		6
Credits		15
Fourth Year		
First Semester		
MKTG 385	Consumer Behavior	3
MKTG 482	Marketing Analytics	3
MKTG electives and/or concentration courses		6
Electives		3
Credits		15
Second Semester		
COB 487	Strategic Management	3
MKTG 485	Marketing Management	3
MKTG electives, concentration courses and/or university electives		9
Credits		15
Total Credits		120

Recommended Schedule for Majors

Marketing majors should follow the course schedule below to complete the final two years of their program. It is possible to deviate from this

Mathematics, B.A.

Program Description

The mathematics major provides a program of study with core fundamentals along with electives that prepare students for a wide variety of careers, including jobs in industry, careers in teaching and further study in graduate school. The core courses provide a solid foundation of critical thinking and mathematical concepts, while the various elective courses provide a breadth of knowledge in relevant topics. Our diverse faculty have research interests in a number of mathematical disciplines, and provide undergraduate research opportunities in those disciplines beyond the course requirements for the major.

Credit by Examination

The Department of Mathematics offers credit by examination for some of the courses taught in the department. Students who want to take an examination must apply to the department head. Upon application students will receive details regarding approval to take the examination.

Degree Requirements

Degree Requirements	Credit Hours
General Education ^{1,2}	41
Foreign Language classes (intermediate level required)	0-12
Philosophy course (in addition to General Education)	3
Major requirements	43-45
University electives	21-35
Total	120

¹ MATH 231 or MATH 235 must be included and students seeking secondary teaching licensure must include PSYC 160.

² The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

Major Requirements

Mathematics requirements depend on whether or not the student is seeking secondary teaching licensure. All students must complete 31-33 credit hours of the following required core mathematics courses and 12 credit hours of mathematics courses beyond the core.

Code	Title	Credits
Core Courses		
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 245	Discrete Mathematics	3
MATH 248	Computers and Numerical Algorithms	4
MATH 300	Linear Algebra ¹	3
MATH 336	Elementary Differential Equations ¹	3
MATH 410	Advanced Calculus I	3
MATH 430	Abstract Algebra I	3
MATH 318	Introduction to Probability and Statistics	4-6

or MATH 229 & MATH 329	Introduction to Applied Statistics Using R [C3QR] and Introduction to Probability and Theoretical Statistics	
Subtotal		31-33
Additional Requirements		
All students must take an additional 12 credit hours of mathematics courses beyond the core. There are two options available for students who are not seeking secondary teaching licensure, and one option available for students seeking secondary teaching licensure. These options are listed below.		12
Total Credits		43-45

¹ MATH 238 may be substituted for MATH 300 and MATH 336 if the student is completing a double major, seeking the computational sciences concentration, or received credit for MATH 238 before declaring a math major.

² MATH 229 and MATH 329 is recommended. For students choosing MATH 318, a course with statistical data analysis (MATH 321, MATH 322, MATH 324, MATH 327 or MATH 354) is recommended.

Additional Requirements for Students Not Seeking Secondary Teaching Licensure

Students not seeking secondary teaching licensure track must complete (in addition to the required core courses) one of the following options, each consisting of 12 credit hours of mathematics courses.

The option chosen and the courses chosen to satisfy an option by a student are made in consultation with the student's advisor and are dependent upon the student's interests and career objectives. Students interested in pursuing graduate studies in mathematics are strongly urged to complete both MATH 411 and MATH 431.

Code	Title	Credits
Choose one of the following options:		12
<i>Option 1:</i>		
9 credits of MATH electives at MATH 310 or above ¹		
MATH 411	Advanced Calculus II	
or MATH 431	Abstract Algebra II	
or MATH 434	Advanced Linear Algebra	
or MATH 431	Introduction to Topology	
<i>Option 2:</i>		
6 credits of MATH electives at MATH 310 or above		
Choose two of the following:		
MATH 440	Fourier Analysis and Partial Differential Equations	
MATH 441	Analysis and Dynamics of Differential Equations	
MATH 448	Numerical Analysis	
MATH 449	Numerical Analysis for Differential Equations	
Or take both of the following courses:		
MATH 426	Probability and Mathematical Statistics I	
MATH 427	Probability and Mathematical Statistics II	
Total Credits		12

¹ No more than six of the nine hours may be MATH 486 and/or MATH 497

Additional Requirements for Students Seeking Secondary Teaching Licensure

Students seeking secondary teaching licensure earn the Bachelor of Arts or Bachelor of Sciences degree and then complete the Master of Arts in Teaching degree.

It is necessary to be admitted to the teacher education program prior to enrolling in pre-professional education courses. For a full description of the program in secondary education, refer to the College of Education, Department of Middle, Secondary and Mathematics Education.

Students seeking secondary teaching licensure must (in addition to the required core courses):

- Complete the 22-24 credit hours that comprise the pre-professional education program in secondary education.
- Complete 12 credit hours of mathematics courses:

Code	Title	Credits
MATH 310	Elementary Number Theory	3
or MATH 315	Real Number System	
MATH 415	History of Mathematics	3
MATH 470	Connections in Mathematics	3
MATH 475	Fundamental Concepts of Geometry	3
Total Credits		12

Recommended Schedule for Mathematics Majors Not Seeking Secondary Licensure

Course	Title	Credits
First Year		
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
Discrete Mathematics		3
General Education requirement		16
Credits		30
Second Year		
MATH 237	Calculus III	4
MATH 248	Computers and Numerical Algorithms	4
MATH 300	Linear Algebra	3
MATH 329	Introduction to Probability and Theoretical Statistics	3
MATH 336	Elementary Differential Equations	3
General Education requirement / Electives		13
Credits		30
Third Year		
General Education requirement / Electives		21
Mathematics required or elective courses ¹		6
MATH 410	Advanced Calculus I	3
or MATH 430	or Abstract Algebra I	
Credits		30
Fourth Year		
Mathematics required or elective courses ¹		9
Electives		18
MATH 410	Advanced Calculus I	3
or MATH 430	or Abstract Algebra I	
Credits		30
Total Credits		120

¹ Mathematics required or elective courses numbered MATH 310 or above

Recommended Schedule for Majors Seeking Secondary Licensure

Course	Title	Credits
First Year		
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 245	Discrete Mathematics	3
General Education requirement		16
Credits		30
Second Year		
MATH 237	Calculus III	4
MATH 248	Computers and Numerical Algorithms	4
MATH 300	Linear Algebra	3
MATH 329	Introduction to Probability and Theoretical Statistics	3
MATH 336	Elementary Differential Equations	3
Pre-professional education requirements		6
General Education		7
Credits		30
Third Year		
MATH 430	Abstract Algebra I	3
Pre-professional education requirements and General Education requirement / Electives		21
Choose two of the following:		6
MATH 310	Elementary Number Theory	
MATH 315	Real Number System	
MATH 410	Advanced Calculus I	
MATH 415	History of Mathematics	
MATH 475	Fundamental Concepts of Geometry	
Mathematics electives ¹		
Credits		30
Fourth Year		
Pre-professional education requirements and General Education requirement / Electives		21
Choose three of the following:		9
MATH 310	Elementary Number Theory	
MATH 315	Real Number System	
MATH 410	Advanced Calculus I	
MATH 415	History of Mathematics	
MATH 470	Connections in Mathematics	
MATH 475	Fundamental Concepts of Geometry	
Mathematics electives ¹		
Credits		30
Total Credits		120

¹ Mathematics electives numbered MATH 310 or above

Mathematics, B.S.

Program Description

The mathematics major provides a program of study with core fundamentals along with electives that prepare students for a wide variety of careers, including jobs in industry, careers in teaching and further study in graduate school. The core courses provide a solid foundation of critical thinking and mathematical concepts, while the various elective courses provide a breadth of knowledge in relevant topics. Our diverse faculty have research interests in a number of mathematical disciplines, and provide undergraduate research opportunities in those disciplines beyond the course requirements for the major.

Credit by Examination

The Department of Mathematics offers credit by examination for some of the courses taught in the department. Students who want to take an examination must apply to the department head. Upon application students will receive details regarding approval to take the examination.

Degree Requirements

Degree Requirements	Credit Hours
General Education ^{1,2}	41
Quantitative Requirement (in addition to General Education) ³	3
Scientific Literacy requirement (in addition to General Education)	3-4
University electives	32-35
Major requirements (listed below)	43-45
Total	120

¹ MATH 231 or MATH 235 must be included and students seeking secondary teaching licensure must include PSYC 160.

² The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

³ The quantitative requirement is automatically fulfilled by completion of the major requirements.

Major Requirements

Mathematics requirements depend on whether or not the student is seeking secondary teaching licensure. All students must complete 31-33 credit hours of the following required core mathematics courses and 12 credit hours of mathematics courses beyond the core.

Code	Title	Credits
Core Courses		
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 245	Discrete Mathematics	3
MATH 248	Computers and Numerical Algorithms	4
MATH 300	Linear Algebra ¹	3
MATH 336	Elementary Differential Equations ¹	3
MATH 410	Advanced Calculus I	3
MATH 430	Abstract Algebra I	3
MATH 318	Introduction to Probability and Statistics	4-6

or MATH 229 & MATH 329	Introduction to Applied Statistics Using R [C3QR] and Introduction to Probability and Theoretical Statistics	
Subtotal		31-33
Additional Requirements		
All students must take an additional 12 credit hours of mathematics courses beyond the core. There are two options available for students who are not seeking secondary teaching licensure, and one option available for students seeking secondary teaching licensure. These options are listed below.		12
Total Credits		43-45

¹ MATH 238 may be substituted for MATH 300 and MATH 336 if the student is completing a double major, seeking the computational sciences concentration, or received credit for MATH 238 before declaring a math major.

² MATH 229 and MATH 329 is recommended. For students choosing MATH 318, a course with statistical data analysis (MATH 321, MATH 322, MATH 324, MATH 327 or MATH 354) is recommended.

Additional Requirements for Students Not Seeking Secondary Teaching Licensure

Students not seeking secondary teaching licensure track must complete (in addition to the required core courses) one of the following options, each consisting of 12 credit hours of mathematics courses.

The option chosen and the courses chosen to satisfy an option by a student are made in consultation with the student's advisor and are dependent upon the student's interests and career objectives. Students interested in pursuing graduate studies in mathematics are strongly urged to complete both MATH 411 and MATH 431.

Code	Title	Credits
Choose one of the following options:		12
Option 1:		
9 credits of MATH electives at MATH 310 or above ¹		
MATH 411	Advanced Calculus II	
or MATH 431	Abstract Algebra II	
or MATH 434	Advanced Linear Algebra	
or MATH 436	Introduction to Topology	
Option 2:		
6 credits of MATH electives at MATH 310 or above		
Choose two of the following:		
MATH 440	Fourier Analysis and Partial Differential Equations	
MATH 441	Analysis and Dynamics of Differential Equations	
MATH 448	Numerical Analysis	
MATH 449	Numerical Analysis for Differential Equations	
Or take both of the following courses:		
MATH 426	Probability and Mathematical Statistics I	
MATH 427	Probability and Mathematical Statistics II	
Total Credits		12

¹ No more than six of the nine hours may be MATH 486 and/or MATH 497

Additional Requirements for Students Seeking Secondary Teaching Licensure

Students seeking secondary teaching licensure earn the Bachelor of Arts or Bachelor of Sciences degree and then complete the Master of Arts in Teaching degree.

It is necessary to be admitted to the teacher education program prior to enrolling in pre-professional education courses. For a full description of the program in secondary education, refer to the College of Education, Department of Middle, Secondary and Mathematics Education.

Students seeking secondary teaching licensure must (in addition to the required core courses):

- Complete the 22-24 credit hours that comprise the pre-professional education program in secondary education.
- Complete 12 credit hours of mathematics courses:

Code	Title	Credits
MATH 310	Elementary Number Theory	3
or MATH 315	Real Number System	
MATH 415	History of Mathematics	3
MATH 470	Connections in Mathematics	3
MATH 475	Fundamental Concepts of Geometry	3
Total Credits		12

Recommended Schedule for Majors Seeking Secondary Licensure

Course	Title	Credits
First Year		
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 245	Discrete Mathematics	3
General Education		16
Credits		30
Second Year		
MATH 237	Calculus III	4
MATH 248	Computers and Numerical Algorithms	4
MATH 300	Linear Algebra	3
MATH 329	Introduction to Probability and Theoretical Statistics	3
MATH 336	Elementary Differential Equations	3
Pre-professional education requirements		6
General Education		7
Credits		30
Third Year		
MATH 430	Abstract Algebra I	3
Pre-professional education requirements and General Education requirement / Electives		21
Choose two of the following:		6
MATH 310	Elementary Number Theory	
MATH 315	Real Number System	
MATH 410	Advanced Calculus I	
MATH 415	History of Mathematics	
MATH 475	Fundamental Concepts of Geometry	
Mathematics electives ¹		
Credits		30
Fourth Year		
Pre-professional education requirements and General Education requirement / Electives		21

Choose three of the following:	9
MATH 310	Elementary Number Theory
MATH 315	Real Number System
MATH 410	Advanced Calculus I
MATH 415	History of Mathematics
MATH 470	Connections in Mathematics
MATH 475	Fundamental Concepts of Geometry
Mathematics electives ¹	
Credits	30
Total Credits	120

¹ Mathematics electives numbered MATH 310 or above

Recommended Schedule for Mathematics Majors Not Seeking Secondary Licensure

Course	Title	Credits
First Year		
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
Discrete Mathematics		3
General Education requirement		16
Credits		30
Second Year		
MATH 237	Calculus III	4
MATH 248	Computers and Numerical Algorithms	4
MATH 300	Linear Algebra	3
MATH 329	Introduction to Probability and Theoretical Statistics	3
MATH 336	Elementary Differential Equations	3
General Education requirement / Electives		13
Credits		30
Third Year		
General Education requirement / Electives		18
Mathematics required or elective courses ¹		9
MATH 410	Advanced Calculus I	3
or MATH 430	or Abstract Algebra I	
Credits		30
Fourth Year		
Mathematics required or elective courses ¹		6
Electives		21
MATH 410	Advanced Calculus I	3
or MATH 430	or Abstract Algebra I	
Credits		30
Total Credits		120

¹ Mathematics required or elective courses numbered MATH 310 or above

Media Arts and Design, B.A.

Program Description

The School of Media Arts and Design incorporates the study of mass media and mediated visual and aural expression, focusing on creative advertising, video and cinema production, interactive design, journalism, film studies, and creative writing. SMAD offers a program of study that prepares students for careers in the media and/or corporate environments that rely on mass media and related areas of expertise, or graduate study in mass communication and related areas.

Admission and Minimum Grades

Special Admission and Retention Requirements

Admission to the Media Arts and Design, B.A. is limited and competitive. There are two pathways to full admission to the media arts and design major.

Direct Admission

Entering first year and transfer students who are awarded a competitive scholarship by the university, the College of Arts and Letters, or by the School of Media Arts and Design will be fully and directly admitted to the media arts and design major. Those scholarships include the Dingledine-Bluestone Scholarships; the Senator Harry S. Byrd Jr.—D. Lathan Mims Renewable Scholarship; and the Alison B. Parker Memorial Renewable Scholarship. Their concentration will be determined by the SMAD Director, in consultation with faculty. Such scholarship holders are exempt from the admission application requirement but must fulfill any conditions attached to their awards.

Admission by Application

Students interested in majoring in the programs offered by the school must apply for a limited number of spaces in the major. An admission committee in the school reviews applications and offers admission based on availability to the most qualified students. A student may apply no more than two times for admission to the school.

September Application Period

The September application process is only for transfer students matriculating to JMU in the current fall semester or the immediately preceding spring or summer semesters and have declared the media arts and design major. These students must be officially admitted to JMU as transfer students and must have earned at least 30 hours of JMU approved post high school college credit. They must be degree-seeking students and must be enrolled in at least 12 hours at JMU during the fall semester. The application is due at 4 p.m. on the third Friday of the fall semester. The exact date will be posted on the SMAD website.

Annual Application Period

Currently enrolled JMU students (including transfer students matriculating in January or in previous semesters) may apply during the annual application period that occurs in the spring semester. Students applying to the major at JMU must be degree-seeking and have completed at least 12 hours or be enrolled in at least 12 hours at JMU. The application is due at 4 p.m. on the third Friday of the spring semester. The exact date will be posted on the SMAD website (<https://www.jmu.edu/smad/index.shtml>).

Process

Students interested in applying for admission to this major must "declare" media arts and design as a major and apply to become a "fully admitted" major. Declaring a media arts and design major does not guarantee the student will be fully admitted into the media arts and design major. Students must declare media arts and design as a major prior to being eligible to enroll in SMAD 101. A declared media arts and design student may take SMAD 101, but the student is restricted from SMAD courses beyond SMAD 101 until being fully admitted into the major.

A complete application to the major includes the following:

- Completion of SMAD 101 or enrollment in SMAD 101. Students must earn a minimum grade of "C-" (minimum 70%) in SMAD 101 or they will have to retake the course. Students get only two attempts to earn a minimum of 70% in the course.
- Completion of the SMAD Admission Test with sections on English grammar and usage, timed writing samples and basic computer knowledge.
- Submission of all required components on the "SMAD Admissions" Canvas course:
 - An application letter/personal essay expressing reasons for interest in your first-choice concentration. If you wish to be considered for a second-choice concentration, submit a separate letter specifically for that concentration. (A student will only be accepted into one concentration.)
 - Unofficial transcript(s). (Transfer students must include unofficial transcripts from previous institutions and JMU Transfer Credit Report.)
 - A SMAD Admissions Questionnaire on Canvas that asks about media creation and experience.
 - A SMAD Contract on Canvas.

Following review by the SMAD admission committee, students will be notified about their acceptance into the major. If accepted, students will be eligible to register for SMAD courses beyond SMAD 101. Students not accepted into the major will be notified and may reapply the following January or a later January. Students may apply to the major no more than two times.

Change of Concentration

Students accepted into the major are assigned to a concentration. Students who wish to change concentrations after being accepted into the major may ask to do so only during the annual application period, which ends at 4 p.m. on the third Friday of the spring semester.

To request a different concentration, the student must submit on the "SMAD Admissions" Canvas course the following:

1. A letter explaining why the student is seeking that change. This letter should be addressed to the Admissions Committee and be no more than two pages in length.
2. A completed Concentration Change Application form, which the student can download from the "SMAD Admissions" Canvas course and fill out after consulting with their advisor and obtaining the advisor's signature. The student must scan and upload the finished form on the Canvas site.
3. The student's unofficial transcripts.

A student seeking a concentration change must request access to the "SMAD Admissions" Canvas course. Directions for doing so are posted on SMAD's website.

If the Admissions Committee approves the student's request, the application will be reviewed by the SMAD director, who will grant final permission to change concentrations. There are no guarantees that the change will be granted, and students who change concentrations will not receive overrides or prerequisite waivers to expedite graduation.

Minimum Grades

If a student enrolled in SMAD 101 is accepted into the major and makes an "F" in the course, the student will be dropped from the major. To be eligible to reapply for the major, the student must re-take SMAD 101 and reapply. If a student enrolled in SMAD 101 is accepted into the major and makes a "D+," "D" or "D-" in the course, the student may continue to take classes in the major; however, the student must retake SMAD 101 the next semester and earn a minimum grade of "C-". Students are limited to taking SMAD 101 twice. Failure to achieve a minimum grade of "C-" after the second time will result in the student being dropped from the major, including being dropped from any SMAD courses for which the student is preregistered, and the student may not reapply for admission to the major.

To graduate with a major in media arts and design, a student must have a minimum grade point average of 2.0 ("C") in the major.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ^{2,3}	0-12
Philosophy course (in addition to General Education courses) ³	3
Major requirements ⁴	48
University electives	16-28
Total	120

¹ The General Education program contains a set of requirements each student must fulfill.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

³ The foreign language and philosophy requirements may be satisfied by courses in the breadth requirement for the major, except for PHIL 120 and PHIL 150.

⁴ A student must complete a minimum of 48 credits for the media arts and design major (42 SMAD credits and 6 non-Media Arts and Design, B.A. breadth credits). In addition to courses in the School of Media Arts and Design, students must complete at least 72 non-SMAD credits.

Major Requirements

Code	Title	Credits
Core Requirements		
SMAD 101	Introduction to Media Arts and Design	3
SMAD 201	Foundations of Visual Communication Design	3

SMAD 202	Foundations of Audio-Visual Storytelling	3
SMAD 203	Foundations of User Experience Design	3
SMAD 204	Foundations of Media Storytelling and Research	3
SMAD 301	The Media Arts: Culture by Design	3
SMAD 400	Senior Assessment in Media Arts and Design	0
SMAD 450	Law of Media Arts and Design	3
Subtotal		21

Breadth Requirement

To be accomplished visual storytellers, students seeking a degree in media arts and design require extra breadth in foreign language and philosophy. Students seeking a Bachelor of Arts degree in media arts and design must complete any foreign language class between course numbers 212 or 232-399. In addition, those students must complete any 100-300 level philosophy course or PHIL 466 or PHIL 468. ¹	6
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Concentration

Media arts and design majors apply for and are assigned to one of four concentrations listed below (creative advertising, digital video and cinema, interactive design, journalism), and must complete all requirements for that concentration. Students are restricted from taking classes in other concentrations unless there is room and permission is granted.	21
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Total Credits	48
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¹ Classes that satisfy the breadth requirements for media arts and design that qualify as satisfying the university's degree requirements will also fulfill the university degree requirements. (PHIL 120 and PHIL 150 do not fulfill the university's degree requirements for philosophy.) In no case may a student double-count a general education class toward the university B.A. degree requirements.

Media Arts and Design Concentrations

Creative Advertising Concentration

This professionally-oriented concentration provides students with the knowledge and skills to create and manage advertising programs emphasizing new media. In addition to the theoretical concepts underlying advertising, students learn strategic creative writing, effective message design and media production. Students also learn strategic planning, implementation and evaluation of new media technologies, and research and analysis of current trends and applications. The program reviews ethical and legal issues involved in the creative advertising process and the use of new media. Students planning careers in creative advertising should obtain a broad liberal arts education to better understand the characteristics that make the diversified communication process effective across various media and organizations.

Code	Title	Credits
Required Courses		
SMAD 242	Introduction to Advertising and New Media	3
SMAD 252	Copywriting for Advertising ¹	3
SMAD 342	Elements of Creative Advertising	3
SMAD 443	Creative Advertising Campaigns	3
Subtotal		12

Applied Skills Courses

Choose one of the following:	3
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SMAD 210	News Reporting and Writing	
SMAD 220	Copy Editing	
SMAD 225	Photojournalism	
SMAD 305	Special Topics in Media Production ²	
SMAD 307	Interactive Design Web I	
SMAD 310	Advanced Reporting and Writing	
SMAD 315	Digital Social Media Advertising	
SMAD 327	User-Centered Research and Usability Evaluation	
SMAD 332	Visual Communication Design	
SMAD 497	Advanced Projects in Media Arts and Design	
Subtotal		3
Critical Analysis Courses		
Choose two of the following:		6
SMAD 321	The Media Arts: Culture by Design in London ³	
SMAD 360	British Media and Society ³	
SMAD 372	Media History	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design ²	
SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations ⁴	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ⁴	
SMAD 466	British Film Adaptations ³	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics	
SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ³	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		6
Total Credits		21

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

² These courses satisfy concentration requirements when the topic is appropriate.

³ These courses are offered as part of the Semester in London program.

⁴ This course is offered in the JMU in LA summer study away program.

Digital Video and Cinema Concentration

The core mission of the digital video and cinema concentration is to prepare students to be visual storytellers. Students will develop a broad understanding of the aesthetics, techniques and technologies that embody the visual storytelling process as well as a critical perspective of how media tools are used to create content, convey information and impact audiences.

The program offers preparation in writing, development, production and post-production in various genres, and an exploration of the continuously evolving commercial and artistic potential of visual media and story. Digital video and cinema students are encouraged to complement their

concentration with a robust liberal arts experience that could include a minor and/or a second major. An internship is also recommended.

Code	Title	Credits
Required Courses		
SMAD 302	Video Production	3
Choose one of the following:		3
SMAD 250	Scriptwriting	
SMAD 251	Screenplay Writing	
SMAD 253	The Shooting Script: The Process of Script to Screen ¹	
Choose one of the following:		3
SMAD 407	Business and Management of Digital Media	
SMAD 410	The Creative Producer ²	
Subtotal		9
Applied Skills Courses		
Choose two of the following:		6
SMAD 303	Post Production	
SMAD 304	Audio Production	
SMAD 305	Special Topics in Media Production	
SMAD 306	Studio Production	
SMAD 312	Cinematography	
SMAD 340	Advanced Screenplay Writing	
SMAD 402	Motion Graphic Design and Production	
SMAD 403	Documentary Production	
SMAD 405	Producing and Directing	
SMAD 497	Advanced Projects in Media Arts and Design	
Subtotal		6
Critical Analysis Courses		
Choose two of the following:		6
SMAD 321	The Media Arts: Culture by Design in London ³	
SMAD 372	Media History	
SCOM 360	British Media and Society ³	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design ⁴	
SMAD 460	Film and Society ⁴	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations ⁴	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ²	
SMAD 466	British Film Adaptations ³	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics	
SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ³	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		6
Total Credits		21

¹ This course is offered in the JMU in New York summer study away program.

² This course is offered in the JMU in LA summer study away program.

³ These courses are offered as part of the Semester in London program.

⁴ These courses satisfy concentration requirements when the topic is appropriate.

Interactive Design Concentration

This concentration serves students interested in digital convergence – the delivery of content via multiple media formats such as text, images, audio, video and interactive websites. Students are involved in content creation and distribution using diverse communications media and are prepared for a variety of roles in media industries. Class work and practical experiences are grounded in online media and provide students with opportunities to develop additional skills using other media formats. At the same time, the program encourages students to obtain a broad liberal arts education so they will understand the theories, design, legalities and applications of convergence in society.

Code	Title	Credits
Required Courses		
SMAD 307	Interactive Design Web I	3
SMAD 317	User Interaction Design	3
SMAD 417	Business and Management of Interactive Design	3
Subtotal		9
Intermediate Applied Skills Course		
Choose one of the following:		3
SMAD 305	Special Topics in Media Production ¹	
SMAD 308	Interactive Design II	
SMAD 315	Digital Social Media Advertising	
SMAD 327	User-Centered Research and Usability Evaluation	
SMAD 332	Visual Communication Design	
Subtotal		3
Advanced Applied Skills Course		
Choose one of the following:		3
SMAD 404	Advanced Interactive Design	
SMAD 408	Converged Media Lab	
SMAD 497	Advanced Projects in Media Arts and Design	
Subtotal		3
Critical Analysis Courses		
Choose two of the following:		6
SMAD 321	The Media Arts: Culture by Design in London ²	
SMAD 360	British Media and Society ²	
SMAD 372	Media History	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design	
SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ³	
SMAD 466	British Film Adaptations ²	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics	
SMAD 472	Media and Politics	

SMAD 473	British Media and Politics ²	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		6
Total Credits		21

¹ These courses satisfy concentration requirements when the topic is appropriate.

² These courses are offered as part of the Semester in London program.

³ This course is offered in the JMU in LA summer study away program.

Journalism Concentration

This professionally-oriented program enables students to develop reporting, writing, editing and production skills needed to work in magazines, newspapers, video/broadcast and new media journalism through course work, internships and other practical experiences. At the same time the program encourages students to obtain a broad liberal arts education so they will understand many of the issues facing contemporary society.

Code	Title	Credits
Required Courses		
SMAD 210	News Reporting and Writing ¹	3
SMAD 309	Video Journalism	3
SMAD 310	Advanced Reporting and Writing	3
SMAD 471	Media Ethics	3
Subtotal		12
Intermediate Applied Skills Course		
Choose one of the following:		3
SMAD 220	Copy Editing	
SMAD 225	Photojournalism	
SMAD 243	Sport Communication Techniques: Broadcasting	
SMAD 305	Special Topics in Media Production	
SMAD 307	Interactive Design Web I	
SMAD 315	Digital Social Media Advertising	
SMAD 327	User-Centered Research and Usability Evaluation	
SMAD 332	Visual Communication Design	
SMAD 403	Documentary Production	
Subtotal		3
Advanced Applied Skills Course		
Choose one of the following:		3
SMAD 409	Electronic News Producing and Editing	
SMAD 421	Feature Magazine Production	
SMAD 422	Multimedia Journalism	
SMAD 497	Advanced Projects in Media Arts and Design (when topic is appropriate)	
Subtotal		3
Critical Analysis Course		
Choose one of the following: ²		3
SMAD 321	The Media Arts: Culture by Design in London ³	
SMAD 360	British Media and Society ³	
SMAD 372	Media History	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design ⁴	

SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ⁵	
SMAD 466	British Film Adaptations ³	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics (already required of Journalism concentrators)	
SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ³	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		3
Total Credits		21

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

² SMAD 471 is required of journalism concentrators. They must select an additional Critical Analysis course

³ These courses are offered as part of the Semester in London program.

⁴ These courses satisfy concentration requirements when the topic is appropriate.

⁵ This course is offered in the JMU in LA summer study away program.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
SMAD 101	Introduction to Media Arts and Design	3
General Education courses		27
Credits		30
Second Year		
SMAD 201	Foundations of Visual Communication Design	3
SMAD 202	Foundations of Audio-Visual Storytelling	3
SMAD 203	Foundations of User Experience Design	3
SMAD 204	Foundations of Media Storytelling and Research	3
General Education courses		14
University electives		4
Credits		30
Third Year		
SMAD 301	The Media Arts: Culture by Design	3
SMAD Concentration courses		12
University electives		15
Credits		30
Fourth Year		
SMAD 400	Senior Assessment in Media Arts and Design	0
SMAD 450	Law of Media Arts and Design	3
SMAD Concentration courses		9
University electives		18
Credits		30
Total Credits		120

Media Arts and Design, B.S.

Program Description

The School of Media Arts and Design incorporates the study of mass media and mediated visual and aural expression, focusing on creative advertising, video and cinema production, interactive design, journalism, film studies, and creative writing. SMAD offers a program of study that prepares students for careers in the media and/or corporate environments that rely on mass media and related areas of expertise, or graduate study in mass communication and related areas.

Admission and Minimum Grades

Special Admission and Retention Requirements

Admission to the School of Media Arts and Design is limited and competitive. There are two pathways to full admission to the media arts and design major.

Direct Admission

Entering first year and transfer students who are awarded a competitive scholarship by the university, the College of Arts and Letters, or by the School of Media Arts and Design will be fully and directly admitted to the media arts and design major. Those scholarships include the Dingledine-Bluestone Scholarships; the Senator Harry S. Byrd Jr.—D. Lathan Mims Renewable Scholarship; and the Alison B. Parker Memorial Renewable Scholarship. Their concentration will be determined by the SMAD Director, in consultation with faculty. Such scholarship holders are exempt from the admission application requirement but must fulfill any conditions attached to their awards.

Admission by Application

Students interested in majoring in the programs offered by the school must apply for a limited number of spaces in the major. An admission committee in the school reviews applications and offers admission based on availability to the most qualified students. A student may apply no more than two times for admission to the school.

September Application Period

The September application process is only for transfer students matriculating to JMU in the current fall semester or the immediately preceding spring or summer semesters and have declared the media arts and design major. These students must be officially admitted to JMU as transfer students and must have earned at least 30 hours of JMU approved post high school college credit. They must be degree-seeking students and must be enrolled in at least 12 hours at JMU during the fall semester. The application is due at 4 p.m. on the third Friday of the fall semester. The exact date will be posted on the SMAD website.

Annual Application Period

Currently enrolled JMU students (including transfer students matriculating in January or in previous semesters) may apply during the annual application period that occurs in the spring semester. Students applying to the major at JMU must be degree-seeking and have completed at least 12 hours or be enrolled in at least 12 hours at JMU. The application is due at 4 p.m. on the third Friday of the spring semester. The exact date will be posted on the SMAD website.

Process

Students interested in applying for admission to this major must "declare" media arts and design as a major and apply to become a "fully admitted"

major. Declaring a media arts and design major does not guarantee the student will be fully admitted into the media arts and design major. Students must declare media arts and design as a major prior to being eligible to enroll in SMAD 101. A declared media arts and design student may take SMAD 101, but the student is restricted from SMAD courses beyond SMAD 101 until being fully admitted into the major.

A complete application to the major includes the following:

- Completion of SMAD 101 or enrollment in SMAD 101. Students must earn a minimum grade of "C-" (minimum 70%) in SMAD 101 or they will have to retake the course. Students get only two attempts to earn a minimum of 70% in the course.
- Completion of the SMAD Admission Test with sections on English grammar and usage, timed writing samples and basic computer knowledge.
- Submission of all required components on the "SMAD Admissions" Canvas course:
 - An application letter/personal essay expressing reasons for interest in your first-choice concentration. If you wish to be considered for a second-choice concentration, submit a separate letter specifically for that concentration. (A student will only be accepted into one concentration.)
 - Unofficial transcript(s). (Transfer students must include unofficial transcripts from previous institutions and JMU Transfer Credit Report.)
 - A SMAD Admissions Questionnaire on Canvas that asks about media creation and experience.
 - A SMAD Contract on Canvas.

Following review by the SMAD admission committee, students will be notified about their acceptance into the major. If accepted, students will be eligible to register for SMAD courses beyond SMAD 101. Students not accepted into the major will be notified and may reapply the following January or a later January. Students may apply to the major no more than two times.

Change of Concentration

Students accepted into the major are assigned to a concentration. Students who wish to change concentrations after being accepted into the major may ask to do so only during the annual application period, which ends at 4 p.m. on the third Friday of the spring semester.

To request a different concentration, the student must submit on the "SMAD Admissions" Canvas course the following:

1. A letter explaining why the student is seeking that change. This letter should be addressed to the Admissions Committee and be no more than two pages in length.
2. A completed Concentration Change Application form, which the student can download from the "SMAD Admissions" Canvas course and fill out after consulting with their advisor and obtaining the advisor's signature. The student must scan and upload the finished form on the Canvas site.
3. The student's unofficial transcripts.

A student seeking a concentration change must request access to the "SMAD Admissions" Canvas course. Directions for doing so are posted on SMAD's website.

If the Admissions Committee approves the student's request, the application will be reviewed by the SMAD director, who will grant final

permission to change concentrations. There are no guarantees that the change will be granted, and students who change concentrations will not receive overrides or prerequisite waivers to expedite graduation.

Minimum Grades

If a student enrolled in SMAD 101 is accepted into the major and makes an "F" in the course, the student will be dropped from the major. To be eligible to reapply for the major, the student must re-take SMAD 101 and reapply. If a student enrolled in SMAD 101 is accepted into the major and makes a "D+," "D" or "D-" in the course, the student may continue to take classes in the major; however, the student must retake SMAD 101 the next semester and earn a minimum grade of "C-". Students are limited to taking SMAD 101 twice. Failure to achieve a minimum grade of "C-" after the second time will result in the student being dropped from the major, including being dropped from any SMAD courses for which the student is preregistered, and the student may not reapply for admission to the major.

To graduate with a major in media arts and design, a student must have a minimum grade point average of 2.0 ("C") in the major.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ^{2,3}	3
Scientific Literacy requirement ^{2,3}	3-4
Major requirements ⁴	48
University electives	24-25
Total	120

¹ The General Education program contains a set of requirements each student must fulfill.

² In addition to course work taken to fulfill General Education requirements.

³ The quantitative and scientific literacy requirements may be satisfied by courses in the breadth requirement for the major.

⁴ A student must complete a minimum of 48 credits for the media arts and design major (42 SMAD credits and 6 non-Media Arts and Design, B.S. breadth credits). In addition to courses in the School of Media Arts and Design, students must complete at least 72 non-SMAD credits

Major Requirements

Code	Title	Credits
Core Requirements		
SMAD 101	Introduction to Media Arts and Design	3
SMAD 201	Foundations of Visual Communication Design	3
SMAD 202	Foundations of Audio-Visual Storytelling	3
SMAD 203	Foundations of User Experience Design	3
SMAD 204	Foundations of Media Storytelling and Research	3
SMAD 301	The Media Arts: Culture by Design	3
SMAD 400	Senior Assessment in Media Arts and Design	0
SMAD 450	Law of Media Arts and Design	3
Subtotal		21
Breadth Requirement		

To be accomplished visual storytellers, students seeking a degree in media arts and design require extra breadth in scientific and quantitative literacy. Students seeking a Bachelor of Science degree in media arts and design must complete two courses (six credits) from any of the following areas: anthropology (ANTH), astronomy (ASTR), biology (BIO), chemistry (CHEM), computer science (CS), economics (ECON), geography (GEOG), geology (GEOL), integrated science and technology (ISAT), intelligence analysis (IA), interdisciplinary science (ISCI), kinesiology (KIN), mathematics (MATH), physics (PHYS), political science (POSC) and sociology (SOCI). ¹	6
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Concentration	
Media arts and design majors apply for and are assigned to one of four concentrations (creative advertising, digital video and cinema, interactive design, journalism), and must complete all requirements for that concentration. Students are restricted from taking classes in other concentrations unless there is room and permission is granted.	21

Total Credits	48
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¹ Classes that satisfy the breadth requirements for media arts and design that qualify as satisfying the university's degree requirements will also fulfill the university degree requirements. In no case may a student double-count a general education class toward the university B.S. degree requirements.

Media Arts and Design Concentrations
Creative Advertising Concentration

This professionally-oriented concentration provides students with the knowledge and skills to create and manage advertising programs emphasizing new media. In addition to the theoretical concepts underlying advertising, students learn strategic creative writing, effective message design and media production. Students also learn strategic planning, implementation and evaluation of new media technologies, and research and analysis of current trends and applications. The program reviews ethical and legal issues involved in the creative advertising process and the use of new media. Students planning careers in creative advertising should obtain a broad liberal arts education to better understand the characteristics that make the diversified communication process effective across various media and organizations.

Code	Title	Credits
Required Courses		
SMAD 242	Introduction to Advertising and New Media	3
SMAD 252	Copywriting for Advertising ¹	3
SMAD 342	Elements of Creative Advertising	3
SMAD 443	Creative Advertising Campaigns	3
Subtotal		12
Applied Skills Courses		
Choose one of the following:		3
SMAD 210	News Reporting and Writing	
SMAD 220	Copy Editing	
SMAD 225	Photojournalism	
SMAD 305	Special Topics in Media Production ²	
SMAD 307	Interactive Design Web I	
SMAD 310	Advanced Reporting and Writing	
SMAD 315	Digital Social Media Advertising	
SMAD 327	User-Centered Research and Usability Evaluation	

SMAD 332	Visual Communication Design	
SMAD 497	Advanced Projects in Media Arts and Design	
Subtotal		3
Critical Analysis Courses		
Choose two of the following:		6
SMAD 321	The Media Arts: Culture by Design in London ³	
SMAD 360	British Media and Society ³	
SMAD 372	Media History	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design ²	
SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations ⁴	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ⁴	
SMAD 466	British Film Adaptations ³	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics	
SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ³	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		6
Total Credits		21

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

² These courses satisfy concentration requirements when the topic is appropriate.

³ These courses are offered as part of the Semester in London program.

⁴ This course is offered in the JMU in LA summer study away program.

Digital Video and Cinema Concentration

The core mission of the digital video and cinema concentration is to prepare students to be visual storytellers. Students will develop a broad understanding of the aesthetics, techniques and technologies that embody the visual storytelling process as well as a critical perspective of how media tools are used to create content, convey information and impact audiences.

The program offers preparation in writing, development, production and post-production in various genres, and an exploration of the continuously evolving commercial and artistic potential of visual media and story. Digital video and cinema students are encouraged to complement their concentration with a robust liberal arts experience that could include a minor and/or a second major. An internship is also recommended.

Code	Title	Credits
Required Courses		
SMAD 302	Video Production	3
Choose one of the following:		3
SMAD 250	Scriptwriting	
SMAD 251	Screenplay Writing	
SMAD 253	The Shooting Script: The Process of Script to Screen ¹	

Choose one of the following:		3
SMAD 407	Business and Management of Digital Media	
SMAD 410	The Creative Producer ²	
Subtotal		9

Applied Skills Courses		
Choose two of the following:		6
SMAD 303	Post Production	
SMAD 304	Audio Production	
SMAD 305	Special Topics in Media Production	
SMAD 306	Studio Production	
SMAD 312	Cinematography	
SMAD 340	Advanced Screenplay Writing	
SMAD 402	Motion Graphic Design and Production	
SMAD 403	Documentary Production	
SMAD 405	Producing and Directing	
SMAD 497	Advanced Projects in Media Arts and Design	
Subtotal		6

Critical Analysis Courses		
Choose two of the following:		6
SMAD 321	The Media Arts: Culture by Design in London ³	
SMAD 372	Media History	
SCOM 360	British Media and Society ³	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design ⁴	
SMAD 460	Film and Society ⁴	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations ⁴	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ²	
SMAD 466	British Film Adaptations ³	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics	
SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ³	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		6

Total Credits **21**

¹ This course is offered in the JMU in New York summer study away program.

² This course is offered in the JMU in LA summer study away program.

³ These courses are offered as part of the Semester in London program.

⁴ These courses satisfy concentration requirements when the topic is appropriate.

Interactive Design Concentration

This concentration serves students interested in digital convergence – the delivery of content via multiple media formats such as text, images, audio, video and interactive websites. Students are involved in content creation and distribution using diverse communications media and are prepared for a variety of roles in media industries. Class work and practical experiences are grounded in online media and provide students

with opportunities to develop additional skills using other media formats. At the same time, the program encourages students to obtain a broad liberal arts education so they will understand the theories, design, legalities and applications of convergence in society.

Code	Title	Credits
Required Courses		
SMAD 307	Interactive Design Web I	3
SMAD 317	User Interaction Design	3
SMAD 417	Business and Management of Interactive Design	3
Subtotal		9
Intermediate Applied Skills Course		
Choose one of the following:		3
SMAD 305	Special Topics in Media Production ¹	
SMAD 308	Interactive Design II	
SMAD 315	Digital Social Media Advertising	
SMAD 327	User-Centered Research and Usability Evaluation	
SMAD 332	Visual Communication Design	
Subtotal		3
Advanced Applied Skills Course		
Choose one of the following:		3
SMAD 404	Advanced Interactive Design	
SMAD 408	Converged Media Lab	
SMAD 497	Advanced Projects in Media Arts and Design	
Subtotal		3
Critical Analysis Courses		
Choose two of the following:		6
SMAD 321	The Media Arts: Culture by Design in London ²	
SMAD 360	British Media and Society ²	
SMAD 372	Media History	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design	
SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ³	
SMAD 466	British Film Adaptations ²	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics	
SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ²	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		6
Total Credits		21

¹ These courses satisfy concentration requirements when the topic is appropriate.

² These courses are offered as part of the Semester in London program.

³ This course is offered in the JMU in LA summer study away program.

Journalism Concentration

This professionally-oriented program enables students to develop reporting, writing, editing and production skills needed to work in magazines, newspapers, video/broadcast and new media journalism through course work, internships and other practical experiences. At the same time the program encourages students to obtain a broad liberal arts education so they will understand many of the issues facing contemporary society.

Code	Title	Credits
Required Courses		
SMAD 210	News Reporting and Writing ¹	3
SMAD 309	Video Journalism	3
SMAD 310	Advanced Reporting and Writing	3
SMAD 471	Media Ethics	3
Subtotal		12
Intermediate Applied Skills Course		
Choose one of the following:		3
SMAD 220	Copy Editing	
SMAD 225	Photojournalism	
SMAD 243	Sport Communication Techniques: Broadcasting	
SMAD 305	Special Topics in Media Production	
SMAD 307	Interactive Design Web I	
SMAD 315	Digital Social Media Advertising	
SMAD 327	User-Centered Research and Usability Evaluation	
SMAD 332	Visual Communication Design	
SMAD 403	Documentary Production	
Subtotal		3
Advanced Applied Skills Course		
Choose one of the following:		3
SMAD 409	Electronic News Producing and Editing	
SMAD 421	Feature Magazine Production	
SMAD 422	Multimedia Journalism	
SMAD 497	Advanced Projects in Media Arts and Design (when topic is appropriate)	
Subtotal		3
Critical Analysis Course		
Choose one of the following: ²		3
SMAD 321	The Media Arts: Culture by Design in London ³	
SMAD 360	British Media and Society ³	
SMAD 372	Media History	
SMAD 373	Media Analysis and Criticism	
SMAD 380	Introduction to Film	
SMAD 398	Critical Studies in Media Arts and Design ⁴	
SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations	
SMAD 464	Contemporary American Film	
SMAD 465	The Entertainment Industry ⁵	
SMAD 466	British Film Adaptations ³	
SMAD 470	New Media and Society	
SMAD 471	Media Ethics (already required of Journalism concentrators)	

SMAD 472	Media and Politics	
SMAD 473	British Media and Politics ³	
SMAD 498	Senior Seminar in Media Arts and Design	
Subtotal		3
Total Credits		21

¹ This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

² SMAD 471 is required of journalism concentrators. They must select an additional Critical Analysis course

³ These courses are offered as part of the Semester in London program.

⁴ These courses satisfy concentration requirements when the topic is appropriate.

⁵ This course is offered in the JMU in LA summer study away program.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
SMAD 101	Introduction to Media Arts and Design	3
General Education courses		27
Credits		30
Second Year		
SMAD 201	Foundations of Visual Communication Design	3
SMAD 202	Foundations of Audio-Visual Storytelling	3
SMAD 203	Foundations of User Experience Design	3
SMAD 204	Foundations of Media Storytelling and Research	3
General Education courses		14
University electives		4
Credits		30
Third Year		
SMAD 301	The Media Arts: Culture by Design	3
SMAD Concentration courses		12
University electives		15
Credits		30
Fourth Year		
SMAD 400	Senior Assessment in Media Arts and Design	0
SMAD 450	Law of Media Arts and Design	3
SMAD Concentration courses		9
University electives		18
Credits		30
Total Credits		120

Middle Grades Education, B.S.

Program Description

Middle Grades Education is an undergraduate licensure program designed to develop content-area teachers for grades 6-8. In this major, students learn the developmental characteristics of students aged 10 to 14; the social, emotional, physical and intellectual abilities and needs of diverse middle school students; and the design of appropriate cross-disciplinary curriculum and instruction while completing multiple field experiences in local middle school classrooms.

This program of study provides students with course offerings and experiences that form the required foundation for licensure in middle education at James Madison University. Students major in middle grades education while completing licensure area course requirements in a discipline closely associated with the desired teaching area. The state-approved licensure areas in the middle grades education major at JMU are English, mathematics, science and social studies. While students must earn licensure in one concentration, it is also possible to work toward licensure in two concentrations. In most cases, licensure courses may be counted toward a minor. Licensure endorsements for specialized teaching areas (e.g., Algebra I, journalism, gifted education) are available as add-on endorsements. Students interested in pursuing any licensure option must consult with a program advisor regarding requirements and scheduling.

For successful completion of the Middle Grades Education major, students must earn a minimum grade of "B-" in all required Middle Grades Education courses in the middle education major (excludes WRTC 103, EDUC 200 and EDUC 175 or PSYC 160) and earn credit in all licensure area content course work to continue in and complete the licensure area.

Students should consult with a middle grades education advisor in their first year to obtain information concerning completion of General Education, major and licensure area requirements. In addition to declaring the middle grades education major, students must apply and meet the requirements for admission to teacher education. Students must be fully admitted to teacher education prior to registering for MSSE 370 and MSSE 371.

Note: Actual teacher licensure requirements may differ from the listed catalog requirements because of changes enacted by the Virginia Department of Education and/or national accrediting agencies. Students are encouraged to confer regularly with their advisor to determine their progress toward graduation.

Admission Requirements

Students must meet specific requirements to be admitted to programs within the College of Education (p. 18). These criteria can be found on the College of Education (p. 18) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement	3
Scientific Literacy Requirement	3
Major Requirements	50
Licensure Area Course Requirements	28
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within a program. EDUC 175 or PSYC 160 are double counted between General Education and major requirements.

Major Requirements

Students are responsible for transportation to field placements at local schools. Before beginning the licensure portion of the professional program, candidates must meet all criteria for admission to teacher education (e.g., minimum GPA of 2.5, completing all required trainings). Candidates must meet all additional requirements of the program before completion. Candidates must submit passing scores on the Praxis Subject Assessment prior to registration for MSSE 420 and prior to beginning MSSE 465.

Code	Title	Credits
Core Requirements		
EDUC 175	Learning, Behavior and Engagement in Context [C5SD]	3
or PSYC 160	Life Span Human Development [C5SD]	
EDUC 200	Foundations of Education in the United States	3
LED 212	Foundations of Literacy for Middle School Teachers	3
LED 440	Disciplinary Literacy in Secondary Education	3
MIED 211	Literacy Field Experience in Middle Grades Education	3
MSSE 210	Diversity in Middle and Secondary Education	3
MSSE 403	Behavior and Classroom Management in a Diverse Society	3
MSSE 370	General Instructional Methods for Grades 6-12	3
MSSE 371	Clinical Experience in Adolescent Education	1
MSSE 415	Content Area Field Experience in Middle Schools	3
MSSE 420	Differentiation for Adolescents	3
MSSE 425	Assessment in Middle and Secondary Education	3
MSSE 450	Internship Seminar	2
MSSE 455	Special Topics in Pedagogical Content Knowledge	3
Subtotal		39
Student Teaching/Internship Requirements		
MSSE 465	Internship in Middle School	8
Subtotal		8
Concentrations		
Choose one of the following concentrations:		3
MSSE 411	English/Language Arts Teaching Methods, Grades 6-8	
MSSE 413	Mathematics Teaching Methods, Grades 6-8	
MSSE 414	Natural Sciences Teaching Methods, Grades 6-8	
MSSE 412	Social Studies Teaching Methods, Grades 6-8	
Subtotal		3
Total Credits		50

Recommended Schedule for Majors

Course	Title	Credits
First Year		
EDUC 175 or PSYC 160	Learning, Behavior and Engagement in Context [C5SD] or Life Span Human Development [C5SD]	3
EDUC 200	Foundations of Education in the United States	3
General Education Madison Foundations		18
B.S. degree requirements		3
Credits		27
Second Year		
LED 212	Foundations of Literacy for Middle School Teachers	3
MIED 211	Literacy Field Experience in Middle Grades Education	3
MSSE 210	Diversity in Middle and Secondary Education	3
MSSE 370	General Instructional Methods for Grades 6-12	3
MSSE 371	Clinical Experience in Adolescent Education	1
General Education		9
B.S. degree requirements		3
Licensure Area Content Courses		9
Credits		34
Third Year		
Remaining General Education		11
Content courses		7
All students seeking licensure:		
MSSE 415	Content Area Field Experience in Middle Schools	3
MSSE 425	Assessment in Middle and Secondary Education	3
LED 440	Disciplinary Literacy in Secondary Education	3
Content Specific Methods Course:		
Choose one of the following options:		3
Option 1: For those seeking English licensure:		
MSSE 411	English/Language Arts Teaching Methods, Grades 6-8	
Option 2: For those seeking social studies licensure:		
MSSE 412	Social Studies Teaching Methods, Grades 6-8	
Option 3: For those seeking mathematics licensure:		
MSSE 413	Mathematics Teaching Methods, Grades 6-8	
Option 4: For those seeking science licensure:		
MSSE 414	Natural Sciences Teaching Methods, Grades 6-8	
Credits		30
Fourth Year		
MSSE 403	Behavior and Classroom Management in a Diverse Society	3
MSSE 420	Differentiation for Adolescents	3
MSSE 450	Internship Seminar	2
MSSE 455	Special Topics in Pedagogical Content Knowledge	3
MSSE 465	Internship in Middle School	8
Licensure-Area Content Courses and/or Electives/Add-on Endorsement		10
Credits		29
Total Credits		120

Modern Foreign Languages, B.A.

Program Description

Students can study world languages and cultures and earn a B.A. degree in Modern Foreign Languages with a major in Arabic, French, German, Italian, or Spanish. They can minor in any of these languages as well as Chinese and Russian; however, a student may not major and minor in the same language. The department also offers instruction in Japanese, Korean and Latin.

To complete a major, students should choose a combination of courses based on the language they are studying. Courses listed without a specific language prefix are offered in all languages. Up to two courses taught in English may be counted toward the total major unit requirement. Course substitutions must be approved by the advisor.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	33
University electives	31-43
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

All language sections share certain core requirements. Up to two courses taught in English may be counted toward the total major unit requirement.

Code	Title	Credits
Language Foundations		
300:	Language for Living	3
301:	Pop Culture Through Language	3
307 or 308:	Culture and Civilization	3
320 (321):	Culture and Communication in Context	3
322:	Language Landscapes	3
Subtotal		15
Exploring Languages and Cultures		
3 courses; choose one course from each group:		9
Group 1: Culture and literature		
Group 2: Linguistics and Language Structures		
Group 3: Professional language		
Subtotal		9
Electives		

Choose 9 credits from the following:	9
Any other courses from the above core categories	
Elective classes in the language not included in the core	
WLC courses or other approved courses related to the language studied	
Subtotal	9
Total Credits	33

Arabic

Code	Title	Credits
Language Foundations		
ARAB 300	Language for Living	3
ARAB 301	Pop Culture through Language	3
ARAB 307 or ARAB 308	A History of Islamicate Civilization 600-1600 A.D. Contemporary Islamicate Civilization	3
ARAB 320	Culture and Communication in Context	3
ARAB 322	Language Landscapes	3
Subtotal		15
Exploring Languages and Cultures		
Choose one course from each of the three groups.		
Group 1: Culture and Literature:		
Choose one of the following:		3
ARAB 335	A Journey through Literature	
ARAB 385	The Arabian Nights	
ARAB 386	Arabic Fables and Short Stories	
ARAB 405	The Arabic Novel	
ARAB 410	Media Arabic	
ARAB 446	Special Topics in Arabic Literature	
ARAB 447	Special Topics in Arabic Civilization and Culture	
ARAB 465	Arabic Cinema	
ARAB 484	Literatures of Islamic Spain and Italy	
Group 2: Linguistic and Language Structures:		
Choose one of the following:		3
ARAB 400	Advanced Arabic Writing and Conversation	
ARAB 411	Colloquial Egyptian Arabic I	
ARAB 412	Colloquial Egyptian Arabic II	
ARAB 413	Colloquial Levantine Arabic I (Jordanian/Palestinian)	
ARAB 414	Colloquial Levantine Arabic II (Syrian/Lebanese)	
ARAB 448	Special Topics in Arabic Linguistics	
LING 311	Study of Language Structures	
LING 312	Bilingualism in the United States	
LING 470	World Language Teaching Methods	
Group 3: Professional Language:		
Choose one of the following:		3
ARAB 330	Business Arabic	
ARAB 371	Advanced Arabic Grammar and Translation	
ARAB 375	Business and Society in the Arab World	
Subtotal		9
Electives		
Choose three additional courses from the following list:		9
Any other courses from the above categories		

Elective ARAB classes not included in the core	
WLC courses or other approved courses related to Arabic	
Total Credits	33

French

Code	Title	Credits
Language Foundations		
FR 300	Language for Living	3
FR 301	Pop Culture through Language	3
FR 307	History of French Culture and Society	3
or FR 308	Contemporary History and Culture in the Francophone World	
FR 320	Culture and Communication in Context	3
FR 322	Language Landscapes	3
Subtotal		15
Exploring Languages and Cultures		
Choose one course from each of the three groups.		
Group 1: Culture and Literature:		
Choose one of the following:		3
FR 335	A Journey through Literature	
FR 386	French Short Stories	
FR 420	Stories and Histories in French Culture	
FR 421	Stories and Histories in Francophone Culture	
FR 422	Voices of Today	
FR 425	Transformative Texts	
FR 446	Special Topics in Francophone Literature	
FR 447	Special Topics in Francophone Culture	
FR 464	French and Francophone Theater	
FR 465	French and Francophone Cinema	
Group 2: Linguistics and Language Structures:		
Choose one of the following:		3
FR 315	Sounds of Speech	
FR 448	Special Topics in Francophone Linguistics	
LING 410	Introduction to Romance Linguistics	
LING 311	Study of Language Structures	
LING 312	Bilingualism in the United States	
LING 470	World Language Teaching Methods	
Group 3: Professional Language:		
Choose one of the following:		3
FR 330	Business French	
FR 375	Business and Society in the Francophone World	
FR 435	Translation Strategies	
Subtotal		9
Electives		
Choose three additional courses from the following list:		9
Any other courses from the above categories		
Elective FR classes not included in the core		
WLC courses or other approved courses related to French		
Subtotal		9
Total Credits		33

German

Code	Title	Credits
Language Foundations		
GER 300	Language for Living	3
GER 301	Pop Culture through Language	3
GER 307	History of Germanic Cultures and Societies	3
or GER 308	Modern German Culture and Society	
GER 320	Culture and Communication in Context	3
GER 322	Language Landscapes	3
Subtotal		15
Exploring Languages and Cultures		
Choose one course from each of the three groups.		
Group 1: Culture and Literature:		
Choose one of the following:		3
GER 335	A Journey through Literature	
GER 386	German Short Stories	
GER 422	Voices of Today	
GER 425	Transformative Texts	
GER 446	Special Topics in German Literature	
GER 447	Special Topics in German Culture	
GER 464	German Theater	
Group 2: Linguistics and Language Structures:		
Choose one of the following:		3
GER 315	Sounds of Speech	
GER 448	Special Topics in German Linguistics	
LING 311	Study of Language Structures	
LING 312	Bilingualism in the United States	
LING 470	World Language Teaching Methods	
Group 3: Professional Language:		
Choose one of the following:		3
GER 330	Business German	
GER 375	Business and Society in Germany	
GER 435	Translation Strategies	
Subtotal		9
Electives		
Choose three additional courses from the following list:		9
Any other courses from the above categories		
Elective GER classes not included in the core		
WLC courses or other approved courses related to German		
Subtotal		9
Total Credits		33

Code	Title	Credits
Language Foundations		
ITAL 300	Language for Living	3
ITAL 301	Pop Culture through Language	3
ITAL 307	History of Italian Culture and Society	3
or ITAL 308	Contemporary Italian Culture and Society	
ITAL 320	Culture and Communication in Context	3
ITAL 322	Language Landscapes	3

Subtotal	15
Exploring Languages and Cultures	
Choose one course from each of the three groups.	
Group 1: Culture and Literature:	
Choose one of the following:	
ITAL 335	A Journey through Literature
ITAL 386	Italian Short Stories
ITAL 421	Stories and Histories: Italian Theatre
ITAL 422	Voices of Today
ITAL 446	Special Topics in Italian Literature
ITAL 447	Special Topics in Italian Culture
ITAL 465	Italian Cinema
Group 2: Linguistics and Language Structures:	
Choose one of the following:	
ITAL 315	Sounds of Speech
LING 410	Introduction to Romance Linguistics
ITAL 448	Special Topics in Italian Linguistics
LING 311	Study of Language Structures
LING 312	Bilingualism in the United States
LING 470	World Language Teaching Methods
Group 3: Professional Language:	
Choose one of the following:	
ITAL 330	Business Italian
ITAL 375	Business and Society in Italy
ITAL 435	Translation Strategies
WLC 456	Translating Italian Literature
Subtotal	9
Electives	
Choose three additional courses from the following list:	
Any other courses from the above categories	
Elective ITAL classes not included in the core	
WLC courses or other approved courses related to Italian	
Subtotal	9
Total Credits	33

Spanish

Code	Title	Credits
Language Foundations		
SPAN 300	Language for Living	3
SPAN 301	Pop Culture through Language	3
SPAN 307	History of Spanish Culture and Society	3
or SPAN 308	Latin American Cultures	
SPAN 320	Culture and Communication in Context	3
or SPAN 321	Culture and Communication in Context (for Heritage Speakers)	
SPAN 322	Language Landscapes	3
Subtotal		15
Exploring Languages and Cultures		
Choose one course from each of the three groups.		
Group 1: Culture and Literature:		
Choose one of the following:		
SPAN 335	A Journey through Literature	3

SPAN 385	Latin American Short Story
SPAN 386	Spanish Short Stories
SPAN 420	Stories and Histories: The Generation of 1898
SPAN 421	Stories and Histories: Latin American Novel
SPAN 422	Voices of Today: Wounds and Words in Post-War Spain
SPAN 423	Voices of Today: Latin American Poetry
SPAN 425	Transformative Texts
SPAN 426	Voices of the Past: Works of the Golden Age
SPAN 446	Special Topics in Spanish Literature
SPAN 447	Special Topics in Spanish-Speaking Cultures
SPAN 464	Spanish Theater
SPAN 465	Spanish-Language Cinema
Group 2: Linguistics and Language Structures:	
Choose one of the following:	
LING 311	Study of Language Structures
LING 312	Bilingualism in the United States
LING 410	Introduction to Romance Linguistics
LING 430	Forensic Linguistics: Language and the Law
LING 470	World Language Teaching Methods
SPAN 311	Contrastive Linguistics
SPAN 310	Introduction to Spanish Linguistics
SPAN 315	Sounds of Speech
SPAN 404	Spanish in the United States
SPAN 448	Special Topics in Spanish Linguistics
Group 3: Professional Language:	
Choose one of the following:	
SPAN 306	Spanish for Social Work
SPAN 312	Translation Competencies
SPAN 330	Business Spanish
SPAN 360	Spanish for Public Safety
SPAN 365	Spanish for Medical Professionals I
SPAN 430	Advanced Business Spanish
SPAN 435	Translation Strategies
SPAN 436	Introduction to Interpretation
SPAN 475	Spanish for Medical Professionals II
SPAN 476	Culture and Medicine in Latin America
SPAN 477	Culture and Medicine in Spain
SPAN 478	Spanish in Healthcare: Policy and Practice in Latinx Health
SPAN 485	Business and Society in Latin America
SPAN 486	Business and Society in Spain
Subtotal	9
Electives	
Choose three additional courses from the following list:	
Any other courses from the above core categories	
Elective SPAN classes not included in the core	
WLC courses or other approved courses related to Spanish	
Subtotal	9
Total Credits	33

Recommended Schedule for Majors

Students are encouraged to begin their WLC course work as soon as possible in their degree plans. The following outline is a sample four-year program. The actual courses a student takes may vary.

Course	Title	Credits
First Year		
Language courses (101-232 or above) ¹		6
General Education requirement and/or elective courses		24
Credits		30
Second Year		
Language courses (101-232 or above) ¹		6
Philosophy course		3
General Education requirement and/or elective courses		21
Credits		30
Third Year		
300- and 400-level courses		18
General Education requirement and/or other electives		12
Credits		30
Fourth Year		
300- and 400-level courses		15
Other electives		15
Credits		30
Total Credits		120

¹ Other courses may be substituted if a student places into higher-level language courses.

Music, B.A.

Program Description

The Bachelor of Arts in Music degree offers students the benefits of both first-rate artistic training and the flexibility to explore the offerings of a comprehensive university experience. In addition to a rigorous core music experience, students in this degree program take several courses outside of music, with the option to add one or more minors or a second major.

The B.A. in Music offers two concentrations (music studies and music industry), both of which consist of a music core of 34 credit hours in addition to the general education program. As part of this music core, students enroll in MUS 195, attending a minimum of 10 recitals per semester for four semesters (totaling 40 concerts) until the requirement is fulfilled. The remaining hours are specified under each concentration below. To be admitted to the program, B.A. in Music students submit an application. Those applying for music industry also submit a portfolio. Students can opt to audition to study voice or an instrument with our faculty if they wish.

Special Admission and Retention Requirements

Admission Requirements

To be considered for undergraduate admission to the music program, all entering first year students, transfer students and previously enrolled students seeking re-admission must complete the following:

- Submit an application for admission to JMU's Office of Admission (<https://www.jmu.edu/admissions/index.shtml>).
- Submit the Undergraduate Music Application through Slideroom (<https://www.jmu.edu/music/admissions/bachelor-of-music-application.shtml>).

Those wishing to audition to study an instrument or voice will find guidelines on the music website.

Students are encouraged to contact the applied faculty on their area of performance expertise with specific questions about auditions. General questions about the music degree programs may be answered by an admissions assistant at (540) 568-3851 or by sending a message to music_admit@jmu.edu.

Application, Exam and Placement Test

No student will be accepted into the music degree programs until an application is successfully completed.

Bachelor of Arts in Music candidates must complete an application which should reveal the student's interest and aptitude for music.

In addition to submitting an application, all students who intend to major in the Bachelor of Arts in Music, emphasis in music industry are required to first complete MUI 221 and then enroll in MUI 230 and MUI 231. To successfully complete MUI 230, prospective majors and minors must submit a portfolio for review, showing evidence of their interest and potential for success in the music industry. The portfolio should represent work completed in the above classes with assistance in resume preparation from University Career Center. Portfolios will be reviewed once each fall and once each spring semester. Following the portfolio review, students who are recommended for the major or minor will be eligible to register for upper level music industry classes. Students who are not recommended for admission to the major or minor may

reapply the following semester. To obtain the schedule of portfolio review sessions, as well as other specific information about the admission process, contact the School of Music, music industry area.

Students who score a 4 or 5 on the Music Theory – Aural Subscore Advanced Placement exam may earn credit for MUS 143.

Retention Policy

Students admitted as music majors must meet school standards published in the School of Music Student Handbook (<https://www.jmu.edu/music/resources/students/handbooks-forms.shtml>) to continue.

Music majors must earn a minimum grade of "C-" in all music courses required for their degree. In situations where courses are sequential, with prerequisites (for example, MUS 143-MUS 144 or MUS 243-MUS 244), if a grade lower than "C-" is earned, it does not constitute fulfillment of the prerequisite.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign language courses (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses) ³	3
Major Requirements	34
Concentration requirements	
Music Industry Concentration	30
Music Studies Concentration	18
University electives ⁴	4-30
Total	124

¹ Some General Education requirements can be fulfilled through required course(s) in the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232), or by placing out of that language through the Department of World Languages and Cultures' placement test.

³ Any PHIL course except PHIL 120 or PHIL 150. Consult the list of courses satisfying Bachelor of Arts degree requirements on the Registrar's website.

⁴ For the Music Studies concentration, a maximum of 8 music credits may be taken as University electives.

Major Requirements

Code	Title	Credits
Core Courses		
Incoming Music Major Assessment Milestone		0
MUS 143	Aural Skills I (is taken twice) ¹	2
MUS 144	Aural Skills II (is taken twice) ²	2
MUS 195	Recital Attendance (is taken four times)	0
MUS 206	Introduction to Global Music [C2VPA]	3
MUS 207	History of Western Music	4
MUS 241	Diatonic and Chromatic Harmony	3
MUS 380	Musicological Perspectives	3

MUS 381	Music Research Lab	1
School of Music Senior Assessment Milestone		0
Subtotal		18
Additional Requirement		
Choose one of the following: ³		3
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
Subtotal		3
Assigned Ensembles		
MUAP 301	Assigned Ensemble (Students should enroll in 1 credit for 4 semesters.)	4
Subtotal		4
Restricted Electives		
Choose 6 credits of the following:		6
MUS 141	Foundations of Music Theory	
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950 ³	
MUS 253	Analysis of Music 1950-Present ³	
MUS 254	Form and Analysis: Popular Music	
MUS 255	Jazz Theory	
MUS 256	Contrapuntal Techniques	
MUS 257	Techniques of Composition ⁴	
MUS 258	Film Music Analysis	
Subtotal		6
Capstone Requirement		
MUS 493	B.A. Capstone Project (must be taken for 3 credit hours)	3
Subtotal		3
Concentrations		
Students select a concentration either in Music Industry or in Music Studies. See below for more information.		18-30
Total Credits		52-64

¹ Students may substitute 1 credit of MUS 243 or MUS 244 for 1 credit of MUS 143 rather than repeating the course.

² Students may substitute 1 credit of MUS 243 or MUS 244 for 1 credit of MUS 144 rather than repeating the course.

³ These courses may not double count for other categories. The course selected to fulfill the "Restricted Electives" category cannot also be used to fulfill the "Additional Requirement" category.

⁴ MUS 257 may be repeated for elective credit, but only 3 credits (i.e. 1 instance of MUS 257) may count toward the "Restricted Electives" category.

Concentrations

Music Industry Concentration

Code	Title	Credits
Music Industry Core		
MUI 221	Survey of Music Industry	3
MUI 230	Music Industry Gateway	0
MUI 231	Legal Aspects of Music Industry	3
MUI 440	Entrepreneurship in Music Industry	3

MUI 492	Internship in Music Industry (is taken for 3 credit hours)	3
Subtotal		12
College of Business Courses		
ACTG 244	Accounting for Non-Business Majors	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
Subtotal		9
Music Industry Electives		
Choose nine credit hours from the following:		9
MUI 315	Songwriting	
MUI 324	Introduction to Audio Devices	
MUI 330	Music Publishing	
MUI 400	Pro Tools: Multitrack Recording and Music Production	
MUI 405	Logic Pro: Synthesis and Sound Design	
MUI 411	Audio Postproduction	
MUI 415	Songwriting II	
MUI 422	Concert Production and Promotion	
MUI 423	Sound Reinforcement	
MUI 430	Artist Management	
MUI 435	Marketing in Music Production	
Subtotal		9
Total Credits		30

Music Studies Concentration

Code	Title	Credits
Students select 18 credits of courses with the following prefixes: MUAP, MUI, MUS, MUED. A maximum of four additional MUAP ensemble credits can be used to fulfill this category.		18
Total Credits		18

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
MUS 143	Aural Skills I	1
MUS 195	Recital Attendance	0
MUS 141	Foundations of Music Theory	3
or MUS 241	or Diatonic and Chromatic Harmony	
Assigned Ensemble		1
Madison Foundations courses		6
Foreign Language or University Elective(s)		3
Credits		14
Spring Semester		
MUS 195	Recital Attendance	0
MUS 143	Aural Skills I	1
or MUS 144	or Aural Skills II	
MUS 206	Introduction to Global Music [C2VPA]	3
Assigned Ensemble		1
Madison Foundations courses		3
Foreign Language or University Elective(s)		3
Restricted Elective		3
Credits		14

Second Year**Fall Semester**

MUS 195	Recital Attendance	0
MUS 207	History of Western Music	4
MUS 144 or MUS 243	Aural Skills II or Aural Skills III	1
MUS 252 or MUS 253	Analysis of Music 1900-1950 or Analysis of Music 1950-Present	3
Assigned Ensemble		1
General Education		3
Foreign Language or University Elective(s)		3
Credits		15

Spring Semester

MUS 195	Recital Attendance	0
MUS 144 or MUS 244	Aural Skills II or Aural Skills IV	1
MUS 380	Musicological Perspectives	3
MUS 381	Music Research Lab	1
Assigned Ensemble		1
General Education		3
Foreign Language or University Elective(s)		3
Restricted Elective		3
Credits		15

Third Year**Fall Semester**

University Degree Requirement: Philosophy Course		3
The Natural World		4
General Education		3
Electives		4
Concentration course(s)		3
Credits		17

Spring Semester

General Education		3
Electives or Concentration course(s)		5
Concentration course(s)		9
Credits		17

Fourth Year**Fall Semester**

MUS 493	B.A. Capstone Project (taken for 1 credit)	1
General Education requirement		6
American and Global Perspectives		4
Electives		3
Electives or Concentration course(s)		3
Credits		17

Spring Semester

MUS 493	B.A. Capstone Project (taken for 2 credits)	2
Restricted Elective		3
General Education		3
University Elective(s) or Concentration course(s)		7
Credits		15
Total Credits		124

Music, B.M.

Program Description

The Bachelor of Music degree offers six specialized concentrations: performance, composition, music industry, music education, jazz studies and music theatre. All B.M. students must complete the General Education program and a 30-credit hour core program of music courses common to all the concentrations' curricula. The remaining hours are specified under each of the various concentrations.

All undergraduate music majors pursuing the Bachelor of Music, regardless of concentration, must perform on the primary instrument in one assigned ensemble each semester when one or more of the following indicators applies:

- They are enrolled for a minimum of 12 credit hours on campus.
- They are enrolled for applied lessons on the primary instrument.
- They have not completed all of the requirements in the music major.
- They are not enrolled in an off-campus internship or in student teaching.

While students are required to participate in only one assigned ensemble each semester, they are encouraged to take advantage of the many performance opportunities offered by the School of Music by participating in more than one ensemble. For information regarding the ensemble audition process, see the Undergraduate Music Student Handbook.

If piano is the primary instrument, see the specific requirements in the Undergraduate Music Student Handbook.

All Bachelor of Music students should enroll in class piano each semester until they meet the keyboard skills requirements appropriate to their specific degree or concentration.

Another vital aspect of the programs is attendance at recitals and concerts. Attending these events contributes to the breadth of students' knowledge of music literature. Hearing performances also allows students to observe performing techniques from the student to the professional levels. Consequently, Bachelor of Music majors must attend a minimum of 10 recitals per semester for six semesters, totaling 60 attendances during their degree studies. Students must continue to register for MUS 195 until they fulfill this requirement.

Objectives

The eight major degree concentrations are each designed to establish a set of skills and a knowledge base necessary for success as a practitioner in specific career areas in the broad field of music. These objectives are achieved through School of Music classes:

- Through core music classes, students will attain a general level of functional musicianship sufficient to begin and sustain a professional career in the music field.
- To gain awareness and basic competency in composition and analysis of the standard forms and styles of western music.
- To gain a broad historical perspective on the development of the forms and styles of western music, as well as diverse world music.
- To develop a knowledge base enabling the placement of music within stylistic and chronological eras through cues that can be aurally identified.

- To develop a minimal ability to use a piano keyboard in the study, analysis and performance of music.
- To learn and practice the basic skills of conducting a musical ensemble and leading a rehearsal.
- Through attendance at musical performances, students will gain awareness and acceptance of a broad variety of music, as well as of the traditional practices of concert musicians through listening to and watching others perform.
- Through specialized classes in each concentration, students will attain skills and attitudes necessary for the establishment and maintenance of a career as a professional musician.
- Students in the Bachelor of Music will take weekly lessons in a primary instrument until they have mastered the skills of performance on that instrument sufficiently well to pass the graduation level for the specific concentration and to successfully complete a senior recital in performance or composition.
- Students perform regularly in a variety of settings, allowing them to gain a variety of professional-level performing experiences. At least one ensemble per semester is required of all Bachelor of Music students until they have completed all the major requirements.
- Students complete specialized classes at the upper levels to learn the skills and more concentrated knowledge bases of the individual concentration areas. These classes may be aimed at developing a broad survey knowledge of music literature or history in a particular segment of the repertoire, or at the discovery and development of skills needed in the field but not necessarily to be mastered through individual practice and performance.
- In some major concentrations, internships are required that put the student into the work world in a supervised off-campus learning activity designed to give practical workplace experience in the field.

Special Admission and Retention Requirements

Admission Requirements

To be considered for undergraduate admission to the music program, all entering first year students, transfer students and previously enrolled students seeking re-admission must complete the following:

- Submit an application for admission to JMU's Office of Admission.
- Submit the Undergraduate Music Application on Slideroom.
- Successfully complete a B.M. audition.

Guidelines may be found on the music website.

Students are encouraged to contact the applied faculty on their area of performance expertise with specific questions about auditions. General questions about the music degree programs may be answered by an admissions assistant at (540) 568-3851 or by sending a message to music_admit@jmu.edu.

Audition, Exam and Placement Test

No student will be accepted into the music degree programs until an audition or interview is successfully completed.

Bachelor of Music candidates are expected to perform the entrance audition on one of the scheduled audition dates. The school does not encourage special appointments and will arrange them only when applicants have serious conflicts with the scheduled auditions.

Because assessing a student's ability through a recorded performance is difficult, only students who live at great distances are encouraged to send audition recordings. All recordings submitted must be of high quality and demonstrate the student's ability.

The audition should reveal the student's highest level of musical attainment. Appropriate literature in at least two varying styles and tempos is suggested so that the school can evaluate the candidate's ability accurately. If you have specific questions regarding appropriate audition literature, please contact the applied professor of your area of performance.

In addition to meeting the audition requirements above, all students who intend to major in the Bachelor of Music, concentration in music industry are required to first complete MUI 221 and then enroll in MUI 230 and MUI 231. To successfully complete MUI 230, prospective majors and minors must submit a portfolio for review, showing evidence of their interest and potential for success in the music industry. The portfolio should represent work completed in the above classes with assistance in resume preparation from the University Career Center. Portfolios will be reviewed once each fall and once each spring semester. Following the portfolio review, students who are recommended for the major or minor will be eligible to register for upper-level music industry classes. Students who are not recommended for admission to the major or minor may reapply the following semester. To obtain the schedule of portfolio review sessions, as well as other specific information about the admission process, contact the School of Music, music industry area.

Students who score a 4 or 5 on the Music Theory – Aural Subscore Advanced Placement exam may earn credit for MUS 143.

All major concentrations in the School of Music require potential candidates to develop additional qualifications before full admission to the concentrations. These gateways are generally attempted at the end of the first or second year in the major. For additional details, see the School of Music's Undergraduate Music Student Handbook.

Retention Policy

Students admitted as music majors must meet school standards published in the School of Music Student Handbook to continue.

Music majors must earn a minimum grade of "C-" in all music courses required for their degree. In situations where courses are sequential, with prerequisites (for example, MUS 143-MUS 144 or MUS 243-MUS 244), if a grade lower than "C-" is earned, it does not constitute fulfillment of the prerequisite.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Core music program courses	30
Major and concentration requirements and electives	43-59
Professional education sequence ²	20
University Electives	0-1
Total	120-128
Jazz Studies Concentration Requirements	127
Music Composition Concentration Requirements	123
Music Education, Instrumental Track Requirements	128 ²

Music Education, Instrumental Track, Subtrack in Jazz Requirements	128 ²
Music Education, Vocal Track Requirements	125 ²
Music Industry Concentration Requirements	125
Music Performance, Instrumental Track	120
Music Performance, Piano Track Requirements	124
Music Performance, Piano Track - Subtrack in Accompanying/Coaching Requirements	124
Music Performance, Vocal Track Requirements	120
Music Theatre Concentration Requirements	123-125

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements in the major may be used to fulfill General Education requirements.

² In addition to the School of Music requirements outlined below, students seeking a concentration in music education must meet all the criteria for admission to the teacher education program and complete the professional education sequence for teaching licensure.

Major Requirements

Code	Title	Credits
Required Courses		
Incoming Music Major Assessment Milestone		0
MUS 143	Aural Skills I (taken twice) ¹	2
MUS 144	Aural Skills II (taken twice) ²	2
MUS 195	Recital Attendance (six semesters)	0
MUS 206	Introduction to Global Music [C2VPA] ³	3
MUS 207	History of Western Music	4
MUS 317	Basic Conducting	2
MUS 380	Musicological Perspectives	3
MUS 381	Music Research Lab	1
School of Music Senior Assessment Milestone		0
Subtotal		17
Keyboard Requirement		
Choose one of the following:		1
MUS 101	Keyboard Skills II ⁴	
MUS 303	Keyboard Skills IV ⁵	
MUS 305	Jazz Keyboard Skills ⁵	
Subtotal		1
Written Music Theory Paths		
Choose one of the following paths:		12
<i>Written Music Theory Path One:</i>		
Choose this path if you have passed the music theory diagnostic exam. None of the courses in this section may be repeated for credit.		
MUS 241	Diatonic and Chromatic Harmony	
MUS 252	Analysis of Music 1900-1950	
or MUS 253	Analysis of Music 1950-Present	
Choose two of the following:		
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
MUS 254	Form and Analysis: Popular Music	

MUS 255	Jazz Theory ⁶
MUS 256	Contrapuntal Techniques
MUS 257	Techniques of Composition ⁷
MUS 258	Film Music Analysis

Written Music Theory Path Two:

Choose this path if you have not successfully passed the music theory diagnostic exam. None of the courses in this section may be repeated for credit.

MUS 141	Foundations of Music Theory	
MUS 241	Diatonic and Chromatic Harmony	
MUS 252	Analysis of Music 1900-1950	
or MUS 253	Analysis of Music 1950-Present	
Choose one of the following:		
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
MUS 254	Form and Analysis: Popular Music	
MUS 255	Jazz Theory ⁶	
MUS 256	Contrapuntal Techniques	
MUS 257	Techniques of Composition ⁷	
MUS 258	Film Music Analysis	
Subtotal		12
Concentrations		
Choose one of the following concentrations:		43-59
Jazz Studies (59 Credits)		
Music Composition (55 Credits)		
Music Education (43-46 + 20 Credits)		
Music Industry (57 Credits)		
Music Performance (51-56 Credits)		
Music Theatre (55-57 Credits)		
Subtotal		43-59
Total Credits		73-89

¹ Students may substitute 1 credit of MUS 243 or MUS 244 for 1 credit of MUS 143 rather than repeating the course.

² Students may substitute 1 credit of MUS 243 or MUS 244 for 1 credit of MUS 144 rather than repeating the course.

³ MUS 206 will count for the Music Core and also for the General Education, Arts and Humanities, Part 2, if desired. There will be a specific music major section of MUS 206 offered once each academic year.

⁴ Required for music industry students and prerequisite for MUS 202-MUS 303. Students whose primary instrument is piano must take MUS 304.

⁵ All music students except music industry, jazz studies and music education jazz subtrack must take MUS 303, which must be passed prior to student teaching. Students in the B.M. jazz studies and music education instrumental jazz subtrack concentrations enroll in MUS 305 instead of MUS 303.

⁶ Students concentrating in Jazz or Music Education Jazz Subtrack must take MUS 255.

⁷ MUS 257 is not open to music composition students.

Additional Information

For a recommended eight-semester outline of curricular requirements for each concentration, see the School of Music's Undergraduate Music Student Handbook.

Concentrations

Jazz Studies Concentration

Coordinator: Charles Dotas

Email: dotascj@jmu.edu

Phone: (540) 568-6180

Code	Title	Credits
Required Courses		
Admission to the Jazz Studies concentration by successful completion of a jazz performance audition no sooner than the end of the first year as an approved music major.		
MUS 345	Small Ensemble Jazz Arranging	3
MUS 346	Large Ensemble Jazz Arranging	3
MUS 356	History of Jazz	3
MUS 395	Junior or Senior Half Recital (presented at the end of the junior year, jazz repertoire)	0
MUS 440	Jazz Improvisation Laboratory II (taken for four semesters for a total of 8 credit hours)	8
MUS 473	Jazz Ensemble Procedures and Techniques	2
MUS 495	Senior Graduation Recital	1
MUAP 301	Assigned Ensemble (one each semester for 8 semesters) ¹	8
MUAP 355	Jazz Chamber Ensemble (taken for six semesters for a total of 6 credit hours in addition to the assigned ensemble requirement)	6
Approved jazz electives ²		5
Subtotal		39
Applied Music Lesson in Primary Instrument		
Applied music study (primary instrument), Level 1-4. Jazz Studies students enroll in 2 credits in the primary instrument for 4 semesters in the first two years for a total of 8.00 credit hours. Students who can demonstrate level 4 competency by the end of their first year are eligible to attempt credit by exam for up to 4 credit hours of primary instrument lessons. Each area has its own procedure for credit by exam and students should consult their primary applied instructor if they wish to pursue this option. Students accepted into the Jazz Studies program may continue applied music study (in addition to applied jazz study) with consent of primary applied instructor.		
MUAP 300	Primary/Major Applied Music Lesson (taken at two credit hours for four semesters for a total of 8 credit hours)	8
Subtotal		8
Applied Music Lesson in Jazz		
Applied music study (Jazz). Two credits of Applied Jazz Study, Level 5-8, for 6 semesters for a total of 12 credit hours; may be taken concurrently with applied music study in the primary instrument. The student must be enrolled in Applied Jazz Study during the semesters of the Half Recital MUS 395 and the Graduation Recital MUS 495.		
MUAP 300	Primary/Major Applied Music Lesson (taken at two credit hours for six semesters for a total of 12 credit hours)	12

Subtotal	12
Total Credits	59

¹ Prior to acceptance into Jazz Studies program, ensemble placement determined by primary instrument applied faculty and Ensemble Committee. Students accepted into the Jazz Studies program must enroll in MUAP 347 (or MUAP 348 with jazz studies coordinator permission) each semester until graduation. Students in the Jazz Studies program are encouraged to continue participation in additional School of Music ensembles.

² Approved electives include: MUS 446, MUS 480 and MUS 485. Students may substitute one ARTS or IART course for up to 3.00 credits of approved jazz electives.

Music Composition Concentration

Coordinator: Jason Haney

Email: haneyjx@jmu.edu

Phone: (540) 568-6664

Code	Title	Credits
Required Courses		
Additional requirements: Admission to this concentration by approval of theory/composition faculty. Students should take both MUS 252 and MUS 253 from the theory menu; MUS 257 is not open to music composition students.		
MUS 256	Contrapuntal Techniques ¹	3
MUS 352	Music Composition (repeatable for a total of 16 credit hours)	16
MUS 435	Instrumentation	3
MUS 445	Orchestration	3
MUS 450	Topics in Music Analysis	3
MUS 480	Advanced Seminar in Musicological Topics	3
or MUS 485	Advanced Jazz Topics Seminar	
MUS 495	Senior Graduation Recital	1
Applied major study		12
MUAP 301	Assigned Ensemble (one assigned ensemble each semester) ²	8
Approved music, ARTS or IART electives ³		3
Total Credits		55

¹ MUS 256 must be taken in addition to the 4 courses required as part of the music theory core.

² Students whose primary instrument is piano must take MUAP 301-0014 (Collaborative Piano) for a minimum of two semesters.

³ Approved music electives may not be fulfilled by additional ensemble credits.

Music Education Concentration

Coordinator: Amy Birdsong

Email: birdsoae@jmu.edu

Phone: (540) 568-2384

The Music Education program is designed primarily for those preparing to teach vocal or instrumental music in public schools. In addition to the School of Music requirements outlined below, students seeking a degree in music education must meet all the criteria for admission to the teacher

education program and complete the professional education sequence for teaching licensure.

Endorsement in Both Vocal and Instrumental Music

Students desiring licensure in both instrumental and vocal music should consult the coordinator of music education to plan their programs.

Teaching Licensure Requirements

Students who want to pursue the music education program must earn acceptance into the teaching licensure program offered by the College of Education.

Professional Education Sequence Requirements

Code	Title	Credits
EDUC 102	Critical Questions in Education [C1CT] ^{1,2}	3
LED 420	Content Area Literacy, K-12	2
MUED 480	Student Teaching (senior year) (must be taken for a total of 12 credits)	12
PSYC 160	Life Span Human Development [C5SD] ^{1,2}	3
Total Credits		20

¹ Minimum grade of "C" required in EDUC 102, PSYC 160 and WRTC 103.

² EDUC 102 and PSYC 160 fulfill General Education requirements.

Music Education, Instrumental Track

(Applied study in winds, brass, percussion or strings)

Students whose primary instrument is piano must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in instrumental ensembles each semester except for two during which they must take MUAP 301-0014 (Collaborative Piano). Performance in band and orchestral ensembles will be on the secondary instrument with one semester on piano permitted as ensemble needs dictate.

Subtrack in Winds, Brass, Percussion, Piano Requirements

Code	Title	Credits
Required Courses		
MUED 200	Small Ensemble for Instrumental Music Education	0
MUED 271	Music Education: A Professional Choice	1
MUED 273	Music Education: Professional Practice	1
MUED 275	Practicum in Music Education (taken twice) ¹	2
MUED 310	Vocal Techniques	1
MUED 371	Beginning Methods and Materials for Instrumental Music	2
MUED 373	Advanced Methods and Materials for Instrumental Music	2
MUED 380	Music in the Elementary School	2
MUED 470	Marching Band Procedures	1
MUS 319	Intermediate Instrumental Conducting	2
MUS 395	Junior or Senior Half Recital ²	0
MUS 442	Instrumental Arranging and Technology	2
Applied music study (primary instrument)		13
MUAP 237	Marching Band (taken twice for a total of 3 credit hours) ³	3
MUAP 301	Assigned Ensemble (one each semester for seven semesters) ⁴	7
Subtotal		39

Instrumental Techniques Classes		
MUED 301	Woodwind Techniques	1
MUED 303	Brass Techniques	1
MUED 305	Percussion Techniques	1
MUED 307	String Techniques	1
MUED 311	Ensemble Techniques	1
Subtotal		5
Restricted Electives		
Choose two credit hours from the following:		2
MUED 277	MUED Teaching Lab	
MUED 472	Survey of String Orchestra Repertoire	
MUED 474	Teaching Popular Music and Guitar in the Classroom	
MUED 482	Orff and Kodaly: Literature, Principles and Practices	
MUS 150	Introduction to Technological Applications in Music	
MUS 473	Jazz Ensemble Procedures and Techniques	
Any ARTS or IART course		
Subtotal		2
Total Credits		46

¹ MUED 275 should be taken twice: once for elementary general music practicum experience and once for secondary music (middle/high) practicum experience.

² Students may substitute MUS 495 for MUS 395.

³ Wind, brass, and percussion students are required to participate in marching band for a minimum of two years, but participation every year is recommended.

⁴ Students whose primary instrument is piano must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in a MUAP 301 assigned instrumental ensemble for five semesters. For two additional semesters they must take MUAP 301-0014 (Collaborative Piano). Performance in band or orchestral ensembles will be on the secondary instrument with one semester on piano permitted as ensemble needs dictate.

Subtrack in Strings, Piano, Guitar Requirements

Code	Title	Credits
Required Courses		
MUED 200	Small Ensemble for Instrumental Music Education	0
MUED 271	Music Education: A Professional Choice	1
MUED 273	Music Education: Professional Practice	1
MUED 275	Practicum in Music Education (taken twice) ¹	2
MUED 310	Vocal Techniques	1
MUED 371	Beginning Methods and Materials for Instrumental Music	2
MUED 373	Advanced Methods and Materials for Instrumental Music	2
MUED 380	Music in the Elementary School	2
MUED 472	Survey of String Orchestra Repertoire	2
MUED 474	Teaching Popular Music and Guitar in the Classroom	2
MUS 319	Intermediate Instrumental Conducting	2
MUS 395	Junior or Senior Half Recital ²	0

MUS 442	Instrumental Arranging and Technology	2
Applied music study (primary instrument)		13
MUAP 301	Assigned Ensemble (one each semester for seven semesters) ²	7
Subtotal		39
Instrumental Techniques Classes		
MUED 301	Woodwind Techniques	1
MUED 303	Brass Techniques	1
MUED 305	Percussion Techniques	1
MUED 307	String Techniques	1
MUED 311	Ensemble Techniques	1
Subtotal		5
Restricted Electives		
Choose two credit hours from the following:		2
MUED 277	MUED Teaching Lab	
MUED 482	Orff and Kodaly: Literature, Principles and Practices	
MUS 150	Introduction to Technological Applications in Music	
MUS 473	Jazz Ensemble Procedures and Techniques	
Subtotal		2
Total Credits		46

¹ MUED 275 should be taken twice: once for elementary general music practicum experience and once for secondary music (middle/high) practicum experience.

² Students may substitute MUS 495 for MUS 395.

³ Students whose primary instrument is piano must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in a MUAP 301 assigned instrumental ensemble for five semesters. For two additional semesters they must take MUAP 301-0014 (Collaborative Piano). Performance in band or orchestral ensembles will be on the secondary instrument with one semester on piano permitted as ensemble needs dictate.

Subtrack in Jazz Requirements

(Applied study in winds, strings, percussion, piano, guitar)

Students whose primary instrument is piano must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in three semesters of MUAP 347 or MUAP 348, and four semesters of a non-jazz instrumental ensemble except for two during which they must elect MUAP 357. Performance in Jazz Ensemble or Jazz Band will be on piano; performance in non-jazz ensembles will be on the secondary instrument with one semester on piano permitted as ensemble needs dictate. Students whose primary instrument is guitar must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in three semesters of MUAP 347 or MUAP 348, and four semesters of a non-jazz instrumental ensemble except for two during which they must elect MUAP 353. Performance in Jazz Ensemble or Jazz Band will be on guitar; performance in non-jazz ensembles will be on the secondary instrument with one semester permitted on guitar as ensemble needs dictate.

Conditional admission to the Subtrack in Jazz is by successful completion of jazz performance audition no sooner than the end of the first year; full admission to the Subtrack in Jazz is upon completion of Sophomore Music Education Interview.

Four semesters of traditional applied music studio study are required, to include a Capstone Performance Project (Area Recital, Double Jury, or other public performance approved by the applied instructor), followed by three semesters of applied jazz study, to include a senior half recital comprising jazz repertoire.

Code	Title	Credits
Required Courses		
MUED 200	Small Ensemble for Instrumental Music Education	0
MUED 271	Music Education: A Professional Choice	1
MUED 273	Music Education: Professional Practice	1
MUED 275	Practicum in Music Education (taken twice) ¹	2
MUED 310	Vocal Techniques	1
MUED 371	Beginning Methods and Materials for Instrumental Music	2
MUED 373	Advanced Methods and Materials for Instrumental Music	2
MUED 380	Music in the Elementary School	2
MUS 319	Intermediate Instrumental Conducting	2
MUS 345	Small Ensemble Jazz Arranging ²	3
MUS 395	Junior or Senior Half Recital ^{3,4}	0
Applied music study (primary instrument)		13
MUAP 355	Jazz Chamber Ensemble	1
MUAP 237	Marching Band (must be taken for a total of 2 credit hours) ⁵	2
MUAP 301	Assigned Ensemble (one each semester for seven semesters) ^{6,7}	7
Subtotal		39
Instrumental Techniques Classes		
MUED 301	Woodwind Techniques	1
MUED 303	Brass Techniques	1
MUED 305	Percussion Techniques	1
MUED 307	String Techniques	1
MUED 311	Ensemble Techniques	1
Subtotal		5
Restricted Electives		
Choose two credit hours from the following:		2
MUED 277	MUED Teaching Lab	
MUED 470	Marching Band Procedures	
MUED 472	Survey of String Orchestra Repertoire	
MUED 474	Teaching Popular Music and Guitar in the Classroom	
MUS 473	Jazz Ensemble Procedures and Techniques	
Any ARTS or IART course		
Subtotal		
Total Credits		46

¹ MUED 275 should be taken twice: once for elementary general music practicum experience and once for secondary music (middle/high) practicum experience.

² MUS 346 may be substituted if student teaching occurs during the fall semester.

³ Half recital to comprise jazz repertoire.

⁴ Students may substitute MUS 495 for MUS 395.

⁵ Students are required to participate in marching band for a minimum of one year. Piano and guitar students will perform on the secondary instrument.

⁶ Students in the Jazz Subtrack are required to participate in four semesters of an assigned traditional wind or string ensemble, and three semesters of one assigned jazz ensemble. Students must participate in at least one assigned ensemble each semester for seven semesters.

⁷ Students whose primary instrument is piano must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in three semesters of an assigned jazz ensemble, and four semesters of an assigned non-jazz instrumental ensemble except for two during which they must elect MUAP 301-0014 (Collaborative Piano). Performance in Jazz Ensemble or Jazz Band will be on piano; performance in non-jazz ensembles will be on the secondary instrument with one semester on piano permitted as ensemble needs dictate. Students whose primary instrument is guitar must pass an interview with music education faculty, an audition on a band or orchestra instrument, and participate in three semesters of an assigned jazz ensemble, and four semesters of a non-jazz assigned instrumental ensemble except for two during which they must elect MUAP 301-0015 (Guitar Ensemble). Performance in Jazz Ensemble or Jazz Band will be on guitar; performance in non-jazz ensembles will be on the secondary instrument with one semester permitted on guitar as ensemble needs dictate.

Music Education, Vocal Track

(Applied study in voice or piano)

Code	Title	Credits
Required Courses		
MUED 201	Small Ensemble for Vocal Music Education	0
MUED 206	Instrumental Music Methods for Vocal Music Education	1
MUED 271	Music Education: A Professional Choice	1
MUED 273	Music Education: Professional Practice	1
MUED 275	Practicum in Music Education (taken twice) ¹	2
MUED 372	General Music Practices	2
MUED 376	Choral Music Materials and Techniques	2
MUED 380	Music in the Elementary School	2
MUS 120	Diction for Singers	1
MUS 121	Diction for Singers II	1
MUS 318	Intermediate Choral Conducting	2
MUS 395	Junior or Senior Half Recital ²	0
MUS 441	Vocal Arranging and Technology	2
MUS 477	Foundations of Vocal Pedagogy	1
Applied major study (voice or keyboard)		12
Applied secondary area (Piano majors take MUAP 205 for 2 credit hours; voice majors take MUS 404 for 1 credit hour.) ³		1-2
MUAP 301	Assigned Ensemble (one each semester for seven semesters) ⁴	7
Subtotal		38-39
Restricted Electives		
Piano majors take 4 credit hours; voice majors take 5 credit hours from the following courses: ⁵		4-5
MUED 277	MUED Teaching Lab	
MUED 471	School Musical, Jazz and Show Choir Procedures	

MUED 474	Teaching Popular Music and Guitar in the Classroom	
MUED 482	Orff and Kodaly: Literature, Principles and Practices	
MUED 485	Teaching Music to Students with Special Needs	
MUS 150	Introduction to Technological Applications in Music	
MUS 473	Jazz Ensemble Procedures and Techniques	
Subtotal		4-5
Total Credits		43

¹ MUED 275 should be taken twice: once for elementary general music practicum experience and once for secondary music (middle/high) practicum experience.

² Students may substitute MUS 495 for MUS 395.

³ Vocal track students must pass the Advanced Keyboard Skills exam.

⁴ Piano track students must take MUAP 301-0014 (Collaborative Piano) for two semesters.

⁵ Students may substitute any ARTS or IART class for *one* of the restricted electives listed here.

Music Industry Concentration

Coordinator: David Cottrell

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Phone: (540) 568-6303

Code	Title	Credits
Required Courses		
ACTG 244	Accounting for Non-Business Majors	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
MUI 221	Survey of Music Industry	3
MUI 230	Music Industry Gateway	0
MUI 231	Legal Aspects of Music Industry	3
MUI 440	Entrepreneurship in Music Industry	3
MUI 492	Internship in Music Industry	3
MUS 395	Junior or Senior Half Recital ¹	0
Applied major study		14
MUAP 301	Assigned Ensemble (one each semester for seven semesters) ²	7
Music industry electives		12
Approved music, ARTS or IART electives ³		3
Total Credits		57

¹ Students may substitute MUS 495 for MUS 395.

² Students whose primary instrument is piano must take MUAP 301-0014 (Collaborative Piano) for a minimum of two semesters.

³ Approved music electives may not be fulfilled by additional ensemble credits.

Music Performance

Music Performance, Instrumental Track

Co-coordinators: Beth Chandler Cahill, Carl Donakowski and Chris Carrillo

Email: chand2be@jmu.edu

Phone: (540) 568-6197

Code	Title	Credits
Required Courses		
Additional requirements: Admission to this concentration by successful completion of performance audition, no sooner than end of first year as approved music major.		
MUS 319	Intermediate Instrumental Conducting	2
MUS 395	Junior or Senior Half Recital	0
MUS 472	Instrumental Pedagogy ¹	1
Choose one of the following:		3
MUS 450	Topics in Music Analysis	
MUS 480	Advanced Seminar in Musicological Topics	
MUS 485	Advanced Jazz Topics Seminar	
MUS 495	Senior Graduation Recital	1
Applied music courses (primary instrument)		22
Chamber ensembles (in addition to the eight-semester major assigned ensemble requirement)		6
MUAP 301	Assigned Ensemble (one each semester for 8 semesters)	8
Advanced music literature or seminar electives		3
Approved music, ARTS or IART electives ²		3
Subtotal		49

Music Theory Elective

Choose one of the following courses that has not previously been taken:		3
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
MUS 254	Form and Analysis: Popular Music	
MUS 255	Jazz Theory	
MUS 256	Contrapuntal Techniques	
MUS 257	Techniques of Composition	
MUS 258	Film Music Analysis	
Subtotal		3
Total Credits		52

¹ MUS 450, MUS 480, or MUS 485 also fulfill the requirement for an advanced music literature elective or the approved music electives.

² Approved music electives may not be fulfilled by additional ensemble credits. A maximum of two credits may be counted in secondary applied music study.

Music Performance, Piano Track

Coordinator: Paulo Steinberg

Email: steinbpr@jmu.edu

Phone: (540) 568-6977

Code	Title	Credits
Required Courses		
Additional requirements: Admission to this concentration by successful completion of performance audition, no sooner than end of first year as approved music major.		
MUS 371	Private Piano Pedagogy	2
MUS 395	Junior or Senior Half Recital	0
MUS 420	Piano Technology I	1

MUS 421	Piano Technology II	1
MUS 450	Topics in Music Analysis ¹	3
MUS 460	Piano Literature I	2
MUS 470	Piano Literature II	2
MUS 471	Piano Literature III	2
MUS 495	Senior Graduation Recital	1
Applied piano study		22
MUAP 301	Assigned Ensemble (one each semester for 8 semesters)	8
Approved music electives ²		3
Approved music, ARTS or IART electives ²		3
Subtotal		50
Seminar		
Choose one of the following:		3
MUS 450	Topics in Music Analysis ¹	
MUS 480	Advanced Seminar in Musicological Topics	
MUS 485	Advanced Jazz Topics Seminar	
Subtotal		3
Music Theory Elective		
The chosen course should be one not taken to fulfill the music theory core requirements. ³		
Choose one of the following:		3
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
MUS 254	Form and Analysis: Popular Music	
MUS 255	Jazz Theory	
MUS 256	Contrapuntal Techniques	
MUS 257	Techniques of Composition	
MUS 258	Film Music Analysis	
Subtotal		3
Total Credits		56

¹ If choosing a second iteration of MUS 450 in this category, the semester and seminar topic must be different than the first required MUS 450.

² Approved music electives may not be fulfilled by additional ensemble credits.

³ Course may not have already been taken as part of the music theory core curriculum. In other words, piano performance students should be selecting a fifth course from the music theory menu of options.

Music Performance, Piano Track - Subtrack in Accompanying/Coaching

Coordinator: Gabriel Dobner

Email: dobnergt@jmu.edu

Phone: (540) 568-6002

Code	Title	Credits
Required Courses		
MUS 120	Diction for Singers	1
MUS 121	Diction for Singers II	1
MUS 371	Private Piano Pedagogy	2
MUS 395	Junior or Senior Half Recital	0

MUS 467	Song Literature	3
MUS 470	Piano Literature II	2
MUS 495	Senior Graduation Recital	1
Applied piano accompanying		9
Applied piano study		13
Applied voice study		2
Approved music electives ¹		3
Approved music, ARTS or IART electives ¹		3
MUAP 301	Assigned Ensemble (one each semester for 8 semesters)	8
Subtotal		48
Language Requirement		
Choose one of the following:		
FR 101	Elementary French I	
GER 101	Elementary German I	
ITAL 101	Elementary Italian I	
Subtotal		3
Conducting Requirement		
MUS 318	Intermediate Choral Conducting	2
or MUS 319	Intermediate Instrumental Conducting	
Subtotal		2
Music Theory Elective		
The chosen course should be one not taken to fulfill the music theory core requirements.		
Choose one of the following:		3
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
MUS 254	Form and Analysis: Popular Music	
MUS 255	Jazz Theory	
MUS 256	Contrapuntal Techniques	
MUS 257	Techniques of Composition	
MUS 258	Film Music Analysis	
Subtotal		3
Total Credits		56

¹ Approved music electives may not be fulfilled by additional ensemble credits.

Music Performance, Vocal Track

Coordinator: Jamison Walker

Email: walke5j@jmu.edu

Phone: (540) 568-6970

Code	Title	Credits
Required Courses		
Additional requirements: Admission to this concentration by successful completion of performance audition, no sooner than end of first year as approved music major.		
MUS 120	Diction for Singers	1
MUS 121	Diction for Singers II	1
MUS 330	Skills for the Working Singer	1
MUS 343	Opera Workshop (taken twice)	2

MUS 395	Junior or Senior Half Recital	0
MUS 477	Foundations of Vocal Pedagogy	1
MUS 478	Styles of Vocal Pedagogy	1
MUS 495	Senior Graduation Recital	1
Applied voice study		16
MUAP 301	Assigned Ensemble (one each semester for 8 semesters) ¹	8
Music, ARTS or IART elective		1
Subtotal		33
Vocal Literature Electives		
Choose two of the following:		6
MUS 465	Opera History and Literature	
MUS 466	Special Topics in Solo Vocal Literature	
MUS 467	Song Literature	
Subtotal		6
Language Requirement		
Elementary Languages I:		
FR 101	Elementary French I	3
GER 101	Elementary German I	3
ITAL 101	Elementary Italian I	3
Elementary Languages II:		
Choose one of the following:		3
FR 102	Elementary French II	
GER 102	Elementary German II	
ITAL 102	Elementary Italian II	
Subtotal		12
Total Credits		51

¹ Students in the Bachelor of Music in Performance with a specialization in Performance - Vocal Track must participate in MUAP 301-0006 (Opera Theater) in an onstage capacity for at least two semesters. They may elect to fill the ensemble requirement in their remaining semesters by taking any of the MUAP 301 assigned vocal ensembles in the school: UCTB, UCSA, Treble Chamber Choir, Chorale, Madison Singers, or Opera Theatre. A minimum of one credit of ensemble must be selected each semester.

Music Theatre Concentration

Coordinator: Joshua Borths

Email: borthsjj@jmu.edu

Phone: (540) 568-6863

Code	Title	Credits
Required Courses		
Additional requirements: Admission to this concentration by successful completion of performance audition, no sooner than end of first year as approved music major.		
MUI 221	Survey of Music Industry	3
MUS 120	Diction for Singers	1
MUS 121	Diction for Singers II	1
MUS/THEA 357	Music Theatre History and Analysis	3
MUS 395	Junior or Senior Half Recital ¹	0
MUS 465	Opera History and Literature	3
THEA 251	Acting I: Basic Acting	3

Applied voice study		16
Approved music, theatre, dance, ARTS or IART electives ²		2
MUAP 301	Assigned Ensemble (one each semester for 8 semesters, as listed below) ³	8
THEA 352	Music Theatre Performance for Theatre and Music Majors	2
Choose one of the following:		3
STAD 171	Performance Production	
STAD 273	Design Aspects of Performance	
THEA 375	Sound Design	
Choose one of the following courses not selected above:		2-3
STAD 171	Performance Production	
THEA 261	Voice for the Stage	
STAD 273	Design Aspects of Performance	
THEA 303	Topics in Theatre	
THEA 323	Youth Theatre	
THEA 324	Theatre for Young Audiences	
THEA 332	Costume and Fashion History	
THEA 336	History, Theory and Practice of Stage Makeup	
THEA 351	Acting II: Intermediate Acting	
THEA 354	Music Theatre Workshop	
Choose two of the following courses not selected above:		4-5
MUS 343	Opera Workshop	
THEA 281	Movement for the Actor	
THEA 303	Topics in Theatre	
THEA 454	Advanced Musical Theatre Performance	
Subtotal		51-53

Dance Classes

Choose two of the following:		4
DANC 140	Elementary Modern Dance (0,4)	
DANC 142	Elementary Ballet (0, 4)	
DANC 144	Ballroom Dance (0, 4)	
DANC 146	Jazz Dance (0, 4)	
DANC 147	Tap Dance (0, 4)	
DANC 240	Intermediate Modern Dance I (0, 4)	
DANC 242	Intermediate Ballet I (0, 4)	
DANC 245	Dance Improvisation (0, 4)	
DANC 246	Intermediate Jazz	
DANC 303	Topics in Dance	
DANC 340	Intermediate Modern Dance II (0, 4)	
DANC 342	Intermediate Ballet II (0, 4)	
DANC 346	Intermediate Jazz II/Musical Theatre Styles	
DANC 440	Advanced Modern Dance (0, 4)	
DANC 442	Advanced Ballet (0, 4)	
DANC 446	Advanced Jazz	
Subtotal		4
Total Credits		55-57

¹ Students may substitute MUS 495 for MUS 395.

² Approved music electives may not be fulfilled by additional ensemble credits.

³ Students in the Bachelor of Music, Music Theatre Concentration must participate in MUAP 301-006 (Opera Theater), in an on-stage

capacity for at least two semesters. They may elect to fill the ensemble requirement in their remaining semesters by taking any of the MUAP 301 assigned vocal ensembles in the school – Men's or Women's Chorus; Treble Chamber Choir; Chorale; Madison Singers; Opera Theatre. A minimum of one credit of ensemble must be selected each semester.

Recommended Schedule for First Year Majors

The first year of study shares a common curriculum for all concentrations. The year is devoted to courses from the basic music core requirements, General Education courses, ensemble participation and applied study in the primary applied area.

Course	Title	Credits
First Year		
First Semester		
MUS 141 or MUS 241	Foundations of Music Theory or Diatonic and Chromatic Harmony	3
MUS 143	Aural Skills I	1
MUS 195	Recital Attendance	0
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Applied music major course		2
Music ensemble course		2
General Education Madison Foundations		3
Choose one of the following: (According to placement decision)		1
MUS 100	Keyboard Skills	
MUS 101	Keyboard Skills II	
MUS 202	Keyboard Skills III	
MUS 303	Keyboard Skills IV	
MUS 305	Jazz Keyboard Skills (required for jazz studies students and music education jazz subtrack students)	
Credits		15
Second Semester		
MUS 195	Recital Attendance	0
MUS 143	Aural Skills I	1
Applied music major course		2
Music ensemble course		2
General Education requirement ¹		9
Choose one of the following:		1
MUS 101	Keyboard Skills II	
MUS 202	Keyboard Skills III	
MUS 303	Keyboard Skills IV	
MUS 305	Jazz Keyboard Skills (required for jazz studies students and music education jazz subtrack students)	
Credits		15
Total Credits		30

¹ MUS 206 and final courses from Madison Foundations

Nursing, B.S.N.

Coordinator: Julie Sutherland

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Program Description

The Bachelor of Science in Nursing (B.S.N.) program prepares students for licensure as entry-level professional nurses. The General Education program engages students in the liberal arts, promotes community and civic engagement, and develops students' abilities to use ethical reasoning in decision-making. The B.S.N. program, including the traditional B.S.N. pathway and the fast flex pathway, builds on the liberal arts to develop discipline-specific knowledge, skills and abilities needed to promote health, prevent disease, manage acute and chronic conditions, and coordinate care to individuals, families and communities across the lifespan in a variety of healthcare and community settings.

Admission and Retention Requirements

The B.S.N. in nursing offers three options for completion: the traditional pathway, the Fast Flex pathway and the R.N.-B.S.N. pathway. Admission to the traditional pathway is limited to 113 students each fall and spring. Admission to the Fast Flex pathway is limited to 30 students each summer. For information on the R.N.-B.S.N. pathway, see the R.N.-B.S.N. catalog page (p. 391).

All applicants must be currently admitted to the university, meet admission criteria, and complete the application to the pathway of choice. Detailed application information is available on the nursing school website (<https://www.jmu.edu/chbs/nursing/index.shtml>).

Application Requirements

Admission to the B.S.N. requires completion of a secondary application. The application process consists of two steps.

Step One: The following are required to apply for admission to the traditional B.S.N. pathway and the Fast Flex pathway. Admission is conditional upon meeting all admission requirements.

- Enrolled at JMU
- B.S.N. declared major (deadlines are October 1 for spring registration and March 1 for summer and fall registration)
 - Change of major requests received after the deadline will be processed the following semester. Students changing their major to Nursing after March 1 of their sophomore year should expect an additional semester(s) to complete the program. The number of additional semesters required to complete the program will depend on the timing of the change to Nursing and the number of summer courses completed.
- The first step for changing your major is to log into MyMadison and visit your Student Center. In the "Other Academic..." dropdown menu you should find options for adding and dropping a major. Once you have completed this process, you will need to schedule an appointment with Anna Cho, Academic advisor for the School of Nursing, also through MyMadison.
- Please note that to apply to the Bachelor of Science in Nursing (BSN) program you must be a full-time student at James Madison University and meet all admission requirements and prerequisites as stated on the B.S.N. website (<https://www.jmu.edu/chbs/nursing/bsn/index.shtml>).

- One full-time semester at JMU completed (by the end of the semester of application)
- 36 credit hours completed (by the end of the semester of application)
- Earn a minimum grade of "C-" in prerequisite courses and maintain a minimum prerequisite GPA of 3.0
 - General Education "Madison Foundations" – one Critical Thinking and one Human Communication course. The average of the grades earned in "Madison Foundations" and "Critical Thinking" will be used in your prerequisite GPA calculation.
 - One of the two biology course sequences: BIO 270/BIO 270L or BIO 290/BIO 290L
 - PSYC 160
- After the criteria and pre-requisite courses are met, the following will be used in applicant consideration:
 - Experiences: ROTC, work experiences, and community service
 - An essay of no more than 500 words to address a prompt that demonstrates students' commitment to the values of the School of Nursing

Admission Requirements

Step Two: Conditional admission is contingent upon the student's successful completion of all remaining admission requirements prior to program entry.

After application and before entry into the traditional or Fast Flex pathway, students must maintain a minimum prerequisite GPA of 3.00, complete 48 credit hours, and earn a minimum grade of "C-" in the remaining prerequisite courses:

- Earn a minimum grade of "C-" in prerequisite courses:
- NUTR 280
- MATH 220
- CHEM 120
- BIO 245/BIO 245L
- Complete remaining biology course: BIO 270/BIO 270L or BIO 290/BIO 290L

Students may elect to repeat up to two courses during their enrollment at JMU on a repeat/forgive basis following university policy. For prerequisite courses used in the admission application, specifically biology, chemistry and statistics, when taken at JMU or other institutions as repeat credit, the nursing program will average the grade of the repeated course for a final grade used in the admission application.

Students on the traditional pathway are full-time and take courses and clinicals during day, evening and night-time hours. Students on the fast flex pathway are full-time and take courses and clinicals during evening, night-time and weekend hours. The associate director for undergraduate programs must approve any deviation from the full-time enrollment pattern as outlined in the catalog. Once admitted, the student must earn a minimum grade of "C-" or "Credit" in each required nursing course to remain in good standing in the nursing program. A grade of "D+/-", "F" or "No Credit" is considered a course failure. A student who receives a failing grade in a nursing course may repeat one nursing course with school approval. A student who receives a second failing grade will be ineligible to continue in the program.

Other academic and related policies in effect for students in the nursing program are outlined on the school website (<https://www.jmu.edu/chbs/nursing/index.shtml>).

Additional Information

Madison Foundations normally is completed during the first year. General Education courses in Arts and Humanities, The Natural World, American and Global Perspectives and Sociocultural and Wellness Area can be scheduled as desired by the student across the first and second years.

Most students can complete general education and prerequisite courses in four semesters, but some students find it necessary to enroll in summer school or an additional year to complete the work satisfactorily.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Prerequisites ²	24
Nursing major requirements	61-62
University electives	6-7
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses that are prerequisites for or required within the major.

² 13 credits of prerequisite courses count towards General Education requirements.

Prerequisites

The following are course prerequisites required for the nursing major. Courses must be completed with a minimum grade of "C-."

Code	Title	Credits
BIO 245	General Microbiology	3
BIO 245L	General Microbiology Laboratory	1
BIO 270	Human Physiology [C3NS] ¹	3
BIO 270L	Human Physiology Laboratory [C3L]	1
BIO 290	Human Anatomy	3
BIO 290L	Human Anatomy Laboratory	1
CHEM 120	Concepts of Chemistry [C3PP] ^{1,2}	3
MATH 220	Elementary Statistics [C3QR] ¹	3
NUTR 280	Nutrition for Wellness	3
PSYC 160	Life Span Human Development [C5SD] ³	3
Total Credits		24

¹ Contained in The Natural World.

² CHEM 131 and CHEM 132 can be substituted for CHEM 120.

³ Contained in Sociocultural and Wellness Area.

Major Requirements

The core curriculum for nursing must be completed in a specific sequence and accounts for 59 credits. Students also take 2-3 credits of restricted electives.

Code	Title	Credits
Required Courses		
NSG 340	Nursing Identity	3
NSG 344	Optimizing Wellness I	3
NSG 344L	Optimizing Wellness I Clinical	1

NSG 345	Health Alterations I	3
NSG 345L	Health Alterations I Clinical	1
NSG 380	Professional Integration I	2
NSG 380L	Professional Integration I Clinical	1
NSG 381	The Evidence Based Clinician	3
NSG 382	Optimizing Wellness II	3
NSG 382L	Optimizing Wellness II Clinical	1
NSG 383	Health Alterations II	3
NSG 383L	Health Alterations II Clinical	1
NSG 384	Professional Integration II	3
NSG 384L	Professional Integration II Clinical	1
NSG 440	The Team Member	3
NSG 441	Optimizing Wellness III	3
NSG 441L	Optimizing Wellness III Clinical	1
NSG 443	Health Alterations III	3
NSG 443L	Health Alterations III Clinical	1
NSG 444	Professional Integration III	3
NSG 444L	Professional Integration III Clinical	1
NSG 445	Evolution as a Leader	3
NSG 446	Optimizing Wellness IV	3
NSG 446L	Optimizing Wellness IV Clinical	1
NSG 480	Health Alterations IV	3
NSG 480L	Health Alterations IV Clinical	1
NSG 481	Professional Integration IV	3
NSG 481L	Professional Integration IV Clinical	1
Subtotal		59

Restricted Electives		
Choose one of the following:		2-3
NSG elective		
REL 371	Religion and Disability	
REL 372	Religion and Medicine, East and West	
REL 380	Religion and Science	
Subtotal		2-3
Total Credits		61-62

Recommended Schedule for the Traditional Pathway

Students may fulfill the restricted electives requirement (2-3 credit hours) any time during their third or fourth year.

Course	Title	Credits
First Semester		
NSG 340	Nursing Identity	3
NSG 344	Optimizing Wellness I	3
NSG 344L	Optimizing Wellness I Clinical	1
NSG 345	Health Alterations I	3
NSG 345L	Health Alterations I Clinical	1
NSG 380	Professional Integration I	2
NSG 380L	Professional Integration I Clinical	1
Credits		14
Second Semester		
NSG 381	The Evidence Based Clinician	3
NSG 382	Optimizing Wellness II	3
NSG 382L	Optimizing Wellness II Clinical	1
NSG 383	Health Alterations II	3

NSG 383L	Health Alterations II Clinical	1	NSG 446	Optimizing Wellness IV	3
NSG 384	Professional Integration II	3	NSG 446L	Optimizing Wellness IV Clinical	1
NSG 384L	Professional Integration II Clinical	1	NSG 480	Health Alterations IV	3
Credits		15	NSG 480L	Health Alterations IV Clinical	1
Third Semester			NSG 481	Professional Integration IV	3
NSG 440	The Team Member	3	NSG 481L	Professional Integration IV Clinical	1
NSG 441	Optimizing Wellness III	3	Credits		15
NSG 441L	Optimizing Wellness III Clinical	1	Total Credits		59
NSG 443	Health Alterations III	3			
NSG 443L	Health Alterations III Clinical	1			
NSG 444	Professional Integration III	3			
NSG 444L	Professional Integration III Clinical	1			
Credits		15			
Fourth Semester					
NSG 445	Evolution as a Leader	3			
NSG 446	Optimizing Wellness IV	3			
NSG 446L	Optimizing Wellness IV Clinical	1			
NSG 480	Health Alterations IV	3			
NSG 480L	Health Alterations IV Clinical	1			
NSG 481	Professional Integration IV	3			
NSG 481L	Professional Integration IV Clinical	1			
Credits		15			
Total Credits		59			

Recommended Schedule for the Fast Flex Pathway

Students may fulfill the restricted electives requirement (2-3 credit hours) any time during their third or fourth year.

Course	Title	Credits
First Year		
Summer Semester		
NSG 340	Nursing Identity	3
NSG 344	Optimizing Wellness I	3
NSG 344L	Optimizing Wellness I Clinical	1
NSG 345	Health Alterations I	3
NSG 345L	Health Alterations I Clinical	1
NSG 380	Professional Integration I	2
NSG 380L	Professional Integration I Clinical	1
Credits		14
Fall Semester		
NSG 381	The Evidence Based Clinician	3
NSG 382	Optimizing Wellness II	3
NSG 382L	Optimizing Wellness II Clinical	1
NSG 383	Health Alterations II	3
NSG 383L	Health Alterations II Clinical	1
NSG 384	Professional Integration II	3
NSG 384L	Professional Integration II Clinical	1
Credits		15
Spring Semester		
NSG 440	The Team Member	3
NSG 441	Optimizing Wellness III	3
NSG 441L	Optimizing Wellness III Clinical	1
NSG 443	Health Alterations III	3
NSG 443L	Health Alterations III Clinical	1
NSG 444	Professional Integration III	3
NSG 444L	Professional Integration III Clinical	1
Credits		15
Second Year		
Summer Semester		
NSG 445	Evolution as a Leader	3

Nursing, R.N. to B.S.N.

Interim Coordinator: Karen Weeks

Email: weekska@jmu.edu

Program Description

The Bachelor of Science in Nursing (B.S.N.) program offers a degree-completion pathway for individuals licensed as a Registered Nurse (R.N.). The R.N. to B.S.N. pathway builds on the liberal arts and previous nursing education to further develop discipline-specific knowledge, skills and abilities. An emphasis is placed on scholarship and evidence-based practice, clinical decision-making, quality and safety, systems-based practice, and population health. By the end of the program, students are prepared to provide comprehensive nursing care to individuals, families and populations and to take on leadership and management roles to promote quality and safety in healthcare.

Admission and Retention Requirements

The R.N. to B.S.N. pathway provides flexible learning opportunities for licensed R.N.'s who have an associate degree or diploma to complete the baccalaureate degree. Classes are offered online to allow schedule flexibility while completing the B.S.N. degree. The R.N. to B.S.N. pathway provides a foundation for professional nursing practice and prepares students for further opportunities such as graduate study and advance practice.

JMU will award R.N. to B.S.N. students up to 37 experiential learning credits for prior learning and practice experience as a R.N. (a minimum of 540 hours), and up to 41 transfer credits toward the General Education requirements for a B.S.N.

Six additional corequisite and thirteen general education credits will be required which may be taken concurrently at a local community college and transferred to JMU for credit. Students will complete 30 credits in the R.N. to B.S.N. program in three semesters of full-time study or five semesters of part-time study, completing a total of 120 semester hours for the baccalaureate degree.

Students should apply to the R.N. to B.S.N. pathway to initiate an academic transcript review. Additional program eligibility criteria include:

- Associate Degree or Diploma in Nursing from an accredited college/university.
- Current unrestricted R.N. license in any state, Washington, D.C., or a U.S. possession or territory or an equivalent credential in another country.
- Employment as an R.N. for a minimum of four months and/or plans to practice concurrently with course work.
- Graduates of foreign nursing schools, who are licensed outside of the United States, are required to pass the Qualifying Exam of the Commission on Graduates of Foreign Nursing Schools (CGFNS) prior to application and include the exam report with their application materials.
- TOEFL minimum score of 81 (internet-based test) or IELTS minimum score of 6.5 for international applicants.
- Cumulative Nursing GPA of 2.5 on a 4-point scale.
- Completion of the General Education courses required for the Bachelor of Science. May be taken concurrently with nursing course work but must be completed prior to graduation.
- Completion of all prerequisite courses with a minimum grade of "C."

- Ability to meet the school's technical standards for nursing practice (posted on the nursing website (<https://www.jmu.edu/chbs/nursing/index.shtml>)).

Admission is contingent on admission to the university.

Once admitted, the student must earn a minimum grade of "C-" or "Credit" in each required nursing course to remain in good standing in the nursing program. A grade of "D+/-", "F", "No Credit" or "WF" (withdrawn failing) is considered a course failure. A student who for the first time receives a failure in a nursing course may, with school approval, repeat the course. A student who receives a failing grade in a nursing course may repeat one nursing course, with school approval. A student who receives a second failing grade in a nursing course will be ineligible to continue in the program.

* Students admitted through and including the Fall 2026 semester will follow the program requirements as listed in the 2024-2025 catalog. Students entering in or after Spring 2027 will follow the curriculum listed here.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	
Arts and Humanities	9
American and Global Perspectives	4
Co-requisite courses: PHIL 262 (ethics)	3
Nursing courses	30
Experiential learning credit given for R.N. qualifications	(up to) 37
Credit for Associate Degree and prerequisites ²	34
Total	120

¹ According to transfer agreements, students are required to take at minimum 13 credits of general education courses in the Arts and Humanities and American and Global Perspectives areas.

² Prerequisite courses include the following JMU or VCCS equivalents: WRTC 103; BIO 270/BIO 270L and BIO 290/BIO 290L; PSYC 160; BIO 245/BIO 245L or CHEM 120.

Major Requirements

Code	Title	Credits
NSG 381	The Evidence Based Clinician	3
NSG 384	Professional Integration II	3
NSG 440	The Team Member	3
NSG 441	Optimizing Wellness III	3
NSG 443	Health Alterations III	3
NSG 444	Professional Integration III	3
NSG 445	Evolution as a Leader	3
NSG 446	Optimizing Wellness IV	3
NSG 480	Health Alterations IV	3
NSG 481	Professional Integration IV	3
Total Credits		30

Recommended Schedule for Majors

Full-Time Sequence of Classes

Course	Title	Credits
First Year		
First Semester		
Fall or Spring		
NSG 381	The Evidence Based Clinician	3
NSG 384	Professional Integration II	3
NSG 441	Optimizing Wellness III	3
NSG 443	Health Alterations III	3
Credits		12
Second Semester		
Fall or Spring		
NSG 440	The Team Member	3
NSG 444	Professional Integration III	3
NSG 445	Evolution as a Leader	3
NSG 480	Health Alterations IV	3
Credits		12
Third Semester		
Summer		
NSG 446	Optimizing Wellness IV	3
NSG 481	Professional Integration IV	3
Credits		6
Total Credits		30

Part-Time Sequence of Classes

Course	Title	Credits
First Semester		
Fall or Spring		
NSG 381	The Evidence Based Clinician	3
NSG 441	Optimizing Wellness III	3
Credits		6
Second Semester		
Fall or Spring		
NSG 384	Professional Integration II	3
NSG 443	Health Alterations III	3
Credits		6
Third Semester		
Fall or Spring		
NSG 440	The Team Member	3
NSG 480	Health Alterations IV	3
Credits		6
Fourth Semester		
Fall or Spring		
NSG 444	Professional Integration III	3
NSG 445	Evolution as a Leader	3
Credits		6
Fifth Semester		
Summer		
NSG 446	Optimizing Wellness IV	3
NSG 481	Professional Integration IV	3
Credits		6
Total Credits		30

Philosophy and Religion, B.A.; Major in Global Religions and Cultures

Program Description

The Global Religions and Cultures major empowers students to analyze diverse worldviews, traditions and value systems. In a world that is increasingly pluralistic, ideologically driven and globally connected, skills in cross-cultural understanding and the interpretation of complex data are more important than ever. Students of many different religious identities, and none, thrive in our courses.

Our students apply the Global Religions and Cultures major to a wide variety of career fields, including business, law, education, health, intelligence analysis, non-profit leadership and more. Employers highly value the practical skills that our students cultivate in analytical writing, articulating nuanced and evidence-based arguments, and engaging with perspectives that are different from their own. The capstone seminar rigorously trains students in research methods, critical thinking, presentation and communication skills, digital information literacy, and ethics. Our students excel in graduate school.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses) ³	3
Major requirements	33
University electives	31-43
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major, such as PHIL 101 Introduction to Philosophy and REL 101 Religions of the World.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

³ PHIL 101 Introduction to Philosophy may fulfill this requirement, if not being used for General Education credit.

Major Requirements

The Global Religions and Cultures major requires 33 credit hours, including three core courses and one capstone seminar. Students majoring in Global Religions and Cultures have the opportunity to create their own custom concentrations, if they choose, by completing four or more courses in a focused area of study. For example, students might choose to concentrate in Christian Studies, Indian Religions, Religion in Africa, Religion and Society, Spirituality and Health, Pre-professional Studies, or another concentration they create with support from their faculty advisor.

Because the academic study of religion is interdisciplinary, students may apply up to six credit hours of coursework from other departments toward the major if the courses include a substantial focus on religion,

with approval from the faculty advisor. Our students often double major in Global Religions and Cultures as well as another field such as communication studies, health sciences, intelligence analysis or public policy. Students may add the major in Global Religions and Cultures at any time during their undergraduate career.

Code	Title	Credits
Core Courses		
PHIL 101	Introduction to Philosophy [C2HQC]	3
REL 101	Religions of the World [C2HQC]	3
Subtotal		6
Major Requirement		
REL 200	How to Study Religion and Spirituality	3
Subtotal		3
Elective Requirements		
Students choose 21 credits of REL electives. Students can create their own custom concentrations, if they choose, by completing a minimum of 12 credit hours in a focused area of study (see below for more information). ¹		21
Subtotal		21
Capstone Requirement		
Students majoring in Global Religions and Cultures must complete one of the following capstone courses. Typically, one capstone is offered per semester. Most students take their capstone course in their junior or senior years.		3
REL 420	Controversies in Religion	
REL 430	Islam and the Nature of Being	
REL 440	Religion and American Culture	
REL 450	Religion and Society	
REL 460	Topics in Ancient Jewish and Early Christian Literature	
REL 470	Women's Experience and Global Christianity	
REL 488	Buddhism in the Modern World	
Total Credits		33

¹ Up to six credit hours of elective requirements may be fulfilled by courses from other disciplines (non-REL-prefix courses) if the courses include a substantial focus on religion, with approval from the faculty advisor.

Custom Concentrations

Students majoring in Global Religions and Cultures have the opportunity to create their own custom concentrations, if they choose, by completing four or more courses (12 credit hours minimum) in a focused area of study. For example, students might choose to concentrate in Christian Studies, Indian Religions, Religion in Africa, Religion and Society, Spirituality and Health, Pre-professional Studies, or others that they create. Students may create any concentration name they choose. Faculty advisors support students in designing custom concentrations based on their interests.

We encourage every student, whether or not they declare a concentration, to develop breadth of expertise by studying at least two different religious traditions or cultural regions over the course of their Global Religions and Cultures major.

Recommended Schedule for Majors

The following outline is a sample four-year program. The actual courses and sequence a student takes may vary. Students may declare a major in Global Religions and Cultures at any point in their careers.

Course	Title	Credits
First Year		
REL 101	Religions of the World [C2HQC]	3
One additional Religion course		3
General Education requirement		15
B.A. requirements / University Elective(s)		9
Credits		30
Second Year		
PHIL 101	Introduction to Philosophy [C2HQC]	3
Two additional Religion courses		6
General Education requirement		12
B.A. degree requirements / University Elective(s)		9
Credits		30
Third Year		
REL 200	How to Study Religion and Spirituality	3
Two additional Religion courses		6
General Education requirement		15
B.A. degree requirements / University Elective(s)		6
Credits		30
Fourth Year		
Religion capstone course		3
Two additional Religion courses		6
B.A. degree requirements / University Elective(s)		21
Credits		30
Total Credits		120

Philosophy and Religion, B.A.; Major in Philosophy

Program Description

The major in philosophy empowers students to analyze diverse worldviews and value systems. In a world that is increasingly pluralistic, ideologically driven and globally connected, skills in critical thinking and cross-cultural understanding are more important than ever. Our students apply this degree to a wide variety of career fields.

Students choose from two concentrations: general philosophy and interdisciplinary philosophy. The interdisciplinary concentration allows students to count relevant courses in another field toward the major and is a particularly good fit for students choosing philosophy as a second major.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	33-36
University electives	28-43
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

Code	Title	Credits
Core Requirements		
All students must take the following courses.		
PHIL 101	Introduction to Philosophy [C2HQC]	3
REL 101	Religions of the World [C2HQC]	3
Subtotal		6
Major Requirements		
PHIL 250	Introduction to Symbolic Logic	3
PHIL 330	Moral Theory	3
PHIL 340	Ancient Greek Philosophy	3
PHIL 341	Modern Philosophy	3
Subtotal		12
Concentrations		
Students select a concentration in general philosophy or in interdisciplinary philosophy. See below for details.		15-18
Total Credits		33-36

Concentrations

Concentration in General Philosophy

In addition to the core and major requirements listed above, students choosing the concentration in General Philosophy must take the following:

Code	Title	Credits
History of Philosophy		
Choose one of the following: ¹		3
PHIL 339	Islamic Philosophy	
PHIL 342	Medieval Philosophy	
PHIL 343	Stoics, Skeptics and Epicureans	
PHIL 344	Existentialism	
PHIL 345	History of Ethics	
PHIL 360	Philosophy of Race	
PHIL 370	American Philosophy	
PHIL/REL 375	Nineteenth-Century Philosophy and Theology	
PHIL 376	Nietzsche	
PHIL 377	Hermeneutics	
PHIL 430	Analytic Philosophy	
PHIL 466	Kant	
PHIL 468	Phenomenology	
Subtotal		3
Metaphysics/Epistemology		
Choose one of the following: ²		3
PHIL/REL 218	Philosophy of Religion	
PHIL 300	Knowledge and Belief	
PHIL 311	Metaphysics	
PHIL 380	Philosophy of Language	
PHIL 392	Philosophy of Mind	
PHIL 394	Self and Identity	
PHIL 396	Philosophy of Physics	
PHIL 397	Philosophy of Space and Time	
PHIL 398	Philosophy of Quantum Theory	
PHIL 410	Philosophy of Science	
Subtotal		3
Philosophy Electives		
Choose nine credit hours of philosophy electives (six credits at the 300 level or above) ³		9
Subtotal		9
Total Credits		15

¹ Depending on the manner in which they are taught, the following topics classes may be counted toward the history requirement if they are approved by petition to the academic unit head: PHIL 390, PHIL 391 and PHIL 460.

² Depending on the manner in which they are taught, the following topics classes may be counted toward the metaphysics/epistemology requirement if they are approved by petition to the academic unit head: PHIL 390 and PHIL 391.

³ Neither PHIL 120 nor PHIL 150 can be used as an elective.

Concentration in Interdisciplinary Philosophy

In addition to the core and major requirements listed above, students choosing the concentration in Interdisciplinary Philosophy must take the following:

Code	Title	Credits
Metaphysics/Epistemology		
Choose one of the following: ¹		3
PHIL/REL 218	Philosophy of Religion	
PHIL 300	Knowledge and Belief	
PHIL 311	Metaphysics	
PHIL 380	Philosophy of Language	
PHIL 392	Philosophy of Mind	
PHIL 394	Self and Identity	
PHIL 396	Philosophy of Physics	
PHIL 397	Philosophy of Space and Time	
PHIL 398	Philosophy of Quantum Theory	
PHIL 410	Philosophy of Science	
Subtotal		3
Philosophy Electives		
Choose six credit hours of philosophy electives (three credits at the 300 level or above) ²		6
Subtotal		6
Cognates		
Cognate of three courses from one or more disciplinary areas outside of philosophy		9
Subtotal		9
Total Credits		18

¹ Depending on the manner in which they are taught, the following topics classes may be counted toward the metaphysics/epistemology requirement if they are approved by petition to the academic unit head: PHIL 390 and PHIL 391.

² Neither PHIL 120 nor PHIL 150 can be used as an elective.

Recommended Schedule for Philosophy Majors

The following outline is a sample four-year program. The actual courses and sequence a student takes may vary.

Course	Title	Credits
First Year		
Introductory courses in major		6
B.A. degree requirements		6
General Education requirement		18
Credits		30
Second Year		
Required courses in major		6
B.A. degree requirements or electives		6
General Education requirement		18
Credits		30
Third Year		
Requirements and electives in philosophy or religion		12
Electives (may be outside of major)		12
General Education requirement		6
Credits		30

Fourth Year	
Three Philosophy courses	9
B.A. degree requirements / University Elective(s)	21
Credits	30
Total Credits	120

Physics, B.A.

Program Description

The Department of Physics and Astronomy offers a multitrack program of study which allows each student flexibility to choose a suitable set of courses that will prepare the student for their chosen career goal. With the assistance of physics faculty advisors each student will select a program of theoretical and experimental courses through which they will gain an understanding of the broad range of general principles which characterize the discipline of physics; and they will acquire the ability to apply theoretical and experimental techniques to solve a wide variety of problems.

We offer courses that focus on multiple facets of physics, including materials science, computational physics, electronics and applied nuclear physics. Students are strongly urged to conduct research projects with our faculty, and often present research at professional conferences and produce co-authored publications. Additionally, there are multiple opportunities within the physics and astronomy department for paid research experience.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major core requirements ³	42
Major concentration requirements ³	13-26
University electives	2-8
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level (typically 232) of the student's chosen language or by placing out of that language through the Department of World Languages and Cultures' placement test.

³ Some of the courses required for the major complete The Natural World of General Education. PHIL 101 is part of Arts and Humanities.

Major Requirements

Code	Title	Credits
Required Courses		
PHYS 105	Foundations of Physics	1
PHYS 106	Foundations of Physics Research	1
PHYS 247	Data Acquisition and Analysis Techniques II	1
PHYS 260	University Physics III	3
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 237	Calculus III	4

CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
PHYS 391	Seminar	1
PHYS 392	Seminar	1
PHYS 491	Seminar	1
PHYS 492	Seminar	1
Choose one of the following pairs of introductory courses:		6
PHYS 140 & PHYS 150	College Physics I [C3PP] and College Physics II	
PHYS 240 & PHYS 250	University Physics I [C3PP] and University Physics II	
Choose one of the following pairs of lab courses:		2
PHYS 140L & PHYS 150L	General Physics Laboratory I [C3L] and General Physics Laboratory II	
PHYS 240L & PHYS 250L	University Physics Laboratory I [C3L] and University Physics Laboratory II	
Choose one of the following sequences:		4
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	
Subtotal		42
Concentrations		
Each student, in consultation with their faculty advisor, will choose one of the following program concentrations:		
Individual Option		
Professional		
Total Credits		55-68

Concentrations

Individual Option Concentration

The individual option is a course of study chosen specifically to match the interest and career plans of the student. This option will allow custom-designed cross disciplinary majors such as the history of physics and physics and the fine arts.

A student electing the individual option must complete the major core requirements of the B.A. in physics and will select a program consisting of a coherent collection of a minimum of 26 additional credits of physics courses numbered above 270, astronomy courses numbered above 301 and courses in related fields. This individualized program must be selected in consultation with a faculty advisor in the department and must be approved by that advisor, the department head and one other faculty member in the department. The student needs to declare this concentration at least 3 semesters prior to graduation.

The individualized program, as approved by the department and accepted by the student, becomes the major requirements for the student.

Students are expected to review progress toward completion of the selected program of study with their faculty advisor.

Professional Concentration, Pre-Medical Preparation Track

Students in this concentration must complete the following courses in addition to core requirements:

Code	Title	Credits
Required Courses		
MATH 238	Linear Algebra with Differential Equations	4
PHYS 340	Mechanics	3
PHYS 350	Electricity and Magnetism	3
Three credits chosen from PHYS courses at the 265-level or higher		3
Total Credits		13

Students are expected to review progress toward completion of the selected program of study with their faculty advisor.

Recommended Schedule for Majors

Students pursuing the B.A. in physics and astronomy with a professional concentration may complete a track in pre-medical preparation. In addition to degree and General Education requirements, students complete the following courses.

Course	Title	Credits
First Year		
PHYS 105	Foundations of Physics	1
PHYS 240	University Physics I [C3PP]	3
PHYS 140L or PHYS 240L	General Physics Laboratory I [C3L] or University Physics Laboratory I [C3L]	1
PHYS 250	University Physics II	3
PHYS 150L or PHYS 250L	General Physics Laboratory II or University Physics Laboratory II	1
PHYS 246	Data Acquisition and Analysis Techniques I	1
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
General Education and/or university electives		4
Credits		30
Second Year		
PHYS 247	Data Acquisition and Analysis Techniques II	1
PHYS 260	University Physics III	3
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations	4
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 242L	Organic Chemistry Laboratory	2
General Education and/or university electives		6
Credits		30
Third Year		
PHYS 340	Mechanics	3
PHYS 350	Electricity and Magnetism	3
PHYS 391	Seminar	0
PHYS 392	Seminar	1
PHYS elective 265 or higher ¹		3
CHEM 361	Biochemistry I	3
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
General Education and/or university electives		9
Credits		30

Fourth Year		
Two PHYS electives 265 or higher		6
PHYS 491	Seminar	0
Additional courses suggested by the Pre-professional Health program		
General Education and/or university electives		24
Credits		30
Total Credits		120

¹ PHYS 326 recommended

Physics, B.S.

Program Description

The Department of Physics and Astronomy offers a multitrack program of study which allows each student flexibility to choose a suitable set of courses that will prepare the student for their chosen career goal. With the assistance of physics faculty advisors each student will select a program of theoretical and experimental courses through which they will gain an understanding of the broad range of general principles which characterize the discipline of physics; and they will acquire the ability to apply theoretical and experimental techniques to solve a wide variety of problems.

We offer courses that focus on multiple facets of physics, including materials science, computational physics, electronics and applied nuclear physics. Students are strongly urged to conduct research projects with our faculty, and often present research at professional conferences and produce co-authored publications. Additionally, there are multiple opportunities within the physics and astronomy department for paid research experience.

Degree Requirements

Degree Requirements	Credit Hours
General Education ^{1,2}	41
Quantitative requirement (in addition to General Education) ²	3
Scientific Literacy requirement (in addition to General Education) ²	3-4
Major core requirements ²	40
Major program concentration requirements	24-44
University electives	0-23
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² Some major core courses simultaneously satisfy degree and some General Education requirements so that a student need not take more than 120 credit hours to satisfy graduation requirements. The quantitative (3 credit hours), scientific literacy (3-4 credit hours) and part of the General Education The Natural World (7 credit hours) requirements are fulfilled with major core courses reducing those three categories by 13-14 credit hours.

Major Requirements

Code	Title	Credits
Core Requirements		
PHYS 105	Foundations of Physics	1
PHYS 106	Foundations of Physics Research	1
PHYS 140	College Physics I [C3PP]	3
or PHYS 240	University Physics I [C3PP]	
PHYS 140L	General Physics Laboratory I [C3L]	1
or PHYS 240L	University Physics Laboratory I [C3L]	
PHYS 150	College Physics II	3
or PHYS 250	University Physics II	
PHYS 150L	General Physics Laboratory II	1

or PHYS 250L	University Physics Laboratory II	
PHYS 247	Data Acquisition and Analysis Techniques II	1
PHYS 260	University Physics III	3
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
PHYS 340	Mechanics	3
PHYS 350	Electricity and Magnetism	3
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 237	Calculus III	4
Subtotal		40

Concentrations

Each student, in consultation with their faculty advisor, will choose one of the following program concentrations:

Applied Physics	
Fundamental Studies	
Individual Option	
Multidisciplinary Astrophysics	
Multidisciplinary Biophysics	
Multidisciplinary Business	
Multidisciplinary Geophysics	
Multidisciplinary Secondary Education	
Multidisciplinary Technical and Scientific Communication (TSC)	
Physics and Engineering Combined Program	
Subtotal	22-44
Total Credits	62-84

Concentrations

Applied Physics Concentration

The applied physics concentration is designed to prepare students for careers in a wide variety of scientific areas including laboratory and industrial settings. Students are strongly encouraged to complete at least one of six areas of interest for completion of degree requirements: applied nuclear physics, computational physics, materials physics, electronics and instrumentation, geophysics and biophysics.

All students in the applied physics concentration must complete the following courses:

Code	Title	Credits
Required Courses (In Addition to Core Requirements)		
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
CS 149	Introduction to Programming	3-4
or MATH 248	Computers and Numerical Algorithms	
Choose one of the following:		4-6
MATH 238	Linear Algebra with Differential Equations	
MATH 300 & MATH 336	Linear Algebra and Elementary Differential Equations	
PHYS 347	Advanced Physics Lab	2
PHYS 360	Analog Electronics	4
PHYS 391	Seminar (must be taken for 0credit hours)	0

PHYS 392	Seminar	1
PHYS 491	Seminar (must be taken for 0 credit hours)	0
PHYS 492	Seminar	1
Subtotal		19-22

Research Courses

Choose 2-6 credit hours from the following. At least two credits must come from research courses. (All students must take a minimum of 2 credits. Honors students completing the thesis take 6 credits).¹

PHYS 398	Independent Study in Physics or Astronomy (repeatable to 4 credits)	
or ASTR 398	Independent Study in Physics or Astronomy	
PHYS 498	Undergraduate Research in Physics or Astronomy (repeatable to 6 credits)	
or ASTR 498	Undergraduate Research in Physics or Astronomy	
ISCI 450	Interscience Research (repeatable to 6 credits)	
PHYS 499	Honors (repeatable) ¹	
Subtotal		2-6

Additional Courses

Complete 9 credits approved by the head of the Department of Physics and Astronomy. Ideally these courses should form a coherent set of courses for the best possible academic experience. See below for some suggestions.

Subtotal		9
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Total Credits **30-37**

¹ Students completing honors theses should take PHYS 499 in both the fall and spring semesters of their senior year for a total of 6 credit hours. Alternatively, students could complete their theses over three semesters (spring of junior year, fall and spring of senior year, with two credits of PHYS 499 each semester, for a total of 6 credit hours).

Suggested Areas of Interest to Complete the 9 Credit Hours Required (Minimum)

To satisfy the requirement of at least 9 additional credit hours approved by the department head, students are strongly encouraged to select their courses from one of the coherent sets of courses to round out their academic experience. Below are suggested areas of interest.

Code	Title	Credits
Applied Nuclear Physics Courses		
PHYS 339	Introductory Nuclear Science	4
PHYS 460	Quantum Mechanics	3
PHYS 333	Introduction to Particle Physics	3
PHYS 338	Nuclear Physics	3
CHEM 450	Nuclear and Radiation Chemistry	3
Biophysics Courses		
PHYS 326	Biophysics	3
PHYS 364	Introduction to Fluid Mechanics	4
PHYS 380	Thermodynamics and Statistical Mechanics	3
PHYS 390	Computer Applications in Physics	3
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 361 or CHEM 361	Biochemistry I	3

BIO 490	Mechanics of Animal Movement	4
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 242L	Organic Chemistry Laboratory	2

Computational Physics Courses

Students pursuing this area of interest should declare a minor in mathematics.

PHYS 341	Non-Linear Dynamics and Chaos	3
PHYS 364 or MATH 364	Introduction to Fluid Mechanics	4
PHYS 367 or MATH 367	Introduction to Acoustics	3
PHYS 380	Thermodynamics and Statistical Mechanics	3
PHYS 390	Computer Applications in Physics	3

Electronics and Instrumentation Courses

PHYS 295	Laboratory Apparatus Design and Construction	1
PHYS 371	Digital Electronics	2
PHYS 372	Microcontrollers and Their Applications (2,4)	2
PHYS 380	Thermodynamics and Statistical Mechanics	3
PHYS 386	Robots: Structure and Theory	3
PHYS 387	Theory and Basic Principles of Mobile Robot Construction and Control	3
PHYS 388	Robots for Humanity	3

Geophysics Courses

Any 300- or 400-level GEOL or PHYS course not listed below (with advisor's approval)

PHYS 337	Solid State Physics	3
PHYS 381	Materials Characterization (Lecture/Lab) (Lecture/Lab)	3
PHYS 440 or GEOL 440	Geophysics (3, 2)	4
ASTR 220	General Astronomy I: The Night Sky, the Solar System and Stars	3
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	4
GEOL 210	Applied Physical Geology [C3NS]	3
GEOG 215	Introduction to Geographic Information Science and Cartography	3
GEOG 216	Earth Observation and Satellite Navigation Systems	3
GEOL 280	Mineralogy (3, 2)	4
GEOL 300	Introduction to Petrology (3, 3)	4
GEOL 307	Planetary Geology (2, 2)	3
GEOL 365	Structural Geology, Tectonics and Climate	4
GEOL 399	Field Geology	3-6
GEOL 442	Field Geophysics	3
GEOL 460	Hydrogeology (2, 2)	3

Materials Physics Courses

PHYS 337	Solid State Physics	3
Choose one of the following:		3-4
PHYS/CHEM/ MATS 375	An Introduction to Materials Science	
PHYS 381	Materials Characterization (Lecture/Lab)	
PHYS 420	Modern Optics	

PHYS/CHEM Lasers and Their Applications to Physical Sciences
455

CHEM 485 Science of the Small
& 485L and Science of the Small Laboratory
or PHYS 485 Science of the Small
and Science of the Small Laboratory
& 485L

Suggested Minors

Students pursuing the Applied Physics Concentration are encouraged to consider the following minors as being particularly effective at augmenting a degree in physics:

- Astronomy Minor (p. 453)
- General Business Minor (p. 498)
- Computer Science Minor (p. 472)
- Geophysics Minor (p. 502)
- Materials Science Minor (p. 524)
- Robotics Minor (p. 560)
- Writing, Rhetoric and Technical Communication Minor (p. 580)
(physics majors in this minor are encouraged to take WRTC 316 and WRTC 350 to fulfill electives)

Fundamental Studies Concentration

The fundamental studies concentration is designed to prepare students for immediate post-baccalaureate employment or for entrance to advanced study in physics or related areas.

Code	Title	Credits
Required Courses (In Addition to Core Requirements)		
CS 149 or MATH 248	Introduction to Programming Computers and Numerical Algorithms	3-4
MATH 300	Linear Algebra	3
MATH 336	Elementary Differential Equations	3
PHYS 347	Advanced Physics Lab	2
PHYS 360	Analog Electronics	4
PHYS 380	Thermodynamics and Statistical Mechanics	3
PHYS 391	Seminar (must be taken for 0 credit hours)	0
PHYS 392	Seminar	1
PHYS 460	Quantum Mechanics	3
PHYS 491	Seminar (must be taken for 0 credit hours)	0
PHYS 492	Seminar	1
Subtotal		23-24
Electives		
Choose one of the following: ¹		3-4
PHYS/ASTR 330	Relativity	
ASTR 480	Astrophysics	
PHYS 333	Introduction to Particle Physics	
PHYS 337	Solid State Physics	
PHYS 338	Nuclear Physics	
PHYS 420	Modern Optics	
PHYS 482	Advanced Theoretical Physics	
Subtotal		3-4
Research Courses		

Choose a minimum of two credit hours of research from this area. 2-6
(All students must take a minimum of 2 credits. Honors students completing the thesis take 6 credits). ²

PHYS/ASTR 398	Independent Study in Physics or Astronomy (repeatable to 4 credits)	
PHYS/ASTR 498	Undergraduate Research in Physics or Astronomy (repeatable to 6 credits)	
ISCI 450	Interscience Research (repeatable to 6 credits)	
PHYS 499	Honors (repeatable) ²	
Subtotal		2-6
Total Credits		28-34

¹ If not enrolled in PHYS 499.

² Students completing honors theses should take PHYS 499 in both the fall and spring semesters of their senior year for a total of 6 credit hours. Alternatively, students could complete their theses over three semesters (spring of junior year, fall and spring of senior year, with two credits of PHYS 499 each semester, for a total of 6 credit hours).

Individual Option Concentration

The individual option is a course of studies chosen specifically to match the interest and career plans of the student. This option will allow custom designed cross disciplinary majors such as chemical physics, as well as majors designed for students whose educational and career goals are not met by the existing concentrations in the major.

A student electing the individual option must complete the core requirements for the B.S. in physics and will select a program consisting of a coherent collection of a minimum of 25 additional credits of physics courses numbered above 270, astronomy courses numbered above 301 and courses in related fields.

This individualized program must be selected in consultation with a faculty advisor in the department, and must be approved by that advisor, the department head and one other faculty member in the department.

The student needs to declare this concentration at least three semesters prior to graduation.

The individualized program, as approved by the department and accepted by the student, becomes the major requirements for that student.

Students are expected to review progress toward completion of the selected program of study with their faculty advisor.

Multidisciplinary Concentrations

The multidisciplinary concentrations are designed to provide a solid foundation in physics for students whose interests extend beyond traditional physics fields. The track of secondary education opens doors to the pursuit of graduate degrees in education and other career paths requiring a technical background.

The following are also required for all multidisciplinary concentrations:

Code	Title	Credits
Required Courses		
Choose one of the following:		4-6
MATH 238	Linear Algebra with Differential Equations	
MATH 300 & MATH 336	Linear Algebra and Elementary Differential Equations	
PHYS 380	Thermodynamics and Statistical Mechanics	3

PHYS 391	Seminar (must be taken for 0 credit hours)	0
PHYS 392	Seminar	1
PHYS 491	Seminar (must be taken for 0 credit hours)	0
PHYS 492	Seminar	1
Subtotal		9-11

Research Courses

Choose a minimum of 2 credits. Honors students completing the thesis take 6 credits.¹

PHYS/ASTR 398	Independent Study in Physics or Astronomy (repeatable to 4 credits)	
PHYS/ASTR 498	Undergraduate Research in Physics or Astronomy (repeatable to 6 credits)	
ISCI 450	Interscience Research (repeatable to 6 credits)	
PHYS 499	Honors (repeatable) ¹	
Subtotal		2-6

Total Credits **11-17**

¹ Students completing honors theses should take PHYS 499 in both the fall and spring semesters of their senior year for a total of six credit hours. Alternatively, students could complete their theses over three semesters (spring of junior year, fall and spring of senior year, with two credits of PHYS 499 each semester, for a total of 6 credit hours).

Multidisciplinary Astrophysics Concentration

Code	Title	Credits
Multidisciplinary Core Requirements		11-17
Astrophysics Requirements		
ASTR 220	General Astronomy I: The Night Sky, the Solar System and Stars	3
ASTR 221	General Astronomy II: Star Systems, the Interstellar Medium, and Cosmology	4
ASTR 320	Astronomical Techniques	3
MATH 248	Computers and Numerical Algorithms	4
Subtotal		25-31
Electives		
Choose one of the following: ¹		3-4
ASTR/PHYS 330	Relativity	
ASTR 480	Astrophysics	
GEOL 307	Planetary Geology (2, 2)	
PHYS 333	Introduction to Particle Physics	
PHYS 338	Nuclear Physics	
PHYS 420	Modern Optics	
PHYS 460	Quantum Mechanics	
Subtotal		3-4
Total Credits		28-35

¹ If not enrolled in PHYS 499.

Multidisciplinary Biophysics Concentration

Code	Title	Credits
Multidisciplinary Core requirements		11-17
Requirements		

CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
PHYS 326	Biophysics	3
PHYS 347	Advanced Physics Lab	2
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4

Choose one of the following options: 4-12

Option 1:

BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	
BIO 240 & 240L	Genetics and Genetics Laboratory	
BIO 370	Human and Animal Physiology	

Option 2:

BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	
BIO 490	Mechanics of Animal Movement	

Option 3:

BIO 445	Neurobiology	
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Option 4:

CHEM 241	Organic Chemistry I	
CHEM 242 & 242L	Organic Chemistry II and Organic Chemistry Laboratory	
CHEM 361 or BIO 361	Biochemistry I	

Total Credits **28-42**

Multidisciplinary Business Concentration

Code	Title	Credits
Multidisciplinary Core Requirements		11-17
Business Requirements		
No more than 27 hours may be taken in the College of Business.		
ACTG 244	Accounting for Non-Business Majors	3
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors	3
ECON 201	Introduction to Microeconomics	3
FIN 345	Finance Fundamentals for Non-Business Majors	3
MGT 305	Organizational Behavior and Human Resource Management	3
MKTG 380	Principles of Marketing	3
PHYS 347	Advanced Physics Lab	2
Total Credits		31-37

Multidisciplinary Geophysics Concentration

Code	Title	Credits
Multidisciplinary Core Requirements		11-17
Geophysics Requirements		
PHYS 347	Advanced Physics Lab	2
GEOL 110 or GEOL 210	Physical Geology [C3NS] Applied Physical Geology [C3NS]	3
Choose one of the following:		3-4
ASTR 220	General Astronomy I: The Night Sky, the Solar System and Stars	
GEOL 307	Planetary Geology (2, 2)	

GEOL 365	Structural Geology, Tectonics and Climate	
GEOL 440	Geophysics (3, 2)	
	or PHYS 440 Geophysics (3, 2)	
Choose at least three credits from a field- or geographic-based course:		3-6
GEOG 215	Introduction to Geographic Information Science and Cartography	
GEOG 216	Earth Observation and Satellite Navigation Systems	
GEOL 399	Field Geology	
GEOL 442	Field Geophysics	
Choose at least nine credits from the following:		9
GEOL 280	Mineralogy (3, 2)	
GEOL 300	Introduction to Petrology (3, 3)	
GEOL 460	Hydrogeology (2, 2)	
PHYS 337	Solid State Physics	
PHYS 360	Analog Electronics	
PHYS 381	Materials Characterization (Lecture/Lab)	
Any other 300- or 400-level geology or physics course (Upon approval from advisor)		3
Subtotal		34-44
Total Credits		34-44

Multidisciplinary Secondary Education Concentration

This concentration is designed for students wishing to pursue secondary education teacher licensure in physics. Students in this concentration should also declare the Pre-Professional Secondary Education (SEED) minor. The SEED minor, in conjunction with the bachelor of science degree in physics with a concentration in multidisciplinary secondary education, prepares students to enter the 5th year master's degree program in education that leads to teacher licensure.

Code	Title	Credits
Multidisciplinary Core Requirements		11-17
Secondary Education Requirements Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	4
MATH 238	Linear Algebra with Differential Equations ¹	4
PHYS 460	Quantum Mechanics	3
Subtotal		30-36
Total Credits		30-36

¹ MATH 300 and MATH 336 taken together may substitute for MATH 238.

Multidisciplinary Technical and Scientific Communication (TSC) Concentration

Code	Title	Credits
Multidisciplinary Core Requirements		11-17
TSC Requirements		
ASTR 480	Astrophysics	3
PHYS 347	Advanced Physics Lab	2

WRTC 300	Professional Editing	3
WRTC 350	Foundations of Technical Communication	3
Physics courses numbered above 300		6
Choose six credits of the following:		6
WRTC 301	Language, Law and Ethics	
WRTC 318	Intercultural Professional Communication	
WRTC 458	Scientific and Medical Communication	
Other upper-level courses with permission		
Subtotal		34-40
Total Credits		34-40

Physics and Engineering Combined Program Concentration

This dual degree program makes it possible for the student to earn a B.S. degree in physics from JMU and a Master of Engineering degree from the University of Virginia. The engineering areas available under this program include biomedical, environmental, transportation, materials science, systems engineering and engineering physics.

During the first three years at JMU, the student must complete 96 credit hours including all JMU General Education requirements, the physics core requirements, MATH 248 or CS 149, and at least eight additional credit hours in physics courses designated by the JMU Department of Physics and Astronomy with a minimum grade average of "B+." In general, these eight additional hours will be chosen from those recommended for the applied physics concentration, but substitutions may be approved by the program advisor, Sean Scully. During the fourth year of study (when the student will be in residence at the University of Virginia), the student will take further courses approved by the JMU Department of Physics and Astronomy for credit toward the Bachelor of Science degree in physics. A total of 24 credit hours of physics or other physics-related courses taken at either school will be required for the JMU Bachelor of Science degree in physics. For further information, consult the head of the Department of Physics and Astronomy.

Code	Title	Credits
Additional Upper Level Courses (Required for the Combined Program)		
CS 149 or MATH 248	Introduction to Programming Computers and Numerical Algorithms	3-4
PHYS 347	Advanced Physics Lab	2
Total Credits		5-6

Recommended Schedule for Majors First Year

Course	Title	Credits
First Year		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
PHYS 105	Foundations of Physics	1
General Education requirement		9
Choose one of the following:		6
PHYS 140 & PHYS 150	College Physics I [C3PP] and College Physics II	

PHYS 240 & PHYS 250	University Physics I [C3PP] and University Physics II	
Choose one of the following pairs of laboratory courses:		2
PHYS 140L & PHYS 150L	General Physics Laboratory I [C3L] and General Physics Laboratory II	
PHYS 240L & PHYS 250L	University Physics Laboratory I [C3L] and University Physics Laboratory II	
Credits		34
Total Credits		34

Third and Fourth Years

During their junior and senior years, students will select courses to complete the specific program concentration which they are following. These course selections will be made with the assistance of a faculty advisor.

Second Year

Applied Physics, Multidisciplinary, and Physics and Engineering Combined Program Concentrations

Course	Title	Credits
Second Year		
MATH 237	Calculus III	4
Choose one of the following:		4-6
MATH 238	Linear Algebra with Differential Equations	
MATH 300 & MATH 336	Linear Algebra and Elementary Differential Equations	
PHYS 247	Data Acquisition and Analysis Techniques II	1
PHYS 260	University Physics III	3
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
CS 149 or MATH 248	Introduction to Programming or Computers and Numerical Algorithms	3-4
General Education requirement		11-15
Credits		30-37
Total Credits		30-37

Fundamental Studies

Course	Title	Credits
Second Year		
CS 149	Introduction to Programming	3
MATH 237	Calculus III	4
MATH 300	Linear Algebra	3
MATH 336	Elementary Differential Equations	3
PHYS 247	Data Acquisition and Analysis Techniques II	1
PHYS 260	University Physics III	3
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
General Education requirement		11-15
Credits		32-36
Total Credits		32-36

Individual Option

Course	Title	Credits
Second Year		
MATH 237	Calculus III	4
Choose one of the following:		4-6
MATH 238	Linear Algebra with Differential Equations	
MATH 300 & MATH 336	Linear Algebra and Elementary Differential Equations	
MATH 248	Computers and Numerical Algorithms	4
PHYS 247	Data Acquisition and Analysis Techniques II	1
PHYS 260	University Physics III	3
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
General Education requirement		11-15
Credits		31-37
Total Credits		31-37

Political Science, B.A.

Coordinator: Scott Hammond
Location: Miller Hall, Room 2127
Email: hammonsj@jmu.edu
Phone: (540) 568-6313

Program Description

For a major in political science, the student must satisfactorily complete a minimum of 39 credit hours, including:

- A 17-hour core introducing students to the conduct of political inquiry in the major areas of the discipline, an awareness of global issues and cultural diversity, and the techniques of original research.
- An 18-hour depth requirement in which students take at least one course from four of the major areas of political inquiry and acquire additional experience in the application of research techniques.
- A four-hour capstone experience designed to bring together knowledge gained in different courses into a coherent whole, to foster a capacity for lifelong learning, to connect the major to experiences outside the university and to provide the opportunity to work individually with a faculty member.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	38-39
University electives	25-38
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

Code	Title	Credits
Core Requirements		
POSC 225	U.S. Government [C4AE]	4
POSC 201	Introduction to Western Political Theory	3
POSC 230	International Relations	3
POSC 240	Comparative Politics	3
POSC 295	Research Methods in Political Science ¹	4
Subtotal		17
Electives		

In addition to the core requirements, political science majors must elect 18 hours of approved courses as shown, including at least one upper-level course from each of the four major areas of political inquiry. Students are encouraged to complete the core requirements before taking elective courses. A political science major may seek permission to count a completed Senior Honors Project as a three-credit elective within the major. Students must consult with the political science program coordinator prior to conducting an honors project to clarify the specific nature of its potential applicability to the major's elective options.

<i>Comparative Politics Requirement</i>		
Choose one of the following:		3
POSC 340	Politics of Development	
POSC 341	Social Movements in the U.S. and Abroad	
POSC 344	Politics of the European Union	
POSC 345	Politics of Western Europe	
POSC 346	Politics of Central and Eastern Europe	
POSC 347	Comparative Public Policy	
POSC 348	The Politics of Cultural Pluralism	
POSC 350	Latin American Politics	
POSC 353	African Politics	
POSC 354	Politics of the Middle East	
POSC 355	East Asian Politics	
POSC 356	Politics of China	
POSC 371	Topics in Comparative Politics	
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation	
POSC/WGSS 383	Women and Politics in Comparative Perspective	
POSC 440	Post-Conflict Societies	
POSC 457	Fates of Empires	

<i>International Relations Requirement</i>		
Choose one of the following:		3
POSC/JUST 331	Human Rights in Theory and Practice	
POSC 361	Topics in International Relations	
POSC 370	U.S. Foreign Policy	
POSC/JUST 372	Ethics and International Politics	
POSC 379	International Laws of War	
POSC/JUST 392	Peace Studies	
POSC 393	Human Security	
POSC 394	International Conflict	
POSC 395	International Law	
POSC 396	International Organizations	
POSC 397	The Politics of International Economic Relations	
POSC 430	International Security and Conflict Management	
POSC 435	International Terrorism	
POSC 458	International Political Analysis	

<i>Political Theory Requirement</i>		
Choose one of the following:		3
POSC 310	Ancient Political Theory: Classical Greek Political Philosophy to Roman Stoicism	

POSC 311	The Ideas and Legacies of Medieval and Renaissance Political Theory
POSC 315	Political Theory: Early Modern to the 19th Century
POSC 316	Contemporary Political Theory
POSC 321	Political Theory and Ideology
POSC 330	American Political Thought
POSC 332	Black Political Thought
POSC 333	American Political Thought II: Civil War to the Present
POSC 334	Political Thought of James Madison
POSC 381	Topics in Political Theory

American Government Requirement

Choose one of the following:	3
POSC 302	State and Local Government
POSC 322	Civic Engagement
POSC 323	Cities and Placemaking
POSC 324	Politics of Human Rights in the United States
POSC 325	Constitutional Law
POSC 326	Civil Rights
POSC 327	Race and Capitalism in the United States
POSC 328	Fair and Affordable Housing Law and Policy
POSC 329	Civil Rights and Criminal Procedures
POSC 351	Topics in American Politics
POSC 360	The Politics of Mental Health
POSC 362	Political Behavior
POSC 364	Political Parties in America
POSC 365	American Political Campaigning
POSC 366	Introduction to the Politics of Race and Ethnicity
POSC 367	U.S. Immigration and Refugee Law
POSC 368	Interest Groups, Lobbying and Public Policy
POSC 369	Elections and Voting Behavior
POSC 375	Latina/o/x Politics: Origins and Futures
POSC 376	Introduction to Black Politics
POSC 378	Borderlands Policy: Latina/o/x Politics in Context
POSC 380	The U.S. Presidency
POSC 382	The Role of Religion in American Politics
POSC 384	Minority Group Politics
POSC 385	The U.S. Congress
POSC 386	The U.S. Judiciary
POSC 387	The Appellate Process and Practice
POSC 472	Media and Politics

Subtotal 12

Additional Electives

Choose two additional electives chosen from the 300 or 400 level. 6
For three of these six credit hours, students may also take one of the following courses:

POSC 200	Global Politics [C4GE]
POSC 300	Politics and Film
POSC 301	The Washington Semester Experience
POSC 391	Topics in Public Policy
POSC 398	Simulations
POSC 401	Topics in Political Science
POSC 491	Washington Semester Internship Readiness

POSC 493	Political Communication Internship
POSC 495	Internship in Political Science
PPA 200	Introduction to Public Policy
PPA 265	Public Administration

Subtotal 6

Senior Capstone

Students should fulfill the senior capstone requirement for each major and minor separately. This means that senior experiences completed for one major or minor cannot normally be counted for another. Students who have questions about a particular situation should see the academic unit head or appropriate academic unit coordinator for clarification.

POSC 492	Senior Seminar in Political Science	3-4
Subtotal		3-4

Total Credits 38-39

¹ Completion of the General Education: The Natural World Quantitative Reasoning and either POSC 201, POSC 225, POSC 230 or POSC 240 are prerequisites for POSC 295. INTA 295 may substitute for POSC 295 with approval by the POSC coordinator.

Recommended Schedule for Majors

Students entering the major having already completed their General Education requirements should contact their advisor for guidance.

The following is an example of a four-year course of study for a student seeking a degree in political science:

Course	Title	Credits
First Year		
POSC 201	Introduction to Western Political Theory	3
POSC 225	U.S. Government [C4AE]	4
World language		6
Madison Foundations		9
General Education requirement including Quantitative Reasoning requirement		9
Credits		31
Second Year		
POSC 230	International Relations	3
POSC 240	Comparative Politics	3
World language		6
B.A. Philosophy requirement		3
General Education requirements		15
Credits		30
Third Year		
POSC 295	Research Methods in Political Science	4
Elective requirements in the four areas of the POSC major		12
Coursework for university electives, second major or minor		13
Credits		29
Fourth Year		
POSC 492	Senior Seminar in Political Science	3
Remaining POSC electives		6
Coursework for university electives, second major or minor		21
Credits		30
Total Credits		120

Psychology, B.A.

Program Description

The Department of Psychology offers both the B.A. and B.S. degrees in psychology. Consistent with its mission, “to educate students in, and contribute to, the science of psychology,” these programs emphasize the scientific study of behavior and mental processes. Students are trained in core areas of psychology as a natural science (e.g., cognition, sensation and perception), psychology as a social science (e.g., abnormal psychology, social psychology), as well as courses in research design, statistical analysis and scientific reasoning.

The curriculum provides a rigorous foundation in the methods and principles of psychological science, preparing students to critically evaluate research. Students have opportunities to engage in empirical research through coursework and faculty-mentored projects, gaining experience with data collection, analysis and/or interpretation. The psychology major prepares students for graduate study in psychology and related disciplines, as well as for a wide range of future careers, including roles in research, data analysis, business and human service professions (e.g., applied behavior analysis, social services and healthcare).

Admission and Progression Requirements

Students who wish to pursue the psychology major must:

- 1. declare the major and
- 2. meet admissions requirements.

Declare Psychology B.A., B.S., or 2nd Major

Date by which you must request to declare the major in Psychology in MyMadison	Semester you will begin as a declared major after your declaration is approved
March 1	Summer of the same calendar year
March 1	Fall of the same calendar year
October 1	Spring of the next calendar year

Requests to declare the major after the deadline will be processed the following semester.

Requirements for Admission to the Major

Students interested in completing the psychology major must meet the department’s progression standards and be fully admitted to the major. The number of semesters required to complete the program will depend on the timing of the student’s admission to psychology and the number of summer courses completed.

Students who meet all of the following criteria will be fully admitted to the major and will be permitted to take psychology major courses. Equivalent courses completed at another university for which the student has earned JMU approved transfer credit are acceptable.

- Complete PSYC 101, with a minimum grade of "C." Equivalent courses completed at other institutions that transfer to JMU are acceptable.
- Complete MATH 205, MATH 220, MATH 231, MATH 235 or another acceptable math course with a minimum grade of "C." Equivalent courses completed at other institutions that transfer to JMU are acceptable.
- Complete any General Education Psychology course (PSYC 101, PSYC 122 or PSYC 160) with a minimum grade of "B." Earning a 4 on

the AP Psychology exam meets this requirement. With departmental approval, grades from other institutions can fulfill this requirement.

- Declare the psychology major on MyMadison.

Declared psychology majors who meet the criteria for admission and register for PSYC 210 or PSYC 212 will be fully admitted to the major. Students not admitted to the major may be admitted in a later semester, once they meet admissions requirements.

Prerequisites

Psychology majors must complete PSYC 101 and two statistics and research methodology courses (PSYC 210-PSYC 211 or PSYC 212-PSYC 213) before enrolling in courses numbered 330 and above. The prerequisites for PSYC 210 and PSYC 212 are any mathematics course numbered 205 or above and PSYC 101. Some psychology courses numbered 330 and above have specific prerequisites. See course listings for details.

Registration and Assessment

During registration, psychology majors will be given priority for course selection. Graduating majors must participate in assessment activities. Assessment information helps the faculty modify the curriculum to meet student needs.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required)	0-12
Philosophy course(s) (in addition to General Education courses)	3
Major requirements	47
University Electives	17-29
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

Major Requirements

The courses listed below are required of all students pursuing a Bachelor of Arts degree, regardless of whether psychology is their first or second major.

Code	Title	Credits
Required Courses		
PSYC 101	General Psychology [C5SD]	3
PSYC 160	Life Span Human Development [C5SD]	3
MATH 220	Elementary Statistics [C3QR]	3
Subtotal		9
Methodology Core		
Choose one of the following options:		8
Option 1:		
PSYC 210	Psychological Measurement and Statistics (offered every semester)	
PSYC 211	Psychological Research Methods (offered every semester)	

Option 2:

PSYC 212	Psychological Research Design and Data Analysis I (offered only during the fall semester)
PSYC 213	Psychological Research Design and Data Analysis II (offered only during the spring semester)

Subtotal	8
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SS Content Core – Psychology as a Social Science

Choose at least three of the following:	9
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PSYC 330	Psychology of Personality
PSYC 335	Abnormal Psychology
PSYC 345	Social Psychology
PSYC 365	Developmental Psychology

Subtotal	9
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NS Content Core – Psychology as a Natural Science

Choose at least three of the following:	9
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PSYC/BIO 375	Sensation and Perception
PSYC 380	Cognitive Psychology
PSYC/BIO 385	Biopsychology
PSYC 390	Psychology of Learning
PSYC/BIO 395	Comparative Animal Behavior

Subtotal	9
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Upper-Level Courses

Earn six credit hours from among the following:	6
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PSYC 400	Advanced Topics in Psychology
PSYC 401	Peer Advising
PSYC 402	Independent Study in Psychology ¹
PSYC 403	Independent Research in Psychology ¹
PSYC 410	Psychology of the Workplace
PSYC 415	Forensic Psychology
PSYC 425	School Psychology
PSYC 427	Tests and Measurements
PSYC 428	Educational Psychology
PSYC 430	Clinical Psychology
PSYC 450	Psychology of Child Abuse and Neglect
PSYC 452	Child Psychopathology
PSYC 461	Advanced Study Abroad in Psychology
PSYC 475	Psychology of Adulthood
PSYC 480	Applied Behavior Analysis

Subtotal	6
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Psychology Electives

Complete 3 credit hours of psychology electives	3
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Subtotal	3
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Capstone Course

Departmental approval is required for students to enroll in more than one psychology capstone course.	
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Choose one of the following:	3
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PSYC 492	History of Psychology
PSYC 495	Field Placement in Psychology
PSYC 497	Senior Seminar
PSYC 499A & PSYC 499B & PSYC 499C	Honors and Honors and Honors

Subtotal	3
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Sociocultural Competency

Choose one of the following: ²	
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PSYC 220	Psychology and Culture
PSYC 308	Health Psychology
PSYC 310	The Psychology of Women and Gender
PSYC 325	Counseling Psychology
PSYC 410	Psychology of the Workplace
Other course sections that meet the sociocultural competency requirement	

Total Credits	47
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¹ Repeatable. Additional credits in PSYC 402 or PSYC 403 could fulfill the three credit hour elective requirement. Any additional PSYC 402 or PSYC 403 hours could fulfill the 120 credit hours needed for graduation.

² One sociocultural competency course is required. Credit for this course may be counted under one of the other major categories.

Recommended Schedule for Majors

The following program sample is intended as a guide. Courses must be taken in the sequence outlined; however, the semester during which a particular course is taken depends on a number of factors, including readiness to take 200-level math, the semester the psychology major is started and other majors or minors the student is completing.

Students are encouraged to meet regularly with their psychology academic advisor to plan their personal course schedule. Because of course sequencing requirements, even with the most compressed program, a minimum of five semesters (or four semesters and summer school) following full admission to the major is required to complete the psychology major.

Course	Title	Credits
First Year		
MATH 220	Elementary Statistics [C3QR]	3
PSYC 101	General Psychology [C5SD]	3
General Education, B.A. degree requirement and/or minor program courses		24
Credits		30
Second Year		
PSYC 160	Life Span Human Development [C5SD]	3
PSYC 210 or PSYC 212	Psychological Measurement and Statistics or Psychological Research Design and Data Analysis I	4
Choose one of the following:		4
PSYC 211	Psychological Research Methods ¹	
PSYC 213	Psychological Research Design and Data Analysis II ²	
General Education, B.A. degree requirement and/or minor program courses and/or Psychology elective courses		19
Credits		30
Third Year		
SS and NS content core courses		9
Minor program courses or electives		21
Credits		30
Fourth Year		
SS and NS content core courses		9
Psychology Upper-level Course		6
Psychology Capstone course		3

Minor program or elective courses	12
Credits	30
Total Credits	120

¹ PSYC 210 is a prerequisite for PSYC 211.
² PSYC 212 is a prerequisite for PSYC 213.

Psychology, B.S.

Program Description

The Department of Psychology offers both the B.A. and B.S. degrees in psychology. Consistent with its mission, “to educate students in, and contribute to, the science of psychology,” these programs emphasize the scientific study of behavior and mental processes. Students are trained in core areas of psychology as a natural science (e.g., cognition, sensation and perception), psychology as a social science (e.g., abnormal psychology, social psychology), as well as courses in research design, statistical analysis and scientific reasoning.

The curriculum provides a rigorous foundation in the methods and principles of psychological science, preparing students to critically evaluate research. Students have opportunities to engage in empirical research through coursework and faculty-mentored projects, gaining experience with data collection, analysis and/or interpretation. The psychology major prepares students for graduate study in psychology and related disciplines, as well as for a wide range of future careers, including roles in research, data analysis, business and human service professions (e.g., applied behavior analysis, social services and healthcare).

Admission and Progression Requirements

Students who wish to pursue the psychology major must

- 1. declare the major and
- 2. meet admissions requirements.

Declare Psychology B.A., B.S., or 2nd Major

Date by which you must request to declare the major in Psychology in MyMadison	Semester you will begin as a declared major after your declaration is approved
March 1	Summer of the same calendar year
March 1	Fall of the same calendar year
October 1	Spring of the next calendar year

Requests to declare the major after the deadline will be processed the following semester.

Requirements for Admission to the Major

Students interested in completing the psychology major must meet the department’s progression standards and be fully admitted to the major. The number of semesters required to complete the program will depend on the timing of the student’s admission to psychology and the number of summer courses completed.

Students who meet all of the following criteria will be fully admitted to the major and will be permitted to take psychology major courses. Equivalent courses completed at another university for which the student has earned JMU approved transfer credit are acceptable.

- Complete PSYC 101, with a minimum grade of "C." Equivalent courses completed at other institutions that transfer to JMU are acceptable.
- Complete MATH 205, MATH 220, MATH 231, MATH 235 or another acceptable math course with a minimum grade of "C." Equivalent courses completed at other institutions that transfer to JMU are acceptable.
- Complete any General Education Psychology course (PSYC 101, PSYC 122 or PSYC 160) with a minimum grade of "B." Earning a 4 on

the AP Psychology exam meets this requirement. With departmental approval, grades from other institutions can fulfill this requirement.

- Declare the psychology major on MyMadison.

Declared psychology majors who meet the criteria for admission and register for PSYC 210 or PSYC 212 will be fully admitted to the major. Students not admitted to the major may be admitted in a later semester, once they meet admissions requirements.

Prerequisites

Psychology majors must complete PSYC 101 and two statistics and research methodology courses (PSYC 210-PSYC 211 or PSYC 212-PSYC 213) before enrolling in courses numbered 330 and above. The prerequisites for PSYC 210 and PSYC 212 are any mathematics course numbered 205 or above and PSYC 101. Some psychology courses numbered 330 and above have specific prerequisites. See course listings for details.

Registration and Assessment

During registration, psychology majors will be given priority for course selection. Graduating majors must participate in assessment activities. Assessment information helps the faculty modify the curriculum to meet student needs.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3
Scientific Literacy requirement (in addition to General Education)	3-4
Major requirements including cognate	50
University electives	22-23
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

Major Requirements

The courses listed below are required of all students pursuing a Bachelor of Science degree.

Code	Title	Credits
Required Courses		
PSYC 101	General Psychology [C5SD]	3
PSYC 160	Life Span Human Development [C5SD]	3
MATH 220	Elementary Statistics [C3QR]	3
Subtotal		9
Methodology Core		
Choose one of the following options:		8
PSYC 210 & PSYC 211	Psychological Measurement and Statistics and Psychological Research Methods	
PSYC 212 & PSYC 213	Psychological Research Design and Data Analysis I and Psychological Research Design and Data Analysis II	

Subtotal	8
SS Content Core – Psychology as a Social Science	
Choose at least three of the following:	9
PSYC 330 Psychology of Personality	
PSYC 335 Abnormal Psychology	
PSYC 345 Social Psychology	
PSYC 365 Developmental Psychology	
Subtotal	9
NS Content Core – Psychology as a Natural Science	
Choose at least two of the following:	6
PSYC/BIO 375 Sensation and Perception	
PSYC 380 Cognitive Psychology	
PSYC 390 Psychology of Learning	
PSYC/BIO 395 Comparative Animal Behavior	
Subtotal	6
Biopsychology	
PSYC/BIO 385 Biopsychology	3
Subtotal	3
Upper-Level Courses	
Choose 6 credits of the following:	6
PSYC 400 Advanced Topics in Psychology	
PSYC 401 Peer Advising	
PSYC 402 Independent Study in Psychology ¹	
PSYC 403 Independent Research in Psychology ¹	
PSYC 410 Psychology of the Workplace	
PSYC 415 Forensic Psychology	
PSYC 425 School Psychology	
PSYC 427 Tests and Measurements	
PSYC 428 Educational Psychology	
PSYC 430 Clinical Psychology	
PSYC 450 Psychology of Child Abuse and Neglect	
PSYC 452 Child Psychopathology	
PSYC 461 Advanced Study Abroad in Psychology	
PSYC 475 Psychology of Adulthood	
PSYC 480 Applied Behavior Analysis	
Subtotal	6
Psychology Electives	
Choose three credit hours of psychology electives	3
Subtotal	3
Capstone Course	
Departmental approval is required for students to enroll in more than one capstone course.	
Choose one of the following:	3
PSYC 492 History of Psychology	
PSYC 495 Field Placement in Psychology	
PSYC 497 Senior Seminar	
PSYC 499A Honors	
& PSYC 499B and Honors	
& PSYC 499C and Honors	
Subtotal	3
Sociocultural Competency	
Choose one of the following: ²	
PSYC 220 Psychology and Culture	

PSYC 308 Health Psychology	
PSYC 310 The Psychology of Women and Gender	
PSYC 325 Counseling Psychology	
PSYC 410 Psychology of the Workplace	
Other course sections that meet the sociocultural competency requirement	
Cognate Requirement	
Students must take one additional scientific literacy courses beyond the B.S. degree requirements and general education. Choose a course from the list of courses meeting the B.S. degree scientific literacy requirement. ³	3
Total Credits	50

¹ Repeatable. Additional credits in PSYC 402 or PSYC 403 could fulfill the three credit hour elective requirement. Any additional PSYC 402 or PSYC 403 hours could fulfill the 120 credit hours needed for graduation.

² One sociocultural competency course is required. Credit for this course may be counted under one of the other major categories.

³ The list of courses that fulfill the B.S. scientific literacy requirement can be found on the registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml).

Recommended Schedule for Majors

The following program sample is intended as a guide. Courses must be taken in the sequence outlined; however, the semester during which a particular course is taken depends on a number of factors, including readiness to take 200-level math, the semester the psychology major is started and other majors or minors the student is completing.

Students are encouraged to meet regularly with their psychology academic advisor to plan their personal course schedule. Because of course sequencing requirements, even with the most compressed program, a minimum of five semesters (or four semesters and summer school) following full admission to the major is required to complete the psychology major.

Course	Title	Credits
First Year		
MATH 220	Elementary Statistics [C3QR]	3
PSYC 101	General Psychology [C5SD]	3
General Education, B.S. degree requirement and/or minor program courses		24
Credits		30
Second Year		
PSYC 160	Life Span Human Development [C5SD]	3
PSYC 210 or PSYC 212	Psychological Measurement and Statistics or Psychological Research Design and Data Analysis I	4
Choose one of the following:		4
PSYC 211	Psychological Research Methods ¹	
PSYC 213	Psychological Research Design and Data Analysis II ²	
General Education, B.S. degree requirement, B.S. psychology cognate, minor program and/or psychology elective courses		19
Credits		30
Third Year		
PSYC 385	Biopsychology ³	3
SS content core courses		9
NS content core courses		6
Minor program courses or electives		12
Credits		30

Fourth Year

Psychology upper-level courses	6
Psychology capstone course	3
Minor program or elective courses	21
Credits	30
Total Credits	120

¹ PSYC 210 is a prerequisite for PSYC 211.

² PSYC 212 is a prerequisite for PSYC 213.

³ PSYC 385 may be taken during the fourth year.

Public Administration, B.S.

Coordinator: Robert Roberts

Email: robertrn@jmu.edu

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Program Description

The major in public administration provides students with a general foundation in public policy, public and nonprofit workplaces and their political, legal, and managerial environments. This major prepares students for professional employment and leadership in government and nonprofit organizations. The program consists of a core of courses offering general knowledge essential for understanding and working in the public and nonprofit arenas. This core provides students an appreciation of the political culture and economic environment of public service work, measurement techniques, and a basic understanding of the policy process. The focus areas of the public administration major then seek to give students a foundation in the nature and practice of public policy and public administration by having them examine the complexity of the adoption and implementation of public policy and the tools used to implement public policy and administer public and nonprofit programs.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ^{2,3}	3
Scientific Literacy requirement ²	3
Major requirements	39
University electives	33-34
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. The number of credit hours necessary to fulfill these requirements may vary.

² In addition to course work taken to fulfill General Education requirements.

³ For this requirement, majors should take a class that satisfies the General Education quantitative reasoning requirement and one 200-level PPA course which are the prerequisites for POSC 295, a required core course in the major.

Major Requirements

Code	Title	Credits
Core Courses		
ECON 200	Introduction to Macroeconomics [C4GE]	3
POSC 225	U.S. Government [C4AE]	4
POSC 295	Research Methods in Political Science	4
PPA 200	Introduction to Public Policy	3
PPA 265	Public Administration	3
Subtotal		17
Public Policy		
Choose three of the following:		9
POSC 347	Comparative Public Policy	
POSC 367	U.S. Immigration and Refugee Law	
POSC 378	Borderlands Policy: Latina/o/x Politics in Context	
POSC 391	Topics in Public Policy	

POSC 397	The Politics of International Economic Relations	
PPA 359	Policy Analysis	
PPA 460	Regionalism and Urban Management	
PPA 461	Education and Social Policy	
PPA 462	Social Welfare and Local Government Policy	
PPA 482	Special Topics in Public Policy	
PPA 484	Environmental Regulatory Policy and Politics	
PPA 494	Internship in Public Policy ¹	
PPA 497	Public Policy Internship in Washington ^{1,2}	
Subtotal		9
Tools of Public Administration		
Choose three of the following:		9
ACTG 244	Accounting for Non-Business Majors	
MGT 365	Fundamentals of Human Resource Management	
PPA 381	Budgetary Process	
PPA 412	Seminar in Intergovernmental Relations	
PPA 415	Legal Environment of Public Administration	
PPA 441	Stability and Recovery Operations	
PPA 469	Philanthropy and Volunteerism	
PPA 470	Nonprofit Management and Leadership	
PPA 471	Financial Management for Public and Nonprofit Organizations	
PPA 472	Contract Management	
PPA 473	Economic and Community Development	
PPA 475	Grant Writing and Project Management	
PPA 480	Ethics and Political Corruption	
PPA 483	Special Topics in Public Management	
PPA 485	Governance for Environmental Sustainability	
PPA 496	Internship in Public Management ¹	
Subtotal		9
Capstone		
PPA 420	Senior Seminar in Public Management	4
or PPA 492	Senior Seminar in Public Policy	
Subtotal		4
Total Credits		39

¹ Students may only apply credit from one internship (PPA 494, PPA 496, or PPA 497) toward the major requirements. However, a second internship can count for a minor or the 120 credits needed to graduate.

² A Washington internship counts as six credits but only three of those credits can count toward the major.

Recommended Schedule for Majors

Students entering the major having already completed their General Education requirements should contact their advisor for guidance.

Students switching into the major from another degree program should expect to take at least four semesters to complete the major.

The following is an example of a four-year course of study for a student seeking a degree:

Course	Title	Credits
First Year		
ECON 200	Introduction to Macroeconomics [C4GE]	3

POSC 225	U.S. Government [C4AE]	4
PPA 200	Introduction to Public Policy	3
PPA 265	Public Administration	3
Madison Foundations		9
General Education requirements, including Quantitative Reasoning requirement		9
Credits		31
Second Year		
Courses that fulfill the Public Policy requirements		6
Courses that fulfill the Tools of Public Administration requirements		6
General Education		18
Consider declaring a second major or minor (optional)		
Consider five-year MPA program and meet with academic advisor to discuss coursework (optional)		
Credits		30
Third Year		
POSC 295	Research Methods in Political Science	4
Courses that fulfill the Public Policy requirements		6
Courses that fulfill the Tools of Public Administration requirements		6
Students may use an internship course to satisfy coursework in the Tools of Public Administration area. The third year is an ideal time to seek out internship opportunities.		6
Continue coursework for second major or minor (optional)		7
Credits		29
Fourth Year		
PPA 420 or PPA 492	Senior Seminar in Public Management or Senior Seminar in Public Policy	4
Courses that fulfill the Public Policy requirements		8
Courses that fulfill the Tools of Public Administration requirements		9
Continue coursework for second major or minor (optional)		9
Credits		30
Total Credits		120

Quantitative Finance Major with a Minor in Economics and Mathematics, B.S.

Program Description

The B.S. degree with a major in quantitative finance is designed to prepare students for careers in financial engineering, structured finance, financial modeling, securitization, actuarial science, financial analysis and portfolio management. The quantitative finance major is classified as a STEM program. The focus of this major is on problem solving in the quantitative areas of finance with an added emphasis on the application of complex securities to a variety of financial situations.

The quantitative finance program, which is a cross disciplinary major with many courses co-listed with the Department of Mathematics and Statistics, is a highly structured program requiring minor fields in both mathematics and economics, though many students choose to double major in mathematics and quantitative finance. If the double major is selected, students are required to take ECON 200, ECON 201 and ECON 331, but are not required to complete the economics minor. Students electing this program should consult with their major advisor as early as possible to identify the appropriate course sequencing. Students electing to double major in mathematics and quantitative finance should also consult with a mathematics advisor as soon as possible.

Students enrolled in James Madison University who wish to change their major to quantitative finance should go to the Student Center in MyMadison to submit a request. Students must be in good academic standing to change their major to quantitative finance and, if they have taken any FIN prefix courses at the time of the change request, must meet the prerequisites for the required courses in the quantitative finance major.

Transfer Credit

Each program in the CoB (p. 15) will accept no more than two courses of transfer credit toward the major except in extraordinary circumstances. The major is defined as the course work required by a major field of study in addition to the lower and upper-level BBA core courses. For a course to be considered for acceptance in the student's major, it must have been completed in an AACSB International-accredited business program, or a similarly accredited program, at a four-year university.

In general, all finance course work must be completed at JMU. Transfer credit for finance courses is awarded only in certain circumstances. In no case will transfer credit be awarded for FIN 480. Contact the department head for more information on transfer credit.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Scientific Literacy requirement ²	3
Quantitative requirement ²	3-4
Major requirements	65
Electives	11
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirement.

Major Requirements

Code	Title	Credits
Required Courses		
COB 241	Financial Accounting	3
FIN 250	Introduction to Quantitative Finance	3
FIN 365	Intermediate Financial Management	3
FIN 371	Principles of Investments	3
FIN 380	Elemental and Derivative Securities Analysis	3
FIN/MATH 395	Mathematical Finance	3
FIN/MATH 405	Securities Pricing	3
FIN 450	Financial Risk Management	3
FIN 480	Seminar in Quantitative Finance Research and Data Analysis	3
Choose one of the following:		3
FIN/MATH 328 Time Series Analysis		
FIN/ECON 372 International Finance and Payments		
FIN 451	Risk Management II	
FIN 455	Global Finance	
FIN 471	Portfolio Management	
FIN 475	Financial Modeling and Risk Analysis	
Subtotal		30
Mathematics Courses		
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations	4
MATH 248	Computers and Numerical Algorithms	4
MATH 318	Introduction to Probability and Statistics	4
Choose one of the following recommended MATH electives:		3
MATH 423 Stochastic Processes		
MATH 424 Statistical Decision Theory		
MATH 426 Probability and Mathematical Statistics I		
MATH 440 Fourier Analysis and Partial Differential Equations		
Subtotal		27
Economics Courses		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
ECON 331	Intermediate Microeconomic Theory	3
ECON 332	Intermediate Macroeconomic Theory	3
MATH 322	Applied Linear Regression	3
or ECON 385 Econometrics		
One upper-level economics elective		3
Subtotal		18
Total Credits		75

Concentrations

Though not required, Quantitative Finance majors may elect a concentration. The concentrations are financial analysis and risk management. Students electing these concentrations will be taking specific FIN electives.

Financial Analysis Concentration

The financial analysis concentration helps prepare students for careers as financial analysts and also Levels I and II of the Chartered Financial Analyst (CFA) exam.

Code	Title	Credits
Required Courses		
FIN 362	Financial Analysis	3
FIN 378	Fixed Income Analysis	3
FIN 380	Elemental and Derivative Securities Analysis	3
Choose one of the following:		3
ACTG 343	Corporate Financial Reporting I	
FIN 434	Fundamentals of Financial Data Science	
FIN 475	Financial Modeling and Risk Analysis	
FIN 420	Financial Planning and Wealth Management	
FIN 455	Global Finance	
Total Credits		12

Risk Management Concentration

Though not required, quantitative finance majors may elect the risk management concentration. Students electing this concentration will be taking specific electives and two additional electives. Students may add this concentration to their program when they progress to FIN 380.

Quantitative finance majors who wish to declare the risk management concentration may do so after successfully completing FIN 250 or any semester following.

A focus in risk management is designed for qualitative finance majors pursuing a more in-depth review of the issues facing organizations and the tools needed to address those uncertainties. In the risk management concentration, students focus on the theory of risk management, risk identification, risk measurement and applications in the form of risk modeling techniques such as Value-at-Risk and Monte Carlo simulations.

Code	Title	Credits
Required Courses		
FIN 380	Elemental and Derivative Securities Analysis	3
FIN 450	Financial Risk Management	3
FIN 451	Risk Management II	3
FIN 475	Financial Modeling and Risk Analysis	3
Total Credits		12

Additional Information

FIN 450 is a requirement for quantitative finance majors and FIN 451 and FIN 475 are possible electives for this major. Students electing this concentration take two courses beyond the typical quantitative finance major.

Quantitative Finance/Mathematics Double Major, B.S.

Program Description

The B.S. degree with a major in quantitative finance is designed to prepare students for careers in financial engineering, structured finance, financial modeling, securitization, actuarial science, financial analysis and portfolio management. The quantitative finance major is classified as a STEM program. The focus of this major is on problem solving in the quantitative areas of finance with an added emphasis on the application of complex securities to a variety of financial situations.

The quantitative finance program, which is a cross disciplinary major with many courses co-listed with the math department, is a highly structured program requiring minor fields in both mathematics and economics, though many students choose to double major in mathematics and quantitative finance. If the double major is selected, students are required to take ECON 200, ECON 201, and ECON 331, but are not required to complete the economics minor. Students electing this program should consult with their major advisor as early as possible to identify the appropriate course sequencing. Students electing to double major in mathematics and quantitative finance should also consult with a mathematics advisor as soon as possible.

Students enrolled in James Madison University who wish to change their major to quantitative finance should go to the Student Center in MyMadison to submit a request. Students must be in good academic standing to change their major to quantitative finance and, if they have taken any FIN prefix courses at the time of the change request, must meet the prerequisites for the required courses in the quantitative finance major.

Transfer Credit

Each program in the CoB will accept no more than two courses of transfer credit toward the major except in extraordinary circumstances. The major is defined as the course work required by a major field of study in addition to the lower and upper-level BBA core courses. For a course to be considered for acceptance in the student's major, it must have been completed in an AACSB International-accredited business program, or a similarly accredited program, at a four-year university.

In general, all finance course work must be completed at JMU. Transfer credit for finance courses is awarded only in certain circumstances. In no case will transfer credit be awarded for FIN 480. Contact the department head for more information on transfer credit.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Scientific Literacy requirement ²	3
Quantitative requirement ²	3-4
Major requirements	78
University electives	8
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² In addition to course work taken to fulfill General Education requirement. Both the quantitative and scientific literacy requirements are fulfilled by courses in the major.

Major Requirements

Code	Title	Credits
Required Courses		
COB 241	Financial Accounting	3
FIN 250	Introduction to Quantitative Finance	3
FIN 365	Intermediate Financial Management	3
FIN 371	Principles of Investments	3
FIN 380	Elemental and Derivative Securities Analysis	3
FIN/MATH 395	Mathematical Finance	3
FIN/MATH 405	Securities Pricing	3
FIN 450	Financial Risk Management	3
FIN 480	Seminar in Quantitative Finance Research and Data Analysis	3
Choose one of the following:		3
FIN 328	Time Series Analysis	
MATH 328	Time Series Analysis	
FIN 372	International Finance and Payments	
ECON 372	International Finance and Payments	
FIN 451	Risk Management II	
FIN 455	Global Finance	
FIN 471	Portfolio Management	
FIN 475	Financial Modeling and Risk Analysis	
Subtotal		30
Economics Courses		
ECON 200	Introduction to Macroeconomics [C4GE]	3
ECON 201	Introduction to Microeconomics	3
ECON 331	Intermediate Microeconomic Theory	3
Subtotal		9
Mathematics Courses		
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations	4
MATH 245	Discrete Mathematics	3
MATH 248	Computers and Numerical Algorithms	4
MATH 318	Introduction to Probability and Statistics	4
MATH 410	Advanced Calculus I	3
MATH 430	Abstract Algebra I	3
Choose one of the following mathematics electives:		3
MATH 411	Advanced Calculus II	
MATH 431	Abstract Algebra II	
MATH 434	Advanced Linear Algebra	
MATH 435	Introduction to Topology	
Choose one of the following recommended mathematics electives:		3
MATH 322	Applied Linear Regression	
MATH 423	Stochastic Processes	
MATH 424	Statistical Decision Theory	
MATH 426	Probability and Mathematical Statistics I	

MATH 440	Fourier Analysis and Partial Differential Equations
Subtotal	39
Total Credits	78

Concentrations

Though not required, Quantitative Finance majors may elect a concentration. The concentrations are financial analysis and risk management. Students electing these concentrations will be taking specific FIN electives.

Financial Analysis Concentration

The financial analysis concentration helps prepare students for careers as financial analysts and also Levels I and II of the Chartered Financial Analyst (CFA) exam.

Code	Title	Credits
Required Courses		
FIN 362	Financial Analysis	3
FIN 378	Fixed Income Analysis	3
FIN 380	Elemental and Derivative Securities Analysis	3
Choose one of the following:		3
ACTG 343	Corporate Financial Reporting I	
FIN 475	Financial Modeling and Risk Analysis	
FIN 434	Fundamentals of Financial Data Science	
FIN 420	Financial Planning and Wealth Management	
FIN 455	Global Finance	
Total Credits		12

Risk Management Concentration

Though not required, quantitative finance majors may elect the risk management concentration. Students electing this concentration will be taking specific electives and two additional electives. Students may add this concentration to their program when they progress to FIN 380.

Quantitative finance majors who wish to declare the risk management concentration may do so after successfully completing FIN 250 or any semester following.

A focus in risk management is designed for qualitative finance majors pursuing a more in-depth review of the issues facing organizations and the tools needed to address those uncertainties. In the risk management concentration, students focus on the theory of risk management, risk identification, risk measurement and applications in the form of risk modeling techniques such as Value-at-Risk and Monte Carlo simulations.

Code	Title	Credits
Required Courses		
FIN 380	Elemental and Derivative Securities Analysis	3
FIN 450	Financial Risk Management	3
FIN 451	Risk Management II	3
FIN 475	Financial Modeling and Risk Analysis	3
Total Credits		12

Secondary Education, B.S.

Program Description

Grades Six through Twelve Licensure Program

The undergraduate major in secondary education provides candidates with course offerings and experiences that form the required foundation for licensure in secondary education through James Madison University. In addition to the secondary education major, students also complete a major in a discipline closely associated with the desired teaching area. The state-approved licensure areas in the secondary education concentration at JMU include English, mathematics, biology, chemistry, physics and history/social sciences. Licensure endorsements for other teaching areas (e.g., Algebra I, journalism and gifted education) are available as add-on endorsements. Students should contact the department of Middle, Secondary and Mathematics Education for information concerning those programs.

For successful completion of the secondary education major, students must earn a minimum grade of "B-" in all required secondary education major courses (excludes EDUC 200, WRTC 103 and EDUC 175 or PSYC 160) and earn credit in all licensure area content coursework. Satisfactory completion of both the secondary education major and the content area major or equivalent are necessary to receive an endorsement recommendation through JMU for a teaching license.

Students should be aware that there is a teacher education admission process that must be completed after a secondary education major is declared. Students must be fully admitted to Teacher Education prior to registering for MSSE 370 and MSSE 371. Students should also note that actual teacher licensure requirements may differ from the catalog requirements listed because of changes enacted by the Virginia Department of Education or accrediting agencies after the catalog copy is approved. Therefore, it is important for students to confer regularly with their secondary education advisors.

Admission Requirements

Students must meet specific requirements to be admitted to programs within the College of Education (p. 18). These criteria can be found on the College of Education (p. 18) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement ²	3
Scientific Literacy Requirement ²	3
Major Requirements	50-51
Additional Content Major Requirements	36-56
Total	125-128

¹ General Education courses must include EDUC 175 or PSYC 160. EDUC 175 or PSYC 160 double counts for major requirements and General Education. This brings the total credits to 125-128.

² Quantitative and scientific literacy requirements may be fulfilled by courses required within the major.

Major Requirements

Students are responsible for transportation to field placements at local schools. Before beginning the licensure portion of the professional

program, candidates must meet all criteria for admission to teacher education (e.g., minimum GPA of 2.5, completing required trainings. Candidates are expected to complete all admission processes early in their sophomore year. Candidates must meet all additional requirements of the program before completion. Candidates must submit passing scores on the Praxis Subject Assessment prior to registration for MSSE 420 and prior to beginning MSSE 475.

Code	Title	Credits
Core Courses		
EDUC 175	Learning, Behavior and Engagement in Context [C5SD]	3
or PSYC 160	Life Span Human Development [C5SD]	
EDUC 200	Foundations of Education in the United States	3
LED 440	Disciplinary Literacy in Secondary Education	3
MSSE 210	Diversity in Middle and Secondary Education	3
MSSE 370	General Instructional Methods for Grades 6-12	3
MSSE 371	Clinical Experience in Adolescent Education	1
MSSE 403	Behavior and Classroom Management in a Diverse Society	3
MSSE 415	Content Area Field Experience in Middle Schools	3
MSSE 420	Differentiation for Adolescents	3
MSSE 425	Assessment in Middle and Secondary Education	3
MSSE 450	Internship Seminar	2
Subtotal		30
Education – Student Teaching / Internship		
MSSE 475	Internship in High School	8
Subtotal		8
Concentrations		
Students should select one of the following concentrations.		12-13
Chemistry, Biology or Physics		
English		
Social Studies		
Mathematics		
Subtotal		12-13
Total Credits		50-51

Concentrations

Students should select one of the following concentrations.

Chemistry, Biology or Physics

Code	Title	Credits
MSSE 414	Natural Sciences Teaching Methods, Grades 6-8	3
MSSE 444	Science Content Methods, Grades 9-12	3
MSSE 445	Content Area Field Experience in Secondary Education, Grades 9-12	3
MSSE 397/ PHYS 302	Science Pedagogy	2
MSSE 495/ BIO 489	Topics in Science Pedagogical Content Knowledge	2
Total Credits		13

English

Code	Title	Credits
LED 254	Literature for Adolescents	3
MSSE 411	English/Language Arts Teaching Methods, Grades 6-8	3
MSSE 441	English/Language Arts Content Methods, Grades 9-12	3
MSSE 445	Content Area Field Experience in Secondary Education, Grades 9-12	3
Total Credits		12

Social Studies

Code	Title	Credits
MSSE 412	Social Studies Teaching Methods, Grades 6-8	3
MSSE 442	Social Studies Content Methods, Grades 9-12	3
MSSE 445	Content Area Field Experience in Secondary Education, Grades 9-12	3
MSSE 455	Special Topics in Pedagogical Content Knowledge	3
Total Credits		12

Mathematics

Code	Title	Credits
MSSE 413	Mathematics Teaching Methods, Grades 6-8	3
MSSE 443	Mathematics Content Methods, Grades 9-12	3
MSSE 445	Content Area Field Experience in Secondary Education, Grades 9-12	3
MSSE 455	Special Topics in Pedagogical Content Knowledge	3
Total Credits		12

Recommended Schedule for Majors

Course	Title	Credits
First Year		
EDUC 175 or PSYC 160	Learning, Behavior and Engagement in Context [C5SD] or Life Span Human Development [C5SD]	3
EDUC 200	Foundations of Education in the United States	3
General Education requirement		21
B.S. degree requirements		3
Credits		30
Second Year		
MSSE 210	Diversity in Middle and Secondary Education	3
MSSE 370	General Instructional Methods for Grades 6-12	3
MSSE 371	Clinical Experience in Adolescent Education	1
General Education requirement		12
B.S. degree requirements		3
Content Major Courses		6
Credits		28
Third Year		
LED 440	Disciplinary Literacy in Secondary Education	3
MSSE 415	Content Area Field Experience in Middle Schools	3
MSSE 425	Assessment in Middle and Secondary Education	3
Remaining General Education		7
Content Major Courses		15
Content Specific Teaching Methods Course (English, social studies, mathematics or science)		3
Credits		34

Fourth Year		
MSSE 403	Behavior and Classroom Management in a Diverse Society	3
MSSE 420	Differentiation for Adolescents	3
MSSE 450	Internship Seminar	2
MSSE 475	Internship in High School	8
Licensure-Area Courses and/or Electives and/or Add-on Endorsement		12
Content Specific Methods Course (English, social studies, mathematics or science)		2-3
Pedagogical Content Knowledge Course		2-3
Content Field Placement Course		1
Credits		33-35
Total Credits		125-127

Special Education, B.S.

Program Description

The Bachelor of Science pre-professional program in special education enables students to become knowledgeable about the characteristics, diagnosis and remediation of children with disabilities accessing the general education curriculum and in adapted curriculum settings. While in this program, students will have the opportunity to work with students with disabilities in elementary, middle and high schools across multiple school districts. They will learn from dedicated and engaged faculty members and have opportunities to participate in various scholarly activities inside and outside of the classroom. Students completing the four-year degree program, when combined with the teacher licensure option, are prepared to serve as teachers of students with disabilities in a variety of educational placements.

The licensure program is designed to prepare resilient, culturally responsive educators who are advocates for children and youth with disabilities that are not only qualified for the complexity of their professional roles but also are reflective problem-solvers.

Students enrolled in the program will complete both special education concentrations (accessing the general curriculum and adapted curriculum). The program includes extensive field experiences. Assessment of candidate performance includes evaluation of performance in individual courses and practicums. Students are required to demonstrate satisfactory performance, which includes a minimum grade of "C" in all major and concentration coursework, a "satisfactory" score on all practicum evaluations (i.e. all indicators on the evaluation are within the satisfactory range) and all key assessments. Key assessments will be used to identify those candidates who are not making satisfactory progress towards advancement to the next sequences of courses and experiences. If progress is unsatisfactory, the candidate will be given one additional attempt on the key assessment. The assessments will be used to advise candidates and develop a plan of action for addressing any concerns that have been identified by the faculty. In some cases, a candidate will be allowed to continue in the next semester, but there will be a plan of action for addressing any concerns that have been identified by the faculty.

Every student is assigned an advisor. Students should plan on consulting their advisor regularly.

It is important for students to understand that the undergraduate major in special education does not lead to teacher-education licensure unless licensure requirements are completed. To qualify for a recommendation for licensure in special education, general curriculum K-12 and adapted curriculum K-12, students must complete licensure requirements in addition to requirements for the special education undergraduate major. These requirements include 18 credit hours of practicum and student teaching experiences. Please note that these credit hour requirements are included in the 122 credit hour requirements for the Bachelor of Science in Special Education degree.

Admission Requirements

Students must meet specific requirements to be admitted to programs within the College of Education (p. 18). These criteria can be found on the College of Education (p. 18) catalog page.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative Requirement ²	3
Scientific literacy requirement ²	3-4
Professional Studies Core	12
Special Education Core (including both concentrations: Adapted and Accessing the General Curriculum K-12)	48
Licensure Requirements (or electives)	18
Total	122

¹ The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major. Either EDUC 175 or PSYC 160, required for the professional studies core, also counts for General Education credit.

² In addition to General Education. Consult the list of courses satisfying Bachelor of Science degree requirements on the Registrar's website (https://www.jmu.edu/registrar/ba_and_bs_degree_courses.shtml).

Major Requirements

Code	Title	Credits
Professional Studies		
PSYC 160 or EDUC 175	Life Span Human Development [C5SD] Learning, Behavior and Engagement in Context [C5SD]	3
EDUC 200	Foundations of Education in the United States	3
EDUC 310	Teaching in a Diverse Society	3
EXED 200	Foundations of Exceptional Education	3
Subtotal		12
Special Education Core		
EDUC 303	Foundations of Curriculum and Instruction	3
EXED 303	Foundations of Classroom and Behavior Management	3
EXED 407	Transitions of Learners with Disabilities	3
EXED 419	Foundations and Theories of Literacy for K-12	3
EXED 474	Assessment and Evaluation in Special Education	3
EXED 475	Program Planning in Special Education	3
EXED 484	Inclusive Practices in Special Education (must be taken for 3 credit hours)	3
EXED 485	Systematic Behavioral Support and Interventions	3
Subtotal		24
Required Adapted Curriculum Concentration Courses		
EXED 342	Characteristics of Learners with Low Incidence Disabilities Accessing the Adapted Curriculum	3
EXED 429	Teaching Learners with Complex Care Needs	3
EXED 436	Communication Language and Literacy for Learners on the Adapted Curriculum	3
EXED 473	Methods of Instruction for Learners Accessing the Adapted Curriculum	3
Subtotal		12
Required General Curriculum Concentration Courses		
EXED 341	Characteristics of Learners with Disabilities Accessing the General Curriculum	3

EXED 405	Alternative Math Instructional Methods and Supports for Students with Disabilities	3
EXED 450	Principles of Specialized Reading Instruction	3
EXED 477	Remedial Methods in Special Education	3
Subtotal		12
Licensure Requirements		
Licensure requirements (18 credits) or electives (for students not pursuing licensure in special education. Will be decided between the student and the university advisor).		
EXED 476	Practicum in Literacy and Assessment (taken once for 2 credits)	2
EXED 478	Supervised Practice in Special Education Methods (taken twice at 2 credits for a total of 4 credits)	4
EXED 480	Student Teaching in Special Education (taken twice for a total of 8 credits)	8
EXED 483	Teachers as Professionals (taken twice at 1 credit for a total of 2 credits)	2
EXED 486	Supervised Clinical Practice with Planning and Methods in Special Education (taken once for 2 credits)	2
Subtotal		18
Total Credits		78

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Fall Semester		
EDUC 200	Foundations of Education in the United States	3
EXED 200	Foundations of Exceptional Education	3
General Education requirement		9
Credits		15
Spring Semester		
EDUC 310	Teaching in a Diverse Society	3
General Education requirement		9
PSYC 160 or EDUC 175	Life Span Human Development [C5SD] or Learning, Behavior and Engagement in Context [C5SD]	3
Credits		15
Second Year		
Fall Semester		
EDUC 303	Foundations of Curriculum and Instruction	3
EXED 303	Foundations of Classroom and Behavior Management	3
General Education requirement		9
Credits		15
Spring Semester		
EXED 341	Characteristics of Learners with Disabilities Accessing the General Curriculum	3
EXED 342	Characteristics of Learners with Low Incidence Disabilities Accessing the Adapted Curriculum	3
EXED 429	Teaching Learners with Complex Care Needs	3
General Education requirement		7
Credits		16
Third Year		
Fall Semester		
EXED 405	Alternative Math Instructional Methods and Supports for Students with Disabilities	3
EXED 419	Foundations and Theories of Literacy for K-12	3
EXED 474	Assessment and Evaluation in Special Education	3
EXED 484	Inclusive Practices in Special Education (must be taken for 3 credit hours)	3

EXED 486	Supervised Clinical Practice with Planning and Methods in Special Education (taken once for 2 credit hours)	2
General Education requirement		3
Credits		17
Spring Semester		
EXED 407	Transitions of Learners with Disabilities	3
EXED 450	Principles of Specialized Reading Instruction	3
EXED 475	Program Planning in Special Education	3
EXED 476	Practicum in Literacy and Assessment (taken once for 2 credit hours)	2
General Education requirement		1
Bachelor of Science courses		6
Credits		18
Fourth Year		
Fall Semester		
EXED 436	Communication Language and Literacy for Learners on the Adapted Curriculum	3
EXED 478	Supervised Practice in Special Education Methods (taken for 2 credit hours)	2
EXED 480	Student Teaching in Special Education (must be taken for 4 credit hours)	4
EXED 483	Teachers as Professionals (must be taken for 1 credit hour)	1
EXED 473 or EXED 477	Methods of Instruction for Learners Accessing the Adapted Curriculum or Remedial Methods in Special Education	3
Credits		13
Spring Semester		
EXED 478	Supervised Practice in Special Education Methods (taken for 2 credit hours)	2
EXED 480	Student Teaching in Special Education (must be taken for 4 credit hours)	4
EXED 483	Teachers as Professionals (must be taken for 1 credit hour)	1
EXED 485	Systematic Behavioral Support and Interventions	3
EXED 473 or EXED 477	Methods of Instruction for Learners Accessing the Adapted Curriculum or Remedial Methods in Special Education	3
Credits		13
Total Credits		122

Social Work, B.S.W.

Program Description

The B.S.W. program prepares social workers who are committed to addressing a broad range of social challenges, enhancing individual, family, group, organization and community well-being, and advocating for empowerment of diverse and vulnerable populations. The B.S.W. program graduates competent social workers who are highly prepared for generalist professional practice as well as graduate education through exemplary commitment to ethical practice; engagement and belonging; human rights; and social progress.

Admission and Retention Requirements

Admission Requirements

Social Work Program

Students may declare a major in social work at any time; however, they must apply for admission to the social work program the semester following completion of SOWK 287 and SOWK 288. For unconditional admittance, students must have a minimum cumulative GPA of 2.0 with no single grade lower than a "C" (2.0) in SOWK 287 and SOWK 288, and have completed the 20 hours of community service work required in SOWK 287. Students are evaluated on the basis of community service and life experiences, academic performance, communication skills, work-related habits, ability to work with others, motivation, value orientation and career plans. Students must complete this process of admission to upper-level courses will be restricted. See the Social Work Handbook or the social work website for guidelines.

Following review, the student will be notified of the application decision in writing. Decisions are to admit, to admit conditionally, not to admit or to defer decision. You will have one opportunity per semester to resubmit the application. If you fail to submit or resubmit a document that is still incorrect, you will be required to wait until the next semester deadline to resubmit. While this could potentially slow your progression in the major, it acknowledges your role of accountability in the process, a quality that will be essential for professional practice.

If admitted conditionally, the conditions for acceptance will be described. If the decision is deferred, the student will be notified in writing as to why. If not admitted, the student may appeal the decision to the head of the Department of Social Work.

Field Practicum Application

Students eligible for the field practicum are seniors who have completed the core social work requirements with no grade lower than a "C" (2.0) in SOWK 287, SOWK 288, SOWK 305, SOWK 317, SOWK 320, SOWK 335, SOWK 465, SOWK 466, SOWK 467, SOWK 493 and IPE 415, who have an overall minimum GPA of 2.0, and who have completed 50 community service hours related to human services after SOWK 287.

During the field practicum, students spend four days a week for one semester completing a minimum of 472 hours of directed field practice. Students seeking admission to SOWK 481 and SOWK 482 must complete a field placement application and interview with the director of field placement. The field placement director, with the assistance of social work faculty members, will determine the acceptance and placement of students.

See the Social Work Student Handbook or the social work website for the field application, documentation of community service guidelines and guidelines for the placement process. Students must have no grade lower

than a "C" (2.0) in SOWK 481, SOWK 482 and SOWK 494, which is taken concurrently with the field practicum.

Degree Requirements

Degree Requirements	Credit Hours
General Education courses ¹	41
Major requirements	50
Electives	29
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

Major Requirements

The minimum requirement for the B.S.W. degree is completion of the General Education requirements, 44 credit hours of core social work courses, prerequisite courses as specified in course descriptions and at least six credit hours of electives in social work at the 300/400 level, for a total of 120 credit hours. If a grade of "C" (2.0) or above is not achieved the first time a social work course is taken, the student may repeat the course only once. All social work majors are expected to abide by the NASW Code of Ethics. Additionally, all majors participate in social work student outcome assessment measures.

Code	Title	Credits
Required Courses		
SOWK 287	Introduction to Social Work	3
SOWK 288	Human Rights and Social Progress	3
SOWK 305	Social Work Research Methods	3
SOWK 317	Skills for Generalist Social Work	3
SOWK 320	Human Behavior in the Social Environment	3
SOWK 335	Social Policy	3
SOWK 465	Social Work Practice in Mezzo Systems	3
SOWK 466	Social Work Practice in Micro Systems	3
SOWK 467	Social Work Practice in Macro Systems	3
SOWK 481	Social Work Field Practicum I	6
SOWK 482	Social Work Field Practicum II	6
SOWK 493	Professional Forum I: Preparation for Field	1
SOWK 494	Professional Forum II: Transition to Professional Practice	3
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	1
Subtotal		44
Electives		
Choose six credit hours of SOWK electives at the 300/400 level. Elective courses are offered on a rotating basis.		6
SOWK 301	Workshops in Social Work	
SOWK 302	Workshops in Social Work	
SOWK 314	Rural Health: An Interprofessional Approach	
SOWK 325	Parent-Child Relationships Across the Lifespan	
SOWK 327	Substance Use Interventions	
SOWK 332	Community Mental Health Practice	
SOWK 338	Issues and Policies in Family Services	
SOWK 340	Violence in Families	

SOWK 342	Child Welfare Services
SOWK 347	Issues Affecting LGBTQIAP+ Individuals and Their Families
SOWK 362	Social Work with Immigrants and Refugees
SOWK 372	Programs and Services for Older Adults
SOWK 375	Grant Writing for Agencies
SOWK 386	Youth Empowerment Strategies (YES)
SOWK 387	Working with Adolescents
SOWK 450	Palliative and Hospice Care: An Interdisciplinary Approach
SOWK 487	Special Topics
SOWK 490	Special Studies in Social Work (can be taken for a total of 6.00 credits)
Subtotal	6
Total Credits	50

Recommended Schedule for Majors

Course	Title	Credits
First Year		
General Education courses and/or university electives ¹		30
Credits		30
Second Year		
SOWK 287	Introduction to Social Work	3
SOWK 288	Human Rights and Social Progress	3
General Education courses and/or university electives ¹		22
Credits		28
Third Year		
SOWK 305	Social Work Research Methods	3
SOWK 317	Skills for Generalist Social Work	3
SOWK 320	Human Behavior in the Social Environment	3
SOWK 335	Social Policy	3
SOWK elective		3
Electives		15
Credits		30
Fourth Year		
SOWK 465	Social Work Practice in Mezzo Systems	3
SOWK 466	Social Work Practice in Micro Systems	3
SOWK 467	Social Work Practice in Macro Systems	3
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	1
SOWK 493	Professional Forum I: Preparation for Field	1
SOWK elective		3
Electives		3
SOWK 481	Social Work Field Practicum I	6
SOWK 482	Social Work Field Practicum II	6
SOWK 494	Professional Forum II: Transition to Professional Practice	3
Credits		32
Total Credits		120

¹ Certain General Education courses may also meet prerequisite requirements for social work courses. Pay close attention to General Education requirements when selecting the following courses: See AAAD 200, ANTH 195, GEOG 200, MATH 220, POSC 225, PSYC 101, PSYC 160, SOCI 110, SOCI 140 and WGSS 200.

Sociology, B.A.

Program Description

The sociology program develops the student’s ability to analyze the social world by using diverse sociological theories and research methods. It stresses the importance of social, cultural, and historical contexts for understanding relationships between and among individuals and the social structures that make up societies. Students will gain the ability to think broadly and critically about the diversity of social organization and practice locally and globally. Students will also develop the ability to take a disciplined and scholarly approach to understanding the world in which we live.

The major in sociology strongly develops skills in critical thinking, research, analysis, writing and communication. Sociology students frequently go on to post-graduate education and/or careers in both public and private employment sectors.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	39
University electives	25-37
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student’s chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures’ placement test.

Major Requirements

To earn the B.A. degree with a major in sociology, students must complete a minimum of 39 credit hours in sociology. Of these credit hours, 18 are required courses; the remaining 21 credit hours are electives. Students must observe the prerequisite sequencing of required courses as shown in the course descriptions.

Code	Title	Credits
Core Courses		
SOCI 110	Social Issues in a Global Context [C4GE]	3
SOCI 140	Microsociology: The Individual in Society [C5SD]	3
SOCI 200	Development of Sociological Thought and Method	3
SOCI 231	Introduction to Social Statistics ¹	3
SOCI 300	Sociological Inquiry	3
SOCI 480	Senior Seminar in Sociology ^{2,3}	3
Subtotal		18
Sociology Electives		

In addition to the 18 credit hours of required courses, students must complete 21 credit hours of electives. Students are encouraged to choose one of the concentrations listed below for partial fulfillment of their elective requirements.	21
Subtotal	21
Total Credits	39

¹ Students may substitute SOCI 231 with MATH 220, PSYC 210 or COB 191, but must take an additional sociology course to complete the required 39 hours of sociology.

² This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

³ Completing a supervised internship with a substantial writing expectation can fulfill the senior seminar capstone requirement. Students must secure their own internship placement before enrolling in the internship course and secure advanced approval from a faculty supervisor. Completing an independent study with a substantial writing and analysis expectation (arranged in advance with a faculty supervisor), or successful completion of the SOCI 499A/SOCI 499B/SOCI 499C sequence, can also fulfill the senior seminar capstone requirement. Students seeking any of these pathways should consult with an advisor or the program coordinator in advance.

Sociology Concentrations

The sociology program encourages majors to select electives that create a coherent program of study suited to their special needs and interests. Such a focus would involve four or more courses from the following concentration groupings:¹

¹ Within any of the defined concentrations students may gain credits toward completing the concentration through certain special topics or other courses. On occasion, courses taken outside the major or university may qualify. For special topics courses in sociology, see the instructor of record for that course. For other questions or possibilities see an advisor or the sociology program coordinator.

Communities, Inequalities and Public Policy Concentration

Code	Title	Credits
SOCI/GERN 280	Social Gerontology	3
SOCI 303	Sociology of Death and Dying	3
SOCI 304	Social Deviance	3
SOCI 309	Capitalism and Alternative Economies	3
SOCI 311	Sociology of the Environment	3
SOCI 318	Sociology of Immigration	3
SOCI 321	Politics in Society	3
SOCI/REL 322	Sociology of Religion	3
SOCI 328	Digital Inequality	3
SOCI 334	Self and Society	3
SOCI 335	Disability and Society	3
SOCI 336	Race and Ethnicity	3
SOCI/WGSS 337	Sociology of Gender	3
SOCI 341	Sociology of Education	3
SOCI 347	Queer Theories of Gender and Inequality	3
SOCI/ANTH 352	Birth, Death, Sex: Exploring Demography	3

SOCI 353	Energy and Society	3
SOCI 354	Social Inequality	3
SOCI 357	Sociology of Disasters	3
SOCI 359	The Sociology of Climate Denial	3
SOCI 360	Social Movements	3
SOCI 367	Sociology of Sexuality	3
SOCI 372	Sociology of the Community	3
SOCI 374	Sociology of Families	3
SOCI 375	Medical Sociology	3
SOCI 382	Qualitative Sociology	3
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	3
SOCI 386	Sociology of Nonprofit Organizations	3

Environments, Technologies and Innovations Concentration

Code	Title	Credits
SOCI 309	Capitalism and Alternative Economies	3
SOCI 311	Sociology of the Environment	3
SOCI 315	Technology and Society	3
SOCI 328	Digital Inequality	3
SOCI 344	Work and Society	3
SOCI 353	Energy and Society	3
SOCI 354	Social Inequality	3
SOCI 357	Sociology of Disasters	3
SOCI 359	The Sociology of Climate Denial	3
SOCI 360	Social Movements	3
SOCI 361	Sociology of Organizations	3
SOCI 367	Sociology of Sexuality	3
SOCI 375	Medical Sociology	3
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	3
SOCI 386	Sociology of Nonprofit Organizations	3

Health, Medicine and Society Concentration

Code	Title	Credits
SOCI/GERN 280	Social Gerontology	3
SOCI 303	Sociology of Death and Dying	3
SOCI 315	Technology and Society	3
SOCI 317	Sociology of the Body	3
SOCI 328	Digital Inequality	3
SOCI 335	Disability and Society	3
SOCI 336	Race and Ethnicity	3
SOCI/WGSS 337	Sociology of Gender	3
SOCI 347	Queer Theories of Gender and Inequality	3
SOCI/ANTH 352	Birth, Death, Sex: Exploring Demography	3
SOCI 354	Social Inequality	3
SOCI 357	Sociology of Disasters	3
SOCI 361	Sociology of Organizations	3
SOCI/WGSS 367	Sociology of Sexuality	3
SOCI 375	Medical Sociology	3

SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	3
SOCI 386	Sociology of Nonprofit Organizations	3

Markets and Cultures Concentration

Code	Title	Credits
SOCI 306	Japanese Society and Culture	3
SOCI 309	Capitalism and Alternative Economies	3
SOCI 315	Technology and Society	3
SOCI 317	Sociology of the Body	3
SOCI 318	Sociology of Immigration	3
SOCI 328	Digital Inequality	3
SOCI 329	Sociology of Sport	3
SOCI 334	Self and Society	3
SOCI 341	Sociology of Education	3
SOCI 344	Work and Society	3
SOCI 358	Sociology of Consumption	3
SOCI 361	Sociology of Organizations	3
SOCI 363	Sociology of Culture	3
SOCI 367	Sociology of Sexuality	3
SOCI 382	Qualitative Sociology	3
SOCI 386	Sociology of Nonprofit Organizations	3

Political and Global Analysis Concentration

Code	Title	Credits
SOCI 304	Social Deviance	3
SOCI 306	Japanese Society and Culture	3
SOCI 309	Capitalism and Alternative Economies	3
SOCI 318	Sociology of Immigration	3
SOCI 321	Politics in Society	3
SOCI 328	Digital Inequality	3
SOCI 335	Disability and Society	3
SOCI 336	Race and Ethnicity	3
SOCI/WGSS 337	Sociology of Gender	3
SOCI 344	Work and Society	3
SOCI 347	Queer Theories of Gender and Inequality	3
SOCI/ANTH 352	Birth, Death, Sex: Exploring Demography	3
SOCI 353	Energy and Society	3
SOCI 354	Social Inequality	3
SOCI 359	The Sociology of Climate Denial	3
SOCI 360	Social Movements	3
SOCI 361	Sociology of Organizations	3
SOCI 363	Sociology of Culture	3
SOCI 375	Medical Sociology	3
SOCI 382	Qualitative Sociology	3
SOCI 386	Sociology of Nonprofit Organizations	3

Recommended Schedule for Majors

Transfer students on a two-year course of study should change "Year" in this sequence to "Semester."

The following is an example of a four-year course of study for a student seeking a degree in sociology:

Course	Title	Credits
First Year		
SOCI 110	Social Issues in a Global Context [C4GE]	3
SOCI 140	Microsociology: The Individual in Society [C5SD]	3
Sociology electives		3
Credits		9
Second Year		
SOCI 200	Development of Sociological Thought and Method	3
SOCI 231	Introduction to Social Statistics	3
Sociology electives		3
Credits		9
Third Year		
SOCI 300	Sociological Inquiry	3
Electives		6
Credits		9
Fourth Year		
SOCI 480	Senior Seminar in Sociology ¹	3
Sociology electives		9
Credits		12
Total Credits		39

¹ Majors take this any time after completing SOCI 300

Sociology, B.S.

Program Description

The sociology program develops the student’s ability to analyze the social world by using diverse sociological theories and research methods. It stresses the importance of social, cultural, and historical contexts for understanding relationships between and among individuals and the social structures that make up societies. Students will gain the ability to think broadly and critically about the diversity of social organization and practice locally and globally. Students will also develop the ability to take a disciplined and scholarly approach to understanding the world in which we live.

The major in sociology strongly develops skills in critical thinking, research, analysis, writing and communication. Sociology students frequently go on to post-graduate education and/or careers in both public and private employment sectors.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific Literacy requirement ²	3-4
Major requirements	39
University electives	32-33
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these courses may double count with major requirements.

² In addition to course work taken to fulfill General Education requirement.

Major Requirements

To earn the B.S. degree with a major in sociology, students must complete a minimum of 39 credit hours in sociology. Of these credit hours, 18 are required courses; the remaining 21 credit hours are electives. Students must observe the prerequisite sequencing of required courses as shown in the course descriptions.

Code	Title	Credits
Core Courses		
SOCI 110	Social Issues in a Global Context [C4GE]	3
SOCI 140	Microsociology: The Individual in Society [C5SD]	3
SOCI 200	Development of Sociological Thought and Method	3
SOCI 231	Introduction to Social Statistics ¹	3
SOCI 300	Sociological Inquiry	3
SOCI 480	Senior Seminar in Sociology ^{2,3}	3
Subtotal		18
Sociology Electives		
In addition to the 18 credit hours of required courses, students must complete 21 credit hours of electives. Students are encouraged to choose one of the concentrations listed below for partial fulfillment of their elective requirements.		21

Subtotal	21
Total Credits	39

¹ Students can substitute SOCI 231 with MATH 220, PSYC 210 or COB 191, but must take an additional sociology course to complete the required 39 hours of sociology.

² This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

³ Completing a supervised internship with a substantial writing expectation can fulfill the senior seminar capstone requirement. Students must secure their own internship placement before enrolling in the internship course and secure advanced approval from a faculty supervisor. Completing an independent study with a substantial writing and analysis expectation (arranged in advance with a faculty supervisor), or successful completion of the SOCI 499A/SOCI 499B/SOCI 499C sequence, can also fulfill the senior seminar capstone requirement. Students seeking any of these pathways should consult with an advisor or the program coordinator in advance.

Sociology Concentrations

The sociology program encourages majors to select electives that create a coherent program of study suited to their special needs and interests. Such a focus would involve four or more courses from the following concentration groupings:¹

¹ Within any of the defined concentrations students may gain credits toward completing the concentration through certain special topics or other courses. On occasion, courses taken outside the major or university may qualify. For special topics courses in sociology, see the instructor of record for that course. For other questions or possibilities see an advisor or the sociology program coordinator.

Communities, Inequalities and Public Policy Concentration

Code	Title	Credits
SOCI/GERN 280	Social Gerontology	3
SOCI 303	Sociology of Death and Dying	3
SOCI 304	Social Deviance	3
SOCI 309	Capitalism and Alternative Economies	3
SOCI 311	Sociology of the Environment	3
SOCI 318	Sociology of Immigration	3
SOCI 321	Politics in Society	3
SOCI/REL 322	Sociology of Religion	3
SOCI 328	Digital Inequality	3
SOCI 334	Self and Society	3
SOCI 335	Disability and Society	3
SOCI 336	Race and Ethnicity	3
SOCI/WGSS 337	Sociology of Gender	3
SOCI 341	Sociology of Education	3
SOCI 347	Queer Theories of Gender and Inequality	3
SOCI/ANTH 352	Birth, Death, Sex: Exploring Demography	3
SOCI 353	Energy and Society	3
SOCI 354	Social Inequality	3
SOCI 357	Sociology of Disasters	3
SOCI 359	The Sociology of Climate Denial	3

SOCI 360	Social Movements	3
SOCI 367	Sociology of Sexuality	3
SOCI 372	Sociology of the Community	3
SOCI 374	Sociology of Families	3
SOCI 375	Medical Sociology	3
SOCI 382	Qualitative Sociology	3
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	3
SOCI 386	Sociology of Nonprofit Organizations	3

Environments, Technologies and Innovations Concentration

Code	Title	Credits
SOCI 309	Capitalism and Alternative Economies	3
SOCI 311	Sociology of the Environment	3
SOCI 315	Technology and Society	3
SOCI 328	Digital Inequality	3
SOCI 344	Work and Society	3
SOCI 353	Energy and Society	3
SOCI 354	Social Inequality	3
SOCI 357	Sociology of Disasters	3
SOCI 359	The Sociology of Climate Denial	3
SOCI 360	Social Movements	3
SOCI 361	Sociology of Organizations	3
SOCI 367	Sociology of Sexuality	3
SOCI 375	Medical Sociology	3
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	3
SOCI 386	Sociology of Nonprofit Organizations	3

Health, Medicine and Society Concentration

Code	Title	Credits
SOCI/GERN 280	Social Gerontology	3
SOCI 303	Sociology of Death and Dying	3
SOCI 315	Technology and Society	3
SOCI 317	Sociology of the Body	3
SOCI 328	Digital Inequality	3
SOCI 335	Disability and Society	3
SOCI 336	Race and Ethnicity	3
SOCI/WGSS 337	Sociology of Gender	3
SOCI 347	Queer Theories of Gender and Inequality	3
SOCI/ANTH 352	Birth, Death, Sex: Exploring Demography	3
SOCI 354	Social Inequality	3
SOCI 357	Sociology of Disasters	3
SOCI 361	Sociology of Organizations	3
SOCI/WGSS 367	Sociology of Sexuality	3
SOCI 375	Medical Sociology	3
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	3
SOCI 386	Sociology of Nonprofit Organizations	3

Markets and Cultures Concentration

Code	Title	Credits
SOCI 306	Japanese Society and Culture	3
SOCI 309	Capitalism and Alternative Economies	3
SOCI 315	Technology and Society	3
SOCI 317	Sociology of the Body	3
SOCI 318	Sociology of Immigration	3
SOCI 328	Digital Inequality	3
SOCI 329	Sociology of Sport	3
SOCI 334	Self and Society	3
SOCI 341	Sociology of Education	3
SOCI 344	Work and Society	3
SOCI 358	Sociology of Consumption	3
SOCI 361	Sociology of Organizations	3
SOCI 363	Sociology of Culture	3
SOCI 367	Sociology of Sexuality	3
SOCI 382	Qualitative Sociology	3
SOCI 386	Sociology of Nonprofit Organizations	3

Political and Global Analysis Concentration

Code	Title	Credits
SOCI 304	Social Deviance	3
SOCI 306	Japanese Society and Culture	3
SOCI 309	Capitalism and Alternative Economies	3
SOCI 318	Sociology of Immigration	3
SOCI 321	Politics in Society	3
SOCI 328	Digital Inequality	3
SOCI 335	Disability and Society	3
SOCI 336	Race and Ethnicity	3
SOCI/WGSS 337	Sociology of Gender	3
SOCI 344	Work and Society	3
SOCI 347	Queer Theories of Gender and Inequality	3
SOCI/ANTH 352	Birth, Death, Sex: Exploring Demography	3
SOCI 353	Energy and Society	3
SOCI 354	Social Inequality	3
SOCI 359	The Sociology of Climate Denial	3
SOCI 360	Social Movements	3
SOCI 361	Sociology of Organizations	3
SOCI 363	Sociology of Culture	3
SOCI 375	Medical Sociology	3
SOCI 382	Qualitative Sociology	3
SOCI 386	Sociology of Nonprofit Organizations	3

Recommended Schedule for Majors

Transfer students on a two-year course of study should change "Year" in this sequence to "Semester."

The following is an example of a four-year course of study for a student seeking a degree in sociology:

Course	Title	Credits
First Year		
SOCI 110	Social Issues in a Global Context [C4GE]	3
SOCI 140	Microsociology: The Individual in Society [C5SD]	3

Sociology elective		3
Credits		9
Second Year		
SOCI 200	Development of Sociological Thought and Method	3
SOCI 231	Introduction to Social Statistics	3
Sociology elective		3
Credits		9
Third Year		
SOCI 300	Sociological Inquiry	3
Electives		6
Credits		9
Fourth Year		
SOCI 480	Senior Seminar in Sociology ¹	3
Sociology electives		9
Credits		12
Total Credits		39

¹ Majors take this any time after completing SOCI 300

Sport and Recreation Management, B.S.

Program Description

The mission of the sport and recreation program is to develop a community of learners through quality education that integrates theory, practice, and personal growth in the hospitality, sport and recreation industries.

We are dedicated to the development of future leaders in sport and management professions through a course of study that maximizes the potential of individuals and society.

Sport and recreation management is committed to providing:

- An outstanding undergraduate program based on the criteria of relevant professional associations, which will enable graduates to be successful in their professional endeavors.
- A program that builds upon the strong liberal studies background provided through General Education.
- The opportunities that challenge students to think critically, to use technology and to appreciate the global community.
- The research and development projects that push back the boundaries of knowledge and promote effective practice in sport and recreation management.

Practicums and Internships

SRM requires its students to complete a practicum and an internship. Both count as academic credit hours. These are opportunities for the student to gain valuable experience by working in his/her field of choice in a sport or recreation-related agency. The opportunities are almost endless.

While enrolled in SRM 482, students may only take one additional course (three or four credit hours). The additional course must be approved in advance by the site supervisor and the director of SRM.

Declaring or Changing Major into Sport and Recreation Management

If you are a current JMU student wishing to declare or change a major you must complete a request through MyMadison. After you complete the request for changing majors into Sport and Recreation Management, you are required to contact a member of the Hart School Advisory Team (Scott Arbogast, arbogasw@jmu.edu; Jason Gochenour, gochenjl@jmu.edu) to complete the declaration process. Visit the Hart School's advising site (<https://www.jmu.edu/hartschool/caps/advising/index.shtml>) to schedule an appointment with either Scott or Jason.

The SRM major within the Hart School has cutoff dates. Students are only admitted twice a year, October 1 and March 1. Any student wishing to take Hart School courses in the spring semester or any summer term session would be required to contact the Hart School advisory team and declare the change of major on or before October 1. Any student wishing to take Hart School courses in the Fall semester would be required to contact the Hart School advisory team and declare the change of major on or before March 1.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement (in addition to General Education)	3
Scientific literacy requirement (in addition to General Education)	3-4
Major requirements	45
General Business Minor or Entrepreneurship Minor ²	18
University Electives	9-10
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² Successful completion of these courses with a 2.0 GPA will qualify the student for a general business minor; however, it is the responsibility of the student to complete the necessary paperwork in the College of Business (p. 15) to apply for the minor.

Major Requirements

Code	Title	Credits
Required Courses		
SRM 202	Foundations of Leadership in Sport and Recreation Management	3
SRM 240	Sport and Recreation Management Seminar	3
SRM 241	Introduction to Sport and Recreation Management	3
SRM 242	Sociology and Psychology of Sport and Recreation	3
SRM 282	Practicum in Sport and Recreation	3
SRM 333	Management in Sport and Recreation	3
SRM 337	Programming and Assessment in Sport and Recreation Management	3
SRM 434	Ethical and Legal Issues in Sport and Recreation Management	3
SRM 435	Sport Marketing and Sales	3
SRM 436	Facilities and Event Management	3
SRM 482	Internship in Sport and Recreation Management	6
Subtotal		36
Elective Courses		
Choose any 9 credits from the following:		9
HM 210	Tourism in a Global Context	
HM 316	Country Club Management	
Any SRM course that is not required in the SRM major		
Any course in the Coaching Education minor (registration is not guaranteed for students who are not declared coaching education minors)		
Subtotal		9
Total Credits		45

Required Minor: 18 Credit Hours

All majors must declare either the General Business Minor (p. 498) or the Entrepreneurship Minor (p. 484) through the College of Business (p. 15).

Recommended Schedule for Majors

Course	Title	Credits
First Year		
General Education		15
Madison Foundations		9
SRM 202	Foundations of Leadership in Sport and Recreation Management	3
SRM 240	Sport and Recreation Management Seminar	3
Credits		30
Second Year		
General Education and/or University Elective(s)		15
SRM 241	Introduction to Sport and Recreation Management	3
SRM 242	Sociology and Psychology of Sport and Recreation	3
SRM 282	Practicum in Sport and Recreation	3
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors	3
ECON 201	Introduction to Microeconomics	3
Credits		30
Third Year		
General Education and University Elective(s)		12
SRM 333	Management in Sport and Recreation	3
SRM 337	Programming and Assessment in Sport and Recreation Management	3
Quantitative Requirement		3
Required Major Elective		3
Selection of two (2) minor courses		6
Credits		30
Fourth Year		
University Elective(s) or Scientific Literacy Requirement		3
SRM 434	Ethical and Legal Issues in Sport and Recreation Management	3
SRM 435	Sport Marketing and Sales	3
SRM 436	Facilities and Event Management	3
SRM 482	Internship in Sport and Recreation Management	6
Select two (2) minor courses		6
Required Major Electives		6
Credits		30
Total Credits		120

Statistics, B.S.

Program Description

The Department of Mathematics and Statistics offers a major in statistics to meet the needs of both the public and the private sectors for graduates with degrees in statistics. Students may choose either a standard statistics major or a statistics major with a data science concentration. The required courses provide a balance of applications and theory, which allows students to prepare for immediate employment or graduate studies by proper selection of the program electives.

Degree Requirements

Degree Requirements	Credit Hours
General Education ^{1,2}	41
Quantitative requirement ²	3
Scientific Literacy requirement (in addition to General Education)	3-4
Major requirements (listed below)	44-46
University electives	29-32
Total	120

¹ MATH 231 or MATH 235 must be included.
² The General Education program contains a set of requirements each student must fulfill. Some requirements may be fulfilled by courses within the major. The quantitative requirement is also met by courses within the major and not included in the 120 credit hour total.

Major Requirements

All majors must complete 22 credits of core courses and an additional 22-24 required credits for the B.S. in Statistics. These additional credit hours may be fulfilled by completing a concentration in data science. Students interested in pursuing graduate studies in statistics are strongly encouraged to double-major in statistics and mathematics.

Code	Title	Credits
Core Courses		
MATH 229	Introduction to Applied Statistics Using R [C3QR] ¹	3
MATH 236	Calculus II	4
MATH 322	Applied Linear Regression	3
MATH 327	Categorical Data Analysis	3
MATH 329	Introduction to Probability and Theoretical Statistics	3
MATH 421	Applied Multivariate Statistical Analysis	3
MATH 426	Probability and Mathematical Statistics I	3
Subtotal		22
Additional Requirements		
Choose one of the following options:		22-24
Additional Required Courses and Restricted Electives		
Data Science Concentration		
Subtotal		22-24
Total Credits		44-46

¹ MATH 220 with a minimum grade of B- may be substituted for MATH 229 if the student received credit for MATH 220 before declaring a statistics major.

Additional Requirements

Students who do not elect to complete a concentration in data science must take the following 22-24 credit hours.

Code	Title	Credits
Additional Required Courses		
MATH 237	Calculus III	4
MATH 321	Analysis of Variance and Experimental Design	3
MATH 428	Statistical Consulting	3
MATH 238	Linear Algebra with Differential Equations	3-4
or MATH 300	Linear Algebra	
MATH 248	Computers and Numerical Algorithms	3-4
or MATH 309	SAS Programming and Data Management	
Subtotal		16-18
Restricted Electives		
Choose six credit hours from the following:		6
MATH 268	Data Analysis and Visualization	
MATH 324	Applied Nonparametric Statistics	
MATH 325	Survey Sampling Methods	
MATH 326	Statistical Quality Control	
MATH 328	Time Series Analysis	
MATH 354	Introduction to Biometrics	
MATH 358	Introduction to Machine Learning	
MATH 368	Statistical Computing	
MATH 410	Advanced Calculus I	
MATH 411	Advanced Calculus II ¹	
MATH 423	Stochastic Processes	
MATH 424	Statistical Decision Theory	
MATH 427	Probability and Mathematical Statistics II	
MATH 429	Research Project in Statistics	
Subtotal		6
Total Credits		22-24

¹ MATH 411 is recommended for students planning to do graduate work in statistics.

Data Science Concentration

Students who declare the concentration in data science must take the following 24 credit hours in addition to the core courses.

Code	Title	Credits
Additional Required Courses		
DATA 200	Introduction to Data	3
MATH 250	Programming for Data Science	3
MATH 251	Database Queries	3
MATH 268	Data Analysis and Visualization	3
MATH 300	Linear Algebra	3
MATH 358	Introduction to Machine Learning	3
Subtotal		18

Electives

Choose six credit hours from the following:		6
MATH 321	Analysis of Variance and Experimental Design	
MATH 324	Applied Nonparametric Statistics	
MATH 325	Survey Sampling Methods	
MATH 326	Statistical Quality Control	
MATH 328	Time Series Analysis	
MATH 340	Operations Research I	
MATH 354	Introduction to Biometrics	
MATH 368	Statistical Computing	
MATH 410	Advanced Calculus I	
MATH 411	Advanced Calculus II ¹	
MATH 423	Stochastic Processes	
MATH 424	Statistical Decision Theory	
MATH 427	Probability and Mathematical Statistics II	
MATH 429	Research Project in Statistics	
MATH 434	Advanced Linear Algebra	
Subtotal		6
Total Credits		24

¹ MATH 411 is recommended for students planning to do graduate work in statistics.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
Madison Foundations		9
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 236	Calculus II	4
MATH 309	SAS Programming and Data Management ¹	3
General Education requirement		11
Credits		30
Second Year		
MATH 237	Calculus III	4
MATH 321	Analysis of Variance and Experimental Design	3
MATH 322	Applied Linear Regression	3
MATH 329	Introduction to Probability and Theoretical Statistics	3
General Education requirement / Electives		17
Credits		30
Third Year		
MATH 300	Linear Algebra ²	3
MATH 327	Categorical Data Analysis	3
General Education requirement / Electives		24
Credits		30
Fourth Year		
MATH 421	Applied Multivariate Statistical Analysis	3
MATH 426	Probability and Mathematical Statistics I	3
MATH 428	Statistical Consulting	3
Electives		21
Credits		30
Total Credits		120

¹ MATH 248. Computers and Numerical Algorithms may be substituted.

² MATH 238. Linear Algebra with Differential Equations may be substituted.

Theatre, B.A.

Program Description

The Bachelor of Arts in Theatre combines a liberal arts education with professionally-focused training.

We believe that theatre artists are makers, creators, devisers, designers, writers, leaders, activists and entertainers. So, at the heart of our program is the belief that every student should be given the training and resources they need to realize a unique vision for their life, education, and creative work.

Our major is the place to train intensively as an actor; join an entirely student-produced, designed, and directed Studio Theatre; write, devise, and produce your own work; train as a designer using state-of-the-art technology; and develop a scholarly understanding of culture and art. Students are admitted to the major based on an audition or interview, or what we call our "gateway" process; and graduates enjoy an excellent record of placement in jobs at professional theatres on Broadway and off-Broadway, in film and television, and as teachers, theatre makers, and leaders in the arts.

Special Admission and Retention Requirements

Admission to the dance, theatre and musical theatre majors are competitive and an audition/interview is required. Please see the School of Theatre and Dance website (<https://www.jmu.edu/theatredance/apply/index.shtml>) for more information.

All School of Theatre and Dance majors must complete the requirements for a Bachelor of Arts degree and must complete the General Education program. Theatre/Musical Theatre majors cannot take THEA 210 to meet the visual and performing arts requirements. Dance majors cannot take DANC 215 to meet the visual and performing arts requirement. Courses from other academic units may not be counted both for the School of Theatre and Dance majors and for another major.

A minimum grade of "C" is required in all courses that apply to the Theatre, B.A.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Theatre major requirements ³	45-48
University electives	16-31
Total	120

¹ Some General Education courses may be fulfilled by requirements within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

³ Includes nine credit hours of theatre electives or 9-15 credit hours of concentration-specific courses.

Major Requirements

All students pursuing a B.A. degree in theatre must complete the core requirements.

Code	Title	Credits
Core Courses		
STAD 171	Performance Production	3
STAD 273	Design Aspects of Performance	3
Dance Requirement		
Choose one of the following:		2
DANC 140	Elementary Modern Dance (0,4)	
DANC 142	Elementary Ballet (0, 4)	
DANC 143	International Folk Dance (0, 4)	
DANC 144	Ballroom Dance (0, 4)	
DANC 146	Jazz Dance (0, 4)	
DANC 147	Tap Dance (0, 4)	
DANC 245	Dance Improvisation (0, 4)	
Subtotal		8
Theatre and Performance		
THEA 211	Performance Analysis	3
THEA 251	Acting I: Basic Acting	3
THEA 315	World Theatre Histories, Texts and Performances I	3
THEA 316	World Theatre Histories, Texts and Performances II	3
THEA 317	World Theatre Histories, Texts and Performances III	3
Subtotal		15
Theatre Practica		
Students may apply 4 credit hours from the following courses towards the major requirements. Students may only apply one credit hour from any given area (scenery, lights, costumes, management, performance and sound) towards the major requirements. Students may only enroll in one 200-level and one 300-level course each semester.		4
THEA 204	Theatre Practicum: Scenery [Block] or THEA 304 Theatre Practicum: Scenery	
THEA 205	Theatre Practicum: Lights [Block] or THEA 305 Theatre Practicum: Lights	
THEA 206	Theatre Practicum: Costumes [Block] or THEA 306 Theatre Practicum: Costumes	
THEA 207	Theatre Practicum: Management [Block] or THEA 307 Theatre Practicum: Management	
THEA 208	Theatre Practicum: Performance [Block] or THEA 308 Theatre Practicum: Performance	
THEA 212	Theatre Practicum: Sound [Block] or THEA 312 Theatre Practicum: Sound	
THEA 309	Theatre Education Practicum	
Subtotal		4
Professional Requirement		
Choose one of the following:		3
THEA 325	Professional Issues in Education	

THEA 330	Professional Issues for Designers and Technicians	
THEA 441	Professional Pathways	
THEA 442 & THEA 499	Senior Seminar in Theatre and Honors in Theatre	
THEA 460	Auditioning and Professional Issues	
Subtotal		3
Performance Requirement		
Choose one of the following:		3
THEA 351	Acting II: Intermediate Acting	
THEA 355	Directing for the Theatre ¹	
Subtotal		3
Design/Technology Requirement		
Choose one of the following:		3
STAD 471	Stage Management	
THEA 271	Technical Theatre	
THEA 331	Technical Costuming	
THEA 332	Costume and Fashion History	
THEA 333	Costume Design	
THEA 374	Stage Lighting	
THEA 375	Sound Design	
THEA 376	Scene Design	
Subtotal		3
Electives or Concentration		
Students select 9 credits of THEA electives not already applied to the major, or choose one of the following concentrations:		
Design/Technology Concentration		
New Work Concentration		
Performance Concentration		
Theatre Education and Teaching Artistry Concentration		
Subtotal		9-12
Total Credits		45-48

¹ Theatre Education students must take THEA 355 to fulfill VA licensure requirements.

Concentrations

Theatre majors may fulfill the requirements for one or more concentrations or complete nine THEA credits not already applied to the major. Students may be admitted into multiple concentrations or may be admitted into the theatre major, but not admitted into a specific concentration. Concentrations are not required. Interviews and/or auditions into concentrations can happen at the same time as interviews and/or auditions into the theatre major. Alternatively, incoming students may be admitted as un-concentrated theatre majors and audition/interview for a concentration later in their first year or in a subsequent year.

Design/Technology Concentration

Code	Title	Credits
Choose two of the following not already applied to the major:		
THEA 333	Costume Design	
THEA 374	Stage Lighting	
THEA 375	Sound Design	
THEA 376	Scene Design	

Choose one of the following not already applied to the major:		3
THEA 271	Technical Theatre	
THEA 331	Technical Costuming	
THEA 332	Costume and Fashion History	
THEA 336	History, Theory and Practice of Stage Makeup	
THEA 473	Advanced Design	
THEA 371	Advanced Technical Theatre	
STAD 471	Stage Management	
THEA 390	Directed Projects ¹	
or THEA 490 Special Studies in Theatre		
Total Credits		9

¹ No more than four hours of practicums (THEA 390, THEA 490, THEA 499) may be applied to the theatre major, and no more than 10 hours of special studies (including practicums) may be applied to the theatre major.

New Work Concentration

Code	Title	Credits
Required Course		
THEA 347	Playwriting	3
Choose two of the following:		6
THEA 303	Topics in Theatre	
THEA 355	Directing for the Theatre	
THEA 365	Dramaturgy	
THEA 447	Advanced Playwriting	
THEA 488	Alternative Performance	
Total Credits		9

Performance Concentration

Code	Title	Credits
THEA 296	Studio Production for Performance Concentrators	1
or THEA 396	Studio Directing, Design and Collaboration for Performance Concentrators	
THEA 452	Acting III: Contemporary Scene Study	3
THEA 453	Acting IV: Approach to Heightened Language	3
THEA 460	Auditioning and Professional Issues	3
Total Credits		10

Theatre Education and Teaching Artistry Concentration

This concentration requires 12 THEA/STAD credits. Students who complete the theatre education and teaching artistry concentration are able to apply for the PreK-12 teaching licensure in theatre program; however, they will need to complete 26 additional credits within the Theatre Teaching Licensure (p. 591) program to be eligible for licensure.

Refer to the Theatre Teaching Licensure (p. 591) page for licensure requirements.

Code	Title	Credits
STAD 322	Equity, Inclusion and Access in Education and Performance	3
THEA 319	Theatre Teaching Methods	3
THEA 323	Youth Theatre	3

THEA 325	Professional Issues in Education	3
Total Credits		12

Recommended Schedule for Majors

Course	Title	Credits
First Year		
STAD 171	Performance Production	3
THEA 211	Performance Analysis	3
THEA 251	Acting I: Basic Acting	3
Practicum (Students will take 1 credit of practicum this year) ¹		1
Dance Requirement		2
General Education (beginning with Madison Foundations) and B.A. language requirements		18
Credits		30
Second Year		
STAD 273	Design Aspects of Performance	3
THEA 315	World Theatre Histories, Texts and Performances I	3
THEA 316	World Theatre Histories, Texts and Performances II	3
THEA 355	Directing for the Theatre	3
Practicum (Students will take 2 credits of practicum this year) ¹		2
Choose one Design / Technology Course		3
General Education and/or B.A. Philosophy/Language Requirement		12
Credits		29
Third Year		
THEA 317	World Theatre Histories, Texts and Performances III	3
Theatre Electives		6
General Education, Concentration or University Electives		21
Credits		30
Fourth Year		
Choose one of the following:		3
THEA 325	Professional Issues in Education	
THEA 330	Professional Issues for Designers and Technicians	
THEA 441	Professional Pathways	
THEA 442 & THEA 499	Senior Seminar in Theatre and Honors in Theatre	
THEA 460	Auditioning and Professional Issues	
Practicum (Students will take 1 credit of practicum this year) ¹		1
Theatre Major Requirements		9
Electives or General Education		18
Credits		31
Total Credits		120

¹ THEA 204 - THEA 308

Theatre, Dance Major, B.A.

Program Description

The School of Theatre and Dance (p. 103) offers a major in dance with a rigorous curriculum that prepares students for a diverse range of career paths. We work with each student to help define and shape their individual path, and endeavor to create a positive and collaborative environment that is inclusive and responsive to the needs and goals of the 21st-century artist. The intensive curriculum prepares dancers to create, perform, produce, analyze and teach the art of dance.

Special Admission and Retention Requirements

Admission to the dance, theatre and musical theatre majors are competitive and an audition/interview is required. Please see the School of Theatre and Dance website (<https://www.jmu.edu/theatredance/apply/index.shtml>) for more information.

All School of Theatre and Dance majors must complete the requirements for a Bachelor of Arts degree and must complete the General Education program. Theatre/Musical Theatre majors cannot take THEA 210 to meet the visual and performing arts requirements. Dance majors cannot take DANC 215 to meet the visual and performing arts requirement. Courses from other academic units may not be counted both for the School of Theatre and Dance majors and for another major.

A minimum grade of "C" is required in all courses that apply to Theatre, Dance Major, B.A.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements (listed below)	48-49
University electives	15-28
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

Code	Title	Credits
Core Courses		
STAD 171	Performance Production	3
STAD 273	Design Aspects of Performance	3
Choose one of the following:		2-3
DANC 143	International Folk Dance (0, 4) ¹	

DANC 303	Topics in Dance	
DANC 325	Dance in Community	
DANC 390	Directed Projects	
THEA 251	Acting I: Basic Acting	
Subtotal		8-9

Dance Courses		
DANC 245	Dance Improvisation (0, 4)	2
DANC 248	History of Dance: Renaissance Through the 20th Century	3
DANC 320	Anatomy and Somatic Studies for the Dancer	3
DANC 345	Dance Composition I (2, 2)	3
DANC 445	Dance Composition II (2, 2)	3
DANC 449	The Dance Professional	3
DANC 479	Methods of Teaching Dance	3
Subtotal		20

Technique Courses		
The student is required to complete 10 credits in modern and four credits in ballet technique, with placement to be assessed in conference with dance faculty. Modern, ballet and jazz technique classes all have an advanced level course in the curriculum. The requirements listed below are the minimum level expected of dance major students. Most majors will take additional technique courses as university electives throughout their study in the dance program.		
DANC 240	Intermediate Modern Dance I (0, 4) (Two semesters for a total of four credits)	4
DANC 340	Intermediate Modern Dance II (0, 4) (Two semesters for a total of four credits)	4
DANC 440	Advanced Modern Dance (0, 4) (One semester for a total of two credits)	2
Choose four credits from the following:		4

DANC 242	Intermediate Ballet I (0, 4)	
DANC 342	Intermediate Ballet II (0, 4)	
Subtotal		14

Ensemble Courses		
DANC 110	Associate Group Dance Repertory I (0, 4)	1
DANC 210	Associate Group Dance Repertory II (0, 4)	1
DANC 211	Contemporary Dance Ensemble Repertory I (0, 4)	2
DANC 311	Contemporary Dance Ensemble Repertory II (0, 4)	2
Subtotal		6

Total Credits	48-49
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¹ DANC 143 fulfills the folk dance requirement for students pursuing dance teacher licensure.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
DANC 110	Associate Group Dance Repertory I (0, 4) (fall)	1
DANC 210	Associate Group Dance Repertory II (0, 4) (spring)	1
DANC 240	Intermediate Modern Dance I (0, 4) (fall and spring, total of 4 credit hours)	4
DANC 242	Intermediate Ballet I (0, 4)	2
DANC 245	Dance Improvisation (0, 4)	2
STAD 171	Performance Production	3
Madison Foundations (three courses)		9

Language courses for B.A. requirement (2 courses required)		6
Credits		28
Second Year		
DANC 143 or DANC 390 or DANC 325	International Folk Dance (0, 4) or Directed Projects or Dance in Community	2
DANC 211	Contemporary Dance Ensemble Repertory I (0, 4) (fall)	2
DANC 248	History of Dance: Renaissance Through the 20th Century	3
DANC 340	Intermediate Modern Dance II (0, 4) (fall and spring, total of 4 credit hours)	4
DANC 342	Intermediate Ballet II (0, 4)	2
DANC 345	Dance Composition I (2, 2)	3
Language courses for B.A. requirement (2 courses required)		6
University or dance electives		7
Credits		29
Third Year		
DANC 311	Contemporary Dance Ensemble Repertory II (0, 4)	2
DANC 320	Anatomy and Somatic Studies for the Dancer	3
DANC 440	Advanced Modern Dance (0, 4)	2
DANC 442	Advanced Ballet (0, 4) (optional)	2
DANC 445	Dance Composition II (2, 2)	3
DANC 479	Methods of Teaching Dance	3
General Education and / or degree requirements		12
University or dance electives		4
Credits		31
Fourth Year		
DANC 449	The Dance Professional	3
University or dance electives		6
General Education		23
Credits		32
Total Credits		120

Theatre, Musical Theatre Major, B.A.

Program Description

The Musical Theatre major provides professionally focused training within the context of a broader university experience. The intensive, hands-on curriculum affords students the opportunity to develop into well-rounded creative artists who are prepared to succeed in the professional arena. We endeavor to create a positive and collaborative environment that is inclusive as well as responsive to the needs and goals of the 21st-century artist.

Special Admission and Retention Requirements

Admission to the dance, theatre and musical theatre majors are competitive and an audition/interview is required. Please see the School of Theatre and Dance website (<https://www.jmu.edu/theatredance/apply/index.shtml>) for more information.

All School of Theatre and Dance majors must complete the requirements for a Bachelor of Arts degree and must complete the General Education program. Theatre/Musical Theatre majors cannot take THEA 210 to meet the visual and performing arts requirements. Dance majors cannot take DANC 215 to meet the visual and performing arts requirement. Courses from other academic units may not be counted both for the School of Theatre and Dance majors and for another major.

A minimum grade of "C" is required in all courses that apply to the Theatre, Musical Theatre Major, B.A.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign Language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements (listed below)	55
University electives	9-21
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some credits may be fulfilled by courses required within the major.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level of the student's chosen language (typically 232) or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

All students pursuing a musical theatre major must complete the core requirements listed below.

Code	Title	Credits
Core Courses		
STAD 171	Performance Production	3
STAD 273	Design Aspects of Performance	3

Choose one of the following:		2
DANC 140	Elementary Modern Dance (0,4)	
DANC 142	Elementary Ballet (0, 4)	
DANC 143	International Folk Dance (0, 4)	
DANC 144	Ballroom Dance (0, 4)	
DANC 147	Tap Dance (0, 4)	
DANC 245	Dance Improvisation (0, 4)	
Subtotal		8
Program Requirements		
THEA 208	Theatre Practicum: Performance [Block]	1
or THEA 308	Theatre Practicum: Performance	
THEA 211	Performance Analysis	3
THEA 251	Acting I: Basic Acting	3
THEA 253	Musical Theatre Laboratory ¹	2
THEA 316	World Theatre Histories, Texts and Performances II	3
THEA 351	Acting II: Intermediate Acting	3
THEA 353	Musical Theatre Performance	3
THEA 357	Music Theatre History and Analysis	3
or MUS 357	Music Theatre History and Analysis	
THEA 454	Advanced Musical Theatre Performance	3
THEA 455	Auditioning for Musical Theatre	3
DANC 146	Jazz Dance (0, 4)	2
DANC 246	Intermediate Jazz	2
DANC 346	Intermediate Jazz II/Musical Theatre Styles	2
MUAP 114	Group Voice for Musical Theatre Concentrators ¹	2
MUAP 214	Private Voice for Musical Theatre Concentrators ²	4
MUS 100	Keyboard Skills	1
MUS 140	Introduction to Music Theory and Aural Skills	4
Subtotal		44
Theatre Practica		
Students may apply 3 credit hours from the following courses towards the major requirements. Students may only apply one credit hour from any given area (scenery, lights, costumes, management and sound) towards the major requirements. Students may only enroll in one 200-level and one 300-level course each semester.		3
THEA 204	Theatre Practicum: Scenery [Block]	
or THEA 304	Theatre Practicum: Scenery	
THEA 205	Theatre Practicum: Lights [Block]	
or THEA 305	Theatre Practicum: Lights	
THEA 206	Theatre Practicum: Costumes [Block]	
or THEA 306	Theatre Practicum: Costumes	
THEA 207	Theatre Practicum: Management [Block]	
or THEA 307	Theatre Practicum: Management	
THEA 212	Theatre Practicum: Sound [Block]	
or THEA 312	Theatre Practicum: Sound	
Subtotal		3
Total Credits		55

¹ Two enrollments of one credit each.

² Four enrollments of one credit each.

Recommended Schedule for Majors

Course	Title	Credits
First Year		
STAD 171	Performance Production	3
THEA 211	Performance Analysis	3
THEA 251	Acting I: Basic Acting	3
THEA 253	Musical Theatre Laboratory (Two enrollments of 1 credit each)	2
DANC 146	Jazz Dance (0, 4)	2
MUAP 114	Group Voice for Musical Theatre Concentrators (Two enrollments of 1 credit each)	2
Dance Core		2
General Education courses, university electives and/or major requirements		13
Credits		30
Second Year		
STAD 273	Design Aspects of Performance	3
THEA 351	Acting II: Intermediate Acting	3
THEA 353	Musical Theatre Performance	3
THEA 357	Music Theatre History and Analysis	3
DANC 246	Intermediate Jazz	2
MUS 100	Keyboard Skills	1
MUAP 214	Private Voice for Musical Theatre Concentrators (Two enrollments of 1 credit each)	2
Practicum		1
General Education courses, university electives and/or major requirements		12
Credits		30
Third Year		
THEA 316	World Theatre Histories, Texts and Performances II	3
THEA 455	Auditioning for Musical Theatre	3
MUS 140	Introduction to Music Theory and Aural Skills	4
MUAP 214	Private Voice for Musical Theatre Concentrators (Two enrollments of 1 credit each)	2
Practicum		3
General Education courses, university electives and/or major requirements		15
Credits		30
Fourth Year		
THEA 454	Advanced Musical Theatre Performance	3
General Education courses, university electives and/or major requirements		27
Credits		30
Total Credits		120

Writing, Rhetoric and Technical Communication, B.A.

Program Description

The Writing, Rhetoric and Technical Communication major emphasizes writing in professional contexts such as

- writing for professional communities such as government, medicine, law, science and environment, and education
- proposal and grant writing
- editing
- online inclusive design
- usability and user experience research
- publications management
- content strategy and management

In addition to offering students the rhetorical tools they need to excel as professional communicators, the B.A. and B.S. programs also prepare graduates for study at the graduate level and for professional programs such as law school.

The B.A. and B.S. programs in WRTC provide a flexible, yet historically and theoretically grounded degree program. The WRTC curriculum teaches students to think critically in ways that cross disciplinary lines and to create in multiple genres and media platforms.

Students work with their WRTC advisor to select a track and design a writing major that fits their unique educational needs and career aspirations. Tracks include Professional Writing and Publishing, Writing for Law and Society, Science and Health Writing, and Inclusive Design. Students also have the option to design their own track.

Degree requirements differ between the B.A. and B.S. programs, and students should maintain regular contact with their WRTC advisor to ensure timely graduation.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Foreign language classes (intermediate level required) ²	0-12
Philosophy course (in addition to General Education courses)	3
Major requirements	39
University electives	25-37
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these credits may be fulfilled by courses within a student's major or minor.

² The foreign language requirement may be satisfied by successful completion of the second semester of the intermediate level (typically 232) of the student's chosen language or by placing out of that language through the Department of World Languages and Cultures' placement test.

Major Requirements

All students must complete 21 hours of core requirements, which lay a foundation for success in upper-level courses and in workplace environments. In addition to the required core courses, students take one community engaged learning course (3 credits), one culturally engaged course (3 credits), and 12 credits of electives in the major.

Code	Title	Credits
Core Requirements		
WRTC 200	Introduction to Studies in Writing, Rhetoric and Technical Communication	3
WRTC 300	Professional Editing	3
WRTC 301	Language, Law and Ethics	3
WRTC 316	Research Methodologies in WRTC	3
WRTC 350	Foundations of Technical Communication	3
WRTC 495	Internship in Writing, Rhetoric and Technical Communication	3
WRTC 496	Capstone in Writing, Rhetoric and Technical Communication	3
Subtotal		21
Work-Based Learning Requirement		
Choose one of the following:		3
WRTC 480	Writing for Business and Industry	
WRTC 484	Writing for Nonprofits	
WRTC 486	Writing in the Community	
WRTC 488	Writing in the Health Sciences	
Subtotal		3
Culturally-Engaged Requirement		
Choose one of the following:		3
WRTC 318	Intercultural Professional Communication	
WRTC/SCOM/ WGSS 420	Feminist Rhetorics	
WRTC 446	Rhetorics of Design and Culture	
WRTC 470	Disability Rhetoric	
Subtotal		3
Electives		
The WRTC major is designed to provide students the flexibility to choose electives that fit their interests and future plans. The major includes five advising tracks: Professional Writing and Publishing; Writing for Law and Society; Science and Health Writing; Inclusive Design; and Build Your Own Track. WRTC advisors can provide students with information about the advising tracks to assist with elective course selection; however, no specific grouping of electives is required to complete the major.		
Choose four of the following courses:		12
WRTC 314	Writing in the Public Sphere	
WRTC 318	Intercultural Professional Communication	
WRTC 326/ SCOM 354	Environmental Communication and Advocacy	
WRTC 328	Practicum	
WRTC 330	Rhetorical Analysis and Criticism	
WRTC 332	Creative Approaches to Digital Storytelling	
WRTC 334	Introduction to Popular Writing	
WRTC 336	Writing Consultation	
WRTC 338	Sound Writing	

WRTC 340	Writing as Leading
WRTC 342	Writing Place
WRTC 351	Visual Rhetoric
WRTC 352	Online Design
WRTC 354	Document Design
WRTC 358	Writing About Science and Technology
WRTC 400	Advanced Editing
WRTC 412	Content Strategy
WRTC 416/ SCOM 465	Rhetoric of Environmental Science and Technology
WRTC 420	Feminist Rhetorics
WRTC 426	Special Topics in Writing, Rhetoric and Technical Communication
WRTC 432	Rhetoric of the Personal Narrative
WRTC 433	Animal Rhetorics
WRTC 434	Advanced Popular Writing
WRTC 450	Digital Rhetoric
WRTC 454	Publication Management
WRTC 456	Usability and User Experience
WRTC 458	Scientific and Medical Communication
WRTC 460	Legal Writing
WRTC 462	Government Writing
Subtotal	12
Total Credits	39

Recommended Schedule for Majors

Students are encouraged to begin their WRTC course work as soon as possible in their degree plans. The following sample program of study illustrates how a WRTC major might earn a B.A. degree.

Course	Title	Credits
First Year		
First Semester		
Foreign language course ¹		3
General Education courses		12
Credits		15
Second Semester		
WRTC 200	Introduction to Studies in Writing, Rhetoric and Technical Communication	3
Foreign language course		3
General Education courses		9
Credits		15
Second Year		
First Semester		
WRTC 300	Professional Editing	3
WRTC 316	Research Methodologies in WRTC	3
Foreign language course		3
B.A. Degree philosophy course		3
General Education courses		3
Credits		15
Second Semester		
WRTC 301	Language, Law and Ethics	3
WRTC 350	Foundations of Technical Communication	3
Foreign language course		3
General Education course		3
University electives		3
Credits		15

Third Year		
First Semester		
WRTC Culturally-Engaged Requirement		3
WRTC electives		3
General Education courses		6
University elective		3
Credits		15
Second Semester		
WRTC Work-Based Learning Requirement		3
WRTC elective		3
General Education courses and/or university electives		9
Credits		15
Fourth Year		
First Semester		
WRTC 495	Internship in Writing, Rhetoric and Technical Communication	3
WRTC Electives		3
University electives		9
Credits		15
Second Semester		
WRTC 496	Capstone in Writing, Rhetoric and Technical Communication	3
WRTC electives		3
University electives		9
Credits		15
Total Credits		120

¹ Completion of an intermediate level foreign language is required for the B.A. degree (usually six hours if begun at the intermediate level) unless the language requirement is satisfied by an exemption test. In that case, university electives may be substituted for additional hours indicated as foreign language courses.

Writing, Rhetoric and Technical Communication, B.S.

Program Description

The Writing, Rhetoric and Technical Communication major emphasizes writing in professional contexts such as

- writing for professional communities such as government, medicine, law, science and environment, and education
- proposal and grant writing
- editing
- online inclusive design
- usability and user experience research
- publications management
- content strategy and management

In addition to offering students the rhetorical tools they need to excel as professional communicators, the B.A. and B.S. programs also prepare graduates for study at the graduate level and for professional programs such as law school.

The B.A. and B.S. programs in WRTC provide a flexible, yet historically and theoretically grounded degree program. The WRTC curriculum teaches students to think critically in ways that cross disciplinary lines and to create in multiple genres and media platforms.

Students work with their WRTC advisor to select a track and design a writing major that fits their unique educational needs and career aspirations. Tracks include Professional Writing and Publishing, Writing for Law and Society, Science and Health Writing, and Inclusive Design. Students also have the option to design their own track.

Degree requirements differ between the B.A. and B.S. programs, and students should maintain regular contact with their WRTC advisor to ensure timely graduation.

Degree Requirements

Degree Requirements	Credit Hours
General Education ¹	41
Quantitative requirement ²	3
Scientific literacy requirement ²	3-4
Major requirements	39
University electives	33-34
Total	120

¹ The General Education program contains a set of requirements each student must fulfill. Some of these credits may be fulfilled by courses within a student's major or minor.

² In addition to course work taken to fulfill General Education requirements.

Major Requirements

All students must complete 21 hours of core requirements, which lay a foundation for success in upper-level courses and in workplace environments. In addition to the required core courses, students take one community engaged learning course (3 credits), one culturally engaged course (3 credits), and 12 credits of electives in the major.

Code	Title	Credits
Core Requirements		
WRTC 200	Introduction to Studies in Writing, Rhetoric and Technical Communication	3
WRTC 300	Professional Editing	3
WRTC 301	Language, Law and Ethics	3
WRTC 316	Research Methodologies in WRTC	3
WRTC 350	Foundations of Technical Communication	3
WRTC 495	Internship in Writing, Rhetoric and Technical Communication	3
WRTC 496	Capstone in Writing, Rhetoric and Technical Communication	3
Subtotal		21
Work-Based Learning Requirement		
Choose one of the following:		3
WRTC 480	Writing for Business and Industry	
WRTC 484	Writing for Nonprofits	
WRTC 486	Writing in the Community	
WRTC 488	Writing in the Health Sciences	
Subtotal		3
Culturally-Engaged Requirement		
Choose one of the following:		3
WRTC 318	Intercultural Professional Communication	
WRTC/SCOM/ WGSS 420	Feminist Rhetorics	
WRTC 446	Rhetorics of Design and Culture	
WRTC 470	Disability Rhetoric	
Subtotal		3
Electives		
The WRTC major is designed to provide students the flexibility to choose electives that fit their interests and future plans. The major includes five advising tracks: Professional Writing and Publishing; Writing for Law and Society; Science and Health Writing; Inclusive Design; and Build Your Own Track. WRTC advisors can provide students with information about the advising tracks to assist with elective course selection; however, no specific grouping of electives is required to complete the major.		
Choose four of the following courses:		12
WRTC 314	Writing in the Public Sphere	
WRTC 318	Intercultural Professional Communication	
WRTC 326/ SCOM 354	Environmental Communication and Advocacy	
WRTC 328	Practicum	
WRTC 330	Rhetorical Analysis and Criticism	
WRTC 332	Creative Approaches to Digital Storytelling	
WRTC 334	Introduction to Popular Writing	
WRTC 336	Writing Consultation	
WRTC 338	Sound Writing	
WRTC 340	Writing as Leading	
WRTC 342	Writing Place	
WRTC 351	Visual Rhetoric	
WRTC 352	Online Design	
WRTC 354	Document Design	
WRTC 358	Writing About Science and Technology	

WRTC 400	Advanced Editing	
WRTC 412	Content Strategy	
WRTC 416/ SCOM 465	Rhetoric of Environmental Science and Technology	
WRTC 420	Feminist Rhetorics	
WRTC 426	Special Topics in Writing, Rhetoric and Technical Communication	
WRTC 432	Rhetoric of the Personal Narrative	
WRTC 433	Animal Rhetorics	
WRTC 434	Advanced Popular Writing	
WRTC 450	Digital Rhetoric	
WRTC 454	Publication Management	
WRTC 456	Usability and User Experience	
WRTC 458	Scientific and Medical Communication	
WRTC 460	Legal Writing	
WRTC 462	Government Writing	
Subtotal		12
Total Credits		39

Recommended Schedule for Majors

Students are encouraged to begin their WRTC course work as soon as possible in their degree plans. The following sample program of study illustrates how a WRTC major might earn a B.S. degree.

Course	Title	Credits
First Year		
First Semester		
General Education courses		15
Credits		15
Second Semester		
WRTC 200	Introduction to Studies in Writing, Rhetoric and Technical Communication	3
B.S. requirements		3
General Education courses		9
Credits		15
Second Year		
First Semester		
WRTC 300	Professional Editing	3
WRTC 316	Research Methodologies in WRTC	3
General Education courses		9
Credits		15
Second Semester		
WRTC 301	Language, Law and Ethics	3
WRTC 350	Foundations of Technical Communication	3
B.S. quantitative requirement		3
General Education courses		6
Credits		15
Third Year		
First Semester		
WRTC Culturally-Engaged Requirement		3
WRTC electives		3
B.S. scientific literacy requirement ¹		3
General Education courses		6
Credits		15
Second Semester		
WRTC work-based learning requirement		3
WRTC electives		3

University electives		9
Credits		15
Fourth Year		
First Semester		
WRTC 495	Internship in Writing, Rhetoric and Technical Communication	3
WRTC electives		3
University electives		9
Credits		15
Second Semester		
WRTC 496	Capstone in Writing, Rhetoric and Technical Communication	3
WRTC elective		3
University electives		9
Credits		15
Total Credits		120

¹ Completion of the B.S. degree requires a student to complete either a natural science or a social science course in addition to those required for the General Education program.

African, African American and Diaspora Studies Minor

Coordinator: Rachel Rhoades

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Phone: 540-568-2337

African, African American and Diaspora Studies Minor Website (<http://www.jmu.edu/africana/undergraduate-minor.shtml>)

Program Description

The minor in African, African American and Diaspora Studies (AAAD) broadens students' world perspectives by enhancing their understanding of the peoples, identities and institutions of Africa and the African Diaspora. The AAAD program employs interdisciplinary approaches to analyze Africa and the African Diaspora in a global context. Courses taken to fulfill program requirements are drawn from several departments in the humanities, social sciences and sciences. Students in the program will examine the key disciplines, concepts, methods and topics in African, African American and Diaspora studies.

The AAAD minor is open to all JMU undergraduate students.

Minor Requirements

The AAAD minor requires:

- Successful completion of 19 credit hours according to the requirements listed below.
- No more than nine hours from a single discipline.

Courses taken to complete the minor may also be used to satisfy the student's major, with the permission of the major advisor, as well as General Education requirements.

Code	Title	Credits
Required Courses		
AAAD 200	Introduction to African, African American and Diaspora Studies [C4GE]	3
AAAD 489	African, African American and Diaspora Studies Senior Research Experience ¹	1
Subtotal		4
Electives		
Choose five of the following, one of which must be at the 400 level:		15
AAAD 300	Special Topics in African, African American and Diaspora Studies	
AAAD 400	Advanced Topics in African, African American and Diaspora Studies	
AAAD 401	Internship in African, African American and Diaspora Studies	
AAAD 488	African American Art	
ANTH 280	Peoples and Cultures of Sub-Saharan Africa	
ANTH 391	Study Abroad	
ANTH 395	Special Topics in Anthropology ²	
ANTH 436	Afro-Latin America	
ART 200	Art Today: Themes in Contemporary Art [C2VPA] ²	
ARTH 210	African Art and Culture in the Humanities	
ARTH 312	African Art: Sub-Saharan	
ARTH 320	Travel Study in Art History ²	

ARTH 418	Modern and Contemporary African Art
ARTH 419	Topics in African Art
ARTH 424	Arts of Ancient Egypt
ARTH 426	Advanced Topics in Cross-Cultural Art ²
ARTH 488	African American Art
ARTH 489	Topics in Art History ²
BIO 426	Topics in Biology ²
BIO 427	Topics in Biology with Laboratory ²
EDUC 310	Teaching in a Diverse Society
EDUC 490	Special Topics in Education ²
ELED 310	Diversity in Elementary Education
ENG 239	Studies in World Literature [C2L] ²
ENG 260	Survey of African American Literature [C2L]
ENG 302	Special Topics in Literature and Language ²
ENG 332	Studies in Popular Genres ²
ENG 358	Oral Literature ²
ENG 359	Studies in African American Literature
ENG 361	African American Fiction
ENG 362	African American Poetry
ENG 377	Introduction to African Literature
ENG 385	Special Topics in Film Study ²
ENG 405	Advanced Studies in Anglophone Literature ^{1,2}
ENG 408	Advanced Studies in African American Literature
ENG 410	Advanced Studies in Author ²
ENG 412	Special Topic Seminar ²
ENG 423	Advanced Studies in Gender and Sexuality in Literature ²
ENG 432	Advanced Studies in African Literature
ENGR 498	Advanced Topics in Engineering ²
GEOG 335	Geography of Africa
GEOG 339	Geography of the Caribbean
GEOG 491	International Studies ²
HIST 263	Africa
HIST 307	The Trans-Atlantic Slave Trade
HIST 341	Selected Themes in World History
HIST 356	African American History since 1865
HIST 391	Travel Studies Seminar
HIST 436	Afro-Latin America
HIST 439	Selected Topics in American History ²
HIST 470	Modern Africa
HIST 489	Selected Topics in World History ^{1,2}
HRD 123	Developing Multicultural Competency for Effective Facilitation
HUM 252	Global Cultures [C2HQC] (African Cultures in the Humanities) ²
HUM 252	Global Cultures [C2HQC] (African Experience)
IB 298	Special Topics in International Business ²
IDLS 382	Contemporary United States ¹
IDLS 385	The Unfinished Journey of People of Color in the U.S.
IDLS 391	Study Abroad ²
IDLS 400	Seminar in Liberal Studies ^{1,2}
ISAT 480	Topics in Integrated Science and Technology ²

JUST 301	Advanced Level Special Topics in Justice Studies ^{1,2}	
JUST 328	Race, Class and Justice	
JUST 352	Global Migration	
JUST 357	Environmental Justice	
MUS 200	Music and Culture [C2VPA] ²	
MUS 356	History of Jazz	
POSC 326	Civil Rights	
POSC 353	African Politics	
POSC 361	Topics in International Relations ²	
POSC 371	Topics in Comparative Politics ^{1,2}	
REL 330	Religions of Africa and the African Diaspora ²	
REL 336	African American Religion	
SCOM 248	Intercultural Communication	
SCOM 301	Feminist Blogging: Writing for Shout Out! JMU	
SCOM 313	Topics in Communication Studies ²	
SCOM 318	Practicum in Communication Studies ²	
SCOM 355	Food as Communication ²	
SCOM 395	Study Abroad Seminar ²	
SMAD 498	Senior Seminar in Media Arts and Design ^{1,2}	
SOCI 336	Race and Ethnicity	
SOCI 354	Social Inequality	
SOCI 391	Study Abroad ²	
STAD 322	Equity, Inclusion and Access in Education and Performance	
THEA 323	Youth Theatre	
THEA 324	Theatre for Young Audiences	
UNST 390	Special Studies in University Studies	
UNST 490	Advanced Special Studies in University Studies ²	
WGSS 300	Special Topics in Women's, Gender and Sexuality Studies ^{1,2}	
WGSS 495	Special Topics in Women's, Gender and Sexuality Studies ^{1,2}	
Total Credits		19

¹ Research must have an African and/or African Diaspora focus.

² Course topic and content must focus on the peoples of Africa and/or the African Diasporas as well as be approved by the program coordinator(s).

Additional Information

Other appropriate courses not listed above may be considered for the minor. Consult with program coordinators.

In addition to taking these courses, students are encouraged to participate in travel or study programs to Africa, the Caribbean and other relevant areas. Students who want to earn credit hours through participation in an accredited travel/study program are encouraged to do so with prior approval of the coordinators.

American Studies Minor

Coordinator: Laura Henigman
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American Studies Minor Website (<http://www.jmu.edu/americanstudies/>)

Program Description

The minor in American studies fosters an understanding of the complexity of U.S., colonial North American and North American indigenous societies through employing a range of disciplinary approaches and topics. Students will enroll in the gateway course AMST 200, and then they will select from a list of courses drawn from a variety of disciplines including literature, history, the fine arts, philosophy and the social sciences.

The minor program in American studies is open to all undergraduate students at JMU.

Minor Requirements

The requirement is the successful completion of 18 hours. In addition to AMST 200, students must take five additional courses from the list below. No more than two courses may be from the same program. See the program coordinator for additional courses that may be substituted.

Code	Title	Credits
Core Requirement		
AMST 200	Introduction to American Studies [C2HQC]	3
Subtotal		3
Electives		
Choose five of the following:		15
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	
ANTH 368	Contemporary American Culture	
ARTH 380	American Art to 1870	
	or ARTH 382 American Art from 1870	
ARTH 484	Art of the Americas	
ENG 342	Early American Literature	
ENG 343	Antebellum American Literature	
ENG 344	Late Nineteenth-Century American Literature	
ENG 352	The American Novel to 1914	
ENG 356	Modern American Novel	
ENG 357	Contemporary American Literature	
ENG 359	Studies in African American Literature	
ENG 360	Introduction to Ethnic American Literature	
ENG 361	African American Fiction	
ENG 362	African American Poetry	
ENG 363	Native American Literature	
ENG 364	Introduction to Asian American Literature	
ENG 367	Latinx Literature	
HIST 303	Virginia's Place in #VastEarlyAmerica	
HIST 304	Native Peoples of the United States	
HIST 307	The Trans-Atlantic Slave Trade	
HIST 313	The History of Latinos in the United States	
HIST 315	Global History of Sport	

HIST 316	The Life and Times of James Madison, 1751-1836
HIST 319	Women at Work in U.S. History
HIST 320	Women in U.S. History
HIST 322	Jim Crow South
HIST 323	The Old South
HIST 326	How Cars Changed Everything
HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies
HIST 336	We the People?: The American Revolution Considered
HIST 338	United States Urban History
HIST 339	Selected Themes in U.S. History
HIST 343	History of Today
HIST 345	Pills and Potions: Histories of U.S. Health Care and Medicine
HIST 347	Advertising America
HIST 350	Virginia
HIST 356	African American History since 1865
HIST 362	American Material Religion
HIST 374	The Southern Plantation
HIST 397	War and Society in U.S. History
IDLS 381	Environmental Thought and Activism
IDLS 382	Contemporary United States
IDLS 385	The Unfinished Journey of People of Color in the U.S.
MUS 203	Music in America [C2VPA]
MUS 356	History of Jazz
PHIL 370	American Philosophy
POSC 330	American Political Thought
POSC 380	The U.S. Presidency
POSC 382	The Role of Religion in American Politics
POSC 384	Minority Group Politics
POSC 385	The U.S. Congress
REL 210	Religion in America
REL 324	Latinx and Latin American Christianity
REL 330	Religions of Africa and the African Diaspora
REL 332	Born Again Religion
REL 334	New Religious Movements
REL 336	African American Religion
SCOM 346	Free Speech in America
SMAD 372	Media History
SOCI 318	Sociology of Immigration
SOCI 336	Race and Ethnicity
SOCI 341	Sociology of Education
SOCI 357	Sociology of Disasters
THEA 485	American Theatre
Subtotal	15
Total Credits	18

Anthropology Minor

Coordinator: Julie Solometo

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Anthropology Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/anthropology.shtml>)

Anthropologists document the diversity of ways humans live and have lived throughout the world. Students in the anthropology minor choose from classes in cultural, linguistic and biological anthropology and archaeology. The wide range of research methods, cultural traditions, and ways of thinking about contemporary issues explored in our courses complement many degree programs, including biology, geography, psychology, justice studies, communication studies, and any discipline that requires intercultural competence.

Minor Requirements

Students enrolled in any degree program may minor in anthropology by completing a minimum of 18 credit hours in anthropology courses, including 9 credit hours of required courses and 9 credit hours of electives of their choosing.

Code	Title	Credits
Required Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
ANTH 196	Biological Anthropology [C3NS]	3
ANTH 197	Archaeology	3
Additional ANTH courses		9
Total Credits		18

Art Minor

Advisor: Allyson Taylor
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Phone: (540) 568-6485
Art Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/art.shtml>)

Program Description

The art minor:

- Broadens the student’s understanding of the value and role of the visual arts in general culture.
- Deepens the student’s appreciation for personal artistic production.
- Enhances the student’s creativity and ability to think independently.

Minor Requirements

A minimum of 18 credit hours is required for the art minor.

Code	Title	Credits
Foundations Courses		
Choose two of the following:		6
FDS 102	Color and Composition	
FDS 104	Observational Drawing	
FDS 106	Form and Space	
FDS 108	Expressive and Expanded Drawing ¹	
Subtotal		6
Studio Art Classes		
Choose two of the following: ²		6
ART 210	Figure Drawing	
ART 220	Introductory Ceramics: Potter’s Wheel	
ART 222	Introductory Ceramics	
ART 230	Introduction to Fibers Processes	
ART 232	Introduction to Fiber Arts: Structures	
ART 240	Introductory Metals	
ART 242	Knifemaking: Exploring Form and Function	
ART 252	Introductory Painting	
ART 260	Introductory Black and White Darkroom Photography	
ART 270	Printmaking: Screenprint	
ART 272	Printmaking: Relief and Intaglio	
ART 274	Printmaking: Lithography	
ART 276	Introductory Book Arts: Materials and Structures	
ART 280	Introductory Sculpture: Idea Material Process	
ART 292	Topics in Art	
Subtotal		6
Intermediate and Advanced Studio Art Classes		
Choose two of the following: ³		6
ART 300	Illustration	
ART 320	Molds and Casting: Ceramics	
ART 322	Surface Development: Ceramics	
ART 330	Intermediate Fiber Processes	
ART 332	Woven Structures	
ART 334	Iterative Structures	

ART 336	Pulp	
ART 352	Intermediate Painting	
ART 360	Digital Photography	
ART 362	Experimental Black and White Photography	
ART 364	Large Format Photography	
ART 375	Letterpress	
ART 376	Intermediate Book Arts: Concept, Content, Form	
ART 380	Conceptual Practice: Sculpture	
ART 392	Topics in Art	
ART 420	Advanced Ceramics	
ART 430	Advanced Fiber Processes	
ART 432	Fiber Installation Art	
ART 434	Collaboration in Fiber Arts	
ART 439	Portfolio Development: Fibers	
ART 440	Advanced Metals	
ART 452	Advanced Painting	
ART 460	Alternative Photographic Processes	
ART 462	Wonder and Illusion: Origins of Photography	
ART 464	The Photograph/Video as Document	
ART 466	Performance for the Lens	
ART 468	Screen-Based Photography/Video	
ART 469	Portfolio Development: Photography	
ART 470	Advanced Printmaking	
ART 480	Expanded Field: Sculpture	
ART 489	Portfolio Development: Sculpture	
ART 492	Topics in Art	
Subtotal		6
Total Credits		18

¹ FDS 108 and most 200-level classes have 100-level prerequisites.
² Students may not use ART 200 to satisfy this category. Most 300-400-level classes have specific 200-level prerequisites.
³ Students may not use ART 493, ART 496 or ART 499 to satisfy this category. These 300-400 level courses cannot count toward the students’ major(s).

Art History Minor

Advisor: David Ehrenpreis
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Art History Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/art-history.shtml>)

Program Description

Students enrolled in any JMU degree program may minor in art history by completing a variety of art history courses. Students may add the minor to their academic program using MyMadison and contact your minor advisor to make sure your plan for progress is in place.

Minor Requirements

Students enrolled in any degree program may minor in art history by completing a minimum of 18 credit hours in art history courses, as the following chart shows. At least six of these credit hours must be 400-level art history courses. The student's minor program is subject to approval by the school director.

Code	Title	Credits
Required Courses		
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	3
ARTH 206	Survey of World Art II: Renaissance to Modern [C2VPA]	3
Art history courses (300-400 level) ¹		12
Total Credits		18

¹ For studio art, graphic design and architectural design majors, required ARTH courses in the art emphasis program may count toward the art history minor. Excludes ARTH 390, ARTH 490, ARTH 495 and ARTH 499.

Asian Studies Minor

Coordinator: Michael Seth

Email: sethmj@jmu.edu

Phone: (540) 568-7801

Asian Studies Minor Website (<https://www.jmu.edu/cal/academics/minors/asian-studies/index.shtml>)

Program Description

The purpose of this cross disciplinary program is to broaden the students' perspective by enhancing their understanding and appreciation of Asian culture and institutions. This program combines the offerings of several academic units, such as anthropology; art, design and art history; economics; English; geography; history; international business; world languages and cultures; political science; and religion and philosophy.

Minor Requirements

The minimum requirement for a minor in Asian studies is 18 credit hours. These 18 hours can include any of the following courses. Special topics courses not listed can be applied to the minor degree with approval of the program coordinator. A maximum of eight hours of Chinese, Japanese or Korean language may be included to satisfy credit hour requirements of the minor.

Code	Title	Credits
Required Courses		
Choose 18 credit hours of the following:		18
ANTH 197	Archaeology	
ANTH 295	People and Cultures of East Asia	
ANTH/SOCI 306	Japanese Society and Culture	
ARTH 430	Far Eastern Art	
CHIN 101	Elementary Chinese I	
CHIN 102	Elementary Chinese II	
CHIN 231	Intermediate Chinese I	
CHIN 232	Intermediate Chinese II	
CHIN 300	Language for Living	
CHIN 400	Advanced Chinese	
ECON 312	Comparative Economic Systems	
ENG 378	Studies in South Asian Literature	
GEOG 334	Geography of East and Southeast Asia	
GEOG 336	Environmental Hazards: A Focus on Southeast Asia	
GEOG 338	Geography of the Philippine Islands: Problems and Possibilities	
GEOG 375	Political Geography	
HUM 200	Great Works [C2L] (Asia)	
HUM 252	Global Cultures [C2HQC] (Cross-Cultural Perspectives – East Asian Humanities)	
HUM 252	Global Cultures [C2HQC] (Cross-Cultural Perspectives – Global South Asia)	
HUM 252	Global Cultures [C2HQC] (Cross Cultural Perspectives – Gandhi, Non-Violence and Global Transformation)	
HIST 274	Modern East Asia	
HIST 311	China to 1600	

HIST 312	Japan since 1600
HIST 371	India
HIST 377	History of Korea
HIST 378	China in the Modern World
IB 298	Special Topics in International Business (Business Environment in China and Southeast Asia)
IB 298	Special Topics in International Business (Chinese Business Operations)
JAPN 101	Elementary Japanese I
JAPN 102	Elementary Japanese II
JAPN 212	Intensive Japanese II
JAPN 231	Intermediate Japanese I
JAPN 232	Intermediate Japanese II
JAPN 300	Language for Living
JAPN 320	Culture and Communication in Context
JAPN 446	Special Topics in Japanese Literature
JAPN 447	Special Topics in Japanese Culture
KOR 101	Elementary Korean I
KOR 102	Elementary Korean II
KOR 231	Intermediate Korean I
KOR 232	Intermediate Korean II
KOR 300	Language for Living
KOR 320	Culture and Communication in Context
KOR 447	Special Topics in Korean Culture
POSC 355	East Asian Politics
POSC 356	Politics of China
REL 213	Introduction to Ancient Chinese Wisdom Traditions
REL 216	(Re)thinking Order and Chaos: An Introduction to Hinduism
REL 217	The Mystery of the True Path: An Introduction to Daoism
REL 288	Introduction to Buddhism
REL 314	Hope or Terror? Gandhi and the Other 9/11
REL 316	Dharma/Adharma: Hindu Ethical Reasoning
REL 388	Tibetan Buddhism
REL 488	Buddhism in the Modern World

Total Credits **18**

Astronomy Minor

Coordinator: Harold Butner
Email: butnerhm@jmu.edu
Phone: (540) 568-5358
Astronomy Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/astromony.shtml>)

Program Description

The Department of Physics and Astronomy (p. 94) offers a minor in astronomy. Students choosing to minor in astronomy enroll in required and elective astronomy and physics courses. Topics covered include appearance and movements of the night sky, stellar evolution and an overview of modern astronomical techniques.

Minor Requirements

The minimum requirement for a minor in astronomy is 21 credit hours selected as follows:

Code	Title	Credits
Required Courses		
ASTR 220	General Astronomy I: The Night Sky, the Solar System and Stars	3
ASTR 221	General Astronomy II: Star Systems, the Interstellar Medium, and Cosmology	4
ASTR 320	Astronomical Techniques	3
Subtotal		10
Physics Lecture Courses		
Choose one of the following:		6
PHYS 140 & PHYS 150	College Physics I [C3PP] and College Physics II	
PHYS 240 & PHYS 250	University Physics I [C3PP] and University Physics II	
Subtotal		6
Physics Laboratory Courses		
Choose one of the following:		2
PHYS 140L & PHYS 150L	General Physics Laboratory I [C3L] and General Physics Laboratory II	
PHYS 240L & PHYS 250L	University Physics Laboratory I [C3L] and University Physics Laboratory II	
Subtotal		2
Elective		
Choose one of the following:		3
ASTR/PHYS 330	Relativity	
ASTR 480	Astrophysics	
GEOL 307	Planetary Geology (2, 2)	
HON 300	Advanced Special Topics in Honors (Life Beyond Earth)	
ASTR 297	Topics in Astronomy (Appropriate topics could include the relativity, cosmology, cosmic rays or other astronomy-related topics with approval of minor advisor.)	

PHYS 297	Topics in Physics (Appropriate topics could include the relativity, cosmology, cosmic rays or other astronomy-related topics with approval of minor advisor.)
PHYS 397	Topics in Physics (Appropriate topics could include the relativity, cosmology, cosmic rays or other astronomy-related topics with approval of minor advisor.)
PHYS 497	Topics in Physics (Appropriate topics could include the relativity, cosmology, cosmic rays or other astronomy-related topics with approval of minor advisor.)
Subtotal	3
Total Credits	21

Biology Minor

Advisor: Julia Stutzman

Email: stutzmjk@jmu.edu

Phone: (540) 568-4018

Biology Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/biology.shtml>)

Program Description

The biology minor offers students a way to explore primary concepts in biology from molecules to ecosystems through four core biology classes with their associated labs. Main topics include cell and molecular biology, organismal biology, biodiversity, genetics, ecology and evolution. Students also choose one or more upper-level level courses in biology within their area of interest. The biology minor is an ideal minor for any student interested in the biological sciences. The biology minor also pairs well with other science and social science majors, or pre-professional health minors.

Special Requirements

Practical hands-on experience in the field and/or laboratory is important content of laboratory-based courses. Transfer credit for courses including laboratories will only be awarded for those having skill and application content comparable to that of JMU courses. Elective biology credits applicable toward the minor are transferred as specific courses or as BIO 426 or BIO 427. A maximum of eight credit hours of first year (100) level BIO courses may be applied to the biology major or minor requirements. Biology and biotechnology majors may not minor in biology.

Minor Requirements

Students completing the biology minor must earn minimum grades of "C-" in BIO 140, BIO 140L, BIO 150 and BIO 150L.

Code	Title	Credits
Required Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	4
Biology elective (300-level and above)		3-4
Total Credits		19-20

Biochemistry and Molecular Biology Minor

Co-Coordinator: Terrie Rife

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Phone: (540) 568-3343

Co-Coordinator: Gina MacDonald

Email: macdongx@jmu.edu

Phone: (540) 568-6852

Biochemistry and Molecular Biology Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/biochemistry-and-molecular-biology.shtml>)

Program Description

The biochemistry and molecular biology minor is open to students not majoring in biotechnology.

Minor Requirements

The following are prerequisites for entry into the biochemistry and molecular biology minor program (some prerequisites are not shown; see the course descriptions).

Prerequisites

Code	Title	Credits
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
Choose one of the following:		2
CHEM 242L	Organic Chemistry Laboratory	
CHEM 287L	Integrated Inorganic/Organic Laboratory I	
CHEM 288L	Integrated Inorganic/Organic Laboratory II	
Total Credits		12

Required Courses

Code	Title	Credits
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 480 & 480L	Advanced Molecular Biology and Advanced Molecular Biology Lab	4
CHEM 361 or BIO 361	Biochemistry I Biochemistry I	3
CHEM 362	Biochemistry II	3
CHEM 366L	Biochemistry Laboratory	2
Choose one of the following:		4-5
CHEM 331 & CHEM 336L	Physical Chemistry I and Applied Physical Chemistry Laboratory	
CHEM 351	Analytical Chemistry	
Total Credits		20-21

Book Arts Minor

Co-Coordinator: Dawn M. McCusker
Email: mcccuskm@jmu.edu
Phone: (540) 568-6500

Co-Coordinator: Karin Tollefson-Hall
Email: tolleflk@jmu.edu
Phone: (540) 568-4304

Book Arts Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/book-arts.shtml>)

Program Description

The minor in book arts is a cross-disciplinary program designed to broaden students' understanding of the value and role of the art of the book in general culture while enhancing written and visual creativity, artistic production, and the ability to think independently.

Minor Requirements

Students enrolled in any degree program may minor in book arts by completing a minimum of 21 credit hours. The student's minor program is subject to the approval of the coordinator of the Book Arts Minor.

A total of nine credit hours may be double counted between the minor and the major.

Code	Title	Credits
Required Courses		
FDS 102	Color and Composition	3
FDS 104	Observational Drawing	3
ART 276	Introductory Book Arts: Materials and Structures	3
ENG 415	Advanced Studies in Textuality and the History of the Book	3
Choose one of the following:		3
ENG 391	Introduction to Creative Writing - Nonfiction	
ENG 392	Introduction to Creative Writing - Poetry	
ENG 393	Introduction to Creative Writing - Fiction	
Subtotal		15
Electives		
Choose two of the following:		6
ART 230	Introduction to Fibers Processes	
ART 232	Introduction to Fiber Arts: Structures	
ART 260	Introductory Black and White Darkroom Photography	
ART 270	Printmaking: Screenprint	
or ART 272	Printmaking: Relief and Intaglio	
or ART 274	Printmaking: Lithography	
ART 280	Introductory Sculpture: Idea Material Process	
ART 336	Pulp	
ART/GRPH 375	Letterpress	
ART/GRPH 376	Intermediate Book Arts: Concept, Content, Form	
ENG 496	Advanced Topics in Creative Writing	
GRPH 101	Digital Media Design I	

Subtotal	6
Total Credits	21

British Communication and Media Minor

Coordinator: Dietrich Maune
Email: maunedx@jmu.edu
Phone: (540) 586-6925
British Communication and Media Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/british-communication-and-media.shtml>)

Program Description

The cross disciplinary British communication and media minor enables students to expand their knowledge of communication and media in Great Britain, and to enhance their appreciation of the impact culture has on communication and media. Students must participate in the JMU Semester in London program (Fall, Spring or Summer) to complete this minor.

Minor Requirements

The minor requires at least 18 credit hours, chosen from courses offered both at JMU and in London. Students may double-count a maximum of six credit hours toward both a major and this minor.

Code	Title	Credits
Required Courses		
SCOM 248	Intercultural Communication ¹	3
SCOM/SMAD/ WRTC 360	British Media and Society	3
Subtotal		6
Elective Courses		
Choose four of the following courses, all of which must be taken in London:		12
SCOM 359	Communication, Diversity and British Popular Culture	12
SMAD 321	The Media Arts: Culture by Design in London ²	
or SCOM 351 Visual Rhetoric		
or ARTH 389 Topics in Art History		
ENG 463	British Film Adaptations ³	3
or SMAD 466 British Film Adaptations		
or SCOM 395 Study Abroad Seminar		
THEA 449	London Theatre ⁴	3
or HUM 200 Great Works [C2L]		
WRTC 314	Writing in the Public Sphere	3
SCOM/SMAD/ POSC 473	British Media and Politics	
IDLS 486	Internship and Field Experience	3
Subtotal		12
Total Credits		18

¹ Must be taken in Harrisonburg at JMU. Regularly offered in fall semesters.
² ARTH 389 when topic is The Media Arts: Culture by Design.
³ SCOM 395 when topic is British Literature and Film.
⁴ HUM 200 when topic is London Theatre.

Additional Information

Appropriate independent studies, directed projects, and Semester in London Faculty Member in Residence courses may be counted as part of the minor with approval by the minor coordinator. Students studying in London during the summer will need to complete one additional course, approved by the minor coordinator, following the student's return to JMU.

Business Analytics Minor

Contact: CIS & Business Analytics Department

Email: CIS&BSAN-Minors@jmu.edu (CIS&BSAN-Minors@jmu.edu)

Phone: (540) 568-3255

Business Analytics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/business-analytics.shtml>)

Business analytics is a technical approach to analyzing problems and making business-related decisions. It uses statistical methods, management science techniques and mathematical modeling to forecast the implications of various choices and identify the best alternatives. Business analytics focuses on the effective use of data and information to provide fact-based insights and drive positive business actions. The minor in business analytics prepares students to solve complex decision problems in a business environment with a combination of quantitative skills and hands-on expertise using current software applications. Students enrolled in the business analytics minor are not permitted to pursue the business analytics concentration of the computer information systems major.

Special Admission Requirements

Admission to the BSAN program is limited and competitive.

To be eligible to apply for admission, students must have completed the following:

- One introductory calculus course (MATH 205, MATH 231, MATH 235, ISAT 151 or equivalent)
- One statistics course (COB 191, MATH 220, MATH 285, MATH 318, ISAT 251 or equivalent)
- One economics course (ECON 200 or ECON 201)
- One business analytics course (COB 291) with a minimum grade of "B-"

In addition, a student must have a minimum of 2 full semesters remaining when starting the minor to complete the course requirements.

A complete application to the minor includes the following:

- A completed BSAN minor application form
- A letter of recommendation from COB 291 instructor

Students may use transfer and/or college credits that have been accepted by JMU to meet competency requirements. Admission is based on an average of a student's highest grades in one course from each competency area and the letter of recommendation. Only grades earned at JMU and verified by the registrar will be used; transfer transcripts will also be reviewed. More emphasis will be placed on grades than on the recommendation letter.

Students seeking to add the BSAN minor must submit an application to the Department of Computer Information Systems and Business Analytics. Applications are accepted every semester. See the website (<https://www.jmu.edu/cob/cis/academics/bsan-minor.shtml>) for the BSAN minor for the application link and current deadlines.

The Business Analytics minor is a competitive program with limited available seats. Admission is not guaranteed.

Minor Requirements

No more than one approved transfer course can be used to complete the requirements of the minor. Transfer courses should be approved in advance.

Code	Title	Credits
Program Requirements		
Choose one of the following:		3-4
COB 191	Business Analytics I	
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	
MATH 220	Elementary Statistics [C3QR]	
MATH 285	Data Analysis	
MATH 318	Introduction to Probability and Statistics	
Subtotal		3-4
Required Courses		
COB 291	Business Analytics II ¹	3
BSAN 390	Statistical Modeling for Business Analytics	3
BSAN 391	Decision Modeling for Business Analytics	3
BSAN 393	Data Mining for Business Analytics	3
or CIS 393	Data Mining for Business Analytics	
Subtotal		12
Elective		
Choose one of the following:		3
BSAN 498	Special Topics in Business Analytics	
CIS 463	Business Intelligence	
ISAT 344	Applied Artificial Intelligence	
ECON 385	Econometrics	
FIN 475	Financial Modeling and Risk Analysis	
MATH 322	Applied Linear Regression	
MKTG 482	Marketing Analytics	
Subtotal		3
Total Credits		18-19

¹ COB 191 and MATH 205 or equivalent are prerequisites for COB 291. A minimum grade of "B-" in COB 291 and junior or senior standing are prerequisites for taking the subsequent required courses (BSAN 390, BSAN 391, and BSAN 393/CIS 393) in the business analytics minor.

Business Spanish Minor

Advisor: Veronica Haun
Email: haunvx@jmu.edu
Phone: 540-568-6851
Business Spanish Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/business-spanish.shtml>)

Program Description

The professional minor in business Spanish is intended for students who wish to consolidate and improve their knowledge of Spanish in several business areas. The main aim of the minor is to teach students how to use specific business terms in Spanish and to handle selling techniques, negotiation strategies, product presentations, commercial activities, marketing and business correspondence in Spanish. Students acquire a greater knowledge of Spanish as well as deeper understanding of Spanish or Hispanic societies and their business cultures.

Minor Requirements

The minor comprises one linguistic component, one elective component, one translation/interpretation component and three profession-specific linguistic components.

Code	Title	Credits
Required Core Courses		
SPAN 300	Language for Living	3
SPAN 330	Business Spanish	3
SPAN 430	Advanced Business Spanish (offered only in spring semesters)	3
Subtotal		9
Business and Society		
Choose one of the following:		3
SPAN 485	Business and Society in Latin America (offered only in fall semesters)	1
SPAN 486	Business and Society in Spain	
Subtotal		3
Translation and Interpretation		
Choose one of the following:		3
SPAN 435	Translation Strategies	3
SPAN 436	Introduction to Interpretation	
SPAN 437	Introduction to Medical Interpretation	
SPAN 438	Introduction to Legal Interpretation	
Subtotal		3
Elective Spanish		
Choose one of the following:		3
SPAN 307	History of Spanish Culture and Society	3
SPAN 308	Latin American Cultures	
SPAN 311	Contrastive Linguistics	
SPAN 320	Culture and Communication in Context	
or SPAN 321 Culture and Communication in Context (for Heritage Speakers)		
SPAN 407	Aspects of Contemporary Spanish Culture and Society	3
SPAN 447	Special Topics in Spanish-Speaking Cultures	
SPAN 476	Culture and Medicine in Latin America	

Subtotal	3
Total Credits	18

¹ Student must contact Business Spanish minor coordinator to receive instructions and complete an end-of-course assignment focused on Latin American business.

Chemistry Minor

Advisor: Scott Lewis

Email: lewissb@jmu.edu

Phone: (540) 568-2873

Chemistry Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/chemistry.shtml>)

Program Description

The minor in chemistry is designed for students majoring in a STEM field who want to gain additional technical competencies in chemistry. The chemistry minor is not open to students majoring in chemistry or biophysical chemistry.

Minor Requirements

The requirements for a minor in chemistry are a minimum of 24 credit hours in chemistry, distributed as follows:

Code	Title	Credits
General Chemistry		
CHEM 131	General Chemistry I [C3PP]	3
CHEM 132	General Chemistry II	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132L	General Chemistry Laboratory	1
Subtotal		8
Organic Chemistry		
Choose one of the following options:		4-8
Option 1:		
CHEM 241 & 241L	Organic Chemistry I and Concepts of Organic Chemistry Laboratory	
Option 2:		
CHEM 241 & CHEM 242 & CHEM 242L	Organic Chemistry I and Organic Chemistry II and Organic Chemistry Laboratory	
Subtotal		4-8
Physical Chemistry		
Choose one of the following sets of courses:		5
CHEM 331 & CHEM 336L	Physical Chemistry I and Applied Physical Chemistry Laboratory	
CHEM 432 & CHEM 438L	Physical Chemistry II and Physical Chemistry Laboratory	
Subtotal		5
Analytical Chemistry		
CHEM 351	Analytical Chemistry	4
Subtotal		4
Electives		
An approved CHEM elective		3
Subtotal		3
Total Credits		24-28

Additional Information

In order to complete this minor, prerequisite courses in mathematics and physics are required. These courses collectively fulfill the nine-credit approved technical elective package for the Bachelor of Science in Engineering.

Christian Studies Minor

Coordinator: Jennifer Aycock

Email: aycockjl@jmu.edu

Phone: (540) 568-6394

Christian Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/christian-studies.shtml>)

Program Description

This minor is designed to provide students with a systematic and comprehensive understanding of the nature and history of Christianity as a religion of global significance. Students begin their studies by exploring the history, beliefs and practices of diverse Christian traditions, from the ancient to the modern. Additional courses address the complex relationship between Christianity and various facets of the contemporary world, such as politics, culture, religious pluralism, terrorism and race.

Minor Requirements

Code	Title	Credits
Required Courses		
REL 202	Introduction to the New Testament	3
REL 240	Introduction to Christianity	3
REL 348	Christianity in Global Context	3
Subtotal		9
Electives		
Choose three courses from the following, at least two of which must be at the 300- or 400-level:		9
REL 201	Introduction to Hebrew Bible / Old Testament	
REL 218	Philosophy of Religion	
REL 324	Latinx and Latin American Christianity	
REL 325	Catholicism in the Modern World	
REL 332	Born Again Religion	
REL 334	New Religious Movements	
REL 336	African American Religion	
REL 341	Early Christian Gospels: Origins, History and Conflict	
REL 342	The Historical Jesus Quest	
REL 343	Paul and the Origins of Christianity	
REL 363	Apocalypticism, Religious Terrorism and Peace	
REL 380	Religion and Science	
REL 450	Religion and Society	
REL 460	Topics in Ancient Jewish and Early Christian Literature	
Subtotal		9
Total Credits		18

Chronic Illness Minor

Coordinator: Christina Lam

Email: russelck@jmu.edu

Phone: (540) 568-7616

Chronic Illness Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/chronic-illness.shtml>)

Program Description

The cross disciplinary minor in chronic illness prepares students from any major to understand and respond to the impacts of chronic illness on the individual, family, health care system and society. Knowledge and strategies to address the prevention and the management of chronic illness across its trajectory will enhance the skills and abilities of students who plan to practice in any profession. The minor is open to all undergraduate students at JMU and requires a minimum of 18 credits with no more than six credits in the student's major counting toward the minor.

Declaration Deadlines

Deadlines for declaring the Chronic Illness minor must be completed in MyMadison by:

- October 1 for fall registration for the spring term.
- March 1 for spring registration for the fall term.

Minor Requirements

Code	Title	Credits
Core Requirements		
Students completing the nursing major must complete the core requirements prior to starting major courses.		
NSG 390	Impact of Chronic Illness	3
NSG 391	Living Successfully with Chronic Illness	3
Subtotal		6
Elective Requirements		
Nursing majors select 6 credits. Non-Nursing majors select 12 credits.		6-12
ANTH 360	Medical Anthropology	
EXED 306	Lifespan Issues for Individuals with Disabilities	
GERN/SOCI 280	Social Gerontology	
GERN 305	Programs and Services for the Elderly	
HTH 255	Chronic Disease and Disabilities ¹	
HTH 351	Health Behavior Change ¹	
IPE 313	Issues and Applications of Family Caregiving: Caregiver's Community Network	
IPE 490	Special Topics in Health and Human Services (when topic is appropriate)	
KIN 424/ NUTR 455	Exercise and Nutrition in Chronic Disease	
NSG 306	Combating Childhood Obesity	
NSG 308	Empowerment and Resilience	
NSG 310	Helping Persons in Pain	
NSG/IPE/ SOWK 314	Rural Health: An Interprofessional Approach	

NSG 318	Nursing, Chronic Illness and Health in a Global Context	
NSG 319	Understanding Sexuality across the Health-Illness Continuum	
NSG 322	Integrative Health Care	
NSG 326	Care and Consideration for Children and Young Adults with Disabilities	
NSG 328	Life, Death, and the In-Between	
NSG 329	Best Practices in Diabetes Care	
NSG 330	Medical Innovations	
NSG 360	HHS in Costa Rica: Interprofessional Perspective	
NSG 361	International Health Perspectives: Malta	
NSG 363	Health Care in Spain: A Cross Cultural Exploration of Illness and Wellness	
NSG 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	
NSG 490	Special Studies (when topic is appropriate)	
NSG 490	Special Studies (Study Abroad Experiences) (Australia, Kenya, Tanzania) (credits vary)	
SOWK 332	Community Mental Health Practice	
Subtotal		6-12
Nursing Major Electives		
Nursing majors must take the following courses:		
NSG 352	Clinical Applications and Reasoning in Nursing Care I	4
NSG 352L	Clinical Applications and Reasoning in Nursing Care I Clinical	2
Subtotal		6
Total Credits		18

¹ Health Sciences majors only

Civic Leadership Minor

Coordinator: Kara Dillard

Email: dillarkn@jmu.edu

Phone: (540) 568-3979

Civic Leadership Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/civic-leadership.shtml>)

Program Description

The minor program will provide students with the civic knowledge, skills and judgment to be engaged citizens that impact the world. It intends to prepare students to be citizen leaders in addressing today's most pressing civic problems. Embedded within the traditional academic fields of all of JMU's colleges, the minor provides the theories and skills needed for trans-disciplinary problem solvers and civic leaders.

Minor Requirements

Code	Title	Credits
Required Course		
CIV 300	Civic Leadership: Theories and Practice	3
Internship/Project		
Students completing the internship or project for three credits must then complete an additional course in the elective categories for a total of 12 elective credits.		
Choose one of the following:		3-6
CIV 400	Civic Leadership Internship	
CIV 401	Civic Leadership Project	
Equivalent Internship or Project within the student's major as approved by Civic Engagement Advisor		
Restricted Electives		
Students must take at least three credits from each of the following three elective categories outlined below.		9-12
Category 1: Critical, Historical, Legal and Political Foundations of Civic Leadership		
Category 2: Civic Science and Technology		
Category 3: Practicing Community Civic Leadership		
Total Credits		18

Category 1: Critical, Historical, Legal and Political Foundations of Civic Leadership

Understanding the critical, historical, legal and political foundations of civil society, civic behaviors and leadership for the common good.

Students must select at least three credits from the following courses:

Code	Title	Credits
ARTH 396	History Beyond the Classroom: Public History	3
or HIST 396	History Beyond the Classroom: Public History	
ARTH 471	Commemoration and Controversy: Public Art in America	3
or HIST 471	Commemoration and Controversy: Public Art in America	
CIV 150	Mechanics of Democracy for Civic Leadership	3
ECON 222	Contemporary Economic Issues and Policy Alternatives	3
ECON 405	Political Economy	3

ECON 426	Theory of Public Choice	3
EDUC 305	Critical Studies in Education, Race, Culture and Politics	3
EDUC 310	Teaching in a Diverse Society	3
GEOG 375	Political Geography	3
IA 375	Political Geography	3
GEOG 476	Sustainable Cities Seminar	3
HIST 343	History of Today	3
PHIL 267	Philosophy of Law	3
PHIL 335	The Individual, The State and Justice	3
POSC 225	U.S. Government [C4AE]	4
POSC 302	State and Local Government	3
POSC 351	Topics in American Politics	3
POSC 363	Hacking for Diplomacy (open to non-POSC majors with Political Science AUH approval)	3
POSC 364	Political Parties in America	3
POSC 369	Elections and Voting Behavior	3
POSC 378	Borderlands Policy: Latina/o/x Politics in Context	3
or LAXC 378	Borderlands Policy: Latina/o/x Politics in Context	
POSC 389	Hacking for National Security and Defense (open to non-POSC majors with Political Science AUH approval)	3
PPA 462	Social Welfare and Local Government Policy	3
PPA 484	Environmental Regulatory Policy and Politics	3
PSYC 220	Psychology and Culture	3
REL 376	Religion and Civic Leadership	3
REL 450	Religion and Society	3
SCOM 342	Argument and Advocacy	3
SCOM 346	Free Speech in America	3
SCOM 352	Communication and Social Movements	3
SCOM 353	American Political Culture and Communication	3
SMAD 450	Law of Media Arts and Design	3
SCOM/POSC/SMAD 472	Media and Politics	3

Category 2: Civic Science and Technology

Understanding the role of science, technology, and how to use science-based design thinking and development to improve civil society and civic life for the common good.

Students must select at least three credits from the following courses:

Code	Title	Credits
BIO 489	Topics in Science Pedagogical Content Knowledge	2
CHEM 481	Literature and Seminar I	1
CIS 424	Computer Security Management	3
CISE 401	Applied Ethics for Environmental Stewardship	3
CS 330	Societal and Ethical Issues in Computing	3
GEOG 325	Environmental Ethics	3
GEOG 376	Urban Geography	3
ISAT 212	Energy Issues in Science and Technology	3
ISAT 271	Technology, Science and Society	3
ISAT 290	Introduction to Systems Thinking for Complex Problems	3

ISAT 440	Cultural, Ethical, Legal and Social Issues in Computing	3
MGT 320	Management of Innovation and Technology	3
MGT 461	Introduction to Supply Chain Management	3
PHYS 388	Robots for Humanity	3
SMAD 470	New Media and Society	3
SOCI 315	Technology and Society	3

Category 3: Practicing Community Civic Leadership

Developing and practicing the skills needed to improve civic life for the common good.

Students must select at least three credits from the following courses:

Code	Title	Credits
ELED 455	Home, School, Communities	3
HON 200	Special Topics in Honors	1-6
MGT 457	Management Consulting Fundamentals	3
MGT 467	Advanced Management Consulting	3
NUTR 485	Community Nutrition	3
POSC 361	Topics in International Relations	3
PPA 475	Grant Writing and Project Management	3
SCOM 318	Practicum in Communication Studies	1-6
SCOM 325	Small Group Communication	3
SCOM 447	Communication Tools for Organizational Challenges	3
SCOM 453	Political Campaign Communication	3
SMAD 210	News Reporting and Writing	3
SOCI 386	Sociology of Nonprofit Organizations	3
FAM 375	Grant Writing for Agencies	3
GERN 375	Grant Writing for Agencies	3
NPS 375	Grant Writing for Agencies	3
SOWK 375	Grant Writing for Agencies	3
STAD 322	Equity, Inclusion and Access in Education and Performance	3
WRTC 314	Writing in the Public Sphere	3
WRTC 340	Writing as Leading	3
WRTC 462	Government Writing	3
WRTC 484	Writing for Nonprofits	3
WRTC 486	Writing in the Community	3

Classical Studies Minor

Co-Coordinator: Eva Carrara

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Phone: (540) 568-4673

Co-Coordinator: Maria Harvey

Email: harveymx@jmu.edu

Classical Studies Minor Website (<https://www.jmu.edu/cal/academics/minors/classical-studies/index.shtml>)

Program Description

The minor in classical studies introduces students to the historical, philosophical and artistic world of the ancient Mediterranean with a focus on Greece, Rome and Byzantium. Students will adopt an interdisciplinary approach to explore the history, literature, art and philosophy of these cultures. Students may also study classical reception by investigating how different cultures define themselves through and against the legacies of Greece and Rome.

Minor Requirements

The requirement for the minor in classical studies is 18 credit hours. Students who take Latin or Greek may apply up to 12 credit hours to the minor. In addition to the list of courses below, students may also count relevant summer study abroad courses of the relevant faculty and the classical studies minor co-coordinators.

Code	Title	Credits
Introductory Course		
Choose one of the following:		3
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	
HIST 101	World History to 1500 [C2HQC]	
HUM 250	Foundations of Western Culture [C2HQC] ¹	
Subtotal		3
Electives		
Choose five of the following:		15
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA] ²	
ARTH 424	Arts of Ancient Egypt	
ARTH 427	Advanced Topics in Pre-Modern Art ¹	
ARTH 497	Art History On-Campus Advanced Internship	
CLAS 100	Latin and Greek Roots of English Words	
CLAS 266	Greek and Roman Classics in Translation	
CLAS 338	Roman Religion	
CLAS 360	Topics in Greek and Roman Culture	
CLAS 490	Special Studies in Classics	
ENG 305	Mythology	
GRK 101	Elementary Greek I	
GRK 102	Elementary Greek II	
GRK 231	Intermediate Greek I	
GRK 232	Intermediate Greek II	
HIST 101	World History to 1500 [C2HQC] ²	
HIST 335	Topics in Applying History ¹	
HIST 365	Topics in Connected History ¹	

HIST 369	Problems in Greek History
HIST 370	Byzantine Empire
HIST 391	Travel Studies Seminar
HIST 467	The Roman Republic
HIST 325	The Roman Empire
HIST 489	Selected Topics in World History
HUM 250	Foundations of Western Culture [C2HQC] ²
ITAL 307	History of Italian Culture and Society
ITAL 437	Studies in Italian Literature ¹
LAT 101	Elementary Latin I
LAT 102	Elementary Latin II
LAT 231	Intermediate Latin I
LAT 232	Intermediate Latin II
PHIL 240	Greek Philosophy in Context
PHIL 340	Ancient Greek Philosophy
PHIL 343	Stoics, Skeptics and Epicureans
PHIL 460	Topics in Classical Philosophy
POSC 310	Ancient Political Theory: Classical Greek Political Philosophy to Roman Stoicism
REL 333	Angels, Demons and Dreams in Ancient Judaism and Christianity
REL 460	Topics in Ancient Jewish and Early Christian Literature
Subtotal	15
Total Credits	18

¹ When topic is on the ancient Mediterranean, Greece, Rome or Byzantium; or with instructor approval.

² If taken as a core requirement for the minor, this class cannot also count as an elective.

Climate Science Minor

Coordinator: Bill Lukens
Email: lukenswe@jmu.edu
Phone: (540) 568-7134

Climate Science Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/climate-science.shtml>)

Program Description

The Climate Science Minor introduces students to the domain knowledge and analytical skills necessary for understanding and investigating Earth's climate system. The program serves as an organizing framework for students interested in climate science and will broaden disciplinary expertise for advanced study, graduate programs and/or employment. The minor is structured in three tiers:

1. Introductory lecture and lab courses serve as an entry point to the program (4 credits total);
2. Content-specific courses build domain knowledge in climate science (minimum of 3 courses);
3. A minimum of one course focused on mathematical and/or data analytical skills provides tools necessary for working with scientific data; and
4. The remaining credit hours to fulfill the 18 credit-hour minimum requirement may consist of additional courses closely aligned with climate science, such as undergraduate research, internships and honors theses, upon approval by the program director(s).

Students are encouraged to discuss with their advisors or the program director(s) how climate science may interface with their primary degree program(s).

Minor Requirements

The Climate Science Minor requires a minimum of 18 credit hours, and no more than six credits from the minor may be used to double-count with a major or another minor.

Code	Title	Credits
Introductory Lecture and Lab		
GEOG 210	Physical Geography (2, 2)	4
or GEOL 115 & 115L	Earth Systems and Climate Change [C3NS] and Earth Systems and Climate Change Lab [C3L]	
Subtotal		4
Content-Specific Courses		
Choose a minimum of three courses from the following:		9-10
GEOG 301	Introduction to Natural Disasters Response	
GEOG 327	Climatology ¹	
GEOG 329	Global Climate Change ¹	
GEOG 330	Weather, Climate and Society	
GEOG 430	Dynamics of the Atmosphere ¹	
GEOL 211	Introduction to Oceanography [C3NS]	
GEOL 320	Meteorology	
GEOL 355	Geochemistry of Natural Waters ¹	
GEOL 406	Paleoclimatology and Paleoceanography ¹	
CHEM 353	Environmental Chemistry ¹	
BIO 354	Global Climate Change and Life: Ecological and Biological Impacts of Climate Variability ¹	

Subtotal 9-10

Analytical Skills

At least one course focused on analytical skills must be taken. 3-4
 Choose at least one course from the follow lists, and note that only one programming course or one geospatial course may be counted toward the 18 credit hour total for the minor.

Programming Courses:

CS 149	Introduction to Programming ¹
MATH 229	Introduction to Applied Statistics Using R [C3QR] ¹

Geospatial Courses:

GEOG 215	Introduction to Geographic Information Science and Cartography
GEOG 216	Earth Observation and Satellite Navigation Systems
GEOL 360	GIS for the Geoscientist ¹

Other Analytical Skills Courses:

GEOL 440	Geophysics (3, 2) ¹
GEOL 489	Quantitative Methods in Geology
GEOG 447	Climates Through Time ¹
ISAT 420	Environmental Analysis and Modeling
MATH 268	Data Analysis and Visualization ¹

Subtotal 3-4

Other Course Options

Students who have not fulfilled the 18 credit hours with the above courses may fill the remaining credit hours with a course or research experience approved by the program director(s). Course content must include a focus on Earth's climate system at any scale, and may include intersections with society, individuals and/or natural systems. Examples of course options include, but are not limited to, the following.¹

Choose 0-2 credit hours of the following: 0-2

GEOG 308	Environment, Nature and Society ²
GEOG 415	Environment, Landscape and Culture ²
GEOG 490	Senior Research or Field Practicum ^{2,3}
GEOG 495	Internship in Geography ³
GEOG 497	Independent Study ³
GEOG 499A	Honors in Geography ³
GEOG 499B	Honors in Geography ³
GEOG 499C	Honors in Geography ³
ENVT 200	Environmental Systems Theory
ENVT 225	Environmental Advocacy and the Psychology of Change
GEOL 398	Topics in Geology ³
GEOL 491	Geological Literature and Research ^{2,3}
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science ^{2,3}
GEOL 499A	Honors in Geology ²
GEOL 499B	Honors in Geology ³
GEOL 499C	Honors in Geology ³
ISAT 490	Capstone Experience I ³
ISAT 491	Senior Capstone Project I ³
ISAT 499A	Senior Honors Thesis I ³
ISAT 499B	Senior Honors Thesis II ³
ISAT 499C	Senior Honors Thesis III ³

ISAT 499D	Honors Thesis ³
Subtotal	0-2
Total Credits	18

¹ Students who have met the 18 credit hour requirement for the minor should speak to their advisor about the possibility of having the "Other Course Option" waived.

² These courses have prerequisites listed.

³ These courses require director approval.

Coaching Education Minor

Coordinator: Lori Gano-Overway

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Phone: (540) 568-7619

Coaching Education Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/coaching-education.shtml>)

Program Description

This minor provides students with the opportunity to gain knowledge and applied experiences to become competent sport coaches who have the tools to create a safe, fair and positive sport environment and promote holistic athlete development. The Coaching Education minor is accredited by the National Committee for Accreditation of Coaching Education (NCACE) and is aligned with the National Standards for Sport Coaches.

Minor Requirements

For successful completion of the minor, students must complete all courses in the minor and have an overall minor GPA of 2.0 or above.

Code	Title	Credits
Required Courses		
ATEP 205	Introduction to Athletic Training	3
KIN 202	Biological Foundations of Kinesiology	3
KIN 205	Introduction to Coaching	3
KIN 353	Developing Mental and Life Skills through Sport and Physical Education	3
KIN 425	Concepts of Strength and Conditioning	3
KIN 450	Principles of Coaching Methods	3
KIN 473	Internship in Coaching ¹	3
Total Credits		21

¹ To enroll in the internship, students must:

- Successfully complete all other minor courses;
- Complete the Coaching Principles course through Human Kinetics and earn certification (part of KIN 205);
- Present evidence of First Aid/CPR/AED certification, coaching certification in a particular sport, Child Abuse & Neglect training certification, and SafeSport certification or similar training; and
- Have a cumulative minor GPA of 2.0 or above.

Communication Studies Minor

Coordinator: Lee Mayfield

Email: mayfiecl@jmu.edu

Phone: (540) 568-3427

Communication Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/communication-studies.shtml>)

Program Description

The minor in Communication Studies is designed to provide students with principles and theories of communication processes that shape our relationships, workplaces and social change. Students have flexibility in choosing courses for the minor to complement many major fields of study and prepare them to be informed and ethical communicators.

Admission to the Minor

Admission to JMU does not guarantee admission to the School of Communication Studies. All students interested in minoring in Communication Studies must apply for a limited number of spaces by enrolling in and successfully completing SCOM 240 and SCOM 240L. To be eligible to register for these classes, students must add the minor in MyMadison by no later than March 1 for summer and fall registration and by October 1 for spring registration. Students who miss the deadline are still encouraged to declare the minor as SCOM will continue to process applications in the order in which they were received if there are still seats available in SCOM 240/SCOM 240L. Declaring the minor by this date does not guarantee availability of courses. The School of Communication Studies reviews performance in the SCOM 240/SCOM 240L class to make decisions about acceptance into the minor each semester.

Admission to the minor is based on availability to the most qualified students as determined by performance in SCOM 240 and SCOM 240L.

Students who are not admitted are permitted one more attempt (an attempt is any time the student receives a grade for the course on the transcript including W, WP, or WF) to successfully meet performance requirements in SCOM 240/SCOM 240L by enrolling the next regular semester or in the summer if the student desires. For students retaking SCOM 240/SCOM 240L (whether repeat or repeat/forgive), the school will look only at their highest grades earned when evaluating their second attempt.

Once admitted into a Communication Studies minor, a student cannot retroactively apply more than nine hours of SCOM courses, including SCOM 240 and SCOM 240L, to their minor (SCOM 121/SCOM 122/SCOM 123 does not count toward those nine hours).

Minor Requirements

Code	Title	Credits
SCOM 240	Introduction to Communication Theory	2
SCOM 240L	Communication Theory Lab	1
15 additional credits of SCOM courses beyond the General Education requirement. A minimum of 9 credits must be at the 300 level and a minimum of 3 credits must be at the 400 level.		15
Total Credits		18

Computational Analytics Minor

Coordinator: Debra Warne

Email: warneda@jmu.edu

Phone: (540) 568-2546

Computational Analytics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/computational-analytics.shtml>)

¹ These courses may require prerequisites that are not required in the minor.

Program Description

Computational analytics utilizes numerical techniques, programming and tools from applied mathematics to analyze and search for optimal solutions to real-world problems. In the modern world, the demand for computational skills and analytical abilities far surpasses the supply. The minor in computational analytics is designed for students who want to gain such expertise. This minor prepares students to solve complex problems using sound mathematical reasoning, quantitative skills and hands-on experience in computer programming. The minor also provides an opportunity to engage with fields currently in great demand such as operations research, mathematics of machine learning, artificial intelligence and data analytics.

The minor in computational analytics is open to any major.

Minor Requirements

Only credits from MATH 231-MATH 232, MATH 235 and MATH 236 may be used to double-count with the minor in mathematics. A student wishing to obtain both minors must fulfill a separate set of courses other than those used for the computational analytics minor for the additional 10 credits needed for the mathematics minor. The computational analytics minor is available to students majoring in mathematics or statistics but courses taken for the computational analytics minor cannot double count with courses beyond the core for the mathematics or statistics major. For example, a mathematics major cannot double count courses taken to fulfill the additional 12 credit hours for the math major (beyond the core) with courses taken to satisfy the six elective credits in the computational analytics minor. A statistics major may double count the required courses in the computational analytics minor toward the statistics major.

Code	Title	Credits
Required Courses		
MATH 231 & MATH 232 or MATH 235	Calculus with Functions I [C3QR] and Calculus with Functions II Calculus I [C3QR]	4-6
MATH 236	Calculus II	4
MATH 248	Computers and Numerical Algorithms	4
Subtotal		12-14
Restricted Electives		
Choose two of the following:		6
MATH 340	Operations Research I	
MATH 440	Fourier Analysis and Partial Differential Equations ¹	
MATH 448	Numerical Analysis	
MATH 449	Numerical Analysis for Differential Equations ¹	
Subtotal		6
Total Credits		18-20

Computer Information Systems Minor

Contact: CIS & Business Analytics Department

Email: CIS&BSAN-Minors@jmu.edu (CIS&BSAN-Minors@jmu.edu)

Phone: (540) 568-3255

Computer Information Systems Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/computer-information-systems.shtml>)

Program Description

The minor in computer information systems is structured to provide students in various disciplines on campus with the opportunity to study business-oriented information systems.

Special Admission Requirements

Admission to the CIS minor is limited and competitive and is based on a student's performance in one course from each of the following seven competency areas:

- One introductory information-systems course (COB 204 or equivalent as determined by the department head)
- One introductory computer programming course (CIS 221, ISAT 252, CS 149, CS 159, IA 241 or equivalent as determined by the department head)
- One calculus course (MATH 205, MATH 231, MATH 235, ISAT 151 or equivalent as determined by the department head)
- One statistics course (COB 191, MATH 220 or equivalent as determined by the department head)
- One critical thinking course (student's Madison Foundations critical thinking course or equivalent as determined by the department head)
- One writing course (WRTC 103 or equivalent as determined by the department head)
- One quantitatively oriented economics/business course (ECON 200, ECON 201, COB 241, COB 242, COB 291 or equivalent as determined by the department head)

In addition, a student must have a minimum of two full semesters remaining when starting the minor to complete the course requirements.

Admission is based on weighted average of student's highest grades in one course from each of the seven competency areas and is subject to space availability of CIS courses.

Students seeking to add the CIS minor must submit an application to the Department of Computer Information Systems and Business Analytics department when they have completed all seven classes on the application. Students enrolled in their final requisite course may apply but decisions will be deferred until final grades are posted. If this course is a transfer course, decisions cannot be made until the JMU registrar's office has received and officially posted the grade. If this is after admission decisions are made, the applicant will need to apply again the following semester. Applications are accepted every semester. See the website (<https://www.jmu.edu/cob/cis/academics/cis-minor.shtml>) for the CIS minor for the application link and current deadlines.

The CIS Minor is a competitive program with limited available seats. Admission is not guaranteed.

Minor Requirements

No more than one approved transfer course can be used to complete the requirements of the CIS minor. Transfer courses should be approved in advance.

Code	Title	Credits
COB 204	Computer Information Systems	3
CIS 221	Principles of Programming ¹	3
CIS 304	Enterprise Architecture	3
CIS 330	Database Design and Application ²	3
CIS 454	Systems Analysis and Design	3
CIS elective		3
Total Credits		18

¹ CS 149 or ISAT 252 or IA 241 with a grade of "C" or better or enrollment in CS 159 may be substituted for CIS 221.

² CS 374 may be substituted for CIS 330. CS 374 is only available to fully admitted computer science majors or minors.

Computer Science Minor

Advisor: Ellen Hedrick

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Phone: (540) 568-7488

Computer Science Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/computer-science.shtml>)

Program Description

The Computer Science minor gives students the opportunity to learn coding, problem-solving and algorithmic thinking, which are skills that complement virtually any major and help make students prepared to engage with emerging technological demands. By completing the minor, students will understand programming fundamental and be able to effectively collaborate with tech teams, bridge communication gaps and contribute effectively on software-related projects.

Admission Requirements

Admission to JMU does not guarantee admission to the Computer Science minor. Admission is limited, but the department will strive to accommodate all interested students who have demonstrated the potential to succeed in CS.

The Computer Science minor begins with a required two-course introductory sequence (CS 149 and CS 159) that is open to any student who meets the course prerequisites. CS 101 and CS 227/MATH 227 are optional courses that are also open to any student who meets the course prerequisites. Students wishing to continue in the CS minor beyond CS 159 must apply for full admission. The application procedure and requirements are available on the CS department site (<https://www.jmu.edu/cise/cs/index.shtml>).

Students may enroll in CS courses beyond CS 149, CS 159 and CS 227/MATH 227 only if they are fully admitted to the CS major or minor.

Students interested in the CS minor must complete the application process described below to request full admission.

Students who wish to apply for full admission to the CS minor must:

- Be enrolled at JMU
- Have declared the CS minor
- Have completed a minimum of 15 credit hours.
- Have completed or be currently enrolled in CS 159.
- Achieve a grade of "B-" or better in CS 159.
- Submit an official application that includes an essay of no more than 250 words to address a prompt that demonstrates students' commitment to the values of the Department of Computer Science.

Additional application and acceptance information:

- Students will be granted full admission to the CS minor as space permits; admission will be based on grades achieved in CS 159 and on faculty evaluation of the students' potential to succeed in the CS minor.
- There are two application cycles, once in the fall for spring admission, and once in the spring for fall admission.
- Students will be notified of their CS minor admission status no later than January 1st for admission for the spring semester, and no later than May 31st for admission for the fall semester.
- Students may apply to the CS major and/or minor two times, in total.

- Students who have attempted CS 149 at JMU may not attempt the course elsewhere (i.e. they must make any second attempt(s) at JMU).
- Non-transfer students must take CS 159 at JMU.
- An applicant's CS and calculus grades will be considered during the admission process, with a strong emphasis on CS grades. Final admission decisions will be made by a committee of CS faculty members.

Incoming transfer students will be placed into appropriate CS courses for their first semester after consultation with a transfer advisor based on placement test performance and other considerations. Students who are placed into CS 149 will be subject to the same eligibility requirements and application process as non-transfer students. Transfer students who are placed into courses other than CS 149 must apply for full admission at the end of their first semester. They will be evaluated by a committee of CS faculty members based on their performance in the first semester as well as the other elements of the admissions application. Transfer students must have credit for, or be taking, CS 159 the semester they apply for full admission to the minor. Transfer students who are denied admission on their first attempt may continue to take appropriate CS courses during a subsequent semester as determined through consultation with a transfer advisor. Admission decisions for the second attempt will be based on the full record of coursework at JMU.

Minor Requirements

Code	Title	Credits
Introductory Courses		
CS 149	Introduction to Programming	3
CS 159	Advanced Programming	3
Subtotal		6
Core Course		
Choose one of the following:		3
CS 240	Algorithms and Data Structures	
CS 261	Computer Systems I	
CS 345	Software Engineering	
Subtotal		3
Electives		
Choose three computer science courses numbered CS 101 or above, excluding CS 280. ¹		9
Total Credits		18

¹ Additional core courses completed (CS 240, CS 261 and CS 345) beyond the one required can be counted for this requirement.

Creative Writing Minor

Coordinator: Greg Wrenn
Email: wrenngj@jmu.edu
Phone: 540-568-3761
Creative Writing Minor Website (<https://www.jmu.edu/english/creative-writing/index.shtml>)

Program Description

The cross disciplinary minor in creative writing is designed to encourage students to develop their writing talents across a number of literary forms and communication contexts.

Course offerings in poetry, fiction, creative nonfiction, scriptwriting, screen- and playwriting give students the opportunity to shape the minor to suit a variety of artistic interests and professional objectives. With the choice of at least one course from a group involving advanced topics, narrative and poetic forms, media criticism and film analysis, students will gain informed perspectives on current issues affecting readers, viewers, writers and their creative works. These courses support the core workshop courses and are vital to competence in the field. Students electing this minor may acquire more information from the creative writing advisor of the Department of English, the School of Media Arts and Design or the School of Theatre and Dance.

Minor Requirements

The minimum requirement for a minor in creative writing is 18 hours. Up to two courses from the student’s major may be double counted toward the minor.

Code	Title	Credits
Required Courses		
Choose four or five core courses from two or more departments:		12-15
ENG 391	Introduction to Creative Writing - Nonfiction	
ENG 392	Introduction to Creative Writing - Poetry	
ENG 393	Introduction to Creative Writing - Fiction	
ENG 493	Advanced Creative Nonfiction	
ENG 494	Advanced Poetry Writing	
ENG 495	Advanced Fiction Writing	
SMAD 250	Scriptwriting	
SMAD 251	Screenplay Writing	
SMAD 340	Advanced Screenplay Writing	
SMAD 498	Senior Seminar in Media Arts and Design (when topic is appropriate) ¹	
THEA 347	Playwriting	
THEA 441	Professional Pathways (when topic is appropriate) ¹	
THEA 447	Advanced Playwriting	
Choose one or two support courses from the following:		3-6
ENG 390	The Environmental Imagination	
ENG 483	Narrative Form	
ENG 484	Poetic Craft and Creativity	
ENG 496	Advanced Topics in Creative Writing	
SMAD 373	Media Analysis and Criticism	
SMAD 463	Film Adaptations	

THEA 481	Theory and Performance Studies
Total Credits	18

¹ Students must check with the professor or creative writing coordinator to see if these courses are appropriate for this minor.

Criminal Justice Minor

Coordinator: Benjamin Meade

Email: meadebd@jmu.edu

Phone: (540) 568-7329

Criminal Justice Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/criminal-justice.shtml>)

Program Description

The cross-disciplinary minor in criminal justice is designed for students who are preparing for careers in law enforcement, corrections, judicial administration or other areas related to the study or management of crime, either directly upon graduation or after further graduate training.

Minor Requirements

The requirement for a minor in criminal justice is 18 credits. Students should allow a minimum of four semesters in which to complete the minor. Students majoring in justice studies may not declare the criminal justice minor.

Code	Title	Credits
Required Courses		
JUST 210	Crime and Criminal Justice	3
or CRJU 215	Introduction to Criminal Justice	
CRJU 300	Ethics in Criminal Justice	3
Subtotal		6
Restricted Electives		
Choose four courses with at least one course from each of the following categories:		12
Criminology and Criminal Behavior:		
JUST 212	Theories of Crime and Criminal Justice	
JUST 307	Justice and Crime	
JUST 309	Global Crime	
JUST 311	Women, Crime and Criminal Justice	
JUST 312	Mass and Serial Murder	
JUST 316	Human Development and Crime	
JUST 317	Victimization of Children	
JUST 318	Sex Offenders	
JUST 320	Organized Crime	
JUST 326	Victimology	
JUST 361	Terrorism	
PSYC 250	Introduction to Abnormal Psychology ¹	
or PSYC 335	Abnormal Psychology	
SOCI 304	Social Deviance	
Criminal Justice Institutions:		
CRJU 335	Law Enforcement	
CRJU 337	Courts and the Judiciary	
CRJU 340	Administration of Justice	
JUST 324	Death Penalty	
JUST 330	Corrections	
JUST 345	Restorative Justice	
Criminal Justice, Law and Procedure:		
CRJU 321	Criminalistics	
CRJU 328	Criminal Procedure	
CRJU 329	Criminal Investigation and Evidence	

JUST 323	Comparative Criminal Justice
JUST 327	Criminal Law
JUST 329	Perspectives on Law
JUST 334	Media and Justice
JUST 397	Legal Research and Information Literacy
Special Topics and Internship:	
The following courses are not required but may also be used for restrictive elective credit:	
CRJU 301	Advanced Level Special Topics in Criminal Justice
CRJU 401	Internship in Criminal Justice ²
Subtotal	12
Total Credits	18

¹ Students majoring in this discipline may double count only one course as credit for the minor and that major.

² This course may be taken only once for credit in the Criminal Justice minor.

Cultural Communication Minor

Coordinator: Reslie Cortés

Email: cortesrx@jmu.edu

Phone: (540) 568-4158

Cultural Communication Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/cultural-communication.shtml>)

Program Description

This minor empowers students to communicate reflectively in a diverse, multicultural and global society. With a distinct focus on critical and cultural communication theories and social justice frameworks, students are equipped with opportunities and tools necessary for understanding culturally complex circumstances, issues and needs that arise in personal relationships, civic settings and other social and mediated contexts. Skills developed in this minor prepare students with confidence to more effectively listen, analyze, adapt and respond to communication contexts in ways that cultivate inclusive environments, cultures and just relations.

Admission to the Minor

Admission to JMU does not guarantee admission to the School of Communication Studies. All students interested in minoring in Cultural Communication must apply for a limited number of spaces by enrolling in and successfully completing SCOM 240 and SCOM 240L. To be eligible to register for these classes, students must add the minor in MyMadison by no later than March 1 for summer and fall registration and by October 1 for spring registration. Students who miss the deadline are still encouraged to declare the minor as SCOM will continue to process applications in the order in which they were received if there are still seats available in SCOM 240/SCOM 240L. Declaring the minor by this date does not guarantee availability of courses. The School of Communication Studies reviews performance in the SCOM 240/SCOM 240L class to make decisions about acceptance into the minor each semester.

Admission to the minor is based on availability to the most qualified students as determined by performance in SCOM 240 and SCOM 240L.

Students who are not admitted are permitted one more attempt (an attempt is any time the student receives a grade for the course on the transcript including W, WP, or WF) to successfully meet performance requirements in SCOM 240/SCOM 240L by enrolling the next regular semester or in the summer if the student desires. For students retaking SCOM 240/SCOM 240L (whether repeat or repeat/forgive), the school will look only at their highest grades earned when evaluating their second attempt.

Once admitted into a Communication Studies minor, a student cannot retroactively apply more than nine hours of SCOM courses, including SCOM 240 and SCOM 240L, to their minor (SCOM 121/SCOM 122/SCOM 123 does not count toward those nine hours).

Minor Requirements

Code	Title	Credits
Required Courses		
SCOM 240	Introduction to Communication Theory	2
SCOM 240L	Communication Theory Lab	1
SCOM 248	Intercultural Communication	3

SCOM 349	Ethnographic Approaches to Communication Studies	3
SCOM 448	Communication, Culture and Identity	3
Subtotal		12
Electives		
Choose two of the following:		6
SCOM/ANTH 305	Language and Culture	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM/WGSS 348	Communication and Gender	
SCOM 352	Communication and Social Movements	
SCOM 355	Food as Communication	
SCOM 356	Immigrant Advocacy	
SCOM/SMAD 357	Youth, Communication and Culture	
SCOM 395	Study Abroad Seminar (only 3 credits may be applied to this requirement)	
SCOM/WGSS/ WRTC 420	Feminist Rhetorics	
Subtotal		6
Total Credits		18

Dance Minor

Coordinator: Javi Padilla
Email: padillji@jmu.edu
Dance Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/dance.shtml>)

Program Description

The JMU dance minor allows students to explore and develop their skills and knowledge in the field of dance while pursuing a major in another subject. The minor combines theoretical coursework along with practical dance training to enable students to begin a journey toward becoming well-rounded artists. Dance minors can select from a diverse menu of options, including ballet, modern, jazz, tap, dance technology, and somatic studies, complemented by required courses in dance history and improvisation.

No audition or portfolio review is required for the dance minor.

Minor Requirements

Code	Title	Credits
Required Course		
DANC 215	The Dance Experience [C2VPA]	3
Subtotal		3
Electives		
Dance technique and elective courses with approval of minor coordinator		12
Electives selected from dance-designated courses and/or dance related courses at the 300- or 400-level with approval of minor coordinator		4
Subtotal		16
Total Credits		19

Data Analytics Minor

Advisor: Karen Barnard-Kubow
Email: barnarkb@jmu.edu
Phone: (540) 568-6225
Data Analytics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/data-analytics.shtml>)

Program Description

The minor in data analytics is designed for students majoring in a STEM field who want to gain additional technical competencies in statistical data analytic methods as it relates to their field of study. This minor will provide students with analytical and technical skills in the ethical use of data in research environments; application of statistical and probabilistic thinking to data analysis; data visualization and data management; and application of machine learning techniques to their own field of study. Although the minor is primarily intended for students majoring in one of the degree programs offered by the College of Science and Mathematics (p. 26), any student that meets program requirements may take the minor.

Minor Requirements

The minor in data analytics requires 18 credit hours. No more than six credits from majors and other minors may be used to double-count toward this minor.

Code	Title	Credits
Required Courses		
DATA 200	Introduction to Data	3
MATH 229	Introduction to Applied Statistics Using R [C3QR]	3
MATH 268	Data Analysis and Visualization	3
DATA/MATH 358	Introduction to Machine Learning	3
Subtotal		12
Restricted Electives		
Choose at least six credit hours from one or more of the following foci:		6
Mathematics/Statistics Focus:		
MATH 248	Computers and Numerical Algorithms	
MATH 300	Linear Algebra	
MATH 309	SAS Programming and Data Management	
MATH 322	Applied Linear Regression	
MATH 327	Categorical Data Analysis	
MATH 328	Time Series Analysis	
MATH 368	Statistical Computing	
MATH 421	Applied Multivariate Statistical Analysis	
MATH 426	Probability and Mathematical Statistics I	
Biological Sciences Focus:		
BIO 204	Viral Genome and Bioinformatics	
BIO/MATH 342	Mathematical Models in Biology	
BIO 346	Laboratory in Bacterial Pathogenomics	
BIO 454/MATH 354	Introduction to Biometrics	
BIO 457	Biological Applications of Geographic Information Systems	
BIO 481	Genomics	
BIO 483	Bioinformatics	

Earth Sciences Focus:	
GEOL 360	GIS for the Geoscientist
GEOL 390	Laboratory Techniques in Geology (2,2)
GEOL 406	Paleoclimatology and Paleoceanography
GEOL 410	Engineering Geology (2, 2)
GEOL 440	Geophysics (3, 2)
GEOL 489	Quantitative Methods in Geology
Physical Sciences Focus:	
PHYS 339	Introductory Nuclear Science
PHYS 344	Advanced Physics Laboratory I
PHYS 345	Advanced Physics Laboratory II
PHYS 346	Advanced Physics Laboratory III
PHYS 386	Robots: Structure and Theory
PHYS 390	Computer Applications in Physics
CHEM 351	Analytical Chemistry
CHEM 352 & 352L	Instrumental Analysis and Instrumental Analysis Laboratory
Subtotal	6
Total Credits	18

Digital Studies Minor

Coordinator: Johannes Burgers

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Phone: (540) 568-6110

Digital Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/digital-studies.shtml>)

Program Description

This cross-disciplinary minor allows students to gain hands-on experience with powerful digital technologies and study the implications of these technologies with a community of thinkers. Courses are welcoming to students of all academic backgrounds and technical skills. Students use digital methods to open up new perspectives in the humanities and social sciences; explore how the digital impacts our everyday lives and local and global communities; and enrich their future careers through experience and practice within the digital landscape. The Digital Studies minor is open to all JMU undergraduate students.

Minor Requirements

The minor requires successful completion of 18 credit hours, including DS 101 Foundations in Digital Studies. DS 101 is not a prerequisite to other Digital Studies courses and can be taken at any time. Students may take no more than 6 credit hours from a single discipline. Up to 6 credits from the student's major can double count for the Digital Studies minor.

Other courses outside of this list may be counted towards the minor with the approval of the minor steering committee on a semester-by-semester basis. A maximum of two courses (6 credit hours) can count in this way. Courses must be approved before the student enrolls.

Code	Title	Credits
Required Core Course		
DS 101	Foundations in Digital Studies	3
Subtotal		3
Electives by Academic Unit		
Choose 15 credit hours from the following. Students may take no more than 6 credits from a single discipline.		15
School of Communication Studies:		
SCOM 313	Topics in Communication Studies (when topic appropriate; must be approved by the minor coordinator before registering) ¹	
SCOM 318	Practicum in Communication Studies (when topic appropriate; must be approved by the minor coordinator before registering) ¹	
Department of English:		
ENG 221	Literature and Ideas [C2L] (when topic appropriate; must be approved by the minor coordinator before registering)	
ENG 302	Special Topics in Literature and Language (Print and Digital Literary Publishing)	
ENG 307	Literature and Ideas (Literature and Ideas: Fuzzy Cartographies)	
ENG 348	Studies in Linguistics and the English Language (when topic appropriate)	
ENG 380	Introduction to Film	

ENG 408	Advanced Studies in African American Literature (when topic Local Black Literature or when appropriate; must be approved by the minor coordinator before registering)
ENG 412	Special Topic Seminar (Video Game Studies)
ENG 417	Advanced Studies in Linguistics and the English Language (when topic appropriate; must be approved by the minor coordinator before registering)
ENG 420	Advanced Studies in Theory and Cultural Studies (when topic appropriate; must be approved by the minor coordinator before registering)
Department of History:	
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas
HIST 407	Digital History
Department of Justice Studies:	
JUST 301	Advanced Level Special Topics in Justice Studies (when topic appropriate; must be approved by the minor coordinator before registering) ¹
JUST 339	Mapping Justice ¹
JUST 365	Justice in Literature, Film and Art ¹
School of Media Arts and Design:	
SMAD 327	User-Centered Research and Usability Evaluation ¹
SMAD 470	New Media and Society ¹
Department of Sociology and Anthropology:	
ANTH 302	The Archaeology of Empires and Rebels
ANTH 303	Andean Archaeology
ANTH 410	Spatial Analysis for Social Research
SOCI 328	Digital Inequality
Department of World Languages and Cultures:	
JAPN 447	Special Topics in Japanese Culture (when topic Japanese Film, TV and Media or when appropriate; must be approved by the minor coordinator before registering)
SPAN 365	Spanish for Medical Professionals I
SPAN 446	Special Topics in Spanish Literature (Spanish Civil War Through Literature and Film) ¹
SPAN 475	Spanish for Medical Professionals II ¹
School of Writing, Rhetoric and Technical Communication:	
WRTC 332	Creative Approaches to Digital Storytelling ¹
WRTC 338	Sound Writing ¹
WRTC 352	Online Design
WRTC 412	Content Strategy ¹
WRTC 426	Special Topics in Writing, Rhetoric and Technical Communication (when topic appropriate; must be approved by the minor coordinator before registering) ¹
WRTC 447	Interaction Design ¹
WRTC 450	Digital Rhetoric ¹
Subtotal	15
Total Credits	18

¹ Course has a prerequisite, which can be waived on a student-by-student basis.

Disability Studies Minor

Co-Coordinator: Daisy Breneman
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Phone: (540) 568-4247

Co-Coordinator: Susan Ghiaciuc
Email: ghiacisx@jmu.edu
Phone: (540) 568-1613

Disability Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/disability-studies.shtml>)

Program Description

The minor in disability studies engages students in an examination of the many ways disability is defined and constructed. Students will build knowledge of the political, rhetorical, social, educational, historical, legal and medical frameworks of disability through both classroom and experiential study in courses drawn from several disciplines. Students will examine the ways systems inhibit and promote full participation of people with disabilities, explore how communities define and reinforce norms and understand disability as an inherent part of diversity and inclusivity.

The minor in disability studies is open to all undergraduate students at JMU.

Minor Requirements

The introductory course DST 100 can be taken the same semester as core courses. More than one core course can be taken to count toward the overall credit requirement. No more than nine hours may be taken from a single discipline. No more than two courses may be double counted for a student's major/minor.

Code	Title	Credits
Required Course		
DST 100	Introduction to Disability Studies	1
Subtotal		1
Core Course		
Choose one of the following: ¹		3
JUST 385	Disability and Justice	
KIN 429	Special Topics in Adapted Physical Education	
SOCI 335	Disability and Society	
SOCI 375	Medical Sociology	
WRTC 470	Disability Rhetoric	
Subtotal		3
Elective Courses		
Choose five of the following:		15
ANTH 360	Medical Anthropology	
ANTH 395	Special Topics in Anthropology (Sex and the Social: The Cultural Politics of Reproduction)	
STAD 322	Equity, Inclusion and Access in Education and Performance	
DANC 325	Dance in Community	
HIST 306	A History of the Body in the West	
JUST 221	Social Justice Theories	
REL 371	Religion and Disability	
REL 372	Religion and Medicine, East and West	

SCOM 370	Introduction to Health Communication
SCOM 371	Disability and Advocacy
SOCI 304	Social Deviance
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness
SRM 242	Sociology and Psychology of Sport and Recreation
SRM 436	Facilities and Event Management
Subtotal	15
Total Credits	19

¹ Additional courses taken from this list will count toward required electives.

Economics Minor

Advisor: College of Business Advising and Support Center (<https://www.jmu.edu/cob/asc/index.shtml>)

Email: cobasc@jmu.edu

Phone: (540) 568-5948

Economics Minor Website (<https://www.jmu.edu/cob/economics/academics/economics-minor.shtml>)

Program Description

The minor in economics will allow you to develop your critical thinking, research, writing and analytical skills as you build a foundation in economic theory and policy. These concrete and marketable skills can complement and add significant value to a wide variety of undergraduate majors and help you to stand out among your peers.

Minor Requirements

Code	Title	Credits
Required Courses		
ECON 200	Introduction to Macroeconomics [C4GE] ¹	3
ECON 201	Introduction to Microeconomics ¹	3
Subtotal		6
Minor Electives		
Take 12 additional credits of ECON courses, at least 6 of which must be at the 300- or 400-level. ²		12
Total Credits		18

¹ Credit in ECON 200 will be granted to students who achieve a grade of 4 or 5 on the Advanced Placement Test in Macroeconomics administered by the Educational Testing Service. Credit in ECON 201 will be granted to students who achieve a grade of 4 or 5 on the Advanced Placement Test in Microeconomics.

² Students may not receive credit toward the minor in economics for both ECON 270 and ECON 370.

Preparation for Graduate Study in Economics

While not formal requirements, students intending to pursue graduate work in economics are encouraged to select as many of the following courses as possible:

Code	Title	Credits
ECON 384	Mathematical Economics	3
ECON 430	Monetary Theory	3
ECON 431	Advanced Microeconomic Theory	3
ECON 432	Advanced Macroeconomics	3
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 238	Linear Algebra with Differential Equations	4
MATH 336	Elementary Differential Equations	3
PHIL 250	Introduction to Symbolic Logic	3

Educational Media Minor

Advisor: Terri Yoder

Email: hrm5xn@jmu.edu

Phone: (540) 568-5361

Educational Media Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/educational-media.shtml>)

Program Description

The educational media minor prepares students for employment in education, business, communications, non-profit and consulting fields where the effective design of information, instruction and media are important. Students who minor in educational media will develop psychological and technological skills intended to enhance their major program of study and prepare them for the workplace. Students who successfully complete the minor will possess skills appropriate for teaching, training, designing instruction and developing related multimedia. Underlying theories and concepts integrated into the course work include but are not limited to the diffusion of information, learning theory, message design, group dynamics and materials evaluation.

Minor Requirements

The educational media minor requires a minimum of 18 credit hours selected in consultation with an advisor for this program.

Code	Title	Credits
Required Courses		
EDUC 175	Learning, Behavior and Engagement in Context [C5SD]	3
or PSYC 160	Life Span Human Development [C5SD]	
LDT 150	Information in Contemporary Society	3
LDT 370	Instructional Technology	3
LDT 372	Visual Literacy	3
LDT 380	Digital Media for Learning	3
LDT 385	Foundations of Instructional Design	3
Total Credits		18

English Minor

Advisor: Breanne Weber
Email: weberbe@jmu.edu
Phone: (540) 568-2500
English Minor Website (<https://www.jmu.edu/english/undergraduate/english-minor.shtml>)

Program Description

The English minor offers students an opportunity for further study in literature, literary history, and criticism beyond the General Education requirement. The minor strengthens skills in written and oral communication and critical thinking, expands students' appreciation of diverse perspectives on our shared world, and may complement and enhance learning in any number of majors, from the humanities and social sciences to finance, computing or the performing arts.

Minor Requirements

Code	Title	Credits
The minimum requirement for a minor in English is 18 credit hours. At least nine hours must be taken in courses at the 300 level or above. A minor in English does not meet requirements for the Secondary Teaching License.		
Total Credits		18

Entrepreneurship Minor

Coordinator: Chris Roeder

Email: roedercc@jmu.edu

Phone: (540) 568-3143

Entrepreneurship Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/entrepreneurship.shtml>)

Program Description

The entrepreneurship minor is designed to introduce students from any discipline to the knowledge, skills and mindset necessary for successful new venture creation. The minor is open to students in any major except management, which offers a track in innovation and entrepreneurship.

Students who successfully complete the entrepreneurship minor will develop the knowledge base and skills needed for new venture creation and management. They will be able to:

- Understand entrepreneurship as a possible career path, its role in economic growth and opportunity, and its associated social benefits.
- Recognize/identify entrepreneurial opportunities that fit the individual student. Includes self-analysis in the context of possible entrepreneurial activities and demonstration of the ability to identify and analyze the attractiveness of potential markets.
- Develop a business model and create a business plan. This includes completing a clear and concise written plan that serves as a blueprint for creation of the proposed venture.
- Apply relevant financial and business principles to assess the need for start-up capital, conduct break even analysis and identify cash flow needed for future growth.
- Identify and evaluate various sources of funding.
- Articulate an effective pitch to gain support from investors, suppliers, customers or potential employees.
- Interact and collaborate with various stakeholders to develop and evaluate ideas, and grow those ideas to fruition.

Special Admission Requirements

Admission to the minor is competitive. To be considered for admission into the entrepreneurship minor students will need to submit an online application. Applications are accepted from September 8th - September 19th in the fall, and from February 8th - February 19th in the spring. The application includes:

- A completed application form.
- An unofficial transcript.
- Responses to essay questions on the applicant's entrepreneurial experience and goals.
- Proof of good academic standing.

Minor Requirements

Code	Title	Credits
Required Courses		
MGT 310	Business Fundamentals for Entrepreneurs ¹	3
or COB 300A	Integrative Business: Management	
MGT 320	Management of Innovation and Technology ²	3
MGT 471	Entrepreneurship	3
MGT 472	New Venture Creation	3
Subtotal		12

Electives

Choose two of the following: ³		6
ACTG 244	Accounting for Non-Business Majors	
FIN 345	Finance Fundamentals for Non-Business Majors	
ISAT 480	Topics in Integrated Science and Technology	
MGT 340	International Management	
MGT 365	Fundamentals of Human Resource Management	
MKTG 380	Principles of Marketing	
WRTC 340	Writing as Leading	
Subtotal		6
Total Credits		18

¹ Non-business majors should take MGT 310. Business majors should take COB 300A.

² Engineering students may substitute ENGR 426 for MGT 320.

³ Students may request permission to count two three-credit electives from their major as their ENTP electives. These courses must be related to innovation and/or entrepreneurship, and they must be approved by the minor coordinator.

All electives must be approved by the coordinator. Students should check with an advisor or the coordinator for the most updated list of suggested electives, as the list can change.

Environmental Humanities Minor

Coordinator: Katey Castellano
Email: castelkm@jmu.edu
Phone: (540) 568-3753
Environmental Humanities Minor Website (<https://www.jmu.edu/environment/humanities.shtml>)

Program Description

This minor allows students to articulate broad historical and cultural views of the complex interactions among humans and the natural and material environment. Students will analyze and produce expressive, creative work that illustrates the complexity of human interaction with the environment.

Minor Requirements

Students select 18 hours as distributed below. No more than three courses can come from a single academic unit in the total hours selected. Students may double-count two courses in their major with the minor. At the discretion of the minor coordinator, science courses may be substituted into the minor.

Code	Title	Credits
Humanities and the Environment		
Choose four of the following, with no more than two from the same discipline:		12
ARCD 330	Materials and Methods	
ENG 221	Literature and Ideas [C2L] (only when the topic is "Environmental Literature")	
ENG 371	Literature and the Environment	
ENG 390	The Environmental Imagination	
ENVT 300	Topics in Environmental Humanities	
ENVT 301	Internship in Environmental Humanities	
ENVT 401	Studio or Independent Study in Environmental Humanities	
GEOG 325	Environmental Ethics	
HIST 304	Native Peoples of the United States	
HIST 315	Global History of Sport	
HIST 344	Magic Mountains, Man-eating Beasts and Unquiet Woods: An Environmental History of British India	
HIST 353	The Global Middle East	
HIST 374	The Southern Plantation	
HIST 427	U.S. Environmental History	
HIST 452	Parks and Recreation: History of National Parks	
HIST 479	Global Environmental History: Mountains, Rivers and Animals in Asia and Africa	
HIST 485	Colonial Knowledge Production in the Middle East and South Asia	
INDU 392	Topics in Industrial Design (only when the topic is "Design and Sustainability")	
REL/ENVT 366	Religion, Animals and the Environment	
SCOM/WGSS 302	Third Wave Ecofeminism	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	

SCOM 465/ WRTC 416	Rhetoric of Environmental Science and Technology
SOCI 311	Sociology of the Environment
SOCI 357	Sociology of Disasters
SOCI 359	The Sociology of Climate Denial
Approved Special Topics Courses	
Approved Internship Programs	
Approved Study Abroad Programs	
Subtotal	12
Humanities/Social Science Elective	
Choose one of the following:	
3	
ANTH 300	The Anthropology of Food
ANTH 307	Climate, Culture and Change
ARTH 380	American Art to 1870 (for landscape, painting, nationalism)
ENG 325	Romantic Literature
ENG 363	Native American Literature
ENG 496	Advanced Topics in Creative Writing (Specific instructor's section only - see minor coordinator)
HIST 326	How Cars Changed Everything
HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies
HIST 405	Travel and Exploration
HIST 443	Modern American Technology and Culture
HIST 453	Patterns of Global History
HIST 469	A History of International Development in the Twentieth Century
HUM 252	Global Cultures [C2HQC] ("Global South Asia" only)
HUMN 301	Introduction to Natural Disasters Response
ISAT 421	Environmental Policy and Regulation
PPA 484	Environmental Regulatory Policy and Politics
SOCI 354	Social Inequality
SOCI 358	Sociology of Consumption
WRTC 342	Writing Place
WRTC 358	Writing About Science and Technology
Subtotal	3
Science Elective	
Choose one of the following: ¹	
3	
ENVT 200	Environmental Systems Theory
GEOG 322	Agricultural Systems
GEOG 341	Wilderness Studies
GEOL 102	Environment: Earth [C3NS]
GEOL 115	Earth Systems and Climate Change [C3NS]
GEOL 211	Introduction to Oceanography [C3NS]
HTH 352	Environmental Health
ISAT 100	Environmental and Energy Sustainability [C3PP]
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]
NUTR 380	Global Nutrition
Subtotal	3
Total Credits	
18	

¹ General Education courses may double count for the minor.

Environmental Information Systems Minor

Coordinator: Steven P. Frysinger

Email: frysinsp@jmu.edu

Phone: (540) 568-2710

Environmental Information Systems Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/environmental-information-systems.shtml>)

Program Description

The cross-disciplinary minor in environmental information systems is designed for undergraduates interested in using computer and information management technology to solve environmental problems and improve environmental stewardship. Some examples of environmental information systems are database systems to track and report hazardous materials in factories, decision support systems to facilitate risk analysis and management, GIS-based natural resource inventory systems and automated business management systems to support and document environmental compliance.

Minor Requirements

The environmental information systems minor requires a minimum of 25 credit hours. Core courses are intended to ensure knowledge of the foundation disciplines. Electives should be chosen in consideration of the student's particular interests within the general field of environmental information systems; the coordinator may approve other elective courses. Students are advised to check prerequisites of listed courses.

Code	Title	Credits
Core Courses		
CS 149	Introduction to Programming ¹	3
ISAT 340	Database Design, Implementation and Management ¹	3
ISAT 320	Fundamentals of Environmental Science and Technology I	4
ISAT 321	Fundamentals of Environmental Science and Technology II	3
ISAT 426	Environmental Information Systems	3
Subtotal		16
Elective Courses		
Choose three of the following:		9
CS 159	Advanced Programming ¹	
GEOG 215	Introduction to Geographic Information Science and Cartography	
GEOG 366	Intermediate Geographic Information Science	
GEOG 385	Digital Image Processing in Remote Sensing	
GEOG 466	GIS and Geographic Databases	
ISAT 341	Machine Learning and Data Science	
ISAT 420	Environmental Analysis and Modeling	
ISAT 447	Interaction Design	
Subtotal		9
Total Credits		25

¹ Or equivalent by permission of director. ISAT 252 or CIS 221 substitute for CS 149; CIS 330 or CS 374 substitute for ISAT 340; CIS 331 substitutes for CS 159.

Environmental Management Minor

Coordinator: Steven P. Frysinger
Email: frysinsp@jmu.edu
Phone: (540) 568-2710
Environmental Management Minor Website (<https://www.jmu.edu/environment/minors/management.shtml>)

Program Description

The cross disciplinary environmental management minor prepares students to apply the principles of environmental science and engineering to contemporary environmental problems in natural resource, industrial and public policy contexts. The minor is particularly suitable for students interested in professional careers in industrial environmental management, natural resources management, and environmental policy and planning.

The environmental management minor strives to develop graduates who can apply science and technology to a broad range of practical environmental problems in professional settings. Students are expected to be literate and competent in the sciences and mathematics underlying environmental problem solving.

Minor Requirements

The environmental management minor requires a total of 29-32 credits, including prerequisite courses. Prerequisite courses may be fulfilled as part of the student's major. Other courses may apply by permission of the coordinator.

Code	Title	Credits
Prerequisites		
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	4
Choose one of the following:		3-4
ISAT 251	Topics in Applied Statistics in ISAT [C3QR]	
MATH 220	Elementary Statistics [C3QR]	
MATH 285	Data Analysis	
MATH 307	Principles of Probability and Statistics	
Subtotal		7-8
Required Courses		
ISAT 320	Fundamentals of Environmental Science and Technology I	4
ISAT 321	Fundamentals of Environmental Science and Technology II	3
CHEM 241	Organic Chemistry I	3
ENVT 400	Capstone Seminar in Environmental Problem Solving	3
Subtotal		13
Concentrations		
Students completing the environmental management minor must concentrate in one of three areas. See below for details.		9-11
Natural Resources		
Industrial Systems		
Environmental Policy		
Total Credits		29-32

Concentrations

Students completing the environmental management minor must concentrate in one of three areas: natural resources, industrial systems or environmental policy. Students should be aware that some of the listed courses may have additional prerequisites.

Natural Resources

Code	Title	Credits
ISAT 424	Natural Resource Management	3
Choose two of the following:		6-8
BIO 402	Forest Ecology	
BIO 457	Biological Applications of Geographic Information Systems	
BIO 459	Freshwater Ecology	
CHEM 354	Environmental Chemistry Field Camp	
CHEM/GEOL 355	Geochemistry of Natural Waters	
GEOG 340	Biogeography	
GEOG 341	Wilderness Studies	
GEOG 342	Management and Protection of Natural Resources	
GEOG 343	Wildlife Management	
GEOL 430	Environmental Soil Science (2, 2)	
ISAT 420	Environmental Analysis and Modeling	
ISAT 425	Environmental Hydrology	
ISAT 429	Sustainability: An Ecological Perspective	
Total Credits		9-11

Industrial Systems

Code	Title	Credits
ISAT 422	Industrial Environmental Management	3
Choose two of the following:		6
HTH 352	Environmental Health	
HTH 450	Epidemiology	
ISAT 423	Environmental Remediation	
ISAT 427	Industrial Hygiene	
ISAT 428	Industrial Ecology	
Total Credits		9

Environmental Policy

Code	Title	Credits
ISAT 421	Environmental Policy and Regulation	3
ISAT 422	Industrial Environmental Management	3
or ISAT 424	Natural Resource Management	
Choose one of the following:		3
ECON 305	Environmental Economics	
ECON 340	Economics of Natural Resources	
GEOG 325	Environmental Ethics	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
HIST 427	U.S. Environmental History	
ISAT 411	Energy Economics and Policy	
ISAT 420	Environmental Analysis and Modeling	
ISAT 423	Environmental Remediation	

PPA 484	Environmental Regulatory Policy and Politics	
SCOM 354	Environmental Communication and Advocacy	
SOCI 311	Sociology of the Environment	
Total Credits		9

Environmental Science Minor

Coordinator: Bruce A. Wiggins

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Phone: (540) 568-6196

Environmental Science Minor Website (<https://www.jmu.edu/environment/minors/science.shtml>)

Program Description

This minor is a cross disciplinary program that any student may elect. Students pursuing programs ranging from the physical, natural or social sciences to education, journalism or business could benefit from this broadly-based curriculum. The program draws from courses that focus on the application of scientific concepts and principles to the understanding of environmental problems and their solutions. The minor draws upon the expertise of faculty in the areas of biology, chemistry, geography, physics and integrated science and technology.

The environmental science minor:

- Provides a scientific background to those students interested in environmental law, environmental economics and environmental sustainability.
- Broadens the student's understanding of how sciences are linked to environmental questions.
- Complements any major by focusing on courses related to environmental issues.

Minor Requirements

The minimum requirement for a minor in environmental science is 24 credit hours taken from the four groups outlined below. Students wishing to complete more than one of the environmental minors (environmental management, environmental science and environmental studies) may receive dual credit for the capstone course (ENVT 400), but may not receive dual credit for any other courses that might be shared by the minors. Pre-approved study abroad and/or internship experiences may be substituted for one or more of the courses listed below.

No more than two courses from a single subject area can count toward the completion of the environmental science minor.

With the pre-approval of the environmental science minor coordinator, international studies, special topics courses, study abroad courses, and/or internship experiences with appropriate content may be counted toward the minor.

Courses

Code	Title	Credits
Group 1. Introduction to Environmental Science		
Choose one of the following:		3-4
BIO 250 & 250L	Quantitative Ecology and Evolution and Quantitative Ecology and Evolution Laboratory	
ENVT 101	Perspectives in Environmental Science [C3PP]	
GEOG 210	Physical Geography (2, 2)	
GEOL 102	Environment: Earth [C3NS]	
GEOL 110	Physical Geology [C3NS]	
GEOL 115	Earth Systems and Climate Change [C3NS]	
ISAT 100	Environmental and Energy Sustainability [C3PP]	

ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	
Subtotal		3-4
Group 2. Advanced Environmental Science Courses		
Choose 15 credit hours from the following:		15
BIO 310	Marine Biology	
BIO 387	Environmental Microbiology	
BIO/GEOL 400	Geology and Ecology of the Bahamas	
BIO 402	Forest Ecology	
BIO 418	Conservation Biology	
BIO 452	Population Ecology	
BIO 453	Microbial Ecology and Evolution	
BIO 454	Introduction to Biometrics	
BIO 457	Biological Applications of Geographic Information Systems	
BIO 459	Freshwater Ecology	
CHEM 353	Environmental Chemistry	
CHEM 354	Environmental Chemistry Field Camp	
CHEM 450	Nuclear and Radiation Chemistry	
ENGR 317	Principles of Sustainable Engineering	
ENGR 472	Environmental Engineering	
ENGR 478	Water Resources Engineering	
ENVT 200	Environmental Systems Theory	
GEOG 215	Introduction to Geographic Information Science and Cartography	
GEOG 216	Earth Observation and Satellite Navigation Systems	
GEOG 310	Environmental Issues	
GEOG 340	Biogeography	
GEOG 365	Cartography and Geospatial Visualization	
GEOG 385	Digital Image Processing in Remote Sensing	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
GEOL 211	Introduction to Oceanography [C3NS]	
GEOL 310	Environmental Impact	
GEOL 311	Mineral Resources and the Environment (2,2)	
GEOL 320	Meteorology	
GEOL 430	Environmental Soil Science (2, 2)	
GEOL/CHEM 355	Geochemistry of Natural Waters	
GEOL 377	Earth Surface Processes (2, 2)	
GEOL 406	Paleoclimatology and Paleooceanography	
GEOL 410	Engineering Geology (2, 2)	
GEOL 442	Field Geophysics	
GEOL 460	Hydrogeology (2, 2)	
ISAT 311	Role of Energy in Modern Society	
ISAT 320	Fundamentals of Environmental Science and Technology I	
ISAT 321	Fundamentals of Environmental Science and Technology II	
ISAT 420	Environmental Analysis and Modeling	
ISAT 423	Environmental Remediation	
ISAT 425	Environmental Hydrology	

ISAT 427	Industrial Hygiene	
ISAT 428	Industrial Ecology	
MATH 321	Analysis of Variance and Experimental Design	
MATH 322	Applied Linear Regression	
MATH 324	Applied Nonparametric Statistics	
MATH 328	Time Series Analysis	
MATH 421	Applied Multivariate Statistical Analysis	
PHYS 215	Energy and the Environment [C3PP]	
Subtotal		15
Group 3. Environmental Studies Courses		
Choose one of the following:		3
ECON 305	Environmental Economics	
ECON 340	Economics of Natural Resources	
ENG 371	Literature and the Environment	
ENG 390	The Environmental Imagination	
GEOG 308	Environment, Nature and Society	
GEOG 325	Environmental Ethics	
HIST 427	U.S. Environmental History	
ISAT 421	Environmental Policy and Regulation	
ISAT 473	Local Agriculture and Farm Internships	
ISAT 474	Political Ecology	
PPA 484	Environmental Regulatory Policy and Politics	
PPA 485	Governance for Environmental Sustainability	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	
SOCI 311	Sociology of the Environment	
WRTC 416/ SCOM 465	Rhetoric of Environmental Science and Technology	
Subtotal		3
Group 4. Capstone Course		
ENVT 400	Capstone Seminar in Environmental Problem Solving ¹	3
Subtotal		3
Total Credits		24-25

¹ All students must complete the capstone course ENVT 400. Students must have completed 15 hours of their environment minor in order to enroll in the capstone.

Environmental Studies Minor

Coordinator: Rob Alexander

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Phone: (540) 568-3771

Environmental Studies Minor Website (<http://www.jmu.edu/environment/studies.shtml>)

The environmental studies minor provides a cross disciplinary education engaging socio-cultural, scientific and technical issues raised by the oft-conflicting needs and desires of globally interacting societies. Designed to complement any major, the goals of the environmental studies minor include:

- To help undergraduates develop an awareness of the cultural, political and scientific aspects of the world's environmental problems.
- To better prepare students for further study at the graduate or professional school level and careers in the expanding field of environmental professions.

Minor Requirements

Overview of Requirements

The requirement for a minor in environmental studies is a minimum of 21 credit hours taken from the three categories:

- **Introduction to Environment and Environmental Science Literacy:** Fundamental to exploring the problems which emerge through the human/nature interaction is an understanding of core scientific concepts and processes that underlie natural systems. Therefore, minors will take an environmental science course that provides a broad survey of core concepts as well as an environmental science course that provides a targeted focus on an environmental science topic.
- **The Socio-Cultural Context of Environmental Studies:** The core of the Environmental Studies minor is exploring the human/nature relationship and the challenges emerging from it through multiple disciplinary lenses, resulting in a truly interdisciplinary environmental studies experience.
- **The Senior Seminar capstone course:** The culminating experience for the environmental studies minor is the combined capstone, co-taught by two faculty members from different fields and joined by the environmental science and environmental management minors. With a small size, opportunities for experiential field trips and engagement with current environmental challenges, ENVT 400 creates an all hands on deck opportunity to demonstrate interdisciplinary competency in environmental studies and to make a difference with real environmental, sustainability, climate justice and resiliency problems.

No more than two courses from a single subject (e.g., GEOG, GEOL, ANTH, ENG) may count toward completion of the minor. Students wishing to complete more than one of the environment minors (environmental management (p. 488), environmental science (p. 490) and environmental studies) may receive dual credit for the capstone course (ENVT 400), but may not receive dual credit for any other courses that might be shared by the minors.

Courses

Code	Title	Credits
Introduction to Environment		
All students seeking the minor must have one broad survey course on the human/nature relationship.		
Choose one of the following:		3-4
BIO 103	Contemporary Biology [C3NS]	
BIO 250	Quantitative Ecology and Evolution	
ENVT 200	Environmental Systems Theory	
GEOG 210	Physical Geography (2, 2)	
GEOL 102	Environment: Earth [C3NS]	
GEOL 110	Physical Geology [C3NS]	
GEOL 115	Earth Systems and Climate Change [C3NS]	
GEOL 210	Applied Physical Geology [C3NS]	
ISAT 100	Environmental and Energy Sustainability [C3PP]	
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	
Subtotal		3-4
Environmental Science Literacy		
The Environmental Literacy requirement ensures that Minors will have at least one course providing depth and focus on at least one environmental science topic.		
Choose one of the following:		3-4
BIO 457	Biological Applications of Geographic Information Systems	
BIO 459	Freshwater Ecology	
GEOG 327	Climatology	
GEOL 211	Introduction to Oceanography [C3NS]	
GEOL 230	Evolution of Earth (3, 2)	
GEOL 310	Environmental Impact	
GEOL 320	Meteorology	
GEOL 430	Environmental Soil Science (2, 2)	
ISAT 320	Fundamentals of Environmental Science and Technology I	
PHYS 215	Energy and the Environment [C3PP]	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
Subtotal		3-4
The Socio-Cultural Context of Environmental Studies		
Choose four courses (a minimum of 12 credit hours) from the following: ¹		12
ECON 305	Environmental Economics	
ECON 340	Economics of Natural Resources	
ENG 221	Literature and Ideas [C2L] (Lit, Nature, Env section only)	
ENG 371	Literature and the Environment	
ENG 390	The Environmental Imagination	
GEOL 310	Environmental Impact	
	or GEOG 310 Environmental Issues	
GEOG 322	Agricultural Systems	
GEOG 325	Environmental Ethics	
GEOG 330	Weather, Climate and Society	
GEOG 427	Water Resources of the World	

HIST 427	U.S. Environmental History	
ISAT 290	Introduction to Systems Thinking for Complex Problems	
ISAT 421	Environmental Policy and Regulation	
ISAT 424	Natural Resource Management	
ISAT 474	Political Ecology	
PPA 484	Environmental Regulatory Policy and Politics	
PPA 485	Governance for Environmental Sustainability	
SCOM/WGSS 302	Third Wave Ecofeminism	
SCOM 354/ WRTC 326	Environmental Communication and Advocacy	
SOCI 311	Sociology of the Environment	
SOCI 357	Sociology of Disasters	
WRTC 416/ SCOM 465	Rhetoric of Environmental Science and Technology	
Approved special topics courses		
Approved internship programs		
Approved study abroad courses		
Subtotal		12
Senior Seminar Capstone Course		
ENVT 400	Capstone Seminar in Environmental Problem Solving	3
Subtotal		3
Total Credits		21-23

¹ Only two courses with the same course subject may be taken.

Ethics Minor

Coordinator: Patrick Fleming

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Phone: (540) 568-4233

Ethics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/ethics.shtml>)

Program Description

The 18-hour minor in ethics offers students an opportunity for deeper study into respected ethical thinkers and how ethical theory from philosophy and global religions can be used to understand our individual and social lives.

Minor Requirements

Code	Title	Credits
Students are to choose six courses total from either of the lists below. At least one of the six must be from the philosophy list, and at least one of the six must be from the religion list. Students majoring in Philosophy or Global Religions and Cultures may double-count two courses (6 credits) from their major toward the minor. Students may also count one course from another department toward the minor, if the course includes a focus on ethics, with approval from the faculty advisor.		18
Total Credits		18

Philosophy Courses

Code	Title	Credits
PHIL 150	Ethical Reasoning [C1CT]	3
PHIL 200	Character and Virtue	3
PHIL 262	Problems in Applied Ethics	3
PHIL 267	Philosophy of Law	3
PHIL/REL 323	Comparative Ethics East and West	3
PHIL 330	Moral Theory	3
PHIL 335	The Individual, The State and Justice	3
PHIL 345	History of Ethics	3
PHIL/WGSS 350	The Philosophy of Feminism	3
PHIL 440	Advanced Moral Philosophy	3
PHIL 495	Philosophy Course Assistantship (if the assistantship is for an ethics course)	1-6

Religion Courses

Code	Title	Credits
REL 306	Gender and Sexuality in Islam	3
REL 314	Hope or Terror? Gandhi and the Other 9/11	3
REL 315	Women and Religion	3
REL 316	Dharma/Adharma: Hindu Ethical Reasoning	3
REL/PHIL 323	Comparative Ethics East and West	3
REL/IA 363	Apocalypticism, Religious Terrorism and Peace	3
REL 366	Religion, Animals and the Environment	3
REL 383	The Global Refugee Crisis and Religion	3
REL 385	Religion and the Law of War	3
REL 450	Religion and Society	3
REL 488	Buddhism in the Modern World	3

Family Studies Minor

Advisor: Nancy Poe
Phone: (540) 568-6955
Email: familystudies@jmu.edu
Family Studies Minor Website (<http://www.jmu.edu/socwork/familystudies.html>)

Program Description

The minor in family studies is designed for undergraduates seeking enhancement of their major, desiring to increase understanding of self and relationships, and seeking to make a positive contribution to society. The minor offers two focuses: Family ecology (FECO) and gerontology (GERO). Both focuses explore a range of family factors, family processes, policies, laws, services and the interrelationship of families and society. The minor will enhance majors in many fields including anthropology, early and middle education, health sciences, management, nursing, psychology, social work and sociology.

Minor Requirements

The family studies minor requires a minimum of 18 credit hours, with no more than six credit hours coming from the student's major area (for example, SOWK majors cannot apply more than six credits of SOWK courses toward the minor). Note that some courses have prerequisites that must be completed before enrollment. Some courses may be restricted to students enrolled in particular majors.

One introductory course is required: either FAM 133 or SOCI 374.

For FECO focus students, six other credit hours are required and must be selected from these areas: Development and Family Processes, and Social and System Impacts on Families. Six additional credits may be selected from these two areas or from approved electives.

For GERO focus students, nine additional credits of core required courses and three credits from approved electives must be completed.

Each focus requires completion of a linked three credit capstone experience: For FECO: FAM 400 and FAM 401; for GERO: FAM 400 and IPE 313—002.

Code	Title	Credits
Core Course		
FAM 133 or SOCI 374	The Contemporary Family Sociology of Families	3
Subtotal		3
Focuses		
Choose one of the following:		15
Family Ecology Focus		
Gerontology Focus		
Subtotal		15
Total Credits		18

Focuses

Family Ecology Focus

Code	Title	Credits
Family Development and Process		
Choose one of the following:		3

EXED 306	Lifespan Issues for Individuals with Disabilities ¹	
FAM 300	Child Development or PSYC 365 Developmental Psychology	
FAM/SOWK 325	Parent-Child Relationships Across the Lifespan	
HTH 372	Human Sexuality	
HTH 471	Health Aspects of Gerontology	
PSYC 450	Psychology of Child Abuse and Neglect or SOWK 346 Violence in Families	
PSYC 475	Psychology of Adulthood	
SCOM 440	Family Communication ²	
SOWK 387	Working with Adolescents	
Subtotal		3
Social and System Impacts on Families		
Choose one of the following:		3
FAM/SOWK 347	Issues Affecting LGBTQIAP+ Individuals and Their Families	
GERN/SOCI 280	Social Gerontology	
GERN/SOWK 372	Programs and Services for Older Adults	
HIST 320	Women in U.S. History	
NSG 326	Care and Consideration for Children and Young Adults with Disabilities	
NSG 390	Impact of Chronic Illness	
SOCI 303	Sociology of Death and Dying	
SOCI/WGSS 337	Sociology of Gender	
SOWK 332	Community Mental Health Practice	
SOWK 338	Issues and Policies in Family Services	
SOWK 342	Child Welfare Services	
SOWK 362	Social Work with Immigrants and Refugees	
Subtotal		3
Electives		
Choose six credit hours of additional courses from Social and System Impacts on Families and/or Family Development and Process clusters or the following:		6
FAM/SOWK 327	Substance Use Interventions	
FAM/NPS/SOWK 375	Grant Writing for Agencies	
FAM/SOWK 386	Youth Empowerment Strategies (YES)	
FAM 487	Special Topics in Family Studies	
FAM 490	Special Studies in Family Studies ³	
HTH/IPE/SOWK 314	Rural Health: An Interprofessional Approach	
IPE 313	Issues and Applications of Family Caregiving: Caregiver's Community Network (FECO focus students limited to one credit section; can be repeated up to three credits)	
Subtotal		6
Frameworks and Focus Capstone Courses		
FAM 400	Theory Survey ⁴	1

FAM 401	Issues and Applications ⁴	2
Subtotal		3
Total Credits		15

¹ Special education non-teaching and chronic illness concentration students only.

² SCOM majors and minors only.

³ Can be taken for a total of six credits.

⁴ Must be taken in same semester

Gerontology Focus

Code	Title	Credits
Foundation Courses		
GERN/SOCI 280	Social Gerontology ¹	3
GERN/SOWK 372	Programs and Services for Older Adults	3
PSYC 160	Life Span Human Development [C5SD]	3
Subtotal		9
Elective Course		
Choose one of the following:		3
BIO 301	Introductory Neuroscience	
FAM/SOWK 325	Parent-Child Relationships Across the Lifespan	
GERN 450	Palliative and Hospice Care: An Interdisciplinary Approach	
GERN 487	Special Topics in Gerontology	
GERN 490	Special Studies in Gerontology	
HTH 378	The Use and Effects of Drugs	
HTH 471	Health Aspects of Gerontology	
IPE 313	Issues and Applications of Family Caregiving: Caregiver's Community Network ²	
IPE/NSG 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach ³	
NSG 310	Helping Persons in Pain	
NSG 322	Integrative Health Care	
NSG 328	Life, Death, and the In-Between	
NSG 391	Living Successfully with Chronic Illness	
SOCI 303	Sociology of Death and Dying	
SOWK/HTH/ IPE 314	Rural Health: An Interprofessional Approach	
Subtotal		3
Frameworks and Focus Capstone Courses		
FAM 400	Theory Survey ⁴	1
IPE 313	Issues and Applications of Family Caregiving: Caregiver's Community Network ^{4,5}	2
Subtotal		3
Total Credits		15

¹ Health Services Administration majors may substitute HSA 290.

² GERO focus students can take IPE 313-001 as an elective for one credit hour.

³ SOWK majors must take IPE 415 concurrent with SOWK 465.

⁴ Must be taken in same semester.

⁵ GERO focus students must take IPE 313-002 for two credit hours for capstone credit.

Film Studies Minor

Coordinator: Kevin J. Reynolds

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Phone: (540) 568-8183

Film Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/film-studies.shtml>)

Program Description

The minor in film studies is designed for students who wish to extend their critical understanding of visual communication and narrative form by studying how movies tell stories, convey information and influence audiences. Because the program is cross disciplinary, it examines films as both art and entertainment, bringing together the literary traditions of English and the communication perspectives of media arts and design. Together, the aim is to explore cinema's pictorial "language," investigate its commercial consequences and evaluate its impact across cultures as a medium for enacting the human story.

Minor Requirements

Six credit hours may be double-counted between the minor and major. Information is available from the film studies advisor of the School of Media Arts and Design or the Department of English.

Code	Title	Credits
Required Course		
ENG/SMAD 380	Introduction to Film	3
Subtotal		3
Electives		
Choose five of the following:		15
ENG 381	History of Global Cinemas	
ENG 382	Race, Class and Gender in Film	
ENG 383	Film Genre	
ENG 384	Film Authorship	
ENG 385	Special Topics in Film Study	
ENG 421	Advanced Studies in Film Theory	
SMAD 460	Film and Society	
SMAD 461	Film as Art	
SMAD 462	Documentary in Film and Television	
SMAD 463	Film Adaptations	
SMAD 464	Contemporary American Film	
SMAD 498	Senior Seminar in Media Arts and Design (when topic is film)	
Subtotal		15
Total Credits		18

General Business Minor

Advisor: College of Business Advising and Support Center (<https://www.jmu.edu/cob/asc/index.shtml>)

Email: cobasc@jmu.edu

Phone: (540) 568-5948

General Business Minor Website (<https://www.jmu.edu/cob/undergraduate/minors/general-business.shtml>)

Program Description

The General Business Minor is designed to provide students with an understanding of fundamental business concepts and practices. This minor is suitable for students in many disciplines who wish to complement their major with a solid foundation in business knowledge. Through a diverse set of courses such as accounting, economics, finance, marketing, management and computer information systems, students can gain insights into making informed business decisions and understanding the broader organizational landscape. With the completion of the minor, students will be able to:

- Demonstrate a broad understanding of core business principles in fields such as finance, management, and marketing
- Read and interpret financial statements
- Develop basic proficiency in using business-related technologies

Admissions

A minor in general business is available to students in specific majors by declaration (see below) and for all other JMU students in good standing via a competitive application process when space is available.

Students who declare one of the following majors are automatically eligible for the general business minor and may declare the minor without submitting an application:

- Chemistry, Biophysical Chemistry Major, B.S. (p. 220) / Chemistry, B.S. (p. 217)
- Engineering (p. 261)
- Health Services Administration (p. 291)
- Hospitality Management (p. 298)
- Physics, B.A. (p. 397) / Physics, B.S. (p. 399)
- Sport and Recreation Management (p. 431)

If a student changes majors, they may be required to apply and be accepted into the minor, regardless of whether or not the student has completed any of the course requirements.

Students are highly encouraged to officially declare the minor prior to taking courses within the minor. Completion of courses in the minor courses does not guarantee nor facilitate acceptance into the minor.

Students with declared majors in any other undergraduate degree program may submit an application for consideration for admission into the general business minor. Admission is limited and competitive based on an applicant's admission packet consisting of:

- An essay describing motivation for earning the general business minor is optional.
- Grades earned in General Education Madison Foundations – Critical Thinking and The Natural World Quantitative Reasoning courses. Only applicants with minimum grades of "C" in both areas are eligible for consideration.

- Students seeking to declare the minor must electronically submit their application packets prior to March 1 or October 1 for admission consideration for the subsequent fall and spring semesters. Applications are accepted year-round but are only reviewed in March and October.

Minor Requirements

The following courses must be successfully completed with a minimum 2.0 grade point average in order to graduate with the general business minor.

Matriculated JMU students may only take one minor course for transfer credit. Students who plan to take a course for transfer credit should review the Office of the Registrar's guidelines and complete the (electronic) Intent to Transfer Credit form, including any approvals, prior to registering for the course elsewhere.

Code	Title	Credits
Required Courses		
ACTG 244	Accounting for Non-Business Majors ¹	3
CIS 204	Computer Information Systems Fundamentals for Non-Business Majors ²	3
ECON 201	Introduction to Microeconomics	3
FIN 345	Finance Fundamentals for Non-Business Majors	3
MGT 305	Organizational Behavior and Human Resource Management ³	3
MKTG 380	Principles of Marketing	3
Total Credits		18

¹ COB 242 may substitute for ACTG 244.

² COB 204 may substitute for CIS 204.

³ Engineering majors may substitute ENGR 426 and ENGR 226 for MGT 305.

General Music Minor

Advisor: Pedro Aponte
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Phone: (540) 568-5367
General Music Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/music.shtml>)

placement assessment. Students may not count MUS 241 for both the fundamentals and the literature and history categories.

Program Description

Open to all JMU undergraduate students, the general music minor develops both the performing and non-performing musician's understanding of music. The minor in music does not qualify a student to teach music in Virginia public schools.

Minor Requirements

The minor requires the successful completion of 18 credits drawn from three areas – fundamentals, literature and history, and electives. Students must earn 50 percent of the required curriculum at JMU.

Code	Title	Credits
Fundamentals		
Choose 3-6 credit hours of the following:		3-6
MUS 131	Fundamentals of Music	
MUS 140	Introduction to Music Theory and Aural Skills	
MUS 141	Foundations of Music Theory ¹	
MUS 241	Diatonic and Chromatic Harmony ¹	
Subtotal		3-6
Literature and History		
Choose 6-9 credit hours of the following:		6-9
MUS 200	Music and Culture [C2VPA]	
MUS 203	Music in America [C2VPA]	
MUS 206	Introduction to Global Music [C2VPA]	
MUS 241	Diatonic and Chromatic Harmony ¹	
MUS 250	Form and Analysis: Baroque and Classical	
MUS 251	Form and Analysis: Romantic	
MUS 252	Analysis of Music 1900-1950	
MUS 253	Analysis of Music 1950-Present	
MUS 254	Form and Analysis: Popular Music	
MUS 255	Jazz Theory	
MUS 256	Contrapuntal Techniques	
MUS 257	Techniques of Composition	
MUS 258	Film Music Analysis	
MUS 356	History of Jazz	
MUS 357	Music Theatre History and Analysis	
Subtotal		6-9
Electives		6
Students must take 6 additional credits of music courses. Prefixes include: MUAP, MUED, MUI, and MUS.		
Subtotal		6
Total Credits		18

¹ MUS 141 and MUS 241 are more advanced courses typically taken by music majors. Students who have prior experience in music theory who may benefit from these courses are encouraged to contact the coordinator of the general music minor to request a

Geography Minor

Advisor: Kimberly Stewart
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Phone: (540) 568-6199
Geography Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/geography.shtml>)

Program Description

A geography minor complements any major by providing a broader perspective and enhancing your understanding of the world.

- You'll gain a deeper understanding of global issues, cultures, and societies.
- You'll develop a broader perspective on how human activities and environmental processes interact across different regions of the world.
- You'll develop strong problem-solving and critical thinking skills and analyze complex spatial patterns, interpret data, and understand the relationships between people and their environment.
- You'll explore diverse topics, such as cultural geography, urban studies, and environmental sustainability.
- You'll broaden your understanding of the world and foster a lifelong appreciation for the complexities of our planet.

Minor Requirements

The minor in geography consists of three required courses and three additional geography courses for a minimum of 19 credit hours.

Code	Title	Credits
Required Courses		
GEOG 210	Physical Geography (2, 2)	4
GEOG 215	Introduction to Geographic Information Science and Cartography	3
GEOG 280	Human Geography: The Cultural Landscape	3
Three additional GEOG courses		9-12
Total Credits		19-22

Geology Minor

Coordinator: Steve Leslie
Email: lesliesa@jmu.edu
Phone: (540) 568-6130
Geology Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/geology.shtml>)

Program Description

Geology is the scientific study of the processes, composition and history of terrestrial planets.

The field has a broad variety of applications, from geographic information systems to environmental science to the examination of other planetary bodies. At JMU, we urge students to go beyond the classroom and into the real world to explore their scientific interests.

Minor Requirements

Code	Title	Credits
The requirement for a minor in geology is a minimum of 18 credit hours of geology approved by the student's geology advisor.		18
Total Credits		18

Geophysics Minor

Coordinator: Shane McGary

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Phone: (540) 568-6130

Geophysics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/geophysics.shtml>)

Program Description

The minor in geophysics is designed to provide adequate training for professional work or graduate school in geophysics in the broad sense.

Minor Requirements

No more than 10 credits from the geophysics minor may be used to double count with a major. The requirements for the geophysics minor are a minimum of 22 credits, including at least one semester each of geology, mathematics and physics, distributed as follows:

Code	Title	Credits
Introductory Course		
Choose one of the following pairs:		4
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	
GEOL 210 & GEOL 110L	Applied Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	
PHYS 240 & PHYS 140L	University Physics I [C3PP] and General Physics Laboratory I [C3L]	
Subtotal		4
Required Courses		
GEOL 440	Geophysics (3, 2)	4
GEOL 442	Field Geophysics	3
MATH 238	Linear Algebra with Differential Equations	4
MATH 248	Computers and Numerical Algorithms	4
Subtotal		15
Elective Courses		
Choose one of the following: ¹		3-4
ASTR 220	General Astronomy I: The Night Sky, the Solar System and Stars	
CHEM 331	Physical Chemistry I	
GEOL 410	Engineering Geology (2, 2)	
GEOL 460	Hydrogeology (2, 2)	
MATH 237	Calculus III	
PHYS 260	University Physics III	
Subtotal		3-4
Total Credits		22-23

¹ Additional elective courses are permitted upon approval from minor advisor.

Note for Geology, B.S. Majors

The B.S. in Geology (p. 281) requires a sequence of math and physics courses. The following sequence of courses will satisfy the major requirements and is strongly recommended for students interested in

geophysics. In particular, MATH 236 is a prerequisite for several courses included in the geophysics minor.

Code	Title	Credits
MATH 235	Calculus I [C3QR]	4
MATH 236	Calculus II	4
PHYS 240	University Physics I [C3PP] (and Lab)	3
PHYS 250	University Physics II (and Lab)	3

Global Religions and Cultures Minor

Advisor: Emily Gravett
Email: graveteo@jmu.edu
Phone: (540) 568-4846
Global Religions and Cultures Minor (<https://www.jmu.edu/academics/undergraduate/minors/religion.shtml>)

Program Description

The Global Religions and Cultures minor equips students with skills in religious literacy: understanding how religion works in people's lives as well as in the larger historical, cultural and sociopolitical forces that shape our world, from religious terrorism to nonviolence, from the suppression of women to the advocacy of civil rights. In this minor, students will study major global religions and their impact on real-world social and political issues, past and present.

Minor Requirements

Code	Title	Credits
Core Course		
REL 101	Religions of the World [C2HQC]	3
Subtotal		3
Religion Electives		
Students choose five additional religion (REL-prefix) courses in order to reach 18 total credit hours for the Global Religions and Cultures minor.		15
Subtotal		15
Total Credits		18

Global Supply Chain Management Minor

Coordinator: Mert Tokman
Email: tokmanmx@jmu.edu
Phone: (540) 568-3020

Coordinator: Rebecca Simmons
Email: simmo2ra@jmu.edu
Phone: (540) 568-2788

Global Supply Chain Management Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/global-supply-chain-management.shtml>)

Program Description

The global supply chain management minor is designed to provide a broad overview of the supply chain management and logistics functions in a global context. The minor is open to COB, ISAT and ENGR majors.

Application and Admission Information

Application deadlines for the global supply chain management minor are October 15th for the fall and March 15th for the spring. Admission is competitive and limited to a maximum of 30-35 students per academic year. Admission is based on performance in the prerequisite course, which requires a minimum grade of "C+", and an essay in application to the minor. More information can be found on the minor's website (<https://www.jmu.edu/cob/undergraduate/minors/global-supply-chain-management/index.shtml>).

Minor Requirements

The minor is comprised of two core courses (MGT 461 and MKTG 462) and four electives that can be chosen from available classes in marketing, management, international business, integrated sciences and engineering.

Prerequisite

Admission to the minor requires a minimum grade of "C+" in one of the following courses:

Code	Title	Credits
COB 300C	Integrative Business: Operations (For COB Majors)	3
ISAT 211	Issues in Modern Production Systems (For CISE Major)	3

Program Requirements

Code	Title	Credits
Core Courses		
(On campus or Study Abroad)		
MGT 461	Introduction to Supply Chain Management	3
MKTG 462	International Logistics	3
Subtotal		6
Electives		
Choose 12 credits according to chosen major. See below for details.		12
Marketing Majors		
Management Majors		
International Business and other COB Majors		

Integrated Science and Technology Majors

Engineering Majors	
Subtotal	12
Total Credits	18

Electives

Marketing Majors

Code	Title	Credits
Choose two of the following:		6
MKTG 450	Business-to-Business (B2B) Marketing	
MKTG 460	Global Marketing	
MKTG 494	Marketing Internship	
Choose two of the following:		6
GEOG 215	Introduction to Geographic Information Science and Cartography	
GEOG 344	Globalization and Development	
ISAT 422	Industrial Environmental Management	
ISAT 426	Environmental Information Systems	
ISAT 427	Industrial Hygiene	
ISAT 428	Industrial Ecology	
ISAT 430	Materials and Manufacturing Processes	
ISAT 431	Advanced Manufacturing Processes and Systems	
ISAT 433	Foundations of Lean and Six Sigma	
ISAT 434	Industrial Informatics	
ISAT 435	Lean Six Sigma Applications	
ISAT 437	Technological Innovation and Sustainable Design	
Total Credits		12

Management Majors

Code	Title	Credits
MGT 325	Project Management	3
MGT 340	International Management	3
Choose two of the following:		6
GEOG 215	Introduction to Geographic Information Science and Cartography	
GEOG 344	Globalization and Development	
ISAT 422	Industrial Environmental Management	
ISAT 426	Environmental Information Systems	
ISAT 427	Industrial Hygiene	
ISAT 428	Industrial Ecology	
ISAT 430	Materials and Manufacturing Processes	
ISAT 431	Advanced Manufacturing Processes and Systems	
ISAT 433	Foundations of Lean and Six Sigma	
ISAT 434	Industrial Informatics	
ISAT 435	Lean Six Sigma Applications	
ISAT 437	Technological Innovation and Sustainable Design	
Total Credits		12

International Business and other COB Majors

Code	Title	Credits
Choose two of the following:		6
IB 494	International Business Internship	
MGT 340	International Management	
MKTG 460	Global Marketing	

Choose two of the following:	6
GEOG 215 Introduction to Geographic Information Science and Cartography	
GEOG 344 Globalization and Development	
ISAT 422 Industrial Environmental Management	
ISAT 426 Environmental Information Systems	
ISAT 427 Industrial Hygiene	
ISAT 428 Industrial Ecology	
ISAT 430 Materials and Manufacturing Processes	
ISAT 431 Advanced Manufacturing Processes and Systems	
ISAT 433 Foundations of Lean and Six Sigma	
ISAT 434 Industrial Informatics	
ISAT 435 Lean Six Sigma Applications	
ISAT 437 Technological Innovation and Sustainable Design	
Total Credits	12

Integrated Science and Technology Majors

Code	Title	Credits
ISAT 330	Sustainable Manufacturing Systems	3
ISAT 331	Cyber-Physical Industry	3
Choose two of the following:		6

GEOG 215	Introduction to Geographic Information Science and Cartography
GEOG 344	Globalization and Development
ISAT 341	Machine Learning and Data Science ¹
or ISAT 344	Applied Artificial Intelligence
ISAT 422	Industrial Environmental Management
ISAT 426	Environmental Information Systems
ISAT 427	Industrial Hygiene
ISAT 428	Industrial Ecology
ISAT 430	Materials and Manufacturing Processes
ISAT 431	Advanced Manufacturing Processes and Systems
ISAT 432	Materials Selection in Product Design
ISAT 433	Foundations of Lean and Six Sigma
ISAT 434	Industrial Informatics
ISAT 435	Lean Six Sigma Applications
ISAT 437	Technological Innovation and Sustainable Design

Total Credits 12

¹ Integrated Science and Technology students may receive credit for either ISAT 341 or ISAT 344, but not both

Engineering Majors

Code	Title	Credits
ENGR 426	Product Development and Entrepreneurial Engineering	3
ENGR 226	Engineering Project Management	3
Choose two of the following:		6

ENGR 317	Principles of Sustainable Engineering
GEOG 215	Introduction to Geographic Information Science and Cartography
GEOG 344	Globalization and Development
ISAT 330	Sustainable Manufacturing Systems ¹
or ISAT 331	Cyber-Physical Industry

ISAT 341	Machine Learning and Data Science ²
or ISAT 344	Applied Artificial Intelligence
ISAT 422	Industrial Environmental Management
ISAT 426	Environmental Information Systems
ISAT 427	Industrial Hygiene
ISAT 428	Industrial Ecology
ISAT 430	Materials and Manufacturing Processes
ISAT 431	Advanced Manufacturing Processes and Systems
ISAT 432	Materials Selection in Product Design
ISAT 433	Foundations of Lean and Six Sigma
ISAT 434	Industrial Informatics
ISAT 435	Lean Six Sigma Applications
ISAT 437	Technological Innovation and Sustainable Design
Total Credits	12

¹ Engineering students may receive credit for either ISAT 330 or ISAT 331, but not both

² Engineering students may receive credit for either ISAT 341 or ISAT 344, but not both

Health Communication Minor

Coordinator: Sharlene Richards

Phone: (540) 568-3586

Email: richarst@jmu.edu

Health Communication Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/health-communication.shtml>)

Program Description

The minor in health communication is designed for students who are interested in working in health care or related fields and are completing coursework in areas such as nursing, health sciences and the pre-professional programs. The minor offers students flexibility in terms of area of focus for the minor, so that they can gain the most applicable knowledge for their future endeavors.

Admission to the Minor

Admission to JMU does not guarantee admission to the School of Communication Studies. All students interested in minoring in Health Communication must apply for a limited number of spaces by enrolling in and successfully completing SCOM 240 and SCOM 240L. To be eligible to register for these classes, students must add the minor in MyMadison by no later than March 1 for summer and fall registration and by October 1 for spring registration. Students who miss the deadline are still encouraged to declare the minor as SCOM will continue to process applications in the order in which they were received if there are still seats available in SCOM 240/SCOM 240L. Declaring the minor by this date does not guarantee availability of courses. The School of Communication Studies reviews performance in the SCOM 240/SCOM 240L class to make decisions about acceptance into the minor each semester.

Admission to the minor is based on availability to the most qualified students as determined by performance in SCOM 240 and SCOM 240L.

Students who are not admitted are permitted one more attempt (an attempt is any time the student receives a grade for the course on the transcript including W, WP, or WF) to successfully meet performance requirements in SCOM 240/SCOM 240L by enrolling the next regular semester or in the summer if the student desires. For students retaking SCOM 240/SCOM 240L (whether repeat or repeat/forgive), the school will look only at their highest grades earned when evaluating their second attempt.

Once admitted into a Communication Studies minor, a student cannot retroactively apply more than nine hours of SCOM courses, including SCOM 240 and SCOM 240L, to their minor (SCOM 121/SCOM 122/SCOM 123 does not count toward those nine hours).

Minor Requirements

Code	Title	Credits
Required Courses		
SCOM 240	Introduction to Communication Theory	2
SCOM 240L	Communication Theory Lab	1
SCOM 370	Introduction to Health Communication	3
Subtotal		6
Electives		
Choose four of the following:		12
SCOM 303	Special Topics in Health Communication	

SCOM 320	Introduction to Interpersonal Communication
SCOM 342	Argument and Advocacy
SCOM 350	Organizational Communication
SCOM 358	Business and Professional Communication Studies
SCOM 440	Family Communication
SCOM 468	Interpersonal Communication in Health Contexts
SCOM 470	Health Communication Campaigns
SCOM 471	Culture and Health Communication
Any SCOM course at the 300 or 400 level	
Subtotal	12
Total Credits	18

Historical Archaeology Minor

Co-coordinator: Julie Solometo
Email: solomejp@jmu.edu
Phone: (540) 568-6171

Co-coordinator: Gabrielle M. Lanier
Email: laniergm@jmu.edu
Phone: (540) 568-3615

Historical Archaeology Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/historical-archaeology.shtml>)

Program Description

The minor is designed for students interested in the sub-field of historical archaeology, a discipline that integrates the research interests and methods of archaeology and history. The minor is designed to complement existing majors in anthropology and history, and it may also be of interest to students in art history and public administration. While guided by the theoretical underpinnings of history and anthropology, the minor in historical archaeology is field- and research-oriented. Students enrolling in the program should anticipate courses that require significant effort outside of the classroom.

Minor Requirements

Two courses (six credits) may be double-counted between the minor and the student’s major.

Code	Title	Credits
Required Courses		
ANTH/HIST 331	Historical Archaeology	3
ANTH 494	Field Techniques in Archaeology	4
HIST/ARTH 396	History Beyond the Classroom: Public History	3
Subtotal		10
Electives		
Choose three of the following courses. Of these, one course must be chosen from the history/anthropology options and one must be from the geology or geography options. ¹		9
ANTH 410	Spatial Analysis for Social Research	
ANTH 455	Archaeology: Methods of Analysis and Interpretation	
HIST/ARTH 394	Introduction to Museum Work	
HIST/ANTH/ ARTH 492	History from Things: Material Culture	
HIST/ARTH 493	Historic Preservation	
GEOL 430	Environmental Soil Science (2, 2)	
GEOL 442	Field Geophysics	
Subtotal		9
Capstone Research Course		
The personalized capstone course can take place over one or two semesters and will require the student to work closely with a faculty advisor. The project will require the student to define a research topic, establish a theoretical and methodological base for the project, gather data, interpret the results and prepare an appropriate statement of findings.		
ANTH/HIST 496	Research Thesis	

Subtotal	3
Total Credits	22

¹ Access to geology and geography courses is competitive and will require some computer experience.

History Minor

Coordinator: Gabrielle Lanier
Email: laniergm@jmu.edu
Phone: (540) 568-3615
History Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/history.shtml>)

Program Description

Historians study the past as a window into the human experience over time. We study it for its own sake, and for the ways it helps us understand our present. History can be a valuable minor alongside any other degree program.

Minor Requirements

Code	Title	Credits
Students select 18 credit hours in history to fulfill the minor. At least one course must be at the 300 or 400 level. No more than 3 credit hours of HIST 000 elective transfer credit may count toward minor requirements.		18
Total Credits		18

Honors Interdisciplinary Studies Minor

Contact: Allison Fagan, Associate Dean
Email: faganae@jmu.edu
Phone: (540) 568-6170
Honors Interdisciplinary Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/honors.shtml>)

Program Description

Students accepted into the Honors College pursue a minor in Honors Interdisciplinary Studies, with the option to complete a Capstone Experience. The requirements of the Honors Interdisciplinary Studies Minor are flexible, and are designed to provide students with a space to synthesize multiple avenues of interest. The mission of the James Madison University Honors College is to develop a diverse community of inquisitive and aspirational learners from all parts of campus through distinctive and challenging educational experiences. Our core values include independent, student-centered learning; interdisciplinary training; collaboration; and active and informed service and leadership.

The interdisciplinary space of the JMU Honors College creates a community of scholars in which students, faculty and staff from across the campus may engage in environments which encourage an exploration of ideas from multiple disciplines, viewpoints and perspectives.

Admissions

Students are admitted to the Honors College by application. All Honors College students pursue a minor in Honors Interdisciplinary Studies. Students are admitted based on the strength of their Honors application essay and grades from previous and current institution(s). Applicants typically have a cumulative unweighted GPA above 3.5 at all previous institutions or a 3.3 at JMU.

To remain in good standing, students must meet the following progression requirements:

1. In their first two semesters in the Honors College, students must satisfactorily complete (a) HON 100 or HON 101 and (b) two additional Honors courses.
2. After their first year in the Honors College, students must complete at least one Honors course in each academic year.
3. After their first year in the Honors College, students must interact with Honors Advising at least once per academic year.
4. Students must maintain a 3.0 GPA overall and a 3.0 GPA in Honors courses.

Students may choose to pursue Honors Scholar and distinction recognitions by completing a capstone pathway.

1. Students pursuing Honors Scholar designations must declare their intent to complete the Honors Senior Portfolio Project or a departmental capstone (e.g. thesis) by the end of the second semester of junior year. Students completing departmental capstones must also submit a full capstone proposal by the end of the second semester of junior year.

Students that do not meet progression requirements are placed on Honors Alert. Students have one academic year to return to good standing. Students on alert must meet with an Honors Academic advisor

each semester. If students do not meet with an Honors Academic Advisor as required, they will lose Honors College eligibility.

Students slated for removal from the Honors College for failure to meet progression standards may petition to remain in the Honors College if they see the program as still well-aligned with their academic goals. Petitions are reviewed and acted on by a committee revised September 2023 that includes Honors College Academic Advisors and the Associate Dean, with possibility of appeal to the Dean of the Honors College.

Honors College Academic Advisors will complete progression reviews for all Honors College students each June. Students will be notified of their standing and will receive information about appropriate resources to complete the minor.

Honors Interdisciplinary Studies Minor Requirements

All students admitted to the Honors College pursue a minor in Honors Interdisciplinary Studies. They may also elect to pursue an honors scholar designation. The honors scholar designation requires the completion of either the honors senior e-portfolio project or a departmental honors capstone. Students who complete a departmental honors capstone project will receive a graduation distinction in the major or minor of their capstone project.

Students must show progression in honors coursework and maintain a GPA of 3.0 in honors courses to remain in good standing in the college.

Students must declare their intended honors scholar pathway or request a waiver for the honors scholar requirements by the first semester of their junior year.

Code	Title	Credits
Introduction to Honors		
Choose one of the following:		1
HON 100	Honors First Year Seminar (for incoming first year students)	
HON 101	Honors Orientation Seminar (for transfer or current JMU students)	
Subtotal		1
Honors Courses		
Choose 12 credit hours of Honors approved courses in General Education, the Honors College or academic unit of choice. ¹		12
Subtotal		12
Civic, Cultural or Global Engagement Course		
Choose three credit hours from the list of civic, cultural or global engagement courses.		3
Subtotal		3
Research or Experiential Learning Course		
Choose three credit hours from the list of Honors research or experiential learning courses.		3
Subtotal		3
Honors Scholar Designation		

The Honors Scholar designation requires students to complete 25 credits, 19 of which come from the Honors Core. Students can earn the Honors Scholar designation by completing the Honors ePortfolio Project Pathway or the Departmental Honors Capstone Pathway. Students who wish to complete the minor without the Honors Scholar designation should contact an Honors College advisor for information on waiving the ePortfolio/capstone project requirement.

Choose one of the following: 6

Senior ePortfolio Project

Completion of the following courses for a total of 3 credits and final completion of the project:

HON 401	Introduction to Honors ePortfolio
HON 402	Honors ePortfolio

Portfolio students must complete an additional 3 credits of Honors approved coursework.

Departmental Honors Capstone Project

Completion of a 499 Honors sequence or departmental approved courses for 6 hours of credit and presentation requirements 6 credit hours. Capstone requirements are defined by the academic unit in consultation with the Honors College. Students will work with their academic unit and their Honors Liaison to define the capstone. May be but is not limited to a thesis, project, or portfolio. Approval from the academic unit in which the capstone is completed is required. Capstone students will earn a graduation designation as an "Honors Scholar with Distinction in (Major or Minor)".

Subtotal 6

Total Credits 25

¹ Some majors have specific courses that have been approved for the Honors Interdisciplinary Studies Minor; please contact an Honors Academic Advisor and your Honors Liaison with specific questions.

List of Civic, Cultural or Global Engagement Courses

Code	Title	Credits
AAAD 200	Introduction to African, African American and Diaspora Studies [C4GE]	3
AAAD 400	Advanced Topics in African, African American and Diaspora Studies	3
ANTH 250	Anthropology of the American Southwest	3
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	3
ANTH 280	Peoples and Cultures of Sub-Saharan Africa	3
ANTH 295	People and Cultures of East Asia	3
ANTH 303	Andean Archaeology	3
ANTH 306	Japanese Society and Culture	3
ANTH 364	U.S./Latin American Borders	3
ARTH 210	African Art and Culture in the Humanities	3
ARTH 424	Arts of Ancient Egypt	3
ARTH 484	Art of the Americas	3
BIO 230	Biology, Gender and Society	3
CIV 300	Civic Leadership: Theories and Practice	3
EDUC 310	Teaching in a Diverse Society	3
ELED 310	Diversity in Elementary Education	3
ENG 223	Literature and Cultures	3

ENG 239	Studies in World Literature [C2L] ("Literatures of Global English" or "African Oral Literature" topics only)	3
ENG 260	Survey of African American Literature [C2L]	3
ENG 360	Introduction to Ethnic American Literature	3
ENG 361	African American Fiction	3
ENG 362	African American Poetry	3
ENG 367	Latinx Literature	3
ENG 368	Women's Literature	3
ENG 369	Feminist Literary Theory	3
ENG 370	Queer Literature	3
ENG 374	Contemporary Global Anglophone Literature (Alternative Modernities)	3
ENG 408	Advanced Studies in African American Literature	3
ENG 423	Advanced Studies in Gender and Sexuality in Literature (Gender, Sexuality and Ubuntu in African Literature)	3
ENG 433	Advanced Studies in Arabic Literature	3
FAM 347	Issues Affecting LGBTQIAP+ Individuals and Their Families	3
FAM 362	Social Work with Immigrants and Refugees	3
GEOG 334	Geography of East and Southeast Asia	3
GEOG 337	Geography of Latin America	3
GEOG 380	Cultural Geography	3
HIST 263	Africa	3
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	3
HIST 307	The Trans-Atlantic Slave Trade	3
HIST 321	Gender and Sexuality in Modern Europe	3
HIST 322	Jim Crow South	3
HIST 353	The Global Middle East	3
HIST 372	Modern Afghanistan	3
HIST 378	China in the Modern World	3
HIST 449	Women and Fascism	3
HIST 459	Gender and Popular Culture	3
HRD 123	Developing Multicultural Competency for Effective Facilitation	3
HUM 252	Global Cultures [C2HQC] (Latin American Cultures)	3
IB 200	Introduction to International Business	3
IDLS 384	Knowledge, Power and Inequality	3
IDLS 385	The Unfinished Journey of People of Color in the U.S.	3
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]	3
PHIL 323	Comparative Ethics East and West	3
PHIL 360	Philosophy of Race	3
POSC 200	Global Politics [C4GE]	3
POSC 326	Civil Rights	3
POSC 344	Politics of the European Union	3
POSC 345	Politics of Western Europe	3
POSC 346	Politics of Central and Eastern Europe	3
POSC 350	Latin American Politics	3
POSC 353	African Politics	3
POSC 354	Politics of the Middle East	3

POSC 355	East Asian Politics	3	AAAD 489	African, African American and Diaspora Studies Senior Research Experience	1
POSC 356	Politics of China	3	ANTH 391	Study Abroad	1-6
POSC 371	Topics in Comparative Politics	3	ANTH 485	Anthropology Course Assistantship	1-6
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation	3	ANTH 490	Special Studies in Anthropology	1-3
POSC 383	Women and Politics in Comparative Perspective	3	ARCD 300	Architect Design Studio III	6
POSC 393	Human Security	3	ARCD 302	Architect Design Studio IV	6
QUST 200	Introduction to LGBTQ+ and Queer Studies	3	ARCD 390	Independent Studies in Architectural Design	1-3
QUST 300	Special Topics in LGBTQ+ / Queer Studies	3	ARCD 490	Independent Studies in Architectural Design	1-3
REL 216	(Re)thinking Order and Chaos: An Introduction to Hinduism	3	ARCD 392	Topics in Architectural Design	3
REL 220	Introduction to Judaism	3	ARCD 400	Architect Design Studio V	6
REL 240	Introduction to Christianity	3	ARCD 402	Architect Design Studio VI	6
REL 250	Introduction to Islam	3	ARCD 491	Studio Assistant	1-3
REL 288	Introduction to Buddhism	3	ARED 480	Student Teaching	8
REL 306	Gender and Sexuality in Islam	3	ARED 490	Field Experiences in Art Education	1-3
REL 307	Knowledge, Identity and Power in Islamic Societies	3	ART 210	Figure Drawing	3
REL 314	Hope or Terror? Gandhi and the Other 9/11	3	ART 320	Molds and Casting: Ceramics	3
REL 336	African American Religion	3	ART 322	Surface Development: Ceramics	3
REL 348	Christianity in Global Context	3	ART 330	Intermediate Fiber Processes	3
REL 371	Religion and Disability	3	ART 352	Intermediate Painting	3
REL 373	Race and Religion	3	ART 360	Digital Photography	3
REL 383	The Global Refugee Crisis and Religion	3	ART 362	Experimental Black and White Photography	3
REL 388	Tibetan Buddhism	3	ART 364	Large Format Photography	3
REL 440	Religion and American Culture	3	ART 376	Intermediate Book Arts: Concept, Content, Form	3
REL 450	Religion and Society	3	ART 380	Conceptual Practice: Sculpture	3
REL 488	Buddhism in the Modern World	3	ART 392	Topics in Art	3
SCOM 248	Intercultural Communication	3	ART 420	Advanced Ceramics	1-3
SCOM 347	Communication, Diversity and Popular Culture	3	ART 440	Advanced Metals	1-3
SCOM 348	Communication and Gender	3	ART 452	Advanced Painting	1-3
SOCI 318	Sociology of Immigration	3	ART 464	The Photograph/Video as Document	3
SOCI 322	Sociology of Religion	3	ART 468	Screen-Based Photography/Video	3
SOCI 336	Race and Ethnicity	3	ART 469	Portfolio Development: Photography	1-3
SOCI 337	Sociology of Gender	3	ART 470	Advanced Printmaking	1-3
SOCI 347	Queer Theories of Gender and Inequality	3	ART 480	Expanded Field: Sculpture	1-3
SOCI 354	Social Inequality	3	ART 490	Independent Study in Art	1-3
SOCI 363	Sociology of Culture	3	ART 491	Studio Assistant	1-3
SOWK 288	Human Rights and Social Progress	3	ART 492	Topics in Art	3
SOWK 347	Issues Affecting LGBTQIAP+ Individuals and Their Families	3	ART 496	Internship in Art	1-8
SOWK 362	Social Work with Immigrants and Refugees	3	ARTH 495	Internship in Art History	1-3
STAD 322	Equity, Inclusion and Access in Education and Performance	3	BIO 370	Human and Animal Physiology	4
WGSS 200	Introduction to Women's, Gender and Sexuality Studies [C5SD]	3	BIO 427	Topics in Biology with Laboratory	1-4
WGSS 300	Special Topics in Women's, Gender and Sexuality Studies (African American Women in/and the Media)	3	BIO 430	Ornithology: Research and Current Advances	4
			BIO 448	Developmental Neurobiology	3
			BIO 455	Plant Physiology	4
			BIO 484	Aquatic Microbiology	3
			BIO 493	Pre-Veterinary Student Internship	2
			BIO 494	Internship in Biology	1-2
			CHEM 351	Analytical Chemistry	4
			CHEM 352	Instrumental Analysis	3
			CHEM 354	Environmental Chemistry Field Camp	3
			CHEM 485 & 485L	Science of the Small and Science of the Small Laboratory	4

List of Honors Research and Experiential Learning Courses

Code	Title	Credits
AAAD 401	Internship in African, African American and Diaspora Studies	1-6

ELED 321	Practicum with a Focus on Learners and Learning	1-3	HRD 485	Development of Materials and Programs	3
ELED 322	Practicum with a Focus on Curriculum Integration and Guiding Behavior	1-3	HTH 408	Health Research Methods	3
ELED 490	Special Studies in Elementary Education	1-3	IB 494	International Business Internship	3
ENG 359	Studies in African American Literature	3	IDLS 395	Topics in IDLS	1-6
ENG 360	Introduction to Ethnic American Literature	3	IDLS 486	Internship and Field Experience	1-6
ENG 371	Literature and the Environment	3	IDLS 490	Independent Study in Liberal Studies	3
ENG 385	Special Topics in Film Study	3	IECE 301	Initial Field Experience in Inclusive Early Childhood Education	1
ENG 390	The Environmental Imagination	3	IECE 321	Initial Field Experience in IECE Part 1	1
ENG 415	Advanced Studies in Textuality and the History of the Book	3	IECE 423	Intermediate Practicum II	2
ENGR 440	Mechatronics	3	IECE 461	Advanced Field Experience in IECE Part 1	3
ENGR 471	Water in Africa	3	INDU 300	Product Design	6
ENGR 472	Environmental Engineering	3	INDU 302	Transportation Studio	6
ENGR 481	Principles of Process Engineering	3	INDU 210	Making Is Thinking	3
ENGR 478	Water Resources Engineering	3	INDU 390	Independent Studies in Industrial Design	1-3
ENGR 490	Advanced Projects in Engineering	1-4	INDU 404	Industrial Design Advanced Special Topics Studio	6
ENGR 498	Advanced Topics in Engineering	3	INDU 490	Independent Studies in Industrial Design	1-3
ENVT 301	Internship in Environmental Humanities	3	INDU 491	Studio Assistant	1-3
EXED 376	Initial Practicum for Special Education Pre-Professional Preparation	1	INDU 492	Topics in Industrial Design	3-6
EXED 476	Practicum in Literacy and Assessment	2	IS 300	Sponsored Learning	1-6
EXED 478	Supervised Practice in Special Education Methods	2	IS 498	Individualized Study Project	3-6
EXED 480	Student Teaching in Special Education	4	ISAT 320	Fundamentals of Environmental Science and Technology I	4
EXED 486	Supervised Clinical Practice with Planning and Methods in Special Education	2	ISAT 330	Sustainable Manufacturing Systems	3
GEOG 490	Senior Research or Field Practicum	3	ISAT 423	Environmental Remediation	3
GEOG 491	International Studies	1-3	ISAT 491	Senior Capstone Project I	1
GEOG 497	Independent Study	3	ISAT 492	Senior Capstone Project II	2
GRPH 300	Illustration	3	ISAT 493	Senior Capstone Project III	2
GRPH 306	Typography II	3	JUST 350	Justice and Globalization	3
GRPH 312	Web Design	3	JUST 399	Justice Research Methods	4
GRPH 313	Interactive Media	3	JUST 401	Internship in Justice Studies	3
GRPH 376	Intermediate Book Arts: Concept, Content, Form	3	JUST 404	Practicum in Community-Based Research	3
GRPH 390	Independent Study in Graphic Design	1-3	JUST 405	Research Assistantship in Justice Studies	1-3
GRPH 392	Topics in Graphic Design	3	MKTG 494	Marketing Internship	3-6
GRPH 406	Advanced Typography	3	MIED 211	Literacy Field Experience in Middle Grades Education	3
GRPH 408	Brand Identity	3	MSCI 315	Advanced Leadership Laboratory I	3
GRPH 410	Graphic Design Portfolio	3	MSCI 415	Senior Leadership Laboratory I	3
GRPH 496	Internship in Graphic Design	1-6	MSCI 480	Internship in Advanced Military Leadership	6
HIST 340	Internship in History	3	MSCI 490	Special Studies in Military Science	1-3
HIST 360	Research Apprenticeship in History	3	NPS 400	Internship/Practicum in Nonprofit Studies	4-6
HIST 391	Travel Studies Seminar	3	NSG 351	Health Assessment	3
HIST 390	Special Studies in History	3	NSG 352	Clinical Applications and Reasoning in Nursing Care I	4
HM 252	Culinary Arts and Menu Management	3	NSG 356	Clinical Applications and Reasoning in Nursing Care II	4
HM 444	Supervisory Hospitality Internship	3	NSG 360	HHS in Costa Rica: Interprofessional Perspective	2-4
HM 419	Napa and Sonoma Wine and Culture	3	NSG 452	Clinical Applications and Reasoning in Nursing Care III	4
HM 490	Special Studies in Hospitality and Tourism Management	1-3	NSG 456	Capstone	4
HON 496	Internship	1-6	NUTR 340	The Science and Culture of Food	3
HRD 380	Performance and Task Analysis in Human Resource Development	3	NUTR 363	Quantity Food Production	4
HRD 400	Human Resource Development Internship	3	NUTR 446	Experimental Foods	3

NUTR 490	Field Experience in Dietetics	3
PHIL 495	Philosophy Course Assistantship	1-6
POSC 493	Political Communication Internship	4
POSC 495	Internship in Political Science	4
POSC 498	Research in Political Science	1
PPA 496	Internship in Public Management	4
PSYC 495	Field Placement in Psychology	4
REL 493	Religion Course Assistantship	3
REL 495	Internship	1-3
SCOM 301	Feminist Blogging: Writing for Shout Out! JMU	3
SCOM 318	Practicum in Communication Studies	1-6
SCOM 390	Directed Projects	2-3
SCOM 490	Special Studies in Communication Studies	1-3
SCOM 495	Internship in Communication Studies ²	3-6
SMAD 390	Directed Projects	2
SMAD 403	Documentary Production	3
SMAD 404	Advanced Interactive Design	3
SMAD 405	Producing and Directing	3
SMAD 408	Converged Media Lab	3
SMAD 421	Feature Magazine Production	3
SMAD 409	Electronic News Producing and Editing	3
SMAD 422	Multimedia Journalism	3
SMAD 443	Creative Advertising Campaigns	3
SMAD 495	Internship in Media Arts and Design	1-4
SMAD 497	Advanced Projects in Media Arts and Design	3
SOCI 391	Study Abroad	1-6
SOCI 485	Sociology Course Assistantship	1-3
SOCI 490	Independent Study in Sociology	1-3
SOCI 492	Internship in Sociology	1-3
SOWK 481	Social Work Field Practicum I	6
SOWK 482	Social Work Field Practicum II	6
SPAN 490	Independent Study in Spanish	3
SRM 482	Internship in Sport and Recreation Management	6
SRM 490	Special Studies in Sport and Recreation Management	1-3
TESL 381	Practicum in TESOL Instruction	3
TESL 385	Practicum in TESOL	1-3
TESL 480	Student Teaching/Internship	5
WGSS 301	Feminist Blogging: Writing for Shout Out! JMU	3
WGSS 492	Internship in Women's, Gender and Sexuality Studies	1-3

Departmental Honors Capstone for Non-Minor Students

- Completion of a 499 Honors sequence or departmental approved courses for 6 hours of credit and presentation requirements 6 Credit Hours.
- Capstone requirements are defined by the academic unit in consultation with the Honors College Students will work with their academic unit and their Honors Liaison to define the capstone. May be but is not limited to a thesis, project, or portfolio.
- Approval from the academic unit in which the capstone is completed is required.
- Capstone students will earn a graduation designation as "With Distinction in (Major or Minor)."

Department-Approved Honors for Non-Minor Students

Current JMU students who are not in the Honors Interdisciplinary Studies minor may pursue independent scholarship or creative activity leading to the completion of a Departmental Honors Capstone Project in their major or minor. These students must declare their interest in completing departmental honors and must obtain Honors College permission and submit the final project to the Honors College.

Upon approval of the major or minor academic unit and the Honors College a graduation designation with distinction in the major or minor is awarded to students completing this project with a grade of "B" or better.

Humanitarian Affairs Minor

Coordinator: Tara Parsons
Email: parsonstn@jmu.edu
Phone: (540) 568-7645
Humanitarian Affairs Minor Website (<http://www.jmu.edu/humanitarian>)

Program Description

The cross disciplinary humanitarian affairs minor provides students with a global perspective on the major humanitarian crises facing the human community today. Students will investigate the impacts of natural disasters, disease, hunger and famine, poverty, conflict, and national and international policies on the welfare of human communities around the world. They will also explore solutions to various humanitarian crises and evaluate the problems of practicing humanitarianism in the world today. Students will become acquainted with the organizations that address these crises and the challenges they face. Finally, students will acquire knowledge and skills that prepare them for careers in governmental and non-governmental organizations that address humanitarian issues. The minor in humanitarian studies is open to all undergraduate students at JMU.

Minor Requirements

Code	Title	Credits
Core Requirements		
HUMN 201	Introduction to Humanitarian Affairs	3
Subtotal		3
Electives		
Choose five of the following courses, with no more than two from any single discipline: 15		
HUMN/GEOG 301	Introduction to Natural Disasters Response	
HUMN 302	Legal Issues in Humanitarian Affairs	
HUMN 350	Special Topics in Humanitarian Affairs	
HUMN/GEOG 360	GIS for Humanitarian Assistance	
HUMN 490	Humanitarian Affairs Field Experience	
ANTH 366	Anthropology of War	
BIO 325	Medical Parasitology	
ECON 270	International Economics	
ENGR 471	Water in Africa	
GEOG 280	Human Geography: The Cultural Landscape	
GEOG 322	Agricultural Systems	
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet	
GEOG 335	Geography of Africa	
GEOG 339	Geography of the Caribbean	
GEOG 345	Geography of Poverty	
GEOG 374	Population Geography	
GEOG 375	Political Geography	
HTH 456	Grant Writing in Health Sciences (Summer Only)	
ISAT 459	Awareness and Understanding of Chemical, Biological and Radiological Weapons of Mass Destruction	
IA 460	All Hazards Response and Management Systems	
JUST 235	Justice in the Global Community	

JUST 341	Gender and Justice	
JUST 345	Restorative Justice	
JUST 351	Building Democracy	
JUST 352	Global Migration	
JUST 353	Justice and Development	
JUST 354	Native Justice and Resistance	
JUST 355	Solutions to Global Poverty	
JUST 356	Refugees and Humanitarian Response	
JUST 357	Environmental Justice	
JUST/POSC 372	Ethics and International Politics	
JUST 373	Rebuilding Post Conflict Societies	
JUST/POSC 392	Peace Studies	
NPS 300	Introduction to Nonprofits	
NUTR 380	Global Nutrition	
POSC 200	Global Politics [C4GE]	
POSC/JUST 331	Human Rights in Theory and Practice	
POSC 340	Politics of Development	
POSC 347	Comparative Public Policy	
POSC 348	The Politics of Cultural Pluralism	
POSC 393	Human Security	
POSC 395	International Law	
POSC 396	International Organizations	
POSC 397	The Politics of International Economic Relations	
REL 385	Religion and the Law of War	
SCOM 331	Communication and Conflict	
SOCI 357	Sociology of Disasters	
SOWK 288	Human Rights and Social Progress	
SOWK 375	Grant Writing for Agencies	
WRTC 314	Writing in the Public Sphere	
WRTC 484	Writing for Nonprofits	
WRTC 486	Writing in the Community	
Subtotal		15
Total Credits		18

Illustration Minor

Coordinator: Rich Hilliard
Email: hilliard@jmu.edu
Phone: (540) 568-6367
Illustration Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/illustration.shtml>)

ART 276	Introductory Book Arts: Materials and Structures
ART 270	Printmaking: Screenprint
Subtotal	3
Total Credits	18

Program Description

The minor in illustration offers students the opportunity to learn how to effectively use both traditional and digital media. The student will learn to create illustrations that clearly communicate messages, narratives and technical information. The minor in illustration is designed to broaden students' understanding of the value and role of illustration in art, design and professional practice.

Students enrolled in any major within the School of Art, Design and Art History may minor in illustration.

Admission Requirements

All students interested in minoring must first complete FDS 102, FDS 104 and FDS 108. Once these courses are complete, the student can apply for the illustration minor with portfolio submission.

Admission is limited and competitive based on an applicant's portfolio consisting of work completed in FDS 102, FDS 104 and FDS 108 with a minimum grade of "B" earned in all three classes.

SADAH Computer Requirements for Illustration Minor

SADAH's programs require all students to have a laptop computer equipped with and capable of running any necessary software. SADAH considers a laptop an essential tool for total participation in classes and content.

SADAH faculty determine the hardware and software requirements annually to ensure students purchase the most up-to-date technology conducive to participating in the curriculum.

For all hardware and software requirements, please see the department's website (<https://www.jmu.edu/sadah/index.shtml>).

Minor Requirements

Students may double count nine credits between their major and the minor.

Code	Title	Credits
Required Courses		
GRPH 200	Image Making	3
ART 210	Figure Drawing	3
ART/GRPH 300	Illustration	3
ART/GRPH 310	Graphic Illustration	3
ART/GRPH 400	Topics in Illustration	3
Subtotal		15
Elective Courses		
Choose three credit hours from the following:		3
GRPH 101	Digital Media Design I	

Integrated Science and Technology Minor

Advisor: Paul Henriksen
Email: henrikpw@jmu.edu
Phone: (540) 568-2755
ISAT Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/integrated-science-and-technology.shtml>)

Program Description

ISAT combines multiple scientific areas into an exciting hands-on, integrated program. Students learn by working directly with technology and applying it to solve real-world problems. Adding a minor enhances your knowledge and skills in different areas and showcases your versatility and adaptability to future employers or graduate schools.

Minor Requirements

The minor in ISAT mirrors the major by including both breadth and depth. The breadth component is satisfied through nine credit hours in Issues in Science and Technology and Instrumentation and Measurement courses. The depth component is satisfied through focused study in a strategic sector.

Students should note that many courses have ISAT prerequisites that are outside the minor (although equivalents to ISAT prerequisite courses will be accepted). In planning a sequence of courses for the minor, students are encouraged to meet with an ISAT advisor to ensure that all needed prerequisites will be met. In addition, a minimum grade of "C-" is required for all ISAT foundation courses that serve as prerequisites for other required ISAT foundation courses.

The minimum requirements for the minor in ISAT are as follows:

Code	Title	Credits
Foundational Courses		
Choose 9-11 credit hours from the following:		9-11
ISAT 112	Environmental Issues in Science and Technology (2,3) [C3PP]	
ISAT 113 & 113L	Biotechnology Issues in Science and Technology [C3NS] and Biotechnology Issues in Science and Technology Laboratory [C3L]	
ISAT 211	Issues in Modern Production Systems	
ISAT 212	Energy Issues in Science and Technology	
ISAT 252	Programming and Problem Solving	
Subtotal		9-11
Sequences		
Choose one of the following sequences:		6-7
Energy:		
ISAT 301	Instrumentation and Measurement in Energy (0,2)	
ISAT 310	Energy Fundamentals I	
ISAT 311	Role of Energy in Modern Society	
Environment and Sustainability:		
ISAT 320	Fundamentals of Environmental Science and Technology I	
ISAT 321	Fundamentals of Environmental Science and Technology II	

Industrial and Manufacturing Systems:		
ISAT 330	Sustainable Manufacturing Systems	
ISAT 331	Cyber-Physical Industry	
Applied Computing:		
ISAT 340	Database Design, Implementation and Management	
ISAT 341	Machine Learning and Data Science	
Applied Biotechnology:		
ISAT 305	Instrumentation and Measurement in Biotechnology (0, 2)	
ISAT 350	Biotechnology for the New Millennium I	
ISAT 351	Biotechnology for the New Millennium II	
Subtotal		6-7
Additional Integrated Science and Technology Course		
Choose one additional Integrated Science and Technology course at the 300 or 400 level		3
Subtotal		3
Total Credits		18-21

Jazz Studies Minor

Advisor: Chuck Dotas

Email: dotascj@jmu.edu

Phone: (540) 568-6180

Jazz Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/jazz-studies.shtml>)

Program Description

The Jazz Studies minor offers interested students an opportunity to explore the basics of jazz style via classroom study and ensemble performance. The minor strengthens skills in small ensemble (combo) and large ensemble (big band) performance, basic music theory, and techniques of jazz improvisation. The Jazz Studies minor expands students' appreciation of this unique, Black American art form.

Open to all undergraduate students at JMU, the jazz studies minor helps students majoring in other disciplines understand and perform the jazz art form.

Minor Requirements

The minor requires the successful completion of 14-15 credit hours in specified music courses and four credit hours in jazz ensembles. Students must earn 50 percent of the required curriculum at JMU.

Code	Title	Credits
Fundamentals		
MUS 240	Jazz Improvisation Laboratory I	2
MUS 356	History of Jazz	3
MUS 440	Jazz Improvisation Laboratory II (must repeat three times for a total of 6 credit hours)	6
Choose one of the following:		3-4
MUS 131	Fundamentals of Music	
MUS 140	Introduction to Music Theory and Aural Skills	
MUS 255	Jazz Theory	
Subtotal		14-15
Ensembles		
Choose one of the following:		4
MUAP 347	Jazz Ensemble (repeatable)	
MUAP 348	Jazz Band	
MUAP 355	Jazz Chamber Ensemble	
Subtotal		4
Total Credits		18-19

Latin American, Latinx and Caribbean Studies Minor

Coordinator: Verónica Dávila Ellis

Email: davilave@jmu.edu

Phone: (540) 568-6128

Advisor: Kristin Wylie

Email: wyliekn@jmu.edu

Phone: (540) 568-5307

Latin American, Latinx and Caribbean Studies Minor Website (<http://www.jmu.edu/laxc>)

Program Description

This cross disciplinary minor helps students to acquire a deeper understanding of the histories, politics and cultures of Latin America, the Caribbean and Latinx communities in the U.S. Participants in the minor are strongly encouraged to pursue course work in languages including Spanish, Portuguese and French, as well as to explore the possibility of studying in a Latin American or Caribbean country for a semester or summer session.

Minor Requirements

Students must select courses in at least three different disciplines, with no more than nine credits in any discipline (e.g., ANTH, HIST). Other special topics classes may be accepted via course directive. Consult the advisor for more information.

Code	Title	Credits
Core Courses		
Choose two of the following:		6
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	
HIST 313	The History of Latinos in the United States	
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]	
LAXC 300	Topics in Latinx Studies ¹	
POSC 350	Latin American Politics	
SPAN 308	Latin American Cultures	
Subtotal		6
LAXC Primary Electives		
Complete at least two and up to four additional courses from the following:		6-12
ANTH 265	Peoples and Cultures of Latin America and the Caribbean	
ANTH 303	Andean Archaeology	
ANTH 364	U.S./Latin American Borders	
ANTH 436	Afro-Latin America	
ARTH 484	Art of the Americas	
ENG 367	Latinx Literature	
GEOG 339	Geography of the Caribbean	
HIST 302	Plazas, Protests and Processions: An Urban History of the Americas	
HIST 307	The Trans-Atlantic Slave Trade	
HIST 313	The History of Latinos in the United States	

HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States
HIST 436	Afro-Latin America
HIST 437	Latin America and Latin Americans through Film: Focus on the Twentieth Century
HIST 444	Revolution and Social Change in Latin America
HIST 447	South America
HIST 448	Gender in Latin America and the Iberian World
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]
LAXC 299	Internship in Latin American, Latinx and Caribbean Studies
LAXC 300	Topics in Latinx Studies ¹
LAXC/POSC 375	Latina/o/x Politics: Origins and Futures
LAXC 490	Independent Study in Latin American, Latinx and Caribbean Studies
LAXC 495	Special Topics in Latin American, Latinx and Caribbean Studies
POSC 350	Latin American Politics
POSC 378	Borderlands Policy: Latina/o/x Politics in Context
REL 324	Latinx and Latin American Christianity
SPAN 308	Latin American Cultures
SPAN 321	Culture and Communication in Context (for Heritage Speakers)
SPAN 385	Latin American Short Story
SPAN 404	Spanish in the United States
SPAN 421	Stories and Histories: Latin American Novel
SPAN 423	Voices of Today: Latin American Poetry
SPAN 476	Culture and Medicine in Latin America
SPAN 485	Business and Society in Latin America
Subtotal	6-12
Global and Ethnic Studies Secondary Electives	
Complete no more than two additional courses from the following:	
ECON 270	International Economics
ECON 312	Comparative Economic Systems
ECON 365	Economic Development
EDUC 310	Teaching in a Diverse Society
ENG 360	Introduction to Ethnic American Literature
HUMN 201	Introduction to Humanitarian Affairs
JUST 328	Race, Class and Justice
JUST 352	Global Migration
JUST 356	Refugees and Humanitarian Response
JUST 357	Environmental Justice
POSC 341	Social Movements in the U.S. and Abroad
POSC 366	Introduction to the Politics of Race and Ethnicity
POSC 367	U.S. Immigration and Refugee Law
POSC 376	Introduction to Black Politics
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation
POSC/WGSS 383	Women and Politics in Comparative Perspective
POSC 397	The Politics of International Economic Relations
REL 450	Religion and Society

SCOM 248	Intercultural Communication	
SOCI 318	Sociology of Immigration	
SPAN 365	Spanish for Medical Professionals I	
SPAN 475	Spanish for Medical Professionals II	
Total Credits		18

¹ LAXC special topics course. Can be repeated once for credit if the topic changes. Only one will count toward the core requirement; the second will count toward required electives.

Leadership in Military Affairs Minor (Air Force)

Email: AFROTC@virginia.edu

Program Description

Grow in your leadership skills as you prepare to be commissioned as an officer in the U.S. Air Force with a Leadership in Military Affairs minor. Study modern military history and gain practical skills in required leadership laboratories each semester as you complete the required courses for commissioning among Air Force ROTC cadets.

Learn from current and non-commissioned officers and civilian employees of the U.S. Air Force who have years of experience in military service as you prepare for your career.

This minor is only available to cadets in the AFROTC program (p. 597).

Minor Requirements

Code	Title	Credits
Required Courses		
AIRS 100	Leadership Laboratory (must be taken for a total of 6 credit hours)	6
AIRS 110	Air Force Heritage and Values I	1
AIRS 120	Air Force Heritage and Values II	1
AIRS 210	Team and Leadership Fundamentals I	1
AIRS 220	Team and Leadership Fundamentals II	1
AIRS 310	Leading People and Effective Communication I	3
AIRS 320	Leading People and Effective Communication II	3
AIRS 410	National Security Affairs/Preparation for Active Duty I	3
AIRS 420	National Security Affairs/Preparation for Active Duty II	3
Total Credits		22

Legal Studies Minor

Coordinator: Melinda Adams

Email: adams2mj@jmu.edu

Phone: (540) 568-3377

Legal Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/legal-studies.shtml>)

Program Description

An understanding of law is vital to an understanding of political, social and economic life and is an important element in becoming an "educated and enlightened citizen." The minor in legal studies offers an interdisciplinary program of study that will give students the opportunity to examine law from a variety of perspectives. The minor in legal studies is not a program in pre-professional legal training; it is designed to serve pre-law students at JMU as well as students who are interested in careers in government, health, technology, business or the non-profit sector. The minor focuses broadly on the skills needed to study the law (logical and ethical reasoning, critical thinking, strong writing, and close reading) as well as the philosophy, praxis and applications of the law writ large. As such, it can serve as a strong complement to many major fields of study or as an area of interest to those who think they may want to pursue law school or a legal career.

Program Goals

- To develop students' critical thinking, communication, and analytical skills
- To introduce students to an interdisciplinary study of law from a variety of perspectives
- To expand and enrich students' understanding of law and the multiple functions it plays in our national (and global) society

Minor Requirements

No more than nine credit hours (three courses) may be taken from a single discipline. Students should check with their major advisors regarding the ability to double count courses between the Legal Studies Minor and their major(s).

Code	Title	Credits
Required Core Course		
PHIL 250	Introduction to Symbolic Logic	3
or PHIL 315	Logic and Legal Reasoning	
Subtotal		3
Electives		
Choose five of the following: ¹		15
CRJU 337	Courts and the Judiciary	
ENG 221	Literature and Ideas [C2L] (when topic is Law and Vengeance)	
ENG 302	Special Topics in Literature and Language (when topic is Literature and Law)	
LING 430	Forensic Linguistics: Language and the Law	
HIST 339	Selected Themes in U.S. History (when topic appropriate)	
HIST 341	Selected Themes in World History (when topic appropriate)	
JUST 300	Perspectives on Comparative Justice	
JUST 301	Advanced Level Special Topics in Justice Studies (when topic is Law and Society)	

JUST 323	Comparative Criminal Justice
JUST 327	Criminal Law
JUST 329	Perspectives on Law
JUST 386	Sexual Orientation, Law and Justice
JUST 397	Legal Research and Information Literacy
PHIL 250	Introduction to Symbolic Logic
PHIL 262	Problems in Applied Ethics
PHIL 267	Philosophy of Law
PHIL 300	Knowledge and Belief
PHIL 310	Symbolic Logic
PHIL 315	Logic and Legal Reasoning
PHIL 330	Moral Theory
PHIL 335	The Individual, The State and Justice
POSC 325	Constitutional Law
POSC 326	Civil Rights
POSC 386	The U.S. Judiciary
POSC 387	The Appellate Process and Practice
REL 383	The Global Refugee Crisis and Religion
SCOM 342	Argument and Advocacy
SCOM 346	Free Speech in America
SMAD 330	New Media Law (limited seat availability for minors) ²
SMAD 370	Mass Communication Law (limited seat availability for minors) ²
SMAD 450	Law of Media Arts and Design (limited seat availability for minors) ²
WRTC 301	Language, Law and Ethics
WRTC 460	Legal Writing
Subtotal	15
Total Credits	18

¹ Students may select no more than two PHIL courses (six credit hours) to fulfill the elective requirement.

Other courses outside of this list may be counted towards the minor with the approval of the minor coordinator(s). A maximum of two courses (six credit hours) can count in this way. Courses must be approved before the student enrolls.

² SMAD 330 and SMAD 370 can count toward the Legal Studies minor with minor coordinator permission.

Linguistics Minor

Coordinator: Jennifer Lang-Rigal

Email: langrijr@jmu.edu

Phone: 540-568-6886

Linguistics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/linguistics.shtml>)

Program Description

The interdisciplinary minor in linguistics offers courses founded on the core concepts of linguistics, methods of linguistic analysis, modern linguistic theories and the application of linguistics to other areas. The program is drawn from a variety of departments and colleges offering diverse courses whose approaches to linguistics range from theoretical linguistics, to anthropological linguistics, to Spanish linguistics and speech pathology. The purpose of this program is to provide the student with a fundamental understanding of human language and the different ways language impacts our lives. This program is relevant for students interested in a variety of professions, including but not limited to: language teaching, interpretation, translation, speech-language pathology and audiology, language policy and planning, language documentation and revitalization, and computer sciences/natural language processing. Students will develop their knowledge of their native and second languages through metalinguistic inquiry, consequently enriching their ability to engage more broadly with the forms and functions of linguistic communication. The linguistics minor will encourage students to appreciate linguistic diversity, think about language-focused discrimination and recognize the importance of language in efforts to encourage inclusive excellence.

Minor Requirements

Students must take six courses (18 credits) from the following list. A maximum of 9 credits hours from a single discipline (e.g., ENG or SPAN) may be earned. One three-credit foreign language course of any level may be counted toward the minor by approval of the minor coordinator. These credits cannot also be counted toward the requirements for the Bachelor of Arts in Modern Foreign Languages (p. 371). One other appropriate course not listed below may be considered for the minor upon consultation with the minor coordinator.

Code	Title	Credits
Choose six of the following:		18
ANTH/SCOM 305	Language and Culture	
ANTH 405	Topics in Linguistic Anthropology	
CSD 200	Introduction to Communication Sciences and Disorders	
ENG 301	Old English Language and Literature	
ENG 303	History of the English Language	
ENG 308	Introduction to Linguistics	
ENG 310	Structure of the English Language	
ENG 348	Studies in Linguistics and the English Language	
ENG 417	Advanced Studies in Linguistics and the English Language	
GER 435	Translation Strategies ¹	
GER 448	Special Topics in German Linguistics ¹	
LING 311	Study of Language Structures	
LING 312	Bilingualism in the United States	

ITAL 315	Sounds of Speech ²	
LING 410	Introduction to Romance Linguistics	
LING 430	Forensic Linguistics: Language and the Law	
LING 470	World Language Teaching Methods	
PHIL 250	Introduction to Symbolic Logic	
PHIL 380	Philosophy of Language	
SCOM 245	Signs, Symbols and Social Interaction	
SPAN 310	Introduction to Spanish Linguistics ³	
SPAN 311	Contrastive Linguistics ³	
SPAN 315	Sounds of Speech ³	
SPAN 404	Spanish in the United States ³	
SPAN 435	Translation Strategies ³	
SPAN 436	Introduction to Interpretation ³	
SPAN 448	Special Topics in Spanish Linguistics ³	
TESL 300	Introduction to Second Language Acquisition	
Total Credits		18

¹ GER courses are taught in German and have a prerequisite of GER 300 or equivalent in a placement test.

² ITAL courses are taught in Italian and have a prerequisite of ITAL 300 or equivalent in a placement test.

³ SPAN courses are taught in Spanish and have a prerequisite of SPAN 300 or equivalent in a placement test.

Logic and Reasoning Minor

Coordinator: Tracy A. Lupher

Email: lupherta@jmu.edu

Phone: (540) 568-6394

Logic and Reasoning Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/logic-and-reasoning.shtml>)

Program Description

Logic and reasoning are central to human inquiry. The minor brings together several disciplines which take logic and reasoning as part of their subject matter, emphasize the core logical elements that are common across disciplines and show how to apply these common logical concepts and tools in different domains.

Minor Requirements

The requirement for a minor in logic and reasoning is 18-19 credits: 12-13 credit hours from a set of core courses and six credit hours from electives.

Code	Title	Credits
Required Logic Courses		
PHIL 250	Introduction to Symbolic Logic	3
PHIL 310	Symbolic Logic	3
or PHIL 326	Introduction to Non-Standard Logics	
Subtotal		6
Mathematics Requirements		
Choose one of the following pairs of courses:		6-7
MATH 235	Calculus I [C3QR]	
MATH 245	Discrete Mathematics	
or		
MATH/CS 227	Discrete Structures I	
CS 327	Discrete Structures II	
Subtotal		6-7
Electives		
Choose two of the following:		6
CS 444	Artificial Intelligence	
MATH 424	Statistical Decision Theory	
PHIL 310	Symbolic Logic ¹	
PHIL 315	Logic and Legal Reasoning	
PHIL 320	Inductive Logic	
PHIL 326	Introduction to Non-Standard Logics ¹	
PHIL 396	Philosophy of Physics	
PHIL 397	Philosophy of Space and Time	
PHIL 398	Philosophy of Quantum Theory	
PHIL 410	Philosophy of Science	
Subtotal		6
Total Credits		18-19

¹ If taken as a required logic course, this course cannot also count as an elective.

Materials Science Minor

Coordinator: Christopher Hughes

Email: hugheswc@jmu.edu

Phone: (540) 568-8069

Materials Science Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/materials-science.shtml>)

Program Description

The educational mission of the Center for Materials Science is to develop and maintain an innovative cross disciplinary and multidisciplinary undergraduate program in materials science that will increase the maturation of students, their research experience and their employment opportunities. The mission includes the integration of undergraduate education with basic and applied research in materials science. Faculty and students focus on problems of interest to industry and government in materials processing, materials characterization, materials applications and thermal sciences including thermal structural interactions and infrared analysis.

Minor Requirements

The minor in materials science includes four major components:

- Choice of an entry-level introductory course in materials science.
- Lecture or laboratory course that emphasizes more specialized areas in materials science.
- Materials science electives that can include all specialized courses.
- Research or an additional materials science lecture or laboratory experience.

Courses for the minor are offered through the departments of chemistry, geology and environmental studies, integrated science and technology, mathematics, and physics.

Code	Title	Credits
Introductory Course		
MATS/PHYS/ CHEM 375	An Introduction to Materials Science	3
or ISAT 430	Materials and Manufacturing Processes	
Subtotal		3
Lecture/Laboratory Course		
PHYS 381	Materials Characterization (Lecture/Lab)	3
or ISAT 432	Materials Selection in Product Design	
Subtotal		3
Material Science Electives		
Choose nine credit hours of the following:		9
GEOL 280	Mineralogy (3, 2)	
GEOL 300	Introduction to Petrology (3, 3)	
ENGR 314	Materials and Intermediate Solid Mechanics	
PHYS 337	Solid State Physics	
PHYS 381	Materials Characterization (Lecture/Lab)	
PHYS 380	Thermodynamics and Statistical Mechanics	
CHEM 331	Physical Chemistry I	
ISAT 432	Materials Selection in Product Design	
CHEM 445	Polymer Chemistry	
CHEM 485 & 485L	Science of the Small and Science of the Small Laboratory	

PHYS 485 & 485L	Science of the Small and Science of the Small Laboratory	
Special Topics in Materials Science registered under:		
CHEM 480	Selected Topics in Chemistry (materials science)	
GEOL 398	Topics in Geology (materials science)	
ISAT 480	Topics in Integrated Science and Technology (i.e., light metals)	
MATS 498	Undergraduate Materials Science Research	
PHYS 497	Topics in Physics (materials science)	
Subtotal		9
Research in Materials Science		
Register for Research in Materials Science under one of the following:		3
CHEM 497	Undergraduate Chemical Research (A, B, C) (in materials science, 2-4 credits)	
GEOL 497	Work-Based Learning Experiences in Geology and Environmental Science (in materials science, 1-3 credits)	
ISAT 491 & ISAT 492 & ISAT 493	Senior Capstone Project I and Senior Capstone Project II and Senior Capstone Project III (in materials science, 6 credits)	
PHYS 498	Undergraduate Research in Physics or Astronomy (in materials science, 2-4 credits)	
MATS 498	Undergraduate Materials Science Research (1-3 credits, repeatable to 6 credits)	
Subtotal		3
Total Credits		18

Mathematics for Elementary and Middle Grades Teachers Minor

Advisor: Alexis Stevens

Email: stevenal@jmu.edu

Phone: (540) 568-6701

Mathematics for Elementary and Middle Grades Teachers Minor Website
(<https://www.jmu.edu/academics/undergraduate/minors/math-for-middle-grades.shtml>)

Program Description

The Mathematics for Elementary and Middle Grades Teachers Minor is open to students not majoring in mathematics or statistics. Each student must obtain prior approval of all courses to be counted in the minor from the advisor or from the department head.

The Department of Mathematics and Statistics (p. 82) offers credit by examination for some of the courses taught in the department. Students who want to take an examination must apply to the department head. Upon application students will receive details regarding approval to take the examination.

Minor Requirements

The minor requires 18 credit hours. At least six hours must be earned within the JMU Department of Mathematics and Statistics.

Code	Title	Credits
Required Courses		
MATH 110	Mathematics for Elementary and Middle School Teachers I [C3QR]	4
MATH 111	Mathematics Seminar for Elementary and Middle School Teachers	1
MATH 210	Mathematics for Elementary and Middle School Teachers II	4
MATH 304	Principles of Algebra	3
MATH 305	Principles of Geometry	3
MATH 307	Principles of Probability and Statistics	3
Total Credits		18

Mathematics Minor

Advisor: Alex Capaldi

Email: capaldaj@jmu.edu

Phone: (540) 568-5102

Mathematics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/mathematics.shtml>)

Program Description

Mathematics forms the foundation of many fields of study; students can support their studies in another major by completing a minor in Mathematics. The minor provides a solid foundation of critical thinking and mathematical concepts while also allowing for a bit of advanced abstract thinking. This collection of skills often appeals to employers and graduate schools. Students majoring in Biology, Biotechnology, Chemistry, Computer Science, Economics, Engineering, Finance and Physics may be particularly interested in a minor in Mathematics because of requirements of those programs.

The mathematics minor is open to students not majoring in mathematics or statistics. Each student must obtain prior approval of all courses to be counted in the minor from the advisor, Alex Capaldi, or from the department head.

Minor Requirements

A minor in mathematics requires a minimum of 18 credit hours. At least six hours must be earned within the JMU Department of Mathematics and Statistics.

Credit by Examination

The Department of Mathematics and Statistics offers credit by examination for some of the courses taught in the department. Students who want to take an examination must apply to the department head. Upon application students will receive details regarding approval to take the examination.

Core Courses

Code	Title	Credits
MATH 231 & MATH 232 & MATH 236 or MATH 235 & MATH 236	Calculus with Functions I [C3QR] and Calculus with Functions II and Calculus II Calculus I [C3QR] and Calculus II	8-10
Mathematics courses numbered MATH 237 or above, excluding mathematics courses numbered 301-309		10
Total Credits		18-20

Medical Humanities Minor

Coordinator: Michael J. Klein
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Medical Humanities Minor Website (<http://www.jmu.edu/medical-humanities>)

Program Description

The cross disciplinary minor in medical humanities provides students with a humanistic and social study of illness, health and the body. Typically, we approach the human body from a medical, diagnostic model. But physicians have long recognized that this approach to health and well-being is not sufficient for understanding our somatic life. Drawing upon concepts found in the disciplines contributing to the minor, courses in the curriculum share the common focus of examining how individuals and institutions articulate and disseminate conceptions of the body, including constituent issues such as race, ethnicity, sexuality and gender.

Minor Requirements

Students select a minimum of 18 credit hours taken from a list of approved courses. No more than six credits from a single subject (e.g. ANTH, ENG, SCOM, SOCI) can count toward the minor.

Code	Title	Credits
Required Courses		
Choose six of the following:		18
ANTH/SOCI 352	Birth, Death, Sex: Exploring Demography	
ANTH 360	Medical Anthropology	
ANTH 376	Anthropology of Reproduction	
ENG 360	Introduction to Ethnic American Literature	
ENG/WGSS 368	Women's Literature	
ENG/WGSS 369	Feminist Literary Theory	
ENG/WGSS 370	Queer Literature	
ENG 423	Advanced Studies in Gender and Sexuality in Literature	
ENG/WGSS 466	Advanced Studies in Women's Literature	
GEOG 323	The Geography of Human Genetics, Infectious Diseases and Diet	
HTH 372	Human Sexuality	
HIST 306	A History of the Body in the West	
HIST 319	Women at Work in U.S. History	
HIST 320	Women in U.S. History	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 448	Gender in Latin America and the Iberian World	
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	
ISAT 456	Ethical, Legal and Social Implications of Biotechnology	
ISAT/WGSS 485	Gender Issues in Science	
PHIL 262	Problems in Applied Ethics	

PHIL/WGSS 350	The Philosophy of Feminism	
PSYC 308	Health Psychology	
REL 315	Women and Religion	
REL 371	Religion and Disability	
REL 372	Religion and Medicine, East and West	
REL 373	Race and Religion	
REL 380	Religion and Science	
SCOM 303	Special Topics in Health Communication	
SCOM/WGSS 348	Communication and Gender	
SCOM 371	Disability and Advocacy	
SCOM 468	Interpersonal Communication in Health Contexts	
SCOM 471	Culture and Health Communication	
SOCI 336	Race and Ethnicity	
SOCI/WGSS 337	Sociology of Gender	
SOCI 375	Medical Sociology	
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	
WRTC/WGSS/SCOM 420	Feminist Rhetorics	
WRTC 458	Scientific and Medical Communication	
WRTC 488	Writing in the Health Sciences	
Total Credits		18

Medical Spanish Minor

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Medical Spanish Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/medical-spanish.shtml>)

Subtotal	3
Total Credits	18

Program Description

For health service professionals, knowledge of medical Spanish is no longer an option but a necessity. Unless they have a reasonable level of Spanish, communication with patients whose only language is Spanish will be difficult. Courses in medical Spanish are therefore hugely relevant today. The minor in medical Spanish is intended for all future health service professionals, including physicians and their assistants, nurses, pharmacists, dentists, physical therapists, occupational therapists, medical laboratory technicians, emergency medical technicians, medical aides, medical interpreters, healthcare industry professionals and psychologists.

Minor Requirements

The minor comprises one linguistic component, one elective component, one translation/interpretation component and three profession-specific linguistic components.

Code	Title	Credits
Language Foundations		
Choose one of the following:		3
SPAN 300	Language for Living	
SPAN 301	Pop Culture through Language	
SPAN 307	History of Spanish Culture and Society	
SPAN 308	Latin American Cultures	
SPAN 320	Culture and Communication in Context	
SPAN 321	Culture and Communication in Context (for Heritage Speakers)	
SPAN 322	Language Landscapes	
Subtotal		3
Medical Spanish Core		
SPAN 365	Spanish for Medical Professionals I	3
Choose two of the following:		6
SPAN 475	Spanish for Medical Professionals II	
SPAN 476	Culture and Medicine in Latin America	
SPAN 477	Culture and Medicine in Spain	
SPAN 478	Spanish in Healthcare: Policy and Practice in Latinx Health	
Subtotal		9
Interpretation Course		
Choose one of the following:		3
SPAN 435	Translation Strategies	
SPAN 436	Introduction to Interpretation	
SPAN 437	Introduction to Medical Interpretation	
SPAN 438	Introduction to Legal Interpretation	
Subtotal		3
Electives		
Choose any 300- or 400-level SPAN course not already taken.		3

Medieval and Renaissance Studies Minor

Coordinator: Mary K. Gayne
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Medieval and Renaissance Studies Minor Website (<https://www.jmu.edu/cal/academics/minors/medren/index.shtml>)

Program Description

This minor focuses on the period from the fall of Rome (5th century C.E.) to 1700. The minor allows students to take courses in art history, English, history, music, philosophy, political science, religion and some languages, including medieval languages such as Old English and Middle English in the original or in translation, and elementary and intermediate courses in Arabic, Greek and Latin.

Minor Requirements

The minor requires 18 credits total from the list below, with the following restrictions. The 18 credit hours must include courses from at least three distinct disciplines. Note that all world language courses (ARAB, FR, GER, ITAL and LAT) fall into the single discipline of world languages and cultures. Other languages may also be counted for credit with the approval of the instructor. Introductory (i.e., 100 and 200 level) modern language courses are not permitted for credit. At least four of the six required classes must be 300- or 400-level classes. It is recommended that students take one introductory course before taking advanced classes.

Other courses with significant Medieval and/or Renaissance content may be counted toward the minor with the approval of the coordinator. In addition, students interested in pursuing a three-credit hour focused individual research project should contact individual faculty members with whom they might wish to work to determine the suitability of this option in their particular cases. Only one such independent study will count toward the minor.

Approval of the minor coordinator will also be required in the following two cases. First, courses with an asterisk will count toward the minor only if their content is focused on the Medieval and/or Renaissance periods. Second, because the content of HUM 200, HUM 250 and HUM 252 may vary, the minor coordinator will determine the appropriate disciplinary category for these courses.

Required Courses

Code	Title	Credits
Introductory		
Only two introductory courses may be counted toward the minor. Choose two of the following:		6
ARTH 205	Survey of World Art I: Prehistoric to Renaissance [C2VPA]	
ENG 235	Survey of English Literature: From Beowulf to the Eighteenth Century [C2L]	
HIST 101	World History to 1500 [C2HQC]	
HUM 200	Great Works [C2L] ¹	
HUM 250	Foundations of Western Culture [C2HQC] ¹	
HUM 252	Global Cultures [C2HQC] ¹	
REL 250	Introduction to Islam	

Subtotal		6
Advanced		
At least four courses must be 300 or 400 level. Choose four of the following:		12
ARTH 313	Masterpieces of Italian Renaissance Art [C2VPA]	
ARTH 426	Advanced Topics in Cross-Cultural Art (Only 3 credit hours toward the minor.) ¹	
ARTH 427	Advanced Topics in Pre-Modern Art ¹	
ARTH 446	Renaissance Art and the East	
ARTH 448	Studies in Leonardo and Michelangelo	
ENG 302	Special Topics in Literature and Language ¹	
ENG 306	The Bible as Literature ¹	
ENG 311	Medieval Literature and Culture	
ENG 313	Studies in Early Modern Literature and Culture	
ENG 316	Early Modern Drama	
ENG 317	Studies in Shakespeare	
ENG 320	Shakespeare on the Page and Stage in London	
ENG 401	Advanced Studies in Medieval Literature	
ENG 402	Advanced Studies in British Literature Before 1700	
ENG 410	Advanced Studies in Author ¹	
ENG 412	Special Topic Seminar ¹	
ENG 415	Advanced Studies in Textuality and the History of the Book ¹	
FR 446	Special Topics in Francophone Literature ¹	
GER 307	History of Germanic Cultures and Societies ¹	
GER 436	Studies in German Literature ¹	
HIST 305	History of Science and Christianity	
HIST 352	The Premodern Middle East: Themes and Intersections	
HIST 373	History of the Ottoman Empire, 1453-1922	
HIST 401	Research Capstone Topics ¹	
HIST 473	Early Modern Islamic Empires	
HIST 482	French History Seminar ¹	
ITAL 307	History of Italian Culture and Society ¹	
ITAL/ENG 437	Studies in Italian Literature ¹	
POSC 311	The Ideas and Legacies of Medieval and Renaissance Political Theory	
PHIL 460	Topics in Classical Philosophy ¹	
PHIL 342	Medieval Philosophy	
REL 304	Art, Religion and Identity in Medieval Sicily	
REL 306	Gender and Sexuality in Islam ¹	
REL 307	Knowledge, Identity and Power in Islamic Societies ¹	
REL/PHIL 339	Islamic Philosophy	
Total Credits		18

¹ Course will count toward the minor only if their content is focused on the Medieval and/or Renaissance periods.

Middle Eastern Communities and Migrations Minor

Coordinator: Shah Mahmoud Hanifi

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Phone: (540) 568-1743

Middle Eastern Communities and Migrations Minor Website (<https://www.jmu.edu/cal/academics/minors/mecm/index.shtml>)

Program Description

This minor concentrates on social and political issues involving Muslim, Christian and Jewish populations in their own right and in relation to one another in the territory between the Nile and Indus rivers during the modern period. The program is also designed to accommodate consideration of other communities including Hindu and other South Asians, Anatolian and Central Asian Turks, and Mediterranean peoples in the larger area stretching from North Africa to Southeast Asia as well as Middle Eastern diaspora communities in Europe and the Americas from the ancient period to the present.

The cross disciplinary orientation of the program emphasizes comparison and a synthesis of local, regional, trans-national and global perspectives. The program provides an intellectual foundation that can be usefully applied and built upon in graduate school, the private sector or government service.

Minor Requirements

Students are required to take 18 credits comprised of six credits of core course work and 12 credits of electives. Students may take no more than nine credits from a single academic unit.

The College of Arts and Letters (p. 13) departments offer a number of courses listed generically, usually at the 300- and 400-level (including disciplinary capstone and methodology courses, often numbered 395) that can also count toward the MECM minor under certain circumstances.

Code	Title	Credits
Foundational Courses		
Choose two of the following:		6
REL 250	Introduction to Islam	
HIST 352	The Premodern Middle East: Themes and Intersections	
HIST 353	The Global Middle East	
Subtotal		6
Electives		
Choose 12 credit hours from the following options. Special topics courses listed below may count toward the minor if they include substantial content relating to the foci and goals of the program. Contact the program coordinator to discuss applying a special topics course to the MECM minor.		12
ARAB 101	Elementary Arabic I	
ARAB 102	Elementary Arabic II	
ARAB 111	Intensive Arabic I	
ARAB 212	Intensive Arabic II	
ARAB 231	Intermediate Arabic I	
ARAB 232	Intermediate Arabic II	
ARAB 301	Pop Culture through Language	
ARAB 322	Language Landscapes	
ARAB 330	Business Arabic	
ARAB 335	A Journey through Literature	
ARAB 340	Intermediate Arabic Conversation	
ARAB 400	Advanced Arabic Writing and Conversation	
ARAB 447	Special Topics in Arabic Civilization and Culture	
ARAB 448	Special Topics in Arabic Linguistics	
ARTH 332	Islamic Art and Architecture	
ARTH 419	Topics in African Art	
ARTH 424	Arts of Ancient Egypt	
ARTH 440	City of Jerusalem	
ARTH 446	Renaissance Art and the East	
ENG 430	Advanced Studies in Comparative Literature	
ENG 433	Advanced Studies in Arabic Literature	
HIST 348	Maps and Mobility in Islamic World History	
HIST 351	War in Afghanistan: Local and Global Perspectives	
HIST 352	The Premodern Middle East: Themes and Intersections	
HIST 353	The Global Middle East	
HIST 372	Modern Afghanistan	
HIST 373	History of the Ottoman Empire, 1453-1922	
HIST 473	Early Modern Islamic Empires	
HIST 485	Colonial Knowledge Production in the Middle East and South Asia	
HUM 252	Global Cultures [C2HQC] (when topic is Islamic Civilization or Turkey: Meeting of East and West)	
PERS 101	Elementary Persian I	
PERS 102	Elementary Persian II	
PERS 231	Intermediate Persian I	
PERS 232	Intermediate Persian II	
POSC 340	Politics of Development	
POSC 354	Politics of the Middle East	
POSC 361	Topics in International Relations	
POSC 371	Topics in Comparative Politics	
REL 201	Introduction to Hebrew Bible / Old Testament	
REL 202	Introduction to the New Testament	
REL 220	Introduction to Judaism	
REL 250	Introduction to Islam	
REL 300	Selected Topics in Religion	
REL 306	Gender and Sexuality in Islam	
REL 307	Knowledge, Identity and Power in Islamic Societies	
REL 383	The Global Refugee Crisis and Religion	
SOCI 311	Sociology of the Environment	
SOCI 321	Politics in Society	
SOCI 360	Social Movements	
WLC 490	Special Studies: World Languages and Cultures in Translation	
Subtotal		12
Total Credits		18

Middle Grades Science Minor

Advisor: Eric Pyle
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Phone: (540) 568-7115
Middle Grades Science Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/middle-grades-science.shtml>)

Program Description

Students in the Middle Grades Science minor will complete requirements at the introductory and intermediate level in each of three content areas: earth and planetary science, physical science, and life sciences. Within each area, multiple options are available, allowing students to apply previously earned credit, such as AP, dual enrollment and/or JMU courses earned prior to declaring the minor. In each area, there is a single course that serves as the preferred option for students with no previous coursework in the area. These courses are denoted with an asterisk.

Minor Requirements

Code	Title	Credits
Earth and Planetary Science		
Choose one of the following:		3-4
ISCI 171	Earth and Planetary Science for Teachers [C3NS] ¹	
GEOL 102	Environment: Earth [C3NS]	
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]	
Subtotal		3-4
Introductory Level: Middle Grades Science Minor		
<i>Physical Science:</i>		
Choose one of the following options:		3-8
Option 1:		
ISCI 172	Physical Science for Teachers [C3PP] ¹	
Option 2:		
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	
or PHYS 240 & 240L	University Physics I [C3PP] and University Physics Laboratory I [C3L]	
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	
<i>Life Science:</i>		
Choose one of the following:		3-4
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	
ISCI 173 ¹	Life and Environmental Science for Teachers [C3L]	
Subtotal		6-12
Intermediate Level: Middle Grades Science Minor		
<i>Earth and Planetary Science:</i>		
GEOL 211	Introduction to Oceanography [C3NS] ¹	3
<i>Physical Science:</i>		
Choose one of the following:		3-4
CHEM 120	Concepts of Chemistry [C3PP]	
PHYS 121	The Physical Nature of Light and Sound (3, 1) [C3PP]	

CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	
PHYS 150 & 150L	College Physics II and General Physics Laboratory II	
PHYS 250 & 250L	University Physics II and University Physics Laboratory II	
<i>Life Science:</i>		
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
Subtotal		10-11
Total Credits		19-27

¹ This is the preferred option for students with no previous coursework in the area.

Military Leadership Minor

Advisor: Craig Gibson

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Military Leadership Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/military-leadership.shtml>)

Program Description

The Army ROTC program recruits, inspires, develops and commissions competent and confident warrior leaders of character as officers in the United States Army, Army Reserves, and Army National Guard upon graduation. As a part of the Professional Military Education requirements for commissioning, students are awarded a minor in Military Leadership through the Military Science course curriculum by learning and developing aspects of leadership, communication, warrior skills and teamwork.

This minor is only available to cadets enrolled in the Army ROTC program.

Minor Requirements

Code	Title	Credits
Basic Military Science Courses		
MSCI 110	Introduction to Leadership and the Army	2
MSCI 115	Basic Leadership Laboratory I	2
MSCI 120	Leadership Development Fundamentals	2
MSCI 125	Basic Leadership Laboratory II	2
MSCI 210	Leadership Styles: Theory and Application	2
MSCI 215	Intermediate Leadership Laboratory I	2
MSCI 220	Developing Leadership Skills	2
MSCI 225	Intermediate Leadership Laboratory II	2
Subtotal		16
Advanced Military Science Courses		
MSCI 310	Leading Small Organizations	3
MSCI 315	Advanced Leadership Laboratory I	3
MSCI 320	Developing Advanced Leader Skills	3
MSCI 325	Advanced Leadership Laboratory II	3
MSCI 410	Seminar Command Management	3
MSCI 415	Senior Leadership Laboratory I	3
MSCI 420	Seminar on Officership Transition	3
MSCI 425	Senior Leadership Laboratory II	3
Subtotal		24
Military History		
MSCI 150	American Military History	3
Subtotal		3
Total Credits		43

Modern European Studies Minor

Coordinator: John A. Scherpereel

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Phone: (540) 568-3933

Modern European Studies Website (<http://www.jmu.edu/mes/minor.shtml>)

Program Description

The minor in modern European studies provides a cross disciplinary understanding of social dynamics in Europe from the Enlightenment (18th century) through the present. Modern European studies minors will be prepared for careers in public, private, non-profit and educational institutions and will be able to apply critical insights on European culture, thought, history and society in their professions.

Minor Requirements

Students must complete 18 credit hours in addition to the language requirement. These 18 credit hours must include six core credits and 12 elective credits. Six elective credits must cover culture and thought, and six elective credits must cover history and society. The minor's 18 credits must be distributed across at least three disciplines and must include at least one history course.

All minors must attain proficiency in French, German, Italian, Spanish or another European Union language at or above the intermediate level (232 or above).

Code	Title	Credits
Core Courses		
Choose two of the following:		6
HIST 382	Europe in the 20th Century	
POSC 344	Politics of the European Union	
POSC 345	Politics of Western Europe	
POSC 346	Politics of Central and Eastern Europe	
Subtotal		6
Culture and Thought Electives		
Choose two of the following:		6
ARTH 313	Masterpieces of Italian Renaissance Art [C2VPA]	
ARTH 314	Masterpieces of Spanish Art [C2VPA]	
ARTH 316	Experiencing Art in London [C2VPA]	
ARTH 360	Nineteenth-Century Art	
ARTH 372	Modern Art from 1900-1945	
ARTH 472	Modern Art Since 1945	
ENG 325	Romantic Literature	
ENG 327	Gothic Literature	
ENG 329	Victorian Literature	
ENG 330	Nineteenth-Century British Novel	
ENG 340	Modern British and Irish Literature	
ENG 341	Contemporary British and Irish Literature	
ENG 375	Irish and Anglo-Irish Literature	
FR 308	Contemporary History and Culture in the Francophone World	
FR 446	Special Topics in Francophone Literature ¹	
GER 307	History of Germanic Cultures and Societies	
GER 308	Modern German Culture and Society	

GER 425	Transformative Texts	
GER 465	German Cinema	
HUM 200	Great Works [C2L] ¹	
ITAL 308	Contemporary Italian Culture and Society	
or ITAL 425	Transformative Texts	
ITAL 390	Italian Language and Culture Abroad	
ITAL 446	Special Topics in Italian Literature ¹	
ITAL 465	Italian Cinema	
ITAL 490	Independent Study in Italian ¹	
PHIL 344	Existentialism	
PHIL 390	Special Topics in Philosophy ¹	
PHIL 468	Phenomenology	
POSC 316	Contemporary Political Theory	
POSC 321	Political Theory and Ideology	
REL/PHIL 375	Nineteenth-Century Philosophy and Theology	
SMAD 466	British Film Adaptations	
SPAN 307	History of Spanish Culture and Society	
SPAN 335	A Journey through Literature	
SPAN 422	Voices of Today: Wounds and Words in Post-War Spain	
SPAN 465	Spanish-Language Cinema	
Subtotal		6
History and Society Electives		
Choose two of the following:		6
GEOG 332	Geography of Europe	
HIST 308	Contemporary Italian History and Culture	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 341	Selected Themes in World History ¹	
HIST 382	Europe in the 20th Century ²	
HIST 388	Germany since 1871	
HIST 398	European Military History	
HIST 462	Human Trafficking in African History	
HIST 478	Eastern Europe	
HIST 486	Europe Since 1914	
HIST 487	World War II	
HIST 488	The Holocaust in Global Context	
HIST 489	Selected Topics in World History ¹	
POSC 344	Politics of the European Union ²	
POSC 345	Politics of Western Europe ²	
POSC 346	Politics of Central and Eastern Europe ²	
POSC 371	Topics in Comparative Politics ^{1,2}	
POSC/SCOM/SMAD 473	British Media and Politics	
SCOM/SMAD/WRTC 360	British Media and Society	
Subtotal		6
Total Credits		18

¹ This course may only be used in the minor when its topic is relevant to the minor. Questions should be directed to the minor coordinator.

² This course may not be double-counted toward the minor.

Modern Foreign Language Minor

Advisors

Phone: (540) 568-6904

Modern Foreign Languages Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/modern-foreign-languages.shtml>)

Program Description

A minor in a modern world language is designed for students in any major who wish to expand their critical understanding of the language and culture of a particular country or region. Students gain skills in cross-cultural competence as they explore the language, material and intangible cultures, history, traditions, film, literature, and social practices of non-English-speaking nations. The minor seeks to provide students with the critical and analytical skills necessary to reflect upon and appreciate these cultures and societies and to encourage them to think on a more global scale. The minor provides a deeper understanding of cross-cultural perspectives and increases students' ability to engage in intercultural communication and act as citizens of the world. It also expands the range of professional opportunities open to them. The minor is designed to be flexible and to encourage students to explore interdisciplinary connections in International Affairs, International Business, Anthropology, History, Health Sciences, Justice Studies and other academic fields.

A minor in a modern world language consists of the specified upper-division courses in a specific language as well as elective courses.

Minor Requirements

Arabic

Code	Title	Credits
Language Foundations		
Choose three of the following:		9
ARAB 300	Language for Living	
ARAB 301	Pop Culture through Language	
ARAB 307	A History of Islamicate Civilization 600-1600 A.D.	
ARAB 308	Contemporary Islamicate Civilization	
ARAB 320	Culture and Communication in Context	
ARAB 322	Language Landscapes	
Subtotal		9
Electives		
Choose three additional courses from the list above or other ARAB, LING, or WLC elective courses related to Arabic language and/or culture (two can be in English).		9
Subtotal		9
Total Credits		18

Chinese

Code	Title	Credits
Language Foundations		
Choose three of the following:		9
CHIN 300	Language for Living	
CHIN 301	Pop Culture through Language	
CHIN 307	Chinese Culture and Civilization	
CHIN 320	Culture and Communication in Context	
CHIN 322	Language Landscapes	

Subtotal	9
Electives	
Choose three additional courses from the list above or other CHIN, LING, or WLC elective courses related to Chinese language and/or culture (two can be in English).	
Subtotal	9
Total Credits	18

French

Code	Title	Credits
Language Foundations		
Choose three of the following:		9
FR 300	Language for Living	
FR 301	Pop Culture through Language	
FR 307	History of French Culture and Society	
FR 308	Contemporary History and Culture in the Francophone World	
FR 320	Culture and Communication in Context	
FR 322	Language Landscapes	
Subtotal		9
Electives		
Choose three additional courses from the list above or other FR, LING, or WLC elective courses related to French language and/or culture (two can be in English).		9
Subtotal		9
Total Credits		18

German

Code	Title	Credits
Language Foundations		
Choose three of the following:		9
GER 300	Language for Living	
GER 301	Pop Culture through Language	
GER 307	History of Germanic Cultures and Societies	
GER 308	Modern German Culture and Society	
GER 320	Culture and Communication in Context	
GER 322	Language Landscapes	
Subtotal		9
Electives		
Choose three additional courses from the list above or other GER, LING, or WLC elective courses related to German language and/or culture (two can be in English).		9
Subtotal		9
Total Credits		18

Italian

Code	Title	Credits
Language Foundations		
Choose three of the following:		9
ITAL 300	Language for Living	
ITAL 301	Pop Culture through Language	
ITAL 307	History of Italian Culture and Society	
ITAL 308	Contemporary Italian Culture and Society	
ITAL 320	Culture and Communication in Context	

ITAL 322	Language Landscapes	
Subtotal		9
Electives		
Choose three additional courses from the list above or other ITAL, LING, or WLC elective courses related to Italian language and/or culture (two can be in English).		9
Subtotal		9
Total Credits		18

Spanish

Code	Title	Credits
Language Foundations		
Choose two of the following:		6
SPAN 300	Language for Living	
SPAN 301	Pop Culture through Language	
SPAN 320	Culture and Communication in Context	
or SPAN 321	Culture and Communication in Context (for Heritage Speakers)	
SPAN 322	Language Landscapes	
Subtotal		6
Culture		
SPAN 307	History of Spanish Culture and Society	3
or SPAN 308	Latin American Cultures	
Literature		
SPAN 335	A Journey through Literature	3
Electives		
Choose any 300- or 400-level SPAN course not already taken.		6
Total Credits		18

Music and Human Services Minor

Coordinator: John Peterson

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Music and Human Services Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/music-human-services.shtml>)

Program Description

This minor will prepare students for vocational and avocational opportunities to use music performance, education, entrepreneurship and community engagement skills to serve diverse populations in human service settings. Students with interest and musical backgrounds are welcome.

Minor Requirements

Requirements for the minor are 18 credits: 15 credit hours from a set of core courses and three credits from electives.

Code	Title	Credits
Core Courses		
MUS 206	Introduction to Global Music [C2VPA]	3
MUS 429	Introduction to Intentional Music	1
MUS 430	Music and Human Services: Experiences and Practicum	2
MUED 431	Psychology of Music	3
MUED 485	Teaching Music to Students with Special Needs	3
Choose one of the following:		3
PSYC 250	Introduction to Abnormal Psychology ¹	
PSYC 335	Abnormal Psychology ²	
Subtotal		15
Electives		
Choose three credit hours from the following:		3
DANC 325	Dance in Community	
EXED 200	Foundations of Exceptional Education	
EXED 303	Foundations of Classroom and Behavior Management	
EXED 306	Lifespan Issues for Individuals with Disabilities	
EXED 443	Assistive Technology Use for Individuals with Disabilities	
MUS 150	Introduction to Technological Applications in Music	
Subtotal		3
Total Credits		18

¹ Access limited; additional sections offered during the summer.

² Only offered to psychology majors or through transfer credit.

Music Industry Minor

Advisor: David Cottrell

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Music Industry Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/music-industry.shtml>)

Program Description

Open to all JMU undergraduate students, the music industry minor provides students majoring in other disciplines with a foundation for exploring career opportunities in the entertainment industry.

Minor Requirements

The minor requires successful completion of MUI 221; MUI 231 and of an additional 12 elective credit hours. Students must earn 50 percent of the required curriculum at JMU.

Code	Title	Credits
Required Courses		
MUI 221	Survey of Music Industry	3
MUI 231	Legal Aspects of Music Industry	3
MUI 230	Music Industry Gateway	0
Subtotal		6
Electives ¹		
Choose 12 credit hours from the following:		12
MUI 324	Introduction to Audio Devices	
MUI 330	Music Publishing	
MUI 400	Pro Tools: Multitrack Recording and Music Production	
MUI 405	Logic Pro: Synthesis and Sound Design	
MUI 315	Songwriting	
MUI 411	Audio Postproduction	
MUI 415	Songwriting II	
MUI 422	Concert Production and Promotion	
MUI 423	Sound Reinforcement	
MUI 430	Artist Management	
MUI 435	Marketing in Music Production	
MUI 440	Entrepreneurship in Music Industry	
MUI 492	Internship in Music Industry	
Subtotal		12
Total Credits		18

¹ Any MUI course may count in this category.

Nonprofit Studies Minor

Advisor: Fred Mayhew

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Phone: (540) 568-2511

Nonprofit Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/nonprofit-studies.shtml>)

Program Description

The nonprofit studies minor prepares students from a variety of disciplines to understand the unique role of nonprofit organizations in American society. Emphasis is placed on history, theory, legal issues and management topics. The minor includes a capstone seminar and a field experience in a nonprofit agency with the focus to be determined in conjunction with the advisor.

Minor Requirements

Code	Title	Credits
Required Courses		
NPS 300	Introduction to Nonprofits	3
NPS 320	Nonprofit Management	3
or PPA 470	Nonprofit Management and Leadership	
NPS 400	Internship/Practicum in Nonprofit Studies ¹	4-6
NPS 450	Nonprofit Studies Capstone Seminar	3
Major elective (discipline specific) ¹		3
Elective ¹		3
Total Credits		19-21

¹ A course to be determined in consultation with the nonprofit studies advisor.

Philosophy Minor

Advisor: Patrick Fleming
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Philosophy Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/philosophy.shtml>)

Program Description

Philosophy is the oldest of the liberal arts, an academic discipline that tackles fundamental questions about the nature of knowledge (epistemology), the nature of reality and what sorts of things exist (metaphysics), what actions and which kinds of people are moral (ethics), the nature of beauty and art (aesthetics), and how to model good reasoning (logic). The Philosophy minor is an excellent choice for students wishing to delve into these topics, as the skills they gain in rational argument analysis and cogent writing will invariably complement any major. Students completing a minor in Philosophy will have the opportunity to study the central figures in various historically important eras (e.g., the Ancient world and the Enlightenment) as well as take upper-division electives focused on specific philosophical areas.

Minor Requirements

Code	Title	Credits
Introductory Course		
PHIL 101	Introduction to Philosophy [C2HQC]	3
Subtotal		3
Historical Requirement		
Choose two of the following:		6
PHIL 340	Ancient Greek Philosophy	
PHIL 341	Modern Philosophy	
PHIL 342	Medieval Philosophy	
PHIL 375	Nineteenth-Century Philosophy and Theology	
Subtotal		6
Electives		
Choose three additional philosophy courses ¹		9
Subtotal		9
Total Credits		18

¹ At least two must be above PHIL 300. Neither PHIL 120 nor PHIL 150 can be used as electives.

Physics Minor for Engineers

Coordinator: Ioana Niculescu

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Phone: (540) 568-2980

Physics for Engineers Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/physics-for-engineers.shtml>)

Program Description

The minor in physics for engineering students is a program that enables JMU undergraduate students majoring in engineering to take a minor in physics. Interested students will take courses in upper-level physics to explore physical phenomena related to their professional development as an engineer.

Minor Requirements

Code	Title	Credits
Physics Minor for Engineers Courses		
PHYS 270 & 270L	Modern Physics and Modern Physics Laboratory	4
PHYS course at the 300-level or above ¹		3-4
Choose one of the following pairs:		6
PHYS 140 & PHYS 150	College Physics I [C3PP] and College Physics II	
PHYS 240 & PHYS 250	University Physics I [C3PP] and University Physics II	
Choose one of the following pairs:		2
PHYS 140L & PHYS 150L	General Physics Laboratory I [C3L] and General Physics Laboratory II	
PHYS 240L & PHYS 250L	University Physics Laboratory I [C3L] and University Physics Laboratory II	
Choose one of the following:		4
ENGR 313	Circuits and Devices	
PHYS 360	Analog Electronics	
Total Credits		19-20

¹ May not include PHYS 360

Physics Minor

Coordinator: Ioana Niculescu

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Physics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/physics.shtml>)

Program Description

The Department of Physics and Astronomy (p. 94) offers a multitrack program of study which allows each student flexibility to choose a suitable set of courses that will prepare the student for their chosen career goal. With the assistance of physics faculty advisors each student will select a program of theoretical and experimental courses through which they will gain an understanding of the broad range of general principles which characterize the discipline of physics; and they will acquire the ability to apply theoretical and experimental techniques to solve a wide variety of problems.

Minor Requirements

The minimum requirement for a minor in physics is 21 credit hours selected as follows:

Code	Title	Credits
Physics Minor Courses		
PHYS 260	University Physics III	3
Choose one of the following pairs:		6
PHYS 140 & PHYS 150	College Physics I [C3PP] and College Physics II	
PHYS 240 & PHYS 250	University Physics I [C3PP] and University Physics II	
Choose one of the following pairs:		2
PHYS 140L & PHYS 150L	General Physics Laboratory I [C3L] and General Physics Laboratory II	
PHYS 240L & PHYS 250L	University Physics Laboratory I [C3L] and University Physics Laboratory II	
Choose 10 credits of the following:		10
Physics courses numbered above 260		
ASTR 320	Astronomical Techniques	
ASTR 480	Astrophysics	
Total Credits		21

Political Communication Minor

Coordinator: Valerie A. Sulfaro

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Phone: (540) 568-3997

Political Communication Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/political-communication.shtml>)

Program Description

The program in political communication is designed for those students wishing to supplement their major programs with an emphasis on communication skills, knowledge and abilities specifically relevant to participation in political environments.

Minor Requirements

All students must complete 25-26 credit hours for the minor. Students must complete a core set of courses, an internship in the field of political communication and a selection of electives. No course substitution will be permitted for any required course.

All political communication minors are required to take a PCOM internship in the Department of Political Science or the Department of Communication Studies. Before enrolling for a PCOM internship, students should have attained junior status, completed the PCOM core requirements and taken at least one upper level PCOM course. Some academic units may have additional prerequisites for their internship course. All SCOM majors with a minor in PCOM are required to take SCOM 495 for their PCOM internship requirement (and to have met the prerequisites for this course). POSC, INTA and PPA majors minoring in PCOM must complete POSC 493 to fulfill the internship requirement. All other majors may take either POSC 493 or SCOM 495 as a means of satisfying their internship. Before pursuing any internship, students must consult with the PCOM coordinator to ensure that the internship is suitable for the minor. All internships must incorporate political science and communication content.

SMAD majors may double count POSC 472/SCOM 472/SMAD 472 for the SMAD major and the PCOM minor.

Code	Title	Credits
Required Courses		
POSC 225	U.S. Government [C4AE]	4
POSC/SCOM/ SMAD 472	Media and Politics	3
SCOM 353	American Political Culture and Communication	3
Subtotal		10
Choose one of the following:		3
POSC 365	American Political Campaigning ¹	
POSC 369	Elections and Voting Behavior ¹	
Choose one of the following:		3
SCOM 352	Communication and Social Movements ²	
SCOM 453	Political Campaign Communication ²	
Choose two of the following:		6
POSC 300	Politics and Film	
POSC 341	Social Movements in the U.S. and Abroad	
POSC 362	Political Behavior	
POSC 364	Political Parties in America	
POSC 368	Interest Groups, Lobbying and Public Policy	

POSC 382	The Role of Religion in American Politics	
POSC 383	Women and Politics in Comparative Perspective	
POSC 384	Minority Group Politics	
POSC 385	The U.S. Congress	
POSC 387	The Appellate Process and Practice	
SCOM 342	Argument and Advocacy	
SCOM 346	Free Speech in America	
SCOM 354	Environmental Communication and Advocacy	
SCOM/WRTC/ WGSS 420	Feminist Rhetorics	
SCOM 431	Legal Communication	
Internship Requirement		
POSC 493	Political Communication Internship ³	3-4
or SCOM 495	Internship in Communication Studies	
Total Credits		25-26

- ¹ If both POSC 365 and POSC 369 are taken, one may be counted as an elective.
- ² If both SCOM 352 and SCOM 453 are taken, one may be counted as an elective.
- ³ POSC, PPA and INTA majors must take POSC 493 for 4 credits. SCOM majors must take SCOM 495 for 3 credits. All other majors must take either POSC 493 or SCOM 495 for 3-4 credits.

Political Science Minor

Coordinator: Scott Hammond
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Political Science Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/political-science.shtml>)

The Political Science minor empowers students to gain an understanding of political processes through coursework in U.S. government and politics, political theory and international politics with an option to concentrate on either international relations or comparative politics. The Political Science minor complements a range of majors inside and outside of the Department of Political Science (p. 95).

Minor Requirements

A student may minor in political science by completing 19 credit hours of approved courses from the following list. Check with the department office for department availability of the minor.

Code	Title	Credits
Required Courses		
POSC 201	Introduction to Western Political Theory	3
POSC 225	U.S. Government [C4AE]	4
Subtotal		7
American Government Elective		
Choose one of the following:		3
POSC 300	Politics and Film	
POSC 301	The Washington Semester Experience	
POSC 302	State and Local Government	
POSC 322	Civic Engagement	
POSC 323	Cities and Placemaking	
POSC 324	Politics of Human Rights in the United States	
POSC 325	Constitutional Law	
POSC 326	Civil Rights	
POSC 327	Race and Capitalism in the United States	
POSC 328	Fair and Affordable Housing Law and Policy	
POSC 329	Civil Rights and Criminal Procedures	
POSC 351	Topics in American Politics	
POSC 360	The Politics of Mental Health	
POSC 362	Political Behavior	
POSC 364	Political Parties in America	
POSC 365	American Political Campaigning	
POSC 367	U.S. Immigration and Refugee Law	
POSC 368	Interest Groups, Lobbying and Public Policy	
POSC 369	Elections and Voting Behavior	
POSC 380	The U.S. Presidency	
POSC 382	The Role of Religion in American Politics	
POSC 384	Minority Group Politics	
POSC 385	The U.S. Congress	
POSC 386	The U.S. Judiciary	
POSC 387	The Appellate Process and Practice	
POSC/SCOM/ SMAD 472	Media and Politics	
Subtotal		3

Upper-Level Elective		
Choose an additional 300-level course listed above or from the following list:		3
PPA 200	Introduction to Public Policy	
PPA 265	Public Administration	
POSC 200	Global Politics [C4GE]	
POSC 310	Ancient Political Theory: Classical Greek Political Philosophy to Roman Stoicism	
POSC 315	Political Theory: Early Modern to the 19th Century	
POSC 316	Contemporary Political Theory	
POSC 321	Political Theory and Ideology	
POSC 330	American Political Thought	
POSC 333	American Political Thought II: Civil War to the Present	
POSC 334	Political Thought of James Madison	
POSC 381	Topics in Political Theory	
POSC 391	Topics in Public Policy	
Subtotal		3
International or Comparative Options		
Choose one of the following options:		6
Option 1:		
POSC 230	International Relations	
Choose one of the following:		
POSC/JUST 331	Human Rights in Theory and Practice	
POSC 361	Topics in International Relations	
POSC 370	U.S. Foreign Policy	
POSC/JUST 372	Ethics and International Politics	
POSC/JUST 392	Peace Studies	
POSC 393	Human Security	
POSC 395	International Law	
POSC 396	International Organizations	
POSC 397	The Politics of International Economic Relations	
POSC 430	International Security and Conflict Management	
POSC 435	International Terrorism	
POSC 458	International Political Analysis	
Option 2:		
POSC 240	Comparative Politics	
Choose one of the following:		
POSC 340	Politics of Development	
POSC 341	Social Movements in the U.S. and Abroad	
POSC 344	Politics of the European Union	
POSC 345	Politics of Western Europe	
POSC 346	Politics of Central and Eastern Europe	
POSC 347	Comparative Public Policy	
POSC 348	The Politics of Cultural Pluralism	
POSC 350	Latin American Politics	
POSC 353	African Politics	
POSC 354	Politics of the Middle East	
POSC 355	East Asian Politics	
POSC 356	Politics of China	

POSC 371	Topics in Comparative Politics	
POSC/WGSS 383	Women and Politics in Comparative Perspective	
POSC 440	Post-Conflict Societies	
Subtotal		6
Total Credits		19

Pre-Athletic Training Minor

Coordinator: Kristen Crawford
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Email: crawf2ka@jmu.edu
Pre-Athletic Training Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-athletic-training-minor.shtml>)

Program Description

The Pre-Athletic Training program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to graduate athletic training programs after completing their Bachelor's degree at James Madison University. The Pre-AT program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Athletic Training (Admission Standards)

Incoming first-year and transfer students can declare pre-athletic training without verification from PPH Advising (p. 158). Current students can declare the pre-athletic training minor through MyMadison without needing further approval.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with athletic training schools of interest to confirm details of admission requirements.

Minor Requirements

Requirements for the pre-athletic training minor are based on the prerequisite courses of 79 Athletic Training Education Programs in 17 states (CT, DE, FL, GA, MA, MD, ME, NC, NH, NJ, NY, PA, SC, TN, VA, VT and WV). Students should review individual athletic training programs' prerequisites to assure completion of all prerequisite course work.

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	4
or BIO 370	Human and Animal Physiology	
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
HTH 210	Medical Terminology	3
KIN 302 & 302L	Exercise Physiology and Exercise Physiology Laboratory	4
NUTR 280	Nutrition for Wellness	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
or PHYS 125 & 125L	Physics with Biological Applications [C3PP] and Physics with Biological Applications Laboratory [C3L]	
PSYC 101	General Psychology [C5SD]	3

Subtotal		33
Biomechanics		
KIN 306 & 306L	Human Biomechanics and Human Biomechanics Laboratory	4
Subtotal		4
Mathematics		
MATH 220 or MATH 229	Elementary Statistics [C3QR] Introduction to Applied Statistics Using R [C3QR]	3
Subtotal		3
Writing and Literature		
WRTC 103	Rhetorical Reading and Writing [C1W]	3
One literature course from (C2L) ENG or HUM		3
Subtotal		6
Total Credits		46

Recommended Electives (Not Required)

Code	Title	Credits
ATEP 205	Introduction to Athletic Training	3
PSYC 160	Life Span Human Development [C5SD]	3

Pre-Dentistry Minor

Coordinator: Sharon Babcock
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Pre-Dentistry Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-dentistry.shtml>)

Program Description

The Pre-Dentistry program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to Doctor of Dental Surgery (DDS) or Doctor of Dental Medicine (DMD) after completing their Bachelor's degree at James Madison University. The Pre-Dentistry program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Dentistry (Admission Standards)

Incoming first-year and transfer students can declare pre-dentistry without verification from PPH Advising (p. 158). Current students can request to declare the pre-dentistry minor through MyMadison but will need approval for the request to be granted. If the student has completed fewer than 13 hours of biology, chemistry, physics and math (BCPM) credits, the request will be granted after verification of course work. If the student has completed 13 hours or more BCPM credits, the request will be granted if the BCPM GPA is 2.90 or greater.

Pre-Dentistry Progression Standards

After completing 13 credits of biology, chemistry, physics and math (BCPM), pre-dentistry students undergo academic reviews twice a year. Pre-dentistry students whose BCPM GPA falls below 2.90 are at high risk of not being admitted to a dental school and will receive targeted advising focused on academic improvement and career exploration. If a student falls below the standard in a subsequent performance review, the pre-dentistry minor will be administratively dropped. Students may re-enter the minor upon meeting the 2.90 BCPM GPA requirement.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with dental schools of interest to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3

CHEM 242L	Organic Chemistry Laboratory	2
CHEM 361	Biochemistry I	3
PHYS 140	College Physics I [C3PP]	3
PHYS 140L	General Physics Laboratory I [C3L]	1
PHYS 150	College Physics II	3
PHYS 150L	General Physics Laboratory II	1
Subtotal		35
Intermediate and Advanced BIO Coursework		
Choose 16 credit hours from subjects recommended by dental schools:		16
BIO 240 & 240L	Genetics and Genetics Laboratory	
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	
BIO 304	Cell and Molecular Biology ¹	
BIO 370	Human and Animal Physiology ¹	
BIO 482	Human Histology ²	
Subtotal		16
Writing and Literature		
WRTC 103	Rhetorical Reading and Writing [C1W]	3
One literature course from (C2L) ENG or HUM		3
Subtotal		6
Mathematics		
Recommend one course in statistics and one course in calculus		6
Subtotal		6
Behavioral (PSYC) or Social Science (SOCI)		
Choose one Behavioral (PSYC) or social science (SOCI) course		3
Subtotal		3
Total Credits		66

¹ BIO 240/BIO 240L is a prerequisite.
² BIO 290/BIO 290L or BIO 370 is a prerequisite.

Pre-Medicine Minor

Co-Coordinators: Katie Fisher, Kristen Crawford and Sharon Babcock

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Pre-Medicine Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-med.shtml>)

Program Description

The Pre-Medicine program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to Doctor of Medicine (M.D.), Doctor of Osteopathic Medicine (D.O.), Doctor of Podiatric Medicine (D.P.M.), Doctor of Chiropractic (D.C.), and Doctor of Naturopathic Medicine (N.D.) programs after completing their bachelor's degree at James Madison University. The Pre-Med program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Medicine (Admission Standards)

Incoming first-year and transfer students can declare pre-medicine without verification from PPH Advising. Current students can request to declare the pre-medicine minor through MyMadison but will need approval for the request to be granted. If the student has completed fewer than 13 hours of biology, chemistry, physics and math (BCPM) credits, the request will be granted after verification of course work. If the student has completed 13 hours or more BCPM credits, the request will be granted if the BCPM GPA is 3.00 or greater.

Access to Pre-Medicine Advising (Progression Standards)

After completing 13 credits of biology, chemistry, physics and math (BCPM), pre-medicine students undergo academic reviews twice a year. Students who maintain a 3.00 BCPM GPA are invited to participate in advising activities including career exploration, experiential planning and application preparation.

Pre-medicine students whose BCPM GPA falls below 3.00 are at high risk of not being admitted to medical school and will receive targeted advising focused on academic improvement and career exploration. If a student falls below the standard in a subsequent performance review, the pre-medicine minor will be administratively dropped. Students may re-enter the minor upon meeting the 3.00 BCPM GPA requirement.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with medical schools of interest to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4

CHEM 131	General Chemistry I [C3PP]	3
CHEM 131L	General Chemistry Laboratory [C3L]	1
CHEM 132	General Chemistry II	3
CHEM 132L	General Chemistry Laboratory	1
CHEM 241	Organic Chemistry I	3
CHEM 242	Organic Chemistry II	3
CHEM 242L	Organic Chemistry Laboratory	2
CHEM 361	Biochemistry I	3
PHYS 140	College Physics I [C3PP]	3
PHYS 140L	General Physics Laboratory I [C3L]	1
PHYS 150	College Physics II	3
PHYS 150L	General Physics Laboratory II	1
Subtotal		35

Intermediate and Advanced BIO Course Work

Choose 12 credit hours from subjects recommended by medical schools and associated with foundational concepts on the MCAT:		12
BIO 240 & 240L	Genetics and Genetics Laboratory	
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	
BIO 304	Cell and Molecular Biology ¹	
BIO 370	Human and Animal Physiology ¹	
Subtotal		12

Writing and Literature

WRTC 103	Rhetorical Reading and Writing [C1W]	3
One literature course from (C2L) ENG or HUM		3
Subtotal		6

Calculus

Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 233	Calculus with Modeling I [C3QR]	
MATH 235	Calculus I [C3QR]	
Subtotal		3-4

Statistics

MATH 220	Elementary Statistics [C3QR]	3
or MATH 229	Introduction to Applied Statistics Using R [C3QR]	
Subtotal		3

Behavioral (PSYC) and Social Sciences (SOCI/ANTH)

Choose three of the following: ²		9
PSYC 101	General Psychology [C5SD]	
PSYC 160	Life Span Human Development [C5SD]	
PSYC 308	Health Psychology	
SOCI 110	Social Issues in a Global Context [C4GE]	
SOCI 140	Microsociology: The Individual in Society [C5SD]	
SOCI 375	Medical Sociology	
SOCI 385	Madness and Society: The Sociology of Mental Health and Illness	
ANTH 360	Medical Anthropology	
Subtotal		9

Total Credits **68-69**

¹ BIO 240/BIO 240L is a prerequisite.

² Three credits must be in PSYC, three credits must be in SOCI and the remaining three credits of your choice.

Recommended Elective (Not Required)

Code	Title	Credits
PHIL 120	Critical Thinking [C1CT]	3

Pre-Occupational Therapy Minor

Coordinator: Kelly Schuhmann
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Phone: (540) 568-6652
Pre-Occupational Therapy Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-occupational-therapy.shtml>)

Program Description

The Pre-Occupational Therapy program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to graduate occupational therapy programs after completing their Bachelor's degree at James Madison University. The Pre-OT program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Occupational Therapy (Admission Standards)

Incoming first-year and transfer students can declare pre-occupational therapy without verification from PPH Advising (p. 158). Current students can request declaration of the pre-occupational therapy minor through MyMadison but will need approval for the request to be granted. If the student has completed fewer than nine hours of the prerequisite courses included in the minor the request will be granted after verification of course work. If the student has completed nine hours or more of the prerequisite courses, the request will be granted if the prerequisite GPA is 2.80 or greater.

Pre-Occupational Therapy Progression Standards

After completing nine hours of the prerequisite courses, pre-occupational therapy students undergo academic reviews twice a year.

Pre-occupational therapy students whose prerequisite GPA falls below 2.80 are at high risk of not being admitted to occupational therapy school and will receive targeted advising focused on academic improvement and career exploration. If a student falls below the standard in a subsequent performance review, the pre-occupational therapy minor will be administratively dropped. Students may re-enter the minor upon meeting the 2.80 prerequisite GPA requirement.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with occupational therapy schools of interest, including JMU, to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
ANTH 195	Cultural Anthropology [C4GE]	3
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	4
or BIO 370	Human and Animal Physiology	
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
HTH 210	Medical Terminology	3

PSYC 101	General Psychology [C5SD]	3
PSYC 160	Life Span Human Development [C5SD]	3
PSYC 250	Introduction to Abnormal Psychology ¹	3
Subtotal		23
Chemistry		
Choose one of the following:		3-4
CHEM 120	Concepts of Chemistry [C3PP]	
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	
Subtotal		3-4
Biomechanics		
Choose one of the following:		4
KIN 306 & 306L	Human Biomechanics and Human Biomechanics Laboratory	
PHYS 125 & 125L	Physics with Biological Applications [C3PP] and Physics with Biological Applications Laboratory [C3L]	
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	
Subtotal		4
Social Science		
SOCI 110 or SOCI 140	Social Issues in a Global Context [C4GE] or Microsociology: The Individual in Society [C5SD]	3
Subtotal		3
Statistics		
MATH 220	Elementary Statistics [C3QR]	3
Subtotal		3
Total Credits		36-37

¹ Psychology majors may take PSYC 335 in place of PSYC 250.

Recommended Electives (Not Required)

Code	Title	Credits
Social Science		
Any SOCI or ANTH course at the 200 level or above.		3
Introductory Biology		
Choose one of the following:		3-4
BIO 103	Contemporary Biology [C3NS]	
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	
Ethics		
Choose 1-3 credit hours from the following:		1-3
HTH 423	Ethics and Critical Thinking in Health	
IPE 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	
NSG 415	Ethical Decision-Making in Healthcare: An Interprofessional Approach	
PHIL 150	Ethical Reasoning [C1CT]	

Pre-Optometry Minor

Coordinator: Jeff Andre
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Phone: (540) 568-6652
Pre-Optometry Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-optometry.shtml>)

Program Description

The Pre-Optometry program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to Doctor of Optometry (OD) programs after completing their Bachelor's degree at James Madison University. The Pre-Optometry program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Optometry (Admission Standards)

Incoming first-year and transfer students can declare pre-optometry without verification from PPH Advising (p. 158). Current students can declare the pre-optometry minor through MyMadison without needing further approval.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with optometry schools of interest to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L]	4
or BIO 370	Human and Animal Physiology	
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
CHEM 241	Organic Chemistry I	3
CHEM 242 & 242L	Organic Chemistry II and Organic Chemistry Laboratory	5
CHEM 361	Biochemistry I	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
PHYS 150 & 150L	College Physics II and General Physics Laboratory II	4
PSYC 101	General Psychology [C5SD]	3

Subtotal	50
Intermediate or Advanced BIO Coursework	
Choose one of the following from subjects recommended by optometry schools:	4
BIO 240 & 240L	Genetics and Genetics Laboratory
BIO 304	Cell and Molecular Biology ¹
BIO 425	Immunology
BIO 482	Human Histology
Subtotal	4
Calculus	
Choose one of the following:	3-4
MATH 205	Applied Calculus [C3QR]
MATH 231	Calculus with Functions I [C3QR]
MATH 233	Calculus with Modeling I [C3QR]
MATH 235	Calculus I [C3QR]
Subtotal	3-4
Statistics	
MATH 220	Elementary Statistics [C3QR]
or MATH 229	Introduction to Applied Statistics Using R [C3QR]
Subtotal	3
Writing and Literature	
WRTC 103	Rhetorical Reading and Writing [C1W]
One literature course from (C2L) ENG or HUM	3
Subtotal	6
Total Credits	66-67

¹ BIO 240 is a prerequisite.

Pre-Pharmacy Minor

Coordinator: Mary Hildebrand

Email: pph@jmu.edu

Phone: (540) 568-6652

Pre-Pharmacy Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-pharmacy.shtml>)

Program Description

The Pre-Pharmacy program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to graduate Doctor of Pharmacy (PharmD) programs after completing their Bachelor's degree at James Madison University. The Pre-Pharmacy program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Pharmacy (Admission Standards)

Incoming first-year and transfer students can declare pre-pharmacy without verification from PPH Advising (p. 158). Current students can request to declare the pre-pharmacy minor through MyMadison but will need approval for the request to be granted. If the student has completed fewer than 10 hours of biology, chemistry and physics (BCP) credits, the request will be granted after verification of course work. If the student has completed 10 or more BCP credits, the request will be granted if the BCP GPA is 2.50 or greater.

Pre-Pharmacy Progression Standards

After completing 10 hours of foundational biology, chemistry and physics course work, pre-pharmacy students undergo academic reviews twice a year.

Pre-pharmacy students whose BCP GPA falls below 2.50 are at high risk of not being admitted to pharmacy school and will receive targeted advising focused on academic improvement and career exploration. If a student falls below the standard in a subsequent performance review, the pre-pharmacy minor will be administratively dropped. Students may re-enter the minor upon meeting the 2.50 BCP GPA requirement.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with pharmacy schools of interest to confirm details of admission requirement.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L] ¹	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory ¹	4
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L] ¹	4
or BIO 370	Human and Animal Physiology	

BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L] ¹	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory ¹	4
CHEM 241	Organic Chemistry I	3
CHEM 242 & 242L	Organic Chemistry II and Organic Chemistry Laboratory	5
CHEM 361	Biochemistry I	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L] ¹	4
or PHYS 125 & 125L	Physics with Biological Applications [C3PP] and Physics with Biological Applications Laboratory [C3L]	
Subtotal		43

Intermediate or Advanced BIO Coursework

Choose one of the following from subjects recommended by pharmacy schools: 3-4

BIO 240 & 240L	Genetics and Genetics Laboratory	
BIO 304	Cell and Molecular Biology ²	
BIO 425	Immunology	
BIO 480	Advanced Molecular Biology ²	
Subtotal		3-4

Calculus

Choose one of the following: 3-4

MATH 205	Applied Calculus [C3QR] ¹	
MATH 231	Calculus with Functions I [C3QR] ¹	
MATH 233	Calculus with Modeling I [C3QR] ¹	
MATH 235	Calculus I [C3QR] ¹	
Subtotal		3-4

Statistics

Choose one of the following: 3

MATH 220	Elementary Statistics [C3QR] ¹	
MATH 229	Introduction to Applied Statistics Using R [C3QR] ¹	
Subtotal		3

Public Speaking

Choose one of the following: 3

SCOM 121	Fundamental Human Communication: Presentations [C1HC]	
SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC]	
SCOM 123	Fundamental Human Communication: Group Presentations [C1HC]	
Subtotal		3

Writing and Literature

WRTC 103 Rhetorical Reading and Writing [C1W] 3

One literature course from (C2L) ENG or HUM 3

Subtotal 6

Behavioral Science

PSYC 101 General Psychology [C5SD] 3

or PSYC 160 Life Span Human Development [C5SD]

Subtotal 3

Social Science		
SOCI 110	Social Issues in a Global Context [C4GE]	3
or SOCI 140	Microsociology: The Individual in Society [C5SD]	
Subtotal		3
Economics		
ECON 201	Introduction to Microeconomics	3
or ECON 200	Introduction to Macroeconomics [C4GE]	
Subtotal		3
Total Credits		70-72

¹ Foundational course for minor.

² BIO 240 is a prerequisite.

Recommended Electives (Not Required)

Code	Title	Credits
CHEM 366L	Biochemistry Laboratory	2
PHYS 150 & 150L	College Physics II and General Physics Laboratory II	4

Pre-Physical Therapy Minor

Co-Coordinator: Kelly Schuhmann and Kristen Crawford
Email: pph@jmu.edu
Phone: (540) 568-6652
Pre-Physical Therapy Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-physical-therapy.shtml>)

Program Description

The Pre-Physical Therapy program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to Doctor of Physical Therapy (D.P.T.) programs after completing their Bachelor's degree at James Madison University. The Pre-Physical Therapy program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Physical Therapy (Admission Standards)

Incoming first-year and transfer students can declare pre-physical therapy without verification from PPH Advising (p. 158). Current students can request declaration of the pre-physical therapy minor (through MyMadison) but will need approval for the request to be granted. If the student has completed fewer than 10 hours of the prerequisite credits, the request will be granted after verification of course work. If the student has completed 10 hours or more of the prerequisite credits, the request will be granted only if the prerequisite GPA is 2.7 or greater.

Pre-Physical Therapy Progression Standards

After completing 10 hours of the prerequisite courses, pre-physical therapy students undergo academic reviews twice a year.

Pre-physical therapy students whose prerequisite GPA falls below 2.70 are at high risk of not being admitted to physical therapy school and will receive targeted advising focused on academic improvement and career exploration. If a student falls below the standard in a subsequent performance review, the pre-physical therapy minor will be administratively dropped. Students may re-enter the minor upon meeting the 2.70 prerequisite GPA requirement.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with physical therapy schools of interest to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L] ¹	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 270 & 270L or BIO 370	Human Physiology [C3NS] and Human Physiology Laboratory [C3L] ¹ Human and Animal Physiology	4

BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory ¹	4
CHEM 131	General Chemistry I [C3PP] ¹	3
CHEM 131L	General Chemistry Laboratory [C3L] ¹	1
CHEM 132	General Chemistry II ¹	3
CHEM 132L	General Chemistry Laboratory ¹	1
MATH 220	Elementary Statistics [C3QR] ¹	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L] ¹	4
PHYS 150 & 150L	College Physics II and General Physics Laboratory II ¹	4
Subtotal		35
Behavioral Science		
Complete at least two of the following:		6
PSYC 101	General Psychology [C5SD] ²	
PSYC 160	Life Span Human Development [C5SD] ²	
PSYC 250	Introduction to Abnormal Psychology ²	
Subtotal		6
Required Elective		
Complete at least one of the following:		3-4
HTH 210	Medical Terminology	
KIN 302 & 302L	Exercise Physiology and Exercise Physiology Laboratory	
Subtotal		3-4
Writing		
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Subtotal		3
Total Credits		47-48

¹ Course grade used for progression standards.
² Course grade in one PSYC course used for progression standards.

Recommended Electives (Not Required)

Code	Title	Credits
Biomechanics		
Choose one of the following:		4
BIO 490	Mechanics of Animal Movement	
KIN 306 & 306L	Human Biomechanics and Human Biomechanics Laboratory	
Calculus		
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 233	Calculus with Modeling I [C3QR]	
MATH 235	Calculus I [C3QR]	

Pre-Physician Assistant Minor

Coordinators: Katie Fisher and Kelly Schuhmann
Email: pph@jmu.edu
Phone: (540) 568-6652
Pre-Physician Assistant Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-physician-assistant.shtml>)

Program Description

The Pre-Physician Assistant program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to graduate Physician Assistant (M.P.A.S.) programs after completing their Bachelor's degree at James Madison University. The Pre-PA program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Physician Assistant (Admission Standards)

Incoming first-year and transfer students can declare pre-physician assistant without verification from PPH Advising (p. 158). Current students can request to declare the minor through MyMadison but will need approval for the request to be granted. If the student has completed fewer than 12 hours of biology and chemistry (BC) credits, the request will be granted after verification of course work. If the student has completed 12 hours or more BC credits, the request will be granted if the BC GPA is 2.70 or greater.

Pre-Physician Assistant Progression Standards

After completing 12 hours of biology or chemistry (BC) coursework, pre-physician assistant students undergo academic reviews twice a year.

Pre-physician assistant students whose BC GPA falls below the standard of a 2.70 are at high risk of not being admitted to physician assistant school and will receive targeted advising focused on academic improvement and career exploration. If a student falls below the standard in a subsequent performance review, the pre-physician assistant minor will be administratively dropped. Students may re-enter the minor upon meeting the 2.70 BC GPA requirement.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with physician assistant schools of interest, including JMU, to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4

BIO 270 & 270L or BIO 370	Human Physiology [C3NS] and Human Physiology Laboratory [C3L] Human and Animal Physiology	4
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
CHEM 241 & 241L	Organic Chemistry I and Concepts of Organic Chemistry Laboratory ¹	4
HTH 210	Medical Terminology	3
Subtotal		39
Biochemistry		
CHEM 260 or CHEM 361	Concepts of Biochemistry Biochemistry I	3
Subtotal		3
Behavioral Science (PSYC)		
Recommend choosing from the following:		6
PSYC 101	General Psychology [C5SD]	
PSYC 160	Life Span Human Development [C5SD]	
PSYC 250	Introduction to Abnormal Psychology	
Subtotal		6
Statistics		
MATH 220 or MATH 229	Elementary Statistics [C3QR] Introduction to Applied Statistics Using R [C3QR]	3
Subtotal		3
Total Credits		51

¹ CHEM 242L may be substituted for CHEM 241L. The CHEM 242L corequisite is CHEM 242.

Recommended Electives (Not Required)

Code	Title	Credits
BIO 304	Cell and Molecular Biology	3
BIO 425	Immunology	3
BIO 482	Human Histology	4
CHEM 260L	Concepts of Biochemistry Laboratory	1
Additional MATH course		3

Pre-Professional Secondary Education (SEED) Minor

Advisor: Amber Loyacano

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Pre-Professional Secondary Education Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-professional-secondary-education.shtml>)

Program Description

The pre-professional secondary education minor is designed to provide candidates with the prerequisite course offerings and experiences that form the required foundation for admission to the Master of Arts in Teaching (M.A.T.) dual degree program. Completion of the M.A.T. program is one of the pathways that can be selected for a recommendation for Virginia licensure in secondary education through JMU. Undergraduate candidates who are planning to pursue licensure to teach in an academic area at the secondary school level in this dual degree pathway should complete the 20-22 credit hour pre-professional secondary education minor.

Candidates should also complete a major or the equivalent in one of the arts and sciences disciplines closely associated with the desired teaching area. The state-approved licensure areas in the secondary education program at JMU include English, mathematics, the natural sciences (biology, chemistry, physics, earth science) and history/social science (social studies). Licensure endorsements for other teaching areas (e.g., algebra) are available as add-on programs. Candidates should contact a secondary education advisor for information concerning those programs.

Candidates should consult with a secondary education advisor early during the first year or as soon, thereafter, as possible to obtain information concerning completing the related General Education and content area requirements, as well as the undergraduate and graduate education requirements.

This secondary education licensure program pathway is an integrated program of undergraduate and graduate dual degree requirements and experiences. It is important for candidates to understand that they must meet the requirements for a baccalaureate degree and that the appropriate undergraduate pre-professional courses (i.e. the pre-professional SEED minor) and experiences must be completed satisfactorily before they will be admitted to the M.A.T. program.

Minor Requirements

Candidates must earn a grade of "B-" or better (unless noted otherwise) in all required pre-professional undergraduate courses in the education program to continue in and complete the pre-professional program. In addition, a minimum undergraduate cumulative GPA of 2.5 is required for admission into the Teacher Education program. A minimum undergraduate cumulative GPA of 2.7 is required for admission to the graduate program. All students applying to the M.A.T. in Education must ensure they complete a minimum of 120 undergraduate credits to receive their bachelor's degree.

If utilizing this pathway to licensure, admission to and the satisfactory completion of the M.A.T. program are required in order to receive a

recommendation through JMU for a teaching license at the secondary school level.

Candidates should note that, at times, actual teacher licensure requirements may differ from the catalog requirements listed for a program because changes enacted by the Virginia Department of Education or accrediting agencies may have occurred after the catalog copy was approved. Therefore, it is important for students to confer with their education advisors on a regular basis.

Prerequisites to the Minor

Students must have completed these prerequisite requirements prior to applying for the initial application to teaching education:

Code	Title	Credits
WRTC 103	Rhetorical Reading and Writing [C1W] ^{1,2}	3
Choose one of the following:		3
PSYC 160	Life Span Human Development [C5SD] ¹	
EDUC 175	Learning, Behavior and Engagement in Context [C5SD] ¹	
Total Credits		6

¹ Minimum grade of "C" is required.

² Students who do not receive a minimum grade of "C" in this course may retake the course or take an equivalent writing course and pass with a minimum grade of "C." Contact the department to determine what course would be most appropriate.

Program Requirements

Code	Title	Credits
Teacher Education Core Course		
Students must complete this prerequisite requirement as part of the initial application to teaching education:		
EDUC 200	Foundations of Education in the United States ¹	3
Subtotal		3
Orientation to the Profession		
Choose one of the following courses, which can be taken prior to applying to teacher education.		1-3
CSM 199	Science Outreach in Schools ¹	
MSSE 101	Orientation to the Profession (fall and spring) ¹	
MSSE 201	Introduction to Social Studies Education Seminar (spring only) ¹	
Subtotal		1-3
Required Upper Level Course		
MSSE 210	Diversity in Middle and Secondary Education	3
Subtotal		3
General Instructional Methods (GIM) and Initial Field Placement (IFP)		
GIM Course:		
MSSE 370	General Instructional Methods for Grades 6-12 ³	3
IFP Courses:		
Choose one of the following (must equal a minimum of 4.00 credits with GIM course):		1-3
MSSE 371	Clinical Experience in Adolescent Education ³	
MSSE 430	Introduction to Mentoring Valley Scholars (fall term start; contact MSME office for more information)	

Subtotal	4-6
Disciplinary Literacy	
LED 440 Disciplinary Literacy in Secondary Education	3
Subtotal	3
Content Specific Teaching Methods, Grades 6-8	
Choose one of the following content specific teaching methods courses:	3
MSSE 411 English/Language Arts Teaching Methods, Grades 6-8 ⁴	
MSSE 412 Social Studies Teaching Methods, Grades 6-8 ^{4,5}	
MSSE 413 Mathematics Teaching Methods, Grades 6-8 ⁴	
MSSE 414 Natural Sciences Teaching Methods, Grades 6-8 ⁴	
Subtotal	3
Field Experience in Middle Grades Education	
MSSE 415 Content Area Field Experience in Middle Schools ⁴	3
Subtotal	3
Total Credits	20-24

¹ Minimum grade of "C" is required.

² Students who do not receive a minimum grade of "C" in this course may retake the course or take an equivalent writing course and pass with a minimum grade of "C." Contact the department to determine what course would be most appropriate.

³ Course enrollment is blocked and requires admission to Teacher Education.

⁴ Course work requires successful completion of MSSE 210, an orientation to the profession course, a general methods course and an initial field placement.

⁵ History or political science majors must enroll in HIST 357 or POSC 457 as a corequisite to MSSE 412 and MSSE 415.

Recommended Schedule for Secondary Education Minor

Course	Title	Credits
First and/or Second Year		
EDUC 175 or PSYC 160	Learning, Behavior and Engagement in Context [C5SD] or Life Span Human Development [C5SD]	3
EDUC 200	Foundations of Education in the United States	3
WRTC 103	Rhetorical Reading and Writing [C1W]	3
Choose one of the following:		1-3
CSM 199	Science Outreach in Schools	
MSSE 101	Orientation to the Profession	
MSSE 201	Introduction to Social Studies Education Seminar	
Credits		10-12
Total Credits		10-12

Additional Information

Students must declare the secondary education minor and be fully admitted into Teacher Education prior to enrolling in the remaining courses. Contact the departmental office to determine the requirements for admission to Teacher Education. The first four courses listed above are co-requisite offerings and must be completed before enrolling in next courses: MSSE 210, MSSE 370, MSSE 371/MSSE 430, which are then prerequisites for the remaining three courses (MSSE 411/MSSE 412/MSSE 413/MSSE 414, MSSE 415, and LED 440).

Course	Title	Credits
Third and/or Fourth Year		
LED 440	Disciplinary Literacy in Secondary Education	3
Choose 4 credits of the following:		4
MSSE 370	General Instructional Methods for Grades 6-12	
MSSE 371 or MSSE 430	Clinical Experience in Adolescent Education or Introduction to Mentoring Valley Scholars	
Choose the content specific teaching methods course:		
MSSE 411	English/Language Arts Teaching Methods, Grades 6-8	
MSSE 412	Social Studies Teaching Methods, Grades 6-8	
MSSE 413	Mathematics Teaching Methods, Grades 6-8	
MSSE 414	Natural Sciences Teaching Methods, Grades 6-8	
MSSE 415	Content Area Field Experience in Middle Schools	3
Credits		10
Total Credits		10

Pre-Veterinary Medicine Minor

Coordinator: Katie Fisher
Email: fishermk@jmu.edu
Phone: (540) 568-6652
Pre-Veterinary Medicine Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/pre-vet-medicine.shtml>)

Program Description

The Pre-Veterinary Medicine program provides guidance on courses, experiential activities, and competencies for students that are preparing to be competitive applicants to Doctor of Veterinary Medicine (D.V.M.) programs after completing their Bachelor's degree at James Madison University. The Pre-Vet Medicine program is a minor that students complete as part of their baccalaureate degree.

Declaring Pre-Veterinary Medicine (Admission Standards)

Incoming first-year and transfer students can declare pre-veterinary medicine without verification from PPH Advising. Current students can declare the pre-veterinary medicine minor through MyMadison without needing further approval.

Additional Information

Pre-professional health programs do not ensure that all prerequisites for admission to health professions schools are met. Students should inquire with veterinary schools of interest to confirm details of admission requirements.

Minor Requirements

Code	Title	Credits
Courses		
BIO 140 & 140L	Foundations of Biology I [C3NS] and Foundations of Biology I Laboratory [C3L]	4
BIO 150 & 150L	Foundations of Biology II and Foundations of Biology II Laboratory	4
BIO 240 & 240L	Genetics and Genetics Laboratory	4
BIO 245 & 245L	General Microbiology and General Microbiology Laboratory	4
CHEM 131 & 131L	General Chemistry I [C3PP] and General Chemistry Laboratory [C3L]	4
CHEM 132 & 132L	General Chemistry II and General Chemistry Laboratory	4
CHEM 241	Organic Chemistry I	3
CHEM 242 & 242L	Organic Chemistry II and Organic Chemistry Laboratory	5
CHEM 361	Biochemistry I	3
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	4
PHYS 150 & 150L	College Physics II and General Physics Laboratory II	4
Subtotal		43
Intermediate and Advanced BIO Course Work		
Choose 13 credit hours from subjects recommended by veterinary schools:		13

BIO 304	Cell and Molecular Biology	
BIO 316	Animal Development	
BIO 320	Comparative Anatomy of Vertebrates	
BIO 325	Medical Parasitology	
BIO 370	Human and Animal Physiology	
BIO 425	Immunology	
BIO 480	Advanced Molecular Biology	
BIO 482	Human Histology	
Subtotal		13
Calculus		
Choose one of the following:		3-4
MATH 205	Applied Calculus [C3QR]	
MATH 231	Calculus with Functions I [C3QR]	
MATH 233	Calculus with Modeling I [C3QR]	
MATH 235	Calculus I [C3QR]	
Subtotal		3-4
Statistics		
MATH 220	Elementary Statistics [C3QR]	3
or MATH 229	Introduction to Applied Statistics Using R [C3QR]	
Subtotal		3
Writing and Literature		
WRTC 103	Rhetorical Reading and Writing [C1W]	3
One literature course from (C2L) ENG or HUM		3
Subtotal		6
Public Speaking		
Choose one of the following:		3
SCOM 121	Fundamental Human Communication: Presentations [C1HC]	
SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC]	
SCOM 123	Fundamental Human Communication: Group Presentations [C1HC]	
Subtotal		3
Total Credits		71-72

Recommended Electives (Not Required)

Code	Title	Credits
CHEM 366L	Biochemistry Laboratory	2
HTH 210	Medical Terminology	3
PSYC 101	General Psychology [C5SD]	3

Public Policy and Administration Minor

Coordinator: Robert Roberts

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Phone: (540) 568-6323

Public Policy and Administration Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/public-policy-and-administration.shtml>)

Program Description

The minor in public policy and administration seeks to give students a foundation in the nature and practice of public policy and public administration. Students have an opportunity to complete an internship as part of the minor.

Minor Requirements

The minor in public policy and administration requires 19 credit hours of course work.

Code	Title	Credits
Core Courses		
POSC 225	U.S. Government [C4AE]	4
PPA 200	Introduction to Public Policy	3
PPA 265	Public Administration	3
Subtotal		10
Public Policy		
Choose one of the following:		3
POSC 302	State and Local Government	
POSC 347	Comparative Public Policy	
POSC 367	U.S. Immigration and Refugee Law	
POSC 368	Interest Groups, Lobbying and Public Policy	
POSC 378	Borderlands Policy: Latina/o/x Politics in Context	
POSC 391	Topics in Public Policy	
PPA 460	Regionalism and Urban Management	
PPA 461	Education and Social Policy	
PPA 462	Social Welfare and Local Government Policy	
PPA 482	Special Topics in Public Policy	
Subtotal		3
Tools of Public Management		
Choose one of the following:		3
MGT 365	Fundamentals of Human Resource Management	
PPA 381	Budgetary Process	
PPA 412	Seminar in Intergovernmental Relations	
PPA 415	Legal Environment of Public Administration	
PPA 469	Philanthropy and Volunteerism	
PPA 470	Nonprofit Management and Leadership	
PPA 471	Financial Management for Public and Nonprofit Organizations	
PPA 472	Contract Management	
PPA 473	Economic and Community Development	
PPA 475	Grant Writing and Project Management	
PPA 483	Special Topics in Public Management	
PPA 496	Internship in Public Management	

Subtotal	3
Elective Course	
Select one additional course from either the Public Policy or Tools of Public Management areas.	3
Subtotal	3
Total Credits	19

Queer Studies Minor

Coordinator: Dawn M. Goode

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Queer Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/queer-studies.shtml>)

Program Description

This is an 18-credit hour interdisciplinary program that focuses not only on LGBTQ+ experiences, histories and perspectives, but more broadly on the social production and regulation of sexuality and gender. Queer studies offers a critique of normative models of sex, gender and sexuality, arguing for the validity and significance of various marginalized identities and practices as well as identifying the mechanisms and structures of power fundamental to systemic oppression. Along with sexism, homophobia and transphobia, queer studies examines racism, capitalism, imperialism and ableism, among other forms of inequality. Students in the minor will acquire knowledge and skills that prepare them for careers in advocacy, communications, health and social services and education.

Minor Requirements

Students are limited to no more than nine (9) credits in a single discipline.

Code	Title	Credits
Required Course		
QUST 200	Introduction to LGBTQ+ and Queer Studies	3
Subtotal		3
Restricted Elective Courses		
Choose two of the following:		6
ENG/WGSS 370	Queer Literature	
ENG/WGSS 423	Advanced Studies in Gender and Sexuality in Literature ¹	
JUST/WGSS 386	Sexual Orientation, Law and Justice	
QUST 300	Special Topics in LGBTQ+ / Queer Studies ²	
SOCI/WGSS 347	Queer Theories of Gender and Inequality	
SOCI/WGSS 367	Sociology of Sexuality	
WGSS 300	Special Topics in Women's, Gender and Sexuality Studies ^{3,4}	
Subtotal		6
Elective Courses		
Choose three of the following. No more than nine total credits may be taken in a single discipline for this minor.		9
EDUC 305	Critical Studies in Education, Race, Culture and Politics ¹	
EDUC 310	Teaching in a Diverse Society	
ENG 402	Advanced Studies in British Literature Before 1700 ^{3,4}	
ENG/WGSS 423	Advanced Studies in Gender and Sexuality in Literature ^{1,5}	
GEOG 377	Feminist Geography	

JUST 301	Advanced Level Special Topics in Justice Studies ^{1,3,4}
JUST 365	Justice in Literature, Film and Art
JUST 385	Disability and Justice
JUST/WGSS 387	Gender Identity, Law and Justice
LAXC 252	Introduction to Latin American, Latinx and Caribbean Studies [C2HQC] ³
LAXC 300	Topics in Latinx Studies ³
POSC 341	Social Movements in the U.S. and Abroad
POSC 377	Global Black Feminist Politics: On Power, Resistance and Transformation
QUST 300	Special Topics in LGBTQ+ / Queer Studies
QUST 400	Advanced Topics in LGBTQ+ / Queer Studies
QUST 410	Independent Research in LGBTQ+ / Queer Studies
QUST 425	Internship in LGBTQ+ / Queer Studies
REL 306	Gender and Sexuality in Islam
SCOM/WGSS 302	Third Wave Ecofeminism
SCOM 313	Topics in Communication Studies ^{3,4}
SOCI/WGSS 347	Queer Theories of Gender and Inequality ⁵
SOCI/WGSS 367	Sociology of Sexuality ⁵
THEA 303	Topics in Theatre (Queer Theory and Performance - must be taken for 3 credit hours) ³
WGSS 200	Introduction to Women's, Gender and Sexuality Studies [C5SD]
WGSS 300	Special Topics in Women's, Gender and Sexuality Studies ^{3,4}
ENG 302	Special Topics in Literature and Language ^{3,4}
WRTC/SCOM/ WGSS 420	Feminist Rhetorics
Subtotal	9
Total Credits	18

¹ Final project must pertain to queer topic or apply queer methodology.

² QUST special topics course. Can be repeated once for credit if topic changes. Only one will count toward the core requirement; the second will count toward required electives.

³ When pertaining to queer topics/approaches, it may count towards minor. Contact minor advisor for approval.

⁴ May be repeated as topic varies (LAXC 252 is not repeatable).

⁵ Can be taken for this category if not already taken in the "select two" from the previous category.

Robotics Minor

Coordinator: Nathan Sprague
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Phone: (540) 568-3312
Robotics Minor Website

Program Description

The cross disciplinary minor in robotics is intended to offer STEM majors and students with an interest in science and technology a fundamental understanding of scientific and technical issues involved in the design, construction and application of robots.

Educational Goals

- Students will have a basic understanding of robot control systems, sensors, motion, circuits and the overall design of robots.
- Students will be able to design and develop autonomous robots and robot control software.
- Students will develop an understanding of how advances in robotics technology can be used in diverse applications.
- Students will learn to work on a cross disciplinary team developing a technical product.

Minor Requirements

Code	Title	Credits
Basic Preparation		
Basic preparation courses consist of an introductory programming requirement, a calculus requirement and a physics requirement. A basic preparation course may be waived by the minor advisor if a student has completed a comparable course or experience.		
<i>Introductory Programming</i>		
Choose one of the following:		3
CS 149	Introduction to Programming	
IA 241	Introduction to Programming and Data Science	
ISAT 252	Programming and Problem Solving	
Subtotal		3
<i>Calculus Requirement</i>		
Choose one of the following:		3-4
MATH 232	Calculus with Functions II	
MATH 235	Calculus I [C3QR]	
ISAT 151	Topics in Applied Calculus in ISAT [C3QR]	
Subtotal		3-4
<i>Physics Requirement</i>		
Choose one of the following. Students who have completed the physical principles area of The Natural World with a course not listed below should meet with the minor advisor.		4
PHYS 140 & 140L	College Physics I [C3PP] and General Physics Laboratory I [C3L]	
PHYS 240 & 240L	University Physics I [C3PP] and University Physics Laboratory I [C3L]	
ISAT 152	Topics in Applied Physics in ISAT	
Subtotal		4
Core Course		
Choose one of the following:		3
CS 354	Introduction to Autonomous Robotics	

PHYS 386	Robots: Structure and Theory	
PHYS 387	Theory and Basic Principles of Mobile Robot Construction and Control	
Subtotal		3
Electives		
Choose at least six credits from the following. At least one elective must be from a different department from the core course.		6
CS 444	Artificial Intelligence	
CS 445	Machine Learning	
ENGR 440	Mechatronics	
ENGR 445	Embedded System Design	
ISAT 331	Cyber-Physical Industry	
ISAT 341	Machine Learning and Data Science	
ISAT 441	Simulation and Modeling	
ISAT 449	Emerging Topics in Applied Data Science	
IT 340	Data Science and Machine Learning	
MATH 238	Linear Algebra with Differential Equations	
MATH 248	Computers and Numerical Algorithms	
MATH 300	Linear Algebra	
MATH 336	Elementary Differential Equations	
MATH 341	Non-Linear Dynamics and Chaos	
PHYS 371	Digital Electronics	
PHYS 372	Microcontrollers and Their Applications (2,4)	
PHYS 388	Robots for Humanity	
Subtotal		6
Robotics Project		
CISE 481	Robotics Project Summary ¹	1
Subtotal		1
Total Credits		20-21

¹ CISE 481 is intended to summarize a capstone robotics project experience. In order to register for CISE 481 students must have completed or be enrolled in an approved capstone, special topics, or independent study course that involves a robotics project. At the discretion of the robotics minor advisor, students may also qualify for CISE 481 by completing an internship, research experience or professional project that entails robotics design and/or development. Consult the minor advisor for more information.

Russian, Eurasian and Slavic Studies Minor

Co-Coordinator: Stephany G. Plecker
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Russian, Eurasian and Slavic Studies Minor Website (<https://www.jmu.edu/cal/academics/minors/russian-studies/index.shtml>)

This minor offers students a broad, cross disciplinary perspective on Russian culture, history, political institutions, economy and geography. It deepens the students' understanding and knowledge of the Russian and non-Russian peoples of the former Soviet Union, and prepares them for careers in teaching, government and international business.

Minor Requirements

This minor requires 18 credits with a minimum of six credits of Russian language. Students must take courses in at least three different disciplines.

Special topics courses in any discipline can be applied to the minor with the approval of the minor coordinator. Students may also earn credit by participating in a study abroad program in Eastern Europe (including Russia), the Caucasus, or Central Asia.

Students who have proficiency in a language of the region not taught at JMU may satisfy the RUS language requirement by taking the AVANT Stamp assessment (<https://www.avantassessment.com/tests/stamp>) and demonstrating proficiency at the Advanced Low level. Other certifications of Advanced Low level proficiency include the ACTFL Oral Proficiency Interview, a 5 on the AP exam, an IB level of 6 or above, or a secondary school transcript where the target language was used for instruction.

Code	Title	Credits
Courses		
Choose 18 credits from the following courses:		18
GEOG 375	Political Geography	
HIST 351	War in Afghanistan: Local and Global Perspectives	
HIST 385	The Russian Empire	
HIST 386	The Soviet Union: A Socialist Experiment	
HIST 474	Stalinism: Approaches, Experiences and Legacies	
HIST 476	Revolutionary Russia	
HIST 478	Eastern Europe	
HUM 200	Great Works [C2L] (Russian Literature in Translation)	
POSC 346	Politics of Central and Eastern Europe	
RUS 101	Elementary Russian I	
RUS 102	Elementary Russian II	
RUS 111	Intensive Russian I	
RUS 212	Intensive Russian II	
RUS 231	Intermediate Russian I	
RUS 232	Intermediate Russian II	

RUS 300	Language for Living	
RUS 301	Pop Culture through Language	
RUS 308	Introduction to Russian Culture	
RUS 315	Sounds of Speech	
RUS 320	Culture and Communication in Context	
RUS 322	Language Landscapes	
RUS 400	Advanced Russian	
WLC 446	Special Topics: World Literatures in Translation (with approval when topic is appropriate)	
WLC 447	Special Topics: World Cultures in Translation (with approval when topic is appropriate)	
Total Credits		18

Science, Technology and Society Minor

Coordinator: Shannon Conley
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Science, Technology and Society Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/science-technology-and-society.shtml>)

Program Description

Science, technology and society (STS) is an internationally recognized field of cross disciplinary study that integrates philosophical, social scientific and humanistic studies to better understand the natural and human-built world. The minor offers students the opportunity to critically examine science, technology and medicine as methods for reasoning about and acting upon the natural world as expressions of human cultures, past and present.

Students learn to scrutinize the ideas, reasoning, values, practices and artifacts embedded in the world they inhabit today. They explore how choices made within various historical, social, economic and political contexts sometimes influence the development of science, technology and medicine. They also see how the adoption and diffusion of ideas, artifacts and techniques can then influence individuals, society, politics and culture. Courses in this minor draw students together from diverse majors across the campus and encourage open inquiry into the role of science and technology in society.

The minor in STS is open to all undergraduate students at JMU. Courses taken to complete the STS minor can also be used to satisfy the student's major and General Education requirements.

Minor Requirements

Code	Title	Credits
Required Courses		
Choose one of the following:		3
HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies	
SOCI 315	Technology and Society	
ISAT 271	Technology, Science and Society	
Subtotal		3
Elective Courses		
Special topics courses not listed can be applied to the minor with prior approval of the program coordinator.		
Choose five of the following from at least four different prefixes:		15
ANTH 360	Medical Anthropology	
CS 330	Societal and Ethical Issues in Computing	
GEOG 322	Agricultural Systems	
GEOG 325	Environmental Ethics	
GEOG/ISAT 429	Sustainability: An Ecological Perspective	
HIST 305	History of Science and Christianity	
HIST 306	A History of the Body in the West	
HIST 326	How Cars Changed Everything	
HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies	

HIST 404	Science and Society in Early Modern Europe
HIST 405	Travel and Exploration
HIST 427	U.S. Environmental History
HIST 443	Modern American Technology and Culture
ISAT 171	Governance of Sociotechnical Systems
ISAT 311	Role of Energy in Modern Society
ISAT 411	Energy Economics and Policy
ISAT 421	Environmental Policy and Regulation
ISAT 456	Ethical, Legal and Social Implications of Biotechnology
ISAT 477	Complex Systems and How They Fail
ISAT 483	Topics in Science, Technology and Society
ISAT 487	Science, Technology and Society Futures Laboratory
PHIL 396	Philosophy of Physics
PHIL 397	Philosophy of Space and Time
PHIL 398	Philosophy of Quantum Theory
PHIL 410	Philosophy of Science
SOCI 311	Sociology of the Environment
SOCI 375	Medical Sociology
WGSS/ISAT 485	Gender Issues in Science
WRTC 358	Writing About Science and Technology
WRTC 416/SCOM 465	Rhetoric of Environmental Science and Technology
WRTC 458	Scientific and Medical Communication
Subtotal	15
Total Credits	18

Social Justice Studies Minor

Coordinator: Dr. Christine Robinson

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Social Justice Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/social-justice-studies.shtml>)

Program Description

The cross-disciplinary minor in Social Justice Studies, administered by the Department of Justice Studies, fosters educated and enlightened citizenship through a foundational curriculum designed for students who are engaged in social justice endeavors and/or preparing for careers and advanced professional or graduate work in relevant fields.

This minor is designed to:

- introduce students to the cross-disciplinary field of social justice, focusing on understanding, analyzing and evaluating scholarship on the origins, persistence and nature of oppression; social justice theories; and diverse, exemplar models of social justice praxis
- provide a curriculum where students may pursue their own topical interests in social justice studies and apply/further their knowledge and skills in social justice studies
- support students' success and resiliency in social justice endeavors (civic or professional)

The minor is open to all undergraduate students at James Madison University except Justice Studies majors whose concentration area is Social Justice Engagement. Students should allow a minimum of four semesters in which to complete the minor.

Minor Requirements

The requirement for completing the minor is 18 credits.

Students will complete one required introductory course (from two options) and then select five additional electives. No more than two courses may be double-counted in the minor from a student's major/minor. The Department of Justice Studies will waive the prerequisite (JUST 200) for SJS minors whose major is not Justice Studies.

Code	Title	Credits
Required Course ¹		
Students must declare the SJS minor to take one of the following two courses:		
JUST 221	Social Justice Theories	3
or JUST 223	Social Justice Interventions and Policies	
Subtotal		3
Electives		
Choose five additional electives from the list below. Electives must represent at least three different departments. Only one sociology course may be selected. ²		
ENG 359	Studies in African American Literature	
ENG 360	Introduction to Ethnic American Literature	
ENG 361	African American Fiction	
ENG 362	African American Poetry	
ENG 363	Native American Literature	
ENG 379	Literature and Empire	
HIST 461	Marxism	

JUST 317	Victimization of Children ¹
JUST 331	Human Rights in Theory and Practice ¹
JUST/WGSS 341	Gender and Justice ¹
JUST 343	Justice and Society ¹
JUST 357	Environmental Justice ¹
JUST 375	Genocide in the 20th Century ¹
JUST 385	Disability and Justice ¹
JUST 386	Sexual Orientation, Law and Justice ¹
JUST 387	Gender Identity, Law and Justice ¹
POSC 341	Social Movements in the U.S. and Abroad
POSC 351	Topics in American Politics (when the topic is approved)
POSC 367	U.S. Immigration and Refugee Law
POSC 371	Topics in Comparative Politics (when the topic is approved)
POSC/WGSS 383	Women and Politics in Comparative Perspective
POSC 384	Minority Group Politics (when the topic is approved)
SCOM 343	Contemporary Rhetorical Theory and Practice
SCOM 348	Communication and Gender
SCOM 352	Communication and Social Movements
SCOM 354/WRTC 326	Environmental Communication and Advocacy
SCOM 355	Food as Communication
SCOM 357	Youth, Communication and Culture
SCOM/WGSS/WRTC 420	Feminist Rhetorics
SCOM 356	Immigrant Advocacy
SCOM 448	Communication, Culture and Identity (for SCOM majors only) ³
SJS 400	Social Justice Studies Internship ⁴
SJS 401	Social Justice Studies Project ⁴
SOCI 318	Sociology of Immigration
SOCI 335	Disability and Society
SOCI/WGSS 337	Sociology of Gender
SOCI 357	Sociology of Disasters
SOCI 360	Social Movements
SOCI/WGSS 367	Sociology of Sexuality
SOWK 288	Human Rights and Social Progress ⁵
THEA 303	Topics in Theatre (when the topic is approved)
WRTC 314	Writing in the Public Sphere
WRTC 426	Special Topics in Writing, Rhetoric and Technical Communication (when the topic is approved)
WRTC 484	Writing for Nonprofits
WRTC 486	Writing in the Community
Subtotal	15
Total Credits	18

¹ The Department of Justice Studies will waive the prerequisite (JUST 200) for SJS minors whose major is not Justice Studies.

- ² Students must declare the Social Justice Studies minor and successfully complete JUST 221 or JUST 223 to take elective courses for the minor.
- ³ This course may be taken by SCOM majors only.
- ⁴ Students should seek advance approval from the Social Justice Studies minor coordinator and may register for this course only through departmental approval.
- ⁵ Open to Social Work majors only.

Social Studies Minor

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Social Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/social-studies.shtml>)

The social studies minor offers a program of integrated study that exposes students to diverse methodologies, philosophies and controversies that define the social sciences. It is a required minor for secondary education students who wish to teach social studies and suggested for those planning to teach middle school. The minor incorporates economics, geography, history and political science and is offered in three forms: one for secondary education with a history major, one for secondary education with a political science major and one for middle education.

To be licensed to teach secondary school social studies, the student must satisfactorily complete requirements for a baccalaureate degree in either history or political science and complete the SS minor that is paired with that major. Students not selecting the secondary education licensure program should declare the middle education form of the minor.

Minor Requirements

Social Studies Minor for History Majors

Code	Title	Credits
Social Studies Minor Core		
HIST 225	U.S. History [C4AE] ¹	4
POSC 225	U.S. Government [C4AE] ¹	4
ECON 200	Introduction to Macroeconomics [C4GE]	3
or ECON 201	Introduction to Microeconomics	
GEOG 200	Geography: The Global Dimension [C4GE]	3
or GEOG 280	Human Geography: The Cultural Landscape	
Subtotal		14
Capstone		
Choose one of the following:		3
HIST 357	Fates of Empires ¹	
POSC 457	Fates of Empires ²	
Subtotal		3
Government and Politics Requirement		6-9

All History majors, whether in the four-year or five-year program, take the following course:

POSC 335	Comparative Politics for Teachers
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Four-Year Program

Students in the four-year program select an American government elective from the following options:

POSC 362	Political Behavior
POSC 364	Political Parties in America
POSC 366	Introduction to the Politics of Race and Ethnicity
POSC 368	Interest Groups, Lobbying and Public Policy
POSC 369	Elections and Voting Behavior
POSC 380	The U.S. Presidency

POSC 382	The Role of Religion in American Politics
POSC 385	The U.S. Congress
POSC 386	The U.S. Judiciary

Five-Year Program

Students in the five-year program select two of the following American government and civic engagement electives:

POSC 336	Civic Engagement for Teachers
POSC 362	Political Behavior
POSC 364	Political Parties in America
POSC 366	Introduction to the Politics of Race and Ethnicity
POSC 368	Interest Groups, Lobbying and Public Policy
POSC 369	Elections and Voting Behavior
POSC 380	The U.S. Presidency
POSC 382	The Role of Religion in American Politics
POSC 385	The U.S. Congress
POSC 386	The U.S. Judiciary

African American History Requirement

African American history is now a state licensure requirement for everyone licensed to teach social studies graduating after May 2023. Therefore all students in this program need to complete a course in African American history and culture. This requirement may double-count towards elective credit in the History major.

HIST 307	The Trans-Atlantic Slave Trade
HIST 322	Jim Crow South
HIST 323	The Old South
HIST 324	Terror and Survival in American Lynching
HIST 329	Frederick Douglass’s America: The U.S. in the Civil War and Reconstruction Era
HIST 350	Virginia
HIST 356	African American History since 1865

Total Credits 26-29

¹ For four-year program: MSSE 412 and HIST 357 spring junior year double count in history and SS.
² For five-year program: MSSE 412 and POSC 457 fall senior year count in SS only.

Social Studies Minor for Political Science Majors

HIST 225 must be completed for General Education credit.

Code	Title	Credits
Social Studies Minor Core		
HIST 225	U.S. History [C4AE] ¹	4
POSC 225	U.S. Government [C4AE] ¹	4
Choose one of the following:		3
ECON 200	Introduction to Macroeconomics [C4GE]	
ECON 201	Introduction to Microeconomics	
GEOG 200	Geography: The Global Dimension [C4GE]	3
or GEOG 280	Human Geography: The Cultural Landscape	
Subtotal		14
Capstone		
HIST 357	Fates of Empires ^{1,2}	3
or POSC 457	Fates of Empires	
Subtotal		3

Virginia History Requirement		
HIST 303	Virginia's Place in #VastEarlyAmerica	3
or HIST 350	Virginia	
Subtotal		3
Connected History Courses		
Choose one of the following:		
HIST 307	The Trans-Atlantic Slave Trade	
HIST 314	Plagues, Saints and Empires: The World before 1500	
HIST 315	Global History of Sport	
HIST 317	Plagues and Pandemics in Modern World History	
HIST 318	Our Better Angels?: The World History of Violence	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 328	Towers, Time and Technology on the Eve of World War I	
HIST 333	Maps, Money, Manufacture and Trade	
HIST 334	A Global History of Nationalism	
HIST 353	The Global Middle East	
HIST 365	Topics in Connected History	
HIST 368	Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States	
HIST 373	History of the Ottoman Empire, 1453-1922	
Subtotal		3
African American History Requirement or History Elective		
African American history is now a state licensure requirement for everyone licensed to teach social studies graduating after May 2023. Therefore all students in this program need to complete a course in African American history and culture. If the course chosen to fulfill this requirement also double counts for either the Virginia History (HIST 350) or Connected History (HIST 307) requirement, then the student must select an additional HIST course at the 300 or 400 level.		
HIST 307	The Trans-Atlantic Slave Trade	
HIST 322	Jim Crow South	
HIST 323	The Old South	
HIST 324	Terror and Survival in American Lynching	
HIST 329	Frederick Douglass's America: The U.S. in the Civil War and Reconstruction Era	
HIST 350	Virginia	
HIST 356	African American History since 1865	
Subtotal		3
Total Credits		26

¹ For four year program: MSSE 412 and POSC 457 spring junior year double count in political science and SS.

² For five year program: MSSE 412 and POSC 457 fall senior year count in SS only.

Social Studies Minor for Middle Education

Code	Title	Credits
Required Courses		
HIST 102	World History since 1500 [C2HQC]	3
HIST 225	U.S. History [C4AE]	4
POSC 335	Comparative Politics for Teachers	3

Subtotal		10
Capstone		
Choose one of the following:		
HIST 357	Fates of Empires ¹	
POSC 457	Fates of Empires ¹	
Subtotal		3
Virginia History Requirement		
HIST 303	Virginia's Place in #VastEarlyAmerica	3
or HIST 350	Virginia	
Subtotal		3
Thinking with Geography and Economics		
Choose one of the following:		
HIST 333	Maps, Money, Manufacture and Trade	
IDLS 395	Topics in IDLS (must have topic: People, Money & Place for three credit hours)	
Subtotal		3
Pre-Modern World History Course		
Choose one of the following:		
ANTH 205	Buried Cities, Ancestral Places: The Rise and Fall of Early Human Societies [C2HQC]	
HIST 101	World History to 1500 [C2HQC]	
HIST 314	Plagues, Saints and Empires: The World before 1500	
REL 101	Religions of the World [C2HQC]	
Other courses with permission		
Subtotal		3
African American History Requirement and Electives		
African American History requirement and/or two social studies electives ³		
African American history is now a state licensure requirement for everyone licensed to teach social studies graduating after May 2023. Therefore all students in this program need to complete a course in African American history and culture. The course chosen to fulfill this requirement may double-count with either the Virginia History or the Social Studies elective requirement, but not both.		
HIST 307	The Trans-Atlantic Slave Trade	
HIST 322	Jim Crow South	
HIST 323	The Old South	
HIST 324	Terror and Survival in American Lynching	
HIST 329	Frederick Douglass's America: The U.S. in the Civil War and Reconstruction Era	
HIST 350	Virginia	
HIST 356	African American History since 1865	
Subtotal		6-7
Total Credits		28-29

¹ Capstone; corequisite with MSSE 412.

² The social studies electives requirement will be waived for students who have declared two content areas for the Middle Education major. The course that fulfills the African American history requirement can also count towards this requirement.
See advisor for this requirement. Generally, courses in HIST/POSC/SOC/ANTH/GEOG/ECON/IDLS offered at the 200-400 level will be approved for this requirement.

Sociology Minor

Coordinator: Benjamin Brewer
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Phone: (540) 568-7391
Sociology Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/sociology.shtml>)

Program Description

Sociologists study the causes and consequences of social inequality. They also explore the social aspects of the environment, technology, politics, health, globalization, social movements, social justice, markets, mass media/popular culture and more.

Minor Requirements

To minor in sociology, a student must complete a minimum of 18 credit hours of sociology course work including three core credit hours and 15 elective credit hours.

Code	Title	Credits
SOCI 200	Development of Sociological Thought and Method	3
Sociology electives		15
Total Credits		18

Spanish-English Translation and Interpretation Minor

Advisor: Marianne Mason

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Phone: (540) 568-6327

Spanish-English Translation and Interpretation Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/spanish-english-translation-and-interpretation.shtml>)

Program Description

Translation and interpretation are two of today's fastest growing professions. An obvious requirement for anyone wishing to enter these professions is the ability to speak two languages fluently. Students who wish to make a positive impact as a translator or interpreter, however, require more than just the ability to speak two languages well. What they need is the linguistic, cultural and technical expertise they can only get from courses designed specifically to prepare them for the demands of these challenging professions. In the minor in translation and interpretation, students take their first steps toward acquiring this expertise.

Minor Requirements

Code	Title	Credits
Language Foundations		
Choose one of the following:		3
SPAN 300	Language for Living	
SPAN 301	Pop Culture through Language	
SPAN 307	History of Spanish Culture and Society	
SPAN 308	Latin American Cultures	
SPAN 320	Culture and Communication in Context	
SPAN 321	Culture and Communication in Context (for Heritage Speakers)	
SPAN 322	Language Landscapes	
Subtotal		3
Interpretation Course		
Choose one of the following:		3
SPAN 436	Introduction to Interpretation	
SPAN 437	Introduction to Medical Interpretation	
SPAN 438	Introduction to Legal Interpretation	
Subtotal		3
Linguistics Course		
Choose one of the following:		3
SPAN 310	Introduction to Spanish Linguistics	
SPAN 311	Contrastive Linguistics	
SPAN 315	Sounds of Speech	
SPAN 404	Spanish in the United States	
SPAN 448	Special Topics in Spanish Linguistics	
Any LING course		
Subtotal		3
Practicum		
Choose one of the following:		3
SPAN 496	Spanish Internship Abroad	
SPAN 497	Spanish Internship	

Subtotal		3
Professional Course		
Choose one of the following:		3
SPAN 306	Spanish for Social Work	
SPAN 312	Translation Competencies	
SPAN 330	Business Spanish	
SPAN 360	Spanish for Public Safety	
SPAN 365	Spanish for Medical Professionals I	
SPAN 430	Advanced Business Spanish	
SPAN 435	Translation Strategies	
SPAN 475	Spanish for Medical Professionals II	
SPAN 476	Culture and Medicine in Latin America	
SPAN 477	Culture and Medicine in Spain	
SPAN 478	Spanish in Healthcare: Policy and Practice in Latinx Health	
SPAN 485	Business and Society in Latin America	
SPAN 486	Business and Society in Spain	
Subtotal		3
Electives		
Choose any 300- or 400-level SPAN course not already taken.		3
Subtotal		3
Total Credits		18

Sport Communication Minor

Coordinator: Jason Mollica

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Sport Communication Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/sport-communication.shtml>)

Program Description

The sport communication minor provides students with a view into sports media and sports organization and offers them the chance to consider the varied stakeholders of sports and how sports intersect with many societal issues. Students develop the skills necessary to produce accurate and engaging sports content, manage strategic communications for athletic organizations and critically analyze the cultural, political and economic impact of sports in society. Students learn the tools to pursue professional roles in a variety of settings, include sports media outlets, collegiate and professional athletic organizations, and public relations firms. Students gain hands-on experience in sports broadcasting, journalism and public relations.

This minor consists of course work offered in communication studies (p. 49), media arts and design (p. 84), math (p. 82) and elective coursework in several other fields for students with an interest in sports media and communication.

Because of the curriculum sequence, it typically takes three to four consecutive semesters to complete the minor. Rising seniors typically cannot complete the minor by graduation. Some required courses are offered only one semester a year.

This minor is open to any student at the university including SCOM majors.

Minor Requirements

Code	Title	Credits
Required Courses		
MATH 105 or MATH 220 or COB 191	Statistical Literacy [C3QR] Elementary Statistics [C3QR] Business Analytics I	3
SCOM 230	Introduction to Sport Communication	3
SCOM 260	Introduction to Public Relations	3
SCOM 365	Sports Public Relations	3
Choose one of the following:		3
SCOM/SMAD 244	Sport Communication Techniques: Writing and Reporting	
SCOM 261	Public Relations Techniques I: Written	
SMAD 210	News Reporting and Writing	
SMAD 243	Sport Communication Techniques: Broadcasting	
Subtotal		15
Practicum Course		
SCOM 312	Sport Communication Minor Practicum	3
Subtotal		3
Elective		
Choose one of the following:		3
KIN 304	Historical and Philosophical Foundations of Sport and Physical Activity	
KIN 329	Social Aspects of Sport	

SCOM 342	Argument and Advocacy	
SCOM 347	Communication, Diversity and Popular Culture	
SCOM/WGSS 348	Communication and Gender	
SCOM 350	Organizational Communication	
SCOM 370	Introduction to Health Communication	
SCOM 412	Sport Communication Internship	
SCOM 464	Special Topics in Sport Communication	
SMAD 295	Practicum in Media Arts and Design ((taken for 3 credits))	
SOCI 329	Sociology of Sport	
Approved special topics course in KIN, SCOM, SMAD or SRM		
Subtotal		3
Total Credits		21

Statistics Minor

Coordinator: Prabhashi Withana Gamage
Email: withanpw@jmu.edu
Phone: (540) 568-1736
Statistics Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/statistics.shtml>)

Program Description

The Statistics Minor program is an interactive and applied educational experience for students from diverse academic backgrounds. It is meticulously crafted to equip students with the skills essential for the job market. This program prioritizes practical knowledge and real-world applications, empowering students with the capacity to effectively analyze data and make informed decisions. By integrating theory with hands-on experience using cutting-edge statistical software, students develop strong problem-solving and critical thinking abilities. This comprehensive training not only enhances students' statistical literacy but also complements their primary areas of study, ensuring they are prepared for both academic and professional challenges, including graduate school and careers in the workforce.

Credit by Examination

The Department of Mathematics and Statistics (p. 82) offers credit by examination for some of the courses taught in the department. Students who want to take an examination must apply to the department head. Upon application students will receive details regarding approval to take the examination.

Minor Requirements

A minor in statistics requires a minimum of 18 credit hours. Students seeking the minor in statistics must complete one of the following options.

Option 1

Not open to students majoring in mathematics. Students completing Option 1 must obtain prior approval of the courses to be counted in the minor from the statistics minor coordinator.

Code	Title	Credits
Required Courses		
MATH 322	Applied Linear Regression	3
Choose one of the following:		3-4
MATH 220	Elementary Statistics [C3QR]	
MATH 229	Introduction to Applied Statistics Using R [C3QR]	
MATH 318	Introduction to Probability and Statistics	
MATH 329	Introduction to Probability and Theoretical Statistics	
Choose four of the following:		12
MATH 268	Data Analysis and Visualization	
MATH 309	SAS Programming and Data Management	
MATH 321	Analysis of Variance and Experimental Design	
MATH 324	Applied Nonparametric Statistics	
MATH 325	Survey Sampling Methods	
MATH 326	Statistical Quality Control	
MATH 327	Categorical Data Analysis	
MATH 328	Time Series Analysis	

MATH 358	Introduction to Machine Learning	
MATH 368	Statistical Computing	
MATH 421	Applied Multivariate Statistical Analysis	
MATH 423	Stochastic Processes	
MATH 424	Statistical Decision Theory	
MATH 426	Probability and Mathematical Statistics I	
MATH 427	Probability and Mathematical Statistics II	
BIO 454	Introduction to Biometrics	
Total Credits		18-19

Option 2

Not open to students majoring in mathematics. Students completing Option 2 must obtain prior approval of the courses to be counted in the minor from the statistics minor coordinator.

Code	Title	Credits
Required Courses		
MATH 236	Calculus II	4
MATH 237	Calculus III	4
MATH 322	Applied Linear Regression	3
MATH 318	Introduction to Probability and Statistics	3-4
or MATH 329		Introduction to Probability and Theoretical Statistics
Choose two of the following:		6
MATH 268	Data Analysis and Visualization	
MATH 321	Analysis of Variance and Experimental Design	
MATH 326	Statistical Quality Control	
MATH 358	Introduction to Machine Learning	
MATH 368	Statistical Computing	
MATH 423	Stochastic Processes	
MATH 426	Probability and Mathematical Statistics I	
Total Credits		20-21

Option 3

The Option 3 minor in statistics is only open to students majoring in mathematics. Up to six hours may be double-counted towards the mathematics major. Students completing Option 3 must obtain prior approval of the courses to be counted in the minor from the statistics minor coordinator.

Code	Title	Credits
Required Courses		
MATH 229	Introduction to Applied Statistics Using R [C3QR] ¹	3
MATH 321	Analysis of Variance and Experimental Design	3
MATH 322	Applied Linear Regression	3
MATH 329	Introduction to Probability and Theoretical Statistics ¹	3
Subtotal		12
Electives		
Choose 6-10 credit hours from the following:		6-10
MATH 268	Data Analysis and Visualization	
MATH 324	Applied Nonparametric Statistics	
MATH 325	Survey Sampling Methods	
MATH 327	Categorical Data Analysis	
MATH 328	Time Series Analysis	
MATH 354	Introduction to Biometrics	

MATH 358	Introduction to Machine Learning	
MATH 368	Statistical Computing	
MATH 421	Applied Multivariate Statistical Analysis	
MATH 426	Probability and Mathematical Statistics I	
Subtotal		6-10
Total Credits		18-22

¹ MATH 318 may be substituted for MATH 229 and MATH 329 by students taking at least 9 hours of elective courses.

Substance Use Disorder Education Minor

Advisor: Katherine Ott Walter
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Phone: (540) 568-8972
Substance Use Disorder Education Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/substance-abuse-prevention.shtml>)

Program Description

This cross-disciplinary substance use disorder education minor provides the student with the foundational knowledge and skills necessary to assist communities in designing evidenced-based prevention/education programs to encourage healthy attitudes and behaviors toward alcohol, tobacco and other drugs. Course work in this minor will help prepare the student for the Certified Prevention Professionals-ATOD exam, which they can take after one year of prevention-related paid work experience and additional prevention-specific training.

Minor Requirements

Students must complete 18 credit hours from the categories outlined below. Students must take one of the Alcohol, Tobacco and Other Drugs (ATOD) specific courses. The remaining 15 credits must be selected from the additional courses listed below, preferably one from each category. No more than three courses may be taken from a single subject area (PSYC, HTH, SOWK, etc.), and no more than three courses can count for both the student's major/additional minor and the substance use disorder education minor.

Code	Title	Credits
Alcohol, Tobacco and Other Drugs (ATOD)		
Choose one of the following:		3
HTH 378	The Use and Effects of Drugs	
PSYC 285	Drugs and Behavior	
SOWK 327	Substance Use Interventions	
Subtotal		3
Electives		
Choose five courses from the list of courses below. No more than three courses may be taken from a single subject area, e.g., PSYC, HTH, SOWK, etc.		15
Prevention, Education and Service Delivery:		
HTH 351	Health Behavior Change	
JUST 221	Social Justice Theories	
JUST 301	Advanced Level Special Topics in Justice Studies ¹	
JUST 345	Restorative Justice	
PSYC 250	Introduction to Abnormal Psychology	
PSYC 308	Health Psychology	
PSYC 335	Abnormal Psychology	
SOWK 386	Youth Empowerment Strategies (YES)	
SOWK 387	Working with Adolescents	
Communication:		
SCOM 370	Introduction to Health Communication	
SCOM 470	Health Communication Campaigns	
WRTC 484	Writing for Nonprofits	
WRTC 486	Writing in the Community	
WRTC 488	Writing in the Health Sciences	

Community Organizing:	
FAM/GERN/ NPS/SOWK 375	Grant Writing for Agencies
HTH 456	Grant Writing in Health Sciences ²
NPS 300	Introduction to Nonprofits
Public Policy and Environmental Change:	
PPA 359	Policy Analysis
PPA 462	Social Welfare and Local Government Policy
WRTC 326/ SCOM 354	Environmental Communication and Advocacy
SOWK 335	Social Policy (SOWK majors only)
HTH 420	Global Perspectives on Alcohol and Drug Policies ²
HTH 422	Public Health and Healthcare Policy
Planning and Evaluation:	
HTH 458	Health Program Planning and Evaluation
Subtotal	15
Total Credits	18

¹ When topics are minor related, to be approved by the coordinator.
² Offered in summer only.

Teaching English to Speakers of Other Languages Minor

Coordinator: Katya Koubek

Email: koubekex@jmu.edu

Phone: (540) 568-6760

TESOL Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/tesol.shtml>)

Program Description

The minor in TESOL is designed for students in various fields who wish to acquire professional knowledge related to teaching English to speakers of other languages. Students will gain skills in cross-cultural competence, understand and implement equitable, evidence-based ways of working with multilingual learners, and become familiar with the processes of first and second language acquisition. Students will acquire a foundational knowledge of effective teaching practices to assist and assess multilingual learners.

Minor Requirements

The TESOL minor requires a minimum of 19 credit hours; 16 credits are required core courses and 3 credits are electives selected in consultation with a program advisor.

Code	Title	Credits
Required Courses		
EDUC 310	Teaching in a Diverse Society	3
LED 430	Development, Assessment and Instruction of Literacy: K-12	3
TESL 300	Introduction to Second Language Acquisition	3
TESL 385	Practicum in TESOL (must be taken for one credit hour)	1
TESL 428	Language Assessment for Multilingual Learners	3
TESL 470	Methods in Teaching English to Speakers of Other Languages	3
Subtotal		16
Electives		
Choose one of the following:		3
EDUC 200	Foundations of Education in the United States	
ENG 308	Introduction to Linguistics	
EXED 303	Foundations of Classroom and Behavior Management	
TESL 310	Pedagogical Grammar	
Subtotal		3
Total Credits		19

Theatre Minor

Advisor: James Ogden
Email: ogdenjv@jmu.edu
Theatre Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/theatre.shtml>)

Program Description

The JMU theatre minor allows students to explore and develop their skills and knowledge in the field of theatre while pursuing a major in another subject. Students have the freedom to explore a wide range of theatre courses according to their interests, all of which will help them explore how theatre performances are created through diverse types of collaboration; how performance forms are related to the cultures they develop out of; and how skills used by artists and audiences also have utility in non-theatrical contexts.

Minor Requirements

Code	Title	Credits
Required Courses		
STAD 171	Performance Production	3
or STAD 273	Design Aspects of Performance	
THEA 210	Introduction to Theatre [C2VPA]	3
or THEA 211	Performance Analysis	
Subtotal		6
Electives		
Choose 15 credit hours from the theatre-designated courses with the approval of the minor advisor.		15
Subtotal		15
Total Credits		21

Urban and Regional Studies Minor

Coordinator: Henry A. Way

Email: wayha@jmu.edu

Phone: (540) 568-2740

Urban and Regional Studies Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/urban-and-regional-studies.shtml>)

Program Description

The minor in urban and regional studies prepares students for careers or graduate training in government. While satisfying the B.A. or B.S. requirements of their chosen major, students may complement that major with a 21 credit hour minor in urban and regional studies.

Minor Requirements

Code	Title	Credits
Required Courses		
GEOG 376	Urban Geography	3
ECON 475	Regional Economics	3
SOCI 372	Sociology of the Community	3
Subtotal		9
Electives		
Electives must be in a discipline other than the student's major.		
Choose 12 credit hours from the following:		12
ECON 326	Public Finance	
ECON 340	Economics of Natural Resources	
GEOG 315	Field Studies in Geography	
GEOL 310	Environmental Impact (A - D) or GEOG 310 Environmental Issues	
GEOL 430	Environmental Soil Science (2, 2)	
POSC 302	State and Local Government	
POSC 495	Internship in Political Science	
SOCI 352	Birth, Death, Sex: Exploring Demography	
SOCI 361	Sociology of Organizations	
Subtotal		12
Total Credits		21

Women's, Gender and Sexuality Studies Minor

Coordinator: Mary Thompson

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Phone: (540) 568-3758

WGSS Minor Website (<http://www.jmu.edu/womenstudies>)

Program Description

This is an 18-credit hour cross disciplinary program that explores the scholarship related to gender, sexuality and women. Women's, gender and sexuality studies is concerned with gender as it intersects with multiple categories such as race/ethnicity, sexuality, economic class and age. This concern is framed by recognition of increased globalization and technological growth impacting social institutions and gendered identities. Students in women's, gender and sexuality studies will develop academic frameworks for understanding gender, diversity and social justice; they will acquire knowledge and skills that prepare them for careers in advocacy and communications.

Minor Requirements

Students may double count a maximum of nine credit hours toward both a major and this minor. Students may count only six credit hours below the 300-level.

Code	Title	Credits
Required Course		
Choose one of the following:		3
WGSS 200	Introduction to Women's, Gender and Sexuality Studies [C5SD]	
WGSS 337	Sociology of Gender	
SOCI 337	Sociology of Gender	
Choose five of the following: ¹		15
ANTH 360	Medical Anthropology	
ANTH 370	Topics in the Anthropology of Gender	
ANTH/WGSS 376	Anthropology of Reproduction	
BIO/WGSS 230	Biology, Gender and Society	
ECON 306	The Economics of Women and the Family	
ENG 327	Gothic Literature	
ENG/WGSS 368	Women's Literature	
ENG/WGSS 369	Feminist Literary Theory	
ENG/WGSS 370	Queer Literature	
ENG 423	Advanced Studies in Gender and Sexuality in Literature	
ENG/WGSS 466	Advanced Studies in Women's Literature	
GEOG 377	Feminist Geography	
GEOG 477	Gendered Geographies	
HIST 319	Women at Work in U.S. History	
HIST 320	Women in U.S. History	
HIST 321	Gender and Sexuality in Modern Europe	
HIST 326	How Cars Changed Everything ²	

HIST 327	Makers, Users, Skeptics and Thieves: Americans and Their Technologies ³
HIST 448	Gender in Latin America and the Iberian World
HIST 449	Women and Fascism
ISAT/WGSS 485	Gender Issues in Science
JUST 311	Women, Crime and Criminal Justice
JUST/WGSS 341	Gender and Justice
JUST 385	Disability and Justice ²
JUST/WGSS 386	Sexual Orientation, Law and Justice
JUST/WGSS 387	Gender Identity, Law and Justice
PHIL/WGSS 350	The Philosophy of Feminism
POSC/WGSS 383	Women and Politics in Comparative Perspective
PSYC 310	The Psychology of Women and Gender
REL 306	Gender and Sexuality in Islam
REL 315	Women and Religion
SCOM/WGSS 301	Feminist Blogging: Writing for Shout Out! JMU
SCOM/WGSS 302	Third Wave Ecofeminism
SCOM/WGSS 348	Communication and Gender
SOCI 336	Race and Ethnicity ²
SOCI/WGSS 347	Queer Theories of Gender and Inequality
SOCI 354	Social Inequality
SOCI/WGSS 367	Sociology of Sexuality
SCOM/WRTC/WGSS 420	Feminist Rhetorics
SPAN 455	Women in Hispanic Literatures
WGSS 300	Special Topics in Women's, Gender and Sexuality Studies ⁴
WGSS 305	Theories and Methods in Women's, Gender and Sexuality Studies
WGSS 315	Transnational Feminisms
WGSS 325	Gender and Violence
WGSS 355	American Women at War
WGSS 400	Issues and Research in Women's, Gender and Sexuality Studies
WGSS/HTH/NSG 417	Women's Global Health and Human Rights
WGSS 490	Independent Studies in Women's, Gender and Sexuality Studies
WGSS 492	Internship in Women's, Gender and Sexuality Studies
WGSS 495	Special Topics in Women's, Gender and Sexuality Studies

Total Credits

18

¹ Any WGSS course may count toward the WGSS minor.

² Final project must take gender as its topic.

³ If research project addresses issue of gender.

⁴ May be repeated as topic varies.

World Literature Minor

Coordinator: Debali Mookerjee-Leonard

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Phone: (540) 568-6744

World Literature Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/world-literature.shtml>)

Program Description

The cross-disciplinary minor in world literature is designed for students who wish to extend their critical understanding of literature and other artistic productions beyond the confines of national borders. By offering a broad yet substantial introduction to literature and other arts from across the globe, the minor provides students with a bridge to understanding literatures, languages, and cultures. It aims to equip students with the critical and analytical skills necessary to reflect upon and appreciate these artistic practices as well as to develop a clear understanding of social context and artistic expression by encouraging them to think outside the local and on a more global level. The minor provides a good foundation for students to meet the challenges of graduate studies in comparative literature, world cinema and cultural studies/history.

Minor Requirements

The minimum requirement for a minor in world literature is 18 hours. Six credit hours may be double-counted between the minor and the major or general education courses. In addition to the gateway course, ENG 239, students can take one 200-level course to count toward the minor. The Department of English and the Department of World Languages and Cultures (WLC) offer a range of courses which will fulfill the minor. WLC offers students the opportunity to study non-English language literatures often in the original language, while all courses in the English department are taught in English or in translation. Students can also take a maximum of two relevant courses from disciplines other than English and foreign languages, with the permission of the program coordinator. Students should consult the coordinator about new courses and appropriateness of the subject of special topics courses. The following courses may include several sections per term—ENG 221, ENG 222, ENG 302, ENG 368, ENG 383, ENG 385, ENG 405, ENG 410, ENG 412, ENG 423, ENG 466, HUM 200—and only certain sections will satisfy the minor requirement. Students electing this minor can acquire more information from the program coordinator regarding courses which will count toward the minor. Courses from disciplines other than English and foreign languages may be considered for credit with the prior approval of the coordinator.

Code	Title	Credits
Required Course		
ENG 239	Studies in World Literature [C2L]	3
Subtotal		3
Elective Courses		
Choose five of the following courses:		15
ARAB 307	A History of Islamicate Civilization 600-1600 A.D.	
ARAB 308	Contemporary Islamicate Civilization	
ARAB 335	A Journey through Literature	
ARAB 385	The Arabian Nights	
ARAB 386	Arabic Fables and Short Stories	
ARAB 446	Special Topics in Arabic Literature	
ENG 221	Literature and Ideas [C2L]	
ENG 222	Literature and Genres [C2L]	

ENG 260	Survey of African American Literature [C2L]
ENG 302	Special Topics in Literature and Language
ENG 305	Mythology
ENG 306	The Bible as Literature
ENG 358	Oral Literature
ENG 360	Introduction to Ethnic American Literature
ENG 361	African American Fiction
ENG 362	African American Poetry
ENG 363	Native American Literature
ENG 364	Introduction to Asian American Literature
ENG 368	Women's Literature
ENG 377	Introduction to African Literature
ENG 378	Studies in South Asian Literature
ENG 379	Literature and Empire
ENG 383	Film Genre
ENG 385	Special Topics in Film Study
ENG 405	Advanced Studies in Anglophone Literature
ENG 408	Advanced Studies in African American Literature
ENG 410	Advanced Studies in Author
ENG 412	Special Topic Seminar
ENG 420	Advanced Studies in Theory and Cultural Studies
ENG 423	Advanced Studies in Gender and Sexuality in Literature
ENG 430	Advanced Studies in Comparative Literature
ENG 432	Advanced Studies in African Literature
ENG 433	Advanced Studies in Arabic Literature
ENG 434	Advanced Studies in Latin American Literature in Translation
ENG/GER 436	Studies in German Literature
ENG/ITAL 437	Studies in Italian Literature
ENG/RUS 438	Studies in Russian Literature
ENG 439	Advanced Studies in Major Authors of Literature in Spanish in Translation
ENG 466	Advanced Studies in Women's Literature
FR 307	History of French Culture and Society
FR 308	Contemporary History and Culture in the Francophone World
FR 335	A Journey through Literature
FR 425	Transformative Texts
FR 386	French Short Stories
FR 420	Stories and Histories in French Culture
FR 446	Special Topics in Francophone Literature
FR 447	Special Topics in Francophone Culture
FR 448	Special Topics in Francophone Linguistics
FR 490	Independent Study in French
GER 307	History of Germanic Cultures and Societies
GER 308	Modern German Culture and Society
GER 335	A Journey through Literature
GER 386	German Short Stories
GER 425	Transformative Texts
GER 446	Special Topics in German Literature
GER 447	Special Topics in German Culture
GER 465	German Cinema

GER 490	Independent Study in German	
HUM 200	Great Works [C2L]	
HUM 252	Global Cultures [C2HQC] ¹	
ITAL 307	History of Italian Culture and Society	
ITAL/HIST 308	Contemporary Italian Culture and Society	
ITAL 335	A Journey through Literature	
ITAL 380	Italian Cuisine	
ITAL 386	Italian Short Stories	
ITAL 425	Transformative Texts	
ITAL 446	Special Topics in Italian Literature	
ITAL 447	Special Topics in Italian Culture	
ITAL 465	Italian Cinema	
ITAL 490	Independent Study in Italian	
JAPN 446	Special Topics in Japanese Literature	
JAPN 447	Special Topics in Japanese Culture	
JAPN 490	Independent Study in Japanese	
RUS 308	Introduction to Russian Culture	
SPAN 307	History of Spanish Culture and Society	
SPAN 308	Latin American Cultures	
SPAN 335	A Journey through Literature	
SPAN 385	Latin American Short Story	
SPAN 386	Spanish Short Stories	
SPAN 407	Aspects of Contemporary Spanish Culture and Society	
SPAN 421	Stories and Histories: Latin American Novel	
SPAN 422	Voices of Today: Wounds and Words in Post-War Spain	
SPAN 423	Voices of Today: Latin American Poetry	
SPAN 425	Transformative Texts	
SPAN 426	Voices of the Past: Works of the Golden Age	
SPAN 446	Special Topics in Spanish Literature	
SPAN 447	Special Topics in Spanish-Speaking Cultures	
SPAN 455	Women in Hispanic Literatures	
SPAN 464	Spanish Theater	
SPAN 465	Spanish-Language Cinema	
SPAN 490	Independent Study in Spanish	
Subtotal		15
Total Credits		18

¹ Global South Asia section only.

Writing, Rhetoric and Technical Communication Minor

Advisor: Karen McDonnell
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Phone: (540) 568-7941

Writing, Rhetoric and Technical Communication Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/writing-rhetoric-and-technical-communication.shtml>)

Program Description

The Writing, Rhetoric and Technical Communication minor complements all majors and is designed for students who wish to extend, enrich, and formalize their training as writers and communicators. WRTC minors can focus their coursework on personal and civic writing, social media and content strategy, user experience (UX), online theory and inclusive design, publication management, editing, technical writing, scientific and medical communication, legal and government writing, and more.

The 18-credit minor in Writing, Rhetoric and Technical Communication is open to all JMU undergraduate students.

Minor Requirements

WRTC minors work through a set of three core courses offered every semester and choose three additional courses from a longlist of electives that offer opportunities for students to deepen their communications skills in applied contexts related to their majors and other areas of interest.

Code	Title	Credits
WRTC 200	Introduction to Studies in Writing, Rhetoric and Technical Communication	3
WRTC 300	Professional Editing	3
WRTC 301	Language, Law and Ethics	3
Choose any three WRTC electives		9
Total Credits		18

Youth Justice Minor

⁵ Social work majors only.

Coordinator: Rita Poteyeva

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Phone: (540) 568-7124

Youth Justice Minor Website (<https://www.jmu.edu/academics/undergraduate/minors/youth-justice.shtml>)

Program Description

The cross disciplinary minor in youth justice is designed for students who are interested in working in the juvenile justice system or in other agencies serving adolescents. Students will become proficient in the latest research on the principles of adolescent development, will understand the impact of trauma and will become familiar with current best practices of preventing and responding to juvenile delinquency.

The minor program is open to all undergraduate students at JMU.

Minor Requirements

The requirement is the successful completion of 18 credit hours. No more than two courses may be double counted for a student's major/minor. No more than nine hours may be taken from a single discipline.

Code	Title	Credits
Required Course		
YTHJ 300	Introduction to Youth Justice	3
Elective Courses		
Choose five of the following:		15
JUST 301	Advanced Level Special Topics in Justice Studies (When topic is relevant for YTHJ program. Consult program coordinator.) ¹	
JUST 316	Human Development and Crime ¹	
JUST 317	Victimization of Children ¹	
JUST 345	Restorative Justice ¹	
PPA 200	Introduction to Public Policy ²	
PSYC 285	Drugs and Behavior ³	
PSYC 365	Developmental Psychology ⁴	
SOCI 336	Race and Ethnicity	
SOCI 341	Sociology of Education	
SOWK 325	Parent-Child Relationships Across the Lifespan ⁵	
SOWK 335	Social Policy ⁵	
SOWK 338	Issues and Policies in Family Services ⁵	
SOWK 340	Violence in Families	
SOWK 342	Child Welfare Services	
SOWK 375	Grant Writing for Agencies	
SOWK 387	Working with Adolescents	
SOWK 487	Special Topics (Substance Abuse Intervention)	
YTHJ 400	Transforming Youth Justice Integrative Seminar	
YTHJ 405	Internship in Transformative Youth Justice Studies	
Total Credits		18

¹ These courses have lower-level prerequisite courses that justice studies majors take early in their careers. The prerequisites are waived for registered youth justice minors.

² Prerequisite POSC 225

³ Prerequisite PSYC 101

⁴ Psychology majors only.

Art Education Professional PK-12 Teaching Licensure

Coordinator: William Wightman

Phone: (540) 568-5144

Email: wightmwh@jmu.edu

ARED 480	Student Teaching (taken twice for a total of 16 Credit Hours)	16
Total Credits		24

¹ PSYC 160 may double-count toward Sociocultural and Wellness Area.

² Minimum grade of "C" required in EDUC 200, PSYC 160 and WRTC 103.

Program Description

Art education professional PK-12 licensure is available to all art history and design majors. In addition to the general education and academic major requirements, students desiring PreK-12 teaching licensure in art must complete 26 credits of additional course work in art education, education and psychology, and 16 credits of student teaching. It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For a complete description of admission and retention policies and procedures for teacher education, refer to the College of Education (p. 18).

Students seeking licensure are encouraged to consult regularly with the program coordinator of art education. The undergraduate degree leading to licensure must include the following minimum requirements in art:

- Nine credits must be earned in art history and art appreciation.
- Six credits must be earned in ceramics and crafts, with a minimum of one course in ceramics.
- Six credits must be earned in three-dimensional media, with one course in sculpture.
- Twelve credits must be earned in four different two-dimensional media.
- Twenty-seven studio credits, with a minimum of six credits in each of three separate studio areas.

Licensure Requirements

The following is a list of the required courses leading to PK-12 art licensure and a suggested sequence of when each may be taken.

Code	Title	Credits
Required Courses		
ART 304	Methods of Art Criticism (spring, second year)	3
ARED 300	Elementary Art Education Methods (fall, third year)	3
ARED 302	Secondary Art Education Methods (spring, third year)	3
ARED 400	Visual Arts Across the Curriculum (fall, fourth year)	3
ARED 490	Field Experiences in Art Education ¹	6
Total Credits		18

¹ ARED 490 repeatable; taken concurrently with ARED 300, ARED 302 and ARED 400 for two credit hours three times for a total of six credit hours.

Required Education and Psychology Courses

Code	Title	Credits
Required Courses		
PSYC 160	Life Span Human Development [C5SD] ^{1,2}	3
EDUC 200	Foundations of Education in the United States ²	3
LED 420	Content Area Literacy, K-12	2

Biology Teaching Licensure

Program Description

Students interested in becoming secondary education biology teachers (grades 9-12) must meet specific curriculum requirements to qualify for biology teacher licensure. Many of these requirements may also count as part of the 40-credit hour Biology, B.S. (p. 204) requirements.

Licensure Requirements

Code	Title	Credits
Required Courses		
BIO 140	Foundations of Biology I [C3NS]	3
BIO 140L	Foundations of Biology I Laboratory [C3L]	1
BIO 150	Foundations of Biology II	3
BIO 150L	Foundations of Biology II Laboratory	1
BIO 240	Genetics	3
BIO 240L	Genetics Laboratory	1
BIO 250	Quantitative Ecology and Evolution	3
BIO 250L	Quantitative Ecology and Evolution Laboratory	1
Subtotal		16
VDOE Anatomy/Physiology Requirement		
Choose one of the following:		4
BIO 260	Foundations of Human Anatomy and Physiology	
BIO 270 & 270L	Human Physiology [C3NS] and Human Physiology Laboratory [C3L] ¹	
BIO 290 & 290L	Human Anatomy and Human Anatomy Laboratory	
BIO 320	Comparative Anatomy of Vertebrates ²	
BIO 370	Human and Animal Physiology ²	
Subtotal		4
VDOE Botany/Plant-Focused Requirement		
Choose one of the following:		3-4
BIO 340	Morphology and Anatomy of Vascular Plants	
BIO 360	Plant Biology	
BIO 364	Human Uses of Plants	
BIO 386	Field Botany	
BIO 402	Forest Ecology	
BIO 455	Plant Physiology	
BIO 486	Systematics of Vascular Plants	
Subtotal		3-4
VDOE Zoology/Animal-Focused Requirement		
Choose one of the following:		3-4
BIO 310	Marine Biology	
BIO 316	Animal Development	
BIO 320	Comparative Anatomy of Vertebrates ²	
BIO 350	Paleobiology ²	
BIO 370	Human and Animal Physiology ²	
BIO 395	Comparative Animal Behavior	
BIO 405	Vertebrate Paleontology ²	
BIO 412	Mammalogy	
BIO 430	Ornithology: Research and Current Advances	
BIO 449	Insect Ecology	
BIO 490	Mechanics of Animal Movement	

Subtotal	3-4
VDOE Earth Science Requirement	
Choose one of the following:	
3-4	
BIO/GEOL 350	Paleobiology ³
BIO/GEOL 405	Vertebrate Paleontology ³
CHEM/GEOL 355	Geochemistry of Natural Waters ⁴
GEOL 110 & 110L	Physical Geology [C3NS] and Physical Geology Laboratory [C3L]
GEOL 115	Earth Systems and Climate Change [C3NS]
GEOL 167	History and Philosophy of the Geosciences
GEOL 210	Applied Physical Geology [C3NS]
GEOL 211	Introduction to Oceanography [C3NS]
Subtotal	3-4

¹ BIO 270 and BIO 270L do not meet the biology major requirements.

² Students must complete unique courses in each category to meet licensure requirements and have a broad preparation in the subject area.

³ This course will also fulfill biology elective hours and the VDOE zoology requirement.

⁴ This course will also meet a chemistry cognate requirement.

Dance Teaching Licensure

Program Description

In addition to general education and theatre and dance requirements, students desiring PreK-12 teaching licensure in dance must complete 19 credits of additional course work in education, psychology, and dance and 12 credits of student teaching. It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For a complete description of admission and retention policies and procedures for teacher education, refer to the College of Education (p. 18).

Licensure Requirements

Students seeking licensure are encouraged to consult regularly with the program coordinator of dance education. The undergraduate degree leading to licensure must include the following minimum requirements in dance:

- Course work in each area: ballet, folk¹, jazz and modern dance. Typically completed as part of the dance major, but may require additional course(s) to fulfill.
- Course work in one area of concentration in ballet, folk, jazz or modern dance beyond entry level.
- Course work in composition, improvisation, and dance arts production.
- Course work in scientific foundations, including human anatomy, kinesiology, and injury prevention and care for dance arts.
- Course work in cultural understanding, including cultural context and dance history.

¹ DANC 143 or DANC 303 (topic: folk dance) fulfill the folk dance requirement.

The following is a list of the required courses leading to PreK-12 dance licensure and a suggested sequence of when each may be taken.

Code	Title	Credits
Required Courses		
STAD 322	Equity, Inclusion and Access in Education and Performance (junior)	3
DANC 315	Injury Prevention and Conditioning	3
DANC 380	Dance Teaching Practicum (prior to student teaching)	3
Subtotal		9
Required Education and Psychology Courses		
EDUC 200	Foundations of Education in the United States (spring, sophomore) ¹	3
PSYC 160	Life Span Human Development [C5SD] (may double count for General Education) ¹	3
LED 420	Content Area Literacy, K-12 (fall, senior)	2
DANC 480	Student Teaching (must be taken for 12 Credit Hours)	12
Subtotal		20
Required Jazz Dance Course		
DANC 146	Jazz Dance (0, 4)	2

Subtotal	2
Total Credits	31

¹ Minimum grade of "C" required in EDUC 200, PSYC 160 and WRTC 103.

Earth Science Teaching Licensure

Program Description

Teaching Licensure Students interested in becoming teachers must meet specific curriculum requirements in their major as part of the undergraduate academic degree. The Earth science major includes state course requirements in astronomy, meteorology and oceanography, to total no less than 32 hours in the Earth sciences (including geology) and a minimum of 16 hours total in physics, chemistry and biology. In addition to the general education and academic major requirements, Earth science majors desiring secondary teacher licensure must be admitted to the pre-professional program in secondary education at the undergraduate level and complete the graduate level Master of Arts in Teaching degree.

It is critical that students seeking teaching licensure consult regularly with both their education advisor and their major advisor to support their progression through the programs. For a full description of the program in secondary teaching, refer to the Department of Middle, Secondary and Mathematics Education (p. 86), in addition to the College of Education (p. 18).

English Teaching Licensure

Students interested in becoming secondary education English teachers (grades 6-12) must meet specific curriculum requirements in order to qualify for English teacher licensure. Many of these requirements may also count as part of the 36 credit hour English major requirements. All students completing their Secondary Education English Teaching Licensure must also fulfill all requirements for the BA English major.

In addition to the General Education and academic major requirements, English majors desiring secondary education English teacher licensure must choose one of the following pathways:

1. English major (p. 265), secondary education (SEED) major (p. 419), admission to COE teacher education program and completion of program requirements, and fulfillment of the English teacher licensure requirements; or
2. English major (p. 265), secondary education (SEED) minor (p. 555), admission to COE teacher education program and completion of program requirements, fulfillment of English teacher licensure requirements, and completion of the graduate level Master of Arts in Teaching degree.

It is critical that students seeking licensure consult regularly with both their education advisor and their major advisor to support their progression through the programs. For a full description of the program in secondary teaching, refer to the Department of Middle, Secondary and Mathematics Education (p. 86), in addition to the College of Education (p. 18).

Licensure Requirements

Code	Title	Credits
Adolescent Literature		
LED 254	Literature for Adolescents	3
English Grammar		
Choose one of the following:		3
ENG 303	History of the English Language	
ENG 308	Introduction to Linguistics	
ENG 309	Traditional English Grammar	
ENG 310	Structure of the English Language	
Pre-1700 (must include Shakespeare)		
Choose one of the following:		3
ENG 317	Studies in Shakespeare	
ENG 319	Teaching Shakespeare	
Gender and Sexuality		
Choose one of the following:		3
ENG 368	Women's Literature	
ENG 369	Feminist Literary Theory	
ENG 370	Queer Literature	
ENG 423	Advanced Studies in Gender and Sexuality in Literature	
ENG 466	Advanced Studies in Women's Literature	
African American Literature		
Choose one of the following:		3
ENG 260	Survey of African American Literature [C2L]	
ENG 361	African American Fiction	
ENG 362	African American Poetry	

ENG 374	Contemporary Global Anglophone Literature	
ENG 359	Studies in African American Literature	
ENG 408	Advanced Studies in African American Literature	
World Literature		
Choose one of the following:		3
ENG 239	Studies in World Literature [C2L]	
ENG 358	Oral Literature	
ENG 374	Contemporary Global Anglophone Literature	
ENG 377	Introduction to African Literature	
ENG 378	Studies in South Asian Literature	
ENG 379	Literature and Empire	
ENG 388	Studies in Global Literatures	
ENG 405	Advanced Studies in Anglophone Literature	
ENG 432	Advanced Studies in African Literature	
ENG 433	Advanced Studies in Arabic Literature	
ENG 434	Advanced Studies in Latin American Literature in Translation	
Writing		
ENG 399	Advanced Writing in Literary Studies	3

History Teaching Licensure

Students interested in becoming secondary social studies teachers have their choice of a 5-year dual degree program (with MAT) or a 4-year undergraduate-only program. Either can be done with the history major (p. 293), choosing the teacher education concentration, a social studies minor (p. 565), and either a major in secondary education (4-year program) (p. 419) or a minor in pre-professional secondary education (p. 555) (5-year program).

Students in this program must also be admitted to the College of Education (p. 18). It is critical that students seeking licensure consult regularly with their education advisor, their social studies advisor, and their major advisor to support their progression through the programs. For a full description of the requirements see the teaching history section of the history department website (<https://www.jmu.edu/history/iss.shtml>).

Media Arts and Design Journalism Teaching Endorsement

Program Description

In addition to the general education and academic major requirements, media arts and design majors desiring to add an endorsement in journalism to a secondary teacher license in another content area must be admitted to teacher education, complete the pre-professional program in secondary education at the undergraduate level and complete the graduate level Master of Arts in Teaching degree.

It is critical that students seeking this endorsement consult regularly with both their education advisor and their major advisor to support their progression through the programs. For a full description of the program in secondary teaching, refer to the Department of Middle, Secondary and Mathematics Education (p. 86).

Modern Foreign Languages Teaching Licensure

Coordinator: Cynthia Chalupa
Email: chalupcs@jmu.edu
Phone: (540) 568-6128

Program Description

In conjunction with the College of Education (p. 18), the department offers programs leading to teaching licensure in French, German and Spanish. In addition to the General Education and academic major requirements, students seeking PreK-12 teaching licensure in French, German or Spanish must be accepted for admission into the teacher education program offered by the College of Education (p. 18) prior to enrolling in professional education courses. In addition to the 33 foreign language credits, an additional 39 credits are required for teaching licensure.

Students interested in teacher licensure in French, German or Spanish must also meet the specific curriculum requirements of their foreign language major as part of the undergraduate academic degree.

For a complete description of admission and retention policies and procedures for teacher education, refer to the College of Education (p. 18). Students seeking licensure are encouraged to consult regularly with their advisor.

Licensure Requirements

Code	Title	Credits
Required Courses		
PSYC 160	Life Span Human Development [C5SD] ¹	3
EDUC 200	Foundations of Education in the United States ¹	3
EXED 303	Foundations of Classroom and Behavior Management	3
MSSE 370	General Instructional Methods for Grades 6-12	3
MSSE 371	Clinical Experience in Adolescent Education	1
MSSE 425	Assessment in Middle and Secondary Education	3
LING 470	World Language Teaching Methods	3
FLED 471	Modern Foreign Language Field Experience	3
FLED 475	Supervised Student Teaching Experience	12
FLED 476	Student Teaching Seminar	3
LED 420	Content Area Literacy, K-12	2
Total Credits		39

¹ Minimum grade of "C" required in EDUC 200 and PSYC 160.

Music Teaching Licensure

Students interested in obtaining teaching licensure in music should see the Music Education Concentration (p. 378).

Theatre Teaching Licensure

Advisor: Dr. Rachel Rhoades

Email: rhoadere@jmu.edu

Theatre Teaching Licensure Website (<https://www.jmu.edu/academics/undergraduate/pre-professional/theater-education.shtml>)

Program Description

In addition to general education (including WRTC 103) and theatre major requirements, students desiring PK-12 licensure in theatre must complete additional course work in education, psychology and theatre. It is necessary to be admitted to the teacher education program prior to enrolling in professional education courses. For a complete description of admission and retention policies and procedures for teacher education, refer to the College of Education (p. 18).

Licensure Requirements

Students seeking licensure are encouraged to consult regularly with the theatre education program advisor. The undergraduate degree leading to licensure must include the following minimum requirements in theatre:

- Theatre education and teaching artistry concentration requirements (12 credits)
- All theatre major core requirements (8 credits)
- All theatre major non-concentration requirements (25 credits)
- Directing (i.e., THEA 355) must be taken to fulfill the major's "performance requirement" (3 credits)

Theatre Education and Teaching Artistry Concentration

Students who complete the theatre education and teaching artistry concentration (p. 435) are able to apply for a PreK-12 teaching licensure in theatre. Please see that page for requirements.

Code	Title	Credits
Required Education and Psychology Courses		
PSYC 160	Life Span Human Development [C5SD] ¹	3
EDUC 200	Foundations of Education in the United States ¹	3
LED 420	Content Area Literacy, K-12	2
THEA 325	Professional Issues in Education ²	3
THEA 480	Student Teaching (must be taken for a total of 12 credits) ²	12
Total Credits		23

¹ Minimum grade of "C" required in EDUC 200, PSYC 160 and WRTC 103, and are prerequisites for teacher education.

² Students must be accepted to the teacher education program through the College of Education (p. 18) to enroll in THEA 325 and THEA 480.

Pre-Law

Coordinator: Melinda Adams

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Email: adams2mj@jmu.edu

Program Description

Students who plan to apply to law school may select their major from a wide range of fields, depending upon their interests. The scope of the law is broad and offers room for individuals of varied educational and intellectual backgrounds. Students should choose courses that provide them with broad informational and cultural preparation and develop their reasoning abilities.

Especially valuable to a pre-law program are courses in:

- Communication, including composition, language and speech, which enable students to express themselves well.
- The humanities and social sciences, which help students to appreciate and perform effectively in their culture and society.
- Logic, mathematics and the natural sciences, which develop skills of fact discrimination, analysis and synthesis.
- Accounting and economics.

Also important to the pre-law program are student organizations like Madison Legal Scholars, Moot Court & Mock Trial Club, and Phi Alpha Delta, which organize law-related activities.

Pre-Theological Studies

Coordinator: Jennifer Aycock

Phone: (540) 568-6394

Email: aycockjl@jmu.edu

Program Description

Students join the pre-theology program to prepare for careers in the service of religious communities. Usually, these careers—such as ministry, religious education, counseling and chaplaincy—require specialized training and graduate degrees from theological seminaries, rabbinical schools or university divinity schools. The pre-theology advisor guides students toward courses and experiential learning at JMU that will provide an academic foundation for graduate studies in theology as well as an opportunity to develop their practical skills or vocational gifts. Students in this program may major in any field, although the Association for Theological Schools recommends substantial pre-professional training in philosophy and religion.

In the Department of Philosophy and Religion (p. 92), pre-theology students will find a robust offering of courses ranging from the interpretation of religious texts to theology and ethics, and from the histories of particular religious traditions to cross-cultural topics in global religion. Class assignments develop students' skills in critical thinking, ethical reasoning, scriptural hermeneutics, and written and oral communication, all of which theological schools highly value. The pre-theology advisor may direct students to relevant courses in foreign languages, social work, non-profit studies, humanitarian affairs, education or justice studies to complement their academic training in religion. Qualified students are encouraged to undertake independent studies or write honors theses in order to explore in greater depth the theological topics most important to them.

The department also offers students the opportunity to receive academic credit for practical supervised field work with social agencies and religious organizations in order to help students explore the particular forms of religious service they would like to pursue. Making connections with faith leaders within their own traditions helps students access larger networks of tradition-specific resources and contacts as they reflect on their goals and decide where to apply for further study in theology. We encourage students to visit various theological schools during the application season and to meet with representatives that the department invites to campus to discuss theology programs with students.

Studies Abroad

Director: Katie Sensabaugh
Email: sensabkb@jmu.edu

Associate Director: Angeline McMullin
Email: mcmullja@jmu.edu

Study Abroad Website (<https://www.jmu.edu/global/abroad/index.shtml>)

Recognizing its responsibility to provide international experiences to JMU students, the university sponsors semesters abroad in Antwerp (p. 594), Florence (p. 595), London (p. 595), Spain (p. 595) and Scotland, as well as numerous short-term programs that take place during JMU breaks (winter, spring break, summer). JMU students experience life in a different culture and gain direct access to the historical, artistic and sociopolitical traditions of the environment while pursuing an approved course of study toward their bachelor's degree.

The programs are open to all students in good academic standing. Applicants should have a minimum 2.0 grade point average.

Semester in Scotland

Students spend four weeks in Edinburgh and four weeks in St. Andrews as part of the Semester in Scotland program. This study abroad opportunity is open to all majors and is only offered during the summer. Classes are taught by JMU professors and faculty members from the University of Edinburgh and University of St. Andrews, two of the best and most prestigious universities in the United Kingdom. Participating students take 13 credits in General Education courses. Course offerings vary each summer but will always cover courses in the Arts and Humanities, The Natural World, American and Global Perspectives and Sociocultural and Wellness Area of JMU's General Education curriculum.

Short-Term Programs

Short-term programs are faculty-led and designed to blend coursework with guest lectures, excursions, and unique cultural experiences. Smaller group sizes foster lasting relationships between students and allow for authentic interactions with locals and fellow travelers. Short-term programs vary in length, location, and area of study to allow for seamless integration into even the most structured academic schedules. With programs in over 50 countries, a short-term program is an excellent way for students to practice a foreign language, make progress toward degree completion, or enroll in general education courses and electives in an immersive format.

Program Cost

The programs' costs differ, but all programs include tuition, housing, course-related travel, instructional events, such as theater, concerts, historical tours and guest lectures, and some social activities. Most programs include a basic food allowance as specified on the website.

All programs exclude the costs of students' travel to and from the program site. Loans and scholarships may be applied, and semester abroad scholarships and assistantships are available. Students enrolled in JMU-sponsored study abroad programs will be covered by a medical insurance policy for the period of time that they are participating in the JMU program.

Interested students can get information about study abroad programs from the Center for Global Engagement.

International Exchange Programs

The university is committed to increasing the opportunities for students' global awareness. Numerous exchange programs with schools and universities abroad are available. JMU's exchange programs are based at prestigious universities around the world, such as Flinders University in Australia; University of Leicester in England; Hiroshima University, Nanzan University and Ritsumeikan University in Japan; Yonsei University in South Korea; the Universidad de Salamanca in Spain; University of Versailles in France; and Malmo University in Sweden. Language requirements and curricula vary.

Exchanges are available to all majors and minors. Students should consult the Center for Global Engagement for application deadlines and further information.

Study at Oxford, Cambridge or St. Andrews

The university also offers the opportunity to study abroad for a semester or a full academic year at three of Great Britain's oldest and most respected universities: the University of Cambridge and the University of Oxford in England and the University of St. Andrews in Scotland. Participants will take part in tutorials across a wide range of topics and live with and share all of the benefits of regular university students. Study at Oxford, Cambridge or St. Andrews is open to all JMU students possessing a 3.3 grade-point average; students do not need to be honors students to participate in this program.

External Abroad Programs

Students may apply to participate in other approved study abroad programs. The university will accept credits earned abroad at approved institutions in accordance with its policy of accepting transfer credits. Approval of proposed study programs must be obtained from the Center for Global Engagement and the head of the academic unit in which the transfer credit will be awarded.

Semester in Antwerp

The Semester in Antwerp Program is an experience-based program, blending classroom theory with field trips to actual European businesses and sites of historic and cultural interest. The totality of the experience will result in a deep understanding of European business practices.

While in Belgium, students attend classes onsite at the University of Antwerp with business faculty who specialize in European Integration and the effects of economic and monetary union. This program is open only to business majors who have been accepted into COB 300 and is offered in fall, spring, and summer. COB 300 is an integrated learning experience consisting of four courses, taken concurrently, which introduces the fundamental conceptual tools of management, finance, operations and marketing in such a way as to establish their mutual relevance and interdependence. No foreign language is needed to participate in this program.

The following courses are offered during the Study Abroad – Antwerp semester, but not all courses may be offered each semester. Students should consult with the Study Abroad advisor for exact courses offered each semester.

Code	Title	Credits
Courses		
COB 300A	Integrative Business: Management	3

COB 300B	Integrative Business: Finance	3
COB 300C	Integrative Business: Operations	3
COB 300D	Integrative Business: Marketing	3
COB 301	European Integration, Culture and History	3

Semester in Florence

The Semester in Florence program is open to all majors and offers students the opportunity to discover Florence, Tuscany, Italy, and broader Europe. The curriculum promotes engaged, hands-on learning in the city that travelers routinely designate "the best city in the world." Courses focus on Italian gastronomy and oenology, the Renaissance tradition, and contemporary European education, business and politics.

In addition to courses on Italian culture, hospitality management and art history, among others, this program gives students the opportunity to complete six credits toward the General Education requirement for arts and humanities and is offered during the fall, spring, and summer. Interested students can also earn substantial credit toward a minor in Modern European Studies.

The following courses are offered during the Study Abroad – Florence semester, but not all courses may be offered each semester. Students should consult with the Study Abroad advisor for exact courses offered each semester.

Code	Title	Credits
Courses		
ARTH 313	Masterpieces of Italian Renaissance Art [C2VPA]	3
ENG 302	Special Topics in Literature and Language	3
HUM 200	Great Works [C2L]	3
HM 252	Culinary Arts and Menu Management	3
HM 361	Italian Culture and Wine	3
HM 362	Italian Gastronomy	3
ITAL 490	Independent Study in Italian	1-4
MKTG 380	Principles of Marketing	3
POSC 371	Topics in Comparative Politics (when topic is appropriate)	3
or SCOM 395	Study Abroad Seminar	
UNST 390	Special Studies in University Studies	1
WLC 447	Special Topics: World Cultures in Translation	3

Semester in London

During the Semester in London, students study both classic and modern literature and then examine those great works in context, exploring how history influenced the present in Europe's most cosmopolitan city. Internships are also available as part of the program. With more than 200 possible intern sites in London, positions are widely varied and open to all majors. This unique opportunity allows students to build essential life and career skills while experiencing the British lifestyle first-hand. This program is offered during the fall, spring and summer. Interested students can also earn substantial credit toward a minor in British Communication and Media.

The following courses are offered during the Study Abroad – London semester, but not all courses may be offered each semester. Students should consult with the Study Abroad advisor for exact courses offered each semester.

Code	Title	Credits
ARTH 316	Experiencing Art in London [C2VPA]	3
ENG 412	Special Topic Seminar (The London Theatre)	3
or HUM 200	Great Works [C2L]	
or THEA 449	London Theatre	
IDLS 486	Internship and Field Experience	3 or 6
SCOM 359	Communication, Diversity and British Popular Culture	3
SCOM 360	British Media and Society	3
or SMAD 360	British Media and Society	
or WRTC 360	British Media and Society	
or HUM 251	Modern Perspectives [C2HQC]	
SCOM 351	Visual Rhetoric	3
or SMAD 321	The Media Arts: Culture by Design in London	
or WRTC 351	Visual Rhetoric	
or ARTH 389	Topics in Art History	
SMAD 466	British Film Adaptations	3
or ENG 463	British Film Adaptations	
or SCOM 395	Study Abroad Seminar	
POSC 472	Media and Politics	3
or SCOM 473	British Media and Politics	
or SMAD 473	British Media and Politics	
WRTC 103	Rhetorical Reading and Writing [C1W]	3
UNST 390	Special Studies in University Studies	1

Semester in Spain

Based in Salamanca, the Semester in Spain program gives students the opportunity to experience the cultural richness of Spain, as their exposure to theaters, concerts, cinemas and other performing arts groups will unveil a new aspect of learning. During spring and summer terms, all courses are taught in Spanish; thus, all applicants must have completed the intermediate level of Spanish (SPAN 232) prior to departure. In addition, during the fall term, students enroll in a 3-credit intensive language class at the level appropriate to their needs during the first two weeks in Salamanca. This requirement permits time for students to adapt to Spanish language and culture before entering regular semester courses.

Spring and summer semesters offer a language immersion opportunity with courses taught in Spanish. For more information on course offerings, see the Academics page. This program also offers an additional 4-week Internship in Valencia, Spain for spring and summer participants. Students will have the opportunity to intern in several industries such as business, education, communication, law, engineering, and non-profit. Fall semester is a special term designed for freshmen and sophomores looking to complete general education requirements. Students will select four of the six general education courses offered, plus an elementary or intermediate Spanish course, and UNST 390. All courses are taught in English.

The following courses are offered during the Study Abroad – Spain semester, but not all courses may be offered each semester. Students should consult with the Study Abroad advisor for exact courses offered each semester.

Spanish Language and Culture Options (Spring, Summer)

Code	Title	Credits
ARTH 314	Masterpieces of Spanish Art [C2VPA]	3
SPAN 300	Language for Living	3
SPAN 301	Pop Culture through Language	3
SPAN 307	History of Spanish Culture and Society	3
SPAN 308	Latin American Cultures	3
SPAN 320	Culture and Communication in Context	3
SPAN 322	Language Landscapes	3
SPAN 330	Business Spanish	3
SPAN 335	A Journey through Literature	3
SPAN 365	Spanish for Medical Professionals I	3
SPAN 435	Translation Strategies	3
SPAN 465	Spanish-Language Cinema	3
SPAN 476	Culture and Medicine in Latin America	3
SPAN 486	Business and Society in Spain	3
SPAN 490	Independent Study in Spanish	3
SPAN 493	Business Spanish Internship	3
SPAN 496	Spanish Internship Abroad	3

Spanish Language and Culture Options (Fall)

Code	Title	Credits
SPAN 101	Elementary Spanish I	3
SPAN 102	Elementary Spanish II	3
SPAN 231	Intermediate Spanish I	3
SPAN 232	Intermediate Spanish II	3
SPAN 496	Spanish Internship Abroad	3

General Education Options (Fall)

Code	Title	Credits
ART 200	Art Today: Themes in Contemporary Art [C2VPA]	3
GEOL 102	Environment: Earth [C3NS]	3
HUM 200	Great Works [C2L]	3
POSC 200	Global Politics [C4GE]	3
SCOM 122	Fundamental Human Communication: Individual Presentations [C1HC]	3
SOCI 140	Microsociology: The Individual in Society [C5SD]	3
UNST 390	Special Studies in University Studies	1
WRTC 103	Rhetorical Reading and Writing [C1W]	3

Air Force Reserve Officers Training Corps

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Mission Statement

The Air Force ROTC program is designed to recruit, educate and commission officer candidates through college campus programs based on Air Force requirements. Units are located at 145 college and university campuses throughout the United States and Puerto Rico. Students from schools near Air Force ROTC host institutions can attend classes through separate crosstown enrollment programs or consortium agreements.

Goals

The Air Force Reserve Officers' Training Corps (AFROTC) program is offered and hosted at the University of Virginia and maintains a crosstown agreement with James Madison University. JMU students will take AFROTC classes at the University of Virginia for JMU credit. AFROTC provides a program of leadership development that prepares cadets for service as commissioned officers in the United States Air and Space Forces. After graduation, they assume active duty positions that are awarded prior to completion of the AFROTC program. To accomplish this, the Department of Air Science recruits, selects, retains and commissions officer candidates as second lieutenants in the United States Air Force and United States Space Force. AFROTC:

- Provides ethics and values education, stressing the Air Force core values of integrity first, service before self and excellence in all we do.
- Prepares cadets to be commissioned as second lieutenants through a curriculum that develops leadership and management skills and provides opportunities to use these skills in a variety of practical applications.
- Provides an understanding of how to lead and influence small organizations, with a strong emphasis on personal integrity, honor and individual responsibility. The techniques of effective leadership and quality management are stressed to achieve mission accomplishment.
- Enhances the cadets' understanding of the role of the military in society and provides an in-depth orientation to history and tradition of the Air Force and how the Air Force and Space Force serves the nation. This goal is reinforced by summer hands-on leadership training and career orientation opportunities at Air Force bases around the country.
- Provides programs and experiences that increase self-confidence, self-discipline, accountability, physical stamina, poise and other traits

essential to the development of a leader of character who is prepared to serve the nation in and out of uniform.

The curriculum and leader development programs of the Department of Air Science are mentally and physically challenging. Cadets learn basic military skills and participate in a physical conditioning program starting in the first year. The program develops cadets progressively from followership to leadership experiences and culminates with the fourth-year class cadets planning, organizing and administering all leadership training for the cadet wing. On-campus instruction is conducted both in the classroom by Air and Space Force officers and in the field environment by both officer and cadet leaders. A demanding summer field training encampment prior to the junior year prepares the cadets for acceptance into the commissioning track.

Air Force ROTC offers four-and three-year tracks. All cadets must complete four credits of AFROTC academics prior to a summer encampment. The three-year track cadets must complete both first and second year academic courses during their first year in ROTC. Students who have Junior ROTC, other-service ROTC, national guard, reserve or active duty military experience may request credit and advanced placement.

Unless the student earns an AFROTC scholarship, there is no service obligation inside the first two years of the four-year program. However, all students who enter into the Professional Officer Course (the last two years) enter into a contractual obligation with the Air Force or Space Force to serve on active duty upon commissioning.

After graduation and commissioning as second lieutenants in the Air Force or Space Force, graduates serve in any number of career fields for a minimum of a four-year active duty service commitment. Interested and qualified students may compete to become Air Force pilots, combat systems officers (CSO), air battle managers (ABM) or remotely piloted aircraft pilots (RPA). Successful pilot candidates serve a 10-year active duty service commitment, and CSO, ABM or RPA candidates incur a six-year active duty service commitment after completion of career field training. All service commitments begin after completion of career field training.

Co-curricular Activities and Organizations

- Arnold Air Society
- Drill Team
- i5 Squadron

AFROTC Scholarships

Merit-based financial scholarships may be offered to highly academically competitive and qualified students already enrolled in the program. Awardees may be offered an AFROTC scholarship for full or partial college tuition, incidental fees and textbook allowances and a monthly subsistence allowance of \$300-\$500 depending on academic year. Scholarship students incur a military obligation after their first year.

Special Admission and Retention Requirements

The Professional Officer Course is normally taken during the junior and senior years. Qualified students pursuing a commission as a second lieutenant are contracted and paid a subsistence allowance of \$300-\$500 per month. Cadets must maintain at least a 2.5 GPA, meet DoD medical

fitness standards and meet Air Force physical fitness and weight control standards.

Program Requirements

The Leadership Laboratory (LLAB) course taken during the first year is a weekly laboratory that touches on the topics of Air Force customs and courtesies, health and physical fitness, and drill and ceremonies. The second year LLAB course provides students with the opportunity to demonstrate fundamental management skills and prepares them for Field Training. The third year LLAB course provides the opportunity to develop fundamental management skills while planning and conducting cadet activities. Finally, the fourth year LLAB course provides the opportunity to use leadership skills in planning and conducting cadet activities. It prepares students for commissioning and entry into the active-duty Air and Space Forces.

The curriculum is divided into two phases:

Phase One: General Military Courses

Code	Title	Credits
AIRS 100	Leadership Laboratory (every semester)	1
AIRS 110	Air Force Heritage and Values I (fall, first year)	1
AIRS 120	Air Force Heritage and Values II (spring, first year)	1
AIRS 210	Team and Leadership Fundamentals I (fall, second year)	1
AIRS 220	Team and Leadership Fundamentals II (spring, second year)	1

Phase Two: Professional Officer Courses

Code	Title	Credits
AIRS 100	Leadership Laboratory (every semester)	1
AIRS 310	Leading People and Effective Communication I (fall, third year)	3
AIRS 320	Leading People and Effective Communication II (spring, third year)	3
AIRS 410	National Security Affairs/Preparation for Active Duty I (fall, fourth year)	3
AIRS 420	National Security Affairs/Preparation for Active Duty II (spring, fourth year)	3

Upon completion of all academic courses, cadets are eligible to earn the Leadership in Military Affairs Minor (p. 520).

Military Science

Program Description

Department of Military Science Website (<https://www.jmu.edu/rotc/index.shtml>)

This non-degree military science curriculum is divided into two phases: basic military science and advanced military science.

The basic course is open to all JMU students. There is no military obligation incurred for taking 100- and 200-level military science courses. This curriculum is designed to help students in the near-term as leaders on campus. The classes will also help students be more effective leaders in the long-term, whether they serve in the military or as leaders in civilian life. Topics addressed include problem solving, critical thinking, problem-solving methods, leadership theory, followership, group cohesion, goal setting, feedback mechanisms, physical fitness and land navigation.

Lessons are taught in a seminar format, emphasizing student discussions and practical exercises. Courses are open to all students with no prerequisites and no military obligation. Students must enroll in both the lab and the classroom instruction.

Placement credit for the basic course may be awarded through multiple programs including: prior military service, basic training, or successful completion of the ROTC Basic Camp training course at Ft. Knox, Ky., during the summer after the sophomore year, or before entering a 2 year graduate program.

The advanced course focuses on instruction and case studies which build leadership competencies and military skills in preparation for future responsibilities as Army officers and successful completion of the ROTC Advanced Camp training course at Ft. Knox, KY., during the summer between the junior and senior year of college. Instruction includes the principles of war, decision-making processes, planning models and risk assessment. Advanced leadership instruction focuses on motivational theory, the role and actions of leaders, and organizational communications. Courses are only open to advanced course-contracted cadets with prerequisites and an 8 year minimum military obligation is incurred upon graduation and commissioning.

Students may be interested in the military leadership minor (p. 532), which requires one additional course in military history.

Special Admission and Retention Requirements

Advanced military science courses are normally taken during the junior and senior years, or during graduate school. Qualified students pursuing a commission as a second lieutenant are contracted and paid a subsistence allowance of \$420 per month for up to 10 months during the school year. Prior to commissioning, each cadet must successfully complete the four-week ROTC Cadet Summer Training (CST) at Ft. Knox, Kentucky. Cadets must maintain at least a 2.8 GPA, meet Department of War medical fitness standards, as well as pass the Army Fitness Test and height/weight control standards.

Program Requirements

Code	Title	Credits
Basic Military Science		
MSCI 110	Introduction to Leadership and the Army	2

MSCI 115	Basic Leadership Laboratory I	2
MSCI 120	Leadership Development Fundamentals	2
MSCI 125	Basic Leadership Laboratory II	2
MSCI 210	Leadership Styles: Theory and Application	2
MSCI 215	Intermediate Leadership Laboratory I	2
MSCI 220	Developing Leadership Skills	2
MSCI 225	Intermediate Leadership Laboratory II	2
Subtotal		16

Advanced Military Science

MSCI 310	Leading Small Organizations	3
MSCI 315	Advanced Leadership Laboratory I	3
MSCI 320	Developing Advanced Leader Skills	3
MSCI 325	Advanced Leadership Laboratory II	3
MSCI 410	Seminar Command Management	3
MSCI 415	Senior Leadership Laboratory I	3
MSCI 420	Seminar on Officership Transition	3
MSCI 425	Senior Leadership Laboratory II	3
Subtotal		24

Total Credits		40
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Accounting (ACTG)

ACTG 244 Accounting for Non-Business Majors

Credits 3 Grading Basis Graded

For non-business majors only. This course introduces basic business and accounting topics such as revenue, investments, expenditures, liabilities, credit, cash management, and taxation. Heavy emphasis is placed on the measurement of operating performance, and interpretation and use of accounting data for organizational decision-making.

Prerequisites: Non-business majors only. Junior standing or higher.

ACTG 302 Introduction to the Profession: Role of Accountants

Credits 1 Grading Basis Graded

This class relies heavily on discussions with practicing accountants in public accounting, industry and government. Topics include career preparation and the role of accountants in business and capital markets. Written assignments are used to enhance communication skills.

Prerequisites: Completion of both COB 241 and COB 242 with minimum grades of "B" in each and COB 300. Corequisites: COB 300.

ACTG 305 Spreadsheet Skills and Data Analysis for Accounting Information

Credits 3 Grading Basis Graded

This course provides students the opportunity to learn and become proficient with spreadsheet and analytics skills expected by today's employers utilizing Microsoft Excel and Power BI analytics software. (May be taken with or after COB 242.)

Prerequisites: Accounting major. Corequisites: COB 242.

ACTG 310 Data Analytics for Accountants

Credits 3 Grading Basis Graded

This course is an introduction to data analytics and the use of data analytics in various accounting settings. Topics include the data analytics mindset, data preparation, data quality, statistical analysis, and data visualization and reporting.

Prerequisites: Accounting major, ACTG 305, minimum grade of "B-" in both COB 241 and COB 242; or permission of the department head.

ACTG 313 Accounting Information Systems

Credits 3 Grading Basis Graded

This course covers the use of accounting systems for the collection, organization, analysis and reporting of accounting data. Topics include: internal controls, documentation of accounting systems, transaction processing cycles, auditing information technology, computer and information systems security, and integration of business functions in the accounting process.

Prerequisites: Accounting majors only. COB 300, ACTG 305, and a minimum grade of "B-" in both COB 241 and COB 242; or permission of the department head.

ACTG 343 Corporate Financial Reporting I

Credits 3 Grading Basis Graded

This course provides a theoretical framework to explain and critically evaluate financial reporting by businesses. In addition to studying the authoritative standards for preparing financial statements, students develop the ability to read, use, analyze and interpret financial statements. Students gain an understanding that managers can shape the financial information communicated to investors and creditors.

Prerequisites: Accounting majors only. COB 300 and a minimum grade of "B-" in both COB 241 and COB 242; or permission of the department head.

ACTG 344 Corporate Financial Reporting II

Credits 3 Grading Basis Graded

This course continues the development of a theoretical framework to read, use, analyze, interpret, explain and critically evaluate financial reporting by businesses. Selected topics include financial instruments, corporate obligations, stockholders' equity and other corporate reporting issues.

Prerequisites: ACTG 343 with a minimum grade of "C-."

ACTG 377 Federal Income Tax Accounting

Credits 3 Grading Basis Graded

This course is designed to introduce students to the federal income tax system, including individual and business entity taxation. Topics include income, exclusions, deductions and property transactions. The course also facilitates development of research skills.

Prerequisites: Accounting majors only. COB 300 and a minimum grade of "B-" in both COB 241 and COB 242; or permission of the department head.

ACTG 410 Auditing and Assurance

Credits 3 Grading Basis Graded

This course provides a study of techniques available for gathering, summarizing, analyzing and interpreting the data presented in financial statements and procedures used in verifying the fairness of the information. The course will also emphasize ethical and legal aspects and considerations.

Prerequisites: ACTG 313 and ACTG 344. Corequisites: ACTG 344.

ACTG 413 Cybersecurity and Accounting Information Protection Strategies

Credits 3 Grading Basis Graded

This course will examine application of knowledge of internal controls over financial reporting to the operational area of cybersecurity. Topics include understanding best practices around IT governance and controls, applying knowledge of internal controls toward preventing cyberattacks of corporate information within ERP systems, evaluating the role of third-party service organizations and their control reports (SOC), and analyzing accounting disclosure strategies surrounding information protection controls and cybersecurity incidents.

Prerequisites: ACTG 313.

ACTG 475 Strategic Cost Accounting and Analytics

Credits 3 Grading Basis Graded

This course involves the study of cost accounting concepts and information used by business organizations to make strategic, organizational and operational decisions. Topics include the role of planning and control in attaining organizational goals and objectives; the relationship among cost structure, cost behavior and operating income; traditional and activity-based costing approaches to product and service costing; differential analysis in decision making; and ethical issues for accountants. Students will also engage in activities designed to enhance technological and communication skills.

Prerequisites: ACTG 343.

ACTG 490 Special Studies in Accounting

Credits 1-6 Grading Basis Graded

Designed to give capable students in accounting an opportunity to do independent study under faculty supervision. Admission only by recommendation of the instructor and permission of the director.

ACTG 491 Advanced Data Analytics and Emerging Technologies in Accounting and Forensics

Credits 3 Grading Basis Graded

This course focuses on advanced data analytics and skills focusing on data wrangling, advanced analysis and advanced visualizations. In addition, this course focuses on emerging technologies used in accounting and the business world as well as the impact of biases on decision making.

Prerequisites: ACTG 310.

ACTG 499A Honors

Credits 1 Grading Basis Graded

ACTG 499B Honors Thesis

Credits 3 Grading Basis Graded

ACTG 499C Honors

Credits 2 Grading Basis Graded

African, African American and Diaspora Studies (AAAD)

AAAD 200 Introduction to African, African American and Diaspora Studies [C4GE]

Credits 3 Grading Basis Graded

An introductory survey of basic theoretical concepts to analyze the Black experience, with special focus on the general historical process common to Africa and the African Diaspora. May be used for general education credit.

AAAD 300 Special Topics in African, African American and Diaspora Studies

Credits 3 Grading Basis Graded

Examination of selected topics of importance to the field of African, African Americans and Diaspora studies.

AAAD 400 Advanced Topics in African, African American and Diaspora Studies

Credits 3 Grading Basis Graded

Selected topics in African, African American and Diaspora studies are studied in depth. This course may be repeated when content changes.

AAAD 401 Internship in African, African American and Diaspora Studies

Credits 1-6 Grading Basis Graded

This internship course provides the student with the opportunity to apply knowledge learned in the classroom in a practical/real-world setting(s). It prepares students for working independently in the field. Any internship experience must be approved by the internship coordinator in advance, and details of supervision and evaluation should be spelled out in advance by the supervising faculty member. If the internship is through an academic unit, it must be approved for credit by the African, African American and Diaspora Studies internship coordinator in advance of the experience.

AAAD 402 Advanced Projects in African, African American and Diaspora Studies

Credits 1-3 Grading Basis Graded

This course provides opportunities for students to work closely with AAAD-affiliated faculty on research projects, fieldwork opportunities or community engagement projects. This course may be repeated for up to 6.00 credits.

AAAD 488 African American Art

Cross-listed with: ARTH 488

Credits 3 Grading Basis Graded

This course examines visual arts produced by people of African descent in the United States from the colonial period until the present. Course themes include debates about the relationship between racial identity and artistic production; the complex interchange between African American art and the cultural traditions of Africa and Europe; black artists' engagement with popular representations of African Americans; and the intersection of race with class, gender and sexuality.

Prerequisites: AAAD 200, ARTH 206 or permission of the instructor.

AAAD 489 African, African American and Diaspora Studies Senior Research Experience

Credits 1 Grading Basis Graded

In this research-oriented experience, students design and complete research projects relevant to their interests in African, African American and Diaspora studies, as well as connect their projects to previous course work and experiences within the AAAD studies minor.

Prerequisites: AAAD 200, senior standing and permission of the instructor.

Air Force ROTC (AIRS)

AIRS 100 Leadership Laboratory

Credits 1 Grading Basis Credit / No Credit

This course is a mandatory laboratory in leadership and followership development for AFROTC cadets. As a complement to the air science classes, this laboratory focuses on applying leadership principles and understanding leaders' responsibilities while emphasizing the benefits of practical experience. JMU students will take AFROTC classes at the University of Virginia for JMU credit. Students must be enrolled in Air Force ROTC to take this course.

Corequisites: Any Air Force ROTC class.

AIRS 110 Air Force Heritage and Values I

Credits 1 Grading Basis Graded

This course introduces the United States Air Force and Air Force Reserve Officers Training Corps. Topics include mission and organization of the Air Force, officership and professionalism, military customs and courtesies, Air Force officer opportunities and communication skills. JMU students will take AFROTC classes at the University of Virginia for JMU credit. AFROTC cadets and students interested in joining Air Force ROTC must also register for AIRS 100, Leadership Laboratory.

Corequisites: AIRS 100.

AIRS 120 Air Force Heritage and Values II

Credits 1 Grading Basis Graded

This course introduces the United States Air Force and Air Force Reserve Officers Training Corps. Topics include mission and organization of the Air Force, officership and professionalism, military customs and courtesies, Air Force officer opportunities and communication skills. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

AIRS 210 Team and Leadership Fundamentals I**Credits 1 Grading Basis** Graded

This course is designed to provide a fundamental understanding of both leadership and team building. There are many layers to leadership, including aspects that don't always jump to mind. Such things include listening, understanding yourself, being a good follower, and problem solving efficiently. Students will apply these leadership perspectives when completing team building activities and discussing things such as conflict management. Students should demonstrate basic verbal and written communication skills. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

AIRS 220 Team and Leadership Fundamentals II**Credits 1 Grading Basis** Graded

This course is designed to provide a fundamental understanding of both leadership and team building. There are many layers to leadership, including aspects that don't always jump to mind. Such things include listening, understanding yourself, being a good follower, and problem solving efficiently. Students will apply these leadership perspectives when completing team building activities and discussing things such as conflict management. Students should demonstrate basic verbal and written communication skills. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

AIRS 310 Leading People and Effective Communication I**Credits 3 Grading Basis** Graded

This course studies leadership, management fundamentals and professional knowledge, Air Force personnel and evaluation systems, leadership ethics, and communication skills required of Air Force junior officers. The class examines Air Force leadership and management situations, using case studies as a means of demonstrating and applying the concepts under consideration. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

AIRS 320 Leading People and Effective Communication II**Credits 3 Grading Basis** Graded

This course studies leadership, management fundamentals and professional knowledge, Air Force personnel and evaluation systems, leadership ethics, and communication skills required of Air Force junior officers. The class examines Air Force leadership and management situations, using case studies as a means of demonstrating and applying the concepts under consideration. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

AIRS 410 National Security Affairs/Preparation for Active Duty I**Credits 3 Grading Basis** Graded

This course examines the national security process, regional studies, advanced leadership ethics and Air Force doctrine. Topics include the military as a profession, officership, military justice, civilian control of the military, preparation for active duty and current issues affecting the military profession. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

AIRS 420 National Security Affairs/Preparation for Active Duty II**Credits 3 Grading Basis** Graded

This course examines the national security process, regional studies, advanced leadership ethics and Air Force doctrine. Topics include the military as a profession, officership, military justice, civilian control of the military, preparation for active duty and current issues affecting the military profession. JMU students will take AFROTC classes at the University of Virginia for JMU credit.

Corequisites: AIRS 100.

American Sign Language (ASL)

ASL 101 Introductory American Sign Language I**Credits 3 Grading Basis** Graded

This course provides an introduction to American Sign Language (ASL) and Deaf culture. Students learn basic person-to-person conversational signing. The course emphasizes development of expressive and receptive skills. Students will increase their knowledge of ASL vocabulary and the syntax, semantics and pragmatics of the language. The course begins to explore cultural issues related to the Deaf community.

ASL 102 Introductory American Sign Language II**Credits 3 Grading Basis** Graded

This is the second course in the progress of introducing students to American Sign Language (ASL) and Deaf culture. The course continues with the development of expressive and receptive skills and of ASL vocabulary and the syntax, semantics and pragmatics of the language. The course continues exploration of cultural issues in the Deaf community.

Prerequisites: ASL 101 or CSD 420; or by permission of the instructor.

ASL 231 Intermediate American Sign Language I**Credits 3 Grading Basis** Graded

This course focuses on intermediate level skills in American Sign Language (ASL) and Deaf culture. Students will increase competencies in person-to-person conversational signing, including expressive and receptive skills, vocabulary, syntax, semantics, and pragmatics.

Prerequisites: ASL 102 or CSD 421; or by permission of the instructor.

ASL 232 Intermediate American Sign Language II**Credits 3 Grading Basis** Graded

This course continues training in American Sign Language at the intermediate level, with focus on more complex sentence types, signs and idioms. The course may consider ASL literary forms such as poetry, theater and story telling, as well as Deaf history and other related topics.

Prerequisites: ASL 231 or equivalent with approval of the department.

American Studies (AMST)

AMST 200 Introduction to American Studies [C2HQC]**Credits 3 Grading Basis** Graded

This interdisciplinary course will highlight the student's role in interrogating the cultural and political function of representations of America in literature, history, philosophy, religion, popular culture, music and art. Students will gain an understanding of why definitions of American identity matter and learn about the contemporary debates that inform the discipline of American Studies today. Questions about the changing role of national studies in the face of globalization are central. This course may be used for general education credit.

AMST 302 Immigrants in American Society**Credits 3 Grading Basis Graded**

Immigrants today come to America from many countries, but because scholarly research had tended to concentrate on only a few of the largest nationality groups, the full range of immigrant experience has yet to be explored and documented. In this cross-disciplinary course, we will give special attention to two groups, Mexicans and Vietnamese, about whom much has been written, as examples of the varying ways that immigrants come to the U.S. and adapt to living here while maintaining important ties to their homelands and cultures. We will also consider the consequences of immigration for both the first and second generations of immigrants as well as for the communities that receive them, including the Central Shenandoah Valley.

Anthropology (ANTH)

ANTH 195 Cultural Anthropology [C4GE]**Credits 3 Grading Basis Graded**

An introduction to the nature of culture and its relationship to language, economics, politics, kinship and other institutions in diverse cultures. The course also provides an overview of the theories, methods and ethical responsibilities involved in the study of cultural systems and ethnographic writing. May be used for general education credit.

ANTH 196 Biological Anthropology [C3NS]**Credits 3 Grading Basis Graded**

An introduction to the origins, evolution and genetic variability of humans and their relationship to nonhuman primates. Examination of the fossil record, the relationship between biology and culture and human genetics are included. Theories and methods used in the study of biological anthropology are also introduced. This course may be used for general education credit.

ANTH 196L Scientific Perspectives in Biological Anthropology [C3L]**Credits 1 Grading Basis Graded**

This course provides a laboratory-based introduction to biological anthropology, covering topics including the origins, evolution and genetic variability of humans and their relationship to nonhuman primates. It is intended to be taken in conjunction with (or after) ANTH 196. May be used for general education credit.
Corequisites: ANTH 196.

ANTH 197 Archaeology**Credits 3 Grading Basis Graded**

An introduction to the goals, methods, and theory of anthropological archaeology. The course examines the variety of techniques archaeologists use to reconstruct the past from material remains. Archaeological ethics and the impact of the past on contemporary society are also considered.

ANTH 201 The Discipline of Anthropology**Credits 1 Grading Basis Graded**

This required course introduces students to the subdisciplines of cultural, biological, linguistic and archaeological anthropology and the logic of their integration within the larger discipline of anthropology. Students will be introduced to current research questions within anthropology and how they are addressed from the perspective of the various subdisciplines. Prerequisites: Major status or permission of the instructor. It is recommended that students have had at least one of the introductory-level ANTH courses (ANTH 195, ANTH 196 or ANTH 197).

ANTH 205 Buried Cities, Ancestral Places: The Rise and Fall of Early Human Societies [C2HQC]**Credits 3 Grading Basis Graded**

This course takes an archaeological and comparative perspective on the origins of human institutions, including art, architecture, religion, centralized political formations and urban life. The development and collapse of early societies in multiple world regions, including Mesopotamia, Egypt, the Indus Valley, Mesoamerica and the Andes will be explored. This course may be used for general education credit.

ANTH 250 Anthropology of the American Southwest**Credits 3 Grading Basis Graded**

This course examines the development of Southwestern societies from early hunter-gatherers to the Native American communities of today. Major issues of anthropological interest, such as the adoption of agriculture, the development of village life, migration and abandonment, the spread of religious "cults," the extent of Mesoamerican influence and the effects of the Spanish conquest are explored.

ANTH 265 Peoples and Cultures of Latin America and the Caribbean**Credits 3 Grading Basis Graded**

Anthropological and historical perspectives on the cultures of Latin America and the Caribbean through such themes as colonialism, nationalism, ethnicity, development, aesthetic traditions, gender, religion and urban and rural resistance movements.

ANTH 280 Peoples and Cultures of Sub-Saharan Africa**Credits 3 Grading Basis Graded**

This is an introductory course emphasizing cultural diversity of sub-Saharan African societies. Basic anthropological concepts are used in analyzing African economics, political systems, marriage patterns and family organization, religious beliefs and the impacts of colonialism and post-colonial development practices.

ANTH 295 People and Cultures of East Asia**Credits 3 Grading Basis Graded**

This introductory course examines the peoples and cultures of the core East Asian countries: China, Japan and Korea. The course is organized around anthropological perspectives on topics such as nationalism, consumption, gender, ethnicity and development but also emphasizes the cultural, social and historical characteristics of various groups in these nations, in addition to important cultural flows within region.

ANTH 300 The Anthropology of Food**Credits 3 Grading Basis Graded**

This course explores anthropological approaches to food production, distribution, preparation and consumption in the contemporary world. Topics include food preferences and taboos, food and the senses, ritual and identity, technological risks, diet and nutrition, cuisine and class and the political economy of food.

Prerequisites: Any lower-level course in anthropology or permission of the instructor.

ANTH 302 The Archaeology of Empires and Rebels**Credits 3 Grading Basis Graded**

This course examines empires and rebels from an archaeological perspective. Students will survey major empires of the past and how they were challenged by acts of resistance, revolts and rebellions. Some of the major empires covered will include the Inca Empire, the Roman Empire, the Qin Empire, the Assyrian Empire and the Aztec Empire. In this discussion-based course, students will review how archaeologists have uncovered state strategies of control as well as resistance to those forms of control by different groups in those empires.

Prerequisites: ANTH 195, ANTH 197 or ANTH 205; or permission of the instructor.

ANTH 303 Andean Archaeology**Credits 3 Grading Basis Graded**

This course provides an overview of Andean archaeology. The time period ranges from the arrival of the first settlers of the Andes to the mighty Inkas. The Andes contain the highest and densest variety of microclimates in the world, ranging from scorching deserts to humid rainforests. A land of contrasts and extremes, it is also the setting for diverse peoples who eked out a living from the land, developed unique political systems and passed down rich symbolic knowledge from one generation to another. The main purpose of the course is to understand these peoples: how they interacted with one another, how they interacted with their environment and how they made society.

ANTH 305 Language and Culture**Cross-listed with: SCOM 305****Credits 3 Grading Basis Graded**

An introduction to linguistic anthropology. Explores the complex relationships between language and culture through topics such as language acquisition and socialization; language, thought and worldview; language and identity; multilingualism; how and why languages change; literacy; and the politics of language use and language ideologies.

ANTH 306 Japanese Society and Culture**Cross-listed with: SOCI 306****Credits 3 Grading Basis Graded**

This introductory course takes a critical and interdisciplinary approach to exploring Japan. We will apply sociological, anthropological, and demographic perspectives to comparatively understand, analyze, and discuss Japanese society and culture. Students will read and discuss issues related to history, socio-demographic change, gender, work, social class, race/ethnicity, family, health care and aging in Japanese society.

ANTH 307 Climate, Culture and Change**Credits 3 Grading Basis Graded**

This course explores the implications of climate change for human cultures. The overall perspective will be historical, but the relevance of past experiences to contemporary circumstances will be under constant discussion. A series of case studies drawn from archaeological and historical records will attest to a range of possible human responses, including examples that involve both naturally- and culturally-induced change. Students will be introduced to the fundamentals of paleoclimatology, such as basic processes that influence climate change, as well as the sources of long-term climate records. Under an anthropological perspective, students will also become acquainted with the range of documented human reactions to climate change and their ramifications. Each case study will be addressed in three parts over the course of two class sessions: lecture/presentation, discussion and supplemental analysis.

ANTH 308 Fundamentals of Forensic Anthropology**Credits 3 Grading Basis Graded**

This course serves as an introduction to the field of forensic anthropology through an examination of the methods used by forensic anthropologists to recover human remains, positively identify unidentified human remains, and evaluate trauma and environmental alterations to bone in forensic investigations. Following an overview of forensic anthropology and explorations of past and current definitions of the field, we will discuss topics foundational to forensic anthropology, including the search for forensic scenes, the recovery of the remains in the field, data collection in the morgue, analysis in the forensic anthropology laboratory, the reconstruction of events surrounding the crime scene and the preparation of the final report.

Prerequisites: ANTH 196 or permission of the instructor.

ANTH 311 Archaeology of Virginia**Credits 3 Grading Basis Graded**

A comprehensive survey of human experience in the Old (Old!) Dominion. Archaeological discoveries in Virginia, ranging from evidence of Ice Age pioneers to colonial-era plantations, will be the basis for charting wider cultural developments in eastern North America. Through these cases, basic archaeological principles, methods and theories will be reinforced. Activities and assignments will involve manipulation of original archaeological data using basic quantitative analysis. The course will also feature profiles of prominent Virginia archaeologists and sites, as well as an overview of the discipline as it is practiced in the state today. ANTH 197 is recommended, though not required.

ANTH 315 Human Evolution**Credits 3 Grading Basis Graded**

Discusses the emergence of the hominids as a lineage distinct from other apes. Provides evidence for the evolution of bipedalism, tool use, hunting/gathering, major increases in brain size, language and material culture and the hypotheses that have been developed to explain the emergence of these characteristics.

Prerequisites: ANTH 196, or BIO 140/BIO 140L and BIO 150/BIO 150L, or permission of the instructor.

ANTH 316 Human Evolutionary Psychology**Credits 3 Grading Basis Graded**

An exploration of human behavior from an evolutionary perspective. Emphasis is placed on the critical evaluation of adaptive hypotheses purported to explain fundamental human behaviors such as survival and mating strategies, reproduction and parenting, kinship and cooperation, dominance and aggression, cultural evolution and religion.

Prerequisites: ANTH 196 or permission of the instructor.

ANTH 317 Primate Evolutionary Ecology**Credits 3 Grading Basis Graded**

This course explores the interface between an organism and its environment, illustrated with examples from the primates. Behaviors related to feeding, moving, grouping and socializing are considered from an evolutionary perspective. Topics to be discussed include feeding ecology, functional anatomy, the ecology of primate social systems, ranging behaviors, community ecology and the role humans play in shaping primate communities.

Prerequisites: ANTH 196 or permission of the instructor.

ANTH 318 The Evolution of Primate Sexuality and Reproduction**Credits 3 Grading Basis Graded**

A survey of non-human primate sexuality from an evolutionary perspective. Emphasis is placed on the diversity of behavioral, anatomical, and physiological aspects of mating and reproduction across the order Primates. Where appropriate, comparisons with human sexuality are made.

Prerequisites: ANTH 196 or permission of the instructor.

ANTH 319 Human Osteology**Credits 3 Grading Basis Graded**

An analysis of the individual bones and teeth that comprise the human skeleton. Emphasis is placed on learning individual bones and teeth as well as the numerous osteological and dental landmarks that characterize them. Applied topics such as bone growth and the analysis of age, sex, stature, pathology and geographic ancestry will also be addressed.

Prerequisites: ANTH 196 or permission of the instructor.

ANTH 320 Plagues and Pandemics**Credits 3 Grading Basis** Graded

Humans have a long history of suffering from and dealing with infectious diseases. This course explores some of the most important infectious diseases in human history, such as COVID-19, influenza, tuberculosis, smallpox, malaria and bubonic plague from an anthropological perspective. In addition to exploring the biological and cultural effects that infectious diseases have had on humans, the course focuses on how human behavior has historically affected the emergence, spread and control of disease. The course emphasizes how the study of disease in the past may benefit people today and in the future.

Prerequisites: ANTH 196 or permission of the instructor.

ANTH 321 Anthropology and Film**Credits 3 Grading Basis** Graded

This course provides an examination of current issues in the study of ethnographic film. The course studies the changing role, practice and methods of film in anthropological research and scholarship. Topics include methods and genres of ethnographic film composition; the scientific status of filmed data; cross-cultural visual literacies; narrative form as social scientific inquiry; film and sensory data; sound, subtitles and translations; "indigenous media;" and audience reception.

Prerequisites: ANTH 195 or permission of the instructor.

ANTH 323 Anthropology and Photography**Credits 3 Grading Basis** Graded

This course explores the anthropological use of photographic data for the description, analysis, communication and interpretation of human behavior. Topics include phenomenological, cross-cultural and historical understanding of still photography; the social life of photographs; visual cultural production and the consumption of photographs; and still photography after colonialism, globalization, and postmodernity.

ANTH 324 Mindfulness and Anthropology**Credits 3 Grading Basis** Graded

Mindfulness provides ways of attending to the world in its present moment in a non-judgmental manner. Students study and practice contemporary mindfulness practices, while applying these techniques to the holistic collection, analysis and presentation of ethnographic data. The course considers recent ethnographic practice and theory in the context of mindfulness, including the anthropology of poetic expression, the senses and embodied experience.

Prerequisites: ANTH 195 or permission of the instructor.

ANTH 327 Ancient North American Civilizations**Credits 3 Grading Basis** Graded

This course studies the emergence of Native American societies prior to historic contact and emphasizes prehistoric developments in the eastern United States.

ANTH 331 Historical Archaeology**Cross-listed with:** HIST 331**Credits 3 Grading Basis** Graded

The course introduces students to the purposes, subject matter, methodology and historical background of the discipline of historical archaeology. Building on research issues and methodologies of anthropological archaeology and history, the multidisciplinary aspects of this field are introduced through field trips, projects, guest lectures, readings and classroom presentations.

ANTH 352 Birth, Death, Sex: Exploring Demography**Cross-listed with:** SOCI 352**Credits 3 Grading Basis** Graded

This course examines fertility (birth) and mortality (death) and their biological and social determinants in cross-cultural and evolutionary/historical frameworks. It focuses on the exploration of the dynamic between the material constraints on and symbolic significance of reproduction, sexuality and death within a cultural context, and the critical examination of population growth as a global "problem," as well as basic demographic methods.

ANTH 360 Medical Anthropology**Credits 3 Grading Basis** Graded

This course takes an anthropological approach to the study of health, illness, and healing; how do different cultural systems and social institutions influence the experience and interpretation of different bodily states? Material covers critical analyses of Western medicine and ethnomedicine in both specific cultural settings and their global circulation. Topics include disease epidemics, illness narratives, public health, suffering, pharmaceuticals, disability and health care systems.

Prerequisites: ANTH 195 or permission of the instructor.

ANTH 364 U.S./Latin American Borders**Credits 3 Grading Basis** Graded

This course examines the experiences of Latin American migrants to the United States. It stresses the cultural expression of those experiences, globalization and its effects on local communities in Latin America, the U.S. responses to migration and migrants.

ANTH 366 Anthropology of War**Credits 3 Grading Basis** Graded

This course examines the causes, conduct, and consequences of warfare in non-state societies using both ethnographic and archaeological data. Case studies drawn from throughout the world are used to examine topics such as the co-evolution of war and society, the impact of colonialism on native warfare, the process of making peace and claims about the biological "inevitability" of war.

ANTH 368 Contemporary American Culture**Credits 3 Grading Basis** Graded

This course analyzes contemporary American society in relation to popular cultural formations and representations. Cultural expressions found in music, literature, theatre, film, television, cyberspace and sports will be examined with respect to the values, sentiments, identity constructions and lived experiences of differentially situated social actors.

ANTH 370 Topics in the Anthropology of Gender**Credits 3 Grading Basis** Graded

This course examines the many ways in which gender is constructed and negotiated in different historical and social contexts. Topics will vary with the instructor to include both cultural and biocultural perspectives.

ANTH 373 Anthropology and the Environment**Credits 3 Grading Basis** Graded

This course examines how anthropologists engage with the environment and intervene in urgent matters of environmental and social justice, focusing on the theoretical and methodological tools used to move beyond nature/culture divides and conceptualize elements of the changing environment. The course explores how these approaches can be productively applied to understand and address contemporary environmental problems.

Prerequisites: ANTH 195 or permission of the instructor.

ANTH 375 Anthropological Theory**Credits 3 Grading Basis Graded**

This course introduces the major social theories and debates that inform anthropological analysis. Special emphasis is given to their analytical or explanatory power for understanding human behavior and our cultural and social worlds, the contexts in which they were produced, and their contributions to ongoing debates. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: ANTH 195 and junior standing.

ANTH 376 Anthropology of Reproduction**Cross-listed with: WGSS 376****Credits 3 Grading Basis Graded**

This course provides students with a critical and cross-cultural perspective on human reproduction. Examining how individuals draw on social and symbolic resources to sort out complicated private decisions, we will discuss how reproductive experiences are embedded in local, national and transnational politics. Topics covered may include: cross-cultural perspectives on childbearing and childlessness, kinship, and the globalization of new reproductive technologies such as in vitro fertilization.

Prerequisites: ANTH 195 or permission of the instructor.

ANTH 377 Space/Culture/Power**Credits 3 Grading Basis Graded**

An introduction to social-scientific approaches to space. We will think critically about how people across cultures shape-and are shaped by-the spaces we occupy. Drawing on examples from around the world we examine the intersections of space, culture, and power through such topics as segregation, maps, architecture, prisons, schools, migration politics and more. The course encourages students to think of space-and its intersections with culture and power-in new and sophisticated ways.

Prerequisites: ANTH 195.

ANTH 390 Topics in Cultural Studies**Cross-listed with: SOCI 390****Credits 3 Grading Basis Graded**

This course explores contemporary culture through a "cultural studies" lens, an interdisciplinary perspective interested in using empirical knowledge to encourage more just human relations. Specific topics of investigation will vary by semester, but each course will cover cultural studies' intellectual history and its application to cultural expressions found in everyday life, film, music and text.

ANTH 391 Study Abroad**Credits 1-6 Grading Basis Graded**

Designed to encourage students to enhance their academic programs through studying abroad. Arrangements must be made with a faculty member who will direct the study with preparatory instructions and final requirements. May be repeated up to 12 credits.

ANTH 395 Special Topics in Anthropology**Credits 3 Grading Basis Graded**

This course is an examination of selected topics which are of current importance to anthropology. May be repeated for credit when course content changes.

ANTH 405 Topics in Linguistic Anthropology**Credits 3 Grading Basis Graded**

This course examines current issues in the anthropology of language. Topics vary by semester, but each course will include hands-on analysis of social interaction and/or investigation of contemporary case studies of language policy, ideologies and use.

ANTH 410 Spatial Analysis for Social Research**Cross-listed with: SOCI 410****Credits 4 Grading Basis Graded**

The course teaches students how to identify and solve anthropological problems with spatial dimensions. Hands-on experience is stressed in the acquisition, analysis and display of spatial data using Geographic Information Systems software. Topics include the mapping of race and ethnicity, the spatial distribution of cultural variables and human modification and use of the landscape.

Prerequisites: ANTH 195, ANTH 196, ANTH 197, SOCI 110 or SOCI 140; or by permission of the instructor.

ANTH 415 Anthropological Genetics**Credits 3 Grading Basis Graded**

This course surveys the theory and methods of evolutionary genetics as applied to human evolution and human diversity. It emphasizes human evolution as illuminated by genetics, as well as the intersection of human genetics with social issues such as racism, bioethics and eugenics.

Prerequisites: ANTH 196.

ANTH 435 Ethnographic Genres and Methods**Credits 4 Grading Basis Graded**

This course explores ethnographic methods and conventions of ethnographic writing through close reading, analysis and production of ethnographic texts. Students develop critical skills in assessing ethnographic practice by examining how ethnographies are shaped by authors' fieldwork experiences, intellectual traditions and theoretical perspectives. Students engage in fieldwork and craft their own ethnographic accounts.

Prerequisites: ANTH 375.

ANTH 436 Afro-Latin America**Cross-listed with: HIST 436****Credits 3 Grading Basis Graded**

Latin America and the Caribbean were the first and largest parts of the Western Hemisphere to be populated by Africans. Afro-Latin America examines cultural formations Africans brought to these regions. Beginning with an overview of the slave trade, it examines the histories of Africans and African-descent people throughout Latin America, as well as contemporary Afro-Latin American culture(s).

ANTH 445 Methods and Research Perspectives in Biological Anthropology**Credits 4 Grading Basis Graded**

This course explores current research topics and methods in biological anthropology. Students will learn to write scientifically and critically evaluate the peer-reviewed literature and will receive training in data collection, analysis, interpretation and presentation.

Prerequisites: ANTH 196 and ANTH 375, or permission of the instructor. Corequisites: ANTH 375.

ANTH 455 Archaeology: Methods of Analysis and Interpretation**Credits 4 Grading Basis Graded**

A review of the nature of inquiry, recent theory, and the means by which archaeologists acquire, analyze and interpret their data. In addition to practical training in methods of analysis used in contemporary practice, students will gain experience in designing, conducting and reporting archaeological research.

Prerequisites: ANTH 195, ANTH 197 and one upper-level archaeology elective.

ANTH 485 Anthropology Course Assistantship**Credits** 1-6 **Grading Basis** Graded

Students participate as course assistants in anthropology.

Assistantships provide students with a sense of what it is like to teach an anthropology course by allowing them to work closely with faculty members through different phases of course preparation, presentation and evaluation. Assistantships also allow for a deeper understanding of course material by providing opportunities for student assistants to lead discussion and to help their peers review the material outside of the classroom. May be repeated up to six credits; only three credits can count toward the major. A student may only serve as a course assistant to the same course twice.

Prerequisites: Students must have junior /senior standing, must have earned a minimum grade of "B" in the course for which they will serve as assistant and may register by faculty invitation only.

ANTH 486 Internship in Anthropology**Credits** 1-6 **Grading Basis** Graded

Designed to encourage students to enhance their academic programs by employing and refining anthropological skills through internships in public or private agencies. Arrangements must be made with a faculty member who will oversee the internship. By permission only. May be repeated up to six credits.

ANTH 490 Special Studies in Anthropology**Credits** 1-3 **Grading Basis** Graded

Course offers students an opportunity to do independent study under staff supervision. Admission only by recommendation of the instructor. More than one repeat requires department head approval. Admission only by recommendation of the instructor. More than one repeat requires department head approval.

ANTH 492 History from Things: Material Culture**Cross-listed with:** ARTH 492, HIST 492**Credits** 3 **Grading Basis** Graded

This course explores the multidisciplinary "field" of material culture studies and the ways of looking at and learning from objects, things, or "stuff," ranging from furniture and teapots to houses and farmyards. It examines how scholars from a variety of disciplines have used material culture in their work. Readings draw from multiple sources, approaches and methodologies, reflecting the interdisciplinarity of the field.

ANTH 494 Field Techniques in Archaeology**Credits** 4-8 **Grading Basis** Graded

Laboratory course directed at teaching students the basic field techniques and procedures of historic and prehistoric archaeology. Classroom lectures will present techniques and relevant aspects of method and theory.

ANTH 496 Research Thesis**Cross-listed with:** HIST 496**Credits** 3 **Grading Basis** Graded

Students will gather, analyze and interpret archaeological/historical data over two semesters. Students will work on a project that demonstrates theory, research design, data gathering and analysis, culminating in a written thesis. The course meets the capstone requirement for the historical archaeology minor but is also available to students in history and anthropology.

Prerequisites: Junior or senior standing only.

ANTH 499A Honors**Credits** 1 **Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

ANTH 499B Honors**Credits** 3 **Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

ANTH 499C Honors**Credits** 2 **Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

Arabic (ARAB)

ARAB 101 Elementary Arabic I**Credits** 3 **Grading Basis** Graded

This course introduces the fundamentals of modern standard Arabic through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

ARAB 102 Elementary Arabic II**Credits** 3 **Grading Basis** Graded

This course builds on the foundations established in ARAB 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension.

Prerequisites: ARAB 101; or by placement exam or permission of the instructor.

ARAB 103 Passport to Arabic: A Survival Kit**Credits** 4 **Grading Basis** Graded

The course is designed to help students speak Arabic with confidence while visiting or living in an Arabic country. Besides gaining basic knowledge in reading and writing in Arabic, students will attain communicative skills to enable them to engage in a variety of activities and situations. Emphasis is on cultural competence and interaction skills. Seniors and ROTC students are encouraged to take this class. This course does not fulfill the B.A. language requirement. Instruction is in Arabic.

ARAB 109 Accelerated Review of Elementary Arabic**Credits** 3 **Grading Basis** Graded

This course reviews elementary Arabic grammar, reading, writing, speaking and listening skills in Arabic, with one hour of work a week in the language laboratory. For students who have had no more than two or three years of Arabic in high school and qualify through the placement exam. Native speakers are excluded.

ARAB 111 Intensive Arabic I**Credits** 6 **Grading Basis** Graded

This course involves the fundamentals of Arabic through intensive listening, speaking, reading and writing. This four-week course is the equivalent of ARAB 101 - ARAB 102.

ARAB 212 Intensive Arabic II**Credits** 6 **Grading Basis** Graded

This course is a thorough review of grammar, vocabulary building, conversation, composition and reading at the intermediate level. This four-week course is the equivalent of ARAB 231-ARAB 232.

Prerequisites: ARAB 102, ARAB 109 or ARAB 111, or permission of the instructor.

ARAB 231 Intermediate Arabic I**Credits 3 Grading Basis Graded**

This course is a thorough review of first-year grammar and vocabulary building. Conversation, composition and reading will be chosen to reach competency at the lower intermediate level Arabic.

Prerequisites: ARAB 102, ARAB 109 or ARAB 111, or permission of the instructor. A student cannot earn credit for ARAB 231 after completing ARAB 232.

ARAB 232 Intermediate Arabic II**Credits 3 Grading Basis Graded**

A thorough review of grammar and vocabulary building, conversation, composition, and reading.

Prerequisites: ARAB 231 or permission of the instructor.

ARAB 300 Language for Living**Credits 3 Grading Basis Graded**

The course focuses on enhancing students' practical communication skills in Arabic, preparing them for everyday interactions in personal, academic and professional contexts. Building on the foundations established in ARAB 232, this course emphasizes real-life language use through topics related to daily life, social events and work. Students will engage in listening, speaking, reading and writing activities designed to improve fluency and cultural understanding. The course prepares students to navigate various real-world situations while deepening their knowledge of Arabic social customs and language norms. Instruction is in Arabic.

Prerequisites: ARAB 212 or ARAB 232 or permission of the instructor.

ARAB 301 Pop Culture through Language**Credits 3 Grading Basis Graded**

This course explores Arabic language and culture through various forms of popular media, including social media, movies, TV series, music and cartoons/caricatures. Students will engage with authentic materials to enhance their understanding of contemporary Arabic language while gaining insights into modern Arab societies and cultural trends. By analyzing these media, students will expand their vocabulary, improve listening comprehension and develop their ability to critically evaluate cultural products. The course is designed for upper-beginner to lower-intermediate students looking to apply their language skills in real-world, culturally relevant contexts. Through discussions, presentations and creative projects, students will strengthen both their linguistic abilities and their cultural literacy. Instruction is in Arabic.

Prerequisites: ARAB 212 or ARAB 232 or permission of the instructor.

ARAB 307 A History of Islamicate Civilization 600-1600 A.D.**Credits 3 Grading Basis Graded**

This course is a study of society, economics, politics, culture and the arts of the Islamicate world from the rise of Islam to the establishment of the gunpowder empires (Ottoman, Safavid, and Mughal). Instruction is in English.

ARAB 308 Contemporary Islamicate Civilization**Credits 3 Grading Basis Graded**

This course is a study of society, economics, politics, culture and the arts of the Islamicate world, with a focus on the Arabic-speaking regions, from 1700 A.D. to the present. Instruction is in English.

ARAB 320 Culture and Communication in Context**Credits 3 Grading Basis Graded**

This Arabic course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence.

The course's essential question is "How do Arabic-speaking peoples communicate on subjects that are at the heart of their societies?"

Instruction is in Arabic.

Prerequisites: ARAB 212 or ARAB 232 or permission of the instructor.

ARAB 322 Language Landscapes**Credits 3 Grading Basis Graded**

This Arabic course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in this culturally and linguistically defined area and explores the diverse cultural landscapes that shape their lives. Instruction is in Arabic.

Prerequisites: ARAB 212 or ARAB 232 or permission of the instructor.

ARAB 330 Business Arabic**Credits 3 Grading Basis Graded**

This course studies commercial and trade vocabulary and customs in conjunction with practice in commercial communication, including letter writing, interviews and interpretations. Instruction is in Arabic.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 335 A Journey through Literature**Credits 3 Grading Basis Graded**

This course provides a thorough analysis of selected passages from important authors from early Arabic and Persian literature up to the beginning of the 18th century AD. The course will cover Arabic poetry and prose and the historical contexts in which they were written. Instruction is in Arabic.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 340 Intermediate Arabic Conversation**Credits 3 Grading Basis Graded**

Course emphasizes oral communication at the intermediate level. Students will use the vocabulary they have learned in the previous Arabic language classes. In addition, students will acquire new vocabulary from in- and out- of class conversational situations.

Prerequisites: ARAB 212 or ARAB 232 or permission of the instructor.

ARAB 371 Advanced Arabic Grammar and Translation**Credits 3 Grading Basis Graded**

This course covers Arabic/English translation applied in several fields. In this course students analyze the main grammatical differences between Arabic and English with the focus on producing accurate and idiomatic translations into both languages.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 375 Business and Society in the Arab World**Credits 3 Grading Basis Graded**

This course provides an introduction to the society, economy and political history from colonialism, through independence and up to the present of the twenty-two countries that are members of the Arab League (Jami'at al-Duwal al-'Arabiyya). The creation and development of the political, health and education systems and the ethnic and religious diversity of each country will be presented as well as the political problems and territorial disputes that they may be involved in. Business practices and the economies will be analyzed with a focus on the imports and exports of each country, the development of agriculture and fishing, the tourism industry (including medical tourism), financial institutions and banking (including Islamic banking and finance), and the oil industry.

Prerequisites: ARAB 232 or permission of the instructor.

ARAB 385 The Arabian Nights**Credits 3 Grading Basis Graded**

This course introduces the collection of popular tales known as "The Arabian Nights" or "The Thousand and One Nights." Readings also include medieval and modern European short stories that are adapted from the "Nights", or which appropriate themes and techniques from the "Nights." Instruction is in English.

ARAB 386 Arabic Fables and Short Stories**Credits 3 Grading Basis Graded**

This course provides an overview of short Arabic literary texts, ranging from the animal fables translated by Ibn al-Muqaffa' in the 8th century to Arabic short stories from the 20th century. Students will read a number of stories in English as well as a selection of texts in classical and modern standard Arabic by some of the most prominent Arabic-writing authors of the last thirteen centuries. Advanced topics in grammar will be addressed as the need arises.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 400 Advanced Arabic Writing and Conversation**Credits 3 Grading Basis Graded**

Discussions and writings deal with topics of current interest.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 405 The Arabic Novel**Credits 3 Grading Basis Graded**

This course provides an introduction to novels written in the Arabic language. The goal is to study the development of the modern Arabic novel from the nineteenth century to the present through a selection of ten texts written by authors originating from a number of different Arabic countries. The texts will be placed within the social and political contexts in which they were written. Additionally, social, religious and historical aspects of modern Arab culture will be explored through the texts. The students will gain some familiarity with the major literary trends, themes and genres of this time period.

ARAB 410 Media Arabic**Credits 3 Grading Basis Graded**

This course is an introduction to the vocabulary and language of Arabic press media and to the history of mass media in the Arabic speaking world. The focus is on print and Internet media, although media broadcasting in other forms is also covered.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 411 Colloquial Egyptian Arabic I**Credits 3 Grading Basis Graded**

This course provides an introduction to Egyptian Colloquial Arabic (ECA), the most widely spoken and understood Arabic dialect. Students will learn the phonological, morphological, syntactic and semantic particularities of ECA, and the major differences between Modern Standard Arabic and ECA, while developing their listening and speaking skills necessary for personal everyday life.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 412 Colloquial Egyptian Arabic II**Credits 3 Grading Basis Graded**

This course is a continuation of Colloquial Egyptian Arabic I. It is the second part of an intensive introduction to Egyptian Colloquial Arabic (ECA). Students will build on previously learned material through scenarios of everyday life.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 413 Colloquial Levantine Arabic I (Jordanian/Palestinian)**Credits 3 Grading Basis Graded**

This course provides an introduction to Jordanian and Palestinian Colloquial Arabic (JCA and PCA), the Arabic dialects spoken in Jordan and Palestine/Israel. Students will learn the phonological, morphological, syntactic and semantic particularities of JCA and PCA, and the major differences between them and Modern Standard Arabic, while developing their listening and speaking skills necessary for personal everyday life.

Prerequisites: ARAB 300 or permission of the instructor.

ARAB 414 Colloquial Levantine Arabic II (Syrian/Lebanese)**Credits 3 Grading Basis Graded**

This course provides an introduction to Syrian and Lebanese Colloquial Arabic (SCA and LCA), the Arabic dialects spoken in Syria and Lebanon. Students will learn the phonological, morphological, syntactic and semantic particularities of SCA and LCA, and the major differences between them and Modern Standard Arabic, while developing their listening and speaking skills necessary for personal everyday life.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 428 Muhammad and the Qur'an**Credits 3 Grading Basis Graded**

This course studies the life of the Prophet Muhammad and the contents of his message as found in the "Sira" (his biography), the Qur'an and the "hadith" (reports concerning his sayings and actions). Instruction is in English.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 429 Introduction to Qur'anic Arabic**Credits 3 Grading Basis Graded**

This course introduces students to the Arabic vocabulary and grammar of the Qur'an and hadith literature. The course provides an intensive overview of Arabic grammar, from basic to advanced topics. Students will gain the ability to read excerpts from the Qur'an and other Islamic religious texts in Arabic.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 446 Special Topics in Arabic Literature**Credits 3 Grading Basis** Graded

This course offers a study of a particular topic in Arabic Literature. It may cover all or a specific Arabic literature genre. This course may be repeated if content changes.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 447 Special Topics in Arabic Civilization and Culture**Credits 3 Grading Basis** Graded

Students will study a particular topic in the civilization and/or culture of Arabic countries. This course may be repeated if content changes.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 448 Special Topics in Arabic Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in Arabic linguistics. Topics could include an introduction to Arabic sociolinguistics and psycholinguistics.

This course may be repeated if content changes.

Prerequisites: ARAB 300, ARAB 301, ARAB 320 or ARAB 322; or by placement exam or permission of the instructor.

ARAB 465 Arabic Cinema**Credits 3 Grading Basis** Graded

This course provides an introduction to the history of Arabic cinema from its inception at the beginning of the twentieth century to the present. Films are chosen for their artistic qualities and their cultural content. Students will be acquainted with contemporary Arabic culture through the study of a number of films from various Arabic countries. The focus will be on the relation of these films to important elements of the history, societies, cultures and literatures of the Arabic-speaking world. Some films are based on novels or novellas; students will compare these literary works and their visual interpretation in the films. Films will be analyzed in terms of their social, political and intellectual contexts.

ARAB 484 Literatures of Islamic Spain and Italy**Credits 3 Grading Basis** Graded

This course introduces the literature of Islamic Sicily (Ar. Saqalliyya or Siqilliyya) and Andalusia (Ar. al-Andalus = Spain and Portugal) from the eighth to the fifteenth century, from the Muslim conquest of the territories to the Swabian rule of southern Italy and the Christian Reconquista of the Iberian Peninsula. This course is intended for students who are interested in Arabic literature and Islamic history and culture, but can also be of interest to students of history, medieval studies, and of Spanish culture and civilization.

ARAB 490 Independent Study in Arabic**Credits 1-4 Grading Basis** Graded

Students undertake independent study in Arabic under the supervision of the instructor.

ARAB 497 Arabic Internship**Credits 1-3 Grading Basis** Graded

This course will provide students with experiential learning opportunities within the Arabic-speaking field. The internship course is designed to provide practical experience in fields such as government, business, industry, health sciences, administration, education, translation or the arts, where they can apply their language skills. Examples of internship work may include working for a company in the student's field of interest abroad; serving as a research assistant for a professor; completing a translation project; or working as an interpreter. A journal, internship report, research paper, bibliography or evaluation from the internship provider is required. Credits earned correspond to the internship hours completed. Up to six credits may be applied as elective credits in the Arabic major or minor.

Prerequisites: One ARAB course at the 300-level or above and permission of the instructor.

Architectural Design (ARCD)

ARCD 200 Architect Design Studio I**Credits 6 Grading Basis** Graded

This studio focuses on the design process through the creation of objects and architectural spaces. Projects involve investigations into syntax and design language, program interpretation, materiality, ritual, use and the constructed order of built space. Emphasis is placed on experimentation, risk and play. Design projects will incorporate constructed drawings, sketching, diagramming, model-building and writing.

Prerequisites: Architectural Design majors only. FDS 102, FDS 104, FDS 106 and FDS 108; ARTH 205, ARTH 206 or equivalent; or by permission of the instructor.

ARCD 202 Architect Design Studio II**Credits 6 Grading Basis** Graded

Design studio building on the fundamentals of ARCD 200. Projects will address both three-dimensional design of spaces and site composition. Projects will include questions of landscape, urbanism, color, material properties and prototypes. Introduction of workshop, digital graphics and photography incorporated in studio work, with an emphasis on fabrication and constructed full-scale objects.

Prerequisites: ARCD 200.

ARCD 208 Portfolio Review**Credits 0 Grading Basis** Credit / No Credit

This course includes a review of initial student work created during the ARCD 200 level courses, including design, concepts and functionality. Portfolio review required to enroll in architectural design courses at 300 level and above.

Corequisites: ARCD 200 or ARCD 202.

ARCD 220 Digital Processes**Credits 3 Grading Basis** Graded

This course will introduce students to principles used in digital processes and representation in architectural production. Technologies to draw three dimensionally on the computer will be considered as a discipline within itself, and students will be instructed to use the machine for design exploration. Various software packages will be utilized during the semester.

Prerequisites: ARCD 200.

ARCD 292 Topics in Architectural Design**Credits 1-3 Grading Basis** Graded

Study of selected topics in architectural design at the introductory level. May be repeated when course content changes. See MyMadison for current topics.

ARCD 300 Architect Design Studio III**Credits 6 Grading Basis** Graded

Intermediate design studio building upon skills of ARCD 200-ARCD 202 sequence. Projects will be of greater complexity, scope and technical requirement. Design studies will incorporate drawing, diagramming, models, digital studies and writing. Projects will have a focus on site development in an urban or exurban context. Case studies will be required.

Prerequisites: ARCD 208.

ARCD 302 Architect Design Studio IV**Credits 6 Grading Basis** Graded

Intermediate design studio building upon skills of ARCD 300. Projects will be an extension of ARCD 300, but with greater complexity, scope and technical requirements. Design studies will incorporate drawing, diagramming, models, digital studies and writing. Course includes expansion of the design role into collaborative teams, interdisciplinary teams, actual clients, service projects and competitions.

Prerequisites: ARCD 300.

ARCD 330 Materials and Methods**Credits 3 Grading Basis** Graded

A lecture course introducing components and materials used in construction and building systems.

Prerequisites: ARCD 208.

ARCD 370 History of Architecture**Credits 3 Grading Basis** Graded

Survey of the evolution of architectural design and architectural thought from ancient to modern times.

Prerequisites: ARTH 206.

ARCD 376 Modern Architecture**Credits 3 Grading Basis** Graded

Survey of architecture from 1851 to the present day. Thematic investigations will include regional, philosophical and technical developments in architectural space. Architects may include Labrouste, Berlage, Wagner and Richardson, through Wright, Mies, Le Corbusier, to the avant-garde Muncutt, Siza, Nouvel and Mockbee.

Prerequisites: ARTH 206.

ARCD 390 Independent Studies in Architectural Design**Credits 1-3 Grading Basis** Graded

Independent activity at the intermediate level, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with the consent of the instructor.

ARCD 392 Topics in Architectural Design**Credits 3 Grading Basis** Graded

Study of selected topics in architectural design at the intermediate level. May be repeated when course content changes. See MyMadison for current topics.

Prerequisites: ARCD 208.

ARCD 400 Architect Design Studio V**Credits 6 Grading Basis** Graded

An upper-level design studio building upon the design rigor and technical craft acquired in the ARCD 300-ARCD 302 sequence. The studio will be organized into a series of short topical projects. Projects will include contemporary research initiatives including the body, landscape, urbanism, materiality and/or social justice.

Prerequisites: ARCD 302.

ARCD 402 Architect Design Studio VI**Credits 6 Grading Basis** Graded

Final upper-level architectural design studio culminating in a thesis project. Individual student directives for the development and execution of a design research project.

Prerequisites: ARCD 400.

ARCD 440 Professional Design Practices**Credits 3 Grading Basis** Graded

This course examines procedures and practices in the architectural design profession.

Prerequisites: ARCD 300.

ARCD 470 Contemporary Design**Credits 3 Grading Basis** Graded

Seminar format class which explores influential ideas in contemporary design, including critical regionalism, postmodernism, minimalism, the neo-avant-garde, green design, Marxism and post-structuralism.

Prerequisites: ARCD 208 and ARTH 206.

ARCD 490 Independent Studies in Architectural Design**Credits 1-3 Grading Basis** Graded

Independent activity, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with consent of the instructor.

ARCD 491 Studio Assistant**Cross-listed with:** ART 491, ARED 491, GRPH 491, INDU 491**Credits 1-3 Grading Basis** Graded

An on-campus program monitored on an individual basis designed to provide practical studio experience in the visual arts. Students will learn safe studio practices and management skills, including material use, inventory control and the proper operation of equipment found within various individual classroom studios.

ARCD 492 Topics in Architectural Design**Credits 3 Grading Basis** Graded

Study of selected topics in architectural design at the advanced level. May be repeated when course content changes. See MyMadison for current topics.

Prerequisites: ARCD 208.

ARCD 496 Internship**Credits 0 Grading Basis** Graded

An off-campus program prepared and monitored on an individual basis. Internships are designed to provide practical experience in professional design settings.

ARCD 499 Honors**Credits 1-6 Grading Basis** Graded

Honors 499 is taken as a three-course sequence. In the first course, the student works with their faculty advisor to generate an idea for an independent Honors research project, including a reading list and annotated bibliography. In the second course, the student completes the research for their project. In the third course, the student prepares an oral and written presentation of their research project results. Repeatable for a maximum of six credits for Honors students.

Prerequisites: ARCD 208.

Art (ART)

ART 105 Introduction to Studio Art

Credits 3 Grading Basis Graded

This course for art majors is focused on the exploration and exchange of ideas related to embarking on a studio career. Contemporary issues and responsibilities faced by emerging artists are emphasized. Students will investigate various visual art disciplines offered by JMU, write an artist's statement, and construct a website and digital portfolio.

Prerequisites: Art Major.

ART 200 Art Today: Themes in Contemporary Art [C2VPA]

Credits 3 Grading Basis Graded

Much of today's art is experiential and socially responsive, where embracing new technology and activism often exists alongside material explorations. For this introduction to contemporary art, students will address how art is created and why, the ways we respond to it, and the role art and artists play in global contexts. Students will explore key concepts from the last decade and will learn to evaluate art from many points of view (aesthetic, formal, historical, cultural) through readings, writings, discussions, experiential learning and art making. This course may be used for general education credit.

ART 210 Figure Drawing

Credits 3 Grading Basis Graded

This introductory course covers problems stressing the fundamental skills, approaches and concepts involved in drawing the human figure.

Prerequisites: FDS 104 and FDS 108.

ART 220 Introductory Ceramics: Potter's Wheel

Credits 3 Grading Basis Graded

Explores the aesthetics, conceptualization and design of functional objects. Investigates tactility and process of realizing form, and the effective use of the wheel as a creative tool. Introduces historic and contemporary approaches, firing techniques and glaze application.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 222 Introductory Ceramics

Credits 3 Grading Basis Graded

This course explores forming techniques for both vessel and sculptural work. It addresses construction concerns such as timing, structure and mass. Conceptual issues of hand-formation and wheel thrown ceramic objects are discussed. It introduces historic and contemporary approaches, firing techniques and glaze application.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 230 Introduction to Fibers Processes

Credits 3 Grading Basis Graded

In this introductory level studio course, students develop skills and cultivate conceptual concerns related to the fibers medium, specifically pertaining to structural forms and embellished or manipulated surfaces. Traditional and experimental materials are used to explore such techniques as crochet, off-loom weaving, rope-making, embroidery, piecing and appliqué. This class is a prerequisite for upper division fibers courses.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 232 Introduction to Fiber Arts: Structures

Credits 3 Grading Basis Graded

In this introductory level studio course, students develop skills and cultivate conceptual concerns related to the fibers medium related to structural forms. Traditional and experimental materials will be used to explore such techniques as knitting, crochet, looping, felting, off-loom weaving and embroidery. Topics will include the historical relevance of fibers; its relationship to ideas such as labor and identity; and what textiles have to offer within the expanded field of contemporary art.

Prerequisites: FDS 102 and FDS 104.

ART 240 Introductory Metals

Credits 3 Grading Basis Graded

This course provides an introduction to designing and executing adornment and functional objects through various fabrication and finishing techniques, and the exploration of metal as a medium of personal aesthetic expression.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 242 Knifemaking: Exploring Form and Function

Credits 3 Grading Basis Graded

Students will explore the complex interplay of form and function through the techniques, materials and concepts of knifemaking.

ART 252 Introductory Painting

Credits 3 Grading Basis Graded

This course provides an introduction to basic materials and techniques in oil or acrylic painting. This class extends previous practice with design and drawing through introductory experiences in painting.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 260 Introductory Black and White Darkroom Photography

Credits 3 Grading Basis Graded

This course offers a creative approach to photography with emphasis on understanding materials and techniques. (Students must provide a fully manual 35mm camera and a light meter which may be built into the camera or separate.)

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 270 Printmaking: Screenprint

Credits 3 Grading Basis Graded

An introduction to the history and techniques of screenprinting. Lectures, demonstrations and projects will involve photographic and nonphotographic stencils, related digital processes and color registration.

Prerequisites: FDS 102 and FDS 104.

ART 271 Introductory Printmaking: A Survey of Analog Printmaking Methods

Credits 3 Grading Basis Graded

This survey course introduces students to the practice of fine art printmaking through relief, intaglio, lithography, silkscreen and related processes. Students develop skills in image construction, editioning and safe studio procedures while translating concepts into printed form. The course emphasizes technical proficiency and creative experimentation within historical and contemporary contexts of printmaking.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 272 Printmaking: Relief and Intaglio**Credits 3 Grading Basis** Graded

An introduction to the history and techniques of relief and intaglio printing. Lectures, demonstrations and projects will involve oil based and water soluble inks, linocut, woodcut, wood engraving, collagraph, drypoint, engraving, line etching, aquatint, softground, color registration and related photographic processes.

Prerequisites: FDS 102 and FDS 104.

ART 274 Printmaking: Lithography**Credits 3 Grading Basis** Graded

An introduction to the history and techniques of lithography printing. Lectures, demonstrations and projects will involve stone and plate lithography, color registration and related photographic processes.

Prerequisites: FDS 102 and FDS 104.

ART 276 Introductory Book Arts: Materials and Structures**Credits 3 Grading Basis** Graded

This course offers students an opportunity to engage in the techniques, structures, tools, and materials used in creating artists' books. A broad range of studio practice will be explored as they examine the relationship of verbal, visual and structural content in books. Students will complete group and individual projects.

Prerequisites: FDS 102 and FDS 104.

ART 280 Introductory Sculpture: Idea Material Process**Credits 3 Grading Basis** Graded

This course introduces sculpture as an expansive and interdisciplinary practice. Rather than beginning with a single material or technique, students start with ideas and themes, then identify the most effective materials and processes to communicate them. Emphasis is placed on context: how works are shaped by placement, proximity, history and cultural references. Through studio production, readings, artist lectures and class discussions, students will develop technical skills, critical thinking and conceptual fluency. The course encourages experimentation, collaboration and cross-disciplinary exploration, positioning sculpture as a vital means of engaging with the world around us.

Prerequisites: FDS 102, FDS 104 and FDS 106; or permission of the instructor.

ART 292 Topics in Art**Credits 1-3 Grading Basis** Graded

Study of selected topics in art at the introductory level. May be repeated when course content changes. See MyMadison for current topics.

ART 300 Illustration**Cross-listed with:** GRPH 300**Credits 3 Grading Basis** Graded

Through demonstrations, theory and practical application, students are introduced to numerous media and illustrative techniques. Encouraged experimentation is tempered by an understanding of problem solving and conceptualization.

Prerequisite: FDS 108.

ART 304 Methods of Art Criticism**Credits 3 Grading Basis** Graded

The practical analysis and interpretation of works of art through oral and written forms. Emphasis is on the practice of art criticism in public settings such as the school classroom, art museum and college art studio.

ART 305 Professional Practices in Studio Art**Credits 3 Grading Basis** Graded

In this course, students consider several aspects of an artist's work that are not related to studio practice, but are nonetheless crucial as a creative practitioner. The course will combine seminar discussions, guest lectures and workshop sessions. Areas to cover include: how to produce effective artist's statements and resumes, marketing and self-promotion; how to use photography to document your artwork; grant-writing strategies; and cover letters for residencies and internships. Students will be expected to review their blog or website including the statements, resumes and photos produced in the course.

Prerequisites: ART 105 and B.F.A. Art majors only.

ART 310 Graphic Illustration**Cross-listed with:** GRPH 310**Credits 3 Grading Basis** Graded

Students will use traditional fine arts and digital media to explore the conceptualization, production and communication of visual information integral to selling products, clarifying and organizing data, explaining abstract concepts, and educating through the creation of imagery for advertising, infographics, and technical and scientific illustrations. Emphasis is placed on development of technical skillsets, personal aesthetic and style.

Prerequisites: FDS 102, FDS 104 and FDS 108.

ART 320 Molds and Casting: Ceramics**Credits 3 Grading Basis** Graded

This course covers mold-making systems and processes for ceramic slip casting and press molding. Conceptual issues of multiples, reproductions and material transformation are discussed. This course is also suitable for students wanting to utilize molds with other materials in their artistic production.

Prerequisites: ART 220 or ART 222.

ART 322 Surface Development: Ceramics**Credits 3 Grading Basis** Graded

In this course, students research and experiment with ceramic materials and finishes, glaze formulation, and application for finishing ceramic artwork. Forming processes may include any combination of the potter's wheel, handbuilding and mold making.

Prerequisites: ART 220 or ART 222 or by permission of the instructor.

ART 324 Potter's Wheel**Credits 3 Grading Basis** Graded

This course explores the aesthetics, conceptualization and design of functional objects using the potter's wheel. It investigates tactility and the process of realizing form and the effective use of the wheel as a creative tool. This course also introduces historic and contemporary approaches, firing techniques and glaze application.

Prerequisites: ART 222.

ART 330 Intermediate Fiber Processes**Credits 3 Grading Basis** Graded

Introduction to surface design techniques such as painting and printing on fabric. Further work may be in this area or in weaving or other fiber techniques. Course is repeatable.

Prerequisites: ART 230 or ART 232 or permission of the instructor.

ART 332 Woven Structures**Credits 3 Grading Basis** Graded

Traditional and experimental use of materials, techniques and structures are used to understand weaving as a cross-disciplinary practice including exploration of floor-loom hand weaving. Students will explore various forms of encoding information and address personal strategies for developing individual art practices. Weaving's role within the larger contemporary art world will be discussed and understandings of what constitutes a woven structure will be challenged.

Prerequisites: ART 230 or ART 232 or permission of the instructor.

ART 334 Iterative Structures**Credits 3 Grading Basis** Graded

This course engages students in spatiality, object creation and physical construction and will supplement artistic practice and design methodologies for sculptural fiber art or experimental design. Strategies for engaging in off loom fiber art structures such as knitting, macrame, crochet and basketry techniques will be explored. Class activities will include demonstrations, lectures, critiques and discussions revolving around iterative design strategies in fiber art.

Prerequisites: ART 230 or ART 232 or permission of the instructor.

ART 336 Pulp**Credits 3 Grading Basis** Graded

Students will use raw, natural and recycled materials to explore the basic techniques, processes and concepts used in papermaking. Pulp materials and fabrication processes will be explored through techniques using traditional mold and deckle, sculptural mold making, and pulp inclusions. Students will gain skills in beating a variety of fibers, sheet forming, sculpting with pulp and pulp painting.

Prerequisites: ART 276.

ART 340 Molds and Casting: Metals**Credits 3 Grading Basis** Graded

This course provides an intermediate exploration of metal as a medium of personal aesthetic expression with a focus on modeling, mold-making and processes for metal casting.

Prerequisites: ART 240.

ART 342 Knifemaking**Credits 3 Grading Basis** Graded

This course provides an intermediate exploration of metal as a medium of personal aesthetic expression with a focus on knifemaking skills and technologies

Prerequisites: ART 240.

ART 344 Welding and Metal Fabrication**Credits 3 Grading Basis** Graded

This course provides an intermediate exploration of metal as a medium of personal aesthetic expression with a focus on micro-arc and laser welding.

Prerequisites: ART 240.

ART 346 Color on Metal**Credits 3 Grading Basis** Graded

This course provides further exploration of metal as a medium of personal aesthetic expression with a focus on expanding students' ability to incorporate color into their metalwork, including stone setting, enameling, powder coating, anodizing, plating and patination processes.

Prerequisites: ART 240.

ART 352 Intermediate Painting**Credits 3 Grading Basis** Graded

Intermediate experiences in materials and techniques in oil, acrylic and nontraditional painting media. This class extends previous experiences introduced in ART 252, with a focus on developing a more personal iconography and content. A variety materials, techniques, surfaces and philosophies of working are discussed in lecture in lecture, demonstration, and in both individual and group critique.

Prerequisites: ART 252.

ART 360 Digital Photography**Credits 3 Grading Basis** Graded

This course provides an intensive exploration of digital photography with an introduction to digital camera techniques, combinations of traditional and digital photographic methods, image manipulation and modes of output.

Prerequisites: ART 260.

ART 362 Experimental Black and White Photography**Credits 3 Grading Basis** Graded

This course provides an intensive exploration of advanced black and white photography using alternative cameras, pinhole, a variety of film speeds and papers, and sequential concepts.

Prerequisites: ART 260.

ART 364 Large Format Photography**Credits 3 Grading Basis** Graded

This course explores medium format and view camera techniques, film exposure and advanced black and white printing.

Prerequisites: ART 260.

ART 372 Mechanical Reproduction(s): Relief, Intaglio and Technology**Credits 3 Grading Basis** Graded

This course investigates the relationship between printmaking and technologies of reproduction, focusing on relief, intaglio and related printmaking methods. Students examine how tools and mechanical processes have shaped both the practice and meaning of print across time. Emphasis is placed on experimentation with traditional and adapted tools to develop hybrid approaches that bridge historical technique and contemporary practice.

Prerequisites: ART 271 or permission of the instructor.

ART 373 Graphic Edition(s): Lithography, Silkscreen and Digital Processes**Credits 3 Grading Basis** Graded

This course examines the role of digital technologies in contemporary printmaking, with a focus on lithography and silkscreen. Students explore how photography, graphic design and other 2D visual practices intersect with digital and analog print processes to produce original editions. Emphasis is placed on experimentation and integration of these methods to develop innovative approaches to layering, sequencing and replication within historical and contemporary visual culture.

Prerequisites: ART 271 or permission of the instructor.

ART 375 Letterpress**Cross-listed with:** GRPH 375**Credits 3 Grading Basis** Graded

This studio course offers students an opportunity to engage in the process and product of letterpress printing through various techniques and conceptual approaches. Instruction focuses on text and image relationships by integrating metal and wood type, and other type-high surfaces. Emphasis will be placed on the acquisition of skills and vocabulary and the creative use of type and image. The course will address the history of letterpress and its contribution to contemporary art and design.

Prerequisites: FDS 102 and FDS 104.

ART 376 Intermediate Book Arts: Concept, Content, Form**Cross-listed with:** GRPH 376**Credits 3 Grading Basis** Graded

This course challenges the student to develop a limited edition hand-made artists' book. The appropriate format for each individual's concepts are identified, adapted, customized, applied, and produced. Content development, book design, integration of various media and the functionality of various bookmaking materials are explored. We will consider the artists' book as a sculptural form and locate it within the broader context of contemporary writing and visual art.

Prerequisites: ART 276.

ART 380 Conceptual Practice: Sculpture**Credits 3 Grading Basis** Graded

This class is an exploration of both theoretical and practical concerns in relation to sculpture. Students will develop methods of thinking and making that are derived from their own ideas and interests. Projects will be generated from sources relevant to contemporary art and culture.

Prerequisites: ART 280.

ART 390 Independent Study in Art**Credits 1-3 Grading Basis** Graded

Independent activity at the intermediate level, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with the consent of the instructor.

ART 392 Topics in Art**Credits 3 Grading Basis** Graded

Study of selected topics in art at the intermediate level. May be repeated when course content changes. See MyMadison for current topics.

ART 400 Topics in Illustration**Cross-listed with:** GRPH 400**Credits 3 Grading Basis** Graded

This course allows for the development of an advanced level class that teaches content not introduced in required illustration courses or expanding on the existing illustration courses. Instructional goals and defined outcomes will be unique to the class topic.

Prerequisite: ART 300 or ART 310, or permission of the instructor.

ART 405 Studio Art Capstone Thesis Seminar**Credits 3 Grading Basis** Graded

This course is designed to be a transition into professional life in the field of art. In this course, students will organize and prepare themselves for graduation and work with faculty on professional concerns beyond the studio. The course prepares students for their senior thesis exhibition and presentation, and aids in developing an e-portfolio and promotional packet (including resume and artist's statement). Students will develop plans to apply for opportunities within the field of art, such as graduate school programs, scholarships, internships, fellowships, residencies and grants beyond JMU.

Prerequisites: ART 105 and ART 305. Corequisites: Any studio art course at the 200-level or above.

ART 420 Advanced Ceramics**Credits 1-3 Grading Basis** Graded

This course provides students with a self-directed, focused course of study with supervision of the instructor. Emphasis is placed on the creation of advanced level artwork through the development of technical skills, knowledge of the process, and the refinement of individual expression using the ceramic medium.

Prerequisites: ART 322 and either ART 320 or ART 324; or permission of the instructor.

ART 429 Portfolio Development: Ceramics**Credits 3 Grading Basis** Graded

This is a self-directed, focused course of study with supervision of the instructor. This course is the culmination of ceramic study, resulting in a body of work suitable for exhibition and that is representative of the student's research and development.

Prerequisites: ART 322 and either ART 320 or ART 324; or permission of the instructor.

ART 430 Advanced Fiber Processes**Credits 1-3 Grading Basis** Graded

A series of fiber projects selected by the student with the approval of the instructor.

Prerequisites: ART 330, ART 332 or ART 334.

ART 432 Fiber Installation Art**Credits 3 Grading Basis** Graded

This studio course focuses on the planning and development of an outside group exhibition or self-directed studio work. It is designed for students engaged in the development of installation-based work. It involves critiques, discussions, demonstrations, field trips and introduction to contemporary artists. It explores space, site, situation, immersive experience, viewer relations and exhibition design. It examines various combinations of fiber-based installation art, including video, sound, performance, social exchange, collaboration and interactivity. It addresses execution and presentation phases of large-scale project development.

Prerequisites: ART 330, ART 332 or ART 334.

ART 434 Collaboration in Fiber Arts**Credits 3 Grading Basis** Graded

The course will examine social practice in fiber arts including art and activism, network art, Social Aesthetics, poststudio practices and Relational Aesthetics. This course will place a strong emphasis on contemporary examples of art and social practice. Through group activities, discussions, student led seminars and participatory projects, the class will work together to address contemporary issues in collaboration.

Prerequisites: ART 330, ART 332 or ART 334.

ART 439 Portfolio Development: Fibers**Credits 3 Grading Basis Graded**

This intensive exploration in fiber arts focuses on a theme or process that results in a cohesive body of work from a self-directed project and a written artist's statement.

Prerequisites: ART 330, ART 332 or ART 334.

ART 440 Advanced Metals**Credits 1-3 Grading Basis Graded**

This course consists of a series of metal arts projects selected by the student with the approval of the instructor. The course is repeatable.

Prerequisites: ART 340, ART 342, ART 344 or ART 346.

ART 449 Portfolio Development: Metals**Credits 3 Grading Basis Graded**

This intensive exploration in metals focuses on a theme or process that results in a cohesive body of work from a self-directed project and a written artist's statement.

Prerequisites: ART 340, ART 342, ART 344 or ART 346.

ART 452 Advanced Painting**Credits 1-3 Grading Basis Graded**

Advanced problems in media selected by the student with the advice of the instructor.

Prerequisites: ART 352.

ART 459 Portfolio Development: Painting**Credits 3 Grading Basis Graded**

This intensive exploration in painting focuses on a theme or process that results in a cohesive body of work from a self-directed project and a written artist's statement.

Prerequisites: ART 352 or ART 452.

ART 460 Alternative Photographic Processes**Credits 3 Grading Basis Graded**

Advanced study in photography focusing on alternative processes and experimental approaches including non-silver 19th century techniques, Polaroid and liquid emulsion, among others.

Prerequisites: ART 360, ART 362 or ART 364.

ART 462 Wonder and Illusion: Origins of Photography**Credits 3 Grading Basis Graded**

This course explores the culture and trends leading to the invention of photography, facilitating the formation of concepts and objects which create wonder.

Prerequisites: ART 360, ART 362 or ART 364.

ART 464 The Photograph/Video as Document**Credits 3 Grading Basis Graded**

This course explores technical, conceptual and theoretical approaches to making documentary-based photographic work using film, digital or video cameras.

Prerequisites: ART 360, ART 362 or ART 364.

ART 466 Performance for the Lens**Credits 3 Grading Basis Graded**

This course explores technical, conceptual and theoretical approaches to making, staging and directing performance-based photographic work using film, digital or video cameras.

Prerequisites: ART 360, ART 362 or ART 364.

ART 468 Screen-Based Photography/Video**Credits 3 Grading Basis Graded**

This course explores technical, conceptual and theoretical approaches to making screen-based photographic projects culminating in a fully developed online presentation.

Prerequisites: ART 362.

ART 469 Portfolio Development: Photography**Credits 1-3 Grading Basis Graded**

This course provides an intensive exploration in photography focusing on a theme or process that results in a cohesive body of work from a self-directed project and a written artist's statement.

Prerequisites: ART 360, ART 362 or ART 364.

ART 470 Advanced Printmaking**Credits 1-3 Grading Basis Graded**

Advanced projects in printmaking to be determined by the student and instructor.

Prerequisites: ART 270, ART 272 or ART 274.

ART 474 Expanded Print Media: Contemporary Printmaking and Interdisciplinary Practices**Credits 3 Grading Basis Graded**

This course explores both traditional and expanded printmaking practices, ranging from 2D editions to installation, time-based processes and socially engaged approaches. Students investigate experimental methods and conceptual strategies that bridge conventional techniques with contemporary innovations, creating dynamic works that engage space, audience and time. Emphasis is placed on developing a cohesive portfolio that reflects individual artistic voice and situates work within the broader field of contemporary print media.

Prerequisites: ART 271 and either ART 372 or ART 373; or permission of the instructor.

ART 479 Portfolio Development: Printmaking and Expanded Media**Credits 3 Grading Basis Graded**

This intensive exploration in printmaking and expanded media focuses on a theme or process that results in a cohesive body of work from a self-directed project and a written artist's statement.

Prerequisites: ART 470 or permission of the instructor.

ART 480 Expanded Field: Sculpture**Credits 1-3 Grading Basis Graded**

This class is an opportunity for students to pursue an individualized body of work. It approaches sculpture as a diverse and expanding field that embraces spatial-object and material-based practices; performance and lens-based media; forms of digital technology and the internet; social and relational practice; and landscape and environmental engagement.

Prerequisites: ART 380.

ART 489 Portfolio Development: Sculpture**Credits 3 Grading Basis Graded**

This intensive exploration in sculpture focuses on a theme or process that results in a cohesive body of work from a self-directed project and a written artist's statement.

Prerequisites: ART 380.

ART 490 Independent Study in Art**Credits 1-3 Grading Basis Graded**

Independent activity, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offerings must be arranged with the instructors who will direct them. Offered only with consent of the instructor.

ART 491 Studio Assistant**Cross-listed with:** ARED 491, GRPH 491, ARCD 491, INDU 491**Credits 1-3 Grading Basis Graded**

An on-campus program monitored on an individual basis designed to provide practical studio experience in the visual arts. Students will learn safe studio practices and management skills, including material use, inventory control and the proper operation of equipment found within various individual classroom studios.

ART 492 Topics in Art**Credits 3 Grading Basis** Graded

This course consists of a study of selected topics in art at the advanced level. The course may be repeated for credit when course content changes. See MyMadison for current topics.

ART 493 Contemporary Art Theory**Credits 3 Grading Basis** Graded

This is a reading, research and discussion seminar designed for upper level undergraduate and first-year graduate level studio art majors addressing historic and contemporary issues surrounding vision, the senses and aesthetics in the theory and practice of art.

ART 496 Internship in Art**Credits 1-8 Grading Basis** Graded

A program prepared and monitored on an individual basis. Internships are designed to provide practical experience in the arts.

Prerequisites: Permission of the instructor and ARTH 394 if in museums and galleries.

ART 499 Honors**Credits 1-3 Grading Basis** Graded

Art and Design Foundations (FDS)

FDS 102 Color and Composition**Credits 3 Grading Basis** Graded

Color and Composition is a foundational course that concentrates on the principles and elements of design, color theory, 2D compositional strategies and problem solving through the use of both traditional and digital media.

FDS 104 Observational Drawing**Credits 3 Grading Basis** Graded

Observational Drawing is an introductory course designed to develop and expand the student's observational abilities and rendering skills. A variety of subject matter and concepts are explored, including linear perspective and the figure.

FDS 106 Form and Space**Credits 3 Grading Basis** Graded

Form and Space is a foundational course that teaches strategies for creating objects and working with spatial considerations. Both traditional methods and time-based/digital tools are introduced.

FDS 108 Expressive and Expanded Drawing**Credits 3 Grading Basis** Graded

Expressive and Expanded Drawing develops the student's skills through advanced methods and approaches to drawing. In addition to an initial review of observational drawing basics, the course focuses on conceptual, expressive, and expanded drawing challenges through traditional and digital media.

Prerequisites: FDS 104.

Art Education (ARED)

ARED 300 Elementary Art Education Methods**Credits 3 Grading Basis** Graded

A study of the aims, philosophy, management, and current methods of art education in elementary school with an emphasis on child growth and development in visual expression. Experiences include developing thematic lessons with art techniques and materials suitable from Kindergarten through grade five with emphasis on instructional art content, lesson plan preparation, teaching strategies, assessment of student learning, and professional dispositions.

Prerequisites: PSYC 160, EDUC 200 and permission of the instructor.

ARED 302 Secondary Art Education Methods**Credits 3 Grading Basis** Graded

The study of art education methods, philosophy and contemporary practices as they relate to public education secondary school and lifelong learning with adults. Experience with concepts and ideas relevant to secondary curriculum with a focus on instructional art content, lesson plan preparation, teaching strategies, assessment of student learning and professional dispositions.

Prerequisites: PSYC 160, EDUC 200 and permission of the instructor.

ARED 392 Topics in Art Education**Credits 3 Grading Basis** Graded

Study of selected topics in art education at the intermediate level. See MyMadison for current topics.

ARED 400 Visual Arts Across the Curriculum**Credits 3 Grading Basis** Graded

Explores ways in which the art teacher can promote relationships among art and other subjects within the public school curriculum. Emphasis will be placed on how art experiences can be used to teach skills and concepts associated with other subjects. Includes a range of intensive non-art classroom field experiences at the middle grade level.

Prerequisites: ARED 300 and ARED 302, EDUC 200, PSYC 160 and permission of the instructor.

ARED 480 Student Teaching**Credits 8 Grading Basis** Credit / No Credit

Enables students to apply, in the public school classrooms and the comprehensive child development programs, those skills and attitudes acquired in all components of teacher education. Under the guidance of university supervisors, students are provided activities designed to familiarize them with the classroom teacher's role. Two sections of student teaching are completed for a total of sixteen credits. Sections must include the elementary level and secondary level.

Prerequisites: ARED 300, ARED 302 and ARED 400, the appropriate education courses, and permission of the coordinator of field experiences.

ARED 490 Field Experiences in Art Education**Credits 1-3 Grading Basis** Graded

Field experiences tied to the appropriate methods course at the elementary, middle, or high school levels under faculty supervision and an assigned cooperating teacher in the public schools. Practicum field experiences are arranged, approved, and directed by the methods course instructor in conjunction with the coordinator of field experiences. Offered only with the consent of the instructor.

Corequisites: Concurrent with the appropriate methods course: ARED 300, ARED 302 or ARED 400.

ARED 491 Studio Assistant**Cross-listed with:** ART 491, GRPH 491, ARCD 491, INDU 491**Credits** 1-3 **Grading Basis** Graded

An on-campus program monitored on an individual basis designed to provide practical studio experience in the visual arts. Students will learn safe studio practices and management skills, including material use, inventory control and the proper operation of equipment found within various individual classroom studios.

ARED 496 Internship in Art Education**Credits** 1-8 **Grading Basis** Graded

A program prepared and monitored on an individual basis. Internships are designed to provide practical experience in the arts.

Art History (ARTH)

ARTH 204 Art, Power and Society [C2VPA]**Credits** 3 **Grading Basis** Graded

Art and buildings today reflect the challenges of both our past and our present. This course asks pressing questions of select, major monuments: What power structures at the local, national and global levels define art and building in society? What images and works of architecture most reflect our current values? How can institutions promote positive change and a re-evaluation of the past towards greater understanding of the present? May be used for general education credit.

ARTH 205 Survey of World Art I: Prehistoric to Renaissance [C2VPA]**Credits** 3 **Grading Basis** Graded

An introduction to the art and architecture of the world from cave painting through European pre-Renaissance art. Includes ancient through medieval art in Europe and the Near East, as well as Asian and African arts. May be used for general education credit.

ARTH 206 Survey of World Art II: Renaissance to Modern [C2VPA]**Credits** 3 **Grading Basis** Graded

An introduction to the art and architecture of the world from the Renaissance through Modern ages. Includes European Renaissance, Baroque, Enlightenment, 19th and 20th centuries, as well as Asian and African arts. May be used for general education credit.

ARTH 210 African Art and Culture in the Humanities**Credits** 3 **Grading Basis** Graded

An interdisciplinary introduction to African art and culture with topics focusing on life ways, music, religion, philosophy, art, literature, and cinema. This course provides a strong background for upper-division course work in the arts of Africa and the African Diaspora, as well as for students pursuing degrees in history and anthropology.

ARTH 289 Topics in Art History**Credits** 3 **Grading Basis** Graded

Study of selected topics in art history, including museum studies, at the introductory level. May be repeated when course content changes. See MyMadison for current topics.

ARTH 300 Art History Seminar**Credits** 3 **Grading Basis** Graded

Exploration of various methodological approaches in the history of art, including connoisseurship, iconography, formalism, psychological studies and interpretations of art and society. Students will examine contrasting interpretations of major works of art. Seminar format.

Prerequisites: ARTH 205, ARTH 206, or permission of the instructor.

ARTH 301 Institutions and Professions in Art History**Credits** 3 **Grading Basis** Graded

This elective art history course prepares students for professions related to the field through exploration of the histories and missions of its important institutions. Organizations at the international, national and local levels that support the arts are considered. While conducting advanced research and writing on a chosen institution, or on an important issue in the field, students are exposed to a wide range of future opportunities in their professional lives.

ARTH 303 History of Design**Credits** 3 **Grading Basis** Graded

This course is an investigation into the domains of graphic, architectural and industrial design.

Prerequisites: ARTH 206.

ARTH 304 History of Photography**Credits** 3 **Grading Basis** Graded

A survey of photography as an art form from its discovery to the present day. Emphasis is on 20th-century developments and recent contemporary trends.

ARTH 305 History of Decorative Arts**Credits** 3 **Grading Basis** Graded

A history of the decorative arts in Europe and America from c. 1200 to c. 1930. This course provides a stylistic and contextual analysis concentrating on domestic furnishings, including textiles, furniture, metals, ceramics, and glass.

Prerequisites: ARTH 205 or ARTH 206.

ARTH 309 Topics in Museum Studies**Credits** 3 **Grading Basis** Graded

This course may include studies of select museums, or categories of museums, their histories and current missions and projects; the history of collecting, as well as the dispersal of collections; the repatriation of artworks; and consideration of significant issues in the field at the intermediate level. Topics vary. See MyMadison for current topics.

ARTH 312 African Art: Sub-Saharan**Credits** 3 **Grading Basis** Graded

A survey of the arts and cultures of sub-Saharan Africa, focusing on the major style areas of the continent. Coverage will include: the historic sites of Nigeria, the Guinea Coast, and central, eastern and southern Africa.

Prerequisites: ARTH 205 or ARTH 206.

ARTH 313 Masterpieces of Italian Renaissance Art [C2VPA]**Credits** 3 **Grading Basis** Graded

A survey of Italian Renaissance painting and sculpture (1280-1550), including the works of Giotto, Donatello, Masaccio, Fra Angelico, Botticelli, Leonardo and Michelangelo. Weekly visits to the Uffizi, San Marco, the Accademia and other Florentine museums. May be used for general education credit.

ARTH 314 Masterpieces of Spanish Art [C2VPA]**Credits** 3 **Grading Basis** Graded

A survey of art in Spain from prehistoric cave painting through 20th-century art. Emphasis is given to 17th-18th century Baroque and modern artists including El Greco, Velasquez, Goya, Gaudi and Picasso. Visits to Altamira, the Alhambra, the Prado, Toledo, Santillana del Mar and other sites. May be used for general education credit.

ARTH 316 Experiencing Art in London [C2VPA]**Credits 3 Grading Basis** Graded

This course offers an introduction to art in London via site visits to visual arts venues, monuments and museums. We will follow the course of art in Britain; its extensive contacts and interactions with the European continent, countries subjected to British colonization, and the world; and examine how these London museums came to acquire certain works. Discussion, analysis and writing are required. May be used for general education credit.

ARTH 320 Travel Study in Art History**Credits 3 Grading Basis** Graded

Art history credit is available to students participating in formal travel study programs with an emphasis on art history. Students maintain a journal with an emphasis on their art history experiences and write a research paper.

ARTH 325 Topics in Non-Western Art and Architecture**Credits 3 Grading Basis** Graded

This course will consider the study of a culture outside of the European and Anglo-American traditions at the intermediate level. Topics vary. See MyMadison for current topics.

ARTH 326 Topics in Cross-Cultural Art**Credits 3 Grading Basis** Graded

This course will consider the study of two or more artistic traditions that intersect at the intermediate level. Topic varies. See MyMadison for current topics.

ARTH 327 Topics in Pre-Modern Art**Credits 3 Grading Basis** Graded

This course will consider the study of a culture or tradition before the Modern Age at the Intermediate level. Topics vary. See MyMadison for current topics.

ARTH 328 Topics in Modern and Contemporary Art**Credits 3 Grading Basis** Graded

This course will consider the study of a culture or tradition in the most recent centuries or decades at the Intermediate level. Topics vary. See MyMadison for current topics.

ARTH 332 Islamic Art and Architecture**Credits 3 Grading Basis** Graded

This course will consider art from the age of the prophet Muhammad through the sixteenth century. The political and cultural contexts in which Islamic art developed will lay the foundations for understanding later traditions. These may include the role of the mosque in Muslim society; calligraphy and illustrated books; palace building and the arts of luxury; and modes of figural representation, including issues of gender.

ARTH 360 Nineteenth-Century Art**Credits 3 Grading Basis** Graded

A study of European art (1750-1890) concentrating on Neoclassicism, Romanticism, Realism, Impressionism and Post-Impressionism in France. Prerequisites: ARTH 206.

ARTH 372 Modern Art from 1900-1945**Credits 3 Grading Basis** Graded

A survey of avant-garde art movements from the late nineteenth century to mid-twentieth century. An analysis of the meanings of modernism and modernity in the context of urbanism, industrialism, colonialism and the world wars.

ARTH 380 American Art to 1870**Credits 3 Grading Basis** Graded

American painting, sculpture, architecture and decorative arts from the Colonial period through 1870. Topics will include Colonial portraiture, African American aesthetics, the definition of folk art, nationalism and landscape painting and the question of American exceptionalism. The course will also introduce students to problems of interpretation in current scholarship.

Prerequisites: ARTH 206.

ARTH 382 American Art from 1870**Credits 3 Grading Basis** Graded

American painting, sculpture, architecture and decorative arts from 1870 to 1945. This course will address topics such as the American Renaissance, the Harlem Renaissance, Chicago School architecture, masculinity in Western American art, notions of decay in turn-of-the-century art and American modernism. It will also introduce methodological debates in current scholarship.

Prerequisites: ARTH 206.

ARTH 389 Topics in Art History**Credits 3 Grading Basis** Graded

Study of selected topics in art history, including museum studies, at the intermediate level. May be repeated when course content changes. See MyMadison for current topics.

ARTH 390 Independent Study in Art History**Credits 1-3 Grading Basis** Graded

Independent activity at the intermediate level, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with the consent of the instructor.

ARTH 394 Introduction to Museum Work**Cross-listed with:** HIST 394**Credits 3 Grading Basis** Graded

A study of the philosophy and practice of museum work including the areas of exhibit design, conservation registration, education and administration. Subject is taught from the perspective of the museum profession and is applicable to diverse disciplines and types of collections.

ARTH 396 History Beyond the Classroom: Public History**Cross-listed with:** HIST 396**Credits 3 Grading Basis** Graded

This course introduces the interdisciplinary subfield of public history as it is practiced today. The course examines the various branches of public history through readings, class discussion, exercises, and occasional guest speakers and field trips, when possible.

ARTH 397 Art History On-Campus Internship**Credits 1-3 Grading Basis** Graded

An on-campus program prepared and monitored on an individual basis. Internships are designed to provide practical experience in museums and galleries.

ARTH 400 Art History Research Portfolio**Credits 0 Grading Basis** Credit / No Credit

This course is required for all art history majors. In both the fall and spring semesters, majors with junior and senior standing enroll in ARTH 400.

ARTH 404 Topics in the History of Photography**Credits 3 Grading Basis** Graded

This course may include studies of major photographers or consideration of thematic issues such as the documentary tradition and the New Deal. Topics vary. See MyMadison for current topics.

ARTH 406 Monticello**Cross-listed with:** HIST 406**Credits 3 Grading Basis** Graded

A seminar on the architecture and material culture of Thomas Jefferson's Monticello. The course will examine the house's design, artwork, decorative arts, mechanical devices, landscape/garden design and Mulberry Row. Topics will include African American artisans at the Monticello joinery, Jefferson's Indian Hall and European and African American domestic life in the Federal Period. Required field trips.

ARTH 408 The Museum: History and Controversies**Cross-listed with:** HIST 408**Credits 3 Grading Basis** Graded

This seminar centers on art museums in the United States. Topics include the historical development of museums, related cultures of display, recent debates on institutional mission and responsibility, and contemporary artists who employ the museum as medium, subject matter or site. Required field trips.

Prerequisites: ARTH 206.

ARTH 409 Topics in Museum Studies**Credits 3 Grading Basis** Graded

This course may include the advanced study of individual case studies or critical issues in the field. Topics vary. See MyMadison for current topics.

ARTH 418 Modern and Contemporary African Art**Credits 3 Grading Basis** Graded

This course examines modern and contemporary art movements in Africa and the African diaspora. Artists, art schools, exhibition strategies and developments in painting, sculpture, photography, video and film illustrate the richness of artistic productions and display, as well as the tensions and triumphs of contemporary Africa and its diaspora. Course work centers on a substantial research paper based on primary source material.

ARTH 419 Topics in African Art**Credits 3 Grading Basis** Graded

This course will deal with the current thematic or methodological issues such as contemporary African arts and artists, arts of the African Diaspora, a particular media (such as architecture or the textile arts), portraiture and identity, the royal arts of Africa, African film and performance, or gender in the arts of Africa. Topic varies. See MyMadison for current topics.

ARTH 424 Arts of Ancient Egypt**Credits 3 Grading Basis** Graded

A study of the arts and culture of Ancient Egypt (c. 3000 B.C. to c. 300 B.C.). This course will focus on the art and architecture of the Old and New Kingdoms and also examine the enduring fascination with this unique artistic heritage from the excavations of Napoleon to the present.

ARTH 425 Advanced Topics in Non-Western Art**Credits 3 Grading Basis** Graded

This course will consider the study of a culture outside of the European and Anglo American traditions at the advanced level. Topics vary. See MyMadison for current topics.

ARTH 426 Advanced Topics in Cross-Cultural Art**Credits 3 Grading Basis** Graded

This course will consider the study of two or more artistic traditions that intersect at the advanced level. Topics vary. See MyMadison for current topics.

ARTH 427 Advanced Topics in Pre-Modern Art**Credits 3 Grading Basis** Graded

This course will consider the study of a culture or tradition before the Modern Age at the advanced level. Topics vary. See MyMadison for current topics.

ARTH 428 Advanced Topics in Modern and Contemporary Art**Credits 3 Grading Basis** Graded

This course will consider the study of a culture or tradition in the most recent centuries or decades at the advanced level. Topics vary. See MyMadison for current topics.

ARTH 430 Far Eastern Art**Credits 3 Grading Basis** Graded

A survey of East Asian art from prehistoric times to 19th century colonialism. Emphasis is on the areas of major production: India, China and Japan, with less attention to such centers as Cambodia, Siam and Korea.

Prerequisites: ARTH 205 or ARTH 206.

ARTH 440 City of Jerusalem**Credits 3 Grading Basis** Graded

A critical examination of medieval and early modern Jerusalem, c. 1000-1800, with a special focus on the city's religious diversity evidenced through its art and architecture. The course will examine Islamic, Christian and Jewish cultural traditions in the city.

ARTH 446 Renaissance Art and the East**Credits 3 Grading Basis** Graded

This seminar explores artistic exchange between the Christian west and competing cultures in the east from c. 1250-1600, focusing on the powers of Italy and their interaction with the Islamic dynasties, the Mamluks of Egypt and the Ottomans in Turkey, as well as the Christian state of Byzantium. Special topics of interest may include palace architecture and imperial ceremony; urban planning; portraiture and caricature; the exchange of luxury goods; and the use of art as a diplomatic tool.

ARTH 448 Studies in Leonardo and Michelangelo**Credits 3 Grading Basis** Graded

Seminar that examines the artworks of Leonardo da Vinci and Michelangelo Buonarroti. Discusses issues such as the artist's creative process, the development of the artist's style, the patron's role in artwork and interrelationships between the artist's visual and literary works.

Prerequisites: ARTH 206.

ARTH 471 Commemoration and Controversy: Public Art in America**Cross-listed with:** HIST 471**Credits 3 Grading Basis** Graded

This course examines the socio-historical, political, cultural, and philosophical dimensions of public art in American society, from the Early Republic to the present day. Topics may include: the nature of public art, its uses and functions, as well as civic and official attitudes towards art in the public sphere. Issues of censorship, propaganda, and the "culture wars" of the 1980s and 90s will be highlighted.

ARTH 472 Modern Art Since 1945**Credits 3 Grading Basis** Graded

A study of the many developments and trends in American and European art since 1945. This course will focus on such movements as abstract expressionism, pop, conceptual art, installation art, video, film and computer art with an understanding of the theoretical basis of those movements and the development of the art market during this period.

ARTH 484 Art of the Americas**Credits 3 Grading Basis** Graded

Art of indigenous peoples in the Americas (Meso, Central, South and/or North) before European contact. The course will examine domestic and state architecture, painting, textiles, ceramics, metalwork, and earthworks within the context of geographic, state, religious and social issues. Other topics include museum display, repatriation and western taxonomies.

ARTH 488 African American Art**Cross-listed with:** AAAD 488**Credits 3 Grading Basis** Graded

This course examines visual arts produced by people of African descent in the United States from the colonial period until the present. Course themes include debates about the relationship between racial identity and artistic production; the complex interchange between African American art and the cultural traditions of Africa and Europe; black artists' engagement with popular representations of African Americans; and the intersection of race with class, gender and sexuality. Prerequisites: AAAD 200, ARTH 206 or permission of the instructor.

ARTH 489 Topics in Art History**Credits 3 Grading Basis** Graded

Study of selected topics in art history at the advanced level. May be repeated when course content changes. See MyMadison for current topics.

ARTH 490 Independent Study in Art History**Credits 1-3 Grading Basis** Graded

Independent activity such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with consent of the instructor.

ARTH 491 Exhibition Seminar**Credits 3 Grading Basis** Graded

Advanced seminar explores museum theory and practice through the collaborative design of an exhibition for an on campus gallery space. The course focuses on developing and researching the exhibition topic, investigating the specific art objects, and preparing critical educational materials or catalogue. The class project also includes mounting the exhibit in a real or virtual installation space. Prerequisites: ARTH 205 or ARTH 206.

ARTH 492 History from Things: Material Culture**Cross-listed with:** ANTH 492, HIST 492**Credits 3 Grading Basis** Graded

This course explores the multidisciplinary "field" of material culture studies and the ways of looking at and learning from objects, things, or "stuff," ranging from furniture and teapots to houses and farmyards. It examines how scholars from a variety of disciplines have used material culture in their work. Readings draw from multiple sources, approaches and methodologies, reflecting the interdisciplinarity of the field.

ARTH 493 Historic Preservation**Cross-listed with:** HIST 493**Credits 3 Grading Basis** Graded

An introduction to the philosophy and techniques of historic preservation, guidelines for restoration, state and National Register forms and procedures, historic architecture, structural analysis, restoration techniques as well as the business aspects of historic preservation projects. Field trips are a major component of the course.

ARTH 495 Internship in Art History**Credits 1-3 Grading Basis** Graded

An off-campus program prepared and monitored on an individual basis. Internships are designed to provide practical experience in the arts. Prerequisites: Permission of the instructor and ARTH 394 if in museums and galleries.

ARTH 497 Art History On-Campus Advanced Internship**Credits 1-3 Grading Basis** Graded

An on-campus, advanced program prepared and monitored on an individual basis. Internships are designed to provide practical experience in museums and galleries.

ARTH 499 Honors**Credits 1-6 Grading Basis** Graded

Arts Entrepreneurship (ARTE)

ARTE 520 Leadership and Community Engagement in the Arts and Design**Credits 3 Grading Basis** Graded

This course explores theories, practices and challenges related to leadership and community engagement in the arts sector. Through the study of interdisciplinary perspectives from art and human centered design, students will examine the role of leadership in fostering innovative and sustainable impact within cultural organizations. Through theoretical frameworks, case studies and practical applications, students will develop the knowledge and skills necessary to lead and facilitate arts-based and design-based initiatives that promote community engagement, social change and cultural impact.

ARTE 530 Business and Marketing in the Arts and Design**Credits 3 Grading Basis** Graded

This course explores the intersection of business principles, marketing strategies and the principles of design within the arts and cultural sector. Through a combination of theoretical knowledge and practical application, this course equips students with the skills and expertise necessary to succeed as leaders and innovators in the business and marketing aspects of the arts and cultural sector.

ARTE 620 Ethics and Law in the Arts and Design**Credits 3 Grading Basis** Graded

This course provides an in-depth exploration of ethical considerations and legal issues that arise in the context of the arts. Through case studies, discussions and practical exercises, students will develop the knowledge and skills necessary to navigate complex ethical dilemmas and legal challenges in the creation, exhibition design and dissemination of artistic works.

ARTE 720 Arts Integration: Interdisciplinary Project Creation**Credits 3 Grading Basis** Graded

This course is a capstone project that integrates earlier course content with the student's own form of interdisciplinary artistic expression. Students will develop a deeper understanding of the practicalities of artistic and design production through the completion of a project that will help them achieve broader career goals, such as a business venture, performance, exhibition or community project. Through experiential learning, goal setting and assessment, students will deepen their understanding of the arts and cultivate their own creative voices. Students will engage in discussions, peer critiques and feedback sessions to foster a supportive and collaborative learning environment.

Astronomy (ASTR)

ASTR 120 The Solar System [C3PP]**Credits 3 Grading Basis** Graded

An introductory course in astronomy, which includes the following topics: motions of celestial objects, eclipses, historical development, the nature of light, telescopes, properties and evolution of the solar system. May be used for general education credit.

ASTR 121 Stars, Galaxies, and Cosmology [C3PP]**Credits 3 Grading Basis** Graded

An introductory course in astronomy which includes the following topics: the Sun, stellar properties, stellar evolution, black holes, the Milky Way, galactic evolution, quasars, cosmology. May be used for general education credit.

ASTR 220 General Astronomy I: The Night Sky, the Solar System and Stars**Credits 3 Grading Basis** Graded

ASTR 220 is the first in a two-course sequence in general astronomy intended for students with a background in physics. Topics covered include: appearance and movements of the night sky; astronomical coordinate systems and timekeeping; seasons, eclipses and planetary configurations; planetary motions and gravitation; fundamental forces; electromagnetic radiation and its detection; content, structure, formation and evolution of solar system; observations and models of the Sun, stellar interior models; stellar magnitudes and spectra, classifications; Hertzsprung-Russell diagram.

Prerequisites: PHYS 140 or PHYS 240.

ASTR 221 General Astronomy II: Star Systems, the Interstellar Medium, and Cosmology**Credits 4 Grading Basis** Graded

ASTR 221 is the second in a two-course sequence in general astronomy intended for students interested in science. Topics covered include: stellar evolution; variability and high-energy phenomena in stars and multiple-star systems; content, structure, and dynamics of the Milky Way; external galaxies, quasars and AGN; large-scale structure and the distance scale of the universe; the Big Bang model and alternative cosmologies, possible geometries and eventual fates of the universe. An observational astronomy laboratory component is part of this course. The lab component will cover basics of telescope set up and operation as well as astronomical coordinate systems.

Prerequisites: ASTR 220.

ASTR 297 Topics in Astronomy**Credits 1-4 Grading Basis** Graded

Topics in astronomy at the second year level. May be repeated for credit when course content changes. Topics selected may dictate prerequisites. Students should consult instructor prior to enrolling for course.

ASTR 301 Searching for Life in the Universe**Credits 3 Grading Basis** Graded

A study of the search for life in the universe, with emphasis on teacher preparation. Topics include how life on earth can guide the search, conditions for life within our solar system, extrasolar planets that may be conducive to life, possible radio communications with other civilizations and technologies necessary for search. Significant time is spent developing student lesson plans.

Prerequisites: ISCI 171 and ISCI 172.

ASTR 320 Astronomical Techniques**Credits 3 Grading Basis** Graded

An overview of modern astronomical techniques with an emphasis on quantitative data collection and analysis. The design and use of various astronomical devices will be covered. Topics will include visible light telescopes and radio telescopes as well as CCD data collection in addition to other current astronomical techniques. Data reduction software will also be addressed.

Prerequisites: ASTR 220 and ASTR 221.

ASTR 330 Relativity**Cross-listed with:** PHYS 330**Credits 4 Grading Basis** Graded

The basic mathematical descriptions of both Einstein's special and general theories of relativity are introduced through specific physical cases and calculus-based lectures. Special relativity is explored through its physical consequences on a variety of phenomena. General relativity is studied by considering the effects of gravity on space-time curvature on astrophysical objects. Simple numerical experiments in both special and general relativity will be conducted as class projects.

Prerequisites: MATH 237 and PHYS 260, or permission of the instructor.

ASTR 398 Independent Study in Physics or Astronomy**Cross-listed with:** PHYS 398**Credits 1-3 Grading Basis** Graded

An individual project related to some aspect of physics or astronomy. Must be under the guidance of a faculty advisor. A student may not earn more than a total of 4 credits for PHYS 398 and ASTR 398.

ASTR 480 Astrophysics**Credits 3 Grading Basis** Graded

An introduction to the problems of modern astronomy and the quantitative application of physical principles to these problems. Topics of study include stellar structure and evolution, the interstellar medium and star formation, cosmic rays, pulsars, galactic structure, extragalactic astronomy and cosmology.

Prerequisites: PHYS 340 and PHYS 380.

ASTR 498 Undergraduate Research in Physics or Astronomy**Cross-listed with:** PHYS 498**Credits 1-4 Grading Basis** Graded

Research in a selected area of physics or astronomy as arranged with a faculty research advisor. A student may not earn more than a total of 6 credits for ASTR 498 / PHYS 498. Prerequisite: Proposal for study must be approved prior to registration.

Athletic Training Education Program (ATEP)

ATEP 205 Introduction to Athletic Training

Credits 3 Grading Basis Graded

This course provides a broad introduction to the profession of athletic training. Lectures focus on the domains of athletic training. Emphasis is placed on basic emergency management as well as injury prevention including environmental issues, strength and conditioning, and selection of equipment.

ATEP 291 Pre-Professional Practicum in Athletic Training

Credits 2 Grading Basis Graded

This course is designed to help students better understand the duties and responsibilities of the athletic trainer. By focusing on psychomotor skills and the application of didactic knowledge, students build a foundation which prepares them for future clinical rotations.

ATEP 306 Athletic Injury Recognition

Credits 3 Grading Basis Graded

This course introduces students to musculoskeletal pathologies common in physically active individuals. Students will demonstrate an understanding of common mechanisms, risk factors and clinical presentation of musculoskeletal pathologies. Students will integrate their medical and anatomical terminology and hands-on skills used in injury evaluation.

Prerequisites: BIO 290/BIO 290L.

Biology (BIO)

BIO 103 Contemporary Biology [C3NS]

Credits 3 Grading Basis Graded

An in-depth exploration of selected biological concepts, connected to current, relevant topics and emphasizing an understanding of science as a way of obtaining knowledge. Not available for major or minor credit in biology. May be used for general education credit. May not be used for major credit.

BIO 110 First Year Seminar in Biology

Credits 1 Grading Basis Credit / No Credit

Orientation to the biology and biotechnology majors for first year students, including freshmen and transfer students. Course content includes introduction to the biology/biotechnology curriculum, academic plan building, time management and study skills, stress and anxiety management, campus student resources, undergraduate research, graduate school considerations, and discussion of biology career options. Course provides university elective credit. Not available for biology or biotechnology major credit. Course is offered credit/no-credit.

BIO 140 Foundations of Biology I [C3NS]

Credits 3 Grading Basis Graded

The foundations of the cellular molecules, structures and processes that sustain life in the context of evolution will be explored. Topics will include structure and function; information flow, storage and exchange; pathways and transformations of energy and matter; and systems biology. May be used for general education credit.

Prerequisites: BIO 140L. Corequisites: BIO 140L.

BIO 140L Foundations of Biology I Laboratory [C3L]

Credits 1 Grading Basis Graded

Students learn the process of science and the basics of experimental design while building scientific writing skills. Students learn foundational techniques of molecular biology (pipetting, DNA extraction, PCR, gel electrophoresis and basic bioinformatics) in the context of a DNA Barcoding project. Students will leave this course with the preparation needed to design their own independent DNA barcoding projects. May be used for general education credit.

Prerequisites: BIO 140. Corequisites: BIO 140.

BIO 150 Foundations of Biology II

Credits 3 Grading Basis Graded

Foundations II focuses on the diversity of life, anatomy and physiology of organisms and ecological organization in an evolutionary context. This class will explore structure and function; how information is transmitted; transformations of energy and matter; and how biological units act as integrated systems. The process of science, interdisciplinary approaches and the relevance of biology to society will be addressed.

Prerequisites: BIO 150L and minimum grade of "C-" in BIO 140 and BIO 140L. Corequisites: BIO 150L.

BIO 150L Foundations of Biology II Laboratory

Credits 1 Grading Basis Graded

Student teams develop and execute semester-long independent research projects using DNA barcoding. Students will learn how to ask scientific questions, explore scientific literature, design scientific studies, analyze and interpret data, and share results using images and scientific writing. This lab emphasizes the development and refinement of research techniques in molecular biology, critical thinking, study design, teamwork and skills in finding, reading and writing scientific literature. Students will leave this course with a deeper understanding of the process of science, an introduction to bioinformatics and molecular biology and improved skills in scientific communication.

Prerequisites: BIO 150 and minimum grade of "C-" in BIO 140 and BIO 140L. Corequisites: BIO 150.

BIO 200 Major Biology Elective

Credits 1-4 Grading Basis Student Option

This course is awarded for lower division transfer elective credit applicable toward the biology major or minor. A maximum of four hours may be transferred and used toward the major or minor. Additional hours may be transferred as BIO 000.

BIO 201 Trelawny Learning Community Seminar I

Credits 1 Grading Basis Graded

The first part of the research skills seminar for first-year students in the Trelawny Learning Community. In addition to lab/field work with faculty or graduate students for 2-3 hours per week, students attend a weekly seminar. Seminar includes research skill and team-building exercises, guest speakers and case studies geared toward a career in science. Faculty and students interact with community members to provide perspectives on the major and research in a scientific network. Corequisites: Membership in the Trelawny Learning Community.

BIO 202 Trelawny Learning Community Seminar II

Credits 2 Grading Basis Graded

The second part of the research skills seminar for first-year students in the Trelawny Learning Community. Students will have already begun research projects during fall semester, and will continue their projects into spring semester while continuing to build research skills. Students will take part in community, campus and social events and gain experience helping to run a research-based event.

Prerequisites: BIO 201. Corequisites: Membership in the Trelawny Learning Community.

BIO 203 Viral Discovery**Cross-listed with:** ISAT 203**Credits 2 Grading Basis** Graded

An exploratory laboratory experience, designed for incoming freshmen. Students will learn about the life cycle and ecology of viruses infecting bacteria. Soil samples will be collected, and techniques for isolation and purification of the viruses will be performed by the students. Isolated viruses will be visualized using electron microscopy. The genomic material will be isolated and prepared for nucleic acid sequencing.

BIO 204 Viral Genome and Bioinformatics**Cross-listed with:** ISAT 204**Credits 2 Grading Basis** Graded

A computer-based laboratory experience, designed for students completing the Viral Discovery course. Students will learn to identify genes in a viral genome, compare the predicted proteins with known proteins in databases, describe the contents of the genome and notate all the relevant information for publication. Students will also research the role of bacteriophages in ecology and evolution.

Prerequisites: ISAT 203 or BIO 203.

BIO 226 Introductory Topics in Biology (Variable)**Credits 1-4 Grading Basis** Graded

Introductory studies in specific areas of biology. May be repeated for credit when course content changes.

BIO 230 Biology, Gender and Society**Cross-listed with:** WGSS 230**Credits 3 Grading Basis** Graded

Students in this course will explore the biology of sex and gender and consider their interactions and interrelationships through a lens that considers intersectional identities and societal expectations. Topics include reproductive biology, sexual differentiation, gender development and expression, and the interactions between science and society that inform gendered bodies and practices.

Prerequisites: Either BIO 103, BIO 150, CHEM 120, CHEM 131, GEOL 110, GEOL 210, or ISAT 112.

BIO 240 Genetics**Credits 3 Grading Basis** Graded

Students will explore how genetic information is utilized throughout the lifetime of the organism. The course focuses on the flow of information and evolutionary principles through the lenses of classical and molecular genetics and genomics.

Prerequisites: BIO 240L and minimum grade of "C-" in BIO 150 and BIO 150L. Corequisites: BIO 240L.

BIO 240L Genetics Laboratory**Credits 1 Grading Basis** Graded

Students will explore both classic and molecular genetic techniques using genetic model organisms. Additionally, students will learn how to analyze large data sets using widely available data bases, data analysis programs and introductory coding.

Prerequisites: BIO 240 and minimum grade of "C-" in BIO 150 and BIO 150L. Corequisites: BIO 240.

BIO 245 General Microbiology**Credits 3 Grading Basis** Graded

A study of the microorganisms, their significance, evolution, structure and function, genetics and the systems in which they exist.

Prerequisites: BIO 245L and BIO 103 or BIO 140 or CHEM 120 or CHEM 131 or ISCI 101 or equivalent. Corequisites: BIO 245L and BIO 103 or BIO 140 or CHEM 120 or CHEM 131 or ISCI 101 or equivalent.

BIO 245L General Microbiology Laboratory**Credits 1 Grading Basis** Graded

This laboratory course is designed to complement and supplement the BIO 245 lecture course.

Prerequisites: BIO 245 and BIO 103 or BIO 140/BIO 140L or CHEM 120 or CHEM 131 or ISCI 101 or equivalent. Corequisites: BIO 245 and BIO 103 or BIO 140/BIO 140L or CHEM 120 or CHEM 131 or ISCI 101 or equivalent.

BIO 246 Microbiology Seminar**Credits 1 Grading Basis** Graded

Seminars, readings and discussions of the primary scientific literature with a focus on the current research in the field of microbiology.

Prerequisites: BIO 140, BIO 140L and declared microbiology concentration.

BIO 250 Quantitative Ecology and Evolution**Credits 3 Grading Basis** Graded

Students will use an interactive e-text application to explore the changing natural world through modeling and simulation in a discovery-based classroom environment. Throughout every unit, the relationship among evolution, ecology and the environment are woven together using current research from primary literature. Evolution is explored in many relevant contexts, including artificial and natural selection of GMOs and of antibiotic resistance. In a module covering the evolution of decision making, students will learn quantitative approaches to game theory, optimal foraging and sexual selection. An understanding of population ecology and evolution is gained as students manipulate models to understand competition and predator-prey dynamics with real case studies of both microbes and endangered species. Climate change is discussed through the use of mathematical models and current and historical data.

Prerequisites: BIO 250L and a minimum grade of "C-" in BIO 150 and BIO 150L or GEOL 110, GEOG 210 or ISAT 112. Corequisites: BIO 250L.

BIO 250L Quantitative Ecology and Evolution Laboratory**Credits 1 Grading Basis** Graded

Hands-on laboratory and field-based experiences which emphasize experimental design and both statistical and quantitative analyses, with an emphasis on oral and written professional communication.

Prerequisites: BIO 250 and a minimum grade of "C-" in BIO 150 and BIO 150L or GEOL 110, GEOG 210 or ISAT 112. Corequisites: BIO 250.

BIO 260 Foundations of Human Anatomy and Physiology**Credits 4 Grading Basis** Graded

An introduction to anatomy (structure) and physiology (function) of the human body and its organ systems which will provide a strong foundation in the relationship of human structure and function. BIO 260 is designed to meet the Virginia Science Content educational guidelines for A&P. Credit may not be earned in both BIO 260 and BIO 270/BIO 270L and BIO 290/BIO 290L.

Prerequisites: BIO 140 and BIO 140L; Students in the following majors and minors are not eligible for this course: Dietetics, Health Sciences, Kinesiology and Nursing and PPH. Credit may not be earned in both BIO 260 and BIO 270/270L and BIO 290/290L.

BIO 270 Human Physiology [C3NS]**Credits 3 Grading Basis Graded**

An introduction to basic physiological principles using humans as the primary organism. Physiological adaptations will be examined at the molecular through organismal levels. Intended for students in health-related fields and The Natural World of the general education program. Not available for biology or biotechnology major credit. Credit for BIO 270/BIO 270L and BIO 290/BIO 290L may be transferred together to JMU only when transferable credits for two semesters of anatomy and physiology (I & II) are completed at the same institution. A single semester transfer, or courses from different institutions, transfer as BIO 000.

Prerequisites: BIO 270L; CHEM 120 or CHEM 131; and MATH 220 or MATH 229 or COB 191. Credit may not be earned in both BIO 270/BIO 270L and BIO 260. Corequisites: BIO 270L; CHEM 120 or CHEM 131; and MATH 220 or MATH 229 or COB 191.

BIO 270L Human Physiology Laboratory [C3L]**Credits 1 Grading Basis Graded**

This laboratory course is conducted to directly observe how the body's systems work and how scientific knowledge is accumulated in conjunction with the BIO 270 lecture course. Organizational and leadership skills are promoted by individual and group laboratory activities. Credit for BIO 270L and BIO 290L may be transferred together to JMU only when transferable credits for two semesters of anatomy and physiology (I & II) lab are completed at the same institution. A single semester transfer, or courses from different institutions, transfer as BIO 000. May be used for general education credit.

Prerequisites: BIO 270, CHEM 120 or CHEM 131, and MATH 220 or MATH 229 or COB 191. Credit may not be earned in both BIO 270/BIO 270L and BIO 260. Corequisites: BIO 270, CHEM 120 or CHEM 131, and MATH 220 or MATH 229 or COB 191.

BIO 290 Human Anatomy**Credits 3 Grading Basis Graded**

A regionally based study of the human body. Emphasis on major body structures and function is reinforced through hands-on examination of prosected cadaver specimens and other anatomical resources during the prerequisite or corequisite laboratory experience. Credit for BIO 270/BIO 270L and BIO 290/BIO 290L; may be transferred together to JMU only when transferrable credits for two semesters of Anatomy & Physiology (I & II) are completed at the same institution. A single semester, or courses from different institutions, transfer as BIO 000.

Prerequisites: BIO 290L. Credit may not be earned in both BIO 290 and BIO 260.

BIO 290L Human Anatomy Laboratory**Credits 1 Grading Basis Graded**

A regionally-based study of the human body. Emphasis on major body structures and function through hands-on examination of prosected cadaver specimens and other anatomical resources. Credit for BIO 270L and BIO 290L may be transferred together to JMU only when transferrable credits for two semesters of anatomy and physiology (I & II) lab are completed at the same institution. A single semester transfer, or courses from different institutions, transfer as BIO 000.

Prerequisites: BIO 290. Credit may not be earned in both BIO 290/BIO 290L and BIO 260. Corequisites: BIO 290.

BIO 301 Introductory Neuroscience**Credits 3 Grading Basis Graded**

This course will examine molecular control of neuronal function. Topics include the structure and function of neuronal excitability, chemical and contact mediated neuronal communication, developing and regenerating nervous systems, sensation and perception, learning and memory formation, repair from neuronal damage, and the neuronal pathways of sensation and motor control. In the context of these subjects, we will review the neuronal alterations that cause some common brain diseases. Prerequisites: BIO 140/BIO 140L and CHEM 132, or by permission of the instructor for PSYC majors.

BIO 302 Bioethics for Biologists**Credits 3 Grading Basis Graded**

In this primarily discussion-based class, students will develop an understanding of the impact of biological and medical advances on society and why biologists and citizens must be aware of the accompanying bioethical issues. The development of the bioethics field will be explored, and biological, medical and technological advances will be described. Students will explore these topics with appropriate bioethics readings, short writing assignments synthesizing the readings, group discussions and in-class presentations selected by the students. Prerequisites: BIO 140 or permission of the instructor.

BIO 304 Cell and Molecular Biology**Credits 3 Grading Basis Graded**

Students will develop a deeper understanding of the molecular mechanisms employed by procaryotic and eucaryotic cells. Topics will include (1) macromolecular structure and function and how macromolecules coordinate to maintain life, (2) molecular mechanisms that regulate the synthesis and flow of genetic information and (3) molecular aspects of homeostasis, bioenergetics and cell signaling. Prerequisites: Minimum grade of "C" in BIO 240/BIO 240L and CHEM 132/CHEM 132L.

BIO 310 Marine Biology**Credits 3 Grading Basis Graded**

An exploration of the functional biology, ecology, and biodiversity of marine organisms. Students will be provided with a general understanding of how the ocean works and how marine organisms operate within their environment. The diversity of marine organisms in the open sea and benthos will be surveyed and a discussion of the numerous threats impacting various marine ecosystems will also be included.

Prerequisites: BIO 250/BIO 250L.

BIO 316 Animal Development**Credits 3 Grading Basis Graded**

This course integrates cell and molecular biology and genetics to understand the processes and mechanisms underlying body plan formation and organ formation in vertebrate animals and insects. The course additionally covers the development of muscle, skeleton and nervous tissues, the postembryonic phenomena of growth, metamorphosis and regeneration, and the developmental basis of evolutionary changes in animal anatomy.

Prerequisites: BIO 240/BIO 240L.

BIO 318 Experimental Vertebrate Development**Credits 2 Grading Basis Graded**

Students design and carry out independent research projects on different aspects of frog or chick development. Skills include manipulating live embryos, raising embryos and larvae from eggs, using wax histology to make microscope slides, immunohistochemistry, 3D reconstruction and morphometry-based analyses, and filming animal behavior. Projects are hypothesis-driven and investigate patterns, mechanisms and behaviors involved in the development of skeleton, lungs, tails, limbs and other organs.

Prerequisites: BIO 140/BIO 140L.

BIO 320 Comparative Anatomy of Vertebrates**Credits 4 Grading Basis Graded**

This course explores the anatomical evolution of vertebrates, including the phylogenetic, developmental and functional significance of structures and systems across the vertebrate family tree. Students will gain key competencies in tree-thinking, specimen dissection, comparing and conveying anatomical form, and evaluating and synthesizing information to generate hypotheses and propose future directions in the field of comparative vertebrate anatomy.

Prerequisites: BIO 150/BIO 150L.

BIO 324 Human Genetics**Credits 3 Grading Basis Graded**

An intermediate genetics course with an emphasis on human biology. Topics include cytogenetics, pedigree analysis, quantitative traits, mutation, epigenetics, genomics and ethical issues raised by developing technologies.

Prerequisites: BIO 240/BIO 240L.

BIO 325 Medical Parasitology**Credits 3 Grading Basis Graded**

This course covers the study and medical implications of parasites that infect humans. Class activities will emphasize parasite morphology, modes of transmission, mechanisms of host entry and infection, niche selection, life cycles, pathogenesis, diagnosis, and treatment and control.

Prerequisites: BIO 240/BIO 240L.

BIO 326 Intermediate Topics in Biology**Credits 1-4 Grading Basis Graded**

This course covers specified areas of biology taught at the 300 level. May be repeated for credit when course content changes. See MyMadison for prerequisites for specific topics.

BIO 327 Biological Illustration**Credits 3 Grading Basis Graded**

Students will consider how the skills of visual art are used in the dissemination of data. Students will learn how to sketch, paint and digitally illustrate microscopic specimens, live and dead plants and animals, bones, cellular processes and scientific methodology. Students will be instructed in different illustration techniques and be encouraged to find their preferred medium to produce a portfolio of images. The course content can be modified to reflect the interests/needs of the students. Students are responsible for obtaining all materials required for completing exercises and projects. Students may also obtain their own biological subjects.

Prerequisites: BIO 150/BIO 150L.

BIO 330 Evolution**Credits 3 Grading Basis Graded**

This course provides an introduction to evolutionary biology, including evidence for evolution, natural selection, genetic drift, sexual selection, phylogenetics, genome evolution, speciation, macroevolution, the history of life on Earth and the evolution of infectious disease.

Prerequisites: BIO 250/BIO 250L.

BIO 340 Morphology and Anatomy of Vascular Plants**Credits 4 Grading Basis Graded**

A detailed study of the comparative morphology and anatomy of tracheophytes.

Prerequisites: BIO 150/BIO 150L or BIO 250/BIO 250L.

BIO 342 Mathematical Models in Biology**Cross-listed with: MATH 342****Credits 3 Grading Basis Graded**

Introduction to dynamical models (discrete and continuous time) applied to biology. Tools of mathematical analysis from linear and nonlinear dynamics will be taught, including stability analysis of equilibrium, as well as appropriate use of software packages. Emphasis will be on model development and interpretation in the context of applications, including effective written and oral presentation.

Prerequisites: MATH 232 or MATH 235 or equivalent.

BIO 346 Laboratory in Bacterial Pathogenomics**Credits 2 Grading Basis Graded**

A course-based-research laboratory course. Students isolate pathogenic bacteria from environmental samples, identify and characterize them, and sequence their whole genomes. Students learn the basics of microbial genomics, including genome assembly, annotation and comparative genomics methods to study mobile genetic elements, virulence, antibiotic resistance, etc. They also learn how these methods are used in the epidemiological tracking of pathogens.

Prerequisites: Minimum grade of "C" in BIO 245/BIO 245L.

BIO 348 Medical Microbiology**Credits 3 Grading Basis Graded**

Students will learn about microorganisms of medical importance, mainly bacteria and viruses. Key topics for each organism include general cell structure, unique structures/functions, epidemiology of the disease that the organism causes, mechanisms of pathogenesis, isolation and identification of the organism, and treatment options.

Prerequisites: Minimum grade of "C" in BIO 245/BIO 245L.

BIO 348L Medical Microbiology Laboratory**Credits 1 Grading Basis Graded**

This laboratory course consists of a series of case studies that require the students to identify the likely causative agent of the symptoms described in the case study.

Prerequisites: BIO 348. Corequisites: BIO 348.

BIO 350 Paleobiology**Cross-listed with: GEOL 350****Credits 4 Grading Basis Graded**

The evolution and ecological structure of the biosphere from the origin of life to the present, emphasizing the evolution and paleobiology of animal life as shown by the fossil record. Lectures discuss methods used to interpret the fossil record and cover topics such as phylogeny and systematics, functional morphology, biostratigraphy, paleoecology, evolution and extinction. Laboratories focus on the major groups of invertebrates that are common in the geologic record.

Prerequisites: GEOL 230, BIO 150/BIO 150L or permission of the instructor.

BIO 354 Global Climate Change and Life: Ecological and Biological Impacts of Climate Variability**Credits 4 Grading Basis** Graded

Global climate change is important to the distribution, diversity, health and survival of organisms. The biota have changed through evolution in part as a response to selection pressures from these variations. Living things can also adjust to change through phenotypic flexibility. This course examines, in a seminar/discussion format, the potential ecological impacts of past and current patterns of climate alteration on organisms.

Prerequisite: "C-" or better in one of the following: BIO 140/BIO 140L, GEOL 110, GEOL 115, ISAT 112, GEOG 210 or CHEM 353.

BIO 357 Galápagos Islands: Biodiversity and Conservation**Credits 3 Grading Basis** Graded

A detailed study of the biodiversity, conservation, biogeography and natural history of the Galápagos Islands.

Prerequisites: BIO 150/BIO 150L.

BIO 360 Plant Biology**Credits 3 Grading Basis** Graded

This course is an introduction to the biology of plants including evolution, diversity, form and function, ecology, and biotechnology.

Prerequisites: BIO 250/BIO 250L with minimum grades of "C-";

Corequisites: BIO 360L.

BIO 360L Plant Biology Laboratory**Credits 1 Grading Basis** Graded

An exploration of plant structure, function, evolution and biological diversity.

Prerequisites: BIO 360. Corequisites: BIO 360.

BIO 361 Biochemistry I**Cross-listed with:** CHEM 361**Credits 3 Grading Basis** Graded

An introduction to the molecules and chemical reactions of living systems. Structure and function of important classes of biomolecules are explored and the relationship of structure to function is stressed. Basic metabolic sequences are discussed.

Prerequisites: CHEM 241 and permission of the instructor.

BIO 364 Human Uses of Plants**Credits 3 Grading Basis** Graded

A survey of past, present and future uses of plants with emphasis on economically important plant families. Issues of cultivated plant origins, biodiversity and germplasm preservation are considered.

Prerequisites: BIO 150/BIO 150L or BIO 250/BIO 250L.

BIO 364L Laboratory in Human Uses of Plants**Credits 1 Grading Basis** Graded

An investigative examination of plants and their constituents with an emphasis on their physiological ecology, adaptations and economic utilization by humans.

Prerequisites: BIO 364. Corequisites: BIO 364.

BIO 370 Human and Animal Physiology**Credits 4 Grading Basis** Graded

Design and function of cellular and organ physiology will be explored in both humans and other animals. The course will begin by using an organ system approach to introduce basic physiological principles and develop understanding of physiological processes which requires the integration of multiple organ systems functioning. The course is intended for biology majors and minors. Class activities will emphasize problem-solving and collaborative and independent learning. The laboratories will utilize computer simulations and animal/human experiments to examine principles of both physiology and scientific investigation.

Prerequisites: CHEM 132 and BIO 240/BIO 240L (minimum grade of "C-" in both lab and lecture) or BIO 304. One semester each of calculus and statistics is recommended.

BIO 375 Sensation and Perception**Cross-listed with:** PSYC 375**Credits 3 Grading Basis** Graded

This course explores the nature and development of human sensory capabilities and processing, and how these affect perception of the environment. A potential list of the variety of systems reviewed includes hearing, vision, smell, taste and touch/pain, as well as phenomena such as the perception of balance.

Prerequisites: Declared Neuroscience concentration; PSYC 101 or PSYC 160; and MATH 220, 229, 285 or 318.

BIO 380 Advanced Microbiology Laboratory Techniques**Credits 1 Grading Basis** Graded

This course will focus primarily on advanced microbiology laboratory techniques. The techniques may include microscopy, staining, autoclave use and preparation of media, maintenance of stock cultures, and spectroscopy. This course will help prepare students for research and advanced microbiology laboratory courses.

Prerequisites: BIO 245 and BIO 245L.

BIO 385 Biopsychology**Cross-listed with:** PSYC 385**Credits 3 Grading Basis** Graded

This course is a survey of the neurological and chemical mechanisms which control behavior. Students will examine the brain and how it processes sensation, perception, cognition, movement, motivation, learning, memory and other behavioral processes of interest to psychologists.

Prerequisites: Declared Neuroscience concentration; PSYC 101 or PSYC 160; and MATH 220, 229, 285 or 318.

BIO 386 Field Botany**Credits 4 Grading Basis** Graded

An in-depth survey of vascular plants in the field with emphasis on identification, diversity of form and function, and ecology. Laboratory topics will include techniques for sampling plant communities, identifying local flora and preserving botanical materials.

Prerequisites: BIO 150/BIO 150L or BIO 250/BIO 250L.

BIO 387 Environmental Microbiology**Credits 3 Grading Basis** Graded

An examination of microbial diversity in the environment and the functional role of microbes in ecosystems around the world.

Prerequisites: Minimum grade of "C-" in BIO 245/BIO 245L.

BIO 387L Environmental Microbiology Laboratory**Credits 1 Grading Basis** Graded

This course is a field-based investigative lab experience exploring the structure and function of microbial communities and applications of microbiology.

Prerequisites: BIO 387. Corequisites: BIO 387.

BIO 395 Comparative Animal Behavior**Cross-listed with:** PSYC 395**Credits 3 Grading Basis** Graded

This course covers aspects of the development, function and evolution of the behavior of nonhuman animals. Topics covered include intraspecies communication, feeding, aggression, territoriality, reproductive behavior and social behavior.

Prerequisites: Psychology majors: A "C-" or better in PSYC 211 or PSYC 213. Biology and biotechnology majors: BIO 250/BIO 250L, and one of the following (with a "C-" or better): MATH 205, MATH 220, MATH 231, MATH 235, MATH 285, MATH 318.

BIO 400 Geology and Ecology of the Bahamas**Cross-listed with:** GEOL 400, BIO 500**Credits 3 Grading Basis** Graded

This course explores the geology and ecology of the shallow-water marine environment by examining the preeminent modern example, the Bahamas platform. The Bahamas provide an excellent model for understanding modern and ancient carbonate and reef deposits and a variety of terrestrial/aquatic habitats. Biological processes are responsible for many of the geological features of the Bahamas, so the course considers the biology/ecology of marine organisms in addition to geological topics.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 211 or a 200- level GEOL or BIO course; at least four hours of additional lab science, at least sophomore status, and permission of the instructor.

BIO 402 Forest Ecology**Cross-listed with:** BIO 502**Credits 4 Grading Basis** Graded

A study of the function, structure and composition of forested ecosystems. The effect of physical geography on the distribution of forest communities will be explored. Issues of forest management and restoration will also be considered. Field laboratory topics will include dendrology and sampling techniques within different forest successional stages.

Prerequisites: BIO 250/BIO 250L.

BIO 404 Evolutionary Analysis**Credits 3 Grading Basis** Graded

An examination of the place of theoretical thought in biology. The concepts of phylogenetic relationships and the mechanisms of organic change as expressed through the principles of organic evolution will be stressed.

Prerequisites: BIO 240/BIO 240L.

BIO 405 Vertebrate Paleontology**Cross-listed with:** GEOL 405**Credits 3 Grading Basis** Graded

A study of the origin and evolution of the vertebrates. Emphasis will be on understanding how the processes of Earth evolution and biological evolution have interacted through time to produce a coherent picture of vertebrate history.

Prerequisites: GEOL 230 or BIO 150/BIO 150L or BIO 250/BIO 250L or permission of the instructor.

BIO 410 Advanced Human Anatomy**Credits 3 Grading Basis** Graded

An advanced study of topics in human anatomy using dissection techniques.

Prerequisites: BIO 290/BIO 290L or BIO 320 and permission of the instructor.

BIO 412 Mammalogy**Credits 4 Grading Basis** Graded

An introduction to the study of mammals, incorporating evolutionary history, general physiology, reproductive biology, systematics, ecology, wildlife management and behavior, followed by review of mammalian taxonomic orders. Lecture is reinforced with laboratory and field experience. Completion of BIO 320 recommended.

Prerequisites: BIO 150/BIO 150L or BIO 250/BIO 250L.

BIO 415 Neuroanatomy and Neurogenic Communication Disorders**Cross-listed with:** CSD 415**Credits 3 Grading Basis** Graded

This course is an introduction to neuroanatomy with emphasis on neurogenic communication disorders.

Prerequisites: CSD 208 or permission of the instructor.

BIO 418 Conservation Biology**Credits 3 Grading Basis** Graded

Conservation biology is the study and management of biodiversity with the aim of protecting species, their habitats and ecosystems from excessive rates of extinction. The course will focus on ecological theory, population dynamics, conservation and restoration strategies, and ethical questions for conservation. The course will also teach students to analyze open-ended questions via case studies.

Prerequisites: BIO 250/BIO 250L with minimum grade of "C-".

BIO 419 Animals and their Environments**Credits 4 Grading Basis** Graded

This course provides an exploration of terrestrial and aquatic animal ecology with a focus on the ecology of vertebrates in a rapidly changing world. Students will investigate how animal ecology is shaped by the environment and how animals in turn shape their biotic and abiotic environments. Lectures will be composed of small group work to synthesize information from primary literature readings and other sources. Field laboratory topics will involve travel to various local ecosystems. Students will build skills with techniques used in ecological studies focused on various local taxa and ecological communities. Students will build transferable skills in experimental design, field techniques, problem solving, data management, manipulation, visualization and analysis via activities embedded in both lecture and laboratory throughout the course. Students will use these skills to collaboratively design, complete and present course-based field research projects.

Prerequisites: BIO 250/BIO 250L with minimum grade of "C-".

BIO 425 Immunology**Credits 3 Grading Basis** Graded

This course studies the molecular and cellular basis of the immune system. Topics include the properties of antigens and immunoglobulins, the development and regulation of humoral and cell-mediated immunity, resistance and immunization to infectious diseases, allergies and autoimmune and immunodeficiency disorders.

Prerequisites: BIO 240/BIO 240L.

BIO 425L Immunology Laboratory**Credits 1 Grading Basis Graded**

This course will introduce students to the theory and application of many of the methods currently used in clinical and research immunology. Laboratory exercises will focus on methods for identifying, quantifying and assessing functional activities of immune cells and molecules. Students will gain experience using experimental animals and in animal cell culture techniques.

Prerequisites: BIO 425. Corequisites: BIO 425.

BIO 426 Topics in Biology**Credits 1-4 Grading Basis Graded**

Specified areas of biology. May be repeated for credit when course content changes. See MyMadison for prerequisites for specific topics.

BIO 427 Topics in Biology with Laboratory**Credits 1-4 Grading Basis Graded**

Laboratory studies in special areas of biology to accompany BIO 426 or stand alone. May be repeated for credit when course content changes.

An accompanying 3-credit BIO 426 course may be a corequisite. See MyMadison for prerequisites for specific topics.

BIO 430 Ornithology: Research and Current Advances**Credits 4 Grading Basis Graded**

Students will design and engage in course-based research projects using modern approaches to ornithology. Classes will investigate current primary literature on ornithological research, recent technological advances and conservation. Lectures and labs will explore the biodiversity of birds, identification by sight and sound, and the ecology, evolution, physiology and behavior of birds. Labs will involve fieldwork, data collection and workshops such as sound analysis and geographic tracking. BIO 430 will be taught concurrently with BIO 530.

Prerequisites: BIO 250/BIO 250L. Credit may not be earned in both BIO 430 and BIO 530.

BIO 432 Light Microscopy**Cross-listed with: BIO 532****Credits 4 Grading Basis Graded**

This course covers the principles behind light microscopy, from the properties of light to the latest technologies in microscopy. Students will get hands-on experience with the different kinds of microscopes, including the confocal microscope. The course also covers fluorescent probes, advanced fluorescence techniques, digital imaging, methods of quantification and figure preparation for publication, with an emphasis on biological applications.

Prerequisites: BIO 240/BIO 240L.

BIO 443 Understanding Techniques in Neuroscience**Credits 3 Grading Basis Graded**

A writing-intensive introduction to basic neurobiological processes that draws from molecular biology, genetics, biochemistry, behavioral biology, electrophysiology, pharmacology, computer science and optics. Topics will include both classic and state-of-the-art techniques. Special emphasis will be placed on framing techniques as tools to answer basic scientific questions in neuroscience. BIO 443 is taught concurrently with BIO 543. Credit may not be earned in both BIO 443 and BIO 543.

Prerequisites: BIO 240/BIO 240L or BIO 250/BIO 250L. Credit may not be earned in both BIO 443 and BIO 543.

BIO 444 Virology**Credits 3 Grading Basis Graded**

A study of the fundamental aspects of both basic and medical virology. Credit may not be earned in both BIO 444 and BIO 544.

Prerequisites: Minimum grade of "C" in each of the following: CHEM 241, BIO 304 and BIO 240/BIO 240L.

BIO 445 Neurobiology**Credits 4 Grading Basis Graded**

Molecular, cellular and network mechanisms underlying behavior will be studied using problem-solving, discussion, lecture and critical reading of the primary literature. Similarities and differences between nervous systems and computers will be explored. Laboratories will utilize contemporary electrophysiology and computer simulation to examine the neurobiology of simple animal model systems.

Prerequisites: BIO 140/BIO 140L or BIO 304.

BIO 446 Experimental Neurobiology**Cross-listed with: BIO 546****Credits 4 Grading Basis Graded**

The experimental basis of neurobiology will be explored in invertebrates and humans using modern neurobiological techniques, such as extracellular, intracellular, electromyogram and voltage clamp recording. The course will revolve primarily around the laboratory activities, with the single lecture section designed to support the laboratory. Students will be expected to complete a project that involves developing a new experimental approach or model system.

Prerequisites: BIO 140/BIO 140L or BIO 304.

BIO 447 Evolution and Ecology of Infectious Disease**Cross-listed with: BIO 547****Credits 3 Grading Basis Graded**

An introduction to the evolution and ecology of pathogenic microorganisms, with an emphasis on the bacteria. Emphasis will be placed on the study, discussion and critique of scientific literature, as well as formal presentation of scientific information and data.

Prerequisites: BIO 245/BIO 245L and BIO 250/BIO 250L.

BIO 448 Developmental Neurobiology**Credits 3 Grading Basis Graded**

Exploration of molecular and cellular mechanisms that determine neuronal identity, migration, synaptogenesis, synaptic plasticity and neural systems organization. Critical and intensive evaluation and communication of primary literature.

Prerequisites: BIO 240/BIO 240L or BIO 250/BIO 250L.

BIO 449 Insect Ecology**Cross-listed with: BIO 549****Credits 4 Grading Basis Graded**

An overview of insect ecology from an evolutionary perspective, focusing on the processes that affect the diversity, distribution and abundance of insects in natural and managed ecosystems. The role of insects as model systems in understanding ecological and evolutionary principles is included, and current literature in the field is highlighted. Field surveys and experiments are emphasized, including general insect collection and identification.

Prerequisites: BIO 250/BIO 250L.

BIO 451 Current Topics in Human Development and Evolution**Cross-listed with: BIO 551****Credits 3 Grading Basis Graded**

Discussion-based course on the developmental changes involved in our evolution from ape-like ancestors; using the science of cloning and embryonic stem cells to prolong human life; the evolutionary theory of aging; the genetic conflict theory of human pregnancy; the developmental basis of variation in human sexuality; and the direct effects of environment on modern human development and evolution.

Prerequisites: BIO 240/BIO 240L.

BIO 452 Population Ecology**Cross-listed with:** BIO 552**Credits 4 Grading Basis** Graded

Theoretical and applied aspects of distribution and abundance, population regulation, interactions between populations and conservation will be studied in selected organisms, including humans. Credit may not be earned in both BIO 452 and BIO 552.

Prerequisites: BIO 250/BIO 250L.

BIO 453 Microbial Ecology and Evolution**Cross-listed with:** BIO 553**Credits 3 Grading Basis** Graded

The ecology of microorganisms will be covered, including those important in human health and in natural environments. Emphasis will be placed on the study and critique of scientific literature. Credit may not be earned in both BIO 453 and 553.

Prerequisites: BIO 245/BIO 245L and BIO 250/BIO 250L.

BIO 454 Introduction to Biometrics**Cross-listed with:** MATH 354**Credits 4 Grading Basis** Graded

This course discusses the role of statistics in biological research and interpretation of biological phenomena. The course will cover topics of sampling, correlation, regression analysis, tests of hypotheses, commonly observed distributions in natural populations, nonparametric tests, goodness-of-fit tests and ANOVA. In order to fully comprehend the statistical analysis of those publications, students will review approximately half a dozen publications from different fields of biology. Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

BIO 455 Plant Physiology**Cross-listed with:** BIO 555**Credits 4 Grading Basis** Graded

An in-depth study of plant function including metabolism, water relations, transport phenomena, growth and development, physiological ecology, and responses to pathogens and abiotic stress. Credit may not be earned in both BIO 455 and BIO 555.

Prerequisites: BIO 240/BIO 240L and CHEM 132.

BIO 456 Ecological Models and Data**Credits 4 Grading Basis** Graded

Emerging issues in health and environmental sciences require an understanding of how to make inferences and decisions in the face of limited data and uncertainty. In the past few decades, advances in computational sciences have opened new ways of using and interpreting data. These advances are fundamentally changing how biologists approach statistics. This course introduces students to the principles of probability and likelihood that underpin these growing areas of statistics, and teaches students how to apply these principles to statistical computing. Examples come from ecological models and data and reflect general principles of working with discrete and continuous data, maximum likelihood and Bayesian estimation techniques, and using fundamental principles of probability to build and interpret mixed and mixture models. Work will be done using the open-source statistics program, R, and exercises will emphasize building skills by working with peers and by trial-and-error, both of which are key features of the open-source computing community.

Prerequisites: MATH 220 or MATH 229 or MATH 318; BIO 250; or permission of the instructor. Corequisites: BIO 250.

BIO 457 Biological Applications of Geographic Information Systems**Credits 4 Grading Basis** Graded

This course will explore the various ways that geographic information systems (GIS) can be used to answer biological questions. Students will use GIS software to study applications in ecology, conservation biology, and environmental biology. No prior GIS experience required.

Prerequisites: BIO 150/BIO 150L or BIO 250/BIO 250L.

BIO 459 Freshwater Ecology**Credits 4 Grading Basis** Graded

Functional relationships and productivity of freshwater communities are examined as they are affected by their physical, chemical and biotic environment. Organisms inhabiting lakes, ponds, rivers, streams and estuaries are studied at the population, community and ecosystem levels.

Prerequisites: BIO 250/BIO 250L and CHEM 132.

BIO 460 Plant Biotechnology**Cross-listed with:** BIO 560**Credits 4 Grading Basis** Graded

Theory and practice of growing isolated plant cells, tissues and organs.

Prerequisites: BIO 150/BIO 150L and CHEM 132.

BIO 467 Human Cardiovascular Physiology**Credits 3 Grading Basis** Graded

An in-depth look at the human cardiovascular system by means of advanced physiology with clinical applications and case-based learning. While the course is focused on the human heart and blood vessels, other body systems will be discussed in the context of their association and integration with the cardiovascular system.

Prerequisites: BIO 370 or minimum grade of "B-" in BIO 270/BIO 270L.

BIO 468 Endocrinology**Cross-listed with:** BIO 568**Credits 3 Grading Basis** Graded

Study of endocrine organs and mechanisms of action of hormones.

Students will be involved in interactive lectures and team-based learning activities to understand endocrine functions and disorders.

Prerequisites: BIO 240/BIO 240L or BIO 270/BIO 270L.

BIO 472 Human Metabolism**Credits 3 Grading Basis** Graded

This course will focus on the cellular physiological mechanisms responsible for regulation of normal human metabolism and place them in the context of the development of chronic disease processes.

Prerequisites: CHEM 241 and BIO 240/BIO 240L or permission of the instructor.

BIO 474 Reproductive Physiology**Credits 3 Grading Basis** Graded

This is an advanced-level vertebrate physiology course in which students will learn reproductive anatomy, physiology, development and behavior of a variety of organisms with humans serving as the foundational species. Innate and manipulated physiological mechanisms controlling reproduction will be examined at the molecular through organismal levels. Intended for students in organismal biology and health-related fields.

Prerequisites: BIO 370 or both BIO 270 and BIO 290.

BIO 477 The Genetics of Cancer**Cross-listed with:** BIO 577**Credits 3 Grading Basis** Graded

Exploration of the genetic and epigenetic factors that drive the evolution of cancer cells, taking into account both inherited and environmental contributions to this process. The cellular mechanisms debilitated or subverted during cancer development will be studied, and student teams will demonstrate their understanding of the material through the diagnosis, genetic characterization and treatment of a hypothetical cancer patient.

Prerequisites: BIO 240/BIO 240L.

BIO 478 Aging and Neurodegeneration**Credits 3 Grading Basis** Graded

This course will examine cellular mechanisms, genetics and environmental factors that underlie the onset and progression of age-related neurodegenerative diseases such as Alzheimer's, Parkinson's, Huntington's and Amyotrophic lateral sclerosis (ALS). This course will build on students' foundational knowledge of cell function, apoptosis and genetics to understand mechanisms that initiate and drive the proliferation neuronal cell death in age-related neurodegenerative diseases. In this course, students also study advancements in research and technology that have led to improved diagnosis and treatment of neurodegenerative disease.

Prerequisites: BIO 240/BIO 240L, and BIO 301 or BIO 370 or BIO 385/ PSYC 385 or BIO 445 or BIO 446.

BIO 480 Advanced Molecular Biology**Credits 3 Grading Basis** Graded

This course is a study of the history, basic principles and ethical considerations in the field of molecular biology. Students will learn about the structure and function of nucleic acids, proteins and other cellular macromolecules to appreciate the power and limitations of current resources and knowledge.

Prerequisites: CHEM 241 and a "C-" or better in BIO 240/BIO 240L.

BIO 480L Advanced Molecular Biology Lab**Credits 1 Grading Basis** Graded

This course is a study of the applied practices in the field of molecular biology. Students will participate in hands-on wet bench and computational experiments analyzing nucleic acids, proteins and other cellular macromolecules to appreciate the power and limitations of current resources and knowledge.

Prerequisites: BIO 480 and CHEM 241 and a "C-" or better in BIO 240/ BIO 240L. Corequisites: BIO 480.

BIO 481 Genomics**Cross-listed with:** BIO 581**Credits 4 Grading Basis** Graded

This advanced biology course is designed for students to learn about the structure and function of genomes, to develop facility in web-based tools and resources, and to appreciate the power and limitations of current resources and knowledge. Focus is on the biological questions that genomics can help to answer. Laboratory exercises will be sequencing and analyzing genomic DNA.

Prerequisites: BIO 240/BIO 240L.

BIO 482 Human Histology**Credits 4 Grading Basis** Graded

Students will examine microscopic structure of cells, tissues and major organ systems of the body. Basic anatomical and physiological function is presented to emphasize the histological significance of the examined organ systems.

Prerequisites: BIO 260, BIO 270/BIO 270L, BIO 290/BIO 290L, or BIO 370.

Credit may not be earned in both BIO 482 and BIO 582.

BIO 483 Bioinformatics**Cross-listed with:** BIO 583**Credits 4 Grading Basis** Graded

Focuses on building databases and computer programs to manage and analyze biological sequence data, and secondarily on theoretical aspects. The overall objective is to learn current information about the intersection of information science and biology, to develop facility in the many web-based tools and resources for further studies and research in genomics/ bioinformatics, and to appreciate the power and limitations of current resources and knowledge.

Prerequisites: BIO 240/BIO 240L.

BIO 484 Aquatic Microbiology**Credits 3 Grading Basis** Graded

This research-embedded course will introduce students to the techniques and experimental design currently used in aquatic microbial ecology through both literature and laboratory procedures. Students will participate in projects that are currently a component of the instructor's research program. BIO 484 will be taught concurrently with BIO 584.

Prerequisites: BIO 245/BIO 245L or equivalent. Students may not earn credit in both BIO 484 and BIO 584.

BIO 486 Systematics of Vascular Plants**Credits 4 Grading Basis** Graded

Study of systematic theory and an overview of the classification and evolution of higher plants, with particular attention to flowering plant families. Techniques for plant identification and collection and for construction of phylogenies will be taught in lab.

Prerequisites: BIO 150/BIO 150L or BIO 250/BIO 250L.

BIO 489 Topics in Science Pedagogical Content Knowledge**Cross-listed with:** MSSE 495**Credits 2 Grading Basis** Graded

Contributes to the further development of pre-service secondary science teachers' pedagogical content knowledge. Students will engage in education research to gain experience and develop instructional practices relevant to the Virginia Standards of Learning. To supplement their understanding of scientific concepts and the process of science, the focus will be on the nature of science encompassing literature from the history, philosophy, and sociology of science.

Prerequisites: BIO 150/BIO 150L.

BIO 490 Mechanics of Animal Movement**Credits 4 Grading Basis** Graded

The interactions of organisms with their physical environment. Concepts from fluid and solid mechanics are applied to biological form and function.

Prerequisites: BIO 150/BIO 150L.

BIO 491 Scientific Writing, Presentation and Critical Thinking**Credits 2 Grading Basis** Credit / No Credit

A discussion-based course for the development of the fundamental thinking, writing and presentation skills necessary to be a successful researcher. This course is required for all Biology Track I and II Honors students in their sophomore year, and is strongly recommended for all Biology majors who are intending to do research. Offered as credit/no credit only.

Prerequisites: BIO 140/BIO 140L.

BIO 492 Mentored Biology Course Assistant**Credits** 0-1 **Grading Basis** Credit / No Credit

Students are trained in the support and function of undergraduate biology courses. Students must contact and make arrangements with the supervising instructor in the term prior to registration. May be repeated when course content and/or student responsibilities within the course change. Offered as credit/no credit only.

Prerequisites: GPA of 2.5 or higher and permission of the instructor.

BIO 493 Pre-Veterinary Student Internship**Credits** 2 **Grading Basis** Credit / No Credit

Students are supervised by veterinarians and lab technicians in diagnostic lab activities at a regional animal health laboratory. A proposal and final presentation are required. Enrollment is limited to 1-2 individuals per term and students are advised to contact the Pre-Veterinary coordinator to be waitlisted. Offered as credit/no credit only. Prerequisites: Completion of BIO 240/BIO 240L, GPA of 2.5 or higher and permission of the instructor.

BIO 494 Internship in Biology**Credits** 1-2 **Grading Basis** Credit / No Credit

Students participate in research or applied biology outside of this university. Students must contact and obtain approval of a supervising instructor at the off campus location and with the department internship coordinator in the term prior to registration. A proposal must be approved prior to registration and a final paper or presentation will be completed. Offered as credit/no credit only.

Prerequisites: Biology or biotechnology major with a minimum of eight biology credit hours and a GPA of 2.5 or greater.

BIO 496 Biological Research**Credits** 0-1 **Grading Basis** Credit / No Credit

Students pursue a lab or field research project in a selected area of biology with a faculty research mentor. This could be research techniques (lab or field), literature review or both. Students must contact and make arrangements with a supervising instructor in the term prior to registration. This course may be repeated, and up to 2 credits may be used to partially satisfy one upper level laboratory requirement for the biology major. This course is offered as credit/no-credit only.

Prerequisites: GPA of 2.5 or greater.

BIO 497 Advanced Biological Research**Credits** 0-1 **Grading Basis** Credit / No Credit

Students pursue a lab or field research project in a selected area of biology with a faculty research mentor. Students must contact and make arrangements with a supervising instructor in the term prior to registration. Students must complete a proposal or form of scientific communication by the end of the semester, such as a conference presentation, Biosymposium presentation, lab presentation or research paper, at the discretion of the instructor. This course may be repeated, and may be used towards the combined 3 credits of research to satisfy one upper level lab requirement for the biology major. This course is offered as credit/no credit only.

Prerequisites: GPA of 2.5 or higher and permission of the instructor.

BIO 499A Senior Project**Credits** 2 **Grading Basis** Graded

Three semester courses taken as Parts A, B and C; 2 credits each. Expectations, requirements and prerequisites are defined in the department senior project policy. Students must contact and make arrangements with a supervising instructor in the term prior to registration.

BIO 499B Senior Project**Credits** 2 **Grading Basis** Graded

Three semester courses taken as Parts A, B and C; 2 credits each. Expectations, requirements and prerequisites are defined in the department senior project policy. Students must contact and make arrangements with a supervising instructor in the term prior to registration.

BIO 499C Senior Project**Credits** 2 **Grading Basis** Graded

Three semester courses taken as Parts A, B and C; 2 credits each. Expectations, requirements and prerequisites are defined in the department senior project policy. Students must contact and make arrangements with a supervising instructor in the term prior to registration.

Biotechnology (BIOT)

BIOT 260 Biotechnology Seminar**Credits** 1 **Grading Basis** Graded

An introduction to biotechnology. Topics will include research opportunities, careers, and current topics in biotechnology. Not available for biology major or minor credit.

BIOT 494 Internship in Biotechnology**Credits** 1-2 **Grading Basis** Graded

Students participate in research or applied biotechnology outside of the university. A proposal must be approved by the biotechnology program's internship coordinator or director prior to registration. A final deliverable, in a format deemed appropriate for the learning experience, will be submitted at the conclusion of the course.

Prerequisites: Biotechnology major with a minimum of eight credit hours and a GPA of 2.5 or higher with instructor permission.

BIOT 495 Techniques in Biotechnology**Credits** 1 **Grading Basis** Credit / No Credit

This course is designed to allow students to participate in a mentored research project with a Biotechnology faculty member. Students are trained in research theory and techniques related to biotechnology research. Students must contact a mentor who will supervise them in the course. Repeatable up to a maximum of six credit hours. Requires permission of supervising instructor and GPA greater than or equal to 2.5.

BIOT 496 Biotechnology Literature Research**Credits** 1 **Grading Basis** Credit / No Credit

This course is designed to allow students to participate in a mentored research project with a Biotechnology faculty member. Students pursue literature research in a selected area of biotechnology. Students must contact a mentor who will supervise them in the course. Repeatable up to a maximum of six credit hours. Requires permission of supervising instructor and GPA greater than or equal to 2.5.

BIOT 497 Research in Biotechnology**Credits** 1 **Grading Basis** Credit / No Credit

This course is designed to allow students to participate in a mentored research project with a Biotechnology faculty member. Students pursue a lab or field research project in a selected area of biotechnology. Students must write a 2-3 page proposal, approved no later than five weeks after the start of the course. For successful completion of BIOT 497, a professional presentation is required before graduation. Repeatable up to a maximum of six credit hours. Requires permission of supervising instructor and GPA greater than or equal to 2.5.

BIOT 499A Honors Capstone Project**Credits 1 Grading Basis** Credit / No Credit

This course is designed to allow students to participate in a mentored research project with a Biotechnology faculty member as part of the Honors program. Three semester courses taken as Parts A, B, and C: 1, 2, and 3 credits each, respectively. Expectations, requirements and prerequisites are defined by the honors program. Students must contact and make arrangements with a supervising instructor in the term prior to registration. Requirements are set by the Honors College.

BIOT 499B Honors Capstone Project**Credits 2 Grading Basis** Credit / No Credit

This course is designed to allow students to participate in a mentored research project with a Biotechnology faculty member as part of the Honors program. Three semester courses taken as Parts A, B, and C: 1, 2, and 3 credits each, respectively. Expectations, requirements and prerequisites are defined by the honors program. Students must contact and make arrangements with a supervising instructor in the term prior to registration. Requirements are set by the Honors College.

BIOT 499C Honors Capstone Project**Credits 3 Grading Basis** Credit / No Credit

This course is designed to allow students to participate in a mentored research project with a Biotechnology faculty member as part of the Honors program. Three semester courses taken as Parts A, B, and C: 1, 2, and 3 credits each, respectively. Expectations, requirements and prerequisites are defined by the honors program. Students must contact and make arrangements with a supervising instructor in the term prior to registration. Requirements are set by the Honors College.

Business (BUS)

BUS 160 Applied Critical Thinking in Business [C1CT]**Credits 3 Grading Basis** Graded

This course develops critical thinking abilities by exploring various business topics. Emphasis is placed on analyzing arguments, evaluating the credibility of sources and claims, and exploring multiple perspectives in business decision-making. Practical exercises and real-world scenarios are used to enhance critical thinking skills needed to effectively navigate today's diverse and complex business landscape. May be used for general education credit.

Prerequisites: Students who have earned credit for a Critical Thinking course must receive permission to enroll in an additional Critical Thinking course. You may not take BUS 160 if you have completed COB 300.

BUS 200 Personal Finance**Credits 3 Grading Basis** Graded

This course covers the fundamentals of personal finance, enabling the student to make better financial decisions both as a student and upon graduation. This course is not part of the finance major requirements and will not be considered a FIN elective.

Prerequisites: ECON 200 and either MATH 155, MATH 205, MATH 231 or MATH 235.

Business Analytics (BSAN)

BSAN 390 Statistical Modeling for Business Analytics**Credits 3 Grading Basis** Graded

This course integrates data analytics with advanced analytical methods from statistics using computer applications. The emphasis is on applying statistical modeling and analysis of data from a variety of business processes to support managerial decision-making and improved predictive capabilities.

Prerequisites: Declared BSAN minor or declared BSAN concentration in CIS major, COB 291 or equivalent with a minimum grade of "B-", and junior or senior standing.

BSAN 391 Decision Modeling for Business Analytics**Credits 3 Grading Basis** Graded

This course addresses a wide range of complex business decision problems through quantitative modeling using management science techniques with appropriate computer applications. Approaches include optimization and sensitivity analysis, multi-objective decision making and simulation risk analysis.

Prerequisites: Declared BSAN minor or declared BSAN concentration in CIS major, and BSAN 390 with a grade of "C" or better.

BSAN 393 Data Mining for Business Analytics**Cross-listed with:** CIS 393**Credits 3 Grading Basis** Graded

This course focuses on using statistical and machine learning techniques in computer applications that allow the extraction of useful, previously unrecognized information from large data sets for predictive purposes. By effectively sifting through databases such as those generated by many businesses, data mining allows the analyst to recognize potentially important patterns and clusters and to target business opportunities.

Prerequisites: Declared BSAN minor or declared BSAN concentration in CIS major, and BSAN 390 with a minimum grade of "C."

BSAN 490 Special Studies in Computer Information Systems or Business Analytics**Cross-listed with:** CIS 490**Credits 1-3 Grading Basis** Graded

An advanced course in information and/or business analytics designed to give qualified students an opportunity to complete independent study under faculty supervision.

Prerequisites: Senior standing, recommendation of the instructor and written approval of the department head prior to registration.

BSAN 498 Special Topics in Business Analytics**Credits 3 Grading Basis** Graded

An advanced course designed to allow exploration of current topics in business analytics. Course content will vary. See advisor for current content.

Prerequisites: Permission of the instructor.

Business Law (BLAW)

BLAW 360 Legal Analysis and Business Persuasion**Credits 3 Grading Basis** Graded

This course addresses the principles of professional argumentation and writing in the context of the business world and the legal services industry. First, students research and write an analytical legal memorandum. Second, students research and prepare a persuasive business proposal that advocates for one business acquisition over other potential targets. Finally, students learn important day-to-day professional analytical and drafting skills.

Prerequisites: COB 300.

BLAW 470 Securities Regulation**Credits 3 Grading Basis Graded**

This course addresses public and private enforcement of U.S. securities laws and regulations from the Great Depression through today. The course provides an overview of the regulation of securities offerings and trading under the Securities Act of 1933 and the Securities Exchange Act of 1934. The course explores the disclosure obligations related to the registration, distribution and trading of investment securities. Related to these issues, the course addresses registration exemptions, disclosure of forward-looking information, insider trading and corporate control acquisitions. The course will also examine the role that the Securities Exchange Commission and private plaintiffs play in shaping these regulations.

Prerequisites: COB 318 and junior standing.

BLAW 480 Estate Planning**Credits 3 Grading Basis Graded**

Legal, financial and practical considerations in the creation, management and conservation of an estate. Includes trust principles and practices; various estate planning tools, including wills, trusts, life insurance and powers of appointment; and pertinent income, estate and gift provisions. Prerequisites: COB 300 and COB 318. Corequisites: COB 300 and COB 318.

BLAW 494 White Collar Crime**Credits 3 Grading Basis Graded**

A study of white collar crime in America, a unique type of criminal activity that primarily affects businesses. The course explores the substance of white collar crime and focuses on the unique elements of various crimes through the study of actual cases. The course also examines how white collar crimes are prosecuted and defended in state and federal courts. Students are introduced to federal and state criminal procedure, substantive defenses, and the use of sentencing guidelines. Prerequisites: COB 300.

BLAW 495 Contract Law, Sales and Secured Transactions**Credits 3 Grading Basis Graded**

A study of the law of contracts, Article Two of the Uniform Commercial Code, product liability, legal liability of accountants, secured transactions and bankruptcy, with emphasis on the role these play in professional and personal decision making. The courts, the legislature and the interaction of these two branches of government in responding to a changing society are studied throughout.

Prerequisites: COB 318 and COB 300.

BLAW 496 The Law of Business Organizations, Negotiable Property Instruments and Property**Credits 3 Grading Basis Graded**

A study of Article Three of the Uniform Commercial Code, agency, partnerships, corporations, securities regulations, real property, trusts and decedents estates with emphasis on the role these play in professional and personal decision making.

Prerequisites: COB 318 and COB 300.

BLAW 497 Legal Aspects of International Business**Credits 3 Grading Basis Graded**

Survey of legal implications of international business dealings including foreign direct sales, distributorship arrangements, licensing of technology and legal aspects of the multi-national corporation. The foreign legal environment, relevant conventions and trade regulations, and the transnational reach of regulatory law will be considered.

Prerequisites: COB 318 and senior standing. Corequisites: COB 318 and senior standing.

BLAW 498 Special Topics in Business Law**Credits 3 Grading Basis Graded**

This course is designed to allow explorations of areas of current topics in business law. Course content will vary by semester and instructor. For current content, consult the advisor.

Prerequisites: COB 300.

CAL Internship Course (CAL)

CAL 495 Work-Based Learning Experience**Credits 1-3 Grading Basis Credit / No Credit**

This course will provide students credit for a directed, on-the-job learning experience in which the student applies learned skills and gains practical knowledge and experience. Course may be repeated for a maximum of three credit hours.

Chemistry (CHEM)

CHEM 111 Preparatory Skills for Chemistry**Credits 1 Grading Basis Graded**

This course offers an in-depth introduction to Chapters 1-3 of the General Chemistry I curriculum. It is designed for students who will enroll in CHEM 131 in the following semester. This flipped class offering will provide videos for students to watch in the morning, and live, virtual classes will be used to review topics of the day where students will work in small groups on daily worksheets. Students are expected to attend live classes every day.

Corequisites: CHEM 131.

CHEM 120 Concepts of Chemistry [C3PP]**Credits 3 Grading Basis Graded**

A one-semester introduction to the fundamental principles, laws and applications of chemistry. Examples relating to the health sciences are emphasized. Not available for major or minor credit in chemistry. May be used for general education credit.

CHEM 130 Metacognitive Activities for eXcellence in Chemistry - a companion course for CHEM 131**Credits 1 Grading Basis Graded**

This course is designed to be a sidecar course to CHEM 131 (General Chemistry I) to support student success and help develop lifelong learning skills. In this class, course content and assignments are all online. Realistically, to do well, students should be able to put in at least three hours per week for the eight weeks of this course.

Corequisites: CHEM 131.

CHEM 131 General Chemistry I [C3PP]**Credits 3 Grading Basis Graded**

The first of a two-course general chemistry sequence for science majors. It is designed to introduce students to basic chemical concepts including atomic structure, periodic properties of the elements, nomenclature, basic stoichiometry, theories related to reactivity and bonding, and the behavior of materials. May be used for general education credit.

Corequisites: CHEM 131L or CHEM 135L.

CHEM 131L General Chemistry Laboratory [C3L]**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement the CHEM 131 lecture course. The laboratory and lecture portions must be taken concurrently. Chemistry majors take CHEM 135L. May be used for general education credit.

Corequisites: CHEM 131.

CHEM 132 General Chemistry II**Credits 3 Grading Basis Graded**

A course designed to examine the mechanisms by which chemists obtain information about reacting systems. Major concepts covered include: chemical reactivity, chemical equilibrium, electrochemistry, thermodynamics and kinetics.

Prerequisites: Minimum grades of "C-" in CHEM 131 and either CHEM 131L or CHEM 135L. Corequisites: CHEM 132L or CHEM 136L.

CHEM 132L General Chemistry Laboratory**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement the CHEM 132 lecture course. The laboratory and lecture portions must be taken concurrently. Chemistry majors take CHEM 136L.

Prerequisites: Grades of "C-" or higher in CHEM 131 and either CHEM 131L or CHEM 135L.

CHEM 135L Special General Chemistry Laboratory I**Credits 1 Grading Basis Graded**

An enriched laboratory course designed primarily for chemistry majors. May be used for general education credit.

Corequisites: CHEM 131.

CHEM 136L Special General Chemistry Laboratory II**Credits 2 Grading Basis Graded**

An enriched laboratory course that includes special topics and experiments not presented in the regular CHEM 132 laboratory.

Prerequisites: Minimum grades of "C-" in CHEM 131, CHEM 132 and either CHEM 131L or CHEM 135L. Corequisites: CHEM 132.

CHEM 241 Organic Chemistry I**Credits 3 Grading Basis Graded**

The major objective for this course is to teach the modern method of scientific problem solving using organic compounds as models. Emphasis will be on the chemical language (nomenclature and terminology), molecular electronic concepts, and stereochemistry followed by a brief survey of chemical reactions/mechanisms. Functional groups studied include alkyl halides, alcohols, alkenes, alkynes, carbonyl additions, carboxylic acids and their derivatives as well as alpha-carbon chemistry leading to the ability to perform multi-step synthetic problems. Credit cannot be earned in both CHEM 241L and CHEM 242L.

Prerequisites: A minimum grade of "C-" in CHEM 132.

CHEM 241L Concepts of Organic Chemistry Laboratory**Credits 1 Grading Basis Graded**

Laboratory work will include training in the techniques of organic chemistry, preparation of compounds and some organic qualitative analysis. Credit cannot be earned in both CHEM 241L and CHEM 242L.

Prerequisites: CHEM 241 and CHEM 132L. Corequisites: CHEM 241.

Credit cannot be earned in both CHEM 241L and CHEM 242L.

CHEM 242 Organic Chemistry II**Credits 3 Grading Basis Graded**

The major objective for this course is to teach the modern method of scientific problem solving using organic compounds as models. Emphasis will be a more advanced study of chemical reactivity and mechanisms of the functional groups leading to applications in multi-step synthetic problems as well as structure elucidation of organic compounds. It is strongly recommended that students take CHEM 242L with CHEM 242 lecture.

Prerequisites: A minimum grade of "C-" in CHEM 241.

CHEM 242L Organic Chemistry Laboratory**Credits 2 Grading Basis Graded**

This course will present laboratory techniques and experiments associated with organic chemistry, including an introduction to synthesis, spectroscopic methods, chromatographic techniques and some qualitative organic analysis. Credit cannot be earned in both CHEM 241L and 242L.

Prerequisites: CHEM 132L and a minimum grade of "C-" in CHEM 241.

Corequisites: CHEM 242. Credit cannot be earned in both CHEM 241L and CHEM 242L.

CHEM 260 Concepts of Biochemistry**Credits 3 Grading Basis Graded**

A brief survey of the principal constituents of living cells, proteins, carbohydrates, lipids and nucleic acids, with emphasis on their synthesis and transformations in vivo. Intermediary metabolism and protein replication will be stressed. Not available for major or minor credit.

Prerequisites: CHEM 241 and either CHEM 241L or CHEM 242L.

CHEM 260L Concepts of Biochemistry Laboratory**Credits 1 Grading Basis Graded**

The laboratory work will comprise experiments demonstrating some of the pertinent reactions including those of analytical value.

Prerequisites: CHEM 260. Corequisites: CHEM 260.

CHEM 270 Inorganic Chemistry I**Credits 3 Grading Basis Graded**

A survey of the chemistry of the elements and modern theories of bonding.

Prerequisites: A minimum grade of "C-" in CHEM 132.

CHEM 280 An Alternative Lower-Division Chemistry Experience**Credits 1-4 Grading Basis Graded**

This course will provide a mechanism for offering a nontraditional, lower division, lecture and/or laboratory course. It will be offered only with the approval of the full-time teaching faculty. No course will be offered more than three times under the 280 designation. Students may repeat CHEM 280 for credit when course content changes.

CHEM 287L Integrated Inorganic/Organic Laboratory I**Credits 2 Grading Basis Graded**

This course provides an enriched, integrated introduction to the laboratory procedures associated with inorganic and organic chemistry. Topics include apparatus design and construction, synthesis, separation methods, spectroscopic analysis and application of computers in the laboratory.

Prerequisites: CHEM 241 and CHEM 132L. Corequisites: CHEM 241.

CHEM 288L Integrated Inorganic/Organic Laboratory II**Credits 2 Grading Basis Graded**

An enriched, integrated introduction to the laboratory procedures associated with inorganic and organic chemistry. Topics include apparatus design and construction, synthesis, separation methods, spectroscopic analysis and application of computers in the laboratory. Prerequisites: A minimum grade of "C-" in CHEM 241 and CHEM 270 and CHEM 287L. Corequisites: CHEM 270.

CHEM 315 Instructional Experiences in Chemistry**Credits 1 Grading Basis Graded**

Students are trained in the support and function of undergraduate chemistry courses. Students must contact and make arrangements with the supervising instructor in the term prior to registration. May be repeated when course content and/or student responsibilities within the course change.

CHEM 325 Chemical Hazards and Laboratory Safety**Credits 1 Grading Basis** Graded

A brief introduction to physical and chemical hazards which may be encountered in a laboratory setting. Methods of personal protection will be emphasized.

CHEM 331 Physical Chemistry I**Credits 3 Grading Basis** Graded

A study of thermodynamics, solutions, kinetics and macromolecules with applications of chemical and biological problems.

Prerequisites: CHEM 132, MATH 236 and PHYS 240.

CHEM 336L Applied Physical Chemistry Laboratory**Credits 2 Grading Basis** Graded

A laboratory course which emphasizes the applied experimental aspects of physical chemistry.

Prerequisites: CHEM 331. Corequisites: CHEM 331.

CHEM 351 Analytical Chemistry**Credits 4 Grading Basis** Graded

The total analysis concept is introduced and developed. This framework encompasses the areas of experiment design, sample collection and treatment, and statistical evaluation of results, as well as standard analysis techniques.

Prerequisites: CHEM 132.

CHEM 352 Instrumental Analysis**Credits 3 Grading Basis** Graded

This course emphasizes the application of instrumental techniques to the quantitative determination of chemical composition. Both instrument theory and practical applications are presented.

Prerequisites: CHEM 351 and MATH 235.

CHEM 352L Instrumental Analysis Laboratory**Credits 2 Grading Basis** Graded

This course will introduce students to the methodology and technology associated with the design and use of chemical instrumentation.

Students perform experiments that illustrate the theoretical principles associated with instrument designs and the application of instruments to the solution of qualitative and quantitative analysis problems.

Corequisites: CHEM 352.

CHEM 353 Environmental Chemistry**Credits 3 Grading Basis** Graded

An introduction to chemical pollution of water in the air, soil, groundwater, streams and drinking water from a chemist's perspective. Topics include sources and methods of assessment, fates and reactions, and transport of chemicals in the environment. This course does not include laboratory activities.

Prerequisites: A minimum grade of "C" in CHEM 241 or permission of the instructor.

CHEM 354 Environmental Chemistry Field Camp**Credits 3 Grading Basis** Graded

Fundamentals of environmental chemistry with laboratory and field trip components. The basic chemical principles of environmental problems are studied. Field trips and laboratory work on real samples are integrated with lecture material.

Prerequisites: CHEM 241 or permission of the instructor.

CHEM 355 Geochemistry of Natural Waters**Cross-listed with:** GEOL 355**Credits 3 Grading Basis** Graded

Study of chemical theory and reactions important in natural water systems. The role of atmospheric, geologic and biological inputs in determining the geochemistry of streams, rivers and oceans.

Prerequisites: CHEM 131 and CHEM 132 or equivalent.

CHEM 361 Biochemistry I**Cross-listed with:** BIO 361**Credits 3 Grading Basis** Graded

An introduction to the molecules and chemical reactions of living systems. Structure and function of important classes of biomolecules are explored and the relationship of structure to function is stressed. Basic metabolic sequences are discussed.

Prerequisites: Grade of "C" or higher in CHEM 241 and permission of the instructor. Completion of CHEM 242 is strongly recommended.

CHEM 362 Biochemistry II**Credits 3 Grading Basis** Graded

A continuation of CHEM 361 including metabolic regulation, protein biosynthesis, analytical methods and isolation of biomolecules.

Prerequisites: CHEM 361 or permission of the instructor.

CHEM 363 Biophysical Chemistry**Credits 3 Grading Basis** Graded

The study of structure-function relationships in biological molecules as well as the theory and applications of biophysical methods used for the analysis of proteins and nucleic acids. Topics may include analytical ultracentrifugation, light scattering, fluorescence, calorimetry, surface plasmon resonance, single molecule approaches, X-ray diffraction, NMR, TEM, AFM, structure prediction, and computational simulations.

Prerequisites: A minimum grade of "C" in CHEM 361.

CHEM 366L Biochemistry Laboratory**Credits 2 Grading Basis** Graded

An introduction to laboratory techniques and experimental approaches associated with modern biochemistry. Isolation and characterization of enzymes and other biomolecules are emphasized.

Prerequisites: CHEM 361 and either CHEM 241L or CHEM 242L or CHEM 287L.

CHEM 367L Biochemistry Laboratory**Credits 2 Grading Basis** Graded

An introduction to laboratory techniques and experimental approaches associated with modern biochemistry. Isolation and characterization of proteins is emphasized.

Prerequisites: Restricted to majors in Chemistry or Biophysical Chemistry, or by permission of the instructor.

CHEM 368L Biophysical Chemistry Laboratory**Credits 2 Grading Basis** Graded

An introduction to experimental approaches used to characterize proteins. Techniques may include NMR, UV-Vis, AFM, circular dichroism and infrared spectroscopy.

Prerequisites: Restricted to majors in Chemistry or Biophysical Chemistry, or by permission of the instructor.

CHEM 375 An Introduction to Materials Science**Cross-listed with:** MATS 375, PHYS 375**Credits 3 Grading Basis** Graded

An introduction to materials science with emphasis on general properties of materials. Topics will include crystal structure, extended and point defects, and mechanical, electrical, thermal and magnetic properties of metals, ceramics, electronic materials, composites and organic materials. Prerequisites: CHEM 131 and PHYS 150 or PHYS 250 or ISAT 212 or permission of the instructor.

CHEM 390 Problems in Chemistry**Credits 1-3 Grading Basis** Graded

A project is undertaken dealing with some aspect of chemistry under the guidance of a faculty advisor. Repeatable for a total of 4 credits.

CHEM 395 Perspectives in Chemistry**Credits 1 Grading Basis** Graded

A description of the technical and nontechnical capabilities expected of a university graduate who enters industry, government or academia is presented. The student is introduced to the various laws governing the chemical industry as well as to the fields of toxicology and environmental health. Experts in various disciplines discuss current topics of concern to the chemistry and biology student.

CHEM 432 Physical Chemistry II**Credits 3 Grading Basis** Graded

A study of atomic and molecular energy levels and structure as interpreted by quantum theory.

Prerequisites: CHEM 132 and MATH 236 and PHYS 250.

CHEM 438L Physical Chemistry Laboratory**Credits 2 Grading Basis** Graded

A laboratory course which emphasizes the application of various physical measurement techniques as a means of obtaining data to test fundamental chemical theory.

Corequisites: CHEM 432.

CHEM 440 Intermediate Organic Chemistry**Credits 3 Grading Basis** Graded

An advanced study of the theory of organic chemistry as applied to chemical reactions and synthetic methods. Such topics as reaction mechanisms, spectroscopy and stereochemistry will be included.

Prerequisites: CHEM 242.

CHEM 445 Polymer Chemistry**Credits 3 Grading Basis** Graded

A study of the synthesis and characterization of macromolecules. Polymer chemistry is discussed in a manner that focuses most attention on the properties of macromolecules that can be understood at the molecular level.

Prerequisites: CHEM 242.

CHEM 445L Polymer Chem Lab**Credits 1 Grading Basis** Graded

A laboratory course focused on the synthesis and characterization of macromolecules. The experiments focus most on the properties of macromolecules so the materials can be understood at the molecular level.

Prerequisites: CHEM 242. Corequisites: CHEM 445.

CHEM 450 Nuclear and Radiation Chemistry**Credits 3 Grading Basis** Graded

A study of the fundamentals of radioactivity in chemistry. Topics include the effects of radiation on matter, measurement of radiation, activation analysis, tracer studies and the nuclear fuel cycle. Applications of radioactive materials and radiation in industry and medicine will be described.

Prerequisites: CHEM 132 and PHYS 250 or permission of the instructor.

CHEM 450L Laboratory for Nuclear and Radiation Chemistry**Credits 1 Grading Basis** Graded

A laboratory course designed to demonstrate the topics covered in CHEM 450.

Prerequisites: CHEM 132 and PHYS 250 or permission of the instructor.

Corequisites: CHEM 450.

CHEM 455 Lasers and Their Applications to Physical Sciences**Cross-listed with:** PHYS 455**Credits 3 Grading Basis** Graded

An introduction to both the theoretical and practical aspects of lasers and their applications in the physical sciences.

Prerequisites: PHYS 270/PHYS 270L, CHEM 331, or permission of the instructor.

CHEM 470 Inorganic Chemistry II**Credits 3 Grading Basis** Graded

A study of selected topics in the field of advanced inorganic chemistry.

Prerequisites: A minimum grade of "C-" in CHEM 270 and CHEM 331.

Corequisites: CHEM 331.

CHEM 480 Selected Topics in Chemistry**Credits 1-4 Grading Basis** Graded

This course is designed to allow an in-depth study of specific topics in chemistry selected according to student and faculty interests.

CHEM 481 Literature and Seminar I**Credits 1 Grading Basis** Graded

Provides instruction in methods of abstracting specific information from the body of chemical literature. Attendance at regularly scheduled department seminars is required.

Prerequisites: CHEM 242 or CHEM 288L.

CHEM 482 Literature and Seminar II**Credits 1 Grading Basis** Graded

Provides practice in preparing and presenting a literature-based seminar and paper on a chemical topic. Attendance at regularly scheduled department seminars is required.

Prerequisites: CHEM 481 or permission of the instructor.

CHEM 485 Science of the Small**Cross-listed with:** PHYS 485**Credits 3 Grading Basis** Graded

The chemistry and physics of materials of the nanometer scale will be explored. Lectures will cover contemporary issues in surfaces, self-assembly and low dimensional materials. The intersection of biology, chemistry, engineering and physics will be explored through nanoscience.

Prerequisites: CHEM 132 and MATH 236; and PHYS 150 or PHYS 250.

Corequisites: PHYS 150 or PHYS 250.

CHEM 485L Science of the Small Laboratory**Cross-listed with:** PHYS 485L**Credits 1 Grading Basis** Graded

The chemistry and physics of materials of the nanometer scale will be explored. Labs will cover contemporary issues in surfaces, self-assembly and low dimensional materials. The intersection of biology, chemistry, engineering and physics will be explored through nanoscience.

Prerequisites: CHEM 132 and MATH 236; and PHYS 150 or PHYS 250.

Corequisites: PHYS 485; PHYS 150 or PHYS 250.

CHEM 494 Internship in Chemistry**Credits 1-2 Grading Basis** Graded

Students participate in research or applied chemistry outside of the university. A proposal must be approved prior to registration, and a final paper will be completed.

CHEM 497 Undergraduate Chemical Research**Credits 1-3 Grading Basis** Graded

Research in a selected area of chemistry, as arranged with and approved by a faculty research advisor the semester prior to registration.

Repeatable up to a total of six credits.

CHEM 499A Honors**Credits 1 Grading Basis** Graded**CHEM 499B** Honors**Credits 2 Grading Basis** Graded**CHEM 499C** Honors**Credits 3 Grading Basis** Graded

Chinese (CHIN)

CHIN 101 Elementary Chinese I**Credits 3 Grading Basis** Graded

This course covers the fundamentals of Mandarin Chinese through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

CHIN 102 Elementary Chinese II**Credits 3 Grading Basis** Graded

This course builds on the foundations established in CHIN 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension.

Prerequisites: CHIN 101 or by permission of the instructor.

CHIN 111 Intensive Chinese I**Credits 6 Grading Basis** Graded

The fundamentals of Chinese through listening, speaking and writing. The course is the equivalent of CHIN 101-CHIN 102.

CHIN 212 Intensive Chinese II**Credits 6 Grading Basis** Graded

The fundamentals of Chinese through intensive listening, speaking, reading and writing at the intermediate level. This four-week course is the equivalent of CHIN 231-CHIN 232.

Prerequisites: CHIN 102 or CHIN 111 or permission of the instructor.

CHIN 231 Intermediate Chinese I**Credits 3 Grading Basis** Graded

A more in-depth study of grammar, vocabulary building, conversation and reading. Includes an introduction to composition.

Prerequisites: CHIN 102, CHIN 111, or permission of the instructor.

CHIN 232 Intermediate Chinese II**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.

Prerequisites: CHIN 231 or permission of the instructor.

CHIN 300 Language for Living**Credits 3 Grading Basis** Graded

This course focuses on enhancing students' practical communication skills in Chinese, preparing them for everyday interactions in personal, academic and professional contexts. Building on the foundations established in CHIN 232, this course emphasizes real-life language use through topics related to daily life, social events and work. Students will engage in listening, speaking, reading and writing activities designed to improve fluency and cultural understanding. The course prepares students to navigate various real-world situations while deepening their knowledge of Chinese social customs and language norms. Instruction is in Chinese.

Prerequisites: CHIN 212 or CHIN 232 or permission of the instructor.

CHIN 301 Pop Culture through Language**Credits 3 Grading Basis** Graded

This course explores Chinese language and culture through various forms of popular media, including wechat, tik-tok, video games, movies, TV series and Chinese pop music. Students will engage with authentic materials to enhance their understanding of contemporary Chinese language while gaining insights into modern Chinese society and cultural trends. By analyzing these media, students will expand their vocabulary, improve listening comprehension, and develop their ability to critically evaluate cultural products. The course is designed for upper-beginner to lower-intermediate students looking to apply their language skills in real-world, culturally relevant contexts. Through discussions, presentations and creative projects, students will strengthen both their linguistic abilities and their cultural literacy. Instruction is in Chinese.

Prerequisites: CHIN 212 or CHIN 232 or permission of the instructor.

CHIN 307 Chinese Culture and Civilization**Credits 3 Grading Basis** Graded

This course offers a general history of China from the earliest records of Chinese civilization through 1911, end of the Chinese dynasties. The course is introductory, has no prerequisites, and assumes no prior knowledge of China or its language. The course approaches China from a wide range of viewpoints, focusing on political, economic, social, religious, philosophical and artistic developments. Instruction is in English.

CHIN 320 Culture and Communication in Context**Credits 3 Grading Basis** Graded

This Chinese course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do Chinese-speaking peoples communicate on subjects that are at the heart of their societies?" Instruction is in Chinese.

Prerequisites: CHIN 212 or CHIN 232 or permission of the instructor.

CHIN 322 Language Landscapes**Credits 3 Grading Basis** Graded

This Chinese course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in these culturally and linguistically defined areas and explores the diverse cultural landscapes that shape their lives. Instruction is in Chinese.

Prerequisites: CHIN 212 or CHIN 232 or permission of the instructor.

CHIN 330 Business Chinese**Credits 3 Grading Basis** Graded

This course is designed for learners who already have a foundational understanding of Mandarin and want to expand their skills in a professional and business context. It bridges the gap between conversational fluency and workplace proficiency, focusing on vocabulary, cultural norms and communication strategies essential for doing business in Chinese-speaking environments.

Prerequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322; or by permission of the instructor. Corequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322.

CHIN 375 Business and Society in China**Credits 3 Grading Basis** Graded

The course is an introduction to business and society in China, the world's second largest economy. The course explores the development of Chinese society in business and economic contexts. It provides students with an understanding of the geography, society, culture and ideology in China. The course is conducted in a research-based approach, through lectures, readings, presentations and discussions. Students will learn the fundamental bases of Chinese society from a historical and global perspective. They will also achieve a better understanding of business activities in China.

Prerequisites: CHIN 232 or permission of the instructor.

CHIN 400 Advanced Chinese**Credits 3 Grading Basis** Graded

The major goal of this course is to help students further develop proficiency in reading and writing based on their competence in listening and speaking. Students are expected to appropriately express their ideas on a wide range of topics and demonstrate reading competence in Mandarin.

Prerequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322; or by permission of the instructor. Corequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322.

CHIN 435 Translation Strategies**Credits 3 Grading Basis** Graded

This course addresses theoretical strategies of translation and applies them in practical contexts using internet-based language resources on a variety of topics to improve students' knowledge of Chinese language and translation strategies.

Prerequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322; or by permission of the instructor. Corequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322.

CHIN 446 Special Topics in Chinese Literature**Credits 3 Grading Basis** Graded

This course will examine a particular topic in Chinese literature. It may cover all or a specific Chinese literature genre. Course may be repeated if content varies.

Prerequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322; or by permission of the instructor. Corequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322.

CHIN 447 Special Topics in Chinese Culture**Credits 3 Grading Basis** Graded

Students will study a particular topic in the culture of Chinese-speaking countries. This course may be repeated if content varies.

Prerequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322; or by permission of the instructor. Corequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322.

CHIN 448 Special Topics in Chinese Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in Chinese linguistics. Topics could include an introduction to Chinese sociolinguistics or psycholinguistics. This course may be repeated if content varies.

Prerequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322; or by permission of the instructor. Corequisites: CHIN 300, CHIN 301, CHIN 320 or CHIN 322.

CHIN 490 Special Studies in Chinese**Credits 1-4 Grading Basis** Graded

Special topics or independent studies in Chinese.

Civic Leadership (CIV)

CIV 150 Mechanics of Democracy for Civic Leadership**Credits 3 Grading Basis** Credit / No Credit

Students will examine how democratic systems function (and sometimes falter) at the local, state and national levels. Through a combination of seminar discussions, deliberation exercises, guest speakers and experiential learning opportunities - including field trips to Richmond and Washington, DC, local government meetings and community forums - students will explore the real-world mechanics that civic leaders must navigate to make democracy work. As part of a residential learning community, students will engage in collaborative projects, practice deliberative dialogue in their residence halls and develop the civic skills necessary for meaningful participation in American democracy. By the end of the semester, students will not only understand the machinery of democratic governance but also see themselves as active contributors to its ongoing maintenance and improvement.

CIV 300 Civic Leadership: Theories and Practice**Credits 3 Grading Basis** Graded

Addressing growing cynicism and distrust in government today calls for active citizen civic leaders. Framed and guided by James Madison's legacy, civic leadership involves learning key skills to foster collaboration across differences, embracing complexity in decision making and in dialogue with others, and valuing pragmatic solutions that advance the common good. In many ways, this is the antithesis of what we see from contemporary national political leaders, making this class important for students wanting to engage deeply and meaningfully improve the civic health and participation of their communities.

CIV 400 Civic Leadership Internship**Credits 3-6 Grading Basis** Graded

Working with the civic leadership minor coordinator and a faculty advisor, students will complete a 3-6 credit civically engaged leadership internship and report. The internship is designed to provide an academic context for performing and analyzing off-campus work in government, public service, and non-profit settings. Utilizing knowledge, theories, dialogue, and collaboration methods and skills gained in CIV 300, students will assess and reflect on their internship experience in real time by writing weekly structured reflections.

Prerequisites: CIV 300 and permission of the instructor. Corequisites: CIV 300.

CIV 401 Civic Leadership Project**Credits 3-6 Grading Basis** Graded

Working with the civic leadership minor coordinator and a faculty advisor, students will complete a 3-6 credit civically engaged research or creative project. This course is designed to provide an academic context for conducting academic research or developing a creative project in civic leadership. Utilizing knowledge, theories, dialogue, and collaboration methods and skills gained in CIV 300, students will design and complete a supervised research project or creative project.

Prerequisites: CIV 300 and permission of the instructor. Corequisites: CIV 300.

Classics (CLAS)

CLAS 100 Latin and Greek Roots of English Words**Credits 3 Grading Basis** Graded

Intensive study of Latin and Greek word-roots, prefixes and suffixes in the forms they take in English words. An English vocabulary-development course for students with no knowledge of Latin or Greek. Does not count toward licensure in Latin.

CLAS 266 Greek and Roman Classics in Translation**Credits 3 Grading Basis** Graded

Discussion of the writings that illustrate the cultural values and intellectual attitudes which constitute the most important legacy of Classical civilization. All readings are in English.

CLAS 338 Roman Religion**Credits 3 Grading Basis** Graded

Roman religion will treat the history and nature of Roman religion from its earliest manifestations in the early Republic through the rise of Christianity and its legalization in A.D. 313 and end in the fifth century AD after an examination of the interactions of paganism and Christianity. It will address the great diversity of pagan cults, the cross-cultural interaction across the Mediterranean and beyond and the variety of cultic practice and thought.

CLAS 360 Topics in Greek and Roman Culture**Credits 3 Grading Basis** Graded

A study of selected topics in the culture of Ancient Greece and Rome. May be repeated for credit with change of topics.

CLAS 490 Special Studies in Classics**Credits 1-4 Grading Basis** Graded

Special topics or independent studies in classics.

College of Business (COB)

COB 110 First-Year Success Seminar**Credits 1 Grading Basis** Graded

This course is designed to assist first year students with the transition from the high school classroom to the college classroom by developing a better understanding of the learning process while acquiring essential academic skills. Conducted in a seminar setting, this course provides a support group by discussing concerns confronting first year students. Throughout the semester students will engage with various guest speakers to learn about available resources and explore business career paths.

Prerequisites: Declared business major; first year students only.

COB 111 Business Learning Community Seminar**Credits 1 Grading Basis** Credit / No Credit

This course introduces students to business careers, academic course work in the James Madison University College of Business, and faculty, staff, and alumni who offer insights for success. Students will learn what is expected to achieve academic success and succeeding in the business world. Students will also engage in the JMU environment and develop leadership and civic responsibility skills. Students will reflect on learned concepts through written and oral assignments, guest lectures, and service learning.

Prerequisites: MadisonBiz Residential Learning Community students only.

COB 191 Business Analytics I**Credits 3 Grading Basis** Graded

The application of data analysis and statistical methods in business. Introduces data visualization, descriptive statistics, probability, sampling, estimation and hypothesis testing. Emphasis is on using spreadsheet tools and functions for statistical analysis to support analytical decision-making in a business environment.

Prerequisites: One of the following: MATH 155, MATH 156, MATH 205, MATH 231, MATH 235, ISAT 151, or sufficient score on the mathematics placement exam. Max of 2 attempts in COB 191.

COB 202 Interpersonal Skills**Credits 3 Grading Basis** Graded

An applied course consisting of experiential exercises followed by class discussion. Cases are used as learning activities where the instructor acts as a facilitator to learning. Essential theory emanates from class discussions with a student-based rather than instructor-based format. Theory and application are intertwined by means of student self-assessment exercises and group discussion.

Prerequisites: Open only to sophomore business majors. Maximum of 2 attempts for COB 202.

COB 204 Computer Information Systems**Credits 3 Grading Basis** Graded

An introduction to computer-based information systems for students with a major in the COB or a CIS minor. Emphasis is placed on the role of computers in business and society, computer hardware and software, analysis, design and implementation of information systems, computer ethics, and collaboration using computers. Students will create databases and collaborate using computer-based tools.

Prerequisites: Maximum of 2 attempts for COB 204.

COB 241 Financial Accounting**Credits 3 Grading Basis** Graded

This course examines the role of financial data in contemporary society; the problems of measuring and reporting income, assets, liabilities and equities; and the interpretation of financial statements.

Prerequisites: Sophomore standing and declared business major. Maximum of 2 attempts for COB 241.

COB 242 Managerial Accounting**Credits 3 Grading Basis** Graded

The attention-directing and problem-solving functions of accounting in relation to current planning and control, evaluation of performance, special decisions and long-range planning.

Prerequisites: COB 241. Maximum 2 attempts of COB 242.

COB 291 Business Analytics II**Credits 3 Grading Basis Graded**

The application of quantitative modeling and analysis to managerial decision-making. Introduces techniques from predictive and prescriptive analytics including linear optimization, decision analysis, queuing, regression, timeseries forecasting, and simulation. Emphasis is on implementing, solving, and interpreting results from spreadsheet models for business applications.

Prerequisites: COB 191 or MATH 220 and MATH 205 or equivalent.

Maximum of 2 attempts for COB 291.

COB 300A Integrative Business: Management**Credits 3 Grading Basis Graded**

COB 300A is the management component of an integrated learning experience consisting of four courses, taken concurrently, which introduces the fundamental conceptual tools of management, finance, operation and marketing in such a way as to establish their mutual relevance and interdependence. Students work in small project teams on tasks designed to require the application in concert of conceptual tools from each of the function areas.

Prerequisites: Completion of all required 100- and 200-level B.B.A. core courses, junior standing, and formal admission to the College of Business.

COB 300B Integrative Business: Finance**Credits 3 Grading Basis Graded**

COB 300B is the finance component of an integrated learning experience consisting of four courses, taken concurrently, which introduces the fundamental conceptual tools of management, finance, operation and marketing in such a way as to establish their mutual relevance and interdependence. Students work in small project teams on tasks designed to require the application in concert of conceptual tools from each of the function areas.

Prerequisites: Completion of all required 100- and 200-level B.B.A. core courses, junior standing, and formal admission to the College of Business.

COB 300C Integrative Business: Operations**Credits 3 Grading Basis Graded**

COB 300C is the operations component of an integrated learning experience consisting of four courses, taken concurrently, which introduces the fundamental conceptual tools of management, finance, operation and marketing in such a way as to establish their mutual relevance and interdependence. Students work in small project teams on tasks designed to require the application in concert of conceptual tools from each of the function areas.

Prerequisites: Completion of all required 100- and 200-level B.B.A. core courses, junior standing, and formal admission to the College of Business.

COB 300D Integrative Business: Marketing**Credits 3 Grading Basis Graded**

COB 300D is the marketing component of an integrated learning experience consisting of four courses, taken concurrently, which introduces the fundamental conceptual tools of management, finance, operation and marketing in such a way as to establish their mutual relevance and interdependence. Students work in small project teams on tasks designed to require the application in concert of conceptual tools from each of the function areas.

Prerequisites: Completion of all required 100- and 200-level B.B.A. core courses, junior standing, and formal admission to the College of Business.

COB 301 European Integration, Culture and History**Credits 3 Grading Basis Graded**

This course is designed to complement the COB 300 A-D and will only be offered as part of the semester in Antwerp program. Students will study European integration in the classroom and visit governmental institutions, historical places and cultural events associated with course content.

Prerequisites: Requires acceptance to the semester in Antwerp program. Cannot be used as an elective to fulfill any major or minor in COB.

Corequisites: Must be taken with COB 300.

COB 318 Legal and Ethical Environment of Business**Credits 3 Grading Basis Graded**

A study of the law as a means of social, political and economic change.

The course focuses on the law in positive leadership development and organizational success. The course has two main goals: To develop leaders who are able to achieve success by understanding the legal and ethical aspects of their business responsibilities and to show how organizations can achieve competitive advantage by reducing legal risk and using the law to create economic value, while also encouraging responsible conduct. The course examines legal and ethical issues that every businessperson should understand to make sound business decisions in a global environment. It provides an overview of the legal system, the nature and uses of law, ethical decision-making frameworks, business organizations and agency, contract law, torts and the law of product liability, criminal law, the regulatory environment, intellectual property, employment law, securities law, and mergers and acquisitions law.

Prerequisites: COB 300. Corequisites: COB 300.

COB 487 Strategic Management**Credits 3 Grading Basis Graded**

Strategic management is designed to be the capstone course for seniors completing their undergraduate studies in the various functional areas of business administration. The course is comprehensive and structured to build on the foundational knowledge students have gained through completing the interdisciplinary COB 300, Integrated Functional Systems, learning experience and from their respective concentrations.

Prerequisites: COB 300, COB 318, senior standing and completion of one full academic semester after completing COB 300.

COB 490 Special Studies in Business Administration**Credits 1-3 Grading Basis Graded**

Designed to give capable students an opportunity to complete a faculty supervised independent study apart from a specific major.

Prerequisites: Permission from the Associate Dean for Academic Affairs.

College of Integrated Science and Engineering (CISE)

CISE 401 Applied Ethics for Environmental Stewardship**Credits 3 Grading Basis Graded**

This course will apply ethical reasoning to questions about the environment and our place in the natural world. Students will study a variety of ethical theories and consider the views of thinkers from the past and contemporary society. Topics will include animal rights, conservation and use of natural resources, and environmental justice with respect to pollution and climate change.

CISE 460 Geospatial Intelligence Certificate Program Capstone Project
Credits 3 Grading Basis Graded

This is a senior-level course in which students demonstrate how the geospatial knowledge and skills learned from earlier courses can be applied to a national security problem. Key elements of the capstone project include: preparation of a problem statement (intelligence question); identifying sources of information; applying technical, communications and analytic principles; understanding situational conditions; preparing actionable recommendations for decision makers; and communicating findings in a professional manner. Repeatable for a maximum of six credit hours.

CISE 480 Special Topics in Integrated Science and Engineering
Credits 1-6 Grading Basis Graded

This is a special topics course designator that can be used for any of a variety of topics that cut across disciplines within the broad realm of integrated science and engineering. Requisites may vary by section.

CISE 481 Robotics Project Summary
Credits 1 Grading Basis Graded

Students will reflect upon their experience gained in a prerequisite supervised robotics project course, a robotics internship, a robotics research project or an equivalent experience. Results will be disseminated to the academic community via a presentation, a report and/or other means.

College of Science and Mathematics (CSM)

CSM 199 Science Outreach in Schools
Credits 1 Grading Basis Credit / No Credit

A service learning course in which students engage in outreach activities in area high schools. Emphasis is placed on planning and implementing effective tutoring or enrichment activities as well as reflection and assessment of outcomes.

CSM 200 Exploration of Topics in STEM
Credits 1-3 Grading Basis Graded

This course provides introduction to and exploration of a variety of STEM-related topics. Course content will vary and could include introduction to STEM curriculum, academic planning, time management and study skills, stress and anxiety management, campus student resources, undergraduate research topics, graduate school considerations, and discussion of STEM career options. May be repeated for credit as course content changes. Students should consult the instructor prior to enrolling for this course. Course provides university elective credit.

CSM 299 General Instructional Methods for Science in Grades 6-12
Credits 2 Grading Basis Graded

This general teaching methods course provides a solid foundation for planning curriculum and instruction for middle and high school students. Topics include engaging adolescent learners, lesson design, instructional strategies and assessment models. Students in this course must have a science major and expressed a strong interest in becoming a secondary science teacher.

Prerequisites: PHYS 397, Topic: Science Pedagogy.

Communication Sciences and Disorders (CSD)

CSD 200 Introduction to Communication Sciences and Disorders
Credits 3 Grading Basis Graded

This course is an introduction to human communication. It surveys mechanisms of communication and clinical populations served by audiologists, speech-language pathologists and relevant professionals such as educators and neuropsychologists. Roles and responsibilities of audiologists and speech-language pathologists are discussed.

CSD 207 Phonetics
Credits 3 Grading Basis Graded

This course is an introduction to the articulation of speech sounds, techniques for speech transcription, including basic transcription skills in International Phonetic Alphabet (IPA), and considerations of phonetic and phonemic systems.

Prerequisites: CSD majors only.

CSD 208 Anatomy and Physiology of the Speech and Hearing Mechanism
Credits 3 Grading Basis Graded

This course offers a detailed study of the anatomy and physiology related to communication sciences and disorders.

Prerequisites: CSD majors only.

CSD 209 Speech and Hearing Science
Credits 3 Grading Basis Graded

This course provides an introduction to the acoustics of speech and hearing. It is also an introduction to physical acoustics, sound generation and transmission, resonance, speech acoustics, and speech perception.

Prerequisites: CSD majors only.

CSD 300 Language Development
Credits 3 Grading Basis Graded

This course is the study of language acquisition, development, structure and function in typically developing children.

Prerequisites: CSD majors only and junior standing;

CSD 207, CSD 208 and CSD 209; or departmental permission.

CSD 301 Audiology
Credits 3 Grading Basis Graded

This course provides an introduction to the symptoms, causes and treatment of hearing disorders. Hearing test instrumentation and interpretation in clinical situations are emphasized.

Prerequisites: CSD majors only and junior standing;

CSD 207, CSD 208 and CSD 209; or departmental permission.

CSD 314 Pediatric Speech and Language Disorders
Credits 3 Grading Basis Graded

This course is an introduction to speech and language disorders in children. An overview of assessment and treatment is presented.

Prerequisites: CSD 300 and junior standing; or departmental permission.

CSD 318 Aural Rehabilitation
Credits 3 Grading Basis Graded

This course offers a focused study on the impact of hearing loss on human communication across the lifespan.

Prerequisites: CSD 301 and junior standing; or departmental permission.

CSD 400 Topics in Speech Production and Perception: Implications for Clinical Populations

Credits 2-3 Grading Basis Graded

This course is a focused study of topics in speech production and perception as related to communication sciences and disorders. The underlying mechanisms associated with these functions for human communication are surveyed and examined with an emphasis on the effects of communicative disorders. The interdisciplinary nature of communication sciences and disorders is highlighted. This course is relevant to a range of fields of study including (but not limited to) speech-language pathology, audiology, speech, language, hearing sciences, psychology, linguistics, health sciences, nursing and occupational therapy.

Prerequisites: PSYC 101, CSD 207, CSD 209 and MATH 220, or permission of the instructor.

CSD 412 Professional Issues and Person-Centered Care in Communication Sciences and Disorders

Credits 3 Grading Basis Graded

This course addresses professional issues including professional conduct, evidence-based practice, ethics, social determinants of health, and considerations for person-, family-, and/or care-partner centered care. Prerequisites: CSD 200, CSD 207, CSD 300, CSD 314 and senior standing; or departmental permission.

CSD 415 Neuroanatomy and Neurogenic Communication Disorders

Cross-listed with: BIO 415

Credits 3 Grading Basis Graded

This course is an introduction to neuroanatomy with emphasis on neurogenic communication disorders.

Prerequisites: CSD 208 and senior standing, or departmental permission.

CSD 416 Adult Neurogenic Communication Disorders

Credits 3 Grading Basis Graded

This course addresses fundamental considerations for communication and swallowing disorders of acquired neurogenic etiology. Evaluation methods and treatment strategies are explored.

Prerequisites: CSD 208, CSD 415 and senior standing; or departmental permission.

CSD 420 Introduction to Sign Language

Credits 3 Grading Basis Graded

Provides an introduction to American Sign Language, the deaf community and English-based signed systems.

CSD 421 Sign Language II

Credits 3 Grading Basis Graded

Focuses on developing conversational skills of students whose core vocabulary and knowledge of the grammar and pragmatics of sign language are basic; distinguishes ASL from English sign systems.

Prerequisites: CSD 420 or permission of instructor.

CSD 470 Methods and Observation in Speech Language Pathology

Credits 3 Grading Basis Graded

This course is an introduction to the clinical process in speech-language pathology via guided observation and participation in laboratory and professional clinical interactions. Practical clinical methodologies are emphasized.

Prerequisites: CSD majors only; CSD 314 and senior standing; or departmental permission.

CSD 471 Methods and Observation in Audiology

Credits 3 Grading Basis Graded

This course is an introduction to the clinical process in audiology via guided observation and participation in laboratory and professional clinical interactions. Practical clinical methodologies are emphasized. Prerequisites: CSD majors only; CSD 301 and senior standing; or departmental permission.

CSD 490 Special Studies in Communication Sciences and Disorders

Credits 1-3 Grading Basis Graded

Provides students opportunity for independent study and/or small class instruction in elective topics.

CSD 499 Honors

Credits 1-6 Grading Basis Graded

See catalog section "Graduation with Honors."

Communication Studies (SCOM)

SCOM 121 Fundamental Human Communication: Presentations [C1HC]

Credits 3 Grading Basis Graded

Study of human communication as a process. Overview of the principles and practices of interpersonal, small group and public communication. Emphasis on examining the role of self-concept, perception, culture, verbal and nonverbal dimensions in the communication process, using power and managing conflict, applying critical listening, practicing audience analysis within informative speech making. Public speaking required. May be used for general education credit. May not be used for major credit.

Prerequisites: If you have already earned credit for SCOM 121, SCOM 122, or SCOM 123, you are not eligible to enroll for another SCOM 121, SCOM 122 or SCOM 123.

SCOM 122 Fundamental Human Communication: Individual Presentations [C1HC]

Credits 3 Grading Basis Graded

Study of human communication as a process. Overview of the principles and practices of interpersonal, small group and public communication. Emphasis on examining the role of self-concept, perception, culture, verbal and nonverbal dimensions in the communication process, using power and managing conflict, applying critical listening, practicing audience analysis within informative speech making. Public speaking required. May be used for general education credit. May not be used for major credit.

Prerequisites: If you have already earned credit for SCOM 121, SCOM 122, or SCOM 123, you are not eligible to enroll for another SCOM 121, SCOM 122 or SCOM 123.

SCOM 123 Fundamental Human Communication: Group Presentations [C1HC]

Credits 3 Grading Basis Graded

Study of human communication as a process. Overview of the principles and practices of communication in small group and public communication contexts. Emphasis on examining the role of self-concept, perception, culture, verbal and nonverbal dimensions in the communication process, using power and managing conflict, applying critical listening, practicing audience analysis, and constructing informative and persuasive group presentations. Public speaking required. May be used for general education credit. May not be used for major credit.

Prerequisites: If you have already earned credit for SCOM 121, SCOM 122, or SCOM 123, you are not eligible to enroll for another SCOM 121, SCOM 122 or SCOM 123.

SCOM 230 Introduction to Sport Communication**Credits 3 Grading Basis** Graded

This class provides a foundation for students who are pursuing a minor in sport communication. The course provides an overview about information management and how it applies to the professional sporting industry. Students will learn about careers in sport information, how an organization communicates with the media and its constituents, and how communication in sport has evolved and continues to evolve.

Prerequisites: Sport communication minor.

SCOM 240 Introduction to Communication Theory**Credits 2 Grading Basis** Graded

This course studies the theories and models that inform students' understanding of human communication processes. Emphasis is placed on the processes of theory building, comparison of theories, and the implications and application of theory to particular contexts. Consideration is given to the role of communication in all human endeavors. This course must be taken concurrently with SCOM 240L.

Prerequisites: Declared SCOM majors or declared SCOM minors only, and SCOM 121, SCOM 122 or SCOM 123, and sophomore standing or above.

Corequisites: SCOM 240L.

SCOM 240L Communication Theory Lab**Credits 1 Grading Basis** Graded

This lab is designed to complement and supplement the SCOM 240 lecture course. Students will discuss, write and/or give presentations related to content covered in SCOM 240. This course must be taken concurrently with SCOM 240.

Prerequisites: Declared SCOM majors or SCOM minors only, and SCOM 121, SCOM 122 or SCOM 123, and sophomore standing or above.

Corequisites: SCOM 240.

SCOM 242 Presentational Speaking**Credits 3 Grading Basis** Graded

This course offers a study of methods for preparing and presenting public speeches. Students consider impromptu and extemporaneous speaking, sales speeches, business presentations and other special occasion speeches. Emphasis is placed on performance and evaluation.

Prerequisites: Fully-admitted SCOM major.

SCOM 244 Sport Communication Techniques: Writing and Reporting**Cross-listed with:** SMAD 244**Credits 3 Grading Basis** Graded

Basic skills of sport writing and reporting are studied and applied. Students gain experience in a variety of sports and learn and apply skills in researching, interviewing, reporting, writing columns and features involving the world of sports.

Prerequisites: SCOM 230.

SCOM 245 Signs, Symbols and Social Interaction**Credits 3 Grading Basis** Graded

This course studies verbal and nonverbal communication as used in human interaction. Consideration is given to the function of symbolic systems in self-concept development, the structuring of reality, and social discourse. Attention is directed toward the use of signs and symbols by different ethnic groups, genders, age groups and geographic groups.

Prerequisites: SCOM 121, SCOM 122 or SCOM 123.

SCOM 248 Intercultural Communication**Credits 3 Grading Basis** Graded

This course explores intercultural theories and concepts within historical and political contexts and examines systems of power implicated in our communication. Exploring communication processes from critical and cultural perspectives encourages students to reflect on cultural knowledge, practices, attitudes, belief systems, and assumptions of self and others in our globalized and technological world. Working through course materials and activities together, we develop intercultural communication skills to cultivate more equitable relations.

Prerequisites: SCOM 121, SCOM 122 or SCOM 123.

SCOM 260 Introduction to Public Relations**Credits 3 Grading Basis** Graded

Study of basic principles and practices of public relations. Consideration given to public relations problems and pragmatic solutions utilizing oral, written and electronic communication media and skills.

SCOM 261 Public Relations Techniques I: Written**Credits 3 Grading Basis** Graded

This course offers a study of writing fundamentals for public relations. Emphasis is placed on practice of effective writing for a variety of media (press releases, public service announcements, brochures, newsletters). Students must be able to use word processing software.

Prerequisites: SCOM 260 and fully-admitted SCOM major or Sport Communication minor.

SCOM 280 Introduction to Communication Research**Credits 3 Grading Basis** Graded

This course is an introduction to the principles, methods and analysis techniques used in the field of communication. Emphasis is placed on a broad-based understanding of the breadth of research in the field. This course includes both qualitative and quantitative research methods, methods of literature review and research article critiques. This course fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Fully-admitted SCOM major.

SCOM 295 Study Abroad Pre-Departure Seminar**Credits 1-3 Grading Basis** Credit / No Credit

This course prepares students participating in short-term study abroad programs in communication studies by exploring important social, historical and cultural content relevant to the study abroad locations and topics.

SCOM 300 Interdisciplinary Special Topics in Communication Studies**Credits 1-3 Grading Basis** Graded

This course is a study of interdisciplinary topics and issues in human communication. Emphasis will be on contemporary theories, research and principles. May be repeated up to six credits.

SCOM 301 Feminist Blogging: Writing for Shout Out! JMU**Cross-listed with:** WGSS 301**Credits 3 Grading Basis** Graded

The goal of this course is to work in a feminist collective to write daily blog posts for the Shout Out! JMU blog. Students will organize the blog, research and write blog posts, publicize the blog, and create dialogue by commenting on one another's blog posts as a means of consciousness raising.

SCOM 302 Third Wave Ecofeminism**Cross-listed with:** WGSS 302**Credits 3 Grading Basis** Graded

This course explores the historically strong association between women and nature, in which the image of Mother Earth is central, and critiques the power-as-domination assumption of our culture shown in the exploitation of women and of the earth itself. Religious, psychological, social, historical and scientific manifestations of this assumption will be examined, along with alternative models of power and responsibility. Students will explore the considerable research on ecofeminism.

SCOM 303 Special Topics in Health Communication**Credits 3 Grading Basis** Graded

Study of current topics and issues in health communication. Course content will vary by faculty. See MyMadison for the topic. May be repeated up to six credits.

SCOM 305 Language and Culture**Cross-listed with:** ANTH 305**Credits 3 Grading Basis** Graded

An introduction to linguistic anthropology. Explores the complex relationships between language and culture through topics such as language acquisition and socialization; language, thought and worldview; language and identity; multilingualism; how and why languages change; literacy; and the politics of language use and language ideologies.

SCOM 306 Special Topics in Organizational Communication**Credits 3 Grading Basis** Graded

Study of topics and issues in organizational communication. Emphasis on contemporary theories, research and principles. May be repeated up to six credits.

Prerequisites: SCOM 240, SCOM 350, and Organizational Communication concentration; or permission of the instructor.

SCOM 312 Sport Communication Minor Practicum**Credits 1-3 Grading Basis** Graded

This course provides approved co-curricular learning experience in sport communication. May be repeated for a maximum of 6.00 credit hours.

Prerequisites: SCOM 230; Sport communication minor.

SCOM 313 Topics in Communication Studies**Credits 1-3 Grading Basis** Graded

This course studies current topics and issues in human communication. Emphasis is placed on contemporary theories, research and principles. Prerequisites: Fully-admitted SCOM major or minor, or permission of the instructor.

SCOM 314 Communication in Romantic Relationships**Credits 3 Grading Basis** Graded

This class acquaints students with relevant theory and research associated with communication in various stages of romantic relationships including initiation, maintenance and dissolution. Emphasis is on honing one's ability to understand communication theory, research and application while enhancing one's communication skills in romantic relationships. Additionally, students will improve their ability to communicate about relationship research through writing, presenting and discussing.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 316 Competitive Speech Team**Credits 1-3 Grading Basis** Graded

This course offers students experience in intercollegiate speech competition. Students have an opportunity to research, prepare, practice and compete in interpretation, public address and limited preparation events. Students will cultivate communication skills while researching, practicing and competing; while listening to and critically evaluating others' speeches; and while meeting and engaging with other competitors, coaches and judges. No more than six hours of practicum credit can be applied to the SCOM major. Students must be on the university's speech team to enroll.

SCOM 317 Debate Team**Credits 1-3 Grading Basis** Graded

This course is a practicum course for students competing in intercollegiate debate competition with the JMU Debate Team. Students are required to practice, prepare and research arguments on the national collegiate debate topic and participate in practice and competition debates. No more than six hours of practicum credit can be applied to the SCOM major. Students must be on the university's Debate Team to enroll.

SCOM 318 Practicum in Communication Studies**Credits 1-6 Grading Basis** Graded

Approved co-curricular activities and/or projects of a practical nature. No more than six hours of practicum credit can be applied to major. Proposals must be submitted to and approved by the course instructor for section and credit hour registration. To receive repeat credit, see school director.

SCOM 319 Bluestone Communications Practicum**Credits 3 Grading Basis** Graded

This is the practicum course for JMU's student-run public relations firm, Bluestone Communications. Students further their theoretical understanding and practical skills in the processes of research, planning, communication/action and evaluation by implementing public relations strategies and tactics for specific organizations. Students need to apply to take this course.

SCOM 320 Introduction to Interpersonal Communication**Credits 3 Grading Basis** Graded

This course provides an introduction to the fundamental theoretical perspectives in interpersonal communication. Emphasis is placed on the effects of verbal and nonverbal messages on continuity and change in personal relationships. Students consider the influence of cultural and social contexts on messages in relationships and develop their communication competence in diverse interpersonal contexts. Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 325 Small Group Communication**Credits 3 Grading Basis** Graded

This course studies the communication processes involved in solving problems when working with others in a small group context. Emphasis is placed on the concepts of roles, norms, leadership and decision-making. Students consider small group factors which influence problem-solving effectiveness.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 330 Special Topics in Interpersonal Communication**Credits 3 Grading Basis** Graded

This course studies current topics and issues in interpersonal communication. Topics and issues may include, but are not limited to aging and lifespan, communication education, computer mediated relations, deception and secrecy, friendship and rivalry, relationship rejuvenation, and social support. This course may be repeated up to six credits.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 331 Communication and Conflict**Credits 3 Grading Basis** Graded

This course considers theories of conflict emerging from the communication discipline and their application to different forms of conflict at all levels of human interaction. This course offers an examination of communication and varied responses to conflict in diverse situations. Emphasis is placed on competencies required for successful management, intervention and transformation of conflict.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 333 Negotiations**Credits 3 Grading Basis** Graded

This course provides an overview of negotiation as a strategy for dealing with conflict.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 338 The Art of Public Debate**Credits 3 Grading Basis** Graded

This course focuses on the practical application and critical examination of debate in the public sphere. The course provides students with experience in researching, constructing and presenting topics for public debate. Additionally, students will be asked to examine pieces of public debate in order to provide them with the tools to critically examine public discourse in a manner consistent with being a productive and active citizen. This course will culminate with a public debate tournament.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 340 Principles and Processes of Interviewing**Credits 3 Grading Basis** Graded

This course studies communication principles and processes in interviewing. Emphasis is placed on interpersonal processes involved in interview structures, goals and question types. Students develop their communication skills in information, persuasive, counseling, health care, employment and performance appraisal interviews.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 341 Persuasion**Credits 3 Grading Basis** Graded

The study of oral communication as a determinant of attitudinal and behavioral change. Emphasis on the various kinds of artistic and nonartistic proofs as they apply to human motivation. Consideration of the application of behavioral research findings to persuasion.

Prerequisites: SCOM 240.

SCOM 342 Argument and Advocacy**Credits 3 Grading Basis** Graded

This course studies the techniques and principles of argument and advocacy. Emphasis is placed on developing, presenting and defending a position on controversial questions. Consideration is given to contemporary theories of public argument.

Prerequisites: Fully-admitted SCOM major or minor or Sport Communication minor.

SCOM 343 Contemporary Rhetorical Theory and Practice**Credits 3 Grading Basis** Graded

Examines contemporary rhetorical theory and practice in relation to specific social, economic and technological changes in the 20th century, with particular emphasis on theoretical frameworks. Students learn about the changing needs of postmodern communicators and how new rhetorical theories have developed to anticipate, respond to, and shape those changes. Also explored is the value of contemporary rhetorical theories for communicative and performative events.

Prerequisites: WRTC 200 or permission of the instructor.

SCOM 344 Oral Interpretation**Credits 3 Grading Basis** Graded

Study and application of theories concerning the oral presentation of various forms of literature including prose, poetry, drama and nonfiction materials. Emphasis on performance.

Prerequisites: Any 100-level SCOM course.

SCOM 345 Nonverbal Communication**Credits 3 Grading Basis** Graded

Study of nonverbal means through which people relate to one another. Consideration of the communicative effects of environment, facial expression, voice, posture, gestures, touch, distance and physical appearance.

Prerequisites: Any 200-level SCOM course.

SCOM 346 Free Speech in America**Credits 3 Grading Basis** Graded

The study of the evolution of freedom of speech in America from Colonial times to the present day. Emphasis on the major periods of development and on the role of courts in defining freedom of speech. Special consideration of contemporary freedom of speech controversies.

SCOM 347 Communication, Diversity and Popular Culture**Credits 3 Grading Basis** Graded

This course explores the representational, institutional and communication practices that contribute to the construction of identities and diversity in popular culture. Students will explore how ideologies are reproduced and how understandings of race, class, gender, ethnicity, sexuality, ability and a range of other intersecting axes of identity are shaped within popular culture artifacts. Emphasis will be placed on critical analysis of current popular culture texts.

Prerequisites: Any 100-level SCOM course.

SCOM 348 Communication and Gender**Cross-listed with:** WGSS 348**Credits 3 Grading Basis** Graded

This course explores the role of communication in the social construction of gender across a variety of contexts. Students will critically examine the interconnectedness of sex, gender, sexuality, race and class toward understanding systems of privilege and oppression in communication practices and social institutions. Focus is on communication as a tool for personal and social change.

Prerequisites: Any 100-level SCOM course.

SCOM 349 Ethnographic Approaches to Communication Studies**Credits 3 Grading Basis** Graded

This course examines the value of ethnographic methodologies and other narrative forms of scholarship for enriching understandings of complex social issues and exploring the potential of communication toward shaping social change. Students will develop literacies in reading and evaluating ethnographic works and interrogating their contributions toward cultivating more just social worlds.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 350 Organizational Communication**Credits 3 Grading Basis Graded**

This course draws on communication theory to analyze various organizational communication practices such as the management of workers, development of organizational culture and interaction with larger systems. Students gain a complex understanding of organizing practices by investigating how historical events have influenced organizational communication and managerial practices at work. Learning is complemented by an experiential learning project. Prerequisites: Fully-admitted SCOM major or minor or Sport Communication minor.

SCOM 351 Visual Rhetoric**Cross-listed with:** WRTC 351, WRTC 351L**Credits 3 Grading Basis Graded**

A study of the rhetorical foundations of visual and verbal arguments in academic disciplines and popular culture. Students will analyze and produce visual and verbal arguments in a variety of rhetorical contexts. Prerequisites: WRTC 103 and any 100-level communication studies course or permission of the instructor.

SCOM 352 Communication and Social Movements**Credits 3 Grading Basis Graded**

This course studies the use of communication in social movements. Emphasis is placed on the types of communication used in social movements and on ways to produce and respond to such messages. This course examines a variety of different social movements within the political process including nonprofit organizations. Prerequisites: Fully-admitted SCOM major or minor or Sport Communication minor.

SCOM 353 American Political Culture and Communication**Credits 3 Grading Basis Graded**

This course studies the functions performed by communication in politics. Emphasis is placed on a variety of communication forms and techniques used by advocates both in campaigning and governing. Consideration is given to contemporary campaigns and the role of communication in their successes and failures. Prerequisites: Fully-admitted SCOM major or minor or Sport Communication minor.

SCOM 354 Environmental Communication and Advocacy**Cross-listed with:** WRTC 326**Credits 3 Grading Basis Graded**

This course is an exploration of the multifaceted aspects of environmental controversies including the rhetoric, advocacy campaigns and decision-making processes that produce and attempt to manage environmental conflict. Emphasis on persuasive efforts by, interest groups, corporations, resource managers, government agencies, scientific experts, politicians and citizens to influence public understanding of environmental issues, adoption of sustainable behavior and lifestyles, and public policy outcomes. Prerequisites: Fully-admitted SCOM major or minor; or environmental sciences, environmental studies, environmental management and environmental humanities minor; or political communication minor.

SCOM 355 Food as Communication**Credits 3 Grading Basis Graded**

This course examines food as symbolic in everyday life, with attention to negotiated understandings of identities, communities, cultures and environments. The course focuses on issues such as how we communicate identity through food consumed and not consumed; how food choices create, shape and reflect communities and cultures; how messages from mass media and government frame, influence and shape food practices; and how food practices intersect, communicate and impact surrounding environments. Prerequisites: Fully-admitted SCOM major or minor or African, African American and Diaspora Studies minor.

SCOM 356 Immigrant Advocacy**Credits 3 Grading Basis Graded**

This course invites students to adopt narrative approaches to communication and identity, and to reflectively learn from the storied and lived experiences of immigrants locally and in the United States. Students will have opportunities to hear from members of local non-profit organizations that serve migrant populations and will work toward developing engaging, creative advocacy of social justice and universal human rights. Prerequisites: Any 100-level SCOM course.

SCOM 357 Youth, Communication and Culture**Cross-listed with:** SMAD 357**Credits 3 Grading Basis Graded**

This course critically examines the complex and evolving interconnections between young people, communication and culture. Grounded in the understanding that there is no one singular lived experience of "youth," the course focuses on: how young people are represented in popular culture artifacts; how cultural beliefs about various young people are constructed by public discourse; how diverse young people navigate popular culture messages; how they are constructed as target markets; and how they use technology to claim space in the broader culture. This course encourages students to examine their own experiences with and relationships to a range of communication artifacts. Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 358 Business and Professional Communication Studies**Credits 3 Grading Basis Graded**

Students investigate the nuance and complexity of communication in modern organizational life. A portion of the class is dedicated to the skills involved in a competitive, successful career search. In addition, students develop the skills to become an ethical and effective organizational citizen. Prerequisites: Fully admitted SCOM major or minor, and junior or senior standing.

SCOM 359 Communication, Diversity and British Popular Culture**Credits 3 Grading Basis Graded**

This course explores the rhetorical dimension of communication practices and texts found in British popular culture. Emphasizing critical analysis, students will examine how issues of diversity are communicated, represented and contested across British media such as television, film, music and digital platforms. The course examines how strategic communication choices construct cultural narratives in a diverse, multicultural world. Emphasis is placed on critical thinking and self-reflexivity. Prerequisites: Admission to the JMU semester in London.

SCOM 360 British Media and Society**Cross-listed with:** HUM 251L, SMAD 360, WRTC 360**Credits 3 Grading Basis** Graded

Study the history, nature and impact of mass media in the United Kingdom. Emphasis on the impact modern media has on society, and society has on media. Consideration of similarities and differences in mass media in the United States and Great Britain. Consideration of the relationship between mass media and the arts. Focus on 20th century mass media in London, one of the world's preeminent and most influential centers. London term only.

SCOM 361 Public Relations Techniques II: Visual**Credits 3 Grading Basis** Graded

Study of visual communication techniques for public relations. Survey of design principles and elements used for developing visually effective messages with an emphasis on publication design and production, photography, and computer-mediated presentations. Students should provide a camera and be familiar with desktop publishing and presentational software.

Prerequisites: SCOM 261 and fully-admitted SCOM major in the public relations concentration.

SCOM 362 Public Relations Theory and Concepts**Credits 3 Grading Basis** Graded

The purpose of this course is to introduce students to the theoretical body of knowledge in public relations. In this course, students will learn core tenants of important theories and concepts and apply these to the ethical practice of public relations. The breadth of theories covered will be useful for any public relations specialization and builds an essential foundation to better understand the research-based character of the public relations process.

Prerequisites: SCOM 260 and fully-admitted SCOM major in the public relations concentration.

SCOM 363 International Public Relations**Credits 3 Grading Basis** Graded

This course explores the special professional challenges and opportunities arising from the dynamic global public relations developments characterizing the beginning of this century, taking into account social, economic, political, legal and cultural factors as well as new media developments.

Prerequisites: SCOM 260 and fully-admitted SCOM major.

SCOM 365 Sports Public Relations**Credits 3 Grading Basis** Graded

This class is designed to provide students with opportunities to explore the profession of sports public relations. In this class, students will discuss various sports communication issues including media relations, community relations, player relations, fan relations, crisis communication, sports social responsibility and more.

Prerequisites: SCOM 260 and fully-admitted SCOM major or Sport Communication minor.

SCOM 367 Advanced Public Relations Writing**Credits 3 Grading Basis** Graded

This course offers advanced public relations students experience in the wide range of writing style and applications that are essential to successfully begin their professional careers. The course focuses on understanding and mastering action-oriented communication methods and best professional practices. This course provides both a conceptual framework and in-depth training in advanced techniques.

Prerequisites: SCOM 261 and fully-admitted SCOM major in the public relations concentration.

SCOM 369 Topics in Engaged Public Relations**Credits 3 Grading Basis** Graded

Students will explore a specific context of public relations and how public relations theories, processes, and skills can be used in the context. During the semester, students will discuss specific publics within the context and apply ethical communication strategies and tactics to engage with those publics. Courses provided could include crisis communication, corporate social responsibility, social media management, fashion public relations, etc. based on the expertise of the instructor.

Prerequisites: SCOM 260 and fully-admitted SCOM major.

SCOM 370 Introduction to Health Communication**Credits 3 Grading Basis** Graded

This course offers an introduction to the study of communication in health and illness contexts. It covers the impact of interpersonal relationships, sociocultural and organizational influences, and media and health campaigns on health beliefs and behaviors. Emphasis is placed on strategies and approaches that promote effective communication in a variety of health contexts.

Prerequisites: Fully-admitted SCOM major or minor or Sport Communication minor.

SCOM 371 Disability and Advocacy**Credits 3 Grading Basis** Graded

This class will go over the history of disability rights rhetoric in the advancement of disability civil rights in the United States. Topics will include (but not be limited to) the three models of disability discourse (moral, medical and social); strategies employed during the passage of the Americans with Disabilities Act (ADA); the ways in which disability civil rights has been coopted by non-disabled folks; how some advocacy strategies created limitations in the enforcement of the ADA; and current discourse surrounding the improvement of disability civil rights in the United States.

Prerequisites: SCOM majors or minors, disability studies minors, medical humanities minors or permission of instructor.

SCOM 381 Rhetorical Research Methods**Credits 3 Grading Basis** Graded

The study of rhetorical and critical research methods in various communication contexts. Emphasis is on developing and applying appropriate standards to consider communicative factors such as audience, tone, and rhetorical strategies. Consideration of criticism of public address, mass media, and other forms of persuasive communication. Students will be expected to conduct an original research project and present findings in a paper and/or oral presentation.

Prerequisites: SCOM 280 with a minimum grade of "C-".

SCOM 383 Quantitative Experimental Research Methods**Credits 3 Grading Basis** Graded

The study of quantitative research methods in various communication contexts. Emphasis is on ability to research literature, develop and evaluate experimental design, and apply and interpret statistical tests. Students will be expected to conduct an original research project and present findings in a paper and/or oral presentation.

Prerequisites: SCOM 280 with a minimum grade of "C-".

SCOM 385 Qualitative Research Methods**Credits 3 Grading Basis Graded**

Study of interpretive approaches to communication research using a variety of qualitative research methods, including field observation, qualitative interviewing, focus groups, narrative analysis and discourse analysis. Students will plan and conduct an exploratory qualitative study, prepare a written research report including a literature review and make a research presentation.

Prerequisites: SCOM 280 with a minimum grade of "C-".

SCOM 386 Survey Research Methods**Credits 3 Grading Basis Graded**

The study of survey research methods in various communication contexts. Emphasis on using survey research methodology in communication audits, public relations problems and public opinion polling and the ability to research literature, develop and evaluate survey questions, and apply and interpret statistical tests. Students will be expected to conduct an original research project and present findings in a paper and/or oral presentation.

Prerequisites: SCOM 280 with a minimum grade of "C-".

SCOM 388 Campaign Research Methods**Credits 3 Grading Basis Graded**

The study of research methods in various communication contexts. Emphasis is on ability to use research methods at various stages of communication planning including background research, primary research, strategic planning, monitoring and evaluation. Students will be expected to conduct an original research project and present findings in a paper and/or oral presentation.

Prerequisites: SCOM 280 with a minimum grade of "C-".

SCOM 390 Directed Projects**Credits 2-3 Grading Basis Graded**

Supervised projects related to any aspect of human communication. Emphasis on original individual or group programs beyond the school's usual curricular or co-curricular offerings. Formal report(s) required for awarding of credit.

SCOM 395 Study Abroad Seminar**Credits 3-6 Grading Basis Graded**

Intensive examination of specialized international communication topics arranged in cooperation with a faculty member. Prior arrangements must be made with the program director.

SCOM 412 Sport Communication Internship**Credits 3 Grading Basis Graded**

This advanced internship course offers students the opportunity to apply and refine their sports communication skills in a professional setting beyond the practicum level. Designed for students who have successfully completed the sports communication practicum, this course emphasizes deeper engagement, strategic communication and increased responsibility in the field. Students will take on more specialized roles, contribute to higher-level projects and demonstrate advanced competencies in areas such as media relations, content creation, digital strategy or event communication. This course may be repeated for a maximum of six credit hours with departmental approval.

Prerequisites: Sport Communication minors only; SCOM 230 and SCOM 312; and permission of the school director.

SCOM 413 Advanced Topics in Communication Studies**Credits 1-6 Grading Basis Graded**

This course offers an in-depth exploration and analysis of a communication-related theory, context, topic or problem, culminating in a research project documented in written, oral, visual and/or multimedia presentations. Course content varies based on faculty expertise.

Prerequisites: Fully admitted SCOM major or minor, and junior or senior standing.

SCOM 414 Attachment Communication**Credits 3 Grading Basis Graded**

This class acquaints students with relevant theory and research associated with the science of attachment, its impact on brain development and its subsequent impact on how we communicate throughout the lifespan. In addition to gaining an in-depth understanding of how attachment impacts communication and relationships throughout life, students will also improve their ability to communicate about attachment research through writing, presenting and discussing.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 420 Feminist Rhetorics**Cross-listed with: WGSS 420, WRTC 420****Credits 3 Grading Basis Graded**

Surveys key women figures in classical and contemporary rhetorical traditions and challenges the strategies used to historicize this tradition from feminist perspectives. Explores diverse feminist rhetorical discourses informed by race, sexual orientation, ethnicity and social class.

Prerequisites: WRTC 103 or equivalent and junior or senior standing; or permission of the instructor.

SCOM 425 Leadership Communication**Credits 3 Grading Basis Graded**

This course promotes understanding and development of organizational leadership through investigation of theory and analysis. Students explore varied and sometimes contradictory models of leadership and learn how to articulate and express ideas that encourage others to advocate for and bring about positive change.

Prerequisites: Fully admitted SCOM major or minor, and junior or senior standing.

SCOM 431 Legal Communication**Credits 3 Grading Basis Graded**

Study of the role of communication in the legal process. Emphasis on communication questions/problems which litigants, lawyers, judges and jurors face. Consideration of legal argument, negotiation, trial advocacy, decision making and communication technologies.

SCOM 440 Family Communication**Credits 3 Grading Basis Graded**

Study of the processes and functions of family communication, including managing dating, marital, parent-child and inter-generational relations. Theoretical and applied examination of communication and cultural processes that define and construct family structures, systems and boundaries. Course also examines histories of family communication as constructed in popular culture.

Prerequisites: SCOM 280 and Fully Admitted SCOM Majors only.

SCOM 441 Oral History**Cross-listed with:** HIST 441**Credits 3 Grading Basis** Graded

This course will explore the theory and practice of oral history. Through a series of readings, students will consider the many promises and challenges of the discipline, including issues related to memory, objectivity, ethics, the law, and technology. Students will also engage in an experiential learning exercise in which they collaborate to produce an oral history project.

SCOM 442 Advanced Topics in Advocacy Studies**Credits 3 Grading Basis** Graded

Advanced topics in advocacy studies is the concentration's capstone. Through case studies, readings, discussions and experiential activities students investigate complex issues in the field of advocacy. Students apply theoretical knowledge and skills as they explore the interaction between advocacy and communication.

Prerequisites: Fully admitted SCOM major or minor, and senior standing.

SCOM 447 Communication Tools for Organizational Challenges**Credits 3 Grading Basis** Graded

Organizations frequently face challenges related to communication, collaboration, training, facilitating meetings and conversations about complex issues, and navigating change. This course equips students with the knowledge and tools to address these challenges and implement communication-based interventions. Through hands on learning, students will identify an organizational challenge and design responses that facilitate positive change and collaboration.

Prerequisites: Fully admitted SCOM major or minor, and junior or senior standing.

SCOM 448 Communication, Culture and Identity**Credits 3 Grading Basis** Graded

This class examines theory and research of cultural communication, and reflexively considers how communication practices construct identities and arrange social relations within diverse contexts and applied settings.

Prerequisites: Any one of the following courses: SCOM 305, SCOM 347, SCOM 348, SCOM 349, SCOM 352, or SCOM 357.

SCOM 450 Advanced Studies in Organizational Communication**Credits 3 Grading Basis** Graded

This capstone course uses case studies, readings, discussions and experiential activities to investigate complex issues that emerge for organizations and their members. Students apply theoretical knowledge and skills to the analysis of communication problems through an applied field study in an organization.

Prerequisites: SCOM 350 and senior standing, or permission of instructor.

SCOM 453 Political Campaign Communication**Credits 3 Grading Basis** Graded

An advanced study of communication techniques, procedures and processes as they relate to political campaign communication. Emphasis upon the design, execution and production of various communication messages. Consideration of the impact and utilization of various technologies in political campaigns.

SCOM 456 Fundraising Campaigns and Donor Motivation**Credits 3 Grading Basis** Graded

This course examines the theories and practices of fundraising campaigns, including campaign planning, fundraising communication, donor research and motivation, branding and message development, stewardship and recognition, special events, planned giving, capital campaigns, program evaluation, and fundraising ethics.

Prerequisites: Fully admitted SCOM major or minor, and junior or senior standing.

SCOM 460 Public Relations Management**Credits 3 Grading Basis** Graded

This course offers intensive study and research of advanced communication management skills, theory and principles using case and field studies. Special attention is given to systematic and ethical management of communication and action affecting an organization's internal and external publics.

Prerequisites: SCOM 260 and fully-admitted SCOM major in the public relations concentration.

SCOM 461 Public Relations Campaigns**Credits 3 Grading Basis** Graded

This is the capstone course for the public relations program of study. Students further their theoretical understanding and practical skills in the processes of research, planning, communication/action and evaluation by conducting campaigns for specific organizations.

Prerequisites: SCOM 362, one advanced research method course (SCOM 381, SCOM 383, SCOM 385, SCOM 386 or SCOM 388), and one public relations context course (SCOM 363, SCOM 365, SCOM 369 or SCOM 460).

SCOM 462 Student-Run Public Relations Firm**Credits 3 Grading Basis** Graded

This is the capstone course for the public relations program of study. Students further their theoretical understanding and practical skills in the processes of research, planning, communication/action and evaluation by implementing public relations strategies and tactics for specific organizations. Students need to apply to take this course.

SCOM 464 Special Topics in Sport Communication**Credits 3 Grading Basis** Graded

This course studies current topics and issues in sports communication. Course content varies based on faculty expertise. This course may be repeated up to six credits.

Prerequisites: Sport Communication minor and SCOM 230.

SCOM 465 Rhetoric of Environmental Science and Technology**Cross-listed with:** WRTC 416**Credits 3 Grading Basis** Graded

This course offers an advanced study of the way the public receives, makes sense of, and influences scientific and technical information about environmental issues. Implications of these processes on environmental policy will be analyzed. Readings and assignments will concentrate on the interactions between technical and public spheres of communication, with an in-depth examination of the way the media facilitates the transfer of information between scientific communities and public audiences.

Prerequisites: WRTC 103 or equivalent and junior or senior standing; or permission of the instructor.

SCOM 468 Interpersonal Communication in Health Contexts**Credits 3 Grading Basis** Graded

This course explores how effective interpersonal communication supports the management of mental and physical health within social networks. Students will examine health-related experiences; the roles of stress, coping and social support; the connection between relationship contexts on health outcomes; and strategies for collaboration in clinical and non-clinical settings.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 470 Health Communication Campaigns**Credits 3 Grading Basis Graded**

This course focuses on the theoretical principles and practical considerations relevant to designing, implementing and evaluating health communication campaigns. Emphasis is placed on the collection and analysis of formative research. The final project for the course involves designing a health campaign targeting a specific audience.

Prerequisites: Fully-admitted SCOM major or minor, and SCOM 370; or permission of the instructor.

SCOM 471 Culture and Health Communication**Credits 3 Grading Basis Graded**

This course explores the relationship between culture and health in different contexts. In particular, the course focuses on the role of communication in understanding diverse health beliefs and engaging in dialogue about our own health beliefs.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 472 Media and Politics**Cross-listed with:** SCOM 472L, SMAD 472, SMAD 472L, POSC 472**Credits 3 Grading Basis Graded**

This course is a study of the media's role in the political system, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories; how media practices and technologies shape political decision making; how political forces influence media institutions; and how public opinion and media narratives are shaped.

Prerequisites: Fully admitted SCOM major or minor (SCOM, CLTC, HTHC).

SCOM 472L Media and Politics**Cross-listed with:** SCOM 472, SMAD 472, SMAD 472L, POSC 472**Credits 3 Grading Basis Graded**

This course is a study of the media's role in the political system, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories; how media practices and technologies shape political decision making; how political forces influence media institutions; and how public opinion and media narratives are shaped.

SCOM 473 British Media and Politics**Cross-listed with:** POSC 473, SMAD 473**Credits 3 Grading Basis Graded**

This course is a study of the media's role in the political system of the United Kingdom, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories as they apply to the British context; how UK media practices and technologies shape political decision making; how political forces influence British media institutions; and how public opinion and media narratives are shaped within a parliamentary democracy.

SCOM 490 Special Studies in Communication Studies**Credits 1-3 Grading Basis Graded**

An independent study for students to pursue individual research under the guidance of faculty. Limited to senior communication studies majors in good standing with permission of school director.

SCOM 495 Internship in Communication Studies**Credits 3-6 Grading Basis Graded**

This course offers credit for the application of communication theory and skills in a directed, on-the-job learning experience. It is open only to communication studies majors who meet specific criteria. Up to six credits may be applied as electives in the communication studies major.

Prerequisites: Fully admitted SCOM major, 12 credits of coursework in SCOM at the 200-level or above, and permission of the instructor.

SCOM 499 Honors in Communication Studies**Credits 1-3 Grading Basis Graded**

In this individualized capstone course, honors students develop and complete a significant scholarly or creative work in Communication Studies. Working independently with faculty mentorship, students will explore a significant topic, integrate relevant theory and methods, and produce a substantive final product such as a written thesis, campaign or presentation. This course may be repeated for up to 6 credits.

Computer Information Systems (CIS)

CIS 204 Computer Information Systems Fundamentals for Non-Business Majors**Credits 3 Grading Basis Graded**

An introduction to computer-based information systems. Emphasis is placed on the role of computers in organizations and society, computer hardware and software, uses of information systems, computer ethics, and collaboration using computers. Students will use typical business applications. This course is designed to fulfill requirements for the General Business minor. Not open to business majors. This course may NOT be substituted for COB 204 by business majors or CIS minors. Not open to students who have completed COB 204 with a passing grade.

Prerequisites: Not open to business majors. Not open to students who have completed COB 204 with a passing grade.

CIS 221 Principles of Programming**Credits 3 Grading Basis Graded**

Students will be required to solve real-world business problems using a programming language. Projects will include the use of control structures (selection and iteration), collections/lists of data, functions and file processing logic. Students will also be introduced to objects and graphical user interfaces.

Prerequisites: Not open to students who have taken CS 149; or ISAT 252 or IA 241 with a minimum grade of "C"; or are taking or have taken CS 159.

CIS 301 Cloud Server Management**Credits 1 Grading Basis Graded**

This is a lab-based course that introduces students to operating systems and server administration in a business environment. Students will learn the basic functions of managing cloud servers through the hands-on use of cloud resources. Additionally, students will acquire hands-on server administration skills in order to better understand the operational and security demands of business applications.

Corequisites: COB majors: COB 300. Non-COB majors: Junior standing and COB 204 or CIS 204.

CIS 304 Enterprise Architecture**Credits 3 Grading Basis Graded**

This course introduces students to key issues of managing enterprise IT solutions. Emphasis will be placed on strategic and systems thinking in planning, modeling and evaluating the enterprise IT architecture. Topics include business process modeling; computer network and cloud architecture modeling; risk assessment; security and disaster recovery planning; emerging technologies; and ethical, financial and global considerations.

Prerequisites: For CIS majors: COB 300. For CIS minors: junior or senior standing. Corequisites: For CIS majors: COB 300.

CIS 311 Analyzing Data in Organizations**Credits 3 Grading Basis Graded**

This course provides an overview of how to work with databases and other data sources in order to access relevant information in a timely and user-friendly manner. It includes discussions of a variety of data representation types, including relational databases, XML documents, and cloud data. Students learn essential database concepts and gain practical experience in querying, reporting, and analyzing data.

Prerequisites: CIS 204 or equivalent knowledge (instructor permission is needed). Open only to Adult Degree Program students.

CIS 312 Systems Planning and Analysis**Credits 3 Grading Basis Graded**

Information systems couple both technical (hardware, software, database, telecom) and socio-organizational (business processes, ethics, knowledge, users, developers) subsystems to create rich and available information for the purpose of optimizing business decisions. This course covers the techniques and common tools employed for planning and analyzing these systems. Emphasis will be placed on the system development life cycle, planning and analysis tools, and professional business writing.

Prerequisites: CIS 204 or equivalent knowledge (instructor permission is needed). Open only to Adult Degree Program students.

CIS 313 Designing for the Web**Credits 3 Grading Basis Graded**

This course is an introduction to the design and development of web pages and web sites. Major topics to be covered include: Hypertext Markup Language (HTML5), Cascading Style Sheets (CSS), the principles of design for user experience, responsive design, and JavaScript.

Prerequisites: CIS 204 or equivalent. Not open to CIS majors or minors.

CIS 320 Principles of Computing Networks**Credits 3 Grading Basis Graded**

This course focuses on the underlying principles of computer networks and how these principles are deployed to provide efficient and secure networks for providing voice, data and video services. Emphasis is placed on understanding basic routing, switching and data aggregation techniques; information security strategies; and understanding how basic information systems applications utilize network services.

Prerequisites: CIS majors and minors: CIS 304. ISAT majors: ISAT 252. CS majors: CS 149. Corequisites: CIS majors and minors: CIS 304.

CIS 330 Database Design and Application**Credits 3 Grading Basis Graded**

A study of the tools and techniques of database analysis and design including the implementation of the design using common database management system models. Not open to students who have taken CS 374.

Prerequisites: For CIS majors: COB 300 and minimum grade of "C" in one of the following: CS 149, CIS 221 or ISAT 252. For CIS minors: Junior or senior standing and minimum grade of "C" in: CS 149, CIS 221 or ISAT 252. Corequisites: COB 300.

CIS 331 Intermediate Computer Programming**Credits 3 Grading Basis Graded**

Study of concepts and techniques used in object-oriented programming for business applications including program specification, design, development, testing, implementation and documentation. Topics include: basic programming structures; method, array and memory analysis; object-oriented principles (encapsulation, inheritance, polymorphism); graphical user interface (GUI) design and database connectivity.

Prerequisites: For CIS majors and minors: CIS 330. Corequisites: For CIS majors and minors: CIS 330.

CIS 354 Advanced Visual Basic Programming**Credits 3 Grading Basis Graded**

Advanced course in Visual Basic programming. Emphasis will be placed on Object-Oriented programming, sequential and random data files and error trapping. Other topics covered will include data access objects, client server, printing in VB and Crystal Reports.

Prerequisites: A minimum grade of "C" in one of the following: CIS 221, CS 149, or ISAT 252 and CIS 330. Corequisites: CIS 330.

CIS 361 Computer Information Systems Internship**Credits 0 Grading Basis Credit / No Credit**

To enable students to gain valuable work experience in a CIS-related field. Requires 300 hours of approved computer information systems work experience. All work sites must be pre-approved.

Prerequisites: CIS major and COB 300.

CIS 366 Web Design and Development**Credits 3 Grading Basis Graded**

This course is an introduction to the design and development of web pages and websites. Major topics to be covered include: Hypertext Markup Language (HTML5), Cascading Style Sheets (CSS), the principles of design for user experience, responsive design and a programming language for web development.

Prerequisites: Declared CIS major or CIS minor and a minimum grade of "C" in CIS 221, CS 149 or ISAT 252. And: CIS majors: COB 300; CIS minors: junior or senior standing.

CIS 393 Data Mining for Business Analytics**Cross-listed with: BSAN 393****Credits 3 Grading Basis Graded**

This course focuses on using statistical and machine learning techniques in computer applications that allow the extraction of useful, previously unrecognized information from large data sets for predictive purposes. By effectively sifting through databases such as those generated by many businesses, data mining allows the analyst to recognize potentially important patterns and clusters and to target business opportunities.

Prerequisites: Declared BSAN minor or declared BSAN concentration in CIS major, and BSAN 390 with a minimum grade of "C."

CIS 420 Advanced Topics in Computing Networks**Credits 3 Grading Basis Graded**

This course presents advanced computer networking topics and concepts. Topics and concepts include network design and configuration, network security practices, and network administration practices. Students will use both network emulation software and networking hardware for assignments and labs.

Prerequisites: CIS 320.

CIS 421 Network Security**Credits 3 Grading Basis Graded**

This course will introduce various topics of cybersecurity and network defense to prevent and investigate malicious attacks and crimes committed with and against computers. Students will be exposed to different aspects of security intrusions and ways to mitigate their impacts.

CIS 424 Computer Security Management**Credits 3 Grading Basis Graded**

Instruction and discussion in the design, development and implementation of a computer security program including legal and ethical considerations.

Prerequisites: CIS 304 and a minimum grade of "C" in one of the following: CIS 221, CS 149 or ISAT 252.

CIS 425 Defensive Cybersecurity**Credits 3 Grading Basis Graded**

This course introduces the concepts of offensive web security through a series of hands-on labs that are built upon real world examples. Doing so allows students to understand the mechanisms of online attacks and learn how to respond to IT security breaches with counter measures.

Prerequisites: CIS 320 or permission of the instructor. Corequisites: CIS 320 or permission of the instructor.

CIS 428 Mobile Computing and Security**Credits 3 Grading Basis Graded**

The development of mobile software applications using current environments and frameworks is the primary objective of the class. Several different development and programming environments and platforms will be included as will the actual deployment of the application to a wireless device. An important aspect of the class will be the security implications of deploying mobile devices.

Prerequisites: A minimum grade of "C" in one of the following: CIS 221, CS 149, or ISAT 252 and CIS 331. Corequisites: CIS 331.

CIS 434 Information Technology Consulting**Credits 3 Grading Basis Graded**

This course investigates the tools used by and skills necessary for information technology consultants. The class will use a team-oriented project approach. Teams will assigned professional consulting firms as manager/mentors and will work with their manager/mentor firm to complete projects that cover each phase of the consulting life cycle.

Prerequisites: Permission of the instructor.

CIS 454 Systems Analysis and Design**Credits 3 Grading Basis Graded**

This course explores the concepts, processes and tools involved in solving contemporary business problems using information systems. Students will focus on developing skills in requirements elicitation, analysis and business modeling in the context of the information systems life cycle.

Prerequisites: Declared CIS major or minor; CIS 304 and either CIS 330 or CS 374. Corequisites: CIS 304 and either CIS 330 or CS 374.

CIS 463 Business Intelligence**Credits 3 Grading Basis Graded**

This course provides a comprehensive discussion of advanced database techniques, data warehousing, online analytical processing (OLAP), data mining, data visualization, decision support systems (DSS), artificial intelligence (AI) methods and other Business Intelligence (BI) topics. Students gain practical experience using contemporary BI tools and technologies, and apply sound design principles for creating intelligent solutions to realistic business problems.

Prerequisites: Grade of "C" in CIS 330.

CIS 464 Information Systems Project Management**Credits 3 Grading Basis Graded**

This course focuses on the concepts, tools, techniques and practices for planning and managing information systems projects, including stakeholder management, project risk and team building. Students gain expertise in using project management software.

Prerequisites: For CIS majors: COB 300, and a minimum grade of "C" in one of the following: CIS 221, CS 149, or ISAT 252. For CIS minors: A minimum grade of "C" in: CIS 221, CS 149, or ISAT 252, and junior standing. Corequisites: For CIS majors: COB 300.

CIS 484 Information Systems Development and Implementation**Credits 3 Grading Basis Graded**

Comprehensive development and implementation of enterprise-level systems using object-oriented methodologies, database driven architectures, systems analysis and design procedures, and project management skills. Topics covered will include advanced programming techniques, database processing, GUI design, object communication and a comprehensive group capstone project.

Prerequisites: For students pursuing the IS Concentration: CIS 331 with a minimum grade of "C"; For all students: CIS 330 with a minimum grade of "C". Corequisites: CIS 454

CIS 490 Special Studies in Computer Information Systems or Business Analytics**Cross-listed with: BSAN 490****Credits 1-3 Grading Basis Graded**

An advanced course in information and/or business analytics designed to give qualified students an opportunity to complete independent study under faculty supervision.

Prerequisites: Senior standing, recommendation of the instructor and written approval of the department head prior to registration.

CIS 498 Special Topics in Computer Information Systems**Credits 1-3 Grading Basis Graded**

An advanced course designed to allow exploration of current topics in computer information systems. Course content will vary. See advisor for current content.

Prerequisites: Permission of the instructor or department. Restricted to CIS majors or minors only.

CIS 499A Honors**Credits 1-6 Grading Basis Graded**

Year course. See catalog section "Graduation with Honors."

Prerequisites: COB 300.

CIS 499B Honors**Credits 3 Grading Basis Graded**

Year course. See catalog section "Graduation with Honors."

Prerequisites: COB 300.

CIS 499C Honors**Credits 2 Grading Basis Graded**

Year course. See catalog section "Graduation with Honors."

Prerequisites: COB 300.

Computer Science (CS)

CS 101 Introduction to Computer Science**Credits 3 Grading Basis Graded**

How to think like a computer scientist. Topics include an overview of the context of computing, computational operations, computational devices, algorithms and data structures, the storage and transmission of data, the presentation of information, and the limits of computing. Students learn about the design and implementation of computational systems, the value of abstraction, problem solving and the ways in which computation impacts society.

CS 149 Introduction to Programming**Credits 3 Grading Basis Graded**

Fundamental problem-solving techniques using a modern programming language. Topics include variables, input/output, decisions, loops, functions, arrays, and objects. Students learn about algorithm development, testing strategies, and software tools. You may only enroll in CS 149 at most twice.

Prerequisites: MATH 155, MATH 156 or sufficient score on the Mathematics Placement Exam. You may only attempt CS 149 two times.

CS 159 Advanced Programming**Credits 3 Grading Basis Graded**

Students use advanced problem-solving strategies to develop algorithms using classes and objects and techniques such as recursion, exceptions and file I/O. This course also focuses on designing small applications and effective testing strategies. You may only enroll in CS 159 at most twice. Prerequisites: A minimum grade of "B-" in CS 149 or equivalent. You may only attempt CS 159 two times.

CS 227 Discrete Structures I**Cross-listed with: MATH 227****Credits 3 Grading Basis Graded**

An introduction to discrete mathematical structures including functions, relations, sets, logic, matrices, elementary number theory, proof techniques, basics of counting, graphic theory, discrete probability, digital logic, finite state machines, integer and floating point representations. Prerequisites: A minimum grade of "B-" in CS 149.

CS 240 Algorithms and Data Structures**Credits 3 Grading Basis Graded**

Students learn how to implement stacks, queues, lists, sets and maps using arrays, linked lists, binary trees, heaps, binary search trees, balanced trees and hashing. Students study graph representations and algorithms. Students analyze recursive and non-recursive algorithms, illustrated by various searching and sorting algorithms. Additional topics may be included as time permits. You may only enroll in CS 240 at most twice.

Prerequisites: Fully admitted Computer Science majors or minors only and grades of "C-" or better in CS 159, CS 227/MATH 227 or MATH 245, and MATH 231 or equivalent. You may only attempt CS 240 two times.

CS 261 Computer Systems I**Credits 3 Grading Basis Graded**

This course provides an introduction to the operation of modern interrupt-driven computer systems. Explores the representation of software and information in binary memory, the primary components of a CPU, systems programming and basic interactions with an Operating System. You may only enroll in CS 261 at most twice.

Prerequisites: Fully admitted Computer Science majors or minors only and minimum grade of "C-" in either CS 227/MATH 227 or MATH 245, and in CS 159. You may only take CS 261 two times.

CS 280 Projects in Computer Science**Credits 1-3 Grading Basis Graded**

Projects or topics in computer science which are of interest to the lower division student. May be repeated for credit when course content changes. Topics may vary.

Prerequisites: Fully admitted Computer Science majors or minors only and students should consult the instructor prior to enrolling for the course.

CS 327 Discrete Structures II**Credits 3 Grading Basis Graded**

This course builds on the theoretical foundation developed in Discrete Structures I. Students study a range of topics from discrete mathematics and theoretical computer science including elementary number theory, matrices, graph theory, automata, computability and complexity theory. Prerequisites: Fully admitted Computer Science majors only and a minimum grade of "C-" in CS 227/MATH 227 or MATH 245, and CS 240.

CS 330 Societal and Ethical Issues in Computing**Credits 3 Grading Basis Graded**

Overview of philosophical and professional ethics, and a survey of societal and ethical issues in computing such as privacy, intellectual property, computer security, computer crime, product liability, and the societal, environmental, and economic impact of computers. Students develop skills in assuming and defending positions on societal and ethical issues through oral presentations and written reports.

Prerequisites: Fully admitted Computer Science majors or minors only.

CS 332 Technical Communication for Computer Science**Credits 3 Grading Basis Graded**

A study of the process of planning, researching, producing and revising technical documents for specific audiences in the computing industry. Document forms studied include definitions, requirements specifications, design specifications, instructions, correspondence, descriptions, plans and reports. Specialized notations and languages are also considered. Prerequisites: Fully admitted Computer Science majors or minors only.

CS 343 Application Development**Credits 3 Grading Basis Graded**

Overview of application software development fundamentals and their use in building stand-alone applications, visualizing and interacting with complex data representations, and controlling the Internet of Things. This course provides an introduction to the technologies used to create modern user interfaces (e.g., web technologies), establishing a common foundation for later application-focused courses.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 159.

CS 345 Software Engineering**Credits 3 Grading Basis Graded**

Study of modern methods and tools for the development of software products. Topics include software development process models, software development tools, design and construction principles and practice, software quality assurance and program management.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 159.

CS 347 Web Development**Credits 3 Grading Basis Graded**

This course covers design and development of World Wide Web applications. Students examine protocols, architectures, development languages and facilities, and issues such as accessibility, usability, security and reliability. Students work with a suite of development tools and languages to construct a client or server application that includes a GUI and a database.

Prerequisites: Fully admitted Computer Science majors or minors and a minimum grade of "C-" in CS 343.

CS 349 Developing Interactive Multimedia**Credits 3 Grading Basis** Graded

Students learn the concepts of multimedia, the issues in designing multimedia to interact effectively with users, the performance and speed issues in designing multimedia and how to implement interactive multimedia applications.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240.

CS 354 Introduction to Autonomous Robotics**Credits 3 Grading Basis** Graded

A hands-on introduction to programming autonomous mobile robots. The focus of this course is on designing robotic systems that navigate independently in complex environments. Specific topics include localization, mapping, kinematics, path planning and computer vision.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240 and either MATH 220 or MATH 229 or MATH 318.

CS 361 Computer Systems II**Credits 3 Grading Basis** Graded

Intermediate exploration of modern interrupt-driven computer systems. Explores models of computation and complex systems, techniques for communication and synchronization of parallel and concurrent software, and the protocols that make up the Internet.

Prerequisites: Fully admitted Computer Science majors only and a minimum grade of "C-" in CS 240 and CS 261.

CS 374 Database Systems**Credits 3 Grading Basis** Graded

An introduction to database design and management with emphasis on data definition, data manipulation and query languages found in modern database management systems.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in either CS 240 or CS 343 or equivalent.

CS 412 Applied Algorithms**Credits 3 Grading Basis** Graded

A practical study of algorithms and their use in problem solving. Algorithms are selected from categories including divide-and-conquer, greedy, backtracking, network flow and dynamic programming. The classes P and NP are introduced and intractable problems discussed. This is a programming intensive course.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 327 and either MATH 220 or MATH 229 or MATH 318. Students may not receive credit for both CS 412 and CS 452.

CS 415 Data Visualization Fundamentals and Techniques**Cross-listed with:** IT 415**Credits 3 Grading Basis** Graded

This course provides an introductory examination of data visualization through lectures, applied learning activities, readings and hands-on experience with current visualization tools. It also introduces a theoretical framework that covers different types of data, including temporal, geospatial, topical and network data, and their analysis and visualization. Students will improve their data and information literacy, strengthen data visualization skills and build analytic abilities relevant to visualization problems. They will also gain an overview of various data types and examine visualizations and workflows that aim to answer "when" (temporal data), "where" (geospatial data), "what" (topical data) and "with whom" (trees and networks) questions. The course is intended for students with little or no prior background in data visualization and is targeted to those interested in using visualization in their own work and future academic studies.

Prerequisites: CS 240 with a minimum grade of "C-" or IT 340.

CS 430 Programming Languages**Credits 3 Grading Basis** Graded

Several actual programming languages are studied in terms of the fundamental principles of computer programming language design, including object-oriented programming, functional programming, concurrent programming and logic programming.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240 and CS 261.

CS 432 Compilers**Credits 3 Grading Basis** Graded

Introduction to the theory and implementation of compilers. Explores the lexical, syntactic, and semantic analysis of high-level language code as well as code generation, optimization techniques, and architectural or operating system considerations. Course work includes a significant programming component.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 327 and CS 361.

CS 442 Logic in Computer Science**Credits 3 Grading Basis** Graded

An exploration of some of the many connections between logic and computing, such as the application of classical and temporal logic in program verification, logic and logic programming, decidability, computability, automatic theorem proving, the computational complexity of logic algorithms, and applications of logic in artificial intelligence. A course in discrete mathematics or logic is recommended as preparation for this course.

Prerequisites: Fully admitted Computer Science majors and minors only and junior or senior standing.

CS 444 Artificial Intelligence**Credits 3 Grading Basis** Graded

Students will study the history, premises, goals, social impact and philosophical implications of artificial intelligence. Students will study heuristic algorithms for large state spaces and learn to develop recursive and non-deterministic algorithms.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240 and either MATH 220, MATH 229 or MATH 318.

CS 445 Machine Learning**Credits 3 Grading Basis Graded**

An introduction to the field of machine learning. Explores key problems and algorithms in machine learning from both a theoretical and an applied perspective. Specific topics include supervised and unsupervised learning, anomaly detection, text mining and experimental design. Students who have completed BSAN 393, ISAT 341 and MATH 358 may also take this course.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 327 and either MATH 220, MATH 229 or MATH 318.

CS 446 Software Analysis and Design**Credits 3 Grading Basis Graded**

Contemporary software analysis and design methods, tools, notations, techniques, processes, principles and practices. Students solve analysis and design problems alone or in teams and present their work to their peers and the instructor.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240 and CS 345.

CS 447 Interaction Design**Credits 3 Grading Basis Graded**

Study of and practice with processes, principles, tools, models and techniques for designing interactions between humans and digital products and systems. Topics include physiological and psychological factors affecting interaction design, interaction design processes, interaction models, styles and paradigms, design notations and representations, prototyping, and interaction design evaluation.

Prerequisites: Fully admitted Computer Science majors or minors and a minimum grade of "C-" in CS 343.

CS 448 Numerical Analysis**Cross-listed with: MATH 448****Credits 3 Grading Basis Graded**

This course is the study and analysis of algorithms used to solve nonlinear equations and systems of linear and nonlinear equations. Students will learn iterative methods for matrices and Newton-type methods as well as numerical differential and integral calculus. Students will perform programming using a high-level language and/or software packages.

Prerequisites: Fully admitted Computer Science majors or minors only; and MATH 248 or consent of instructor.

CS 449 Numerical Analysis for Differential Equations**Cross-listed with: MATH 449****Credits 3 Grading Basis Graded**

Study and analysis of numerical techniques to solve ordinary and partial differential equations, including Euler, Runge-Kutta, Picard, finite-difference and finite-element methods. Programming using a high-level language and/or software packages.

Prerequisites: Fully admitted Computer Science majors or minors only and MATH 237, MATH 238 or MATH 336, and MATH 248.

CS 450 Operating Systems**Credits 3 Grading Basis Graded**

Introduction to the design and implementation of modern operating systems. Explores fundamental concepts of operating systems, memory management, virtualization, resource allocation, file systems and system protection mechanisms. Course work includes a significant programming component.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 361.

CS 452 Design and Analysis of Algorithms**Cross-listed with: MATH 452****Credits 3 Grading Basis Graded**

An introduction to algorithm design techniques and the mathematical analysis of algorithms. Techniques covered include divide-and-conquer, greedy, backtracking, network flow and dynamic programming. The classes P and NP are introduced and intractable problems are discussed. This is a proof-intensive course. Students may not receive credit for both CS 412 and CS 452.

Prerequisites: A grade of "C-" or better in CS 327 and either MATH 220 or MATH 229 or MATH 318. Students may not receive credit for both CS 412 and CS 452.

CS 455 Advanced Computer Networking**Credits 3 Grading Basis Graded**

An in-depth analysis of networking protocols and tools to capture, analyze and synthesize network traffic. Includes significant programming projects that simulate the operation of network protocols. Introduces the design principles and implementation of client-server network applications.

Prerequisites: Fully admitted computer science majors or minors only and a minimum grade of "C-" in CS 361.

CS 456 Computer Architecture**Credits 3 Grading Basis Graded**

Introduction to the design and implementation of modern CPU architectures. Explores hardware-based parallel execution, quantitative performance evaluation, I/O interfacing techniques and hardware descriptor languages. Course work includes a significant programming component.

Prerequisites: Fully admitted computer science majors or minors only and a minimum grade of "C-" in CS 361.

CS 457 Information Security**Credits 3 Grading Basis Graded**

This course covers the basic issues of information system security. The roles of planning, management, policies, procedures and personnel in protecting the confidentiality, integrity and availability of information are described. Specific threats (malicious code, network attacks and hostile content) and widely used countermeasures (access control, mechanisms, firewalls and intrusion detection systems) are also discussed.

Prerequisites: Fully admitted Computer Science majors or minors only. CS 361 and a minimum grade of "C-" in either MATH 220 or MATH 229 or MATH 318. Corequisites: CS 361.

CS 458 Cyber Defense**Credits 3 Grading Basis Graded**

A hands-on, lab-based learning experience in which the students engage in a series of mini projects to perform security assessment, penetration testing and hardening of networked systems. Students also participate in a cyber defense exercise.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grades of "C-" in CS 457 and CS 361.

CS 462 Network Applications Development**Credits 3 Grading Basis Graded**

Design and implementation of network-based applications using languages and architectures such as sockets, JAVA, TL1 and CORBA. Concepts in distributed processing, including synchronization of interprocess communication and management of replicated data. Analysis of performance issues related to distributed applications.

Prerequisites: Fully admitted Computer Science majors or minors only and CIS 361.

CS 470 Parallel and Distributed Systems**Credits 3 Grading Basis** Graded

Introduction to parallel and distributed systems. Explores shared memory, cluster, grid, peer-to-peer, and cloud computing models along with parallel software patterns, distributed file systems and performance considerations. Course work includes a significant programming component.

Prerequisites: Fully admitted computer science majors or minors only and a minimum grade of "C-" in CS 361.

CS 480 Selected Topics in Computer Science**Credits 1-3 Grading Basis** Graded

Topics in computer science which are of interest but not otherwise covered in the regular computer science offerings of the department. Offered only with the approval of the department head; may be repeated for credit when course content changes. Topics selected may dictate further prerequisites; students should consult the instructor prior to enrolling for course.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 159. Topics selected may dictate further prerequisites; students should consult the instructor prior to enrolling for course.

CS 482 Selected Topics in Information Security**Credits 1-3 Grading Basis** Graded

Topics in information security. Offered only with the approval of the department head; may be repeated for credit when course content changes. Topics selected may dictate further prerequisites; students should consult the instructor prior to enrolling for course.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240 and CS 261.

CS 488 Computer Graphics Applications**Credits 3 Grading Basis** Graded

This course develops a computer graphics application package based on standard graphics functions as well as attributes of a graphical user interface. It includes experience in applying interactive computer graphics techniques to industrial problems.

Prerequisites: Fully admitted Computer Science majors or minors only and a minimum grade of "C-" in CS 240 and CS 261.

CS 497 Independent Study**Credits 1-3 Grading Basis** Graded

An advanced course to give independent study experience under faculty supervision. May be taken multiple times for credit, but no more than three credits may be used in the computer science program graduation requirements.

Prerequisites: Fully admitted Computer Science majors or minors only and junior standing, and permission of the program coordinator.

CS 499A Honors**Credits 1 Grading Basis** Graded**CS 499B** Senior Honors Thesis**Credits 3 Grading Basis** Graded**CS 499C** Senior Honors Thesis**Credits 2 Grading Basis** Graded

Continuing Education (CE)

CE 490 Special Studies in Continuing Education**Credits 1-3 Grading Basis** Graded

This course is designed to allow exploration of current topics of interest including various trends and issues in a given field of study.

Prerequisites: Continuing Education Students Only

Criminal Justice (CRJU)

CRJU 215 Introduction to Criminal Justice**Credits 3 Grading Basis** Graded

An introduction to the development of the American criminal justice system from early English beginnings to the present in its three dimensions: police, courts and corrections.

Prerequisites: Admission to the criminal justice minor.

CRJU 300 Ethics in Criminal Justice**Credits 3 Grading Basis** Graded

This class offers an overview of ethical issues in the various branches of the criminal justice system and reviews approaches to establishing and using ethical practices.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 301 Advanced Level Special Topics in Criminal Justice**Credits 3 Grading Basis** Graded

This course provides an advanced level examination of topics that are of current interest in the field of criminal justice. The class may be repeated for credit when course content changes.

Prerequisites: CRJU 215 or JUST 210.

CRJU 321 Criminalistics**Credits 3 Grading Basis** Graded

This course introduces students to crime scene investigation and the major disciplines of modern forensic science. Topics include an examination of the historical background of forensic science in the criminal justice system, an assessment of general principles of the current practice of forensic science, examination of the role of expert testimony and likely interaction(s) of the forensic scientist with other individuals and components of the criminal justice system.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 325 Criminology**Credits 3 Grading Basis** Graded

Study of the extent, causes and possible deterrents to crime including murder, assault, white-collar offenses and organized crime with attention to the role of the victim and policy implications.

Prerequisites: CRJU 215.

CRJU 328 Criminal Procedure**Credits 3 Grading Basis** Graded

This course studies the criminal justice process from arrest through appeal with emphasis upon the rights of the accused including due process, the right to counsel, search and seizure, and the privilege against self-incrimination.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 329 Criminal Investigation and Evidence**Credits 3 Grading Basis** Graded

This course covers characteristics, legal aspects, organizational objectives, theories and systematic procedure of criminal investigation. It includes a survey of the investigative function, interviewing witnesses, interrogation, physical evidence, the investigation of common serious offenses and the principles of evidence, including the legal rules controlling the presentation of evidence in court.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 335 Law Enforcement**Credits 3 Grading Basis** Graded

This course provides students with an overview of the practice of law enforcement and the legal and social issues associated with this work in the United States.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 337 Courts and the Judiciary**Credits 3 Grading Basis** Graded

This class offers students an in-depth introduction to the workings of the court system in the United States.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 340 Administration of Justice**Credits 3 Grading Basis** Graded

This course is designed to identify unique challenges to administrators of criminal justice organizations. The structures, functions and processes in the administration of criminal justice organizations is examined. Topics of interest include a variety of public management theories, the role of leadership, and communication as it relates to criminal justice organizations.

Prerequisites: Admission to the Criminal Justice minor and either CRJU 215 or JUST 210.

CRJU 401 Internship in Criminal Justice**Credits 3 Grading Basis** Graded

This course allows students to receive academic credit for work experienced in an agency or organization related to the criminal justice minor. Students should consult the director of the criminal justice minor for assistance in arranging approved internships.

Prerequisites: CRJU 215.

Cross Disciplinary Studies (CDS)

CDS 301 Special Topics**Credits 1-3 Grading Basis** Graded

This course allows instructors working through recognized university centers or institutes to offer an examination of current topics that are cross disciplinary in nature and not covered elsewhere in the curriculum. The course may be repeated for credit when course content changes.

CDS 401 Internship**Credits 1-3 Grading Basis** Graded

This course allows students to receive academic credit for work experienced in a recognized JMU center or institute. Internships must be approved in advance by the center director and follow the guidelines established by the participating center or institute. Internship is granted at the discretion of the center director.

Prerequisites: Junior or senior standing only.

Dance (DANC)

DANC 110 Associate Group Dance Repertory I (0, 4)**Credits 1 Grading Basis** Graded

Introduction to group dance experiences through rehearsal, performance, dance technique training and technical theatre practice.

Prerequisites: Dance majors only and concurrent enrollment in a dance technique course.

DANC 140 Elementary Modern Dance (0,4)**Credits 2 Grading Basis** Graded

Emphasis on modern dance technique and fundamentals of improvisation and choreography. May be repeated for credit.

DANC 142 Elementary Ballet (0, 4)**Credits 2 Grading Basis** Graded

Fundamentals of ballet technique, basic vocabulary and combinations. May be repeated for credit.

DANC 143 International Folk Dance (0, 4)**Credits 2 Grading Basis** Graded

Traditional folk dance steps and international folk dances. Emphasis on dances from eastern and western Europe, Mexico and Israel. May be repeated for credit.

DANC 144 Ballroom Dance (0, 4)**Credits 2 Grading Basis** Graded

Ballroom dance skills including steps, styling and leading, and following for American and Latin ballroom dance forms. May be repeated for credit.

DANC 146 Jazz Dance (0, 4)**Credits 2 Grading Basis** Graded

Fundamentals of jazz technique, basic vocabulary and combinations. May be repeated for credit.

Prerequisites: School of Theatre and Dance majors and minors only.

DANC 147 Tap Dance (0, 4)**Credits 2 Grading Basis** Graded

Fundamentals of tap dance, basic vocabulary and combinations. May be repeated for credit.

Prerequisites: School of Theatre and Dance majors and minors only.

DANC 210 Associate Group Dance Repertory II (0, 4)**Credits 1 Grading Basis** Graded

Practice in group dance experiences through rehearsal, performance, dance technique training and technical theatre practice.

Prerequisites: Dance majors only and concurrent enrollment in a dance technique course.

DANC 211 Contemporary Dance Ensemble Repertory I (0, 4)**Credits 2 Grading Basis** Graded

Introduction to a modern dance ensemble with performance, choreographic and technical theatre experiences. May be repeated for credit.

Prerequisites: Dance majors only and concurrent enrollment in a dance technique course.

DANC 212 Virginia Repertory Dance Company I (0, 4)**Credits 2 Grading Basis** Graded

Rehearsal, performance and technical theatre experiences in a modern dance company. May be repeated for credit.

Prerequisites: Dance majors only and concurrent enrollment in a dance technique course.

DANC 215 The Dance Experience [C2VPA]**Credits 3 Grading Basis** Graded

This course is an exploration of dance as an art form. It emphasizes learning about and experiencing a broad spectrum of dance forms through lecture, discussion, movement exploration, and attending and viewing dance performances. The course considers the historic and cultural significance of dance as well as the components of a dance performance: dancers, choreographic elements, production design (costuming, lighting, scenic elements, etc.) and audience. The course reveals how dance has informed the human experience throughout history. It does this by placing dance's "embodied way of knowing" within social, artistic, religious, political and cultural contexts. It provides an opportunity for the student to experience an appreciation for the dynamic, varied and complex role that dance has played in human societies over time, and understand how it relates to other art forms. May be used for general education credit.

DANC 240 Intermediate Modern Dance I (0, 4)**Credits 2 Grading Basis** Graded

Intermediate skills in modern dance technique. May be repeated for credit.

Prerequisites: None for School of Theatre and Dance majors and dance minors. Non-majors: DANC 140 or permission of the instructor.

DANC 242 Intermediate Ballet I (0, 4)**Credits 2 Grading Basis** Graded

Intermediate skills in ballet technique. May be repeated for credit.

Prerequisites: School of Theatre and Dance majors and dance minors only, or DANC 142 or permission of the instructor.

DANC 245 Dance Improvisation (0, 4)**Credits 2 Grading Basis** Graded

Development of individual, group and environmental awareness; extension of individual movement vocabulary; and theory and exploration of the interrelationships of the visual and theatre arts through structured improvisation.

Prerequisites: School of Theatre and Dance majors and dance minors only.

DANC 246 Intermediate Jazz**Credits 2 Grading Basis** Graded

Intermediate skills in jazz dance technique, vocabulary and movement combinations. May be repeated for credit.

Prerequisites: None for School of Theatre and Dance majors and dance minors. Non-majors: DANC 146 or permission of the instructor.

DANC 247 Intermediate Tap**Credits 2 Grading Basis** Graded

Intermediate skills in tap dance technique, vocabulary and models of sequencing. May be repeated for credit.

Prerequisites: None for School of Theatre and Dance majors and dance minors. Non-majors: DANC 147 or permission of the instructor.

DANC 248 History of Dance: Renaissance Through the 20th Century**Credits 3 Grading Basis** Graded

A survey of dance history in Western civilization from the Renaissance to the present. Emphasis is on the dance idioms of ballet and modern.

DANC 303 Topics in Dance**Credits 1-3 Grading Basis** Graded

Study of selected timely topics in dance. May be repeated when course content changes. See MyMadison for current topic.

DANC 311 Contemporary Dance Ensemble Repertory II (0, 4)**Credits 2 Grading Basis** Graded

Intermediate-level experiences in performance, choreography and technical theatre with a modern dance ensemble. May be repeated for credit. Prerequisite: DANC 211 or the equivalent and concurrent enrollment in a dance technique course.

Prerequisites: DANC 211 or the equivalent and concurrent enrollment in a dance technique course.

DANC 312 Virginia Repertory Dance Company II (0, 4)**Credits 2 Grading Basis** Graded

Advanced rehearsal, performance and technical theatre experience in a modern dance company. May be repeated for credit.

Prerequisites: DANC 212 or the equivalent and concurrent enrollment in a dance technique course.

DANC 315 Injury Prevention and Conditioning**Credits 3 Grading Basis** Graded

This course is designed to educate the dancer and dance education student on specific injury prevention and conditioning principles to improve overall dancer health and wellness. A combination of lecture and movement-based labs will cover all aspects that may contribute to injury risk in dancers and allows recognition of these risks in the educational or performance setting.

Prerequisites: Dance majors, dance minors and STAD majors; or permission of the instructor.

DANC 320 Anatomy and Somatic Studies for the Dancer**Credits 3 Grading Basis** Graded

An introduction to the structure and function of the human body, along with an overview of contemporary bodywork theories and techniques essential to the education of dancers. Emphasis will be on the application of knowledge within the context of dance technique.

Prerequisites: Sophomore, junior and senior dance majors only.

DANC 325 Dance in Community**Credits 3 Grading Basis** Graded

This course introduces students to the use of creative movement and dance in community settings. Emphasis is on concepts and skills utilized in designing and implementing movement experiences for diverse populations.

DANC 340 Intermediate Modern Dance II (0, 4)**Credits 2 Grading Basis** Graded

Modern dance technique for an accelerated intermediate level. May be repeated for credit.

Prerequisites: DANC 240 or permission of the instructor.

DANC 342 Intermediate Ballet II (0, 4)**Credits 2 Grading Basis** Graded

Ballet technique on an accelerated intermediate level. May be repeated for credit.

Prerequisites: DANC 242 or permission of the instructor.

DANC 345 Dance Composition I (2, 2)**Credits 3 Grading Basis** Graded

Introductory studies of dance composition with focus on the learning and development of choreographic techniques and styles. Compositional movement studies and solo/duet dances will be utilized for exploring and developing the student's individual creativity.

Prerequisites: Dance majors only.

DANC 346 Intermediate Jazz II/Musical Theatre Styles**Credits 2 Grading Basis** Graded

Intermediate Jazz II / Musical Theatre Styles is a continuation of the jazz dance techniques in the dance program at the upper-intermediate level. The primary focus of this class will be on the study and training of historic and contemporary musical theatre jazz movement and its cultural influences. The course may be repeated for credit.

Prerequisites: DANC 246 or permission of the instructor.

DANC 380 Dance Teaching Practicum**Credits 3 Grading Basis** Graded

This course provides pre-service teachers with hands-on teaching experience prior to full-time student teaching. The course provides the opportunity for dance education students to observe certified teachers, plan lessons, reflect on effective teaching strategies, analyze the complexities of teaching and demonstrate professional behaviors that promote student learning. Course work includes topics on classroom management, assessment and effective teaching strategies.

DANC 390 Directed Projects**Credits 1-3 Grading Basis** Graded

Supervised projects related to the study of dance. Credit given for original, individual or group programs beyond the usual course offerings in dance. A suitable completed project or report is required before credit will be given. Teaching assistantships are also registered as directed projects. May be repeated multiple times.

Prerequisites: Dance majors only.

DANC 411 Contemporary Dance Ensemble Repertory III (0, 4)**Credits 2 Grading Basis** Graded

Advanced-level experiences in performance, choreography and technical theatre with a modern dance ensemble. May be repeated for credit.

Prerequisites: DANC 311 or the equivalent and concurrent enrollment in a dance technique course.

DANC 412 Virginia Repertory Dance Company III (0, 4)**Credits 2 Grading Basis** Graded

Advanced rehearsal, performance and technical theatre experience in a modern dance company for the professional level dancer. May be repeated for credit.

Prerequisites: DANC 312 or the equivalent and concurrent enrollment in a dance technique course.

DANC 440 Advanced Modern Dance (0, 4)**Credits 2 Grading Basis** Graded

Modern dance technique on an advanced level. May be repeated for credit.

Prerequisites: DANC 340 or the equivalent.

DANC 442 Advanced Ballet (0, 4)**Credits 2 Grading Basis** Graded

Ballet technique on an advanced level. May be repeated for credit.

Prerequisites: DANC 342 or the equivalent.

DANC 445 Dance Composition II (2, 2)**Credits 3 Grading Basis** Graded

Dance composition study involving the development and use of complex choreographic structures with emphasis on creating small and large group dance compositions, as well as the exploration of contemporary choreographic styles and techniques.

Prerequisites: DANC 345 or permission of the instructor.

DANC 446 Advanced Jazz**Credits 2 Grading Basis** Graded

A continuation of the dance technique in the dance program. Advanced skills in jazz technique with special emphasis on the development of performance skills. May be repeated for credit.

DANC 449 The Dance Professional**Credits 3 Grading Basis** Graded

Introduction to the professional dance world through lectures, discussions and research. Emphasis will be placed on the preparation of skills and materials necessary to pursue dance as a career. Contemporary dance trends and issues will also be explored.

Prerequisites: Senior dance majors only.

DANC 479 Methods of Teaching Dance**Credits 3 Grading Basis** Graded

An exploration of teaching creative movement in elementary school and dance in secondary school. Topics include creative movement, dance education and arts integration. This course considers lesson planning, classroom management and assessment in middle and high school.

Prerequisites: Junior and senior dance majors only.

DANC 480 Student Teaching**Credits 3-12 Grading Basis** Credit / No Credit

Enables students to apply, in the public school classrooms and the comprehensive child development programs, those skills and attitudes acquired in all components of teacher education. Under the guidance of university supervisors, students are provided activities designed to familiarize them with the classroom teacher's role.

Prerequisites: PSYC 160 and EDUC 200.

DANC 490 Special Studies in Dance**Credits 1-3 Grading Basis** Graded

Special studies are independent studies for students to pursue individual research under the guidance of a faculty advisor. Special studies differ from directed projects in that they are more research-based than project-based, although some student mainstage designs might also be undertaken as special studies. Based on the agreed upon workload, this course may be assigned credit hours of 1-3 credits.

DANC 495 Internship**Credits 3-6 Grading Basis** Graded

Internships may be registered for credit at the discretion of the individual faculty advisor. Overrides for internships must be completed no later than four weeks prior to the internship start date. Based on the agreed upon workload, this course may be assigned credit hours of 3-6 credits.

DANC 499 Honors**Credits 1-3 Grading Basis** Graded

Repeatable for a maximum of 6 credits.

Data Sciences (DATA)

DATA 200 Introduction to Data**Credits 3 Grading Basis** Graded

An interdisciplinary introduction to how data is organized, managed, visualized and analyzed across STEM disciplines. Ethical collection and use of data, with real-world test cases. Fundamentals of the tools and applications used in the analysis of data sets.

DATA 358 Introduction to Machine Learning**Cross-listed with:** MATH 358**Credits 3 Grading Basis** Graded

This is an introductory course on machine learning with focus on regression and classification methods. The objective is to familiarize students with some basic models and algorithms for machine learning and prepare them for research or industry application of machine learning techniques. Several software libraries and publicly available datasets will be used to illustrate the application of these techniques. The emphasis will be on machine learning algorithms and application, with some broad explanation of the underlying principles.

Prerequisites: MATH 229, or MATH 220 with permission of the instructor.

Digital Studies (DS)

DS 101 Foundations in Digital Studies**Credits 3 Grading Basis** Graded

This course uses hands-on work with digital tools and techniques to answer questions in the humanities and social sciences. It examines the histories of major digital projects and methods; introduces students to a range of resources, spaces, and practitioners within the College of Arts and Letters and across JMU; and considers the impact of digital technologies as complex, socially situated and political tools through which humans make meaning. This course is required for the Digital Studies minor, but is open to all.

Disability Studies (DST)

DST 100 Introduction to Disability Studies**Credits 1 Grading Basis** Credit / No Credit

This course is required for the disability studies minor. It provides students with an overview of the ways disability is positioned within a number of intellectual frameworks, including the political, rhetorical, social, educational, historical and legal.

Prerequisites: Declaration of the disability studies minor.

Early Childhood Education (ECED)

ECED 304 Diversity, Equity and Justice in Early Childhood Education**Credits 3 Grading Basis** Graded

Considers current problems and issues in early childhood education.

Prerequisites: Fully accepted into Teacher Education; minimum grades of "C" or better in ECED 372, ECED 444, ECED 445 and LED 355. Corequisites: ECED 412, ECED 461 and EXED 325.

ECED 310 Diversity in Early Childhood Education**Credits 3 Grading Basis** Graded

This course guides students in critically examining their own perspectives regarding diversity, seeks to expand students' awareness and understanding of individuals and groups representing differences, and explores pedagogical issues and practices essential for embracing the whole community of learners and families.

ECED 372 Introduction to Early Childhood Education**Credits 3 Grading Basis** Graded

Introductory study of the role of the teacher, the role of the learner and the developing child as the basis for designing programs and developing curriculum for children 0-8 with different abilities and from various cultures.

Prerequisites: Applied to Teacher Education (01 Flag).

ECED 412 Natural and Social Sciences for Young Children**Credits 3 Grading Basis** Graded

Study of content, processes, teaching methods and materials for teaching science and social studies in the early childhood classroom. Knowledge of cognitive development as applied to the selection of content in method will be examined.

Prerequisites: Fully accepted into Teacher Education (02 Flag);

Minimum grades of "C" in ECED 372, ECED 444, ECED 445 and LED 355.

Corequisites: ECED 304, ECED 461 and EXED 325.

ECED 441 Practicum in Child Development**Credits 1 Grading Basis** Graded

This course is a supervised field experience in which students work in an early childhood laboratory classroom with pre-kindergarten age children. It emphasizes applications of age-appropriate guidance strategies for facilitating children's total development, including children with diverse needs. Observational strategies for assessing growth and progress are developed.

Prerequisites: A minimum grade of "C" in each of ELED 271, ECED 372 and LED 366; Corequisites: ECED 442, ECED 443 and LED 436.

ECED 442 The Young Child**Credits 3 Grading Basis** Graded

This course integrates child development knowledge and theories, academic content knowledge, and age/stage appropriate guidance strategies for teaching children pre-kindergarten through grade three. Emphasis on observational/assessment strategies and inquiry processes related to young children's growth and development as a basis for teaching decisions.

Prerequisites: Applied to Teacher Education (01 Flag); a minimum grade of "C" in ECED 372. Corequisites: ECED 443, ECED 445 and LED 456.

ECED 443 Practicum in Primary Grades**Credits 1 Grading Basis** Graded

This field experience supports the study of child development and emerging literacy through direct observation and interactions with children in a classroom setting. Candidates will examine and reflect on their personal attitudes, assumptions and behaviors toward students and their families who represent diverse class, cultural and linguistic backgrounds. Students will complete 80 hours over the course of the full semester (16 weeks).

Prerequisites: Applied to Teacher Education (01 Flag), A minimum grade of "C" in ECED 372. Corequisites: ECED 442, ECED 445 and LED 456.

ECED 444 Mathematics Content and Pedagogy for Early Childhood Education I**Credits 3 Grading Basis** Graded

This is the first of two courses that helps prospective early childhood teachers (PreK-3) develop mathematical content and pedagogical knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-3. Emphasis is on children's mathematical learning in the following mathematical strands: number and number sense, computation and estimation, and patterns, functions and algebra.

Prerequisites: Applied to Teacher Education (01 Flag). Corequisites: ECED 372 and LED 355.

ECED 445 Mathematics Content and Pedagogy for Early Childhood Education II**Credits 3 Grading Basis** Graded

This is the second of two courses that helps prospective early childhood teachers (PreK-3) develop mathematical content and pedagogical knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-3. Emphasis is on children's mathematical learning of geometry, measurement, data analysis and probability.

Prerequisites: Applied to Teacher Education (01 Flag); a minimum grade of "C" in ECED 372. Corequisites: ECED 304, ECED 442, ECED 443 and LED 456.

ECED 461 Integrated Day Practicum**Credits 1 Grading Basis** Graded

This all-day, primary-grade practicum allows students to demonstrate their educational decision-making skills through planning, implementing and evaluating appropriate activities for children of diverse interests, needs and abilities. Strategies to assess learning, guide behavior, work with professionals and involve families are applied in this practicum and accompanying seminar.

Prerequisites: Fully accepted into Teacher Education (02 Flag); minimum grades of "C" in ECED 372, ECED 444, ECED 445 and LED 355. Corequisites: ECED 304, ECED 412 and EXED 325.

ECED 480 Student Teaching ECED**Credits 8 Grading Basis** Credit / No Credit

Enables students to apply in public school classrooms those understandings, skills and attitudes acquired in all components of teacher education. Under the guidance of university supervisors and cooperating teachers, students engage in full-time teaching responsibilities.

Prerequisites: Meet all requirements for student teaching (03 Flag); minimum grades of "C" in ECED 304, ECED 372, ECED 442, ECED 443, ECED 444, ECED 445, LED 355 and LED 456.

ECED 481 Fieldwork in Family and Community**Credits 2 Grading Basis** Credit / No Credit

This fieldwork is designed to provide support for students and reinforces skills and concepts learned during the education program which are being applied during student teaching. Particular attention is given to school and family/community unity.

Prerequisites: Minimum grades of "C" in ECED 372, ECED 441, ECED 442, ECED 443, ECED 454, ECED 461, LED 366, LED 436, and ECED 444. Corequisites: ECED 480.

ECED 490 Special Studies in Early Childhood Education**Credits 1-3 Grading Basis** Graded

Designed to give capable students opportunities to complete independent research on educational problems under faculty guidance. The plan for the study must be presented to the department head in prescribed form for approval prior to registration.

Prerequisites: Applied to Teacher Education (01 Flag).

ECED 498 Capstone in Early Childhood Education**Credits 3 Grading Basis** Graded

The teaching seminar is a space for students to reflect on their experiences, grapple with how to best meet the needs of diverse learners and contemplate their evolving identity as a teacher. Through this course, students will recognize the vital role that families and communities play in shaping and supporting children. Students will become a resource for themselves, their peers, their grade level team, their students and the families. As students transition from being a student to a professional, they will continue to develop not only their content knowledge and pedagogy, but also their skills and dispositions.

Corequisites: ECED 480.

ECED 499 Honors Project**Credits 1 Grading Basis** Graded

Prerequisites: Teacher Ed Program 02 Flag.

Economics (ECON)

ECON 122 Introduction to an Economic Topic**Credits 1 Grading Basis** Graded

A one-week course offered during winter term or summer that introduces an economic topic/issue that is currently capturing significant interest in today's world.

ECON 200 Introduction to Macroeconomics [C4GE]**Credits 3 Grading Basis** Graded

Behavior of systems at the national and international levels. Topics include the methodology of economics as a social science, supply and demand, definition and measurement of important macroeconomic variables, and theoretical models of growth, inflation, interest rates, unemployment, business cycles, stabilization policy, exchange rates and the balance of payments. May be used for general education credit.

Prerequisites: Not open to students who are enrolled in or who have received credit for ECON 332. Maximum of 2 attempts for ECON 200.

ECON 201 Introduction to Microeconomics**Credits 3 Grading Basis** Graded

Topics covered include supply and demand, consumer choice, economics of the firm and industry, production costs, distribution theory, international trade, public policy, comparative economic systems and the philosophy of economics. This course is not open to students who are enrolled in or who have received credit for ECON 331.

Prerequisites: Not open to students who are enrolled in or who have received credit for ECON 331. Maximum of 2 attempts for ECON 201.

ECON 222 Contemporary Economic Issues and Policy Alternatives**Credits 3 Grading Basis** Graded

Application of elementary economic theory to current economic issues. Special emphasis is placed on public policy alternatives.

Prerequisites: ECON 200 and ECON 201.

ECON 270 International Economics**Credits 3 Grading Basis** Graded

A survey of the relationships among national economies, including trade theory, trade policy, international monetary relations and the balance of payments.

Prerequisites: ECON 200 and ECON 201. Not open to students who are enrolled in or have already received credit in ECON 370 or ECON 372.

ECON 284 Introduction to Quantitative Analysis in Economics**Credits 3 Grading Basis** Graded

Students will learn key mathematical and statistical concepts used in economics. Topics will include algebra of summation, optimization, probability theory, joint probability distribution, conditional expectations, hypothesis testing and an introduction to data analysis using statistical software. Concepts will be emphasized using economic examples. Does not count as COB 291 for COB 300.

Prerequisites: MATH 205, MATH 231, MATH 235 or equivalent; and either COB 191, MATH 220, MATH 318 or equivalent.

ECON 300 Special Topics in Economics**Credits 3 Grading Basis** Graded

Examination of special topics in theoretical or applied economics not covered in the current economics curriculum. Specific topics to be determined by the instructor.

Prerequisites: ECON 200 and ECON 201.

ECON 302 History of Economic Thought**Credits 3 Grading Basis** Graded

Major contributions within the history of economic thought are studied in relation to both the historical circumstances within which they arose and the role each played in shaping contemporary, competing economic doctrines.

Prerequisites: ECON 200 and ECON 201.

ECON 305 Environmental Economics**Credits 3 Grading Basis** Graded

An analysis of the problems of the environment, their causes and alternative proposed methods of solution. Air and water pollution will be stressed as case studies of environmental problems.

Prerequisites: ECON 201.

ECON 306 The Economics of Women and the Family**Credits 3 Grading Basis** Graded

Examines facts and theories pertaining to the various economic roles of women in America. The economics of marriage, divorce and childbearing are examined as are empirical and theoretical explanations of occupational and wage differentials between the sexes.

Prerequisites: ECON 201.

ECON 310 Economic History of the United States**Credits 3 Grading Basis** Graded

A survey of the economic growth and development of the United States from Colonial times to the present.

Prerequisites: ECON 200 and ECON 201.

ECON 312 Comparative Economic Systems**Credits 3 Grading Basis** Graded

An examination of the distinguishing characteristics, institutions and performances of the various types of major economic systems in the world today.

Prerequisites: ECON 200 and ECON 201.

ECON 325 Money and Banking**Cross-listed with:** FIN 325**Credits 3 Grading Basis** Graded

The purpose of this course is to examine the economic role of money, banking and monetary policy within current institutional settings and under alternative theories explaining the interrelationships between money, the financial system and economic activity.

Prerequisites: ECON 200 and ECON 201.

ECON 326 Public Finance**Credits 3 Grading Basis** Graded

Introduction to the field of public finance including theories and principles of taxation, government expenditure, public debt, and fiscal administration. Studies interrelationships between federal, state and local finance, shifting and incidence of tax, and the burden of public debt.

Prerequisites: ECON 200 and ECON 201.

ECON 327 Game Theory**Credits 3 Grading Basis** Graded

Examines interdependent decision making in economics and other social sciences, and covers both non-cooperative and cooperative games. Topics may include applications of game theory to industrial organization, law and economics, public choice, political economy, evolutionary biology, international affairs and theories of justice.

Prerequisites: ECON 200 and ECON 201 or permission of the instructor.

ECON 328 Behavioral Economics**Credits 3 Grading Basis** Graded

An introductory course in behavioral economics that combines insights from psychology with economics. Using data from field and lab experiments, we will examine the explanatory power of the standard economic model and learn new frameworks that incorporate psychologically realistic assumptions grounded in observed behavior. Among the topics studied are decision-making under uncertainty, intertemporal choice and others regarding preferences. The course will conclude by examining implications this approach to economic analysis has for public policy.

Prerequisites: ECON 201.

ECON 329 Health Economics**Credits 3 Grading Basis** Graded

This course explores the structure, conduct, and performance of hospitals, physician practices, pharmaceutical and medical device manufacturers, and long-term care facilities. The demand for healthcare and insurance is presented using economic theories of risk and uncertainty. Economic evaluation of healthcare treatments and contemporary policies are examined.

Prerequisites: ECON 200 and ECON 201.

ECON 331 Intermediate Microeconomic Theory**Credits 3 Grading Basis** Graded

Intermediate analysis of the determination of price, resource allocation and product distribution in a free enterprise economy.

Prerequisites: Minimum grade of "C" in the following courses: ECON 200 and ECON 201; and either ECON 284, COB 291, or MATH 236.

ECON 332 Intermediate Macroeconomic Theory**Credits 3 Grading Basis** Graded

Intermediate-level analysis of the major approaches to the determination of economic aggregates, with emphasis given to structuring a common analytic framework.

Prerequisites: Minimum grade of "C" in the following courses: ECON 200 and ECON 201; and either ECON 284, COB 291, or MATH 236.

ECON 340 Economics of Natural Resources**Credits 3 Grading Basis** Graded

Emphasizes availability of exhaustible resources and optimum utilization rates. Examines questions of intertemporal allocation and costs of growth.

Prerequisites: ECON 200 and ECON 201.

ECON 345 Industrial Organization**Credits 3 Grading Basis Graded**

An examination of contemporary U.S. industrial concentration both in the aggregate and within particular industries with emphasis on public policy implications. Alternative theories of the firm are considered in relation to different market structures.

Prerequisites: ECON 200 and ECON 201.

ECON 360 Labor Economics**Credits 3 Grading Basis Graded**

Study of the economics of labor markets. Attention is given to the structure and operation of labor markets, wage determination, employment, unions, and contemporary labor problems and policies.

Prerequisites: ECON 200 and ECON 201.

ECON 365 Economic Development**Credits 3 Grading Basis Graded**

A study of the characteristics of under-development, theories of economic development and the underlying causes for varying standards of living among the world's people. Considerable time will be spent on studying social and cultural factors that influence economic growth and their potential effect on the economic progress of the less-developed countries.

Prerequisites: ECON 200 and ECON 201.

ECON 370 International Trade and Trade Policies**Credits 3 Grading Basis Graded**

An examination of the classical and modern theories of international trade, the effects of such trade on the domestic economy, the effects of barriers to free trade and an appraisal of U.S. commercial policy since 1948.

Prerequisites: ECON 200 and ECON 201.

ECON 372 International Finance and Payments**Cross-listed with: FIN 372****Credits 3 Grading Basis Graded**

Examines international financial markets, instruments and institutions; determination of spot and forward exchange rates, interest arbitrage, hedging and speculation; and alternative policies for achieving equilibrium in international payments.

Prerequisites: ECON 200 and ECON 201.

ECON 382 Urban Economics**Credits 3 Grading Basis Graded**

A detailed examination of the economic aspects of urbanization with emphasis on metropolitan land use and location theory. Urban problems considered include housing, poverty, labor markets and municipal finances.

Prerequisites: ECON 200 and ECON 201.

ECON 384 Mathematical Economics**Credits 3 Grading Basis Graded**

Course will provide basic mathematical tools for microeconomic and macroeconomic analysis. Topics covered include linear algebra, univariate and multivariate differentiation, economic interpretation of the derivatives, constrained and unconstrained optimization, expected utility model, and dynamic analysis of economic models.

Prerequisites: Minimum grade of "C" in the following courses: ECON 200 and ECON 201; and either ECON 284, COB 291, or MATH 236.

ECON 385 Econometrics**Credits 3 Grading Basis Graded**

This course covers the application of single- and multi-variable linear regression models to economic outcomes, with emphasis on hypothesis testing, causal effect estimation and data analyses using statistical software.

Prerequisites: Minimum grade of "C" in the following courses: ECON 200 and ECON 201; and either ECON 284, COB 291, or MATH 236.

ECON 394 Economics Internship**Credits 3 Grading Basis Credit / No Credit**

Academic credit for an approved internship experience. Registration for the course must be concurrent with the internship. An application showing how all requirements for the internship will be met must be approved prior to registration. May be taken on a credit/no credit basis only.

Prerequisites: ECON 200 and ECON 201.

ECON 399 Career Preparation in Economics**Credits 0 Grading Basis Credit / No Credit**

Students participate in a number of activities designed to enhance career preparedness and job search outcomes in economics as approved by the economics program (including attending workshops and registering to be informed of job opportunities).

Prerequisites: ECON 331, ECON 332, ECON 385 and senior standing.

ECON 400 Advanced Topics in Economics**Credits 3 Grading Basis Graded**

Examination of special topics in theoretical or applied economics not covered in the current economics curriculum. Specific topics to be determined by the instructor.

Prerequisites: ECON 331 and ECON 332 or permission of the instructor.

ECON 405 Political Economy**Credits 3 Grading Basis Graded**

Evaluation and critique of mainstream and nontraditional economic paradigms. The interaction of economics and politics in the United States as it affects the distribution of wealth and domestic and international economic policies.

Prerequisites: ECON 200, ECON 201, and junior or senior standing.

ECON 426 Theory of Public Choice**Credits 3 Grading Basis Graded**

Examines the justification for and nature of public sector activity in a market-based mixed economy. Emphasis is placed on theories of market failure, voting models, conditions of production and provision in the public sector, and models of bureaucratic behavior.

Prerequisites: ECON 326 or ECON 331.

ECON 427 Experimental Economics**Credits 3 Grading Basis Graded**

This course focuses on using incentivized experiments, whereby subjects make financially-motivated decisions in controlled environments, to identify causality, test and develop theories, and explore behavioral phenomena related to economic issues. Students will study and apply the methods and data analysis tools commonly used in experimental economics to a broad range of economic topics. Established experimental results from previous and current economic research will also be thoroughly examined.

Prerequisites: ECON 327 or ECON 331.

ECON 430 Monetary Theory**Credits 3 Grading Basis Graded**

Examines alternative theories of the relationships between money, interest rates, price levels, employment and output in order to assess the effectiveness of monetary policy for economic stabilization.

Prerequisites: ECON 332.

ECON 431 Advanced Microeconomic Theory**Credits 3 Grading Basis Graded**

Examines theories of general equilibrium and the distribution of income, welfare economics, capital theory and information theory.

Prerequisites: ECON 331.

ECON 432 Advanced Macroeconomics**Credits 3 Grading Basis Graded**

This course studies macroeconomics at an advanced level. Topics covered in this course may include short-run dynamics of an economy, theory of economic fluctuations, models of long-run economic growth, role of fiscal and monetary policies, and comparative macroeconomic analysis of the U.S. and other economies.

Prerequisites: ECON 332.

ECON 437 Economic Growth**Credits 3 Grading Basis Graded**

Economic growth is at the heart of economics: improving standards of living in the long run. Students will study models of economic growth, their applications and the empirical evidence across countries. Particular emphasis will be placed on recent advances in the area. Students will apply data analysis methods commonly used in economic growth to a broad range of topics.

Prerequisites: ECON 332.

ECON 455 Economics of Regulated Industries**Credits 3 Grading Basis Graded**

A study of the rationale, methods and impact on industry behavior of government regulations including public utility regulation and antitrust policies relating to monopoly and competition in the United States.

Prerequisites: ECON 385 or permission of the instructor; and ECON 331 or ECON 345 (to use ECON 345, students must have permission of the ECON 455 instructor).

ECON 475 Regional Economics**Credits 3 Grading Basis Graded**

A study of local and subnational economics viewed as integral parts of a unified system. Emphasis will be given to the basic economic forces associated with regional growth and decline and related public policy considerations.

Prerequisites: ECON 200 and ECON 201.

ECON 483 Applied Time Series Analysis**Credits 3 Grading Basis Graded**

This is an advanced level class in Econometrics with an emphasis on time series techniques. This course covers a broad range of time series methods commonly used in research and industry for generating short-term forecasts for important economics and financial variables. Some of the topics covered are time series decomposition methods, smoothing methods, ARIMA, ARCH/GARCH and VAR.

Prerequisites: A minimum grade of "B-" in ECON 385.

ECON 485 Advanced Econometrics**Credits 3 Grading Basis Graded**

Theory and application of statistical techniques to study empirical relationships among economic variables. This course will cover a range of econometric techniques to estimate economic models including panel data, limited dependent variable, and simultaneous equation models. In addition, the course will cover the assembly and analysis of requisite data sets, and the preparation and presentation of empirical results.

Prerequisites: Minimum grade of "B-" in ECON 385.

ECON 487 Economic Consulting**Credits 3 Grading Basis Graded**

A research-oriented, senior-level course that provides students an opportunity to integrate theoretical knowledge, quantitative techniques and writing skills through research on a set of simulated consulting projects.

Prerequisites: ECON 331, ECON 332, ECON 385 and senior standing.

ECON 488 Senior Capstone Seminar in Economics**Credits 3 Grading Basis Graded**

This course is a writing-intensive seminar offering a student the opportunity to integrate many of their undergraduate studies in economics. Its substantive content will emphasize applying the methods of theoretical and empirical analyses employed by all economists. The seminar will be structured so as to contain embedded assessment measures of the learning objectives specified by the department of economics, including those related to command of basic economic theory and of quantitative methods used in quantifying empirical relationships and testing hypothesis.

Prerequisites: Senior standing and minimum grade of "C" in the following courses: ECON 331, ECON 332 and ECON 385.

ECON 490 Special Studies in Economics**Credits 1-3 Grading Basis Graded**

Designed to give capable students in economics an opportunity to complete independent study under faculty supervision. Admission by recommendation of the instructor and written permission of the director of economics prior to registration. May not be used toward fulfillment of the 400-level requirement for a major in economics.

ECON 499A Honors**Credits 1-3 Grading Basis Graded**

Year course. See catalog section "Graduation with Honors."

ECON 499B Honors**Credits 3 Grading Basis Graded****ECON 499C Honors****Credits 2 Grading Basis Graded**

Education (EDUC)

EDUC 100 The Study of the Future: An Interdisciplinary Approach**Credits 3 Grading Basis Graded**

Introduces the students to an interdisciplinary study of the future within the context of education. Various topic areas, such as population, science/technology, lifestyle, economics, international relations, energy and religion will be explored in terms of future trends and how education responds to these trends and their impacts.

EDUC 102 Critical Questions in Education [C1CT]**Credits 3 Grading Basis** Graded

This course provides an overview of the U.S. American public education system. Specific emphasis will be placed on the role, funding and history of education as they relate to social justice, equity and the diversity of U.S. American society. May be used for general education credit.

Prerequisites: If you have already completed, or are currently registered for, a Critical Thinking (Madison Foundations) class, then you must secure permission to receive credit for a second Critical Thinking class.

EDUC 165 Social Foundations of Exceptionalities**Credits 3 Grading Basis** Graded

This course examines the social, cultural, and educational contexts - through the lifespan - of individuals with exceptionalities. Specific focus will be given to the study of historical perspectives, models, theories, philosophies and trends in the field of special education. In addition, the role of culture, environment, family and individual differences will be explored.

EDUC 175 Learning, Behavior and Engagement in Context [C5SD]**Credits 3 Grading Basis** Graded

This course examines development and sociocultural dynamics, informed by 21st century skills, with a retrospective focus on how past developmental experiences impact present and future learning. Students will apply knowledge and research about human development to identify best practices for learning and resilience. This course may be used for general education credit.

EDUC 200 Foundations of Education in the United States**Credits 3 Grading Basis** Graded

This course addresses understanding of American education, philosophical approaches to education, the history of education, and the socio-cultural changes in U.S. society with characteristics of today's schools and with current educational issues.

EDUC 279 Comparative Intercultural Education**Credits 3 Grading Basis** Graded

Students will study and compare educational goals, practices, policies and beliefs in countries and cultures around the world. Content is grounded in social, cultural, historical and economic processes that are influenced and influence cultural understandings and systems of education.

Prerequisites: EDUC 200.

EDUC 303 Foundations of Curriculum and Instruction**Credits 3 Grading Basis** Graded

This is an introductory course that examines the historical, sociological, philosophical, and psychological perspectives that influence and support the curricular and instructional structures of general education. Throughout the course the implementation of those structures through a systematic process of planning for effective classroom instruction and evaluation will be examined. Connections to how they reflect an understanding of various models of learning and cognition and their implications for educational practice will be established.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 200. Corequisites: B.I.S./SPED: EXED 341, EXED 342 and EXED 474.

EDUC 305 Critical Studies in Education, Race, Culture and Politics**Credits 3 Grading Basis** Graded

This course engages students in critical studies of educational outcomes and practices for minority populations in contemporary and historical contexts.

EDUC 310 Teaching in a Diverse Society**Credits 3 Grading Basis** Graded

This course will examine how personal and professional values, attitudes, beliefs, and behaviors affect teaching and learning. The pre-service teachers will develop an understanding of similar unique characteristics of Pre-K to 12 grade students and their families, including culture, race, ethnicity, heritage language and learning abilities, gender socialization, and sexual orientation.

EDUC 370 Instructional Technology**Cross-listed with:** LDT 370**Credits 3 Grading Basis** Graded

In this course, learners will establish a foundational understanding of technology and media as a means to empower pedagogical practice and set conditions for learning. The course is intended to help develop awareness of, and appreciation for, a range of technical skills and reflective practices that may be applied in daily life and in future course work and careers. Learners will create an ePortfolio of digital media developed in this course.

EDUC 401 Problems in Education**Credits 1-3 Grading Basis** Graded

Workshop experiences for the development and training of teachers. Prerequisites: EDUC 200.

EDUC 413 Digging into History and Education at James Madison's Montpelier**Credits 3 Grading Basis** Graded

Students will learn about, reflect on and critique the ways in which systemic racism influences American society, U.S. education systems and their own lives. The course is organized to promote race reflection through critical explorations of the ways race and racism are inscribed within places and spaces. Students will critically engage with course texts and materials, unearth artifacts on an archaeological dig at Montpelier, explore numerous exhibits at Montpelier and other local historic sites in and around Orange County, Virginia, and confront structural racism while planning anti-racist K-12 curriculum.

EDUC 455 Perspectives in Environmental Education**Credits 3 Grading Basis** Graded

This course illuminates the role of educators in presenting environmental topics in learning communities. Activities will include principles for formal classrooms and other educational settings, such as museums, parks and zoos. In a local classroom /learning environment, participants will collaborate with practicing educators to teach learners about local environmental topics.

EDUC 479 Undergraduate Teaching Assistantship in Education**Credits 2 Grading Basis** Graded

Undergraduate students in education majors will serve as teaching assistants in large-format education courses. Students will perform essential duties, including facilitating classroom discussions, maintaining accurate class records and assisting the instructor with class lectures.

Prerequisites: EDUC 200.

EDUC 480 Student Teaching**Credits 6-12 Grading Basis** Credit / No Credit

Enables students to apply, in the public school classrooms and the comprehensive child development programs, those skills and attitudes acquired in all components of teacher education. Under the guidance of university supervisors, students are provided activities designed to familiarize them with the classroom teacher's role.

Prerequisites: PSYC 160 and EDUC 200.

EDUC 490 Special Topics in Education**Credits** 1-4 **Grading Basis** Graded

This course offers an in-depth examination of selected topics which are of current importance in the field of education. This course may be repeated for credit when course content changes.

Prerequisites: Junior or senior standing and consent of the instructor.

EDUC 499A Honors**Cross-listed with:** EDUC 499B, EDUC 499C**Credits** 1-6 **Grading Basis** Graded

Independent research topic initiated and completed by qualified upper-division students. See catalog descriptions entitled "Graduation with Distinction" and "Graduation with Honors."

EDUC 499B Honors**Cross-listed with:** EDUC 499A, EDUC 499C**Credits** 1-6 **Grading Basis** Graded

Independent research topic initiated and completed by qualified upper-division students. See catalog descriptions entitled "Graduation with Distinction" and "Graduation with Honors."

EDUC 499C Honors**Cross-listed with:** EDUC 499A, EDUC 499B**Credits** 1-6 **Grading Basis** Graded

Independent research topic initiated and completed by qualified upper-division students. See catalog descriptions entitled "Graduation with Distinction" and "Graduation with Honors."

Elementary Education (ELED)

ELED 200 Introduction to Global Education**Credits** 3 **Grading Basis** Graded

This course is an introduction to global issues shaping education policy and practice around the world. Students are invited to examine their own identities and actions and their implications for global issues at the local and transnational levels. Course readings, co-curricular activities and instruction are designed to promote an attainable global perspective in order to support the development of global competencies in future course taking and educational praxis. A portfolio of Global Education assignments, essays and curriculum materials will be established.

Prerequisites: Declared ELED Global Concentration.

ELED 208 Child Development: Birth Through Adolescence**Credits** 3 **Grading Basis** Graded

Skills for observing, recording and interpreting the behavior of children ages three through 12 will be developed so that adult intervention and guidance is appropriate and meaningful.

ELED 270 Global Diversity in Elementary Education**Credits** 3 **Grading Basis** Graded

This course will examine how personal and professional values, attitudes, beliefs and behaviors affect teaching and learning. The pre-service teachers will develop an understanding of unique characteristics of Pre-K to 12th grade students and their families, including culture, race, ethnicity, heritage, language, learning abilities, gender socialization and sexual orientation. In this globalized version of the course, students will also develop global competencies by examining local and global issues, engaging in open and critical conversations with oneself and others, and acting for collective well-being and sustainable development. The course integrates the United Nations Sustainable Development Goals (SDGs) to foster a deeper understanding of global challenges and their relevance to elementary education.

Prerequisites: Declared ELED global concentration and ELED 200.

Corequisite: ELED 272 and ELED 278.

ELED 271 Practicum in Early and Elementary Education**Credits** 1-2 **Grading Basis** Graded

Field experiences in grades PK-6 will provide for extensive observation and experience with learners in elementary classrooms and the opportunity to assist teachers in daily activities. Through direct observation and interactions with children in classroom settings, candidates will examine and reflect on how children develop and learn, exploring their own personal attitudes, assumptions and behaviors towards students and their families are influenced by class, cultural and linguistic backgrounds.

Corequisites: ELED 272.

ELED 272 Introduction to Early and Elementary Education**Credits** 3 **Grading Basis** Graded

This course is designed to introduce elementary education students to teaching children in grades PK-6. Based on developmental and learning theories, the course focuses on building relationships with and among learners, establishing learning environments, guiding behavior, and providing instruction and assessment. Through observations and participation in practicum, seminars, workshops and lectures, students will construct an understanding of how teachers meet the developmental and learning needs of children by planning, implementing and assessing appropriate curriculum and instruction.

Prerequisites: ELED or B.I.S./ELED major: Teacher Ed flag 01.

Corequisites: ELED majors: ELED 208 and ELED 310, or ELED 270 and ELED 278.

ELED 278 Child Development in a Global Context: Birth Through Adolescence**Credits** 3 **Grading Basis** Graded

This course introduces a lifespan approach to human growth and development with emphasis on the social, physical, emotional and cognitive domains. Students will examine key theories, educational philosophies and research methods in the social sciences and child development. They will also develop skills for observing, recording and interpreting children's behavior to apply appropriate and meaningful guidance. In addition, the course emphasizes the professional development of future educators by fostering reflective and culturally responsive practice. Global perspectives are integrated throughout the course, encouraging students to consider cultural diversity, environmental sustainability and global interconnectedness in their understanding of children's growth and learning. Students will also connect local classroom practices to broaden global citizenship values and strengthen cross-cultural competence as part of their preparation for teaching in diverse contexts.

Prerequisites: Declared ELED global concentration and ELED 200.

Corequisites: ELED 270 and ELED 272.

ELED 310 Diversity in Elementary Education**Credits** 3 **Grading Basis** Graded

This course will examine how personal and professional values, attitudes, beliefs and behaviors affect teaching and learning. The pre-service teachers will develop an understanding of unique characteristics of Pre-K to 12th grade students and their families, including culture, race, ethnicity, heritage, language, learning abilities, gender socialization and sexual orientation.

Prerequisites: ELED or B.I.S./ELED major. Corequisites: ELED majors: ELED 208 and ELED 272.

ELED 321 Practicum with a Focus on Learners and Learning**Credits 1-3 Grading Basis** Graded

This field experience, taken in the fall of junior year, provides candidates with a classroom of students and a mentor teacher with whom to practice building relationships with learners and conduct extensive observations of learners and learning, as well as practice the teaching of content related to their concurrent methods courses (either language arts, mathematics and science OR language arts, mathematics and social studies) with small groups of students. Accompanying monthly seminars will focus on reflecting on themes and experiences in the field and making content connections to methods courses.

Prerequisites: ELED or B.I.S./ELED major; Teacher Ed flag 02. ELED majors: ELED 272. Corequisites: ELED majors: ELED 332 and ELED 333, or ELED 372 and ELED 373.

ELED 322 Practicum with a Focus on Curriculum Integration and Guiding Behavior**Credits 1-3 Grading Basis** Graded

This field experience, taken in the spring of junior year, provides candidates with a classroom of students and a mentor teacher with whom to practice the teaching of content related to their concurrent methods courses (either language arts, mathematics and science OR language arts, mathematics and social studies) with small groups of students as well as in full class contexts. The accompanying monthly seminar explores the integration and construction of meaningful curriculum in elementary education contexts and in classroom management as well as reflection along with field-based partners.

Prerequisites: Accepted to teacher education.

ELED 332 Children and Science**Credits 3 Grading Basis** Graded

This course is a study of content, processes, pedagogy and materials for teaching science in the elementary classroom. Knowledge of cognitive development as applied to the selection of content and methodology for elementary learners will be examined.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 272.

Corequisites: ELED majors: ELED 321, ELED 333 and LED 300.

ELED 333 Children and Mathematics I**Credits 3 Grading Basis** Graded

The first of two courses that provides students with knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-6. This course will focus on mathematical instruction connected the following mathematical strands: Number and number sense, computation and estimation, and patterns, functions and algebra.

Prerequisites: ELED 272. Corequisites: ELED 321, ELED 332 and LED 300.

ELED 334 Children and Social Studies**Credits 3 Grading Basis** Graded

This course focuses on the content, processes, pedagogy and materials for teaching social studies in the elementary classroom. Knowledge of cognitive development as applied to the selection of content, methods, materials and strategies for organizing the learning environment for elementary learners will be examined.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 321 and LED 300. Corequisites: ELED majors: ELED 322, ELED 343 and LED 350.

ELED 343 Children and Mathematics II**Credits 3 Grading Basis** Graded

This is the second course in a series providing students with knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-6. Emphasis is on children's mathematical learning about geometry, measurement, data analysis and probability.

Prerequisites: ELED 321 and LED 300. Corequisites: ELED 322; ELED 334 and LED 350, or ELED 374, ELED 375 and LED 370.

ELED 372 Children and Science in a Global Context**Credits 3 Grading Basis** Graded

This course is a study of content, processes, pedagogy and materials for teaching science in the elementary classroom. Knowledge of cognitive development as applied to the selection of content and methodology for elementary learners will be examined. Students explore methods for fostering scientific literacy within a global context, emphasizing interconnected challenges such as sustainability and public health. The course also examines diverse cultural perspectives on scientific knowledge and practices to create more inclusive classrooms.

Prerequisites: Declared ELED global concentration and ELED 272.

Corequisites: ELED 321, ELED 373 and LED 300.

ELED 373 Children and Mathematics I in a Global Context**Credits 3 Grading Basis** Graded

This is the first of two courses that provides students with knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-6. This course will focus on mathematical instruction connected to the following mathematical strands: number and number sense, computation and estimation, and patterns, functions and algebra. In this globalized version of the course, students will also develop global competencies by examining how mathematics is taught differently around the world, and engaging in open and critical conversations with oneself and others to foster a deeper understanding of different mathematical strategies and how to teach them.

Prerequisites: Declared ELED global concentration and ELED 272.

Corequisites: ELED 321, ELED 372 and LED 300.

ELED 374 Children and Social Studies in a Global Context**Credits 3 Grading Basis** Graded

This course focuses on the content, processes, pedagogy and materials for teaching globally-aware social studies in the elementary classroom. Knowledge of cognitive development as applied to the selection of inclusive and global content, culturally sustaining methods, materials, and strategies for organizing the learning environment for elementary learners will be examined.

Prerequisites: Declared ELED global concentration, ELED 321 and LED 300. Corequisites: ELED 322, ELED 343 and LED 370.

ELED 375 Seminar in Global Education Praxis I**Credits 1 Grading Basis** Graded

In this course students will explore the needs and challenges of elementary school learners and reflect on what they've learned about teaching young children in a global context. Throughout the semester, students will be prompted to reflect on their experiences with teaching in a global context. Reflections will focus on understanding the complexities inherent in teaching as well as their own personal beliefs, strengths and areas for growth. Students will revisit and expand upon the portfolio begun in ELED 200.

Prerequisites: Declared ELED global concentration, ELED 321 and LED 300. Corequisites: ELED 322, ELED 343, ELED 374 and LED 370.

ELED 400 Elementary Education Final Practicum**Credits 1 Grading Basis Graded**

Upon completion of this course, teacher candidates will be able to build upon the unique strengths and contributions of K-6 students in order to create and teach lessons in various subject areas as well as reflect on that process.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 322 and ELED 343. Corequisites: ELED majors: ELED 480.

ELED 433 Children and Mathematics: Number, Operations, Algebraic and Geometric Reasoning**Credits 3 Grading Basis Graded**

The first of two courses that provides students with knowledge, skills and understanding of design and implementation for effective, developmentally appropriate mathematics instruction for grades PreK-6. Emphasis is on children's mathematical learning and pre-numerical stages through the acquisition of advanced numerical processes and operations and connections to geometric and algebraic reasoning.

Prerequisites: MATH 107, MATH 108, MATH 207 and LED 366.

Corequisites: LED 436, ELED 322, ELED 332 and ELED 334.

ELED 444 Mathematics Content and Pedagogy for Elementary Education I**Credits 3 Grading Basis Graded**

This is the first of two courses that helps prospective elementary education teachers (PreK-6) develop mathematical content and pedagogical knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-6. Emphasis is on children's mathematical learning in the following mathematical strands: number and number sense, computation and estimation, and patterns, functions and algebra.

Prerequisites: Applied to Teacher Education (01 Flag).

ELED 445 Mathematics Content and Pedagogy for Elementary Education II**Credits 3 Grading Basis Graded**

This is the second of two courses that helps prospective elementary teachers (PreK-6) develop mathematical content and pedagogical knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for grades PreK-6. Emphasis is on children's mathematical learning of geometry, measurement, data analysis and probability.

Prerequisites: Applied to Teacher Education (01 Flag); ELED 444.

ELED 450 Planning Instruction and Assessment**Credits 3 Grading Basis Graded**

Elementary education teacher candidates will learn the process of planning, implementing curriculum and using instructional strategies and assessments that promote and monitor students' understanding of content beyond memorization. Students will design units of instruction that utilize evidence-based practices for active engagement strategies, specifically for providing an inclusive environment for a diverse student population. Students will gain the knowledge and skills necessary to use assessments that allow for data-driven instructional decisions and inform subsequent teaching and learning.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 322 and ELED 343. Corequisites: ELED majors: ELED 485.

ELED 455 Home, School, Communities**Credits 3 Grading Basis Graded**

Study of the role of the teacher in relating to and working with families, school personnel and communities. Methods of involving stakeholders and providing effective communication among groups are emphasized. Resources for supporting parents and engaging community stakeholders are examined.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 450.

Corequisites: ELED majors: LED 462, or ELED 475 and LED 473.

ELED 475 Seminar in Global Education Praxis II**Credits 1 Grading Basis Graded**

In this course, students will explore challenges and opportunities in developing a globally focused praxis for teaching content and skills to diverse elementary school learners. Students will integrate learning in teaching placements with content from education courses taken in the ELED major. Throughout the semester, students will be prompted to reflect on their experiences with teaching in a global context. Reflections will focus on understanding the complexities of planning curricula that expand students' worldviews and promote critical global citizenship. Students will revisit and expand upon the portfolio begun in ELED 200 and expanded in ELED 375.

Prerequisites: Declared ELED global concentration and ELED 450.

Corequisites: ELED 455 and LED 473.

ELED 480 Student Teaching in Elementary Education**Credits 8 Grading Basis Credit / No Credit**

This internship provides a context in which to demonstrate and integrate the skills, knowledge and dispositions that are essential for success as an elementary classroom teacher. Under the guidance of a university supervisor, students are expected to reflect on the teaching role as they assume the full responsibilities of a teacher in grades PreK-6. Students majoring in elementary education must complete this course twice for a total of 16 credit hours.

Prerequisites: ELED or B.I.S./ELED major; Teacher Ed flag 03. ELED majors: ELED 322 and ELED 343. Corequisites: ELED majors: ELED 400.

ELED 485 Guiding Student Behavior: Responsive Classroom Management**Credits 3 Grading Basis Graded**

This course presents an overview to classroom management for the early childhood and elementary classroom teacher. It includes an historical look at the guidance tradition, a review of traditional approaches to classroom management and a focus away from punishment toward proactive strategies. Best practices in classroom management will be presented from an inclusive perspective for teachers of all children.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 322 and ELED 343. Corequisites: ELED majors: ELED 450. Also open to B.I.S./ELED majors.

ELED 490 Special Studies in Elementary Education**Credits 1-3 Grading Basis Graded**

Designed to give students opportunities to complete independent research on educational problems under faculty guidance. The plan for the study must be presented to the department head in prescribed form for approval prior to registration.

Prerequisites: Accepted to teacher education.

Engineering (ENGR)

ENGR 101 First Year Engineering Seminar

Credits 1 Grading Basis Graded

This seminar course introduces the engineering curriculum and career options to first-year students and describes how various elements of the curriculum and available electives relate to the goals and objectives of the program. This course also contextualizes the engineering profession with practical examples to help students develop strategies for success. The activities in the course are aimed at assisting students in successfully making the transition from high school to college.

ENGR 112 Engineering Skills and Decisions

Credits 3 Grading Basis Graded

This project-based course equips first-year engineering students with fundamental skills needed to solve engineering problems. The course emphasizes processes associated with making evidence-based decisions in engineering fields. This course covers the following topics: processes of decision making, engineering documentation, data analysis, visualization and engineering orthographic drawing. Students are permitted a maximum of two attempts for this course in consideration for full admission to the Engineering major (excluding final letter grades of W, WP, and WF).

Prerequisites: Students are permitted a maximum of two attempts in ENGR 112 (excluding grades of W, WP and WF).

ENGR 212 Statics and Introductory Solid Mechanics

Credits 3 Grading Basis Graded

This course introduces the fundamental and governing principles of systems of forces acting on rigid bodies, beams and columns as well as the analysis of truss, frame and machine structures. Topics include external and internal equilibrium of rigid bodies; systems of force and moment vectors; centroids; friction; loading that results in uniaxial, shear, flexure and torsion stresses and strains; and strength of materials. ENGR 212 and ENGR 212L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, MATH 236, PHYS 240 and a minimum grade of "C-" in ENGR 112. Corequisites: ENGR 212L.

ENGR 212L Statics and Introductory Solid Mechanics Laboratory

Credits 1 Grading Basis Graded

This laboratory course is designed to complement and supplement ENGR 212 course topics with experiential learning through experiments, projects and in-depth problem solving. ENGR 212 and ENGR 212L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, MATH 236, PHYS 240 and a minimum grade of "C-" in ENGR 112. Corequisites: ENGR 212.

ENGR 226 Engineering Project Management

Credits 3 Grading Basis Graded

This course introduces the fundamentals of project management. Students will identify project requirements, develop a work breakdown structure and network diagram, and apply resources to develop a project schedule. Knowledge is demonstrated through proper application of the decision tools available to project managers, such as determining a project's current status for cost, schedule and performance. Students will analyze why a project's current status is not according to plan, develop an action plan to resolve cost and scheduling challenges, and project final cost, completion date and performance. A computer-based project management system is used for managing projects. Completion of this course fulfills the requirement for 23 formal education hours for the Certified Associate in Project Management (CAPM) credential.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 231.

ENGR 231 Engineering Design I: Product Design and Development

Credits 3 Grading Basis Graded

This project-based course provides students with an overview of the engineering design process with a focus on product development. Topics include understanding and detailing customer requirements, exploring the engineering solution space, down-selecting conceptual design solutions and developing low-fidelity prototypes. Additional topics include force analysis and equilibrium in the context of an interdisciplinary project experience. This course includes significant team and project-based components and provides opportunities to build interpersonal communication skills that lead to effective problem solving, idea generation, design embodiment and decision making. Students are permitted a maximum of two attempts for this course in consideration for full admission to the Engineering major (excluding final letter grades of W, WP, and WF).

Prerequisites: Engineering majors only and a minimum grade of "C-" in ENGR 112. Students are permitted a maximum of two attempts of ENGR 231 (excluding grades of W, WP and WF). Corequisites: MATH 232 or MATH 235 or MATH 236; PHYS 240.

ENGR 232 Engineering Design II: Human-Centered Design and Systems Analysis

Credits 3 Grading Basis Graded

This project-based course provides students with an overview of the engineering design process with a focus on designing sustainable engineering systems. Specifically, students will conduct economic, environmental and social tradeoff analysis to articulate the impacts of an engineering solution. Additional topics include engineering circuits in the context of an interdisciplinary project experience. This course includes significant team and project-based components and provides opportunities to reinforce interpersonal communication skills that lead to effective problem solving, idea generation, design embodiment and decision making.

Prerequisites: Fully admitted to the Engineering major, MATH 236, PHYS 240 and a minimum grade of "C-" in ENGR 231. Corequisites: PHYS 250.

ENGR 290 Projects in Engineering

Credits 1-4 Grading Basis Graded

Lower-division students can pursue independent study or projects in engineering. This course may be repeated for credit when content changes. Projects or topics selected may dictate prerequisites. Students should consult the instructor prior to enrolling in the course.

Prerequisites: Permission of the instructor.

ENGR 296 Study Abroad with Engineering I**Credits 1-3 Grading Basis Graded**

This course is designed to provide lower-division students with the opportunity to explore engineering challenges and solutions occurring in international settings. Students seeking credit must secure the approval of the department head and a faculty supervisor who will provide the academic structure, assignments and student evaluation.

Prerequisites: Permission of the instructor.

ENGR 297 Engineering Research**Credits 1-3 Grading Basis Graded**

Lower-division students pursue research in a selected area of engineering. Student must make arrangements with a supervising instructor prior to registration.

Prerequisites: Permission of the instructor.

ENGR 298 Topics in Engineering**Credits 1-3 Grading Basis Graded**

This course is designed to provide lower-division students with the opportunity to explore engineering topics in greater depth. The specific topic of interest may dictate prerequisites. Students should consult the instructor prior to enrolling in the course. This course may be repeated for credit when the topic changes.

Prerequisites: Permission of the instructor.

ENGR 301 Transfer Year Engineering Bridge Course**Credits 3 Grading Basis Graded**

This project-based course provides transfer students with an introduction to the JMU Engineering program. The purpose is to familiarize transfer students with aspects of the engineering curriculum that typically do not transfer from outside programs. Topics include project management, product design and development, and human centered design. Additional topics in engineering documentation, visualization, engineering orthographic drawing and shop training are reinforced in the context of a team-based engineering project.

Prerequisites: Engineering majors only and permission of the department head.

ENGR 311 Fluid Mechanics**Credits 3 Grading Basis Graded**

This course focuses on principles applicable to fluid mechanics of incompressible flow. Topics include fluid statics; mass, energy and momentum balances; as well as the principles of internal and external flows. Wide-ranging applications of these principles to fluid systems across engineering disciplines are emphasized. ENGR 311 and ENGR 311L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, MATH 237 and a minimum grade of "C-" in ENGR 212. Corequisites: ENGR 311L.

ENGR 311L Fluid Mechanics Laboratory**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement ENGR 311 course topics with experiential learning through experiments, projects and in-depth problem solving. ENGR 311 and ENGR 311L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, MATH 237 and minimum grade of "C-" in ENGR 212. Corequisites: ENGR 311.

ENGR 312 Thermodynamics and Heat Transfer**Credits 3 Grading Basis Graded**

This course introduces the fundamental principles of thermodynamics and heat transfer in engineering systems. Topics include mass and energy balance concepts and incorporates the laws of thermodynamics in addition to the fundamental mechanisms of 1-dimensional heat transfer. ENGR 312 and ENGR 312L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, CHEM 132 and MATH 237. Corequisites: ENGR 312L.

ENGR 312L Thermodynamics and Heat Transfer Laboratory**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement ENGR 312 course topics with experiential learning through experiments, projects and in-depth problem solving. ENGR 312 and ENGR 312L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, CHEM 132 and MATH 237. Corequisites: ENGR 312.

ENGR 313 Circuits and Devices**Credits 3 Grading Basis Graded**

This course presents the fundamentals of circuit analysis and the measurement of physical phenomena. Circuit analysis topics include Kirchhoff's laws, 1st and 2nd order circuits, and AC systems. Instrumentation principles and common measurement devices are discussed. ENGR 313 and ENGR 313L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, MATH 237 and PHYS 250. Corequisites: ENGR 313L.

ENGR 313L Circuits and Devices Laboratory**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement ENGR 313 course topics with experiential learning through experiments, projects and in-depth problem solving. ENGR 313 and ENGR 313L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, MATH 237 and PHYS 250. Corequisites: ENGR 313.

ENGR 314 Materials and Intermediate Solid Mechanics**Credits 3 Grading Basis Graded**

This course extends into intermediate mechanics of materials topics and additionally explores the fundamentals of materials science, materials engineering and material failure theory. Topics include process-structure-property relationships, crystalline structures, an introduction to materials processing, plane stress transformations, principal stresses, and various ductile and brittle failure theories. ENGR 314 and ENGR 314L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, CHEM 131, MATH 237 and a minimum grade of "C-" in ENGR 212. Corequisites: ENGR 314L.

ENGR 314L Materials and Intermediate Solid Mechanics Laboratory**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement ENGR 314 course topics with experiential learning through experiments, projects and in-depth problem solving. ENGR 314 and ENGR 314L must be taken concurrently.

Prerequisites: Fully admitted to the Engineering major, CHEM 131, MATH 237 and a minimum grade of "C-" in ENGR 212. Corequisite: ENGR 314.

ENGR 315 Computational Methods in Engineering**Credits 3 Grading Basis** Graded

This course is an introduction to applied data analysis techniques for modern engineering systems that focus on modeling, mathematical and statistical analysis, data visualization, and programming techniques that can be used in engineering decision-making. Students will create programs to import data from engineering systems and conduct automated analysis to render appropriate decisions. Integrated development tools will be used to maintain programming environments and allow for code management and collaboration.

Prerequisites: Fully admitted to the Engineering major, MATH 236 and a minimum grade of "C-" in ENGR 112.

ENGR 317 Principles of Sustainable Engineering**Credits 3 Grading Basis** Graded

This course provides a foundation in evaluating sustainably engineered systems. This course focuses on natural resources, environmental systems, energy, climate and single factor analysis of common indicators of sustainability. The material presented supports an understanding of the environmental, social and economic impacts of design and technology.

Prerequisites: Fully admitted to the Engineering major, CHEM 132 and MATH 236; or permission of the instructor.

ENGR 321 Engineering Practicum I**Credits 1 Grading Basis** Graded

The engineering practicum supports the engineering project experience by facilitating student team construction and development of engineering project management artifacts. This first course in the multi-course sequence provides an opportunity to explore the projects available to the student, engage with fellow project applicants in team formation activities, and be assigned to one of the projects. At the culmination of this course, students will have been formed into project teams and will perform project initiation activities.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 232. Corequisites: ENGR 226 or ENGR 301.

ENGR 323 Engineering Practicum II**Credits 1 Grading Basis** Graded

The engineering practicum supports the engineering project experience by facilitating student team construction and development of engineering project management artifacts. This second course in the multi-course sequence focuses on the implementation of engineering project management practices as relevant to the continuation of the engineering projects established in the preceding course.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 321. Corequisites: ENGR 324.

ENGR 324 Engineering Project I**Credits 1 Grading Basis** Graded

The engineering project experience entails a process of devising a system, component or process to meet desired needs and specifications within constraints. This first course builds on initial project selection and team formation to expand into engineering project scoping, planning and execution.

Prerequisites: Fully admitted to the Engineering major, CHEM 132, PHYS 250 and MATH 237; MATH 238 or both MATH 300 and MATH 336; and a minimum grade of "C-" in each of ENGR 212 and ENGR 321. Corequisites: ENGR 323.

ENGR 331 Engineering Design III: Design Thinking and Fabrication Methods**Credits 3 Grading Basis** Graded

This course provides students with an overview of the engineering design process with a focus on design thinking principles. This course emphasizes methods and practices in problem exploration, stakeholder analysis, requirements identification and writing, conceptual design, and select topics in embodiment and detailed design. Students will apply various design tools and methods, including modeling and prototyping using various fabrication techniques to engage in iterative testing and refinement of engineering solutions.

Prerequisites: Fully admitted to the Engineering major, MATH 237 and PHYS 250; and a minimum grade of "C-" in each of ENGR 212, ENGR 226 and ENGR 232.

ENGR 345 Foundations of Computer Engineering**Credits 3 Grading Basis** Graded

This course focuses on fundamental design and analysis techniques required for engineers working in the broad computer engineering and computer systems field. Topics covered include information representation, Boolean algebra, combinational and sequential circuit design, microprocessor organization, and peripheral interfaces. Student designs will be tested and validated.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 112; or permission of the instructor.

ENGR 360 Structural Analysis**Credits 3 Grading Basis** Graded

This course extends from topics introduced in statics and introductory mechanics into structural engineering analysis. Topics include analysis of statically determinate and indeterminate structures by consistent deformation; drawing load, shear, moment and deflected shape diagrams for beams and frames; and calculating deformations using manual- and computational-methods.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 212. Corequisites: ENGR 314.

ENGR 387 Foundations of Brewery Engineering and Systems**Credits 3 Grading Basis** Graded

This is the first course in a multi-course sequence focusing on the topic of brewery engineering and systems. This course introduces the field of brewery engineering inclusive of terminology, technology and techniques.

This course is the starting point for a focus on brewery engineering and could be beneficial for a student interested in an engineering or operations career in the brewed and distilled beverage industry.

Prerequisites: Fully admitted to the Engineering major and CHEM 131.

ENGR 401 Seminar on Engineering Workforce Success**Credits 1 Grading Basis** Graded

This seminar course makes connections between the engineering curriculum and near-future engineering profession to senior-year students. This course contextualizes the engineering profession with practical examples to help students develop strategies for success. The activities in the course are aimed at assisting students in successfully making the transition from college to the engineering workforce.

ENGR 421 Engineering Practicum III**Credits 1 Grading Basis Graded**

The engineering practicum supports the engineering project experience by facilitating student team construction and development of engineering project management artifacts. This third course in the multi-course sequence focuses on the implementation of engineering project management practices as relevant to the finalization of the engineering projects established in the preceding course.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in both ENGR 323 and ENGR 324. Corequisites: ENGR 422.

ENGR 422 Engineering Project II**Credits 1 Grading Basis Graded**

The engineering project experience entails a process of devising a system, component or process to meet desired needs and specifications within constraints. This second course builds on engineering project scoping, planning and execution to expand into engineering project analysis, improvement and evaluation.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in both ENGR 323 and ENGR 324. Corequisites: ENGR 421.

ENGR 424 Engineering Capstone**Credits 2 Grading Basis Graded**

The engineering project experience entails a process of devising a system, component or process to meet desired needs and specifications within constraints. This final course builds on engineering project analysis, improvement and evaluation to expand into engineering project justification, delivery and documentation.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in each of ENGR 421 and ENGR 422.

ENGR 426 Product Development and Entrepreneurial Engineering**Credits 3 Grading Basis Graded**

This course provides the fundamental and governing principles of management of technology including general business functions of management, marketing, finance, accounting and operations; system analysis skills; and additional topics in project management. Students develop an understanding and appreciation for the importance of technology and innovation in organizations.

Prerequisites: Fully admitted to the Engineering minimum grade of "C-" in ENGR 226.

ENGR 440 Mechatronics**Credits 3 Grading Basis Graded**

Mechatronics is the synergetic integration of mechanical, electrical, computer and control systems to create intelligent systems. This course explores mechatronics with an emphasis on sensors and automation. Students will learn and apply theoretical and practical knowledge of how sensors work and interface with systems to enable automation. Additional topics covered include industrial controller programming, actuators, signal conditioning and interface circuits, and industrial robotics.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 313.

ENGR 445 Embedded System Design**Credits 3 Grading Basis Graded**

This project-based course prepares students to design novel computing systems that involve sensing, actuation and networking. Various topics will be covered including component selection, microprocessor peripheral interfaces, energy efficiency and power-aware design, and wireless communication methods. Students will design, test and fabricate their own embedded computing system to collect and analyze biometric data.

Prerequisites: Fully admitted to the Engineering major; CS 261 or a minimum grade of "C-" in ENGR 345; or permission of the instructor.

ENGR 452 Mechanisms in Motion: Applied Kinematics**Credits 3 Grading Basis Graded**

This course introduces the elements of mechanisms and machines and the underlying physical principles of mechanical systems that generate and convey mechanical motion. Topics include linkages, cams and gears. An included design studio allows for the use of engineering software and prototyping processes for the design and fabrication of mechanized elements.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 212.

ENGR 453 Machine Design**Credits 3 Grading Basis Graded**

This course addresses and applies fundamental mechanics topics to the analysis of mechanical components as well as conventions associated with the design of mechanical components. Topics include fatigue life, thermal expansion and stress concentrations. Mechanical components addressed include pressure vessels, springs, gears, bearings, mechanical drive systems and threaded fasteners. Conventions associated with typical assemblies, including threaded fasteners and fits, are included.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 314.

ENGR 457 Electromechanical Systems Analysis**Credits 3 Grading Basis Graded**

This course focuses on the concepts of systems thinking and analysis for complex engineered systems. Students will develop basic knowledge and tools to identify a system, decompose it into parts, define interactions, perform analysis and apply control measures if necessary. Application of computational tools and mathematical modeling will be emphasized.

Prerequisites: Fully admitted to the Engineering major and ENGR 312.

ENGR 461 Structural Systems Analysis**Credits 3 Grading Basis Graded**

This course extends from the foundational principles of structural analysis. Topics include matrix methods of structural analysis, finite element techniques and stability considerations. Advanced manual and computational analytical techniques are used to assess deformations, internal forces and reactions in structural systems.

Prerequisites: Fully admitted to the Engineering major; MATH 238 or MATH 300; and a minimum grade of "C-" in ENGR 360.

ENGR 462 Foundations of Structural Engineering Design**Credits 3 Grading Basis Graded**

This course focuses on structural design, loading and modeling of real-world structures. Topics include building and construction codes and requirements, reading and interpretation of construction documents, determination of load paths, structural modeling, and design of basic structural elements for various structural materials. Mechanics of materials and structural analysis methods are used to develop appropriate loading requirements.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in ENGR 212. Corequisites: ENGR 360.

ENGR 463 Concrete Infrastructure**Credits 3 Grading Basis Graded**

This course introduces the design and evaluation of Portland-cement based concrete infrastructure, particularly for its engineering material and mechanical properties. Topics include concrete mix design, handling and placement of concrete, behavior of concrete under various types of loading and environment, and design principles of reinforced concrete structures. An included laboratory component incorporates physical evaluation of fresh and hardened concrete properties.

Prerequisites: Fully admitted to the Engineering major and a minimum grade of "C-" in both ENGR 314 and ENGR 360.

ENGR 466 Mechanical Vibrations and Structural Dynamics**Credits 3 Grading Basis Graded**

This course introduces the fundamental concepts of dynamics, mechanical vibrations and structural dynamics. It covers single- and multi-degree-of-freedom systems, free and forced vibrations, damping and vibration control. The course focuses on practical applications in mechanical and structural systems.

Prerequisites: Fully admitted to the Engineering major, a minimum grade of "C-" in ENGR 314 and a minimum grade of "C-" in either MATH 238 or MATH 336.

ENGR 471 Water in Africa**Credits 3 Grading Basis Graded**

This course has a three-part focus: cross-cultural training, promoting health in developing countries and using appropriate technologies for eradicating water-related illnesses. Project teams use course content as the foundation for developing and implementing service projects. This course is a service-learning course and addresses issues of access to water in West Africa.

Prerequisites: Fully admitted to the Engineering major and CHEM 132; or permission of the instructor.

ENGR 472 Environmental Engineering**Credits 3 Grading Basis Graded**

This course is for engineering and environmental science students interested in environmental engineering. Water, wastewater and air treatment are emphasized. Students must be proficient in mathematics, chemistry and thermal sciences. Quantitative relationships are derived for characterizing water quality, designing biological and physical treatment processes, and modeling environmental systems. Systems are described by mass and energy balances that relate pollutant removal efficiency to process input parameters.

Prerequisites: Fully admitted to the Engineering major and CHEM 132; or permission of the instructor.

ENGR 477 Sustainable Systems Analysis**Credits 3 Grading Basis Graded**

This course provides a foundation in evaluating sustainable engineering systems. This course focuses on industrial design and systems and multi-factor analysis of common indicators of sustainability. The material presented furthers the understanding of the environmental, social and economic impacts of design and technology.

Prerequisites: Fully admitted to the Engineering major and ENGR 317.

ENGR 478 Water Resources Engineering**Credits 3 Grading Basis Graded**

This course provides an introduction to basic engineering principles used in water resources and management. Hydrologic and hydraulic processes will be investigated using the fundamentals of fluid mechanics. Specific emphasis will be placed on water sources flows, distribution and stream restoration principles.

Prerequisites: Fully admitted to the Engineering major and ENGR 311.

ENGR 481 Principles of Process Engineering**Credits 3 Grading Basis Graded**

This course introduces the basic principles and techniques used in process engineering. The major emphasis of the course is on understanding materials and energy flows in simple and complex process systems. Quantitative models and equilibrium concepts will be developed and applied to assess product quality, economics, safety and environmental issues. This course should be of interest to students looking toward a career or graduate studies in chemical, environmental, biochemical, petrochemical and manufacturing engineering.

Prerequisites: Fully admitted to the Engineering major and CHEM 131; or permission of the instructor.

ENGR 487 Brewery Engineering and Systems Practicum**Credits 3 Grading Basis Graded**

This is the component course in a multi-course sequence focusing on the topic of brewery engineering and systems. This course provides hands-on instruction in the operation of the Madison Academic Brewery and completing a practicum at a partner brewery. This course complements the previous courses in that the theory and engineering of beer making is experientially explored. This practical course approach could be beneficial for a student interested in an engineering or operations career in the brewed and distilled beverage industry. Enrollment is controlled by an application process in which a student must be aged 21 years or older. Prerequisites: Fully admitted to the Engineering major, ENGR 387 and permission of the instructor.

ENGR 490 Advanced Projects in Engineering**Credits 1-4 Grading Basis Graded**

This course supports independent study or advanced projects in engineering which are of interest to the upper-division student. This course may be repeated for credit when course content changes. Projects or topics selected may dictate prerequisites. Students should consult the instructor prior to enrolling in the course.

Prerequisites: Permission of the instructor.

ENGR 496 Study Abroad with Engineering II**Credits 1-3 Grading Basis Graded**

This course is designed to provide upper-division students with the opportunity to explore engineering challenges and solutions occurring in international settings. Students seeking credit must secure the approval of the department head and a faculty supervisor who will provide the academic structure, assignments and student evaluation.

Prerequisites: Permission of the instructor.

ENGR 497 Advanced Engineering Research**Credits 1-3 Grading Basis Graded**

Upper-division students pursue advanced research in a selected area of engineering. Student must make arrangements with a supervising instructor prior to registration.

Prerequisites: Permission of the instructor.

ENGR 498 Advanced Topics in Engineering**Credits 3 Grading Basis Graded**

This course is designed to provide upper-division students with the opportunity to explore advanced engineering topics in greater depth. The specific topic of interest may dictate prerequisites. Students should consult the instructor prior to enrolling in the course.

Prerequisites: Permission of the instructor.

ENGR 499A Engineering Honors I**Credits 1 Grading Basis Graded**

This is the first course in a three-course sequence. Students generate an idea for and write a proposal for an independent research project that meets the requirements set forth by the Honors program and the Department of Engineering. Students must identify and analyze an engineering-based problem, identify potential solutions, recommend an approach, and prepare a written proposal.

Prerequisites: Fully admitted to the Engineering major and permission of the engineering honors director or department head.

ENGR 499B Engineering Honors II**Credits 3 Grading Basis** Graded

This is the second course in a three-course sequence. Students complete the research for an independent research project that meets the requirements set forth by the Honors program and the Department of Engineering derived from the project described in the proposal from ENGR 499A.

Prerequisites: Fully admitted to the Engineering major and ENGR 499A; or permission of the engineering honors director or department head.

ENGR 499C Engineering Honors III**Credits 2 Grading Basis** Graded

This is the third course in a three-course sequence. Students prepare an oral and written presentation of their results for an independent research project that meets the requirements set forth by the Honors program and the Department of Engineering. Students complete and present (in written and oral form) the project described in their proposal from ENGR 499A.

Prerequisites: Fully admitted to the Engineering major and ENGR 499B.

English (ENG)

ENG 221 Literature and Ideas [C2L]**Credits 3 Grading Basis** Graded

This course will take a thematic approach to understanding and analyzing literature by examining multiple texts that engage with a common course theme or idea concerned with one or more aspects of human experience. This course will introduce students to methods for studying literature through thematically focused analysis. The organizing principle of this course will be thematic, but the course will also engage with form and the cultural aspects of the literature studied. This is a writing-infused course. Specific topics will vary. This course may be used for general education credit.

ENG 222 Literature and Genres [C2L]**Credits 3 Grading Basis** Graded

This course will take a genre-based or formalist approach to understanding and analyzing literature by examining multiple texts that belong to a genre, subgenre and/or other category of literary and/or artistic creation. This course will introduce students to methods for studying literature through analysis focused on the forms and conventions of the selected texts. The organizing principle of this course will be the analysis of literary form, but the course will also engage with the thematic and cultural aspects of the literature studied. This is a writing-infused course. Specific topics will vary. This course may be used for general education credit.

ENG 223 Literature and Cultures**Credits 3 Grading Basis** Graded

This course will take a cultural and contextual approach to understanding and analyzing literature by examining multiple texts that develop from, represent or respond to human experiences of specific times, places, histories and/or identities. This course will introduce students to methods for studying literature through culturally and contextually focused analysis. The organizing principle for this course will be contextual analysis, but the course will also engage with form and the thematic aspects of the literature studied. This is a writing-infused course. Specific topics will vary.

ENG 235 Survey of English Literature: From Beowulf to the Eighteenth Century [C2L]**Credits 3 Grading Basis** Graded

A general survey presented chronologically. May be used for general education credit.

ENG 236 Survey of English Literature: Eighteenth Century to Modern [C2L]**Credits 3 Grading Basis** Graded

A general survey presented chronologically. May be used for general education credit.

ENG 239 Studies in World Literature [C2L]**Credits 3 Grading Basis** Graded

Introduction to masterpieces of world literature with emphasis on non-Western literature. (May be focused regionally or topically.) May be used for general education credit.

ENG 247 Survey of American Literature: From the Beginning to the Civil War [C2L]**Credits 3 Grading Basis** Graded

A general survey presented chronologically. May be used for general education credit.

ENG 248 Survey of American Literature: From the Civil War to the Modern Period [C2L]**Credits 3 Grading Basis** Graded

A general survey presented chronologically. May be used for general education credit.

ENG 260 Survey of African American Literature [C2L]**Credits 3 Grading Basis** Graded

Survey of literature by African American authors from the 18th century to the present. May be used for general education credit.

ENG 294 Internship in English**Credits 1-3 Grading Basis** Graded

Internships in English provide students with opportunities for professionalization through work experience in a variety of fields. Regular reports, reflection exercises, self- and employer evaluations, and a portfolio project are among the assignments. A maximum of 3 credits may be counted towards the English major degree requirements. Prerequisites: English major or minor and approval of the internship director.

ENG 299 Introduction to Literary Studies**Credits 3 Grading Basis** Graded

In this course, students practice the skills of literary analysis, including how to generate, articulate and support viable arguments about literary meaning, and how to structure these arguments for specific audiences. Writing and revision will form central components of this course. Students will refine and apply their knowledge of specialized terms, concepts and critical methodologies for literary analysis. This course prepares students for upper-level work in literary studies. Prerequisites: Declared English major or minor; or permission of the instructor.

ENG 301 Old English Language and Literature**Credits 3 Grading Basis** Graded

An introduction to the Old English language through selected readings in poetry and prose.

ENG 302 Special Topics in Literature and Language**Credits 3 Grading Basis** Graded

Study of a particular literary or linguistics topic. May be repeated for credit when course content changes but not more than once, except with the approval of the department head.

ENG 303 History of the English Language**Credits 3 Grading Basis** Graded

Introduction to the historical study of English including its Indo-European origins. May be repeated for credit when course content changes.

ENG 305 Mythology**Credits 3 Grading Basis Graded**

Study of the nature and meaning of myths specific to a particular culture or comparative myths as interpreted and reinterpreted in significant works of ancient and modern literature.

ENG 306 The Bible as Literature**Credits 3 Grading Basis Graded**

Study of Hebrew and Christian scripture as literary and cultural texts, which have influenced subsequent literature and culture.

ENG 307 Literature and Ideas**Credits 3 Grading Basis Graded**

Literary treatment of an organizing theme or approach appropriate to the 300-level. The topic may be framed as part of the human experience or within a tradition of studies in the humanities. Course content may include pertinent readings from other disciplines. Possible topics: Disease in literature, literature and the law, literature and psychology. May be repeated as course topic changes.

ENG 308 Introduction to Linguistics**Credits 3 Grading Basis Graded**

Introduction to the study of the various subfields of linguistics, including questions about the nature and use of language in general, with the English language as the primary example.

ENG 309 Traditional English Grammar**Credits 3 Grading Basis Graded**

Introduction to traditional grammar, probing its logic, system and history, with an examination of modern applications of conventional rules.

ENG 310 Structure of the English Language**Credits 3 Grading Basis Graded**

This course studies the structure of the English language from a modern, linguistic perspective. The course focuses on the unconscious principles/rules and details of native speaker competence in the grammar of English and on applying this knowledge to a wide range of examples in the English language.

ENG 311 Medieval Literature and Culture**Credits 3 Grading Basis Graded**

Studies in the literature and culture of the Middle Ages through selected Old English, Norse/Icelandic, Middle English, Old Irish, French, German, Latin and Arabic texts in translation.

ENG 313 Studies in Early Modern Literature and Culture**Credits 3 Grading Basis Graded**

Students will study non-dramatic poetry and prose produced in England during the sixteenth century and roughly the first half of the seventeenth century.

ENG 316 Early Modern Drama**Credits 3 Grading Basis Graded**

Major works of British dramatists, excluding Shakespeare, from 1550-1660.

ENG 317 Studies in Shakespeare**Credits 3 Grading Basis Graded**

Students will study selected dramas by Shakespeare; nondramatic work may be considered.

ENG 319 Teaching Shakespeare**Credits 3 Grading Basis Graded**

A study of Shakespeare's plays, with emphasis on pedagogical techniques for teaching drama in the classroom

ENG 320 Shakespeare on the Page and Stage in London**Credits 3 Grading Basis Graded**

Students will study the plays of Shakespeare currently in production in London and England with special emphasis on the productions of the Royal Shakespeare Company and the National Theatre. Course can be substituted for ENG 317 but may not be taken for credit in addition.

ENG 321 Restoration and Eighteenth-Century British Prose and Poetry**Credits 3 Grading Basis Graded**

A study of the poetry and prose (other than the novel) from the Age of Satire, including works by Dryden, Swift, Pope and Johnson.

ENG 322 Restoration and Eighteenth-Century British Drama**Credits 3 Grading Basis Graded**

A study of British drama in the eighteenth century.

ENG 325 Romantic Literature**Credits 3 Grading Basis Graded**

A study emphasizing selected works of Romantic literature. Attention given to critical theories, intellectual and cultural movements, or poetic forms.

ENG 327 Gothic Literature**Credits 3 Grading Basis Graded**

A study of the origins, influence and transformations of Gothic fiction from the 18th century to the present.

ENG 329 Victorian Literature**Credits 3 Grading Basis Graded**

Study of British literature of the Victorian period with primary emphasis on poetry and nonfiction prose.

ENG 330 Nineteenth-Century British Novel**Credits 3 Grading Basis Graded**

The development of the British novel in the nineteenth century and the study of representative works.

ENG 331 Studies in Poetry**Credits 3 Grading Basis Graded**

A study of select poetic works. Specific time periods or genres studied may vary. Course may be repeated as topic changes.

ENG 332 Studies in Popular Genres**Credits 3 Grading Basis Graded**

A study of a select popular or specialized genre, such as speculative fiction, detective fiction or the graphic novel. Course may be repeated as topic changes.

ENG 334 Contemporary Drama**Credits 3 Grading Basis Graded**

Drama from 1960 to the present.

ENG 335 Studies in Children's Literature**Credits 3 Grading Basis Graded**

A study of children's literature, such as a survey of children's literature, young adult literature, African American children's literature, children's fantasy fiction or the picture book, or a study of a particular author or series. Course may be repeated as topic changes.

ENG 340 Modern British and Irish Literature**Credits 3 Grading Basis Graded**

Literature from Britain and Ireland, 1900 to 1945.

ENG 341 Contemporary British and Irish Literature**Credits 3 Grading Basis Graded**

Literature from Britain and Ireland, from 1945 to the present.

ENG 342 Early American Literature**Credits 3 Grading Basis** Graded

Significant genres, writers and literary movements of the seventeenth and eighteenth centuries.

ENG 343 Antebellum American Literature**Credits 3 Grading Basis** Graded

American literature of the early nineteenth century.

ENG 344 Late Nineteenth-Century American Literature**Credits 3 Grading Basis** Graded

American literature of the late nineteenth century.

ENG 347 Playwriting**Cross-listed with:** THEA 347**Credits 3 Grading Basis** Graded

Study of the process of writing plays. Consideration of plot, character, thematic material, conflict and dramatic structure. Emphasis on individual writing assignments.

Prerequisites: Sophomore standing or permission of the instructor.

ENG 348 Studies in Linguistics and the English Language**Credits 3 Grading Basis** Graded

Broad study of a particular subfield of linguistics, focusing on the English language, or an application of various linguistic theories and methodologies to the study of literature. Some possible areas of study include semantics, pragmatics and discourse, non-literal language, lexicography, English dialects and variation, language and humor, and linguistic approaches to literature. Course may be repeated as topic changes.

ENG 352 The American Novel to 1914**Credits 3 Grading Basis** Graded

A study of the development of the American novel from its beginnings to the modern period.

ENG 355 Southern Literature**Credits 3 Grading Basis** Graded

Southern authors, especially those of the twentieth century.

ENG 356 Modern American Novel**Credits 3 Grading Basis** Graded

The American novel from 1914 to 1945.

ENG 357 Contemporary American Literature**Credits 3 Grading Basis** Graded

A study of contemporary American literature written since 1945.

ENG 358 Oral Literature**Credits 3 Grading Basis** Graded

This course is a study of oral literature, which may be organized by theme, geography or genre. The course examines the social, political and artistic reasons for the creation and popularity of this literature. May be repeated for credit when course content changes.

ENG 359 Studies in African American Literature**Credits 3 Grading Basis** Graded

A study of selected works of African American literature in multiple genres. May be repeated for credit when course content changes.

ENG 360 Introduction to Ethnic American Literature**Credits 3 Grading Basis** Graded

An overview of the literary production of one group or a variety of ethnic and immigrant American writers, including but not limited to Native American, Asian American, Mexican American, Indian American, Caribbean American, and Latino American literatures. Examines the experience of historically marginalized groups in the United States with a particular emphasis on identity, nationality, tradition, and language.

ENG 361 African American Fiction**Credits 3 Grading Basis** Graded

Selected works of fiction by major African American writers.

ENG 362 African American Poetry**Credits 3 Grading Basis** Graded

Selected works of poetry by major African American writers. May be repeated for credit when course content changes.

ENG 363 Native American Literature**Credits 3 Grading Basis** Graded

A study of Native American communities' and authors' literatures in North America, with emphasis on the relationship of that literature to their traditions and historical experiences.

ENG 364 Introduction to Asian American Literature**Credits 3 Grading Basis** Graded

The course is designed to introduce students to a variety of writings by Asian American authors. It will explore the historical formation of Asian American identities; issues of immigration, citizenship and race; and the aesthetic forms of representation explored by Asian American writers and artists.

ENG 366 Contemporary Critical Theory**Credits 3 Grading Basis** Graded

Study of the major debates in current critical discourse.

ENG 367 Latinx Literature**Credits 3 Grading Basis** Graded

This course is a study of literature by U.S. Latinx authors. The course explores the way writers from a variety of racial, ethnic and cultural backgrounds who identify as Latinx use the written word in service of narrating counter-histories, identity formation, and aesthetic and imaginative engagement with the world.

ENG 368 Women's Literature**Cross-listed with:** WGSS 368**Credits 3 Grading Basis** Graded

A study of literature by women.

ENG 369 Feminist Literary Theory**Cross-listed with:** WGSS 369**Credits 3 Grading Basis** Graded

An intensive study of a variety of feminist critical approaches and their applications to literature.

ENG 370 Queer Literature**Cross-listed with:** WGSS 370**Credits 3 Grading Basis** Graded

An exploration of texts and issues in literature written by and about lesbian, gay, bisexual, transgender and queer writers, including critical and theoretical issues as well as questions of canon. Text studied may include fiction, poetry, drama, essays and memoirs written primarily, but not exclusively, in the 20th century.

ENG 371 Literature and the Environment**Credits 3 Grading Basis** Graded

A critical examination of literature's representation of the interconnections between human beings, non-human beings and the environment.

ENG 374 Contemporary Global Anglophone Literature**Credits 3 Grading Basis** Graded

This course examines contemporary literature from many parts of the English-speaking world, with special attention to writers' attempts to assess the global legacies of colonialism in the contemporary moment, as well as what it means to participate in a self-consciously global literary culture today.

ENG 375 Irish and Anglo-Irish Literature**Credits 3 Grading Basis** Graded

A study of the works of Irish or Anglo-Irish writers.

ENG 376 Introduction to Scottish Literature**Credits 3 Grading Basis** Graded

An overview of Scottish literature, with an emphasis on fiction, from the Romantic period to the present. Emphasis on the problems of nation, identity and the politics of language and tradition in "minority literature."

ENG 377 Introduction to African Literature**Credits 3 Grading Basis** Graded

An introduction to African literature, tracing its changes over time. Examination of African literary theoretical concepts, literary genres (both oral and written), as well as an examination of Africa's contribution to familiar genres of written and oral literature.

ENG 378 Studies in South Asian Literature**Credits 3 Grading Basis** Graded

A study of selected works of South Asian literature.

ENG 379 Literature and Empire**Credits 3 Grading Basis** Graded

The course is designed as an overview of writings from regions of the world that were formerly colonized by Britain. It examines the colonial, nationalist and postcolonial shaping of individual and collective identities through literature; the intersections of race, gender and nation; the crafting of a new idiom in English in response to both political and literary histories; and the significance of choices of genre and form.

ENG 380 Introduction to Film**Cross-listed with:** SMAD 380**Credits 3 Grading Basis** Graded

An introduction to the study of film as an aesthetic practice, including formal and industrial aspects of film analysis, theoretical approaches to film and writing and research methodologies of film and media studies.

ENG 381 History of Global Cinemas**Credits 3 Grading Basis** Graded

A historical and worldwide survey of major film movements from the silent era to the present-day, including Asian, European, African, American, Middle Eastern, Australasian and transnational cinemas. Focus will be placed on the aesthetics, politics and sociocultural contexts of global cinemas, with a focus on films that explore shifting ideologies and cultural identities in times of social tumult. Students will investigate how film movements worldwide respond to issues of geopolitics, globalization, colonization, nation-building, modernization, underdevelopment, migration and social marginalization.

ENG 382 Race, Class and Gender in Film**Credits 3 Grading Basis** Graded

This course explores specific topics within the study of race, class, gender and their intersections in national and/or global cinemas. The course uses key films and readings to introduce students to interrogate key historical and theoretical issues informing cinematic constructions of race, class and/or gender. The course will familiarize students with evolving discourses of racial, class and sexual identities in national and/or global cinemas, with a particular focus on the ways in which specific films and media industries reinforce and/or challenge dominant discourses on race, class and gender identity.

ENG 383 Film Genre**Credits 3 Grading Basis** Graded

Study of film genre through the consideration of one or more film genres.

ENG 384 Film Authorship**Credits 3 Grading Basis** Graded

Study of film authorship through the consideration of one or more directors.

ENG 385 Special Topics in Film Study**Credits 3 Grading Basis** Graded

Study of a particular topic in film. May be repeated for credit when course content changes but not more than once, except with the approval of the department head.

ENG 386 Historical Survey of British and Irish Literatures**Credits 3 Grading Basis** Graded

This course provides an historical survey of British and Irish literatures, broadly conceived, spanning multiple historical periods with attention to continuities and changes in literary practices and concerns over time. Specific topics and texts covered may vary by instructor.

ENG 387 Historical Survey of American Literatures**Credits 3 Grading Basis** Graded

This course provides an historical survey of American literatures, spanning multiple historical periods with attention to continuities and changes in literary practices and concerns over time. Specific topics and texts covered may vary by instructor.

ENG 388 Studies in Global Literatures**Credits 3 Grading Basis** Graded

This course will be organized topically or regionally depending on the instructor, with attention to continuities and changes in literary practices and concerns across space and time, as well as acts of translation and adaptation that take place within and across literary cultures. Specific topics and texts covered may vary by instructor. The course may include other cultural texts or media. The course may be repeated for credit if the topic changes.

ENG 390 The Environmental Imagination**Credits 3 Grading Basis** Graded

An introductory creative writing class that provides a sampling of American environmental writing, with emphasis on the creative intelligence of the writer's imagination, process and craft.

ENG 391 Introduction to Creative Writing - Nonfiction**Credits 3 Grading Basis** Graded

A basic workshop in reading and writing works of creative nonfiction.

ENG 392 Introduction to Creative Writing - Poetry**Credits 3 Grading Basis** Graded

A basic workshop in reading and writing poetry.

ENG 393 Introduction to Creative Writing - Fiction**Credits 3 Grading Basis** Graded

A basic workshop in reading and writing fiction. May be repeated for credit when course content changes.

ENG 396 Advanced Composition**Credits 3 Grading Basis** Graded

Extensive exercises in expository writing, with emphasis on rhetorical types of composition, designed to develop sophistication of style in the student's writing.

ENG 399 Advanced Writing in Literary Studies**Credits 3 Grading Basis** Graded

In this course, students refine the analysis and writing skills they learned in ENG 299. Students will also develop research skills, specifically in locating, analyzing and assessing secondary sources about linguistic, literary, artistic and/or cultural texts. They will integrate appropriate critical perspectives and research into a variety of assignments with well-defined audiences while simultaneously cultivating their own original arguments and writing style.

Prerequisites: For English majors and minors only: ENG 299.

ENG 401 Advanced Studies in Medieval Literature**Credits 3 Grading Basis** Graded

Advanced literary and philological study of non-British Medieval or British Medieval texts written in cross-cultural dialogue with those written circa 500-1480 c.e. in Old Irish, Old Norse/Icelandic, Old French, Old and Middle High German, Old Castilian, Classical and Medieval Latin, and Arabic. Readings in the original or in translation. Topics may be determined by period or geography, culture or politics, theme or genre. Course may be repeated as topic changes.

ENG 402 Advanced Studies in British Literature Before 1700**Credits 3 Grading Basis** Graded

British literatures written prior to 1700, both within and without the British isles. Topics may include Anglo-Saxon and Anglo-Norman literature, vernacular literature, neo-Latin literature, Chaucer, late-medieval literature, Tudor and early modern literature, Shakespeare, Milton, and seventeenth-century literature. May be repeated as course topic changes.

ENG 403 Advanced Studies in British Literature After 1700**Credits 3 Grading Basis** Graded

Advanced study of British literatures dating from 1700 to the present. Topics may focus on a particular period of literature (eighteenth century, Romantic, Victorian, Modernist, or contemporary), or topics may focus on a genre or theme that engages multiple periods. Course may be repeated as topic changes.

ENG 405 Advanced Studies in Anglophone Literature**Credits 3 Grading Basis** Graded

Anglophone (English-speaking) literature from around the globe (including the Caribbean, Canada, Ireland, Australasia, Africa or India), in which identification with a particular nation, colonial status or imperial power is problematic or no longer applicable. Topic may focus on a particular theme or event, genre, time frame, formal or stylistic trend, author or set of authors, issue or problem. May be repeated as topic changes.

ENG 407 Advanced Studies in American Literature**Credits 3 Grading Basis** Graded

Study of literature(s) of the United States and North America, from the Colonial Period through the 21st Century. May include writing in multiple genres: narrative prose, poetry, drama, nonfiction. Topics may be historically delimited or thematically organized; specific sections may focus on a group of authors, a literary movement, a historical moment, or a broad theoretical question. Course may be repeated as topic changes.

ENG 408 Advanced Studies in African American Literature**Credits 3 Grading Basis** Graded

Study of literature produced by African Americans, from the Colonial Period through the 21st century. May include writing in multiple genres: narrative prose, poetry, drama, nonfiction. Topics may be historically delimited or thematically organized; specific sections may focus on a group of authors, a literary movement, a historical moment, or a broad theoretical question. Course may be repeated as topic changes.

ENG 410 Advanced Studies in Author**Credits 3 Grading Basis** Graded

Study of the works of one (or two) British, American or Anglophone writers. May be repeated for credit when course content changes.

ENG 412 Special Topic Seminar**Credits 3 Grading Basis** Graded

Study of a literary school, movement, genre or some other significant literary or linguistic topic. May be repeated for credit when course content changes.

ENG 413 Advanced Studies in Literature and Ideas**Credits 3 Grading Basis** Graded

Advanced study of the literary treatment of an organizing theme, which may be framed broadly as part of the human experience or within a tradition of studies in the humanities. Course content may include pertinent readings from other disciplines. Possible themes: love, death, nature, evil, the comic spirit. May be repeated as course topic changes.

ENG 414 Advanced Studies in Genre**Credits 3 Grading Basis** Graded

Advanced study of works drawn from a specific literary or film genre or subgenre or a small, related set of (sub)genres. May be repeated as course topic changes.

ENG 415 Advanced Studies in Textuality and the History of the Book**Credits 3 Grading Basis** Graded

Detailed literary, bibliographical, political and cultural analysis of the material features of texts as physical objects. Topics may include the relation between a book's physical features and its intellectual contexts; the production, dissemination and receptions of texts; the history of manuscript, print and digital text technologies; the material history of reading and of literacy; and so forth. May be repeated as course topic changes.

ENG 417 Advanced Studies in Linguistics and the English Language**Credits 3 Grading Basis** Graded

This course provides advanced study of a particular topic in English linguistics or in English language studies. Topics may include particular subfields of linguistics or linguistic theories, applications of linguistic theory to literary studies or other related fields, or specific structural, historical, and cultural aspects of the English language. May be repeated as topic changes.

ENG 420 Advanced Studies in Theory and Cultural Studies**Credits 3 Grading Basis** Graded

Advanced study of a topic or debate within contemporary critical theoretical or cultural studies discourses in the humanities. Course may be repeated as topic changes.

ENG 421 Advanced Studies in Film Theory**Credits 3 Grading Basis** Graded

A study of major theories in film and media studies. Foundational schools of thought will be explored in terms of how they contribute to intellectual understandings of the relationship between media and modernity, ranging from classical film theory, political modernism, feminist and queer film theory, postcolonial film theory, to new media studies and postmodernism.

ENG 423 Advanced Studies in Gender and Sexuality in Literature**Cross-listed with:** WGSS 423**Credits 3 Grading Basis** Graded

Advanced study of a topic using a gender and sexuality studies approach to literary texts. This course will explore how gender and sexuality and their representation in literature are shaped by social, cultural, historical, and political contexts. Course may be repeated as topic changes. Course may be repeated as topic changes.

ENG 430 Advanced Studies in Comparative Literature**Credits 3 Grading Basis** Graded

Comparative study of selected world literature. May be repeated as course topic changes.

ENG 432 Advanced Studies in African Literature**Credits 3 Grading Basis** Graded

A study of selected works by African writers, focused by theme, geography or genre. May be repeated for credit when content varies.

ENG 433 Advanced Studies in Arabic Literature**Credits 3 Grading Basis** Graded

A study of Arabic writers. May be repeated for credit when content varies.

ENG 434 Advanced Studies in Latin American Literature in Translation**Credits 3 Grading Basis** Graded

This course will study Latin American literature in translation. The course will focus on the work of major Spanish-American authors. May be repeated as course content changes.

ENG 436 Studies in German Literature**Cross-listed with:** GER 436**Credits 3 Grading Basis** Graded

A study of selected works of German literature. Instruction is in English. May be repeated for credit when course content changes.

ENG 437 Studies in Italian Literature**Cross-listed with:** ITAL 437**Credits 3 Grading Basis** Graded

A study of selected works of Italian literature. Instruction is in English. May be repeated for credit when content varies.

ENG 438 Studies in Russian Literature**Cross-listed with:** RUS 438**Credits 3 Grading Basis** Graded

A study of selected works of Russian literature. Instruction is in English. May be repeated for credit when course content changes.

ENG 439 Advanced Studies in Major Authors of Literature in Spanish in Translation**Credits 3 Grading Basis** Graded

This course will study the work of both Peninsular and Latin American authors in translation. The course will focus on major Spanish-speaking authors and their work, both in Latin America and in Spain. May be repeated as course content changes.

ENG 447 Advanced Playwriting**Cross-listed with:** THEA 447**Credits 3 Grading Basis** Graded

An advanced workshop with emphasis on developing full-length dramatic material.

Prerequisites: THEA 347 or ENG 347 and junior standing.

ENG 463 British Film Adaptations**Cross-listed with:** SMAD 466**Credits 3 Grading Basis** Graded

This course studies the process of adapting British literature into feature films. Consideration is given to the original literary work, as well as to the changes undergone in its adaptation to film.

ENG 463L Film Adaptations**Cross-listed with:** SMAD 463**Credits 3 Grading Basis** Graded

This course is the study of the process of adapting literature into feature films. Consideration is given to the original literary work, as well as to the changes undergone in its adaptation to film.

Prerequisites: SMAD 301; for non-majors: ENG 381 and admission to the creative writing minor.

ENG 466 Advanced Studies in Women's Literature**Cross-listed with:** WGSS 466**Credits 3 Grading Basis** Graded

Advanced study of women's literary achievements in several cultural and historical contexts. May be focused by theme. May be repeated as course content varies.

Prerequisites: ENG 368/WGSS 368 or ENG 369/WGSS 369.

ENG 483 Narrative Form**Credits 3 Grading Basis** Graded

The study, development and practice of narrative craft.

Prerequisites: ENG 393 or permission of the instructor.

ENG 484 Poetic Craft and Creativity**Credits 3 Grading Basis** Graded

The study, development and practice of poetic craft.

Prerequisites: ENG 392 or permission of the instructor.

ENG 490 Special Studies in English**Credits 3 Grading Basis** Graded

Independent study for students with high academic standing. Students may select work in (1) a literary type, period, or author; (2) imaginative writing; or (3) linguistics. May be repeated for credit when course content changes.

ENG 493 Advanced Creative Nonfiction**Credits 3 Grading Basis** Graded

An advanced workshop in the writing of creative non-fiction narrative, with emphasis on point of view, form and style.

Prerequisites: ENG 391 or permission of the instructor.

ENG 494 Advanced Poetry Writing**Credits 3 Grading Basis** Graded

An advanced workshop with emphasis on developing sound poetic form, voice and vision.

Prerequisites: ENG 392 or permission of the instructor.

ENG 495 Advanced Fiction Writing**Credits 3 Grading Basis** Graded

An advanced workshop with emphasis on developing sound narrative prose form, style and vision. May be repeated for credit when course content changes.

Prerequisites: ENG 393 or permission of the instructor.

ENG 496 Advanced Topics in Creative Writing**Credits 3 Grading Basis** Graded

Study of a specific and concentrated aspect of creative writing. Topics will vary semester to semester. May be repeated for credit when the topic changes.

Prerequisites: ENG 391, ENG 392 or ENG 393, as appropriate to course content.

ENG 499A Honors Project**Credits 2 Grading Basis** Credit / No Credit

See catalog section "Graduation with Honors."

ENG 499B Honors Project**Credits 2 Grading Basis** Graded

ENG 499C Honors Project
Credits 2 Grading Basis Graded

Environment (ENVT)

ENVT 101 Perspectives in Environmental Science [C3PP]
Credits 3 Grading Basis Graded

This course offers an introductory view of essential themes in environmental science. It integrates the study of the natural processes that take place on Earth. Emphasis is placed on understanding the development of the physical and chemical principles and their application in understanding environmental issues in the modern world. May be used for general education credit.

Prerequisites: One of the following: MATH 103, MATH 110, MATH 205, MATH 220, MATH 231 or MATH 235.

ENVT 200 Environmental Systems Theory
Credits 3 Grading Basis Graded

This course explores three aspects of understanding the environment. First, the kind of problem the environment is and the thinking strategies that will best yield insights and understanding. Second, how humans create and/or respond to environmental issues and crises. Third, examination of past environmental changes and how humans have been affected by and responded to those changes. Final synthesis explores what we can and cannot do practically to respond to future changes. Does not satisfy elective credit or count as credit for geology or earth science majors.

ENVT 225 Environmental Advocacy and the Psychology of Change
Credits 3 Grading Basis Graded

This is a summer study abroad course in Malta about environmental advocacy. Examines sense of place, nature connectedness and cultural influences on environmental advocacy and environmental movements through interaction with a variety of NGOs and government organizations. Explores the psychology of change and elements of cooperation and collaboration that strengthen efforts to protect the environment and engage citizens in their natural world.

ENVT 300 Topics in Environmental Humanities
Credits 3 Grading Basis Graded

Special topics in the humanities and arts that enhance the study of environmental humanities and are not currently taught in the regular curriculum.

ENVT 301 Internship in Environmental Humanities
Credits 3 Grading Basis Graded

This course provides the student with practical experience in areas pertinent to the study of environmental humanities. Any internship experience must be approved by the coordinator in advance, and details of supervision and evaluation should be spelled out in advance. The internship may also be through an academic unit, but must be approved for credit by the environmental humanities coordinator in advance of the experience.

ENVT 302 Internship in Environment
Credits 1-4 Grading Basis Graded

This course provides the student with practical experience in areas pertinent to the study of environmental studies, sciences or management. Any internship experience must be approved by the coordinator in advance, and details of supervision and assessment should be spelled out in advance. May be taken for a maximum of six credits.

Prerequisites: Minor coordinator permission.

ENVT 303 Topics in Environment
Credits 3 Grading Basis Graded

Examination of selected topics that are of current importance for interdisciplinary environmental studies, sciences and management including: environmental problem solving, sustainability, addressing climate change, environmental resiliency and more. May be repeated for credit when course content changes.

ENVT 366 Religion, Animals and the Environment
Cross-listed with: REL 366

Credits 3 Grading Basis Graded

This course analyzes the ways in which various religions, especially Judaism and Christianity, understand the relationship of human beings to the environment and non-human animals. We will explore how religious worldviews translate into human impact on the non-human biosphere. Given the global reach of American environmental policies and lifestyles, and the prominent role that biblical interpretation has played in shaping American attitudes and policy, special emphasis will be placed on historical interpretations of the Bible, ranging from human dominionism to human responsibility in caring for the earth, its creatures and its ecosystems. The course explores the complexities involved in using religious positions, interpretations of biblical texts and apocalyptic arguments to inform both environmentalism and policy debates related to the environment. Students will gain a basic introduction to methods of biblical scholarship and familiarity with the general state of scientific consensus on environmental challenges.

ENVT 400 Capstone Seminar in Environmental Problem Solving
Credits 3 Grading Basis Graded

Integrates perspectives from three environment programs: environmental management, environmental science and environmental studies. The course is team taught using a case-study approach to environmental issues, emphasizing teamwork and student initiative. Topics vary. Students wishing to complete more than one of the environmental minors (environmental studies, environmental science or environmental management) may receive dual credit.

Prerequisites: Completion of 15 hours in declared environment minor or permission of the instructor.

ENVT 401 Studio or Independent Study in Environmental Humanities
Credits 3 Grading Basis Graded

Students may propose to engage in a creative, expressive or scholarly project either through their home department or using the ENVT identifier. Studio or independent study must be approved by the Environmental Humanities coordinator in advance of the experience.

ENVT 403 Independent Study in Environment
Credits 1-4 Grading Basis Graded

Students propose to engage in a scholarly or applied project about interdisciplinary environmental problem-solving in coordination with the relevant environmental minor coordinator (studies, sciences, management). Proposals must include a topic or key question, a learning plan with readings and other content, an appropriately sized productive output, and a faculty supervisor. Must be approved by the environmental minor coordinator. May be repeated for maximum of six credit hours.

Prerequisites: Minor coordinator permission.

Environmental Management (ENVM)

ENVM 480 Selected Topics in Environmental Management

Credits 1-4 Grading Basis Graded

Topics in environmental management which are of interest to the upper division student but not otherwise covered in the regular course offerings. Offered only with the approval of the director. May be repeated for credit when course content changes. Students should consult the instructor prior to enrolling.

Prerequisites: Junior or senior standing in environmental management program. Topic selected may dictate additional prerequisites.

Exceptional Education (EXED)

EXED 200 Foundations of Exceptional Education

Credits 3 Grading Basis Graded

This course examines the social, cultural and educational contexts - through the lifespan - of individuals with exceptionalities. Specific focus will be given to the study of historical perspectives, models, theories, philosophies and trends in the field of special education. Legislative and judicial mandates and current regulations related to individuals with ELNs and the 'rights and responsibilities' of various stakeholders as they relate to exceptionality will be stressed. In addition, the role of culture, environment, family and individual differences will be explored.

EXED 202 Field Experiences in Special Education

Credits 3 Grading Basis Credit / No Credit

Provides students with supervised experiences with persons with disabilities. Placements are made in various settings, including schools, institutions and recreational programs.

Prerequisites: EXED 200.

EXED 203 Contemporary Issues in Special Education

Credits 1 Grading Basis Graded

This course investigates current issues and controversies in the field of exceptional education. This course is designed to support the student's understanding of how sociological, familial, political and cultural issues are affecting the field of special education. It will also challenge students to apply this knowledge to their practice.

Prerequisites: EXED 200.

EXED 300 Educational Technology for Students with Disabilities

Credits 1 Grading Basis Graded

An introduction to instructional technology for persons with disabilities. The role of assistive technology in the educational process is investigated. Students are exposed to a variety of instructional programs and equipment. Federal and state guidelines, interdisciplinary team functioning, and program as well as equipment selection are addressed.

Prerequisite: EXED 200.

EXED 302 Mentoring Children and Youth with Mild Disabilities

Credits 2 Grading Basis Graded

The course will provide students with the knowledge and skills to engage in mentoring of children and youth with learning disabilities and attentional disorders. Students will focus on increasing their understanding of self-awareness related to living with a disability, effective compensatory learning strategies and self-advocacy skills. Because of the purpose of this course is to increase self-awareness and mentoring skills related to understanding disabilities, it is open only to students who are registered with the Office of Disability Services.

EXED 303 Foundations of Classroom and Behavior Management

Cross-listed with: EXED 512

Credits 3 Grading Basis Graded

This course provides students with an understanding of and skills to apply classroom and behavior management techniques and interventions, including techniques that promote emotional well-being and that teach and maintain behavioral conduct. These skills are consistent with the norms, standards and rules of the educational environment. Diverse approaches for classroom and behavior management based on behavioral, cognitive, affective, social and ecological theory and practice will be learned.

Prerequisites: Minimum grade of "C" in requisites. SPED majors:

EXED 200. B.I.S./SPED: EXED 484. Corequisites: B.I.S./SPED: EXED 405 and EXED 475.

EXED 305 Characteristics of Acquired Brain Injury

Credits 3 Grading Basis Graded

EXED 305 provides an overview of acquired brain injury - its causes, complications, consequences and considerations for survivors, caregivers and families. Instruction will emphasize rehab treatment philosophies and approaches, cultural competencies, gender and sexuality issues, military populations, pediatrics and adolescents, and families. Students will learn how to apply knowledge of brain injury sequelae to future healthcare settings and educational learning environments.

Prerequisites: EXED 200.

EXED 306 Lifespan Issues for Individuals with Disabilities

Credits 3 Grading Basis Graded

This course examines how issues such as legal mandates and policies, self advocacy, family involvement, educational services, transition and interagency collaboration impact individuals with disabilities and their families from birth through postsecondary life. The students will be challenged to compare and analyze needs and services available and accessed by individuals with disabilities.

Prerequisites: EXED 200.

EXED 309 Supporting Community Access for Adults with Disabilities

Credits 3 Grading Basis Graded

This course provides students with an understanding of and experience working with young adults with disabilities who are exiting Special Education eligibility. The goal is to support these young adults in learning to access the JMU community and gain experience with nondisabled same-age peers.

EXED 310 Survey of Emotional Disturbance

Credits 3 Grading Basis Graded

This course is a detailed study of the characteristics, diagnosis, treatment, assessment and education of individuals with emotional/behavioral disorders. Medical, psychological, behavioral and environmental causes are presented as well as therapeutic interventions, educational resources and instructional strategies.

Prerequisites: EXED 200.

EXED 320 Survey of Learning Disabilities

Credits 3 Grading Basis Graded

This course is a detailed study of the theories, characteristics, etiology and needs of individuals with learning disabilities including ADHD. Focus will be on causation and terminology as well as historical perspectives and current trends related to practices in identification and treatment of learning disabilities.

Prerequisites: EXED 200.

EXED 325 Inclusive Practices in Early Childhood**Credits 3 Grading Basis** Graded

This course is designed to introduce students to the issues and trends in the inclusive education of young children. It will provide the historical, philosophical, social and legal background for current practices in the field. This class will also engage students in synthesizing and analyzing evidence-based practices in field.

EXED 330 Survey of Intellectual Disabilities**Credits 3 Grading Basis** Graded

This course is a detailed study of the characteristics, diagnosis, treatment and education of individuals with intellectual disabilities. Medical aspects and implications for support needs are addressed as well as educational settings, resources and instructional techniques designed to facilitate integration for individuals with intellectual disabilities.

Prerequisites: EXED 200.

EXED 339 The Role of the Speech-Language Pathologist in Public Schools**Credits 3 Grading Basis** Graded

How a speech/language pathologist manages multiple schools; schedules across general education instruction; screens; assesses; uses diagnostic tools for evaluation and placement of students based on the educational impact of the speech-language impairment; attends special education and school meetings with an array of teachers, therapists and parents; completes divisional and state paperwork and; provides therapy based on each student's Individualized Education Program (IEP) will be covered.

Prerequisites: EXED 200.

EXED 341 Characteristics of Learners with Disabilities Accessing the General Curriculum**Credits 3 Grading Basis** Graded

The course covers historical, ethical and legal aspects involving the education of children with disabilities. Definitions and foundational characteristics of teaching children with disabilities accessing the general curriculum are featured. Identification, education and evaluation of academic skills and behavior of students with disabilities are presented. How family, cultural, socioeconomic and environment impact education are discussed. Professional behavior standards are examined. Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 303. B.I.S./SPED: EXED 200. Corequisites: SPED majors: EXED 342 and EXED 429. B.I.S./SPED: EXED 303, EXED 342 and EXED 474.

EXED 342 Characteristics of Learners with Low Incidence Disabilities Accessing the Adapted Curriculum**Credits 3 Grading Basis** Graded

This course provides students with an understanding of the current research and practices supporting the unique characteristics - including learning and behavioral - of students with low-incidence disabilities accessing the adapted curriculum. These students include those with autism spectrum disorder, developmental delay, intellectual disability and traumatic brain injury. This course will address age span issues; cultural, ethnic and socioeconomic factors; cognitive functioning; emotional and behavioral adjustment; medical aspects and considerations; and social and language development.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 303. B.I.S./SPED: EXED 200. Corequisites: SPED majors: EXED 341 and EXED 429. B.I.S./SPED: EXED 303, EXED 341 and EXED 474.

EXED 375 Overview Study of Autism Spectrum Disorders**Credits 3 Grading Basis** Graded

This course provides an overview of the current issues involved in working with people who have been identified as having an autism spectrum disorder. Areas addressed will include learning characteristics; current research and factors involved with causation, assessment and diagnosis; positive behavioral supports; social skills development; sensory processing, motor planning and sensory integration; and communication and language development.

Prerequisites: EXED 200.

EXED 376 Initial Practicum for Special Education PreProfessional Preparation**Credits 1 Grading Basis** Graded

A practicum experience that will provide an opportunity to observe the roles of special education teachers who are working with students with disabilities who are accessing the general curriculum. Practicum students will have the opportunity to observe instruction at both elementary and secondary levels. They will be able to observe classroom management and instruction in math, reading and writing. Some introductory level instruction opportunities may be provided.

Prerequisites: EXED 200.

EXED 401 Issues in Exceptional Education**Credits 1-3 Grading Basis** Graded

This course considers current problems and issues in special education as they relate to the professional education of teachers. The course may be repeated if the topic varies.

Prerequisites: EXED 200.

EXED 402 Positive Behavior Supports and Individualized Interventions for Young Children**Credits 1-3 Grading Basis** Graded

This course is designed for early childhood educators (ECEs) who may work with young children with challenging behaviors. This course focuses on basic principles of behavior as means to help ECEs understand the function of behavior, or the reasons why behavior occurs. This course focuses on proactive strategies as a means to prevent challenging behavior, measuring behaviors using data collection and implementing behavior interventions that are aligned with behavior function. This course emphasizes a positive approach to behavior management aligned with best practices in teaching social emotional skills to young children. This course is open only to students with an applied associate degree in early childhood from a Virginia community college.

Prerequisites: EDUC 200 and PSYC 160.

EXED 405 Alternative Math Instructional Methods and Supports for Students with Disabilities**Credits 3 Grading Basis** Graded

This course prepares preservice special education teachers to identify, remediate and support individuals with math disabilities and challenges through the use of dynamic assessment and evidence-based instructional methods. In this course, a review of the skills, concepts and processes related to the major math content area as outlined in the Virginia SOL will be provided. The course will provide an in-depth examination of math deficits, supports and alternative instructional methods related to each math content area.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 341. B.I.S./SPED: EXED 484. Corequisites: SPED majors: EXED 419, EXED 474, EXED 484 and EXED 486. B.I.S./SPED: EXED 303 and EXED 475.

EXED 407 Transitions of Learners with Disabilities**Credits 3 Grading Basis** Graded

This course provides a detailed study of the transitions throughout the educational experience of students with disabilities, including transitions between the Individuals with Disabilities Education (IDEA, 2004) Part C and Part B services, elementary school through middle school, middle school through high school, and high school to postsecondary environments. Focus is given to the services and skills needed to support successful integration into and functioning within the routines and activities of these settings.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 419. B.I.S./SPED: EXED 485. Corequisites: SPED majors: EXED 450, EXED 475 and EXED 476. B.I.S./SPED: EXED 436.

EXED 411 Applied Behavior Analytic Services as a Paraprofessional**Credits 3 Grading Basis** Graded

This course is designed to provide the student with the necessary instruction to apply for licensure as a Registered Behavior Technician (RBT). This course will provide the student with the skills needed to implement behavior analytic services in a variety of settings (schools, homes, clinics) with individuals across the lifespan. The course will address all items required for RBT training, including measurement, assessment, skill acquisition, behavior reduction, documentation and reporting, professional conduct, and scope of practice.

Prerequisites: EXED 200.

EXED 416 Overview and Assessment of Autism Disorders**Credits 3 Grading Basis** Graded

This course is designed to provide an overview of the current issues involving working with children who have been identified as having an autism spectrum disorder. Areas covered in-depth will include learning characteristics, current research and factors involved with causation, assessment and diagnosis. We will discuss positive behavioral supports; social skills development; sensory processing, motor planning and sensory integration; and communication and language development as these will be covered in much greater depth in other courses. A range of institutional methodologies and techniques will be emphasized throughout the course.

Prerequisites: EXED 200.

EXED 417 Communication, Language and Sensory Issues of Learners with ASD and/or on the Adapted Curriculum**Credits 3 Grading Basis** Graded

This course is designed to provide an in-depth study of the current issues involved in working with learners who have been identified as having an autism spectrum disorder and/or being on the adapted curriculum. The course will review learning characteristics, current research and factors involved with causation, assessment and diagnosis, and positive behavioral supports to set the stage. The course will explore social skill development; sensory processing, motor planning and sensory integration; and communication and language development. The course will consider a range of instructional methodologies and techniques for providing instruction, support and generalization of skills in these areas.

Prerequisites: EXED 416.

EXED 418 Functional Behavioral Assessment and Developing Intervention Plans for Challenging Behaviors**Credits 3 Grading Basis** Graded

This course is designed to provide an in-depth look at the behavioral challenges those with a disability in the autism spectrum might face and display. Areas addressed will include behavioral characteristics, current research and factors related to behavioral challenges in this population, positive behavioral supports, Functional Behavioral Plan Development, implementation and monitoring. We will cover data collection in relation to assessment and monitoring behaviors. We will review social skills development; sensory processing, motor planning and sensory integration; and communication and language development as these are covered in much greater depth in other courses. A range of institutional methodologies and techniques will be emphasized throughout the course.

Prerequisites: EXED 416 and EXED 417.

EXED 419 Foundations and Theories of Literacy for K-12**Credits 3 Grading Basis** Graded

This course provides a comprehensive exploration of the science of literacy, focusing on how children and adolescents acquire and develop literacy skills across a variety of learners in K-12 classrooms. Grounded in evidence-based research, the course examines the cognitive and neurological processes involved in literacy development and acquisition, including an in-depth look at the reading brain and how it adapts to written language. Students will engage with foundational frameworks such as the Simple View of Reading, which emphasizes the dual importance of decoding and language comprehension, and will analyze the five pillars of literacy-phonemic awareness, phonics, fluency, vocabulary and comprehension. Additionally, the course will explore Scarborough's Reading Rope, illustrating the complex, interwoven strands that contribute to skilled reading. The course also emphasizes the role of assessment in guiding instruction and monitoring progress, equipping educators with tools to evaluate literacy skills effectively. By the end of the course, students will be prepared to apply scientifically grounded literacy practices to support all learners in achieving reading proficiency.

Prerequisites: Minimum grade of "C" in requisites. EXED 341.

Corequisites: SPED majors: EXED 405, EXED 474, EXED 484 and EXED 486. B.I.S./SPED: EXED 429 and EXED 484.

EXED 429 Teaching Learners with Complex Care Needs**Credits 3 Grading Basis** Graded

This course is designed to give special educators exposure to working with children (birth through 21) who have disabilities that include significant medical issues. The role and function of special educators as team members are addressed. The role of related technology will receive extensive attention, as will issues of etiology and remediation.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 303. B.I.S./SPED: EXED 342. Corequisites: SPED majors: EXED 341 and EXED 342. B.I.S./SPED: EXED 484 and EXED 419.

EXED 431 Assistive Technology for Individuals with Sensory Impairments**Cross-listed with:** EXED 531**Credits 3 Grading Basis** Graded

This course is designed to heighten the awareness of participants to specific technology and resources available to enhance and improve the abilities of individuals with sensory impairments to succeed in school, daily living activities and employment. This course is delivered via a distance education format.

Prerequisites: EXED 435. Corequisites: EXED 435.

EXED 432 Braille Code**Credits 3 Grading Basis Graded**

This course provides instruction in the development, use and application of the Braille literary code and its implications for educational/literacy programs for students with visual disabilities. Students will develop the skills to read and write contracted and uncontracted Braille, while acquiring instructional methodologies for teaching children who are blind to read and write. Sources of Braille materials for educational purposes are identified. This course is delivered via a distance education format. Prerequisites: EXED 435. Corequisites: EXED 435.

EXED 434 Curriculum and Assessment for Students with Visual Impairments**Credits 3 Grading Basis Graded**

This course provides students with knowledge and understanding of the educational assessment of students with visual impairments and additional disabilities including deaf-blindness. Students practice assessing and planning educational programs for students with visual impairments. Also covered in this course are assessment technology for students with visual impairments; determination of learning needs and appropriate learning media; and the relationship of assessment, IEP development, and placement. This course is delivered via a distance education format.

Prerequisites: EXED 435. Corequisites: EXED 435.

EXED 435 Characteristics of Students with Visual Impairments**Credits 1 Grading Basis Graded**

This course provides an overview of the characteristics of and services to persons with visual impairments, including the impact of visual impairments on infants' and children's growth and development, child and adolescent emotional and social development, and family interaction patterns. It considers the educational, conceptual, psycho-social and physical implications of a visual impairment. This course is delivered via a distance education format.

EXED 436 Communication Language and Literacy for Learners on the Adapted Curriculum**Credits 3 Grading Basis Graded**

This course provides an in-depth study of the current issues involved in working with learners who have been identified as having an autism spectrum disorder and other developmental disabilities who participate in the adapted curriculum. The communication, sensory and literacy needs of students with autism spectrum disorders (ASD) and developmental disabilities participating in the adapted curriculum will be explored. Assessment and instruction in social skills, sensory processing, communication and language development, and literacy instruction (including reading and writing) will be explored. A range of instructional methodologies and techniques for providing instruction, support and generalization of skills in these areas will be considered. This course will focus on the importance of relying on evidence-based practices and where to find information about EBPs.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 475. B.I.S./SPED: EXED 485. Corequisites: SPED majors: EXED 478. B.I.S./SPED: EXED 407.

EXED 440 Classroom Management and Professional Collaboration**Credits 3 Grading Basis Graded**

A focus on techniques used to manage the behavior of students. Emphasized are strategies used to prevent inappropriate behavior from occurring and/or worsening. Other interventions are taught such as techniques for working with others (e.g., parents, teachers and administrators) who may provide behavior management assistance to teachers and administrators.

EXED 441 Functional Applications of Low Tech Assistive Technology**Credits 2 Grading Basis Graded**

This course will focus on functional applications of low-technology solutions within the areas of self-care; mobility and transfer; communication; stability and support; sports, recreation, and leisure; and academic and work environments. The course will include exploration and opportunities to design and create low-tech devices for children and adults.

Prerequisites: EXED 300.

EXED 443 Assistive Technology Use for Individuals with Disabilities**Credits 3 Grading Basis Graded**

This course is designed to enhance students' awareness and understanding of the range of assistive technologies available and their instructional implications for individuals with disabilities. Laboratory and demonstration experiences will enable students to select and utilize devices and software in settings serving individuals with disabilities.

Prerequisites: EXED 200.

EXED 450 Principles of Specialized Reading Instruction**Credits 3 Grading Basis Graded**

The course focuses on acquisition and development of reading skills for students with disabilities. Content includes characteristics of students with reading disabilities; informal assessment; relationship of oral language to reading; stages of reading development; research-based methods; principles of specialized reading instruction and collaboration to support reading development.

Prerequisites: Minimum grade of "C" in requisites. EXED 419.

Corequisites: SPED majors: EXED 407, EXED 475 and EXED 476, B.I.S./SPED: EXED 473, EXED 477 and EXED 478.

EXED 455 Collaborative Teaching for Learners with Disabilities**Credits 3 Grading Basis Graded**

Students in this course will gain knowledge and practice skills in consultation, case management and collaboration with individuals, families, educators, related service providers and other human service professionals. An overview of collaborative processes, collaborative models for supporting the education of students with disabilities and for effective management of paraprofessionals will be studied.

Prerequisites: EXED 200.

EXED 460 Differentiation of Instruction and Academic Collaboration**Credits 3 Grading Basis Graded**

This course assists preservice teachers in using their understanding of exceptional learners and learning to accommodate the diversity of students in the general education classroom. In addition, preservice teachers will explore the roles of teachers and how general and special education teachers collaborate to meet the needs of exceptional students. Teaching education students only.

EXED 465 Perspectives of Early Childhood Special Education**Credits 3 Grading Basis Graded**

This course is designed to provide the student with an overview of educational programming and service delivery for children with developmental delays and/or disabilities, ages 0 to 5. Particular attention is given to federal legislation, historical perspective and current recommended practice in programming educational services for young children with delays and/or disabilities.

EXED 473 Methods of Instruction for Learners Accessing the Adapted Curriculum**Credits 3 Grading Basis** Graded

Methods for remediation and adaptation of instruction in academic and functional skills for students with moderate/severe disabilities accessing the adapted curriculum. Emphasizes instructional approaches such as task analysis, systematic instruction, functional/daily living skills and general curriculum access using adapted content standards. A practicum placement provides field experience to practice skills learned in this course.

Prerequisites: Minimum grade of "C" in requisites. EXED 475.

Corequisites: SPED majors: EXED 436, EXED 478, EXED 480 and EXED 483. B.I.S./SPED: EXED 450, EXED 477 and EXED 478.

EXED 474 Assessment and Evaluation in Special Education**Credits 3 Grading Basis** Graded

This course was designed to provide study and application of the foundations of assessment and evaluation related to management of instruction and behavior of individuals with exceptional learning needs (ELN). The course emphasizes issues and skills in selection, administration, interpretation, and use of a variety of assessment tools and techniques in the stages of the decision-making process for management of instruction and behavior. Application of this new knowledge and skill will be through case studies and direct assessment. Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 341. B.I.S./SPED: EXED 200. Corequisites: SPED majors: EXED 405, EXED 419, EXED 484 and EXED 486. B.I.S./SPED: EXED 303, EXED 341 and EXED 342.

EXED 475 Program Planning in Special Education**Credits 3 Grading Basis** Graded

This course was designed for student exploration of practical skills and strategies in development and use of programming to meet the academic and behavioral needs of students with disabilities accessing the K-12 general curriculum. Skills will be applicable in a variety of placement settings and service delivery models. Included is a detailed study of purposes and procedures involved in the development of individualized education programs and the selection or design of curriculum-based assessment to plan and evaluate instruction in academics, social behaviors and life skills.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 419. B.I.S./SPED: EXED 484. Corequisites: SPED majors: EXED 407, EXED 450 and EXED 476. B.I.S./SPED: EXED 303 and EXED 405.

EXED 476 Practicum in Literacy and Assessment**Credits 2 Grading Basis** Graded

This practicum is designed to provide a structured supervised experience assessing learning, planning and delivery of instruction to students with disabilities accessing the general education curriculum, and gathering data to make decisions about the effectiveness of instruction. Application of skills in planning direct instruction, creating instructional materials, collecting performance data, managing behavior and developing social skills will be emphasized.

Prerequisites: Minimum grade of "C" in requisites. EXED 419.

Corequisites: EXED 407, EXED 450 and EXED 475.

EXED 477 Remedial Methods in Special Education**Credits 3 Grading Basis** Graded

This course is designed to provide students with knowledge about intensive remediation methods for students with disabilities who are significantly below grade level. The students will examine intensive instructional methods that address not only reading, writing and math, but also metacognitive processes and support skills.

Prerequisites: Minimum grade of "C" in requisites. EXED 475.

Corequisites: SPED majors: EXED 478, EXED 480 and EXED 483. B.I.S./SPED: EXED 450, EXED 473 and EXED 478.

EXED 478 Supervised Practice in Special Education Methods**Credits 2 Grading Basis** Graded

This course provides field experience opportunities to practice the special education instructional methods developed in EXED 474 and previous courses.

Prerequisites: Minimum grade of "C" in requisites. EXED 475.

Corequisites: SPED majors: EXED 473 or EXED 477. B.I.S./SPED: EXED 450, EXED 473 and EXED 477.

EXED 480 Student Teaching in Special Education**Credits 4 Grading Basis** Credit / No Credit

This student teaching experience in the area of special education enables the teacher candidate to apply the education interventions and management competencies acquired in previous courses and fieldwork. The teacher candidate will participate in varied experiences related to the licensure area(s) within special education they are seeking. The teacher candidate has the opportunity to develop their professional skills and ethical understanding of the profession of teaching and how special education fits within this profession. Over the course of the placements, the teacher candidate will assume a range of responsibilities for planning and implementing multiple individualized education programs for students with disabilities. They will experience the many roles of a special education teacher and how collaboration is a key tool in each of those roles. Due to the mentoring of a cooperating special education teacher, the teacher candidate should feel free to take risks in trying new methods and developing their skills. The course is taken on a credit or no credit evaluation criterion. It is repeatable for up to 12 credits and must be taken twice for initial special education licensure.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 475. B.I.S. majors/concentration in SPED: EXED 478. Corequisites: SPED majors: EXED 478.

EXED 483 Teachers as Professionals**Credits 1 Grading Basis** Graded

This seminar is designed to accompany the student teaching experiences. Student teachers will have the opportunity to reflect on their skills, problem-solve school and classroom experiences, increase awareness of the need to continue career-long learning, discuss linkages between program coursework and student teaching experience, explore their own skill level, and plan for their on-going professional development. Seminars in fall and spring will have differentiated content designed to link with the curriculum content being covered in that semester, as well as to the student teacher's experience.

Prerequisites: Minimum grade of "C" in requisites. EXED 475.

Corequisites: EXED 480.

EXED 484 Inclusive Practices in Special Education**Credits 3 Grading Basis** Graded

This course is a detailed study of inclusive practices for students with disabilities. This course is part 1 of our Special Education Teaching Methods. The course emphasizes instructional approaches such as task analysis, accommodations, collaborative and co-teaching, along with case management. Focus is on specific understanding quality instruction at Tier 1 and Tier 2 levels in reading, math, written language, study skills and some life skills which would be applied in general education settings and small groups. A co-requisite practicum provides field experience opportunities to practice skills learned in this and previous courses.

Prerequisites: Minimum grade of "C" in requisites. EXED 341.

Corequisites: SPED majors: EXED 405, EXED 419, EXED 474, and EXED 486. B.I.S./SPED: EXED 419 and EXED 429.

EXED 485 Systematic Behavioral Support and Interventions**Credits 3 Grading Basis** Graded

This course is designed to provide application and evaluation of group management techniques and individual interventions that teach and maintain emotional, behavioral and social skills. Students will study various systematic behavioral interventions to support the behavior and learning of individuals with disabilities accessing the general education curriculum including positive behavioral supports, functional assessments of behavior and teaching social skills. A variety of data collection procedures to inform their professional practice will be examined. Students will be applying evidence-based practices related to positive behavior supports, assessment of challenging behaviors, and data collection to inform instructional practice and document student response to interventions that improve behavior such that academic learning can occur.

Prerequisites: Minimum grade of "C" in requisites. SPED majors: EXED 475. B.I.S./SPED: EXED 478. Corequisites: SPED majors: EXED 478. B.I.S./SPED: EXED 480.

EXED 486 Supervised Clinical Practice with Planning and Methods in Special Education**Credits 2 Grading Basis** Graded

This course provides field experience opportunities to practice the special education instructional methods related to inclusion and assessment being taught in EXED 484 and EXED 474. All previous coursework is expected to be applied as necessary.

Prerequisites: Minimum grade of "C" in requisites. EXED 341.

Corequisites: EXED 405, EXED 419, EXED 474 and EXED 484.

EXED 487 Fostering Positive Classroom Behaviors**Credits 3 Grading Basis** Graded

This course will provide some foundational knowledge related to individual behavior interventions but is designed to focus on providing application and evaluation of group management techniques and individual interventions that teach and maintain emotional, behavioral and social skills. Students will study various systematic behavioral interventions to support the behavior and learning of individuals with disabilities of all levels including positive behavioral supports, functional assessments of behavior and teaching social skills. A variety of data collection procedures to inform their professional practice will be examined. Students will apply evidence-based practices related to Positive Behavior Supports, assessment of challenging behaviors and data collection to inform instructional practice and document student response to interventions that improve behavior such that academic learning can occur.

Prerequisites: Individualized Study majors with concentration in special education.

EXED 490 Special Studies in Special Education**Credits 1-3 Grading Basis** Graded

This course is designed to allow the student to complete independent study under faculty supervision. May be repeated for credit when course content changes.

Prerequisites: EXED 200 and permission of the academic unit head.

Family (FAM)

FAM 133 The Contemporary Family**Credits 3 Grading Basis** Graded

Concepts of variations in forms and lifestyles of families. Consideration is given to the family life cycle and the interdependency between the family and society.

FAM 300 Child Development**Credits 3 Grading Basis** Graded

A study of the factors influencing the physical, cognitive, social and emotional growth of the young child. Emphasis is given to the importance of family relations and development of observational skills.

Prerequisites: PSYC 101, PSYC 160 or equivalent.

FAM 325 Parent-Child Relationships Across the Lifespan**Cross-listed with:** SOWK 325**Credits 3 Grading Basis** Graded

This course focuses on giving and receiving familial attention, care and support over the lifespan and across generations. Combines developmental and ecosystems frameworks to explore family life as the primary arena in which people learn and establish social behavior and interpersonal relationships, in conjunction with extrafamilial connections.

Prerequisites: FAM 133 or SOCI 374.

FAM 327 Substance Use Interventions**Cross-listed with:** SOWK 327**Credits 3 Grading Basis** Graded

Explore types, effects and prevalence of substance use in the U.S. Impacts of use on individuals, families, communities and society is considered through biopsychosocial, economic and legal/political lenses. Dual foci include developmental risks and outcomes for individuals and families throughout the life course, and social policy debate and legislation. Includes a survey of intervention approaches and treatment modalities across the span of self-help, medical treatment, programs in behavioral health and court-mandated models.

FAM 347 Issues Affecting LGBTQIAP+ Individuals and Their Families**Cross-listed with:** SOWK 347**Credits 3 Grading Basis** Graded

Operating from a cultural humility framework, this course focuses on practice, policy, and historical and social issues impacting LGBTQIAP+ individuals and their families. This course utilizes a strengths-based perspective to identify protective factors and explore resiliency within the LGBTQIAP+ community in the face of social stigma, marginalization and oppression. The course encourages students' critical self-reflection of personal and cultural value systems and how these may impact understanding of LGBTQIAP+ populations.

FAM 362 Social Work with Immigrants and Refugees**Cross-listed with:** SOWK 362**Credits 3 Grading Basis** Graded

This course explores the scope and complexity of global/international migration and its impacts on national policy and social service structures and organizational response. It provides a framework to understand transmigration and acculturation, to analyze aspects of current immigration law, and to examine the role of various service systems to address the needs of immigrant and refugee individuals and families integrating into local communities. Service-learning in an agency that serves the local immigrant community is required.

FAM 375 Grant Writing for Agencies**Cross-listed with:** GERN 375, NPS 375, SOWK 375**Credits 3 Grading Basis** Graded

Emphasizing active learning, this course teaches the basics of grant and proposal writing. Efficient research, persuasive prose and the importance of relationships are stressed. Private and corporate philanthropy and government grants are examined.

FAM 386 Youth Empowerment Strategies (YES)**Cross-listed with:** SOWK 386**Credits 3 Grading Basis** Graded

Students learn to use group activities that include the creative arts, low ropes, and self-discovery in youth empowerment. The goal is to help youth build life skills and make informed decisions. Prior to beginning work with youth, students complete 25 hours of training.

FAM 400 Theory Survey**Credits 1 Grading Basis** Graded

This seminar provides students with several key theoretical frameworks that span the family ecology and gerontology concentrations of the minor and will inform and support their capstone experiences.

Prerequisites: FAM 133 or SOCI 374. FECO focus: 9 cr from the focus, JR/SR standing. GERO focus: PSYC 160, SOCI 280/GERN 280, GERN 305, and JR/SR standing. Coreqs: FECO focus: FAM 401. GERO focus: IPE 313.

FAM 401 Issues and Applications**Credits 2 Grading Basis** Graded

The course emphasizes the application of knowledge through the development and implementation of a substantial, theory-based, in-depth individualized project. Students may adapt the project to fit their career objectives and/or plans for graduate study.

Prerequisites: FAM 133 or SOCI 374, and at least 9 additional credits from the family ecology track, not including FAM 400. Corequisites: FAM 400.

FAM 487 Special Topics in Family Studies**Credits 3 Grading Basis** Graded

Examination of selected topics that are of current importance to family studies. Course may be repeated for credit.

Prerequisites: FAM 133 or SOCI 374.

FAM 490 Special Studies in Family Studies**Credits 1-3 Grading Basis** Graded

The course is designed to give students in family studies an opportunity to complete independent study under faculty supervision. Course may be repeated for credit.

Prerequisites: FAM 133 or SOCI 374 and two additional courses in the family studies minor, or permission of the instructor.

Finance (FIN)

FIN 201 Introduction to Financial Markets and Careers**Credits 1 Grading Basis** Credit / No Credit

This class provides an introduction of terminology in financial markets and basic functions of various financial products and instruction in the preparation of professional etiquette and business communication. Discussions are held with industry experts on roles of finance professionals in business, government and capital markets.

FIN 250 Introduction to Quantitative Finance**Credits 3 Grading Basis** Graded

The purpose of this course is to provide a broad introduction to the markets and instruments of engineered finance. The focus of the course is to expose students to the properties and uses of the array of non-traditional financial instruments that are increasingly trading in both the exchange and over-the-counter markets.

Prerequisites: Minimum grade of "C" in MATH 235, MATH 236, ECON 200 and ECON 201.

FIN 299 Topics in Finance**Credits 1-3 Grading Basis** Credit / No Credit

This is a sophomore level course that covers finance-related topics of interest such as introductory financial markets or finance career options. Topics may vary according to the instructor. May be taken for a maximum of six credits if the topics vary.

FIN 301 Principles of Finance**Credits 3 Grading Basis** Graded

The purpose of this course is to provide a foundation in the principles and tools of finance, which include financial analysis, the time value of money, capital budgeting and capital structure. Open to students with a B.S. or B.A. economics major with a concentration in finance as well as students majoring in health sciences.

Prerequisites: Placement in B.S. or B.A. Economics major with a concentration in Financial Economics or placement in Health Sciences. Junior standing and COB 241.

FIN 310 Principles of Real Estate**Credits 3 Grading Basis** Graded

The purpose of this course is to assist students in developing a foundation in real estate principles, apply finance and economic principles to valuation and evaluation of real estate and analyze contracts and financing in residential and commercial real estate transactions.

Prerequisites: Minimum grade of "C" in both FIN 360 and COB 300B.

FIN 325 Money and Banking**Cross-listed with:** ECON 325**Credits 3 Grading Basis** Graded

The purpose of this course is to examine the economic role of money, banking and monetary policy within current institutional settings and under alternative theories explaining the interrelationships between money, the financial system and economic activity.

Prerequisites: ECON 200 and ECON 201.

FIN 328 Time Series Analysis**Cross-listed with:** MATH 328**Credits 3 Grading Basis** Graded

Regression and exponential smoothing methods for forecasting nonseasonal and seasonal time series, stochastic processes, Box-Jenkins' autoregressive and moving average models.

Prerequisites: MATH 238 or MATH 300; and MATH 318 or MATH 329.

FIN 345 Finance Fundamentals for Non-Business Majors**Credits 3 Grading Basis Graded**

Offers an introduction to the basic principles in financial decision-making, providing an overview of the role of the forms of business, taxation and financing of a business enterprise. This course develops the basic financial skills and tools necessary for financial analysis, capital investment project analysis and valuation. Not recommended for students seeking admission to MBA programs. Students will not receive credit for this course if they also take FIN 301 or COB 300B.

Prerequisites: ACTG 244 or COB 242, junior standing (60 hours) and a cumulative 2.0 GPA in all courses taken at JMU. Restricted to non-college of business majors.

FIN 355 International Financial Management**Credits 3 Grading Basis Graded**

The purpose of this course is to provide a comprehensive examination of the investing and financing decisions of a multinational business entity. Particular emphasis is on global financial markets and instruments, exchange-rate risk management, short-term and long-term financing for multinational firms and asset-liability management in an international environment.

Prerequisites: COB 300B. International business majors only.

FIN 360 Analytical Methods in Finance**Credits 3 Grading Basis Graded**

The purpose of this course is to introduce the finance major to quantitative methods in finance, as applied to financial instruments and capital markets. Emphasis is placed in the theoretical determination of asset prices, risk and return, as well as the estimation and analysis of returns and asset prices.

Corequisites: Not permitted for quantitative finance majors. COB 300; a minimum grade of "C" in COB 291 (finance, international business and economics B.B.A. majors); FIN 301; a minimum grade of "C" in ECON 385 (economics B.A. and B.S. majors).

FIN 362 Financial Analysis**Credits 3 Grading Basis Graded**

This course is designed to prepare the finance major to use and interpret economic and accounting information that is essential in financial analysis and valuation.

Prerequisites: Minimum grade of "C" in COB 300B, FIN 360, COB 241 and COB 242.

FIN 365 Intermediate Financial Management**Credits 3 Grading Basis Graded**

The purpose of this course is to provide an in-depth study of the theories of capital structure, long-term financing decisions, working capital management and current topics such as mergers and bankruptcy. Computer applications.

Prerequisites: Minimum grade of "C" in COB 300B and "C" in FIN 360 (finance majors); minimum grade of "C" in FIN 250 (quantitative finance majors).

FIN 370 Real Estate Finance**Credits 3 Grading Basis Graded**

The purpose of this course is to offer a comprehensive examination of the decision-making processes involved in purchasing and financing real assets. The focus is on the cost of funds and optimal financial structure of complex real estate projects.

Prerequisites: Minimum grade of "C" in COB 300B and FIN 360.

FIN 371 Principles of Investments**Credits 3 Grading Basis Graded**

The purpose of this course is to provide an investor's view of the operation of the capital markets. This course covers the theories and practice of investments, including analysis of financial instruments and real assets and their effective combination into portfolios.

Prerequisites: Minimum grade of "C" in COB 300B and "C" in FIN 360 (finance majors); minimum grade of "C" in FIN 250 (quantitative finance majors); minimum grade of "C" in FIN 301 and "C" in FIN 360 (economics B.S. and B.A. majors).

FIN 372 International Finance and Payments**Cross-listed with: ECON 372****Credits 3 Grading Basis Graded**

Examines international financial markets, instruments and institutions; determination of spot and forward exchange rates, interest arbitrage, hedging and speculation; and alternative policies for achieving equilibrium in international payments.

Prerequisites: ECON 200 and ECON 201.

FIN 375 Madison Investment Fund Management**Credits 3 Grading Basis Graded**

The course is an accelerated introduction to the investment environment with a focus on fundamental analysis of stocks, overlay strategies to enhance portfolio performance, and familiarization with the trading and pricing of the wide variety of instruments in the modern financial marketplace. The course is limited to members of the Madison Investment Fund.

Prerequisites: FIN 371 and department head permission with membership in the Madison Investment Fund.

FIN 378 Fixed Income Analysis**Credits 3 Grading Basis Graded**

The purpose of this course is to examine the market for and the price/yield determinants of various fixed income securities including Treasury debt, corporate bonds, agency debt, municipal bonds and mortgage and asset-backed securities. Topics include securitization, the term structure of interest rates and portfolio management strategies such as duration, convexity and immunization.

Prerequisites: FIN 371.

FIN 380 Elemental and Derivative Securities Analysis**Credits 3 Grading Basis Graded**

The purpose of this course is to undertake an in-depth examination of the broad array of financial instruments traded in contemporary finance markets that are available for financing, investing and managing financial price risk.

Prerequisites: Minimum grade of "C" in COB 300B and "C" in FIN 360 (finance majors); minimum grade of "C" in FIN 250 (quantitative finance majors); minimum grade of "C" in FIN 301 and "C" in FIN 360 (economics B.S. and B.A. majors).

FIN 395 Mathematical Finance**Cross-listed with: MATH 395****Credits 3 Grading Basis Graded**

This course presents an overview of the role of mathematical concepts in financial applications. Topics include continuous time finance, optimization, numerical analysis and applications in asset pricing.

Prerequisites: MATH 237 and FIN 380.

FIN 405 Securities Pricing**Cross-listed with:** MATH 405**Credits 3 Grading Basis** Graded

A quantitative treatment of the theory and method of financial securities pricing to include an examination of closed form pricing models such as the Black-Scholes and its various derivatives as well as numerical solution techniques such as binomial methods.

Prerequisites: FIN 395 or MATH 395.

FIN 420 Financial Planning and Wealth Management**Credits 3 Grading Basis** Graded

This course introduces the student to financial planning and wealth management. The course is designed for students who may want to pursue a career in financial planning and related professional designations. In addition to the financial planning process and working with clients, this course covers the topics of insurance, cash management, debt management, investment management, taxes and estate planning.

Prerequisites: FIN 371.

FIN 434 Fundamentals of Financial Data Science**Credits 3 Grading Basis** Graded

An introduction to data analysis and programming in finance. Emphasis is placed on problem identification, the selection and application of tools and techniques to address problems, and data analysis.

Prerequisites: "C" or better in FIN 360 or FIN 250.

FIN 450 Financial Risk Management**Credits 3 Grading Basis** Graded

The purpose of this course is to explore forward contracts, futures, swaps and options, which are the basic building blocks for creating financial risk management programs for companies subject to financial risks such as changes in exchange rates, commodity price fluctuations and changes in interest rates.

Prerequisites: FIN 371 and a minimum grade of "B-" in FIN 380.

FIN 451 Risk Management II**Credits 3 Grading Basis** Graded

The purpose of this course is to present an intermediate treatment of the theory and applications of market, credit, liquidity and operational risk management. It builds upon concepts introduced in previous course work to present the theoretical constructs underlying risk management, as well as the quantitative skills required for risk analysis and the implementation of risk management techniques.

Prerequisites: FIN 450.

FIN 455 Global Finance**Credits 3 Grading Basis** Graded

The purpose of this course is to focus on the analysis of major international financial management issues and risk faced by businesses operating in global markets. The emphasis is on the management of foreign exchange transactions, operating and translation exposures by multinational corporations, the functions of various currency hedging instruments, and the application of international funding and investment techniques.

Prerequisites: Minimum grade of "C" in COB 300B and FIN 360.

FIN 460 Commercial Banking**Credits 3 Grading Basis** Graded

The purpose of this course is to study the objectives, functions, policies, organizational practices and government regulation of commercial banks. An intensive study is undertaken of the asset and liability structure of commercial banks. Special emphasis is placed on how banks are adapting to the changes in their operating and regulatory environments.

Prerequisites: Minimum grade of "C" in COB 300B and FIN 360.

FIN 471 Portfolio Management**Credits 3 Grading Basis** Graded

The purpose of this course is to cover the application of investment concepts within a case format. The course focuses on investment management, bringing together economics, capital markets and valuation to form a basis for decision making in financial asset selection, risk/reward analysis, portfolio selection and formation.

Prerequisites: FIN 371.

FIN 475 Financial Modeling and Risk Analysis**Credits 3 Grading Basis** Graded

The purpose of this course is to introduce students to practical methods used to identify, quantify, predict, value, diversify and manage risk in the financial environment. Students use sensitivity analysis, Monte Carlo and Latin Hypercube simulations, bootstrapping, time series forecasting and dynamic optimization techniques as applied to capital budgeting and structure, pro forma financial statements, multi-objective portfolio allocation, discounted cash flow analysis and real options.

Prerequisites: FIN 365 and FIN 371.

FIN 480 Seminar in Quantitative Finance Research and Data Analysis**Credits 3 Grading Basis** Graded

The purpose of this course is to develop the research skills of quantitative finance students. The primary focus of the course is on empirically-oriented student-initiated research. Students are guided through the process of financial research with a particular emphasis on the analysis of real-world data. Presentation of original research is expected.

Prerequisites: FIN 405/ MATH 405.

FIN 488 Advanced Financial Policy**Credits 3 Grading Basis** Graded

The financing of a new business enterprise or expansion and the financial condition of existing firms is examined within a case format. The objective of this examination is to elicit a policy decision which effectively addresses the issues identified in the case.

Prerequisites: FIN 360 and FIN 365 and senior Finance majors.

FIN 490 Special Studies in Finance**Credits 1-3 Grading Basis** Graded

Designed to give capable students in finance an opportunity to complete independent study under faculty supervision. Admission by recommendation of the instructor and permission of the director.

FIN 498 Special Topics in Finance**Credits 1-3 Grading Basis** Graded

The purpose of this course is to provide an opportunity for students to explore areas of current topical interest or to exploit special situations. Course content will vary. For current course content, consult with your advisor.

Prerequisites: FIN 250 or FIN 360.

FIN 499A Honors**Credits 1 Grading Basis** Graded

FIN 499B Honors
Credits 3 Grading Basis Graded

FIN 499C Honors
Credits 2 Grading Basis Graded

Fine Arts (ARTS)

ARTS 100 Introduction to the Visual and Performing Arts
Credits 1 Grading Basis Credit / No Credit

An exploration of the concept and practice of creativity, the commonalities that exist in all arts endeavors, the artist as citizen and the concurrent responsibilities, and the almost infinite possibilities opened in the market place by the study of the arts. The course includes guest lectures by resident and visiting artist/scholars and field experiences, such as attending exhibits and performances with pre- and post-event discussions.

Prerequisites: Must be admitted to the arts learning community and in residence in Wayland Hall during the semester in which the course is taken.

ARTS 300 Arts Administration Internship
Credits 1-3 Grading Basis Graded

The internship experience provides students with an opportunity to explore career interests while applying knowledge and skills learned in the classroom in a professional work setting. The experience also helps students gain a clearer sense of what they still need to learn and provides an opportunity to build professional networks.

ARTS 400 Advanced Arts Administration Internship
Credits 1-3 Grading Basis Graded

The internship experience provides students with an in-depth opportunity to apply knowledge and skills learned in the classroom in a professional work setting. The experience provides an opportunity to build professional networks and develop a resume for a career in arts administration.

Foreign Language Education (FLED)

FLED 370 Modern Foreign Language Assessment
Credits 3 Grading Basis Graded

This course is designed to prepare prospective foreign language teachers to create and evaluate both formal and informal learner assessments and to prepare candidates to meet future teacher evaluation criteria.

FLED 471 Modern Foreign Language Field Experience
Credits 3 Grading Basis Graded

Provides practical classroom experience in elementary, middle and high school settings to preK-12 Foreign Language students under the supervision of an inservice teacher and a clinical professor. Students engage in classroom activities that provide an opportunity for them to practice the strategies and concepts learned in the methods courses. Corequisites: LING 470.

FLED 475 Supervised Student Teaching Experience
Credits 6 Grading Basis Credit / No Credit

Participants will experience the full range of conditions and tasks expected of a teacher for students in grades preK-12. They will be expected to develop and demonstrate competencies in teaching with the supervision and support of experienced teachers. Students must register for two eight-week blocks during the same semester for a total of twelve credits.

Prerequisites: All required courses for licensure and approval for student teaching through the teacher education program. Corequisites: FLED 476.

FLED 476 Student Teaching Seminar
Credits 3 Grading Basis Graded

A seminar designed to promote reflective decision making among teacher candidates during their student teaching experiences. Teacher candidates will engage in case analysis and complete the Teacher Work Sample as their culminating senior project. Corequisites: FLED 475.

French (FR)

FR 101 Elementary French I
Credits 3 Grading Basis Graded

This course introduces the fundamentals of French through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension. Instruction is in French.

FR 102 Elementary French II
Credits 3 Grading Basis Graded

This course builds on the foundations established in FR 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension. Instruction is in French.

Prerequisites: FR 101 or by placement exam.

FR 103 Passport to French: A Survival Kit
Credits 3 Grading Basis Graded

This course teaches students basic communication skills and cultural competence needed for studying, traveling or working in a French-speaking country. Students will practice typical communicative contexts using authentic materials and gain insight into the real-life situations of French speakers. This course has no prerequisite and is designed for those who have had little or no previous experience with French. It does not count towards the French major or minor or fulfill the B.A. language requirement. Instruction is in French and English.

FR 109 Accelerated Review of Elementary French
Credits 3 Grading Basis Graded

This course reviews elementary reading, writing, speaking and listening skills in French. This course is for students who qualify through the placement exam. Instruction is in French.

Prerequisites: Permission of the instructor or by placement exam score.

FR 111 Intensive French I
Credits 6 Grading Basis Graded

This course covers the fundamentals of French through intensive listening, speaking, reading and writing. The course is the equivalent of FR 101 and FR 102. Instruction is in French.

FR 212 Intensive French II
Credits 6 Grading Basis Graded

This course covers the fundamentals of French through listening, speaking, reading and writing. This course is the equivalent of FR 231 and FR 232. Instruction is in French.

Prerequisites: FR 102, FR 109 or FR 111; or by placement exam or permission of the instructor.

FR 231 Intermediate French I
Credits 3 Grading Basis Graded

This course provides a thorough review of grammar, vocabulary building, conversation, composition and reading. Instruction is in French.

Prerequisites: FR 102, FR 109 or FR 111; or by placement exam or permission of the instructor.

FR 232 Intermediate French II**Credits 3 Grading Basis Graded**

This course provides a thorough review of grammar, vocabulary building, conversation, composition and reading at the advanced intermediate level. Instruction is in French.

Prerequisites: FR 231; or by placement exam or permission of the instructor.

FR 300 Language for Living**Credits 3 Grading Basis Graded**

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in French language. It is intended for students who have begun to develop advanced skills and are interested in continued training in language proficiency through reinforcing foundations of everyday language communication and cultural knowledge. Instruction is in French.

Prerequisites: FR 212 or FR 232 or per placement exam score.

FR 301 Pop Culture through Language**Credits 3 Grading Basis Graded**

This class examines popular culture in France and the Francophone world in a variety of forms and media: history, politics, literature, sport, music, cinema, fashion, comics and media. By focusing on pop culture and different media, this course teaches the critical analysis of textual and visual imagery in their historical and cultural context. Students will be able to learn the main pop culture references that are recognized nationally and internationally. Instruction is in French.

Prerequisites: FR 212 or FR 232 or per placement exam score.

FR 307 History of French Culture and Society**Credits 3 Grading Basis Graded**

This course studies the social, economic, political and artistic development of France from the Middle Ages to 1900. Instruction is in French.

Prerequisites: FR 212 or FR 232.

FR 308 Contemporary History and Culture in the Francophone World**Credits 3 Grading Basis Graded**

This course is a study of life and culture in the Francophone world, with emphasis on the 20th century. Instruction is in French.

Prerequisites: FR 212 or FR 232.

FR 315 Sounds of Speech**Credits 3 Grading Basis Graded**

This course is designed to help students improve their competency in French oral expression. Students will be listening and reacting to audio and visual documents, transcribing, analyzing, and acting out daily situations using authentic materials in French. The class involves an empirical aspect based on exercises relying on phonetics, phonology, prosody, pronunciation and intonation. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 320 Culture and Communication in Context**Credits 3 Grading Basis Graded**

This French course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do French and Francophone people communicate on subjects that are at the heart of their society?" Instruction is in French.

Prerequisites: FR 212 or FR 232 or per placement exam score.

FR 322 Language Landscapes**Credits 3 Grading Basis Graded**

This French course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in these culturally and linguistically defined areas and explores the diverse cultural landscapes that shape their lives. Instruction is in French.

Prerequisites: FR 212 or FR 232 or per placement exam score.

FR 330 Business French**Credits 3 Grading Basis Graded**

This course studies commercial and technical vocabulary and trade customs in conjunction with practice in the art of commercial communication including interviews, letter writing and simultaneous interpretation. Instruction is in French.

Prerequisites: FR 212 or FR 232.

FR 335 A Journey through Literature**Credits 3 Grading Basis Graded**

The purpose of this course is to acquaint students with French and Francophone literature through the study of excerpts of canonical texts ranging from the 10th to the 21st century. Different genres will be studied: philosophy, essay, poetry, theater, novel, short story and song, always presented within an historical, political, social and artistic framework. In this class, students will also learn strategies, using models and tools (such as the methodology for textual analysis and the study of rhetorical terms) to read and analyze any text better. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 365 French for Global Health**Credits 3 Grading Basis Graded**

This course examines issues in global health throughout the French-speaking world, with particular emphasis on relevant terminology and vocabulary, moments of both challenge and opportunity throughout the current Francophone world, and real-life preparation for international professional opportunities in the global health and humanitarian job fields. Instruction is in French.

Prerequisites: FR 212 or FR 232. Corequisites: FR 212 or FR 232.

FR 375 Business and Society in the Francophone World**Credits 3 Grading Basis Graded**

This course explores the development of the Francophone societies in the historical, political and economic context. In this course several aspects will be investigated: agricultural, textile, fashion and wine industries; motion picture, music and media industries; and import and export products. In addition, this course will include the study of banking and financial institutions, and health and education systems. Instruction is in French.

Prerequisites: FR 212 or FR 232. Corequisites: FR 212 or FR 232.

FR 380 French Cuisine**Credits 3 Grading Basis Graded**

This course aims at making students confident about one of the most appreciated aspects of Francophone culture: Francophone cuisine. Students will engage in the art of food in theory and practice. In addition to learning how to properly prepare some of the best French recipes and analyze them in French, students will get a better understanding of the history of the Francophone gastronomic traditions. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 386 French Short Stories**Credits 3 Grading Basis Graded**

The purpose of this course is to acquaint students with different types of French short stories and the various narrative techniques writers use to address issues of plot, character, place and theme. Students will acquire familiarity with methodologies and tools that will sharpen their understanding of the genre and their ability to analyze it. Students will engage with the course material through academic and/or creative writing. Instruction and readings are in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 390 French Language and Culture Abroad**Credits 1-3 Grading Basis Graded**

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in French literature and culture. It is intended for students who have begun to master advanced skills in interpretive, interpersonal and presentational communication and are interested in examining nuances of the French language and culture while studying abroad.

Prerequisites: FR 232 or by permission of the instructor.

FR 420 Stories and Histories in French Culture**Credits 3 Grading Basis Graded**

This class uses literature, stage performances, comics, series, movies and advertisements to study the specificity of French humor. In this class, we'll study how French humor is used as an outlet to broadcast sharp political commentary on the constantly evolving social situation. While exploring the specificity of French humor from the 15th century to today, students will learn a palette of strategies for the critical assessment of creative works in various formats and identifying and analyzing iconic comedic acts while learning about the history, politics and social context of the time. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 421 Stories and Histories in Francophone Culture**Credits 3 Grading Basis Graded**

This class focuses on the literature of Francophone cultures through the ages. Performances, comics, series, movies and advertisements serve as subjects of analysis. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322; or by placement exam or permission of the instructor. Corequisites: FR 300, FR 301, FR 320 or FR 322.

FR 422 Voices of Today**Credits 3 Grading Basis Graded**

This course provides an exploration of key literary works in France during the 20th century. Emphasis is placed on the intersection of French history, society and culture, focusing on the postwar period (1945-) and the role of historical memory in shaping both the artistic expressions and the collective consciousness of France. This course explores topics such as cultural memory, identity, globalization, multiculturalism, climate change, digitalization and social justice. The topics, literary works and genres will vary according to the instructor's expertise. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 425 Transformative Texts**Credits 3 Grading Basis Graded**

The purpose of this course is to acquaint students with historical periods in French literature and culture through the study of key influential texts. Analyses of these texts (i.e. essays, novels, plays, films, paintings and comics) will be grounded in key political, scientific and artistic movements. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322; or by placement exam or permission of the instructor. Corequisites: FR 300, FR 301, FR 320 or FR 322.

FR 435 Translation Strategies**Credits 3 Grading Basis Graded**

This course includes learning about both the theory and the practice of translation. Students will learn some basic electronic tools and word processing skills for translators and practice several types of translation, including direct translation, inverse translation and back translation.

A variety of examples and texts will be used such as professional and technical texts, idiomatic expressions, comics, songs, poetry and theater. Instruction is in French and in English.

Prerequisites: FR 212 or FR 232.

FR 446 Special Topics in Francophone Literature**Credits 3 Grading Basis Graded**

This course is a study of a particular topic in Francophone literature. It may cover one or more genre. Course may be repeated if content varies. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322; or by placement exam or permission of the instructor. Corequisites: FR 300, FR 301, FR 320 or FR 322.

FR 447 Special Topics in Francophone Culture**Credits 3 Grading Basis Graded**

Students will study a particular topic in the culture of Francophone countries. This course may be repeated if content varies. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322; or by placement exam or permission of the instructor. Corequisites: FR 300, FR 301, FR 320 or FR 322.

FR 448 Special Topics in Francophone Linguistics**Credits 3 Grading Basis Graded**

Students will study a particular topic of French linguistics within the Francophone world. Topics could include an introduction to French sociolinguistics and psycholinguistics. This course may be repeated if content varies. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322; or by placement exam or permission of the instructor. Corequisites: FR 300, FR 301, FR 320 or FR 322.

FR 464 French and Francophone Theater**Credits 3 Grading Basis Graded**

In this class, we work on refining French language skills to facilitate further study of French language and culture. To this end, the course will focus on the reading and studying of full unabridged plays from various centuries as well as staging and performing selected scenes. In class, students will engage in improvisations and games with classmates to study all aspects of theater. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 465 French and Francophone Cinema**Credits 3 Grading Basis** Graded

This course is designed to help students improve their French conversational and writing skills by discussing a variety of topics as they are portrayed in cinema. It integrates language acquisition with cultural immersion through film. This course is both an elementary introduction to film studies (learning to critically read a film) and an opportunity for extensive practice in speaking and writing in French. Emphasis is placed on building practical vocabulary and enhancing comprehension, as well as oral and written expression. Through the study of selected films, students will strengthen their listening, speaking, reading and writing skills, while gaining a deeper understanding of French and Francophone cultures. Each film will provide a foundation for exploring thematic, cultural and linguistic content, offering authentic contexts for language practice and discussion. Instruction is in French.

Prerequisites: FR 300, FR 301, FR 320 or FR 322.

FR 490 Independent Study in French**Credits 3 Grading Basis** Graded

This course allows students an opportunity to complete independent studies under faculty supervision. Instruction is in French.

Prerequisites: One FR course at the 300-level or above and permission of the instructor.

FR 497 French Internship**Credits 3 Grading Basis** Graded

This course will provide students with experiential learning opportunities within the Francophone field. The internship course is designed to provide practical experience in fields such as government, business, industry, health sciences, administration, education, translation or the arts, where they can apply their language skills. Examples of internship work may include working for a company in the student's field of interest abroad; serving as a research assistant for a professor; completing a translation project or working as an interpreter. A journal, internship report, research paper, bibliography or evaluation from the internship provider is required. Credits earned correspond to the internship hours completed. Up to six credits may be applied as electives in the French major or minor.

Prerequisites: One FR course at the 300-level or above and permission of the instructor.

FR 499A Honors**Credits 1 Grading Basis** Graded**FR 499B Honors Project****Credits 3 Grading Basis** Graded**FR 499C Senior Honors Project****Credits 2 Grading Basis** Graded

Geography (GEOG)

GEOG 161 Geospatial Tools and Techniques**Credits 1-6 Grading Basis** Graded

An introduction to the use of geospatial tools, such as geographic information systems (GIS), global positioning systems (GPS) and remote sensing, applied to a variety of areas, including cultural geography, environmental science, ecology, geology and public planning.

GEOG 200 Geography: The Global Dimension [C4GE]**Credits 3 Grading Basis** Graded

This course promotes global understanding through the study of humans, their institutions and processes, and the resulting interactions between humans and the environment. The course will include the study of western and non-western peoples and their social, cultural, political and economic relationships. May be used for general education credit.

GEOG 202 Introduction to Your Geographical Future**Credits 1 Grading Basis** Graded

This course gives an introduction to the Geography program, its curriculum and faculty research. It provides students with an understanding of the discipline of Geography and their potential skills, knowledge, and expertise as geographers. Students learn about research and major course opportunities, as well as preparing for their career and further academic study. Students also get the opportunity to plan their curricular options through the Geography major.

GEOG 210 Physical Geography (2, 2)**Credits 4 Grading Basis** Graded

This introductory course is an examination of systems and processes that influence patterns of Earth's atmosphere, biotic communities, soils and landforms at multiple spatial and temporal scales. Included are classroom and laboratory experiences that are geared toward investigating interrelationships among atmospheric conditions, Earth's natural surface characteristics and human-induced modifications of Earth's features.

GEOG 215 Introduction to Geographic Information Science and Cartography**Credits 3 Grading Basis** Graded

This course introduces the fundamental principles of geographic information systems, including various data models and their spatial relationships. Students learn to make maps and to process and analyze spatial data. The course also explores the use of many types of geospatial data. Students learn how these data can be used to aid visualization and decision-making.

GEOG 216 Earth Observation and Satellite Navigation Systems**Credits 3 Grading Basis** Graded

An introduction to satellite and aerial imagery, satellite navigation, and computer fundamentals. Students will learn the science behind these technologies, and lab exercises will give students hands-on experience with satellite navigation systems and interpreting and processing many types of satellite and aerial imagery. Environmental and human applications will be featured.

GEOG 230 Spatial Thinking and Problem Solving**Credits 3 Grading Basis** Graded

Introduction to the critical thinking skills associated with problems with inherent spatial components. Identification of the spatial elements of a given problem, the data requirements for addressing that problem, collections/acquisitions and organization of data, and use of geographic information systems to explore spatial patterns relevant to the problem of interest.

Prerequisites: GEOG 215 with a "C" or better, GEOG 216 with a "C" or better and an introductory course in Statistics (ISAT 251, MATH 220 or equivalent) or permission of the instructor.

GEOG 280 Human Geography: The Cultural Landscape**Credits 3 Grading Basis Graded**

This course introduces the geographical sub-discipline of human geography. Framed by the spatial dimensions of social science and drawing from critical, qualitative, and quantitative approaches, the course analyzes a range of global and local processes within human geography. Topics include, but are not limited to, culture, migration, resource use, geopolitics, religious diversity, colonization, demographics, urbanization, sustainability, agricultural systems and globalization. Students will utilize key theoretical and conceptual tools to explore these subject areas through the main sub-fields of economic, cultural and political geography, as well as in an introduction to human-environment interactions.

GEOG 301 Introduction to Natural Disasters Response**Cross-listed with: HUMN 301****Credits 3 Grading Basis Graded**

This course is designed to give students an overview of the various types of natural disasters; a look at the world regions that are most vulnerable to each type of disaster; and, a preview of disaster planning, management, relief and response as related to natural disasters.

GEOG 305 History and Philosophy of Geography**Credits 3 Grading Basis Graded**

Topics from the classical period to the modern period include 20th century theories and paradigm shifts involving cultural geography, physical geography, human-environment traditions, regional geographies and modeling. Diverse philosophies such as quantitative /positivist, qualitative/humanistic, social theory, and GIS are viewed for their contributions to the discipline of geography.

Prerequisites: A grade of "C" or better in GEOG 210 and GEOG 280, and junior standing or permission of the instructor.

GEOG 308 Environment, Nature and Society**Credits 3 Grading Basis Graded**

This course focuses on the interaction and integration of human and physical environments. It uses geographical approaches to examine relationships between people and the natural world, and it explores the origins and implications of the perceived separation of humans and nature. At a variety of scales, and using case studies, issues of resource use, environmental and human vulnerability, global and local environmental change, landscape modification, and human impacts on natural systems are explored. Along with the close analysis of spatial patterns and processes in the natural world, questions of politics, economics, culture, perception, justice and ethics may be considered in the examination of human-environment interactions.

Prerequisites: GEOG 210 with a "C" or better and GEOG 280 with a "C" or better.

GEOG 310 Environmental Issues**Credits 1-4 Grading Basis Graded**

Courses cover environmental issues such as air pollution, forest and wildlife management, water, resource management, soils and land use, and energy and the environment (among other topics). Courses examine the interface between humans and environmental systems while addressing the impact of social, economic and political systems and activities on the environment. May be repeated as course content changes.

GEOG 315 Field Studies in Geography**Credits 3 Grading Basis Graded**

This course exposes students to the methods and techniques commonly used by geographers while conducting fieldwork. The course will cover identifying and defining a researchable project, designing and testing data collection methods, and different methods of collecting, recording and presenting data. Students will also become familiar with various types of field equipment.

GEOG 320 Land Use Geography**Credits 3 Grading Basis Graded**

This course investigates the dynamics of land use and land cover change, including the policies that are created to influence and respond to change. Students will learn about numerous land use types, such as commercial and residential land, forest and grassland, farmland, and land set aside for resource extraction. The course addresses human drivers of land use change and the effects of change on environmental factors such as climate, air and water quality, and biodiversity. Students will also examine spatial patterns by learning methods for detection, monitoring, and analysis.

Prerequisites: GEOG 308 with a minimum grade of "C."

GEOG 322 Agricultural Systems**Credits 3 Grading Basis Graded**

This course covers four distinct areas: the foundation of agriculture, the nature and distribution of soils on a global basis; the history of agriculture from the original selection of domestic crops to the 20th century; modern industrial agriculture and trade; and alternatives to chemical and energy intensive agriculture in the 21st century.

Prerequisites: GEOG 308 with a minimum grade of "C."

GEOG 323 The Geography of Human Genetics, Infectious Diseases and Diet**Credits 3 Grading Basis Graded**

Throughout history, infectious diseases have profoundly affected human populations. Using a case study approach, this course will give students an opportunity to investigate social and historical aspects of infectious disease as well as the microbiology, genetics, biochemistry and medical aspects of human infectious diseases and diet.

GEOG 325 Environmental Ethics**Credits 3 Grading Basis Graded**

Examines the basic principles of resource use including geographic, economic, social and political processes. Explores concepts underlying such issues as resource consumption and conservation, environmental perception, resource and environmental conflict, population growth and control, carrying capacity, and the evolution of the environmental movement.

GEOG 327 Climatology**Credits 3 Grading Basis Graded**

The systematic study of the atmosphere with emphasis on such phenomena as temperature, pressure, humidity, air masses and fronts; the occurrence of these phenomena on a global basis; and a detailed survey of the worldwide distribution of climate types.

Prerequisites: GEOG 210 with a "C" or better or GEOL 320.

GEOG 329 Global Climate Change**Credits 3 Grading Basis Graded**

This course examines the interrelationship of the physical nature of the climate system, climate variability and change, and human activities. Case studies (El Nino, global warming, and stratospheric ozone depletion) are used to investigate how climate affects society, how human activities affect climate, and how non-climatic issues complicate our understanding of the relationship between climate and society. We consider how projections of future weather and climate affect decision-making.

Prerequisites: GEOG 210 with a "C" or better.

GEOG 330 Weather, Climate and Society**Credits 3 Grading Basis Graded**

This course adopts a case study approach to explore a series of environmental hazards (e.g., ozone depletion, El Niño, abrupt climate change) as examples of risk management. For each case, students will learn the scientific principles underlying the hazard and examine how social and economic policies can be developed and implemented to mitigate the perceived risk.

GEOG 331 Geography of Virginia**Credits 3 Grading Basis Graded**

The course will examine the human and physical geography of the development of modern-day Virginia, providing an overview of its prehistory, then tracing its development from the beginning of the seventeenth century through the present. The course will include an analysis of Virginia's population, resources, and regional landscapes as they have been influenced by physical, cultural, historical, and economic factors. The relationship of Virginia to the rest of the world will also be examined.

GEOG 332 Geography of Europe**Credits 3 Grading Basis Graded**

Geographic assessment of regional and national characteristics of the European nations.

GEOG 334 Geography of East and Southeast Asia**Credits 3 Grading Basis Graded**

A survey of the physical and cultural environments of China, Taiwan, Japan, the Koreas, Indochina and the countries of Southeast Asia. Topics covered include weather and climate, physiography, natural resources, population characteristics, political systems, aspects of the economy and the role that each country plays on the regional and world stage.

GEOG 335 Geography of Africa**Credits 3 Grading Basis Graded**

An introduction to the regional geography of Sub-Saharan Africa that examines the physical geography of the continent, the historical roots of its present political geography, the consequences of its colonial past on communities and cultures as well as its natural resources. Students will examine continental issues such as resource management, food production, hunger, disease patterns and management of wildlife.

GEOG 336 Environmental Hazards: A Focus on Southeast Asia**Credits 3 Grading Basis Graded**

This course will focus on interactions between earth surface systems and social environments to demonstrate the complexity of natural hazards, and particularly those of Southeast Asia. Through regional geography, students will learn differing adaptation mechanisms of societies and cultures unique to the area and some of the natural hazards within the region.

GEOG 337 Geography of Latin America**Credits 3 Grading Basis Graded**

A study of countries in Latin America which includes their physical landforms, weather and climate, biogeography, natural resource base, attitudes toward the physical environment, characteristics of the economy, the current political role in international activities, and population characteristics that include growth rate, distribution, migration and ethnicity.

GEOG 338 Geography of the Philippine Islands: Problems and Possibilities**Credits 3 Grading Basis Graded**

Exploration of the Philippines focuses on poverty, environmental conservation, resource exploitation and ecosystem degradation in upland and marine environments. Topics include population dynamics, political pressure and instability, and urban challenges. The future of the country is investigated on all geographic scales with regard to its role in a globalized world economy.

GEOG 339 Geography of the Caribbean**Credits 3 Grading Basis Graded**

This course is designed to give students a general geographical overview of the islands states and territories surrounded by the Caribbean Sea. Students will study physical landforms, weather and climate, environmental issues, population characteristics, history, local and regional politics, and economic aspects of political units in the region.

GEOG 340 Biogeography**Credits 3 Grading Basis Graded**

This course emphasizes geographical biogeography and is as an advanced physical geography class. Included are analyses of spatial patterns of biota from local to global scales and examinations of the systems and processes that result in spatial and temporal patterns of species existence and diversity, community composition, energy pathways, adaptive traits, and human influences on biotic systems and processes.

Prerequisites: GEOG 210 with a "C" or better.

GEOG 341 Wilderness Studies**Credits 3 Grading Basis Graded**

Wilderness issues pertaining to legislation, legal mandates and management are examined. Human impacts due to overuse or conflicting uses are studied, as are the philosophical aspects of wilderness ethics. Practical wilderness knowledge, needed to safely and responsibly enjoy wilderness, is also studied. This course may include an overnight field activity.

GEOG 342 Management and Protection of Natural Resources**Credits 3 Grading Basis Graded**

This course provides a managerial perspective for protection and management of natural resources. A systems approach for applied management strategies is provided for aquatic, terrestrial, threatened and endangered ecosystems. Topics include application of state, federal, international laws, regulations, policies and guidelines. Students develop management plans and explore jurisdictional resource protection issues. Prerequisites: Junior or senior standing only.

GEOG 343 Wildlife Management**Credits 3 Grading Basis Graded**

An introductory discussion of applied management strategies for wildlife species and their ecological requirements is provided relative to human influences. Management techniques that are useful for determining population or health status are demonstrated for select vertebrate species. The evolution of wildlife laws, policies, and management strategies are addressed to provide relevant awareness into the appropriate concepts of wildlife management.

Prerequisites: Junior or senior standing only.

GEOG 344 Globalization and Development**Credits 3 Grading Basis Graded**

An overview of the classification of economic activities, the factors involved in the location of various types of economic activities and the regional variation in the standard of living associated with economic development. Additional topics include regional economic growth and types of economic systems and development perspectives, the roles that politics and demographics play in the economic development of a country, and the globalization of economic activities.

GEOG 345 Geography of Poverty**Credits 3 Grading Basis Graded**

This course provides a geographical perspective on poverty faced by communities and countries of the world today. The focus is on how poverty is defined, measured and mapped, the causes and impacts of poverty, theories for ending poverty and organizations that work to address poverty. It includes a geographical study of communities and countries that have successfully alleviated extreme poverty.

GEOG 348 Indigenous Geographies**Credits 3 Grading Basis Graded**

This course introduces indigenous geographic representations. Topics include territorial sovereignty, traditional resource use, sustainable development, and protection of sacred sites. Students will explore the central geographic practice of cartography, which has taken on new meaning as cultural identification for Indigenous peoples. The use of new technologies to represent traditional understandings of Earth is also explored.

Prerequisites: GEOG 280 or permission of the instructor.

GEOG 350 Topics in Geography**Credits 1-3 Grading Basis Graded**

Examination of geographic topics that are of current interest. Can be repeated as course content changes.

Prerequisites: GEOG 200 or GEOG 210 or GEOG 215 or GEOG 280, or permission of the instructor.

GEOG 355 Global Navigation Satellite Systems**Credits 3 Grading Basis Graded**

This course provides fundamental and advanced instruction in global navigation satellite systems (GNSS) technology. Topics include the history and development of GNSS, system operation, the integration of satellite and ground components, applications, and visualization. Active learning allows students to operate different models of receivers, minimize field error, gather and differentially correct data, and import the finished project into a GIS for further analysis. Students will form teams to design, execute and present a formal project to a client.

GEOG 360 GIS for Humanitarian Assistance**Cross-listed with: HUMN 360****Credits 3 Grading Basis Graded**

In responding to humanitarian crises, governments and aid organizations must deploy aid workers, deliver essential services, set up temporary settlements, and distribute items such as water and food that are needed for survival. Spatial analysis and maps are critical to the success of these efforts. In this course, students learn the basics of Geographic Information Systems (GIS) for humanitarian assistance and learn how relief organizations use GIS in their work.

Prerequisites: GEOG 215.

GEOG 365 Cartography and Geospatial Visualization**Credits 3 Grading Basis Graded**

This course examines the fundamentals of visualizing spatial data in static and dynamic environments. Students will learn about cartographic design, thematic cartographic techniques and geographic visualization. Students will develop a broad portfolio of reference and thematic maps.

Prerequisites: A minimum grade of "C" in either GEOG 161 or GEOG 215.

GEOG 366 Intermediate Geographic Information Science**Credits 3 Grading Basis Graded**

An in-depth course in geographic information science and its role in technology and society. Spatial databases and descriptive data will be created and implemented into various geographic information systems. Advanced operations will be used to explore the analysis capabilities of geographic information systems.

Prerequisites: GEOG 215 with a "C" or better or GEOG 161 and permission of the instructor.

GEOG 367 Programming for Geographic Information Science**Credits 3 Grading Basis Graded**

Geographic information science relies on programming for effective geospatial analysis for large datasets and the automation of resource-intensive analytical methods. This course advances the knowledge of programming for geographic information systems. In addition to providing a solid foundation in Python programming language, the course introduces several libraries and applications that are commonly used for working with geospatial data in GIS software.

Prerequisites: A minimum grade of "C" in either GEOG 161 or GEOG 215; a minimum grade of "C" in GEOG 216; or permission of the instructor.

GEOG 374 Population Geography**Credits 3 Grading Basis Graded**

This advanced human geography course spatially investigates the human population, with emphases on public policy, inequality and current events throughout. Major topics include population measurement, distribution and structure, social stratification, fertility, health, aging and mortality, environment and resource dynamics, migration across spatial and temporal scales of analysis, and community geography.

GEOG 375 Political Geography**Cross-listed with: IA 375****Credits 3 Grading Basis Graded**

Geopolitical conflicts and issues are examined. Concepts such as territoriality, nationalism, religious and ethnic struggle, environmental degradation, and freedom and justice are discussed in the context of political unrest. Significant geopolitical theories, and social and economic processes are explored.

GEOG 376 Urban Geography**Credits 3 Grading Basis Graded**

This course offers a study of the city in its geographic setting, giving perspective of modern urban problems, the origin and growth of cities, and the influence of location on city functions. This course also looks at the internal structure of cities and the influence of the internal structure on its population groups.

GEOG 377 Feminist Geography**Credits 3 Grading Basis Graded**

This course introduces the field and practice of feminist geography with emphasis on understanding gender and other social categories such as race, class and sexuality through a geographic perspective. Concepts and topics examined include the relationship of such identities to spatial politics, culture, knowledge and social justice.

Prerequisites: GEOG 200 or GEOG 210 or GEOG 215 or GEOG 280, or permission of the instructor.

GEOG 380 Cultural Geography**Credits 3 Grading Basis Graded**

Introduction to cultural geography with emphasis on diversity of language, religion and folklore, as well as culture traits and practices and their historical diffusion. Ties to livelihood, the rural-urban continuum and demographic change are explored, as are foci on philosophy, power, race, class and gender. Exploitation and sustainability will be introduced as dimensions of cultural and environmental interaction.

GEOG 385 Digital Image Processing in Remote Sensing**Credits 3 Grading Basis Graded**

This is an intermediate course on imagery and other type of data acquired by satellites, aircraft and remote piloted aircraft systems (drones). Image processing techniques will be emphasized, and topics covered will include data collection and selection, geometric and radiometric correction, image transformations, image classification and feature extraction, change detection, photogrammetry, as well as specialized techniques for hyperspectral and thermal imagery, and radar and LiDAR data. Applications across multiple areas of geography will be discussed.

Prerequisites: GEOG 216 with a "C" or better.

GEOG 390 Practicing Geographic Science**Credits 1 Grading Basis Graded**

Professional development for students in geography. Educational opportunities and career options. Library and literature research skills.

Prerequisites: Junior standing.

GEOG 406 Forest Inventory: A Geospatial Approach**Credits 3 Grading Basis Graded**

This course teaches forest measurement at the tree, plot, stand, and forest levels. It includes heights, stem diameters, volumes and biomasses of individual trees; fixed and variable radius plots; basal area estimates; sampling designs, and stand and stock table construction. Students learn how geospatial technologies can be used to collect data on forests, make forest inventories more accurate and precise, and assist in performing forest-related analyses and visualizations.

Prerequisites: A minimum grade of "C" in either GEOG 161 or GEOG 215; a minimum grade of "C" in GEOG 216; MATH 220, ISAT 251 or COB 191; or permission of the instructor.

GEOG 410 Geography and Film**Credits 3 Grading Basis Graded**

This course is concerned with the intersection of geography and film. An always-present undercurrent focuses on films whose location and/or culture are an essential backdrop in the cinematic experience.

Prerequisites: GEOG 308 or permission of the instructor.

GEOG 415 Environment, Landscape and Culture**Credits 3 Grading Basis Graded**

This seminar analyzes human-environment interactions as shaped by cultural perceptions, past events and ecological processes. Place-based and interdisciplinary, landscape studies focus on the unique circumstances of a location and shift temporal and geographic scales to achieve broader understandings of cultural and ecological adaptation and resilience. Methods of historical ecology and field research are introduced. Regional projects underscore readings and research assignments.

Prerequisites: GEOG 308 or permission of the instructor.

GEOG 427 Water Resources of the World**Credits 3 Grading Basis Graded**

Humans face water challenges that can be understood through the lens of geography. Students will study different physical aspects of Earth's water, including the hydrologic cycle, climate controls of precipitation, river basin hydrology, water quality and availability, and aquatic ecosystems. The course investigates human dimensions of water such as economics, disease, management and planning, conflict, and the future of water resources.

Prerequisites: GEOG 210 or permission of the instructor.

GEOG 429 Sustainability: An Ecological Perspective**Cross-listed with: ISAT 429****Credits 3 Grading Basis Graded**

This course examines present global environmental impacts and efforts made to change production and consumption patterns toward those that reduce impact on ecosystems or promote increased ecosystems health. The focus lies in understanding the basic resources of productivity including soils, agricultural systems, agroforestry, forestry and aquatic environments and applying solutions on a personal and community level.

Prerequisites: GEOG 308 or permission of the instructor.

GEOG 430 Dynamics of the Atmosphere**Credits 3 Grading Basis Graded**

This course provides an integrated view of atmospheric science with a focus on the structure, evolution and dynamics of mid-latitude weather systems. Topics include the theoretical development of conceptual models and analysis techniques for synoptic waves, cyclones, fronts, jets and regional precipitation events.

Prerequisites: Either GEOG 327, GEOG 329, or GEOG 330; or permission of the instructor.

GEOG 440 Advanced Biogeography**Credits 3 Grading Basis Graded**

This course emphasizes physical and human-oriented processes that influence Earth's biodiversity. It addresses how human constructs influence what is known about biodiversity and the conservation of biodiversity. It introduces human challenges that are causing a decline in biodiversity and resultant impacts on human quality of life. The course builds on knowledge from prerequisites and will be taught through a current biogeography theme such as biodiversity in a changing world, applied island biogeography, biodiversity conservation at multiple scales or other pertinent themes.

Prerequisites: GEOG 308 and GEOG 340.

GEOG 447 Climates Through Time**Credits 3 Grading Basis** Graded

This course is a geographic exploration of basic controls of climate, the world's climate regions, global circulation models and the worldwide collection of data. Students will investigate climate change for various places on the surface of the Earth by analyzing atmospheric data from online sources. They will also explore historic patterns of extreme weather events through review of material in the scientific literature and in news reports. Finally, students will investigate what scientists are predicting for the future for the places under study. As a project-driven course, students will formally present their research at the end of the semester.

Prerequisites: GEOG 210 and GEOG 327 or permission of the instructor.

Corequisites: GEOG 327.

GEOG 464 Advanced Cartography**Credits 3 Grading Basis** Graded

This course focuses on learning innovative mapping techniques and exploring different forms of traditional and interactive map design. It addresses the mapping styles and techniques of expert cartographers and focuses on advanced techniques for map composition, feature symbology, terrain representation and web mapping. With more time dedicated to each project, the course will be conducted in a workshop setting for students to share their ideas with one another and get more in-depth feedback as they move forward.

Prerequisites: GEOG 365 or permission of the instructor.

GEOG 465 Topics in GIS**Credits 3 Grading Basis** Graded

The course examines varying topical issues in geographic information science. The course may be repeated as course topics vary.

Prerequisites: GEOG 366 or permission of the instructor.

GEOG 466 GIS and Geographic Databases**Credits 3 Grading Basis** Graded

An introduction to the creation, use, and management of digital spatial data used by industry and government. Integration of large spatial data sets into the geographic information system, data management and data exchange and the geodetic transformation of data sets are emphasized. Digital elevation models, land use data, population data, digital topographic map and street network data will be used.

Prerequisites: GEOG 366 or permission of the instructor.

GEOG 467 GIS Project Management**Credits 3 Grading Basis** Graded

An introduction to geographic information systems (GIS) project management. Basic project management techniques will be applied by defining, designing, implementing and documenting a geographic information system.

Prerequisites: GEOG 366 or permission of the instructor.

GEOG 469 Applications of Geographic Information Systems**Credits 3 Grading Basis** Graded

The course advances the knowledge of GIS in theory and practice by focusing on specific application areas. Spatial databases and complex attribute data will be created, and GIS modeling techniques will be used to solve problems relevant to the specified topical area. The course may be repeated once for additional credit when the topic changes.

Prerequisites: GEOG 366 or permission of the instructor.

GEOG 470 Senior Seminar in Environmental Conservation, Sustainability and Development**Credits 3 Grading Basis** Graded

This capstone seminar integrates the student's previous class experiences to provide a holistic exploration of linkages between environmental conservation and human development status and strategies through in-depth analysis of compelling human-environment issues. Topics vary by semester and include environmental politics, global perspectives on population, sustainable communities and global biodiversity. For majors and minors only.

Prerequisites: Senior standing and GEOG 308.

GEOG 476 Sustainable Cities Seminar**Credits 3 Grading Basis** Graded

This seminar explores ways to make cities more environmentally, socially and economically sustainable. Students will study key urban sustainability principles, examine global case studies, undertake local fieldwork and pursue projects developing interventions in real-world city settings. The course emphasizes a geographical approach in addressing urban human and environmental challenges.

Prerequisites: GEOG 308 or permission of the instructor.

GEOG 477 Gendered Geographies**Credits 3 Grading Basis** Graded

This course examines the gendered nature of local, global and everyday geographies. Concepts and theories explored emphasize how gender and other social categories such as race, class and sexuality intersect with geographic concepts of space, place and scale. Specific topics may vary by semester and emphasize human, human-environment or other geographic foci.

Prerequisites: Senior standing and either GEOG 308 or GEOG 377, or permission of the instructor.

GEOG 485 Advanced Topics in Remote Sensing**Credits 3 Grading Basis** Graded

This course allows students to explore a variety of advanced remote sensing techniques and applications. As part of the course, students will complete one or more projects in which they use remotely sensed data to help address geography-related problems.

Prerequisites: GEOG 385 or permission of the instructor.

GEOG 487 Advanced Cloud-Based Remote Sensing**Credits 3 Grading Basis** Graded

This is an advanced course on image processing using cloud-based remote sensing. The course covers basic processing techniques such as PCA, Tasseled-cap and temporal compositing, as well as advanced techniques such as spectral mixture analysis, time series analysis and multi-sensor fusion. The course emphasizes the utilization of large quantities of remote sensing data and cloud-based processing tools for environmental applications. Examples will be given from a range of remote sensing systems and applications, with a focus on land remote sensing for monitoring environmental changes. Previous experience in coding is recommended.

Prerequisites: A minimum grade of "C" in either GEOG 161 or GEOG 215; and a minimum grade of "C" in GEOG 216; GEOG 385 recommended.

GEOG 490 Senior Research or Field Practicum**Credits 3 Grading Basis** Graded

Working with a research advisor, student completes an internship, a study abroad program, or project research. Student delivers interim progress reports and an annotated bibliography or other relevant research products. May be repeated once.

Prerequisites: GEOG 390 and permission of research advisor.

GEOG 491 International Studies**Credits** 1-3 **Grading Basis** Graded

Student will make arrangements for the international experience. A research project or work-study project will be designed by the student and faculty member prior to departure. The research or work will be carried out in the country of travel. May not be taken for capstone credit. Project must be approved by GS faculty. May be repeated for credit.

GEOG 495 Internship in Geography**Credits** 3-6 **Grading Basis** Graded

Practical experience within a public agency, non-profit or private business utilizing geographic methodology. Work experience will be supervised by an official of the business or agency and a faculty member. Written reports are required. Can be taken for a total of six credits. May not be taken for capstone credit.

GEOG 496 Senior Thesis**Credits** 2 **Grading Basis** Graded

Student completes an independent research project, either alone or within an investigative team, to identify and analyze a geographic problem or phenomenon, and provides a written report and public presentation on the problem analysis and solution. Taken during final semester of the GS program.

Prerequisites: GEOG 490 and senior standing.

GEOG 497 Independent Study**Credits** 3 **Grading Basis** Graded

Student performs an independent research project, either alone or within an investigative team, to identify and analyze a problem from a geographic perspective. May be taken for repeated credit. May not be taken for capstone credit.

GEOG 499A Honors in Geography**Credits** 1 **Grading Basis** Graded**GEOG 499B** Honors in Geography**Credits** 3 **Grading Basis** Graded**GEOG 499C** Honors in Geography**Credits** 2 **Grading Basis** Graded

Geology (GEOL)

GEOL 102 Environment: Earth [C3NS]**Credits** 3 **Grading Basis** Graded

A study of geological processes causing global change and their impact on human thought. The relationship between some geological processes and life on the Earth is also considered. Not available for major or minor credit in geology. May be used for general education credit.

GEOL 110 Physical Geology [C3NS]**Credits** 3 **Grading Basis** Graded

A systematic study of earth materials and the internal and external processes that affect earth structure and landforms. Topics include the genesis/properties of rocks and minerals, plate tectonics, and the agents of change that drive surface processes and land form development. The laboratory and lecture portions of GEOL 110 must be taken concurrently. May be used for general education credit.

Corequisites: GEOL 110L.

GEOL 110L Physical Geology Laboratory [C3L]**Credits** 1 **Grading Basis** Graded

This laboratory course is designed to complement and supplement the GEOL 110 course. The laboratory and lecture portions must be taken concurrently. May be used for general education credit.

Corequisites: GEOL 110 or permission of the instructor.

GEOL 115 Earth Systems and Climate Change [C3NS]**Credits** 3 **Grading Basis** Graded

This course explores cycles, trends and abrupt events in the Earth system. Analyses of the geologic record and global climate models provide perspective for understanding paleoclimate and future climate changes, including global warming. Current hypotheses for causes of climate change are evaluated, including plate tectonics, orbital cyclicity, variations in the sun's strength and human activities. The two reoccurring questions of this class are: What are Earth's climate stories? How do we know? May be used for general education credit.

GEOL 115L Earth Systems and Climate Change Lab [C3L]**Credits** 1 **Grading Basis** Graded

In this course students will explore interactions between Earth's climate and humans through the framework of Earth systems science. Students will work with data and models to learn about the components of the climate system, climate change in the past and projections for climate in the future. In addition, students will examine the important roles that the cycling of carbon, water and energy have in the Earth's climate system. May be used for general education credit.

Prerequisites: GEOL 115. Corequisites: GEOL 115.

GEOL 167 History and Philosophy of the Geosciences**Credits** 3 **Grading Basis** Graded

As an introductory experience in the Bachelor of Arts in Earth Science, students will be inculcated in the philosophy of geosciences as an interdisciplinary medium for extending classical science viewpoints to complex earth systems. Students will study the geosciences as distinct among sciences, establishing relevance and value of earth science literacy in professional and personal settings.

GEOL 210 Applied Physical Geology [C3NS]**Credits** 3 **Grading Basis** Graded

A problem-based study of earth materials and the processes that affect earth structure and landforms. Topics include plate tectonics, the genesis/properties of rocks and minerals, and agents of change that drive surface processes and landform development. Quantitative problem-solving skills will be applied to case studies that address 3D visualization and time-based processes, such as earth materials, solid earth and surface processes, natural hazards and engineering applications. May be used for general education credit.

GEOL 211 Introduction to Oceanography [C3NS]**Credits** 3 **Grading Basis** Graded

An introduction to the oceanography of coastal environs including barrier islands, estuaries and tidal marshes. The physical, geological and biochemical characteristics of coastal waters will be discussed in the context of the economic and social pressures brought to bear on these areas by an increasing global population. Cannot receive credit for both GEOL 211 and GEOL 401. May be used for general education credit.

GEOL 230 Evolution of Earth (3, 2)**Credits** 4 **Grading Basis** Graded

An introduction to the evidence, methods and assumptions used by scientists to unravel the Earth's origin and history. Emphasis on rock analysis/interpretation, modern and ancient processes of mountain building, origin and evolution of life and the history of the North American continent.

Prerequisites: GEOL 110L or GEOL 115L.

GEOL 280 Mineralogy (3, 2)**Credits 4 Grading Basis Graded**

A comprehensive study of minerals, including crystallography, mineral chemistry, x-ray diffraction, mineral optics with thin section recognition using petrographic microscope and hand specimen identification of both silicate and non-silicate minerals.

Prerequisites: GEOL 110L or GEOL 115L. Corequisites: CHEM 131.

GEOL 291 Writing and Communicating in the Geosciences**Credits 1 Grading Basis Graded**

This course prepares students for independent research by providing them the fundamental skills in literature searches, writing, critical reading and communication in the geosciences. Must take prior to senior year.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 102.

GEOL 300 Introduction to Petrology (3, 3)**Credits 4 Grading Basis Graded**

Igneous and metamorphic processes are explained using crystallization theory, phase diagrams, thermodynamics and geochemistry.

The laboratory study of rocks, their chemical and mineralogical signatures, and their geologic origins will be explored.

Prerequisites: GEOL 110L or GEOL 115L, and GEOL 280 and CHEM 131; or permission of the instructor.

GEOL 301 Earth Science for Teachers**Credits 4 Grading Basis Graded**

Earth science content is blended with a systems approach to provide pre-service teachers with an understanding of how the Earth works, as well as strategies for teaching it. Major content themes include reconstructing the geologic history of the mid-Atlantic, exploring the interaction of living things and the environment, and predicting how matter and energy circulate in the earth system.

GEOL 302 Sedimentary Petrology**Credits 1 Grading Basis Graded**

This is a course about sedimentary rocks in hand sample and thin section. With the polarizing microscope, students will study thin sections of outcrop and core samples. They will identify common minerals, textures and fabrics. They will identify the four components of all sedimentary rocks (framework grains, cements, matrix, pore types) and will make qualitative and quantitative measurements at a fundamentally different scale than is possible with just a field study of sedimentary rocks.

Prerequisites: GEOL 300 or permission of the instructor. Corequisites: GEOL 300 or permission of the instructor.

GEOL 307 Planetary Geology (2, 2)**Credits 3 Grading Basis Graded**

This course surveys currently developing ideas in planetology including origin of the planets, meteorites and planetary interiors. Also included are geologic processes and land forms on the moon and terrestrial planets, their modification under various planetary environments, and analogies to familiar earth land forms. Includes laboratory.

Prerequisites: GEOL 110L or GEOL 115L.

GEOL 310 Environmental Impact**Credits 1-4 Grading Basis Graded**

Focuses on a selected environmental realm. The course will examine the interface between human activities and environmental systems. It will address the impacts of social, economic and political activities on the environment. A-Atmosphere (air pollution); B-Biosphere (vegetation/wildlife); C-Hydrosphere (water); D-Lithosphere (geologic hazards/land issues).

GEOL 311 Mineral Resources and the Environment (2,2)**Credits 3 Grading Basis Graded**

This course will focus on specific environmental problems that arise from mineral resources exploitation or utilization either (or both) in generating the problem or restoring it. Emphasis will be given to mineral characterization and use, environmental aspects of mine extraction, weathering reactions on buildings, artifacts, and monuments that constitute collective cultural heritage. The relationship between minerals and human health will be studied as a special case where the environment includes the human body itself. Ethical implications on mineral extraction will be considered from the point of view of environmental protection. Examples will be provided of analytical, experimental approaches applied to characterize mineral resources. Prerequisites: GEOL 280 or CHEM 132 or GEOL 367 or permission of the instructor.

GEOL 320 Meteorology**Credits 3 Grading Basis Graded**

A survey of the science of weather including weather forecasting, weather maps and related atmospheric processes. Emphasis is placed on the dynamic aspects of meteorology and the interrelationships of atmospheric phenomena with land masses and the world ocean.

GEOL 330 Sedimentology, Stratigraphy and Paleoclimatology**Credits 4 Grading Basis Graded**

This course is a study of the origin and distribution of sedimentary rocks within a tectonic and climatic framework. Laboratory and field exercises explore sedimentation, sedimentary petrology, facies analysis, weathering and stratigraphy with the goal of discerning Earth's records of past environments and climates.

Prerequisites: GEOL 110L or GEOL 115L; GEOL 230; or permission of the instructor.

GEOL 350 Paleobiology**Cross-listed with: BIO 350****Credits 4 Grading Basis Graded**

The evolution and ecological structure of the biosphere from the origin of life to the present, emphasizing the evolution and paleobiology of animal life as shown by the fossil record. Lectures discuss methods used to interpret the fossil record and cover topics such as phylogeny and systematics, functional morphology, biostratigraphy, paleoecology, evolution and extinction. Laboratories focus on the major groups of invertebrates that are common in the geologic record.

Prerequisites: GEOL 230, BIO 150/BIO 150L or permission of the instructor.

GEOL 355 Geochemistry of Natural Waters**Cross-listed with: CHEM 355****Credits 3 Grading Basis Graded**

Study of chemical theory and reactions important in natural water systems. The role of atmospheric, geologic and biological inputs in determining the geochemistry of streams, rivers and oceans.

Prerequisites: CHEM 131 and CHEM 132 or equivalent.

GEOL 360 GIS for the Geoscientist**Credits 3 Grading Basis Graded**

This course is designed to help students master practical geospatial skills helpful for an entry-level job or graduate school in the geosciences.

GEOL 365 Structural Geology, Tectonics and Climate**Credits 4 Grading Basis** Graded

This course examines deformation-related structures within Earth's crust, and the mechanical principles involved in their formation. Students explore the feedback(s) between tectonics, structural geology and climate. This course includes a lecture and a laboratory component, with two to three field trips throughout the semester.

Prerequisites: GEOL 110L or GEOL 115L; GEOL 230; or permission of the instructor.

GEOL 367 Genesis of Solid Earth Materials (3, 2)**Credits 4 Grading Basis** Graded

This course addresses the natural relationship between minerals and the rocks they make up. Using the concept of mineralizing environments, illustrated by classic examples, students will investigate minerals through the processes of mineral genesis and associated rock types. This approach provides insight and predictive value for natural conditions in which specific minerals and rocks occur. Not acceptable for B.S. in geology.

Prerequisites: GEOL 110L or GEOL 115L.

GEOL 377 Earth Surface Processes (2, 2)**Credits 3 Grading Basis** Graded

The interrelationships among climate, landscapes, soils and bedrock geology are examined using the mid-Atlantic region as a conceptual laboratory. Course instruction includes lecture, laboratory and field trip meetings. The processes of rock weathering and erosion and soil formation are reinvestigated. Topographic maps and aerial photography are examined for landforms and landscape evolution.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 210.

GEOL 380 Genetic Mineralogy (2,2)**Credits 3 Grading Basis** Graded

A study of mineral genesis. Emphasis is directed toward mineralogical environments, mineral associations and the geology/mineralogy of classical localities. An appreciation of mineral value and aesthetics is incorporated throughout the course.

Prerequisites: GEOL 280.

GEOL 387 Stratigraphy, Structure and Tectonics (3, 2)**Credits 4 Grading Basis** Graded

Examination of how stratigraphic, structural and tectonic principles control the character and distribution of rocks. Study of principles, regional patterns in sedimentary rocks and stresses that deform rocks are explored in laboratory and field exercises. Topics and techniques are discussed within the framework of the 1.2 billion-year geologic history of the VA region and its connection with tectonic processes throughout the rest of the world.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 230.

GEOL 388 Advanced Structure, Stratigraphy, Tectonics (3,2)**Credits 4 Grading Basis** Graded

Advanced concepts in structural geology (analyses of rheology, stress & strain, deformation fabrics, and chronologic constraints) applied to the tectonic evolution of orogens and basins. This will include exploration of the mechanisms controlling the stratigraphic record (auto- and allogenic mechanisms and sequence theory) of basins, as well as methods of dating the stratigraphic record (biofacies and biostratigraphy, geochronology and chronostratigraphy, and magneto and chemostratigraphy).

Prerequisites: GEOL 387.

GEOL 390 Laboratory Techniques in Geology (2,2)**Credits 3 Grading Basis** Graded

An elective course for science majors. A study of the basic theories and techniques of laboratory methods and instrumentation. Implementation and application of techniques to geological problems.

Prerequisites: GEOL 280.

GEOL 398 Topics in Geology**Credits 1-4 Grading Basis** Graded

Topics in geology at the advanced level. May be repeated for credit when course content changes. Topics selected may determine prerequisites. Students should consult the instructor prior to enrolling in the course.

GEOL 399 Field Geology**Credits 3-6 Grading Basis** Graded

Field Geology is the capstone course for geology majors. Students learn to recognize and interpret a wide variety of rocks and structures, as well as geomorphic, hydrologic, and other geologic features. Methods of mapmaking, data recording, and report preparation are emphasized. Projects from one to five days' duration are conducted in regions where igneous, metamorphic, and sedimentary rocks, surficial deposits, and karst features are well-exposed.

Prerequisites: GEOL 300 and GEOL 387 or permission of the instructor.

GEOL 400 Geology and Ecology of the Bahamas**Cross-listed with:** BIO 400, BIO 500**Credits 3 Grading Basis** Graded

This course explores the geology and ecology of the shallow-water marine environment by examining the preeminent modern example, the Bahamas platform. The Bahamas provide an excellent model for understanding modern and ancient carbonate and reef deposits and a variety of terrestrial/aquatic habitats. Biological processes are responsible for many of the geological features of the Bahamas, so the course considers the biology/ecology of marine organisms in addition to geological topics.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 211 or a 200- level GEOL or BIO course; at least four hours of additional lab science, at least sophomore status, and permission of the instructor.

GEOL 401 Oceanography for Teachers**Credits 3 Grading Basis** Graded

A comprehensive study of the world's oceans and the interrelationships among physical, chemical, biological and geological oceanography for pre- and in-service teachers. Special emphases on Virginia coastal oceanography, the National Ocean Literacy Principles and the integration of pedagogy applicable to K-12 instruction. Includes a field trip to the Virginia coast. Credit may not be earned in both GEOL 211 and GEOL 401.

GEOL 405 Vertebrate Paleontology**Cross-listed with:** BIO 405**Credits 3 Grading Basis** Graded

A study of the origin and evolution of the vertebrates. Emphasis will be on understanding how the processes of Earth evolution and biological evolution have interacted through time to produce a coherent picture of vertebrate history.

Prerequisites: GEOL 230, BIO 150/BIO 150L or permission of the instructor.

GEOL 406 Paleoclimatology and Paleoceanography**Credits 3 Grading Basis** Graded

Investigate the methodologies and data used to reconstruct Earth's climate history. Emphasis will be placed on the marine sediment records of the Cenozoic through detailed lecture, lab activities and course-based original research. Case studies include the Paleocene Eocene Thermal Maximum, the glaciation of Antarctica, Milankovitch cyclicity and Northern Hemisphere glaciation.

Prerequisites: GEOL 211, GEOL 230 or BIO 354, or permission of the instructor.

GEOL 410 Engineering Geology (2, 2)**Credits 3 Grading Basis** Graded

Study of the applications of geology to engineering practice. Topics include soil mechanics, foundations, engineering classification of soils, slope stability and mineral aggregates.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 210 or GEOG 210, and MATH 231 or MATH 235 or equivalent.

GEOL 430 Environmental Soil Science (2, 2)**Credits 3 Grading Basis** Graded

The origin, distribution, and chemical, biological, and physical properties of soil are introduced. Processes responsible for soil properties are emphasized. Field trips highlight the sustainability of soils, their distribution across the Shenandoah Valley and their role in biogeochemical cycles.

Prerequisites: GEOL 110L or GEOL 115L or GEOL 210, or permission of the instructor may be granted for students with four hours of a lab course.

GEOL 440 Geophysics (3, 2)**Cross-listed with:** PHYS 440**Credits 4 Grading Basis** Graded

This course provides a survey of geophysical methods, with joint attention on near-surface and solid earth applications. Topics include seismology, heat flow, gravity, magnetism, electrical methods, ground penetrating radar and geophysical aspects of plate tectonics. Labs focus on practical experience with data acquisition, reduction and interpretation and are a combination of field, classroom and computational activities.

Prerequisites: GEOL 110L or GEOL 115L, or PHYS 140 and PHYS 150, or PHYS 240 and PHYS 250; or permission of the instructor.

GEOL 442 Field Geophysics**Credits 3 Grading Basis** Graded

This course focuses on collection of geophysical data in the field and interpretation, analysis and technical reporting afterward. Case studies discussed include applications to geology, archaeology and engineering. Students will get hands-on experience with geophysical equipment and an understanding of how and where these tools can be applied.

Topics include ground penetrating radar, electrical resistivity, magnetism, seismic refraction and location (GPS and/or total station) data.

Prerequisites: GEOL 110 or GEOL 115L or GEOL 210 or ANTH 197 or permission of the instructor.

GEOL 460 Hydrogeology (2, 2)**Credits 3 Grading Basis** Graded

Basic concepts of subsurface water as a part of the hydrologic cycle. Topics include storativity and permeability in porous media, principles of flow, computer applications, groundwater exploration, and mapping and environmental aspects of groundwater.

Prerequisites: GEOL 110L or GEOL 115L.

GEOL 477 Contemporary Issues in the Geosciences**Credits 4 Grading Basis** Graded

As a capstone experience, this course serves as an opportunity for students to view issues of the Earth system from an Earth-based perspective. Building on previous course work in the major (physical geology, meteorology, oceanography, etc.), students will investigate such issues as global warming, population and sustainable development and environmental ethics. Particular emphasis is placed upon the Earth's perspective from a historical viewpoint.

Prerequisites: GEOL 211, GEOL 320, GEOL 367 and GEOL 377.

GEOL 489 Quantitative Methods in Geology**Credits 3 Grading Basis** Graded

An introduction to the mathematical methods and statistical techniques that are employed by scientists in the disciplines of geochemistry, geophysics, hydrology and the petroleum/mineral industry. The course provides the quantitative skills necessary to manipulate geological data.

GEOL 491 Geological Literature and Research**Credits 2 Grading Basis** Graded

Provides advanced instruction in literature research to meet the BA Earth Science and BS Geology research requirements. Activities include the identification of a literature-based research problem, literature research techniques, critical reading and discussion, and the preparation of individual review papers on each student's research topic.

Prerequisites: GEOL 291 or permission of the instructor.

GEOL 494 Internship in Geology**Credits 1-3 Grading Basis** Graded

Student conducts a research or applied project in geology outside of the university. Requires an approved proposal prior to registration and a final report at the culmination of the project.

Prerequisites: Minimum of eight credit hours in geology, GEOL 291 and a geology GPA of 2.5 or higher.

GEOL 497 Work-Based Learning Experiences in Geology and Environmental Science**Credits 1-3 Grading Basis** Graded

This course provides an undergraduate work-based research or internship course in one of the fields of geology. It is open to advanced students who have adequate preparation.

Prerequisites: GEOL 291 or permission of the instructor.

GEOL 499A Honors in Geology**Credits 1 Grading Basis** Graded

Prerequisites: GEOL 291 and 3.25 GPA or higher.

GEOL 499B Honors in Geology**Credits 2 Grading Basis** Graded

Prerequisites: GEOL 291 and 3.25 GPA or higher.

GEOL 499C Honors in Geology**Credits 3 Grading Basis** Graded

Prerequisites: GEOL 291 and 3.25 GPA or higher.

German (GER)

GER 101 Elementary German I**Credits 3 Grading Basis** Graded

This course covers the fundamentals of German through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

GER 102 Elementary German II**Credits 3 Grading Basis Graded**

This course builds on the foundations established in GER 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension.

Prerequisites: GER 101 or by placement exam.

GER 103 Passport to German: A Survival Kit**Credits 3 Grading Basis Graded**

This course teaches students basic communication skills and cultural competence needed for studying, traveling or working in German-speaking countries. Students will practice typical communicative contexts using authentic materials and gain insight into the real-life situations of German speakers. This course has no prerequisite and is designed for those who have not had previous German instruction. It does not count towards the German major or minor or fulfill the B.A. language requirement. Instruction is in German and English.

GER 111 Intensive German I**Credits 6 Grading Basis Graded**

This course covers the fundamentals of German through intensive listening, speaking, reading and writing. This course is the equivalent of GER 101-GER 102.

GER 212 Intensive German II**Credits 6 Grading Basis Graded**

The fundamentals of German through intensive listening, speaking, reading and writing at the intermediate level. This four-week course is the equivalent of GER 231-GER 232.

Prerequisites: GER 102, GER 111, or per placement exam score.

GER 231 Intermediate German I**Credits 3 Grading Basis Graded**

A thorough review of grammar and vocabulary building, conversation, composition and reading.

Prerequisites: GER 102, GER 111, or per placement exam score.

GER 232 Intermediate German II**Credits 3 Grading Basis Graded**

A thorough review of GER 231 grammar and vocabulary building. Conversation, composition and reading will be chosen to reach competency at the advanced intermediate level.

Prerequisites: GER 231 or per placement exam score.

GER 300 Language for Living**Credits 3 Grading Basis Graded**

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in German language. It is intended for students who have begun to develop advanced skills and are interested in continued training in language proficiency through reinforcing foundations of everyday language communication and cultural knowledge. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 301 Pop Culture through Language**Credits 3 Grading Basis Graded**

The course is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in the German language and culture. It is intended for students who have begun to develop advanced skills in interpretive, interpersonal and presentational communication and are interested in delving deeper into the study of German-speaking countries, their histories and their cultures. The course will focus on popular culture in the German-speaking world including music, film, television, literature and social media, and analyze how these reflect and shape cultural identities and trends. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 307 History of Germanic Cultures and Societies**Credits 3 Grading Basis Graded**

This course provides an overview of major developments and figures in central European cultural history from the Germanic tribes to 1871. This course will explore texts, architecture, music, philosophy and works of art in order to analyze how past political and cultural history informs German and European collective identity today. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 308 Modern German Culture and Society**Credits 3 Grading Basis Graded**

This course examines German culture and history from 1871 to the present. The course is centered on questions of German identity and how it has been influenced by historical events and cultural shifts. Students will examine cultural artifacts from the areas of art, literature and film created during or with reference to the historical period studied. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 315 Sounds of Speech**Credits 3 Grading Basis Graded**

This course introduces students to German linguistics, with special emphasis on German phonetics. Students will engage with theoretical concepts and practical applications to understand the phonological structures and sounds that make up the German language. The course also includes units on sociolinguistics and German dialects. Class activities and assignments aim to enhance overall language proficiency, including listening, reading, speaking and pronunciation skills. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 320 Culture and Communication in Context**Credits 3 Grading Basis Graded**

This German course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do German-speaking peoples communicate on subjects that are at the heart of their societies?" Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 322 Language Landscapes**Credits 3 Grading Basis Graded**

This German course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in these culturally and linguistically defined areas and explores the diverse cultural landscapes that shape their lives. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 330 Business German**Credits 3 Grading Basis Graded**

This course studies commercial and trade vocabulary and customs in conjunction with practice in commercial communication, including letter writing, interviews and interpretation. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by placement exam or permission of the instructor.

GER 335 A Journey through Literature**Credits 3 Grading Basis Graded**

This course introduces students to German-language literature through the study of selected short texts written from the Middle Ages to the present. Different literary genres will be studied including short stories, fairy tales, poetry, songs, literary essays, drama and the graphic novel. In this class, students will improve their reading comprehension, as well as develop the vocabulary and interpretive strategies needed to read and analyze any text better. Instruction is in German.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor. Corequisites: GER 300, GER 301, GER 320 or GER 322.

GER 375 Business and Society in Germany**Credits 3 Grading Basis Graded**

Germany, "The Land of Ideas," known for its engineering and technology, is a leader in innovation and research in every field. It is the strongest economic power in Europe. The course explores German society in its historical, political and economic contexts. Several aspects will be investigated: the EU, Globalization, immigration, environment, family policies, current affairs and the latest trends in the society. Instruction is in German.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 380 German Cuisine**Credits 3 Grading Basis Graded**

This course is designed to introduce students to the rich culinary traditions in Germany. Students will familiarize themselves with the diversity in German cuisine through readings and presentations and prepare dishes that represent various regions of Germany. Instruction is in German.

Prerequisites: GER 212 or GER 232; or by permission of the instructor.

GER 386 German Short Stories**Credits 3 Grading Basis Graded**

A study of literary German short prose from the 18th century to the present, including a selection of fables, fairy tales, modernist parables, aphorisms, narratives and short novellas. Authors studied may include Brecht, Kafka, Hoffmann, Ebner-Eschenbach, Schnitzler, Kleist, Bachmann, Borchert and Mann, among others. Instruction is in German.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 390 German Language and Culture Abroad**Credits 1-3 Grading Basis Graded**

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in German literature and culture. It is intended for students who have begun to master advanced skills in interpretive, interpersonal and presentational communication and are interested in examining nuances of the German language and culture while studying abroad.

Prerequisites: GER 232 or permission of the instructor.

GER 400 Advanced German**Credits 3 Grading Basis Graded**

This course focuses on advanced German language and its use in real-life contexts. Discussions deal with current events.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 422 Voices of Today**Credits 3 Grading Basis Graded**

This course provides a study of contemporary German-language literature and media, focusing on cultural production since the 1990s. This course explores topics such as cultural memory, identity, German reunification, globalization, multiculturalism, climate change, digitalization and social justice. Instruction is in German.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 425 Transformative Texts**Credits 3 Grading Basis Graded**

This course offers a study of the works of major German writers of the 20th century. Instruction is in German.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor. Corequisites: GER 300, GER 301, GER 320 or GER 322.

GER 435 Translation Strategies**Credits 3 Grading Basis Graded**

In this course, students will develop linguistic competencies required in translation, including reading comprehension, summary writing, text analysis and use of mono- and bilingual dictionaries. Students will learn some basic electronic tools and word processing skills for translators and practice several types of translation, including direct translation, inverse translation and back translation. This course fulfills the College of Arts and Letters writing-intensive requirement for the major. Instruction is in English and German.

Prerequisites: GER 212 or GER 232; or by permission of the instructor.

GER 436 Studies in German Literature**Cross-listed with: ENG 436****Credits 3 Grading Basis Graded**

A study of selected works of German literature. Instruction is in English. May be repeated for credit when course content changes.

GER 446 Special Topics in German Literature**Credits 3 Grading Basis Graded**

This course studies a particular topic in German literature. It may cover all or a specific German literature genre. This course may be repeated if content varies.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 447 Special Topics in German Culture**Credits 3 Grading Basis Graded**

Students will study a particular topic in the civilization and/or culture of Germany. Course may be repeated if content varies.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 448 Special Topics in German Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in German linguistics. Topics could include an introduction to German sociolinguistics and psycholinguistics. Course may be repeated if content varies.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 464 German Theater**Credits 3 Grading Basis** Graded

This course offers a study of German-language theater from the 18th-century to the present. Students will examine the cultural, historical and political events that influenced stage productions of the day. Instruction is in German.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 465 German Cinema**Credits 3 Grading Basis** Graded

This course analyzes German cinema from the 1920s through the present. Emphasis will be on the relations between the German film and certain seminal periods in German history.

Prerequisites: GER 300, GER 301, GER 320 or GER 322; or by permission of the instructor.

GER 490 Independent Study in German**Credits 1-4 Grading Basis** Graded

Allows students an opportunity to complete independent studies under faculty supervision.

GER 497 German Internship**Credits 1-3 Grading Basis** Graded

This course will provide students with experiential learning opportunities within the German-speaking field. The internship course is designed to provide practical experience in fields such as government, business, industry, health sciences, administration, education, translation or the arts, where they can apply their language skills. Examples of internship work may include working for a company in the student's field of interest abroad; serving as a research assistant for a professor; completing a translation project or working as an interpreter. A journal, internship report, research paper, bibliography or evaluation from the internship provider is required. Credits earned correspond to the internship hours completed. Up to six credits may be applied as electives in the German major or minor.

Prerequisites: One GER course at the 300-level or above and permission of the instructor.

Gerontology (GERN)

GERN 280 Social Gerontology**Cross-listed with:** SOCI 280**Credits 3 Grading Basis** Graded

An interdisciplinary introduction to the study of aging. The course provides an overview of issues surrounding aging in contemporary society: personal, familial, communal and societal.

GERN 305 Programs and Services for the Elderly**Credits 3 Grading Basis** Graded

A review of the programs and services provided for the elderly in the public and private sectors of America. Observations and participation in local programs for the elderly will be required.

Prerequisites: GERN 280/SOCI 280.

GERN 372 Programs and Services for Older Adults**Cross-listed with:** SOWK 372**Credits 3 Grading Basis** Graded

An examination of social policies and responses to aging persons from historical and current perspectives in the United States. A survey of programs and services developed for older adults in the public and private sectors at individual, group, and community levels is examined, with consideration of professional skills, competencies, and credentials necessary for ethical and competent program implementation. Engaged learning through observations and participation in local programs serving older adults is required.

Prerequisites: GERN 280/SOCI 280.

GERN 375 Grant Writing for Agencies**Cross-listed with:** FAM 375, NPS 375, SOWK 375**Credits 3 Grading Basis** Graded

Emphasizing active learning, this course teaches the basics of grant and proposal writing. Efficient research, persuasive prose and the importance of relationships are stressed. Private and corporate philanthropy and government grants are examined.

GERN 450 Palliative and Hospice Care: An Interdisciplinary Approach**Cross-listed with:** SOWK 450**Credits 3 Grading Basis** Graded

An interdisciplinary approach to assessment and care planning of older adults experiencing need for palliative and/or hospice support. Familial, cultural, legislative and social implications of programs and service delivery.

GERN 487 Special Topics in Gerontology**Credits 3 Grading Basis** Graded

Examination of selected topics in gerontology that are of current importance in the field of gerontology. Course may be repeated for credit.

GERN 490 Special Studies in Gerontology**Credits 1-3 Grading Basis** Graded

Independent study in gerontology under faculty supervision. Limited to gerontology minors. Can be repeated for credit.

Prerequisites: GERN 280/SOCI 280 and GERN 305.

Graphic Design (GRPH)

GRPH 101 Digital Media Design I**Credits 3 Grading Basis** Graded

An introduction to graphic design using industry standard software. Students will explore fundamental concepts, terminology and media that relate to the creation and presentation of graphic design projects.

Prerequisites: FDS 102 and FDS 104.

GRPH 200 Image Making**Credits 3 Grading Basis** Graded

An exploration of strategies for conceptualizing, analyzing and solving design problems through image making. Emphasis is placed on graphic presentation of ideas and the creative process through a variety of mediums and materials.

Prerequisites: FDS 102.

GRPH 201 Digital Media Design II**Credits 3 Grading Basis** Graded

A hands-on approach to digital media and production using leading edge software for web and interactive design. Concepts and content include visual language for screens and devices, exploration of time-based media, audio/video integration, user feedback tools and digital portfolios.

Prerequisites: GRPH 101.

GRPH 206 Typography I**Credits 3 Grading Basis** Graded

An introduction to the study of letter forms for their aesthetic and communicative value. Typographic fundamentals of alphabet history, type classification, printing technology effects, font design, visual space, hierarchy and grid systems will be explored.

Prerequisites: GRPH 101.

GRPH 208 Portfolio Review**Credits 0 Grading Basis** Credit / No Credit

Portfolio review required to enroll in graphic design courses at upper division standing. May be repeated once for pass/fail standing.

Prerequisites: GRPH 101, GRPH 200 and GRPH 206. Corequisites: GRPH 206.

GRPH 292 Topics in Graphic Design**Credits 1-3 Grading Basis** Graded

This course is a study of selected topics in graphic design at the introductory level. May be repeated when course content changes. See MyMadison for current topics.

GRPH 300 Illustration**Cross-listed with:** ART 300**Credits 3 Grading Basis** Graded

Through demonstrations, theory and practical application, students are introduced to numerous media and illustrative techniques. Encouraged experimentation is tempered by an understanding of problem solving and conceptualization.

Prerequisites: GRPH 208.

GRPH 301 Digital Media Design III**Credits 3 Grading Basis** Graded

Through demonstrations and practical application, students are introduced to digital techniques used in design applications. Experimentation is encouraged through problem solving and conceptualization, filtered through the lens of contemporary digital production. Students will be challenged with executing effective design solutions through creative thinking and image creation as it applies to graphic design.

Prerequisites: GRPH 208.

GRPH 304 Package Design**Credits 3 Grading Basis** Graded

Through theory, demonstrations and practical application, students learn to design in three-dimensions. Focus will be placed on aesthetics, as well as the form and function of a product's housing.

Prerequisites: GRPH 208.

GRPH 306 Typography II**Credits 3 Grading Basis** Graded

An intermediate study of typography for its aesthetic and communicative value. Context effects on legibility and readability, type as image, type in sequence and typographic systems will be explored.

Prerequisites: GRPH 208.

GRPH 310 Graphic Illustration**Cross-listed with:** ART 310**Credits 3 Grading Basis** Graded

Students will use traditional fine arts and digital media to explore the conceptualization, production and communication of visual information integral to selling products, clarifying and organizing data, explaining abstract concepts, and educating through the creation of imagery for advertising, infographics, and technical and scientific illustrations. Emphasis is placed on development of technical skillsets, personal aesthetic and style.

Prerequisites: FDS 102, FDS 104 and FDS 108.

GRPH 312 Web Design**Credits 3 Grading Basis** Graded

Intermediate level web design and front-end development through theory and practical application. Assignments will focus on the unique form, content and structures associated with designing for a device-agnostic web. Creative iteration and digital design are emphasized in tandem with changing technology.

Prerequisites: GRPH 201 and GRPH 208.

GRPH 313 Interactive Media**Credits 3 Grading Basis** Graded

Intermediate level visual interface design, motion graphics, and interaction design through theory, demonstration and practical application. Assignments will focus on digital craft, creativity, and presentation. Emphasis on digital and interactive media, and functionality, usability, and user experience.

Prerequisites: GRPH 201 and GRPH 208.

GRPH 340 Poster Design**Credits 3 Grading Basis** Graded

Through theory and practical application, students learn to design for the poster realm. Focus will be placed on aesthetics, as well as form and function. Encouraged experimentation is tempered by an understanding of problem solving and conceptualization.

Prerequisites: GRPH 208.

GRPH 375 Letterpress**Cross-listed with:** ART 375**Credits 3 Grading Basis** Graded

This studio course offers students an opportunity to engage in the process and product of letterpress printing through various techniques and conceptual approaches. Instruction focuses on text and image relationships by integrating metal and wood type, and other type-high surfaces. Emphasis will be placed on the acquisition of skills and vocabulary and the creative use of type and image. The course will address the history of letterpress and its contribution to contemporary art and design.

Prerequisites: FDS 102, FDS 104, and GRPH 208.

GRPH 376 Intermediate Book Arts: Concept, Content, Form**Cross-listed with:** ART 376**Credits 3 Grading Basis** Graded

This course challenges the student to develop a limited edition hand-made artists' book. The appropriate format for each individual's concepts are identified, adapted, customized, applied, and produced. Content development, book design, integration of various media and the functionality of various bookmaking materials are explored. We will consider the artists' book as a sculptural form and locate it within the broader context of contemporary writing and visual art.

Prerequisites: ART 276 and GRPH 208.

GRPH 380 Wayfinding, Signage, Exhibit Design**Credits 3 Grading Basis** Graded

This course examines graphic design as applied to the built environment. Environmental graphic design applications include but are not limited to wayfinding and directional signage, exhibition graphics, large-format graphics and pop up displays (e.g. storefronts, showrooms, etc.), user interface design, branding, and retail design.

Prerequisites: GRPH 208.

GRPH 390 Independent Study in Graphic Design**Credits** 1-3 **Grading Basis** Graded

Independent activity at the intermediate level, such as research or studio practice, under faculty supervision. Projected studies in Graphic Design must be arranged with the instructors who will direct them. Offered only with the consent of the instructor.

Prerequisites: GRPH 208.

GRPH 392 Topics in Graphic Design**Credits** 3 **Grading Basis** Graded

Study of selected topics in graphic design at the intermediate level. May be repeated when course content changes. See MyMadison for current topics.

Prerequisites: GRPH 208.

GRPH 400 Topics in Illustration**Cross-listed with:** ART 400**Credits** 3 **Grading Basis** Graded

This course allows for the development of an advanced level class that teaches content not introduced in required illustration courses or expanding on the existing illustration courses. Instructional goals and defined outcomes will be unique to the class topic.

Prerequisites: GRPH 208 and either GRPH 300 or GRPH 310, or permission of the instructor.

GRPH 406 Advanced Typography**Credits** 3 **Grading Basis** Graded

An advanced study of typography for its aesthetic and communicative value. Topics of research, narrative, information design, format organization and production techniques will be explored. Solutions reflect advanced text and image integration knowledge through complex and variable structure development.

Prerequisites: GRPH 306.

GRPH 408 Brand Identity**Credits** 3 **Grading Basis** Graded

Exploration of visual identities utilizing a holistic, systems approach to design. Introduces business strategies and design techniques associated with brand development. Emphasis is placed on methods of thinking, research and implementation.

Prerequisites: GRPH 306.

GRPH 410 Graphic Design Portfolio**Credits** 3 **Grading Basis** Graded

This course examines the business of graphic design, employment strategies and freelance opportunities. Focus is placed on solidification of the portfolio for employment and/or graduate school. This course is strongly recommended for the last semester prior to graduation.

Prerequisites: GRPH 408 and GRPH 492; or permission of the instructor.

GRPH 490 Independent Studies in Graphic Design**Credits** 1-3 **Grading Basis** Graded

Independent activity, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with consent of the director.

Prerequisites: GRPH 208.

GRPH 491 Studio Assistant**Cross-listed with:** ART 491, ARED 491, ARCD 491, INDU 491**Credits** 1-3 **Grading Basis** Graded

An on-campus program monitored on an individual basis designed to provide practical studio experience in the visual arts. Students will learn safe studio practices and management skills, including material use, inventory control and the proper operation of equipment found within various individual classroom studios.

GRPH 492 Topics in Graphic Design**Credits** 3 **Grading Basis** Graded

Students study selected topics in graphic design at the advanced level. This course may be repeated when course content changes. See MyMadison for current topics.

Prerequisites: GRPH 306 and GRPH 312; or permission of the instructor.

GRPH 496 Internship in Graphic Design**Credits** 1-6 **Grading Basis** Graded

An off-campus program prepared and monitored on an individual basis. Internships provide practical experience in professional graphic design settings.

Prerequisites: Permission of the instructor and GRPH 208.

GRPH 497 Graphic Design Senior Capstone**Credits** 1-6 **Grading Basis** Graded

Advanced level exploration of graphic design principles, methods, applications and professional practice. Independent research projects are emphasized. Students will produce and refine a professional graphic design portfolio suitable for employment and/or application to graduate school. This course is strongly recommended for the last semester prior to graduation.

Prerequisites: GRPH 312 and GRPH 408; or permission of the instructor.

GRPH 499 Honors**Credits** 1-6 **Grading Basis** Graded

Prerequisites: GRPH 208.

Greek (GRK)

GRK 101 Elementary Greek I**Credits** 3 **Grading Basis** Graded

Designed to provide a reading knowledge of Classical Greek as well as New Testament koine. Greek life, thought and culture are also discussed. Especially recommended for history, English, and philosophy and religion majors.

GRK 102 Elementary Greek II**Credits** 3 **Grading Basis** Graded

Designed to provide a reading knowledge of Classical Greek as well as New Testament koine. Greek life, thought and culture are stressed. Especially recommended for science, English and philosophy majors.

Prerequisites: GRK 101 or equivalent.

GRK 231 Intermediate Greek I**Credits** 3 **Grading Basis** Graded

An intensive reading course. Selections from Classical Greek writers and/or the New Testament.

Prerequisites: One year of college Greek or equivalent. A student cannot earn credit for GRK 231 after completing GRK 232.

GRK 232 Intermediate Greek II**Credits** 3 **Grading Basis** Graded

An intensive reading course. Selections from Classical Greek writers and/or the New Testament.

Prerequisites: GRK 231 or permission of the instructor.

Health Sciences (HTH)

HTH 100 Personal Wellness [C5W]

Credits 3 Grading Basis Graded

Emphasizes lifestyle behaviors contributing to health promotion and disease prevention. General areas affecting health status are identified. Suggestions are made as to how health-related behaviors, self-care and individual decisions contribute to health and influence dimensions of wellness. May be used for general education credit.

HTH 150 Introduction to Health Sciences

Credits 2 Grading Basis Graded

This course orients students in the major to foundational expectations and requirements for successful completion of the Health Science major and Health Studies concentration. The course introduces conceptual ideas such as a wide array of health careers, importance of gaining early and regular field experiences, and writing and research basics.

HTH 204 Emergency Health Care

Credits 3 Grading Basis Student Option

A survey of various dimensions of the legal aspects of emergency care, cardiorespiratory emergencies, hemorrhage control, wounds, shock, heat injuries and other health emergencies. Selected American Red Cross and American Heart Association certifications available.

HTH 210 Medical Terminology

Credits 3 Grading Basis Graded

Study of terms that relate to body systems, anatomical structures, medical processes and procedures, and a variety of diseases/disorders that afflict human organisms.

HTH 231 Population Health Determinants

Credits 3 Grading Basis Graded

Many factors combine together to affect the health of individuals and communities. In an effort to advance public health, health professionals must be cognizant not only of behaviors which may compromise or promote health, but also the interaction of social and physical factors which impact health outcomes.

HTH 245 Foundations of Infectious Disease

Credits 3 Grading Basis Graded

An overview of the incidence, prevalence, causation, and prevention of the major infectious diseases which are currently of concern in the twenty-first century. Major signs and symptoms of the diseases as well as treatment will be reviewed. The course will also cover the body's defense system and the principles of disease occurrence.

Prerequisites: HTH 150. Corequisites: HTH 150.

HTH 255 Chronic Disease and Disabilities

Credits 3 Grading Basis Graded

An overview of the incidence, prevalence, causation and prevention of chronic diseases and disabilities across the lifespan. Major signs and symptoms of the diseases and disabilities as well as treatment will be reviewed.

Prerequisites: HTH 150. Corequisites: HTH 150.

HTH 314 Rural Health: An Interprofessional Approach

Cross-listed with: IPE 314, NSG 314, SOWK 314

Credits 3 Grading Basis Graded

Students study, observe and participate in interprofessional assessment, planning and delivery of community-based primary health care in partnership with residents and agencies of a host rural county. Learning activities will emphasize rural culture, rural health care and interprofessional practice.

HTH 315 Peer Leadership in Health Sciences

Credits 2 Grading Basis Credit / No Credit

Undergraduate students are provided a leadership opportunity to serve as a peer leader for a selected class. Teaching and learning occur under faculty mentorship. The student and faculty member collaborate to ensure proper levels of rigor and support, identify specific goals, discuss reflections of experiences, and make connections to future professions. Serving as a peer leader allows students to see college teaching from a new perspective. Course may be repeated for a maximum of 4 credits.

Prerequisites: Application and/or interview with supervising faculty member must be complete; student must have previously completed the course earning a "B" or higher; junior or senior status; and permission of the instructor.

HTH 320 Statistical Methods for Health Science Research

Credits 3 Grading Basis Graded

This course reviews statistical concepts and techniques with special reference to health science applications and issues. It also reinforces the logic of statistical decision making for health and medical research concerns.

Prerequisites: MATH 220 or COB 191.

HTH 351 Health Behavior Change

Credits 3 Grading Basis Graded

This course will identify individual and group characteristics that influence health behaviors and the features of programs designed to change behavior that makes them more or less effective. Students will develop the skills necessary to formulate health-modification programs. Prerequisites: HTH 231.

HTH 352 Environmental Health

Credits 3 Grading Basis Graded

An investigation of environmental factors and their effects on the health of the individual, community and society.

HTH 354 U.S. and Global Health Care Systems

Credits 3 Grading Basis Graded

The course provides an overview of the U.S. health care system by examining the organization, financing, and delivery of health services. The class will include current issues in health care policy, the future of health care systems in America, and a comparative global analysis of health care systems.

HTH 355 HIV/AIDS: A Global Perspective

Credits 1 Grading Basis Graded

Discussion includes theories of origins, statistics and characteristics of the causative pathogen, incubation, illness patterns, transmission, prevention and treatment of HIV/AIDS and other STDs. HIV/AIDS and other STDs in relation to prisons, children, schools, global concerns, health care systems and legal factors will be considered.

HTH 368 Global Health

Credits 3 Grading Basis Graded

This course provides students with an opportunity to critically explore and examine a wide range of social, economic, cultural and demographic themes in global health. Focus is given to the various ways health and health conditions (possibly including the ecology and etiology of major infections and chronic diseases, and the social and economic consequences of changes in mortality and health) are conceptualized, valued, prevented and treated in different contexts around the world.

HTH 370 Child and Adolescent Health**Credits 3 Grading Basis** Graded

This course will discuss the CDC priority health risk categories and the Healthy People Objectives for the Nation as they relate to child and adolescent health. In addition, the CDC model for Coordinated School Health Programs will be explored to determine specific strategies for health improvement.

HTH 372 Human Sexuality**Credits 3 Grading Basis** Graded

An in-depth study of sexuality across the lifespan. Emphasis is placed on the development of sexuality with attention given to the psychological, physiological, ethical and socio-cultural implications.

HTH 378 The Use and Effects of Drugs**Credits 3 Grading Basis** Graded

A study of the use and pharmacological properties of popular legal and illegal drugs and their effects on the health of individuals and society.

HTH 380 Health Communication**Credits 3 Grading Basis** Graded

The course will provide an introduction to health communication theory, research and practice. Students will learn how to design, communicate and evaluate effective health promotion messages.

Prerequisites: HTH 351. Corequisites: HTH 351.

HTH 389 Practicum in Health Sciences**Credits 1-3 Grading Basis** Graded

Selected practicum experiences which provide students with supervised practicum experiences.

HTH 390 Special Studies in Health Science**Credits 1-2 Grading Basis** Graded

An independent study for students to pursue individual projects or research related to a selected topic of current importance to the Health Science field. Students will be under the supervision of a faculty member. Course may be repeated if the topic varies for a maximum of 4 credits.

HTH 391 Special Topics in Health Sciences**Credits 3 Grading Basis** Graded

The courses offered under this title are designed to allow exploration of current topics in health science not covered elsewhere in the curriculum. Course content will vary by term and instructor. Course may be repeated for credit when course content changes.

HTH 408 Health Research Methods**Credits 3 Grading Basis** Graded

This course will present an overview of research methods within public health, emphasizing the steps involved in the research process. Methodological issues covered will include the ethics of health studies research, qualitative and quantitative research designs, operationalization of concepts, measurement of variables, and techniques of sampling, data collection and analysis.

Prerequisites: HTH 320 and HTH 351.

HTH 417 Women's Global Health and Human Rights**Cross-listed with:** NSG 417, WGSS 417**Credits 3 Grading Basis** Graded

An international and human rights approach providing an overview of health issues within the context of a woman's life cycle. Attention will be given to critical issues of women's health such as access to health care and gender-based violence. Such issues as sexuality, nutrition, diseases affecting women, violence, harmful traditional practices, and sex trafficking will be discussed.

HTH 420 Global Perspectives on Alcohol and Drug Policies**Credits 3 Grading Basis** Graded

This course will study drug policy from a public health perspective. Students will examine global alcohol and other drug policies assessing their impact at the regional, state, national and international levels. Emphasis will be given to the contemporary phases of alcohol and other drug policies and the war on drugs.

HTH 422 Public Health and Healthcare Policy**Credits 3 Grading Basis** Graded

This course will provide a framework for understanding and analyzing a range of health policy issues such as accessibility, cost and quality of healthcare; disaster preparedness; the safety of our food, water, environment and medications; and the right to make individual decisions about our personal health. The course will explore the U.S. policy-making system, civic engagement, the essential issues in health policy and how health policies impact the work of health professionals.

Prerequisites: HTH 231 and HTH 354.

HTH 423 Ethics and Critical Thinking in Health**Credits 3 Grading Basis** Graded

This course explores ethical issues facing the public health and healthcare systems in the United States. Key ethical frameworks are used to analyze contemporary public health and healthcare issues and the overlap and distinctions between them. The course will review fundamental and emerging topics (individual rights vs. the common good, genetics, health promotion and disease prevention) and challenge students to think critically about them. Student will play an active role in presenting and evaluating ethical concepts and conflicts to facilitate class discussion.

Prerequisites: HTH 351 and HTH 354. Corequisites: HTH 354.

HTH 441 Biomechanics for the Healthcare Professional**Credits 3 Grading Basis** Graded

This course is designed to introduce the student to a variety of biomechanical concepts and applications as related to the health professions. Specific attention will be given to the biomechanical aspects of the musculoskeletal system.

Prerequisites: BIO 290/BIO 290L.

HTH 450 Epidemiology**Credits 3 Grading Basis** Graded

A study of the causation and prevention of the major diseases that affect the quality of an individual's life. Practical skills utilized by practicing epidemiologists are emphasized.

Prerequisites: HTH 320 and HTH 351; or senior health services administration major.

HTH 456 Grant Writing in Health Sciences**Credits 3 Grading Basis** Graded

Increasingly, social service and health providers are required to have grant writing skills and/or be responsible for obtaining external funds to support their positions, fund projects or promote programs. This course will provide the overview of grantsmanship skills. These skills include locating funding sources, improving collaboration skills, understanding program development and preparing grant proposals.

Prerequisites: HTH 351 or permission of the instructor. Corequisites: HTH 351.

HTH 458 Health Program Planning and Evaluation**Credits 3 Grading Basis Graded**

This lecture and laboratory course introduces students to principles and techniques employed to plan, implement and evaluate health promotion programs. Students will conduct a campus or community event and compile a formal report.

Prerequisites: HTH 351; or health services administration major and HSA 358; or senior substance use disorder education minor.

HTH 471 Health Aspects of Gerontology**Credits 3 Grading Basis Graded**

Promotion of health in the aged; physiological aspects of the aging process; community, state and federal health programs and services for the aged.

HTH 490 Advanced Special Studies in Health Science**Credits 3 Grading Basis Graded**

An advanced opportunity to conduct research where the student will apply their knowledge of Health Science to a specific research project or area. Students will be under the supervision of a faculty member. Course may be repeated if the topic varies for a maximum of six credits.

Prerequisites: Senior standing.

HTH 495 Internship in Health Organizations**Credits 3 Grading Basis Graded**

Full-time directed field experience in a health organization. Opportunity provided to work in an appropriate setting. Student furnishes off-campus living and traveling expenses.

Prerequisites: Permission of the instructor and a 2.5 grade point average.

HTH 499A Honors**Credits 1 Grading Basis Graded**

The purpose of the first semester honors capstone project is to develop a project outline, select an advisor and two readers, and submit the proposal with the completed signature page to the Honors College by the deadline.

Corequisites: HTH 320 and HTH 408.

HTH 499B Honors**Credits 3 Grading Basis Graded**

The purpose of the second semester honors capstone project is to complete an introduction to the research topic; complete a thorough literature review and a methodology section of a research paper; complete IRB training; complete an IRB application; and begin data collection.

Prerequisites: HTH 499A . Corequisites: HTH 499A.

HTH 499C Honors**Credits 2 Grading Basis Graded**

The third semester of the honors capstone project is designed for students to complete their projects. They should conduct data analyses if appropriate, and write summary, conclusions and recommendations (the equivalent of Chapters 5 and 6). In addition, students will prepare a research poster for the honors symposium.

Prerequisites: HTH 499B. Corequisites: HTH 499B.

Health Services Administration (HSA)

HSA 290 Gerontology for Health Services Administration**Credits 3 Grading Basis Graded**

This course provides an introduction to the study of aging from a multidisciplinary perspective including the biological, psychological and sociological aspects of aging. Emphasis will be placed on theoretical and practical application of course content to careers in health services administration.

Prerequisites: Fully admitted HSA majors only.

HSA 358 Health Services Administration Cornerstone**Credits 3 Grading Basis Graded**

This course provides an introduction to management functions, tasks and roles as they are carried out in health services organizations.

Discussion of emerging issues affecting the management of health services organizations is provided. This course uses the case method of analysis to develop critical thinking skills.

Prerequisites: Fully admitted HSA major and HTH 354.

HSA 360 Health Care Marketing**Credits 3 Grading Basis Graded**

This course introduces the role, functions and tasks of health care marketing. Attention is devoted to understanding basic marketing principles; using oral, written and visual electronic communications media; and developing marketing plans for health care organizations.

Prerequisites: Fully admitted HSA majors only.

HSA 363 Health Economics**Credits 3 Grading Basis Graded**

This course explains how economic forces affect the health service sector and how economic tools can be used to assess and improve health industry performance. Students examine issues related to efficiency, effectiveness, value and behavior in the production and consumption of health and healthcare.

Prerequisites: Fully admitted HSA major, ECON 201 and HTH 320.

HSA 365 Healthcare Ethics and the Law**Credits 3 Grading Basis Graded**

This course provides an overview of health ethics and health law for students majoring in health services administration. Students will address the major principles facing a health service professional in the delivery of health services. Particular attention will be paid to development of methodologies for ethical decision-making.

Prerequisites: Fully admitted HSA major and HTH 354.

HSA 366 Health Politics and Policy**Credits 3 Grading Basis Graded**

This course introduces students to health policy at the local, state and national levels. Students will explore the policy-making process from agenda-setting through implementation. External determinants and stakeholders will be identified to examine their influence on policy. Emphasis will be given to how health policy impacts health service organizations and the delivery of healthcare.

Prerequisites: Fully admitted HSA major and HTH 354.

HSA 367 Comparative International Health Systems**Credits 3 Grading Basis Graded**

This course concentrates on various approaches used to provide public and personal health services around the world. Strengths and weaknesses of different health care systems in terms of availability, accessibility, acceptability, cost effectiveness, continuity and quality of services will be considered.

Prerequisites: Fully admitted HSA major and HTH 354.

HSA 385 Professional Development Seminar**Credits 1 Grading Basis Graded**

This course is a career development seminar for health services administration students. An array of health administrators from varied health care organizations helps expand students' understanding of the health administration field. This course is designed for junior-level students with an expectation of a major concentration in health services administration.

Prerequisites: Fully admitted HSA majors only.

HSA 390 Special Topics in Health Services Administration**Credits 1-3 Grading Basis Graded**

A course for students to examine a selected topic or critical issue of current importance to the field of health services administration. Topics will vary. Courses offered under this title are designed to allow for exploration of current topics in health services administration not covered elsewhere in the curriculum. Students will work under the supervision of a faculty member. Course may be repeated for a maximum of six credits if the topic varies.

HSA 452 Hospital Organization and Administration**Credits 3 Grading Basis Graded**

This course examines the organization and operation of community hospitals in the U.S. Specific attention is devoted to management's role in internal operations and in external relationships with the community and other stakeholders. Discussion of emerging issues affecting the management of hospitals and hospital systems is provided.

Prerequisites: Fully admitted HSA major, HTH 354 and HSA 358 or permission of the instructor.

HSA 454 Internship in Health Organizations**Credits 3 Grading Basis Graded**

Full-time directed field experience in a health organization (320 hours). Opportunity provided to work in an appropriate setting. Student furnishes off-campus living and traveling expenses.

Prerequisites: Fully admitted HSA major, HSA 385, HSA 464 and two of: HSA 452, HSA 455, HSA 456 or permission of the instructor. Corequisites: HSA 385, HSA 464 and two of: HSA 452, HSA 455, HSA 456 or permission of the instructor.

HSA 455 Long Term Care Organization and Administration**Credits 3 Grading Basis Graded**

This senior-level course addresses health services administration students about the health care operations across the continuum of long-term care. Course emphasis will be placed on the delivery, financing, regulation and management of post-acute and long-term care services in the community and institutional settings. Other relevant topics will include the long-term care workforce, quality improvement and current trends in long-term care policy. This course will consist of subject lectures, guest speakers, site visits and case studies.

Prerequisites: Fully admitted HSA major and HSA 358.

HSA 456 Principles of Ambulatory Care Operations**Credits 3 Grading Basis Graded**

This course provides an in-depth investigation of the operational requirements in ambulatory care including medical group practices, community and hospital-based clinics; urgent care centers; and ambulatory surgery facilities. Areas of focus will include staffing, budgeting, facility management, facility design, information technology, regulatory requirements and innovation.

Prerequisites: Fully admitted HSA major, HTH 354 and HSA 358 or permission of the instructor.

HSA 462 Managed Care**Credits 3 Grading Basis Graded**

The growing influence of managed health care on the organization and delivery of health services in the United States is addressed in this course. Structural and operational characteristics of managed care organizations and plans, including HMOs, PPOs and other plans are explored, as are the implications of managed care plans for the management of hospitals and other health care organizations.

Prerequisites: Fully admitted HSA major and HSA 358.

HSA 463 Healthcare Quality Management and Analytics**Credits 3 Grading Basis Graded**

The purpose of this course is to investigate the evolution of healthcare quality improvements and current initiatives in the industry, and to develop the knowledge and skills necessary to plan and conduct healthcare quality improvement activities across various healthcare organizations. Emphasis will be made on current industry quality procedures and anticipated governmental regulatory initiatives that impact health care organizations' operations and reimbursement based on quality outcomes. Students will explore the value of healthcare analytics and its role in tracking, identifying, analyzing and shaping proactive, safe, effective, quality strategic initiatives.

Prerequisites: Fully admitted HSA major, HTH 354 and HSA 358 or permission of the instructor.

HSA 464 Healthcare Finance**Credits 3 Grading Basis Graded**

This course provides a general understanding of the fundamental concepts and tools of healthcare organizational financial management, as applied to healthcare organizations. The course examines payment sources and reimbursement arrangements; the public and private financing of healthcare service organizations from both a theoretical and practical perspective; and discusses emerging trends in the healthcare industry that affect financial decision-making.

Prerequisites: Fully admitted HSA major, FIN 345, HTH 320 and HSA 358.

HSA 480 Health Services Administration Capstone**Credits 3 Grading Basis Graded**

This capstone course integrates the student's knowledge and skills obtained throughout the health services administration curriculum. Students examine a variety of topics, culminating in the identification, critique and development of salient recommendations on current trends and issues affecting our health care system. The course utilizes a multidisciplinary, multi-system approach examining solutions from organizational behavior, human resources, strategic planning, business and organizational leadership.

Prerequisites: HSA 452, HSA 455 and HSA 456. Corequisites: HSA 452, HSA 455 and HSA 456.

HSA 490 Independent Studies in Health Services Administration**Credits 1-3 Grading Basis Graded**

An independent study for students to pursue independent research or experiential learning opportunity in health services administration under the guidance of a faculty advisor. Students will work under the supervision of a faculty member. Course may be repeated for a maximum of six credits if the topic varies.

History (HIST)

HIST 101 World History to 1500 [C2HQC]

Credits 3 Grading Basis Graded

A survey of important historical developments from prehistoric times to 1500. Emphasis is given to the rise and decline of great world civilizations and their lasting contributions to humanity. May be used for general education credit.

HIST 102 World History since 1500 [C2HQC]

Credits 3 Grading Basis Graded

A survey of important historical developments from 1500 to the present. Emphasis is given to the growth of nationalism, the development of colonialism, and to world events, problems, and conflicts of the present century. May be used for general education credit.

HIST 150 Critical Issues in Recent Global History [C1CT]

Credits 3 Grading Basis Graded

This course examines issues in recent history as a means to introduce, develop and enhance critical thinking skills and to supplement writing, oral communication, library and computing skills objectives for General Education Madison Foundations. A seminar format allows for careful examination of issues in both oral and written formats. The course emphasizes the development and articulation of well-reasoned arguments in organized and grammatically acceptable prose. May be used for general education credit. May not be used for major credit.

HIST 225 U.S. History [C4AE]

Credits 4 Grading Basis Graded

A survey of U.S. history from the Colonial period to the present, emphasizing the development of American civic life, the involvement of the U.S. in world affairs and the cultural richness of the American people. This course stresses the analysis and interpretation of primary sources. May be used for general education credit.

HIST 239 Topics in History

Credits 3 Grading Basis Graded

The study of selected topics in history at the introductory level.

HIST 263 Africa

Credits 3 Grading Basis Graded

Emphasis is placed on the social and cultural aspects, as well as the emerging role the continent plays in contemporary world history.

HIST 274 Modern East Asia

Credits 3 Grading Basis Graded

This course is an introduction to modern historical experiences of East Asia, particularly China, Japan and Korea. In addition to overviews of each of these countries, the course will focus on several topics illustrating both the unity and diversity of East Asia: perceptions of each other, the philosophical tradition of Confucianism, the role of imperialism and nationalism, revolution, reform and the future of the region in the twenty-first century.

HIST 299 Putting History Skills to Work

Credits 1 Grading Basis Credit / No Credit

This course is aimed at helping our majors transition to careers using history. Students will learn about where to look for a job, how to get it and how to thrive in it. Students will hear about the career paths of history majors. Students will produce a job-market oriented product such as a cover letter, a resume or other product that would help them in searching for a post-college job.

Prerequisites: History major.

HIST 300 Historical Thinking

Credits 3 Grading Basis Graded

A seminar that introduces students to the methodological questions central to the discipline of history. Students learn how to recognize and appreciate the diversity of historical interpretation; how to identify, locate and interrogate secondary and primary sources; how to integrate critical analysis of evidence into writing; and how to employ different modes to present historical knowledge. Open to all students but required of history majors.

HIST 302 Plazas, Protests and Processions: An Urban History of the Americas

Credits 3 Grading Basis Graded

There is no group of people in the world more urban-minded than Latin Americans. Historically, cities in this region played an all-encompassing role that included administration as well as the reproduction of capital and responsibility for virtually all cultural activities. This class explores that history alongside plans for further urban development, cultural activities and architectural design.

HIST 303 Virginia's Place in #VastEarlyAmerica

Credits 3 Grading Basis Graded

This course will examine the history of early America from the colonial to the early national period. Topics will include the clash of African, European and Native American cultures, the regionalization of the American colonies, the growth of American slavery, and the creation of an American character and politics.

HIST 304 Native Peoples of the United States

Credits 3 Grading Basis Graded

A survey of the experiences of Native Peoples, including Native Hawaiians and Native Alaskans, from the pre-contact period to the present, with emphasis on traditional lifeways, impact of Anglo Americans, cultural persistence and revivals, activism, and contemporary challenges to Native communities and identities.

HIST 305 History of Science and Christianity

Credits 3 Grading Basis Graded

Over the last 2000 years, there have been recurring controversies over the proper relationship between science and Christianity. This class uses case studies such as Galileo, Darwin and creationism to explore the larger cultural context that gave life to the controversies. In the process, we'll examine changing ideas of what counts as science, how to interpret the Bible, and who gets to decide.

HIST 306 A History of the Body in the West

Credits 3 Grading Basis Graded

This course views the human body as a historical artifact whose physical appearance and social, cultural and political meanings reflect the historical contexts of specific times and places. The emphasis is on the perspectives of Europeans and their descendants, inside Europe and beyond it.

HIST 307 The Trans-Atlantic Slave Trade

Credits 3 Grading Basis Graded

This course explores the origins, processes and outcomes of the infamous trade. By studying participants' lives in Africa, Europe, Latin America and North America, the course helps students understand people's inhumanity to each other and the ways in which slavery and the trade in slaves forever altered the development of the Atlantic world.

HIST 308 Contemporary Italian History and Culture

Credits 3 Grading Basis Graded

A study of Italian society, economics, politics and the arts from 1814 to the present.

HIST 309 French History since 1648**Credits 3 Grading Basis** Graded

A survey of important historical developments in France from 1648 to the present. It explores how complex historical legacies in French society define and shape the experience of "being French" and how different groups and citizens work with and against each other in a collective effort to define the early modern and modern French experience. Points of focus include economy, society, culture and religion, state, politics, and borders.

HIST 311 China to 1600**Credits 3 Grading Basis** Graded

A survey of Chinese history from its earliest origins to the fall of the Ming Empire in the 17th century. The course incorporates translated primary materials and modern scholarship to provide a general picture of Chinese state formation, socio-economic changes, and literature and arts during the three millennia. It will also discuss the teachings and intellectual impacts of Confucianism, Daoism, and Buddhism in East Asia through the ages.

HIST 312 Japan since 1600**Credits 3 Grading Basis** Graded

A survey of Japanese history from the end of the Warring States Period to the present. It first introduces Tokugawa society in the 17th and 18th centuries and then examines the Meiji period in detail, focusing on political and cultural changes, the rise of ultra-nationalism, the military expansion before WWII, and the course of the war. The last part of the course will discuss the American occupation, postwar developments, and the last two decades in Japan.

HIST 313 The History of Latinos in the United States**Credits 3 Grading Basis** Graded

Currently the largest minority group in the U.S., Latinos have diverse sociological and historical origins, yet today in the U.S. are thought of as a monolithic racial and ethnic group. Starting with the Spanish and Portuguese colonial enterprise and continuing through current debates, this class will provide an introduction to and overview of the broad scholarship on Latino and Hispanic identity in the United States, examining issues of coercion, migration, refugees, labor, family networks, religion, war, economics, state building, politics, racialization, international disputes and globalization.

HIST 314 Plagues, Saints and Empires: The World before 1500**Credits 3 Grading Basis** Graded

This course will look at themes and patterns in world history before 1500. Topics will include practical issues, such as ways to feed people and develop trading networks; different ways to run increasingly complicated societies, from small nomadic groups to cities to empires; patterns of disease and epidemics and environmental changes; the changing roles of ideas and religions and education; varying ideas about gender roles and class structures; and many ways people tried to integrate different peoples, cultures and ideas into a unified society.

HIST 315 Global History of Sport**Credits 3 Grading Basis** Graded

An interpretive study of the role of sports globally. Time periods and specific nations will vary depending on the faculty member teaching the course, but all will consider sport across time and place.

HIST 316 The Life and Times of James Madison, 1751-1836**Credits 3 Grading Basis** Graded

An overview of the major political, philosophical, social and literary events that helped shape the world of the founders. James Madison's life will provide the framework for the course and emphasis will be given to his important role during this era.

HIST 317 Plagues and Pandemics in Modern World History**Credits 3 Grading Basis** Graded

Epidemics captivate and terrify us. Americans and others around the world are obsessed with epidemic diseases and their social, cultural, political and economic effects. But why are we so fascinated by epidemics? How have humans responded to epidemics historically, and have those responses varied by place or evolved over time? This course seeks to answer these questions and others by examining the history of epidemic diseases in modern world history, using selected case studies to investigate the social, cultural, political and economic effects of disease outbreaks.

HIST 318 Our Better Angels?: The World History of Violence**Credits 3 Grading Basis** Graded

An interdisciplinary and comparative exploration of human violence. The course uses theoretical studies and empirical research to address big questions about the kinds, causes and consequences of violence in human societies from the deep past to the present. Students examine everyday and extraordinary forms of violence, debate if and how violence has declined across time, and learn to use historical methodologies to better understand conflict and coercion in today's world.

HIST 319 Women at Work in U.S. History**Credits 3 Grading Basis** Graded

This course examines the experiences of women employed in a variety of industries and workplaces in twentieth-century U.S. history. Topics include women's employment opportunities, experiences of sexism and other biases, collective action to change working conditions, innovation and entrepreneurship of women, and popular culture portrayals of gender in workplaces.

HIST 320 Women in U.S. History**Credits 3 Grading Basis** Graded

PeopleSoft Course ID 005864 Grading Basis GRD A survey of the role of women in the United States from the Colonial period to the present. Attention is given to contributions of the ordinary women, the Women's Rights movements, the impact of women on reform and political movements, and the changing status of women in society.

HIST 321 Gender and Sexuality in Modern Europe**Credits 3 Grading Basis** Graded

This class examines categories of gender, masculinity and femininity, as well as patriarchy and sexuality in the context of Modern Europe. The course will cover the significance of Republican Motherhood, the birth of feminism, the effects of empire, war and dictatorship on gender and sexuality, and the contradictions of sexuality and morality in Europe from the 18th to 20th centuries. With an emphasis on transnationalism, the course gives special attention to issues of gender and sexuality in England, France, Germany and the former Soviet Union. These same topics will be discussed in the context of European Empire.

HIST 322 Jim Crow South**Credits 3 Grading Basis** Graded

During the era of Jim Crow—a period that spanned the years from the 1890s to the civil rights movement of the 1950s and 1960s—state laws, municipal ordinances and social customs racially segregated public and private life across the American South. Jim Crow, as both legal framework and social practice, shaped the everyday lives of three generations of Southerners, both black and white. This class examines the complexities of segregation—its legal boundaries, its social and cultural peculiarities, and its violent enforcement. It especially focuses on the political actions that challenged and eventually dismantled it.

HIST 323 The Old South**Credits 3 Grading Basis Graded**

Economic, cultural and social history of the antebellum South, 1790-1860. The region's political history will serve as a supporting part of the course.

HIST 324 Terror and Survival in American Lynching**Credits 3 Grading Basis Graded**

This course focuses on the history and politics of racial violence in the United States since Emancipation. It explores what it means to bear witness to these difficult, disturbing, suppressed, and yet critical events that continue to define the present world that we inhabit. The course thus asks that we encounter disturbing, violent and racially sensitive material. But we do so out of a sincere conviction that we cannot find a path to restitution, reconciliation and repair without first acknowledging, confronting and reckoning with the crimes of the past.

HIST 325 The Roman Empire**Credits 3 Grading Basis Graded**

This course is an introduction to the history of Roman imperialism and the Roman Empire (ca. 60 BCE - ca. 600 CE). Students will approach the past both chronologically and thematically, with an emphasis on cultural, social, military and political history, and on daily life and lived experiences throughout the Roman empire. Students will study ancient texts and artifacts and become familiar with major scholarly views and debates about the Roman empire.

HIST 326 How Cars Changed Everything**Credits 3 Grading Basis Graded**

This course uses the automobile as a window into 20th century American life. It examines the influence of automobility on patterns of work and leisure; on struggles over gender, race and ethnicity; on individualism, consumerism and government regulation. It also surveys mass automobility's effects on our physical and natural environments and looks at future prospects of automobility in the information age.

HIST 327 Makers, Users, Skeptics and Thieves: Americans and Their Technologies**Credits 3 Grading Basis Graded**

Americans -individuals and groups- have used technology to do many things; some good, some bad. Never has it been neutral. This course explores the complex and changing relationship between technology and American society from Native American canoes in the 17th century to Facebook trolling in the early 21st Century, paying particular attention to technology's role in America's dynamic gender, race and workplace relationships along the way.

HIST 328 Towers, Time and Technology on the Eve of World War I**Credits 3 Grading Basis Graded**

This course traces the intellectual, technical, mechanical and scientific developments that had a profound effect on the ways people saw and understood their world from 1880-1914. The course explores peoples' relationships to the environment; state structures; and turn-of-the-century revolutionary political, economic and cultural ideas with a focus on Europe, the US and the impact of "western" technological change on the world.

HIST 329 Frederick Douglass's America: The U.S. in the Civil War and Reconstruction Era**Credits 3 Grading Basis Graded**

The Civil War and Reconstruction are arguably the central events in United States History. But in the last several decades, the ways in which historians and other Americans interpret and remember the Civil War era have radically changed. This course critically examines these developments through the eyes of Frederick Douglass, a radical abolitionist and self-proclaimed "woman's rights man," who profoundly shaped Americans' views on slavery, abolition and civil rights for Black and female Americans.

HIST 331 Historical Archaeology**Cross-listed with: ANTH 331****Credits 3 Grading Basis Graded**

The course introduces students to the purposes, subject matter, methodology and historical background of the discipline of historical archaeology. Building on research issues and methodologies of anthropological archaeology and history, the multidisciplinary aspects of this field are introduced through field trips, projects, guest lectures, readings and classroom presentations. Prerequisites: Prerequisite: ANTH 197.

HIST 332 History of Twentieth-Century Spain**Credits 3 Grading Basis Graded**

This course will trace the twentieth-century political and social history of Spain including the Second Republic, the Spanish Civil War, the Franco regime, and the transition to democracy in 1975. The course will pay special attention to Franco's dictatorship, the role of women, the Catholic Church, as well as Spain's relationship to the rest of Europe.

HIST 333 Maps, Money, Manufacture and Trade**Credits 3 Grading Basis Graded**

This is an interdisciplinary class designed to integrate perspectives from various disciplines, especially history, geography and economics, into a coherent account of an increasingly globalized world. Particular attention will be paid to map construction and use (both historical and contemporary) and the relationship between economic ideas and world events. The class will focus on a variety of case studies over the last millennium.

HIST 334 A Global History of Nationalism**Credits 3 Grading Basis Graded**

This course examines the modern origins of nationalism, its historical development as a political and economic theory, and its consequences, including ethnic cleansing and genocide, in a variety of case studies in order to answer these and other questions. The course will consider questions such as: What is nationalism? Do nations create nationalism or does nationalism create nations? Why does the modern world vest sovereignty in the nation? This course is intended to fulfill requirements for the International Affairs major.

HIST 335 Topics in Applying History**Credits 3 Grading Basis Graded**

This course guides students in applying historical thinking and the use of evidence outside the classroom. The course meets the Applying History requirement. Topics and approaches will vary by instructor. Course can be repeated for credit with different instructors and themes.

HIST 336 We the People?: The American Revolution Considered**Credits 3 Grading Basis** Graded

This course will advance understanding of the political, economic, social, cultural and intellectual history of the events that led to the formation of the United States as an independent democratic republic. This dynamic course considers the causes and consequences of the American Revolution, and examines questions such as: How high were taxes really? Why was throwing tea in the harbor the height of political intrigue? Why does every politician claim they are doing what the Founding Fathers would have wanted - and are all those other folks just wrong?

HIST 337 Creating the United States Constitution**Credits 3 Grading Basis** Graded

This course examines the intellectual and political negotiations that produced the United States Constitution. It explores the contemporary understandings of the pressing political crisis that confronted the U.S. in the period immediately following the Revolution and that demanded the prompt revision of the continental government. The bulk of the semester is devoted to a simulation of the Philadelphia Convention of 1787, utilizing an innovative pedagogy known as Reacting to the Past.

HIST 338 United States Urban History**Credits 3 Grading Basis** Graded

In this course students explore the history of urban spaces in the United States by investigating American cities using a broad chronological, geographical, and thematic framework. Drawing from an array of primary and secondary sources, students will become familiar with the central themes of urban history, and also have the opportunity to produce their own narratives.

HIST 339 Selected Themes in U.S. History**Credits 3 Grading Basis** Graded

Selected themes are studied in depth. See MyMadison for current classes. Course may be repeated when content changes.

HIST 340 Internship in History**Credits 3 Grading Basis** Graded

Provides students with practical experience in using historical skills in a public or private entity. Periodic student reports and seminar meetings required. This course may be repeated with permission of department head.

Prerequisites: HIST 300, junior or senior standing, and permission of the history internship coordinator.

HIST 341 Selected Themes in World History**Credits 3 Grading Basis** Graded

Selected themes are studied in depth. Course may be repeated when content changes. Only courses with significant content outside of Europe will count toward the world history requirement. See MyMadison and the history department website for information on current classes.

HIST 342 Selected Themes in European History**Credits 3 Grading Basis** Graded

Selected themes are studied in depth. See MyMadison and the history department website for information on current classes. Course may be repeated when content changes.

HIST 343 History of Today**Credits 3 Grading Basis** Graded

In this course, students will use history to help understand the present. Whether it's applying historical questions to contemporary settings, leveraging historical skills to analyze current events or offering historical context to divisive issues, history can aid in thinking through the problems of today and even the possibilities of tomorrow.

HIST 344 Magic Mountains, Man-eating Beasts and Unquiet Woods: An Environmental History of British India**Credits 3 Grading Basis** Graded

This course examines the history of British India during the 19th and 20th centuries in relation to how the British colonial state interacted with India's natural environment. Students will learn how the British attempted to understand and control India's hilltops, forests and wildlife in effort to govern more efficiently, profitably and effectively. They will learn that this was not without challenges, however, as colonized subjects spoke up and acted out against British authority.

HIST 345 Pills and Potions: Histories of U.S. Health Care and Medicine**Credits 3 Grading Basis** Graded

This course explores the history of attempts to heal Americans from disease over four centuries. The course will consider the complex interplay of scientific ideas about disease with the social and cultural contexts that shaped the experiences of people in accessing health care. Students will consider the increasing success with which medicine has healed the body, and the development of the medical professions from the late eighteenth century to the present. The course will also consider issues of public health as opposed to individual medical practice. Governments' and corporations' inevitable influence on public health measures, as the example of cigarettes shows, will be addressed. The influence of race, class and sex on how bodies are viewed and healthcare is received will be discussed.

HIST 346 Law and Order: British India**Credits 3 Grading Basis** Graded

This course covers the new British criminal code that redefined what constituted a crime and identified who was a criminal in the 19th century. Students will examine the nature of elite classes and the character of their relationship with subordinate classes. In addition, students will examine crime statistics and crime reports that provide information about behavior and experiences of people at the bottom of colonial society.

HIST 347 Advertising America**Credits 3 Grading Basis** Graded

This course provides an analysis of advertising's roles in U.S. society from the late nineteenth century through the present. The course uses advertisements and other primary sources to consider how the advertising industry has shaped and reflected social norms, including ideas about gender, race, class, religion, sexuality and age. It will also consider the influence of advertising on art, entertainment and politics.

HIST 348 Maps and Mobility in Islamic World History**Credits 3 Grading Basis** Graded

Islam arose out of established patterns of commercial mobility and cultural exchange, but it also produced new local and global movements and circulations that are a primary concern of this course. The annual hajj pilgrimage to Mecca, pastoral and nomadic seasonal migrations, inland and port city market relations, imperial capital supply systems, and Islamic travel literature anchor the mobility pillars of the course. While Islam generated new kinds of migrations and interactions, it also produced new cartographic structures and sensibilities built around new geographies, sciences and world views. European colonialism and nationalist movements in the Islamic world generated additional historical layers of mapping pressures, possibilities and realities. Islamic, European and national maps in the Muslim world form the cartographic pillars of the course.

HIST 349 History, Site, Memory**Credits 3 Grading Basis** Graded

This course explores the interplay between memory, history and place. Through readings, research projects and discussion, this course explores how recent trends are shaping the way we approach place and historical memory.

HIST 350 Virginia**Credits 3 Grading Basis** Graded

An interpretive survey of the history of Virginia from its Colonial beginnings to the present time.

HIST 351 War in Afghanistan: Local and Global Perspectives**Credits 3 Grading Basis** Graded

This course offers critical perspectives on war in Afghanistan from historical and contemporary local and global perspectives. This course considers the philosophy of war and the Anglo-Afghan Wars (1839-1919). The course then deals with U.S.-Afghan relations during the Cold War, and covers the post-9/11 period. In the context of war in Afghanistan, the course provides critical analyses of democracy, inequality, neoliberalism, power and race.

HIST 352 The Premodern Middle East: Themes and Intersections**Credits 3 Grading Basis** Graded

A thematic introduction to the premodern Middle East, the course emphasizes the political, social and religious developments that have shaped the region's diverse communities and connected them to world history.

HIST 353 The Global Middle East**Credits 3 Grading Basis** Graded

This course positions the Middle East in global cultural processes since the rise of Islam. It focuses on linguistic, commercial, religious, imperial and national development and interaction in the region between the Nile and Indus Rivers. The first part of the course considers Arab, Persian and Turkish cultural and historical contributions to the rise and expansion of Islam from roughly 600-1500. The second part of the course examines the Ottoman, Safavid and Mughal empires with an emphasis on literary and artistic developments until about 1800. The third part examines European colonialism and nationalist activity in Palestine and Iran, and projections of American interests through educational exchange and development programs from the Cold War through the present.

HIST 356 African American History since 1865**Credits 3 Grading Basis** Graded

A survey of the experience and changing status of African Americans in the United States from Reconstruction to the present, emphasizing the strengthening of social and cultural institutions; Afro-American leadership; the impact of segregation; the Great Migration; labor, protest and cultural movements; pan-Africanism; the Civil Rights Movement; and contemporary issues.

HIST 357 Fates of Empires**Cross-listed with:** POSC 457**Credits 3 Grading Basis** Graded

Fates of Empire looks at several different historical empires to gain an interdisciplinary perspective on what makes empires succeed or fall apart. The course will apply a variety of theoretical approaches from history, political science, economics and geography to consider each case study and the broader concept of empire. This class is particularly aimed at students in the social studies teacher licensure programs, to help them learn how to apply the history in middle and high school classrooms.

Corequisites: For students with a Social Studies Minor, this course must be taken with MSSE 412.

HIST 358 Modern Africa, ca. 1800-1990s**Credits 3 Grading Basis** Graded

This course examines selected critical themes in the history of modern Africa. It explores historical transitions and challenges in making societies, economies and states. The thematic focus transcends traditional chronological boundaries to enable students to understand continuities and disruptions, the historical trajectories of Africa's complex situation, and its position in the world.

HIST 359 History and Public Policy**Credits 3 Grading Basis** Graded

History provides a powerful way to think through complicated dilemmas common for policy analysts, practitioners and advocates. This course will consider how history can shape approaches to current U.S. and global policy problems. No previous experience in history is required.

HIST 360 Research Apprenticeship in History**Credits 3 Grading Basis** Graded

Provides students with advanced research and writing opportunities. Student learning contract must be approved before a student can enroll. Periodic student reports and seminars required. Open to history majors only.

HIST 362 American Material Religion**Credits 3 Grading Basis** Graded

This course explores material religion in the history of the United States. Building on the insight that religions are embodied, material forms of practice, the course investigates the place of food, clothing, objects, buildings and landscapes in an array of religions throughout U.S. history.

HIST 365 Topics in Connected History**Credits 3 Grading Basis** Graded

The Connected History requirement offers classes that think historically about big-picture issues in a global context. Classes might consider regions or the globe. Topics and areas of the world will change with each instructor. Students may repeat this class for credit with different themes and instructors.

HIST 366 Topics in Public History**Credits 3 Grading Basis** Graded

This course is organized around a particular public history subject, theme or approach. Topics will vary by instructor. The course can be repeated for credit when the topic changes.

HIST 368 Blurred Borders: A Cultural History of the Caribbean, Latin America and the United States**Credits 3 Grading Basis** Graded

This course explores the political, social and diplomatic relations between the United States, the Caribbean and Latin America. In each of these regions, Europeans, Africans and indigenous peoples came together in unequal colonial systems after 1492. The course will analyze cultural expressions such as theater, literature, music, film, poetry, essays and diplomatic correspondence to examine the interconnectedness of these regions and explore how the identities of these territories are shaped in connection to one another.

HIST 369 Problems in Greek History**Credits 3 Grading Basis Graded**

This course introduces students to the cultural, political, military, social, economic and intellectual history of the ancient Greek world by centering discussions and explorations of major controversies and theories that have shaped modern understandings of the history of the Greek world and its neighbors. Students will survey the broader history of ancient Greece, ca. 3000-30 BCE as the background to a focus on narrower themes like, but not limited to, civil war, environmental history, gender, identity and colonization. The course is a mixture of lectures, discussions, and collaborations in class. Students will read primary sources and modern scholarship.

HIST 370 Byzantine Empire**Credits 3 Grading Basis Graded**

A survey of the political, economic, military and religious history of the Byzantine Empire, 330-1453.

HIST 371 India**Credits 3 Grading Basis Graded**

A survey of the history of the Indian subcontinent from antiquity to the present. The course stresses the arrival of Islam, the impact of Western colonization, the struggle for independence, and the problems and achievements of nationhood in the post-Colonial era.

HIST 372 Modern Afghanistan**Credits 3 Grading Basis Graded**

This course addresses how the geographical space defined by the Hindu Kush section of Himalayan Mountain range and bordered by the Amu Darya or Oxus River in the north and Indus River in the east-southeast became known as Afghanistan. It examines repeated attempted imperial transformations of this decentralized frontier space to a bordered, centralized territorial state format from the eighteenth through the twenty-first centuries. It explores successive Mughal, Safavid, British, Russian and American policies, wars and occupations, as well as local adaptation, collaboration and resistance to those repeated imperial initiatives and agendas. Thematically, it engages questions about empire, state formation, social change, cultural resilience, warfare, technology and the environment.

HIST 373 History of the Ottoman Empire, 1453-1922**Credits 3 Grading Basis Graded**

This course surveys the history of the Ottoman Empire from its roots in medieval Anatolia through its demise in World War One. We will examine the persons, events, institutions, and ideas associated with imperial state formation, military expansion, cultural elaboration, political reformism, and the evolution of modern identities.

HIST 374 The Southern Plantation**Credits 3 Grading Basis Graded**

This course explores the plantations of the American South, both physical and imagined spaces. It challenges students to consider the spatial boundaries of the plantation, offering an examination of African trading posts, American slave markets and small courthouse towns along with the dwellings, outbuildings, fields and gardens of antebellum commercial slave agriculture. This broader spatial history framework uses a number of historical subfields, particularly architectural, environmental and public history, exposing students to a variety of scholarly perspectives. The class incorporates a wealth of primary material from diverse authors and points of view, including slave narratives, government records, travel accounts, letters, novels, paintings and architectural drawings, in order to demonstrate that the same physical landscape can be perceived differently depending on the viewer's perspective and experience. The course also includes on-site visits to two historic plantations.

HIST 377 History of Korea**Credits 3 Grading Basis Graded**

A survey of Korean history from its earliest times to the present day. It is designed to develop an understanding in Korea, its historical tradition and the place of Korea in the larger narrative of East Asia and world history.

HIST 378 China in the Modern World**Credits 3 Grading Basis Graded**

This course is an exploration of China's encounters with the modern world and the ways in which China has, and has not, changed as consequence of those encounters. Topics include the impacts of both Western and Japanese imperialisms; participation in international systems; adaptations of Christianity, democracy and communism; and the resulting upheavals in Chinese society.

HIST 382 Europe in the 20th Century**Credits 3 Grading Basis Graded**

This course is a survey of European history covering the late-imperial era, the world wars, the Cold War and the dynamics of European integration. Emphasis will be given to political, social, economic and cultural developments. Upon completing the course, students will be able to demonstrate knowledge of major movements, figures and events in twentieth-century European history.

HIST 383 Early England**Credits 3 Grading Basis Graded**

A survey of English history from the earliest times to the late 17th century. Particular attention is given to the rise of Parliament and the growth of limited monarchy.

HIST 385 The Russian Empire**Credits 3 Grading Basis Graded**

This course examines the history of Russia as a world power under tsarist rule, including the development of the autocracy, the adoption of Orthodoxy, and the establishment and abolition of serfdom. It explores attempts to reform Russia from above and below and investigates the relationships between state, society and the masses. It also considers the impact of gender, empire and revolution on imperial Russian history.

HIST 386 The Soviet Union: A Socialist Experiment**Credits 3 Grading Basis Graded**

In 1917, a party of revolutionaries established the world's first socialist state. Although inspired by the goals and lessons of the Enlightenment and the French Revolution, the lived experience of Soviet socialism was characterized by enormous suffering, violence and death. Yet during the more than 70 years of its existence, the Soviet Union also accomplished much that was good, even extraordinary. This course examines the main features, successes and failures of the Soviet socialist experiment from its revolutionary origins in the 1860s through its collapse in 1991.

HIST 387 The History of Ideas in the Modern World**Credits 3 Grading Basis Graded**

This course will examine the history of ideas from a range of disciplines in the nineteenth and twentieth centuries and how they have affected Western and global political, social and cultural trends. Instead of merely surveying ideas, the course will trace specific schools of thought over time, emphasizing the close analysis and contextualization of thinkers and ideas.

HIST 388 Germany since 1871**Credits 3 Grading Basis Graded**

A survey of German history during the Second Reich, World War I, the Weimar Republic, the Third Reich and the post-World War II periods of Cold War and Detente. Emphasis is given to political, diplomatic and military affairs, although social, economic and cultural developments are included.

HIST 389 Revolution and Radical Change in Modern Europe**Credits 3 Grading Basis** Graded

This course covers the turbulent history of revolution and radical political change in modern Europe, beginning with the French Revolution of 1789 and ending with the overthrow of the communist dictatorships in Central-Eastern Europe near the end of the twentieth century. Students will gain in-depth knowledge of the causes, chronologies and outcomes of modern Europe's many political revolutions. We will also examine how revolutions have-at times-generated tremendous social, economic and/or cultural change.

HIST 390 Special Studies in History**Credits 3 Grading Basis** Graded

This course is designed to give capable students in history an opportunity to complete independent study under faculty supervision.

HIST 391 Travel Studies Seminar**Credits 3 Grading Basis** Graded

Designed to encourage the student to augment the regular academic program through independent investigation including organized travel-study. Prearrangements must be made with a designated faculty member who will direct the study with preparatory instructions and final requirements.

HIST 394 Introduction to Museum Work**Cross-listed with:** ARTH 394**Credits 3 Grading Basis** Graded

A study of the philosophy and practice of museum work including the areas of exhibit design, conservation registration, education and administration. Subject is taught from the perspective of the museum profession and is applicable to diverse disciplines and types of collections.

HIST 396 History Beyond the Classroom: Public History**Cross-listed with:** ARTH 396**Credits 3 Grading Basis** Graded

This course introduces the interdisciplinary subfield of public history as it is practiced today. The course examines the various branches of public history through readings, class discussion, exercises, and occasional guest speakers and field trips, when possible.

HIST 397 War and Society in U.S. History**Credits 3 Grading Basis** Graded

This course surveys war and armed conflicts in U.S. history from the colonial period to the post-9/11 era, emphasizing the impact of war on American society, politics and culture as well as the development of military institutions.

HIST 398 European Military History**Credits 3 Grading Basis** Graded

A survey of European military history (including Russia/Soviet Union) from the Hellenistic period through the 1982 Falklands-Malvinas War. The evolution of strategic doctrine and military institutions, their effect upon European society and their role in European imperialism will be emphasized.

HIST 401 Research Capstone Topics**Credits 3 Grading Basis** Graded

The 400-level history capstones provide students with an integrative intellectual experience building on the skills and habits of mind that have been cultivated during their program. These capstones focus on a culminating project (e.g., a research paper, historiography paper, museum exhibit, website, teaching unit, podcast, briefing paper, op-ed, etc.). Students immerse themselves in a topic and make it their own. Topics and research projects will change with each instructor. Students may repeat the course for credit if taken with different instructors.

HIST 402 Workshop in Colonial American Life**Credits 3 Grading Basis** Graded

A comparative study of life in colonial and Revolutionary America. Required field trips may include Jamestown, Colonial Williamsburg, Philadelphia, Charleston and Lancaster County, PA. Published sources, lectures, films and two multi-day field trips are used. Supplemental fee required.

HIST 403 Workshop in Civil War Virginia**Credits 3 Grading Basis** Graded

This workshop examines the impact of the Civil War upon Virginia and its citizens. It explores the secession crisis, the revolution in firepower that forced changes in battlefield tactics and war aims, and the development of "hard war." A four-day battlefield tour will reinforce ideas discussed in the classroom. Supplemental fee required.

HIST 404 Science and Society in Early Modern Europe**Credits 3 Grading Basis** Graded

Examines the connections between knowledge of the natural world and other aspects of European societies between 1500 and 1700. Topics may include the scientific revolution (Copernicus, Galileo and Newton); medicine, anatomy, and ideas of disease; exploration, commerce and natural history; technology and empire; alchemy, astrology, and the boundaries of science; and comparisons between science in Europe and in other areas of the world.

HIST 405 Travel and Exploration**Credits 3 Grading Basis** Graded

This class is about travel and exploration in world history, using specific episodes to examine motives, consequences and the experience of travel. In studying long-distance trade, pilgrimages, voyages of exploration and discovery, and even tourism, we will look at the logistics of travel, attempts to map the world, and the difficulties people had in interpreting what they found.

HIST 406 Monticello**Cross-listed with:** ARTH 406**Credits 3 Grading Basis** Graded

A seminar on the architecture and material culture of Thomas Jefferson's Monticello. The course will examine the house's design, artwork, decorative arts, mechanical devices, landscape/garden design and Mulberry Row. Topics will include African American artisans at the Monticello joinery, Jefferson's Indian Hall and European and African American domestic life in the Federal Period. Required field trips.

HIST 407 Digital History**Credits 3 Grading Basis** Graded

This course will provide an introduction to digital history. It explores some of the ways in which digital technologies can change how we research, write, document, exhibit, produce, and think about history. Students should not expect to become an expert in any single technology, but will develop a familiarity with a wide range of tools and applications and will have the chance to create their own digital history project.

HIST 408 The Museum: History and Controversies**Cross-listed with:** ARTH 408**Credits 3 Grading Basis** Graded

This seminar centers on art museums in the United States. Topics include the historical development of museums, related cultures of display, recent debates on institutional mission and responsibility, and contemporary artists who employ the museum as medium, subject matter or site. Required field trips.

Prerequisites: ARTH 206.

HIST 409 Early Modern Atlantic History**Credits 3 Grading Basis Graded**

This course focuses on the history of the North and South American Atlantic World from the "Age of Exploration" to the "Age of Democratic Revolutions." It traces European exploration of and expansion into the South and later North Atlantic beginning in the fourteenth century, and concludes with the assertion of sovereignty and independence by European colonies throughout the extended Atlantic littoral.

HIST 410 Storytelling from the Historical Record**Credits 3 Grading Basis Graded**

This course focuses on how to turn sources from the historical record into stories people want to read. Students will learn how to conceptualize their research into a well-framed story with a plot, narrative arc and controlling idea. We will work on strategies for creating suspense and narrative tension. We will learn some of the key elements of character development, including how to build a cast of protagonists, antagonists, witnesses and supporting players. Throughout the course, we will read the works of historians who excel as storytellers. Significant class time will be devoted to workshoping drafts of our own stories. By the end of the semester, students will complete their own work of creative nonfiction.

HIST 411 Colonial and Revolutionary North America**Credits 3 Grading Basis Graded**

This seminar explores the history of North America from the 16th through the 18th centuries. Thematic emphasis will vary from semester to semester.

HIST 412 Early American Cultural History**Credits 3 Grading Basis Graded**

This course will consider the experiences of peoples of early North America through the lens of cultural history. Topics may include: how people ate, dressed, bathed (or didn't) and thought about politics, consumption, health and manners. These seemingly routine life practices are a window into fundamental ways of understanding the world.

HIST 413 The Anglo-American Constitutional Tradition, 1603-1791**Credits 3 Grading Basis Graded**

This course surveys Anglo-American political and constitutional traditions. Emphasizes the evolution of 17th- and 18th-century British constitutionalism, its transferal to the British North American colonies, and the development of the first national and state constitutions in the United States.

HIST 420 Mobs, Taxes, War and Laws: The American Revolution**Credits 3 Grading Basis Graded**

An interpretive study of the political, economic, social and cultural history of the United States from the French and Indian War through the Federalist period.

HIST 422 Spaces and Places of the Early United States**Credits 3 Grading Basis Graded**

This seminar explores the history of the early United States from the Constitution to the Civil War through original primary source research and scholarly studies that illuminate the organization, meaning and influence of its spaces and places.

HIST 425 Civil War and Reconstruction**Credits 3 Grading Basis Graded**

A study of the background, development, personalities and aftermath of the Civil War. Special attention is given to the coming of the war and different explanations of its causes and to the policies and significance of Reconstruction, with varying interpretations thereof.

HIST 426 Great Black Migration of the Twentieth Century**Credits 3 Grading Basis Graded**

Although African Americans had been leaving the South since the days of slavery, it was not until the twentieth century that they migrated on a massive scale to begin new lives in the urban North. From about 1910 to 1970, millions of African American migrants embarked on millions of individual journeys that resulted in a dramatic redistribution of the African American population. This Great Black Migration, as it came to be called, shifted the center of African American social, economic, political and cultural life from the rural South to the urban North and West. Students will explore the Great Migration and its wide impact on African American life, culture and politics, through a variety of sources, including historical scholarship, literature, film, oral history, memoir, journalism and other sources contemporary to the period.

HIST 427 U.S. Environmental History**Credits 3 Grading Basis Graded**

An interpretive study of the development of environmental thought in the United States. Emphasis is given to philosophies of nature, land and resource usage and conservation, the environmental movement and organizations, environmental activism and radicalism, landscape restoration, and environmental mitigation and protection.

HIST 435 History of Immigration to the United States**Credits 3 Grading Basis Graded**

The United States is a land populated by migrants. Most residents of the U.S. today trace their ancestry to Latin America, Europe, Asia or Africa. The U.S. has been both the largest receiving destination of immigrants in the last two hundred years and the inventor of modern border restrictions. This course surveys patterns of migration and immigrant experience from colonial times to the present. Students will consider migration to the United States as part of the evolving global labor market and colonial expansion in the modern world. Students will learn about the relationship of immigration to nation-building, national expansion, war, and the production and reproduction of national identity. Students will also learn about the history of the legal regulation of immigration, the political debates surrounding the role of immigration in society and the experience of immigrants in settling and negotiating life in a new society. Students will explore immigration history by exploring a rich array of primary source material: fiction, memoirs, government documents, oral history interviews, films, maps and more.

HIST 436 Afro-Latin America**Cross-listed with: ANTH 436****Credits 3 Grading Basis Graded**

Latin America and the Caribbean were the first and largest parts of the Western Hemisphere to be populated by Africans. Afro-Latin America examines cultural formations Africans brought to these regions. Beginning with an overview of the slave trade, it examines the histories of Africans and African-descent people throughout Latin America, as well as contemporary Afro-Latin American culture(s).

HIST 437 Latin America and Latin Americans through Film: Focus on the Twentieth Century**Credits 3 Grading Basis Graded**

This course will provide students with the tools they need to be skilled visual readers as well as to link national and international representations of Latin America to their appropriate historical, social, cultural and political contexts.

HIST 438 Workshop in Public and Local History**Credits 3 Grading Basis** Graded

Selected historical topics relating to the Shenandoah Valley and surrounding region are studied in depth. Students will undertake primary research and collaborate on final project. See MyMadison for current classes.

Prerequisites: HIST 225.

HIST 439 Selected Topics in American History**Credits 3 Grading Basis** Graded

Selected topics are studied in depth. See MyMadison for current topic. Course may be repeated for credit when content changes.

HIST 440 The History Museum**Cross-listed with:** HIST 540**Credits 3 Grading Basis** Graded

An exploration of the history, evolution, and function of history museums. Readings and discussions cover the history and genealogy of the modern museum; exhibits and the influence of other forms of display such as world's fairs and department stores; ethics, mission, and administration; collections management and conservation; education and interpretation; emerging technologies; historical memory and controversy in museums; the role of the community; and museums on a global stage.

HIST 441 Oral History**Cross-listed with:** SCOM 441**Credits 3 Grading Basis** Graded

This course will explore the theory and practice of oral history. Through a series of readings, students will consider the many promises and challenges of the discipline, including issues related to memory, objectivity, ethics, the law, and technology. Students will also engage in an experiential learning exercise in which they collaborate to produce an oral history project.

HIST 442 Selected Topics in European History**Credits 3 Grading Basis** Graded

Selected topics are studied in depth. See MyMadison for current topic. Course may be repeated for credit when content changes.

HIST 443 Modern American Technology and Culture**Credits 3 Grading Basis** Graded

This seminar examines the sociotechnical history of twentieth century America. It employs several analytical frameworks to examine the complex relationship between social and technological change, casting particular attention on the mass production ethos, the social meanings of everyday household technologies, the nuclear age, the space age, countercultural technology and the high tech age.

HIST 444 Revolution and Social Change in Latin America**Credits 3 Grading Basis** Graded

This seminar will explore why revolutions were a major feature of the Latin American landscape throughout the modern era and how they contributed to changes in society. In a typical semester the course will explore the lives of leaders such as Che and Emiliano Zapata and investigate the causes and consequences of revolutionary actions in Cuba, Mexico and Nicaragua.

HIST 445 In the Year 2000: America at the Turn of the Century**Credits 3 Grading Basis** Graded

Students will focus on historical research and writing skills through the thematic study of American culture and politics in and around the year 2000. Students will read, listen, watch and analyze turn-of-the-century sources, engage in a series of practical exercises to explore a variety of historical methods and complete a semester-long research project.

HIST 447 South America**Credits 3 Grading Basis** Graded

An examination of nineteenth and twentieth-century South America by emphasizing recent historiographies of the region. The class draws from social and cultural history to explore themes such as gender, race and ethnicity, nation-building, and historical memory.

HIST 448 Gender in Latin America and the Iberian World**Credits 3 Grading Basis** Graded

This course is designed to introduce students to critical issues, theories and methods of gender history through the study of the history of Latin America and the broader Iberian world. Students will study select peoples and cultures of Latin America and the Iberian Peninsula exploring how they lived and understood gender and sexuality during the pre-colonial, colonial and/or modern eras.

HIST 449 Women and Fascism**Credits 3 Grading Basis** Graded

This course offers a comparative understanding of fascism and women with a focus on Europe, including Nazi Germany, Fascist Italy and Francoist Spain. We will also discuss fascist movements and right-wing women in other European countries and in Latin America. The course will uncover the origins of fascism and the rise of the fascist party and the women's branch.

HIST 451 Murderers, Arsonists and Thieves: Crime and Criminality in World History**Credits 3 Grading Basis** Graded

Who defines crime? Who decides punishment? Who is punished? What does punishment look like? What happens to those who are punished after they have been punished? This course discusses these questions and more from a historical perspective, using examples from around the world. Students will learn soon enough that few forms of "crime" were that unique or particular, since economic, social, cultural and political trends produced similar patterns across national boundaries.

HIST 452 Parks and Recreation: History of National Parks**Credits 3 Grading Basis** Graded

What are national parks? Why were they formed? How were they formed? Were people living in areas that became national parks? What happened to them? Who visits these parks? This course discusses these questions and more from a historical perspective, turning to works on national parks from around the world. Students will explore how national parks are best understood as "transnational parks," established through processes of learning, support, exchange and even pressure that transcended national boundaries.

HIST 453 Patterns of Global History**Credits 3 Grading Basis** Graded

This course introduces students to the literature, concepts, themes and methodology of global history, a subfield of history that seeks to compare experiences across regional, area, cultural and temporal boundaries, to look at cross-cultural interactions and to examine large-scale patterns that have shaped history on a global scale.

HIST 454 Innocent Babies and Gray Heads: Ideas about Age in American History

Credits 3 Grading Basis Graded

This course will consider the influence of ideas about chronological age and the way these expectations have shaped life and public policy throughout American history. Come discover how American ideas about childhood, adulthood and old age have changed over four centuries. The course will investigate the wide variety of ideas about what chronological age tells us about who people are and what they are capable of. Boy kings, child brides, protected and idealized pastoral childhoods - all have been possible and promoted at times in American life. So, too, have both respect for old age and revulsion at the old existed. The course will consider how society decided how old is old enough to drink and how old is old enough to be drafted. This course will examine questions such as: What makes an adult anyway? Why were teenagers once considered adults? Why is 65 now retirement age?

HIST 458 Modern European Intellectual History

Credits 3 Grading Basis Graded

This upper-level seminar considers important thinkers and major trends in philosophical, social, and aesthetic thought in nineteenth- and twentieth-century Europe. Instead of merely surveying a series of ideas and thinkers, the course will trace the development of ideas across times and cultures by undertaking careful readings of key texts.

HIST 459 Gender and Popular Culture

Credits 3 Grading Basis Graded

This course evaluates the history of U.S. popular culture as a force shaping and reflecting gender norms. It will analyze creators' and consumers' use of popular culture to promote stereotypes and to challenge them, with attention to ways that messages about gender, race, class, religion, sexuality and age intertwined. Students' original research will focus on the creation and reception of media including magazines, film and television.

HIST 461 Marxism

Credits 3 Grading Basis Graded

A study of the most significant ideas concerning politics, society, economics and philosophy which shaped Communism and Marxist varieties of Socialism.

HIST 462 Human Trafficking in African History

Credits 3 Grading Basis Graded

This seminar examines key themes in the history of human trafficking in the entire sweep of African history: the trans-Saharan slave trade, the Indian Ocean slave trade, the Atlantic slave trade, systems of African slavery and slave trade, and religions and slave trade. Students will recognize that the history of human trafficking as a significant global economic factor defies racial boundaries, connecting water bodies and continents, and develop familiarity with competing historiographical interpretations, debates and theories. Students will have opportunities to appreciate the historical roles of human trafficking in transforming societies, cultures, politics, religions, and people's perceptions and worldviews.

HIST 463 Pandemics and Epidemics in African History

Credits 3 Grading Basis Graded

This course explores the histories of epidemics and pandemics over the entire sweep of African history using a continent-wide perspective from the earliest recorded pestilence to the rise and expansion of successive Ebola epidemics. The course offers students the opportunity to understand how human interaction with the African environment - one of the major themes in African history - affected pandemics unfolding and how pandemics shaped African demography and settlement patterns, economies, societies, cultures and politics.

HIST 467 The Roman Republic

Credits 3 Grading Basis Graded

In this course, students will investigate the history of the Roman Republic (ca. 500 - 90 BCE), with an emphasis on society and culture during the wars with Carthage (3rd to late 2nd century BCE). Topics studied include the formation of Roman identities and the processes of cultural change during the critical transitional period of Rome from city to imperial power. Students will consider archaeological evidence from Italy and beyond and will read against the ancient sources to find the voices of ancient non-Romans, women and enslaved people.

HIST 469 A History of International Development in the Twentieth Century

Credits 3 Grading Basis Graded

This seminar considers major trends in the history of international development since World War II, focusing on American development theories, institutions, and programs but also considering case studies of aid programs worldwide. The course will trace the history of international development by undertaking careful readings and discussions of primary and secondary texts from a variety of disciplines, including history, economics, sociology, anthropology and political science.

HIST 470 Modern Africa

Credits 3 Grading Basis Graded

A historical analysis of the major themes in modern African history since the colonial era.

HIST 471 Commemoration and Controversy: Public Art in America

Cross-listed with: ARTH 471

Credits 3 Grading Basis Graded

This course examines the socio-historical, political, cultural, and philosophical dimensions of public art in American society, from the Early Republic to the present day. Topics may include: the nature of public art, its uses and functions, as well as civic and official attitudes towards art in the public sphere. Issues of censorship, propaganda, and the "culture wars" of the 1980s and 90s will be highlighted.

HIST 472 Military, Militarism and Warfare in African History

Credits 3 Grading Basis Graded

This seminar course examines the historical role of the military, militarism and warfare in shaping African politics, economies, states, societies and cultures. The course will be organized thematically recognizing significant turning points in the history of Africa. Key themes of scholarly debate about the role of the military, militarism and warfare in the history of Africa, emphasizing developments since the beginning of the nineteenth century, will inform class discussions and assignments.

HIST 473 Early Modern Islamic Empires

Credits 3 Grading Basis Graded

This seminar surveys and compares Islamic imperial formations from the 14th Century through World War I, focusing on the Ottoman, Safavid, and Mughal empires that flourished in the global age of early modernity.

HIST 474 Stalinism: Approaches, Experiences and Legacies

Credits 3 Grading Basis Graded

This course will examine the lived experience of Stalinism from a variety of perspectives. Its goals are to question the applicability of the concept of totalitarianism to Stalin's Soviet Union and to explore the ways that different approaches to the study of history (political, social, cultural) have shaped our understanding of Soviet socialism under Stalin.

HIST 475 Modern Russia

Credits 3 Grading Basis Graded

A study of Russia from the 1917 Revolution to the present. Readings and discussion will emphasize significant political, economic, social and cultural developments.

HIST 476 Revolutionary Russia**Credits 3 Grading Basis** Graded

This course examines the causes, key events and consequences of the Russian revolutionary movement that culminated in the overthrow of the autocracy and establishment of the world's first socialist state in 1917. It explores both Western and Soviet historians' interpretations of the revolution and compares revolutionary experiences across classes, genders and nationalities in different parts of the Russian empire.

HIST 478 Eastern Europe**Credits 3 Grading Basis** Graded

A study of the lands between Germany and Russia, from the Baltic to the Balkans. Emphasis is on the Hapsburg Empire and its successor states, the origins of the World Wars, the post-World War II communist governments, and the cultural and intellectual contributions of the Eastern European people.

HIST 479 Global Environmental History: Mountains, Rivers and Animals in Asia and Africa**Credits 3 Grading Basis** Graded

Environmental history cultivates critical thinking beyond the confines of single-nation states and their respective geographies and chronologies to consider longer stretches of time and wider territorial scales of analysis. This course is organized around the study of the Himalayas, Hindu Kush and other mountain ranges including the Zagros, Taurus and Atlas where forests and forest products are the primary foci. In addition to Nile and Indus river histories, the course also considers the history of water more generally from the perspective of ancient hydraulic technologies in Mesopotamia to the geopolitics of mega-dams along the Mekong River. The course will address animal histories and cultures, primarily those associated with camels, horses and elephants while paying lesser attention to donkeys, mules, goats, sheep, dogs and yak. The general concern throughout will be human interaction with the natural world.

HIST 482 French History Seminar**Credits 3 Grading Basis** Graded

Broad introduction to a particular aspect of early modern, revolutionary or modern French history that is characterized by extensive historical debate. See instructor for thematic focus. Students develop knowledge of historical content and of the historiography/methodological approaches, conduct independent research and present findings in writing and in formal research colloquia. Students may repeat seminar for credit if topics differ.

HIST 485 Colonial Knowledge Production in the Middle East and South Asia**Credits 3 Grading Basis** Graded

This course examines modern colonialism and imperialism, particularly their cultural, intellectual and scientific dimensions. Thematic foci include systems of imperial representation including Orientalism, and linguistic, cartographic, archaeological, urban, rural and social scientific transformations under colonial regimes of knowledge and power. Colonial and imperial relationships in nineteenth-century British India and Egypt, twentieth-century Palestine and Israel, and the twenty first-century US and Afghanistan will be examined.

HIST 486 Europe Since 1914**Credits 3 Grading Basis** Graded

An interpretive study of European history from World War I to the post-Cold War era, with special emphasis on the revolutions of 1917-19, the rise of totalitarianism, the origins of the Second World War, the Cold War and the continuing crisis of values.

HIST 487 World War II**Credits 3 Grading Basis** Graded

An examination of the origins, conduct and immediate aftermath of World War II in Europe and Asia. Attention is given to Japan's Pacific War, Hitler's war in Europe and the ultimate victory of the allies. The major military campaigns are discussed as are collaboration, resistance, and the War Crimes Trials.

HIST 488 The Holocaust in Global Context**Credits 3 Grading Basis** Graded

Introduces students to the most significant accomplishments and debates of recent Holocaust scholarship, emphasizing how historical memory of the Holocaust has been created and has evolved over time. Analyzes the historical causes and development of the Holocaust, as well as its cultural, political, and scholarly resonance in the post-1945 world.

HIST 489 Selected Topics in World History**Credits 3 Grading Basis** Graded

Selected topics are studied in depth. Course may be repeated when content changes. Only courses with significant content outside of Europe will count toward the world history requirement. See MyMadison and the history department website for information on current classes.

HIST 492 History from Things: Material Culture**Cross-listed with:** ANTH 492, ARTH 492**Credits 3 Grading Basis** Graded

This course explores the multidisciplinary "field" of material culture studies and the ways of looking at and learning from objects, things, or "stuff," ranging from furniture and teapots to houses and farmyards. It examines how scholars from a variety of disciplines have used material culture in their work. Readings draw from multiple sources, approaches and methodologies, reflecting the interdisciplinarity of the field.

HIST 493 Historic Preservation**Cross-listed with:** ARTH 493**Credits 3 Grading Basis** Graded

An introduction to the philosophy and techniques of historic preservation, guidelines for restoration, state and National Register forms and procedures, historic architecture, structural analysis, restoration techniques as well as the business aspects of historic preservation projects. Field trips are a major component of the course.

HIST 496 Research Thesis**Cross-listed with:** ANTH 496**Credits 3 Grading Basis** Graded

Students will gather, analyze and interpret archaeological/historical data over two semesters. Students will work on a project that demonstrates theory, research design, data gathering and analysis, culminating in a written thesis. The course meets the capstone requirement for the historical archaeology minor but is also available to students in history and anthropology.

Prerequisites: Junior or senior standing only.

HIST 499A Honors**Credits 1 Grading Basis** Graded

The purpose of the 1-credit HIST499A honors capstone course - to be taken during the spring of a student's junior year - is to select a committee consisting of a director and two readers, develop a project proposal in consultation with the director, and submit the application and proposal with the completed signature page to the Honors College by the deadline.

HIST 499B Honors**Credits 3 Grading Basis** Graded

An original research project for History majors who want to graduate with distinction.

Prerequisites: HIST 499A.

HIST 499C Honors**Credits 2 Grading Basis** Graded

HIST 499C is the final segment of the 3-semester sequence of HIST499A, HIST499B, and HIST499C. In HIST499C, students complete their History Honors Capstone Project.

Prerequisites: HIST 499A and HIST 499B.

Honors (HON)

HON 100 Honors First Year Seminar**Credits 1 Grading Basis** Graded

As an introductory experience in the Honors College, students will be oriented to Honors activities and goals, high impact university learning practices, seminar requirements and areas of emphasis sequences, and leadership, service, and research opportunities. Students will examine their personal and educational goals and participate in ethical reflection that expresses itself in scholar-citizenship and community engagement.

HON 101 Honors Orientation Seminar**Credits 1 Grading Basis** Graded

This course is an introductory experience in the Honors College for current JMU students entering as new Honors Scholars in the second semester of their first year and beyond. Newly admitted Honors Scholars will be oriented to Honors activities and goals; high impact university learning practices; curricular requirements; and leadership, service and research opportunities. Students will examine their personal and educational goals and participate in ethical reflection that expresses itself in scholar-citizenship and community engagement.

HON 200 Special Topics in Honors**Credits 1-6 Grading Basis** Graded

Unique interdisciplinary courses designed specifically for the Honors College, these courses explore a range of complex topics that deal with contemporary issues in society, multicultural and comparative studies, and advanced applications in business and the natural and social sciences. Seminars are designed to be flexible small classes that may reflect unique, sometimes experimental, styles of teaching. May be repeated when topic changes.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 300 Advanced Special Topics in Honors**Credits 1-6 Grading Basis** Graded

Unique interdisciplinary courses designed specifically for the Honors College, these courses explore a range of complex topics that deal with contemporary issues in society, multicultural and comparative studies, and advanced applications in business and the natural and social sciences. Seminars are designed to be flexible small classes that may reflect unique, sometimes experimental, styles of teaching. HON 300 seminars are not recommended for first year students. May be repeated when topic changes.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 321 Leadership I**Credits 3 Grading Basis** Graded

This Honors Seminar is the first course within the Leadership Area of Emphasis. Students will be introduced to the meaning, study and practice of leadership through the examination and evaluation of leadership styles and behavior. Students will analyze the way in which leadership principles and practices have been and can be applied to their lives. Through this study, students will develop their own personal leadership potential.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 322 Leadership II**Credits 3 Grading Basis** Graded

This honors seminar offers a variety of individualized and/or small group experiential learning opportunities in association with the leadership area of emphasis.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 323 Leadership III**Credits 1-3 Grading Basis** Graded

This course offers an intense independent study opportunity for students to work one-on-one or in small groups with faculty mentors on a project of their design. The specifics of these offerings will be designed by faculty mentors and the area of emphasis honors student(s). Objectives, goals and deliverables should be an extension and follow-up to activities associated with an experiential area of emphasis course. Repeatable for a maximum of three credits.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 331 Global Studies I**Credits 3 Grading Basis** Graded

This Honors Seminar is the first course within the Global Studies Area of Emphasis. We will examine how people study and perceive similarities, differences and interdependencies among human societies through the exploration of social sciences, arts, humanities, health, education, environmental and developmental studies. Course components will promote global citizenship through discussion, the introduction of new concepts and critical thinking related to current issues and case studies.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 332 Global Studies II**Credits 3 Grading Basis** Graded

This honors seminar offers a variety of individualized and/or small group experiential learning opportunities in association with the global studies area of emphasis.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 333 Global Studies III**Credits 1-3 Grading Basis** Graded

This practicum course offers an intense independent study opportunity for students to work one-on-one or in small groups with faculty mentors on a project of their design. The specifics of these offerings will be designed by faculty mentors and the area of emphasis honors student(s). Objectives, goals and deliverables should be an extension and follow-up to activities associated with an experiential area of emphasis course. Repeatable for a maximum of three credits.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 341 Scientific Research I**Credits 3 Grading Basis Graded**

This Honors Seminar is the first in the Research Area of Emphasis series. The course is designed as an introduction to the nature of scientific inquiry and what it means to be a research scientist and effective communicator. Course objectives will be met through the collective investigation of a "big" problem and the professional communication of the resulting findings. In doing so, students will gain scientific literacy skills to support future research efforts.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 342 Scientific Research II**Credits 3 Grading Basis Graded**

This honors seminar offers a variety of individualized and/or small group experiential learning opportunities in association with the research area of emphasis.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 343 Scientific Research III**Credits 1-3 Grading Basis Graded**

Practicum courses offer an intense independent study opportunity for students to work one-on-one, or in small groups, with faculty mentors on a project of their design. The specifics of these offerings will be designed by faculty mentors and the area of emphasis honors student(s). Objectives, goals and deliverables should be an extension and follow-up to activities associated with an experiential area of emphasis course. Repeatable for a maximum of three credits.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 351 Service I**Credits 3 Grading Basis Graded**

This Honors Seminar is the first course within the Service Area of Emphasis. Engaged citizens make differences in the quality of life in local, national and global communities. Upon studying a wide spectrum of contemporary issues, engaged citizens take action. This seminar provides opportunities for students to combine their intellectual pursuits with civic engagement and discourse, thereby empowering them to become engaged participants in tomorrow's global society.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 352 Service II**Credits 3 Grading Basis Graded**

This honors seminar offers a variety of individualized and/or small group experiential learning opportunities in association with the service area of emphasis.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 353 Service III**Credits 1-3 Grading Basis Graded**

This practicum course offers an intense independent study opportunity for students to work one-on-one, or in small groups, with faculty mentors on a project of their design. The specifics of these offerings will be designed by faculty mentors and the area of emphasis honors student(s). Objectives, goals and deliverables should be an extension and follow-up to activities associated with an experiential area of emphasis course. Repeatable for a maximum of three credits.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 361 Creativity I**Credits 3 Grading Basis Graded**

The course explores basic concepts of creativity across the disciplines and cultures. Course content includes the study and analysis of creative expression, the application of theories and conceptual frameworks, and the various modes of creative cognition.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 362 Creativity II**Credits 1-3 Grading Basis Graded**

This honors seminar offers a variety of individualized and/or small group experiential learning opportunities in association with the creativity area of emphasis.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 363 Creativity III**Credits 1-3 Grading Basis Graded**

This practicum course offers an intense independent study opportunity for students to work one-on-one or in small groups with faculty mentors on a project of their design. The specifics of these offerings will be designed by faculty mentors and the area of emphasis honors student(s). Objectives, goals and deliverables should be an extension and follow-up to activities associated with an experiential area of emphasis course. Repeatable for a maximum of three credits.

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 401 Introduction to Honors ePortfolio**Credits 1-2 Grading Basis Graded**

Students choosing the Honors Portfolio option must work with a faculty advisor to create a portfolio showcasing their substantial scholarship or creative work and demonstrating expert skill, knowledge, imagination and/or craftsmanship in their chosen major or minor. This course is designed to build coherence around three main principles: making reflection an integral part of learning (metacognition), making connections across courses and programs (integrative learning), and highlighting the social component of learning (social pedagogies).

Prerequisites: Enrollment in the Honors College or permission of the Honors College dean.

HON 402 Honors ePortfolio**Credits 1-2 Grading Basis Graded**

Students choosing the Honors Portfolio option must work with a faculty advisor to create a portfolio showcasing their substantial scholarship or creative work and demonstrating expert skill, knowledge, imagination and/or craftsmanship in their chosen major or minor. The HON 402 course provides students the opportunity to concentrate on ePortfolio completion and synthesis.

Prerequisites: HON 401.

HON 450 Honors Independent Study**Credits 1-6 Grading Basis Graded**

This course provides an opportunity to support projects designed for individualized learning experiences for honors students.

HON 496 Internship**Credits 1-6 Grading Basis Graded**

An experiential learning opportunity connected to the student's professional long-term goals. May include directed readings, journaling and completion of a final reflective paper. May be repeated for a total of nine credits.

HON 499A Honors Senior Project**Credits 1-3 Grading Basis Graded**

This is a three semester course, offered as parts A, B and C for one, three and two credits per semester, respectively. Enrollment is restricted to those honors students whose senior projects are not discipline specific. This could include certain collaborative projects or experientially-based projects.

HON 499B Honors Senior Project**Credits 1-3 Grading Basis Graded**

This is a three semester course, offered as parts A, B, and C for one, three and two credits per semester, respectively. Enrollment is restricted to those honors students whose senior projects are not discipline specific. This could include certain collaborative projects or experimentally based projects.

HON 499C Honors Senior Project**Credits 1-3 Grading Basis Graded**

This is a three semester course, offered as parts A, B, and C for one, three and two credits per semester, respectively. Enrollment is restricted to those honors students whose senior projects are not discipline specific. This could include certain collaborative projects or experimentally based projects.

Hospitality Management (HM)

HM 100 First-Year Seminar**Credits 1 Grading Basis Graded**

This course is designed to assist first year hospitality majors with the transition into an industry-specific college classroom by developing a better understanding of the learning process while acquiring essential skills. Conducted in a seminar setting, this course provides a support group by discussing concerns confronting students new to this major. Throughout the semester students will engage with various guest speakers to learn about available resources and explore hospitality career paths.

Prerequisites: Hospitality Management major or permission of the program director.

HM 101 Lecture Series in Hospitality Leadership and Innovation**Credits 1 Grading Basis Credit / No Credit**

This lecture series class is geared toward an open forum for students to interact and discuss issues related to hospitality leadership and innovation. The course consists of one hour of lecture or discussion per week, with opportunities to interact with professionals outside of class.

HM 111 Overview of Hospitality and Tourism Management**Credits 3 Grading Basis Graded**

This course exposes students to the areas of lodging, food and beverage, tourism and entertainment management, special events and meeting planning, and club and resort management. Emphasis is on hospitality industry scope, organization and economic impact; includes familiarization with industry terminology and individual and business contributors to the field of hospitality and tourism management.

Prerequisites: Hospitality Management major or permission of the program director.

HM 112 Foundations of Leadership of Hospitality Management**Credits 3 Grading Basis Graded**

This course provides an introduction to leadership in Hospitality Management and the Mega Industry. The primary focus will be leadership theory and skill application with a focus on personal awareness.

Prerequisites: Hospitality Management major or permission of the program director.

HM 201 Foundations of Hospitality, Sport and Recreation Management**Credits 3 Grading Basis Graded**

An introduction to the basis for the professions that make up the School of Hospitality, Sport and Recreation Management. A focus on these professions in governmental, voluntary, private and commercial settings is incorporated. Both the economical significance and the professional preparation for success in the industry is both introduced and practiced.

HM 203 Foundations of Ethics and Law in Hospitality, Sport and Recreation Management**Credits 3 Grading Basis Graded**

An introduction to ethics and law within the sport, hospitality and recreation industry. The ethical portion introduces students to select theories of ethics, ethical issues and an ethical decision-making model. The legal portion introduces students to basic legal terminology and concepts while concentrating on negligence and employment issues.

HM 210 Tourism in a Global Context**Credits 3 Grading Basis Graded**

This course examines tourism as a global industry and human activity that promotes and facilitates understanding of historical and cultural values, and of international institutions that characterize the broader global system.

HM 221 Special Event Management: Le Gourmet**Cross-listed with: SRM 221****Credits 3 Grading Basis Graded**

This course is designed to provide students with a practical hands-on application of event planning for non-profit fundraising. In this course students will actively plan and execute the Hart School's major annual fundraising event, Le Gourmet.

HM 222 Hospitality Professional Work Experience**Credits 3 Grading Basis Graded**

This applied hospitality course consists of experiential exercises followed by class discussion and actual work experience as an employee within the hospitality field. Role-playing and cases are used as learning activities where the instructor acts as a facilitator to learning. Debriefing is used extensively as a way of creating essential theory. Students must successfully complete ServSafe Food Protection Manager Certification and Training for Intervention Procedures (TIPS). Students complete approximately 100 hours of hospitality-related practical experience. Prerequisites: Hospitality Management major or permission of the program director.

HM 244 Disney Internship**Credits 3-6 Grading Basis Credit / No Credit**

The paid, semester-long internship program allows students to live just outside Walt Disney World Resort in Florida or the Disneyland Resort in California while earning college credit at Disney parks and resorts through a combination of on-the-job experience, shadowing and seminars.

Prerequisites: Permission of the hospitality management program director.

HM 251 Cost Control and Budgeting**Credits 3 Grading Basis Graded**

Introduction to cost control, budgeting and financial analysis as it relates to the hospitality industry. Students work as a team to create a detailed business plan using MS Excel.

Prerequisites: Hospitality Management major or permission of the program director.

HM 252 Culinary Arts and Menu Management**Credits 3 Grading Basis** Graded

This course covers application of basic food preparation and menu development. The focus is on preparing students to understand gastronomy and communicate with culinary professionals. This course includes the development of effective menus, plate presentation, preparation methods, flavor development and food service trends. Lab fee applies. Uniform required.

Prerequisites: HM 222 or permission of program director.

HM 261 Hospitality Data Analytics**Credits 3 Grading Basis** Graded

This course introduces students to the fundamental concepts and practical applications of data in the hospitality industry. Students will explore how data is generated, structured and interpreted, with a focus on deriving meaningful insights for decision-making in hospitality management. Core topics include data literacy, essential analytical techniques and an introduction to artificial intelligence fundamentals. By the end of the course, students will gain the knowledge and skills needed to confidently apply data-driven approaches across other hospitality management courses and professional settings.

HM 298 Special Studies in Hospitality Management**Credits 3 Grading Basis** Graded

A special studies course designed to explore areas of current topical concern in the lodging, food and beverage, travel and tourism, and entertainment industries. May be used for general education credit. Course content will vary.

Prerequisites: Hospitality Management major or permission of the program director.

HM 310 Practical Work Experience**Credits 0 Grading Basis** Credit / No Credit

Required 600 hours of approved hospitality and tourism work experience. Credit/no credit only. All work sites must be approved.

Prerequisites: Minimum of 45 credit hours complete, HM 201, HM 203, HM 111 and HM 222. Corequisites: HM 252.

HM 316 Country Club Management**Credits 3 Grading Basis** Graded

This course is an application of business concepts to the private equity club and full-service resort industry. Industry cases are used to facilitate discussion of similarities and differences among private equity clubs, full service resorts and other hospitality business in the areas of culture, asset management and operations.

Prerequisites: Junior standing

HM 321 Introduction to Event Management**Credits 3 Grading Basis** Graded

This course is designed to explore conferences, conventions, expositions, meetings and special events as they relate to the responsibilities of a planner, selection criteria for host venues, legal and ethical issues, negotiating process, program design, budgeting, contracts, marketing, logistics and evaluation. Practical hands-on experience through projects may occur.

Prerequisites: HM 251.

HM 322 Lodging, Clubs and Resort Operations and Technology**Credits 3 Grading Basis** Graded

This course offers an in-depth look at full-service hotels, clubs and resorts through the eyes of a general manager. The course will focus on operations, engineering, housekeeping, uniformed services, front office, reservations and revenue management. Different hospitality technology platforms and software programs will be used to expose students to hospitality technology.

Prerequisites: HM 251.

HM 333 Internship in Hospitality Management**Credits 3 Grading Basis** Credit / No Credit

Students will complete 200 hours of approved hospitality work experience. CR/NC only. All worksites must be pre-approved. In addition, students must validate 600 hours of professional work experience in the services industry as a prerequisite to this course.

Prerequisites: HM 222 and HM 251 and permission of the instructor.

HM 361 Italian Culture and Wine**Credits 3 Grading Basis** Graded

An introduction to the fundamentals of wine making, wine tasting and glossary of terms provide a framework for visual, olfactory and gustative analysis. The historical value of wine, together with its cultural, economic and social meaning in Italy, are explored. Course taught in Florence, Italy. Lab fee applies. Student must be enrolled in JMU in Florence Program.

HM 362 Italian Gastronomy**Credits 3 Grading Basis** Graded

This course is designed to teach students the applied approach to match wine and food from different parts of the world using flavors, textures and components present in food and wine in complementing strategies. Emphasis will be placed on menu planning, cooking methods and tasting wines with food in a formal dining room. Course offered in Florence, Italy. Lab fee applies. Student must be enrolled in JMU in Florence Program.

HM 371 Profit Optimization and Revenue Management**Credits 3 Grading Basis** Graded

In this course, students will apply and analyze data to effectively manage rates and capacity applied in many service industries, including hospitality. The focus of this course is to help students learn how to apply revenue management principles to maximize profitability in lodging, clubs, and the food and beverage sectors.

Prerequisites: HM 321 and HM 322.

HM 411 Hospitality Law**Credits 3 Grading Basis** Graded

The course focuses on the application of the law to the hospitality industry including rights and obligations of guests and lodging, food service, club, event management and association operators. The identification of potential legal problems and formulation of preventive measures to limit/prevent liability are emphasized.

Prerequisites: HM 111 and junior standing.

HM 414 Beverage Management and Marketing**Credits 3 Grading Basis** Graded

The course is designed to enhance knowledge in the identification and evaluation of beverages typically served in hospitality establishments. Special attention is given to alcoholic and non-alcoholic beverages with regard to price/quality relationships; channels of distribution and marketing; trends and current issues faced by the industry; and service ethics.

Prerequisites: Must be 21 years of age or older and declared HM major, HM 444 or permission of director.

HM 417 Advanced Event Management**Credits 3 Grading Basis** Graded

This course examines event planning and management within a project management framework. The course is designed to provide hands-on experience in event planning. Students will work to plan and execute an event from start to finish with instructor oversight. Students will demonstrate, by executing an event, the core competencies required of a successful event manager.

Prerequisites: HM 321, HM 322, HM 371 and permission of the instructor or director.

HM 419 Napa and Sonoma Wine and Culture**Credits 3 Grading Basis** Graded

Napa and Sonoma are the premier wine growing regions in the United States. The wines influence wines across the US and around the world. The ability to impressively learn about the wine, food, and culture that influence the region can only be fully accomplished by visiting. Students are able to visit wineries and speak with owners, visit growers and speak with the farmers, visit nurseries and discuss varietals.

Prerequisites: HM Major, 21 years old the first day of class and permission of the director.

HM 421 Hospitality Ethics**Credits 3 Grading Basis** Graded

Ethical issues and actions that have occurred, or are currently occurring, in the hospitality industry are examined from multiple perspectives. Examining recent scandals, real-world scenarios, news stories and common ethical dilemmas will enable a student to recognize an ethical dilemma, understand the components of the dilemma and make an ethical decision.

Prerequisites: HM 111 and junior standing.

HM 422 Hospitality Human Resources Management**Credits 3 Grading Basis** Graded

This course examines the identification and exploration of the information needs of the hospitality manager in making policy and personnel decisions. Different philosophies and processes for locating, attracting, hiring and training a qualified staff are examined. Emphasis is placed on the work environment within the service industry. Employment law will be emphasized as a part of the course.

Prerequisites: HM 321, HM 322 and HM 333; or permission of the director.

HM 442 Hospitality Seminar**Credits 3 Grading Basis** Graded

This senior capstone course is designed to assist in transitioning from being a university student to holding a leadership role within the hospitality industry. The interactive course draws upon the umbrella curricular goals of the hospitality management program in communication, ethics, finance and leadership. Senior assessment may also occur.

Prerequisites: HM 333.

HM 444 Supervisory Hospitality Internship**Credits 3 Grading Basis** Credit / No Credit

Students will complete 200 hours of approved hospitality supervisory experience. All work sites must be pre-approved.

Prerequisites: Senior standing and HM 333.

HM 490 Special Studies in Hospitality and Tourism Management**Credits 1-3 Grading Basis** Graded

Designed to give capable students in hospitality and tourism management an opportunity to complete independent study under faculty supervision.

HM 498 Special Topics in Hospitality and Tourism Management**Credits 3 Grading Basis** Graded

This course is designed to allow explorations of areas of current topical concern, or to exploit special situations. Course content will vary. For current course content, consult the advisor.

HM 499 Honors**Credits 1-6 Grading Basis** Graded

Year course. See catalog section "Graduation with Honors."

Human Resource Development (HRD)

HRD 100 Human Resource Development Leadership Lab**Credits 2 Grading Basis** Graded

Hands-on practicum of leadership strategies and techniques designed to give each student a better appreciation for the dynamics of leadership in intimate, physically challenging and stressful environments, both indoors and out. Students operate in teams which are formed and reorganized on a continuous basis, surrounded by peers at several levels of leadership experience and training. Collaborative learning is enhanced when students apply what they learn in class by describing relevant lessons learned through experiences outside the classroom. The focus of this course is to provide students with the opportunity to lead and follow in an observed setting and receive constant feedback and mentoring on their demonstrated leadership skills. Students learn through leading as well as through a critical reflection, inquiry, dialogue and group interaction. Everyone is responsible for contributing to the learning process.

HRD 101 Introduction to Leadership**Credits 1 Grading Basis** Graded

An introduction to: various leadership styles and their effect on organizations; insight into the leaders' roles and responsibilities within the context of the organization; character- and values-based leadership; basic leadership actions; the importance of self improvement in the areas of time management, health and fitness, goal setting, academic accomplishment and communications; group dynamics, and the development of interpersonal skills.

Corequisites: HRD 100.

HRD 123 Developing Multicultural Competency for Effective Facilitation**Credits 3 Grading Basis** Graded

This course will provide students with a deeper understanding of social identities, and they will gain the necessary skills to facilitate training, programs and dialogues focusing on areas of diversity, multiculturalism, inclusion and access. Students enrolled in this course will be prepared to become a diversity educator in the Diversity Education Empowerment Program sponsored by the Center for Multicultural Student Services at James Madison University.

HRD 200 Introduction to Workplace Learning**Credits 3 Grading Basis** Graded

The purpose of this course is to cover the emergence and advancement of the field of human resource development and provide essential skills for the process of planning, implementing and evaluating workplace learning.

Prerequisites: Must be declared as an Individualized Study major.

HRD 201 Leadership Styles Theory and Application**Credits 2 Grading Basis** Graded

Explores the dimensions of creative and innovative leadership strategies and styles by examining team dynamics and two historical leadership theories that form the basis of the leadership framework (trait and behavior theories). Students practice aspects of personal motivation and team building in the context of planning, executing and assessing team exercises and participating in leadership skills labs. Focus is on continued development of the knowledge of leadership values and attributes through an understanding of institutional structures, duties and responsibilities of organizational/institutional leaders, and leadership in small organizations. Case studies provide tangible context for learning leadership skills, values, actions and attributes as they apply to a contemporary setting.

Prerequisites: HRD 100 and HRD 101.

HRD 240 Introduction to Human Resource Development**Credits 3 Grading Basis** Graded

An introduction to the role and scope of human resource development with particular emphasis on required competencies for HRD professionals. Critical moral and ethical issues are introduced.

HRD 245 Leadership in Organizational Settings**Credits 3 Grading Basis** Graded

An examination of the principles of leadership and their application within organizational settings. Emphasis will be placed on the critical appraisal of the facets of leadership through the use of cases and readings.

HRD 250 Leadership in the Workplace**Credits 3 Grading Basis** Graded

Leadership, motivation and management: how do they relate to workplace learning? This course explores such themes and examines various leadership theories. In addition, learners will develop an understanding of the competencies and skills needed to succeed in a leadership role in HRD.

Prerequisites: Must be declared as an Individualized Study major.

HRD 300 Designing Workplace Learning**Credits 3 Grading Basis** Graded

Through authentic workplace scenarios and case studies, adult learners will analyze jobs and tasks to identify performance gaps or improvement needs. Applying systems thinking, students will design and develop effective HRD solutions, such as detailed performance improvement plans or targeted training programs, tailored to address specific workplace challenges. This course empowers learners to make strategic HRD recommendations that enhance workforce effectiveness and align with organizational objectives.

Prerequisites: Must be declared as an Individualized Study major.

HRD 380 Performance and Task Analysis in Human Resource Development**Credits 3 Grading Basis** Graded

Provides the basic skill level for students in the area of performance analysis and subsequent assessment. Particular emphasis is placed on actual analysis and assessment situations with application to program and curricular design.

Prerequisites: Declared HRD minor or Adult Degree Program students and HRD 240, HRD 245 and LDT 370. Corequisites: LDT 370.

HRD 400 Human Resource Development Internship**Credits 3 Grading Basis** Graded

A structured internship experience designed to provide students with the opportunity to contribute within an actual work setting the role and scope of human resource development efforts. Minimum 400-hour experience approved by the program coordinator. The minor does not provide internships.

HRD 425 Learning in the Workplace**Credits 3 Grading Basis** Graded

The purpose of this course is to develop adult learners by focusing on adult learning theories, cognitive and noncognitive learning processes, and diverse cultures and systems of knowing and learning in the workplace.

Prerequisites: Must be declared as an Individualized Study major.

HRD 475 Seminar in Leadership**Credits 3 Grading Basis** Graded

Designed to provide senior-level students with the opportunity to face some of the dilemmas of leadership. Students will be encouraged to critically examine leaders and organizations to draw from real events the ambiguities of leadership. Course will involve extensive reading, case work and a range of guest speakers.

Prerequisites: HRD 245.

HRD 480 Learning in Adulthood**Credits 3 Grading Basis** Graded

This course is a study of the learning processes of the adult learner with an emphasis on adaptations of the instructional process to accommodate the differences inherent in the adult learning environment. Practical applications to actual adult learning situations are included.

Prerequisites: Declared HRD minor or Adult Degree Program student; and LDT 370.

HRD 485 Development of Materials and Programs**Credits 3 Grading Basis** Graded

This course is designed to provide students with the basic skills necessary to design and develop performance-based training programs and courses. Emphasis will be placed on the actual design and development of training materials while working with an actual client.

Prerequisites: Declared HRD minor or Adult Degree Program student; and LDT 370 and HRD 380.

Humanitarian Affairs (HUMN)

HUMN 201 Introduction to Humanitarian Affairs**Credits 3 Grading Basis** Graded

A geographical overview of poverty, armed conflict, hunger, disease, and natural disasters and how they can lead to humanitarian crises. It includes a study of human rights along with a look at international efforts to address, and international organizations that deal with, humanitarian crises.

HUMN 301 Introduction to Natural Disasters Response**Cross-listed with:** GEOG 301**Credits 3 Grading Basis** Graded

This course is designed to give students an overview of the various types of natural disasters; a look at the world regions that are most vulnerable to each type of disaster; and, a preview of disaster planning, management, relief and response as related to natural disasters.

HUMN 302 Legal Issues in Humanitarian Affairs**Credits 3 Grading Basis** Graded

This course prepares students interested in humanitarian response with basic information on the legal issues of humanitarian law. It covers principles of domestic and international law relevant to the practice of humanitarian response, including international humanitarian law, the law of armed conflict, international human rights law, international criminal law and other legal topics relevant to humanitarian affairs practitioners. Prerequisites: Junior standing.

HUMN 331 Human Rights in Theory and Practice**Cross-listed with:** JUST 331, POSC 331**Credits 3 Grading Basis** Graded

This course will explore the nature and value of human rights by investigating some major debates over their status and meaning, and by examining some of the ways people have tried to secure human rights in practice.

Prerequisites: Humanitarian Affairs minors only.

HUMN 350 Special Topics in Humanitarian Affairs**Credits 1-3 Grading Basis** Graded

Focused, in-depth study of specific areas or subjects related to Humanitarian Affairs. Topics will provide background and skills important for humanitarian work. Can be repeated as course content changes.

HUMN 360 GIS for Humanitarian Assistance**Cross-listed with:** GEOG 360**Credits 3 Grading Basis** Graded

In responding to humanitarian crises, governments and aid organizations must deploy aid workers, deliver essential services, set up temporary settlements, and distribute items such as water and food that are needed for survival. Spatial analysis and maps are critical to the success of these efforts. In this course, students learn the basics of Geographic Information Systems (GIS) for humanitarian assistance and learn how relief organizations use GIS in their work.

Prerequisites: GEOG 215.

HUMN 490 Humanitarian Affairs Field Experience**Credits 3 Grading Basis** Graded

The HUMN 490 course offers students an opportunity to gain experience and practical skills, preferably in an "international setting," either in the U.S. or abroad, and to apply knowledge and skills acquired through the Humanitarian Affairs program. This course must be completed in a setting approved by the student's advisor and the humanitarian affairs minor committee.

Prerequisites: HUMN 201 and junior status.

Humanities (HUM)

HUM 200 Great Works [C2L]**Credits 3 Grading Basis** Graded

An intensive examination of great literary works that focus on key issues of knowledge and reality, meaning and purpose, ethics, and aesthetics. Discussion, analysis and intensive writing are required. Texts will vary by section and instructor. May be used for general education credit.

HUM 200L Great Works - London [C2L]**Credits 3 Grading Basis** Graded

An intensive examination of great literary works that focus on key issues of knowledge and reality, meaning and purpose, ethics, and aesthetics. Discussion, analysis and intensive writing are required. Texts will vary by section and instructor. Course is taught in London as part of Study Abroad Program. May be used for general education credit.

HUM 250 Foundations of Western Culture [C2HQC]**Credits 3 Grading Basis** Graded

This course is a study of the roots of our Western tradition in Greek, Roman, Medieval or Renaissance culture. Students examine the interrelationships among history and literary works; the fine arts; philosophical and religious thought and intellectual contexts. Content will vary depending on section and instructor. May be used for general education credit.

HUM 251 Modern Perspectives [C2HQC]**Credits 3 Grading Basis** Graded

An interdisciplinary study within the modern period of arts and humanities. Students will examine the interrelationships among history and the arts, philosophy, religion and the intellectual ideas of the time. Topics will vary by section. May be used for general education credit.

HUM 251L British Media and Society**Cross-listed with:** SMAD 360, WRTC 360, SCOM 360**Credits 3 Grading Basis** Graded

Study the history, nature and impact of mass media in the United Kingdom. Emphasis on the impact modern media has on society, and society has on media. Consideration of similarities and differences in mass media in the United States and Great Britain. Consideration of the relationship between mass media and the arts. Focus on 20th century mass media in London, one of the world's preeminent and most influential centers. London term only.

HUM 252 Global Cultures [C2HQC]**Credits 3 Grading Basis** Graded

This course is an interdisciplinary study of cultures or regions commonly referred to as non-Western. Students examine diverse responses to enduring human questions at the intersection of local and global cultures. The course offers ways to critically engage comparative views on global cultures and in so doing broaden students' understanding of those cultures and their own. Sections may address themes that span multiple cultural spaces (such as sports or the environment) or focus on regions. Section topics vary by instructor. Only approved topics may count for general education arts and humanities credit.

Inclusive Early Childhood Education (IECE)

IECE 300 Issues and Trends in Inclusive Early Childhood Education**Credits 3 Grading Basis** Graded

This course is designed to introduce students to the issues and trends in the inclusive education of infants, toddlers and young children.

This course will discuss the historical, philosophical, social and legal background for current practices in the field, evidence-based and recommended practices in early childhood inclusive education, and how to use this information in your professional practice as an early childhood educator.

Prerequisites: Minimum grade of "C" in requisites. IECE majors: EXED 200. Corequisites: IECE majors: IECE 303 and IECE 321. B.I.S./IECE: IECE 303, IECE 321 and IECE 322.

IECE 301 Initial Field Experience in Inclusive Early Childhood Education
Credits 1 Grading Basis Credit / No Credit

This practicum supports IECE 300. Students will further their understanding of the issues and trends impacting young children and their families in our community and evaluate their own perspectives and skills as they pertain to working with young children and families from diverse backgrounds, with diverse abilities, and in diverse settings. Prerequisites: Admission to teacher education pre-professional licensure program. Corequisites: IECE 300 and IECE 303

IECE 303 Development of Young Children Birth Through Age 8
Credits 3 Grading Basis Graded

This course provides students with an understanding of the development of infants, toddlers and young children with and without exceptionalities. Skills for observing, recording and interpreting the behavior of the young child as a basis for adult intervention and guidance are developed. Prerequisites: Minimum grade of "C" in requisites. IECE majors: EXED 200. Corequisites: IECE majors: IECE 300 and IECE 321. B.I.S./IECE: IECE 300, IECE 321 and IECE 322.

IECE 321 Initial Field Experience in IECE Part 1
Credits 1 Grading Basis Graded

This first course of the initial field experience provides candidates opportunities to use their knowledge of child development to observe and assess children and then plan meaningful learning environments and experiences for those children in a preschool setting. Students in the IECE concentration in the Bachelors of Individualized Study will complete this practicum setting within their current work environment. Prerequisites: Minimum grade of "C" in requisites. IECE majors: EXED 200. Corequisites: IECE majors: IECE 300 and IECE 303. B.I.S./IECE: IECE 300, IECE 303 and IECE 322.

IECE 322 Curriculum and Instruction in Inclusive Early Childhood Education
Credits 3 Grading Basis Graded

This course explores, analyzes and evaluates curriculum and methodology related to the design and management of a nurturing, supportive and challenging inclusive learning environment for children ages birth - 8 years. Emphasis is on the physical environment, design and selection of curricular components, the role of play in the curriculum, skills for professional intervention and interaction, and use of technology to facilitate young children's learning. Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 300. Corequisites: IECE majors: IECE 331, IECE 355 and IECE 466. B.I.S./IECE: IECE 300, IECE 303 and IECE 321.

IECE 324 Assessment of the Young Child
Credits 3 Grading Basis Graded

This course provides students with an understanding of how to use different assessment techniques and strategies to drive educational decision-making for young children with and without disabilities. Students will be introduced to the different purposes of assessment (e.g., instructional planning, delivery, progress monitoring, eligibility) and apply informal and formal assessment techniques that align with the purposes. Students will learn to link assessment to instructional and individualized education plan goals, choose and design valid and reliable assessments, and use assessment results to make educational decisions and provide formative and summative feedback to students. Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 450. B.I.S./IECE: IECE 429. Corequisites: IECE majors: IECE 431, IECE 460 and IECE 462. B.I.S./IECE: IECE 456, IECE 460 and IECE 461.

IECE 331 Initial Field Experience in IECE Part 2
Credits 1 Grading Basis Graded

This second initial field experience is designed to support all courses by giving students experience in an early elementary classroom. Students will observe young children, collect data, assist classroom teachers and interact appropriately with the individuals within the learning environment. Students will analyze the school environments for access for all young children. Prerequisites: Minimum grade of "C" in requisites. IECE majors: EXED 200. Corequisites: IECE majors: IECE 300 and IECE 303. B.I.S./IECE: IECE 300, IECE 303 and IECE 322.

IECE 355 Foundations of Inclusive Emergent and Early Literacy for Inclusive Early Childhood Education (IECE)
Credits 3 Grading Basis Graded

This is the first course in a two-course sequence designed to provide pre-service inclusive early childhood teachers with the foundation of literacy development. This course explores both the science and art of teaching emergent and early literacy, providing a strong foundation in literacy development for children from birth to age 5. Grounded in theory and research, the course examines how young children acquire early literacy skills and the dynamic, reciprocal nature of literacy learning between parents, educators and students. Students will investigate key concepts in early literacy acquisition, including the importance of oral language development, phonological awareness, print concepts and early writing. The course emphasizes evidence-based instructional strategies and developmental perspectives, equipping educators to support and enhance the emergence and continued growth of literacy in diverse learners. Through practical applications and reflective practice, participants will learn to create responsive, engaging and developmentally appropriate literacy environments that foster lifelong learning.

IECE 421 Intermediate Field Experience in IECE Part 1
Credits 2 Grading Basis Graded

This first intermediate field experience is designed to support methods courses by giving students experience in an early elementary classrooms. Students will observe young children, collect data, assist classroom teachers and interact appropriately with the individuals within the learning environment. Students will analyze the school environments for access by all young children. Students are required to receive satisfactory experience in this field placement to continue in the program. Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 322. B.I.S./IECE: IECE 300. Corequisites: IECE majors: IECE 450, IECE 456 and IECE 464. B.I.S./IECE: IECE 355, IECE 464 and IECE 466.

IECE 423 Intermediate Practicum II
Credits 2 Grading Basis Credit / No Credit

This second intermediate field experience provides candidates opportunities to use their knowledge of child development and assessment to create learning environments and experiences for children. Candidates will learn more about the adult's role in supporting children and managing behavior.

IECE 429 Teaching Early Childhood Learners with Complex Care Needs (IECE)**Credits 3 Grading Basis Graded**

This course is designed to give early childhood special educators exposure to working with children (birth through eight) who have disabilities that include significant medical issues. The role and function of early childhood special educators as team members are addressed both in IEPs and IFSPs (early intervention). The role of related technology, assistive technology and Augmentative and Alternate Communication (AAC) will receive extensive attention, as will issues of etiology and remediation. This course is taken while students are student teaching; therefore, there will be direct application of skills being learned. Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 470. B.I.S./IECE: IECE 466. Corequisites: IECE majors: IECE 471. B.I.S./IECE: IECE 462.

IECE 431 Intermediate Field Experience in IECE Part 2**Credits 2 Grading Basis Graded**

This second intermediate field experience is designed to support methods courses by giving students experience in an early elementary classroom. Students will observe young children, collect data, assist classroom teachers and interact appropriately with the individuals within the learning environment. Students will analyze the school environments for access for all young children.

Prerequisites: Minimum grade of "C" in requisites. IECE 450. Corequisites: IECE 324, IECE 460 and IECE 462.

IECE 450 Contemporary Family Issues in Inclusive Education**Credits 3 Grading Basis Graded**

This course will examine how students' own cultural values shape their interactions with children and families as well as provide concrete, practical strategies for effective and culturally competent interactions with children and families.

Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 322. B.I.S./IECE: IECE 324. Corequisites: IECE majors: IECE 421, IECE 456 and IECE 464. B.I.S./IECE: IECE 470.

IECE 456 Foundations of Literacy (PK-3) for Inclusive Early Childhood Education (IECE)**Credits 3 Grading Basis Graded**

This is the second course in a two-course sequence designed to provide pre-service inclusive early childhood teachers with the foundation of literacy development. This course provides a comprehensive exploration of the science of literacy, focusing on how children acquire and develop literacy skills across a variety of learners in PK-3 classrooms. Grounded in evidence-based research, the course examines the cognitive and neurological processes involved in literacy development and acquisition, including an in-depth look at the reading brain and how it adapts to written language. Students will engage with foundational frameworks such as the Simple View of Reading, which emphasizes the dual importance of decoding and language comprehension, and will analyze the five pillars of literacy - phonemic awareness, phonics, fluency, vocabulary and comprehension. Additionally, the course will explore Scarborough's Reading Rope, illustrating the complex, interwoven strands that contribute to skilled reading. The course also emphasizes the role of assessment in guiding instructional decision making and monitoring progress, equipping educators with tools to evaluate literacy skills effectively. By the end of the course, students will be prepared to apply scientifically grounded literacy practices to support all learners in achieving reading proficiency in the primary literacy classroom (PK-3).

Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 322. B.I.S./IECE: IECE 429. Corequisites: IECE majors: IECE 421, IECE 450 and IECE 464. B.I.S./IECE: IECE 324, IECE 460 and IECE 461.

IECE 460 Instructional Practices in Numeracy**Credits 3 Grading Basis Graded**

This course helps prospective early childhood teachers develop the knowledge, skills and understanding to design and implement effective, developmentally appropriate mathematics instruction for young children, birth to age eight, with and without exceptionalities. Emphasis is on appropriate mathematical content, teaching strategies and manipulative materials from a developmental perspective with special emphasis on adaptations designed to meet the needs of all children.

Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 450. B.I.S./IECE: IECE 429. Corequisites: IECE majors: IECE 324, IECE 431 and IECE 462. B.I.S./IECE: IECE 324, IECE 456 and IECE 461.

IECE 461 Advanced Field Experience in IECE Part 1**Credits 2-3 Grading Basis Graded**

The advanced field experience provides candidates opportunities to use their knowledge of child development to plan meaningful learning experiences with kindergartners and first graders. Candidates will learn the teacher's role in supporting children to construct understandings about the natural and social sciences, social studies, literacy and to use mathematical thinking. Candidates will begin implementing instructional strategies learned in all methods courses. This course precedes student teaching in the fall.

Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 324. B.I.S./IECE: IECE 429. Corequisites: IECE majors: IECE 470. B.I.S./IECE: IECE 324, IECE 456 and IECE 460.

IECE 462 Instructional Practices in Natural Sciences for Young Children**Credits 3 Grading Basis Graded**

This course is designed to prepare highly qualified inclusive early childhood teachers who will engage all students in scientific content and processes through the effective use of evidence-based practices.

Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 450. B.I.S./IECE: IECE 466. Corequisites: IECE majors: IECE 324, IECE 431 and IECE 460. B.I.S./IECE: IECE 429.

IECE 464 Instructional Practices in Social Studies for Young Children**Credits 3 Grading Basis Graded**

This course provides students with the knowledge, skills and understanding to design and implement effective social science programs for all young children, birth to age eight. Focus is on appropriate social science content, teaching strategies and materials from a developmental perspective with a special emphasis on adaptations designed to meet the needs of children with disabilities. Students will use technology to support access to the learning environment and curriculum.

Prerequisites: Minimum grade of "C" in requisites. IECE majors: IECE 322. B.I.S./IECE: IECE 300. Corequisites: IECE majors: IECE 421, IECE 450 and IECE 456. B.I.S./IECE: IECE 355, IECE 421 and IECE 466.

IECE 466 Managing Classrooms and Guiding Behavior**Credits 3 Grading Basis Graded**

This course examines research and professional literature on effective strategies for guiding young children's behavior and managing groups. This course uses experiences in IECE 331 as a foundation for reflection, dialogue and development of a personal philosophy of classroom management.

Prerequisites: Minimum grade of "C" in requisites. IECE 300. Corequisites: IECE majors: IECE 322, IECE 331 and IECE 355. B.I.S./IECE: IECE 355, IECE 421 and IECE 464.

IECE 470 Individualized Behavior Intervention for Young Children**Credits 3 Grading Basis** Graded

This course is designed to provide application and evaluation of group management techniques and individual interventions that teach and maintain emotional, behavioral and social skills. Students will study various systematic behavioral interventions to support the behavior and learning of individuals with disabilities accessing the general education curriculum including positive behavioral supports, functional assessments of behavior and teaching social skills. A variety of data collection procedures to inform their professional practice will be examined. Students will be applying evidence-based practices related to positive behavior supports, assessment of challenging behaviors and data collection to inform instructional practice and document student response to interventions that improve behavior such that academic learning can occur.

Prerequisites: Minimum grade of "C" in requisites. IECE 324. Corequisites: IECE majors: IECE 461. B.I.S./IECE: IECE 450.

IECE 471 Advanced Field Experience in IECE Part 2**Credits 2 Grading Basis** Graded

The advanced field experience provides candidates opportunities to use their knowledge of child development to plan meaningful learning experiences with second and third graders. Candidates will learn the teacher's role in supporting children to construct understandings about the natural and social sciences, social studies, literacy and to use mathematical thinking. Candidates will begin implementing instructional strategies learned in all methods courses. This course precedes student teaching in the spring.

Prerequisites: Minimum grade of "C" in requisites. IECE 470. Corequisites: IECE 429.

IECE 480 Student Teaching in Inclusive Early Childhood Education**Credits 5-6 Grading Basis** Credit / No Credit

This student teaching experience provides students with opportunities to apply, in learning environments for young children, the knowledge, skills and dispositions acquired throughout their teacher licensure program. Under the guidance of university supervisors and professionally licensed cooperating teachers, the student teacher gradually assumes full responsibility as the responsible inclusive early childhood educator. Students majoring in inclusive early childhood education must complete this course twice.

Prerequisites: Minimum grade of "C" in requisites. IECE 324. Corequisites: IECE majors: IECE 481.

IECE 481 Seminar in Inclusive Early Childhood Education**Credits 1-3 Grading Basis** Graded

This seminar accompanies IECE 480 (Student Teaching) and provides students with the opportunity to reflect on, evaluate and improve their experiences as pre-professional educators of young children. Students will evaluate and build their own resiliency, identify ethical dilemmas, and engage in joint problem-solving and ethical decision-making related to their student teaching experience. Students will enhance their abilities to collaborate with families and other service providers.

Prerequisites: Minimum grade of "C" in requisites. IECE 324. Corequisites: IECE 480.

IECE 490 Special Topics in Inclusive Early Childhood Education (IECE)**Credits 1-4 Grading Basis** Graded

This course provides an in-depth examination of selected topics which are of current importance in the field of education. May be repeated for credit when course content changes.

Prerequisites: Junior or senior standing and permission of the instructor.

Independent Scholars (IND)

IND 200 Interdisciplinary Scholarship: Introduction to the Independent Scholars Major at JMU

Credits 3 Grading Basis Graded

IND 200 is designed to introduce students to the Independent Scholars major, and to the concepts of interdisciplinary study. Students will explore interdisciplinary research in a range of contemporary areas of study, including scientific, environmental, political, social, and cultural perspectives. Students will also gain familiarity with the possibilities for individualized study at JMU. Successful completion of the course will involve the development of proposals for individualized curricula.

IND 499A Independent Scholars Senior Project**Credits 1 Grading Basis** Graded

This is a three-semester course sequence involving independent research based on the student-designed curriculum, to be completed by all independent scholars majors. The course is offered as parts A, B and C for one, three, and two credits per semester, respectively.

IND 499B Independent Scholars Senior Project**Credits 3 Grading Basis** Graded

This is a three-semester course sequence involving independent research based on the student-designed curriculum, to be completed by all independent scholars majors. The course is offered as parts A, B and C for one, three, and two credits per semester, respectively.

IND 499C Independent Scholars Senior Project**Credits 2 Grading Basis** Graded

This is a three-semester course sequence involving independent research based on the student-designed curriculum, to be completed by all independent scholars majors. The course is offered as parts A, B and C for one, three, and two credits per semester, respectively.

Individualized Study (IS)

IS 200 Individualized Study Major Program Development

Credits 3 Grading Basis Graded

This introductory course is designed to prepare students for transition into higher education programs. Specific content includes focusing a concentration, creating an individualized program, technology in higher education, accessing career resources, career decision making skills, self-awareness, life planning and identifying college level experiential learning. Students will apply the principles of academic writing, use technology and available resources to successfully navigate within higher education, and demonstrate effective communication skills to include critical thinking and analysis.

Prerequisites: Individualized Study majors only.

IS 203 Portfolio Development Workshop**Credits 1 Grading Basis** Credit / No Credit

A short course designed to support students for transition into higher education programs. Specific content includes identifying college-level experiential learning, documenting learning outcomes achieved from experience, documenting learning, determining credit request and organizing a portfolio assessment.

Prerequisites: Individualized Study majors only.

IS 205 Orientation to Learning Services**Credits** 1-6 **Grading Basis** Graded

A short orientation course designed to prepare students for transition into higher education programs. Specific content includes using the university library, accessing university support services, managing time, developing appropriate study techniques, participating in cultural and workplace diversity exercises, implementing goal-setting strategies, and participating in wellness, nutrition, and stress management programs.

IS 206 Researching, Writing and Document Prior Learning**Credits** 3 **Grading Basis** Graded

This course provides an opportunity for students to analyze and articulate college-level learning they have acquired outside a classroom context. Specific content includes identifying college-level learning, writing narrative evaluations of learning, integrating multiple sources of learning and disciplinary theory, producing and arranging sufficient documentation of learning, and organizing a portfolio for assessment. At the end of the class, students who have completed all of the assignments will have the components of a portfolio ready for submission for credit evaluation. May not be repeated.

Prerequisites: Individualized Study majors only and successful completion of IS 200.

IS 250 Service Learning**Credits** 1-6 **Grading Basis** Graded

Leadership, citizenship, and professional competencies may be acquired through community service experiences. Documented service learning competence will be assessed by the Center for Leadership, Service, and Transition and credit awarded, as appropriate.

Prerequisites: IS 203

IS 270 Selected Topics**Credits** 1-6 **Grading Basis** Graded

In-depth study of selected topics with current importance and interest to lower division students that are not otherwise covered in the regular course offerings of academic departments/schools. Course content will vary.

IS 290 Special Studies**Credits** 1-6 **Grading Basis** Graded

Designed to give students an opportunity to do lower-division independent study in selected interdisciplinary areas under the supervision of a faculty member in the appropriate department/school.

IS 300 Sponsored Learning**Credits** 1-6 **Grading Basis** Graded

This course provides a structured learning activity related to a student's area of study and sponsored by an employer, volunteer agency or other appropriate organization. This course functions like a student-organized internship. The course materials guide the reflective process. This course may be repeated for a maximum of six credits.

IS 480 Cooperative Studies**Credits** 1-6 **Grading Basis** Graded

Two or more upper-division students may elect to study cooperatively in a selected area of current importance and interest under the supervision of a faculty member in the appropriate department/school.

IS 490 Special Studies**Credits** 1-6 **Grading Basis** Graded

Designed to give students an opportunity to do upper-division independent study in selected interdisciplinary areas under the supervision of a faculty member in the appropriate department/school.

IS 498 Individualized Study Project**Credits** 3-6 **Grading Basis** Graded

In this course, a student undertakes an in-depth study of an interdisciplinary topic directly related to the student's area of concentration. A final oral report is required. This course may be repeated for a maximum of six credits.

IS 499 Honors**Credits** 6 **Grading Basis** Graded

Multiple-semester course.

Industrial Design (INDU)

INDU 200 Design Matters**Credits** 3 **Grading Basis** Graded

As modern challenges grow more complex and ambiguous, design thinking has emerged as a powerful method for addressing them across business, social and cultural contexts. While traditionally linked to visual elements - like products, fashion and interiors - design now includes the creation of services, systems and strategies. More than a static concept, it is an active process focused on developing meaningful solutions. Through this course, students will learn the fundamentals of the design thinking methodology by applying creative problem-solving techniques such as research, ideation and prototyping.

INDU 202 Industrial Design Studio**Credits** 3 **Grading Basis** Graded

This course takes the fundamentals of INDU 200, INDU 210 and INDU 220 and combines them into industrial design students' first individual holistic design endeavor. Projects will build on the basic skills of design thinking, visual communication and digital tools with deep investigations into concept development, research methods and aesthetics. This course will train students to use and build technical skill sets (sketching, prototyping, presentation) while developing creative skill sets for design process, research and concept development. The expectation is that students will establish a design process (built on learned techniques) that works for them and can be applied over and over. This course will challenge students to make design decisions at every level, in the interest of successful design.

Prerequisites: INDU 200, INDU 210 and INDU 220. Corequisites: INDU 208.

INDU 208 Portfolio Review**Credits** 0 **Grading Basis** Credit / No Credit

Portfolio review required to enroll in industrial design courses at upper division standing. May be repeated once for pass/fail standing.

Prerequisites: INDU 200, INDU 210 and GRPH 101. Corequisites: INDU 202, INDU 220 and GRPH 101.

INDU 210 Making Is Thinking**Credits 3 Grading Basis** Graded

This studio course will introduce students to historic, contemporary and emerging materials through making. This course introduces students to wood shop equipment, machine shop equipment, soft goods lab equipment, thermoforming and the laser cutter. Quality, craftsmanship and shop safety are emphasized. The skills developed will provide students with the opportunity to create professional prototypes and visual tools to aid in their presentation/design development. Materials covered include: wood, metal, plastics, polymers, textiles and ceramics. Through a historic survey of product design, manufacturing will be introduced but is not the main focus of this course. Students will engage in a variety of written assignments and hands-on making projects. Student success in this course requires combination learning consisting of visual, aural, read/write and kinesthetic (VARK) learning in an observational setting.

Prerequisites: FDS 102, FDS 104 and FDS 106. Corequisites: INDU 200.

INDU 212 Materials and Processes for Manufacturing**Credits 3 Grading Basis** Graded

This lecture course will build on history, contemporary and emerging materials through the manufacturing processes of both small scale and industrial scale products. Materials covered include: wood, metal, plastics, polymers, textiles and ceramics. Manufacturing processes covered will be expanded from historic to future technologies used in a variety of product design development. Students will engage in a variety of written assignments and field trips to local manufacturing facilities. Student success in this course requires combination learning consisting of visual, aural, read/write and kinesthetic (VARK) learning in an observational setting.

Prerequisites: INDU 210. Corequisites: INDU 202, INDU 208 and INDU 222.

INDU 220 Visual Communication for Industrial Design I**Credits 3 Grading Basis** Graded

This course introduces core design visualization techniques that enable students to express design thinking and generate more creative concepts through sketching. By translating a variety of ideas into visual form, students enhance communication within the design process. Because products ultimately exist in three dimensions, students also learn technical drafting to represent their envisioned manufacturing methods and facilitate prototyping. This course cultivates the ability to conceptualize and depict three-dimensional products in 2D, reinforcing foundational skills essential for industrial design practice.

Prerequisites: FDS 102, FDS 104 and FDS 106. Corequisites: INDU 200.

INDU 222 Visual Communication for Industrial Design II**Credits 3 Grading Basis** Graded

This course expands on the foundational skills acquired in INDU 220, focusing on advanced CAD modeling, photorealistic rendering, visual storytelling and digital fabrication. Students refine or revisit prior projects to develop complex geometries and realistic lighting and materials using advanced rendering tools such as KeyShot. The course also explores AI-powered concept generation to broaden creative ideation. Additionally, students will be introduced to digital fabrication techniques, including 3D printing and VR/AR modeling, to extend their ability to prototype and visualize designs in immersive environments. By the end, students will have produced professionally polished visuals and narrative-driven presentations that showcase strong design communication skills.

Prerequisites: INDU 200, INDU 210 and INDU 220.

INDU 292 Topics in Industrial Design**Credits 1-3 Grading Basis** Graded

Study of selected topics in industrial design at the introductory level. May be repeated when course content changes. See MyMadison for current topics.

INDU 300 Product Design**Credits 6 Grading Basis** Graded

This course expands on foundations learned in INDU 200 and INDU 202 and emphasizes a research-driven design process as it applies to the design of objects, including commercial products, packaging design, furniture and more. Students will use quantitative and qualitative research methods and analyze market segments and/or users.

Prerequisites: INDU 208.

INDU 302 Transportation Studio**Credits 6 Grading Basis** Graded

A study of transportation system design utilizing a multi-disciplinary effort. Students will engage with civil engineers, city planners, urban designers, community members and various stakeholders to understand transportation needs and develop creative solutions.

Prerequisites: INDU 208.

INDU 360 Professional Practice in Industrial Design**Credits 3 Grading Basis** Graded

This course prepares students for entry into the professional field of industrial design by focusing on the development of a comprehensive design portfolio and essential career-readiness skills. Students will learn to curate, present and communicate their work effectively, while also gaining insight into professional expectations, ethics, job search strategies and the business of design. Emphasis will be placed on critique, refinement and preparation for employment, freelance or graduate study. Course topics may include the development of professional contacts, interviewing skills, portfolio development and the preparation of résumés, press releases and design contracts. Students develop a comprehensive package to prepare for employment based on improved skill sets and capabilities. In-class learning is augmented by site visits to manufacturing facilities, design offices in the area and remotely via Zoom.

Prerequisites: INDU 202. Corequisites: One 300- or 400-level INDU course.

INDU 390 Independent Studies in Industrial Design**Credits 1-3 Grading Basis** Graded

Independent activity at the intermediate level, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with the consent of the instructor.

INDU 392 Topics in Industrial Design**Credits 3 Grading Basis** Graded

Study of selected topics in art, art history, graphic design, interior design, or industrial design at the intermediate level. May be repeated when course content changes. See MyMadison for current topics.

INDU 400 Systems and Services**Credits 6 Grading Basis** Graded

This studio course aims to engage students in a systems based approach to problem solving. Students will design and test concepts and solutions that focus on interaction and experience. The course emphasizes problem framing, system and service prototyping, graphic design and illustration, user interaction/user experience, and end-user engagement.

Prerequisites: INDU 208.

INDU 402 Social Innovation and Social Entrepreneurship**Credits 6 Grading Basis** Graded

This advanced studio explores design problems and solutions ranging from sanitation and poverty to environmental degradation and access to basic human services. Students will work individually and on teams to target specific areas of need and develop proposals to create products, systems and services to address those needs while generating revenue. Prerequisites: INDU 208.

INDU 404 Industrial Design Advanced Special Topics Studio**Credits 6 Grading Basis** Graded

Study of selected topics in industrial design at the advanced level. May be repeated when course content changes.

INDU 406 Senior Thesis Studio**Credits 6 Grading Basis** Graded

Industrial design students with senior standing submit a proposal focused on developing a semester long studio project.

Prerequisites: INDU 208 and senior standing; or permission of the instructor.

INDU 408 Swing Studio Option**Credits 6 Grading Basis** Graded

Students may take a six credit hour studio in ARCD or the equivalent in any one other SDAH studio disciplines (i.e., ART 240 and ART 340). The ID program coordinator must approve proposals for swing studio options in advance. Art and design elective courses taken for the ID advanced studio requirement may not be double-counted as swing studios.

Prerequisites: INDU 208.

INDU 490 Independent Studies in Industrial Design**Credits 1-3 Grading Basis** Graded

Independent activity, such as research or studio practice, under faculty supervision. Projected studies in any area of the school's offering must be arranged with the instructors who will direct them. Offered only with the consent of the director.

INDU 491 Studio Assistant**Cross-listed with:** ART 491, ARED 491, GRPH 491, ARCD 491**Credits 1-3 Grading Basis** Graded

An on-campus program monitored on an individual basis designed to provide practical studio experience in the visual arts. Students will learn safe studio practices and management skills, including material use, inventory control and the proper operation of equipment found within various individual classroom studios.

INDU 492 Topics in Industrial Design**Credits 3-6 Grading Basis** Graded

Study of selected topics in art, art history, graphic design, interior design, or industrial design at the advanced level. May be repeated when course content changes. See MyMadison for current topics.

Prerequisites: INDU 208.

INDU 496 Internship in Industrial Design**Credits 1-8 Grading Basis** Graded

An off-campus program prepared and monitored on an individual basis. Internships are designed to provide practical experience in the arts.

Information Technology (IT)

IT 101 Introduction to Information Technology**Credits 3 Grading Basis** Graded

This introduction to the field of information technology is designed to prepare students for a successful career, with an emphasis on developing skills with computers, hardware and software, problem-solving, troubleshooting, studying, and time management. The course also compares information technology with related fields such as computer science, computer engineering and information systems. Topics include networking, databases, computing systems and platforms, cybersecurity, web and mobile systems, and application programming.

IT 160 Data Structures and Advanced Programming**Credits 3 Grading Basis** Graded

This course covers applied data structures, introductory algorithm analysis and advanced programming techniques in the Python programming language. Topics include dictionaries, objects, sets, multisets, stacks, queues and priority queues, as well as their applications in the information technology discipline.

Prerequisites: IT major and a minimum grade of "B-" in CS 149.

IT 201 Computational Structures and Logic**Credits 3 Grading Basis** Graded

This course introduces discrete mathematical concepts and their applications within the IT discipline. The structures studied may include functions, relations, sets, logic, matrices, elementary number theory, basics of counting, graph theory, discrete probability and digital logic. Students will explore the fundamental principles, theorems and techniques related to these structures and apply them to practical applications in the field of IT.

Prerequisites: IT major and a minimum grade of "B-" in CS 149.

IT 203 Information Security and Privacy**Credits 3 Grading Basis** Graded

This course teaches computer security principles and incident prevention and management. It covers the information assurance dimensions of availability, integrity, authentication, confidentiality and non-repudiations to ensure secure transmission, storage and processing of information.

Prerequisites: Fully admitted to the IT major; MATH 231 or MATH 235 or equivalent; and a minimum grade of "C" in IT 160 and IT 201.

IT 212 Digital Electronics**Credits 3 Grading Basis** Graded

This course introduces students to digital systems, logic design and the underlying hardware of computer systems. Topics include number systems and binary representations, digital logic gates, various combinational logic blocks, and sequential circuits as well as an introduction to programmable logic devices and basic computer architecture. Design and analysis of circuits will be covered in theory as well as practiced in the laboratory using appropriate industry tools and equipment.

Prerequisites: Fully admitted to the IT major; MATH 231 or MATH 235 or equivalent; and a minimum grade of "C" in IT 160 and IT 201.

IT 215 Introduction to Telecommunications and Computer Networking
Credits 3 Grading Basis Graded

This course introduces students to telecommunications and computer networking concepts. Students learn interactively and collaboratively how networked applications and services are implemented and secured using wireless and wired networks, network services, and protocols. Topics include command-line administration, virtualization, packet analysis, physical/network/transport addresses, the internet, network services and network applications. Students will perform hands-on exercises integrating and troubleshooting networked applications, network techniques, methods and services.

Prerequisites: Fully admitted to the IT major; MATH 231 or MATH 235 or equivalent; and a minimum grade of "C" in IT 160 and IT 201.

IT 240 Database Administration

Credits 3 Grading Basis Graded

This course provides an introduction to database administration focused on securing and maintaining modern database systems in various hosting environments. Through hands-on exercises, students gain knowledge of database deployment and administration, including backup and recovery.

Prerequisites: Fully admitted to the IT major and a minimum grade of "C" in IT 160 and IT 201.

IT 280 Projects in Information Technology

Credits 1-4 Grading Basis Graded

This course will cover new topics in Information Technology that are of interest to students but not otherwise covered in the curriculum. May be repeated for credit when the course content changes. This course differs from IT 480 in that it is intended for first year, sophomore and junior level students.

Prerequisites: First year, sophomore or junior standing IT students only. Topics selected may dictate further prerequisites; students should consult the instructor prior to enrolling for the course.

IT 301 Web Technologies

Credits 3 Grading Basis Graded

This course provides an introduction to web technologies including languages, the client-server architecture, networking, database concepts and web services. Students learn how to develop web applications that are dynamic, secure and usable.

Prerequisites: IT 240 and IT 215.

IT 302 Societal and Ethical Issues in Information Technology

Credits 3 Grading Basis Graded

This course is a survey of professional, societal and ethical issues in information technology. Topics include professional responsibilities, privacy, intellectual property, risk identification and evaluation, IT governance, economic and availability factors of IT work, and environmental issues. Students develop skills in assuming and defending positions on societal and ethical issues through oral presentations and written reports.

Prerequisites: IT 203.

IT 311 Operating Systems Administration

Credits 3 Grading Basis Graded

This course introduces students to configuring, deploying and administering computer operating systems. Topics include system configuration, file systems, security, administration, network interfacing, multitasking, multiuser systems, device driver installation and analysis of operating system performance. Students will perform hands-on exercises administering an operating system.

Prerequisites: IT 212.

IT 313 Community Projects

Credits 3 Grading Basis Graded

Students will apply information technology to projects from community partners, including education, government, business and industry. Students will use the skills they have developed to address real-world problems requiring technology solutions. Students will be divided into teams and each team will be responsible for one part of the semester-long project.

Prerequisites: IT 301 and a minimum grade of "C-" in IT 333.

IT 333 Advanced Networking for Information Technology

Credits 3 Grading Basis Graded

This course presents a practical study of the underlying principles of networking and how these principles are deployed to provide efficient and secure networks for providing voice, data, video and mobility applications. Through hands-on exercises, students gain knowledge of network components, network protocols, network interfaces, network architectures/technologies, network facilities (wired and wireless), network services, network analysis and troubleshooting, network management and policies, network security, and networked applications.

Prerequisites: A minimum grade of "C" in IT 201 or CS 227, and a minimum grade of "C" in IT 215.

IT 334 Computer Cyber Crime, Forensics and Auditing

Credits 3 Grading Basis Graded

This course provides students an introduction to computer crime, relevant laws, agencies and standards. Topics include auditing, logging, forensics and related software, as well as legal principles such as chain of evidence, electronic document discovery, eavesdropping and entrapment. Students will perform hands-on exercises with forensics tools.

Prerequisites: IT 302.

IT 340 Data Science and Machine Learning

Credits 3 Grading Basis Graded

This course provides an overview of data science and machine learning for the Information Technology field. Students will learn how to extract data from sensing devices and use machine learning techniques for data analysis. Students will also learn about skills and tools available for data science.

Prerequisites: IT 240 and either MATH 220 or MATH 232 or MATH 235.

IT 341 Mobile Applications

Credits 3 Grading Basis Graded

This course is a project-based introduction to fundamental principles of application development for the mobile device environment, emphasizing design and programming paradigms that stem from the use of mobile devices. Topics include user interfaces, event-based programming, inter-process communications, networking, mobile-specific capabilities and performance in a resource-restricted environment. Students will perform hands-on exercises to develop and deploy a mobile application.

Prerequisites: IT 240 and a minimum grade of "C-" in IT 333.

IT 342 Server Administration

Credits 3 Grading Basis Graded

This course will introduce students to the management of web servers. Topics include administration activities and domains for computing systems, including configuration, deployment, performance analysis, backup and recovery. Students will perform hands-on exercises administering a web server.

Prerequisites: IT 311 and a minimum grade of "C-" in IT 333.

IT 347 Interactive Computing Systems**Credits 3 Grading Basis Graded**

This course examines design and evaluation of effective user interfaces of a computing system, beginning with principles for designing an IT product for user interaction separate from interactive software development. The development process includes iterative life cycle management, systems analysis, design, usability specifications, design representation techniques, prototyping, formative user-based evaluation and connectivity with other IT designs. The course emphasizes an integrative and cross-disciplinary approach with a primary emphasis on usability methods and the user interaction development process.

Prerequisites: IT 301.

IT 415 Data Visualization Fundamentals and Techniques**Cross-listed with: CS 415****Credits 3 Grading Basis Graded**

This course provides an introductory examination of data visualization through lectures, applied learning activities, readings and hands-on experience with current visualization tools. It also introduces a theoretical framework that covers different types of data, including temporal, geospatial, topical and network data, and their analysis and visualization. Students will improve their data and information literacy, strengthen data visualization skills and build analytic abilities relevant to visualization problems. They will also gain an overview of various data types and examine visualizations and workflows that aim to answer "when" (temporal data), "where" (geospatial data), "what" (topical data) and "with whom" (trees and networks) questions. The course is intended for students with little or no prior background in data visualization and is targeted to those interested in using visualization in their own work and future academic studies.

Prerequisites: CS 240 with a minimum grade of "C-" or IT 340.

IT 433 Internship in Information Technology**Credits 1-3 Grading Basis Graded**

Students participate in an internship related to information technology outside of the university.

Prerequisites: IT 311, a grade of "C-" in IT 333 and prior approval from IT faculty.

IT 435 Applied Cryptography**Credits 3 Grading Basis Graded**

This course presents techniques in cryptography with analysis of its historical and societal consequences and an emphasis on meeting the needs of the information technology field. Topics include elementary combinatorics and number theory such as modular arithmetic and prime numbers, classical ciphers and accompanying attacks, and rotary machines as well as modern encryption schemes including public channel cryptography, secret-key ciphers, hash functions, side-channel attacks and post-quantum cryptography.

Prerequisites: IT 203 and IT 201.

IT 443 Cloud Computing and IoT Networks**Credits 3 Grading Basis Graded**

This course provides an introduction to the network architectures of cloud computing and the Internet of Things (IoT). Topics include software and hardware components including sensors and detectors for cloud and IoT computing as well as their applications. Students will perform hands-on exercises using IoT applications.

Prerequisites: A minimum grade of "C-" in IT 333.

IT 444 Capstone Project Design**Credits 1 Grading Basis Graded**

This is the first course in a two-course sequence. Students will work in groups to propose an IT-based solution to address a problem or need in the community.

Prerequisites: IT 313.

IT 445 Capstone Project Implementation**Credits 3 Grading Basis Graded**

In this second course in a two-course sequence, students will work in groups to implement and integrate their design, manage the project as a team, and prepare an oral and written report on the work.

Prerequisites: IT 444.

IT 460 Advanced Cybersecurity**Credits 3 Grading Basis Graded**

This course focuses on the underlying principles of cybersecurity as applied to information and infrastructure network systems and services. Emphasis is placed on gaining hands-on knowledge in network access control, user authentication and authorization, intrusion and prevention systems, and network boundary protection. The course provides a hands-on introduction to the principles of cybersecurity penetration testing.

Prerequisites: A minimum grade of "C-" in IT 333.

IT 461 Internetworking**Credits 3 Grading Basis Graded**

This course will introduce students to technologies and protocols typically used for interconnection of networks. Interconnections of networks deals with routing and establishing an end-to-end information transfer path among end nodes (clients and servers).

Prerequisites: A minimum grade of "C-" in IT 333.

IT 462 Network Applications Development**Credits 3 Grading Basis Graded**

In this course, students will learn about the design and implementation of network-based applications, focusing on both the application layer and the transport layer of the TCP/IP protocol stack. Students will learn about the design and implementation of network-based applications using languages and architectures such as sockets, Java and Python. Students learn concepts in distributed processing, including synchronization of interprocess communication and management of replicated data.

Students analyze performance issues related to distributed applications.

Prerequisites: A minimum grade of "C-" in IT 333.

IT 465 Wireless Networking and Security**Credits 3 Grading Basis Graded**

This course provides an introduction to wireless networking and wireless LAN security and forensics. Radio frequency fundamentals are introduced with emphasis on applications and services. Hands-on network configurations and analysis tools for wireless LAN are introduced and exercised with emphasis on network performance, security and forensic applications.

Prerequisites: A minimum grade of "C-" in IT 333.

IT 480 Selected Topics in Information Technology**Credits 1-3 Grading Basis Graded**

This course will cover topics in information technology that are of interest but not otherwise covered in the regular information technology offerings of the department. May be repeated for credit when course content changes.

Prerequisites: Junior or senior standing only. Topics selected may dictate further prerequisites; students should consult the instructor prior to enrolling for the course.

IT 497 Independent Study**Credits 1-3 Grading Basis Graded**

This is an advanced course to give independent study experience under faculty supervision. May be taken multiple times for credit, but no more than three credits may count toward the Information Technology major requirements.

Prerequisites: Junior or senior standing and permission of the program coordinator.

IT 499 Honors Capstone Implementation II**Credits 2 Grading Basis Graded**

This is the third course in a three-course sequence including IT 444 and IT 445. Students are divided into several groups or work individually; each group solves an IT-related problem or builds an IT application that help to solve a certain problem in the IT community.

Prerequisites: IT 444 and instructor permission. Corequisites: IT 445.

Integrated Science and Technology (ISAT)

ISAT 100 Environmental and Energy Sustainability [C3PP]**Credits 3 Grading Basis Graded**

This course explores scientific and technical issues important to environmental and energy sustainability. Students study fundamental chemistry and physics and then apply this knowledge to better understand air quality, water quality, and conventional and alternative energy processes. The class also explores the societal impacts of our energy choices and the potential impact we as individuals can have through personal initiative. May be used for general education credit.

ISAT 112 Environmental Issues in Science and Technology (2,3) [C3PP]**Credits 4 Grading Basis Graded**

This course integrates the study of environmental chemistry, ecology and sustainability science to achieve a conceptual and quantitative understanding of environmental issues including air pollution, climate change, ozone depletion, acid deposition, plastics, waste and biodiversity. This course may be used for general education credit.

ISAT 113 Biotechnology Issues in Science and Technology [C3NS]**Credits 3 Grading Basis Graded**

This course is an applied introduction to biological systems, as an entry point for the understanding and analysis of biotechnological solutions. Topics include the way in which features of biochemical structure and function, information flow within cells, genetic inheritance, metabolism, and immune responses can be used to solve problems related to disease, food and the environment. May be used for general education credit. Corequisites: ISAT 113L.

ISAT 113L Biotechnology Issues in Science and Technology Laboratory [C3L]**Credits 1 Grading Basis Graded**

Students will build practical and scientific literacy skills in the context of a life science wet lab. Descriptive and inquiry-based activities will include elements of study design, protocol implementation, and data collection and analysis. Recordkeeping and analytical skills will develop through in-lab documentation and comprehensive reports. May be used for general education credit.

Prerequisites: ISAT 113. Corequisites: ISAT 113.

ISAT 150 Algebra Essentials**Credits 1 Grading Basis Graded**

This course provides review and practice in algebra concepts that are needed to successfully complete ISAT 151. Various mathematical models, including trigonometric, are also reviewed. The course is designed for students who possess a basic understanding of algebra but are not proficient in its application.

Corequisites: ISAT 151.

ISAT 151 Topics in Applied Calculus in ISAT [C3QR]**Credits 4 Grading Basis Graded**

This course introduces the concepts of differential and integral calculus and ordinary differential equations to model real-world applications in science, business, technology and economics. This course includes a computer laboratory component emphasizing modeling and numerical methods. Course assumes familiarity with algebra and trigonometry. May be used for general education credit.

ISAT 151L Topics in Applied Calculus Computer Laboratory**Credits 1 Grading Basis Graded**

This course is the computer laboratory portion of ISAT 151: Topics in Applied Calculus in ISAT. It is intended for students who already have AP credit or calculus lecture credit. Students will use numerical methods to solve mathematical modeling and calculus problems with Microsoft Excel. Students will study linear, polynomial, exponential, logarithmic, S-curve, and trigonometric models in business and the physical and natural sciences. Permission of instructor or department required.

ISAT 152 Topics in Applied Physics in ISAT**Credits 4 Grading Basis Graded**

This course introduces topics in general physics including one- and two-dimensional motion, mechanics, energy, waves, electricity, magnetism, optics, lasers, and early quantum theory. Vectors, algebra, and differential and integral calculus, are used to model physical system behavior. Laboratory experiments and computer exercises enhance understanding of the concepts.

Prerequisites: ISAT 151 or permission of the instructor.

ISAT 160 Problem Solving Approaches in Science and Technology [C1CT]**Credits 3 Grading Basis Graded**

This course examines issues in modern science and technology as a means to introduce, develop and enhance critical thinking and problem-solving skills. Current scientific and technological research and applications will be introduced to reinforce problem solving, instruction in systems thinking and critical inquiry. The course provides opportunities for using both oral and written communication in a variety of learning activities. May be used for general education credit.

Prerequisites: If you have already completed, or are currently registered for, a Critical Thinking (Madison Foundations) class, then you must secure permission to receive credit for a second Critical Thinking class.

ISAT 171 Governance of Sociotechnical Systems**Credits 3 Grading Basis Graded**

Solutions to human problems are mediated by economic and political institutions, which in turn help to shape technology and science. The course covers basic political and economic concepts, institutions and processes as they relate to American and international science and technology.

ISAT 180 Topics in Integrated Science and Technology**Credits** 1-4 **Grading Basis** Credit / No Credit

Special topics in integrated science and technology which are of interest to the entry-level student. May be repeated for credit when course content changes. Students should consult the instructor prior to enrolling for the course.

ISAT 181 Student Research Project**Credits** 1-6 **Grading Basis** Graded

Research project on a science and technology topic of interest, as arranged with a faculty research advisor. Projects will include an assessment of the non-technical issues that surround the technical problem.

ISAT 190 Introduction to Holistic Problem Solving in the ISAT Major**Credits** 1 **Grading Basis** Graded

This course introduces holistic problem solving through the lens of the ISAT major. Skills such as effective teaming, information literacy, systems thinking, stakeholder identification and complex problem analysis are developed.

ISAT 203 Viral Discovery**Cross-listed with:** BIO 203**Credits** 2 **Grading Basis** Graded

An exploratory laboratory experience, designed for incoming freshmen. Students will learn about the life cycle and ecology of viruses infecting bacteria. Soil samples will be collected, and techniques for isolation and purification of the viruses will be performed by the students. Isolated viruses will be visualized using electron microscopy. The genomic material will be isolated and prepared for nucleic acid sequencing.

ISAT 204 Viral Genome and Bioinformatics**Cross-listed with:** BIO 204**Credits** 2 **Grading Basis** Graded

A computer-based laboratory experience, designed for students completing the Viral Discovery course. Students will learn to identify genes in a viral genome, compare the predicted proteins with known proteins in databases, describe the contents of the genome and notate all the relevant information for publication. Students will also research the role of bacteriophages in ecology and evolution.

ISAT 211 Issues in Modern Production Systems**Credits** 3 **Grading Basis** Graded

This course explores the principles of making, designing, integrating and improving production systems by asking the following questions: What is the right way to transform materials and who (or what) determines that? How do political, environmental, social, technological, legal/regulatory, ethical and economic factors impact the way individuals and organizations make, design, integrate and improve? How do production systems affect our ecosystem at different scales, ranging from artisanal craft methods to complex global supply chains and continuous production? Advanced manufacturing, continuous improvement and sustainable production systems will be explored. Includes a lab component.

ISAT 212 Energy Issues in Science and Technology**Credits** 3 **Grading Basis** Graded

Introduction to scientific and economic concepts relevant to energy. Concepts are taught within the context of three or four themes, e.g. residential energy efficiency, renewable energy sources, "make-or-buy" fossil energy application and space power systems. Themes may change from year to year reflecting contemporary issues and opportunities to link with industry and government agencies.

Prerequisites: ISAT 152 or Permission of the Instructor.

ISAT 251 Topics in Applied Statistics in ISAT [C3QR]**Credits** 3 **Grading Basis** Graded

This course introduces statistical thinking - the discipline and methods for collecting, analyzing, and interpreting data for making decisions, doing science, and understanding our world. Topics covered include an introduction to data analysis methods, probability and chance, statistical reasoning and inference, and experimental design. The course includes a laboratory component emphasizing hands-on analysis of data taken from a variety of applications in ISAT. May be used for general education credit.

Prerequisites: Sophomore standing.

ISAT 252 Programming and Problem Solving**Credits** 3 **Grading Basis** Graded

This course is an introduction to computational thinking and formal logic. Students create software to solve problems in applied science, business and engineering taking social context into account. Programming paradigms include procedural, object-oriented and event-driven. Emphasis is placed on effective analysis, planning, documentation, communication and teamwork in professional software development settings.

Prerequisites: Sophomore standing.

ISAT 271 Technology, Science and Society**Credits** 3 **Grading Basis** Graded

This course introduces the social aspects of technology and science. It covers social science methods and related philosophical and ethical analyses. Students learn how the practice of science relates to the human-built world and why critical evaluations of science and technology policies are important.

Prerequisites: ISAT 171.

ISAT 280 Projects in ISAT**Credits** 1-4 **Grading Basis** Graded

Projects or special topics in integrated science and technology which are of interest to the lower-division student. May be repeated for credit when course content changes. Projects or topics selected may dictate prerequisites. Students should consult the instructor prior to enrolling for the course.

ISAT 290 Introduction to Systems Thinking for Complex Problems**Credits** 3 **Grading Basis** Graded

Students will develop skills to apply systems thinking methodology to investigate, define and describe complex problems. They will build a recognition and appreciation of common causal structures that influence the evolution of a problem over time. Structures include interactions between cultural, institutional, natural and technological factors. Collaboration, integration of diverse knowledge sources and project management skills will be emphasized.

Prerequisites: ISAT 190 or permission of the instructor and ISAT 212.

ISAT 300 Applied Computing, Instrumentation and Measurement**Credits** 3 **Grading Basis** Graded

This course covers the fundamental nature of measurement in the practice of science and technology. It examines how computing technology is applied to the collection, storage, processing and presentation of data. In addition to discussions about how and why measurements are taken, the uncertainty involved in all measurements and the statistical methods used for analysis will be covered. Students will learn about measurement systems and how they can be networked using wireless and wireline technology for data acquisition. This course includes a project-based lab component focusing on hands-on integration and troubleshooting of networked measurement systems, including computation methods for data analysis and data visualization.

Prerequisites: ISAT 252 and either ISAT 251 or MATH 220.

ISAT 301 Instrumentation and Measurement in Energy (0,2)**Credits 1 Grading Basis** Graded

Instrumentation is used to acquire data from representative systems that include mechanical, thermal, solar, chemical and nuclear energy. Students analyze the data to enhance understanding of these forms of energy. Energy transport processes are also characterized. Computer-based data acquisition is emphasized.

Prerequisites: ISAT 212 and ISAT 300.

ISAT 302 Instrumentation and Measurement of the Environment (0,2)**Credits 1 Grading Basis** Graded

Traditional and contemporary analytical laboratory and field techniques used in environmental quality monitoring are surveyed. Emphasis is placed on understanding the physical, chemical and biological basis of these techniques. Hands-on laboratory and field work will be emphasized, in addition to quality control/assurance of environmental data.

ISAT 303 Production Systems Automation Lab**Credits 1 Grading Basis** Graded

The role, function and implementation of sensors, actuators and controls in open and closed loop control systems are explored and implemented in constant process flexible manufacturing systems. Programmable logic controllers (PLCs) are emphasized. The synthesis of automation technologies is realized by exploring physical mechatronic systems via hands-on lab experiences using real world equipment.

ISAT 305 Instrumentation and Measurement in Biotechnology (0, 2)**Credits 1 Grading Basis** Graded

This course provides a hands-on experience of the techniques and instrumentation used in the modern biotechnology laboratory. Topics include aseptic techniques for establishing microbial cultures, detection and analysis of recombinant DNA molecules, protein purification, SDS gel electrophoresis and the use of PCR technology for genetic analysis.

Prerequisites: ISAT 113 and ISAT 113L; or BIO 240.

ISAT 310 Energy Fundamentals I**Credits 3 Grading Basis** Graded

This course covers the integration of fundamental concepts from physics, chemistry, mathematics and engineering within the context of energy applications. Principles governing energy transformations, transport and conversion, including laws of thermodynamics, chemical and nuclear reactions and thermal science.

Prerequisites: ISAT 212.

ISAT 311 Role of Energy in Modern Society**Credits 3 Grading Basis** Graded

This course covers the role of energy in the U.S. and world economies. Geology of energy-valued natural resources: size, quality and economics of domestic and world resource base. Models for energy use by different sectors. The role of energy in global climate change; other energy-related environmental concerns; and the implications for national and international security will be studied.

Prerequisites: ISAT 212.

ISAT 320 Fundamentals of Environmental Science and Technology I**Credits 4 Grading Basis** Graded

This course provides the student with a basic understanding of environmental processes, pollution and control technologies. Grounded in ecology and systems thinking, ISAT 320 integrates classroom learning, field-based studies and laboratory analysis of field samples to explore local aquatic and terrestrial environments and contextualize them in broader scientific knowledge.

Prerequisites: ISAT 112.

ISAT 321 Fundamentals of Environmental Science and Technology II**Credits 3 Grading Basis** Graded

This course builds on the student's basic understanding of environmental pollution, processes and control technologies in the context of systems thinking and sustainable practices. Content includes solid and hazardous waste and its management, the principles of environmental risk assessment, air quality analysis and management, drinking water and wastewater treatment, and life cycle analysis.

Prerequisites: ISAT 320 or permission of the instructor.

ISAT 330 Sustainable Manufacturing Systems**Credits 3 Grading Basis** Graded

Introduces and applies principles of production planning and control with an emphasis on sustainable production and environmentally conscious manufacturing. Topics include scheduling, inventory control, facility layout, capacity planning, forecasting and lean manufacturing.

Prerequisites: ISAT 211 and ISAT 251 or permission of the instructor.

ISAT 331 Cyber-Physical Industry**Credits 3 Grading Basis** Graded

This course provides an in-depth treatment and understanding of manufacturing automation and control technologies, material transportation and identification, cyber physical systems, and human factors and ergonomics in manufacturing. Topics include industrial control systems, hardware components for automation and process control, computer numerical control, industrial robotics, programmable logic control (PLC), material transport systems, storage systems, automatic identification and data capture, critical infrastructure, smart cities, cognition, decision making, biomechanics at work, and human computer interaction.

Prerequisites: ISAT 211.

ISAT 340 Database Design, Implementation and Management**Credits 3 Grading Basis** Graded

This course provides an introduction to database technology as a component of information systems. The course will introduce students to the design and management of databases through relational database management systems, including the use of Structured Query Language (SQL), forms and reports. Current developments in the role and deployment of a database as part of the application stack will be covered.

Prerequisites: ISAT 252 or permission of the instructor.

ISAT 341 Machine Learning and Data Science**Credits 3 Grading Basis** Graded

Explores the data science and machine learning landscapes by developing and implementing programs capable of learning from data extracted from several knowledge domains. Students will use production ready tools that facilitate model development and deployment to a variety of computing platforms.

Prerequisites: Junior standing and ISAT 340 or CS 149.

ISAT 344 Applied Artificial Intelligence**Cross-listed with:** ISAT 444**Credits 3 Grading Basis** Graded

This course provides a hands-on introduction to building practical AI systems. Topics include: knowledge representation (propositional and predicate logic), search (local, global, uninformed and informed), planning, machine learning and natural language processing. Programming projects will involve designing intelligent agents that can play various board games using Prolog and Python. The course also discusses societal impacts of current and future AI systems and review state, national and international regulations designed to regulate the development and use of AI systems.

Prerequisites: CS 159 or ISAT 340

ISAT 350 Biotechnology for the New Millennium I**Credits 3 Grading Basis** Graded

This course covers the scientific foundations and historical development of biotechnology. Specific topics include living system nanotechnology; cell structure and function; origins of genetic engineering; and recombinant DNA technology.

Prerequisites: ISAT 113/ISAT 113L.

ISAT 351 Biotechnology for the New Millennium II**Credits 3 Grading Basis** Graded

This course is a continuation of ISAT 350 and describes applications of biotechnology in agriculture, industry and medical science as well as associated social, ethical and philosophical issues. Topics include study of an emerging infectious disease, energy transduction in living systems and novel applications of biotechnology.

Prerequisites: ISAT 113, or BIO 240/BIO 240L and BIOT 260 or permission of the instructor.

ISAT 390 Holistic Problem-Solving Workshop I**Credits 3 Grading Basis** Graded

In this course, students will practice holistic problem-solving approaches to analyze and propose solutions for complex problems that involve a mix of scientific, technological and societal elements. The course will use a variety of case studies that represent real-world problems that significantly impact society. For each case study, students will explore the system dynamics from which the problem emerges and evaluate both short- and long-term consequences of possible solutions.

Prerequisites: ISAT 290 and ISAT 271.

ISAT 391 Holistic Problem-Solving Workshop II**Credits 3 Grading Basis** Graded

This course continues the practice of holistic problem-solving initiated in ISAT 390. In this follow-up course, students will operate in a self-guided manner to analyze and propose solutions for complex problems that involve a mix of scientific, technological, and societal elements. Students will explore the system dynamics from which the problem emerges and evaluate both short- and long-term consequences of possible solutions.

Prerequisites: ISAT 390.

ISAT 401 Advanced Computer-Based Instrumentation I**Credits 3 Grading Basis** Graded

This is largely a laboratory course in which students build and program their own instruments. Topics include programming techniques for real-time instrumentation programming; buffered analog and digital input and output; timing considerations; passive analog filters and active analog filters; digital-filtering techniques; and real-time programming issues.

Prerequisites: ISAT 252 and ISAT 300.

ISAT 410 Sustainable Energy Development**Credits 3 Grading Basis** Graded

This course is concerned with science and the applications of solar and other renewable technologies, e.g., solar thermal electric, photovoltaics, wind power, biomass-derived alcohols, solar hydrogen and ocean thermal energy conversion Energy storage systems and materials, combined renewable conventional systems for peaking and load management and alternative energy sources for transportation will be studied.

Prerequisites: ISAT 310 or permission of the instructor.

ISAT 411 Energy Economics and Policy**Credits 3 Grading Basis** Graded

This course is concerned with methods for analyzing the economics, environmental and societal benefits of energy technologies. Topics include optimization techniques, utility planning and finance, cost-benefit techniques, discounting for time and risk, econometric models and input-output analysis. The role of government in determining energy costs supply and markets will be considered.

Prerequisites: ISAT 311.

ISAT 413 Options for Energy Efficiency**Credits 3 Grading Basis** Graded

This course makes detailed examination of new technologies to increase the efficiency of energy conversion, transportation systems and end-use technologies. Examples include MHD, combined-cycle systems, advanced nuclear reactors, intelligent transportation systems, high-efficiency lighting, energy management and utilization of low-temperature heat. Consideration is made of the socioeconomic and governmental barriers to energy efficiency.

Prerequisites: ISAT 310 and ISAT 311.

ISAT 414 Energy Fundamentals II**Credits 3 Grading Basis** Graded

Introduction to the sciences of fluid mechanics and heat transfer and the physical laws governing the mechanical behavior of liquids and gasses. Conservation of mass, energy and momentum. Discussion of heat transfer by one-dimensional conduction, convection and radiation. Fluid statics, internal and external fluid flow, pipe networks and heat exchanger analysis.

Prerequisites: ISAT 310 or permission of the instructor.

ISAT 416 International Energy Studies**Credits 3 Grading Basis** Graded

Study-abroad course examining international energy problems and providing team-oriented project experiences. Addresses energy issues associated with economic and social development. Project participation, tours and meetings with local experts illustrate energy-related problems that are compared with those in the United States.

Prerequisites: ISAT 212 and ISAT 300.

ISAT 420 Environmental Analysis and Modeling**Credits 3 Grading Basis** Graded

This course explores applications of mathematical techniques and computer models to the environmental field. The course introduces the principles underlying environmental analysis and modeling, including limitations and pitfalls. Several environmental models and analytical systems are then discussed and applied, using a variety of computational platforms.

Prerequisites: ISAT 321.

ISAT 421 Environmental Policy and Regulation**Credits 3 Grading Basis** Graded

This course will familiarize students with basic environmental laws and regulations. The course discusses the purpose of environmental policy, the role of environmental economics in policy decisions and the policy instruments available to environmental regulators. Current federal and state statutes affecting waste disposal, air quality and water quality are discussed.

Prerequisites: ISAT 321 or permission of the instructor.

ISAT 422 Industrial Environmental Management**Credits 3 Grading Basis Graded**

This course addresses environmental issues faced by industry, including such topics as waste management, chemical inventories, pollution prevention and discharge permitting. Industrial ecology is introduced as an approach to the development of a sustainable industrial society, including treatment of life cycle analysis, design for environment, environmentally conscious manufacturing and ISO14000.

ISAT 423 Environmental Remediation**Credits 3 Grading Basis Graded**

This course will examine chemical, physical, economic and regulatory aspects of the remediation of contaminated soil and groundwater. Topics include chemical properties of major contaminants, environmental site assessments, remediation design, and current and emerging remediation technologies and their limitations in soil and groundwater restoration. Prerequisites: ISAT 321.

ISAT 424 Natural Resource Management**Credits 3 Grading Basis Graded**

This course focuses on how resource management decisions affect the human and natural communities involved. Topics include definition and importance of natural resources, resource management styles and policies, and planning for resource conservation. Conservation biology is introduced as a tool for developing sustainable resource use policies. Mandatory weekend field trip. Prerequisites: ISAT 321.

ISAT 425 Environmental Hydrology**Credits 3 Grading Basis Graded**

This course integrates the study of surface water and ground water hydrology and examines current technologies used to assess the behavior and quality of water in the environment. Topics include monitoring and management at the watershed level, the influence of wetlands on water quality and the impact of current regulations. Prerequisites: ISAT 320 or permission of the instructor.

ISAT 426 Environmental Information Systems**Credits 3 Grading Basis Graded**

This course provides students with practical experience applying advanced environmental information systems technologies to environmental problems. Students will employ such technologies as decision support systems, geographic information systems, expert systems, relational databases, multimedia systems, and modeling and simulation. This course is often offered in a study-abroad format.

ISAT 427 Industrial Hygiene**Credits 3 Grading Basis Graded**

This course provides an introductory survey of the field of industrial hygiene. Chemical hazards are addressed first, focusing on respiratory and dermal exposures, followed by a treatment of physical hazards including sound, radiation and temperature. The course includes industrial case studies illustrating administrative and engineering controls in common use.

ISAT 428 Industrial Ecology**Credits 3 Grading Basis Graded**

Industrial Ecology, the 'science of sustainability', seeks to encourage the development of a sustainable industrial society. This course introduces and examines this relatively new field of inquiry and practice. We address various practical topics which are associated with Industrial Ecology, including Life Cycle Assessment, Design for Environment, and Environmentally Conscious Manufacturing.

ISAT 429 Sustainability: An Ecological Perspective**Cross-listed with: GEOG 429****Credits 3 Grading Basis Graded**

This course examines present global environmental impacts and efforts made to change production and consumption patterns toward those that reduce impact on ecosystems or promote increased ecosystems health. The focus lies in understanding the basic resources of productivity including soils, agricultural systems, agroforestry, forestry and aquatic environments and applying solutions on a personal and community level. Prerequisites: ISAT 320 or permission of the instructor.

ISAT 430 Materials and Manufacturing Processes**Credits 3 Grading Basis Graded**

The purpose of this class is to introduce chemical and physical properties of materials (metals, ceramics, polymers, composites, sustainable materials), manufacturing processes (casting, finishing, molding, forming, joining, and machining) and automated design and manufacturing. The topics include the following: material properties and product attributes; engineering materials; solidification processes; processing of metals and ceramics; metal forming and sheet metalworking; metal removal processes; property enhancing & surface processing operations; joining and assembly processes. Prerequisites: ISAT 211.

ISAT 431 Advanced Manufacturing Processes and Systems**Credits 3 Grading Basis Graded**

This course provides an introduction to processes, systems and sustainability improvement in advanced manufacturing. Topics include advanced manufacturing processes and systems, such as additive manufacturing, nano/micro-manufacturing, digital manufacturing, industrial sustainability analysis and systems thinking in sustainable manufacturing. Prerequisites: ISAT 330 and ISAT 251.

ISAT 432 Materials Selection in Product Design**Credits 3 Grading Basis Graded**

This course deals with the interplay among requirements specifications, design, economics, environment, energy, materials selection, fabrication, manufacturing cost and product life cycle considerations. Students will complete a product design case study from requirements to final design concept, including translating requirements to design, selecting appropriate materials and choosing between design and production concepts using qualitative and quantitative analytical methods. Prerequisites: ISAT 330 and junior standing or permission of the instructor.

ISAT 433 Foundations of Lean and Six Sigma**Credits 3 Grading Basis Graded**

This course addresses the questions: What is Lean? What is Six Sigma? How are they different, why do many organizations combine them and how should they be applied in complex industrial/supply chain systems? Students will learn to evaluate process stability, map and model business processes, select and apply appropriate analytical methods, and implement and document the improved process. Prerequisites: ISAT 330 or admission to the global supply chain management minor.

ISAT 434 Industrial Informatics**Credits 3 Grading Basis Graded**

Hands-on application of advanced analytical methods selected from topics in supply chain management, forecasting, measurement systems analysis (MSA), quality and process improvement, marketing research and analytics, design of experiments (DOE), and reliability modeling. Data acquisition from enterprise databases and industrial sources and analysis leveraging big data are addressed.

Prerequisites: ISAT 211 and ISAT 251 or MATH 220 or admission to the global supply chain management minor.

ISAT 435 Lean Six Sigma Applications**Credits 3 Grading Basis Graded**

This course focuses on real-world applications of Lean Six Sigma in production and supply chain environments with an emphasis on current events. Students will learn advanced analytical methods and explore the implications of just-in-time production in an increasingly connected and complex global environment.

Prerequisites: ISAT 433 or permission of the instructor.

ISAT 437 Technological Innovation and Sustainable Design**Credits 3 Grading Basis Graded**

This course is a holistic approach to the nature of technological innovation as an intrinsic dynamic of human society, with an explicit focus on understanding why innovations succeed or fail and the compelling need for sustainable design. The critical relationship between technical innovation and the design process is explored through case studies that illustrate sustainability-inspired models of technology such as biomimicry, design for environment, appropriate technology, zero waste production systems and human-centered design.

Prerequisites: ISAT 211 and junior standing or permission of the instructor.

ISAT 440 Cultural, Ethical, Legal and Social Issues in Computing**Credits 3 Grading Basis Graded**

An exploration of the cultural, ethical, legal and social issues inherent in computing. This course is designed to strengthen students' proclivity and ability to consider and communicate about the reciprocal impacts of computing technology on people and the environment, particularly as it pertains to their roles in designing, developing, procuring, implementing or maintaining computing technology within an organization.

Prerequisites: ISAT 271 or permission of the instructor.

ISAT 441 Simulation and Modeling**Credits 3 Grading Basis Graded**

The purpose of this class is to introduce students to the use of models to understand, analyze and improve systems in several areas of science and technology through computer simulation modeling techniques.

Prerequisites: ISAT 211.

ISAT 444 Applied Artificial Intelligence**Cross-listed with:** ISAT 344**Credits 3 Grading Basis Graded**

This course provides a hands-on introduction to building practical AI systems. Topics include: knowledge representation (propositional and predicate logic), search (local, global, uninformed and informed), planning, machine learning and natural language processing. Programming projects will involve designing intelligent agents that can play various board games using Prolog and Python. The course also discusses societal impacts of current and future AI systems and review state, national and international regulations designed to regulate the development and use of AI systems.

Prerequisites: ISAT 252, CS 149, CIS 221 or IA 241.

ISAT 445 Software Engineering and Agile Development**Credits 3 Grading Basis Graded**

Study of means for the development and maintenance of high quality software products delivered on time and within budget. Topics include requirements analysis and specification, software design, implementation, testing, maintenance, project management, ethics, and the responsibilities of software engineering professionals. Focuses primarily on agile software development.

Prerequisites: ISAT 340.

ISAT 447 Interaction Design**Cross-listed with:** WRTC 447**Credits 3 Grading Basis Graded**

Study of and practice with processes, principles, tools, models, and techniques for designing interactions between humans and digital products and systems. Topics include physiological and psychological factors affecting interaction design, interaction design process, interaction models, styles, and paradigms, design notations and representations, prototyping, and interaction design evaluation.

ISAT 449 Emerging Topics in Applied Data Science**Credits 3 Grading Basis Graded**

This course will cover selected emerging topics in applied data science of interests (e.g., automation, data privacy and security, data science in the cloud, artificial neural network and deep learning, natural language processing, text mining). It will help students develop the meta-cognitive skills needed to become proficient in using and identifying new technologies (including programming languages, architectures, modalities, etc.) that may best contribute to solving a problem in an environment in which technologies are changing briskly.

Prerequisites: ISAT 341 or permission of the instructor.

ISAT 451 Biotechnology in Industry and Agriculture**Credits 3 Grading Basis Graded**

This course illustrates the applications of biotechnology in agriculture and industry, linking scientific discoveries to business and manufacturing practices. Topics include pharmaceutical product development, genetic engineering in agriculture, biotechnology in food processing and regulatory issues.

Prerequisites: ISAT 330 or ISAT 350 or BIO 240 and BIOT 260 or permission of the instructor.

ISAT 452 Medical Biotechnology**Credits 3 Grading Basis Graded**

This course will explore the way in which modern biotechnology can be used as a tool to define and solve problems related to human health. Insights will come through the review and critical analysis of molecular evidence, modeling of the human body and the drug discovery pipeline as complex systems, and an identification and appreciation for stakeholder perspectives.

Prerequisites: ISAT 351 or BIO 240 or permission of the instructor.

ISAT 454 Computer Applications in Biotechnology**Credits 3 Grading Basis Graded**

Students learn how complex biological molecules support and regulate processes in living systems, through building interactive computer models of protein and nucleic acid structure and function. The course requirements include written and oral presentations and creation of Web pages.

Prerequisites: ISAT 351 or BIO 240 and BIOT 260 or permission of the instructor.

ISAT 455 Regulatory Issues Biotechnology**Credits 3 Grading Basis Graded**

This course examines the policies and guidelines of federal government agencies that regulate the use of recombinant DNA technology, genetically engineered organisms and the manufacturing of biotechnology products. Issues of product safety, product labeling, physical and biological containment, environmental release, and mammalian cloning are presented.

Prerequisites: ISAT 350 or BIO 240 and BIOT 260 or permission of the instructor.

ISAT 456 Ethical, Legal and Social Implications of Biotechnology**Credits 3 Grading Basis Graded**

The ethical, legal, and social implications of the field of biotechnology and its applications are explored in this course. Students will analyze at an in-depth level the social impacts and ethical implications of human subjects and biological materials research, cloning, human genetic engineering, and transgenic agricultural crops.

Prerequisites: ISAT 171 and ISAT 271 or BIOT 260 or permission of the instructor.

ISAT 459 Awareness and Understanding of Chemical, Biological and Radiological Weapons of Mass Destruction**Cross-listed with: IA 459****Credits 3 Grading Basis Graded**

This course introduces awareness, science and societal impact of weapons of mass destruction (WMD) agents. Students study the development of vaccines and therapeutic and diagnostic drugs used in the detection and treatment of these agents. The course consists of lectures and safety training sessions that introduce tactical and logistical techniques used against chemical, biological and radiological WMD.

ISAT 473 Local Agriculture and Farm Internships**Credits 4 Grading Basis Graded**

The objectives for this course include understanding local ecology and its impacts on farming, as well as how farming impacts local ecology; practicing diversified farming techniques; understanding how small-scale farms operate as businesses; examining localization and slow-food movements and recognizing the impacts of globalized or industrial food and fiber production; identifying the strengths and limitations of small-scale farming.

ISAT 474 Political Ecology**Credits 3 Grading Basis Graded**

This seminar focuses on political ecology as a framework and body of work that examines environmental change, particularly notions of degradation, through analyses of complex social, economic, political and ecological interrelationships. Course materials raise and address questions of how people study and have perceived ways in which human societies and various 'environments' shape one another over time, as well as the contexts in which 'development' efforts are constructed, implemented and, sometimes, abandoned.

ISAT 477 Complex Systems and How They Fail**Credits 3 Grading Basis Graded**

Interdisciplinary study of complex system operation, interdependencies, and failure focusing on real-world critical infrastructure systems (e.g., electric power, telecommunications, and health) system response to natural and human-induced hazards, including cascading effects. Examination of risk management strategies including technical and policy solutions.

ISAT 480 Topics in Integrated Science and Technology**Credits 1-4 Grading Basis Graded**

Topics in integrated science and technology which are of interest to the upper-division student but not otherwise covered in the regular course offerings. Offered only with the approval of the program coordinator. May be repeated for credit when course content changes. Students should consult the instructor prior to enrolling for the course. Topic selected may dictate prerequisite.

Prerequisites: Junior or senior standing only.

ISAT 483 Topics in Science, Technology and Society**Credits 3 Grading Basis Graded**

This course engages students in key theories, methods and approaches from the field of science, technology and society (STS) and its related fields (history, philosophy, law, sociology, anthropology, political science, economics and communication) to address selected problems at the intersection of science, technology and society.

Prerequisites: ISAT 171 and ISAT 271.

ISAT 485 Gender Issues in Science**Cross-listed with: WGSS 485****Credits 3 Grading Basis Graded**

An interdisciplinary course that looks at the scientific process, science practitioners, and science students through the lens of gender analysis. Students read literature, lead discussions, perform experiments, and analyze both data and processes to address the effects of educational systems on the preparation and careers of scientists, the influence of politics and culture on scientific inquiry, and the effects of critiques grounded in gender analyses on understanding the scientific process.

ISAT 487 Science, Technology and Society Futures Laboratory**Credits 1-3 Grading Basis Graded**

In the STS futures lab students deepen their knowledge of key theories and methods of the field of science, technology and society. Students conduct independent research on a topic under close mentorship of faculty and within an engaged community of students who are interested in creatively and critically reflecting on the roles of science and technology within society. The futures lab focuses particularly on hands-on practices of anticipating and ethically reasoning about plausible sociotechnical futures and developing skills in cross-disciplinary collaboration and communication. Students also have the opportunity to engage with experts in diverse disciplinary domains through facilitating workshops with invited guest experts to the lab and conducting interviews. Additionally, students can participate in faculty research, develop research publications and participate in conference presentations. The course is repeatable for a maximum of six credit hours.

Prerequisites: ISAT 171 or ISAT 271 or declared STS minor.

ISAT 490 Capstone Experience I**Credits 1 Grading Basis Graded**

This is the first of a four-course sequence. Students perform an independent research project, either alone or within an investigative team, to identify and analyze a science- or technology-based problem (broadly defined), culminating in a public presentation at the ISAT Senior Symposium. In this course, students will identify an appropriate problem to investigate for their Capstone experience and assemble a Team to develop their Capstone Proposal.

ISAT 491 Senior Capstone Project I**Credits 1 Grading Basis** Graded

First course in a three-course sequence. Student generates an idea for and writes a proposal for an independent or team-based research project. Student must identify and analyze a science- or technology-based problem (broadly defined), identify potential solutions, recommend an approach, and prepare a written proposal.

ISAT 492 Senior Capstone Project II**Credits 2 Grading Basis** Graded

Second course in a three-course sequence. Student performs the bulk of the research needed for an independent research project, either alone or within an investigative team, to address a technologically based problem.

ISAT 493 Senior Capstone Project III**Credits 2 Grading Basis** Graded

Third course in a three-course sequence. Student finishes the research and prepares an oral and a written report on the work (either alone or within an investigative team), addressing a technologically based problem and developing alternative solutions.

Prerequisite: ISAT 492

ISAT 495 Technology in Our World: Better by Design**Credits 3 Grading Basis** Graded

Students explore the importance of design in technology and engineering and contrast the design process with the scientific method. This includes evaluating functional requirements, ergonomics, usability, cost, risk and environmental impact. Students complete a design project integrating these factors to address a real-world problem.

ISAT 499A Senior Honors Thesis I**Credits 1 Grading Basis** Graded

First course of a three-course sequence. Student generates an idea for and writes a proposal for an independent research project that meets the requirements set forth by the Honors program. Student must identify and analyze a science- or technology-based problem (broadly defined), identify potential solutions, recommend an approach, and prepare a written proposal. Fulfills same requirements as ISAT 491.

ISAT 499B Senior Honors Thesis II**Credits 1 Grading Basis** Graded

Second course in a three-course sequence. Student begins the research necessary for an independent research project that meets the requirements set forth by the Honors program. Student pursues the approach described in his or her proposal from ISAT 499A. Fulfills same requirements as ISAT 492.

ISAT 499C Senior Honors Thesis III**Credits 2 Grading Basis** Graded

Third course in a three-course sequence. Student completes the research for and prepares an oral and written presentation of their results for an independent research project that meets the requirements set forth by the Honors program. Student completes and presents (in written and oral form) the project described in his or her proposal from ISAT 499A. Fulfills same requirements as ISAT 493.

ISAT 499D Honors Thesis**Credits 2 Grading Basis** Graded

Intelligence Analysis (IA)

IA 150 Introduction to Intelligence Analysis**Credits 3 Grading Basis** Graded

This course presents a holistic understanding of intelligence analysis: its core definition, domains where it is practiced, how it is done by both individuals and teams, relevant reasoning and technological methods, the significance of communication skills, the broader context that shapes the process, and relevant principles of professional ethics and legal compliance. This course is required for students majoring in intelligence analysis and is recommended for those considering the major.

Prerequisites: First year students and sophomores only; others by permission of the program.

IA 175 Introduction to Intelligence Analysis Domains**Credits 3 Grading Basis** Graded

This course provides a knowledge-based introduction to the domains where intelligence analysts work. These domains include, but are not limited to: national security, military, homeland security, law enforcement, competitive, private sector security and technical specialties such as geospatial and cyber. The course also introduces the larger topics in the field, the security problems and - to a lesser extent - what an intelligence analyst in each field does.

Prerequisites: IA 150 or permission of the instructor.

IA 200 Introduction to National Security Intelligence**Credits 3 Grading Basis** Graded

Intelligence analysis is a complex, dynamic process that includes determining the intelligence needs, data collection, pre-processing, analysis and production of the customer's product. This is an introduction to the history, structure and practices of the national security intelligence community (IC). The course is team-oriented, project-based and grounded in the relevant legal and ethical context.

Prerequisites: Either a minimum grade of "C" in IA 150 or current enrollment in IA 150.

IA 240 Technology Applications in a Networked World**Credits 3 Grading Basis** Graded

Students develop broad technical competencies of current and future technology applications, such as databases, spreadsheets and Internet-of-things. They explore technical-socio-ethical aspects of the cyber world. This hands-on and team-based course supports development of working technical knowledge for accessing, evaluating, modeling and analyzing information and data for problem solving. Students develop a systematic appreciation of computing, network and cyber security.

Prerequisites: Either a minimum grade of "C" in IA 150 or current enrollment in IA 150.

IA 241 Introduction to Programming and Data Science**Credits 3 Grading Basis** Graded

The course introduces students to data analysis through computer programming. The course will focus on key elements from data science that are critical to the intelligence analyst such as manipulating, processing, cleaning and crunching data. The course will serve as a practical, modern introduction to scientific computing in Python, tailored for data-intensive applications. This course will introduce the Python language and libraries students need to effectively solve a broad set of data analysis problems. The course is hands-on and places emphasis on practical case studies with a broad scope of techniques useful for the data analyst.

Prerequisites: IA 240.

IA 250 Research Design and Writing for Intelligence Analysis**Credits 3 Grading Basis Graded**

In this course, students will learn how intelligence analysts can produce assessments using a research design framework and communicate those assessments effectively in writing. This includes how intelligence analysts develop research (intelligence) questions, conduct research, evaluate sources, choose methodologies, develop assessments and present those assessments in the form of an argument with an emphasis on precision, clarity and brevity in the written format.

Prerequisites: IA 150 and IA 200.

IA 261 Hypothesis Testing**Credits 3 Grading Basis Graded**

Examines hypothesis testing in national, military, counter and competitive intelligence. By comparing alternate theories in terms of their explanatory power and predictive success, students will learn the most relevant methods for integrating facts into unified theories, assessing theories, and properly qualifying and reevaluating theories to compensate for risk and uncertainty.

Prerequisites: Either a minimum grade of "C" in IA 150 or current enrollment in IA 150.

IA 280 Selected Project in Intelligence Analysis**Credits 3 Grading Basis Graded**

This course will examine projects of interest to lower-division students in intelligence analysis not otherwise offered in regular course offerings. They are offered only with the approval of the program director and they may be repeated when course content changes. Students should consult with the instructor prior to enrolling in the course.

Prerequisites: Junior or senior standing only.

IA 300 Seminar on Issues in Intelligence Analysis**Credits 3 Grading Basis Graded**

This course will focus on important issues in the theory and practice of intelligence analysis. Students will learn about intelligence analysis by examining its purpose, history, methods, processes, context of existing challenges and recommendations for improvement.

Prerequisites: IA 200.

IA 312 Causal Analysis**Credits 3 Grading Basis Graded**

Examines causal analysis in national, military, counter- and competitive intelligence. By assessing a factor's amount and kind of efficacy, students will learn the most reliable methods for distinguishing between relevant/irrelevant events and factors, identifying and excluding "pseudo-causes," and anticipating higher order effects of a causal process.

Prerequisites: IA 261.

IA 313 Counterfactual Reasoning**Credits 3 Grading Basis Graded**

Examines counterfactual reasoning in national, military, counter- and competitive intelligence. By analyzing alternate scenarios and their consequences, students will learn the most relevant methods for employing creative thinking in generating, developing, and assessing possibilities; substantiating "after-action" reports, and structuring futures analysis.

Prerequisites: IA 312.

IA 314 Strategy Assessment (Rational Decision Theory)**Credits 3 Grading Basis Graded**

Examines strategy assessment in national, military, counter, and competitive intelligence. By applying probabilities and goals to potential threats and opportunities (short and long-term), students will learn the most relevant methods for formulating and evaluating possible courses of action, and projecting and explaining actions by assessing an agent's strategic interests and circumstances.

Prerequisites: IA 313.

IA 340 Data Mining, Modeling and Knowledge Discovery**Credits 3 Grading Basis Graded**

Data mining is the nontrivial extraction of previously unknown and potential useful information from (large) data sets to help explain current behaviors and anticipate future outcomes. Students will apply data mining and knowledge discovery methods to data sets from business, industry and government. The course is team oriented, project based and grounded in the relevant legal and ethical context.

Prerequisites: IA 241.

IA 341 System Dynamics Modeling, Simulation and Analysis**Credits 3 Grading Basis Graded**

System dynamics analysis is a perspective and a set of conceptual and computing tools to help us understand the structure and dynamics of complex systems. This course will apply system dynamics analysis to complex systems (problems) that involve the interplay of physical and social-political factors. The course is team-oriented, project-based and grounded in the relevant legal and ethical context.

Prerequisites: IA 312.

IA 342 Visualization Methods, Technologies and Tools for Intelligence Analysis**Credits 3 Grading Basis Graded**

Data visualization presents laboratory or simulation data or the results from sensors out in the field in a way that aids reasoning about and hypothesis building in complex data sets. This course will apply data visualization technologies and tools to timely data sets from business, industry and government. The course is team oriented, project based and grounded in the relevant legal and ethical context.

Prerequisites: IA 340 or ISAT 252.

IA 343 Introduction to Artificial Intelligence and Machine Learning for Intelligence Analysis**Credits 3 Grading Basis Graded**

This course introduces artificial intelligence, machine learning and generative AI, focusing on their applications in intelligence analysis. Students will learn to build, train and deploy machine learning models while exploring the latest generative AI techniques, such as prompt engineering, vector databases, retrieval-augmented generation (RAG) and diffusion models. These techniques will be applied to analyze social media data, text messages and images, addressing real-world intelligence challenges. Experience with Python is required.

Prerequisites: IA 241.

IA 360 Terrorism and Counterterrorism**Credits 3 Grading Basis Graded**

This course will address key concepts in the study of terrorism, to include definitions, history, justifications, explanations and strategies. It will also address the effectiveness of different kinds of counterterrorism strategies and policies and the role that intelligence analysis plays in the process.

Prerequisites: IA 200.

IA 363 Apocalypticism, Religious Terrorism and Peace**Cross-listed with:** REL 363**Credits 3 Grading Basis** Graded

This course traces apocalypticism from its ancient Jewish and Christian roots to its contemporary manifestations in religious groups around the world. Since apocalypticism is a worldview that cuts across religious traditions, the course covers a variety of religious groups. The last half of the course focuses on the complex relationships between apocalyptic thinking and religious terrorism and entails an independent research project.

IA 365 Cyber Intelligence**Credits 3 Grading Basis** Graded

Cyber intelligence is the acquisition of information to identify and analyze cyber adversaries' capabilities, intentions and activities that can be used to support proactive decision-making. Students will explore common cyber security terminology (lexicon), frameworks and methodologies used to analyze cyber security threats in organizations. Using case studies, they will collect and analyze open source information to identify and monitor evolving threats and understand how adversaries operate.

IA 370 Advanced Competitive Intelligence**Credits 3 Grading Basis** Graded

Competitive intelligence is developing actionable foresight regarding competitive dynamics that can be used to enhance the speed and quality of decision-making. This course teaches students to think and act strategically as a CI professional and to develop students' capacity to think strategically about a company, its business position, how it can gain sustainable competitive advantage and how its strategy can be implemented and executed successfully with the support of CI. Prerequisites: IA 175.

IA 371 Foundations of Anti-Money Laundering**Credits 3 Grading Basis** Graded

Money laundering involves masking the source of criminally derived financial proceeds so funds appear legitimate. Money laundering permeates almost every regulated market. In the global economy, contracting companies, casinos, banks and other financial institutions are required to implement specific structural arrangements to spot red flags and other concerning indicators to reduce money laundering. Students learn about anti-money laundering and counter-terrorism financing (AML/CTF), applied skills about risk assessment, illicit cash flows, and Report and Country AML/CTF Evaluation Report. Students also gain an opportunity to learn about legislative and regulatory initiatives, laws, and organizations at the micro and macro levels (regional, national and international) as they pertain to AML/CTF. Overall, students gain introductory skills on detection, preventative measures and investigating AML/CTF activities. As part of the course evaluation and to afford students practical opportunities, student will work alongside real-world clients.

Prerequisites: IA 150 or permission of the instructor.

IA 375 Political Geography**Cross-listed with:** GEOG 375**Credits 3 Grading Basis** Graded

Geopolitical conflicts and issues are examined. Concepts such as territoriality, nationalism, religious and ethnic struggle, environmental degradation, and freedom and justice are discussed in the context of political unrest. Significant geopolitical theories, and social and economic processes are explored.

IA 405 Ethics, Law and Intelligence Analysis**Credits 3 Grading Basis** Graded

This course will examine ethical and legal issues raised in the practice of intelligence analysis. It will draw on philosophical ethical theories and reasoning to explicate the issues addressed, and will explore the relevant constitutional and other legal constraints on the practice of intelligence analysis, particularly issues of information privacy, civil liberties and limitations on government action.

Prerequisites: IA 314.

IA 440 Intelligence Analysis Capstone Project Preparation**Credits 3 Grading Basis** Graded

This course will focus on preparing students for developing team projects in the IA Capstone Seminar. Students will individually identify, analyze, plan and report on a feasible capstone seminar project. Students will then organize teams and develop plans to complete a subset of the most promising projects in the Capstone Seminar.

Prerequisites: Senior standing in the IA program and IA 314.

IA 450 Capstone Project in Intelligence Analysis**Credits 3 Grading Basis** Graded

Building on the Seminar on Issues in Intelligence Analysis students will complete and present solutions for team-based intelligence community or competitive intelligence IA projects. Students will produce written and oral technical reports/briefs of their results.

Prerequisites: IA 440.

IA 459 Awareness and Understanding of Chemical, Biological and Radiological Weapons of Mass Destruction**Cross-listed with:** ISAT 459**Credits 3 Grading Basis** Graded

This course introduces awareness, science and societal impact of weapons of mass destruction (WMD) agents. Students study the development of vaccines and therapeutic and diagnostic drugs used in the detection and treatment of these agents. The course consists of lectures and safety training sessions that introduce tactical and logistical techniques used against chemical, biological and radiological WMD.

IA 460 All Hazards Response and Management Systems**Credits 3 Grading Basis** Graded

Incident Command Systems enables incident management by integration of facilities, equipment, personnel, procedures, and communications operating in a common organizational structure. ICS is used to organize near and long-term operations for a spectrum of emergencies, small to complex incidents, natural and man-made. ICS is used by government, many private and nongovernmental organizations in 5 major functional areas: command, operations, planning, logistics, and finance and administration.

IA 480 Selected Topics in Intelligence Analysis**Credits 3 Grading Basis** Graded

This course will examine topics of interest to upper-division students in intelligence analysis not otherwise offered in regular course offerings. They are offered only with the approval of the program director and they may be repeated when course content changes. Students should consult with the instructor prior to enrolling in the course.

Prerequisites: Junior or senior standing only.

IA 495 Internship in Intelligence Analysis**Credits 3 Grading Basis** Graded

Provides students with opportunities to earn credit for experiential learning as an analyst in a professional internship setting in a variety of fields where they can acquire knowledge and skills consistent with the IA program's learning objectives. Students engage in a period of employment (minimum 120 hours for 3 credits) as an analyst during which time program learning objectives must be addressed.

Prerequisites: IA 250.

IA 499A Honor Thesis I**Credits 1 Grading Basis** Graded

This is an honors thesis proposal course. It is the first of the three-course sequence intended to satisfy the requirements for the honors program as well as the Intelligence Analysis program. Students will develop a thesis proposal, including objectives, audience, literature review, timeline and deliverables. The goal of the course is for the student to develop a compelling and complete senior thesis proposal.

Prerequisites: Junior standing; must be in the honors program.

IA 499B Honor Thesis II**Credits 3 Grading Basis** Graded

This is an honors thesis development course. It is the second of the three-course sequence intended to satisfy the requirements for the honors program as well as the Intelligence Analysis program. Most of the course will be jointly administered/taught with IA 440: Senior Seminar in Intelligence Analysis. In addition, students may be required to meet with the instructor to discuss progress on his or her project.

Prerequisites: Senior Standing and Honor Student.

IA 499C Honors Thesis III**Credits 3 Grading Basis** Graded

This is an honors thesis completion course. It is the third of the three-course sequence intended to satisfy the requirements for the honors program as well as the Intelligence Analysis program. Most of the course will be jointly administered/taught with IA 450, Capstone Project in Intelligence Analysis. In addition, students may be required to meet with the instructor to discuss progress on his or her project.

Prerequisites: Senior Standing and Honor Student.

Interdisciplinary Art (IART)

IART 480 Arts Education for Diverse Learners**Credits 3 Grading Basis** Graded

This course is designed to provide students an understanding of disability categories, special education law and models of disability as they apply to the arts and broader settings. Universal Design for Learning, building collaborative relationships with colleagues, serving disability communities, assistive technology and instructional strategies also will be discussed. This course requires a community service project that will be discussed and developed in class.

Interdisciplinary Liberal Studies (IDLS)

IDLS 200 Interdisciplinary Scholarship**Credits 3 Grading Basis** Graded

This course is designed to introduce students to diverse methods, concepts and ways of knowing. Students will engage with multiple disciplines and perspectives, including scientific, environmental, political, social and cultural areas of inquiry. At the completion of the course, students will demonstrate an ability to integrate diverse sources of information in the design of an interdisciplinary academic plan.

IDLS 300 Individualized Major Workshop**Credits 3 Grading Basis** Graded

This workshop course is designed to provide students in the Independent Scholars and General Studies majors with methodological competencies linked to major learning objectives across their curriculum, including independent research methods, preparing presentations, data visualization, working with teams and enhancing intellectual creativity.

Prerequisites: IDLS 200.

IDLS 350 Literacy and Society**Credits 3 Grading Basis** Graded

An exploration and analysis of societal literacy practices as viewed through cognitive, cultural, class, workplace, and technological lenses.

Prerequisites: WRTC 103.

IDLS 380 American Indian and Indigenous Cultures**Credits 3 Grading Basis** Graded

This course examines culture change among American Indian and Indigenous peoples in North America, with a focus on the lived experiences, voices, distinct ways of knowing and unique forms of social and cultural expression of these populations from pre-Columbian to contemporary times. Issues including settler colonialism, Indigenous knowledge, identity, belonging, citizenship and sovereignty, myth-making, gender, health and well-being, creativity, and protest and resistance are examined to gain insight to both unique historical trajectories and allow comparison of broad-based cultural dynamics. This course integrates the multiple, often diverging, perspectives of historians, novelists, educators, activists, anthropologists, film directors, scientists and more in the study of American Indian and Indigenous culture history.

Prerequisites: IDLS major, Liberal Studies minor or permission of the instructor.

IDLS 381 Environmental Thought and Activism**Credits 3 Grading Basis** Graded

An exploration and analysis of the progression of environmental thought and activism in the United States with a focus on activism, climate, conservation, ecojustice, economic, environmental protection, literary, philosophical, political, preservation, resource, wilderness and wildlife issues.

Prerequisites: IDLS major, Liberal Studies minor or permission of the instructor.

IDLS 382 Contemporary United States**Credits 3 Grading Basis** Graded

An exploration and analysis the United States at the present moment, with a focus on activist, cultural, economic, educational, environmental, foreign policy, gender, identity, political, social, social justice and sexual identity issues.

Prerequisites: IDLS major, Liberal Studies minor or permission of the instructor.

IDLS 384 Knowledge, Power and Inequality**Credits 3 Grading Basis** Graded

This course examines the evolutionary origins, historical trajectory and contemporary dynamics of inequality as a product of power relations rooted in human cultural knowledge and practice, with emphasis on modes of power defined by gender, age, sexual violence, education and cultural sovereignty, ethnic and racial identification, globalization and economic inequality, political polarization, and mobility and immigration. Examples primarily from North America from pre-Columbian to contemporary times and the perspectives of anthropologists, historians, fiction writers and poets, artists, educators, human rights activists, memoirists, politicians, economists and more are studied to untangle the dialectic of cultural-cognitive "ways of knowing" and the embodied experience of power and inequality.

IDLS 385 The Unfinished Journey of People of Color in the U.S.

Credits 3 Grading Basis Graded

This course explores and analyzes the experiences, cultures and current challenges of people who identify as Asian American, Black, Latinx, Native Alaskan, Native American, Native Hawaiian and as multiethnic and multiracial.

IDLS 391 Study Abroad

Credits 1-6 Grading Basis Graded

Credit for academically-grounded, interdisciplinary study abroad. Students seeking credit must secure the approval of the department head and a faculty supervisor who will provide the academic structure, assignments, and student evaluation.

IDLS 395 Topics in IDLS

Credits 1-6 Grading Basis Graded

Examination of selected interdisciplinary topics of importance to Teacher Education content areas. May be taken for a maximum of six credit hours toward the major.

IDLS 400 Seminar in Liberal Studies

Credits 3 Grading Basis Graded

Capstone seminar for IDLS studies in the Humanities/Social Sciences content area. Studies will apply different disciplinary perspectives to a single topic. Course requirements will emphasize superior written and oral communication skills, and the integration and application of content area knowledge to the teaching environment. Students must have completed their lower division Core requirements and be within one course of completing the concentration area. Education minors should be in their third semester of their teacher education program.

IDLS 486 Internship and Field Experience

Credits 1-6 Grading Basis Graded

IDLS credit for academically-grounded internships and field experiences. Students seeking credit must secure the approval of the department head for the use of academic structures, assignments, and evaluation plans provided by qualified internship or field experience supervisors. This course may be taken for a maximum of 12 credit hours.

IDLS 490 Independent Study in Liberal Studies

Credits 3 Grading Basis Graded

Individualized projects in interdisciplinary liberal studies.

IDLS 498 Interdisciplinary Senior Project

Credits 1-6 Grading Basis Graded

This course leads to completion of a senior project. The ultimate scholarly product will utilize oral and written communication, critical thinking and problem-solving skills, and synthesize disciplinary content from the various academic areas included in the student's individualized plan of study.

IDLS 499A Honors

Credits 1-6 Grading Basis Graded

IDLS 499B Honors

Credits 1-6 Grading Basis Graded

IDLS 499C Honors

Credits 1-6 Grading Basis Graded

International Affairs (INTA)

INTA 295 CrossNational Research Skills

Credits 4 Grading Basis Graded

Students learn how to conduct research from theory formulation through data collection and hypothesis testing in the field of international affairs. Special emphasis on research and computer literacy. Prerequisites: Completion of the General Education: The Natural World: Quantitative Reasoning area and either POSC 230 or POSC 240.

INTA 301 The Washington Semester Experience: Global Affairs

Credits 3 Grading Basis Graded

Part of the "Global Affairs" Washington Semester program, the course entails intensive study of a global theme. The theme's dynamics will be explored in varied settings: localities, nation-states, global geographic regions, and international organizations (both governmental and non-governmental). The course provides outlets for engagement with policy actors and institutions based in Washington, D.C., as well as for individual and group experiential learning activities.

INTA 489 Senior Seminar in International Affairs

Credits 3-4 Grading Basis Graded

This is the capstone course in the international affairs major. It provides an interdisciplinary overview of the fields within international affairs and an opportunity for students to complete individual research projects on international problems. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Senior INTA majors.

International Business (IB)

IB 200 Introduction to International Business

Credits 3 Grading Basis Graded

A survey course to introduce students to the theory and practice of international business. The focus is on providing the skills and knowledge to successfully manage operations within the global environment. Not open to international business majors who have earned credit or are enrolled in COB 300.

Prerequisites: Not open to international business majors who have earned credit or are enrolled in COB 300.

IB 294 Internship Abroad

Credits 3 Grading Basis Credit / No Credit

A course providing students an opportunity to work in an organization abroad in order to gain insight into the practical side of modern business operations.

IB 298 Special Topics in International Business

Credits 3 Grading Basis Graded

The course is designed to allow exploration of current topics in international business. Course content will vary. See the program director for current content.

IB 480 International Business Theory and Policy

Credits 3 Grading Basis Graded

The course is designed to serve as an application of theory for business students to allow them to put the total picture of international business together.

Prerequisites: IB major, senior standing.

IB 490 Special Studies in International Business**Credits** 1-3 **Grading Basis** Graded

Designed to give capable students in international business an opportunity to complete an independent study under faculty supervision. Prerequisites: 2.8 GPA, recommendation of the instructor and approval of the director prior to registration.

IB 494 International Business Internship**Credits** 3 **Grading Basis** Graded

A course providing students an opportunity to work in and with an organization in order to gain insight into the practical side of modern international business operations.

Prerequisites: IB major, completion of 85 credit hours and COB 300, minimum cumulative GPA of 3.0, and approval of the director of International Business program prior to registration.

IB 498 Special Topics in International Business**Credits** 3 **Grading Basis** Graded

An advanced course designed to allow exploration of current topics in international business. Course content will vary. See the program director for current content.

Prerequisites: COB 300 and permission of the instructor.

IB 499A Honors**Credits** 1 **Grading Basis** Graded**IB 499B** Honors**Credits** 3 **Grading Basis** Graded**IB 499C** Honors**Credits** 1-3 **Grading Basis** Graded

Interprofessional Education (IPE)

IPE 201 Health Professionals in Diverse Communities**Credits** 1-3 **Grading Basis** Graded

This course provides an introduction to skills in professionalism and interprofessional collaboration in addressing local and global health challenges. Students will examine social determinants of health in diverse communities and learn through reflective practice as they interact with health professionals and faculty.

IPE 202 Health Care Service in Diverse Communities**Credits** 1-3 **Grading Basis** Graded

In this course students will examine interprofessional perspectives on complex local and global health issues and apply skills in professionalism, integration, collaboration and reflection to community-based, experiential service learning.

IPE 313 Issues and Applications of Family Caregiving: Caregiver's Community Network**Credits** 1-2 **Grading Basis** Graded

Students are assigned a CCN family managing caregiving responsibilities for an elder member (age 60+ years) and provide planned contact and support. Case strategies include: weekly visits, phone calls and self-care practices. This course is intended to benefit all participants (family caregivers, their care-receiving members and students) by offering an academic and service-focused experience that validates the common need for human connection, understanding and respect. Can be repeated for a maximum of three credits.

IPE 314 Rural Health: An Interprofessional Approach**Cross-listed with:** HTH 314, NSG 314, SOWK 314**Credits** 3 **Grading Basis** Graded

Students study, observe and participate in interprofessional assessment, planning and delivery of community-based primary health care in partnership with residents and agencies of a host rural county. Learning activities will emphasize rural culture, rural health care and interprofessional practice.

IPE 391 Introduction to Informatics for Health Care Professionals**Credits** 1 **Grading Basis** Graded

Electronic health records and health apps exemplify how health information technology (IT) is changing health care today. Students will learn how health IT tools and big data can improve communication between health providers, support better patient outcomes, and reduce spending on health care. Designed with an emphasis on interprofessional collaborative practice, students will learn about their own and others' professional roles and responsibilities in achieving health goals using health IT.

IPE 401 Workshops in Interprofessional Education and Practice (IPE)**Credits** 0 **Grading Basis** Credit / No Credit

HHS workshops offer a time-limited and concentrated focus on issues that are specific to interprofessional education and/or practice.

IPE 415 Ethical Decision-Making in Healthcare: An Interprofessional Approach**Cross-listed with:** NSG 415**Credits** 1 **Grading Basis** Student Option

Health care ethics is a shared, relevant concern among health and human service disciplines and is an ideal vehicle for students from different fields to learn about one another's disciplines and to participate in interprofessional team analysis, discussion, and problem-solving. We have designed an integrated cross-disciplinary learning experience for students interested in the complex, real-world dilemmas encountered in practice. The readings and activities will emphasize ethical, legal, moral and spiritual issues and principles for practice within the context of communicating in interprofessional teams.

IPE 490 Special Topics in Health and Human Services**Credits** 0-4 **Grading Basis** Graded

This course involves topics of special interest in the area of health and human services but is open to all students. The focus of specific courses is identified for specific offerings. Courses are offered based on faculty and student interest.

Interscience Research (ISCI)

ISCI 101 Physics, Chemistry and the Human Experience [C3PP]**Credits** 3 **Grading Basis** Graded

A survey of the fundamental concepts, principles and ideas of chemistry and physics. Particular emphasis is placed on understanding the development of these principles and their application in understanding the world around us. May be used for general education credit. Prerequisites: One of the following: MATH 103, MATH 105, MATH 110, MATH 205, MATH 220, MATH 229, MATH 231 or MATH 235. Corequisites: One of the following: MATH 103, MATH 105, MATH 110, MATH 205, MATH 220, MATH 229, MATH 231 or MATH 235.

ISCI 104 Scientific Perspectives [C3L]**Credits** 1 **Grading Basis** Graded

A study of topics selected to allow students to participate in mathematical and scientific problem solving approaches to knowledge. May be used for general education credit. Prerequisite or corequisite as indicated on MyMadison.

ISCI 171 Earth and Planetary Science for Teachers [C3NS]**Credits 3 Grading Basis** Graded

This course provides university-level foundations of earth and planetary science for future pk-8 teachers. Content aligns with various teacher competencies, and includes such topics as the formation and evolution of the earth and the earth's solar system, the characteristics of stars, planets, asteroids, and comets, and how earth and planetary science knowledge and technologies function with social context. Hands-on, experiential inquiry will be integrated into the course, as will an exploration of such methods as observation, classification, comparison, measurement, data interpretation, mathematical analysis, inference, prediction and hypothesis testing. Normally open to COE majors only, but other students may request admission by special permission. May be used for general education credit.

ISCI 172 Physical Science for Teachers [C3PP]**Credits 3 Grading Basis** Graded

This course provides university-level foundations of physical science for future pk-8 teachers. Content aligns with various teacher competencies, and includes such topics as matter, conservation of mass and energy, chemical structures and bonds, coordinate systems and their use in describing motion and force, thermodynamics, light, sound, magnetism and electricity, and how physical science knowledge and technologies function with social context. Hands-on, experiential inquiry will be integrated into the course, as will an exploration of such methods as observation, classification, comparison, measurement, data interpretation, mathematical analysis, inference, prediction and hypothesis testing. Normally open to COE majors only, but other students may request admission by special permission. May be used for general education credit.

ISCI 173 Life and Environmental Science for Teachers [C3L]**Credits 3 Grading Basis** Graded

This course provides university-level foundations of life science for future preK-8 teachers. Content aligns with various teacher competencies, and will prepare students to be scientifically literate. Topics will focus on biological structure and function; information flow, storage and exchange; transformations of energy and matter; evolution; and the interconnection and interactions of living systems. The course will engage students in the learning of foundational concepts as well as the process of science and the relevance of biology to society. Normally open to COE majors only, but other students may request admission by special permission. May be used for general education credit.

ISCI 450 Interscience Research**Credits 1-4 Grading Basis** Graded

An investigative experience spanning more than one field of science which may require supervision by multiple faculty members from different disciplines. Students must get prior approval for this course from each of the supervising faculty members and the department head of their program.

Prerequisites: Junior standing.

ISCI 499 Honors**Credits 1-4 Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

Italian (ITAL)

ITAL 101 Elementary Italian I**Credits 3 Grading Basis** Graded

This course introduces the fundamentals of Italian through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

ITAL 102 Elementary Italian II**Credits 3 Grading Basis** Graded

This course builds on the foundations established in ITAL 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension.

Prerequisites: ITAL 101; or by placement exam or permission of the instructor.

ITAL 103 Passport to Italian: A Survival Kit**Credits 4 Grading Basis** Graded

This course provides students with basic communication skills for survival while traveling or studying in Italy. Emphasis is placed on cultural competence and interaction skills. This course does not fulfill the B.A. language requirement. Students who have previously studied Italian are required to take the placement test. Instruction is in Italian.

Prerequisites: Senior standing or permission of the instructor.

ITAL 104 Italian for Spanish and Italian Heritage Speakers**Credits 4 Grading Basis** Graded

This course is designed specifically for heritage Italian speakers and Spanish-speaking learners, building on their existing linguistic knowledge. The course focuses on filling knowledge gaps and strengthening literacy, providing a tailored learning experience that reconnects heritage speakers with their roots while enabling Spanish speakers to gain fluency in Italian more effectively. There is one hour a week of asynchronous online work.

ITAL 111 Intensive Italian I**Credits 6 Grading Basis** Graded

This course covers the fundamentals of Italian through listening, speaking, reading and writing. The course is the equivalent of ITAL 101 and ITAL 102.

ITAL 212 Intensive Italian II**Credits 6 Grading Basis** Graded

This course covers the fundamentals of Italian through listening, speaking, reading, and writing. The course is the equivalent of ITAL 231 and ITAL 232.

Prerequisites: ITAL 102 or ITAL 111; or by placement exam or permission of the instructor.

ITAL 231 Intermediate Italian I**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.

Prerequisites: ITAL 102 or ITAL 111; or by placement exam or permission of the instructor.

ITAL 232 Intermediate Italian II**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.

Prerequisites: ITAL 231; or by placement exam or permission of the instructor.

ITAL 300 Language for Living**Credits 3 Grading Basis** Graded

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in the Italian language and culture. It is intended for students who have begun to develop advanced skills in interpretive, interpersonal and presentational communication and are interested in delving deeper into the study of Italian-speaking countries, their histories and their cultures. Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232; or by placement exam or permission of the instructor.

ITAL 301 Pop Culture through Language**Credits 3 Grading Basis** Graded

This course explores the world of Italian pop culture, examining its impact on contemporary Italian society. We will delve into Italian media, analyze their influence on Italian society and culture, and study the role of music, arts, comics, etc. in shaping cultural trends. Through music videos, film clips and media analysis, students will gain insight into how pop culture reflects Italy's social, historical, political and cultural transformations. This course offers a lens to understand the modern Italian identity and its global resonance. Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232; or by placement exam or permission of the instructor.

ITAL 307 History of Italian Culture and Society**Credits 3 Grading Basis** Graded

In this course, students will study the history and the arts of Italy from the Etruscans to 1814. Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232; or by placement exam or permission of the instructor.

ITAL 308 Contemporary Italian Culture and Society**Credits 3 Grading Basis** Graded

This course studies Italian society, economics, politics and the arts from 1814 to the present. Instruction in Italian.

Prerequisites: ITAL 212 or ITAL 232, or by permission of the instructor.

ITAL 315 Sounds of Speech**Credits 3 Grading Basis** Graded

This course introduces students to Italian linguistics, with special emphasis on Italian phonetics. Students will engage with theoretical concepts and practical applications to understand the phonological structures and sounds that make up the Italian language. Class activities and assignments aim to enhance overall language proficiency, including listening, reading, speaking and pronunciation skills. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320, ITAL 322 or equivalent.

ITAL 320 Culture and Communication in Context**Credits 3 Grading Basis** Graded

This Italian course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do Italian-speaking peoples communicate on subjects that are at the heart of their societies?" Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232; or by placement exam or permission of the instructor.

ITAL 322 Language Landscapes**Credits 3 Grading Basis** Graded

This Italian course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in these culturally and linguistically defined areas and explores the diverse cultural landscapes that shape their lives. Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232; or by placement exam or permission of the instructor.

ITAL 330 Business Italian**Credits 3 Grading Basis** Graded

This course studies commercial and technical vocabulary and trade customs in conjunction with practice in the art of commercial communication, including interviews, letter writing and simultaneous interpretation. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 335 A Journey through Literature**Credits 3 Grading Basis** Graded

The purpose of this course is to acquaint students with Italian literature through the study of excerpts of canonical texts ranging from the 12th to the 21st century. Different genres will be studied: philosophy, essay, poetry, theater, novel, short story and song, always presented within an historical, political, social and artistic framework. In this class, students will also learn strategies, using models and tools (such as the methodology for textual analysis and the study of rhetorical terms), to read and analyze any text better. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320, ITAL 322 or equivalent.

ITAL 375 Business and Society in Italy**Credits 3 Grading Basis** Graded

This course studies Italian business, economy and politics, and the influence the Italian society has on them.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 380 Italian Cuisine**Credits 3 Grading Basis** Graded

This course is designed to introduce the student to the rich Italian culinary tradition. Through readings and presentations, students will familiarize themselves with the diversity present in Italian cuisine and prepare the dishes that best represent various regions of Italy. Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232, or by permission of the instructor.

ITAL 386 Italian Short Stories**Credits 3 Grading Basis** Graded

This course studies literary Italian short stories from the Middle Ages to the 21st century, including a selection of texts by prominent authors such as Giovanni Boccaccio, Giambattista Basile, Giovanni Verga, Grazia Deledda, Gabriele D'Annunzio, Luigi Pirandello, Natalia Ginzburg, Italo Calvino and Ada Negri.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 390 Italian Language and Culture Abroad**Credits** 1-3 **Grading Basis** Graded

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in Italian literature and culture. It is intended for students who have begun to master advanced skills in interpretive, interpersonal and presentational communication and are interested in examining nuances of the Italian language and culture while studying abroad.

Prerequisites: ITAL 212 or ITAL 232, or by permission of the instructor.

ITAL 400 Advanced Conversation**Credits** 3 **Grading Basis** Graded

Discussions deal with topics of current interest. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 408 Italian American Culture**Credits** 3 **Grading Basis** Graded

This course offers an interdisciplinary exploration of Italian American culture, focusing on the narratives, traditions and experiences that define the Italian American identity. Through the study of texts, music, visual arts, culinary traditions, iconic figures and communities, students will examine key themes such as immigration, assimilation, cultural conflict and the search for identity within a multicultural context. Instruction is in Italian.

Prerequisites: ITAL 212 or ITAL 232, or by permission of the instructor.

ITAL 421 Stories and Histories: Italian Theatre**Credits** 3 **Grading Basis** Graded

This course studies Italian theatre from the Renaissance to the 21st century, including a selection of texts by prominent authors such as Niccolò Machiavelli, Torquato Tasso, Carlo Goldoni, Alessandro Manzoni, Luigi Pirandello and Dario Fo.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 422 Voices of Today**Credits** 3 **Grading Basis** Graded

This course studies the works of major Italian writers from the 19th century to the present. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 425 Transformative Texts**Credits** 3 **Grading Basis** Graded

This course studies Italian poetry from the Middle Ages to the 21st century, including a selection of texts by prominent authors such as Dante Alighieri, Francesco Petrarca, Ludovico Ariosto, Gaspara Stampa, Torquato Tasso, Ugo Foscolo, Giacomo Leopardi, Giovanni Pascoli, Giuseppe Ungaretti, Eugenio Montale, Mario Luzi and Alda Merini.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 435 Translation Strategies**Credits** 3 **Grading Basis** Graded

In this course, translation will be practiced as the 5th foreign language skill and will also be used as a pedagogic tool in language learning. A general and brief theoretical study will provide students with an understanding of what is involved in the translation process. The main focus of the course will be practical, from a brief review of the main translation strategies to the practice of different types of translation. Students learn how to use the main translation strategies, different types of translations and interpretation, and translation to develop and improve linguistic and cultural competences in Italian. Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 437 Studies in Italian Literature**Cross-listed with:** ENG 437**Credits** 3 **Grading Basis** Graded

A study of selected works of Italian literature. Instruction is in English. May be repeated for credit when content varies.

ITAL 446 Special Topics in Italian Literature**Credits** 3 **Grading Basis** Graded

This course studies a particular topic in Italian literature. It may cover all or a specific Italian literature genre. This course may be repeated if content varies.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 447 Special Topics in Italian Culture**Credits** 3 **Grading Basis** Graded

Students will study a particular topic in the culture of Italy. Course may be repeated if content varies.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 448 Special Topics in Italian Linguistics**Credits** 3 **Grading Basis** Graded

Students will study a particular topic in Italian linguistics. Topics could include an introduction to Italian sociolinguistics and psycholinguistics. Course may be repeated if content varies.

Prerequisites: ITAL 300 or permission of the instructor.

ITAL 465 Italian Cinema**Credits** 3 **Grading Basis** Graded

This course studies Italian cinema from the early days to the present, with a focus on its relation to key features of Italian history, society and cultures. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 480 Advanced Italian Language Lab**Credits** 1-2 **Grading Basis** Graded

This language lab is aimed at complementing 300- and 400-level Italian content courses. In addition to enrolling into each upper-level content course, students may be required to enroll in the correspondent advanced language lab. The language lab may be taught by the same instructor of the content course or by a different instructor. In the language lab, the students will work on the linguistic component of the course by practicing reading advanced literary/specialized texts in the target language and speaking and/or writing about the contents of the course. The lab will focus on technical/literary lexicon, advanced grammar, and pragmatic and stylistic features.

Prerequisites: ITAL 300 or permission of the instructor.

ITAL 490 Independent Study in Italian**Credits** 1-4 **Grading Basis** Graded

The course allows students an opportunity to complete independent studies under faculty supervision. Instruction is in Italian.

Prerequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322; or by placement exam or permission of the instructor. Corequisites: ITAL 300, ITAL 301, ITAL 320 or ITAL 322.

ITAL 497 Italian Internship**Credits** 1-3 **Grading Basis** Graded

This course will provide students with experiential learning opportunities within the Italian-speaking field. The internship course is designed to provide practical experience in fields such as government, business, industry, health sciences, administration, education, translation or the arts, where they can apply their language skills. Examples of internship work may include working for a company in the student's field of interest abroad; serving as a research assistant for a professor; completing a translation project; or working as an interpreter. A journal, internship report, research paper, bibliography or evaluation from the internship provider is required. Credits earned correspond to the internship hours completed. Up to six credits may be applied as electives in the Italian major or minor.

Prerequisites: One ITAL course at the 300-level or above and permission of the instructor.

ITAL 499A Senior Honors Project**Credits** 1 **Grading Basis** Graded**ITAL 499B** Senior Honors Project**Credits** 3 **Grading Basis** Graded**ITAL 499C** Senior Honors Project**Credits** 2 **Grading Basis** Graded

Japanese (JAPN)

JAPN 101 Elementary Japanese I**Credits** 3 **Grading Basis** Graded

This course introduces the fundamentals of Japanese through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

JAPN 102 Elementary Japanese II**Credits** 3 **Grading Basis** Graded

This course builds on the foundations established in JAPN 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension.

Prerequisites: JAPN 101 or by placement exam.

JAPN 111 Intensive Japanese I**Credits** 6 **Grading Basis** Graded

The fundamentals of Japanese through listening, speaking, reading and writing. The course is the equivalent of JAPN 101-JAPN 102.

JAPN 212 Intensive Japanese II**Credits** 6 **Grading Basis** Graded

The fundamentals of Japanese through listening, speaking, reading and writing. The course is the equivalent of JAPN 231-JAPN 232.

Prerequisites: JAPN 102, JAPN 111 or permission of the instructor.

JAPN 231 Intermediate Japanese I**Credits** 3 **Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition, and reading.

Prerequisites: JAPN 102, JAPN 111 or permission of the instructor.

JAPN 232 Intermediate Japanese II**Credits** 3 **Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition, and reading.

Prerequisites: JAPN 231 or permission of the instructor.

JAPN 300 Language for Living**Credits** 3 **Grading Basis** Graded

This course focuses on enhancing students' practical communication skills in Japanese, preparing them for everyday interactions in personal, academic and professional contexts. Building on the foundations established in JAPN 232, this course emphasizes real-life language use through topics related to daily life, social events and work. Students will engage in listening, speaking, reading and writing activities designed to improve fluency and cultural understanding. The course prepares students to navigate various real-world situations while deepening their knowledge of Japanese social customs and language norms. Instruction is in Japanese.

Prerequisites: JAPN 212 or JAPN 232.

JAPN 301 Pop Culture through Language**Credits** 3 **Grading Basis** Graded

This course explores Japanese language and culture through various forms of popular media, including manga, anime, video games, movies, TV series and J-Pop music. Students will engage with authentic materials to enhance their understanding of contemporary Japanese language while gaining insights into modern Japanese society and cultural trends. By analyzing these media, students will expand their vocabulary, improve listening comprehension and develop their ability to critically evaluate cultural products. The course is designed for upper-beginner to lower-intermediate students looking to apply their language skills in real-world, culturally relevant contexts. Through discussions, presentations and creative projects, students will strengthen both their linguistic abilities and their cultural literacy. Instruction is in Japanese.

Prerequisites: JAPN 212 or JAPN 232.

JAPN 320 Culture and Communication in Context**Credits** 3 **Grading Basis** Graded

This Japanese course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do Japanese-speaking peoples communicate on subjects that are at the heart of their societies?" Instruction is in Japanese.

Prerequisites: JAPN 212 or JAPN 232.

JAPN 322 Language Landscapes**Credits 3 Grading Basis** Graded

This Japanese course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the people living in this culturally and linguistically defined area and explores the diverse cultural landscapes that shape their lives. Instruction is in Japanese.

Prerequisites: JAPN 212 or JAPN 232.

JAPN 340 Practical Japanese Conversation**Credits 3 Grading Basis** Graded

This course will focus on improving students' verbal communication skills in Japanese. Students will learn and practice basic skills necessary in dealing with various practical situations by expressing their thoughts while engaging in dialogues in Japanese with others. Students will use the vocabulary, expressions and grammatical points covered in previous Japanese language courses. New vocabulary, expressions and grammatical points as well as some basic communication strategies in conversations (asking for clarification of the meaning of what someone said, showing the listener that one is thinking, etc.) will also be introduced as needed. In the process, students will learn to speak in a proper manner in Japanese society. They will also be directed to pay more attention to speaking Japanese with proper speed and pronunciation compared to lower-level classes.

Prerequisites: JAPN 212 or JAPN 232.

JAPN 446 Special Topics in Japanese Literature**Credits 3 Grading Basis** Graded

This course studies a particular topic in Japanese literature. It may cover all or a specific Japanese literature genre. This course may be repeated if content varies.

Prerequisites: JAPN 300, JAPN 301, JAPN 320 or JAPN 322; or by permission of the instructor.

JAPN 447 Special Topics in Japanese Culture**Credits 3 Grading Basis** Graded

Students will study a particular topic in the culture of Japan. Course may be repeated if content varies.

JAPN 448 Special Topics in Japanese Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in Japanese linguistics. Topics could include an introduction to Japanese sociolinguistics or psycholinguistics. Course may be repeated if content varies.

Prerequisites: JAPN 300, JAPN 301, JAPN 320 or JAPN 322; or by permission of the instructor.

JAPN 490 Independent Study in Japanese**Credits 1-4 Grading Basis** Graded

This course allows students to pursue independent study in Japanese under the supervision of the instructor.

Justice Studies (JUST)

JUST 100 Justice Studies Proseminar**Credits 1 Grading Basis** Credit / No Credit

This course is designed to help justice studies majors explore opportunities for internships, careers, graduate and professional education, and civic and community engagement. The course focuses on developing skills in resume writing, interviewing and networking techniques, job and internship search skills, and investigating graduate and law school opportunities.

Prerequisites: JUST 200 and junior or senior status, or by permission.

JUST 200 Introduction to Justice Studies**Credits 3 Grading Basis** Graded

This course introduces students to the justice studies major and the interdisciplinary field of justice studies. Students will explore classical and contemporary conceptions and theories of justice and their application to public issues as well as broad and holistic approaches to justice including social, global and criminal justice.

Prerequisites: Justice studies majors only.

JUST 210 Crime and Criminal Justice**Credits 3 Grading Basis** Graded

This course provides an introduction to the nature of the crime problem in the United States, including patterns of victimization and offending and the ways in which the criminal justice system responds to these behaviors.

Prerequisites: Justice studies majors and criminal justice minors only; for justice studies majors: JUST 200 and admission to the major; for criminal justice minors: admission to the minor.

JUST 212 Theories of Crime and Criminal Justice**Credits 3 Grading Basis** Graded

This course provides an in-depth exploration of theoretical perspectives pertaining to the two central realms of criminological inquiry: crime and the response to crime (criminal justice). Both classic and contemporary perspectives are examined. The course will examine why people commit crime, why crime occurs, why it differs across groups and the objectives underlying crime control policy.

Prerequisites: Justice studies major: JUST 200. Criminal justice minors: JUST 210 or CRJU 215.

JUST 221 Social Justice Theories**Credits 3 Grading Basis** Graded

This course serves as a theoretical introduction to the social justice track of the Justice Studies major. It includes an examination of the major concepts regarding inequality. How do we define and create inequality? Can we rid ourselves of inequality or should we accept it as a necessary element in society?

Prerequisites: JUST 200.

JUST 223 Social Justice Interventions and Policies**Credits 3 Grading Basis** Graded

This course is an introduction to the study and practice of social justice. Students will learn to identify and analyze historical and contemporary forms of oppression as well as a diverse range of social justice interventions, including policymaking, community development, collective action and other approaches that aim to end oppression and further social justice.

Prerequisites: For Justice Studies majors: JUST 200; for minors in Social Justice Studies: declaration of the SJS minor and permission of the SJS minor coordinator.

JUST 225 Justice and U.S. Society [C4AE]**Credits 4 Grading Basis** Graded

This course introduces the student to the concept and reality of justice in the United States. It is a broad-based, interdisciplinary consideration of justice: What it is, what it means, and how it intersects with society and social institutions in the United States. Philosophical and theoretical underpinnings of the notion of justice and the historical context of justice in American society will be considered. May be used for general education credit. May not be used for major credit.

JUST 235 Justice in the Global Community**Credits 3 Grading Basis** Graded

A survey of different definitions of justice relating to the operation and development of a global community in international affairs.

Prerequisites: JUST 200.

JUST 237 Dilemmas in Global Justice**Credits 3 Grading Basis** Graded

This course applies the concepts and frameworks of global justice to examine a selection of the world's greatest challenges, placing global affairs in historical, geographical and political contexts.

Prerequisites: JUST 200.

JUST 300 Perspectives on Comparative Justice**Credits 3 Grading Basis** Graded

This course provides an overview of the various systems of justice existent in different cultures throughout the globe. The emphasis here is on legal systems and the ways in which they are used to promote—or not promote—justice. The course also includes a discussion of such topics as war, peace, terrorism, political and social oppression and the ways in which these phenomena are either created by or ameliorated by justice systems.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 301 Advanced Level Special Topics in Justice Studies**Credits 3 Grading Basis** Graded

This course provides an examination of topics which are of current interest in the field of justice studies. The class may be repeated for credit when course content changes.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 307 Justice and Crime**Credits 3 Grading Basis** Graded

The purpose of this course is to provide a critical and interdisciplinary examination of the relationship between justice and criminal behavior. Drawing from social psychology and criminology, this course will examine various conceptions of justice, the implications that (in)justice has for the perpetration of crime and how principles of justice operate in the criminal justice system. In addition, this course will briefly investigate miscarriages of justice and the implications they have for offending behavior.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 308 Social Psychology of Justice**Credits 3 Grading Basis** Graded

This course examines various pressing social problems including, but not limited to, lethal policing, hate crimes, racial profiling and prejudice to illustrate how social psychology affects our understanding of justice. Students will examine how others - ranging from small friendship groups to entire societies - affect an individual and how the individual affects others by learning about various social psychological concepts, such as attributions of blame, stereotypes, attitudes, conformity and altruism. Throughout the course, students will identify the implications individual perceptions and interactions with others have for justice, crime, criminal justice and social justice.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 309 Global Crime**Credits 3 Grading Basis** Graded

This course explores crimes of the powerful from a global criminological perspective by examining historical and current cases of governmental, corporate and political crime. The complex relationships that will be investigated include how the interests of states, corporations and international financial institutions intersect in the commission of different types of global crimes.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 311 Women, Crime and Criminal Justice**Credits 3 Grading Basis** Graded

This class provides an overview of the ways in which women are involved in the phenomenon of crime: as victims, as offenders and as employees in the criminal justice system.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 312 Mass and Serial Murder**Credits 3 Grading Basis** Graded

This course will critically examine the phenomenon of multicide in society, including mass, serial and spree murder. This course will include: analyzing historical and contemporary examples of multicide; investigating the biological, psychological and sociological correlates of multicide offending; and examining media constructions and assessing their impact on public perceptions.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 316 Human Development and Crime**Credits 3 Grading Basis** Graded

This course examines how psychological research and theory shed light on the development of criminal careers, the factors that protect children and adolescents from becoming criminals, how being a victim of crime influences well-being, and the efficacy of rehabilitation. Special attention will be paid to the knowledge base on delinquency and childhood/adolescent victimization.

Prerequisites: PSYC 101; For justice studies majors: JUST 200 and one additional 200 level JUST course, excluding JUST 225. For Criminal Justice minors: JUST 210 or CRJU 215. For Youth Justice minors: YTHJ 300.

JUST 317 Victimization of Children**Credits 3 Grading Basis** Graded

This course provides an overview of patterns, causes and remedies for the various victimizations of children in the United States and throughout the world (abductions, child abuse, sexual exploitation, etc.).

Prerequisites: Justice studies major, criminal justice minor or youth justice minor; for majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; for Youth Justice minors: YTHJ 300.

JUST 318 Sex Offenders**Credits 3 Grading Basis Graded**

This course is designed to introduce students to the issue of sex offenders in society. Students will consider sexual values in American society and how they relate to the development, thought patterns and behavior of individuals who sexually violate others. Students will learn about theories of sex offending, the effects of sex offending on victims and society, and strategies for treating and managing sex offenders.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 320 Organized Crime**Credits 3 Grading Basis Graded**

This course is designed to give an overview of issues associated with Organized Crime. Organized Crime is an increasingly global phenomenon, and as such the class focuses not only on the situation in the United States, but the rest of the world as well. Attention is also given to the 'businesses' of organized crime (e.g., drug trafficking, counterfeiting), the law enforcement responses to Organized Crime, and the role/depiction of Organized Crime in popular culture.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 323 Comparative Criminal Justice**Credits 3 Grading Basis Graded**

A comparative study of criminal justice systems derived from the major world legal systems. The relevant background factors, government, laws, law enforcement, courts, corrections and youthful offenders are examined in each representative country studied. Multinational criminal justice organizations and special issues are addressed.

Prerequisites: Justice studies major or criminal justice minor. For justice studies majors: JUST 200 and one other 200 level JUST course, excluding JUST 225; for criminal justice minors: JUST 210; or permission of the instructor.

JUST 324 Death Penalty**Credits 3 Grading Basis Graded**

This course provides students with a broad survey of the death penalty as penal sanction and the controversies and issues which surround it. Key topical areas covered are history and foundations, legal landscape, execution and death penalty processes, contemporary issues including innocence, cost, discrimination and deterrence, and perspectives and voices surrounding the death penalty.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 326 Victimology**Credits 3 Grading Basis Graded**

This course provides an overview of various perspectives (social, psychological, legal, etc.) on the experience of victimization. Explanations of the phenomenon are discussed in the context of responses to various types of victimization.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 327 Criminal Law**Credits 3 Grading Basis Graded**

Study of substantive criminal law including common law sources and elements of various criminal offenses, justification and defenses.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 328 Race, Class and Justice**Credits 3 Grading Basis Graded**

This course provides students with an overview of contemporary justice issues in a comparative perspective. It includes an introduction to case-studies, comparative research methods and cross-national comparisons of justice issues concerning race and class.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 329 Perspectives on Law**Credits 3 Grading Basis Graded**

In this course, a broad array of perspectives on law and legal processes in the United States are examined. Students will examine perspectives from the realms of jurisprudence, philosophy, sociology, psychology, economics, anthropology and literature, among others, and will consider the intersection of these realms with law, legal processes, legal evolution and development, and the legal professions.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 330 Corrections**Credits 3 Grading Basis Graded**

This course examines the history, philosophy, policies and problems of the treatment of violators by the police, courts and correctional institutions.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 331 Human Rights in Theory and Practice**Cross-listed with: POSC 331, HUMN 331****Credits 3 Grading Basis Graded**

This course will explore the nature and value of human rights by investigating some major debates over their status and meaning, and by examining some of the ways people have tried to secure human rights in practice.

Prerequisites: Justice studies majors only, JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 333 Communities and Courts**Credits 3 Grading Basis Graded**

This course immerses students in procedures and processes at the local Circuit and General District Courts on Court Square in Harrisonburg. Students gather information and evidence on contemporary legal proceedings and outcomes in these spaces while investigating Circuit Court archives for contextual connections. Simultaneously, at a macro level, students examine the complex relationships between community safety, individual rights, court actors, community members and multifaceted conceptualizations of justice. Students' time is split between courses on campus and in local courtrooms observing and collecting data from court proceedings and engaging with the local judiciary.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 334 Media and Justice**Cross-listed with: SMAD 334****Credits 3 Grading Basis Graded**

This course will examine media constructions of justice. Students will be required to critically analyze the portrayal of justice issues in various media forms including television, internet and film. Attention will be given to the accuracy of such portrayals and whether they have any broader social implications in regards to how we view complex justice issues. Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 339 Mapping Justice**Credits 3 Grading Basis Graded**

This course examines maps and practices of mapping to analyze distributions of justice and injustice across space. Using case studies, multimedia content and hands-on exercises spanning the fields of criminal, global and social justice, students learn the fundamentals of geospatial analysis while designing their own digital maps and interactive web sites.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 341 Gender and Justice**Cross-listed with: WGSS 341****Credits 3 Grading Basis Graded**

This course is an interdisciplinary examination of the causes, structure and consequences of gender oppression. Consistent with the social justice track of the major, notions of fairness, justice, and equality with respect to gendered social, political and economic relations will be examined.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 343 Justice and Society**Credits 3 Grading Basis Graded**

This course focuses on the various intra- and interpersonal processes that underlie situations related to (in)justice at the individual and group levels. Theories, empirical research and real-world examples of social perception (e.g., prejudice and discrimination), social influence (e.g., attitudes, conformity) and social relations (e.g., altruism, aggression) related to (in)justice will be examined.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 345 Restorative Justice**Credits 3 Grading Basis Graded**

In this course, the restorative justice paradigm will be examined at multiple levels including consideration in the international, local community and criminal justice contexts. Both the underlying principles and practical applications of restorative justice will be explored. The related concept of community justice will also be addressed.

Prerequisites: Justice studies major, criminal justice minor or youth justice minor; for majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; for Youth Justice minors: YTHJ 300.

JUST 350 Justice and Globalization**Credits 3 Grading Basis Graded**

Globalization is the phenomenon of interconnectedness of economic, political and cultural activities across different parts of the globe. The class will survey several aspects of globalization through the lens of justice and injustice. Class time will be devoted to discussion and student presentations.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 351 Building Democracy**Credits 3 Grading Basis Graded**

This course will review historic and modern transitions to democracy, explore democracy as a 'just' form of government, examine correlates with its strengths and weaknesses, and provide an overview on how to build a quality democracy.

Prerequisites: JUST 200 and one track foundation course.

JUST 352 Global Migration**Credits 3 Grading Basis Graded**

This course surveys human migration across the Earth, throughout history, and in relation to justice. Students analyze interdisciplinary academic literature on migration, settlement and diasporas while studying flows of people across various historical, geographical and political contexts, from prehistoric to contemporary and from global to local.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 353 Justice and Development**Credits 3 Grading Basis Graded**

This course examines introduces the concept of justice as a standard for evaluating strategies for political, economic and social development in the contemporary international system.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 354 Native Justice and Resistance**Credits 3 Grading Basis Graded**

This course covers a wide array of topics relevant to the world's indigenous peoples including shared injustices and grievances, methods indigenous groups have taken to overcome justice inequalities, complexities of dual justice systems and the decolonization of legal and social systems.

Prerequisites: JUST 200 and one other 200-level JUST course.

JUST 355 Solutions to Global Poverty**Credits 3 Grading Basis** Graded

This course will analyze the root causes of global poverty and critically examine a range of approaches designed to improve conditions for the world's poor. Topics include nutrition and agriculture, healthcare, education, gender equality, water and sanitation, environmental management, and microfinance. Throughout the course, students will read policy documents, work with poverty-related data and analyze a variety of real-world case studies to illustrate key ideas.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 356 Refugees and Humanitarian Response**Credits 3 Grading Basis** Graded

This course will examine the practical and ethical challenges facing refugees and humanitarian actors in the contemporary world. We will survey the causes of forced migration, explore global trends in displacement and humanitarian aid and analyze the laws and institutions that facilitate (or inhibit) refugee protection. We will consider the challenges that refugees face throughout their journey, from border crossings, to refugee camps, to resettlement communities, as well as the practical and ethical tradeoffs that humanitarian actors must weigh in providing assistance and protection. The course will include a community engagement component related to refugee resettlement in Harrisonburg. Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 357 Environmental Justice**Credits 3 Grading Basis** Graded

This course provides students with an interdisciplinary introduction to environmental justice. Emphasizing how contemporary environmental issues are profoundly rooted in social, political, and economic conditions, students will apply principles and conceptions of justice to ecological challenges and sustainability efforts in local, national, and global contexts.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 361 Terrorism**Credits 3 Grading Basis** Graded

This course is a survey of the evolution of the phenomenon of terrorism and an analysis of its causes, strategies and forms. This course also discusses the consequences of terrorism in terms of government responses to terrorist threats (homeland security and counter-terrorism) and the economic and political costs of such responses. An interdisciplinary approach linking history, sociology, criminology and political science is used to make sense of this multifaceted phenomenon.

Prerequisites: Justice studies major or criminal justice minor; for justice studies majors: JUST 200 and one other 200-level JUST course, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215; or permission of the instructor.

JUST 365 Justice in Literature, Film and Art**Credits 3 Grading Basis** Graded

This class provides an opportunity for students to consider the ways in which issues related to the Justice fields are expressed in literature, film, and/or art and the ways in which such media may influence or create our conceptions of justice.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 372 Ethics and International Politics**Cross-listed with:** POSC 372**Credits 3 Grading Basis** Graded

This course investigates the significance of ethical questions in the theory and practice of contemporary international politics, introducing a variety of normative approaches that shape the issues of peace and conflict, morality and justice in global affairs. Practical case studies will also be used to address issues of policy relevance, with particular attention paid to the American experience.

Prerequisites: INTA or POSC majors, or JUST majors who have completed JUST 235.

JUST 373 Rebuilding Post Conflict Societies**Credits 3 Grading Basis** Graded

This course examines the social, economic, security and political problems faced in rebuilding societies that are emerging from a period of intense and prolonged conflict. This course presents an overview of the scope of this multidimensional challenge as well as strategies that have been used to address them. In the course of doing so, evaluation standards are discussed that can be used to measure the success, failure and justness of the newly emerging political system.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 374 War and Justice**Cross-listed with:** POSC 374**Credits 3 Grading Basis** Graded

This course is an empirical and normative investigation of the relationship between war and justice. Empirically, it examines the causes of war and the ways in which wars end. From a normative perspective it raises the questions of when are wars just; how should a just war be fought; and what is a just peace.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 375 Genocide in the 20th Century**Credits 3 Grading Basis** Graded

This course is an interdisciplinary examination of the various definitions, causes, structure and consequences of genocide. We will study some of the major 20th century genocides (Indigenous peoples, Armenia, USSR, Jewish Holocaust, Cambodia, Bosnia/Kosovo, Rwanda, and Darfur) using the following conceptual schemes: social death, structural violence, oppression, and mass murder.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 376 Global Social Movements**Credits 3 Grading Basis** Graded

This course introduces justice studies students to the origins, development and outcomes of global social movements. It involves a comprehensive approach, which entails an in-depth examination of the main theories explaining the rise of social movements, as well as an empirical focus on global dynamics of popular mobilization. Special attention will be devoted to the examination of the global justice movement.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 377 Global Futures**Credits 3 Grading Basis Graded**

Explores possible future directions that the global system may take in selected issue areas such as energy, democratization, food supplies, and infectious diseases. Emphasis is on active learning strategies, introducing the analytical tools used for analysis of international trends and addressing questions of how to respond in a manner that promotes global justice.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 385 Disability and Justice**Credits 3 Grading Basis Graded**

An interdisciplinary exploration of justice and disability which examines issues such as the historical and theoretical contexts of disability, the social and rhetorical construction of disability, disability legislation and public policy, disability and the criminal justice system, and disability rights movements and inclusion as they relate to larger civil and human rights movements in the US & globally. The course highlights the importance of disability in broader conversations about justice.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 386 Sexual Orientation, Law and Justice**Cross-listed with:** WGSS 386**Credits 3 Grading Basis Graded**

This course explores considerations of justice in shaping law and policy related to sexual orientation. The concept of sexual orientation is a relatively recent development in academic, legal, medical and other policy contexts in the United States and globally. Topics addressed in the course include the dehumanization, criminalization, medicalization and oppression of sexual orientation minorities, and legal and regulatory interventions and movements for social justice, criminal justice reform and human rights.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

JUST 387 Gender Identity, Law and Justice**Cross-listed with:** WGSS 387**Credits 3 Grading Basis Graded**

This course explores considerations of justice in shaping law and policy related to gender identity. The concept of gender identity is a relatively recent development in academic, legal, medical, and other policy contexts in the United States and globally. Topics addressed in the course include the dehumanization, criminalization, medicalization, and oppression of transgender, non-binary, and gender-expansive populations and legal and regulatory interventions and movements for gender self-determination, social justice, criminal justice reform, and human rights.

Prerequisites: JUST major prereqs: JUST 200 & one other 200-level JUST course, excluding JUST 225. SJS minor: JUST 221 or 223. WGSS minor: WGSS 200 & permission of minor coordinator. QUST minor: permission of instructor.

JUST 392 Peace Studies**Cross-listed with:** POSC 392**Credits 3 Grading Basis Graded**

A study of the evolution, theory and practice of peace studies. The course focuses on how we wage and resolve conflict, how we effect social change, and how we provide for security through nonviolent means.

Prerequisites: JUST 200 and one other 200-level JUST course.

JUST 397 Legal Research and Information Literacy**Credits 3 Grading Basis Graded**

This course will provide a thorough insight into how to study law. Students will acquire fundamental legal skills such as how to read a case and identify and differentiate ratio decidendi and obiter dicta, understand the codification of law in a common law jurisdiction, and learn how to accurately research legal topics utilizing and citing primary and secondary legal sources. The course will focus on domestic (U.S.) legal research, involving both state and federal jurisdictions, in addition to investigating international examples and legal instruments.

Prerequisites: For justice studies majors: JUST 200 and one other 200-level JUST class, excluding JUST 225; for criminal justice minors: JUST 210 or CRJU 215.

JUST 399 Justice Research Methods**Credits 4 Grading Basis Graded**

This course provides students with the tools necessary for conducting independent research in the area of Justice Studies. Both quantitative and qualitative methods are covered. A term project, in which the skills covered in the course are applied, is a significant part of the class.

Prerequisites: JUST 200, one other 200-level JUST course (excluding JUST 225), MATH 220 and one 300-level JUST course.

JUST 400 Senior Seminar in Justice Studies**Credits 3 Grading Basis Graded**

This is the capstone course of the Justice Studies major. Students are expected to enter JUST 400 with a proposed area of study, and will spend the semester in developing a thesis. Class discussion and review of individual projects, along with oral presentation of work, are integral parts of the course. Students are expected to produce a piece of original scholarship related to their study in the major.

Prerequisites: JUST 200, JUST 399 and senior JUST major.

JUST 401 Internship in Justice Studies**Credits 3 Grading Basis Graded**

In this course, students integrate professional experiences as interns with justice studies coursework. In addition to working with an agency or organization, students will produce academic work which examines the connections between their area of study and the context of the internship. By application only.

Prerequisites: JUST 200, JUST 399, minimum GPA (in major and cumulative) of 2.0, and permission of instructor.

JUST 402 Advanced Research in Justice Studies**Credits 3 Grading Basis Graded**

This course is designed to give students a context in which to pursue advanced research in a Justice Studies context. It is especially appropriate for students interested in graduate study in the field.

Prerequisites: JUST 200 and JUST 399, or permission of the instructor.

JUST 403 Nelson Institute Seminar**Credits 3 Grading Basis Graded**

Students enrolled in this course will be assigned to consider a contemporary problem Justice Studies. Working individually and in teams, students will first study the problem, and then propose workable solutions.

Prerequisites: Junior or senior standing only.

JUST 404 Practicum in Community-Based Research**Credits 3 Grading Basis** Graded

This course provides students with an intensive, collaborative community-based research experience working in partnership with a community organization or group. Students will integrate and apply principles and conceptions of justice and methods of scholarly inquiry to a community-identified problem. This course may be taken only once for credit.

Prerequisites: JUST 200 and JUST 399, or permission of the instructor.

JUST 405 Research Assistantship in Justice Studies**Credits 1-3 Grading Basis** Graded

Students will work as research assistants under faculty supervision. Students may take this course only by invitation from a faculty member in justice studies and may register for this course only through departmental approval. No student may take more than three credits of JUST 405.

Prerequisites: JUST 200 and two other 200-level JUST courses, excluding JUST 225; one 300-level JUST course; invitation of the instructor; and permission of the department head.

JUST 499 Honors**Credits 1-3 Grading Basis** Graded

Year course. An independent research topic initiated and completed by qualified majors wishing to graduate with distinction.

Kinesiology (KIN)

KIN 100 Lifetime Fitness and Wellness (2,2) [C5W]**Credits 3 Grading Basis** Graded

This course is designed to help students adopt and maintain the behaviors associated with an active and healthy lifestyle. Through this course students will learn the importance of maintaining wellness through a physically active lifestyle. Through lectures and labs, students study and develop the behavioral patterns consistent with the current knowledge base in fitness and wellness. May be used for general education credit.

KIN 120 Mindfulness for Health and Wellness**Credits 1 Grading Basis** Graded

This course is designed to provide students with an introduction to formal and informal mindfulness practices. Students will study and practice multiple mindfulness techniques while building a personal resource pool to help develop a daily habit of mindfulness to increase overall personal wellness. Instructional methods will involve a combination of online lectures and applied exercises.

KIN 199 Practice in Kinesiology**Credits 1-3 Grading Basis** Credit / No Credit

This course is designed to introduce students to basic and beginner topics in kinesiology and movement. Topics will be chosen each semester and reflect current literature and best practice. This course may be repeated once for credit (for a maximum of 6 credits) when course content changes.

KIN 201 Introduction to Kinesiology as a Profession**Credits 2 Grading Basis** Graded

This course is designed to help students better understand: the breadth and depth of the discipline of kinesiology and the possible professional career applications that fall beneath the kinesiology discipline umbrella. Prerequisites: First-year or sophomore standing.

KIN 202 Biological Foundations of Kinesiology**Credits 3 Grading Basis** Graded

This course is designed to offer an introduction to the biological scientific foundations within the discipline of kinesiology. This survey course includes coverage of topics including applied anatomy and physiology, biomechanics, and exercise physiology.

Prerequisites: Kinesiology majors, coaching education minors, pre-athletic training minors, pre-physical therapy minors, or pre-occupational therapy minors only.

KIN 205 Introduction to Coaching**Credits 3 Grading Basis** Graded

Through experiential learning activities and discussions, students are introduced to the role of the sport coach as well as the professional, interpersonal and intrapersonal knowledge needed to effectively coach.

Prerequisites: KIN 202. Corequisites: KIN 202.

KIN 221 Skill Laboratory: Rhythmic and Dance Activities**Credits 3 Grading Basis** Graded

The skill laboratories are designed for participants who will be in the role of educating others in a particular area of movement. The skill laboratory, Rhythmic & Dance Activities, provides a) basic movement skills, b) analysis of movement, c) developmentally appropriate progressions & teaching ideas, d) curriculum development, and e) micro-teaching opportunities.

KIN 222 Skill Laboratory: Teaching Fitness and Wellness in Schools**Credits 3 Grading Basis** Graded

The skill laboratories are designed for participants who will be in a role of educating others in a particular area of movement. The skill laboratory, Teaching Fitness & Wellness in the Schools, provides a) basic movement skills, b) analysis of movement, c) developmentally appropriate progressions & teaching ideas, d) curriculum development, and e) micro-teaching opportunities.

KIN 223 Skill Laboratory: Individual and Lifetime Activities**Credits 3 Grading Basis** Graded

The skill laboratories are designed for participants who will be in a role of educating others in a particular area of movement. The skill laboratory, Individual & Lifetime Activities, provides: a) basic movement skills, b) analysis of movement, c) developmentally appropriate progressions & teaching ideas, d) curriculum development, and e) micro-teaching opportunities.

KIN 224 Skill Laboratory: Court and Field Games**Credits 3 Grading Basis** Graded

The skill laboratories are designed for participants who will be in a role of educating others in a particular area of movement. The skill laboratory, Court & Field Games, provides: a) basic movement skills, b) analysis of movement, c) developmentally appropriate progressions & teaching ideas, d) curriculum development, and e) micro-teaching opportunities.

KIN 225 Skill Laboratory: Wilderness and Adventure Education**Credits 3 Grading Basis** Graded

The skill laboratories are designed for participants who will be in a role of educating others in a particular area of movement. The skill laboratory, Wilderness & Adventure Education, provides: a) basic movement skills, b) analysis of movement, c) developmentally appropriate progressions & teaching ideas, d) curriculum development, and e) micro-teaching opportunities.

KIN 302 Exercise Physiology**Credits 3 Grading Basis** Graded

This course is designed to help the student explore and understand the physiological changes that occur during an acute bout of exercise and as a result of chronic physical training. Students will study the role various (e.g. cardiovascular, respiratory, nervous, neuro-endocrine, and renal etc.) systems play in maintaining homeostasis during physical activity. In addition, the physiology of physical performance under a range of environmental conditions will also be examined.

Prerequisites: Fully admitted to the exercise science concentration.

Corequisites: KIN 302L.

KIN 302L Exercise Physiology Laboratory**Credits 1 Grading Basis** Graded

This course involves laboratory experiences related to core concepts of exercise physiology.

Prerequisites: Fully admitted to the exercise science concentration.

Corequisites: KIN 302.

KIN 303 Motor Development and Learning**Credits 3 Grading Basis** Graded

This course provides understanding of motor development from early childhood through adulthood. The focus is on the constraints to development and the interaction between the environment, task and learner. The course provides an understanding of the learning processes underlying performance. Emphasis is given to the application in both teaching and coaching settings.

Prerequisites: Fully admitted to the Kinesiology exercise science concentration or declared to the Kinesiology PHETE concentration.

KIN 304 Historical and Philosophical Foundations of Sport and Physical Activity**Credits 3 Grading Basis** Graded

This course includes an introductory analysis of selected historical periods and philosophical theory concerning the development of ideas about sport and physical activity from Ancient Greece and Rome to contemporary United States cultures.

KIN 305 Psychological Foundations of Sport and Performance**Credits 3 Grading Basis** Graded

This course focuses on exploring psychological concepts related to participation and performance in sport and exercise contexts. This course is not open to students who have previously earned credit in KIN 353 or KIN 426.

Prerequisites: Fully admitted to the exercise science concentration or declared to the Kinesiology PHETE concentration. Not open to students who have previously earned credit in KIN 353 or KIN 426.

KIN 306 Human Biomechanics**Credits 3 Grading Basis** Graded

This course involves studies of anatomical, physical and mechanical factors, as these factors affect human movement.

Prerequisites: Fully admitted to the exercise science concentration.

Corequisites: KIN 306L.

KIN 306L Human Biomechanics Laboratory**Credits 1 Grading Basis** Graded

This laboratory course is designed to complement and supplement the lecture course KIN 306. The course will focus on enhancing the students' laboratory experiences in biomechanics.

Prerequisites: Fully admitted to the exercise science concentration.

Corequisites: KIN 306.

KIN 309 School Health Content**Credits 3 Grading Basis** Graded

An overview of select topics in health education content necessary to implement a comprehensive health education program, geared toward child and adolescent populations.

KIN 312 The Profession of Teaching Health and Physical Education**Credits 3 Grading Basis** Graded

This course introduces the roles of the teacher and the learner and the pedagogical content knowledge of health and physical education. An in-depth examination of the unique position and qualifications of the specialist in physical education and health. Systematic observations will occur.

KIN 321 Exercise Testing and Prescription**Credits 3 Grading Basis** Graded

This course will examine the principles of exercise testing and prescription as they apply to fitness, performance and health. The role of health-related components of fitness in performance and health of apparently healthy adults will be examined.

Prerequisites: Fully admitted to the exercise science concentration.

Corequisites: KIN 321L.

KIN 321L Exercise Testing and Prescription Laboratory**Credits 1 Grading Basis** Graded

This laboratory course is designed to complement and supplement the lecture course KIN 321. The course will focus on enhancing and providing practical opportunities to apply the students' exercise testing skills and prescription development knowledge.

Prerequisites: Fully admitted to the exercise science concentration.

Corequisites: KIN 321.

KIN 329 Social Aspects of Sport**Credits 3 Grading Basis** Graded

This class is designed to familiarize students with the concepts of sport as they relate to individuals and the larger society in which they live and the ways in which this relationship is reciprocal, with both people impacting the world of sport and sport impacting individual, social, cultural, economic and political developments. Students learn the tools needed to think in a critical way about sports in society and make informed choices about sports and participation.

KIN 353 Developing Mental and Life Skills through Sport and Physical Education**Credits 3 Grading Basis** Graded

This course explores current sport psychology theories, models and concepts. Students examine mental and life skills and apply the information to their coaching and/or teaching context. Not open to students who have previously earned credit in KIN 305 or KIN 426.

Prerequisites: Fully admitted to the exercise science concentration, declared to the Kinesiology PHETE concentration or coaching education minors. Permission of the instructor. Not open to students who have previously earned credit in KIN 305 or KIN 426.

KIN 381 Practicum in Exercise Science**Credits 3 Grading Basis** Graded

This practicum in exercise science provides the student with supervised experience in their field of study. This course may be repeated in different settings. Instructor permission is required.

Prerequisites: Fully admitted to the exercise science concentration; major GPA of 2.0 or higher; and permission of instructor.

KIN 410 School Health Content for PHETE**Credits 3 Grading Basis Graded**

An overview of selected topics in health content required for teacher candidates preparing to teach health education in public schools. Special emphasis will be on issues relevant to teaching these topics in schools. Prerequisites: KIN 309.

KIN 411 Measurement and Evaluation in Kinesiology**Credits 3 Grading Basis Graded**

The administration and interpretation of measurement and evaluation procedures in kinesiology and recreation.

KIN 417 Adapted Physical Education**Credits 3 Grading Basis Graded**

This course explores the disability categories listed in IDEA and ensures the acquisition of content knowledge and instructional strategies to successfully implement inclusive and adapted physical education programs. Practicum experience is included. Prerequisites: Acceptance to PHETE program.

KIN 418 Assessment in Elementary Physical Education**Credits 3 Grading Basis Graded**

This course is an introductory study of developmentally appropriate, authentic and formal assessment techniques unique to elementary physical education. Prerequisites: Acceptance to PHETE program.

KIN 420 Instructional Methods in Elementary Physical Education**Credits 3 Grading Basis Graded**

This course introduces students to effective teaching strategies and allows application through peer microteaching and teaching students in a school setting. Students will apply the affective, cognitive, and psychomotor principles in class management, unit and lesson planning, and instructional techniques. Principles of self-evaluation and reflective teaching are reinforced.

Prerequisites: Acceptance to PHETE program.

KIN 421 Elementary Curriculum in Physical Education**Credits 3 Grading Basis Graded**

This course is designed to explore the theory and application of games, dance, and gymnastic activities compatible with the developmental characteristics of elementary children and the educational objectives of elementary schools. Students will engage in a practicum experience within an elementary school.

Prerequisites: Acceptance to PHETE program.

KIN 422 Advanced Principles of Group Exercise Leadership**Credits 3 Grading Basis Graded**

Theories, principles and procedures involved with leading group fitness classes. Students will gain knowledge and experience in class design and choreography development. Programming for children, older adults, pregnancy and aquatic exercise will be included. Practical and written assessments will occur throughout the semester. Participation in group exercise classes is a requirement. Students will have the opportunity to enroll in the AFAA Primary Group Exercise Certification.

Prerequisites: KIN 100, KIN 202, or permission of the instructor.

KIN 424 Exercise and Nutrition in Chronic Disease**Cross-listed with: NUTR 455****Credits 3 Grading Basis Graded**

This interdisciplinary course examines the evidence-based relationship between exercise and diet patterns and behaviors, and major chronic diseases. Students will understand the epidemiology and pathogenesis of prevailing chronic diseases. Emphasis will be placed on developing exercise and dietary strategies to reduce the burden of chronic disease. Prerequisites: Fully admitted to the exercise science concentration; KIN 302 and KIN 321.

KIN 425 Concepts of Strength and Conditioning**Credits 3 Grading Basis Graded**

This course examines theory and application of coaching concepts in strength/conditioning training, including program design, testing and specific techniques for the physical development of athletes. It is designed for students interested in working with athletic populations. This course also prepares students for NSCA certification. Prerequisites: Fully admitted to the exercise science concentration or coaching education minor; KIN 100 or HTH 100; KIN 202.

KIN 426 Psychological Foundations of Exercise and Health Behavior**Credits 3 Grading Basis Graded**

This course will focus on the theoretical and practical applications of behavior change related to healthy lifestyles with an emphasis on physical activity. In addition, course content will include a detailed investigation into the psychological and environmental factors associated with adoption and maintenance of healthy behaviors including a regular physical activity program. This course is not open to students who have previously earned credit in KIN 305 or KIN 353. Prerequisites: Fully admitted to the exercise science concentration; KIN 302. Not open to students who have previously earned credit in KIN 305 or KIN 353.

KIN 428 Advanced Topics in Exercise Science and Leadership**Credits 3 Grading Basis Graded**

This course is designed to allow students to study specific topics in exercise science and leadership. Topics will be chosen each semester and reflect current research in exercise science related to clinical exercise physiology, human performance and methodologies or trends in exercise science/leadership. May be repeated once (maximum 6 credits) for credit when course content changes.

Prerequisites: KIN 302 and KIN 302L.

KIN 429 Special Topics in Adapted Physical Education**Cross-listed with: KIN 515****Credits 3 Grading Basis Graded**

This course provides an in depth look into specific areas within the field of adapted physical education and adapted physical activity. This application-based course provides hands-on experiences that will allow students to work with individuals with disabilities in a variety of settings. Graduate students will also be asked to analyze and provide feedback on the instructional behaviors of the undergraduate student instructors.

KIN 430 Exercise Across the Lifespan**Cross-listed with: KIN 542****Credits 3 Grading Basis Graded**

An advanced course in exercise science that examines aspects of physical activity and exercise as they relate to specific phases of the human lifespan. This course will address the importance of physical activity for children, pregnant women, and older adults and will include an in-depth study of aspects of exercise physiology that are unique to these populations.

KIN 431 Environmental Exercise Physiology**Credits 3 Grading Basis Graded**

An advanced course in exercise physiology that examines how human physiological systems respond and adapt to exercise performed in challenging environmental conditions (i.e. high altitude, diving, hot, cold).
Prerequisites: KIN 302 and KIN 302L.

KIN 432 Physiology of Endurance Performance**Credits 3 Grading Basis Graded**

This course aims to provide students with an understanding of how to utilize endurance training to maximize athletic performance. Various factors that limit endurance performance and examine methods to overcome these factors will be explored. Emphasis will be placed on the role of seasonal periodization and program planning to maximize the long-term development of athletes. Supplemental factors that may also aid performance, such as equipment, nutrition, and ergogenic aids will be discussed.

Prerequisites: KIN 302 and KIN 302L.

KIN 433 Exercise, Cardiovascular Disease and ECG Assessment**Credits 3 Grading Basis Graded**

This course is designed to provide students with a foundational information related to electrocardiography, diagnostic exercise testing and the effects of exercise on specific cardiovascular diseases.

Prerequisites: KIN 302.

KIN 440 Exercise Immunology**Credits 3 Grading Basis Graded**

Students will develop an in-depth understanding of the human immune system and how it is impacted by exercise. Specific focus will be placed on examining the relationship between immune function, exercise and risk of infections. Additional aims will be to explore physiological and molecular mechanisms that determine immune function and how they are impacted by acute and chronic exercise.

Prerequisites: KIN 302.

KIN 441 Strength and Conditioning II: Advanced Training Methods and Concepts**Credits 3 Grading Basis Graded**

This course focuses on advanced training techniques and program design for athletes as well as utilizing sport science to monitor the adaptational process. This includes physical test selection, administration, scoring and interpretation of results. The practical portion of the course includes demonstrating and coaching peers through advanced training techniques as well as administering and performing physical tests. Designed for students interested in working with athletic populations and preparing to sit for NSCA-CSCS certification examination.

Prerequisites: KIN 425 and permission of the instructor.

KIN 442 Strength and Conditioning III: Philosophy, Leadership and Managerial Responsibilities**Credits 3 Grading Basis Graded**

This course focuses on the practical, managerial and leadership elements critical to strength and conditioning professionals' everyday operations. This includes developing mission and purpose statements, creating policies and procedures, determining facility design and maintenance, hiring/managing staff, and leading by example. Discussions center around those entering the strength and conditioning profession or a similarly related field. Designed for students interested in a career in strength and conditioning and/or working with athletic populations and preparing to sit for NSCA-CSCS certification examination.

Prerequisites: KIN 425 and permission of the instructor.

KIN 443 Advanced Respiratory Physiology in Exercise Science**Credits 3 Grading Basis Graded**

This course will provide an in-depth exploration of respiratory system function and its critical role in exercise and performance. Through a combination of laboratory data and current research, students will evaluate cardiorespiratory performance and apply physiological concepts to exercise training.

Prerequisites: Fully admitted to the Exercise Science Concentration and KIN 302.

KIN 450 Principles of Coaching Methods**Credits 3 Grading Basis Graded**

This course will focus on methods of teaching sport skills, organizing and running effective practices, evaluating athletes, and developing season plans based on long-term athlete development. Developing team culture, working with multiple stakeholders and creating a safe practice environment are also included. Overall, students will further develop core competencies in coaching practice and will gain practical experience in the field.

Prerequisites: KIN 205.

KIN 473 Internship in Coaching**Credits 3 Grading Basis Graded**

This course offers a sequence of selected experiences in coaching, which provides the student with supervised experience to develop their coaching effectiveness.

Prerequisites: Coaching education minor, successful completion of all courses in the minor, minor GPA of 2.0 or higher and permission of the instructor.

KIN 474 Practicum in Sport Communications**Credits 3 Grading Basis Credit / No Credit**

A sequence of selected practicum experiences which provides the student with supervised practicum experience in Sport Communications.

KIN 480 Student Teaching in Physical Education**Credits 8 Grading Basis Credit / No Credit**

A supervised teaching experience at the elementary school setting that provides teacher candidates with opportunities to experience the classroom environment, grow professionally, and develop their pedagogical skills.

Prerequisites: Acceptance into student teaching.

KIN 481 Internship in Exercise Science**Credits 4-12 Grading Basis Graded**

This professional experience in exercise science affords students the opportunity to apply theory and methodology in their field of study under qualified supervision from the cooperating agency and the university. May be repeated for a maximum of 12 credits.

Prerequisites: Cumulative GPA of 2.0 or higher; KIN 381 or permission of the instructor.

KIN 490 Special Studies in Kinesiology and Recreation**Credits 1-3 Grading Basis Graded**

Designed to give superior students in physical education an opportunity to complete independent study and/or research under faculty supervision.

KIN 499A Honors**Credits 1 Grading Basis Graded**

Students will work with a faculty member to conceptualize a research project and then compose a proposal of the project with anticipated methods.

KIN 499B Honors**Credits 3 Grading Basis** Graded

Students will carry out the data collection for the proposed project under the guidance of their faculty mentor.

KIN 499C Honors**Credits 2 Grading Basis** Graded

Students will compose the final project report, in manuscript format and according to departmental guidelines.

Korean (KOR)

KOR 101 Elementary Korean I**Credits 3 Grading Basis** Graded

This course covers the fundamentals of Korean through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

KOR 102 Elementary Korean II**Credits 3 Grading Basis** Graded

This course builds on the foundations established in KOR 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension.

Prerequisites: KOR 101; or by placement exam or permission of the instructor.

KOR 231 Intermediate Korean I**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition, and reading.

Prerequisites: KOR 102 or permission of the instructor. A student cannot earn credit for KOR 231 after completing KOR 232.

KOR 232 Intermediate Korean II**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition, and reading.

Prerequisites: KOR 231 or permission of the instructor.

KOR 300 Language for Living**Credits 3 Grading Basis** Graded

This course is designed to enhance students' practical communication skills in Korean, preparing them for everyday interactions in personal, academic and professional contexts. Building on the foundations established in KOR 232, this course emphasizes real-life language use through topics related to daily life, social events and work. Students will engage in speaking, listening, reading and writing activities designed to improve fluency and cultural understanding. The course prepares students to navigate various real-world situations while deepening their knowledge of Korean social customs and language norms.

Prerequisites: KOR 232; or by placement exam or by permission of instructor.

KOR 308 Contemporary South Korean Culture and Society**Credits 3 Grading Basis** Graded

This course is designed to introduce students to Korean culture and society with a focus on major social and historical changes related to contemporary Korean culture. The course will explore diverse themes relevant to understanding Korea today, such as a brief history of Korea, modernization, multiculturalism, gender and education issues, lifestyle as well as traditional and contemporary popular culture. This course will combine lectures with analysis and discussions of various media such as texts, literature, films, newspaper articles as well as audio-visual artifacts including Korean news, films, TV dramas and pop songs in class. Instruction is in English.

KOR 320 Culture and Communication in Context**Credits 3 Grading Basis** Graded

This course is designed to develop students' reading and writing skills through the exploration of Korean cultural themes. Students will engage with short, accessible texts and visual materials that introduce traditional and contemporary aspects of Korean society. The course emphasizes formal and informal written communication, helping students gain confidence in expressing themselves in written Korean while gaining insights into social structures, media and cultural practices. Through reading, writing and discussions, students will improve their cultural literacy and written communication skills.

Prerequisites: KOR 232; or by placement exam or by permission of instructor.

KOR 386 Korean Short Stories**Credits 3 Grading Basis** Graded

This course is designed to enhance students' reading comprehension and to explore the unique Korean cultural and social worldviews through folktales. Each reading will introduce useful phrase and idioms, as well as diverse themes of stories including filial piety, morality, greed and love. The course also uses audio and visual materials such as story recordings, paintings, maps and short films. Students will engage in a variety of activities such as discussions, presentations and compositions to achieve effective learning outcomes. Readings are in Korean. Instruction is in Korean and English.

Prerequisites: KOR 232; or by placement exam or by permission of instructor.

KOR 446 Special Topics in Korean Literature**Credits 3 Grading Basis** Graded

This course studies a particular topic in Korean literature. It may cover all or a specific Korean literature genre. This course may be repeated if content varies.

Prerequisites: KOR 300 or KOR 320; or by permission of the instructor.

KOR 447 Special Topics in Korean Culture**Credits 3 Grading Basis** Graded

Students will study a particular topic in the culture of Korea. Course may be repeated if content varies. Instruction is in English.

KOR 448 Special Topics in Korean Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in Korean linguistics. Topics could include an introduction to Korean sociolinguistics or psycholinguistics. Course may be repeated if content varies.

Prerequisites: KOR 300 or KOR 320; or by permission of the instructor.

KOR 490 Independent Study in Korean**Credits 1-4 Grading Basis** Graded

The course allows students an opportunity to pursue independent study in Korean under faculty supervision.

Latin (LAT)

LAT 101 Elementary Latin I**Credits 3 Grading Basis** Graded

This is an introductory course for students who intend to acquire only a reading knowledge of classical and medieval Latin. It includes systematic study of the fundamentals of grammar.

LAT 102 Elementary Latin II**Credits 3 Grading Basis Graded**

This is the second part of an introductory course for students who intend to acquire only a reading knowledge of classical and medieval Latin. It includes systematic study of the fundamentals of grammar.
Prerequisites: LAT 101.

LAT 231 Intermediate Latin I**Credits 3 Grading Basis Graded**

An introduction to Latin literature. The further study of Latin grammar and the elements of Latin prosody are also presented.
Prerequisites: LAT 102 or by permission of the instructor.

LAT 232 Intermediate Latin II**Credits 3 Grading Basis Graded**

An introduction to Latin literature. The further study of Latin grammar and the elements of Latin prosody are also presented.
Prerequisites: LAT 231 or by permission of the instructor.

LAT 490 Independent Study in Latin**Credits 1-4 Grading Basis Graded**

This course allows students to pursue individual study in Latin under the supervision of the instructor.

Latin American, Latinx and Caribbean Studies (LAXC)

LAXC 252 Introduction to Latin American, Latinx and Caribbean Studies [C2HQC]

Credits 3 Grading Basis Graded

An introductory interdisciplinary survey of basic theoretical concepts to analyze the Latin American, Caribbean and U.S. Latinx experience, with special focus on how historical processes in Latin America and the Caribbean gave rise to specific and varied cultural production and forms in the region. May be used for general education credit.

LAXC 299 Internship in Latin American, Latinx and Caribbean Studies

Credits 1-3 Grading Basis Graded

Provides Latin American, Latinx and Caribbean Studies minors with work experience in career fields they are interested in pursuing. A journal, internship report, research paper, bibliography and evaluation from the intern provider are required. May count as an elective for the LAXC minor.

LAXC 300 Topics in Latinx Studies

Credits 3 Grading Basis Graded

This course introduces students to key issues in the interdisciplinary field of Latinx Studies, which examines the communities, cultures, histories and experiences of people of Latin American descent in the United States. The topic for each semester will be announced on MyMadison. Course topics will highlight the historical and contemporary issues that shape the political, social and cultural practices, as well as the lived experiences of Latinxs in the United States. The course will encompass various approaches to the field of Latinx studies, potentially including media studies, political science, literature, history, anthropology, sociology, communication and more. LAXC 300 will begin with a brief introduction that asks "What is Latinx Studies?" The course will then examine a range of topics including colonialism, race, ethnicity, class, gender, sexuality, migration, language, media, representation, participation and resistance. Students will critically engage with written text and various media that encourage them to examine Latinx communities in all their diversity and complexity.
Prerequisites: LAXC minors; or by permission of the instructor.

LAXC 375 Latina/o/x Politics: Origins and Futures

Cross-listed with: POSC 375**Credits 3 Grading Basis Graded**

The course will focus on the Latina/o/x populations in the United States and how the category of 'Latino' came into existence. Furthermore, students will learn how Latina/o/x identity has evolved over time as well as become politically salient within 21st century politics. The goal is for the student to develop a broad, conceptual understanding of how the term 'Latino' came into existence as well as how individuals who identify as 'Latina/o/x' have politically organized throughout history.
Prerequisites: POSC 225.

LAXC 378 Borderlands Policy: Latina/o/x Politics in Context

Cross-listed with: POSC 378**Credits 3 Grading Basis Graded**

This course is centered on the policies surrounding the U.S. and Mexico border, with a focus on the effect these policies have had on Latina/o/x populations. Students will examine peer-reviewed studies of immigration policy, public opinion and political rhetoric, as well as individual accounts of the immigrant experience. By the end of the course, students will have an increased understanding of how border policy has historically evolved and the Latino/a/x population reactions across time.
Prerequisites: POSC 225.

LAXC 490 Independent Study in Latin American, Latinx and Caribbean Studies

Credits 3 Grading Basis Graded

Designed to give capable students in Latin American, Latinx and Caribbean Studies an opportunity to complete independent study under faculty supervision.

LAXC 495 Special Topics in Latin American, Latinx and Caribbean Studies

Credits 3 Grading Basis Graded

In-depth examination of selected topics of current importance to the field of Latin American, Latinx and Caribbean Studies. May be repeated for credit when course content changes.

Learning, Design and Technology (LDT)

LDT 150 Information in Contemporary Society

Credits 3 Grading Basis Graded

This course critically examines how data and information impact individuals and communities in contemporary society, with a focus on how information technologies mediate what and how we learn and how we interact. Recognizing that individuals and communities need, create and analyze information to function in society, learners will examine the impact of new technologies on topics such as democracy, diversity and social justice. This course is designed to encourage reflection by inviting learners into a deeper discourse about the ethical role of information in society. Note: Will not count as a social science course for teacher licensure.

LDT 370 Instructional Technology**Cross-listed with:** EDUC 370**Credits 3 Grading Basis** Graded

In this course, learners will establish a foundational understanding of technology and media as a means to empower pedagogical practice and set conditions for learning. The course is intended to help develop awareness of, and appreciation for, a range of technical skills and reflective practices that may be applied in daily life and in future course work and careers. Learners will create an ePortfolio of digital media developed in this course.

Prerequisites: Human resources development minors, educational media minors or adult degree program students only.

LDT 372 Visual Literacy**Credits 3 Grading Basis** Graded

This course will foster the ability to recognize, critique and edit visual elements and representations for learning. Learners will select appropriate digital tools and techniques, apply proven principles and theories in practice, and give and receive constructive feedback to improve visual design through iteration. Learners will be expected to design original, user-centered and aesthetically pleasing digital visuals grounded in effective design practice and learning theories.

Prerequisites: PSYC 160 or EDUC 175, and LDT 370. Corequisites: LDT 370.

LDT 380 Digital Media for Learning**Credits 3 Grading Basis** Graded

In this course, learners will develop an intermediate understanding of the tools, principles, practices and theoretical underpinnings of designing and creating digital media for learning.

Prerequisites: LDT 370 and LDT 372. Corequisites: LDT 372.

LDT 385 Foundations of Instructional Design**Credits 3 Grading Basis** Graded

This capstone course will facilitate the application of the knowledge, skills and attitudes developed in earlier course work in the educational media minor. Learners will apply their technical, design and development skills to solve authentic problem(s) that require an understanding of how people learn. This will involve using digital media and instructional design processes that lead to the completion of a capstone project.

Prerequisites: PSYC 160 or EDUC 175, LDT 370, LDT 372 and LDT 380. Corequisites: LDT 380.

Early, Elementary, and Literacy Education (LED)

LED 212 Foundations of Literacy for Middle School Teachers**Credits 3 Grading Basis** Graded

This course serves as a starting point for learning about teaching middle school students. It assists future teachers in developing knowledge of literacy and how to form relationships with middle school adolescents, serving as an introduction for preservice teachers to the foundations of reading and writing development and the elements of balanced literacy instruction in the middle grades. Through reading, writing and field applications across content areas, preservice teachers explore literacy engagement and diversity.

Corequisites: Middle education majors must take MIED 211.

LED 240 Children's Literature**Credits 3 Grading Basis** Graded

The study of a variety of children's literature and the practices, principles and procedures for selecting and evaluating works for children, giving consideration to their motivational and developmental effects.

LED 254 Literature for Adolescents**Credits 3 Grading Basis** Graded

A study of literature which has been written for or appeals to young adults and adolescents, including practices, principles and procedures for evaluating and making literary materials appealing to adolescents.

LED 300 Reading Instruction and Assessment for Diverse Learners PK-6**Credits 3 Grading Basis** Graded

This course will provide PK-6 preservice teachers with the foundations of reading development, assessment techniques, and instructional strategies. This course in combination with LED 350 and LED 462 comprise the literacy instruction for prospective elementary education teachers.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 272.

Corequisites: ELED majors: ELED 321; ELED 332 and ELED 333, or ELED 372 and ELED 373.

LED 350 Writing Instruction, Assessment and Technology for Diverse Learners PK-6**Credits 3 Grading Basis** Graded

This course involves instruction and discussion of the writing process, development and assessment. This course, in combination with LED 300 and LED 462, comprise the literacy instruction for prospective elementary education teachers.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 321 and LED 300. Corequisites: ELED majors: ELED 322, ELED 334 and ELED 343.

LED 355 Inclusive Early Literacy Development and Acquisition**Credits 3 Grading Basis** Graded

This course is designed to provide the foundation of literacy development for children (birth-age 6), as well as instruction to support and enhance the emergence and continued development of early literacy learning.

Prerequisites: IECE major or B.I.S. major with a concentration in IECE or ECED. Corequisites: For B.I.S. students with a concentration in ECED: ECED 444 and ECED 372. For IECE B.S. and B.I.S. with a concentration in IECE: IECE 331, IECE 322 and IECE 466.

LED 366 Early Literacy Development and Acquisition**Credits 3 Grading Basis** Graded

This course will provide preservice teachers with an understanding of the foundations of early literacy development, and instructional strategies and assessment techniques which support the acquisition of literacy.

Prerequisites: Teacher Ed Program 02 Flag.

LED 370 Writing Instruction, Assessment and Technology for Diverse Global Learners (PK-6)**Credits 3 Grading Basis** Graded

This course explores global perspectives on writing development and instruction for multilingual and culturally diverse learners in PK-6 classrooms. Students will examine instructional strategies, assessment practices and digital tools that promote equitable and inclusive writing growth. Emphasis is placed on valuing learners' linguistic and cultural identities while applying research-based approaches across diverse educational contexts.

Prerequisites: Declared ELED global concentration, ELED 321 and LED 300. Corequisites: ELED 322, ELED 343, ELED 374 and ELED 375.

LED 401 Problems in Literacy Education**Credits 3 Grading Basis** Graded

Considers current problems and issues in reading education as these problems and issues relate to the professional education of preservice teachers. Prerequisite: Permission from the coordinator of the program where the student is enrolled.

Prerequisites: Teacher Ed Program 02 Flag.

LED 420 Content Area Literacy, K-12**Credits 2 Grading Basis** Graded

This course is designed for preservice teachers and will provide an introduction to the foundations of reading and balanced literacy instruction for students in kindergarten through grade 12. Through reading and writing across content areas, preservice teachers will explore literacy engagement, diversity and special needs.

Prerequisites: Applied to Teacher Education (01 Flag).

LED 430 Development, Assessment and Instruction of Literacy: K-12**Credits 3 Grading Basis** Graded

This course is designed to provide preservice teachers with a foundation in language and literacy development, instructional strategies, and assessment techniques that support the acquisition and development of literacy across the curriculum in diverse classrooms grades K-12.

Prerequisites: SPED major or B.I.S. Majors with a SPED concentration; minimum grades of "C" or better in EXED 303 and EDUC 303.

Corequisites: EXED 484.

LED 435 Literacy Development and Instruction for English Language Learners**Credits 3 Grading Basis** Graded

Prepares for literacy instruction of English Language Learners beyond the emergent stage. Content includes assessment and instruction, particularly comprehension instruction for individuals, small groups, and whole class instruction. Students examine heterogeneous classroom structure and plan instruction for the diversity of abilities, personalities, cultures, languages, and all individual learners in every educational setting. A practicum accompanies this course.

Corequisites: TESL 383.

LED 436 Literacy Learning in the Elementary Grades**Credits 3 Grading Basis** Graded

This course will provide preservice teachers with an understanding of developmentally appropriate instructional strategies and assessment techniques to help all students in elementary grades become literate using reading, writing, listening and speaking in strategic and authentic ways.

Prerequisites: Grade of "C" or better in LED 366.

LED 440 Disciplinary Literacy in Secondary Education**Credits 3 Grading Basis** Graded

This course will provide preservice teachers in secondary education with an understanding of how to create productive contexts for literacy-based learning. Particular areas of emphasis include selecting reading materials, understanding literacy development and facilitating individual student engagement.

Prerequisites: SEED majors, SEED minors, and middle education majors: MSSE 370 and either MSSE 371 or MSSE 430.

LED 456 Inclusive Literacy Learning in the Primary Grades**Credits 3 Grading Basis** Graded

This course is the second of a two-course undergraduate sequence covering the development of the reading process, methods for teaching reading in elementary classrooms, as well as assessment and intervention for students at risk for or with identified disabilities. Participants will develop an understanding of the reading process and reading disabilities. Emphasis will be on the use of effective practices for primary, secondary and tertiary instruction for children across the lifespan.

Prerequisite: ECED or IECE, ECED: min grade of "C" ECED 372 and LED 355 & applied to teacher ed, for IECE, Min grade of "C" IECE 303,300,321; Corequisites: ECED: ECED 442 & 445, IECE BS: IECE 421, IECE 464, IECE 450

LED 462 Disciplinary Literacy for Diverse Classrooms PK-6**Credits 3 Grading Basis** Graded

This capstone literacy course for pre-service PK-6 teachers will build on the foundation of literacy developed in LED 300 and LED 350 to enable students to integrate reading and writing instruction across the content areas and learn how to differentiate literacy to meet the needs of all learners in a classroom.

Prerequisites: ELED or B.I.S./ELED major. ELED majors: ELED 450.

Corequisites: ELED 455.

LED 472 Literacy Assessment and Instruction in the Content Areas for the Middle Grades**Credits 3 Grading Basis** Graded

This course will introduce preservice teachers to the relationship between literacy assessment practices and instructional design for teaching reading and writing in content area classrooms. Using case study methodology, preservice teachers will explore individual students' literacy strengths, areas that need development, and specific instructional strategies.

Prerequisites: LED 212 and MIED 211.

LED 473 Disciplinary Literacy for Diverse Classrooms in PK-6 Global Contexts**Credits 3 Grading Basis** Graded

This capstone literacy course for pre-service PK-6 teachers will build on the foundation of literacy developed in LED 300 and LED 370 to enable students to integrate reading and writing instruction across the content areas and learn how to differentiate literacy to meet the needs of all learners in a classroom. In addition, the course emphasizes global education perspectives by encouraging future teachers to integrate diverse texts, cultural contexts and world issues into literacy instruction, preparing students to engage as responsible and informed global citizens.

Prerequisites: Declared ELED global concentration and ELED 450.

Corequisites: ELED 455 and ELED 475.

LED 490 Special Studies in Literacy**Credits 1-3 Grading Basis** Graded

Designed to give capable students, under faculty guidance, an opportunity to engage in the independent study of educational problems. Plan for the study must be approved by the faculty advisor and the coordinator of the program in which the student is enrolled.

Learning, Technology and Leadership Education (LTLE)

LTLE 332 Video Production**Credits 1 Grading Basis** Graded

A basic understanding of the principles of video equipment along with skills in the use, set up and operation of editing applications.

LTLE 375 Selected Topics in Media**Credits 1-3 Grading Basis** Graded

An in-depth study of a narrowly defined topic or practice in media. May be repeated for credit when course content changes.

LTLE 377 Augmented/Virtual Reality and 360 Media: Innovation in Immersive Technology for Education

Credits 3 Grading Basis Graded

This course is an applied exploration of immersive technology, focused on applications to training and education. Virtual, augmented, mixed reality, 360° media, 3D scanners and other emerging related technology will be covered. The course is intended for novices to the field of AR/VR and immersive media production. Students will develop unique applications, media and more through interdisciplinary team-based projects for real clients, focused on areas of immersive technology in education.

LTLE 470 Diversity and International Human Resource Development

Credits 3 Grading Basis Graded

The course prepares students for meeting the challenges presented by globalization and workforce diversity in HRD. Topics include the impact of increasing globalization and diversity on HRD and the workplace, cross-cultural communications, conflict resolution in diversity settings, global diversity management, ethics, and cross-culture leadership competencies. It examines and explores theories and techniques for dealing with institutional 'isms' (e.g., multiculturalism, sexism, ageism). Prerequisites: HRD 240 and HRD 245.

LTLE 490 Special Study in LTLE

Credits 1-3 Grading Basis Graded

Designed to provide in-depth and up-to-date exposure to the topics/issues pertinent to human resource development and allow students to explore topics of special interest in human resource development.

Linguistics (LING)

LING 311 Study of Language Structures

Credits 3 Grading Basis Graded

Contrastive linguistics introduces students of any language to the study of linguistics using English as the language of communication for all class-related activities. This course uses linguistics to talk about the language, i.e., use metalinguistic speech to describe what is different and what is similar between languages (e.g., English and Spanish). This course teaches how speakers use their second language, switch between the two languages and make sense of the way other people are speaking in order to avoid making grammatical mistakes, cultural misunderstandings and misinterpreting intentions that can get "lost in translation." Students will gain a deeper and fuller understanding of their second language and the way languages work in general.

LING 312 Bilingualism in the United States

Credits 3 Grading Basis Graded

This course introduces students of any language to the sociolinguistic aspects of bilingualism using examples of bilingual communities in the United States for study. It focuses on bilingualism as an individual phenomenon (i.e., how do languages develop and interact in the mind of a bilingual speaker) and as a social phenomenon (i.e., how do groups of bilinguals interact in a community and how does their language shape social structures and those social structures shape their bilingualism). While we will study the use of multiple languages in the US, this course uses English as the language of communication for all class-related activities.

LING 410 Introduction to Romance Linguistics

Credits 3 Grading Basis Graded

This course takes a historical approach to the study of Romance languages, the linguistic evolution from Latin to Romance, with a focus on three of the modern Romance languages, Spanish, Italian and French. The language of instruction is English. Students will study examples of sound change, morphological change, syntactical change and lexical development. The course will introduce linguistic geography, language contact, the relationship of language and culture, the role of dialects, the creation of the national standards, minority languages, and the expansion of Romance languages beyond Europe.

Prerequisites: SPAN 232, FR 232, ITAL 232 or LAT 232; or a pre-approved course in linguistics, upon approval of the instructor. Pre-approved courses: ANTH 305 or SCOM 305; ANTH 405; ENG 308, 310, 348, 417; PHIL 380; SCOM 245; SPAN 310, 311, 404, 448.

LING 430 Forensic Linguistics: Language and the Law

Credits 3 Grading Basis Graded

This course in forensic linguistics will provide students with a deeper understanding of contexts in which language and the law intersect, such as the analysis of police interrogations/interviews, linguistic evidence, police statements, legal statutes and interpreted mediated events. The course will also examine language use (oral and textual) with vulnerable populations (e.g., children, persons with cognitive and intellectual disabilities), non-native speakers of English, African American Vernacular English (AAVE) and the use of linguistic profiling.

Prerequisites: ANTH 305/SCOM 305, ANTH 405, ENG 308, ENG 310, ENG 348, ENG 417, PHIL 380, SCOM 245, SPAN 310, SPAN 311, SPAN 404, or SPAN 448. Courses that are not included on the list require pre-approval.

LING 470 World Language Teaching Methods

Credits 3 Grading Basis Graded

This course is designed for world language education majors and other students who may be interested in pursuing a teaching career or learning about world language teaching. The course focuses on current methods of teaching world languages (and EFL) and their rationales, including national and state standards, language acquisition research, and theories of second language learning. The course also focuses on teaching practices and materials that can be used later in the field. Course activities and assignments will allow students to apply the theoretical knowledge in analytical and creative ways.

Management (MGT)

MGT 305 Organizational Behavior and Human Resource Management

Credits 3 Grading Basis Graded

This course provides students with an overview of topics necessary for effective management of people, including psychological and sociological theories around how people behave in organizations, as well as the policies and practices that organizations can implement to leverage the skills and talents of their employees most effectively for organizational success.

Prerequisites: Non-COB major and sophomore standing.

MGT 310 Business Fundamentals for Entrepreneurs**Credits 3 Grading Basis Graded**

This course requires students to develop basic literacy in the language of business and will include foundational concepts related to forms of business ownership, management, marketing, operations and financial resources. It is designed for students from non-business majors who enroll in the entrepreneurship minor.

Prerequisites: Open only to non-business majors who have been accepted into the entrepreneurship minor.

MGT 320 Management of Innovation and Technology**Credits 3 Grading Basis Graded**

This course focuses on technological innovation as a primary source of achieving and sustaining competitive advantage for established firms as well as start-ups. Students explore the current challenges facing business leaders in their attempts to build firms that can consistently and successfully bring technological innovations to market. The course provides students with opportunities to integrate creativity, innovation and problem-solving processes. Innovation management includes topics such as technology strategy, fit between product strategy and business unit strategy, the psychology of discovery and invention, the entrepreneurial mindset, managing innovation processes, and teams and protecting innovation.

Prerequisites: For management majors: COB 300; for entrepreneurship minors: MGT 310 or COB 300; or departmental permission.

MGT 325 Project Management**Credits 3 Grading Basis Graded**

This course focuses on the knowledge and skills needed to select, plan, schedule, evaluate, control, and complete a variety of complex projects that help achieve organizational goals. Students learn foundational concepts and current project management frameworks. Applying project management tools and techniques to workplace situations using project management software is emphasized.

Prerequisites: COB 300 and Management majors; or departmental permission.

MGT 340 International Management**Credits 3 Grading Basis Graded**

A comparative analysis of management styles and organizational effectiveness across cultural boundaries and within other political, legal and economic environments.

Prerequisites: COB 300 and management or international business major.

MGT 357 Evidence-Based Decision Making**Credits 3 Grading Basis Graded**

This course provides students with the opportunity to use research and data analysis to explore managerial and organizational situations, and to develop and practice empirically based decision making skills. Students will learn about evidence based management, an approach to decision making based on the best obtainable evidence and unbiased organizational facts. The course will help students to develop the knowledge and skills needed to identify, access, interpret, and use factual information and empirical evidence to be more effective decision makers and managers.

Prerequisites: COB 300 and Management majors; or departmental permission.

MGT 365 Fundamentals of Human Resource Management**Credits 3 Grading Basis Graded**

A study of employer-employee relationships in business and industry including personnel policies and methods; selection, placement, training and promotion of employees; and recent trends in employment practices.

Prerequisites: COB 300 for management majors, or PPA 265 for PPA majors; or departmental permission. Corequisites: COB 300 for management majors, or PPA 265 for PPA majors.

MGT 370 Quality Management**Credits 3 Grading Basis Graded**

The application of statistics to the control of quality in planning, organizing, motivating and controlling quality. Application of models and distributions in statistical quality control, process capability, control charts, acceptance sampling and reliability. Emphasis is on the design, implementation and management of quality throughout the firm.

Prerequisites: COB 300 and junior standing.

MGT 375 Business Ethics and Social Responsibility**Credits 3 Grading Basis Graded**

This course explores the nature of moral values, moral judgments, and ethical decision and behaviors in modern business organizations. Alternative perspectives of right and wrong will be considered. A highly interactive course, students will analyze and discuss ethics-related current events, case studies, real-world scenarios, and common ethical dilemmas in order to improve skills in recognition, understanding, and decision-making related to business ethics and social responsibility at various organizational levels. This course is open to students in any COB major provided they meet the course prerequisites.

Prerequisites: COB 300 and management major, or departmental permission.

MGT 390 Organizational Leadership**Credits 3 Grading Basis Graded**

This course provides a foundation for understanding and applying leadership knowledge and skills at the individual, dyadic, team, and organizational levels. It enables students to better understand themselves and others, and utilize that knowledge to more effectively lead and manage in organizational settings. The course incorporates theory, cases, assessment tools, empirical evidence, and application-based exercises. Contemporary leadership theories and related frameworks involving values, attitudes, motivation, ethics, and group processes are used to foster leadership development.

Prerequisites: COB 300 and Management majors; or departmental permission.

MGT 405 Topics in Management for Non-majors**Credits 3 Grading Basis Graded**

This course is designed to allow exploration of areas of current topical or unique interest or to leverage special situations. Course content will vary. Consult your advisor for specific topics each semester when the course is offered. Course may be repeated for credit when topics vary.

Prerequisites: Cumulative 2.0 GPA in all courses taken at JMU, open only to non-business majors.

MGT 457 Management Consulting Fundamentals**Credits 3 Grading Basis** Graded

This course provides an overview of business analysis, the consulting industry and the analytical practices used in the consulting process. Students explore the business analysis process, learning to identify and communicate the need for organizational change and define solutions that maximize organizational value. Also, students examine the nature and purpose of consulting, explore careers related to consulting, learn consulting frameworks and methods and develop an understanding of the consultant-client relationship.

Prerequisites: MGT 357.

MGT 460 Employment Law**Credits 3 Grading Basis** Graded

This course provides students with an in-depth understanding of federal regulations, court decisions, and public labor policies that influence employment decisions, employee and labor relations, and employee safety and health. Includes historical evolution of federal legislation relating to employment, the labor movement, and the emergence of public sector bargaining. Explores the impact of regulations, court cases, and union contracts on general business operations.

Prerequisites: COB 300 and MGT 365.

MGT 461 Introduction to Supply Chain Management**Credits 3 Grading Basis** Graded

Introduction to Supply Chain Management provides the student with a foundational knowledge of the content and processes of supply chains, with an emphasis on the strategic nature of supply chain management decisions. This course offers a survey of critical supply chain topics such as design, sourcing, networks, risk management, transportation, revenue generation, and sustainability. Selected topic areas will be evaluated through the lens of geographic information systems (GIS) using ArcGIS.

Prerequisites: COB 300 (for Management majors), or COB 300 and admission to the Global Supply Chain Management minor (for all other COB Majors), or ISAT 211 and admission to the Global Supply Chain Management minor (for CISE majors).

MGT 467 Advanced Management Consulting**Credits 3 Grading Basis** Graded

This is an applied course where students will work in teams on consulting projects for businesses or organizations. Students apply knowledge and skills developed in MGT 357 and MGT 457. Students engage in a comprehensive consulting experience designed to build competence and confidence in analysis, problem-solving and managing client relations. Grades will be determined largely by the quality of client deliverables.

Prerequisites: MGT 457.

MGT 470 Acquiring and Developing Human Capital**Credits 3 Grading Basis** Graded

This course provides an in-depth study and application of staffing concepts and practices, including how to identify, place, and develop the best candidates for a given position. Topics include workforce planning, recruitment, selection, training and development of employees, and succession planning.

Prerequisites: MGT 365.

MGT 471 Entrepreneurship**Credits 3 Grading Basis** Graded

This course provides an examination of entrepreneurship and its impact and significance in the global economy. Students will learn theory and then apply it through experiential exercises, case studies and projects. This course explores topics such as the entrepreneurial process, idea generation methodology, sources of funding, growth, succession planning and exit strategies. Students engage in activities such as feasibility analysis, selection of appropriate entry and growth strategies, and firm valuation techniques.

Prerequisites: For management majors: COB 300; for entrepreneurship minors: MGT 310 or COB 300; for all others: Departmental permission.

MGT 472 New Venture Creation**Credits 3 Grading Basis** Graded

This experiential course explores the context and entrepreneurial process involved in creating new ventures. Students work in multidisciplinary teams to recognize and evaluate opportunities, develop feasible business models and demonstrate high-performance team practices. Working with experienced entrepreneurs, teams develop and apply financing strategies for the prelaunch and launch of new ventures.

Prerequisites: MGT 320 and MGT 471; or departmental permission.

MGT 475 Managing and Retaining Human Capital**Credits 3 Grading Basis** Graded

This course provides an in-depth study of two major aspects of HR policy and practice: performance management and compensation systems. Students participate in applied exercises and projects that focus on best practices in performance management and performance appraisal systems. The design of compensation and benefits systems is studied as a means to effectively retain workers.

Prerequisites: MGT 365 and Management major.

MGT 480 Organizational Structure and Change**Credits 3 Grading Basis** Graded

Examines the theory and research underlying the design of complex organizations. Takes a macro approach to the study of organizations, placing particular emphasis upon the interaction between an organization and its environment and the impact that the environment has on organizational design, structure and processes.

Prerequisites: COB 300 and management major, or departmental permission.

MGT 481 Negotiation and Dispute Resolution**Credits 3 Grading Basis** Graded

The purpose of this course is to develop an understanding and effective improvement of participants' skills in the areas of both business and interpersonal negotiations. Through case-based exercises, attention will be given to various strategies for negotiation including distributive, integrative, intra-organizational, and multi-lateral bargaining. Additional applications include how negotiation skills can serve as tools for conflict resolution in the workplace and broader life experiences.

Prerequisites: COB 300 and senior standing.

MGT 490 Special Studies in Management**Credits 1-3 Grading Basis** Graded

Designed to give capable students in management an opportunity to complete independent study under faculty supervision.

Prerequisites: Management major and senior standing (90 hours); recommendation of the instructor and written permission of the director prior to enrollment.

MGT 494 Management Internship**Credits 3 Grading Basis** Graded

A course providing an opportunity to work in and with local industry to gain insight into the real side of modern management.

Prerequisites: Management major and permission of the internship coordinator prior to enrollment.

MGT 498 Special Topics in Management**Credits 1-3 Grading Basis** Graded

This course is designed to allow explorations of areas of current topical concern or to exploit special situations. Course content will vary. For current course content consult your advisor.

Prerequisites: COB 300 and management major; or departmental permission.

MGT 499A Honors**Credits 1 Grading Basis** Graded

Year course. See catalog description entitled "Graduation with Distinction" and "Graduation with Honors."

MGT 499B Senior Honors Project**Credits 3 Grading Basis** Graded

Year course. See catalog description entitled "Graduation with Distinction" and "Graduation with Honors."

MGT 499C Honors Thesis**Credits 2 Grading Basis** Graded

Year course. See catalog description entitled "Graduation with Distinction" and "Graduation with Honors."

Marketing (MKTG)

MKTG 110 Discovering Marketing as a Profession**Credits 1 Grading Basis** Graded

This course introduces students to the diverse field of marketing. Students will explore select marketing concepts, from a broad perspective, in order create a plan for career development in the marketing field. This course is suitable for any student curious about marketing as a profession.

MKTG 331 Creative Marketing Design: Digital Vector Techniques**Credits 1 Grading Basis** Graded

Using industry-standard tools, students will learn to apply digital vector techniques to effectively enhance marketing strategies. They will integrate product details and calls to action into marketing materials for improved customer engagement and conversion rates. Students will also create a diverse portfolio of assets like logos, infographics and advertisements, relevant to both traditional and digital marketing, giving students a well-rounded skillset in creative marketing design.

Prerequisites: Marketing Majors only.

MKTG 332 Creative Marketing Design: Visual Editing and Document Styling**Credits 2 Grading Basis** Graded

Students will acquire vital skills in visual editing and document styling by using industry-standard tools to create impactful branding and marketing materials. They will build a versatile portfolio through hands-on projects that cater to diverse audiences at different buyer's journey stages-encompassing awareness, consideration, decision-making and loyalty-building. The course empowers students to apply visual marketing and design principles to transform their creative vision into real-world marketing applications.

Prerequisites: Marketing Majors only.

MKTG 380 Principles of Marketing**Credits 3 Grading Basis** Graded

As a discipline that focuses on creating, communicating and delivering value for the consumer, marketing is essential for the success of both for-profit and nonprofit organizations and individual ideas. Students learn the key elements of marketing such as developing and distributing products, branding, pricing, and designing promotions. Students then use their understanding of the key concepts and tools to solve real-world marketing problems.

Prerequisites: Junior standing.

MKTG 384 Integrated Marketing Communications**Credits 3 Grading Basis** Graded

Integrated marketing communications includes advertising, sales promotions, packaging, public relations, publicity, personal selling, direct marketing and event sponsorship. Students will be involved in creating, planning, implementing and evaluating client-oriented projects by developing an integrated marketing communication campaign.

Prerequisites: COB 300 or permission of the academic unit head. Open to Marketing and International Business majors only.

MKTG 385 Consumer Behavior**Credits 3 Grading Basis** Graded

This course deals with the behavioral science concepts of individual and group behavior of consumers. The course will stress the application of buyer behavior research to marketing management.

Prerequisites: COB 300 or permission of the academic unit head. Open to Marketing and International Business majors only.

MKTG 386 Services Marketing**Credits 3 Grading Basis** Graded

Application of marketing principles to the services sector. The course focuses on review of customer demand for and assessment of services; the employee/customer interface; services operation management; review of the services marketing mix; and development of marketing plans for service organizations.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 388 Omnichannel Retailing**Credits 3 Grading Basis** Graded

This course examines retailing from an omnichannel approach that includes both online and offline strategies and tools. Strategies to create an integrated, cohesive approach that allows consumers to move between channels seamlessly will be evaluated.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 405 Survey Research**Credits 3 Grading Basis** Graded

This course covers techniques and principles, skills and activities that are required to conduct an effective survey project. The course will cover survey planning, survey methods, sampling, survey instrument design, data collection and analysis, and survey reporting. Survey findings are linked to future marketing decision making.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 410 Social Media Marketing**Credits 3 Grading Basis** Graded

The class covers the foundations of social media including the predominantly used social media platforms, when and how to use each effectively, and a glimpse into the social psychology behind interactions within and across social media platforms.

Prerequisites: MKTG 384.

MKTG 430 Professional Selling**Credits 3 Grading Basis Graded**

This course provides an understanding of many aspects of personal selling including preparing for selling, selling techniques and the role of selling in our society.

Prerequisites: COB 300 or permission of the academic unit head. Open to Marketing and International Business majors only.

MKTG 450 Business-to-Business (B2B) Marketing**Credits 3 Grading Basis Graded**

This course analyzes the policies and procedures in marketing to business buyers. The course provides emphasis on special problems connected with the segmentation and target marketing, purchase, distribution, promotion and development of business-to-business goods and services.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 455 Sales Leadership**Credits 3 Grading Basis Graded**

Sales leadership guides students through the various leadership styles and the adoption of the appropriate leadership style for various contexts. The use of sales management skills, coaching and mentoring in a professional sales environment are covered. The course includes experiential use of these skills by providing the opportunity to lead a team of students enrolled in MKTG 430.

Prerequisites: COB 300 and MKTG 430, or permission of the academic unit head.

MKTG 460 Global Marketing**Credits 3 Grading Basis Graded**

Examines marketing in international environments, including foreign entry, local marketing in individual countries and global or standardized marketing across many countries. Emphasis is placed on cultural, economic and strategic variables in deciding how to enter and compete in various markets.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 462 International Logistics**Credits 3 Grading Basis Graded**

The goal of the course is to tie the international logistics concepts and tools to real world situations, preparing students for dynamic decision-making and problem solving. For marketing majors, this elective would complement such courses as global marketing, business-to-business marketing, retail management and marketing analytics and provide students with an understanding of one of the key elements of marketing: place/physical distribution, from an international perspective.

Prerequisites: COB 300 (for Marketing majors), or COB 300 and admission to the Global Supply Chain Management minor (for all other COB majors), or ISAT 211 and admission to the Global Supply Chain Management minor (for CISE majors).

MKTG 465 Customer Relationship Management**Credits 3 Grading Basis Graded**

This course focuses on customer relationship management (CRM), which views customers as assets of the firm and the goal is to create value for both customers and businesses. Students will gain experience using analytics and technology to make data-driven decisions that improve sales and marketing strategies.

Prerequisites: MKTG 430.

MKTG 466 Advanced Professional Selling**Credits 3 Grading Basis Graded**

The purpose of this class is to build on the selling skills first learned in MKTG 430. It will focus on enhancing existing skills and learning new ones in order to even better prepare students for a successful career in sales. In addition, students may be selected to compete in regional or national sales competitions during the semester.

Prerequisites: MKTG 430.

MKTG 470 Strategic Internet Marketing**Credits 3 Grading Basis Graded**

Studies the culture and demographics of the Internet and examines online business strategies. Students will learn the hardware and software tools necessary for Internet commerce, identify appropriate target segments, develop product opportunities, price structures and distribution channels over the Internet and execute marketing strategy in computer-mediated environments.

Prerequisites: COB 300 and MKTG 384 or permission of the academic unit head.

MKTG 477 Digital Marketing Practicum**Credits 3 Grading Basis Graded**

With an applied focus, this course introduces students to some of the most important and fastest growing sectors in digital marketing. Students apply marketing theories in a uniquely applied manner as they become active learners involved in online marketing campaigns, facing real pressures similar to those in the professional workplace (i.e., account management, client relationships, financial constraints, market competition, time limitations, technology, etc.). Student teams will plan and execute paid search, display advertising, paid social or other campaign types for actual clients. Throughout their campaigns, students continually make decisions based on real world data and market conditions.

Prerequisites: MKTG 384 or permission of the academic unit head.

MKTG 480 Product Development and Management**Credits 3 Grading Basis Graded**

The process of developing new products will be developed and explored. The marketing tasks which are unique to this operation will be investigated. An understanding of the marketing management of products throughout their life cycles will complete the course.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 482 Marketing Analytics**Credits 3 Grading Basis Graded**

This course focuses on the use of information technology and marketing metrics to increase marketing productivity. Students learn how to evaluate marketing strategies and performance using database queries and statistical analysis. Information technologies are applied in market segmentation and target marketing, lifetime value analysis and RFM (recency, frequency and monetary value) analysis.

Prerequisites: COB 300 or permission of the academic unit head.

MKTG 485 Marketing Management**Credits 3 Grading Basis Graded**

Case studies are used to develop analytical and decision-making skills. Knowledge gained from previous course work is applied to actual circumstances faced by marketing managers in private, public, profit and not-for-profit organizations. Extensive preparation of case materials outside of class provides the basis for case presentations and discussion of case situations in class.

Prerequisites: COB 300, MKTG 384, MKTG 385 and senior standing required.

MKTG 490 Special Studies in Marketing**Credits** 1-3 **Grading Basis** Graded

Designed to give capable students in marketing an opportunity to complete independent study under faculty supervision.

Prerequisites: GPA of 2.8, instructor recommendation and academic unit head approval prior to registration.

MKTG 494 Marketing Internship**Credits** 3-6 **Grading Basis** Graded

A course providing an opportunity to work in and deal with local industry to gain insight into the realities of modern business.

Prerequisites: COB 300, minimum cumulative GPA of 2.80, senior standing, recommendation of the internship coordinator and approval of the academic unit head prior to registration.

MKTG 498 Special Topics in Marketing**Credits** 1-3 **Grading Basis** Graded

This course is designed to allow explorations of areas of current topical concern or to exploit special situations. Course content will vary. For current course content consult your advisor.

MKTG 499A Honors**Credits** 1 **Grading Basis** Graded**MKTG 499B** Senior Honors Project**Credits** 2 **Grading Basis** Graded**MKTG 499C** Senior Honors Project**Credits** 3 **Grading Basis** Graded

Materials Science (MATS)

MATS 375 An Introduction to Materials Science**Cross-listed with:** CHEM 375, PHYS 375**Credits** 3 **Grading Basis** Graded

An introduction to materials science with emphasis on general properties of materials. Topics will include crystal structure, extended and point defects, and mechanical, electrical, thermal and magnetic properties of metals, ceramics, electronic materials, composites and organic materials. Prerequisites: CHEM 131 and PHYS 150 or PHYS 250 or ISAT 212 or permission of the instructor.

MATS 498 Undergraduate Materials Science Research**Credits** 1-3 **Grading Basis** Graded

Research in a selected area of materials science arranged with and approved by a faculty research advisor. Study proposal must be approved by research advisor and director of Center for Materials Science prior to registration.

Math Education (MAED)

MAED 430 Teaching Mathematics in Grades K-12**Credits** 3 **Grading Basis** Graded

This course provides students with knowledge, skills, and understanding to design and implement effective mathematics instruction (K-12). Focus is on the content, methods, and materials developmentally appropriate for mathematics instruction for a range of students, including diverse and special needs children. The course is based on the Principles and Standards for School Mathematics from NCTM.

Prerequisites: For B.S. SPED majors and B.I.S. majors with a concentration in SPED; minimum grades of "C" or better in EXED 303 and EDUC 303. Corequisites: EXED 474, EXED 484 and LED 430.

Mathematics (MATH)

MATH 103 The Nature of Mathematics [C3QR]**Credits** 3 **Grading Basis** Graded

Topics such as geometry, computing, algebra, number theory, history of mathematics, logic, probability, statistics, modeling and problem solving intended to give students insight into what mathematics is, what it attempts to accomplish and how mathematicians think. May be used for general education credit.

Prerequisites: Math Majors are not eligible to enroll in the course.

MATH 105 Statistical Literacy [C3QR]**Credits** 3 **Grading Basis** Graded

This course covers the interpretation and evaluation of numerical information in context. Students learn how to identify strengths and limitations of information sources and methods; how uncertainty affects data collection and analysis; and the role of hypothesis testing and confidence intervals in research. Not open to majors in mathematics or statistics. May be used for general education credit. Not open to students who have previously earned credit in courses requiring MATH 105 competency except with consent of the Mathematics and Statistics department head.

Prerequisites: This course is not open to students who have completed MATH 220 with a grade of C- or higher, or to majors in mathematics or statistics.

MATH 107 Fundamentals of Mathematics I [C3QR]**Credits** 3 **Grading Basis** Graded

MATH 107 and MATH 108, along with MATH 207, form a sequence that covers the topics of sets, logic, numeration systems, development of real numbers, number operations, number theory, geometry, measurement, algebra, functions, probability and data analysis. Sequence is required for early childhood, elementary, or middle school teacher licensure. May be used for general education credit.

MATH 108 Fundamentals of Mathematics II**Credits** 3 **Grading Basis** Graded

MATH 107 and MATH 108, along with MATH 207, form a sequence that covers the topics of sets, logic, numeration systems, development of real numbers, number operations, number theory, geometry, measurement, algebra, functions, probability and data analysis. Sequence is required for early childhood, elementary, or middle school teacher licensure. Prerequisites: MATH 107 with a grade of "C-" or better and a passing score on Basic Skills Test.

MATH 110 Mathematics for Elementary and Middle School Teachers I [C3QR]**Credits** 4 **Grading Basis** Graded

MATH 110 and MATH 210 form a sequence that is designed for future elementary and middle school teachers to develop mathematical knowledge for teaching by deeply examining concepts in mathematics that include number sense, number systems (whole numbers, integers, rational numbers and real numbers), meanings and properties of operations, ratio and proportion, functions and algebra, number theory, geometry, measurement, probability, and statistics. Problem-solving, reasoning, representations, connections and communication form the foundation through which learning will take place. Priority registration will be given to Inclusive Early Childhood Education, Elementary Education, Middle Grades Education, and Special Education majors. Other majors may enroll during open enrollment. MATH 110 may be used for general education credit in quantitative reasoning.

MATH 111 Mathematics Seminar for Elementary and Middle School Teachers**Credits 1 Grading Basis** Graded

This course is designed for future elementary and middle school teachers to develop mathematical knowledge for teaching by deeply examining concepts in mathematics including: number sense, number systems (whole numbers, integers, rational numbers and real numbers), and meaning and properties of operations. These mathematical concepts will be explored through the lens of the Standards of Mathematical Practice. Problem solving, reasoning, representation, connections and communication form the foundation through which learning will take place.

Prerequisites: Inclusive Early Childhood Education, Elementary Education, Middle Grades Education or Special Education majors or Mathematics for Middle Grades Education minor; and minimum grade of "C-" in MATH 110.

MATH 135 Elementary Functions**Credits 4 Grading Basis** Graded

Algebraic, exponential, logarithmic and trigonometric functions; matrices and matrix solutions to systems of linear equations; vectors. Not open to students who have previously earned credit in MATH 155, MATH 156, MATH 205 or MATH 235, except with the consent of the department head.

You are not eligible to enroll into this course if you have received credit for MATH 155, MATH 156, MATH 205, or MATH 235.

MATH 155 College Algebra**Credits 3 Grading Basis** Graded

Polynomial, rational, exponential and logarithmic functions and applications, systems of equations and inequalities, sequences. An appropriate ALEKS placement score is required in order to enroll.

Prerequisites: Demonstrated Proficiency in Algebra through the Math Placement Test. A test is required to determine placement in MATH 155. Not eligible if you have earned credit for MATH 135, MATH 156, MATH 205, MATH 231, MATH 232 or MATH 235.

MATH 156 College Algebra**Credits 3 Grading Basis** Graded

Covers same topics as MATH 155. MATH 156 will meet five times a week for students requiring more instructional time.

Prerequisites: Demonstrated Proficiency in Algebra through the Math Placement Test. Not eligible if you have earned credit for MATH 135, MATH 155, MATH 205, MATH 231, MATH 232, or MATH 235.

MATH 167 Topics in Mathematics**Credits 1-3 Grading Basis** Graded

Topics or projects in mathematics which are of interest to the lower-division student. May be repeated for credit when course content changes. Topics or projects selected may dictate prerequisites. Students should consult the instructor prior to enrolling for this course.

MATH 199 Algebra /Precalculus Gateway**Credits 1 Grading Basis** Graded

Review of fundamental mathematics required to be successful in MATH 205 or MATH 231, including graphs of functions, factoring, simplifying, solving equations and inequalities, and exponential/logarithmic/trigonometric functions. Self-paced study with required proctored tests. Students needing more instruction should register for MATH 155 instead.

Corequisites: MATH 205 or MATH 231 with appropriate calculus placement score.

MATH 205 Applied Calculus [C3QR]**Credits 3 Grading Basis** Graded

This course is a one semester overview of calculus for students who do not intend further coursework in calculus. Topics include: limits; derivatives; application of derivatives in business, the social sciences and the natural sciences; integrals; and the Fundamental Theorem of Calculus. Throughout, the focus is on algebraic, exponential and logarithmic functions. May be used for general education credit. Prerequisites: MATH 135 or MATH 155 or MATH 156 or sufficient score on the mathematics placement exam. Not open to mathematics or physics majors or to students who have already earned credit in MATH 232 or MATH 235.

MATH 206 Introductory Calculus II**Credits 3 Grading Basis** Graded

Topics from integral calculus with applications to the social, behavioral or life sciences, and business or management.

Prerequisites: MATH 205. Not open to computer science, mathematics or physics majors, or to students who have already earned credit in MATH 236. Not recommended for chemistry majors.

MATH 207 Fundamentals of Mathematics III**Credits 3 Grading Basis** Graded

A continuation of topics listed in the MATH 107-MATH 108 description will be covered. The MATH 107-MATH 108-MATH 207 sequence fulfills the requirements for licensure of prospective early childhood, elementary, or middle school teachers.

IDLS major and MATH 108 are the prerequisites.

MATH 210 Mathematics for Elementary and Middle School Teachers II**Credits 4 Grading Basis** Graded

MATH 110 and MATH 210 form a sequence that is designed for future elementary and middle school teachers to develop mathematical knowledge for teaching by deeply examining concepts in mathematics that include number sense, number systems (whole numbers, integers, rational numbers and real numbers), meanings and properties of operations, ratio and proportion, functions and algebra, number theory, geometry, measurement, probability, and statistics. Problem solving, reasoning, representations, connections and communication form the foundation through which learning will take place. The MATH 110/ MATH 210 sequence is restricted to those seeking teaching licensure in the PreK-8 range.

Prerequisites: Inclusive Early Childhood Education, Elementary Education, Middle Grades Education or Special Education majors only. Must earn "C-" or better in MATH 110.

MATH 220 Elementary Statistics [C3QR]**Credits 3 Grading Basis** Graded

Descriptive statistics, frequency distributions, sampling, estimation and testing of hypotheses, regression, correlation and an introduction to statistical analysis using computers. May be used for general education credit.

Prerequisites: MATH 105 with a grade of "C-" or better or sufficient score on the Mathematics Placement Exam.

MATH 227 Discrete Structures I**Cross-listed with:** CS 227**Credits 3 Grading Basis** Graded

An introduction to discrete mathematical structures including functions, relations, sets, logic, matrices, elementary number theory, proof techniques, basics of counting, graphic theory, discrete probability, digital logic, finite state machines, integer and floating point representations.

Prerequisites: A minimum grade of "B-" in CS 149.

MATH 229 Introduction to Applied Statistics Using R [C3QR]**Credits 3 Grading Basis** Graded

An introduction to applied statistics that is more in-depth than MATH 220 and is encouraged for students majoring or minoring in STEM fields. Descriptive statistics, data collection, frequency distributions, sampling distributions and the Central Limit Theorem, estimation, confidence intervals, testing of hypotheses, simple linear regression, correlation, and chi-square test of independence are covered. Emphasis is on the analysis and interpretation of real data in context. Uses the R programming language. May be used for general education credit.

Prerequisites: Sufficient statistics and calculus placement scores. The calculus score requirement may be waived for students with "C-" or better in MATH 231. Not open to students with credit in MATH 220, MATH 220H, MATH 318, MATH 329 or equivalent.

MATH 231 Calculus with Functions I [C3QR]**Credits 3 Grading Basis** Graded

MATH 231 and MATH 232 form a two-semester sequence that combines a first university course in calculus with review of functions. The sequence offers rigorous coverage of calculus that can serve as a foundation for later courses typical for science, engineering or mathematics majors, while reviewing the properties of algebraic, trigonometric, exponential and logarithmic functions necessary to support that coverage. Topics in MATH 231 include: properties of functions, graphing, properties of polynomials, limits, continuity, differentiation and applications of the derivative. Each topic is treated in the context of algebraic functions. This course may be used for general education credit. Not open to students who have already earned credit in MATH 234 or MATH 235.

Prerequisites: MATH 155, MATH 156 or sufficient score on the Mathematics Placement Exam.

MATH 232 Calculus with Functions II**Credits 3 Grading Basis** Graded

MATH 231 and MATH 232 form a two-semester sequence that combines a first university course in calculus with review of functions. It offers rigorous coverage of calculus that can serve as a foundation for later courses typical for science, engineering or mathematics majors, while reviewing the properties of algebraic, trigonometric, exponential and logarithmic functions necessary to support that coverage. Topics in MATH 232 include: properties and differentiation of trigonometric, exponential and logarithmic functions; L'Hôpital's Rule; Riemann sums; integrals; and the Fundamental Theorem of Calculus.

Prerequisites: MATH 231 with a grade of "C-" or better. Not open to students with credit in MATH 234 or MATH 235.

MATH 233 Calculus with Modeling I [C3QR]**Credits 3 Grading Basis** Graded

MATH 233 and MATH 234 form a sequence that combines first-semester calculus with pre-calculus for students with skills not strong enough for MATH 235. Calculus is motivated by static and dynamic models in the context of scientific applications. MATH 233 includes limits, continuity, summation, and the definite integral as accumulation of change and as the limit of Riemann sums. May be used for general education credit.

Prerequisites: MATH 155 or MATH 156 or Appropriate Placement Score. Not open to students who have already earned credit for MATH 232 or MATH 235.

MATH 234 Calculus with Modeling II**Credits 3 Grading Basis** Graded

MATH 233 and MATH 234 form a sequence that combines first-semester calculus with pre-calculus for students with skills not strong enough for MATH 235. Calculus is motivated by static and dynamic models in the context of scientific applications. MATH 234 includes differentiation and antidifferentiation, including trigonometric functions, as well as the fundamental theorem of calculus.

Prerequisites: MATH 233 with a grade of "C-" or better. Not open to students who have already earned credit for MATH 235.

MATH 235 Calculus I [C3QR]**Credits 4 Grading Basis** Graded

Calculus I is a first university course in calculus. It offers rigorous coverage that can serve as a foundation for later courses typical for science, engineering or mathematics majors. Topics include: limits, continuity, derivatives, applications of derivatives, Riemann sums, integrals and the Fundamental Theorem of Calculus. Each topic is treated in the context of algebraic, trigonometric, exponential and logarithmic functions. This course may be used for general education credit.

Prerequisites: Sufficient score on the Mathematics Placement Exam. MATH 235 is not open to students who have already earned credit in MATH 232 or MATH 234.

MATH 236 Calculus II**Credits 4 Grading Basis** Graded

Calculus II continues the development of differential and integral calculus. It offers rigorous coverage that can serve as a foundation for later courses typical for science, engineering or mathematics majors. Topics include: applications of the definite integral; integration techniques; improper integrals; sequences and series; and Taylor series. Prerequisites: MATH 232 or MATH 234 or MATH 235 with a grade of "C-" or better.

MATH 237 Calculus III**Credits 4 Grading Basis** Graded

Vectors. Multivariate calculus.

Prerequisites: MATH 236 with a minimum grade of "C-."

MATH 238 Linear Algebra with Differential Equations**Credits 4 Grading Basis** Graded

Matrices; determinants; vector spaces; linear transformations; eigenvalues and eigenvectors; separable, exact and linear differential equations; and systems of linear differential equations.

Prerequisites: MATH 236. Not open to students with credit in MATH 300 or MATH 336 without departmental permission.

MATH 245 Discrete Mathematics**Credits 3 Grading Basis** Graded

Logic, set theory, relations and functions, mathematical induction and equivalent forms, recurrence relations, counting techniques.

Prerequisites: MATH 236. Corequisites: MATH 236.

MATH 248 Computers and Numerical Algorithms**Credits 4 Grading Basis** Graded

Programming in a high-level computer language. Applications of numerical algorithms to problems basic to areas such as mathematics, the sciences, and economics and finance. This course is not open to students who have previously earned credit in MATH/CS 448.

Prerequisites: MATH 236. Corequisites: MATH 236 and permission of the instructor. This course is not open to students who have previously earned credit in MATH 448/CS 448.

MATH 250 Programming for Data Science**Credits 3 Grading Basis** Graded

This course is an introduction to programming for data science using a modern programming language.

MATH 251 Database Queries**Credits 3 Grading Basis** Graded

This course provides an introduction to relational database queries. Students will learn about organizing, cleaning, summarizing and transforming data in preparation for exploration, visualization and analysis.

MATH 268 Data Analysis and Visualization**Credits 3 Grading Basis** Graded

This course is intended to give students hands-on experience on the basics of the data analysis techniques using R. It will focus on data management, data visualization, data exploration using descriptive data summaries, the basics of model building and the basics of inference. Topics include data management (creating data frames, splitting data, efficient data search); exploratory data analysis; data visualization; major distributions (univariate and multivariate); numerical summaries and outlier detections; and the basics of dimension reduction using principle component analysis and exploratory factor analysis.

Prerequisites: MATH 220 or MATH 229 or (both MATH 318 and permission of the instructor).

MATH 285 Data Analysis**Credits 4 Grading Basis** Graded

Topics include experimental and survey design, distributions, variation, chance, sampling variation, computer simulation, bootstrapping, estimation and hypothesis testing using real data generated from classroom experiments and large databases.

Prerequisites: MATH 206 or, MATH 236 or permission of the instructor. Not open to students who have already earned credit in MATH 220, MATH 229, MATH 318 or MATH 329.

MATH 286 Independent Study**Credits 1-3 Grading Basis** Graded

Independent study in mathematics under faculty supervision.

MATH 297 Undergraduate Research**Credits 1-4 Grading Basis** Graded

Students pursue research in a selected area of mathematics and/or statistics. Student must make arrangements with a supervising instructor prior to registration. Course may be repeated.

MATH 300 Linear Algebra**Credits 3 Grading Basis** Graded

Students will explore vector spaces, linear transformations, matrices, determinants, systems of linear equations, and eigenvalues and eigenvectors.

Prerequisites: MATH 232 or MATH 235; not open for credit to students with credit in MATH 238.

MATH 304 Principles of Algebra**Credits 3 Grading Basis** Graded

This course is designed for future elementary and middle school teachers to develop mathematical knowledge for teaching by deeply examining concepts in mathematics including: expressions, equations and inequalities, functions, systems of equations and inequalities, and some calculus connections. Problem solving, reasoning, representation, connections and communication form the foundation through which learning will take place.

Prerequisites: A minimum grade of "C-" or better in MATH 210.

MATH 305 Principles of Geometry**Credits 3 Grading Basis** Graded

This course is designed for future elementary and middle school teachers to develop mathematical knowledge for teaching by deeply examining concepts in mathematics including: strategies for proving geometric arguments, properties of and relationships among geometric figures (particularly right triangles leading to basic trigonometric functions), geometric transformations, and geometric constructions. Problem solving, reasoning, representation, connections and communication form the foundation through which learning will take place.

Prerequisites: A minimum grade of "C-" or better in MATH 210.

MATH 306 Principles of Analysis**Credits 3 Grading Basis** Graded

This course covers sequences, discrete calculus and difference equations, derivatives and integrals, and concepts of differential equations and applications.

Prerequisites: A minimum grade of "C-" or better in MATH 210.

MATH 307 Principles of Probability and Statistics**Credits 3 Grading Basis** Graded

This course is designed for future elementary and middle school teachers to develop mathematical knowledge for teaching by deeply examining concepts in mathematics including: experimental and theoretical probability, graphical representations, measures of center, measures of spread, variation, permutations and combinations. Problem solving, reasoning, representation, connections and communication form the foundation through which learning will take place.

Prerequisites: A minimum grade of "C-" or better in MATH 210.

MATH 309 SAS Programming and Data Management**Credits 3 Grading Basis** Graded

Use of statistical software to manage, process and analyze data. Writing of statistical programs to perform simulation experiments.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 310 Elementary Number Theory**Credits 3 Grading Basis** Graded

Properties of integers and prime numbers, divisibility, congruence, residues and selected topics.

Prerequisites: MATH 245 or CS 327 or permission of the instructor.

MATH 315 Real Number System**Credits 3 Grading Basis** Graded

A development of the real number system through a systematic approach to the natural numbers, integers, rationals and irrationals.

Prerequisites: MATH 245 or CS 327 or permission of the instructor.

MATH 318 Introduction to Probability and Statistics**Credits 4 Grading Basis** Graded

This class covers descriptive statistics, counting, probability axioms, discrete and continuous univariate random variables, expected values of random variables and sums of independent random variables, and sampling distributions. It also covers the Central Limit Theorem, single and two sample inference for proportions and means, chi-square test of independence, simple linear regression and correlation.

Prerequisites: MATH 236. Not open to students with credit in MATH 229 or MATH 329.

MATH 321 Analysis of Variance and Experimental Design**Credits 3 Grading Basis** Graded

Introduction to basic concepts in statistics with applications of statistical techniques including estimation, test of hypothesis, analysis of variance and topics in experimental design.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 322 Applied Linear Regression**Credits 3 Grading Basis** Graded

Introduction to basic concepts and methods in regression analysis and the application of these models to real-life situations.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 324 Applied Nonparametric Statistics**Credits 3 Grading Basis** Graded

Methods of analyzing data from non-normal populations including binomial tests, contingency tables, use of ranks, Kolmogorov-Smirnov type statistics and selected topics.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 325 Survey Sampling Methods**Credits 3 Grading Basis** Graded

Theory and practice of sampling including stratified random samples, discussion of simple random samples, cluster sampling, estimating sample size, ratio estimates, subsampling, two-state sampling and analysis of sampling error.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 326 Statistical Quality Control**Credits 3 Grading Basis** Graded

Uses and concepts of probability and sampling procedures. Acceptance sampling by attributes and variables, Shewhart concepts of process control, control chart process capability studies, reliability and life testing. Design of sampling plans.

Prerequisites: MATH 318.

MATH 327 Categorical Data Analysis**Credits 3 Grading Basis** Graded

Exact inference for population proportions, comparison of population proportions for independent and dependent samples, two- and three-way contingency tables, Chi-square tests of independence and homogeneity, Chi-square goodness-of-fit tests and Poisson and logistic regression.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 328 Time Series Analysis**Cross-listed with:** FIN 328**Credits 3 Grading Basis** Graded

Regression and exponential smoothing methods for forecasting nonseasonal and seasonal time series, stochastic processes, Box-Jenkins' autoregressive and moving average models.

Prerequisites: MATH 238 or MATH 300; and MATH 318 or MATH 329.

MATH 329 Introduction to Probability and Theoretical Statistics**Credits 3 Grading Basis** Graded

This course covers counting, probability axioms, discrete and continuous univariate random variables, expected values of random variables and sums of independent random variables. It also discusses theoretical underpinnings of sampling distributions and the Central Limit Theorem, computer simulation, two-sample estimation and inference for proportions and means, and estimation and inference including verification of assumptions in simple linear regression. Uses the R programming language.

Prerequisites: MATH 236 and "C-" or better in MATH 229. MATH 220 with a grade of "B-" or better may be substituted for MATH 229 by permission if student has suitable experience with R. Not open to students with credit in MATH 318.

MATH 336 Elementary Differential Equations**Credits 3 Grading Basis** Graded

Development of techniques for obtaining, analyzing and graphing solutions to differential equations, with emphasis on first and second order equations.

Prerequisites: MATH 236; Not open for credit to students with credit in MATH 238.

MATH 337 Methods of Applied Calculus**Credits 3 Grading Basis** Graded

Laplace transforms, power series and their application to differential equations. Vector differential and integral calculus; parametric curves; coordinate systems; line, surface and volume integrals; and gradient, divergence, and curl including the theorems of Green, Stokes and Gauss.

Prerequisites: MATH 237; and MATH 238 or MATH 336.

MATH 340 Operations Research I**Credits 3 Grading Basis** Graded

This course will cover linear and nonlinear optimization with an emphasis on applications in the sciences, economics and social sciences.

Techniques studied include the simplex, Newton, and Lagrange methods and Kuhn-Tucker theory. Software packages will be used to implement these methods.

Prerequisites: MATH 248 or permission of the instructor.

MATH 341 Non-Linear Dynamics and Chaos**Cross-listed with:** PHYS 341**Credits 3 Grading Basis** Graded

Introductory study of nonlinear dynamics and chaos intended primarily for upper-level undergraduates in science or mathematics. Topics include stability, bifurcations, phase portraits, strange attractors, fractals, and selected applications of nonlinear dynamics in pure and applied science. Computers may be utilized for simulations and graphics.

Prerequisites: MATH 238 or (MATH 300 and MATH 336); and MATH 248.

MATH 342 Mathematical Models in Biology**Cross-listed with:** BIO 342**Credits 3 Grading Basis** Graded

Introduction to dynamical models (discrete and continuous time) applied to biology. Tools of mathematical analysis from linear and nonlinear dynamics will be taught, including stability analysis of equilibrium, as well as appropriate use of software packages. Emphasis will be on model development and interpretation in the context of applications, including effective written and oral presentation.

Prerequisites: MATH 232 or MATH 235 or equivalent.

MATH 353 Graph Theory**Credits 3 Grading Basis** Graded

Graphs and their applications. Possible topics include trees, Euler paths and Hamiltonian circuits, planar graphs, digraphs, adjacency matrices, connectivity and coloring problems.

Prerequisites: MATH 245 or CS 327 or permission of the instructor.

MATH 354 Introduction to Biometrics**Cross-listed with:** BIO 454**Credits 4 Grading Basis** Graded

This course discusses the role of statistics in biological research and interpretation of biological phenomena. The course will cover topics of sampling, correlation, regression analysis, tests of hypotheses, commonly observed distributions in natural populations, nonparametric tests, goodness-of-fit tests and ANOVA. In order to fully comprehend the statistical analysis of those publications, students will review approximately half a dozen publications from different fields of biology.

Prerequisites: MATH 220, MATH 229, MATH 318 or equivalent.

MATH 358 Introduction to Machine Learning**Cross-listed with:** DATA 358**Credits 3 Grading Basis** Graded

This is an introductory course on machine learning with focus on regression and classification methods. The objective is to familiarize students with some basic models and algorithms for machine learning and prepare them for research or industry application of machine learning techniques. Several software libraries and publicly available datasets will be used to illustrate the application of these techniques. The emphasis will be on machine learning algorithms and application, with some broad explanation of the underlying principles.

Prerequisites: MATH 229, or MATH 220 with permission of the instructor.

MATH 360 Complex Variables**Credits 3 Grading Basis** Graded

Introduction to algebraic properties of complex numbers, analytic functions, harmonic functions, mappings of elementary functions, contour integration, series, residues, and poles and conformal mappings. Emphasis on computations and applications to fluid and heat flow.

Prerequisites: MATH 237.

MATH 364 Introduction to Fluid Mechanics**Cross-listed with:** PHYS 364**Credits 4 Grading Basis** Graded

Introduces the student to the application of vector calculus to the description of fluids. The Euler equation, viscosity and the Navier-Stokes equation will be covered.

Prerequisites: MATH 237 and PHYS 260, or permission of the instructor.

MATH 367 Introduction to Acoustics**Cross-listed with:** PHYS 367**Credits 3 Grading Basis** Graded

This course represents an introduction to sound, hearing, and vibration. Architectural, biological and environmental acoustics will also be discussed. Students will develop an ability to use mathematical models and experimental techniques to study problems in acoustics and to transfer this knowledge to analogous situations. They will also develop an ability to conduct a semester-long research or expository project and present it in written and oral form to an audience of peers.

Prerequisites: MATH 236.

MATH 368 Statistical Computing**Credits 3 Grading Basis** Graded

This course is an intensive introduction to statistical computing with R. It intends to develop core computing skill to complement traditional course work in statistics and lay the groundwork for further study for data analytics and data mining. The class will cover the basics of programming in R; visualization of multivariate data; generating random variables; bootstrap, jackknife, and permutation methods; Monte Carlo methods in inference; and Monte Carlo integration and variable reduction.

MATH 385 Data and AI for Defense/Diplomacy**Credits 3 Grading Basis** Graded

Data and AI for Defense/Diplomacy is a project-based course that tackles mission-driven real world problems from the Departments of State and Defense, the intelligence community and other federal departments and agencies. Students use the Lean Launch Pad Methodology, mathematical modeling, operations research, data science and AI to develop implementable minimal viable products and potential solutions. The course provides students the opportunity to address critical problems facing the U.S. government today, addressing current national security and foreign policy problems. Students engage with stakeholders from a range of government agencies, academia and industry.

Prerequisites: MATH 248 or permission of the instructor.

MATH 395 Mathematical Finance**Cross-listed with:** FIN 395**Credits 3 Grading Basis** Graded

This course presents an overview of the role of mathematical concepts in financial applications. Topics include continuous time finance, optimization, numerical analysis and applications in asset pricing.

MATH 405 Securities Pricing**Cross-listed with:** FIN 405**Credits 3 Grading Basis** Graded

A quantitative treatment of the theory and method of financial securities pricing to include an examination of closed form pricing models such as the Black-Scholes and its various derivatives as well as numerical solution techniques such as binomial methods.

Prerequisites: FIN 395 or MATH 395.

MATH 410 Advanced Calculus I**Credits 3 Grading Basis** Graded

This course covers limits, continuity, differentiation, sequences, series, integration and selected topics.

Prerequisites: MATH 236; and MATH 238 or MATH 300; and MATH 245 or CS 327; or instructor permission.

MATH 411 Advanced Calculus II**Credits 3 Grading Basis** Graded

Limits, continuity, differentiation, sequences, series, integration and selected topics.

Prerequisites: MATH 410.

MATH 415 History of Mathematics**Credits 3 Grading Basis** Graded

This course covers topics in the history of mathematics spanning ancient times to the present.

Prerequisites: MATH 245 or CS 327 or consent of the instructor.

MATH 421 Applied Multivariate Statistical Analysis**Credits 3 Grading Basis** Graded

Multivariate statistical methods with applications. Topics include canonical correlation, clustering, discriminant analysis, factor analysis, multivariate analysis of variance, multiple regression, multidimensional scaling and principal component analysis.

Prerequisites: MATH 300 or MATH 238 and MATH 321 or MATH 322.

MATH 423 Stochastic Processes**Credits 3 Grading Basis** Graded

Sequences and classes of random variables. Applications to physical, biological, social and management sciences. Topics include Markov chains, branching processes, the Poisson process, queuing systems and renewal processes.

Prerequisites: MATH 238 or MATH 300; and MATH 318 or MATH 329.

MATH 424 Statistical Decision Theory**Credits 3 Grading Basis** Graded

Development and use of probability and statistics for strategic decision making with applications. Topics include decision flow diagrams, analysis of risk and risk aversion, utility theory, Bayesian statistical methods, the economics of sampling, sensitivity analysis and collective decision making.

Prerequisites: MATH 318 or MATH 329.

MATH 426 Probability and Mathematical Statistics I**Credits 3 Grading Basis** Graded

Derivations and proofs of probability theorems, discrete and continuous univariate and multivariate random variables, conditional distributions, mathematical expectations, functions of random variables, moment generating functions, properties and derivation of estimators including the method of moments and maximum likelihood estimation.

Prerequisites: MATH 318 or MATH 329.

MATH 427 Probability and Mathematical Statistics II**Credits 3 Grading Basis** Graded

Limiting distributions, sampling theory and distributions, theory and applications of estimation and hypothesis testing.

Prerequisites: MATH 426.

MATH 428 Statistical Consulting**Credits 3 Grading Basis** Graded

Training and experience in statistical consulting emphasizing oral and written communication, interview, report writing and presentation skills. Participate in significant interdisciplinary consulting project that will require meeting with the client, creating reports summarizing the client's problem and an analysis performed by the students, and explanation of results using language that can be understood by the client. Students are required to meet with clients outside of class meeting times.

Prerequisites: MATH 318 or MATH 329; MATH 321, at least junior standing and MATH 322. Corequisites: MATH 322.

MATH 429 Research Project in Statistics**Credits 1-3 Grading Basis** Graded

Experience in the design, data collection, and analysis for a survey or experiment.

MATH 430 Abstract Algebra I**Credits 3 Grading Basis** Graded

An introduction to groups, rings and fields.

Prerequisites: MATH 238 or MATH 300; and MATH 245 or CS 327; or permission of the instructor.

MATH 431 Abstract Algebra II**Credits 3 Grading Basis** Graded

An introduction to groups, rings and fields.

Prerequisites: MATH 430.

MATH 434 Advanced Linear Algebra**Credits 3 Grading Basis** Graded

A proof-based linear algebra course covering such topics as vector spaces, linear transformations and matrices, eigenvalues and eigenvectors, inner product spaces, and canonical forms.

Prerequisites: MATH 238 or MATH 300; and MATH 245 or CS 327; or permission of the instructor.

MATH 435 Introduction to Topology**Credits 3 Grading Basis** Graded

This course covers metric spaces, limits, continuous maps and homeomorphisms, connectedness, compact topological spaces, and applications.

Prerequisites: MATH 238 or MATH 300; and MATH 245 or CS 327.

MATH 440 Fourier Analysis and Partial Differential Equations**Credits 3 Grading Basis** Graded

Elementary applied partial differential equations, the heat equation, Laplace's equation, the wave equation; Fourier series and boundary value problems. Both theory and problem-solving will be included.

Prerequisites: MATH 238 or MATH 336.

MATH 441 Analysis and Dynamics of Differential Equations**Credits 3 Grading Basis** Graded

Analysis of qualitative properties and dynamics of linear and non-linear ordinary differential equations, including topics such as existence, uniqueness, phase portraits, stability and chaos, with applications in the sciences.

Prerequisites: MATH 238 or both MATH 300 and MATH 336; and MATH 245 or CS 327 or MATH 440 or permission of the instructor.

MATH 448 Numerical Analysis**Cross-listed with:** CS 448**Credits 3 Grading Basis** Graded

This course is the study and analysis of algorithms used to solve nonlinear equations and systems of linear and nonlinear equations.

Students will learn iterative methods for matrices and Newton-type methods as well as numerical differential and integral calculus. Students will perform programming using a high-level language and/or software packages.

Prerequisites: MATH 248 or permission of the instructor.

MATH 449 Numerical Analysis for Differential Equations**Cross-listed with:** CS 449**Credits 3 Grading Basis** Graded

Study and analysis of numerical techniques to solve ordinary and partial differential equations, including Euler, Runge-Kutta, Picard, finite-difference and finite-element methods. Programming using a high-level language and/or software packages.

Prerequisites: MATH 237; MATH 238 or MATH 336; and MATH 248.

MATH 452 Design and Analysis of Algorithms**Cross-listed with:** CS 452**Credits 3 Grading Basis** Graded

An introduction to algorithm design techniques and the mathematical analysis of algorithms. Techniques covered include divide-and-conquer, greedy, backtracking, network flow and dynamic programming. The classes P and NP are introduced and intractable problems are discussed. This is a proof-intensive course. Students may not receive credit for both CS 412 and CS 452.

Prerequisites: A grade of "C-" or better in CS 327 and either MATH 220 or MATH 229 or MATH 318. Students may not receive credit for both CS 412 and CS 452.

MATH 470 Connections in Mathematics**Credits 3 Grading Basis** Graded

This course is a mathematics capstone course primarily for mathematics majors seeking secondary teaching licensure. It consists of a review of some mathematical topics in the high school curriculum, such as trigonometry, calculus and statistics, highlighting the interconnectedness of advanced mathematics to the secondary curriculum.

Prerequisites: MATH 245, MATH 238 or MATH 300, and MATH 318 or MATH 329. Corequisites: One of MATH 410, MATH 430 or MATH 475.

MATH 475 Fundamental Concepts of Geometry**Credits 3 Grading Basis** Graded

Origin and development of Euclidean and other geometries including axiomatic systems, mathematical proof and special topics from incidence geometry.

Prerequisites: MATH 245 or consent of the instructor.

MATH 485 Selected Topics**Credits 1-4 Grading Basis** Graded

Topics in advanced mathematics or statistics which are not covered in the regularly offered courses. Offered only with approval of the department head; may be repeated for credit when course content changes.

MATH 486 Independent Study**Credits** 1-3 **Grading Basis** Graded

Independent study in mathematics under faculty supervision. Only offered with the consent of the department head.

MATH 497 Undergraduate Research**Credits** 1-4 **Grading Basis** Graded

Students pursue advanced research in a selected area of mathematics and/or statistics. Student must make arrangements with a supervising instructor prior to registration. Offered only with consent of the department head. Repeatable up to 6 credits.

MATH 499A Honors Thesis**Credits** 1-3 **Grading Basis** Graded

This is the first of a 3-course sequence completed over three semesters for a total of 6 credits. The distribution of the 6 credits will be determined by the project advisor.

MATH 499B Honors Thesis**Credits** 1-3 **Grading Basis** Graded

This is the second of a 3-course sequence completed over three semesters for a total of 6 credits. The distribution of the 6 credits will be determined by the project advisor.

MATH 499C Honors Thesis**Credits** 1-3 **Grading Basis** Graded

This is the third of a 3-course sequence completed over three semesters for a total of 6 credits. The distribution of the 6 credits will be determined by the project advisor.

Media Arts and Design (SMAD)

SMAD 101 Introduction to Media Arts and Design**Credits** 3 **Grading Basis** Graded

This course studies the historical and ongoing evolution of media industries, technology and media storytelling, as well as effects on audience members and society. The course emphasizes media career paths and prepares students for applying to the SMAD program. Prerequisites: Formal declaration for admission to the SMAD major.

SMAD 150 Mediated Communication: Issues and Skills [C1CT]**Credits** 3 **Grading Basis** Graded

Study of how mediated communication shapes the content, meaning and impact of spoken, written and pictorial messages. Emphasis on the skills required to integrate speech, text and imagery into mediated presentations. Consideration of issues involving the critical evaluation of mass-mediated communication, their effectiveness and influence. May be used for general education credit. May not be used for major credit. Prerequisites: If you have already completed, or are currently registered for, a Critical Thinking (Madison Foundations) class, then you must secure permission to receive credit for a second Critical Thinking class.

SMAD 201 Foundations of Visual Communication Design**Credits** 3 **Grading Basis** Graded

Study of the basic computer operating systems and the principles and practices of graphic production for digital and interactive media. Focus on tools and techniques used to create graphic content for diverse media delivery systems. Prerequisites: Admission to the SMAD major.

SMAD 202 Foundations of Audio-Visual Storytelling**Credits** 3 **Grading Basis** Graded

Audiovisual storytelling is an experiential learning course designed to give students a foundation in creating and producing engaging stories/content utilizing a varied range of audiovisual communication/media tools and formats for distribution on multiple delivery platforms including, but not limited to, television, interactive, film, documentaries, news, commercials, radio, podcasts and web.

Prerequisites: SMAD 201. Corequisites: SMAD 201.

SMAD 203 Foundations of User Experience Design**Credits** 3 **Grading Basis** Graded

This course is a study of the aesthetic principles and practices of web and interactive narrative design, with an emphasis on visual communication in an interactive media context.

Prerequisites: SMAD 201. Corequisites: SMAD 201.

SMAD 204 Foundations of Media Storytelling and Research**Credits** 3 **Grading Basis** Graded

This course is the study of story development across diverse media forms to enable effective, accurate and ethical storytelling. Emphasis will be on identifying diverse media forms' story goals and objectives, and on research techniques and tools to locate, identify and assess information accuracy. Attention to recognizing and evaluating media story accuracy and effective use of language.

Prerequisites: Admission to SMAD major.

SMAD 210 News Reporting and Writing**Credits** 3 **Grading Basis** Graded

This course is the study and practice of the fundamentals of news writing, including news-gathering techniques and news style. Emphasis is placed on coverage of meetings, events and breaking stories. Consideration of writing across platforms including print, broadcast and online. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Fully admitted SMAD majors in the Creative Advertising or Journalism concentrations and SMAD 204; for Sports Communication minors, declaration of the Sports Communication minor; or permission of the instructor. Corequisites: SMAD 204.

SMAD 220 Copy Editing**Credits** 3 **Grading Basis** Graded

The study and practice of editing stories and visuals for publication. Focus on style, factuality, writing problems, headlines and page design. Other editing roles considered.

Prerequisites: Journalism conc: SMAD 210, Creative Advertising conc: SMAD 210 or SMAD 252; or permission of instructor. Corequisites: Journalism conc: SMAD 210, Creative Advertising conc: SMAD 210 or SMAD 252; or permission of instructor.

SMAD 225 Photojournalism**Credits** 3 **Grading Basis** Graded

The study of visual information gathering for print media including photojournalism. Emphasis on photographic techniques and print media layout. Consideration of new visual technologies including the use of computers for electronic photo editing and design. Students must provide their own camera.

Prerequisites: Fully admitted SMAD majors and SMAD 201; or permission of the instructor.

SMAD 242 Introduction to Advertising and New Media**Credits 3 Grading Basis** Graded

This course introduces fundamentals of advertising in the new media environment. Students examine how organizations use new media to engage with consumers to achieve advertising objectives. Focus is on hands-on practice in the basic skills and processes needed to prepare multi-platform narratives. Students develop a clear understanding of what makes advertising effective across traditional and new media and develop professional research skills.

Prerequisites: Fully admitted SMAD major and Creative Advertising concentration; SMAD 204, or permission of instructor. Corequisites: SMAD 204.

SMAD 243 Sport Communication Techniques: Broadcasting**Credits 3 Grading Basis** Graded

Study and practice of broadcast and A/V techniques applied in a variety of sport settings.

Prerequisites: SCOM 230.

SMAD 244 Sport Communication Techniques: Writing and Reporting**Cross-listed with:** SCOM 244**Credits 3 Grading Basis** Graded

Basic skills of sport writing and reporting are studied and applied. Students gain experience in a variety of sports and learn and apply skills in researching, interviewing, reporting, writing columns and features involving the world of sports.

Prerequisites: SCOM 230.

SMAD 250 Scriptwriting**Credits 3 Grading Basis** Graded

The study of the principles and practices of writing scripts for commercial, noncommercial and corporate media applications. Emphasis on preparing dramatic and information forms for broadcast or recording. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Fully admitted SMAD major; for non-majors, admission to Creative Writing minor; or permission of instructor.

SMAD 251 Screenplay Writing**Credits 3 Grading Basis** Graded

Introductory study of the principles and practices of screenwriting. Emphasis is placed on the basic narrative structures underlying cinematic story-telling and the development of a short film script. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Fully admitted SMAD major; for non-majors, admission to Creative Writing minor; or permission of instructor.

SMAD 252 Copywriting for Advertising**Credits 3 Grading Basis** Graded

This course covers the principles and practices of developing creative copy for advertising. Students will examine how to solve client problems by writing persuasive stories. Students will focus on writing scripts for TV and radio commercials, print ads, headlines, taglines, jingles, web copy, out of home ads, new media, and social media. Emphasis will be on editing, researching and writing creative briefs and strategic plans.

Prerequisites: Admission to the SMAD major and creative advertising concentration and SMAD 204 and SMAD 242. Corequisites: SMAD 204, SMAD 242.

SMAD 253 The Shooting Script: The Process of Script to Screen**Credits 3 Grading Basis** Graded

In this course, students will polish screenplays and develop them into a shooting script for video production. The work in this class involves creating a screenplay that the students will actually shoot and produce. This is a different process than simply writing a script. For example, if the script contains a scene at a swimming pool, but they do not have access to a pool as a location, students must rework the scene to make it work with available locations. The course will only be offered as part of the JMU/SVA summer study abroad program in New York. The class will satisfy the digital video and cinema concentration writing requirement and satisfy the prerequisite for SMAD 340.

Prerequisites: Fully admitted SMAD major and admission to the JMU/SVA summer study abroad program in New York.

SMAD 295 Practicum in Media Arts and Design**Credits 1 Grading Basis** Graded

Freshmen and sophomores participating in cocurricular media activities may receive one hour of credit for field work at The Breeze, WMRA-FM, the Madison Video Productions or other university-related media outlets. Students are limited to one practicum per semester. Application procedures will be available from the school prior to registration.

SMAD 301 The Media Arts: Culture by Design**Credits 3 Grading Basis** Graded

This course is the study of how mediated communication molds perception and influences cultural change. Emphasis will be placed on how language and imagery, sound and music are combined in current media to create meaning. The course will consider emerging media and their implications for cultural design.

Prerequisites: SMAD 201, SMAD 202, SMAD 203, and SMAD 204; or permission of the instructor. Corequisites: SMAD 202, SMAD 203, SMAD 204.

SMAD 302 Video Production**Credits 3 Grading Basis** Graded

Principles and practices of video production and editing. Focus on production planning, visual composition, lighting, recording sound and sequencing of shots. Emphasis on single camera videography and editing for broadcast, non-broadcast, and multimedia applications.

Prerequisites: SMAD 201, SMAD 202, and SMAD 203, or permission of the instructor. Corequisites: SMAD 203 or permission of the instructor.

SMAD 303 Post Production**Credits 3 Grading Basis** Graded

Principles and practices of video editing. Focus on the techniques, aesthetics, and strategies of editing video. Emphasis on technical standards of ingest, storage, non-linear editing, multi-layer compositing, and transcoding required for effective program output and multi-format distribution.

Prerequisites: Fully Admitted SMAD majors and SMAD 302, or permission of the instructor.

SMAD 304 Audio Production**Credits 3 Grading Basis** Graded

Study of digital sound production, and digital sound-for-picture production. Emphasis on advanced theories and applications.

Prerequisites: SMAD 302 or permission of instructor. Corequisites: SMAD 303 or permission of instructor.

SMAD 305 Special Topics in Media Production**Credits 3 Grading Basis** Graded

The study of current topics and issues in media arts and design. Emphasis on contemporary themes of immediate concern.

Prerequisites: Admission to the SMAD major.

SMAD 306 Studio Production**Credits 3 Grading Basis Graded**

The study of principles and practices of multi-camera studio production. Focus on visual composition, audio, lighting and communication in a studio environment. Emphasis on technical and aesthetic demands of multi-camera studio production.

Prerequisites: SMAD 302 or SMAD 309.

SMAD 307 Interactive Design Web I**Credits 3 Grading Basis Graded**

Study of the principles of creating effective communication for the World Wide Web. Emphasis on the techniques used to design and integrate diverse media elements. Focus on the creation and manipulation of text, graphics, audio and video for the web. Consideration of interface design and web delivery issues.

Prerequisites: SMAD 201, SMAD 202 and SMAD 203. Corequisites: SMAD 202.

SMAD 308 Interactive Design II**Credits 3 Grading Basis Graded**

This course will study the principles and practices for creating effective interactive media. An emphasis is placed on user-centered design of the interactive experience. This course includes animation and scripting for online and fixed media.

Prerequisites: SMAD 307 or permission of the instructor. Corequisites: SMAD 307.

SMAD 309 Video Journalism**Credits 3 Grading Basis Graded**

This course covers principles and practices in electronic journalism including information gathering, news writing, camera operations and editing techniques. Emphasis will be placed on the use of sound and video to produce hard news, feature news and long-form perspective-based stories for broadcast and online. Students will examine the internet's impact and the differences between traditional and online media outlets in constructing stories with video and sound.

Prerequisites: SMAD 201, SMAD 202, SMAD 203, SMAD 204, and SMAD 210; or permission of the instructor. Corequisites: SMAD 203, SMAD 204, and SMAD 210.

SMAD 310 Advanced Reporting and Writing**Credits 3 Grading Basis Graded**

Students will study and practice in-depth or specialized reporting roles and techniques, story genres, advanced writing methods and pursuing publication. The course will include reading exemplary writers. Group and instructor critiques will be used.

Prerequisites: SMAD 204 and SMAD 210; or permission of the instructor.

SMAD 312 Cinematography**Credits 3 Grading Basis Graded**

This course explores the core technical and artistic elements of cinematography including camera operation, framing, composition and color theory. Students will also gain experience in the responsibilities of the cinematographer, camera assistant and gaffer, including set etiquette and pre-production tasks. Through hands-on workshops and projects, students will practice camera setups, lighting, blocking, adjustments and troubleshooting scenes. The course will also cover lens choices, camera movement and framing techniques to craft dynamic visuals that support narrative and emotional depth.

Prerequisites: Fully Admitted SMAD majors and SMAD 302, or permission of the instructor.

SMAD 315 Digital Social Media Advertising**Credits 3 Grading Basis Graded**

This course focuses on using social media strategically to create value for a brand or organization. Students will create effective consumer-targeted content across the ever-changing digital social media landscape with emphasis on strategic collaboration, execution and measurement of social media efforts. Students will learn how to brand themselves using social media. By the end of this course, students will demonstrate a proficiency in social media strategy and social media brand management.

Prerequisites: Creative Adv, Digital Video Cinema, and Jour conc: SMAD 201 and SMAD 202. Interactive Des conc: SMAD 201 and SMAD 203. Corequisites: Creative Adv, Digital Video and Cinema, and Journalism conc: SMAD 202. Interactive Design conc: SMAD 203.

SMAD 317 User Interaction Design**Credits 3 Grading Basis Graded**

This course is a study of web and mobile interaction design principles including goal driven design, usability, interactive design strategy, wireframing, and web and mobile app prototype development.

Prerequisites: SMAD 201, SMAD 202 and SMAD 203. Corequisites: SMAD 202.

SMAD 321 The Media Arts: Culture by Design in London**Credits 3 Grading Basis Graded**

This course is a study of how mediated communication molds perception and influences cultural change. Emphasis is placed on how language, imagery, sound and music are combined in current media to create meaning. Consideration is given to emerging media and their implications for cultural design. The focus of this course is on British media and culture.

Prerequisites: Admission to the JMU semester in London.

SMAD 327 User-Centered Research and Usability Evaluation**Credits 3 Grading Basis Graded**

This course provides an in-depth examination of theories, methods and tools used by professionals in user-centered research. Students will learn through the process of completing a research project, from selecting a research topic through reporting findings and practical implications. In addition, the course enables students to pursue more advanced studies in user experience in capstone courses by equipping them with critical analytical and hands-on skills in a user-centered research process.

Prerequisites: SMAD 201 and SMAD 203.

SMAD 330 New Media Law**Credits 3 Grading Basis Graded**

An in-depth investigation of business and legal aspects of new media production, publication and distribution, Examination of legal issues affecting the new media industry including: intellectual property laws, torts contracts and licensing. Discussion of current policies and practices affecting new media development and review of pertinent legal reference materials.

Prerequisites: Fully admitted SMAD major; SMAD 201, SMAD 202, SMAD 203, and SMAD 301; and junior or senior standing; or permission of the instructor.

SMAD 332 Visual Communication Design**Credits 3 Grading Basis Graded**

An introduction to the theories and methods of design for visual communication. Students will learn various techniques needed to produce the design materials that go into print and electronic publications.

Prerequisites: Fully admitted SMAD majors and SMAD 201; or permission of the instructor.

SMAD 334 Media and Justice**Cross-listed with:** JUST 334**Credits 3 Grading Basis** Graded

This course will examine media constructions of justice. Students will be required to critically analyze the portrayal of justice issues in various media forms including television, internet and film. Attention will be given to the accuracy of such portrayals and whether they have any broader social implications in regards to how we view complex justice issues.

Prerequisites: Admission to SMAD major.

SMAD 340 Advanced Screenplay Writing**Credits 3 Grading Basis** Graded

Advanced study of the principles of screenplay writing for feature films and television movies. Emphasis is placed on the structure of narrative film.

Prerequisites: Fully admitted SMAD majors or declared creative writing minors, and SMAD 251 or SMAD 253; or permission of instructor.

SMAD 342 Elements of Creative Advertising**Credits 3 Grading Basis** Graded

This course introduces fundamentals of advertising message design and content production for traditional and new media. It examines common questions associated with producing content (i.e., functions and basics of design for print, broadcast and online venues). Students will focus on characteristics of effective messages across media and developing professional production skills to produce diverse creative pieces. The course emphasizes ethical and legal issues.

Prerequisites: SMAD 201, SMAD 202, SMAD 203, SMAD 204, SMAD 242, and SMAD 252; or permission of instructor. Corequisites: SMAD 203.

SMAD 357 Youth, Communication and Culture**Cross-listed with:** SCOM 357**Credits 3 Grading Basis** Graded

This course critically examines the complex and evolving interconnections between young people, communication and culture. Grounded in the understanding that there is no one singular lived experience of "youth," the course focuses on: how young people are represented in popular culture artifacts; how cultural beliefs about various young people are constructed by public discourse; how diverse young people navigate popular culture messages; how they are constructed as target markets; and how they use technology to claim space in the broader culture. This course encourages students to examine their own experiences with and relationships to a range of communication artifacts.

Prerequisites: Admission to the SMAD major.

SMAD 360 British Media and Society**Cross-listed with:** HUM 251L, WRTC 360, SCOM 360**Credits 3 Grading Basis** Graded

Study the history, nature and impact of mass media in the United Kingdom. Emphasis on the impact modern media has on society, and society has on media. Consideration of similarities and differences in mass media in the United States and Great Britain. Consideration of the relationship between mass media and the arts. Focus on 20th century mass media in London, one of the world's preeminent and most influential centers. London term only.

SMAD 370 Mass Communication Law**Credits 3 Grading Basis** Graded

Principles and case studies in communication law, constitutional guarantees, libel, privacy, contempt, copyright and governmental regulatory agencies. Emphasis on recent cases and their effects on mass communication.

Prerequisites: Fully admitted SMAD major; SMAD 201, SMAD 202, SMAD 203, and SMAD 301; and junior or senior standing; or permission of the instructor.

SMAD 372 Media History**Credits 3 Grading Basis** Graded

The study of the history of the media in relation to American politics and society. Emphasis on key periods in the development of journalism and the mass media, and the role of the journalist in society.

Prerequisites: Fully admitted SMAD major and junior or senior standing; or permission of the instructor.

SMAD 373 Media Analysis and Criticism**Credits 3 Grading Basis** Graded

This course is the study and practice of journalistic and scholarly criticism of the mass media. Emphasis is placed on interpretive writing about television, film and popular music. Students will consider contemporary analytical methods for researching mass-mediated culture.

Prerequisites: SMAD majors or creative writing minors; junior or senior standing. For SMAD majors: SMAD 301 or ENG 380/SMAD 380; or permission of the instructor.

SMAD 380 Introduction to Film**Cross-listed with:** ENG 380**Credits 3 Grading Basis** Graded

An introduction to the study of film as an aesthetic practice, including formal and industrial aspects of film analysis, theoretical approaches to film and writing and research methodologies of film and media studies.

Prerequisites: Admission to the SMAD major.

SMAD 390 Directed Projects**Credits 2 Grading Basis** Graded

Supervised projects related to the study of any of the communication media. Credit given for original individual or group programs beyond the school's usual cocurricular activities. A suitable completed project or report is required before credit can be awarded.

SMAD 395 Advanced Practicum in Media Arts and Design**Credits 1 Grading Basis** Graded

Juniors and seniors participating in cocurricular media activities may receive one hour of credit for field work at The Breeze, Curio, WMRA-FM, the Madison Video Productions or other university-related media outlets. Students are limited to one practicum per semester. Application procedures will be available from the school prior to registration.

Prerequisite: Permission of the instructor and the media operation involved.

SMAD 398 Critical Studies in Media Arts and Design**Credits 3 Grading Basis** Graded

This course covers special issues of contemporary media and their contents. Emphasis is placed on the analysis of emerging issues in content production, reception and media effects. Focus is on the critical methods used to examine such issues. Consideration will be given to how particular methods may be applied in international as well as American media contexts.

Prerequisites: Fully admitted SMAD majors with junior or senior standing and SMAD 201, SMAD 202, SMAD 203, SMAD 204, and SMAD 301; or permission of instructor.

SMAD 400 Senior Assessment in Media Arts and Design**Credits 0 Grading Basis** Credit / No Credit

Students participate in testing, interviews, project reviews and other assessment activities as approved by the School of Media Arts and Design. Grades will be assigned on a credit/no-credit basis.

Prerequisites: Senior standing and one of the following SMAD 330 or SMAD 370 or SMAD 450; or permission of the school director.

Corequisites: SMAD 330 or SMAD 370 or SMAD 450.

SMAD 402 Motion Graphic Design and Production**Credits 3 Grading Basis** Graded

Study of the principles of motion design. Focuses on integrating media (video, greenscreen, photos, vector art, type) within compositing software to tell a story, convey a complex concept or artistically present an idea.

Emphasis on graphic design and motion theory, typography for video, logo animation, and show package design.

Prerequisites: SMAD 303 or permission of the instructor.

SMAD 403 Documentary Production**Credits 3 Grading Basis** Graded

This course explores the key components of documentary filmmaking and production. The course will introduce students to a professional level of documentary filmmaking, and will focus on enhancing nonfiction storytelling skills. The course will explore funding, festival and outreach strategies.

Prerequisites: SMAD 302 or SMAD 309.

SMAD 404 Advanced Interactive Design**Credits 3 Grading Basis** Graded

Study of advanced techniques in interactive media for online and fixed media delivery. Emphasis on emerging technologies and professional development strategies. Includes the creation of an interactive portfolio.

Prerequisites: Fully admitted SMAD major, SMAD 308 and SMAD 317; or permission of the instructor.

SMAD 405 Producing and Directing**Credits 3 Grading Basis** Graded

Study of the principles and practices of producing and directing programs for video and cinema. Emphasis on the producer's and director's contribution to creative development and production. Focus on the director's responsibilities, including directing talent, visualization, production, and project management. Additional focus on the producer's responsibilities, including creative development, production management, financial management, promotion, and distribution.

Prerequisites: SMAD 302 and senior standing, or permission of the instructor.

SMAD 407 Business and Management of Digital Media**Credits 3 Grading Basis** Graded

Study of the principles and practices of managing digital media production. Emphasis on financial and personnel management, client relationships, and the structures and practices of digital media industries. Consideration of new trends in production and distribution. Attention to career preparation and development.

Prerequisites: Fully admitted SMAD majors, senior standing and SMAD 302; or permission of the instructor.

SMAD 408 Converged Media Lab**Credits 3 Grading Basis** Graded

This course is an advanced study in converged media production.

This senior production studio class will focus on creative solutions for complex communication problems in print, video and web media. The client-based projects will require creative brainstorming, teamwork, project management, and analysis and criticism in addition to production and distribution. An emphasis is also placed on portfolio development.

Prerequisites: SMAD 307, SMAD 317 and either SMAD 305, SMAD 315, SMAD 327 or SMAD 332; or permission of instructor.

SMAD 409 Electronic News Producing and Editing**Credits 3 Grading Basis** Graded

Study of the processes and responsibilities of producing news for electronic distribution. Students will research, write and produce stories for distribution over-the-air, on cable and over the Internet. Emphasis on legal and ethical issues in news coverage.

Prerequisites: Fully admitted SMAD majors, SMAD 309 and SMAD 310; or permission of the instructor.

SMAD 410 The Creative Producer**Credits 3 Grading Basis** Graded

Study of the principles and practices of producing for television. Emphasis on programming (entertainment news to late-night talk, sketch comedy, reality TV, drama, and sitcom), and the producing process (moving a project from idea to completion). Focus on budget management, and all phases of production management. Consideration of industry practices to help prepare students to enter the entertainment industry.

Prerequisites: Fully admitted SMAD majors, senior standing and SMAD 302; or permission of the instructor.

SMAD 417 Business and Management of Interactive Design**Credits 3 Grading Basis** Graded

This course provides a study of the principles and practices of developing and managing interactive media projects for businesses and non-profits. Emphasis will be placed on user needs, organizational objectives, research, analytics, project proposals, project management, budgeting, content strategy and testing. Principles of effective leadership of human resources are also covered.

Prerequisites: Fully admitted SMAD major, SMAD 307 and SMAD 317; or permission of the instructor.

SMAD 421 Feature Magazine Production**Credits 3 Grading Basis** Graded

A practical course in which students produce a feature magazine published regularly by the journalism area. Each student is assigned a specific staff position. Students learn to use the latest computer pagination technology in the design and production of the magazine.

Prerequisites: SMAD 309 and SMAD 310, or permission of the instructor.

SMAD 422 Multimedia Journalism**Credits 3 Grading Basis** Graded

Course provides an advanced study and practice of multimedia journalism. Focus on storytelling with platforms including text, video, audio, photo and social media aggregation. This is a hands-on course producing multimedia packages involving idea generation, content production, website design and management.

Prerequisites: Journalism concentrators: SMAD 309 and SMAD 310; creative advertising concentrators: SMAD 225 and SMAD 342; or permission of the instructor.

SMAD 443 Creative Advertising Campaigns**Credits 3 Grading Basis Graded**

Study how to develop a creative advertising campaign across a range of traditional and new media. Examine the creative process including conceptualizing, brainstorming, idea generation, copywriting, and design layout. Emphasis on using creativity as a foundation to develop and sell the big idea. Focus on the stages of creative inspiration, concepts of the creative process and roles/functions of creative professionals in advertising, culminating in a creative advertising campaign.

Prerequisites: SMAD 203 and SMAD 342, or permission of the instructor.

SMAD 450 Law of Media Arts and Design**Credits 3 Grading Basis Graded**

This course provides an in-depth investigation of principles and case studies in constitutional guarantees and laws concerning both traditional mass and new media. Emphasis is placed on the First Amendment, the current U.S. legal structure, intellectual property, privacy, libel and the intersection of technology and the law.

Prerequisites: Fully admitted SMAD majors or legal studies minors; junior or senior standing. For SMAD majors: SMAD 301; or permission of the instructor.

SMAD 460 Film and Society**Credits 3 Grading Basis Graded**

Students will examine the cultural impact of the film industry as an institution of mass communication. Emphasis is placed on the interrelationship among social, economic and technological factors influencing the creation and consumption of motion pictures. Consideration is given to particular films as indicators of cultural change.

Prerequisites: Fully admitted SMAD majors or film studies minors; junior or senior standing. For SMAD majors: SMAD 301 or ENG 380/SMAD 380; or permission of instructor.

SMAD 461 Film as Art**Credits 3 Grading Basis Graded**

This course is an advanced study of how movies tell stories in visual terms. Emphasis is placed on the historical evolution and design of the moving image. Consideration will be given to the latest imaging technologies affecting film narrative.

Prerequisites: Fully admitted SMAD majors or film studies minors; junior or senior standing. For SMAD majors: SMAD 301 or ENG 380/SMAD 380; or permission of instructor.

SMAD 462 Documentary in Film and Television**Credits 3 Grading Basis Graded**

This course is the study of content, style, technique and effect of representative samples of the documentary form. Consideration is given to informational and persuasive elements.

Prerequisites: Fully admitted SMAD majors or film studies minors; junior or senior standing. For SMAD majors: SMAD 301 or ENG 380/SMAD 380; or permission of instructor.

SMAD 463 Film Adaptations**Cross-listed with: ENG 463L****Credits 3 Grading Basis Graded**

This course is the study of the process of adapting literature into feature films. Consideration is given to the original literary work, as well as to the changes undergone in its adaptation to film.

Prerequisites: Junior or senior standing. Fully admitted SMAD majors: SMAD 201, SMAD 202, SMAD 203, SMAD 204, and either SMAD 301 or SMAD 380. Non-SMAD majors: creative writing or film studies minors only; or permission of the instructor.

SMAD 464 Contemporary American Film**Credits 3 Grading Basis Graded**

This course is the study and analysis of modern American films of the past twenty years. Focus is placed on storytelling technique, cinematic elements and thematic content. Films will be examined against the backdrop of their specific historical, political and sociocultural context. Prerequisites: Fully admitted SMAD majors or film studies minors; junior or senior standing. For SMAD majors: SMAD 301 or ENG 380/SMAD 380; or permission of instructor.

SMAD 465 The Entertainment Industry**Credits 3 Grading Basis Graded**

Los Angeles is the entertainment capital of the world, including motion pictures, television, radio, advertising, music, gaming, publishing, theatre, dance, broadcasting and more. This course addresses the business and the critical aspects of the industry from the perspective of current entertainment professionals: producers, directors, actors, screenwriters, musicians, entertainment lawyers, TV writers, production managers, animators, cinematographers, marketers, talent agents, dancers, comedians and more. It emphasizes the critical analysis of careers in the entertainment industry and critical aspects of the business, from the idea through the exhibition. On a practical level, students will discover the extraordinary variety of careers within the entertainment industry and the importance of networking. Method of instruction: lecture, discussion and guest speakers.

Prerequisites: Admission to the JMU in LA program, junior or senior standing and SMAD 301; or permission of the instructor.

SMAD 466 British Film Adaptations**Cross-listed with: ENG 463****Credits 3 Grading Basis Graded**

This course studies the process of adapting British literature into feature films. Consideration is given to the original literary work, as well as to the changes undergone in its adaptation to film.

SMAD 470 New Media and Society**Credits 3 Grading Basis Graded**

This course is the study of the key issues arising from the role of information and information technology in organizations and society. Emphasis is on the spheres of social life in which new media technologies play a role: politics, community, economics, culture, personal and global contexts. Focus is on a wide range of perspectives and on both theoretical and empirical research to examine new media's role in social capital and community.

Prerequisites: Fully admitted SMAD majors or WRTC majors; junior or senior standing. For SMAD majors: SMAD 301.

SMAD 471 Media Ethics**Credits 3 Grading Basis Graded**

This course is the study and discussion of ethical principles and professional codes as applied to mass media production, covering journalism, advertising, public relations and entertainment, in all media forms. Emphasis is on case studies.

Prerequisites: Fully admitted SMAD majors, junior or senior standing and SMAD 301.

SMAD 472 Media and Politics**Cross-listed with: SCOM 472, SCOM 472L, SMAD 472L, POSC 472****Credits 3 Grading Basis Graded**

This course is a study of the media's role in the political system, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories; how media practices and technologies shape political decision making; how political forces influence media institutions; and how public opinion and media narratives are shaped.

SMAD 472L Media and Politics**Cross-listed with:** SCOM 472, SCOM 472L, SMAD 472, POSC 472**Credits 3 Grading Basis** Graded

This course is a study of the media's role in the political system, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories; how media practices and technologies shape political decision making; how political forces influence media institutions; and how public opinion and media narratives are shaped.

SMAD 473 British Media and Politics**Cross-listed with:** POSC 473, SCOM 473**Credits 3 Grading Basis** Graded

This course is a study of the media's role in the political system of the United Kingdom, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories as they apply to the British context; how UK media practices and technologies shape political decision making; how political forces influence British media institutions; and how public opinion and media narratives are shaped within a parliamentary democracy.

SMAD 490 Special Studies**Credits 3 Grading Basis** Graded

An independent study for students to pursue individual research under the guidance of a faculty advisor.

Prerequisites: Senior fully admitted SMAD majors in good standing and permission of the school director.

SMAD 495 Internship in Media Arts and Design**Credits 1-4 Grading Basis** Graded

This internship is an off-campus program prepared and monitored on an individual student basis. Internships are designed to provide practical experience in media arts and design careers. This course may be repeated up to four credits.

SMAD 497 Advanced Projects in Media Arts and Design**Credits 3 Grading Basis** Graded

This advanced course will enable students to collaborate to create diverse media projects. Projects may be designed to focus on one or many concentrations, and are intended to provide a culminating media creation experience.

Prerequisites: Fully admitted SMAD major and junior or senior standing; or permission of the instructor.

SMAD 498 Senior Seminar in Media Arts and Design**Credits 3 Grading Basis** Graded

This course is the study and research in the history and philosophy of the function of mass communication in society. Consideration is given to topics relating to problems of communication systems, technological freedoms and press freedom and responsibilities.

Prerequisites: Fully admitted SMAD majors, junior or senior standing and SMAD 301.

SMAD 499 Honors Senior Project**Credits 1-3 Grading Basis** Graded

This course enables honors students to develop, research and produce media-related creative and/or scholarly independent study honors senior projects. This course may be repeated for up to 6 credits.

Middle Education (MIED)

MIED 211 Literacy Field Experience in Middle Grades Education**Credits 3 Grading Basis** Graded

Literacy-based field placement provides middle school preservice students with opportunities to observe and work with children and teachers in middle school settings (grades 6-8). Students will explore the complex nature of language acquisition and reading including skills in phonemic and other phonological awareness, phonics, fluency, vocabulary development, and comprehension strategies for adolescent learners. Additional skills shall include proficiency in writing strategies, as well as the ability to foster appreciation of a variety of fiction and nonfiction text and independent reading for adolescent learners. Students will have first-hand opportunities to learn about middle level education in the context of real classrooms and provide instruction to young adolescents.

Corequisites: LED 212.

Middle and Secondary Education (MSSE)

MSSE 101 Orientation to the Profession**Credits 1 Grading Basis** Graded

This course provides information about preparation programs and careers for teachers of the middle and high school levels. Students participate in reflective activities for making personal and professional decisions about teaching and also engage in a service learning project. Students are responsible for transportation to field placements at local schools.

MSSE 201 Introduction to Social Studies Education Seminar**Credits 3 Grading Basis** Graded

The course will introduce students to social studies education through exploration of questions including: What is social studies education? Why teach social studies? What is the role of social studies content in the world of a teacher? What makes a social studies educator? Students will learn through class discussions, field based classroom observations and guest speakers. Students should come to class prepared to challenge their preconceived notions surrounding social studies education.

MSSE 210 Diversity in Middle and Secondary Education**Credits 3 Grading Basis** Graded

This course is designed to introduce teacher candidates to the adolescent diversity they will encounter in the secondary classroom. Candidates will explore cultural, ethnic, racial, gendered and linguistic identity including various exceptionalities, as related to adolescents, in order to understand the impact of students' development and backgrounds on classroom experiences and academic learning. Prerequisites: Submitted application to teacher education.

MSSE 220 Measurement and Analysis for Education**Credits 3 Grading Basis** Graded

This is an introductory course in measurement and analysis in education, exploring statistical methods designed primarily for undergraduate preservice teachers in the liberal arts. The class presents basic measurement and statistical concepts with an emphasis on the logic of inferential statistics applicable to school-related professionals. While we assume you have no previous statistical background, we expect you to be proficient in arithmetic and basic algebra. The primary goal of this course is for you to build a solid foundation in measurement and analysis methods used in educational research in order to prepare you to read, understand and interpret testing and measurement data. A secondary goal is to become a better consumer (critical thinker) and advocate of and for educational research.

MSSE 370 General Instructional Methods for Grades 6-12**Credits 3 Grading Basis** Graded

This general teaching methods course provides a solid foundation for planning curriculum and instruction for middle and high school students. Topics include engaging adolescent learners, lesson design, instructional strategies and assessment models.

Prerequisites: SEED majors or minors: Admission to teacher education; MIED majors: MIED 211 and LED 212 and admission to teacher education. Corequisites: MSSE 371 (or MSSE 430 for Valley Scholars mentors).

MSSE 371 Clinical Experience in Adolescent Education**Credits 1 Grading Basis** Graded

This course provides an opportunity for students to work with adolescent learners through a service-learning component. Students will explore the needs and challenges of adolescent learners, be placed in a local middle or high school, explore the art of teaching, and practice engaging with students in a variety of ways. Throughout the semester, students will be prompted to reflect on their experiences in order to better understand the complexities inherent in teaching as well as their own personal beliefs, strengths and areas for growth.

Prerequisites: SEED majors: Admission to teacher education; SEED minors: Admission to teacher education; MIED majors: Admission to teacher education, MIED 211 and LED 212. Corequisites: MSSE 370.

MSSE 397 Science Pedagogy**Cross-listed with:** PHYS 302**Credits 2 Grading Basis** Graded

An introduction to the science of learning and how it informs teaching and learning in laboratory and field-based natural sciences. Special emphasis is given to conceptual change, metacognition and effective teaching strategies based on learning science, science discourse and argumentation, and safety.

MSSE 403 Behavior and Classroom Management in a Diverse Society**Credits 3 Grading Basis** Graded

This course is designed to provide students with an understanding of and skill to apply classroom and behavior management techniques and interventions, including techniques that promote emotional well-being and teach and maintain behavioral conduct and skills consistent with norms, standards and rules of the educational environment. Diverse approaches for classroom and behavior management based upon behavioral, cognitive, affective, social and ecological theory and practice will be learned.

Prerequisites: Admission to teacher education and MSSE 370.

MSSE 411 English/Language Arts Teaching Methods, Grades 6-8**Credits 3 Grading Basis** Graded

This course covers specific instructional and assessment techniques and methods targeted for the middle school learner for middle and secondary teachers in their respective discipline areas. Competencies to be developed will include discipline-specific planning strategies, instructional models, assessment of student learning, etc.

Prerequisites: MSSE 370. Corequisites: MSSE 415.

MSSE 412 Social Studies Teaching Methods, Grades 6-8**Credits 3 Grading Basis** Graded

This course covers specific instructional and assessment techniques and methods targeted for the middle school learner for middle and secondary teachers in their respective discipline areas. Competencies to be developed will include, but are not limited to, discipline-specific planning strategies, instructional models and assessment of student learning.

Prerequisites: MSSE 370. Corequisites: MSSE 415. For students with a Social Studies minor: MSSE 415, and HIST 357 or POSC 457.

MSSE 413 Mathematics Teaching Methods, Grades 6-8**Credits 3 Grading Basis** Graded

This course covers specific instructional and assessment techniques and methods targeted for the middle school learner for middle and secondary teachers in their respective discipline areas. Competencies to be developed will include discipline-specific planning strategies, instructional models, assessment of student learning, etc.

Prerequisites: MSSE 370. Corequisites: MSSE 415.

MSSE 414 Natural Sciences Teaching Methods, Grades 6-8**Credits 3 Grading Basis** Graded

Specific instructional and assessment techniques and methods targeted for the middle school learner for middle and secondary teachers in their respective discipline areas. Competencies to be developed will include discipline-specific planning strategies, instructional models, assessment of student learning, etc.

Prerequisites: MSSE 370. Corequisites: MSSE 415.

MSSE 415 Content Area Field Experience in Middle Schools**Credits 3 Grading Basis** Graded

This course allows students to put into practice many of the skills and theories learned in foundational education coursework. It is designed to introduce the diverse roles and responsibilities that a teacher has in the middle school classroom, while also giving students the opportunity to identify the perspectives of the students they will be teaching. Throughout this field experience, the host teacher is the primary instructor as students "learn on the job" while being supported by a university supervisor.

Prerequisites: For MIED and SEED majors: MSSE 370; For SEED minors: MSSE 370 and MSSE 371. Corequisites: MSSE 411, MSSE 412, MSSE 413 or MSSE 414.

MSSE 420 Differentiation for Adolescents**Credits 3 Grading Basis** Graded

This course is designed to assist teacher candidates in meeting the diverse needs of adolescent learners. Participants will gain an increased understanding of how adolescents differ in their approaches to learning and how a responsive classroom can address these needs. Specific topics addressed include developing a positive classroom environment, using ongoing assessment to identify student needs, utilizing a variety of grouping strategies, designing instruction and assessment at an appropriate level of challenge, and engaging students in equally respectful tasks.

Prerequisites: Admission into teacher education Flag 03, MSSE 370 and MSSE 425.

MSSE 425 Assessment in Middle and Secondary Education**Credits 3 Grading Basis** Graded

This course is designed to prepare prospective teachers to create and evaluate a range of assessment strategies. Candidates will examine classroom assessment as a system and construct valid and reliable instruments for each part of that system. Candidates will also learn to analyze assessment results with the goal of gauging student progress and adjusting instruction appropriately.

Prerequisites: Admission to teacher education and MSSE 370.

MSSE 430 Introduction to Mentoring Valley Scholars**Credits 3 Grading Basis** Graded

This course is designed to prepare and support JMU students as they begin a mentoring relationship with students in the area schools. JMU students will experience guest speakers and activities to assist them in their role helping area adolescents set and maintain the trajectory of their lives toward a college experience. By establishing positive relationships and modeling desirable characteristics of a college student, JMU students will provide support and guidance for the adolescent learners with which they engage.

MSSE 435 Mentoring for Valley Scholars Continued**Credits 1-3 Grading Basis** Graded

This course is designed to support JMU students as they continue to mentor adolescent learners in area schools. Through readings, activities and experiences, JMU students will continue to build the skills first introduced in MSSE 430. By fostering healthy relationships and modeling desirable characteristics of a college learner, JMU students will provide support and guidance for adolescent students transitioning from 8th to 9th grade.

Prerequisites: MSSE 430.

MSSE 441 English/Language Arts Content Methods, Grades 9-12**Credits 3 Grading Basis** Graded

Research findings regarding the multiple dimensions of teaching in English/language arts will be used to identify the most effective writing, creative arts and cognitive teaching methods for teaching students 9th-12th grade. The first half of the course will focus on the writing process (i.e. narrative and persuasive) and cognitive models of instruction, and the second half of the course will focus on literature, research and writing (multi-genre project).

Prerequisites: MSSE 411. Corequisites: MSSE 445.

MSSE 442 Social Studies Content Methods, Grades 9-12**Credits 3 Grading Basis** Graded

The aim of this course is to build, expand and draw from the social studies methods explored in MSSE 412. This course gives students the opportunity to continue to consider, analyze and critically examine the details of their teaching career and the purpose of social studies education. Examples of some of the questions examined are: What is "good" teaching? What is the purpose of social studies education? What should students learn? What knowledge is most valuable? How will I meet the needs of diverse students? What are diverse methods of delivering instruction? How does one's belief about the purpose of education shape their pedagogy?

Prerequisites: MSSE 412. Corequisites: MSSE 445.

MSSE 443 Mathematics Content Methods, Grades 9-12**Credits 3 Grading Basis** Graded

This course is designed to address mathematics-specific methods of instruction for prospective high school mathematics teachers (grades 9-12). Students will develop techniques for planning and assessing student-centered mathematics instruction. Through this course, students will become mathematics teachers who can (1) support their practice and sustain their own learning over time; (2) reflect upon their practice, evaluate it and make changes when necessary; (3) collaborate with their colleagues to improve instruction; and (4) stay informed of current innovations and trends in the field.

Prerequisites: MSSE 413. Corequisites: MSSE 445.

MSSE 444 Science Content Methods, Grades 9-12**Credits 3 Grading Basis** Graded

This course is designed to help the student evaluate and consume research findings about science teaching that identify the most effective instructional strategies for teaching science to students in grades 6-12. Emphasis will be on developing plans for employing the strategies and making appropriate instructional decisions based on instructional goals, the learner and available resources.

Prerequisites: MSSE 414. Corequisites: MSSE 449.

MSSE 445 Content Area Field Experience in Secondary Education, Grades 9-12**Credits 3 Grading Basis** Graded

This course continues to build the foundation for students to develop one's instructional practice to be successful within a high school classroom context. This practicum provides the necessary experiences for which students develop the pedagogical content knowledge and professional dispositions needed for success in the student teaching placement.

Prerequisites: MSSE 411, MSSE 412, MSSE 413, or MSSE 414; and MSSE 415. Corequisites: MSSE 441, MSSE 442, MSSE 443 or MSSE 444.

MSSE 449 Science Field Experience, Grades 9-12**Credits 3 Grading Basis** Graded

This practicum is designed to provide the necessary experience in order for students to be successful in their student teaching placement and support major components of their science course work.

Prerequisites: MSSE 415. Corequisites: MSSE 444.

MSSE 450 Internship Seminar**Credits 2 Grading Basis** Graded

The internship seminar is designed to promote reflective decision-making by students engaged in their student teaching experience. The seminar setting offers students an opportunity to discuss their internship experiences in detail and reflect on the implications of that experience for their future practice. Teacher candidates will have the opportunity to demonstrate and articulate their growth as professionals as well as the effects of their teaching on student learning.

Prerequisites: Approved to student teach. Corequisites: For MIED majors: MSSE 465; For SEED majors: MSSE 465 or MSSE 475.

MSSE 455 Special Topics in Pedagogical Content Knowledge**Credits 3 Grading Basis** Graded

This course will examine connections between theory and practice, and the relationship between knowing content and considering to whom that content is delivered. Pedagogy, the art and science of teaching, is enacted in varied ways and is specific to various disciplines. In this course, learners will collaborate in order to learn research-based practices that support student learning and the development of dispositional frameworks.

Prerequisites: Approved to student teach. Corequisites: For MIED majors: MSSE 465; For SEED majors: MSSE 465 or MSSE 475.

MSSE 465 Internship in Middle School**Credits 8 Grading Basis** Credit / No Credit

During this internship, JMU teacher candidates will experience the full range of conditions and tasks expected of a teacher for students in grades 6-8. Candidates will be expected to develop and demonstrate competencies in teaching with the supervision and support of experienced teachers. Furthermore, they will demonstrate mastery of the university's eleven conceptual framework competencies.

Prerequisites: Approved to student teach. Corequisites: MSSE 450.

MSSE 475 Internship in High School**Credits 8 Grading Basis** Credit / No Credit

During this internship, JMU teacher candidates will experience the full range of conditions and tasks expected of a teacher for students in grades 9-12. Candidates will be expected to develop and demonstrate competencies in teaching with the supervision and support of experienced teachers. Furthermore, they will demonstrate mastery of the university's eleven conceptual framework competencies.

Prerequisites: Approved to student teach. Corequisites: MSSE 450.

MSSE 490 Special Studies in Education**Credits 1-3 Grading Basis** Graded

A supervised, individualized study of one or more issues and/or problems in middle and/or secondary education. Prerequisite: The faculty advisor and the program coordinator must approve the plan for the study.

MSSE 495 Topics in Science Pedagogical Content Knowledge**Cross-listed with:** BIO 489**Credits 2 Grading Basis** Graded

Contributes to the further development of pre-service secondary science teachers' pedagogical content knowledge. Students will engage in education research to gain experience and develop instructional practices relevant to the Virginia Standards of Learning. To supplement their understanding of scientific concepts and the process of science, the focus will be on the nature of science encompassing literature from the history, philosophy, and sociology of science.

Prerequisites: CHEM 131; PHYS 140 or 240; and MSSE 414.

Military Science (MSCI)

MSCI 110 Introduction to Leadership and the Army**Credits 2 Grading Basis** Graded

This course offers an introduction to various leadership styles and their effect on organizations; an introduction to the organization of the U.S. Army, its roles and missions, customs, and traditions; effective writing and oral presentation techniques; orienteering and the use of lensatic compass; and principles of physical conditioning.

Corequisites: MSCI 115.

MSCI 115 Basic Leadership Laboratory I**Credits 2 Grading Basis** Graded

In this course, Cadets are introduced to the basic elements of Army organizational structure, military leadership and officership. The lab complements classroom instruction in MSCI 110, focusing on basic soldiering skills, Army customs and traditions, rank structure, and the roles of commissioned and non-commissioned officers. Cadets participate in hands-on activities such as drill and ceremony, team-building exercises, map reading, first aid and confidence training. These labs are designed to instill discipline, build camaraderie and develop essential life skills like time management and goal setting. The experience serves as a launching pad for future leadership development, emphasizing the Army values and the importance of teamwork in achieving mission success.

Corequisites: MSCI 110.

MSCI 120 Leadership Development Fundamentals**Credits 2 Grading Basis** Graded

This course provides an introduction to the basic principles of leadership. Students develop a personal leadership style and learn effective writing and speaking skills. The course provides an introduction to leadership in small tactical units, and covers the fundamentals of first aid, land navigation and marksmanship, and the principles of personal conditioning.

Prerequisites: MSCI 110. Corequisites: MSCI 115.

MSCI 125 Basic Leadership Laboratory II**Credits 2 Grading Basis** Graded

In this course, Cadets continue building foundational leadership and soldiering skills introduced in the fall. The lab complements MSCI 120 and emphasizes practical application of basic military concepts. Cadets engage in hands-on training in areas such as land navigation, drill and ceremony, time management, and communication techniques. They also begin learning the military problem-solving process and developmental counseling. The lab environment fosters teamwork, discipline and confidence, helping Cadets develop interpersonal skills and leadership styles while reinforcing Army values and customs.

Prerequisites: MSCI 110 and MSCI 115. Corequisites: MSCI 120.

MSCI 150 American Military History**Credits 3 Grading Basis** Graded

An extensive look into the major engagements of the United States military from the Colonial period through the present. Particular attention will be paid to developing an understanding of the factors that have helped shape the unique way America wages war. This course considers the role and use of armed forces in relation to the social, cultural, political, economic, and technological development of the United States. It will not only address such themes as wartime strategy, operational tactics, and combat technology, but also the impact of warfare on society and on the remembrances of ordinary men and women in uniform. Significant attention will be given to the evolution of civil-military relations, the advent of professionalism in the military, the non-military uses of the military, and the military's role as a tool for global power.

MSCI 210 Leadership Styles: Theory and Application**Credits 2 Grading Basis** Graded

This course studies individual leadership styles and techniques, using historical case studies to introduce the branches and specialties of the Army. The focus of this course is communication and goal setting in small organizations, effective writing and public speaking, and performance and evaluation as the leader of a five-member team required to accomplish tasks in a practical setting. Students also serve a member of a 10-person team responsible for accomplishing practical military and team-building tasks in a field setting.

Prerequisites: MSCI 110, MSCI 115, MSCI 120 and MSCI 125.

Corequisites: MSCI 215.

MSCI 215 Intermediate Leadership Laboratory I**Credits 2 Grading Basis** Graded

In this course, Cadets deepen their understanding of leadership and military operations through hands-on training and practical exercises. This lab complements MSCI 210, which explores ethical decision-making in combat scenarios and the responsibilities of Army leaders. Cadets begin applying leadership principles in field settings, focusing on battle drills, land navigation and combat decision-making. The lab environment encourages teamwork, initiative and critical thinking, preparing Cadets to lead small units and enhancing their readiness for more advanced ROTC training in future semesters.

Prerequisites: MSCI 110, MSCI 115, MSCI 120 and MSCI 125.

Corequisites: MSCI 210.

MSCI 220 Developing Leadership Skills**Credits 2 Grading Basis** Graded

This course focuses on analyzing missions to determine specified and implied tasks; organizing and conducting a briefing; preparing a concise written directive (operations order); and detailing how a team will accomplish its tasks. Performance and evaluation of Cadets as the leader of a five person team during a 44-hour, off-campus leadership practicum and evaluation of physical fitness preparedness for attendance at Cadet Profession Development Training are also components of this course.

Prerequisites: MSCI 210 and MSCI 215. Corequisites: MSCI 225.

MSCI 225 Intermediate Leadership Laboratory II**Credits 2 Grading Basis** Graded

In this course, Cadets deepen their understanding of Army doctrine and team development through practical, field-based leadership exercises. This lab supports MSCI 220, which focuses on applying Army values, teamwork and the warrior ethos in the context of the law of land warfare and military service philosophy. Cadets engage in squad-level operations, case studies and exercises that emphasize leading and following within small units. Training includes troop leading procedures, land navigation and introductory battle drills, with MSII Cadets beginning to take on more advanced responsibilities and leadership roles in preparation for the transition to the advanced course.

Prerequisites: MSCI 210 and MSCI 215. Corequisites: MSCI 220.

MSCI 280 Internship in Military Leadership**Credits 6 Grading Basis** Graded

MSCI students may elect to participate in a 4-week leadership internship. The internship develops leadership skills through exposure to interactive personal and group exercises. Students will learn how and when to lead others and the importance of teamwork in accomplishing a goal.

MSCI 310 Leading Small Organizations**Credits 3 Grading Basis** Graded

Ethical dilemmas in leadership - when organizational goals conflict with member welfare; the role of policy and standard procedures in organizations; continued practice in effective writing and oral presentations; performance as a leader of a 10 person team in practical exercises; evaluation as an instructor of a period of instruction for 30 subordinates; evaluation as a leader of a 10 to 30 person team in a 44 hour, off-campus leadership practicum.

Prerequisites: MSCI 110, MSCI 120, MSCI 210 and MSCI 220; or placement credit as determined by the professor of military science; Corequisites: MSCI 315.

MSCI 315 Advanced Leadership Laboratory I**Credits 3 Grading Basis** Graded

This laboratory is a critical component of the advanced course, designed to prepare Cadets for leadership roles in complex operational environments. At this stage, Cadets take on more responsibility by planning, coordinating and executing training events, often serving as squad leaders or platoon sergeants. The lab emphasizes tactical decision-making, mission planning and team leadership in simulated field environments, reinforcing skills such as land navigation, patrolling and operations orders. Through hands-on experience and peer leadership, Cadets refine their ability to lead under pressure, communicate effectively and apply Army leadership principles in preparation for the cadet summer training and future commissioning as Army officers.

Prerequisites: MSCI 220 and MSCI 225 and permission of the department. Corequisites: MSCI 310.

MSCI 320 Developing Advanced Leader Skills**Credits 3 Grading Basis** Graded

Planning for the unexpected in organizations under stress; developing alternate courses of action which are viable with the given constraints; how to delegate tasks and supervise subordinate leaders; avoiding micro-management - the when and how of providing guidance and direction; time management - the "backward planning process"; evaluation of a 10 to 30 person team within tactical and nonactical settings; analysis of assigned tasks, preparation of a written directive to accomplish these tasks and oral presentation to others in a concise briefing.

Corequisites: MSCI 325.

MSCI 325 Advanced Leadership Laboratory II**Credits 3 Grading Basis** Graded

In this course, Cadets transition from foundational leadership and tactical skills to more advanced applications in small unit operations. The focus shifts to mastering Army small unit tactics, battle drills and troop leading procedures, with Cadets taking on leadership roles in planning and executing tactical exercises such as patrol lanes, variable lanes and squad-level missions. Labs emphasize terrain analysis, time management and decision-making under pressure, simulating real-world battlefield conditions. This phase builds on MSCI 310 and MSCI 315, and is designed to prepare Cadets for success at advanced camp and eventual commissioning as Army officers.

Prerequisites: MSCI 310 and MSCI 315. Corequisites: MSCI 320.

MSCI 410 Seminar Command Management**Credits 3 Grading Basis** Graded

Studies in advanced leadership and management; how to assess organizational cohesion and develop strategies to improve it; examine and utilize components of the army's training cycle; methods of instruction, planning and conducting of classes to be taught at leadership lab; conduct of oral After-Action Reviews and preparation of written After-Action Reports; the counseling process for improving individual and group performance of subordinates.

Corequisites: MSCI 415.

MSCI 415 Senior Leadership Laboratory I**Credits 3 Grading Basis** Graded

In this course, Cadets begin transitioning from being mentored to becoming mentors and trainers themselves. The lab complements MSCI 410, which focuses on applying military case studies to decision-making in combat, crisis counseling and Army training management. Cadets take on battalion-level staff roles, plan and lead training events, and apply the Military Decision Making Process (MDMP), Army Writing Style, and Composite Risk Management. Leadership labs emphasize practical exercises in counseling, training meetings, operational graphics and mission essential task lists. This semester is designed to prepare Cadets for their future roles as commissioned officers by instilling the skills and mindset needed to lead effectively in dynamic and complex environments.

Prerequisites: MSCI 320 and MSCI 325. Corequisites: MSCI 410.

MSCI 420 Seminar on Officership Transition**Credits 3 Grading Basis** Graded

Introduction to Army management systems - personnel and logistics; the military's legal system and its application at the company level; the officer-NCO relationship; The Officer Efficiency Report and its support form; establishment of project time lines with milestones, goals and In-Progress Reviews; book review on a required leadership book from chief of staff Army required reading list.

Corequisites: MSCI 425.

MSCI 425 Senior Leadership Laboratory II**Credits 3 Grading Basis** Graded

In this course, Cadets focus on transitioning into their roles as future commissioned officers by applying advanced leadership principles in complex operational environments. This lab complements MSCI 420 and emphasizes ethical decision-making, understanding the law of land warfare and managing interactions with non-governmental organizations, civilians and host nation entities. Cadets serve in battalion staff positions and assist in mentoring and instructing junior Cadets (MS I-III), reinforcing their leadership and management skills. Key topics include supply and maintenance operations, military law, equal opportunity, sexual harassment prevention, financial planning and military social etiquette. The lab is designed to refine the professional competencies necessary for successful transition to active duty or reserve service as Second Lieutenants.

Prerequisites: MSCI 410 and MSCI 415. Corequisites: MSCI 420.

MSCI 480 Internship in Advanced Military Leadership**Credits 6 Grading Basis** Graded

Upper-level MSCI students are required to participate in a 5-week leadership internship. The internship provides practical application of leadership principles, techniques, styles and responsibilities. Each student receives a performance evaluation as a leader of a 10-30 person group for every accomplished task.

MSCI 490 Special Studies in Military Science**Credits 1-3 Grading Basis** Graded

Designed to give students an opportunity to do upper-division independent study in selected areas under the supervision of a faculty member in the Military Science department.

Applied Music (MUAP)

MUAP 113 Small Group Lessons**Credits 1 Grading Basis** Graded

Two hours per week. May be repeated. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 114 Group Voice for Musical Theatre Concentrators**Credits 1 Grading Basis** Graded

First level voice class for Musical Theatre concentrations in the School of Theatre and Dance. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 115 Group Voice for Theatre/Dance Majors**Credits 1 Grading Basis** Graded

May be repeated for up to four credits. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 200 Secondary/Minor Applied Music Lesson**Credits 1-3 Grading Basis** Graded

This course consists of applied weekly lessons for non-majors and secondary instruments. See MyMadison for details. May be repeated for credit. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 205 Small Group Voice for Keyboard Music Majors**Credits 2 Grading Basis** Graded

A practical introduction to singing technique and musicianship designed for future choral music teachers and accompanists. Concludes with Vocal Proficiency Examination. Limited to maximum of eight students. May be repeated for credit. Permission to register must be obtained from the coordinator of the respective applied area.

Prerequisites: Sophomore standing in music or permission of the instructor.

MUAP 214 Private Voice for Musical Theatre Concentrators**Credits 1 Grading Basis** Graded

Private voice lessons for musical theatre concentrators in the School of Theatre and Dance. Permission to register must be obtained from the coordinator of the respective applied area.

Prerequisites: Musical Theatre concentration or Musical Theatre majors only.

MUAP 234 University Chorus: Tenor/Bass (UCTB)**Credits 1 Grading Basis** Graded

The James Madison University Chorus: Tenor/Bass (UCTB) provides an opportunity for singers to explore choral repertoire for lowered and mixed voices from a variety of genres and time periods. This ensemble is open to all JMU tenor and bass singers without audition. The director reserves the right to limit membership because of balance or space considerations.

MUAP 235 Treble Chamber Choir**Credits 1 Grading Basis** Graded

This is an advanced level chamber chorus for women and male countertenors interested in a small choral ensemble experience. Open to all majors, and auditioned at the beginning of every semester, this choir frequently performs off campus as well. Contact the director of choral activities for more information. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 236 University Chorus: Soprano/Alto (UCSA)**Credits 1 Grading Basis** Graded

The James Madison University Chorus: Soprano/Alto (UCSA) ensemble provides an opportunity for singers to explore choral repertoire for treble and mixed voices from a variety of genres and time periods. This ensemble is open to all JMU soprano and alto singers without audition. The director reserves the right to limit membership because of balance or space considerations.

MUAP 237 Marching Band**Credits 1-2 Grading Basis** Graded

The Marching Band will perform music and drill which is artistically structured. The repertoire will be representative of all styles of music. The Marching Band is required for two years of wind and percussion majors in the music education degree program. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 238 Concert Band**Credits 1 Grading Basis** Graded

Open to all interested participants. A wide variety of music is utilized to acquaint the student with different types of band literature. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 239 Symphonic Band**Credits 1 Grading Basis** Graded

The JMU Symphonic Band is a select group of brass, woodwind and percussion students who are dedicated to the performance of both traditional and contemporary band literature. The ensemble performs music of all periods and is open to any university student by audition. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 240 Pep Band**Credits 1 Grading Basis** Graded

Pep Band is an elective ensemble that serves as a spirited athletic band for JMU men's and women's home and post-season basketball games. The unit will rehearse weekly and perform at the Atlantic Union Bank Center during both regular and post-season games. Students must have instrumental music experience. The class only accepts the following instruments: alto sax, tenor sax, trumpet, mellophone, baritone, tuba (sousaphone) and rhythm section (drumset, guitar and electric bass). Wind players are non-auditioned; rhythm section members must audition in the fall semester prior to registering.

MUAP 300 Primary/Major Applied Music Lesson**Credits 1-3 Grading Basis** Graded

This course consists of applied weekly lessons on the primary instrument for music majors. Students must have successfully passed an audition for the studio for which they register. See MyMadison for details. May be repeated.

Prerequisites: Music majors only.

MUAP 301 Assigned Ensemble**Credits 1 Grading Basis** Graded

This course fulfills the assigned ensemble requirement for all music majors. Music majors taking Applied Lessons (MUAP 300) are assigned an ensemble by audition. Music majors not taking applied lessons (MUAP 300) may enroll in a non-auditioned ensemble for assigned ensemble credit, or they may audition for one of the auditioned ensembles. Course can be repeated for credit.

MUAP 340 Chorale**Credits 1 Grading Basis** Graded

A highly selected, 60-voice mixed choir that performs repertoire from the Renaissance to the contemporary era, both sacred and secular. There are opportunities to perform on and off campus. Membership is by audition. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 341 Madison Singers**Credits 1 Grading Basis** Graded

Madison singers is a highly select choral chamber ensemble that performs the best of representative choral music from the Renaissance to the present. The ensemble is open to any university student by audition in the fall of each academic year. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 343 Opera Theatre**Credits 1-2 Grading Basis** Graded

The preparation and public performance of grand opera, light opera and musicals. Work will include coaching of both music and acting. Credit may vary with permission of instructor depending on the amount of time commitment. Admission is by audition only. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 344 Chamber Orchestra**Credits 1 Grading Basis** Graded

Open to all university students. Membership is determined by audition. Music written for chamber orchestra from all periods is studied and performed.

MUAP 345 Symphony Orchestra**Credits 1 Grading Basis** Graded

The JMU Symphony Orchestra is a participating member of the American Symphony Orchestra League. Membership is determined by audition and is open to all university students. Literature performed is from the standard symphonic repertoire.

MUAP 346 Wind Symphony**Credits 1 Grading Basis** Graded

The JMU Wind Symphony is a highly select group of brass, woodwind and percussion students who are dedicated to the performance of wind orchestra music and chamber music for winds. The ensemble performs music from all periods and is open to any university student by audition in the fall of each academic year. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 347 Jazz Ensemble**Credits 1 Grading Basis** Graded

Instrumental music performance ensemble of the standard "big band" instrumentation whose repertoire reflects jazz styles from the swing era to contemporary jazz. Admission is by audition. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 348 Jazz Band**Credits 1 Grading Basis** Graded

Instrumental ensemble, performing the standard and contemporary repertoire of American music with emphasis on the jazz idiom. Jazz band is open to all JMU students by audition. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 349 Monticello Strings Faculty and Students Ensemble**Cross-listed with:** MUAP 649**Credits 1 Grading Basis** Graded

A joint string ensemble of string faculty and students, with a mission to provide mentoring, hands-on and shared experiences in learning, rehearsing, and performing string chamber orchestra repertoire from the Baroque to Contemporary periods. In addition, the ensemble focuses on creative and scholarly endeavors, collaboration between faculty and students, and university outreach with performances on and off campus. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 350 String Ensemble**Credits 1 Grading Basis** Graded

A rehearsal and performance ensemble where students of similar technical and musical ability are grouped in traditional string or piano trios, quartets and quintets, and present public performances of important compositions from the chamber music literature of all historical periods. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 351 Woodwind Ensemble**Credits 1 Grading Basis** Graded

Woodwind ensembles consisting of quintets or other smaller and larger combinations, limited to specially selected personnel through auditions. Concerts and other performances are prepared from a variety of literature from the classical period to the present. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 352 Brass Band**Credits 1 Grading Basis** Graded

Open to all university students by audition. A select brass and percussion ensemble limited to the standard instrumentation of the "British-style" brass band. The ensemble performs literature of all styles from the extensive brass band tradition. The band typically participates in the annual NABBA championships. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 353 Guitar Ensemble**Credits 1 Grading Basis** Graded

The performance of guitar music from Renaissance to 20th century for duos, trios and quartets. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 354 Percussion Ensemble**Credits 1 Grading Basis Graded**

The study and preparation for public performance of percussion ensemble literature. The ensemble is open to all university students by audition. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 355 Jazz Chamber Ensemble**Credits 1 Grading Basis Graded**

Open by audition to students demonstrating a high degree of skill in jazz improvisation. Concerts and other performances are prepared from a variety of literature from early jazz to the present day. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 357 Collaborative Piano**Credits 1 Grading Basis Graded**

A course in basic collaborative skills and style characteristics for pianists. Students will be assigned to vocal or instrumental studios and have the opportunity to perform in master class and private coaching sessions.

Prerequisites: A minimum grade of "C-" in MUS 303 or MUS 304.

MUAP 360 Opera/Music Theatre Orchestra**Credits 1 Grading Basis Graded**

This ensemble will serve as the accompanying ensemble for staged productions within the School of Music and the School of Theatre and Dance. Enrollment in this course will be open to all students at the university by audition. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 362 Brass Chambers Ensembles**Credits 1 Grading Basis Graded**

Brass chamber ensembles consisting of quartets, quintets or larger combinations, limited to specially selected personnel through auditions. Concerts and other performances are prepared and presented from literature spanning the Renaissance to the present day. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 364 Camerata Strings**Credits 1 Grading Basis Graded**

Camerata Strings is a select instrumental string ensemble that performs a broad range of string ensemble literature from the 1600s to the present. The ensemble is open to any student of the university through competitive auditions held at the start of each spring semester. Permission to register must be obtained from the coordinator of the respective applied area.

MUAP 380 Early Music Ensembles**Credits 1 Grading Basis Graded**

The Early Music Ensembles at JMU are vocal and/or instrumental ensembles dedicated to the historically-informed performance of early music (music composed before ca. 1700) Repertoire includes major works primarily of the Renaissance and early Baroque era. Membership is by audition or invitation. May be repeated for credit.

MUAP 381 Bach Aria Group**Credits 1 Grading Basis Graded**

The Bach Aria Group is a select vocal/instrumental ensemble dedicated to the historically-informed performance of arias and small ensembles from the Cantatas, Oratorios and Passions of J.S Bach. Repertoire is specifically limited to this material. Membership is voluntary, and by audition or invitation. May be repeated for credit. This ensemble will not fulfill the music major requirement for a minimum of one ensemble per semester. Permission to register must be obtained from the coordinator of the respective applied area.

Music (MUS)

MUS 100 Keyboard Skills**Credits 1 Grading Basis Graded**

A beginning level keyboard skills class for music majors. Includes basic techniques, sight reading, elementary repertoire and functional theory-related keyboard skills.

Prerequisites: Music majors only.

MUS 101 Keyboard Skills II**Credits 1 Grading Basis Graded**

An elementary, but not beginning, level keyboard skills class for music majors. Includes basic techniques, sight reading, elementary repertoire and functional theory-related keyboard skills.

Prerequisites: Music majors only. MUS 100 or placement by audition/interview.

MUS 120 Diction for Singers**Credits 1 Grading Basis Graded**

The study of English and Italian pronunciation using the phonetic alphabet with emphasis on the performance of song literature in each language. Open to music majors and minors or by permission of instructor.

Prerequisites: Music majors only.

MUS 121 Diction for Singers II**Credits 1 Grading Basis Graded**

The study of French and German pronunciation using the phonetic alphabet with emphasis on the performance of song literature in each language.

Prerequisites: Music majors or minors only.

MUS 131 Fundamentals of Music**Credits 3 Grading Basis Graded**

Covers basic information in music theory, basic ear-training and sight-singing skills, and basic keyboard skills designed for the non-music major.

MUS 140 Introduction to Music Theory and Aural Skills**Credits 4 Grading Basis Graded**

Designed for non-music majors, music minors, musical theater majors, and students wishing to take songwriting, this course introduces students to the fundamental materials of music, both in written and aural form. Topics covered include: notation, rhythm and meter, key signatures, scales, intervals, triads, seventh chords, lead-sheet symbols, cadence, basic harmonic progressions, and form in popular music. This course may not be taken as a substitute for MUS 141 or MUS 143/MUS 144.

MUS 141 Foundations of Music Theory**Credits 3 Grading Basis** Graded

In this course students will master elements of musical literacy, including such concepts as rhythm and meter, clef reading, scales, key signatures, intervals, triads, and seventh chords. Placement in this course is determined by diagnostic exam and recommendation from the music theory and composition area in the School of Music.

MUS 143 Aural Skills I**Credits 1 Grading Basis** Graded

Study of ear training and sight singing involving diatonic materials. Includes melodic, harmonic and rhythmic dictation; singing with moveable Do and Do-based minor solmization; and improvisation in classical and blues styles.

MUS 144 Aural Skills II**Credits 1 Grading Basis** Graded

Continuation of MUS 143. Study of ear training and sight singing involving diatonic and chromatic materials. Includes melodic, harmonic and rhythmic dictation of excerpts involving tonicization and modulation; solo, ensemble and accompanied singing with moveable Do and Do-based minor solmization; and identification, description and discussion of small forms in a variety of genres.

Prerequisites: MUS 143.

MUS 150 Introduction to Technological Applications in Music**Credits 1 Grading Basis** Graded

This course is designed to provide students with an overview of multiple applications of technology in music teaching. Topics to be covered include, but are not limited to: notation software, instructional software (including creative, practice, and performance), MIDI, productivity applications, and digital audio/recording.

MUS 195 Recital Attendance**Credits 0 Grading Basis** Credit / No Credit

Required of all music majors for the first six semesters of their enrollment as a major. Successful completion of the course requires attendance at a specified number of approved recitals/concerts each semester. For complete course guidelines, please refer to the Undergraduate Music Student Handbook.

MUS 200 Music and Culture [C2VPA]**Credits 3 Grading Basis** Graded

This course focuses on the cultural and historical contexts of music, primarily of the European and American musical traditions, past and present. It is designed to increase the student's perceptual ability in listening to music and to encourage an interest in both familiar and unfamiliar music. Specific musical focus will vary by section. May be used for general education credit. May not be used for music major credit.

MUS 202 Keyboard Skills III**Credits 1 Grading Basis** Graded

An intermediate level course in functional keyboard skills for music majors. Includes sight reading, transposition, harmonization, improvisation, playing by ear and accompanying techniques.

Prerequisites: Music Majors only. MUS 101 or placement by audition/interview.

MUS 203 Music in America [C2VPA]**Credits 3 Grading Basis** Graded

Knowledge and skills to increase the student's perceptual ability in music listening with a survey of American music; examining relationships between popular and classical music styles. May be used for general education credit. May not be used for major credit.

MUS 204 History of Rock**Credits 3 Grading Basis** Graded

History of rock is a music appreciation course designed to develop an understanding of both musical and cultural roots of rock music and the ability to hear a direct relationship between those roots and current popular music. A major component of the course is a survey of the history of 20th century American popular music.

MUS 206 Introduction to Global Music [C2VPA]**Credits 3 Grading Basis** Graded

A survey of various world music traditions, including those of Asia, the Pacific, Europe, Africa and the Americas. The course will focus on aesthetics, musical forms and styles, and the relationship between music and other arts. Emphasis will be placed on historical, religious, and cultural events and their influence on the creation and development of music. May be used for general education credit.

MUS 207 History of Western Music**Credits 4 Grading Basis** Graded

Students in this course will explore the historical and cultural contexts of European and American art music from antiquity to the present, and will study a wide variety of musical repertoire as they do so.

MUS 220 Assessment Tests - School of Music**Credits 0 Grading Basis** Non-Graded Component

Required, non-credit School of Music Assessment Test course which is to be taken before the Bachelor of Music degree is received, normally in the final spring semester of senior year.

Prerequisites: Music majors with senior standing.

MUS 240 Jazz Improvisation Laboratory I**Credits 2 Grading Basis** Graded

Presents the fundamentals of improvisation in both jazz and popular musical styles. The class emphasis will be on creative work although some theory and chord nomenclature will be taught. Both vocal and instrumental musicians will be permitted to enroll, including both the general student and the music major.

Prerequisites: Music major or jazz minor or permission of the instructor.

MUS 241 Diatonic and Chromatic Harmony**Credits 3 Grading Basis** Graded

This course investigates harmony and phrase structure in common-practice works. Students will develop an awareness for common harmonic progressions and syntax and an understanding of such chromatic harmonic practices as tonicization and modulation. Students should complete the music theory placement test before enrolling in this course. Please contact the theory area coordinator to take the placement test.

MUS 243 Aural Skills III**Credits 1 Grading Basis** Graded

This course is a continuation of MUS 144. Students continue their study of sight singing and ear training with diatonic and chromatic materials through exercises in melodic, harmonic and rhythmic dictation, and singing of melodic exercises with solmization.

Prerequisites: MUS 144.

MUS 244 Aural Skills IV**Credits 1 Grading Basis** Graded

This course is a study of sight singing and ear training with modal and atonal materials through melodic, harmonic and rhythmic dictation exercises and singing of melodic exercises with solmization.

Prerequisites: MUS 144.

MUS 250 Form and Analysis: Baroque and Classical**Credits 3 Grading Basis** Graded

This course has two main goals: To familiarize students with Baroque-era imitative forms and Classical-era forms, including binary, ternary, sonata and concerto, among others; and to continue students' study of chromatic harmony.

Prerequisites: A grade of "C-" or higher in MUS 241.

MUS 251 Form and Analysis: Romantic**Credits 3 Grading Basis** Graded

This course has two main goals: To familiarize students with Romantic-era developments in forms such as binary, ternary, sonata and concerto, among others; and to continue students' study of chromatic harmony.

Prerequisites: A grade of "C-" or higher in MUS 241.

MUS 252 Analysis of Music 1900-1950**Credits 3 Grading Basis** Graded

An investigation of music from the first half of the twentieth century. Students will gain an understanding of compositional techniques and formal processes in atonal and centric music and an awareness of compositional trends during the time period.

Prerequisites: A grade of "C-" or higher in MUS 141 or MUS 241.

MUS 253 Analysis of Music 1950-Present**Credits 3 Grading Basis** Graded

An investigation of music from the second half of the twentieth century and the beginning of the twenty-first century. Students will gain an understanding of compositional techniques and formal processes and an awareness of compositional trends during the time period.

Prerequisites: A grade of "C-" or higher in MUS 141 or MUS 241.

MUS 254 Form and Analysis: Popular Music**Credits 3 Grading Basis** Graded

An investigation of popular music in the twentieth and twenty-first centuries. Students will develop an understanding of formal, harmonic, metric, rhythmic and timbral aspects of popular music in a variety of eras as well as an awareness of its historical trends.

Prerequisites: A grade of "C-" or higher in MUS 141 or MUS 241.

MUS 255 Jazz Theory**Credits 3 Grading Basis** Graded

This course introduces students to the basic elements of jazz melodic, rhythmic and harmonic structure and style. Written analytical techniques and aural skills development will be emphasized.

Prerequisites: MUS 241.

MUS 256 Contrapuntal Techniques**Credits 3 Grading Basis** Graded

Study of counterpoint through analysis and composition. Although the course emphasizes eighteenth-century counterpoint, other kinds of counterpoint (such as sixteenth- or twentieth-century) may also be covered.

Prerequisites: A grade of "C-" or higher in MUS 241.

MUS 257 Techniques of Composition**Credits 3 Grading Basis** Graded

This course will familiarize students with various techniques of contemporary (i.e. 20th-21st century) composition, including working with various pitch materials, rhythmic devices, formal considerations, writing for various instruments, voices and ensembles, and more.

Students will apply these techniques to compositional etudes throughout the semester, listen to relevant repertoire and create larger projects that will allow for more individual exploration. Private lessons with a teaching assistant and/or composition faculty will be included. A class recital will be held at the end of the semester, if possible. This course may be used to fulfill part of the core music theory requirement. It may be repeated, as the content may vary, but only one instance would count toward the core requirement. Any further instances would be eligible for elective credit only.

Prerequisites: MUS 241 or permission of instructor. Students enrolled in MUS 352 or declared as composition concentrators are not eligible for this course.

MUS 258 Film Music Analysis**Credits 3 Grading Basis** Graded

This course will familiarize students with various techniques of film music from the past to the present, including functional/non-functional harmonies, phrase constructions, forms, instrumentations and more. Students will apply their knowledge to several analytical tasks such as transcriptions, short analyses and critiques, watch relevant films, and write final term papers that will allow for more individual exploration.

Prerequisites: A grade of "C-" or higher in MUS 241.

MUS 303 Keyboard Skills IV**Credits 1 Grading Basis** Graded

A moderately advanced course in functional keyboard skills for music majors. Includes sight reading, transposition, harmonization, improvisation, playing by ear and accompanying techniques. Designed to prepare music major students for required Keyboard Proficiency Examination.

Prerequisites: Music majors only. MUS 202 or placement by audition/interview.

MUS 304 Keyboard Skills IV for Pianists**Credits 1 Grading Basis** Graded

This is a keyboard skills course for music majors whose primary instrument is piano. It includes sight reading, transposition, harmonization, improvisation, playing by ear and accompanying techniques. This course is designed to prepare music major students for the required Keyboard Proficiency Examination.

Prerequisites: Music majors only.

MUS 305 Jazz Keyboard Skills**Credits 1 Grading Basis** Graded

This course introduces and develops the keyboard skills necessary to construct piano accompaniments in the jazz idiom using a progression of chord symbols or a lead sheet. Successful completion of this course is required in order to enroll in upper level courses in the Jazz Studies major.

Prerequisites: MUS 101 or instructor permission.

MUS 317 Basic Conducting**Credits 2 Grading Basis** Graded

Designed to acquaint the student with the fundamental elements of conducting technique, such as beat patterns, cueing, expression, transposition and score reading, with an emphasis on applying these techniques in practical conducting experiences involving vocal and instrumental forces.

Prerequisites: MUS 241 and MUS 144 or junior standing.

MUS 318 Intermediate Choral Conducting**Credits 2 Grading Basis** Graded

Consists of the further application of basic conducting skills learned in MUS 317 to the choral situation with emphasis on baton techniques, score reading, and preparation and introduction to choral literature.
Prerequisites: MUS 317.

MUS 319 Intermediate Instrumental Conducting**Credits 2 Grading Basis** Graded

A continuation of basic conducting to develop skills in baton technique with emphasis on advanced literature for public school use. Included are score sight-reading skills and literature selection.
Prerequisites: MUS 317.

MUS 330 Skills for the Working Singer**Credits 1 Grading Basis** Graded

This course introduces a brief survey of skills required to develop a career in performance and help students navigate professional environments post-graduation. Topics include entrepreneurship skills, non-profit arts organizations and the evolving musical landscapes of the working world.
Prerequisites: Declared voice performance concentration.

MUS 343 Opera Workshop**Credits 1 Grading Basis** Graded

In this course, students develop performance skills through the introduction and application of acting and movement techniques in opera. Beginning with the exploration of various methods and pedagogies, students will work both individually and as a class to build their own acting technique which can be applied to any theatrical setting. Classroom work will consist of ensemble building, character development, staged presentations, stylistic exploration, period movement and creative exercises.

MUS 345 Small Ensemble Jazz Arranging**Credits 3 Grading Basis** Graded

This course introduces students to the techniques of arranging for two-horn, three-horn, and four-horn jazz ensembles. Students will study the classic repertoire of small jazz groups between 1930 and the present day, and create and record small ensemble arrangements in various styles.
Prerequisites: MUS 255, MUS 305 or permission of the instructor.

MUS 346 Large Ensemble Jazz Arranging**Credits 3 Grading Basis** Graded

This course introduces students to the techniques of arranging for large jazz ensemble ("big band"). Students will study representative works of seminal big band composers active between 1930 and the present day, and create and record a full big band arrangement.
Prerequisites: MUS 255, MUS 305, MUS 345, or permission of the instructor.

MUS 352 Music Composition**Credits 2-3 Grading Basis** Graded

Individual/seminar instruction in composition. Compositions are performed publicly. May be repeated.
Prerequisites: MUS 241.

MUS 356 History of Jazz**Credits 3 Grading Basis** Graded

A study of American jazz with particular emphasis on its practices with reference to principal performers and composers of jazz-style periods.

MUS 356M History of Jazz - Montreal**Credits 3 Grading Basis** Graded**MUS 357** Music Theatre History and Analysis**Cross-listed with:** THEA 357**Credits 3 Grading Basis** Graded

Survey of musical theatre genres, composers, lyricists, performers, directors and choreographers in America from 1750 to present. Includes practical study of the format of the libretto and musical score in relationship to the major musical theatre genres. Consideration of how music theatre developed from and reflected the cultural, social and political landscape of its time.
Prerequisites: Music majors with concentration in music theatre only or permission of the instructor.

MUS 371 Private Piano Pedagogy**Credits 2 Grading Basis** Graded

This course will cover the different approaches to piano pedagogy, the pros and cons of precollege-level methodologies and materials, and how to teach students of various ages and abilities. Observed teaching is also a component of this course.

MUS 380 Musicological Perspectives**Credits 3 Grading Basis** Graded

Students in this course will employ critical musicological perspectives as they explore a series of issues and concepts (socio-cultural, technical, ethical, theoretical, ethnographic, philosophical, etc.) of vital importance to musicians. Students will approach these issues via a wide variety of musical case studies taken from diverse genres, regions, periods and styles as they build upon knowledge gained in previous courses.
Prerequisites: MUS 206 and MUS 207.

MUS 381 Music Research Lab**Credits 1 Grading Basis** Graded

Students in this course will work to improve their music-specific research and writing skills.
Prerequisites: MUS 206 and MUS 207.

MUS 395 Junior or Senior Half Recital**Credits 0 Grading Basis** Credit / No Credit

Presentation of a half recital, representative in quality of passing performance level 6 or 7 as set by the pertinent performance area faculty. Memorization is required in those areas that traditionally expect it. Course graded on credit/no credit basis.
Prerequisites: Junior standing and approval of the performance faculty of the performer's area.

MUS 404 Advanced Keyboard Skills**Credits 1 Grading Basis** Graded

Designed for vocal music majors. The emphasis is on achieving fluency in reading vocal accompaniments, SATB open scores and incorporating solo voice parts with accompaniments in preparation for the Advanced Keyboard Skills Exam. This is a group class, which can be repeated.
Prerequisites: MUS 303.

MUS 420 Piano Technology I**Credits 1 Grading Basis** Graded

This course is an introduction to the acoustical and mechanical design and history of the piano. Students will learn the theory of tuning and temperaments and will be introduced to the procedures and techniques used in regulating and voicing pianos.

MUS 421 Piano Technology II**Credits 1 Grading Basis** Graded

Students will practice tuning pianos under the supervision of the instructor. This will allow for practical application of the skills learned in MUS 420. Required for students in the piano performance concentration. Other students may take the course as space allows.
Prerequisites: MUS 420.

MUS 429 Introduction to Intentional Music**Credits 1 Grading Basis** Graded

This course explores intentional uses of music to achieve outcomes with atypical populations. Students will observe intentional music activities in varied settings, study research-based applications of intentional music, and develop intentional music activities for case study populations. They will also reflect on their own musical identities, and those of potential clients, to develop skills in communication and planning.

MUS 430 Music and Human Services: Experiences and Practicum**Credits 2 Grading Basis** Graded

This course integrates psychological and cultural functions of music with activities and goals of human service organizations. Representatives of human services organizations are invited to explain their interventions with diverse populations. Students will be offered practicum experiences in these agencies. Through relating these experiences to the class, in seminar fashion, interdisciplinary practices will be illustrated.

MUS 435 Instrumentation**Cross-listed with:** MUS 535**Credits 3 Grading Basis** Graded

In-depth practicum on idiomatic writing for orchestral instruments both individually and in an ensemble setting. Students are introduced to the characteristics and limitations of each instrument with written exercises, listening and score study examples from significant orchestral works, demonstrations by representative instrumentalists and small-scale scoring projects.

Prerequisites: MUS 241 and MUS 352 (at least one semester), or permission of the instructor.

MUS 440 Jazz Improvisation Laboratory II**Credits 2 Grading Basis** Graded

This course presents intermediate to advanced improvisation skills in the jazz idiom alone. There is an emphasis on the theoretical analysis of chord progressions as well as on creative musical application. The course concludes by introducing some advanced musical improvisation concepts. This course may be repeated for credit.

MUS 440M Jazz Improvisation Lab II - Montreal**Credits 2 Grading Basis** Graded

This course presents intermediate to advanced improvisation skills in the jazz idiom alone. There is an emphasis on the theoretical analysis of chord progressions as well as on creative musical application. The course concludes by introducing some advanced musical improvisation concepts.

Summer SA Canada Montreal

MUS 441 Vocal Arranging and Technology**Credits 2 Grading Basis** Graded

This course examines arranging for vocal ensembles. Included will be fundamental concepts of orchestration developed with the use of appropriate technologies.

Prerequisites: MUS 144 and MUS 241.

MUS 442 Instrumental Arranging and Technology**Credits 2 Grading Basis** Graded

This course covers arranging for various instrumental ensembles, including fundamental concepts for orchestration through appropriate technology.

Prerequisites: MUS 244.

MUS 445 Orchestration**Cross-listed with:** MUS 545**Credits 3 Grading Basis** Graded

Survey of modern orchestration techniques building on the foundations established in MUS 435. Students will gain familiarity with current practice and nuances of scoring for large instrumental ensembles through examples from traditional and contemporary literature and small-scale scoring assignments. Students will focus on the preparation and production of a professional-quality score and orchestral parts leading to a reading session with the JMU Symphony Orchestra at the end of the term.

Prerequisites: MUS 435 or permission of the instructor.

MUS 446 Jazz Composition**Credits 2 Grading Basis** Graded

An introduction to techniques of jazz composition. Students will study classic jazz compositions and create original compositions utilizing various harmonic techniques.

Prerequisites: MUS 255 and MUS 305.

MUS 450 Topics in Music Analysis**Credits 3 Grading Basis** Graded

Analytical investigation of musical examples in a variety of styles. Topics vary. See department to request information about the topic for a given semester. Course may be repeated for different topics.

Prerequisites: MUS 244 and grade of "C-" or higher in any four of the following: MUS 141, MUS 241, MUS 250, MUS 251, MUS 252, MUS 253, MUS 254, MUS 256, MUS 257 or MUS 258.

MUS 460 Piano Literature I**Credits 2 Grading Basis** Graded

A survey of baroque and classical literature for the piano encompassing solo and concerto repertoire. An examination of literature for the clavichord, harpsichord and pianoforte.

MUS 465 Opera History and Literature**Credits 3 Grading Basis** Graded

A survey study of the history of opera. Consideration will be given to the chronological development of all forms of music theatre with an emphasis on style characteristics through aural identification.

Prerequisites: Any four of the following music theory courses: MUS 141, MUS 241, MUS 250, MUS 251, MUS 252, MUS 253, MUS 254 or MUS 256 or permission of the instructor. Students are encouraged to take MUS 250 or MUS 251 prior to this class.

MUS 466 Special Topics in Solo Vocal Literature**Credits 3 Grading Basis** Graded

This course is a study of selected topics in solo vocal literature. Topics vary each semester. The course may be repeated for credit when the topic changes.

Prerequisites: MUS 207 and any four of the following music theory courses: MUS 141, MUS 241, MUS 250, MUS 251, MUS 252, MUS 253, MUS 254, MUS 255, MUS 256, MUS 257 or MUS 258, or permission of the instructor.

MUS 467 Song Literature**Credits 3 Grading Basis** Graded

A survey of vocal art-song literature in Western culture.

Prerequisites: Any four of the following music theory courses: MUS 141, MUS 241, MUS 250, MUS 251, MUS 252, MUS 253, MUS 254 or MUS 256 or permission of the instructor.

MUS 470 Piano Literature II**Credits 2 Grading Basis** Graded

A survey of romantic, impressionistic and 20th-century literature for the piano, including solo and concerto repertoire, with an emphasis on stylistic trends of the 20th century.

MUS 471 Piano Literature III**Credits 2 Grading Basis** Graded

A survey of piano literature from the middle of the 20th century to the present day. In addition to studying the most well-known works, the class will explore and evaluate piano music by composers who may be considered underrepresented and marginalized by reasons of ethnicity, gender, political or sexual identity.

MUS 472 Instrumental Pedagogy**Credits 1 Grading Basis** Graded

Presentations of instrumental methods, solo and ensemble literature related to the instrumental performer's own major area. Private instruction approaches and techniques are also considered with particular reference to the beginning player.

MUS 473 Jazz Ensemble Procedures and Techniques**Credits 2 Grading Basis** Graded

This course addresses all aspects of instrumental jazz instruction (big band and small group). Teaching philosophies, rehearsal techniques and resource materials will be examined; the syllabus includes opportunities to observe and rehearse jazz groups. Enrollment is not limited to traditional jazz instrumentation. Required for the jazz studies concentration and recommended as an elective for music education students.

MUS 477 Foundations of Vocal Pedagogy**Credits 1 Grading Basis** Graded

This course offers an in-depth exploration of classical vocal pedagogy, tailored for aspiring vocal instructors and singers. Students will delve into vocal anatomy, historical techniques and effective teaching methodologies. Key topics include vocal technique, vocal mechanics, repertoire selection, instruction methods and performance practice. Interactive lectures and practical workshops will equip students with the skills to inspire and guide future vocalists. Ideal for those with a foundational understanding of vocal performance, this course refines pedagogical skills and fosters a deep appreciation for classical singing.

MUS 478 Styles of Vocal Pedagogy**Credits 1 Grading Basis** Graded

Embark on an 8-week journey into modern and musical theater vocal pedagogy, designed for vocal instructors and performers eager to master contemporary techniques and teaching strategies. This course covers essential aspects of modern vocal performance and pedagogy, and key topics include vocal techniques, contemporary styles, repertoire development, teaching approaches and performance skills. Through interactive sessions and hands-on workshops, students will gain practical experience and develop a personalized teaching philosophy. This course is perfect for those with a basic understanding of vocal performance seeking to enhance their pedagogical skills and passion for modern and musical theater.

Prerequisites: A minimum grade of "C-" in MUS 477.

MUS 480 Advanced Seminar in Musicological Topics**Credits 3 Grading Basis** Graded

An intensive study of a single topic in musicology or ethnomusicology. Topics change each semester, and may include studies of a specific musical issue (performance practice, etc.), a single composer's music, a single musical genre (the string quartet, etc.), or music at a specific time and/or place (music in fin de siècle Paris, music in modern South Africa, the second Viennese school, etc.). Fulfills the College of Visual and Performing Arts writing-intensive requirement for the major. May be repeated for credit.

Prerequisites: MUS 380 and MUS 381.

MUS 485 Advanced Jazz Topics Seminar**Credits 3 Grading Basis** Graded

An intensive study of a single topic in jazz studies. Topics change each semester, and may include studies of a specific musical issue (performance practice, etc.), a single composer's or performer's music (Duke Ellington, Miles Davis, etc.), a single musical genre (the development of big band style, etc), or a sociological study (jazz in Europe, jazz and American culture, etc.). May be repeated for credit.

Prerequisites: MUS 356 or permission of the instructor.

MUS 490 Special Studies in Music**Credits 1-3 Grading Basis** Graded

Designed to give superior music students an opportunity to complete independent study under faculty supervision.

MUS 493 B.A. Capstone Project**Credits 1-3 Grading Basis** Graded

This course is intended for students pursuing the Bachelor of Arts in Music. Students enroll in this course at the end of their degree program, working closely with a faculty mentor and a committee selected by the student in consultation with their mentor to complete their project. Repeatable. Must be taken for a total of three credit hours.

MUS 495 Senior Graduation Recital**Credits 1 Grading Basis** Graded

Presentation of a full recital, with the quality of performance to meet standards for admission to graduate school master's of music programs. Memorized recital presentations will be required in those areas which traditionally demand them.

Prerequisites: Senior standing.

MUS 497 Senior Project in Theory**Credits 1 Grading Basis** Graded

Analytical paper or other topic approved by the theory-composition area. Enrollment is for students who want to do graduate work in music theory.

MUS 498 Selected Topics in Music**Credits 1-3 Grading Basis** Graded

Courses in music or music education which are of a topical nature. This may be repeated.

MUS 499 Honors**Credits 1-3 Grading Basis** Graded

This is a year long course.

Music Education (MUED)

MUED 200 Small Ensemble for Instrumental Music Education

Credits 0 Grading Basis Credit / No Credit

This course will fulfill the NASM requirement for instrumental music education majors to gain experiences in small ensembles. Students enrolled in this course may complete the requirement through a variety of experiences approved by their academic advisor, including participation in String Ensembles, Camerata Strings, Collegium Musicum, Woodwind Ensembles, Guitar Ensemble, Percussion Ensembles, Jazz Chamber Ensembles, Collaborative Piano and Brass Chamber Ensembles.

MUED 201 Small Ensemble for Vocal Music Education

Credits 0 Grading Basis Credit / No Credit

This course will fulfill the NASM program requirement for vocal music education majors to gain experiences in small ensembles. Students enrolled in this course may complete the requirement through a variety of experiences approved by their academic advisor, including participation in Treble Chamber Choir, Men's and Women's Chamber Choirs, Bach Aria Group, Madison Singers, Collegium Musicum or Opera Theatre.

MUED 206 Instrumental Music Methods for Vocal Music Education

Credits 1 Grading Basis Graded

Provides vocal track music education majors with experiences, methods, and techniques for instrumental music instruction and a fundamental knowledge of and proficiency on woodwind, brass, percussion and string instruments.

MUED 271 Music Education: A Professional Choice

Credits 1 Grading Basis Graded

Students engage with ethical, philosophical and pragmatic issues surrounding their choice to pursue professional licensure in PK-12 music education. This course includes observation of learning and teaching in elementary, middle, and high school music classrooms, as well as an introduction to the application of National Core Music Standards, Virginia Standards of Music for Learning, JMU's Conceptual Framework and JMU's Eight Key Questions. Course assignments and projects contribute to continued development of a digital portfolio.

MUED 273 Music Education: Professional Practice

Credits 1 Grading Basis Graded

Second course in the sophomore music education sequence. Builds on the foundation of philosophy, history and psychology of music teaching established in the first semester of the sequence (MUED 271) with an emphasis on the application of foundational knowledge to planning and leading instruction. Continued observation in PreK-12 and other settings with opportunities to teach.

Prerequisites: MUED 271 or permission of instructor.

MUED 275 Practicum in Music Education

Credits 1 Grading Basis Graded

This course will provide students with practicum experience in elementary general music and secondary ensemble and/or secondary general music in a VA public or private school, paired with a cooperating teacher. Students will be expected to observe and practice their craft site once per week for the semester they are enrolled. Students will have practical applications through assisting with rehearsals, lessons and/or teach an activity during a class or rehearsal.

Prerequisites: MUED 271 and MUED 273, or permission of instructor.

MUED 277 MUED Teaching Lab

Credits 1 Grading Basis Graded

This course actively engages MUED students in the recruitment, planning of and engagement with PK-12 students from the community in music making activities. Students will work with faculty and graduate assistants regarding every step of the process of communication, logistics, programming and teaching.

Prerequisites: MUED 271 and MUED 273. Corequisites: MUED 273.

MUED 301 Woodwind Techniques

Credits 1 Grading Basis Graded

Instruction in the basic skills of playing and teaching standard woodwind instruments in a heterogeneous class situation. Various methods for woodwind teaching will be studied, and materials used in public school teaching will be examined and performed.

Prerequisites: Sophomore standing.

MUED 303 Brass Techniques

Credits 1 Grading Basis Graded

Instruction in the basic skills of playing and teaching standard brass instruments in a heterogeneous class situation. Various methods for brass teaching will be studied, and materials used in public school teaching will be examined and performed.

Prerequisites: Sophomore standing.

MUED 305 Percussion Techniques

Credits 1 Grading Basis Graded

Instruction in the basic skills of playing and teaching standard percussion instruments in a heterogeneous class situation. Various methods of percussion teaching will be studied, and materials used in public school teaching will be examined and performed.

Prerequisites: Sophomore standing.

MUED 307 String Techniques

Credits 1 Grading Basis Graded

Instruction in the basic skills of playing and teaching string instruments. Instruction will be on violin, viola, cello, and bass in a heterogeneous class situation. Various methods for string teaching will be studied, and materials used in public school teaching will be examined and performed.

Prerequisites: Sophomore standing.

MUED 310 Vocal Techniques

Credits 1 Grading Basis Graded

Class instruction designed to acquaint the instrumental (nonvoice) major with fundamentals of vocal and choral techniques including posture, breath support, basic vocal production, physiological functions of the vocal mechanism, singer's diction, vocal exercises, and individual as well as ensemble performances.

Prerequisites: Music majors only.

MUED 311 Ensemble Techniques

Credits 1 Grading Basis Graded

This class focuses on honing students' secondary instrument playing skills learned in techniques courses. This course instructs students on further rehearsal skills and potential related practicum experiences in the context of a laboratory ensemble setting.

Prerequisites: MUED 271, MUED 273, MUED 301, MUED 303, MUED 305, MUED 307 and MUS 317.

MUED 371 Beginning Methods and Materials for Instrumental Music

Credits 2 Grading Basis Graded

Methods and materials for beginning through intermediate instrumental music students. Administrative concerns are included.

Prerequisites: MUED 273, full admission to teacher education and level 3 in major applied area.

MUED 372 General Music Practices**Credits 2 Grading Basis** Graded

Focuses on broad preparation for teaching the general music courses now found at both middle and high school levels.

Prerequisites: MUED 273, full admission to teacher education and level 3 in major applied area.

MUED 373 Advanced Methods and Materials for Instrumental Music**Credits 2 Grading Basis** Graded

Learning experiences related to the career needs of school instrumental music teachers are analyzed, discussed and practiced. Planning and teaching skills are presented for beginning, intermediate and advanced level students.

Prerequisites: MUS 317 and MUED 271.

MUED 376 Choral Music Materials and Techniques**Credits 2 Grading Basis** Graded

This course is an in-depth summary of the pedagogy, principles, methodology and resources in choral music education, with special emphases on high school choral teaching and community outreach. This course is designed to aid students in synthesizing their cumulative music education learning into a defined and refined skill set in advance of student teaching and in preparation for a music teaching career. An embedded practicum experience at an area retirement center is an integral component of this course. The level of this intergenerational choir parallels that of a 9th-grade choral ensemble; as such it provides an ideal, hands-on opportunity to teach choral music, leading to a culminating concert.

Prerequisites: MUS 317 and MUED 273, full admission to teacher education and Level 4 in major applied area.

MUED 380 Music in the Elementary School**Credits 2 Grading Basis** Graded

The general music program in the elementary school presented for future music specialists, K-6. Focus is on the synthesis of current philosophy, learning theories and educational practices for teaching elementary school music. Preparation for organizing music curricula and daily lesson plans is included.

Prerequisites: MUED 273, full admission to teacher education and level 3 in major applied area.

MUED 431 Psychology of Music**Cross-listed with:** MUED 531**Credits 3 Grading Basis** Graded

This course examines the topic of music psychology, and explores current findings and related subtopics, including those from brain sciences. Students will observe the practical application of research findings via practicum placements. Students will reflect on, and synthesize their content knowledge of the topic with their personal field experience knowledge for the purpose of making connections to their own career goals for future employment (e.g. music classroom and music therapy settings).

MUED 470 Marching Band Procedures**Credits 1 Grading Basis** Graded

This course covers the skills and knowledge needed to organize, administer, plan and teach marching band shows, including shows for various competitions, parades, football, basketball and festival events; and techniques for developing both marching and playing style through a functional method of fundamental drills.

Prerequisites: MUED 271 or permission of instructor.

MUED 471 School Musical, Jazz and Show Choir Procedures**Credits 2 Grading Basis** Graded

Covers skills and concepts needed to organize, administer, plan, teach and perform in musicals, jazz choirs and show choirs. Highlights techniques of commercial and theatrical vocal style, fundamentals of producing a musical and choreography for the show choir.

Prerequisites: MUS 317 and MUED 271.

MUED 472 Survey of String Orchestra Repertoire**Credits 2 Grading Basis** Graded

An examination of concert repertoire for string and full orchestra appropriate for performance by students in grades seven through 12. The course will include a study of evaluation and selection of music appropriate for a specific ensemble.

Prerequisites: MUED 271 or permission of instructor.

MUED 474 Teaching Popular Music and Guitar in the Classroom**Credits 2 Grading Basis** Graded

This methods course prepares music educators to teach guitar, modern band, alternative strings and other nontraditional courses and ensembles in school settings. Students develop guitar pedagogy fundamentals and chordal literacy while integrating music technology. Group music making includes jamming, beat-making and songwriting. Grounded in culturally responsive and sustaining pedagogy, the course emphasizes practical teaching strategies using rock, blues, pop and roots music repertoire to engage all skill levels and learners. Through hands-on guitar performance, ensemble work and technology integration, participants experience curriculum from both student and teacher perspectives, developing skills to attract and retain larger numbers of engaged music students in secondary programs. No prior guitar experience required.

MUED 480 Student Teaching**Credits 6-12 Grading Basis** Credit / No Credit

Enables students to apply, in the public school classrooms and the comprehensive child development programs, those skills and attitudes acquired in all components of teacher education. Under the guidance of university supervisors, students are provided activities designed to familiarize them with the classroom teacher's role.

Prerequisites: PSYC 160 and EDUC 200, appropriate methods courses, and permission of the coordinator of field experiences.

MUED 482 Orff and Kodaly: Literature, Principles and Practices**Credits 1 Grading Basis** Graded

Students will study the repertoire employed in the Orff and Kodaly approaches to music education. They will develop skills and understanding of the principles related to these approaches. Work with peers and school children will provide the opportunity to develop teaching skill. May be repeated for credit.

MUED 485 Teaching Music to Students with Special Needs**Cross-listed with:** MUED 585**Credits 3 Grading Basis** Graded

Students will become familiar with trends affecting students with special needs, and introduced to diverse variations in language ability, assessment, inclusion, and music education. Students will collaborate online through discussion forums regarding case study and video presentations. Students will become knowledgeable in language, motor, visual, and social development, connecting current research and theory to classroom practice. Students will also critically assess inclusive school settings.

Music Industry (MUI)

MUI 221 Survey of Music Industry

Credits 3 Grading Basis Graded

An overview of the recording, entertainment and performing arts industries including an examination of the historical, aesthetic and commercial developments of the music industry in the United States.

MUI 230 Music Industry Gateway

Credits 0 Grading Basis Credit / No Credit

Music industry gateway is required to enroll in 300- and 400-level music industry courses. May be repeated once.

MUI 231 Legal Aspects of Music Industry

Credits 3 Grading Basis Graded

An examination of the legal issues affecting the performing arts, recording and music publishing fields. Topics include music rights and licensing, Performing arts unions and guilds, artist representation and contractual relationships.

Prerequisites: MUI 221 and music major, or instructor permission.

MUI 315 Songwriting

Credits 3 Grading Basis Graded

An introduction to form, lyric development and melodic structure of contemporary songwriting for commercial entertainment applications. The course will include examination of leadsheet writing, demo production, copyright protection and publishing of commercial songs.

Prerequisites: Music majors /minors: MUI 230 and MUI 231; or permission of the instructor. Music industry minors: MUI 230 and MUI 231 and MUS 140; or permission of the instructor.

MUI 324 Introduction to Audio Devices

Credits 3 Grading Basis Graded

Introduction to electronic and digital devices utilized in the sound recording industry.

Prerequisites: PHYS 121, MUI 230 and MUI 231; or permission of the instructor.

MUI 330 Music Publishing

Credits 3 Grading Basis Graded

This course will offer a comprehensive overview of the music publication industry. The focus will be on, but not limited to, mainstream popular music. It will also consider songwriter/publisher relations, self-publication, as well as an examination of art, religious and educational music publishing.

Prerequisites: MUI 230 and MUI 231; or permission of the instructor.

MUI 400 Pro Tools: Multitrack Recording and Music Production

Credits 3 Grading Basis Graded

An introduction to contemporary multi-track recording studio techniques utilizing Pro Tools. Students will be introduced to recording studio design, psycho-acoustics, miking techniques, Musical Instrument Digital Interface and the mixing console.

Prerequisites: MUI 230, MUI 231 and MUI 324; or permission of the instructor.

MUI 405 Logic Pro: Synthesis and Sound Design

Credits 3 Grading Basis Graded

An introduction to analog synthesis, digital synthesis, an overview of conditions and events that led to the development of MIDI, a study of the MIDI protocol itself, as well as Apple Certified instruction in music production utilizing Logic Pro.

Prerequisites: MUI 230, MUI 231 and MUI 324; or permission of the instructor.

MUI 411 Audio Postproduction

Credits 3 Grading Basis Graded

An examination of music and sound used in the broadcast and entertainment media from an intensive, hands-on engineering perspective. The course will feature multiple post-production assignments to be completed in our on-campus Studio B, as well as field trips to off-campus post-production studios.

Prerequisites: MUI 230, MUI 324, and MUI 400; or instructor permission.

MUI 415 Songwriting II

Credits 3 Grading Basis Graded

This course is a continuation in the study of form, lyric development and melodic structure of contemporary songwriting for commercial entertainment applications. This course will also take into account some basic music business aspects important in the music industry directly related to song writers.

Prerequisites: Music majors /minors: MUI 230 and MUI 231; or permission of the instructor. Music industry minors: MUI 230 and MUI 231 and MUS 140; or permission of the instructor.

MUI 422 Concert Production and Promotion

Credits 3 Grading Basis Graded

Study of the presentation of cultural and commercial entertainment in the form of concert events from artistic, technical and business viewpoints. The roles of the cultural impresario and concert promoter in contemporary society are examined.

Prerequisites: MUI 230 and MUI 231; or permission of the instructor.

MUI 423 Sound Reinforcement

Credits 3 Grading Basis Graded

An introduction to the history, equipment, skills and business of sound reinforcement. The technique of contemporary sound engineers are examined and experienced by the use of audio amplification systems to design and use for public address and musical performance.

Prerequisites: MUI 230, MUI 231 and MUI 324; or permission of the instructor. MUI 400 is recommended.

MUI 430 Artist Management

Credits 3 Grading Basis Graded

This course will evaluate the function of musician/recording artist representatives in the music industry. Focus of discussions will include artist development from early career stages to concert tours, unions, recording companies, personal appearances, contractual agreements, etc.

Prerequisites: MUI 230 and MUI 231; or permission of the instructor.

MUI 435 Marketing in Music Production

Credits 3 Grading Basis Graded

Examination of the process of studio production, manufacturing, promotion and distribution of contemporary recordings. Record release programs for independent and major label-controlled products are analyzed.

Prerequisites: MUI 230 and MUI 231; or permission of the instructor.

MUI 440 Entrepreneurship in Music Industry

Credits 3 Grading Basis Graded

The study of business aspects of the music industry managing, marketing, finance and sales. Students develop a comprehensive music business plan. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: MUI 230 and MUI 231; or permission of the instructor.

MUI 492 Internship in Music Industry**Credits** 3-6 **Grading Basis** Graded

A supervised off-campus cocurricular learning activity designed to provide practical experience in the music industry.

Prerequisites: MUI 221, MUI 230 and MUI 231; or permission of the instructor.

Nonprofit Studies (NPS)

NPS 300 Introduction to Nonprofits**Credits** 3 **Grading Basis** Graded

An introduction to the development of the nonprofit sector in the American context exploring history, theories, legal issues, governance and ethical considerations. Global nonprofits are also explored. Provides a foundation for subsequent work in the nonprofit studies minor.

Prerequisites: Sophomore standing.

NPS 320 Nonprofit Management**Credits** 3 **Grading Basis** Graded

A study of organizational and management functions in the nonprofit sector. Examination of the unique role of volunteers, boards and public relations in the nonprofit environment.

Prerequisites: NPS 300 and junior standing.

NPS 375 Grant Writing for Agencies**Cross-listed with:** FAM 375, GERN 375, SOWK 375**Credits** 3 **Grading Basis** Graded

Emphasizing active learning, this course teaches the basics of grant and proposal writing. Efficient research, persuasive prose and the importance of relationships are stressed. Private and corporate philanthropy and government grants are examined.

NPS 400 Internship/Practicum in Nonprofit Studies**Credits** 4-6 **Grading Basis** Graded

Supervised internship/practicum in experience in a nonprofit organization setting that allows experiential learning and practice experiences. A research or applied paper, learning journal and presentation based on the experience are required. 4 credits (160 hours in agency), 6 credits (240 hours in agency).

Prerequisites: NPS 300 and the discipline specific elective.

NPS 450 Nonprofit Studies Capstone Seminar**Credits** 3 **Grading Basis** Graded

The capstone seminar is designed to integrate and apply knowledge from the student's major and the Nonprofit Studies minor. A substantial, individualized project will strengthen the student's capabilities in research and/or applied knowledge, information access, and self-directed learning.

Prerequisites: NPS 300, NPS 320, NPS 400, and discipline elective.

Related elective may be taken concurrently.

NPS 487 Special Topics in Nonprofit Studies**Credits** 3 **Grading Basis** Graded

Examination of selected topics in nonprofit studies that are of current importance in the nonprofit arena. Course may be repeated for credit.

Prerequisites: NPS 300.

NPS 490 Special Studies in Nonprofit Studies**Credits** 1-3 **Grading Basis** Graded

This course is designed to provide capable nonprofit studies minors an opportunity to complete independent study under faculty supervision. Course may be repeated for credit.

Prerequisites: NPS 300, NPS 320 and one additional course in the minor; or permission of the instructor.

Nursing (NSG)

NSG 300 Strategies for Success in Nursing**Credits** 1 **Grading Basis** Graded

This course introduces freshman and transfer nursing majors to the nursing curriculum and electives in other disciplines that relate to the goals and objectives of the program. This course will include instruction on campus student resources, time management and study skills, development of a clear academic plan, and nursing career options.

NSG 302 Behavioral and Mental Health in Children and Adolescents**Credits** 2 **Grading Basis** Graded

This course will explore the state of mental and behavioral health of children and adolescents. It will examine the neuropathophysiological nature of mental and behavioral illnesses, health promotion, and care of children and adolescents.

NSG 306 Combating Childhood Obesity**Credits** 1 **Grading Basis** Graded

This course will examine the current evidence and issues related to obesity in children and adolescents in the United States. Emphasis is placed on the health-related outcomes of childhood obesity for individuals, families and communities. Analysis of individual and public health approaches to reduce the prevalence of childhood obesity will be discussed.

NSG 307 Transcultural Concepts in Nursing**Credits** 2 **Grading Basis** Graded

This course examines concepts of cultural awareness and culturally competent care with individuals, families and communities. The influence of cultural beliefs and practices on health care that are specific to identified cultures is emphasized.

NSG 308 Empowerment and Resilience**Credits** 1 **Grading Basis** Graded

This elective course explores traumatic events and the ongoing psychological and social stress responses that result. Empowerment and resilience conceptual premises will be examined as elements to overcoming trauma and/or traumatic events. Historical traumatic community events will be used to highlight strategies of recognition and resilience and provide a micro to macro level perspective.

NSG 309 Trauma and the Continuum of Care**Credits** 1 **Grading Basis** Graded

This course explores adult health alterations that develop as a result of a trauma. Students will apply the nursing process to analyze psychological, physical, social and secondary stress as a result of trauma care situations. Students will analyze a plan of care that engages an interdisciplinary team approach to the care process. Students will develop an understanding of the rehabilitation process and long term outcomes of the traumatically injured.

NSG 310 Helping Persons in Pain**Credits** 2 **Grading Basis** Graded

This course open to all students from all majors, is an examination of pain, its impact on people, causes, treatments and the role of health professionals. Emphasis is placed on understanding how people experience pain and its effect on quality of life.

NSG 314 Rural Health: An Interprofessional Approach**Cross-listed with:** IPE 314, HTH 314, SOWK 314**Credits 3 Grading Basis** Graded

Students study, observe and participate in interprofessional assessment, planning and delivery of community-based primary health care in partnership with residents and agencies of a host rural county. Learning activities will emphasize rural culture, rural health care and interprofessional practice.

NSG 317 History of Nursing**Credits 1 Grading Basis** Graded

An elective nursing course that explores fundamental aspects of nursing history including pertinent nursing founders and leaders as well as examination of the many influences that have shaped the nursing profession. Review of nursing within its historical context provides an opportunity to consider changes for the future.

NSG 318 Nursing, Chronic Illness and Health in a Global Context**Credits 2 Grading Basis** Graded

Non-communicable diseases (NCDs/ a.k.a. chronic illnesses) are universal. The course provides students with an introduction to global health care systems models, and concepts and their influence on health disparities and delivery of healthcare, focusing on NCDs. Additionally, students will examine ways that nurses and nursing can meet the challenges of addressing NCDs at a global level.

NSG 319 Understanding Sexuality across the Health-Illness Continuum**Credits 2 Grading Basis** Graded

This course examines how illness affects sexuality (physiologically, psychologically and culturally) across the lifespan. Sources of interventions and support for issues of sexuality and chronic illness will be addressed.

NSG 321 Faith Community Nursing**Credits 1 Grading Basis** Graded

This course is based on the accepted philosophy and practice of Faith Community Nursing and examines the roots of health and healing found in many religious traditions. Critical thinking strategies, such as Socratic questioning, are used to analyze the spiritual dimension of health and healing for the practitioner as well as clients they serve. Students explore the practice of nursing in faith communities.

NSG 322 Integrative Health Care**Credits 2 Grading Basis** Graded

This course examines the principles, practices, and outcomes of complementary therapies and alternative healing that are widely used in the general population. The integration of alternative and conventional health practices will be examined. Ethical, legal and professional issues will be explored.

NSG 324 R.N./B.S.N. Strategies for Success**Credits 1 Grading Basis** Graded

This course is designed to provide strategies for academic success in an online learning environment for students in the R.N. to B.S.N. program. Students apply concepts of scholarly writing and identify research. An emphasis is placed on competence with technology and working collaboratively within an online environment.

NSG 325 Concepts in Aging**Credits 3 Grading Basis** Graded

This online course examines evidence-based knowledge needed to provide holistic care for an aging population. Issues related to legal and ethical aspects of care, health promotion, chronic health conditions and special issues related to gerontological nursing care are examined. Prerequisites: Admission to R.N.-B.S.N. program.

NSG 326 Care and Consideration for Children and Young Adults with Disabilities**Credits 1 Grading Basis** Graded

Open to students from any major. Students gain insight into the lives of children and young adults with disabilities, discuss how having dependents with disabilities affects the entire family, and provide care for children/young adults with disabilities. Disabling conditions will be investigated. Learning activities include service learning through caring for a child/young adult with disabilities, guest speakers and discussion. Students must provide their own transportation to service-learning sites off campus.

NSG 328 Life, Death, and the In-Between**Credits 1 Grading Basis** Graded

This course focuses on preparing the student to give patient-centered end-of-life care. Using a variety of learning methods, the student will examine theories and care models, and will discuss current topics surrounding death and dying, including social, cultural, ethical, spiritual and legal issues.

NSG 329 Best Practices in Diabetes Care**Credits 2 Grading Basis** Graded

The student will develop a basic understanding of the current practices related to diabetes care and the impact of a diabetes diagnosis on the individual, family and community. The content is centered around the American Association of Diabetes Educators (AADE) 7 Self-Care Behaviors: healthy eating, being active, monitoring, taking medications, problem solving, healthy coping and reducing risks.

NSG 330 Medical Innovations**Credits 3 Grading Basis** Graded

In this course, pre-professional health, pre-nursing and engineering students will work in interprofessional teams to develop and communicate solutions to a community health challenge. Students will use innovative MakerSpace technologies to produce a tangible product aimed at improving the challenge while also exploring technology's potential to cause harm. In addition to instruction of the technologies, students will be taught problem solving, collaboration, and leadership skills.

NSG 333 Health Assessment**Credits 3 Grading Basis** Graded

This course examines the principles and techniques of nursing assessments focusing on health history taking, review of systems and physical examination techniques for individuals across the lifespan. Clinical decision making is emphasized using a holistic approach that takes into account physical, genetic, cultural, psychological, social, spiritual and environmental variables that impact health. Prerequisites: Admission to R.N.-B.S.N. program.

NSG 340 Nursing Identity**Credits 3 Grading Basis** Graded

This course introduces concepts of professional nursing practice, informatics and evidence-based research within the health care delivery system. Students will develop a professional nursing identity and understand their role within the health care system. Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways.

NSG 344 Optimizing Wellness I**Credits 3 Grading Basis** Graded

Students learn how to complete health assessments of individuals and populations. Selected concepts will be used to introduce nursing clinical judgment and decision-making to implement holistic health interventions. Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways. Corequisites: NSG 344L, NSG 380 and NSG 380L.

NSG 344L Optimizing Wellness I Clinical**Credits 1 Grading Basis** Graded

Students learn how to complete health assessments of individuals and populations. Selected concepts will be used to introduce nursing clinical judgment and decision-making to implement holistic health interventions. This is the clinical component to accompany NSG 344.

Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways. Corequisites: NSG 344 and NSG 380.

NSG 345 Health Alterations I**Credits 3 Grading Basis** Graded

This is the first course in a sequence in which students will learn foundational assessment techniques, nursing knowledge and interventions. Students will be introduced to and develop professional communication styles and clinical decision making to promote patient centered care across the lifespan.

Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways. Corequisites: NSG 345L, NSG 380 and NSG 380L.

NSG 345L Health Alterations I Clinical**Credits 1 Grading Basis** Graded

This is the first course in a sequence where students will learn foundational assessment techniques, nursing knowledge and interventions across various clinical settings. Students will be introduced to and develop professional communication styles and clinical decision making to promote patient centered care across the lifespan. This is the clinical component to accompany NSG 345.

Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways. Corequisites: NSG 345, NSG 380 and NSG 380L.

NSG 350 Foundations of Nursing**Credits 2 Grading Basis** Graded

This course provides an overview of foundational principles of professional nursing practice. Students will be introduced to the evolution of nursing, basic nursing theory and knowledge, and beginning concepts. This course promotes self-analysis and socialization into the role of the professional nurse.

Prerequisites: Formal acceptance into the nursing major.

NSG 351 Health Assessment**Credits 3 Grading Basis** Graded

This course develops knowledge and skills necessary to gather, organize, and present relevant health data that includes wellness and illness considerations across the life cycle. Emphasis is placed on systematic strategies, frameworks, and skills used to conduct both comprehensive and need-specific health assessments for individuals in the context of their family and community.

Prerequisites: Formal acceptance into the nursing major.

NSG 352 Clinical Applications and Reasoning in Nursing Care I**Credits 4 Grading Basis** Graded

Through didactic learning, students learn theories, rationale, and principles underlying the application of acute care skills in nursing practice. In the laboratory, students will practice and demonstrate mastery of selected skills. In the clinical setting, students will apply knowledge through clinical reasoning in planning and facilitating nursing care for patients and their families.

Prerequisites: Formal acceptance to the nursing program is the prerequisite. Corequisites: NSG 352L.

NSG 352L Clinical Applications and Reasoning in Nursing Care I Clinical**Credits 2 Grading Basis** Credit / No Credit

Through didactic learning, students learn theories, rationale, and principles underlying the application of acute care skills in nursing practice. In the laboratory, students will practice and demonstrate mastery of selected skills. In the clinical setting, students will apply knowledge through clinical reasoning in planning and facilitating nursing care for patients and their families.

Prerequisites: Formal acceptance into the nursing major. Corequisites: NSG 352.

NSG 353 Pathophysiology and Pharmacology**Credits 4 Grading Basis** Graded

This course is a comprehensive examination of the principles of pathophysiology and pharmacology for nurses. Emphasis will be placed on concepts and rationales necessary for clinical decision making and nursing care of patients with select disease processes.

Prerequisites: Formal acceptance into the nursing major.

NSG 354 The Art and Science of Nursing**Credits 2 Grading Basis** Graded

This course is designed to provide an overview of current issues relevant to the art and science of the practicing nurse. This course will provide the student with a concentrated focus on the role of the professional nurse and the nursing profession. The course explores nursing theory, health care models of practice, diversity issues, as well as legal and ethical realities within the healthcare delivery system.

NSG 355 Women's Health**Credits 3 Grading Basis** Graded

This course promotes synthesis of concepts and principles utilized in health promotion, risk reduction and critical reasoning in the management of women's health care. Areas of focus include women's health issues, perinatal care of mothers and infants, and gynecological health. Clinical experiences provide students with opportunities to apply evidence based practice for women/newborn/family units of diverse cultural backgrounds.

Prerequisites: NSG 350, NSG 351, NSG 352, NSG 352L, NSG 353.

Corequisites: NSG 355L.

NSG 355L Women's Health Clinical**Credits 1 Grading Basis** Credit / No Credit

This course promotes synthesis of concepts and principles utilized in health promotion, risk reduction and critical reasoning in the management of women's health care. Areas of focus include women's health issues, perinatal care of mothers and infants, and gynecological health. Clinical experiences provide students with opportunities to apply evidence based practice for women/newborn/family units of diverse cultural backgrounds.

Prerequisites: NSG 350, NSG 351, NSG 352, NSG 352L, NSG 353.

Corequisites: NSG 355.

NSG 356 Clinical Applications and Reasoning in Nursing Care II**Credits 4 Grading Basis Graded**

This course focuses on pathophysiologic and pharmacologic concepts and principles of nursing process, health promotion, risk reduction, clinical decision making, and collaborative management of care for adults experiencing moderate to severe health alterations. Students will apply concepts, theories and skills in the nursing care of adults.

Prerequisites: NSG 350, NSG 351, NSG 352, NSG 352L, NSG 353.

Corequisites: NSG 356L.

NSG 356L Clinical Applications and Reasoning in Nursing Care II Clinical**Credits 2 Grading Basis Credit / No Credit**

This course focuses on pathophysiologic and pharmacologic concepts and principles of nursing process, health promotion, risk reduction, clinical decision making, and collaborative management of care for adults experiencing moderate to severe health alterations. Students will apply concepts, theories and skills in the nursing care of adults.

Prerequisites: NSG 350, NSG 351, NSG 352, NSG 352L, NSG 353.

Corequisites: NSG 356.

NSG 357 Psychiatric Mental Health Nursing**Credits 3 Grading Basis Graded**

This course examines the pathophysiology, psychosocial manifestations, psychopharmacological and psychiatric mental health nursing treatment of selected mental illnesses. Analysis of the role and practice of psychiatric mental health nursing both as a nursing specialty and as an integral facet of general nursing are emphasized.

Prerequisites: NSG 350, NSG 351, NSG 352, NSG 352L, NSG 353.

Corequisites: NSG 357L.

NSG 357L Psychiatric Mental Health Nursing Clinical**Credits 1 Grading Basis Credit / No Credit**

This course examines the pathophysiology, psychosocial manifestations, psychopharmacological and psychiatric mental health nursing treatment of selected mental illnesses. Analysis of the role and practice of psychiatric mental health nursing both as a nursing specialty and as an integral facet of general nursing are emphasized.

Prerequisites: NSG 350, NSG 351, NSG 352, NSG 352L and NSG 353.

Corequisites: NSG 357.

NSG 360 HHS in Costa Rica: Interprofessional Perspective**Credits 2-4 Grading Basis Credit / No Credit**

In this course students from nursing and other health-related majors will explore the health needs and the health and human services delivery system in Costa Rica, as well as the history and culture of this Central American country. Students will examine issues that impact health in the United States and Costa Rica and the roles of health care professionals in the delivery of healthcare. This immersion experience includes home stay with a host family and Spanish language study.

Prerequisites: Nursing (B.S.N. and R.N. to B.S.N.) and other health-related majors only. Corequisites: R.N. to B.S.N. students only: NSG 367 and NSG 469.

NSG 361 International Health Perspectives: Malta**Credits 4 Grading Basis Graded**

In this course, students will be immersed in the history and culture of communities within Malta to enhance knowledge and understanding of healthcare challenges unique to this population. Students will examine how culture, lifestyle and other social determinants of health impact the health and well-being of the Maltese people. An emphasis will be placed on understanding health care challenges unique to the Maltese.

Prerequisites: Declared nursing or health-related major.

NSG 363 Health Care in Spain: A Cross Cultural Exploration of Illness and Wellness**Credits 4 Grading Basis Graded**

In this course students will explore the health care system of Tarragona, Spain, through a partnership with the faculty of nursing at Universitat Rovira I Virgili. Through an international lens, students will learn about the Spanish nursing profession and related health fields, as well as the unique history, language and culture of this Catalan region of Spain. Students will explore the ways that Spanish health professionals address health needs in both acute and community care settings in the context of the national health system, and they will compare and contrast this with their knowledge of the U.S. health system approach. This immersion experience includes home stay with a Spanish nursing student and Spanish language study. Minimal competence in Spanish preferred.

Prerequisites: Declared nursing or health-related major.

NSG 367 R.N. to B.S.N. Community Clinical Costa Rica**Credits 1 Grading Basis Credit / No Credit**

This practicum for R.N.-B.S.N. students transitions practice into the B.S.N. role through mentored clinical experiences at selected community sites. Emphasis is on collaborative nursing care with individuals, families and groups within the community. Experiences include concepts of health promotion and disease prevention and management of acute or chronic illness.

Prerequisites: Admission to the R.N. to B.S.N. program or accepted VCCS co-enrollment student. Corequisites: NSG 360 and NSG 469.

NSG 368 Chronic Illness: Treatment Strategies for Substance Use Disorders**Credits 2 Grading Basis Graded**

This course examines the urgent need for all nurses to have a better understanding of substance use disorders and available treatment strategies to help address the current national healthcare crisis related to substance abuse and deaths attributed to opioids. Nurses interact with individuals dealing with an array of chronic health issues including the use and abuse of psychoactive substances. This course will help prepare nurses who are often first to recognize problems related to substance use disorders in hospitals and their communities.

Prerequisites: NSG 357 and NSG 357L. Corequisites: NSG 357 and NSG 357L.

NSG 380 Professional Integration I**Credits 2 Grading Basis Graded**

In this course students will demonstrate knowledge, skills, attitudes and behaviors of nursing practice. This course integrates and evaluates foundational concepts and competencies obtained in courses in the first semester of the B.S.N.

Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways. Corequisites: NSG 340, NSG 344, NSG 344L, NSG 345, NSG 345L and NSG 380L.

NSG 380L Professional Integration I Clinical**Credits 1 Grading Basis Graded**

In this course students will demonstrate knowledge, skills, attitudes and behaviors of nursing practice. This course integrates and evaluates foundational concepts and competencies obtained in the courses in the first semester of the B.S.N. This is the clinical component to accompany NSG 380.

Prerequisites: Admission into the B.S.N. through the traditional or fast flex pathways. Corequisites: NSG 344, NSG 344L, NSG 345, NSG 345L and NSG 380.

NSG 381 The Evidence Based Clinician**Credits 3 Grading Basis Graded**

Students will examine the convergence of their personal values and beliefs and the practice of nursing to actively shape their professional identity. Students will explore the contributions of evidence based practice and nursing research in improving nursing practice.

Prerequisites: Traditional and fast flex students: NSG 340; students on the R.N.-B.S.N. pathway: admission into the program.

NSG 382 Optimizing Wellness II**Credits 3 Grading Basis Graded**

This course emphasizes mental health nursing as a facet of generalist nursing practice and prepares students to deliver safe and patient-centered care to promote the well-being and mental health of individuals.

Prerequisites: NSG 344, NSG 344L, NSG 345, NSG 345L, NSG 380 and NSG 380L. Corequisites: NSG 382L, NSG 384 and NSG 384L.

NSG 382L Optimizing Wellness II Clinical**Credits 1 Grading Basis Graded**

This course prepares nursing professionals to deliver safe and patient-centered care to promote the well-being and mental health of individuals. This course emphasizes mental health nursing as a facet of generalist nursing practice and will introduce selected mental illnesses. This is the clinical component to accompany NSG 382.

Prerequisites: NSG 344, NSG 344L, NSG 345, NSG 345L, NSG 380 and NSG 380L. Corequisites: NSG 382, NSG 384 and NSG 384L.

NSG 383 Health Alterations II**Credits 3 Grading Basis Graded**

In this second course of this sequence, students will analyze individual health alterations with an emphasis on supportive care of exacerbation of chronic illness across the lifespan. Students will utilize clinical decision making and communication strategies to provide ethical, holistic, patient-centered care.

Prerequisites: NSG 344, NSG 344L, NSG 345, NSG 345L, NSG 380 and NSG 380L. Corequisites: NSG 383L, NSG 384 and NSG 384L.

NSG 383L Health Alterations II Clinical**Credits 1 Grading Basis Graded**

In this second course of this sequence, students will analyze individual health alterations with an emphasis on supportive care and exacerbation of chronic illness across the lifespan. Students will utilize clinical decision making and communication strategies to provide ethical, holistic, patient centered care. This is the clinical component to accompany NSG 383.

Prerequisites: NSG 344, NSG 344L, NSG 345, NSG 345L, NSG 380 and NSG 380L. Corequisites: NSG 383, NSG 384 and NSG 384L.

NSG 384 Professional Integration II**Credits 3 Grading Basis Graded**

In the second course of this sequence, students will integrate knowledge, skills, attitudes and behaviors of nursing practice. This course integrates and evaluates concepts and competencies required to provide holistic, patient-centered care.

Prerequisites: Traditional and fast flex students: NSG 344, NSG 344L, NSG 345, NSG 345L, NSG 380 and NSG 380L; students on the R.N.-B.S.N. pathway: admitted to the program. Corequisites: Traditional and fast flex students: NSG 384L.

NSG 384L Professional Integration II Clinical**Credits 1 Grading Basis Graded**

In the second course of this sequence, students will integrate knowledge, skills, attitudes and behaviors of nursing practice. This course integrates and evaluates concepts and competencies required to provide holistic, patient-centered care. Program milestone competencies will be measured using various methods of evaluation. This is the clinical component to accompany NSG 384.

Prerequisites: NSG 344, NSG 344L, NSG 345, NSG 345L, NSG 380 and NSG 380L. Corequisites: NSG 384.

NSG 390 Impact of Chronic Illness**Credits 3 Grading Basis Graded**

This course will explore core concepts of chronic illness across the lifespan from an interdisciplinary perspective. Epidemiology, economics, ethics, culture, family and policy will be emphasized. These topics and concepts will be related to model(s) of chronic care.

NSG 391 Living Successfully with Chronic Illness**Credits 3 Grading Basis Graded**

This course will examine models and strategies that aid individuals to live successfully with chronic illness. An interdisciplinary evidence-based approach will be used to investigate how outcomes may be improved through the individual's integration of lifestyle changes within the context of culture and family.

NSG 393 Power-based Personal Violence**Credits 1 Grading Basis Graded**

This course introduces students to the roots of violence, including the political, cultural, social and economic structures that perpetuate violence, and explores approaches to changing those structures in order to reduce or end violence. Students will think critically about the local and global impact of violence and how it intersects with other forms of oppression, and they will achieve an understanding of these issues that will be useful intellectually and personally.

NSG 415 Ethical Decision-Making in Healthcare: An Interprofessional Approach**Cross-listed with: IPE 415****Credits 1 Grading Basis Student Option**

Health care ethics is a shared, relevant concern among health and human service disciplines and is an ideal vehicle for students from different fields to learn about one another's disciplines and to participate in interprofessional team analysis, discussion, and problem-solving. We have designed an integrated cross-disciplinary learning experience for students interested in the complex, real-world dilemmas encountered in practice. The readings and activities will emphasize ethical, legal, moral and spiritual issues and principles for practice within the context of communicating in interprofessional teams.

NSG 417 Women's Global Health and Human Rights**Cross-listed with: HTH 417, WGSS 417****Credits 3 Grading Basis Graded**

An international and human rights approach providing an overview of health issues within the context of a woman's life cycle. Attention will be given to critical issues of women's health such as access to health care and gender-based violence. Such issues as sexuality, nutrition, diseases affecting women, violence, harmful traditional practices, and sex trafficking will be discussed.

NSG 440 The Team Member**Credits 3 Grading Basis Graded**

Students will use evidence-based communication and teamwork strategies to enhance interprofessional practice and advocacy for improved patient outcomes.

Prerequisites: Traditional and fast flex students: NSG 340 and NSG 381; students on the R.N.-B.S.N. pathway: admission to the program.

NSG 441 Optimizing Wellness III**Credits 3 Grading Basis Graded**

This course will emphasize health promotion, risk reduction and care of individuals and their support systems across the lifespan. Students will work collaboratively to implement evidence-based interventions.

Prerequisites: Traditional and fast flex students: NSG 382, NSG 382L, NSG 383, NSG 383L, NSG 384 and NSG 384L; students on the R.N.-B.S.N. pathway: admission to the program. Corequisites: Trad and fast flex students: NSG 441L, NSG 444 and NSG 444L.

NSG 441L Optimizing Wellness III Clinical**Credits 1 Grading Basis Graded**

This course will emphasize health promotion, risk reduction and care of individuals and their support systems across the lifespan. Students will work collaboratively to implement evidence-based interventions. This is the clinical component to accompany NSG 441.

Prerequisites: NSG 382, NSG 382L, NSG 383, NSG 383L, NSG 384 and NSG 384L. Corequisites: NSG 441, NSG 444 and NSG 444L.

NSG 443 Health Alterations III**Credits 3 Grading Basis Graded**

For the third course in this sequence, students will integrate concepts of complex health alterations with previous nursing knowledge. Students will synthesize and prioritize patient data to provide ethical, holistic patient-centered care. A focus will be placed on the nurse as a member of the interprofessional team.

Prerequisites: Traditional and fast flex students: NSG 382, NSG 382L, NSG 383, NSG 383L, NSG 384 and NSG 384L; students on the R.N.-B.S.N. pathway: admission to the program. Corequisites: Trad and fast flex students: NSG 443L, NSG 444 and NSG 444L.

NSG 443L Health Alterations III Clinical**Credits 1 Grading Basis Graded**

For the third course in this sequence students will integrate concepts of complex health alterations with previous nursing knowledge. Students will synthesize and prioritize patient data to provide ethical, holistic patient-centered care. A focus will be placed on the nurse as a member of the interprofessional team. This is the clinical component to accompany NSG 443.

Prerequisites: NSG 382, NSG 382L, NSG 383, NSG 383L, NSG 384 and NSG 384L. Corequisites: NSG 443, NSG 444 and NSG 444L.

NSG 444 Professional Integration III**Credits 3 Grading Basis Graded**

In this course, students will critique and reflect on the knowledge, skills, attitudes and behaviors of nursing practice. This course integrates and evaluates concepts and competencies required to provide patient centered care for patients with complex health needs.

Prerequisites: Traditional and fast flex students: NSG 382, NSG 382L, NSG 383, NSG 383L, NSG 384 and NSG 384L; students on the R.N.-B.S.N. pathway: admission to the program. Corequisites: Traditional and fast flex students: NSG 444L.

NSG 444L Professional Integration III Clinical**Credits 1 Grading Basis Graded**

In this course, students will critique and reflect on the knowledge, skills, attitudes and behaviors of nursing practice. This course integrates and evaluates concepts and competencies required to provide patient centered care for patients with complex health needs. This is the clinical component to accompany NSG 444.

Prerequisites: NSG 382, NSG 382L, NSG 383, NSG 383L, NSG 384 and NSG 384L. Corequisites: NSG 444.

NSG 445 Evolution as a Leader**Credits 3 Grading Basis Graded**

This course examines the role of the nurse as a leader influencing health policy, health systems and organizations through advocacy to improve health outcomes. Students will use knowledge of transformational leadership, self-awareness and collaboration skills to develop their capacity for leadership.

Prerequisites: Traditional and fast flex students: NSG 340, NSG 381 and NSG 440; students on the R.N.-B.S.N. pathway: admission to the program.

NSG 446 Optimizing Wellness IV**Credits 3 Grading Basis Graded**

In this course, students will explore issues impacting health to empower populations to promote, protect, maintain and restore health. Students will work collaboratively to implement evidence-based interventions and improve health outcomes.

Prerequisites: Traditional and fast flex students: NSG 441, NSG 441L, NSG 443, NSG 443L, NSG 444 and NSG 444L; students on the R.N.-B.S.N. pathway: admission to the program. Corequisites: Trad and fast flex students: NSG 446L, NSG 481 and NSG 481L.

NSG 446L Optimizing Wellness IV Clinical**Credits 1 Grading Basis Graded**

In this course, students will explore issues impacting health to empower populations to promote, protect, maintain and restore health. Students will work collaboratively to implement evidence-based interventions and improve health outcomes. This is the clinical component to accompany NSG 446.

Prerequisites: NSG 441, NSG 441L, NSG 443, NSG 443L, NSG 444 and NSG 444L. Corequisites: NSG 446, NSG 481 and NSG 481L.

NSG 450 Nursing Inquiry and Research Methods**Credits 3 Grading Basis Graded**

This course will enhance students' ability to locate and critically appraise evidence-based guidelines and nursing research to determine their applicability to nursing practice. Students critically examine the links among nursing research, theory and practice. Students will develop an understanding of the underlying paradigms and activities within quantitative and qualitative research.

Prerequisites: Formal acceptance into the nursing major.

NSG 451 Child Health**Credits 3 Grading Basis Graded**

This course promotes the development of knowledge, skills and the ability to care for children including those with acute and chronic illnesses/conditions. Learning will focus on the unique healthcare needs of children with emphasis on family centered care. Students will apply knowledge through clinical reasoning in planning and facilitating nursing care for children and families.

Prerequisites: NSG 355, NSG 355L, NSG 356, NSG 356L, NSG 357 and NSG 357L. Corequisites: NSG 451L.

NSG 451L Child Health Clinical**Credits 1 Grading Basis Credit / No Credit**

This course promotes the development of knowledge, skills and the ability to care for children including those with acute and chronic illnesses/conditions. Learning will focus on the unique healthcare needs of children with emphasis on family centered care. Students will apply knowledge through clinical reasoning in planning and facilitating nursing care for children and families.

Prerequisites: NSG 355, NSG 355L, NSG 356, NSG 356L, NSG 357, NSG 357L. Corequisites: NSG 451.

NSG 452 Clinical Applications and Reasoning in Nursing Care III**Credits 4 Grading Basis Graded**

This course focuses on the integration of complex pathophysiologic and pharmacologic concepts and principles for adults experiencing moderate to severe health alterations. Students will apply the nursing process to promote health and safety, to augment clinical reasoning and clinical decision making, and to integrate interprofessional collaboration in the care of patients.

Prerequisites: NSG 352, NSG 352L, NSG 356 and NSG 356L.

NSG 452L Clinical Applications and Reasoning III Clinical**Credits 2 Grading Basis Credit / No Credit**

This course focuses on the integration of complex pathophysiologic and pharmacologic concepts and principles for adults experiencing moderate to severe health alterations. Students will apply the nursing process to promote health and safety, to augment clinical reasoning and clinical decision making, and to integrate interprofessional collaboration in the care of patients.

Prerequisites: NSG 352, NSG 352L, NSG 356 and NSG 356L. Corequisites: NSG 452.

NSG 453 Population-Centered Care in the Community**Credits 3 Grading Basis Graded**

Students will engage in the systematic assessment of individuals and vulnerable populations in community settings and work collaboratively to lead the development and implementation of evidence-based, population-focused community health nursing interventions. The impact of increasing societal and cultural changes across the life span will be analyzed. Students will incorporate theoretical concepts of community-based health promotion and disease prevention of vulnerable populations into their nursing practice.

Prerequisites: NSG 355, NSG 355L, NSG 356, NSG 356L, NSG 357, NSG 357L, NSG 451, NSG 451L, NSG 452 and NSG 452L. Corequisites: NSG 453L, NSG 454 and NSG 456.

NSG 453L Population-Centered Care in the Community Clinical**Credits 2 Grading Basis Credit / No Credit**

Students will engage in the systematic assessment of individuals and vulnerable populations in community settings and work collaboratively to lead the development and implementation of evidence-based, population-focused community health nursing interventions. The impact of increasing societal and cultural changes across the life span will be analyzed. Students will incorporate theoretical concepts of community-based health promotion and disease prevention of vulnerable populations into their nursing practice through clinical experiences.

Prerequisites: NSG 355, NSG 355L, NSG 356, NSG 356L, NSG 357, NSG 357L, NSG 451, NSG 451L, NSG 452 and NSG 452L. Corequisites: NSG 453, NSG 454, NSG 455 and NSG 456.

NSG 454 Transition to Practice**Credits 3 Grading Basis Graded**

This course enhances clinical reasoning in the context of communication and teamwork, patient-centered care, evidence-based practice, and quality improvement. Students will explore pathways for licensure and prepare for a career in nursing. Current issues in leadership, risk management, health policy, health economics and political action will be explored. An emphasis will be placed on the role of the nurse in advocating for quality and safety in healthcare teams.

Prerequisites: NSG 355, NSG 355L, NSG 356, NSG 356L, NSG 357, NSG 357L, NSG 450, NSG 451, NSG 451L, NSG 452 and NSG 452L. Corequisites: NSG 453 and NSG 453L.

NSG 455 Nursing Informatics**Credits 3 Grading Basis Graded**

This course explores nursing informatics and technology applications in health care. Emphasis is on preparing entry-level nurses with core nursing informatics competencies. A major theme is the use of information systems and technologies to improve the quality and safety of patient care in a changing health care environment. Students will develop their nursing informatics knowledge and skills through reading, discussions, exploration and utilization of electronic modalities.

Prerequisites: Formal acceptance into the nursing major; NSG 350, NSG 354 and NSG 450.

NSG 456 Capstone**Credits 4 Grading Basis Graded**

The capstone experience emphasizes continued synthesis of and competence in the nursing process, critical thinking, prioritization of care and clinical skills in order to provide safe, high quality nursing care under the direct supervision of faculty and clinical nurse preceptors.

Prerequisites: NSG 355, NSG 355L, NSG 356, NSG 356L, NSG 357, NSG 357L, NSG 450, NSG 451, NSG 451L, NSG 452 and NSG 452L. Corequisites: NSG 453 and NSG 453L.

NSG 460 Healthcare Informatics**Credits 3 Grading Basis Graded**

This course focuses on the nature and functions of present and future applications of health care informatics. Emphasis is on preparing current and future health care professionals to plan, design, collaborate with other health care disciplines, and utilize healthcare informatics for effective health care delivery, health organizational management, and improved client outcomes.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 461 Pathophysiology and Pharmacology**Credits 3 Grading Basis Graded**

This course, offered to RN-BSN program students, provides an examination of complex physiologic responses and clinical sequel in major body systems in relation to pathologic processes. Emphasis is placed upon physiologic compensation and defense responses. Pharmacologic management of pathology is investigated.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 462 Issues in Contemporary Nursing Practice**Credits 3 Grading Basis Graded**

This course examines contemporary issues related to the profession of nursing. Historical and modern day influences that impact the profession of nursing are explored. Students are introduced to resources that support the individual nurse and the profession of nursing to navigate current issues and create resolutions. Multiple asynchronous course activities are utilized to engage the adult learner.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 463 Professional Role Transition**Credits 3 Grading Basis Graded**

This online course expands the students' current knowledge of concepts related to nursing theory, nursing image and professional role development at the B.S.N. level. Emphasis will be placed upon leadership and management skill development at the personal level.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 464 Introduction to Nursing Research**Credits 3 Grading Basis Graded**

This online course will focus on the study of research methods that generate quantitative and qualitative data. Students will examine the research process with an emphasis on critique of research methodologies and application of research findings to nursing practice.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 466 Community Health Practicum**Credits 1 Grading Basis Credit / No Credit**

This practicum, for R.N.-B.S.N. students, transitions practice into the B.S.N. role through mentored clinical experiences at selected community sites. Emphasis is on collaborative nursing care with individuals, families and groups within the community. Experiences include concepts of health promotion and disease prevention and management of acute or chronic illness.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 469 Caring for the Public's Health: Community Health Nursing**Credits 4 Grading Basis Graded**

This course provides RN-BSN students a perspective of professional nursing at the community level of practice. Course content will provide an overview of specific issues and societal concerns that affect community health nursing practice including historical impact of public health, epidemiology, health promotion and disease prevention, vulnerable populations; communicable disease risk and prevention; and the diversity of the role of the community health nurse.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 471 Leadership and Management in Health Care**Credits 3 Grading Basis Graded**

This online course explores the role of the nurse as a leader in professional health care delivery through an examination of healthcare organizations, leadership theories and management styles, organizational change, conflict resolution, fiscal and economic issues, and professional political responsibility.

Prerequisites: Admission to R.N.-B.S.N. program.

NSG 480 Health Alterations IV**Credits 3 Grading Basis Graded**

In the fourth course of this sequence students will evaluate patient data and clinical decision-making to promote holistic, high-quality care across the health care continuum and lifespan. There is a focus on facilitating professional communication amongst individuals, family and various teams to support patient outcomes.

Prerequisites: Traditional and fast flex students: NSG 441, NSG 441L, NSG 443, NSG 443L, NSG 444 and NSG 444L; students on the R.N.-B.S.N. pathway: admission to the program. Corequisites: Trad and fast flex students: NSG 480L, NSG 481 and NSG 481L.

NSG 480L Health Alterations IV Clinical**Credits 1 Grading Basis Graded**

In the fourth course of this sequence, students will evaluate patient data and clinical decision-making to promote holistic, high-quality care across the healthcare continuum and lifespan. There is a focus on facilitating professional communication among individuals, families and various teams to support patient outcomes. This is the clinical component to accompany NSG 480.

Prerequisites: NSG 441, NSG 441L, NSG 443, NSG 443L, NSG 444 and NSG 444L. Corequisites: NSG 480, NSG 481 and NSG 481L.

NSG 481 Professional Integration IV**Credits 3 Grading Basis Graded**

In this course, students will demonstrate competency of the student learning outcomes of the B.S.N. curriculum. This course integrates and evaluates concepts and competencies across the curriculum.

Prerequisites: Traditional and fast flex students: NSG 441, NSG 441L, NSG 443, NSG 443L, NSG 444 and NSG 444L; students on the R.N.-B.S.N. pathway: admission to the program. Corequisites: Traditional and fast flex students: NSG 481L.

NSG 481L Professional Integration IV Clinical**Credits 1 Grading Basis Graded**

In this course, students will demonstrate competency of the student learning outcomes of the B.S.N. curriculum. This course integrates and evaluates concepts and competencies across the curriculum. This is the clinical component to accompany NSG 481.

Prerequisites: NSG 441, NSG 441L, NSG 443, NSG 443L, NSG 444 and NSG 444L. Corequisites: NSG 481.

NSG 490 Special Studies**Credits 1-6 Grading Basis Graded**

Study of selected topics in nursing. No prerequisites.

NSG 497 Independent Study**Credits 1-3 Grading Basis Graded**

Independent study in nursing under faculty supervision. Prerequisite: Permission of the department head.

NSG 499A Honors Thesis I**Credits 1 Grading Basis Graded****NSG 499B Honors Thesis II****Credits 2-3 Grading Basis Graded****NSG 499C Honors Thesis III****Credits 2 Grading Basis Graded****NSG 499D Honors Thesis IV****Credits 1 Grading Basis Graded**

Nutrition (NUTR)

NUTR 240 Basic Food Preparation Skills**Credits 2 Grading Basis Graded**

Students will become proficient in a conventional kitchen setting by learning and practicing basic culinary skills, cooking techniques and food safety. This course serves as a foundation for highly technical foods-related courses in the dietetics major.

Prerequisites: Dietetics major.

NUTR 280 Nutrition for Wellness**Credits 3 Grading Basis Graded**

Students will study the impact of nutrition on wellness by learning nutrients, their functions in the human body, food sources and appropriate intake levels. Controversies surrounding use of various nutrients for improvement of health and well-being will be discussed.

NUTR 295 Foundations of Nutrition Practice**Credits 2 Grading Basis** Graded

An introduction to the profession of dietetics, credentialing processes in nutrition/dietetics, careers available in the field, and some basic skills needed for the profession.

NUTR 340 The Science and Culture of Food**Credits 3 Grading Basis** Graded

This course explores scientific principles and techniques of food preparation while examining diversity in food patterns and composition. A laboratory experience provides an opportunity to apply scientific principles of food preparation while preparing nutritious and culturally diverse foods.

Prerequisites: Dietetics major and CHEM 131 or equivalent.

NUTR 350 Emerging Research in Nutrition and Dietetics**Credits 3 Grading Basis** Graded

This evidence-based course exposes students to emerging topics related to nutrition and dietetics. Students will develop skills to critically evaluate and interpret current research presented in various formats (e.g., peer-reviewed journal articles, poster presentations, oral scientific presentations) and discuss implications for professional practice and advancement of the field.

Prerequisites: Dietetics major.

NUTR 363 Quantity Food Production**Credits 4 Grading Basis** Graded

The principles of quantity food production and service are studied.

Prerequisites: Dietetics major and NUTR 340.

NUTR 380 Global Nutrition**Credits 3 Grading Basis** Graded

A study of food habits from around the world and their contributions to nutritional adequacy. Factors affecting global food-consumption behaviors including sociocultural practices, religion, health beliefs, agricultural practices, economics, politics and education will be explored.

NUTR 382 Sports Nutrition**Credits 3 Grading Basis** Graded

A study of the relationship of nutrition and athletic performance.

Identification of the effects of age, sex, body build, environment, and state of health on energy needs and energy sources during physical activity.

Prerequisites: NUTR 280.

NUTR 384 Clinical Nutrition I**Credits 3 Grading Basis** Graded

This course introduces nutrition as a disease therapy and the role of the clinical dietitian as a member of the health care team. Topics covered include nutrition screening and assessment, medical records documentation, basic dietary modifications, and patient/family counseling.

Prerequisites: Dietetics major and NUTR 340.

NUTR 385 Nutrition Throughout the Life Cycle**Credits 3 Grading Basis** Graded

A study of the nutritional needs throughout the life cycle and the development of food habits. Nutrition assessment and nutrition education from prenatal health through infancy, childhood, adolescence, adulthood and old age are emphasized.

Prerequisites: Dietetics major.

NUTR 394 Culinary Nutrition Therapy**Credits 3 Grading Basis** Graded

This course integrates the art of cooking and food preparation with the science of health and nutrition through a lecture and lab format. Principles of cooking are integrated with medical nutrition therapy interventions and therapeutic coaching (considered culinary nutrition therapy) to optimize nutrition and health for disease prevention and for populations with chronic disease conditions while considering social, behavioral and biological factors associated with food choice and consumption.

Prerequisites: NUTR 340 and NUTR 385. Corequisites: NUTR 384.

NUTR 395 Introduction to Patient Care in Dietetics**Credits 2 Grading Basis** Graded

A study of the concepts of patient care in dietetics, skills needed for Medical Nutrition Therapy, and the dietitian's role on the health care team.

Prerequisites: Dietetics Majors and NUTR 280.

NUTR 446 Experimental Foods**Credits 3 Grading Basis** Graded

An introduction to research in foods. Different techniques of food preparation are studied and evaluated for most acceptable methods to obtain standard food products.

Prerequisites: Dietetics major, NUTR 340, CHEM 241 and MATH 220.

NUTR 455 Exercise and Nutrition in Chronic Disease**Cross-listed with:** KIN 424**Credits 3 Grading Basis** Graded

This interdisciplinary course examines the evidence-based relationship between exercise and diet patterns and behaviors, and major chronic diseases. Students will understand the epidemiology and pathogenesis of prevailing chronic diseases. Emphasis will be placed on developing exercise and dietary strategies to reduce the burden of chronic disease.

Prerequisites: BIO 270/BIO 270L and NUTR 280.

NUTR 465 Management in Dietetics**Credits 3 Grading Basis** Graded

Application of management concepts, theories and principles to dietetics with a focus on the work environments (clinical and foodservice) in which registered dietitians must effectively practice.

Prerequisites: Dietetics major.

NUTR 482 Nutrition and Metabolism**Credits 3 Grading Basis** Graded

This course is a study of the nutrients, their roles in intermediary metabolism, the effects of genetic errors in metabolism, nutritional deficiencies and means of assessing nutritional status. Agencies and programs concerned with nutrition and health and current trends in nutrition research are emphasized.

Prerequisites: Dietetics major, BIO 290/BIO 290L, and MATH 220.

Corequisites: BIO 290.

NUTR 484 Clinical Nutrition II**Credits 3 Grading Basis** Graded

A study of the use of diet in preventing illness and as a means of treating disease. Emphasis is given to patient education.

Prerequisites: Dietetics major, NUTR 384 and NUTR 482.

NUTR 485 Community Nutrition**Credits 3 Grading Basis** Graded

A study of human nutrition and health problems from a community perspective, programs and policies related to nutrition at local, state, and federal levels including preventive nutrition or wellness, and approaches and techniques for effective application and dissemination of nutrition knowledge in the community.

Prerequisites: Dietetics major.

NUTR 486 Counseling Skills for Dietetics**Credits 3 Grading Basis** Graded

This course will cover principles, theories and techniques of nutrition counseling for dietetics practice.

Prerequisites: Dietetics major, NUTR 384 and NUTR 385.

NUTR 490 Field Experience in Dietetics**Credits 3 Grading Basis** Graded

Students participate in field experience relating to their major area of dietetics and their career goals under the coordination of a dietetics faculty member. On-the-job supervision will be provided by the participating hospital dietitians. Application for enrollment must be completed through the course instructor in the fall semester prior to the summer in which it will be taken.

Prerequisites: Dietetics major, NUTR 384 and NUTR 385.

NUTR 495 Senior Seminar in Dietetics**Credits 3 Grading Basis** Graded

This course focuses on communicating nutrition-related research for specific audiences and introduces the application process for pursuing graduate-level education.

Prerequisites: Dietetics major.

NUTR 496 Special Studies in Nutrition/Dietetics**Credits 1-3 Grading Basis** Graded

This course is designed to give the student in dietetics an opportunity to complete independent study, professional conference participation, and/or research under faculty supervision. Prerequisite: Permission of the coordinator of the dietetics program.

NUTR 499A Honors**Credits 1 Grading Basis** Graded

In this course students will propose their honors thesis research project.

NUTR 499B Honors**Credits 3 Grading Basis** Graded

In this course students will complete a comprehensive review of literature and a formal plan for their honors thesis research project. They may also collect data if appropriate during this semester.

NUTR 499C Honors**Credits 2 Grading Basis** Graded

In this course students will complete research for their honors thesis project, write results, discussion and conclusions sections of the paper, and submit the final paper to the research committee and the Honors Program.

Persian (PERS)

PERS 101 Elementary Persian I**Credits 3-4 Grading Basis** Graded

The fundamentals of Persian through listening, speaking, reading and writing. Practice in pronunciation and development of comprehension. One hour's work a week in the language laboratory. Students will receive no credit for the course if they have had two or more years of the language in high school.

PERS 102 Elementary Persian II**Credits 3-4 Grading Basis** Graded

The fundamentals of Persian through a higher level of listening, speaking, reading and writing. Practice in pronunciation and development of comprehension. One hour's work a week in the language laboratory. If a student has had two or more years of the language in high school they will not receive credit for the course.

Prerequisites: PERS 101.

PERS 111 Intensive Persian I**Credits 6 Grading Basis** Graded

The fundamentals of Persian through listening, speaking, reading and writing. The 14-week course is the equivalent of PERS 101-PERS 102.

PERS 231 Intermediate Persian I**Credits 3 Grading Basis** Graded

A thorough review of first year grammar and vocabulary building.

Conversation, composition and reading will be chosen to reach competency at the lower intermediate level.

Prerequisites: PERS 102.

PERS 232 Intermediate Persian II**Credits 3 Grading Basis** Graded

A thorough review of PERS 231 grammar and vocabulary building.

Conversation, composition and reading will be chosen to reach competency at the advanced intermediate level.

Prerequisites: PERS 231 or by permission of the instructor.

PERS 490 Special Studies in Persian**Credits 1-4 Grading Basis** Graded

Special topics or independent studies in Persian.

Philosophy (PHIL)

PHIL 101 Introduction to Philosophy [C2HQC]**Credits 3 Grading Basis** Graded

An introduction to the basic problems and concepts of philosophy - the nature of the self, ethics, theories of knowledge, philosophy of religion, etc., as revealed in the writings of the major philosophers. May be used for general education credit.

PHIL 103 Introductory Topics in Philosophy**Credits 1-6 Grading Basis** Graded

Course on various topics in philosophy and/or ethics. May be taken multiple times for credit.

PHIL 120 Critical Thinking [C1CT]**Credits 3 Grading Basis** Graded

An introduction to the techniques for analyzing and evaluating information in everyday experience. The functions of language will be discussed. Techniques for judging the strengths of arguments and the probable truth of the arguments' premises will be examined. This course does not meet the philosophy requirement for the B.A. degree. May be used for general education credit. May not be used for major credit.

Prerequisites: If you have already completed, or are currently registered for, a Critical Thinking (Madison Foundations) class, then you must secure permission to receive credit for a second Critical Thinking class.

PHIL 150 Ethical Reasoning [C1CT]**Credits 3 Grading Basis** Graded

An introduction to the principles and techniques of critical thinking in ethics, including analysis of arguments and fallacies, ethical theories, and applications of moral principles to moral issues. This course does not meet the philosophy requirement for the B.A. degree. May be used for general education credit. May not be used for major credit.

Prerequisites: If you have already completed, or are currently registered for, a Critical Thinking (Madison Foundations) class, then you must secure permission to receive credit for a second Critical Thinking class.

PHIL 200 Character and Virtue**Credits 3 Grading Basis** Graded

An examination of philosophical perspectives on the nature and importance of character and virtue.

PHIL 210 Philosophy through Film**Credits 3 Grading Basis** Graded

This course combines feature length films and classic philosophical writings as points of departure for considering perennial philosophical questions such as: What is real? (Metaphysics) How can I know? (Epistemology) What is of value? (Morality).

PHIL 218 Philosophy of Religion**Cross-listed with:** REL 218**Credits 3 Grading Basis** Graded

This course offers an intensive examination of religion from the standpoint of philosophical thinking with particular emphasis on the way philosophers view such problems as the existence of God, evil, immortality, religious language, etc.

PHIL 220 Philosophy and Science Fiction**Credits 3 Grading Basis** Graded

This course will explore the philosophical aspects of science fiction. Topics may include the nature of time, time travel, free will, consciousness, personal identity, possible worlds, artificial intelligence, quantum theory or philosophical skepticism.

PHIL 240 Greek Philosophy in Context**Credits 3 Grading Basis** Graded

This course will involve exploration of the intellectual world of the Ancient Greeks, with particular emphasis on the philosophical, historical, and literary works produced during the period. The course will include lectures and site visits throughout Greece.

PHIL 250 Introduction to Symbolic Logic**Credits 3 Grading Basis** Graded

An introduction to the languages and techniques of propositional logic and first-order quantification theory.

PHIL 262 Problems in Applied Ethics**Credits 3 Grading Basis** Graded

Ethical theories are used to analyze contemporary moral issues in areas such as business and health care. Course content varies.

PHIL 267 Philosophy of Law**Credits 3 Grading Basis** Graded

Examination of the philosophical issues raised by the law, including the nature, foundation, and limits of the law, theories of its interpretation, and the fundamental interests it aims to protect.

PHIL 285 Philosophy, Art, and Literature**Credits 3 Grading Basis** Graded

This course will study artistic works (literary or otherwise) for their philosophical content. Related issues in the philosophy of art for example, the nature of tragedy, theories of interpretation may also be considered.

PHIL 290 Philosophy Teaching Practicum: Doing Philosophy in K-12 Schools**Credits 3 Grading Basis** Graded

This course involves a collaboration between undergraduate students in the Department of Philosophy and Religion and students in local schools. JMU undergraduate students will read texts and discuss ideas in the philosophy of pedagogy as well as meet weekly with local K-12 students. During these meetings, JMU students will discuss philosophical ideas with K-12 students, using storybooks and various activities (e.g., drawing, photographing, building and exploring) to generate discussions.

PHIL 300 Knowledge and Belief**Credits 3 Grading Basis** Graded

An extensive examination of theories of knowledge and philosophical problems concerning knowledge and belief.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 305 Seminar in Aesthetics**Credits 3 Grading Basis** Graded

Readings and discussions in the persistent philosophical problems of the arts, centering on consideration of the work of art, the artist and the audience.

Prerequisites: ART 200, ARTH 205, ARTH 206 or PHIL 101.

PHIL 310 Symbolic Logic**Credits 3 Grading Basis** Graded

The study and application of the principles and techniques of modern deductive logic to natural language. Also, examination of the properties of formal systems and of the logical implications and paradoxes of language.

Prerequisites: PHIL 250 or instructor consent.

PHIL 311 Metaphysics**Credits 3 Grading Basis** Graded

Examination of central questions regarding the fundamental nature of reality. Possible topics: universals and particulars, possibility and necessity, identity of objects over time and puzzles of material constitution, the problem of free will and determinism, and the nature of time.

Prerequisites: One other PHIL course or permission of the instructor.

PHIL 315 Logic and Legal Reasoning**Credits 3 Grading Basis** Graded

Application of symbolic logic (first-order logic with identity) to legal language and deductive legal argument. Will include close logical analysis of at least one of the following: Supreme Court brief, Supreme Court decision, Supreme Court oral argument.

PHIL 320 Inductive Logic**Credits 3 Grading Basis** Graded

Introduction to inductive logic and the philosophical problems it raises. This course may include topics such as the traditional problem of induction; the Goodman paradox and the new riddle of induction; the probability calculus and kinds of probability; Mill's methods of experimental inquiry and the nature of causality, abduction (inference to the best explanation); and confirmation theory.

PHIL 323 Comparative Ethics East and West**Cross-listed with:** REL 323**Credits 3 Grading Basis** Graded

In an increasingly globalized world, ethical decisions often involve considering diverse systems of beliefs about how one ought to live. In this course, students will think about ethical questions from cross-cultural perspectives, including perspectives of Buddhism, Confucianism, Aristotelianism and utilitarianism. By analyzing arguments made in Eastern and Western philosophical and religious traditions, students will begin to see similarities - such as the use of logical reasoning and the undesirability of pain - as well as fundamental differences. Case studies and examinations of contemporary issues make the course content relevant to the present day.

PHIL 325 Philosophy of Mathematics**Credits 3 Grading Basis** Graded

This course is a general survey of topics in the philosophy of mathematics. Topics may include the realism/antirealism debate, structuralism in mathematics, mathematical explanation, mathematical conceptions of infinity, the limits of mathematics or philosophical implications of set theory.

PHIL 326 Introduction to Non-Standard Logics**Credits 3 Grading Basis** Graded

This course will examine some non-standard logics and some of the philosophical issues they raise. Topics may include modal logic, intuitionistic logic, temporal logic or many valued logics.

PHIL 330 Moral Theory**Credits 3 Grading Basis** Graded

An examination, at the intermediate level, of both classical and contemporary moral theories. Critical analysis of the normative and meta-ethical issues these theories raise.

Prerequisites: PHIL 101 or permission of instructor.

PHIL 335 The Individual, The State and Justice**Credits 3 Grading Basis** Graded

Mid-level class in political philosophy. Will read classic and/or contemporary texts in philosophy influential on political thought. Focus may be on views of the justification for, and role of, the state. Consideration may also be given to the proper relationship of individuals and the state, political freedom and autonomy.

Prerequisites: PHIL 101 or permission of instructor.

PHIL 339 Islamic Philosophy**Cross-listed with:** REL 339**Credits 3 Grading Basis** Graded

This course will explore a variety of philosophical perspectives from across the Islamic world, employing methodological tools from both philosophy and religious studies. Themes may include: What is knowledge and how is it acquired? What does it mean to be human? What is the relationship between reason and other human faculties? What is the nature of time and history? What do we really mean when we talk about God? Philosophers covered may include: al-Kindi (d. 873), Ibn Sina (d. 1037), Ibn 'Arabi (d. 1240), Mulla Sadra (d. 1636), Muhammad Iqbal (d. 1938) and Fatema Mernissi (d. 2015).

PHIL 340 Ancient Greek Philosophy**Credits 3 Grading Basis** Graded

This course traces philosophical problems raised by the pre-Socratics, Parmenides, Heraclitus, Pythagoras and the Sophists, through their treatment by Plato and Aristotle. Emphasis is placed on selected writings of Plato and Aristotle.

Prerequisites: One other PHIL class or permission of the instructor.

PHIL 341 Modern Philosophy**Credits 3 Grading Basis** Graded

A selective survey of major issues and thinkers in Western philosophy from Descartes to Kant.

PHIL 342 Medieval Philosophy**Credits 3 Grading Basis** Graded

A survey of the major issues and thinkers of the medieval philosophical world with emphasis on the philosophical writings of those within the Western tradition, such as (e.g.) Augustine, Anselm, Aquinas, and Scotus. Prerequisites: One other PHIL class or permission of the instructor.

PHIL 343 Stoics, Skeptics and Epicureans**Credits 3 Grading Basis** Graded

This course will focus on philosophy from the Hellenistic and Roman periods (4th C. BCE through 5th C. CE), with a special emphasis on Epicureanism, Stoicism and Skepticism.

Prerequisites: One other PHIL course or permission of the instructor.

PHIL 344 Existentialism**Credits 3 Grading Basis** Graded

An examination of existentialism and its major spokesmen including such authors as Kierkegaard, Nietzsche, Sartre, Camus, Marcel and Heidegger.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 345 History of Ethics**Credits 3 Grading Basis** Graded

This course covers topics in ethics through the lens of influential figures in the history of philosophy. Possible topics include the foundations of morality, moral judgment, moral motivation, freedom of the will and the nature of our obligations to others.

PHIL 350 The Philosophy of Feminism**Cross-listed with:** WGSS 350**Credits 3 Grading Basis** Graded

An intermediate-level examination of philosophical problems in feminist theory and feminist contributions to philosophy.

PHIL 355 Philosophical Foundations of Psychoanalysis**Credits 3 Grading Basis** Graded

Rather than considering psychoanalysis as a field of research within psychology, or a form of therapeutic or clinical practice, this course will survey canonical works in psychoanalysis, including but also beyond Freud and Lacan, in an attempt to identify and explicate the philosophical core of psychoanalysis. As part of this survey of psychoanalysis, we will consider psychoanalytically informed discussions of politics, ideology, art, cinema, cultural phenomena and literature.

PHIL 360 Philosophy of Race**Credits 3 Grading Basis** Graded

This course examines some major approaches to the philosophical study of race and racism. Possible topics include the ontology of race and racism, the ethics and politics of identity, just and unjust forms of race-based discrimination, intersectionality, and the morality of race-based social policies, such as reparations.

PHIL 365 French Political Philosophy**Credits 3 Grading Basis** Graded

The middle of the twentieth century in France was a rich period of intellectual production, as well as a moment of intense social upheaval and political turmoil. This course studies the development of political philosophy in the 1950s in France. In particular, the course focuses on the way in which philosophical movements in France like existentialism and poststructuralism became increasingly sutured to their historical context and were thus compelled to critically engage with this historical situation - i.e., to confront capitalism, colonialism and patriarchy as philosophical problems. This course examines the relationship between these philosophical movements and the development of Marxism, postcolonial theory and feminism in French philosophy.

Prerequisites: PHIL 101 or permission of instructor.

PHIL 370 American Philosophy**Credits 3 Grading Basis** Graded

A study of the main philosophical ideas in America, especially pragmatism, with particular emphasis being given to Pierce, James, Royce, Dewey and Whitehead.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 375 Nineteenth-Century Philosophy and Theology**Cross-listed with:** REL 375**Credits 3 Grading Basis** Graded

A selected study of 19th-century thought, with emphasis on controversies concerning the nature and limits of reason, the ultimate meaning of history, and the inner meaning and social significance of religion. Pertinent thinkers include Hegel, Marx, Schleiermacher, Kierkegaard, Nietzsche and others.

PHIL 376 Nietzsche**Credits 3 Grading Basis** Graded

An examination of Nietzsche's philosophy based on close study of one or more of his major works. The course may also more briefly examine one or more thinkers who especially influenced Nietzsche (for example Schopenhauer) or who were strongly influenced by him (for example Foucault).

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 377 Hermeneutics**Cross-listed with:** REL 377**Credits 3 Grading Basis** Graded

This course will examine the main features of hermeneutics with particular emphasis on its contemporary perspectives. Discussion will focus on such themes as human understanding and human finitude, the nature of history and tradition, linguisticity and textuality of experience. Readings may address Gadamer, Ricoeur, Schleiermacher, and Dilthey.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 380 Philosophy of Language**Credits 3 Grading Basis** Graded

An examination of the central issues in contemporary (mainly 20th century) philosophy of language. Potential topics to be covered include meaning, reference, the nature of language and the nature of truth. Potential philosophers to be examined include Mill, Frege, Russell, Kripke, Lewis and Grice.

Prerequisites: PHIL 250 and one other course in philosophy course or permission of the instructor.

PHIL 390 Special Topics in Philosophy**Credits 3 Grading Basis** Graded

Topics for this intermediate-level course may be drawn from any area or period of philosophy chosen by the instructor. The course is designed primarily for philosophy majors and minors, but any suitably prepared student may take the course with permission of the instructor.

Prerequisites: One other PHIL course or permission of the instructor.

PHIL 391 Advanced Special Topics in Philosophy**Credits 3 Grading Basis** Graded

Topics for this advanced course may be drawn from any area or period of philosophy chosen by the instructor. The course is designed primarily for philosophy majors and minors, but any suitably prepared student may take the course with the permission of the instructor. May be repeated for credit.

Prerequisites: Six hours of philosophy or consent of the instructor.

PHIL 392 Philosophy of Mind**Credits 3 Grading Basis** Graded

An examination of competing theories of the intrinsic nature of mental states and mental processes, including careful consideration of questions concerning the relation between the mind and the physical world.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 394 Self and Identity**Credits 3 Grading Basis** Graded

This course will explore the philosophical aspects of personal identity, with particular emphasis on the metaphysics of the self. Other issues considered may include the nature of the soul, the status of self-knowledge and the ethical aspects of personhood.

Prerequisites: One other PHIL course or permission of the instructor.

PHIL 396 Philosophy of Physics**Credits 3 Grading Basis** Graded

This course examines the philosophical foundations of physics. Some of the philosophical issues explored in the course may include how various theories in physics impact metaphysics, ontology and/or epistemology.

Topics may include the nature of space and time, special relativity, general relativity, quantum theory, classical mechanics, thermodynamics, and/or cosmology.

PHIL 397 Philosophy of Space and Time**Credits 3 Grading Basis** Graded

This course will survey debates about the nature of space and time.

Topics may include Zeno's paradoxes, time travel, relationism vs. substantivalism, classical accounts of space and time, and relativistic space time.

PHIL 398 Philosophy of Quantum Theory**Credits 3 Grading Basis** Graded

Quantum theory describes atoms and particles and is one of the most empirically successful physical theories. However, quantum theory seems to have revolutionary conceptual implications for metaphysics and epistemology. This course offers an introduction to philosophical problems raised by quantum theory. Topics may include the measurement problem, quantum entanglement, different interpretations of quantum mechanics, the Einstein-Podolsky-Rosen (EPR) paradox, and Bell's theorem.

PHIL 410 Philosophy of Science**Credits 3 Grading Basis** Graded

This course surveys a number of topics about the nature of science.

The topics may include the problem of distinguishing science from pseudoscience, the nature of scientific explanation, the notion of progress in science, and the realism and anti-realism debate.

PHIL 430 Analytic Philosophy**Credits 3 Grading Basis** Graded

An examination of the origins and development of contemporary philosophical analysis, with special attention given to the nature and uses of language as well as logical structures of confirmation and explanation.

Prerequisites: PHIL 101, PHIL 250 or permission of the instructor.

PHIL 440 Advanced Moral Philosophy**Credits 3 Grading Basis** Graded

A critical examination, at the advanced level, of recent or historical work on normative and/or metaethical issues in moral philosophy.

Prerequisites: PHIL 101, PHIL 330 or permission of the instructor.

PHIL 460 Topics in Classical Philosophy**Credits 3 Grading Basis** Graded

An advanced study of major issues in or the writings of one, or more thinkers in ancient Greece through the western medieval period. May be repeated for credit with change of topics.

Prerequisites: PHIL 340 and Junior Standing.

PHIL 466 Kant**Credits 3 Grading Basis Graded**

This course examines the revolutionary theory of knowledge and critique of traditional metaphysics presented in Kant's Critique of Pure Reason, followed by some study of the novel approach to metaphysics adopted in his Critique of Practical Reason (and based in part on his Groundwork of the Metaphysics of Morals). The course may also briefly explore Kant's contributions to other important debates or Kant's impact on other major thinkers.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 468 Phenomenology**Credits 3 Grading Basis Graded**

A study of phenomenological investigation into the fundamental structures and conditions of conscious experience – meaning-laden, first person experience of objects, of events, of one's self, of other persons, of one's world, and so forth. The course will devote special attention to the work of Husserl and of Heidegger. Additional authors may be studied, such as Sartre, Merleau-Ponty or Levinas.

Prerequisites: PHIL 101 or permission of the instructor.

PHIL 490 Special Topics in Philosophy**Credits 3 Grading Basis Graded**

Designed to give capable students an opportunity to complete independent study in philosophy under faculty supervision. May be repeated for credit.

PHIL 495 Philosophy Course Assistantship**Credits 1-6 Grading Basis Graded**

Students participate as course assistants in philosophy. Assistantships provide students with a sense of what it is like to teach a philosophy course by allowing them to work closely with faculty members through different phases of course preparation, presentation and evaluation. Assistantships may also provide opportunities for student assistants to lead discussion and to help their peers review the material outside of the classroom.

PHIL 497 Ethical Reasoning Educator Practicum**Credits 1 Grading Basis Credit / No Credit**

Students participate in activities related to ethical reasoning on campus. Students receive training in writing ethical reasoning scenarios and leading discussions. The practicum also provides opportunities for peer-to-peer learning.

PHIL 499A Honors Thesis**Credits 1-3 Grading Basis Graded****PHIL 499B Honors Thesis****Credits 1-3 Grading Basis Graded****PHIL 499C Honors Thesis Prep****Credits 1-3 Grading Basis Graded**

Physics (PHYS)

PHYS 105 Foundations of Physics**Credits 1 Grading Basis Graded**

This course introduces beginning physics majors to the study of physics in the department of Physics and Astronomy. Students become acquainted with current research topics in physics and astronomy. The course includes an overview of library methods and programming skills used during physics course work. Students engage in dialogues regarding diversity, equity and inclusion within the physics community and the evolving relationship between physicists and society.

PHYS 106 Foundations of Physics Research**Credits 1 Grading Basis Credit / No Credit**

This introductory practicum course is designed to expose beginning physics majors to a variety of research topics studied and methodologies used across the department of physics and astronomy. The course emphasizes direct involvement in the cutting-edge research of faculty members and their groups, with active contribution to the advancement of scientific knowledge. Students will begin to develop skills in scientific organization, collaboration, data acquisition, analysis and communication of results to general audiences in written and oral formats.

PHYS 121 The Physical Nature of Light and Sound (3, 1) [C3PP]**Credits 4 Grading Basis Graded**

A study of the physical properties of light and sound waves. Topics include production, propagation and spectral analysis of waves. Applications to be covered include musical instruments, sound reproduction, room acoustics, optical instruments (cameras, projectors, lasers), and color in art and nature. The course will include outside-of-class experiential activities. May be used for general education credit.

PHYS 125 Physics with Biological Applications [C3PP]**Credits 3 Grading Basis Graded**

A study of fundamental physical principles pertinent to life processes. Areas include mechanics, thermal energy, fluids, sound, electricity and optics. Realistic biological examples are used to illustrate the relationship between physics and the life sciences. A working knowledge of algebra and trigonometry is required. May be used for general education credit. Corequisites: PHYS 125L.

PHYS 125L Physics with Biological Applications Laboratory [C3L]**Credits 1 Grading Basis Graded**

A laboratory course designed to complement the PHYS 125 lecture course. Laboratory exercises explore the application of physics to living systems. May be used for general education credit. Corequisites: PHYS 125.

PHYS 140 College Physics I [C3PP]**Credits 3 Grading Basis Graded**

The first semester of a non-calculus sequence in general physics. Topics include principles of mechanics, thermal properties of matter, wave motion and sound. A working knowledge of algebra and trigonometry is required. May be used for general education credit.

PHYS 140L General Physics Laboratory I [C3L]**Credits 1 Grading Basis Graded**

This laboratory course is designed to complement and supplement the PHYS 140 lecture course. PHYS 140L may be used for general education credit.

Prerequisites: PHYS 140. Corequisites: PHYS 140.

PHYS 150 College Physics II**Credits 3 Grading Basis Graded**

The second semester of a non-calculus sequence in general physics. Topics include electric charges, circuits, magnetism, optics, atomic and nuclear physics. Prerequisites: PHYS 140 (Must attain a grade of C- or higher).

PHYS 150L General Physics Laboratory II**Credits 1 Grading Basis Graded**

These laboratory courses are designed to complement and supplement the PHYS 140-PHYS 150 lecture courses.

Prerequisites: PHYS 140 and 140L and PHYS 150. Corequisites: PHYS 150.

PHYS 215 Energy and the Environment [C3PP]**Credits 3 Grading Basis** Graded

Energy use, sources and trends; fossil fuels, heat-work conversions, thermodynamic restrictions and electric power production; nuclear fission reactors and fusion energy; solar energy and technologies; alternative energy sources; energy storage; energy conservation; issues of waste and safety. Environmental, social and economic aspects will be discussed. Not open to ISAT majors scheduled to take ISAT 212 as part of their degree requirements. May be used for general education credit. Prerequisites: One college course in science and one in mathematics.

PHYS 240 University Physics I [C3PP]**Credits 3 Grading Basis** Graded

Kinematics, dynamics, energy and momentum conservation, oscillatory motion, fluid mechanics and waves. May be used for general education credit.

Corequisites: MATH 232 or MATH 235.

PHYS 240L University Physics Laboratory I [C3L]**Credits 1 Grading Basis** Graded

This laboratory course is designed to complement and supplement the PHYS 240 lecture courses. PHYS 240L may be used for general education credit.

Prerequisites: PHYS 240. Corequisites: PHYS 240.

PHYS 246 Data Acquisition and Analysis Techniques I**Credits 1 Grading Basis** Graded

This laboratory supplements the PHYS 240 lecture course by establishing the experimental basis of physics. Topics include conception, design and performance of experiments in physics emphasizing data acquisition, analysis of experimental data, and the handling of experimental uncertainties.

Prerequisites: PHYS 240.

PHYS 247 Data Acquisition and Analysis Techniques II**Credits 1 Grading Basis** Graded

This intermediate laboratory course is designed to prepare students for the advanced laboratory sequence and undergraduate research in physics or astronomy. The course emphasizes the conceptualization, design and execution of experiments in physics. Students will develop skills in the techniques of acquiring and analyzing data, performing error analysis, drawing conclusions from experiments, and presenting results in both written and oral form.

Prerequisites: PHYS 150L or PHYS 250L.

PHYS 250 University Physics II**Credits 3 Grading Basis** Graded

Electric forces, fields and potentials; capacitance, dielectrics, resistance and DC circuits; magnetic fields, induced electric fields, inductance and AC circuits; geometrical optics, interference, diffraction and polarization.

Prerequisites: PHYS 240 with a grade of "C-" or higher is the prerequisite. Corequisites: MATH 236.

PHYS 250L University Physics Laboratory II**Credits 1 Grading Basis** Graded

This laboratory course is designed to complement and supplement the PHYS 250: University Physics II lecture courses.

Prerequisites: PHYS 240L, PHYS 240, PHYS 250. Corequisites: PHYS 250.

PHYS 260 University Physics III**Credits 3 Grading Basis** Graded

The course will cover oscillations and wave dynamics, special relativity, and rotational kinematics and dynamics.

Prerequisites: "C" or better in PHYS 150 or PHYS 250. Corequisites: MATH 237.

PHYS 270 Modern Physics**Credits 3 Grading Basis** Graded

A course in modern physics, consisting of a discussion of the experimental basis for and fundamental principles of quantum physics, with applications to atomic structure and nuclear physics.

Prerequisites: "C" or better in PHYS 150 or PHYS 250, or consent of the instructor. Corequisites: MATH 237 and PHYS 270L.

PHYS 270L Modern Physics Laboratory**Credits 1 Grading Basis** Graded

This lab course introduces students to the experiments that revolutionized the field of physics around the turn of the 20th century. Topics include the properties of electrons, atoms, molecules and photons and their interactions. Students will utilize advanced data taking and analysis techniques and expand their programming skills.

Corequisites: PHYS 270.

PHYS 295 Laboratory Apparatus Design and Construction**Credits 1 Grading Basis** Graded

An introduction to the design and fabrication of laboratory apparatus using machine tools.

Prerequisites: PHYS 250 or permission of the instructor.

PHYS 297 Topics in Physics**Credits 1-4 Grading Basis** Graded

Topics in physics at the second-year level. May be repeated for credit when course content changes. Topics selected may dictate prerequisites. Students should consult instructor prior to enrolling for course.

PHYS 302 Science Pedagogy**Cross-listed with:** MSSE 397**Credits 2 Grading Basis** Graded

An introduction to the science of learning and how it informs teaching and learning in laboratory and field-based natural sciences. Special emphasis is given to conceptual change, metacognition and effective teaching strategies based on learning science, science discourse and argumentation, and safety.

PHYS 326 Biophysics**Credits 3 Grading Basis** Graded

Physical models are used to explain biological systems. Topics from biology include cell division, replication, transcription, and translation of DNA, protein folding, and molecular motors. Physics topics include entropy and free energy, diffusion, statistical mechanics of two state systems. Experimental tools for biophysics are also discussed..

Prerequisites: PHYS 150 or PHYS 250.

PHYS 330 Relativity**Cross-listed with:** ASTR 330**Credits 4 Grading Basis** Graded

The basic mathematical descriptions of both Einstein's special and general theories of relativity are introduced through specific physical cases and calculus-based lectures. Special relativity is explored through its physical consequences on a variety of phenomena. General relativity is studied by considering the effects of gravity on space-time curvature on astrophysical objects. Simple numerical experiments in both special and general relativity will be conducted as class projects.

Prerequisites: MATH 237 and PHYS 260, or permission of the instructor.

PHYS 333 Introduction to Particle Physics**Credits 3 Grading Basis** Graded

This is an introduction to current themes and ideas which confront the fundamental nature of matter and interactions. The most widely accepted theory, the Standard Model, will be explored. Possible extension, beyond the Standard Model physics, will be discussed. Basic properties such as charge, mass, and lepton number will be examined within these frameworks. Experiments that illuminate the basic nature of matter and ideas such as symmetry and quantum physics will be reviewed and assessed.

Prerequisites: PHYS 270 and PHYS 270L.

PHYS 337 Solid State Physics**Credits 3 Grading Basis** Graded

A study of the forces between atoms, crystal structure, lattice vibrations and thermal properties of solids, free electron theory of metals, band theory of solids, semiconductors and dielectrics.

Prerequisites: PHYS 270 and PHYS 270L.

PHYS 338 Nuclear Physics**Credits 3 Grading Basis** Graded

An introduction to the study of the atomic nucleus. Topics covered include static nuclear properties and movements, the force between nucleons, the deuteron, nucleon scattering, isospin, nuclear structure, radioactivity, decay kinematics and selection rules, fission, and fusion.

Prerequisites: PHYS 270 and PHYS 270L.

PHYS 339 Introductory Nuclear Science**Credits 4 Grading Basis** Graded

An introduction to nuclear science that will provide a solid foundation for experimental work in applied nuclear physics. Detection of ionizing radiation, as it applies to nuclear physics, will be additionally covered in the laboratory-component of the course. Topics include concepts of radioactive decays, radiation transport and interaction with matter, basics of radiation detection devices, dosimetry, radiation therapy, X-ray production, and fission nuclear reactors.

Prerequisites: PHYS 270 and PHYS 270L.

PHYS 340 Mechanics**Credits 3 Grading Basis** Graded

Application of the fundamental laws of mechanics to particles and rigid bodies. Topics include statics, dynamics, central forces, oscillatory motion and generalized coordinates.

Prerequisites: PHYS 260 and either MATH 238 or MATH 300 or MATH 336. Corequisites: PHYS 260 and either MATH 238 or MATH 300 or MATH 336.

PHYS 341 Non-Linear Dynamics and Chaos**Cross-listed with:** MATH 341**Credits 3 Grading Basis** Graded

Introductory study of nonlinear dynamics and chaos intended primarily for upper-level undergraduates in science or mathematics. Topics include stability, bifurcations, phase portraits, strange attractors, fractals, and selected applications of nonlinear dynamics in pure and applied science. Computers may be utilized for simulations and graphics.

Prerequisites: MATH 238 or (MATH 300 and MATH 336); and MATH 248 or CS 149.

PHYS 342 Mechanics II**Credits 3 Grading Basis** Graded

A continuation of PHYS 340 including Lagrangian dynamics, rigid body motion and the theory of small oscillations.

Prerequisites: PHYS 340.

PHYS 344 Advanced Physics Laboratory I**Credits 1 Grading Basis** Graded

The first course in a three-course laboratory sequence. A set of advanced laboratory experiences in which students are introduced to experimentation in several areas of physics while gaining experience in experiment design, data analysis, formal report writing and presentations.

Prerequisites: PHYS 247. Corequisites: PHYS 345.

PHYS 345 Advanced Physics Laboratory II**Credits 1 Grading Basis** Graded

This is the second course in a three-course laboratory sequence. A set of advanced laboratory experiences in which students are introduced to experimentation in several areas of physics while gaining experience in experiment design, data analysis, formal report writing and presentations.

Prerequisites: PHYS 344. Corequisites: PHYS 344.

PHYS 346 Advanced Physics Laboratory III**Credits 1 Grading Basis** Graded

This is the third course in a three-course laboratory sequence. A set of advanced laboratory experiences in which students are introduced to experimentation in several areas of physics while gaining experience in experiment design, data analysis, formal report writing and presentations.

Prerequisites: PHYS 344 and PHYS 345.

PHYS 347 Advanced Physics Lab**Credits 2 Grading Basis** Graded

In this advanced laboratory, students perform experiments from several areas of physics.

Prerequisites: PHYS 247.

PHYS 350 Electricity and Magnetism**Credits 3 Grading Basis** Graded

A study of the electrostatic field, the magnetic field, direct and alternating currents, and electromagnetic waves.

Prerequisites: PHYS 250 and MATH 237.

PHYS 360 Analog Electronics**Credits 4 Grading Basis** Graded

DC and AC circuits, spectral and pulse circuit response, semiconductor physics, and simple amplifier and oscillator circuits.

Prerequisites: PHYS 250 or permission of the instructor.

PHYS 364 Introduction to Fluid Mechanics**Cross-listed with:** MATH 364**Credits 4 Grading Basis** Graded

Introduces the student to the application of vector calculus to the description of fluids. The Euler equation, viscosity and the Navier-Stokes equation will be covered.

Prerequisites: MATH 237 and PHYS 260, or permission of the instructor.

PHYS 367 Introduction to Acoustics**Cross-listed with:** MATH 367**Credits 3 Grading Basis** Graded

This course represents an introduction to sound, hearing, and vibration. Architectural, biological and environmental acoustics will also be discussed. Students will develop an ability to use mathematical models and experimental techniques to study problems in acoustics and to transfer this knowledge to analogous situations. They will also develop an ability to conduct a semester-long research or expository project and present it in written and oral form to an audience of peers.

Prerequisites: MATH 236.

PHYS 371 Digital Electronics**Credits 2 Grading Basis** Graded

Transistors, integrated circuits, logic families, gates, latches, decoders, multiplexers, multivibrators, counters and displays.

Prerequisites: PHYS 150 or PHYS 250 with a grade of "C" or better or permission of the instructor.

PHYS 372 Microcontrollers and Their Applications (2,4)**Credits 2 Grading Basis** Graded

Microcontrollers, their instructions, architecture and applications.

Prerequisites: PHYS 371 or permission of the instructor.

PHYS 375 An Introduction to Materials Science**Cross-listed with:** CHEM 375, MATS 375**Credits 3 Grading Basis** Graded

An introduction to materials science with emphasis on general properties of materials. Topics will include crystal structure, extended and point defects, and mechanical, electrical, thermal and magnetic properties of metals, ceramics, electronic materials, composites and organic materials.

Prerequisites: CHEM 131 and PHYS 150 or PHYS 250 or ISAT 212 or permission of the instructor.

PHYS 380 Thermodynamics and Statistical Mechanics**Credits 3 Grading Basis** Graded

A treatment of the thermal properties of matter from both macroscopic and microscopic viewpoints. Topics include the laws of thermodynamics, heat, work, internal energy, entropy, elementary statistical concepts, ensembles, classical and quantum statistics and kinetic theory. Approximately equal attention will be given to thermodynamics and statistical mechanics.

Prerequisites: PHYS 270 and PHYS 270L.

PHYS 381 Materials Characterization (Lecture/Lab)**Credits 3 Grading Basis** Graded

A review of the common analytical techniques used in materials science related industries today, including the evaluation of electrical, optical, structural and mechanical properties. Typical techniques may include Hall Effect, scanning probe microscopy, scanning electron microscopy, ellipsometry and x-ray diffraction.

Prerequisites: CHEM 375 /PHYS 375/MATS 375 or ISAT 431.

PHYS 386 Robots: Structure and Theory**Credits 3 Grading Basis** Graded

An introduction to the study of autonomous robotic platforms. Topics include robot structure, propulsion systems, robot kinematics, sensors used in robotics, and sensor integration. The course combines lectures with laboratory activities in which students will get hands-on experience in designing, building, programming, and testing autonomous robotic platforms.

Prerequisites: Completion of CS 149 or ISAT 252 and MATH 232 or MATH 235 or ISAT 151 and ISCI 101 and ISCI 104 or PHYS 240/PHYS 240L or ISAT 152.

PHYS 387 Theory and Basic Principles of Mobile Robot Construction and Control**Credits 3 Grading Basis** Graded

The course is designed to familiarize the student with the basic elements of mobile robots. The course will provide students the opportunity to work with engineers and experts in the field of robotics to build a mobile robot while addressing the issues of control and principles of operation. The course will take advantage of the 4VA telepresence system to bring together experts and students from outside locations to supplement local instruction.

Prerequisites: Completion of CS 149 or ISAT 252 and MATH 232 or MATH 235 or ISAT 151 and ISCI 101 and ISCI 104 or PHYS 240/PHYS 240L or ISAT 152.

PHYS 388 Robots for Humanity**Credits 3 Grading Basis** Graded

Interdisciplinary team based course in which real world problems will be analyzed and tackled using robotics. May be repeated for credit when course content changes.

Prerequisites: PHYS 387 or completion of CS 149 or ISAT 252 and MATH 232 or MATH 235 or ISAT 151 and ISCI 101 and ISCI 104 or PHYS 240/PHYS 240L or ISAT 152.

PHYS 390 Computer Applications in Physics**Credits 3 Grading Basis** Graded

Applications of automatic computation in the study of various physical systems. Problems are taken from mechanics of particles and continua, electromagnetism, optics, quantum physics, thermodynamics and transport physics.

Prerequisites: MATH 248 or CS 149, PHYS 240, PHYS 250 and six additional credit hours in major courses in physics, excluding PHYS 360, PHYS 371 and PHYS 372.

PHYS 391 Seminar**Credits 0-1 Grading Basis** Graded

Participation in the department seminar program.

Prerequisites: Junior or senior standing and consent of the instructor.

PHYS 392 Seminar**Credits 1 Grading Basis** Graded

Participation in the department seminar program.

Prerequisites: Junior or senior standing and consent of the instructor.

PHYS 397 Topics in Physics**Credits 1-4 Grading Basis** Graded

Topics in physics at intermediate level. May be repeated for credit when course content changes. Topics selected may dictate prerequisites. Students should consult instructor prior to enrolling for course.

PHYS 398 Independent Study in Physics or Astronomy**Cross-listed with:** ASTR 398**Credits 1-3 Grading Basis** Graded

An individual project related to some aspect of physics or astronomy. Must be under the guidance of a faculty advisor. A student may not earn more than a total of 4 credits for PHYS 398 and ASTR 398.

PHYS 420 Modern Optics**Credits 3 Grading Basis** Graded

A study of the kinematic properties and physical nature of light including reflection, refraction, interference, diffraction, polarization, coherence and holography.

Prerequisites: PHYS 260, PHYS 270, PHYS 270L and MATH 237.

PHYS 440 Geophysics (3, 2)**Cross-listed with:** GEOL 440**Credits 4 Grading Basis** Graded

This course provides a survey of geophysical methods, with joint attention on near-surface and solid earth applications. Topics include seismology, heat flow, gravity, magnetism, electrical methods, ground penetrating radar and geophysical aspects of plate tectonics. Labs focus on practical experience with data acquisition, reduction and interpretation and are a combination of field, classroom and computational activities. Prerequisites: GEOL 110L or GEOL 115L, or PHYS 140 and PHYS 150, or PHYS 240 and PHYS 250; or permission of the instructor.

PHYS 455 Lasers and Their Applications to Physical Sciences**Cross-listed with:** CHEM 455**Credits 3 Grading Basis** Graded

An introduction to both the theoretical and practical aspects of lasers and their applications in the physical sciences.

Prerequisites: PHYS 270/PHYS 270L, CHEM 331, or permission of the instructor.

PHYS 460 Quantum Mechanics**Credits 3 Grading Basis** Graded

Principles and applications of quantum mechanics. Topics include wave packets and the uncertainty principle, the Schrodinger equation, one-dimensional potentials, operators and eigenvectors, three-dimensional motion and angular momentum, and the hydrogen atom.

Prerequisites: PHYS 340.

PHYS 482 Advanced Theoretical Physics**Credits 3 Grading Basis** Graded

This course is a continuation of topics from mechanics, thermodynamics and statistical Mechanics, electricity and magnetism, and quantum mechanics. The course may include topics from mechanics (Lagrangian dynamics, rigid body motion), electricity and magnetism (solution of Maxwell's equations in the presence of matter), thermodynamics and statistical mechanics (dynamical statistical mechanics) and quantum mechanics (beyond the hydrogen atom).

Prerequisites: PHYS 340 and PHYS 350; or PHYS 460; or permission of instructor.

PHYS 485 Science of the Small**Cross-listed with:** CHEM 485**Credits 3 Grading Basis** Graded

The chemistry and physics of materials of the nanometer scale will be explored. Lectures will cover contemporary issues in surfaces, self-assembly and low dimensional materials. The intersection of biology, chemistry, engineering and physics will be explored through nanoscience. Prerequisites: CHEM 132 and MATH 236; and PHYS 150 or PHYS 250.

Corequisites: PHYS 150 or PHYS 250.

PHYS 485L Science of the Small Laboratory**Cross-listed with:** CHEM 485L**Credits 1 Grading Basis** Graded

The chemistry and physics of materials of the nanometer scale will be explored. Labs will cover contemporary issues in surfaces, self-assembly and low dimensional materials. The intersection of biology, chemistry, engineering and physics will be explored through nanoscience. Prerequisites: CHEM 132 and MATH 236; and PHYS 150 or PHYS 250.

Corequisites: PHYS 485; PHYS 150 or PHYS 250.

PHYS 491 Seminar**Credits 0-1 Grading Basis** Graded

Principal course activities are participation in the departmental assessment program and attendance at departmental seminars.

Prerequisites: PHYS 392.

PHYS 492 Seminar**Credits 1 Grading Basis** Graded

Principal course activities are participation in the departmental assessment program and attendance at departmental seminars.

Prerequisites: PHYS 392.

PHYS 497 Topics in Physics**Credits 1-4 Grading Basis** Graded

Topics in physics at the advanced level. May be repeated for credit when course content changes. Topics selected may determine prerequisites. Students should consult instructor prior to enrolling for course.

PHYS 498 Undergraduate Research in Physics or Astronomy**Cross-listed with:** ASTR 498**Credits 1-4 Grading Basis** Graded

Research in a selected area of physics or astronomy as arranged with a faculty research advisor. A student may not earn more than a total of 6 credits for ASTR 498 / PHYS 498. Prerequisite: Proposal for study must be approved prior to registration.

PHYS 499 Honors**Credits 1-3 Grading Basis** Graded

Participation in this course must be approved during the second semester of the junior year. For details, see catalog section entitled "Graduation with Distinction."

Political Science (POSC)

POSC 200 Global Politics [C4GE]**Credits 3 Grading Basis** Graded

An exploration of political, social and economic issues and structures existing within and between states in the contemporary global community. Students are introduced to alternative approaches to analyzing these issues in diverse cultures and political settings. May be used for general education credit.

POSC 201 Introduction to Western Political Theory**Credits 3 Grading Basis** Graded

A general survey of Western political theory from Plato to Marx, including the foundations of political inquiry and the search for order and freedom.

POSC 225 U.S. Government [C4AE]**Credits 4 Grading Basis** Graded

An examination of institutions, processes and intellectual concepts which structure American political activity. The interaction of the political system with the changing American society and America's changing role in world affairs are also treated. The course provides an introduction to quantitative methodology. May be used for general education credit.

POSC 230 International Relations**Credits 3 Grading Basis** Graded

A survey of the field of international relations including consideration of the elements of national power, foreign policy, diplomacy, propaganda, foreign aid, war, international law and international organization.

POSC 240 Comparative Politics**Credits 3 Grading Basis** Graded

A comparative study of selected political systems. Emphasis is on the structure of government, the political process and the conditions which either promote or constrain political change and stability.

POSC 295 Research Methods in Political Science**Credits 4 Grading Basis** Graded

Students learn how to conduct original research from theory formulation through data collection and hypothesis testing. Special emphasis on research and computer literacy.

Prerequisites: Completion of the General Education: The Natural World: Quantitative Reasoning area and either POSC 201, POSC 225, POSC 230 or POSC 240.

POSC 300 Politics and Film**Credits 3 Grading Basis** Graded

This course examines the relationship between politics and film, broadly construed. The ability of film to inform and promote agendas on key political issues, and the way in which films reflect the world of ideas and political culture in which they are created will be explored in substantive areas that may include: social and political change, human rights and justice, the portrayal of political processes and institutions, and foreign policy.

POSC 301 The Washington Semester Experience**Credits 3 Grading Basis** Graded

A study of the manner in which the policy making process is conducted on the federal level. The function of political and governmental institutions in establishing public policy is examined through readings and observation. Enrollment in Washington Semester program required.

POSC 302 State and Local Government**Credits 3 Grading Basis** Graded

A study of state and local government in the United States with particular focus on Virginia. Emphasis is placed on an understanding of the framework, functions, and problems of state and local governments.

POSC 310 Ancient Political Theory: Classical Greek Political Philosophy to Roman Stoicism**Credits 3 Grading Basis** Graded

This course is a study of the foundations of Western political theory as established by Plato and Aristotle, and developed by subsequent thinkers. Prerequisites: POSC 201.

POSC 311 The Ideas and Legacies of Medieval and Renaissance Political Theory**Credits 3 Grading Basis** Graded

This course is a study of the ideas, principles and legacy of political theory as developed within the Middle Ages and the Renaissance. Among others, and with some variation, works by St. Augustine, Alfarabi, Maimonides, St. Thomas Aquinas, Dante, Marsiglio of Padua, Christine de Pizan, Marsiglio Ficino and Machiavelli will be read. Other figures may also be covered, time permitting. Prerequisites: POSC 201.

POSC 315 Political Theory: Early Modern to the 19th Century**Credits 3 Grading Basis** Graded

A study of political theory from Hobbes and Locke to Hegel, Green and other 19th century thinkers. The course will examine such ideas as freedom, political obligation, justice, progress, ethics and politics, and the relationship between the individual and the human polity.

POSC 316 Contemporary Political Theory**Credits 3 Grading Basis** Graded

An examination of political thinkers and their ideas from the end of the 19th century to the present. Special emphasis will be placed on the writings of Hannah Arendt, Jurgen Habermas, John Rawls and other contemporary thinkers who continue to engage in the pursuit of political inquiry.

POSC 321 Political Theory and Ideology**Credits 3 Grading Basis** Graded

A study of the relationship between normative political theory and ideology, emphasizing the philosophic foundations of modern political thought and its relationship to the emergence of various ideological positions in the 19th and 20th centuries. Includes a study of liberalism, conservatism, socialism, anarchism, nationalism, fascism, feminism, environmentalism and others.

POSC 322 Civic Engagement**Credits 3 Grading Basis** Graded

This course investigates the importance of civic engagement for individuals and communities, long-term trends in civic behaviors and the implications of those trends. It explores traditional ways of preparing young people to be active and informed participants in civic and political life, as well as institutional, structural and political barriers to doing so. New, promising approaches to civic learning geared toward historically marginalized populations are also examined. Prerequisites: POSC 225 or permission of the instructor.

POSC 323 Cities and Placemaking**Credits 3 Grading Basis** Graded

This course examines urban spaces and encourages students to think critically about place through the lens of several different academic disciplines. Topics include the importance of places cultivating feelings of community attachment and why it matters; designing places that promote informal social interactions and connectivity; celebrating cultural identities and confronting complicated histories; implications of transportation alternatives; creating places that are safe without being sterile; and common mistakes of the modern era. Prerequisites: POSC 225 or permission of the instructor.

POSC 324 Politics of Human Rights in the United States**Credits 3 Grading Basis** Graded

The purpose of this course is to introduce students to the study of human rights within a U.S. political context. Course participants will examine the impact of federal, state and local legislation, the influence of US political actors, the work of advocacy organizations along with the responses from government institutions (the bureaucracy) and the American judicial system. The course will further student understanding of the complexities of human rights controversies and how consideration of these rights is conducted in practice in American society today.

POSC 325 Constitutional Law**Credits 3 Grading Basis** Graded

A study of the legal aspects of the American democratic system. The development of the Constitution will be explored and case studies used to portray important events and changes.

POSC 326 Civil Rights**Credits 3 Grading Basis** Graded

An examination of the judicial interpretation of civil rights in America with emphasis on freedom of speech, due process of law and equal protection under the 14th Amendment.

POSC 327 Race and Capitalism in the United States**Credits 3 Grading Basis** Graded

What is the relationship between race and capitalism? How have individuals, institutions and nation-states employed racial categories and hierarchies to generate and distribute wealth and limited resources? This course introduces students to the conjunctive study of American racial politics and American political economy. Specifically, this course introduces students to the concept of "racial capitalism," which posits that race and capitalism are deeply intertwined, mutually constituted institutions. Students will become familiar with foundational literature and recent scholarship in the field that applies theories of racial capitalism to American politics and culture.

POSC 328 Fair and Affordable Housing Law and Policy**Credits 3 Grading Basis** Graded

This class will explore housing law and policy. Topics include private landlord tenant laws, public and other federally subsidized housing, fair housing and the history of racial segregation, fair housing and the accommodation of physical and mental disabilities, tenant screening and evictions based upon criminal records, domestic violence and housing, sub-prime mortgage lending, foreclosures, manufactured-homes, the eviction crisis, and trends to create more affordable housing.

POSC 329 Civil Rights and Criminal Procedures**Credits 3 Grading Basis** Graded

This course introduces students to the issues and processes involved in protecting the personal liberties of criminal suspects, focusing on the 4th, 5th and 6th Amendments. Topics include search and seizure, arrests, use of force, interrogation and confession, right to counsel, pretrial identification, the Exclusionary Rule and other remedies for constitutional violations.

POSC 330 American Political Thought**Credits 3 Grading Basis** Graded

A study of the development and significance of political ideas that have influenced American society and government.

Prerequisites: POSC 201.

POSC 331 Human Rights in Theory and Practice**Cross-listed with:** JUST 331, HUMN 331**Credits 3 Grading Basis** Graded

This course will explore the nature and value of human rights by investigating some major debates over their status and meaning, and by examining some of the ways people have tried to secure human rights in practice.

Prerequisites: Political science and international affairs majors only.

POSC 332 Black Political Thought**Credits 3 Grading Basis** Graded

Throughout the entire history of Black life and contact with what we call "The West," Blacks throughout the African diaspora have interrogated, analyzed and problem-solved questions of oppression, freedom and justice. How, exactly, did Black people, to paraphrase W.E.B. DuBois, become "a problem?" What role do politics, history, law, economics and culture play in problem-solving racial oppression and securing freedom and justice? How do interrelated social dynamics (such as race, ethnicity, class, gender, sexuality, etc.) shape and reflect Black political demands and mobilization? What is the most appropriate lens through which Black intellectuals, both organic and academically credentialed, should analyze and interrogate the persistence of anti-black violence and discrimination? In short, how do Black people finally get free? In this course, we will examine some of the most salient of these reflections and the multiple traditions of Black political thought they represent. Students will develop their ability to critically assess disparate articulations of the relationship between domination, freedom and justice in Black political thought from the 18th century to the present. In doing so, students will develop a well-rounded view of this expansive literature while interrogating their own political and philosophical commitments.

POSC 333 American Political Thought II: Civil War to the Present**Credits 3 Grading Basis** Graded

This course explores American political thought from Reconstruction (ca. 1866) to the present day, drawing largely on primary sources, including writings by political leaders and activists, political theorists, philosophers, public intellectuals and religious thinkers.

Prerequisites: POSC 201.

POSC 334 Political Thought of James Madison**Credits 3 Grading Basis** Graded

This course examines the political thought of James Madison (1751-1836). Madison was a profound political thinker who read voraciously, thought deeply, and wrote eloquently and carefully about many of the leading political questions of his day - and ours. In this class, we will read, understand and critically assess what Madison contributed to these political dialogues and more generally to the American political tradition.

Prerequisites: POSC 201.

POSC 335 Comparative Politics for Teachers**Credits 3 Grading Basis** Graded

This course examines the core themes, concepts and debates in the subfield of Comparative Politics and how they apply to politics in selected countries around the world. The course emphasizes those themes and cases of greatest use to students pursuing careers in education and explores approaches to teaching this content in a variety of classroom settings.

POSC 336 Civic Engagement for Teachers**Credits 3 Grading Basis** Graded

This course serves future social studies teachers. It investigates the importance of civic engagement for individuals and communities, long-term trends in civic behaviors, and the implications of those trends. It explores traditional ways of preparing young people to be active and informed participants in civic and political life, as well as institutional, structural and political barriers to doing so. New, promising approaches to civic learning geared toward historically marginalized populations are also examined.

Prerequisites: POSC 225 or permission of the instructor.

POSC 340 Politics of Development**Credits 3 Grading Basis Graded**

A comparative study of the processes of political development in the developing nations of the Middle East, Africa, Asia and Latin America. Attention is given to the special problems confronting these nations and their implications for the global systems.

POSC 341 Social Movements in the U.S. and Abroad**Credits 3 Grading Basis Graded**

In recent decades, social movements have mobilized people concerned about issues ranging from the rights of ethno-racial minorities, women, sexual minorities and immigrants to the environment, human rights and world peace. This course will examine the origins, modes of action and impact of such movements. We will apply the comparative method to analyze social movements in the United States and abroad, investigating how different socioeconomic and political contexts shape social movements based on common issues. The course will emphasize how social movements emerge and function within and alongside existing structures of formal politics.

POSC 344 Politics of the European Union**Credits 3 Grading Basis Graded**

The course offers an in-depth consideration of the political development of the European Union, the EU policy-making process, and contemporary issues that confront European leaders and citizens.

POSC 345 Politics of Western Europe**Credits 3 Grading Basis Graded**

This course involves comparative analysis of the development and dynamics of political regimes in western Europe. Attention is given to political institutions, political participation, public policy, and political and economic trends since 1945.

POSC 346 Politics of Central and Eastern Europe**Credits 3 Grading Basis Graded**

This course involves comparative analysis of the development and dynamics of political regimes in central and eastern Europe. Attention is given to pre-communist, communist, and post-communist politics and to explaining political and economic trends since 1989.

POSC 347 Comparative Public Policy**Credits 3 Grading Basis Graded**

A study of public policy formation and implementation in selected advanced industrial and Third World nations.

POSC 348 The Politics of Cultural Pluralism**Credits 3 Grading Basis Graded**

This course examines the various manifestations of cultural pluralism, a situation that occurs when multiple ethnic, religious, and/or linguistic groups coexist within a single state. This course considers different institutional and policy approaches to coping with cultural pluralism.

POSC 350 Latin American Politics**Credits 3 Grading Basis Graded**

A comparative study of the political institutions, processes and current issues in the Latin American states, and an analysis of their importance in regional and global relations.

POSC 351 Topics in American Politics**Credits 3 Grading Basis Graded**

In-depth exploration of specialized topics in the area of American politics. The topic for each semester will be announced on MyMadison.

POSC 353 African Politics**Credits 3 Grading Basis Graded**

A comparative study of the institutions and social, economic, and global processes that affect contemporary African states. Political developments explored include the construction and transformation of postcolonial states, ethnic conflict, economic crisis and reform, and regime change.

POSC 354 Politics of the Middle East**Credits 3 Grading Basis Graded**

This course involves comparative analysis of political institutions, social dynamics and economic processes in the contemporary Middle East. The course also focuses on the ways that global developments affect and are affected by Middle Eastern states and peoples.

POSC 355 East Asian Politics**Credits 3 Grading Basis Graded**

A study of the political systems of the major countries of East Asia, including Japan, China and Korea. Issues discussed include political development and democratization movements in the People's Republic of China, the Republic of China and the Republic of Korea.

POSC 356 Politics of China**Credits 3 Grading Basis Graded**

This undergraduate course provides an introduction to key institutions, transformations, and issues in contemporary Chinese politics.

POSC 360 The Politics of Mental Health**Credits 3 Grading Basis Graded**

In what sense is mental health political, and how can political science contribute to discussions around mental health? In this course, we will investigate how mental illness and mental health have been defined over time and examine historical developments in mental health policy from the 19th century to the post-Covid era. We will then turn our attention to how politics and mental health affect one another. How do mental health diagnoses (particularly clinical depression and anxiety) affect political attitudes and political behavior? In turn, how do political events and public policy outcomes affect individuals' propensity for anxiety or depression? The course will conclude by considering the mental health of politicians themselves, the status of mental health care today and factors that will shape access, treatment and outcomes into the future.

Prerequisites: POSC 225.

POSC 361 Topics in International Relations**Credits 3 Grading Basis Graded**

In-depth exploration of specialized topics in the field of international affairs. The topic for each semester will be announced on MyMadison.

POSC 362 Political Behavior**Credits 3 Grading Basis Graded**

A study of how citizens acquire politically relevant attitudes and how these attitudes influence their political behavior. The effects of the mass media on voting behavior are also considered. Emphasis is placed on U.S. voting behavior but behavior in other nations is also covered.

Prerequisites: POSC 225.

POSC 363 Hacking for Diplomacy**Credits 3 Grading Basis Graded**

Hacking for Diplomacy is a project-orientated course that takes real problems from the Department of State, the intelligence community and other federal departments and agencies and has students use the Lean Launch Pad Methodology to test problem understanding and potential solutions. The course provides students the opportunity to serve as resource to help address critical problems facing the U.S. government today. Participation in this course promotes engagement between students and stakeholders from a range of government agencies and provides hands-on opportunity to solve real national security and foreign policy problems. A network of such courses across the U.S. is providing a system that can develop solutions that match government end users' in weeks.

Prerequisites: POSC 230 and POSC 240.

POSC 364 Political Parties in America**Credits 3 Grading Basis Graded**

This course will focus solely on the central role of political parties in the United States. Even though parties are not mentioned in the U.S. Constitution and the framers despised them, they quickly formed and have come to dominate our political environment. Fully understanding political parties is essential for the political science student.

Prerequisites: POSC 225 or permission of the instructor.

POSC 365 American Political Campaigning**Credits 3 Grading Basis Graded**

Study of modern day political campaigning with emphasis on campaign structure, strategy, and the relationship between candidates and political consultants. The course assesses the consequences of the changing nature of political campaigns for democracy in the United States.

Prerequisites: POSC 225.

POSC 366 Introduction to the Politics of Race and Ethnicity**Credits 3 Grading Basis Graded**

This course is centered on the politics of race and ethnicity within the United States of America. More specifically, it discusses how the U.S. government has and continues to shape patterns of racial inequality through public policy. By the end of the course, students will develop a broad, conceptual understanding of how racism is woven into the fabric of our society.

Prerequisites: POSC 225.

POSC 367 U.S. Immigration and Refugee Law**Credits 3 Grading Basis Graded**

This course provides a historical context to the current conversations regarding immigrant integration, immigration policy and immigration reform. The course will focus on the development of our nation's immigration policies and the role that the social construction of race and gender has played in shaping decisions regarding immigration and citizenship. The course will also focus on international and forced migration, including the laws that govern those seeking refuge and asylum, and asylum policy in the U.S.

Prerequisites: POSC 225.

POSC 368 Interest Groups, Lobbying and Public Policy**Credits 3 Grading Basis Graded**

An analysis of the mobilization and lobbying activities of interest groups in the American public policy system, with an emphasis on the origin and maintenance of interest organizations; diversity and bias in the interest group system; lobbying influence and policy advocacy strategies, tactics, and targets; and the regulation of lobbying and related campaign finance activities.

Prerequisites: POSC 225 or permission of the instructor.

POSC 369 Elections and Voting Behavior**Credits 3 Grading Basis Graded**

The electoral process is one of the primary means by which Americans can hope to influence public officials, and thus public policy, in the United States. Elections provide voters an opportunity to hold political officials and political parties responsible for their actions in office; ideally, elections as a means of accountability should serve to keep office holders responsive to the public's needs. In reality, however, there are many impediments that may preclude the democratic system from functioning as effectively as it should, such as low levels of voter knowledge and voter turnout, changes in primary election scheduling, and even mathematical complications posed by the Electoral College system. Only by studying elections can we understand the complexity of our political environment, and how factors beyond voter preferences and candidate agendas can shape election outcomes in the United States.

Prerequisites: POSC 225.

POSC 370 U.S. Foreign Policy**Credits 3 Grading Basis Graded**

An investigation of the processes for making foreign policy, underlying premises influencing specific policies and substance of American foreign policy.

Prerequisites: POSC 230.

POSC 371 Topics in Comparative Politics**Credits 3 Grading Basis Graded**

In-depth exploration of specialized topics in the area of comparative politics. The topic for each semester will be announced on MyMadison.

POSC 371L Special Studies London**Credits 3 Grading Basis Graded**

An introduction to English government and law. Outings include visits to the Houses of Parliament, trials, and other events of political and legal import.

POSC 372 Ethics and International Politics**Cross-listed with: JUST 372****Credits 3 Grading Basis Graded**

This course investigates the significance of ethical questions in the theory and practice of contemporary international politics, introducing a variety of normative approaches that shape the issues of peace and conflict, morality and justice in global affairs. Practical case studies will also be used to address issues of policy relevance, with particular attention paid to the American experience.

Prerequisites: INTA or POSC majors, or JUST majors who have completed JUST 235.

POSC 374 War and Justice**Cross-listed with: JUST 374****Credits 3 Grading Basis Graded**

This course is an empirical and normative investigation of the relationship between war and justice. Empirically, it examines the causes of war and the ways in which wars end. From a normative perspective it raises the questions of when are wars just; how should a just war be fought; and what is a just peace.

Prerequisites: JUST 200 and one other 200-level JUST course, excluding JUST 225.

POSC 375 Latina/o/x Politics: Origins and Futures**Cross-listed with:** LAXC 375**Credits 3 Grading Basis** Graded

The course will focus on the Latina/o/x populations in the United States and how the category of 'Latino' came into existence. Furthermore, students will learn how Latina/o/x identity has evolved over time as well as become politically salient within 21st century politics. The goal is for the student to develop a broad, conceptual understanding of how the term 'Latino' came into existence as well as how individuals who identify as 'Latina/o/x' have politically organized throughout history.

Prerequisites: POSC 225.

POSC 376 Introduction to Black Politics**Credits 3 Grading Basis** Graded

This course is an introduction to how Black political scientists and Black communities have developed and articulated a Weltanschauung or worldview that situates their political thought, behavior, productions, leadership and pathways to liberation from historical, contemporary, transnational and intersectional frameworks. Topics discussed in this course include political organizing for social justice; political thought and philosophy; formal and informal political leadership; race, gender, class and sexuality; electoral politics; and more.

POSC 377 Global Black Feminist Politics: On Power, Resistance and Transformation**Credits 3 Grading Basis** Graded

This course explores how anti-Blackness, racialization, transphobia, misogynoir, violence and more impact the everyday lives of Black women and gender expansive communities from a transnational Black feminist lens.

POSC 378 Borderlands Policy: Latina/o/x Politics in Context**Cross-listed with:** LAXC 378**Credits 3 Grading Basis** Graded

This course is centered on the policies surrounding the U.S. and Mexico border, with a focus on the effect these policies have had on Latina/o/x populations. Students will examine peer-reviewed studies of immigration policy, public opinion and political rhetoric, as well as individual accounts of the immigrant experience. By the end of the course, students will have an increased understanding of how border policy has historically evolved and the Latino/a/x population reactions across time.

Prerequisites: POSC 225.

POSC 379 International Laws of War**Credits 3 Grading Basis** Graded

This course takes a close look at key topics in international laws of war, also known as international humanitarian law (IHL). It is intended to be a survey of international laws of war. It is assumed that students are coming to the course with no or little background in this area of the law (either through study or prior experience), or in related fields. Building upon a foundation of assigned readings on core IHL topics, this course will examine complex and controversial topics such as drone warfare, human rights in counterterrorism and the obligations of military personnel. Students will acquire knowledge of IHL by discovering its rules applicable to practical cases taken from contemporary practice.

POSC 380 The U.S. Presidency**Credits 3 Grading Basis** Graded

A study of the institution of the American presidency focusing on the sources, bases and character of the power required by the president for effective executive action. Relationships of the presidency to foreign affairs, Congress, the public, party structure and the administrative establishment will also be considered.

Prerequisites: POSC 225.

POSC 381 Topics in Political Theory**Credits 3 Grading Basis** Graded

In-depth exploration of specialized topics in the area of political theory. The topic for each semester will be announced on MyMadison.

POSC 382 The Role of Religion in American Politics**Credits 3 Grading Basis** Graded

An examination of the role religion has played and continues to play in American politics. Besides providing an overview of how religion has influenced electoral, legislative, and judicial outcomes, the course will provide an in-depth examination of particular aspects of religion's role in political life.

Prerequisites: POSC 225 or permission of the instructor.

POSC 383 Women and Politics in Comparative Perspective**Cross-listed with:** WGSS 383**Credits 3 Grading Basis** Graded

A study of the causes and consequences of women's political marginalization in the United States and abroad. The course examines socioeconomic and political dimensions of gender inequality, exploring how women have worked through social movements, electoral politics, and public policy initiatives to overcome obstacles to their political empowerment.

POSC 384 Minority Group Politics**Credits 3 Grading Basis** Graded

This course examines the role of minority groups in American Politics. Attention is given to five groups (African Americans, Asian Americans, Native Americans, Latinos and women) that for reason of race or gender have faced institutional discrimination and political domination in the United States.

POSC 385 The U.S. Congress**Credits 3 Grading Basis** Graded

Study of the legislative process will concentrate on the operation of Congress with regard to such matters as its rules and procedure; relationships to the presidency, the bureaucracy, pressure groups and the courts; and a discussion of its current problems.

Prerequisites: POSC 225.

POSC 386 The U.S. Judiciary**Credits 3 Grading Basis** Graded

An investigation of the American court system. The course focuses on the role of the judiciary in American politics, the difference between judicial and other political and bureaucratic decision-making processes, the selection of judges, the decisions made by judges and other actors interacting with the courts, and the impact of court decisions on American society.

POSC 387 The Appellate Process and Practice**Credits 3 Grading Basis** Graded

An examination of the Supreme Court appellate process in the United States from its beginnings with the lower court opinion through oral argument. Class is conducted as a seminar in which students are expected to engage actively. Students study substantive areas of constitutional law and the common law system of precedent.

POSC 389 Hacking for National Security and Defense**Credits 3 Grading Basis** Graded

This is a project-oriented course that takes real problems from the Department of Homeland Security, the Department of Defense and other federal agencies and has students use design thinking and entrepreneurial start-up principles to test problem understanding and potential solutions. The course provides students the opportunity to serve as a resource to help address critical problems facing the U.S. government today. Participation in this course promotes engagement between students and stakeholders from a range of government agencies and provides a hands-on opportunity to solve real national security and defense problems.

Prerequisites: POSC 230 and POSC 240.

POSC 391 Topics in Public Policy**Credits 3 Grading Basis** Graded

In-depth exploration of specialized topics in the area of public policy. The topic for each semester will be announced on MyMadison.

POSC 392 Peace Studies**Cross-listed with:** JUST 392**Credits 3 Grading Basis** Graded

A study of the evolution, theory and practice of peace studies. The course focuses on how we wage and resolve conflict, how we effect social change, and how we provide for security through nonviolent means.

POSC 393 Human Security**Credits 3 Grading Basis** Graded

This course examines the concept of security in a globalized world, paying particular attention to threats to individuals and groups in the post-Cold War international system. Topics include the transition from state security to individual security, human development, human rights, military intervention, responsibility to protect, and global governance. Prerequisites: POSC 230.

POSC 394 International Conflict**Credits 3 Grading Basis** Graded

A study of the causes and consequences of violent political conflict between and within states. The course emphasizes the theoretical explanations of conflict initiation, escalation, and termination, with applications to both individual cases and broader patterns of conflictual behavior.

Prerequisites: POSC 230.

POSC 395 International Law**Credits 3 Grading Basis** Graded

Examination of the role of international law in world politics. Particular attention will be given to the effects of international law on patterns of international exchange and interaction. Case study and other forms of political analysis will be used.

POSC 396 International Organizations**Credits 3 Grading Basis** Graded

Study of the evolution and role of contemporary international organizations in the larger context of world politics. Emphasis on the ways in which the changing patterns of political power influence the processes and effectiveness of such organizations.

POSC 397 The Politics of International Economic Relations**Credits 3 Grading Basis** Graded

A study of the political dynamics and implications of international economic relations.

POSC 398 Simulations**Credits 3 Grading Basis** Graded

Application of concepts and insights learned in the classroom to contemporary policy problems and practical activities. The topic of this course will vary from offering to offering. The exact courses required will vary with the subject matter of the simulation.

POSC 401 Topics in Political Science**Credits 1-3 Grading Basis** Graded

This course introduces students to a novel topic in political science or public affairs. Content will vary from offering to offering.

POSC 430 International Security and Conflict Management**Credits 3 Grading Basis** Graded

This course examines major threats to international security in the post-Cold War world. Topics include the changing global security environment, proliferation of weapons of mass destruction, terrorism and ethnic conflict. The effectiveness of economic sanctions, deterrence, international organizations, preventative war and other tools in dealing with these threats in critically examined.

POSC 435 International Terrorism**Credits 3 Grading Basis** Graded

Systematic study of political terrorism with emphasis upon the destabilizing effect that it has upon the international community.

POSC 440 Post-Conflict Societies**Cross-listed with:** POSC 540**Credits 3 Grading Basis** Graded

An examination of the political dynamics of social, economic, security and governance problems faced in rebuilding societies that are emerging from a period of intense conflict. Emphasis on institutional and behavioral challenges brought about by conflict; strategies employed by domestic and international actors to promote democratic, just and secure societies after conflict; and standards that can be used to assess the success, failure and durability of a newly emerging political system.

POSC 457 Fates of Empires**Cross-listed with:** HIST 357**Credits 3 Grading Basis** Graded

Fates of Empire looks at several different historical empires to gain an interdisciplinary perspective on what makes empires succeed or fall apart. The course will apply a variety of theoretical approaches from history, political science, economics and geography to consider each case study and the broader concept of empire. This class is particularly aimed at students in the social studies teacher licensure programs, to help them learn how to apply the history in middle and high school classrooms.

Corequisite: MSSE 412.

POSC 458 International Political Analysis**Credits 3 Grading Basis** Graded

An examination of techniques and principles for the analysis of future political conditions and future government decisions.

POSC 472 Media and Politics**Cross-listed with:** SCOM 472, SCOM 472L, SMAD 472, SMAD 472L**Credits 3 Grading Basis** Graded

This course is a study of the media's role in the political system, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories; how media practices and technologies shape political decision making; how political forces influence media institutions; and how public opinion and media narratives are shaped.

POSC 473 British Media and Politics**Cross-listed with:** SCOM 473, SMAD 473**Credits 3 Grading Basis** Graded

This course is a study of the media's role in the political system of the United Kingdom, concentrating on the dynamic and interdependent relationships among journalists, politicians, the public and public policy. The course explores media and politics theories as they apply to the British context; how UK media practices and technologies shape political decision making; how political forces influence British media institutions; and how public opinion and media narratives are shaped within a parliamentary democracy.

Prerequisites: Admission to the JMU semester in London.

POSC 490 Senior Tutorial in Political Science**Credits 4 Grading Basis** Graded

A research-oriented tutorial designed to integrate student's prior knowledge and strengthen lifelong learning skills.

Prerequisites: Senior standing.

POSC 491 Washington Semester Internship Readiness**Credits 1 Grading Basis** Credit / No Credit

This course offers practical preparation for students admitted to Washington Semester internships. Students learn how to improve and tailor resumes and cover letters, articulate skills and interests, interview effectively, and network professionally. The course introduces D.C.'s internship landscape, search tools and timelines, and professional norms. Assignments include a targeted resume, cover letter(s), references list, LinkedIn profile refinement, writing sample selection and a curated list of internships with a search plan. Permission of the instructor is required for enrollment.

POSC 492 Senior Seminar in Political Science**Credits 3-4 Grading Basis** Graded

This research-oriented senior seminar provides an overview of the discipline of political science and the different approaches to research in the field. A major research project will strengthen the research, information access and lifelong learning capacities of the student. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Senior standing and POSC 295.

POSC 493 Political Communication Internship**Credits 3-6 Grading Basis** Graded

This course will provide students with experiential learning opportunities in policymaking, campaigning, media, legislative politics and lobbying. A research paper relating the internship to the student's academic work will enhance the learning experience.

Prerequisites: Junior or senior standing; POSC 225; and either SCOM 472/ SMAD 472/POSC 472 or SCOM 353.

POSC 495 Internship in Political Science**Credits 3-6 Grading Basis** Graded

This course provides students with opportunities for experiential learning in a legislative, policy-making, campaign, constituency, interest group or criminal justice organization. A research paper related to the internship and a presentation based on the experience are required. No more than three credit hours can be counted toward the political science major.

Permission of the instructor is required to enroll.

POSC 495W Internship in Political Science - Washington Semester**Credits 6 Grading Basis** Graded

Provide Washington Semester participant with opportunities for experiential learning in a Washington, D.C.-based legislative, policy making, campaign, constituency, interest group or criminal justice organization. Requirements include 360 work hours, a research paper related to the internship, a career report, a daily log, and regular meetings with faculty-in-residence. Prerequisite: Junior or senior standing, 15 credits of political science, public administration or political communication, and successful application to Washington Semester program.

POSC 498 Research in Political Science**Credits 1 Grading Basis** Graded

Research in a selected area of political science as arranged with a faculty sponsor. Research outline must be approved by faculty sponsor and department head the semester before registration. Course may be repeated.

POSC 499A Honors**Credits 1-3 Grading Basis** Graded**POSC 499B** Honors**Credits 3 Grading Basis** Graded**POSC 499C** Honors**Credits 2 Grading Basis** Graded

Portuguese (PORT)

PORT 101 Elementary Portuguese I**Credits 3-4 Grading Basis** Graded

The fundamentals of Portuguese through listening, speaking, reading and writing. Practice in pronunciation and development of comprehension.

One hour's work a week in the language laboratory. Students will receive no credit for the course if they have had two or more years of the language in high school.

PORT 102 Elementary Portuguese II**Credits 3-4 Grading Basis** Graded

The fundamentals of Portuguese through a higher level of listening, speaking, reading and writing. Practice in pronunciation and development of comprehension. One hour's work a week in the language laboratory.

Students will receive no credit for the course if they have had two or more years of the language in high school.

Prerequisites: PORT 101.

PORT 231 Intermediate Portuguese I**Credits 3 Grading Basis** Graded

A thorough review of first year grammar and vocabulary building.

Conversation, composition and readings will be chosen to reach competency at the lower intermediate level Portuguese.

Prerequisites: PORT 102 or by permission of the instructor.

PORT 232 Intermediate Portuguese II**Credits 3 Grading Basis** Graded

A thorough review of PORT 231 grammar and vocabulary building.

Conversation, composition and readings will be chosen to reach competency at the advance intermediate level.

Prerequisites: PORT 231 or by permission of the instructor.

PORT 300 Portuguese Grammar and Communication**Credits 3 Grading Basis** Graded

Intensive training in grammatical structures and their application to oral and written communication. Instruction is in Portuguese. Fulfills the College of Arts and Letters writing-intensive requirement for possible International Affairs majors and/or IBUS majors.

Prerequisites: PORT 232 or by permission of the instructor.

PORT 320 Portuguese Oral and Written Communication**Credits 3 Grading Basis** Graded

Intensive training in the use of modern, everyday Portuguese with emphasis on conversation and composition. Readings in Portuguese will provide a context for discussion and writing.

Prerequisites: PORT 300.

PORT 490 Special Topics in Portuguese**Credits 1-4 Grading Basis** Graded

Special topics or independent studies in Portuguese.

Psychology (PSYC)

PSYC 100 Interpersonal Skills for Resident Advisors**Credits 1 Grading Basis** Graded

Designed to give resident advisor trainees understanding of interpersonal relations. Cannot be used as a psychology major elective.

PSYC 101 General Psychology [C5SD]**Credits 3 Grading Basis** Graded

A study of the nervous system, sensation, perception, consciousness, learning, memory, language, intelligence, motivation, emotion, life span development, personality, psychopathology, psychotherapy, social psychology and the scientific method. May be used for general education credit.

PSYC 105 Orientation to the Major**Credits 1 Grading Basis** Graded

This course will provide psychology majors with knowledge and tools to get the most out of the major. Topics include an introduction to the psychology major at JMU, strategies for discerning between credible and non-credible information, professional (e.g., student/professor) communication and campus learning support resources. This course is only available for students who have not yet completed the two-semester methodology sequence.

Prerequisites: Psychology major and PSYC 101.

PSYC 106 Preparing for Graduate School with a Psychology Major**Credits 1 Grading Basis** Graded

This course will guide students through graduate school selection and application processes. Coverage includes graduate training opportunities across sub-disciplines in psychology and appropriate preparation to pursue such opportunities. Topics include participating in research and fieldwork experiences, preparing for and taking the GREs, selecting appropriate programs, and applying to graduate schools (including letters of recommendation, interviews, and writing curriculum vitae and personal statements).

Prerequisites: Psychology major and PSYC 101.

PSYC 107 Preparing for a Job with a Psychology Major**Credits 1 Grading Basis** Graded

Students will focus on the skills psychology majors can develop to prepare for a variety of careers. Students will discuss what types of jobs will be available to them upon graduation with an undergraduate degree in psychology. Topics will include resume-building, cover letter writing, interviewing and leveraging undergraduate experiences when applying for a job.

Prerequisites: Psychology major and PSYC 101.

PSYC 122 The Science of Vision and Audition [C3NS]**Credits 3 Grading Basis** Graded

A study of human interaction with light and sound waves. Topics include physiological and perceptual mechanisms for processing light and sound, along with connections to real-world applications (e.g., human factors and careers within vision science and audition). Includes activities designed to provide students with in-depth, hands-on experience with course topics. May be used for general education credit.

PSYC 160 Life Span Human Development [C5SD]**Credits 3 Grading Basis** Graded

An introduction to human development. Emphasis is on life span processes within physical, emotional, cognitive, psychosexual, social, personality and moral development. May be used for general education credit.

PSYC 180 Introduction to Behavior Analysis**Credits 3 Grading Basis** Graded

Students will learn the fundamental principles, procedures, and concepts of behavior analysis, how they can be used to explain behavior, and how interventions based on these principles can be used to improve their own lives and the lives of others.

PSYC 200 Topics in Psychology**Credits 1-3 Grading Basis** Graded

Exploration of an important psychological topic. The topics for each semester will be announced on MyMadison and the departmental website.

Prerequisites: PSYC 101.

PSYC 202 Directed Studies in Psychology**Credits 1-3 Grading Basis** Graded

Designed to give capable students an opportunity to complete directed study in an area of psychology under faculty guidance. Experiences may include practica, serving as a teaching assistant, directed readings or similar experience.

Prerequisites: PSYC 101 and a written plan for the directed study must be submitted to the department head for approval one week prior to registration.

PSYC 203 Directed Research in Psychology**Credits 1-3 Grading Basis** Graded

An introductory opportunity to assist a faculty member with a research project. Students may participate in any or all phases of research, including completion of background research, study design, collection of data, data analysis, and interpretation. Specific requirements of project are determined by the instructor.

Prerequisites: PSYC 101 and a written plan for the directed study must be submitted to the department head for approval one week prior to registration.

PSYC 210 Psychological Measurement and Statistics**Credits 4 Grading Basis Graded**

This course provides an introduction to statistical techniques used by psychologists in measuring behavior. Fundamental measures and theory of descriptive and inferential statistics will be discussed. The use of computers for data analysis will be introduced.

Prerequisites: Psychology majors only; PSYC 101 with a minimum grade of "C"; and MATH 205, MATH 220, MATH 231 or MATH 235 with a minimum grade of "C."

PSYC 211 Psychological Research Methods**Credits 4 Grading Basis Graded**

This course provides an introduction to the application of scientific methodology to investigate psychological phenomenon. Through lecture and laboratory, attention is given to choosing research questions, developing hypotheses, designing and conducting research, describing, analyzing and evaluating data, and effectively communicating research findings.

Prerequisites: PSYC 101 and PSYC 210 with a minimum grade of "C-".

PSYC 212 Psychological Research Design and Data Analysis I**Credits 4 Grading Basis Graded**

The PSYC 212-PSYC 213 course sequence introduces the logic of pursuing a scientific approach in psychology and covers descriptive, correlational, experimental, and quasi-experimental approaches. It also covers the statistical tools associated with these methods (namely, descriptive statistics, correlation, regression, t-tests, and ANOVA), and it introduces the basics of inferential statistics and hypothesis testing.

Prerequisites: Psychology majors only; PSYC 101 with a minimum grade of "C"; and MATH 205, MATH 220, MATH 231 or MATH 235 with a minimum grade of "C."

PSYC 213 Psychological Research Design and Data Analysis II**Credits 4 Grading Basis Graded**

The PSYC 212-PSYC 213 course sequence introduces the logic of pursuing a scientific approach in psychology and covers descriptive, correlational, experimental, and quasi-experimental approaches. It also covers the statistical tools associated with these methods (namely, descriptive statistics, correlation, regression, t-tests, and ANOVA), and it introduces the basics of inferential statistics and hypothesis testing.

Prerequisites: PSYC 212 with a minimum grade of "C-".

PSYC 220 Psychology and Culture**Credits 3 Grading Basis Graded**

The study of human psychology is incomplete without taking into account the cultural, historical and social factors involved in human functioning. This course considers the ethnic and cultural variations that exist in human behavior, thought and action. This course meets the sociocultural competency requirement for the psychology major.

Prerequisites: Psychology majors only and PSYC 101. Civic leadership minors by departmental permission.

PSYC 235 Psychology of Adjustment**Credits 3 Grading Basis Graded**

A study of the process and dynamics of the well-integrated personality and the practical application of adjustment theories and behavior change techniques to enhance personal awareness and self-development.

PSYC 250 Introduction to Abnormal Psychology**Credits 3 Grading Basis Graded**

An introduction to the field of Abnormal Psychology, for the non-psychology major. This course will examine methods of defining psychological normality and abnormality and the classification, causes and treatment of abnormal behavior. This course cannot be used for psychology major credit.

Prerequisites: PSYC 101 or PSYC 160. A student cannot earn credit for PSYC 250 after completing PSYC 335.

PSYC 261 Study Abroad in Psychology**Credits 1-4 Grading Basis Graded**

This course involves a study abroad experience including content and application of psychology skills and knowledge in a particular cultural context. Specific content varies based on study abroad location and content focus of the course. Some sections of the course fulfill the sociocultural awareness requirement for the psychology major.

Prerequisites: Three credit hours in psychology and permission of the instructor.

PSYC 270 Foundations of Learning and Cognition for Education**Credits 3 Grading Basis Graded**

This course introduces fundamental principles of cognition and learning as applied to educational practice. It provides a foundation for understanding multiple perspectives and levels of analysis applied to individual learning in educational settings.

Prerequisites: PSYC 101 or PSYC 160.

PSYC 285 Drugs and Behavior**Credits 3 Grading Basis Graded**

An introduction to the pharmacological effects of psychoactive drugs. This course will examine the neural mechanisms and behavioral effects of common substances such as caffeine and nicotine, drugs of abuse and pharmaceuticals that are used to treat mental disorders.

Prerequisites: PSYC 101.

PSYC 301 Psychology Peer Advising Training**Credits 2 Grading Basis Graded**

Introductory training in academic advising, career development, and basic counseling techniques. Not to be used for psychology major credit. Application guidelines available on the psychology website and in the Psychology Peer Advising office.

Prerequisites: PSYC 101, junior standing and permission of the instructor.

PSYC 308 Health Psychology**Credits 3 Grading Basis Graded**

This course deals with personality and its relation to health and illness behaviors. Topics include psychological factors involved in control and helplessness, conflict management, cardiovascular disorders, cancer, pain, substance abuse and other psychophysiological related factors. Course meets sociocultural requirement for the psychology major.

Prerequisites: Psychology majors only and PSYC 101. Medical humanities, pre-medicine, and substance use and disorder education minors by departmental permission.

PSYC 310 The Psychology of Women and Gender**Credits 3 Grading Basis Graded**

This course examines research and theory regarding the abilities and behaviors of women and the changing roles of women. Consideration is given to biological, developmental and societal determinants of sex and gender. This course meets the sociocultural requirement for the psychology major.

Prerequisites: Psychology majors only and PSYC 101. Justice studies majors and women, gender, and sexuality studies minors by departmental permission.

PSYC 325 Counseling Psychology**Credits 3 Grading Basis** Graded

This is a basic counseling theories and skills course designed for students interested in human service and mental health fields. This course meets the sociocultural requirement for the psychology major. Prerequisites: Psychology major and PSYC 101.

PSYC 330 Psychology of Personality**Credits 3 Grading Basis** Graded

Essential elements of leading theories of personality with an emphasis on implications of these theories for human behavior. Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 335 Abnormal Psychology**Credits 3 Grading Basis** Graded

This course for the psychology major critically examines psychological normality and abnormality and the classification, causes and treatment of abnormal behavior. Students learn classification and diagnosis, explore social and multicultural issues relating to diagnosis and discuss research in the field.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213. A student cannot earn credit for PSYC 335 after completing PSYC 250.

PSYC 345 Social Psychology**Credits 3 Grading Basis** Graded

The study of how an individual's behavior, feelings and thoughts are influenced by other people. Topics include attitude formation and change, social perception, attraction, altruistic and antisocial behavior, conformity, leadership and group dynamics, and applications of social psychology to other fields.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 365 Developmental Psychology**Credits 3 Grading Basis** Graded

Psychological aspects of growth, development and behavior from birth through adolescence.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 375 Sensation and Perception**Cross-listed with:** BIO 375**Credits 3 Grading Basis** Graded

This course explores the nature and development of human sensory capabilities and processing, and how these affect perception of the environment. A potential list of the variety of systems reviewed includes hearing, vision, smell, taste and touch/pain, as well as phenomena such as the perception of balance.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 380 Cognitive Psychology**Credits 3 Grading Basis** Graded

This course explores the nature and development of human attention, memory, language and thinking processes. An information processing approach to the study of human cognition is emphasized.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 385 Biopsychology**Cross-listed with:** BIO 385**Credits 3 Grading Basis** Graded

This course is a survey of the neurological and chemical mechanisms which control behavior. Students will examine the brain and how it processes sensation, perception, cognition, movement, motivation, learning, memory and other behavioral processes of interest to psychologists.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 390 Psychology of Learning**Credits 3 Grading Basis** Graded

Basic principles of learning and conditioning with a consideration of extinction, reinforcement, generalization, discrimination, transfer, concept formation and verbal learning.

Prerequisites: "C-" or better in PSYC 211 or PSYC 213.

PSYC 395 Comparative Animal Behavior**Cross-listed with:** BIO 395**Credits 3 Grading Basis** Graded

This course covers aspects of the development, function and evolution of the behavior of nonhuman animals. Topics covered include intraspecies communication, feeding, aggression, territoriality, reproductive behavior and social behavior.

Prerequisites: Psychology majors: A "C-" or better in PSYC 211 or PSYC 213. Biology and biotechnology majors: BIO 250/BIO 250L, and one of the following (with a "C-" or better): MATH 205, MATH 220, MATH 231, MATH 235, MATH 285, MATH 318.

PSYC 400 Advanced Topics in Psychology**Credits 1-3 Grading Basis** Graded

Exploration of a significant psychological topic in depth. The topics for each semester will be announced in the Schedule of Classes.

Prerequisites: One SS content course and one NS content course.

PSYC 401 Peer Advising**Credits 2 Grading Basis** Graded

Supervised practicum in academic and career development and peer advising. May be taken twice for up to four credit hours toward the psychology major (400-level elective).

Prerequisites: One SS content course and one NS content course.

PSYC 402 Independent Study in Psychology**Credits 1-4 Grading Basis** Graded

An opportunity to apply classroom learning to practical problems and to expand the scope of knowledge in psychology to areas not emphasized in the coursework we offer. May include service learning, internship, directed readings, serving as a teaching assistant, or a combination of these activities.

Prerequisites: PSYC 211 or PSYC 213 and a written plan for the directed study must be submitted to the department head for approval one week prior to registration.

PSYC 403 Independent Research in Psychology**Credits 1-4 Grading Basis** Graded

An advanced opportunity to conduct research with a faculty member where students apply their knowledge of Psychology to a specific research project or area. Students may work individually with the instructor or as part of a team that includes several students. Student projects include the development of an evaluated product (e.g., poster presentation, presentation). Specific requirements of project are determined by the instructor.

Prerequisites: PSYC 211 or PSYC 213 and a written plan for the directed study must be submitted to the department head for approval one week prior to registration.

PSYC 410 Psychology of the Workplace**Credits 3 Grading Basis** Graded

This course is a survey of the applications of psychological principles in the workplace. Emphasis is on topics such as research and methods, personnel decisions, training, attitudes, motivation, leadership, teams, and sociocultural issues in the workplace. Other topics of current interest will also be covered. Course will fulfill sociocultural awareness requirement.

Prerequisites: One SS content course and one NS content course.

PSYC 415 Forensic Psychology**Credits 3 Grading Basis Graded**

The application of psychological principles and techniques to the law, the criminal justice system, law enforcement and criminal behavior.

Prerequisites: PSYC 335.

PSYC 425 School Psychology**Credits 3 Grading Basis Graded**

Applications of psychological principles in school settings, including roles and activities of school psychologists, standards, trends, and issues of treatment and evaluation.

Prerequisites: One SS content course and one NS content course.

PSYC 427 Tests and Measurements**Credits 3 Grading Basis Graded**

Standardized psychological tests of mental ability, achievement, aptitude and personality with a review of statistical procedures necessary for interpretation of test results.

Prerequisites: One SS content course and one NS content course.

PSYC 428 Educational Psychology**Credits 3 Grading Basis Graded**

The application of the basic psychological principles of development, learning, cognition, measurement, and social interactions to educational settings. This course examines how psychological theory and research impacts the teaching of reading, writing, science, and mathematics. Students may not count both PSYC 270 and PSYC 428 for psychology major credit.

Prerequisites: One SS content course and one NS content course.

PSYC 430 Clinical Psychology**Credits 3 Grading Basis Graded**

An introduction to the field of clinical psychology including a review of the major theoretical models, psychometrics, psychiatric diagnosis and treatment strategies.

Prerequisites: PSYC 335 and one NS content course.

PSYC 450 Psychology of Child Abuse and Neglect**Credits 3 Grading Basis Graded**

Review of current psychological literature on child abuse and neglect including identification, etiology, treatment, prevention and legal aspects. Family violence issues are also discussed.

Prerequisites: One SS content course and one NS content course.

PSYC 452 Child Psychopathology**Credits 3 Grading Basis Graded**

The causes, symptoms and classification of childhood psychological disorders.

Prerequisites: PSYC 335 or PSYC 365.

PSYC 461 Advanced Study Abroad in Psychology**Credits 1-4 Grading Basis Graded**

This course involves a study abroad experience including advanced content and application of psychology skills and knowledge in a particular cultural context. Specific content varies based on study abroad location and content focus of the course. Some sections of the course fulfill the Sociocultural Awareness requirement for the psychology major.

Prerequisites: One SS content course and one NS content course.

PSYC 475 Psychology of Adulthood**Credits 3 Grading Basis Graded**

The physical, social and psychological factors faced by adults and their progression through the life span.

Prerequisites: One SS content course and one NS content course.

PSYC 480 Applied Behavior Analysis**Credits 3 Grading Basis Graded**

This course focuses on how environmental events influence behavior, and behavior analytic strategies by which behavior may be changed. The emphasis of the course is on the knowledge and skills necessary to plan, develop, and implement interventions for behavior problems in a variety of setting including, but not limited to, business and industry, education, and health and human services.

Prerequisites: PSYC 390.

PSYC 492 History of Psychology**Credits 3 Grading Basis Graded**

The history of psychology as reflected through the individuals, theories and experimental investigation of the discipline. Special emphasis is placed upon relating the current state of psychology to its historical development.

Prerequisites: Must complete at least two courses each from social science and natural science core.

PSYC 495 Field Placement in Psychology**Credits 4 Grading Basis Graded**

Supervised practicum in a counseling, industrial or human service agency. Orientation to agency's service, policies, personnel and professional ethics is provided. The course meets the requirement as a capstone course or as a psychology elective.

Prerequisites: PSYC 211 and junior or senior standing.

PSYC 497 Senior Seminar**Credits 3 Grading Basis Graded**

A seminar course that will require students to integrate theories, research and/or methods from several areas of psychology and/or related disciplines. Topics will vary from semester to semester. May be taken as a capstone course or psychology elective. Up to six credit hours can be used in the psychology major.

Prerequisites: Must complete at least two courses each from social science and natural science core.

PSYC 499A Honors**Credits 2 Grading Basis Graded****PSYC 499B Honors****Credits 2 Grading Basis Graded****PSYC 499C Honors****Credits 2 Grading Basis Graded**

Public Policy and Administration (PPA)

PPA 200 Introduction to Public Policy**Credits 3 Grading Basis Graded**

The course introduces students to the nature, dynamics and substance of public policy. Selected policy issues in the United States will be examined through the use of case studies. Foreign and global influences on U.S. policy-making will also be analyzed. Issues will vary across course sections over time.

PPA 265 Public Administration**Credits 3 Grading Basis Graded**

An introductory survey of the principles, functions and processes of public administration with specific emphasis on the political aspects and environment of bureaucracies, and the how and why of policy-making within an administrative system. Organizational structure, personnel, budgeting, public relations and government values, traditions, and objectives are analyzed.

PPA 359 Policy Analysis**Credits 3 Grading Basis** Graded

Study of public policy analysis. Delivers to students rational and alternative techniques for analyzing public policy while providing them opportunities to develop analytical skills.

Prerequisites: PPA 200.

PPA 381 Budgetary Process**Credits 3 Grading Basis** Graded

An examination of the political planning and strategies of Congress and federal agencies in the budgetary process; politics of budgetary reform; state and local budgetary politics; and intergovernmental impacts on budgeting.

Prerequisites: Junior standing and PPA 265.

PPA 412 Seminar in Intergovernmental Relations**Credits 3 Grading Basis** Graded

Study of the relations between the several levels of government in the United States. Political, fiscal, legal, regulatory and administrative relations as they have evolved within federal and state constitutional frameworks will be examined.

Prerequisites: Junior standing and PPA 265.

PPA 415 Legal Environment of Public Administration**Credits 3 Grading Basis** Graded

Examination of the basic constitutional framework of American public administration. Examines legal constraints imposed on public administrators by law and judicial oversight. Emphasis placed on legal issues affecting public employees. Also examines the basics of public procurement law.

Prerequisites: PPA 265.

PPA 420 Senior Seminar in Public Management**Credits 4 Grading Basis** Graded

Study of the management of public agencies from the executive viewpoint. Management control of public agencies will be explored including establishment of goals, policies, organizational structure and output of services. Case studies illustrate administrative behavior and managerial operations in local, regional, state and federal agencies. Does not count as part of the political science major or minor. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: PPA 265, POSC 295 and senior standing.

PPA 441 Stability and Recovery Operations**Credits 3 Grading Basis** Graded

A study of policies and practices in stability and recovery operations. Conflict and post-conflict zones motivate international and domestic efforts to promote security, humanitarian assistance, economic stabilization and development, the rule of law, and participatory governance. This course develops management and policy design skills relevant to the pursuit of this complex agenda via case studies at the international, national and organizational levels.

Prerequisites: PPA 200 and PPA 265.

PPA 460 Regionalism and Urban Management**Cross-listed with:** PUAD 560**Credits 3 Grading Basis** Graded

A study of the problems and management of urbanization and inter-jurisdictional externalities from a regional perspective. Regionalism will be examined as an approach to solve these problems.

Prerequisites: PPA 200.

PPA 461 Education and Social Policy**Credits 3 Grading Basis** Graded

A study of the development and implementation of education policy in the United States at the national, state, and local levels. Students will be introduced to major issues in contemporary education policy and the evaluation of alternative policies advanced by subgroups of the population. Educational equity and its links to social and economic goals will be examined.

Prerequisites: PPA 200.

PPA 462 Social Welfare and Local Government Policy**Cross-listed with:** PUAD 562**Credits 3 Grading Basis** Graded

A study of the interaction of social welfare policy and local governance in theory and in practice. Students examine state and local government and community-based responses to urban problems from a policy and management perspective. Particular attention is paid to interagency and community collaboration as a way to enhance social service delivery.

Prerequisites: PPA 200.

PPA 469 Philanthropy and Volunteerism**Cross-listed with:** PUAD 569**Credits 3 Grading Basis** Graded

An examination of the role of philanthropy and volunteerism in the nonprofit sector in the United States and globally. Topics discussed include: strategic giving; components of fundraising management and planning; volunteerism as a resource for organizations; board and staff responsibilities in resource acquisition; and monitoring and performance assessment.

Prerequisites: PPA 265.

PPA 470 Nonprofit Management and Leadership**Cross-listed with:** PUAD 570**Credits 3 Grading Basis** Graded

This course is designed to introduce students to the fundamentals of governance, accountability, trusteeship and executive leadership in nonprofit sector organizations. The course will examine critically the principal models, processes and practices used in managing nonprofit organizations and the relationships between the governing boards and executive leaders of such organizations.

Prerequisites: PPA 265.

PPA 471 Financial Management for Public and Nonprofit Organizations**Cross-listed with:** PUAD 571**Credits 3 Grading Basis** Graded

This course provides a foundation in financial management, enabling students to understand and apply financial information as competent entry-level financial analysts in the public or nonprofit sector. This course will provide students with a strong financial vocabulary in budgeting and finance to be able to communicate and interpret financial language. Students will work with several cases that allow them to use the tools learned in the course.

Prerequisites: PPA 265.

PPA 472 Contract Management**Cross-listed with:** PUAD 572**Credits 3 Grading Basis** Graded

The purpose of this course is to provide a broad overview of the theory behind and practical application of contract management. As agencies across government (federal, state and local) expand the use of contracting billions of taxpayer dollars are transferred into the private sector to conduct public business. This trend is not going away; therefore it is essential that public administrators be effective at managing and overseeing contracts.

Prerequisites: PPA 265.

PPA 473 Economic and Community Development**Cross-listed with:** PUAD 573**Credits 3 Grading Basis** Graded

Study of the theory and practice of economic development and community planning. Topics include human capital development, infrastructure development, regionalism, public-private partnerships.

PPA 475 Grant Writing and Project Management**Cross-listed with:** PUAD 575**Credits 3 Grading Basis** Graded

An examination of the practical skills needed for writing successful grant proposals as well as for carrying out grant-funded projects. Proposal development features include needs assessments, work plans, and evaluation programs. The foundations of project management examined include staffing, workflow and expense monitoring, stakeholder relationships and demonstrating results.

Prerequisites: PPA 265.

PPA 480 Ethics and Political Corruption**Credits 3 Grading Basis** Graded

This course examines the ethical dimensions of public service as experienced in governmental and nonprofit organizations. Course content explores governmental, organizational and individual-level efforts to promote ethical behavior and to deter corruption.

Prerequisites: PPA 265.

PPA 482 Special Topics in Public Policy**Cross-listed with:** PUAD 582**Credits 3 Grading Basis** Graded

The course will examine a special topic in public policy. The course may be repeated for credit with a change in the subject matter offered.

Prerequisites: PPA 265.

PPA 483 Special Topics in Public Management**Cross-listed with:** PUAD 583**Credits 3 Grading Basis** Graded

The course will examine a special topic in public and/or nonprofit management. The course may be repeated for credit with a change in the subject matter offered.

Prerequisites: PPA 265.

PPA 484 Environmental Regulatory Policy and Politics**Credits 3 Grading Basis** Graded

A study of environmental politics and the policies that environmental advocacy has produced. Topics include the dynamics of policy construction, various substantive policy issues and the prospects for environmental justice and sustainability.

Prerequisites: PPA 200 or permission of instructor.

PPA 485 Governance for Environmental Sustainability**Credits 3 Grading Basis** Graded

This course is an examination of the collaborative management necessary to address complex environmental sustainability issues facing communities at the local, state and national levels. Topics covered include systems thinking, interest-based negotiation, stakeholder analysis and environmental conflict resolution.

Prerequisites: PPA 265.

PPA 492 Senior Seminar in Public Policy**Credits 4 Grading Basis** Graded

This research-oriented seminar provides an overview of public policy studies and the different approaches to research in this field. A major research project will strengthen the research, information access and lifelong learning capacities of the student. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: Senior standing, POSC 295, and PPA 359.

PPA 494 Internship in Public Policy**Credits 3 Grading Basis** Graded

This course provides students with opportunities for experiential learning in organizations engaged with public policy formulation and analysis. A research paper related to the internship is required. Permission of the instructor is required to enroll.

PPA 496 Internship in Public Management**Credits 3-6 Grading Basis** Graded

This course provides students with opportunities for experiential learning in a governmental or nonprofit organization. A research paper and a presentation based on the experience are required. Permission of the instructor is required to enroll.

PPA 497 Public Policy Internship in Washington**Credits 6 Grading Basis** Graded

This course provides students with opportunities for experiential learning in Washington D.C. with public and nonprofit organizations engaged with public policy formulation and analysis. A research paper related to the internship is required.

Prerequisites: Junior or senior standing, 15 credits of relevant coursework and successful application to the Washington Semester Program.

Queer Studies (QUEST)

QUEST 200 Introduction to LGBTQ+ and Queer Studies**Credits 3 Grading Basis** Graded

This course introduces students to the academic disciplines of LGBTQ+ and Queer Studies, interdisciplinary fields that analyze structures of power and privilege in relation to categories and norms about gender, sex, and sexuality. Students will interrogate both the ways that culture regulates these identity categories and the ways in which they influence social institutions, artistic expressions and political discourses, particularly in relation to LGBTQ+ people.

QUEST 300 Special Topics in LGBTQ+ / Queer Studies**Credits 3 Grading Basis** Graded

Examination of selected topics of importance to the field of LGBTQ+ / Queer studies. May be repeated for credit when course content changes.

QUEST 400 Advanced Topics in LGBTQ+ / Queer Studies**Credits 3 Grading Basis** Graded

In-depth examination of selected topics of current importance to the field of LGBTQ+ / Queer studies. May be repeated for credit when course content changes.

Prerequisites: QUEST 200 and permission of the instructor.

QUEST 410 Independent Research in LGBTQ+ / Queer Studies**Credits 3 Grading Basis** Graded

Designed to give students in LGBTQ+ / Queer studies an opportunity to complete in-depth study under faculty supervision. Admission by recommendation of the instructor and permission of the Queer Studies minor coordinator.

QUEST 425 Internship in LGBTQ+ / Queer Studies**Credits 1-3 Grading Basis** Graded

Provides the student with practical experience in employing and refining LGBTQ+ / Queer studies concepts in a public or private agency, under faculty supervision. Approval by QUEST minor coordinator required.

Prerequisites: QUEST 200 and approval by the Queer Studies minor coordinator.

Religion (REL)

REL 101 Religions of the World [C2HQ]

Credits 3 Grading Basis Graded

An investigation of the world's major religions which will give attention to their origin, history, mythology and doctrines. May be used for general education credit.

REL 102 Religion, Spirituality and the Meaning of Life [C2HQ]

Credits 3 Grading Basis Graded

Where do we come from, and what is our place in the universe? What gives life happiness, purpose and meaning? How can we make sense of religious diversity, and how can we live well together in the midst of our differences? This course is a cross-cultural introduction to some of the ways people across the world have asked, answered and lived out these questions. May be used for general education credit.

REL 103 Introductory Topics in Religion and Theology

Credits 1-6 Grading Basis Graded

Course on various topics in religious studies and/or theological studies. May be taken multiple times for credit.

REL 200 How to Study Religion and Spirituality

Credits 3 Grading Basis Graded

This course will provide an introduction to some of the most influential thinkers, common methodologies, important debates and key concepts in the academic study of religion. The course will also consider recent challenges to this field of study. Emphasis will be on applying insights to real-world examples.

Prerequisites: REL 101.

REL 201 Introduction to Hebrew Bible / Old Testament

Credits 3 Grading Basis Graded

A study of selected texts in the Hebrew Bible or Old Testament using the methods of biblical studies, which examines texts in their original historical settings. No prior familiarity with the Bible is needed. Includes introductions to religious texts and cultures in ancient Egypt, Mesopotamia, and the Levant. We will also consider how contemporary readers use the Bible to support a range of policy issues and social platforms as we reflect on how a biblical studies approach might inform these positions.

REL 202 Introduction to the New Testament

Credits 3 Grading Basis Graded

This course discusses the literature of the New Testament in light of historical, social and religious conditions from which it emerged. Particular attention is given to historical issues related to Jesus and the origins of Christianity.

REL 210 Religion in America

Credits 3 Grading Basis Graded

The purpose of this course is to offer students the opportunity to explore the broad contours of the intersection of religion with other important facets of American society, such as politics and law, civic space and social activism, social identities, intellectual life, and the arts and media. It will consider the relationship of religion to the constructing of an American identity rooted in ideas of pluralism, tolerance, equality, freedom of conscience, democracy and secularism.

REL 213 Introduction to Ancient Chinese Wisdom Traditions

Credits 3 Grading Basis Graded

This course is an introduction to early Chinese thought. The word "thought" is used instead of "philosophy" or "religion" because the Chinese intellectual tradition did not distinguish between philosophy and religion. We will examine how philosophical, religious and literary elements are all uniquely deployed in texts such as the Analects of Confucius, the Dao De Jing and other classics. Our approach will involve analysis of language, arguments, cultural context and authorial intent.

REL 216 (Re)thinking Order and Chaos: An Introduction to Hinduism

Credits 3 Grading Basis Graded

This course introduces Hinduism through the concepts of order (dharma), chaos or disorder (non-dharma), and ethic of responsibility (karma), and their relationships to the pursuits of pleasure and material well-being and the quest for liberation. It makes sense of chaos and discovers the lurking irrationality of order, and thereby suggests a creative role of chaos in order. This course also challenges the category of religion and the universalism associated with Western religious mythologies.

REL 217 The Mystery of the True Path: An Introduction to Daoism

Credits 3 Grading Basis Graded

Beginning with the earliest classics and continuing to the present, the course examines Daoism from historical, literary, philosophical and religious perspectives. It explores the difficulties categorizing Daoism as a "philosophy" or "religion," and how it still informs political, philosophical and religious thought in China. First-person exercises in and out of class will yield a holistic and lived understanding of the Daoist tradition.

REL 218 Philosophy of Religion

Cross-listed with: PHIL 218

Credits 3 Grading Basis Graded

This course offers an intensive examination of religion from the standpoint of philosophical thinking with particular emphasis on the way philosophers view such problems as the existence of God, evil, immortality, religious language, etc.

REL 220 Introduction to Judaism

Credits 3 Grading Basis Graded

This course will offer an examination of the beliefs, practices and historical development of the various forms of Judaism represented in America today: Orthodox, Reform, Conservative and Reconstructionist.

REL 240 Introduction to Christianity

Credits 3 Grading Basis Graded

This course provides a broad survey of the Christian tradition, from its origins in the Mediterranean to its contemporary expressions in diverse communities. In addition to its historical development, the course will consider Christian belief, ritual, moral practice and social engagement. Major intellectual and theological traditions will be addressed through the study of foundational texts. Students will gain a working knowledge of major church communions as well as minor and marginalized movements.

REL 250 Introduction to Islam**Credits 3 Grading Basis** Graded

What constitutes Islam and how has it been defined throughout history? Who represents and speaks for Islam? How have we come to our Islamic studies archive and how does it impact how we view Islam? Students will discover the different ways Muslims and non-Muslims have answered these questions, in different places, times and social contexts. Based on the most recent research and original source material, the course will cover major themes of Islamic religious thought including the Qur'an and its interpretation, the intellectual history of Islam, Sufism, Islamic law, and Muslim reform and revivalism. Students will explore how these components have manifested in diverse cultures throughout Islam's history. In addition to gaining a solid understanding of the problems associated with the study of Islam, successful completion of this course should also equip students with the tools required to analyze broader theoretical issues pertinent to the study of religion. A further goal of this course is to provide a strong theoretical foundation in religious studies so that students are able to think and speak about religion in an informed and insightful manner. The scope of this course is not limited to Islam, as the example of Islam will be used as a case study to wrestle with broader intellectual issues relating to the intersection among religion, history and politics.

REL 288 Introduction to Buddhism**Credits 3 Grading Basis** Graded

This course explores the historical origins of Buddhism, the central tenets of Buddhist thought, the major meditative practices Buddhists employ, and the relationship between meditation and social action in contemporary Buddhist communities.

REL 300 Selected Topics in Religion**Credits 3 Grading Basis** Graded

Selected topics in religion are studied in depth. See MyMadison for current topic. Course may be repeated for credit when content changes.

REL 304 Art, Religion and Identity in Medieval Sicily**Credits 3 Grading Basis** Graded

This course will explore Sicily's rich medieval history through the lenses of religion, art and identity. Bringing together primary source texts, contemporary theory and material evidence, we will analyze how the island's Jews, Christians and Muslims competed over a shared intellectual, artistic and material heritage to construct their identities. Taught on-site in Palermo, Sicily.

REL 305 Black Skin, Divine Realities? Exploring Islam in Africa**Credits 3 Grading Basis** Graded

Legal savants and Sufi sages, jinn wranglers and revolutionaries, African Muslims past and present have developed rich traditions of religious learning, community and practice. This course will explore a few of these traditions, focusing primarily on Islam in sub-Saharan Africa from the medieval period to today. Through a critical attention to both primary and secondary sources, a plurality of African Muslim perspectives will be analyzed while grappling with the thorny theoretical questions they provoke (e.g., Can we speak of a distinctly African Islam? If so, what do we mean? And what are the stakes of asking this question?)

REL 306 Gender and Sexuality in Islam**Credits 3 Grading Basis** Graded

This course will investigate how Muslims across space and time have imagined gender and sexuality. How have Islamic societies mapped the possibilities of sexed bodies and gendered performances? In what ways have they cultivated and disciplined human desire? What boundaries or spectra have they negotiated and contested? What modes of subjectivity and agency have their worlds made possible? Throughout the semester, students will develop a self-reflexive approach to their own ethical ideals, exploring how a critical engagement with diverse Islamic perspectives can illuminate the complexities of gender and sexuality both in the past and today.

REL 307 Knowledge, Identity and Power in Islamic Societies**Credits 3 Grading Basis** Graded

This course will explore how careful attention to questions of knowledge and identity can illuminate the complexity of Muslim perspectives and societies. Philosophers and poets, esotericists and encyclopedists, artists and artisans, reformers and renunciants: Muslims from Detroit to Damascus have wielded a vast array of discourses and practices to fashion themselves and the world(s) around them. Throughout the course, students will investigate these ideas and strategies as dynamic and contingent human products, inextricable from urgent social and political concerns. Students will analyze how knowledge is shaped by relationships of power and molded by institutions, values and practices that privilege some individuals and exclude others. At the same time, students will examine how knowledge production can reorient relationships and identities, opening up new possibilities for action.

REL 314 Hope or Terror? Gandhi and the Other 9/11**Credits 3 Grading Basis** Graded

Humanity has a double legacy, symbolized in two 9/11s, one at each end of the twentieth century, carrying opposite messages. The 9/11 of 2001, when terrorist Osama bin Laden attacked America; and the less remembered, the 9/11 of 1906, when the apostle of peace Mahatma Gandhi launched a new way of waging conflict that can lead humanity from the mire of hatred into the clear land of peace. This course tells the story of the older 9/11: the story of Satyagraha, the way of nonviolence and truth.

REL 315 Women and Religion**Credits 3 Grading Basis** Graded

Students will explore the diverse relationships between women and the world's major religions (Hinduism, Buddhism, Judaism, Christianity and Islam), both in the past and today. Topics will include women's roles, rights, practices, bodies, sexualities, clothing, representations and more. Feminist approaches for critiquing religious histories, texts and traditions will also be considered.

REL 316 Dharma/Adharma: Hindu Ethical Reasoning**Credits 3 Grading Basis** Graded

What values are advanced in Hindu religious and ethical teachings? How do Hindu texts and traditions define and teach the good life and virtuous living? Is moral responsibility conditioned by caste, class, gender or other socioeconomic factors? These are some of the questions that will be considered in this study on Hindu modes of moral reasoning. This course will also make sense of the structure of chaos (adharma) and the chaos of structure (dharma) and thereby suggest that adharma might somewhat make more sense than dharma.

REL 322 Sociology of Religion**Cross-listed with:** SOCI 322**Credits 3 Grading Basis** Graded

This course is a sociological analysis of the nature, function, and structure of religion. Survey of the relationship between religion and society: the social nature of religious phenomena, the interaction between religious beliefs and practices and other arenas of secular societies, the social functions of religion, and the way religion changes and is changed by secular society.

REL 323 Comparative Ethics East and West**Cross-listed with:** PHIL 323**Credits 3 Grading Basis** Graded

In an increasingly globalized world, ethical decisions often involve considering diverse systems of beliefs about how one ought to live. In this course, students will think about ethical questions from cross-cultural perspectives, including perspectives of Buddhism, Confucianism, Aristotelianism and utilitarianism. By analyzing arguments made in Eastern and Western philosophical and religious traditions, students will begin to see similarities - such as the use of logical reasoning and the undesirability of pain - as well as fundamental differences. Case studies and examinations of contemporary issues make the course content relevant to the present day.

REL 324 Latinx and Latin American Christianity**Credits 3 Grading Basis** Graded

The election of Pope Francis to the papacy reflects the growing influence of Latin America upon world Christianity. This course will provide a historical and thematic overview of the changes in the history of Latin American Christianity. It moves chronologically and geographically, from colonial Latin America to Hispanic immigrant communities. Topics covered include race and colonialism, the Cuban Revolution, the Second Vatican Council, social justice and Christianity, the rise of Pentecostalism, labor movements and inequality, and U.S. Hispanic political influence.

REL 325 Catholicism in the Modern World**Credits 3 Grading Basis** Graded

Study of the variety of responses by contemporary Catholic theologians and philosophers to key elements in Christian doctrine and practice. Topics include Vatican II; scripture, tradition and modern scholarship; Jesus and Christology; contemporary Catholic spirituality; moral issues in the church; and ecumenism.

REL 330 Religions of Africa and the African Diaspora**Credits 3 Grading Basis** Graded

An examination of the character of religious traditions, beliefs and liturgical practices of African and African-Diaspora communities. Both primary (historical and anthropological) and secondary sources examined.

REL 332 Born Again Religion**Credits 3 Grading Basis** Graded

Evangelical Protestantism has played a vital role in shaping American religious history. The religious and social allegiances of evangelicalism are quite diverse, however. Evangelicals also maintain a paradoxical relationship with American society, functioning both as powerful insiders and vocal outsiders. This course is designed to introduce students to the history of evangelicalism, its religious patterns and its negotiations with contemporary American culture.

REL 333 Angels, Demons and Dreams in Ancient Judaism and Christianity**Credits 3 Grading Basis** Graded

What did early Jews and Christians believe about angels, demons, and dreams? How did they understand heaven, hell, the afterlife, illness and healing, and the end of the world? The course introduces students to a broad range of writings (both within and outside of the Bible) that ancient Jews and Christians considered to be their inspired Scriptures in the late Persian, Hellenistic, and Roman eras (ca. 500 BCE-200 CE). We will utilize interdisciplinary methodologies from biblical studies to explore Jewish and Christian apocalypses, biblical retellings, magical texts, and other genres of literature, as well as archaeological artifacts and remains, as we weigh the evidence for early Jewish and Christian beliefs about the afterlife, heaven, hell, God, Satan, angels, illness and healing, dreams and visions, the soul, magic, revealed secrets, and the end time. A major unit will focus on what archaeology and an ancient library can tell us about the members of the Dead Sea Scrolls community and their struggles with Hellenistic and Roman cultures, which culminated in war. The course is ideal for any students interested in Biblical Studies, ancient Mediterranean studies, Classical Studies and mythology.

REL 334 New Religious Movements**Credits 3 Grading Basis** Graded

America has proven to be fertile soil for the development of new religious traditions. It has encouraged religious pluralism, and has allowed hundreds of new communities to establish themselves as important elements of society. This course will consider some of the representative new religions in America by examining their histories, beliefs, and practices.

REL 336 African American Religion**Credits 3 Grading Basis** Graded

The purpose of this course is to familiarize the student with the broad contours of the African American religious experience, and its history, practices, and communities by examining the intersection of race and religion in America. Through the close reading of primary texts and increased familiarity with significant scholarly literature, students will gain a basic understanding of the fundamental categories in African American religion.

REL 339 Islamic Philosophy**Cross-listed with:** PHIL 339**Credits 3 Grading Basis** Graded

This course will explore a variety of philosophical perspectives from across the Islamic world, employing methodological tools from both philosophy and religious studies. Themes may include: What is knowledge and how is it acquired? What does it mean to be human? What is the relationship between reason and other human faculties? What is the nature of time and history? What do we really mean when we talk about God? Philosophers covered may include: al-Kindi (d. 873), Ibn Sina (d. 1037), Ibn 'Arabi (d. 1240), Mulla Sadra (d. 1636), Muhammad Iqbal (d. 1938) and Fatema Mernissi (d. 2015).

REL 341 Early Christian Gospels: Origins, History and Conflict**Credits 3 Grading Basis** Graded

Matthew, Mark, Luke, and John are the familiar "canonical" gospels. In the early centuries of Christianity other gospels circulated: the Gospel of Peter, "Q", Infancy Gospels and the so-called Gnostic gospels. This course examines the origins of the canonical and non-canonical gospels, the historical and theological factors at work in the emergence of the canonical gospels to a position of primacy, and the struggles within early Christianity to define its authoritative tradition.

REL 342 The Historical Jesus Quest**Credits 3 Grading Basis Graded**

A study of the attempts of scholarship since the Enlightenment to discover the so-called historical Jesus. Includes discussion of historiographical problems raised in past and present scholarly "quests" for the historical Jesus, analysis of the relationship between memory and the Jesus traditions, and practice in situating the Gospel materials in the social and political context of ancient Roman Palestine.

REL 343 Paul and the Origins of Christianity**Credits 3 Grading Basis Graded**

Some scholars argue that the Apostle Paul was the "real" founder of Christianity, others that he was the faithful interpreter of the Jesus traditions to the Greco-Roman world. After reconstructing the historical course of Paul's life and journeys from the available sources, the course will analyze selected Pauline epistles, sent to early Christian communities, in order to reconstruct his teaching and ethics and to assess his significance for the origins of Christianity.

REL 344 Gospel of John**Credits 3 Grading Basis Graded**

Of the four New Testament gospels, the Gospel of John is the most mystical, philosophical and, in some respects, the most challenging to understand and interpret. It is both a deeply Jewish work and a deeply Christian work. It is a textual encounter zone between ancient Judaism and emergent Christianity. This course focuses on a close reading of the entire Gospel that situates it within the wider narrative world of ancient Judaism and observes its re-centering of that narrative world around the figure of Jesus.

REL 348 Christianity in Global Context**Credits 3 Grading Basis Graded**

Christianity plays a vital role in many societies around the world. Though often treated as a mostly western or European religion, it was in fact a global religion first and foremost. This course examines Christianity from that global perspective. What does Christianity look like around the world? How have indigenous cultures fashioned their own versions of Christianity in the modern world? Is there unity in the diversity of these global Christianities?

REL 363 Apocalypticism, Religious Terrorism and Peace**Cross-listed with: IA 363****Credits 3 Grading Basis Graded**

This course traces apocalypticism from its ancient Jewish and Christian roots to its contemporary manifestations in religious groups around the world. Since apocalypticism is a worldview that cuts across religious traditions, the course covers a variety of religious groups. The last half of the course focuses on the complex relationships between apocalyptic thinking and religious terrorism and entails an independent research project.

REL 365 Religion and Film**Credits 3 Grading Basis Graded**

In this course, students will explore the appearance of the major world religions (Christianity, Judaism, Islam, Buddhism and Hinduism) as well as religious themes (such as redemption or the afterlife) in the medium of film. Students will learn to critique cinematic representations and misrepresentations of religion and to analyze their impact on viewers, as well as to consider how films may provide rituals and other experiences that serve a religious function.

REL 366 Religion, Animals and the Environment**Cross-listed with: ENVT 366****Credits 3 Grading Basis Graded**

This course analyzes the ways in which various religions, especially Judaism and Christianity, understand the relationship of human beings to the environment and non-human animals. We will explore how religious worldviews translate into human impact on the non-human biosphere. Given the global reach of American environmental policies and lifestyles, and the prominent role that biblical interpretation has played in shaping American attitudes and policy, special emphasis will be placed on historical interpretations of the Bible, ranging from human dominionism to human responsibility in caring for the earth, its creatures and its ecosystems. The course explores the complexities involved in using religious positions, interpretations of biblical texts and apocalyptic arguments to inform both environmentalism and policy debates related to the environment. Students will gain a basic introduction to methods of biblical scholarship and familiarity with the general state of scientific consensus on environmental challenges.

REL 371 Religion and Disability**Credits 3 Grading Basis Graded**

What is disability and how is it constructed? What have the major world religions, on their own and in conversation, decided is 'normal' and what they have decided is 'other'? How do they view and treat those individuals/groups they have deemed disabled? How do people with disabilities create their own meaning within and beyond their own religion? This course will investigate the nature of disability in five of the world's religions (Hinduism, Buddhism, Judaism, Christianity and Islam). While we will become familiar with how disability intersects with and across each of these religions, the heart of this course will be deep explorations of specific case studies as well as personal reflections on the students' and others' intersectional identities in the context of various communities.

REL 372 Religion and Medicine, East and West**Credits 3 Grading Basis Graded**

What is the relationship between religion and medicine? How has the idea of healing been influenced by religious faith, historically and philosophically? This course explores a broad range of issues at the intersection of religion and science, from the question of what counts as "proof" to the existential crises that sickness can provoke. Historical perspective is balanced with case studies and current events to show how theoretical concerns affect real-world scenarios. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

REL 373 Race and Religion**Credits 3 Grading Basis Graded**

Both race and religion are interwoven into who we are and how we inhabit our various worlds, even if we don't think of these as important or relevant parts of our lives. This class will explore the complicated and long-standing intersection of these two concepts, wondering: What do they each mean? How have these terms functioned throughout history? How do race and religion interact? How has one informed or shaped our understanding of the other? When differing racial and religious identities converge for individuals and communities, what happens? With the foundation provided by the course and the time together, students will also have the opportunity to explore their own various identities as well as a case study of the students' choosing that involves an intersection of race and religion.

REL 374 Religion and Mental Health**Credits 3 Grading Basis Graded**

What counts as a "mental illness" and who gets to decide? Is demonic possession real? If someone hears voices are they sick, or could they be speaking with a supernatural power? What are the right roles of religion and spirituality when it comes to mental health? Do the answers to these questions remain constant throughout history and across cultures? This course will investigate these issues, along with many others that are directly relevant to our personal experience of suffering and joy, and the future of how we understand mental health.

REL 375 Nineteenth-Century Philosophy and Theology**Cross-listed with: PHIL 375****Credits 3 Grading Basis Graded**

A selected study of 19th-century thought, with emphasis on controversies concerning the nature and limits of reason, the ultimate meaning of history, and the inner meaning and social significance of religion. Pertinent thinkers include Hegel, Marx, Schleiermacher, Kierkegaard, Nietzsche and others.

REL 376 Religion and Civic Leadership**Credits 3 Grading Basis Graded**

Throughout American history, religion has been central to how U.S. citizens and residents have thought about and engaged with issues of public concern. This course will pay particular attention to the Civil Rights Movement, where students will gain an introductory knowledge of key figures, competing theological and political themes, and social groups. Students will become equipped to understand how religion has shaped public discourse, dialogue across divides and our public square itself. They will have the opportunity to analyze a combination of primary and secondary sources, with special attention to the archives and personal papers of major players within the wider historiography of the Civil Rights Movement.

REL 377 Hermeneutics**Cross-listed with: PHIL 377****Credits 3 Grading Basis Graded**

This course will examine the main features of hermeneutics with particular emphasis on its contemporary perspectives. Discussion will focus on such themes as human understanding and human finitude, the nature of history and tradition, linguisticity and textuality of experience. Readings may address Gadamer, Ricoeur, Schleiermacher, and Dilthey.

REL 380 Religion and Science**Credits 3 Grading Basis Graded**

This course will provide an historical survey of the relationship between religion and the sciences; offer overviews of scientific and theological theory; examine the development of theory formation; focus on issues in astronomy, physics and biology; explore the ethical implications of scientific and religious theories; and trace developments.

REL 383 The Global Refugee Crisis and Religion**Credits 3 Grading Basis Graded**

Focused on the contemporary refugee crisis, this course investigates the intersection of religion with human experiences of displacement, migration, resettlement, and welcome. We also analyze the place of religion in refugee policy debates as well as the role of faith-based organizations in refugee relief and resettlement.

REL 385 Religion and the Law of War**Credits 3 Grading Basis Graded**

The law of war, or international humanitarian law (IHL), is the body of law that regulates the conduct of armed conflict in order to preserve basic human dignity and reduce suffering. Although IHL was invented in modern Europe, this body of law is rooted in ethical and religious traditions that are thousands of years old and that span diverse cultures. In this course, students analyze the relationship between the military ethics of IHL and the ethical systems of religious traditions such as Hinduism, Buddhism, Judaism, Christianity, Islam and select indigenous traditions. Students also explore how the International Committee of the Red Cross engages with religious communities in order to strengthen humanitarianism across the globe.

REL 388 Tibetan Buddhism**Credits 3 Grading Basis Graded**

This course explores the teachings of Tibetan Buddhism as they are practiced and experienced in communities on the Tibetan Plateau. Student research projects investigate the globalization of Tibetan Buddhism, especially through the public life of the Dalai Lama, as well as Tibetan Buddhism's relationship to topics like nonviolence, compassion, mindfulness, interfaith dialogue, psychology and politics.

REL 420 Controversies in Religion**Credits 3 Grading Basis Graded**

This course serves as a senior seminar (capstone) for majors in Global Religions and Cultures, centered around various controversies in religion. Original student research, writing and oral presentation are the centerpieces of this religion capstone course. Specific course themes vary by year and instructor.

REL 430 Islam and the Nature of Being**Credits 3 Grading Basis Graded**

This course serves as a senior seminar (capstone) for majors in Global Religions and Cultures, centered around Islam and the nature of being. Original student research, writing and oral presentation are the centerpieces of this religion capstone course. Specific course themes vary by year and instructor.

REL 440 Religion and American Culture**Credits 3 Grading Basis Graded**

This course serves as a senior seminar (capstone) for majors in Global Religions and Cultures, centered around the subject of religion in American culture. Original student research, writing and oral presentation are the centerpieces of this religion capstone course. Specific course themes vary by year and instructor.

REL 450 Religion and Society**Credits 3 Grading Basis Graded**

An advanced exploration of diverse ways in which one or more of the world's religions have engaged questions of significant social impact. Specific course themes vary by year and instructor. Original student research, writing and oral presentation are the centerpieces of this religion capstone course. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

REL 460 Topics in Ancient Jewish and Early Christian Literature**Credits 3 Grading Basis Graded**

An in-depth examination, using critical academic methods, of the historical, literary, and cultural dimensions of selected texts from the literatures of Ancient Judaism and Early Christianity. Fulfills the College of Arts and Letters writing-intensive requirement for the major. Prerequisites: REL 201 or REL 202; or permission of the instructor.

REL 470 Women's Experience and Global Christianity**Credits 3 Grading Basis** Graded

This course serves as a senior seminar (capstone) for majors in Global Religions and Cultures, centered around the subject of women's experiences in global Christianity. Original student research, writing and oral presentation are the centerpieces of this religion capstone course. Specific course themes vary by year and instructor.

REL 488 Buddhism in the Modern World**Credits 3 Grading Basis** Graded

This capstone course in the study of religion applies knowledge of Buddhist thought and practice to contemporary global expressions of Buddhism. A key theme of the course is the tension between tradition and modernity in Buddhist communities, especially in light of post-colonialism and globalization. Specific course themes and content rotate by year. Original student research, writing and oral presentation are the centerpieces of this capstone experience. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

REL 490 Special Studies in Religion**Credits 3 Grading Basis** Graded

This course is designed to give capable students the opportunity to complete independent study in religion under faculty supervision. May be repeated for credit.

REL 491 Religion and Society in Ghana**Credits 3 Grading Basis** Graded

The Ghana Summer 4-week program offers students lecture series and experiences that are interdisciplinary and enhance religious and cultural literacy and skills. While Ghanaian society is culturally and religiously pluralistic, the country has experienced relatively limited violence that emanates from religious differences. The purpose of the summer lecture series is to provide students opportunities to observe local professors and religious practitioners discuss the role and place of religion in Ghanaian society-both traditional and modern.

REL 493 Religion Course Assistantship**Credits 3 Grading Basis** Graded

Students participate as course assistants in religion. Assistantships provide students with a sense of what it is like to teach a religion course by allowing them to work closely with faculty members through different phases of course preparation, presentation and evaluation. Assistantships may also provide opportunities for student assistants to lead discussion and to help their peers review the material outside the classroom.

Prerequisites: Senior standing.

REL 495 Internship**Credits 1-3 Grading Basis** Graded

Gives the structured opportunity to gain practical knowledge and experience while serving the community. Students apply skills in religious literacy, intercultural communication, research, writing and more.

Prerequisites: Global Religions and Cultures major or minor; junior or senior standing. Other students may enroll with the permission of the Academic Unit Head.

REL 496 Religion and Society in Ghana: Internship**Credits 3 Grading Basis** Graded

The Ghana Summer Internship offers students lectures and experiences that are interdisciplinary and enhance religious and cultural literacy and skills. While Ghanaian society is culturally and religiously pluralistic, the country has experienced relatively limited violence that emanates from religious differences. The purpose of the summer internship in religion and society is to provide students opportunities to observe, interact and engage with the leadership and agencies that have ensured the nation's peaceful religious environment.

REL 499A Honors Thesis**Credits 1 Grading Basis** Graded

An original research project for majors in the religion program who want to graduate with distinction.

REL 499B Honors Thesis**Credits 2 Grading Basis** Graded

An original research project for Global Religions and Cultures majors who want to graduate with distinction.

REL 499C Honors Thesis**Credits 3 Grading Basis** Graded

An original research project for Global Religions and Cultures majors who want to graduate with distinction.

Russian (RUS)

RUS 101 Elementary Russian I**Credits 3 Grading Basis** Graded

This course covers the fundamentals of Russian through listening, speaking, reading and writing. This course provides practice in pronunciation and development of comprehension.

RUS 102 Elementary Russian II**Credits 3 Grading Basis** Graded

This course builds on the foundations established in RUS 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension. Prerequisites: RUS 101 or by placement exam.

RUS 111 Intensive Russian I**Credits 6 Grading Basis** Graded

The fundamentals of Russian through listening, speaking, reading and writing. The course is the equivalent of RUS 101- RUS 102.

RUS 212 Intensive Russian II**Credits 6 Grading Basis** Graded

The fundamentals of Russian through listening, speaking, reading and writing. The course is the equivalent of RUS 231-232.

Prerequisites: RUS 102, RUS 111, or by placement exam

RUS 231 Intermediate Russian I**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.

Prerequisites: RUS 102, RUS 111, or by placement exam

RUS 232 Intermediate Russian II**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.

Prerequisites: RUS 231 or by placement exam.

RUS 300 Language for Living**Credits 3 Grading Basis** Graded

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in the Russian language. It is intended for students who have begun to develop advanced skills and are interested in continued training in language proficiency through reinforcing foundations of everyday language communication and cultural knowledge. Instruction is in Russian. Fulfills the College of Arts and Letters writing-intensive requirement for the major.

Prerequisites: RUS 232, RUS 212 or by placement exam.

RUS 301 Pop Culture through Language**Credits 3 Grading Basis** Graded

The course is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in the Russian language and culture. It is intended for students who have begun to develop advanced skills in interpretive, interpersonal and presentational communication and are interested in delving deeper into the study of Russian-speaking countries, their histories and their cultures. The course will focus on popular culture in the Russian-speaking world including music, film, television, literature and social media, and analyze how these reflect and shape cultural identities and trends. Instruction is in Russian.

Prerequisites: RUS 232, RUS 212 or by placement exam.

RUS 308 Introduction to Russian Culture**Credits 3 Grading Basis** Graded

A study of Russian life and culture and the outstanding contributions of Russian-speaking peoples. Instruction is in English.

RUS 315 Sounds of Speech**Credits 3 Grading Basis** Graded

This course introduces students to Russian linguistics, with special emphasis on Russian phonetics. Students will engage with theoretical concepts and practical applications to understand the phonological structures and sounds that make up the Russian language. Class activities and assignments aim to enhance overall language proficiency, including listening, reading, speaking and pronunciation skills. By the end of the course, students will have a solid foundation in Russian phonetics and demonstrate greater confidence, clarity and fluency in spoken Russian. Instruction is in Russian.

Prerequisites: RUS 300, RUS 301, RUS 320 or RUS 322; or by permission of the instructor.

RUS 320 Culture and Communication in Context**Credits 3 Grading Basis** Graded

This Russian course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do Russian-speaking peoples communicate on subjects that are at the heart of their societies?" Instruction is in Russian.

Prerequisites: RUS 232, RUS 212 or by placement exam.

RUS 322 Language Landscapes**Credits 3 Grading Basis** Graded

This Russian course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in these culturally and linguistically defined areas and explores the diverse cultural landscapes that shape their lives. Instruction is in Russian.

Prerequisites: RUS 232, RUS 212 or by placement exam.

RUS 390 Russian Language and Culture Abroad**Credits 1-3 Grading Basis** Graded

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in Russian literature and culture. It is intended for students who have begun to master advanced skills in interpretive, interpersonal and presentational communication and are interested in examining nuances of the Russian language and culture while studying abroad.

Prerequisites: RUS 212 or RUS 232, or permission of the instructor.

RUS 400 Advanced Russian**Credits 3 Grading Basis** Graded

This course focuses on conversation. Discussions deal with topics of current interest. Instruction is in Russian.

Prerequisites: RUS 300, RUS 301, RUS 320 or RUS 322; or by permission of the instructor.

RUS 438 Studies in Russian Literature**Cross-listed with:** ENG 438**Credits 3 Grading Basis** Graded

A study of selected works of Russian literature. Instruction is in English. May be repeated for credit when course content changes.

RUS 446 Special Topics in Russian Literature**Credits 3 Grading Basis** Graded

This course studies a particular topic in Russian literature. It may cover all or a specific Russian literature genre. This course may be repeated if content varies.

Prerequisites: RUS 300, RUS 301, RUS 320 or RUS 322; or by permission of the instructor.

RUS 447 Special Topics in Russian-Speaking Cultures**Credits 3 Grading Basis** Graded

This course studies a particular topic in the culture of Russian-speaking regions. This course may be repeated if content varies.

Prerequisites: RUS 300, RUS 301, RUS 320 or RUS 322; or by permission of the instructor.

RUS 448 Special Topics in Russian Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in Russian linguistics. Topics could include an introduction to Russian sociolinguistics or psycholinguistics. This course may be repeated if content varies.

Prerequisites: RUS 300, RUS 301, RUS 320 or RUS 322; or by permission of the instructor.

RUS 490 Independent Study in Russian**Credits 1-4 Grading Basis** Graded

This course allows students an opportunity to complete independent study in Russian under faculty supervision.

RUS 499A Honors Project**Credits 1 Grading Basis** Graded

This is a three semester course, offered as parts A, B, and C for 1, 3, 2 credits per semester.

RUS 499B Honors Project**Credits 3 Grading Basis** Graded

This is a three semester course, offered as parts A, B, and C for 1, 3, 2 credits per semester.

RUS 499C Honors Project**Credits 2 Grading Basis** Graded

This is a three semester course, offered as parts A, B, and C for 1, 3, 2 credits per semester.

School of Theatre and Dance (STAD)

STAD 100 Theatre and Dance Colloquium

Credits 0 Grading Basis Credit / No Credit

Department colloquium; work in progress presented, viewed and discussed by student body, faculty and guests. Professionals in the field may hold master classes. May fulfill other purposes as needed.

Prerequisites: School of Theatre and Dance majors only.

STAD 171 Performance Production

Credits 3 Grading Basis Graded

An introduction to the methods of the production of scenery, properties, costumes, lighting, sound and performance management for theater and dance performance. Instruction in the skills required for the operation of lighting and sound equipment will be taught. Students are required to complete a main stage running crew assignment as a component of this course.

Prerequisites: School of Theatre and Dance majors and minors and School of Music majors only.

STAD 250 The Collaborative Artist: Sophomore Studio

Credits 3 Grading Basis Graded

The course explores the creative and collaborative processes within the development of performance events. Emphasis is placed upon the role of the artist as a social and artistic force in society. Studio-based creative explorations along with the viewing and discussion of live and recorded performances are integral to the course in order for students to investigate, experience, and devise collaboratively within the context of various performance forms across multiple disciplines and cultures.

Prerequisites: School of Theatre and Dance majors only and sophomore standing, or permission of the instructor.

STAD 273 Design Aspects of Performance

Credits 3 Grading Basis Graded

This course studies the interrelationship between the playwright, director, choreographer, dancer and actor with the scenic, lighting, sound, projections and costume designers. It introduces the basic documents of the design process, and surveys current and historical trends in performance design. No artistic or technical skills are necessary.

STAD 303 Special Topics in Theatre and Dance

Credits 1-3 Grading Basis Graded

Study of current topics and issues in performing arts. Emphasis on contemporary themes of immediate concern.

STAD 322 Equity, Inclusion and Access in Education and Performance

Credits 3 Grading Basis Graded

An exploration of practices in theatre and dance education and performance that support equity, inclusion and access. This course looks at the intersections of arts education, art-making and performance, applied theatre, and community engaged practice.

Prerequisites: School of Theatre and Dance majors, Disability Studies minors, and African, African American and Diaspora Studies minors only.

STAD 471 Stage Management

Credits 3 Grading Basis Graded

Study and analysis of stage management. Consideration given to the methods and strategies for successful stage management for theater, dance and other performances. Emphasis on developing management and organizational skills.

Prerequisites: STAD 171.

Social Justice Studies (SJS)

SJS 400 Social Justice Studies Internship

Credits 3 Grading Basis Graded

This course provides students an opportunity to gain professional and/or practical experience utilizing and developing knowledge and skills in social justice studies in government agencies, non-profit organizations, or other public or private employment or service organizations.

Prerequisites: Declaration of the minor in Social Justice Studies; JUST 221 or JUST 223; approval of the Social Justice Studies minor coordinator; and permission of the department.

SJS 401 Social Justice Studies Project

Credits 3 Grading Basis Graded

This course provides students an opportunity to design and/or complete an independent or collaborative research or creative project in social justice studies, under faculty supervision.

Prerequisites: Declaration of the minor in Social Justice Studies; JUST 221 or JUST 223; approval of the Social Justice Studies minor coordinator; and permission of the department.

Social Work (SOWK)

SOWK 287 Introduction to Social Work

Credits 3 Grading Basis Graded

This course provides an overview of social work as a profession with emphasis on various settings, roles, values and ethics, and professionals' work with diverse populations. It focuses on practical experiences designed to familiarize students with the dynamics of the profession.

SOWK 288 Human Rights and Social Progress

Credits 3 Grading Basis Graded

This course introduces foundational frameworks of social oppression and human rights and how they relate to societal challenges. Appreciation for human differences, lived experiences and cultural competency are emphasized. Processes such as advocacy actions that address social problems are explored.

SOWK 301 Workshops in Social Work

Credits 0-3 Grading Basis Credit / No Credit

Detailed study of a topic of interest in social work. May be repeated for credit.

SOWK 302 Workshops in Social Work

Credits 1-3 Grading Basis Graded

Detailed study of a topic of interest in social work. May be repeated for credit.

SOWK 305 Social Work Research Methods

Credits 3 Grading Basis Graded

Study of qualitative and quantitative methods in social work research. Demonstrating scientific and ethical research processes including formulation of research questions, selecting a design, collecting data, analysis and interpretation of data, and exposure to evaluation of practice. Diversity and inclusion are highlighted in the research process. Prerequisites: MATH 220 or SOCI 231; Minimum grade of "C" in SOWK 287 and SOWK 288.

SOWK 314 Rural Health: An Interprofessional Approach**Cross-listed with:** IPE 314, HTH 314, NSG 314**Credits 3 Grading Basis** Graded

Students study, observe and participate in interprofessional assessment, planning and delivery of community-based primary health care in partnership with residents and agencies of a host rural county. Learning activities will emphasize rural culture, rural health care and interprofessional practice.

SOWK 317 Skills for Generalist Social Work**Credits 3 Grading Basis** Graded

This course examines phases in the social work process, with emphasis on engagement within a strengths-based generalist framework with individuals, families, groups and communities. It provides an introduction to supervisory feedback to develop capacity for reflective application of professional ethics and values, knowledge of human behavior, and key interpersonal skills necessary for collaborative, empowering and culturally responsive practice. Students develop technical writing skills in session planning and case documentation.

Prerequisites: Minimum grade of "C" in SOWK 287 and SOWK 288 or permission of the department. Corequisites: SOWK 320 and junior standing.

SOWK 320 Human Behavior in the Social Environment**Credits 3 Grading Basis** Graded

This course presents theoretical frameworks of human behavior and the social environment at individual, family, group, organizational and community system levels. The application of the frameworks to social work processes of engagement, assessment, intervention and evaluation is explored. The course integrates and expands prerequisite knowledge in psychological and socio-cultural sciences related to the dimensions of human differences in shaping human experience.

Prerequisites: PSYC 101, PSYC 160, and minimum grades of "C" in SOWK 287 and SOWK 288; and one of SOCI 101 or SOCI 110 or SOCI 140; and one of AAAD 200, ANTH 195, GEOG 200, GEOG 280, SOCI 336, SOCI 354 or WGSS 200. Corequisites: SOWK 317.

SOWK 325 Parent-Child Relationships Across the Lifespan**Cross-listed with:** FAM 325**Credits 3 Grading Basis** Graded

This course focuses on giving and receiving familial attention, care and support over the lifespan and across generations. Combines developmental and ecosystems frameworks to explore family life as the primary arena in which people learn and establish social behavior and interpersonal relationships, in conjunction with extrafamilial connections. Prerequisites: SOWK 287 and SOWK 288.

SOWK 327 Substance Use Interventions**Cross-listed with:** FAM 327**Credits 3 Grading Basis** Graded

Explore types, effects and prevalence of substance use in the U.S. Impacts of use on individuals, families, communities and society is considered through biopsychosocial, economic and legal/political lenses. Dual foci include developmental risks and outcomes for individuals and families throughout the life course, and social policy debate and legislation. Includes a survey of intervention approaches and treatment modalities across the span of self-help, medical treatment, programs in behavioral health and court-mandated models.

SOWK 332 Community Mental Health Practice**Credits 3 Grading Basis** Graded

This course introduces students to definitions, historical developments and factors in the social environment that have shaped understanding of mental illness in the United States and the responses of policy makers and practitioners to address community mental health needs. Service delivery approaches, ranging from those that focus on risk reduction, recovery and community integration/inclusion to those that focus on responding to public behavior and social control, are considered from the perspective of social work values and ethical principles. Students will be challenged to ponder questions of self-determination, social responsibility, justice, and the dignity and worth of persons in life situations that include the experience and impacts of mental illness.

SOWK 335 Social Policy**Credits 3 Grading Basis** Graded

This course examines the formulation and consequences of social policy in the context of contemporary social, cultural, demographic, political and economic conditions. Students develop skills in policy analysis, critique and change advocacy in U.S. society with emphasis on policy practice at the organizational, local, state and national levels. Prerequisites: POSC 302 or POSC 225; Minimum grade of "C" in SOWK 287 and SOWK 288.

SOWK 338 Issues and Policies in Family Services**Credits 3 Grading Basis** Graded

Examination of historical and philosophical approaches to family policy. Evolution of family-related social policies in the United States is contrasted with those of selected foreign countries with the view toward a national family policy.

SOWK 340 Violence in Families**Credits 3 Grading Basis** Graded

Examination of violence in the family, including spouse, sibling, elder and child abuse. Studies the social and cultural patterns and etiology of family violence. Examines programs and services for the abused and the abuser including shelters, support systems and counseling.

SOWK 342 Child Welfare Services**Credits 3 Grading Basis** Graded

Study of the basic child welfare services along the continuum of prevention and support; assessment and identification of child abuse and neglect; evidence-based interventions; and family reunification and support. Analysis of legal framework and court services as well as ethical standards that guide practice. Utilizes a resilience and trauma-informed framework to assess practices in the child welfare system as well as collaboration with other systems involved with vulnerable children and families.

SOWK 347 Issues Affecting LGBTQIAP+ Individuals and Their Families**Cross-listed with:** FAM 347**Credits 3 Grading Basis** Graded

Operating from a cultural humility framework, this course focuses on practice, policy, and historical and social issues impacting LGBTQIAP+ individuals and their families. This course utilizes a strengths-based perspective to identify protective factors and explore resiliency within the LGBTQIAP+ community in the face of social stigma, marginalization and oppression. The course encourages students' critical self-reflection of personal and cultural value systems and how these may impact understanding of LGBTQIAP+ populations.

SOWK 362 Social Work with Immigrants and Refugees**Cross-listed with:** FAM 362**Credits 3 Grading Basis Graded**

This course explores the scope and complexity of global/international migration and its impacts on national policy and social service structures and organizational response. It provides a framework to understand transmigration and acculturation, to analyze aspects of current immigration law, and to examine the role of various service systems to address the needs of immigrant and refugee individuals and families integrating into local communities. Service-learning in an agency that serves the local immigrant community is required.

SOWK 372 Programs and Services for Older Adults**Cross-listed with:** GERN 372**Credits 3 Grading Basis Graded**

An examination of social policies and responses to aging persons from historical and current perspectives in the United States. A survey of programs and services developed for older adults in the public and private sectors at individual, group, and community levels is examined, with consideration of professional skills, competencies, and credentials necessary for ethical and competent program implementation. Engaged learning through observations and participation in local programs serving older adults is required.

Prerequisites: GERN 280/SOCI 280.

SOWK 375 Grant Writing for Agencies**Cross-listed with:** FAM 375, GERN 375, NPS 375**Credits 3 Grading Basis Graded**

Emphasizing active learning, this course teaches the basics of grant and proposal writing. Efficient research, persuasive prose and the importance of relationships are stressed. Private and corporate philanthropy and government grants are examined.

SOWK 386 Youth Empowerment Strategies (YES)**Cross-listed with:** FAM 386**Credits 3 Grading Basis Graded**

Students learn to use group activities that include the creative arts, low ropes, and self-discovery in youth empowerment. The goal is to help youth build life skills and make informed decisions. Prior to beginning work with youth, students complete 25 hours of training.

SOWK 387 Working with Adolescents**Credits 3 Grading Basis Graded**

Examination of service delivery systems and programs from a social work "person-in-environment" perspective that considers social, cultural, familial, and personal risk and protective factors. Theories of adolescent development from the physical, psychological and social sciences inform investigation of issues and experiences associated with youths being referred for services including behavioral health/mental illness issues, trauma, substance use, suicide, family disruption, sexual development, school difficulties, runaway and abuse/neglect/exploitation.

SOWK 450 Palliative and Hospice Care: An Interdisciplinary Approach**Cross-listed with:** GERN 450**Credits 3 Grading Basis Graded**

An interdisciplinary approach to assessment and care planning of older adults experiencing need for palliative and/or hospice support. Familial, cultural, legislative and social implications of programs and service delivery.

SOWK 465 Social Work Practice in Mezzo Systems**Credits 3 Grading Basis Graded**

This course emphasizes the application of social work values, theories and skills with small groups. Engagement, assessment, planning intervention strategies, resource utilization and evaluation are examined in the context of experiential learning. Attention is given to how demographic backgrounds, power dynamics, as well as other forms of difference impact mezzo systems.

Prerequisites: Social work majors with

senior standing and minimum grades of "C"

in SOWK 305, SOWK 317, SOWK 320 and SOWK 335. Corequisites:

IPE 415, SOWK 466, SOWK 467 and SOWK 493.

SOWK 466 Social Work Practice in Micro Systems**Credits 3 Grading Basis Graded**

This course accentuates social work values, ethics, theories, perspectives, models and skills in evidence-informed, client-centered, generalist social work practice with individuals. Students learn and apply the phases of the planned change process with clients, including engagement, assessment, intervention and evaluation. Learning is supported through development of a simulated case, including video-recorded client sessions and supervisory meetings with the instructor.

Prerequisites: Social work major with

senior standing and minimum grades of "C"

in SOWK 305, SOWK 317, SOWK 320 and SOWK 335. Corequisites:

IPE 415, SOWK 465, SOWK 467 and SOWK 493 .

SOWK 467 Social Work Practice in Macro Systems**Credits 3 Grading Basis Graded**

The application of multidisciplinary theoretical frameworks, social work values and skills to organizational, community and policy practice is emphasized. Engagement, assessment, planning intervention strategies, resource utilization and evaluation are examined. Attention is given to the impacts of human differences and power differentials on macro systems. Community-based projects for organizations, communities and policies are utilized.

Prerequisites: Social work major with

senior standing and minimum grades of "C"

in SOWK 305, SOWK 317, SOWK 320 and SOWK 335. Corequisites:

IPE 415, SOWK 465, SOWK 466 and SOWK 493.

SOWK 481 Social Work Field Practicum I**Credits 6 Grading Basis Graded**

Offers students an opportunity to gain a broad knowledge of the basic functions, services and roles of the agency as related to actual social work practice, as well as a specific knowledge of practical intervention skills necessary to carry on effective social work practice. The field experience is the application of knowledge and skill components drawn from previous courses.

Prerequisites: Social work majors with senior standing. Minimum grade of "C" in SOWK 465, SOWK 466, SOWK 467 and SOWK 493. Corequisites: SOWK 482 and SOWK 494.

SOWK 482 Social Work Field Practicum II**Credits 6 Grading Basis Graded**

This course offers the student an opportunity to build upon previous field experience by having more responsibility and tasks designed to expand their practice skills in social work.

Prerequisites: Social work majors with senior standing. Corequisites:

SOWK 481 and SOWK 494.

SOWK 487 Special Topics**Credits 3 Grading Basis** Graded

Examination of selected topics of social work practice which are of current importance in the social work profession. Course may be repeated for credit.

SOWK 490 Special Studies in Social Work**Credits 1-3 Grading Basis** Graded

This course is restricted to majors in social work. The course provides students an opportunity to complete independent studies under faculty supervision. Course may be repeated for credit.

SOWK 493 Professional Forum I: Preparation for Field**Credits 1 Grading Basis** Graded

This course provides an orientation to the pedagogy and process of the field practicum. Students become acclimated to the placement process as well as the expectations and requirements of professional social work practice in field sites and includes a series of trainings on topics and issues needed prior to entering their field placement.

Prerequisites: Minimum grade of "C" in SOWK 305, SOWK 317, SOWK 320 and SOWK 335; admission to the social work program; completion of 50 pre-field service hours; senior standing. Corequisites: SOWK 465, SOWK 466 and SOWK 467.

SOWK 494 Professional Forum II: Transition to Professional Practice**Credits 3 Grading Basis** Graded

This course is the synthesis of coursework throughout the curriculum in developing and demonstrating competency as a generalist social work practitioner. Focal content areas are selected based upon results of program assessment trends. Key areas of professional identity and development are presented: professional credentialing, "presentation of self," employment-seeking, recognizing and managing professional burn-out, and self-care. Senior outcome assessment is integral to this course. Prerequisites: Social work majors with senior standing and minimum grades of "C" in SOWK 465, SOWK 466, SOWK 467, and SOWK 493. Corequisites: SOWK 481 and SOWK 482.

SOWK 499A Honors**Credits 1 Grading Basis** Graded

Prerequisites: Social Work Majors only.

SOWK 499B Honors**Credits 3 Grading Basis** Graded

Prerequisites: Social Work Majors only.

SOWK 499C Honors**Credits 2 Grading Basis** Graded

Prerequisites: Social Work Majors only.

Sociology (SOCI)

SOCI 101 Introductory Sociology**Credits 3 Grading Basis** Graded

Provides students with an understanding of the structure and processes of modern societies and their historical antecedents. Explores the universality of the social experience by addressing such topics as culture, socialization, social interaction, bureaucracy, norms and diversity, social inequality, social institutions, modernization, technology and social change, world views, values and behavior.

SOCI 104 The Sociology of Diners, Drive-Ins and Dives**Credits 1 Grading Basis** Graded

The purpose of this course is to think sociologically about commercial hangouts. We will focus our attention on sites of public food and beverage consumption, including diners, drive-ins and dives, as well as coffee shops, ice cream parlors, food trucks, breweries, wine bars, fine-dining, food courts, dining halls and fast-food restaurants. Students will be assigned diverse readings on public life and undertake a series of field studies of commercial hangouts in their area. A major theme running through the readings and participatory assignments will be social process, collective action and symbolic interaction. The one-week, asynchronous online course culminates in a final paper synthesizing the readings and students' first-hand observations.

SOCI 110 Social Issues in a Global Context [C4GE]**Credits 3 Grading Basis** Graded

This course introduces the discipline of sociology from a macrosociological perspective, emphasizing large-scale changes in social organization and institutions. We examine the global forces that shape societies, and the historical, political, social, cultural and economic origins of contemporary social problems. We consider competing theoretical models used in the study of social change as well as the conceptual and methodological challenges in analyzing societies different from one's own. May be used for general education credit.

SOCI 140 Microsociology: The Individual in Society [C5SD]**Credits 3 Grading Basis** Graded

This course introduces the discipline of sociology and the subfield of microsociology. We examine the mutually constitutive relationship between the individual and society. Questions addressed include: How does society influence how we think, feel, believe, act, and interact with others? What influences the self, social identity, shared social meanings, social roles, and one's position in society? How do we, as individuals and as members of social groups, recreate, contest, and change society? May be used for general education credit.

SOCI 200 Development of Sociological Thought and Method**Credits 3 Grading Basis** Graded

This course is a foundation course for sociology majors. Topics will include the historical development of the discipline with an emphasis on the social and philosophical forces that influenced the development of sociology. Main sociological traditions will be introduced including the critical, naturalistic, and interpretive paradigms and sociological analysis from these perspectives.

Prerequisites: SOCI 101, SOCI 110 or SOCI 140; or permission of instructor.

SOCI 231 Introduction to Social Statistics**Credits 3 Grading Basis** Graded

Introduction to the techniques for collecting, describing, analyzing and presenting sociological data.

SOCI 280 Social Gerontology**Cross-listed with:** GERN 280**Credits 3 Grading Basis** Graded

An interdisciplinary introduction to the study of aging. The course provides an overview of issues surrounding aging in contemporary society: personal, familial, communal and societal.

SOCI 300 Sociological Inquiry**Credits 3 Grading Basis Graded**

This course provides a systematic introduction to various modes of sociological investigation, including positivism, interpretivism and critical analysis. Students learn to evaluate, critique and design original sociological inquiries with special attention to how sociological inquiry is guided by different philosophical and theoretical commitments.

Prerequisites: Sociology major and SOCI 200.

SOCI 301 Ethnicity, Race and Nationality: Identity Construction from a Cross-Cultural Perspective**Credits 2 Grading Basis Graded**

This course explores the concepts of identity, ethnicity and nationality from a cross-cultural perspective. Drawing upon theories from multiple social scientific disciplines, this course examines how identity is defined and how societies use such constructions for, among other things, nation-building. Students learn to apply the sociological imagination for understanding in-group and out-group interactions cross-culturally. Theoretical readings are supplemented with case studies from developed and developing countries across different time periods.

SOCI 303 Sociology of Death and Dying**Credits 3 Grading Basis Graded**

Investigation of current American orientations toward death and dying with emphasis also given to the social organization of death and dying.

SOCI 304 Social Deviance**Credits 3 Grading Basis Graded**

Course offers students a wide range of explanations of deviance. Topics considered are the functions, social definitions, societal reactions and political aspects of deviance as characteristic of all societies. Deviant attributes as well as acts are considered.

SOCI 306 Japanese Society and Culture**Cross-listed with: ANTH 306****Credits 3 Grading Basis Graded**

This introductory course takes a critical and interdisciplinary approach to exploring Japan. We will apply sociological, anthropological, and demographic perspectives to comparatively understand, analyze, and discuss Japanese society and culture. Students will read and discuss issues related to history, socio-demographic change, gender, work, social class, race/ethnicity, family, health care and aging in Japanese society.

SOCI 307 Sociology of the Life Course**Credits 3 Grading Basis Graded**

This course introduces students to theories, methods and research based on the life course paradigm. A life course perspective focuses on intersections of individual lives, historical context, social change and social structures including diverse family systems. Based on this framework students will look at how life trajectories are influenced by those systems in the broader social structure and historical context. Students will also look at how prior life stages can positively or negatively influence later life outcomes of individuals. Students will have an opportunity to apply the life course perspective to discuss research and to examine how life changes (marriage, divorce, job, childbirth, death, etc.) are influenced by social factors and contexts.

SOCI 309 Capitalism and Alternative Economies**Credits 3 Grading Basis Graded**

This course examines the structure and dynamics of capitalism as a social system and considers the affordances and limitations of this mode of organizing production, consumption and the satisfaction of human needs. Building on this analysis, the course then critically examines a wide array of alternative or non-capitalist organizations and enterprises emerging from the "new," "solidarity" or "social" economy movements, with particular attention paid to the role of commerce and markets in these arrangements.

SOCI 311 Sociology of the Environment**Credits 3 Grading Basis Graded**

This course will introduce students to the central debates that currently preoccupy environmental sociology and political ecology. Emphasis is placed on the importance of sociological, historical and cultural modes of inquiry for understanding: socio-ecological change/crisis, environmental justice/injustice, eco-technological change and the politics of "nature".

SOCI 312 Public Life Studies**Credits 2 Grading Basis Graded**

This course gives students the opportunity to gain a deeper understanding of everyday life in public spaces by learning to observe how these spaces are designed, used and enjoyed. Regardless of whether a student has prior experience with qualitative research or simply loves to "people watch," all will gain a deeper understanding of public life studies as a type of research. Opportunities to practice direct observation and documentation skills in a public setting are embedded in the course.

SOCI 315 Technology and Society**Credits 3 Grading Basis Graded**

Through an analysis of various issues, problems and case studies, this course will explore the interactions between science, technology and society. The course will examine connections of specific technologies to science, cultural values, social and economic interests, and questions regarding progress.

SOCI 317 Sociology of the Body**Credits 3 Grading Basis Graded**

This course explores the complex relationship between the body and society and challenges the premise that our bodies are strictly physiological entities. Instead, the course explores how the body serves as a canvas upon which sociocultural norms, values and habits are inscribed. These embodied experiences shape how we perceive and experience the continuity of ourselves and the world. Drawing from multiple theories, methodologies and case studies, the course explores themes such as embodiment and identity, the culture of sensation, technology and the body, body politics, and more.

SOCI 318 Sociology of Immigration**Credits 3 Grading Basis Graded**

This course explores the patterns and processes defining immigration around the world: the reasons for migration, the types of migration, and the way it affects the sending and receiving societies. The course investigates the social forces that affect immigrants and the traits that impact their relocation experience; and how immigration creates new identities for those successful in their quest, contributing to greater cultural diversity of receiving societies.

SOCI 321 Politics in Society**Credits 3 Grading Basis Graded**

The relationship between society and politics, the nature and distribution of social power, political participation, political thought and politics as a vehicle of social change are explored.

SOCI 322 Sociology of Religion**Cross-listed with:** REL 322**Credits 3 Grading Basis** Graded

This course is a sociological analysis of the nature, function, and structure of religion. Survey of the relationship between religion and society: the social nature of religious phenomena, the interaction between religious beliefs and practices and other arenas of secular societies, the social functions of religion, and the way religion changes and is changed by secular society.

SOCI 328 Digital Inequality**Credits 3 Grading Basis** Graded

This course examines how social inequality shapes and is reproduced and challenged through digital information and communication technologies. Students will explore how the use of these technologies by consumers, corporations and governments plays an increasingly prominent role in how inequality is structured and experienced in contemporary society.

SOCI 329 Sociology of Sport**Credits 3 Grading Basis** Graded

This course examines sport as a social institution. It explores the origins and evolution of modern sporting practices and their connections to other social institutions and dimensions of social change. In surveying multiple issues connected to sport and society, particular emphasis will be placed on the production and reproduction of social inequalities through sporting practice and consumption.

SOCI 334 Self and Society**Credits 3 Grading Basis** Graded

This course examines the dialectical relationship between self and society. Two primary questions guide the course: How am I, as a social being, created by society? And, in turn, how is society created and sustained by me? In contrast to biological or psychological explanations, this course provides a distinctly sociological analysis of the self. Drawing from multiple perspectives in sociology, the course focuses on the social character of the self, the process of its acquisition and the social influences that continually shape it. Related topics include the social construction of reality, identity construction, emotions, power and inequality, symbolic interaction, ethnomethodology, and impression management.

SOCI 335 Disability and Society**Credits 3 Grading Basis** Graded

This class focuses on the social and cultural aspects of disability in the US and around the world. It explores the meaning of the concept 'disability'; its relation to other social markers such as race, gender, and sexual orientation; how it is experienced by those occupying the category; and major policy approaches to dealing with disability.

SOCI 336 Race and Ethnicity**Credits 3 Grading Basis** Graded

This course examines the social construction of race and ethnicity around the world and how they influence social processes, institutions, change and ideology. The course will include discussions concerning the intersection of race and ethnicity with other aspects of social inequality such as class, gender, sexuality and nationality in contemporary society.

SOCI 337 Sociology of Gender**Cross-listed with:** WGSS 337**Credits 3 Grading Basis** Graded

This course covers foundational sociological theory, research and examination of the lived realities of sex and gender in society. Using a sociological lens, the course explores the interactions, identities and institutions that are implicated in the social construction of gender and gender inequality. A core emphasis is on how individuals learn and "do" gender, how various institutions (e.g., school, work, media and family) produce and perpetuate gender inequality, how gender inequality intersects with other systems of inequality in social life, and movements for gender liberation and justice.

SOCI 341 Sociology of Education**Credits 3 Grading Basis** Graded

Examination of sociological theories and research on education, emphasizing stratification, socialization, organization, and relationship between schooling, family, community, and work. Focus on cross-cultural approaches to education.

SOCI 344 Work and Society**Credits 3 Grading Basis** Graded

This course examines the nature and meaning of work under various social and historical conditions. This includes such things as the relationship of work organization to life chances and personal experience, the place of work in social theory, the organization of occupations, occupational socialization and commitment, and how the nature of work changes in relation to local and global contexts.

SOCI 347 Queer Theories of Gender and Inequality**Cross-listed with:** WGSS 347**Credits 3 Grading Basis** Graded

This course looks at contemporary approaches to the culture and social structure of gender and gender binaries, as they intersect with power, knowledge, science/biology and inequality. Method involves examining and challenging the cultural classification systems that inform knowledge, human action and social institutions. Requires college level reading, writing and intellectual engagement.

SOCI 352 Birth, Death, Sex: Exploring Demography**Cross-listed with:** ANTH 352**Credits 3 Grading Basis** Graded

This course examines fertility (birth) and mortality (death) and their biological and social determinants in cross-cultural and evolutionary/historical frameworks. It focuses on the exploration of the dynamic between the material constraints on and symbolic significance of reproduction, sexuality and death within a cultural context, and the critical examination of population growth as a global "problem," as well as basic demographic methods.

SOCI 353 Energy and Society**Credits 3 Grading Basis** Graded

This course focuses on the role of energy in the maintenance of human organization, and explores continuity and change in how energy is produced and consumed from the individual to societal levels. It explores issues around resource extraction and depletion, renewable and alternative energy, climate change, and the environmental and social effects of traditional and emerging forms of energy production and use.

SOCI 354 Social Inequality**Credits 3 Grading Basis Graded**

Course covers the systems of stratification and inequality in the United States including race, class, gender, religion, sexuality, ethnicity, and nationality. Discussion will center on their role in providing rationales for oppression and discrimination in society and their relationship to the distribution of power and ideological control.

SOCI 357 Sociology of Disasters**Credits 3 Grading Basis Graded**

This course aims to familiarize students with the sociological study of the causes and consequences of disasters. The course takes a broad view of the social science literature on disasters, but largely employs a case study approach. In keeping with sociology's focus on the causes and consequences of social stratification, the relationship between disasters and patterns of inequality will be a central theme throughout the course.

SOCI 358 Sociology of Consumption**Credits 3 Grading Basis Graded**

Consumption encompasses themes that range from identity construction to the macro processes of cultural globalization. As consumption becomes more integral to society, it is becoming more central to various disciplines. This course situates scholarly work from this nascent interdisciplinary field of consumption studies within the context of contemporary social, cultural, and economic issues.

SOCI 359 The Sociology of Climate Denial**Credits 3 Grading Basis Graded**

This course explores the social scientific research on what has been a highly effective campaign to inject doubt and uncertainty into the American public's understanding of human-caused climate change and the extent of scientific consensus on this issue. The course examines the key institutional actors involved in this effort and the diverse set of interests they represent. Students will also explore other denial campaigns with the goal of generating insights useful for challenging climate denial and denialism more generally.

SOCI 360 Social Movements**Credits 3 Grading Basis Graded**

Introduction to the study and analysis of social movements in the United States as agents of social and ideological change. Emphasis is given to movements which have goals of extending and/or protecting rights of individuals and groups in the face of increasing industrialization, urbanization and centralization of power.

SOCI 361 Sociology of Organizations**Credits 3 Grading Basis Graded**

Study of formal organizations primarily in contemporary society. Emphasis is given to the social-historical context that has given rise to and perpetuates the bureaucracy as a form of social organization, and to the study of the structure and dynamics of contemporary formal organizations such as businesses, universities, governments, etc.

SOCI 363 Sociology of Culture**Credits 3 Grading Basis Graded**

This course examines sociological perspectives about values, norms, symbols, rituals and expressions. Course content includes classic perspectives on the relation between culture and institutions as well as the work of contemporary analysts who have developed, revised, and/or challenged these classic positions. Students will learn to apply these perspectives to their own analyses of culture.

SOCI 367 Sociology of Sexuality**Cross-listed with: WGSS 367****Credits 3 Grading Basis Graded**

This course examines sociological theory and research on sexual behaviors, identities, cultures and social movements, investigating how sexuality is shaped by society and its social institutions. In addition, the course examines how sociological research on sexuality is conducted, how society shapes the sociological study of sexuality, the unique ethical concerns and methodological challenges in researching sexuality and the place of sociology in shaping discourse and social policy on relevant social issues.

SOCI 372 Sociology of the Community**Credits 3 Grading Basis Graded**

This course examines the community as a social form. Considered are its function, social definitions, formative processes, development and systems of change. This survey may include, but not be limited to, examination of community studies research and community advocacy for social justice.

SOCI 374 Sociology of Families**Credits 3 Grading Basis Graded**

Covers the basic concepts and theories in marriage and the family; looks at basic issues in modern family life; examines changes in family functions and in the various stages of the family life cycle and discusses the future of the family in contemporary society.

SOCI 375 Medical Sociology**Credits 3 Grading Basis Graded**

An introduction to the field of medical sociology that examines the salient issues in the field and related theoretical perspectives. These two focuses are important in understanding the ability of humans to live to capacity. Attention is given to health-care programs in developing countries as well as modern industrial societies.

SOCI 382 Qualitative Sociology**Credits 3 Grading Basis Graded**

This course introduces students to qualitative research methods, including participant observation, interviewing, and content analysis. Students will read examples of qualitative sociology and learn how to design and conduct a qualitative research study. In addition to reviewing methodological, theoretical, and ethical issues, students will produce an independent qualitative research project.

Prerequisites: SOCI 200.

SOCI 385 Madness and Society: The Sociology of Mental Health and Illness**Credits 3 Grading Basis Graded**

This course will explore the role that social and cultural factors play in the occurrence, diagnosis, experience, and treatment of mental illness. It will compare sociological perspectives to those of biology and psychology. The course will examine the intersection of mental health systems with other systems, such as the broader health care and the criminal justice systems. Finally, it will critically analyze psychiatry, policy, and popular culture depictions of the mentally ill.

SOCI 386 Sociology of Nonprofit Organizations**Credits 3 Grading Basis** Graded

Focusing primarily on the United States, this course examines the historical and theoretical frameworks that have shaped today's nonprofit organizations. Students will gain a practical understanding of the various types of nonprofit organizations (NPOs) in order to fully grasp their significance as drivers of social change. Special focus will be given to the meaning of civil society, philanthropy as a social construct, stakeholders, professionalization, and the opportunities and challenges facing NPOs in an increasingly interconnected world.

SOCI 390 Topics in Cultural Studies**Cross-listed with:** ANTH 390**Credits 3 Grading Basis** Graded

This course explores contemporary culture through a "cultural studies" lens, an interdisciplinary perspective interested in using empirical knowledge to encourage more just human relations. Specific topics of investigation will vary by semester, but each course will cover cultural studies' intellectual history and its application to cultural expressions found in everyday life, film, music and text.

SOCI 391 Study Abroad**Credits 1-6 Grading Basis** Graded

Designed to encourage students to enhance their academic programs through studying abroad. Arrangements must be made with a faculty member who will direct the study with preparatory instructions and final requirements. May be repeated up to 12 credits.

SOCI 395 Special Topics in Sociology**Credits 3 Grading Basis** Graded

Examination of selected topics which are of current importance in sociology. May be repeated for credit when course content changes.

SOCI 410 Spatial Analysis for Social Research**Cross-listed with:** ANTH 410**Credits 4 Grading Basis** Graded

The course teaches students how to identify and solve anthropological problems with spatial dimensions. Hands-on experience is stressed in the acquisition, analysis and display of spatial data using Geographic Information Systems software. Topics include the mapping of race and ethnicity, the spatial distribution of cultural variables and human modification and use of the landscape.

Prerequisites: ANTH 195, ANTH 196, ANTH 197, SOCI 110 or SOCI 140; or by permission of the instructor.

SOCI 480 Senior Seminar in Sociology**Credits 3 Grading Basis** Graded

The integration of previous class experience the student has had during the undergraduate years. Fulfills the College of Arts and Letters writing-intensive requirement for the major. Can be repeated as course content changes.

Prerequisites: SOCI 300.

SOCI 485 Sociology Course Assistantship**Credits 1-3 Grading Basis** Graded

Assistantships provide students with a sense of what it is like to teach a sociology course by allowing them to work closely with faculty members through different phases of course preparation, presentation and evaluation. Assistantships also allow for a deeper understanding of course material by providing opportunities for student assistants to lead discussion and to help their peers review the material outside of the classroom. Prerequisites: Students must have junior /senior standing, must have earned a grade of B or better in the course for which he/she will serve as assistant, and may register by faculty invitation only. May be repeated up to six credits; only three credits can count toward the major. A student may only serve as a course assistant to the same course twice. Prerequisites: Junior or senior standing only.

SOCI 490 Independent Study in Sociology**Credits 1-3 Grading Basis** Graded

Designed to give capable students in sociology an opportunity to complete independent study under supervision. More than one repeat requires department head approval.

SOCI 492 Internship in Sociology**Credits 1-3 Grading Basis** Graded

Provides the student with practical experience in employing and refining sociological skills in a public or private agency under faculty supervision. May be repeated up to six credits.

SOCI 499A Honors**Credits 1 Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

Prerequisites: Declared sociology major.

SOCI 499B Honors**Credits 3 Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

Prerequisites: Declared sociology major.

SOCI 499C Honors**Credits 1-2 Grading Basis** Graded

An independent research topic initiated and completed by qualified senior majors who want to graduate with distinction.

Prerequisites: Declared sociology major.

Spanish (SPAN)

SPAN 101 Elementary Spanish I**Credits 3 Grading Basis** Graded

The fundamentals of Spanish through listening, speaking, reading and writing. Practice in pronunciation and development of comprehension.

SPAN 102 Elementary Spanish II**Credits 3 Grading Basis** Graded

This is the second course in the progress of learning the fundamentals of Spanish through listening, speaking, reading and writing. Practice in pronunciation and development of comprehension.

Prerequisites: SPAN 101 or per placement exam score.

SPAN 103 Passport to Spanish: A Survival Kit**Credits 3 Grading Basis** Graded

This course teaches students basic communication skills and cultural competence needed for studying, traveling or working in a Spanish-speaking country. Students will practice typical communicative contexts using authentic materials and gain insight into the real-life situations of Spanish speakers. This course has no prerequisite and is designed for those who have had little or no previous experience with Spanish. It does not count towards the Spanish major or minor or fulfill the B.A. language requirement. Instruction is in Spanish and English.

SPAN 111 Intensive Spanish I**Credits 6 Grading Basis** Graded

The fundamentals of Spanish through listening, speaking, reading and writing. The course is the equivalent of SPAN 101-SPAN 102.

SPAN 212 Intensive Spanish II**Credits 6 Grading Basis** Graded

The fundamentals of Spanish through listening, speaking, reading and writing. The course is the equivalent of SPAN 231-SPAN 232.
Prerequisites: SPAN 102 or SPAN 111 or per placement exam score.

SPAN 231 Intermediate Spanish I**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.
Prerequisites: SPAN 102 or SPAN 111, or per placement exam score. A student cannot earn credit for SPAN 231 after completing SPAN 232.

SPAN 232 Intermediate Spanish II**Credits 3 Grading Basis** Graded

A thorough review of grammar, vocabulary building, conversation, composition and reading.
Prerequisites: SPAN 231 or per placement exam score.

SPAN 300 Language for Living**Credits 3 Grading Basis** Graded

This advanced communication and culture course is designed to bridge basic language study with upper-level coursework in the Spanish language and culture. It is intended for students who have begun to develop advanced skills in interpretive, interpersonal and presentational communication and are interested in delving deeper into the study of Spanish-speaking countries, their histories and their cultures. Instruction is in Spanish.
Prerequisites: SPAN 212 or SPAN 232, or per placement exam score.

SPAN 301 Pop Culture through Language**Credits 3 Grading Basis** Graded

This class examines popular culture in Spanish-speaking cultures in a variety of forms and media: history, politics, literature, sport, music, cinema, fashion, comics and media. By focusing on pop culture and different media, this course teaches the critical analysis of textual and visual imagery in their historical and cultural context. Students will be able to learn the main pop culture references that are recognized nationally and internationally. Instruction is in Spanish.
Prerequisites: SPAN 212 or SPAN 232, or per placement exam score.

SPAN 306 Spanish for Social Work**Credits 3 Grading Basis** Graded

Cross-cultural competence in social work requires a keen understanding of language and culturally sensitive interactions. This course provides specialized language education to increase cultural sensitivity with diverse Spanish-speaking populations and their corresponding environments. Instruction is in Spanish.
Prerequisites: SPAN 212 or SPAN 232.

SPAN 307 History of Spanish Culture and Society**Credits 3 Grading Basis** Graded

This course examines the history of Spanish culture and society from the Romans until the present. It offers a review of historical and contemporary figures. Instruction is in Spanish.
Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 307S History of Spanish Culture and Society - Spain**Credits 3 Grading Basis** Graded

A history of Spanish culture and society from the Romans until nowadays. Review of historical and contemporary figures. Instruction is in Spanish.
Prerequisites: SPAN 300.

SPAN 308 Latin American Cultures**Credits 3 Grading Basis** Graded

This course studies the geographical, historical and cultural development of Latin America from pre-Columbian times to the present. Instruction is in Spanish.
Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 310 Introduction to Spanish Linguistics**Credits 3 Grading Basis** Graded

This course will introduce students to the fundamental components of linguistics (phonology, morphology, syntax, pragmatics and semantics) using data from the Spanish language. The topics and approach of this course consider the social and dialectal varieties of Spanish and the change of Spanish over time, including the present use of Spanish in the United States. The course requires no previous knowledge of linguistics, but presupposes familiarity with Spanish at the 300 credit level or higher.
Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 311 Contrastive Linguistics**Credits 3 Grading Basis** Graded

In this course students analyze the main grammatical differences between Spanish and English with the focus on producing accurate and idiomatic translations into both languages.
Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 312 Translation Competencies**Credits 3 Grading Basis** Graded

In this course, students will develop linguistic competencies required in translation, including reading comprehension, summary writing, text analysis and use of mono- and bilingual dictionaries. Students will learn some basic electronic tools and word processing skills for translators, and practice several types of translation, including direct translation, inverse translation and back translation. Instruction is in Spanish.
Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 315 Sounds of Speech**Credits 3 Grading Basis** Graded

This course introduces students to Spanish linguistics, with special emphasis on Spanish phonetics. Students will engage with theoretical concepts and practical applications to understand the phonological structures and sounds that make up the Spanish language. Class activities and assignments aim to enhance overall language proficiency, including listening, reading, speaking and pronunciation skills. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 320 Culture and Communication in Context**Credits 3 Grading Basis** Graded

This Spanish course focuses on building proficiency in the language with an emphasis on oral and written expression to facilitate further study of the culture. Students will learn how to use the appropriate language in culturally specific contexts and develop intercultural competence. The course's essential question is "How do Spanish-speaking peoples communicate on subjects that are at the heart of their societies?" Instruction is in Spanish.

Prerequisites: SPAN 212 or SPAN 232; or by placement exam or permission of the instructor. Corequisites: SPAN 212 or SPAN 232.

SPAN 321 Culture and Communication in Context (for Heritage Speakers)**Credits 3 Grading Basis** Graded

This course takes the place of SPAN 320 for the heritage speaker of Spanish (someone who has learned Spanish from family members and/or at home), with mostly oral (and not written) experience. It provides an opportunity for the student to expand their existing proficiency and to develop more formal language skills. Emphasis will be placed on reading, writing and presenting. Topics of current Latino issues will keep students interested and engaged.

Prerequisites: SPAN 212 or SPAN 232; or by placement exam or permission of the instructor. Corequisites: SPAN 212 or SPAN 232.

SPAN 322 Language Landscapes**Credits 3 Grading Basis** Graded

This Spanish course introduces students to the regions or countries where the language is spoken, the features of the area, and the ways of life there. The course examines the identities, customs and events that influence the lives of people living in these culturally and linguistically defined areas and explores the diverse cultural landscapes that shape their lives. Instruction is in Spanish.

Prerequisites: SPAN 212 or SPAN 232; or by placement exam or permission of the instructor. Corequisites: SPAN 212 or SPAN 232.

SPAN 330 Business Spanish**Credits 3 Grading Basis** Graded

This course presents basic business and commercial Spanish vocabulary and terminology. It focuses on trade customs and commercial communication, including interviews, letter writing and simultaneous interpretation. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 330S Business Spanish-Spain**Credits 3 Grading Basis** Graded

This course presents basic business and commercial Spanish vocabulary and terminology. It focuses on trade customs and commercial communication, including interviews, letter writing and simultaneous interpretation. Instruction is in Spanish.

Prerequisites: SPAN 300.

SPAN 335 A Journey through Literature**Credits 3 Grading Basis** Graded

This course is designed to prepare students in literary analysis of the novel as well as short stories, poetry and drama. All necessary terminology will be studied. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 335S A Journey through Literature - Spain**Credits 3 Grading Basis** Graded

This course is designed to prepare students in literary analysis of the novel as well as short stories, poetry and drama. All necessary terminology will be studied. This course is mandatory for all Spanish majors before taking any other literature class. Instruction is in Spanish.

Prerequisites: SPAN 300.

SPAN 360 Spanish for Public Safety**Credits 3 Grading Basis** Graded

This course presents basic Spanish legal vocabulary and terminology. It emphasizes the practical application of the Spanish language in routine and high-risk law enforcement situations and explores cultural issues that are important for law enforcement personnel. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 365 Spanish for Medical Professionals I**Cross-listed with:** SPAN 365S**Credits 3 Grading Basis** Graded

This course focuses on the basic Spanish language and vocabulary required in the fields of medicine and health sciences. Students practice their oral Spanish skills in various simulated medical situations in preparation for future professional application in the medical environment. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 365S Spanish for Medical Professionals I**Cross-listed with:** SPAN 365**Credits 3 Grading Basis** Graded

This course focuses on the basic Spanish language and vocabulary required in the fields of medicine and health sciences. Students practice their oral Spanish skills in various simulated medical situations in preparation for future professional application in the medical environment. Instruction is in Spanish.

Prerequisite: SPAN 300.

SPAN 385 Latin American Short Story**Credits 3 Grading Basis** Graded

In this course, students read and analyze representative short stories from Latin America. Students report on selected authors. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 385S Latin American Short Story - Spain**Credits 3 Grading Basis** Graded

Reading and analysis of representative short stories from Latin America. Student reports on selected authors. Instruction is in Spanish.

Prerequisites: SPAN 335.

SPAN 386 Spanish Short Stories**Credits 3 Grading Basis** Graded

In this course, students study literary Spanish short stories from the Middle Ages to the present, including a selection of texts by prominent authors such as Don Juan Manuel, Cervantes, Alarcón, Bécquer, Zorrilla and Clarín. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 390 Spanish Language and Culture Abroad**Credits 1-3 Grading Basis** Graded

This is an advanced communication and culture course designed to bridge basic language study with upper-level coursework in Spanish literature and culture. It is intended for students who have begun to master advanced skills in interpretive, interpersonal and presentational communication and are interested in examining nuances of the Spanish language and Spanish-speaking cultures while studying abroad.

Prerequisites: SPAN 212 or SPAN 232, or by permission of the instructor.

SPAN 400 Advanced Conversation**Credits 3 Grading Basis** Graded

Discussions deal with topics of current interest. Instruction is in Spanish.

Prerequisites: SPAN 320 or SPAN 321.

SPAN 404 Spanish in the United States**Credits 3 Grading Basis** Graded

This course will study the present situation of Spanish in the United States, how Spanish has been shaped by social forces in the US and how Spanish, in turn, shapes these forces. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 407 Aspects of Contemporary Spanish Culture and Society**Credits 3 Grading Basis** Graded

This course will study the history, culture and society of Spain during the 20th and 21st centuries. Emphasis will be placed on the changes in Spain after Franco. Instruction in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or by placement exam or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 409 Latinx Music Cultures**Credits 3 Grading Basis** Graded

This course explores the heterogeneity of Latina/o/e/x identities and cultures through music from the 20th and 21st centuries. Study of Latina/o/e/x music will be framed by cultural studies through concepts such as hybridity, transculturation, border identities, cultural imperialism and colonialism. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 420 Stories and Histories: The Generation of 1898**Credits 3 Grading Basis** Graded

This course examines major Spanish writers who engaged with their nation's defeat in the 1898 Spanish-American War. Students will read, analyze and discuss works from Azorín, Pío Baroja, Valle-Inclán, Antonio Machado, Maeztu and Unamuno. This group of intellectuals varied in their writing styles but shared a broad commitment to the cultural and aesthetic renewal of Spain. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 421 Stories and Histories: Latin American Novel**Credits 3 Grading Basis** Graded

In this course, students will read and analyze representatives of Latin American novelists of the 20th century. Students report on selected authors. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 422 Voices of Today: Wounds and Words in Post-War Spain**Credits 3 Grading Basis** Graded

This course provides an exploration of key artistic and literary works in Spain during the 20th century. Emphasis is placed on the intersection of Spanish history, society and culture, focusing on the Post War period, Franco's dictatorship (1939-75) and the role of historical memory in shaping both the artistic expressions and the collective consciousness of Spain. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 423 Voices of Today: Latin American Poetry**Credits 3 Grading Basis** Graded

In this course, students will read and analyze representatives of Latin American poets. Students will report on selected authors. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 425 Transformative Texts**Credits 3 Grading Basis** Graded

The purpose of this course is to acquaint students with historical periods in Spanish literature and culture through the study of key influential texts. Analyses of these texts (i.e. essays, novels, plays, films, paintings and comics) will be grounded in key political, scientific and artistic movements. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 426 Voices of the Past: Works of the Golden Age**Credits 3 Grading Basis** Graded

This course offers a study of literary genres and their development through the Spanish Golden Age, culminating in Cervantes. This may include works by Cervantes or Fernando de Rojas. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 322 or SPAN 335.

SPAN 430 Advanced Business Spanish**Credits 3 Grading Basis** Graded

This course presents advanced business and commercial Spanish vocabulary and terminology. By focusing on topics such as the origin of business and companies, company constitution and organization, human resources, labor law and commercial law, marketing, products, services, and the sale of consumer goods, students compare the business environment of Latin America with that of the United States. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 435 Translation Strategies**Credits 3 Grading Basis** Graded

This course discusses aspects of translation theory such as skopos, translation loss, translation gain and language bias. Students learn several translation strategies at the phrase and sentence levels and practice these strategies thoroughly before investigating terminology, terminology mining and terminology management. Instruction is in English and Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 436 Introduction to Interpretation**Credits 3 Grading Basis** Graded

This course presents an overview of the interpreting profession. It introduces students to the main techniques they need to become competent interpreters in the various sectors of interpreting. Instruction is given in English and Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 437 Introduction to Medical Interpretation**Credits 3 Grading Basis** Graded

The course fulfills the program content standards, instructional methods standards and programmatic standards recommended by the National Council on Interpreting in Health Care's (NCIH) National Standards for Healthcare Interpreter Training Programs. The curriculum is designed to meet the examination requirements of both the NCIH and the Certification Commission for Healthcare Interpreters (CCHI) which administers a national certification program for all healthcare interpreters. The areas covered in the course include: basic concepts of medical interpreting; the Code of Ethics and National Standards; best practices that support the Code of Ethics and National Standards; understanding the role of culture and cultural competence in interpreting; identifying appropriate situations for advocacy and "cultural brokering;" building interpreter skills and knowledge (positioning, note-taking, memory use, etc.); and practicing the modes of interpretation: consecutive, simultaneous and sight translation/interpreting. The focus of the course is to build proficiency in the three modes of interpreting and to learn and apply the ethical standards of the profession in healthcare interpreted-mediated settings. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 438 Introduction to Legal Interpretation**Credits 3 Grading Basis** Graded

The course will provide students with knowledge of the US legal system and the interpreting process. The areas covered include knowledge of the structure of the courts (federal and state systems), stages of the legal process (e.g., pre-trial proceedings, trials and post-trial proceedings), ethical principles and standards of professional conduct of interpreters, and the ability to apply this knowledge in practical case scenarios.

Students will develop basic skills in the three modes of interpreting: simultaneous, consecutive and sight interpreting used in legal settings. Students will also learn basic note-taking skills for the enhancement of memory recall during consecutive interpretation. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 446 Special Topics in Spanish Literature**Credits 3 Grading Basis** Graded

This course offers a study of a particular topic in Spanish literature. It may cover all or a specific Spanish literature genre. This course may be repeated if the content varies. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 447 Special Topics in Spanish-Speaking Cultures**Credits 3 Grading Basis** Graded

Students will study a particular topic in the culture of Hispanic countries. This course may be repeated if the content varies. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 448 Special Topics in Spanish Linguistics**Credits 3 Grading Basis** Graded

Students will study a particular topic in Spanish linguistics. Topics could include an introduction to Spanish sociolinguistics and psycholinguistics. This course may be repeated if the content varies. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 455 Women in Hispanic Literatures**Credits 3 Grading Basis** Graded

This course studies women in literature in the Hispanic world. The focus is on women authors, female characters in literature or both. The course may include works from Spain or Latin America from any time period. Students examine feminist literary criticism, canon formation and other critical topics. Emphasis may vary according to the instructor. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 464 Spanish Theater**Credits 3 Grading Basis** Graded

This course focuses on refining Spanish language skills to facilitate further study of Spanish language and culture. To this end, the course will focus on studying of plays from various centuries and staging and performing selected scenes. Students will engage in improvisations and games with classmates to study all aspects of theater. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 465 Spanish-Language Cinema**Credits 3 Grading Basis** Graded

This course studies the structure of the cinema and vocabulary related to film. The course will cover topics in Spain and Latin America. Instruction in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 475 Spanish for Medical Professionals II**Credits 3 Grading Basis** Graded

This course provides future medical professionals with further practice in Spanish in the medical context. Students learn advanced medical vocabulary and anatomical terminology, develop their reading comprehension skills, and acquire greater fluency through student presentations and classroom discussions on the latest medical techniques and advances. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 476 Culture and Medicine in Latin America**Credits 3 Grading Basis** Graded

This course enables students to acquire greater linguistic proficiency and cultural competence in the medical context through classroom discussions and reading comprehension. Students analyze how different kinds of inequalities (e.g., gender, racial, economic, etc.) affect health care in Latin American countries; examine popular and religious beliefs applied to medicine in Latin America; and research a medical challenge or success in a specific Latin American or Caribbean country. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 477 Culture and Medicine in Spain**Credits 3 Grading Basis** Graded

This course enables students to acquire greater linguistic proficiency and cultural competence in the medical context through classroom discussions and reading comprehension. Students analyze historical and contemporary medical discoveries and discuss medical protocols and practices in the Spanish medical system. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 478 Spanish in Healthcare: Policy and Practice in Latinx Health**Credits 3 Grading Basis** Graded

In this course students learn about the health and language policies of the US for Spanish-speaking patients. Some of the topics that will be discussed are how language affects health, health policy in the United States, language concordance in healthcare, and the role of medical interpreters for the Latinx community health. Students will work in groups for a final project to develop an academic poster. Instruction is in Spanish and English.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 485 Business and Society in Latin America**Credits 3 Grading Basis** Graded

This course explores the development of Latin American society in the business and economic contexts and investigates areas such as importation and exportation, the health and education systems, banking and financial institutions, agriculture, and the textile, fashion, wine, motion picture, music, and media industries. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 486 Business and Society in Spain**Credits 3 Grading Basis** Graded

This course investigates several sectors of the Spanish economy, including the food industry, industrial port activity, importation and exportation, the petrochemical industry, the leisure industry, and the wine making industry. Students tour several important multinational companies in Salamanca, Spain, and attend presentations given in Spanish by staff of those companies. This course is taught in Salamanca, Spain. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 490 Independent Study in Spanish**Credits 3 Grading Basis** Graded

This course allows students an opportunity to complete independent studies under faculty supervision. Instruction is in Spanish.

Prerequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322; or permission of the instructor. Corequisites: SPAN 300, SPAN 301, SPAN 320, SPAN 321 or SPAN 322.

SPAN 490T Special Studies in Spanish Art - Spain**Credits 3 Grading Basis** Graded

A study of the art and architecture of Spain from medieval times through the present. Concentration on specific artists, as well as general movements in the history of Spanish art.

SPAN 493 Business Spanish Internship**Credits 3 Grading Basis** Graded

This course gives students the opportunity to practice the theoretical knowledge, linguistic skills and cross-cultural competencies learned in SPAN 330. With permission of the business Spanish minor coordinator, students work as volunteers or interns in the public service, private industry or business management sector, locally or abroad. Faculty-supervised course work required for completion: monthly reports (3), final paper (reflections/lessons learned) and presentation, and performance evaluation by agency or organization.

Prerequisites: SPAN 330.

SPAN 496 Spanish Internship Abroad**Credits 3 Grading Basis** Graded

This course will provide students with experiential learning opportunities within the Hispanic field while participating in a study abroad program.

The internship course is designed to provide practical experience in fields such as government, business, industry, health sciences, nursing, administration, education, translation, non-profit or the arts, where they can apply their language skills. Examples of internship work may include working for a company in the student's field of interest abroad, completing a translation project or working as an interpreter. A journal, internship report, research paper, reflection or evaluation from the internship provider is required. Credits earned correspond to the internship hours completed. This course may be applied as an elective in the Spanish major or minor.

Prerequisites: One SPAN course at the 300-level or above and permission of the instructor.

SPAN 497 Spanish Internship**Credits 1-3 Grading Basis** Graded

This course will provide students with experiential learning opportunities within the Hispanic field. The internship course is designed to provide practical experience in fields such as government, business, industry, health sciences, administration, education, translation or the arts, where they can apply their language skills. Examples of internship work may include working for a company in the student's field of interest abroad; serving as a research assistant for a professor; completing a translation project or working as an interpreter. A journal, internship report, research paper, bibliography or evaluation from the internship provider is required. Credits earned correspond to the internship hours completed. Up to six credits may be applied as electives in the Spanish major or minor.

Prerequisites: SPAN 300.

SPAN 499A Honors**Credits 1 Grading Basis** Graded**SPAN 499B** Senior Honors Project**Credits 3 Grading Basis** Graded**SPAN 499C** Senior Honors Project**Credits 2 Grading Basis** Graded

Sport and Recreation Management (SRM)

SRM 202 Foundations of Leadership in Sport and Recreation Management**Credits 3 Grading Basis** Graded

An introduction to the basis for the professions that make up Sport and Recreation Management. A focus on these professions in governmental, voluntary, private and commercial settings is incorporated. Finally both the economical significance and the professional preparation for success in the industry are introduced and practiced.

Prerequisites: SRM major or permission of the director.

SRM 221 Special Event Management: Le Gourmet**Cross-listed with:** HM 221**Credits 3 Grading Basis** Graded

This course is designed to provide students with a practical hands-on application of event planning for non-profit fundraising. In this course students will actively plan and execute the Hart School's major annual fundraising event, Le Gourmet.

SRM 240 Sport and Recreation Management Seminar**Credits 3 Grading Basis** Graded

This course is for students to learn the business skills required for success in the sport and recreation management industry. Students will evaluate and analyze multiple critical thinking approaches and conflict management and negotiation skills, as well as customer service models, entrepreneurial theories and practices, and concepts within the industries of sport and recreation.

Prerequisites: SRM majors only.

SRM 241 Introduction to Sport and Recreation Management**Credits 3 Grading Basis** Graded

Introduces the sport and recreation management professions in governmental, voluntary, private, public, and commercial settings. Outlines development of sport and recreation and the evolution of the mega-leisure industry. Overviews professional preparation in sport and recreation management.

Prerequisites: SRM majors only.

SRM 242 Sociology and Psychology of Sport and Recreation**Credits 3 Grading Basis** Graded

The primary purpose of this course is to investigate sport and recreation related activities and services from a sociological and psychological perspective. The focus will be on activity through the lifespan and using theory and current issues from both disciplines to aid the practitioner in their interactions with participants and constituents and with the development and management of sport and recreation related activities and services.

Prerequisites: Declared SRM major or disabilities studies minor.

SRM 282 Practicum in Sport and Recreation**Credits 3 Grading Basis** Credit / No Credit

A sequence of selected practicum experiences which provides the student with supervised practicum experience in Sport and Recreation Management.

Prerequisites: SRM 241.

SRM 298 Special Studies in Sport and Recreation Management**Credits 3 Grading Basis** Graded

A special studies course designed to explore areas of current topical concern in sport and recreation environments.

Prerequisites: SRM major and permission of the director.

SRM 330 History of Sport and Recreation**Credits 3 Grading Basis** Graded

This course covers the development of sports and recreational activities and their significance in American life from colonial times to the present. It is designed both to supplement students' understanding of American history and to examine the role that sports and recreational activities have played in shaping contemporary society.

SRM 333 Management in Sport and Recreation**Credits 3 Grading Basis** Graded

This course will provide students with the knowledge to apply the management principles and theories to specific professional organizations in the sport and recreation industry. Sport and recreation management applications covered include administration principles for specific organizations, human resource management, fiscal management, marketing, and risk management.

Prerequisites: SRM 241.

SRM 334 Introduction to Sport Media**Credits 3 Grading Basis** Graded

Examination of the knowledge and skills required for the business of sports communications, including strategic and personal communications, leadership, publishing, advertising, public relation and crisis management. The course also examines sport communications from a sociological and legal perspective and the emergence of online sport communication and the new sport media.

Prerequisites: SRM 241.

SRM 337 Programming and Assessment in Sport and Recreation Management**Credits 3 Grading Basis** Graded

This course will examine the basics of sport and recreation programming and assessment. Students will be presented with tools and strategies for developing and assessing programs and evaluating their outcomes. Students will learn how to program for agencies of various sizes with an understanding of the importance of recognizing service population needs.

Prerequisites: SRM 241.

SRM 434 Ethical and Legal Issues in Sport and Recreation Management**Credits 3 Grading Basis** Graded

This course is designed to introduce students to current ethical and legal issues of concern to professionals in sport, recreation and leisure studies. Students will examine the impact of these issues on organizational and managerial policies and decision-making.

Prerequisites: SRM 333.

SRM 435 Sport Marketing and Sales**Credits 3 Grading Basis** Graded

This course will examine how promotional activities and sales efforts are closely intertwined and impact the success or failure of the sport and leisure industry. Particular emphasis will be placed on ticket sales and sport sponsorship.

Prerequisites: SRM 333.

SRM 436 Facilities and Event Management**Credits 3 Grading Basis** Graded

This course is designed to explore the principles of planning, design, and management of selected sport, recreation, and exercise facilities. The course will also cover the planning and management of special events. This will include budgeting, design, staffing, evaluation/assessment, crowd management and relative human resource management.

Prerequisites: SRM 333.

SRM 438 Human Resources in Sport and Recreation Management**Credits 3 Grading Basis** Graded

An overview of human resource management and development in sport and recreation environments. This course will provide an introduction to administrative practices involving employer and employee relationships and an application of these concepts to day to day management duties. Course focus will be policies, legal issues, recruiting, hiring, job analysis, creating position descriptions, evaluation, training, development and supervision of staff members and volunteers.

Prerequisites: SRM 333.

SRM 482 Internship in Sport and Recreation Management**Credits 6 Grading Basis** Credit / No Credit

This full-time professional experience affords the opportunity to apply theory and methodology under qualified supervision from the cooperating agency and the university. While enrolled in this course, students may take up to one additional course without permission and a second course with permission. The second additional course must be approved in advance by the SRM internship coordinator. All internship sites must be approved by the SRM practicum and internship coordinator.

Prerequisites: SRM 282, SRM 333 and 72 credit hours completed.

SRM 490 Special Studies in Sport and Recreation Management**Credits 1-3 Grading Basis** Graded

Designed to give capable students in Sport and Recreation Management an opportunity to complete independent study under faculty supervision.

Prerequisites: SRM major and permission of the director.

SRM 498 Special Topics in Sport and Recreation Management**Credits 3 Grading Basis** Graded

This course is designed to allow explorations of areas of current topical concern, or to exploit special situations. Course content will vary. For current course content consult your advisor or the SHSRM Director.

Prerequisites: SRM major and permission of the director.

SRM 499A Honors**Credits 1 Grading Basis** Graded

See catalog description entitled "Graduation with Distinction" and "Graduation with Honors."

SRM 499B Honors**Credits 3 Grading Basis** Graded

See catalog description entitled "Graduation with Distinction" and "Graduation with Honors."

SRM 499C Honors**Credits 2 Grading Basis** Graded

See catalog description entitled "Graduation with Distinction" and "Graduation with Honors."

Swahili (SWA)

SWA 101 Elementary Swahili I

Credits 3-4 Grading Basis Graded

The fundamentals of Swahili through listening, speaking, reading and writing. Practice in pronunciation and development of comprehension. One hour's work a week in language laboratory. Students will receive no credit for the course if they have had two or more years of the language in high school.

SWA 102 Elementary Swahili II

Credits 3-4 Grading Basis Graded

This course builds on the foundations established in SWA 101 through listening, speaking, reading and writing activities. This course provides practice in pronunciation and development of comprehension. There is one hour a week of work in the language laboratory. If student has had two or more years of the language in high school they will not receive credit for the course.

SWA 111 Intensive Swahili

Credits 6 Grading Basis Graded

The fundamentals of Swahili through listening, speaking, reading and writing. The course is the equivalent of SWA 101-SWA 102.

SWA 212 Intensive Swahili II

Credits 6 Grading Basis Graded

The fundamentals of Swahili through listening, speaking, reading and writing. The course is the equivalent of SWA 231-SWA 232. Prerequisites: SWA 102 or SWA 111, or by permission of the instructor.

SWA 231 Intermediate Swahili I

Credits 3 Grading Basis Graded

A thorough review of first year grammar and vocabulary building. Conversation, composition and readings will be chosen to reach competency at the lower intermediate level Swahili. Prerequisites: SWA 102 or SWA 111, or by permission of the instructor.

SWA 232 Intermediate Swahili II

Credits 3 Grading Basis Graded

A thorough review of Swahili grammar and vocabulary building. Conversation, composition and readings will be chosen to reach competency at the advance intermediate level. Prerequisites: SWA 231 or permission of the instructor.

SWA 300 Grammar and Communication

Credits 3 Grading Basis Graded

Intensive training in grammatical structures and their application to oral and written communication. Instruction is in Swahili. Fulfills the College of Arts and Letters writing-intensive requirement for the major. Prerequisites: SWA 232 or SWA 212, or by permission of the instructor.

SWA 320 Swahili Oral and Written Communication

Credits 3 Grading Basis Graded

Intensive training in the use of modern, everyday Swahili with emphasis on conversation and composition. Readings in Swahili will provide a context for discussion and writing. Prerequisites: SWA 300.

SWA 490 Special Studies in Swahili

Credits 1-4 Grading Basis Graded

Special topics or independent studies in Swahili.

Teaching English to Speakers of Other Languages (TESL)

TESL 300 Introduction to Second Language Acquisition

Credits 3 Grading Basis Graded

This course provides an overview of first and second language acquisition through the lens of education. Students are guided through an in-depth inquiry process designed to develop researcher-practitioners, an approach that is particularly critical to the promotion of educational justice.

TESL 310 Pedagogical Grammar

Credits 3 Grading Basis Graded

This course develops skills in teaching grammar and syntax in the context of writing. Special focus will be on working with English language learners and the ability to model and teach editing skills and organization of writing using predominant text structures in the content areas.

TESL 381 Practicum in TESOL Instruction

Credits 3 Grading Basis Graded

The course provides supervised field experiences in working with multilingual learners at both elementary and secondary school settings. Preservice teachers will demonstrate competencies related to integrating language and content instruction, and differentiating according to multilingual learners' needs and language proficiency levels.

Prerequisites: TESL 300.

TESL 382 Practicum in Second Language Acquisition

Credits 1 Grading Basis Graded

This one credit practicum provides an opportunity for students to work directly with multilingual learners under the guidance of a university supervisor to identify, recognize and apply principles of English as a new language in educational contexts. Content of the practicum will directly align with topics covered in TESL 300.

Corequisites: TESL 300.

TESL 383 Practicum in TESOL Literacy Development

Credits 1 Grading Basis Graded

This one credit practicum provides an opportunity for students to work directly with English language learners to plan and implement oral language, reading, and writing strategies in language arts and content instruction that are directly related to topics covered in the LED 435 course.

Prerequisites: LED 366. Corequisites: LED 435.

TESL 384 Practicum in Literacy Development

Credits 1 Grading Basis Graded

This one credit practicum provides for pre-service teachers to work directly with students in grades PK-12 (as determined by the LED 366 instructor) on literacy development under the supervision of a mentor teacher and a University Supervisor. They plan and implement literacy strategies in language arts and content instruction directly related to LED 366 course material. Most pre-service teachers will assist Prek-5 students.

Corequisites: LED 366.

TESL 385 Practicum in TESOL

Credits 1-3 Grading Basis Graded

This practicum provides an opportunity for students to apply instructional methods and practices for teaching English to speakers of other languages (TESOL) to meet the linguistic needs of multilingual learners under the supervision of a university supervisor. Content and site locations will vary depending upon the specific focus of the practicum.

TESL 421 Foundations of ESL PK-12 Collaboration**Credits 3 Grading Basis** Graded

This course is a detailed study of ESL co-teaching and other collaborative practices. The course explores the rationale for teacher collaboration to support ESL instruction; presents current, classroom-based, practitioner-oriented research studies related to co-teaching, co-planning, curriculum alignment, teacher professional development and additional collaborative practices; and offers authentic teacher reflections and recommendations on collaboration and co-teaching.

Prerequisites: TESOL majors only.

TESL 425 Cross-Cultural Education**Credits 3 Grading Basis** Graded

The course provides students with knowledge of the effects of socio-cultural variables in an instructional setting.

TESL 428 Language Assessment for Multilingual Learners**Credits 3 Grading Basis** Graded

This course introduces students to a variety of formative and summative assessment practices for multilingual learners. Students will become familiar with equitable assessment practices that aim to minimize cultural and linguistic biases and promote asset-based approaches.

Prerequisites: TESL 300.

TESL 470 Methods in Teaching English to Speakers of Other Languages**Cross-listed with:** TESL 570**Credits 3 Grading Basis** Graded

This course introduces students to instructional methods and practices for teaching English to speakers of other languages (TESOL) to meet the linguistic needs of multilingual learners. Students will develop competences in selecting culturally responsive materials, integrating language and content instruction, and differentiating according to multilingual learners' language proficiency levels.

Prerequisites: TESL 300. Corequisites: TESOL minor: TESL 385; TESOL major: TESL 381.

TESL 480 Student Teaching/Internship**Credits 5 Grading Basis** Credit / No Credit

Enables students to apply skills and attitudes acquired in all components of their professional education preparation. Students design and implement instructional activities under the guidance of teaching professionals.

Prerequisites: PSYC 160, EDUC 310, EDUC 200, pre-professional studies and TESOL core courses. Corequisites: TESL 481.

TESL 481 Student Teaching Seminar in TESOL**Credits 1 Grading Basis** Graded

This seminar is designed to accompany the student teaching experiences. Its purpose is to provide opportunities for student teachers to process and reflect on their experiences that occur in the classroom and school and to strengthen their professional growth. Broad topic areas include understanding of different instructional practices and implementation of a variety of lesson plans with appropriate behavior management and assessment strategies to meet the needs of English learners. Emphasis is placed on guiding student teachers from theory to the world of the professional educator by exploring their own skill levels and plans for their ongoing professional development.

Prerequisites: TESOL majors only. Corequisites: TESL 480.

TESL 482 Professional Development, Partnership and Advocacy Seminar**Credits 1 Grading Basis** Graded

Students examine opportunities for professional development from professional associations, universities and other organizations across PK-16. Strategies to build partnerships with colleagues, families and communities are presented. Important social and political issues affecting education of majority and minority students and models of advocacy for students and their families are presented.

Prerequisites: TESL 300.

TESL 490 Special Studies in TESOL**Credits 1-3 Grading Basis** Graded

The course is designed to allow the student to complete an independent study in a TESOL-related area under faculty supervision.

Theatre (THEA)

THEA 204 Theatre Practicum: Scenery [Block]**Credits 1 Grading Basis** Graded

This practicum in scenery provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance in a block-schedule format. Students will work in the scene shop and build skills in power tool usage, theatrical painting techniques and sourcing materials. Upon completion, students will be able to participate in the functions of a professional-level scenic and/or properties shop.

THEA 205 Theatre Practicum: Lights [Block]**Credits 1 Grading Basis** Graded

This practicum in lighting provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance in a block-schedule format. Students will work in the electrics shop and build skills in hanging and focusing lights, striking, cabling, and maintenance. Upon completion, students will be able to participate in the functions of a professional-level electrics shop.

THEA 206 Theatre Practicum: Costumes [Block]**Credits 1 Grading Basis** Graded

This practicum in costumes provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance in a block-schedule format. Students will work in the costume shop and build skills in alterations, costume crafts, costume stock and show organization. Upon completion, students will be able to participate in the functions of a professional-level costume shop.

THEA 207 Theatre Practicum: Management [Block]**Credits 1 Grading Basis** Graded

This practicum in management provides 35 hours of direct, hands-on experience working on event management and arts administration in the School of Theatre and Dance in a block-schedule format. Students will work with faculty advisors and build skills in stage management, company management, house management and administration. Upon completion, students will be able to participate in the office and management functions of a professional arts organization.

THEA 208 Theatre Practicum: Performance [Block]**Credits 1 Grading Basis** Graded

This practicum in performance provides students direct, hands-on experience working as an actor or performer in the School of Theatre and Dance in a block-schedule format. Eligibility requires having been cast in a faculty or guest directed production. Upon completion, students will understand the expectations and standards required of professional actors and performers.

THEA 210 Introduction to Theatre [C2VPA]**Credits 3 Grading Basis** Graded

Study of the theatre as an art form. Emphasis on introducing students to a broad spectrum of theatrical activity and opinion. Consideration of the components that comprise a theatre event including acting, directing, design, costuming, lighting and playwriting. May be used for general education credit. May not be used for major credit.

THEA 211 Performance Analysis**Credits 3 Grading Basis** Graded

Plays are examined as texts for performance. Theories of performance and methods for the analysis of performances in and out of the theater are studied.

Prerequisites: Theatre and musical theatre majors and theatre minors, American studies minors and creative writing minors only.

THEA 212 Theatre Practicum: Sound [Block]**Credits 1 Grading Basis** Graded

This practicum in sound provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance in a block-schedule format. Students will work in the sound shop and learn proper techniques for setting up, testing and operating equipment for live sound and recording. Upon completion, students will be able to participate in the functions of a professional-level sound shop.

THEA 251 Acting I: Basic Acting**Credits 3 Grading Basis** Graded

A study of basic acting as a performance experience. Emphasis on fundamentals of performance including concentration, transitions, interaction and the structuring of action.

THEA 253 Musical Theatre Laboratory**Credits 1 Grading Basis** Graded

Introduction to issues related to Musical Theatre through music and text analysis, vocal technique, and practice of the repertoire. Topics relating to vocal and physical health, professional practice, and musical theatre resources will also be covered. May be repeated once for credit.

Prerequisites: Musical theatre majors only.

THEA 261 Voice for the Stage**Credits 3 Grading Basis** Graded

Theories and exercises designed to equip the student with proper speech for the stage through a fundamental understanding of phonetics and articulation.

Prerequisites: THEA 251.

THEA 271 Technical Theatre**Credits 3 Grading Basis** Graded

Study of the technical aspects of stage production. Emphasis upon practical experience in the use of stage and shop facilities. Consideration of the physical theater and stage, construction, painting and rigging of scenery as applied to theatrical production.

Prerequisites: STAD 171.

THEA 281 Movement for the Actor**Credits 3 Grading Basis** Graded

Students in this course develop and refine movement skills for stage performance. Through a study of approaches, such as Viewpoints, Animal Work, and Laban Techniques, students develop methods of artistic expression using their body. Through movement, students discover new connections to speech and text and improve the condition, flexibility, and responsiveness of their physical instrument. Students also develop the means for making effective physical choices in the creation of a performance.

Prerequisites: THEA 251.

THEA 295 Studio Production**Credits 1 Grading Basis** Graded

This course is reserved for students working on student productions through the experimental theatre program in the studio theatre, which may include positions such as cast, crew, dramaturgs, stage managers, designers and assistant designers.

THEA 296 Studio Production for Performance Concentrators**Credits 1 Grading Basis** Graded

This course is reserved for performance concentrators working on a studio production in a non-performance capacity, including as an assistant designer, designer, assistant technical director, electrician, head carpenter, assistant stage manager, stage manager or dramaturg, to expand student understanding of, and proficiency in, multiple areas of theatre practice. Actors, directors and playwrights are not eligible for this course.

Prerequisites: Theatre major and performance concentration.

THEA 303 Topics in Theatre**Credits 1-3 Grading Basis** Graded

Study of current topics and issues in theatre. Emphasis on contemporary themes of immediate concern.

THEA 304 Theatre Practicum: Scenery**Credits 1 Grading Basis** Graded

This practicum in scenery provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance. Students will work in the scene shop and build skills in power tool usage, theatrical painting techniques and sourcing materials. Upon completion, students will be able to participate in the functions of a professional-level scenic and/or properties shop.

THEA 305 Theatre Practicum: Lights**Credits 1 Grading Basis** Graded

This practicum in lighting provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance. Students will work in the electrics shop and build skills in hanging and focusing lights, striking, cabling, and maintenance. Upon completion, students will be able to participate in the functions of a professional-level electrics shop.

THEA 306 Theatre Practicum: Costumes**Credits 1 Grading Basis** Graded

This practicum in costumes provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance. Students will work in the costume shop and build skills in alterations, costume crafts, costume stock and show organization. Upon completion, students will be able to participate in the functions of a professional-level costume shop.

THEA 307 Theatre Practicum: Management**Credits 1 Grading Basis** Graded

This practicum in management provides 35 hours of direct, hands-on experience working on event management and arts administration in the School of Theatre and Dance. Students will work with faculty advisors and build skills in stage management, company management, house management and administration. Upon completion, students will be able to participate in the office and management functions of a professional arts organization.

THEA 308 Theatre Practicum: Performance**Credits 1 Grading Basis** Graded

This practicum in performance provides students direct, hands-on experience working as an actor or performer in the School of Theatre and Dance. Eligibility requires having been cast in a faculty or guest directed production. Upon completion, students will understand the expectations and standards required of professional actors and performers.

THEA 309 Theatre Education Practicum**Credits 1 Grading Basis** Credit / No Credit

This course provides 35 hours of direct, hands-on experience working as a teaching artist. Students will practice observation, teaching in individual and small groups, and lesson planning and facilitation in community settings. Upon completion, students will be able to contribute to work as an arts educator in non-school settings.

Prerequisites: THEA 319, THEA 323 and STAD 322.

THEA 312 Theatre Practicum: Sound**Credits 1 Grading Basis** Graded

This practicum in sound provides 35 hours of direct, hands-on experience working on a faculty or guest production in the School of Theatre and Dance. Students will work in the sound shop and learn proper techniques for setting up, testing and operating equipment for live sound and recording. Upon completion, students will be able to participate in the functions of a professional-level sound shop.

THEA 315 World Theatre Histories, Texts and Performances I**Credits 3 Grading Basis** Graded

Study of world theatre from its Origins to the Seventeenth Century. The class explores how theatre has been created and consumed in diverse settings around the globe, with particular emphasis given to dramatic texts and performance traditions, as well as critical, literary, and performance theories. Includes study of theatre architecture and production design; political, social, intellectual, and aesthetic contexts of performances, and the theory and history of acting across time and place.

Prerequisites: THEA 211.

THEA 316 World Theatre Histories, Texts and Performances II**Credits 3 Grading Basis** Graded

This course is a study of world theatre from the seventeenth to early twentieth century. The class explores how theatre has been created and consumed in diverse settings around the globe, with particular emphasis given to dramatic texts and performance traditions, as well as critical, literary and performance theories. Includes study of theatre architecture and production design; the political, social, intellectual and aesthetic contexts of performance; and the theory and history of acting across time and place.

Prerequisites: THEA 211.

THEA 317 World Theatre Histories, Texts and Performances III**Credits 3 Grading Basis** Graded

This course is a study of world theatre from the early twentieth century to the present. The course explores how theatre has been created and consumed in diverse settings around the globe, with particular emphasis given to dramatic texts and performance traditions, as well as critical, literary and performance theories. Includes study of theatre architecture and production design; the political, social, intellectual and aesthetic contexts of performances; and the theory and history of acting across time and place.

Prerequisites: THEA 211, THEA 315 and THEA 316.

THEA 319 Theatre Teaching Methods**Credits 3 Grading Basis** Graded

An exploration of teaching drama and theatre with children and youth in K-12 school settings. Topics include: discipline-based theatre education, drama-based pedagogy, arts integration, and youth theatre workshops/ programs and classrooms with a focus on community-building and social justice. This course helps students expand their skills set in lesson/ workshop planning and facilitation, behavior management, as well as theatre for youth development with K-12 students.

THEA 323 Youth Theatre**Credits 3 Grading Basis** Graded

An exploration of teaching artistry, devised theatre and applied drama/theatre with youth, directing the young performer and building educational programs in schools. With an emphasis on teaching, directing and arts administration, this course examines opportunities for learning through theatre-making and theatre performance.

Prerequisites: School of Theatre and Dance majors and theatre minors only and sophomore standing, or permission of the instructor.

THEA 324 Theatre for Young Audiences**Credits 3 Grading Basis** Graded

An exploration of professional theatre practices in Theatre for Young Audiences (TYA), including Theatre for the Very Young (TVY) and Theatre for Teens. With an emphasis on theatre-making, this course examines historical contexts of TYA, investigates theatre companies and practitioners and identifies applications for education, outreach and engagement.

Prerequisites: School of Theatre and Dance majors only and sophomore standing, or permission of the instructor.

THEA 325 Professional Issues in Education**Credits 3 Grading Basis** Graded

An advanced study of drama/theatre education focusing on professional practices, contemporary issues in education and pedagogy in practice. In addition to course work, students work closely with a cooperating teacher in a local school to observe and reflect, support instruction, and develop and facilitate lessons.

Prerequisites: THEA 319 or permission of the instructor.

THEA 330 Professional Issues for Designers and Technicians**Credits 3 Grading Basis** Graded

This course is designed to provide a working comprehension of the business dynamics involved in being a working artist: portfolio and interview preparation, how to be your own marketing director, accountant, agent, publicist, assistant and financial strategist.

Prerequisites: STAD 273 and one of THEA 271, THEA 332, THEA 333, THEA 336, THEA 374, THEA 376, or STAD 471; or permission of the instructor.

THEA 331 Technical Costuming**Credits 3 Grading Basis** Graded

Introduction to the technical aspects involved in building complete costume ensembles for the stage, screen and dance. Emphasis placed on experiencing unique problems in production situations, including specialized costume and accessory construction, ornamentation, fabric treatment, difficulties and alternatives in presenting historical clothing, and renovating and exploiting available materials.

Prerequisites: STAD 171 and THEA 206 or THEA 306, or permission of the instructor.

THEA 332 Costume and Fashion History**Credits 3 Grading Basis** Graded

A survey of Western world costume from 4000 B.C. to the present as it reflects the sociocultural and socioeconomic aspects of the times. Emphasis on the evaluation of historic costume and fashion in relation to architecture, sculpture, painting and other art forms.

THEA 333 Costume Design**Credits 3 Grading Basis** Graded

The study of basic costume design techniques for performance. Emphasis on costuming in terms of the total production concept including directorial approach, setting and lighting design. Consideration of the process of costuming a production from first production meetings to opening.

THEA 336 History, Theory and Practice of Stage Makeup**Credits 3 Grading Basis** Graded

Study of the history and practice of makeup for ceremonial and theatrical presentations of selected major cultures, from ancient Egypt to the present. Consideration given to makeup as a reflection of the social organization of each culture. Emphasis on makeup as an important element in the history of design aesthetics as well as a vital part of the performing arts.

THEA 347 Playwriting**Cross-listed with:** ENG 347**Credits 3 Grading Basis** Graded

Study of the process of writing plays. Consideration of plot, character, thematic material, conflict and dramatic structure. Emphasis on individual writing assignments.

Prerequisites: Sophomore standing or permission of the instructor.

THEA 351 Acting II: Intermediate Acting**Credits 3 Grading Basis** Graded

Study of the fundamental theories and methods of acting. Emphasis on laboratory experience in the preparation of scenes. Consideration of various acting techniques through performance with maximum individual on-stage instruction.

Prerequisites: THEA 251 and sophomore standing.

THEA 352 Music Theatre Performance for Theatre and Music Majors**Credits 2 Grading Basis** Graded

Introductory course exploring the fundamentals of song analysis technique. Emphasis on solo song preparation. Basic music skills and singing ability are highly recommended. This course is designed for actors and singers who are just beginning to develop their musical theatre repertoire and their capacity to integrate singing and acting techniques.

Prerequisites: THEA 251 and a theatre (School of Theatre and Dance) major or music theatre (School of Music) concentration, or permission of the instructor.

THEA 353 Musical Theatre Performance**Credits 3 Grading Basis** Graded

This introductory course explores the fundamentals of song analysis technique in the preparation of musical theatre repertory for performance. Emphasis is placed on solo song preparation. Basic music skills and singing ability are highly recommended.

Prerequisites: THEA 251 and THEA 253.

THEA 354 Music Theatre Workshop**Credits 1-4 Grading Basis** Graded

The preparation of musical theatre repertory for public performance. Emphasis on the application of the tools acquired in THEA 251, THEA 353, and other theatre, music, and dance skills classes. Students may participate as a performer, or as a member of the production/ artistic team.

THEA 355 Directing for the Theatre**Credits 3 Grading Basis** Graded

Study of the principles, problems and techniques of play direction. Emphasis on historical and modern theories. Techniques of direction considered as applied to the stage and cinematography.

Prerequisites: THEA 211 and THEA 251.

THEA 357 Music Theatre History and Analysis**Cross-listed with:** MUS 357**Credits 3 Grading Basis** Graded

Survey of musical theatre genres, composers, lyricists, performers, directors and choreographers in America from 1750 to present. Includes practical study of the format of the libretto and musical score in relationship to the major musical theatre genres. Consideration of how music theatre developed from and reflected the cultural, social and political landscape of its time.

Prerequisites: Musical Theatre majors only or permission of the instructor.

THEA 365 Dramaturgy**Credits 3 Grading Basis** Graded

Study of the skills and techniques of the dramaturg, including strategies for production dramaturgy and new work dramaturgy of both theatre and dance. Emphasis on communication, adaptation, research and analysis. Consideration of best practices for the dramaturg's collaborations with directors, choreographers, playwrights, performers, designers and audience members.

Prerequisites: THEA 211.

THEA 371 Advanced Technical Theatre**Credits 3 Grading Basis** Graded

Advanced study of the technical aspects of stage production. Emphasis upon contemporary scenographic techniques. Consideration of construction, decoration, rigging and touring problems in theatre production.

Prerequisites: THEA 271.

THEA 372 Scene Painting**Credits 3 Grading Basis** Graded

An introduction to the foundational techniques of scene painting, presented through a series of practical projects designed to provide students with the skills to achieve the core processes of scene painting.

THEA 373 Drafting for the Stage**Credits 3 Grading Basis** Graded

An introduction to the foundational techniques of drafting for the stage, presented through a series of 8 practical drafting projects designed to provide students the skills to produce ground plans, elevations and sections in both pencil and CAD drafting environments.

THEA 374 Stage Lighting**Credits 3 Grading Basis** Graded

Study and analysis of stage lighting. Consideration given to basic elements of electricity, electrical control and circuitry, reflection, refraction, and color. Emphasis on the lighting design and aesthetics of a theatrical production.

THEA 375 Sound Design**Credits 3 Grading Basis** Graded

Study and analysis of the aural environment for theatre. Emphasis placed on sound design process and the methods and tools available to sound designers. Consideration given to the various mediums of sound design through in class projects.

THEA 376 Scene Design**Credits 3 Grading Basis** Graded

Study and analysis of the visual elements of theatrical production. Emphasis on the principles and elements of design, drafting and rendering. Consideration of the aesthetics of stage design through specific applied projects.

THEA 382 Contemporary Theatre**Credits 3 Grading Basis** Graded

Study of post-1968 world theatre with particular attention to English and American drama. Emphasis is placed on significant drama of the last twenty-five years with discussion of trends in theatrical production and dramatic writing in recent decades.

Prerequisites: THEA 211.

THEA 390 Directed Projects**Credits 1-3 Grading Basis** Graded

Supervised projects related to the study of theater. Credit given for original, individual or group programs beyond the usual course offerings in theater. A suitable completed project or report is required before credit can be awarded.

THEA 395 Studio Directing, Design and Collaboration**Credits 1-3 Grading Basis** Graded

This course supports student-produced work in the studio theatre and student application of past coursework to the development of fully realized productions. This course is open to any non-cast member working on a student production through the experimental theatre program and is required for student directors, producers, playwrights, stage managers and lead designers. Students will learn advanced collaboration skills; skills necessary to initiate and plan creative processes; best practices for pre-production, stage management, and rehearsal; effective facilitation of production meetings; technical proficiencies; and the articulation of an individual artistic point of view. May be repeated for a maximum of nine credit hours.

THEA 396 Studio Directing, Design and Collaboration for Performance Concentrators**Credits 1-3 Grading Basis** Graded

This course is reserved for performance concentrators working on a studio production in a non-performance capacity, including as an assistant designer, designer, assistant technical director, electrician, head carpenter, assistant stage manager, stage manager or dramaturg, to expand student understanding of, and proficiency in, multiple areas of theatre practice. Actors, directors and playwrights are not eligible for this course. Students will learn advanced collaboration skills; skills to initiate and plan creative processes; best practices for pre-production, stage management, and rehearsal; effective facilitation of production meetings; technical proficiencies; and the articulation of an individual artistic point of view.

Prerequisites: Theatre major and performance concentration.

THEA 441 Professional Pathways**Credits 3 Grading Basis** Graded

This course provides a context for each student to identify and foster their personal and professional goals. The course is intentionally flexible to invite students to self-select how they want to engage the course's aims and content. Students address individual interests and explore topics of their choosing through a project each student conceives, designs and undertakes, ranging, for example, from project management to grant writing to starting a business as a freelance artist. Students also explore current trends and developments in the field.

Prerequisites: Theatre majors only and senior standing; second-semester juniors may enroll with instructor permission.

THEA 442 Senior Seminar in Theatre**Credits 1 Grading Basis** Graded

A single semester seminar for students in the honors program to be taken concurrently with THEA 499C. Students spend the semester completing their honors thesis project. The course may also include portfolio development, career preparation, practices in the professional theatre and other theatre-related topics of interest.

Prerequisites: Theatre and musical theatre majors only and senior standing. Corequisites: THEA 499.

THEA 447 Advanced Playwriting**Cross-listed with:** ENG 447**Credits 3 Grading Basis** Graded

An advanced workshop with emphasis on developing full-length dramatic material.

Prerequisites: THEA 347 or ENG 347 and junior standing.

THEA 449 London Theatre**Credits 3 Grading Basis** Graded

Study of London theater. Consideration given to current productions of classic and contemporary works. Emphasis on production elements including acting, directing, design, writing and economic considerations.

THEA 451 Meisner Technique Intensive**Credits 3 Grading Basis** Graded

Intensive study and practice of the Meisner Technique. Emphasis on the foundational steps of the technique to develop advanced listening, imaginative, and interactive skills as a means for fostering effective habits for creative freedom and improvisational spontaneity. Practice in fully doing, truthful personal investment, and honest availability to oneself and a partner within heightened circumstances.

Prerequisites: THEA 351.

THEA 452 Acting III: Contemporary Scene Study**Credits 3 Grading Basis** Graded

Study of advanced acting technique through the analysis, discussion, and presentation of contemporary scenes. Emphasis on developing tools to improve the pursuit of an intention, partnering, and moment-to-moment work.

Prerequisites: THEA 351 and junior standing.

THEA 453 Acting IV: Approach to Heightened Language**Credits 3 Grading Basis** Graded

Advanced study of script analysis and performance technique in multiple genres exploring heightened dramatic text. Exploration will be through individual and group assignments. Emphasis on practical tools including scansion, imagery work, and the translation of text into active physical choices.

Prerequisites: THEA 452.

THEA 454 Advanced Musical Theatre Performance**Credits 3 Grading Basis** Graded

This course is a continuation of THEA 353, emphasizing more complex problems in musical theatre performance including duets, trios and musical scenes.

Prerequisites: THEA 353.

THEA 455 Auditioning for Musical Theatre**Credits 3 Grading Basis** Graded

This is an advanced study of auditioning technique for professional musical theatre and understanding of contemporary trends in the field. Emphasis is placed on developing appropriate musical theatre audition repertory. Unions, audition formats, performance opportunities, routes to a professional career, and agents and managers will also be covered.

Prerequisites: Musical theatre major, THEA 353 and THEA 454.

THEA 460 Auditioning and Professional Issues**Credits 3 Grading Basis** Graded

Study of advanced audition techniques for the stage. Emphasis on the selection, scoring, and performance of monologues. Consideration of professional issues in regard to graduate-level study or professional work in performance.

Prerequisites: Theatre major performance concentrators and THEA 453.

THEA 466 Media Performance**Credits 3 Grading Basis** Graded

Study of the principles and techniques of dramatic performance for the camera and microphone. Emphasis on the adaptation of each student's performance and production skills to the recording and filming of narrative works. Consideration of the differing problems and potentials of stage, video and film presentations.

Prerequisites: THEA 452.

THEA 473 Advanced Design**Credits 3 Grading Basis** Graded

Advanced study in design for performance through theoretical production planning. Instruction in presentation methods necessary to communicate scenery, costume, sound and lighting designs for performance. Emphasis is placed on the collaborative interaction between the production designers and the director or choreographer towards the creation of unified design.

Prerequisites: STAD 273 and one of the following: THEA 333, THEA 374, THEA 375 or THEA 376, or permission of the instructor.

THEA 480 Student Teaching**Credits 3-12 Grading Basis** Credit / No Credit

Enables students to apply, in the public school classrooms and the comprehensive child development programs, those skills and attitudes acquired in all components of teacher education. Under the guidance of university supervisors, students are provided activities designed to familiarize them with the classroom teacher's role.

Prerequisites: PSYC 160 and EDUC 200.

THEA 481 Theory and Performance Studies**Credits 3 Grading Basis** Graded

Twentieth and twenty-first century theories of theater and performance are studied and performance traditions outside of Europe are examined.

Prerequisites: THEA 211.

THEA 485 American Theatre**Credits 3 Grading Basis** Graded

Study and analysis of the American theatre experience as presented in the dramatic literature of the country. Emphasis on basic American themes. Consideration of plays, playwrights and performers significant to the development of American theatre.

Prerequisites: THEA 211.

THEA 488 Alternative Performance**Credits 3 Grading Basis** Graded

Focus on key figures of the avant-garde and experimental theatre from the late 20th and early 21st centuries. Explores the motivations, concepts and creative techniques behind innovations in post-dramatic and devised theatre. As part of developing theoretical and historical understanding, students practice different methods of devising, collaborating and conceptualizing theatrical performance, culminating in an original presentation of student work.

THEA 490 Special Studies in Theatre**Credits 1-3 Grading Basis** Graded

An independent study for students to pursue individual research under the guidance of a faculty advisor. Department consent required to enroll.

THEA 495 Internship Theatre**Credits 3-6 Grading Basis** Graded

A faculty-arranged, prepared and monitored off-campus internship program designed to provide practical experience in theatre for students preparing for careers in those areas. Department consent required to enroll.

THEA 499 Honors in Theatre**Credits 1-3 Grading Basis** Graded

Repeatable to a maximum of six credits. Offered fall and spring. Students enrolled for three credits are required to attend class meetings of THEA 441. Department consent required to enroll.

University Studies (UNST)

UNST 102 Career and Academic Planning**Credits 1 Grading Basis** Graded

A course designed to prepare students for career and academic planning. Specific content includes accessing career and academic resources, career decision-making skills, and self awareness.

UNST 110 New Dukes 101**Credits 1 Grading Basis** Graded

New Dukes 101 is a one-credit class designed for incoming students to prepare for success at JMU, in their community and throughout their life. Core features of this seminar will help students: understand their reasons for being in college; develop relationships with peers, faculty and staff; build skills for navigating the transition to college, learning strategies, time management and overcoming challenges; and develop plans for succeeding in college and in their career.

Prerequisites: First year or first year transfer students only; or permission of the instructor.

UNST 120 Student Success Strategies**Credits 1 Grading Basis** Graded

This course is designed for students wanting to better accomplish their goals at JMU. Core features of this seminar will help students: understand their reasons for being in college; develop relationships with peers, faculty and staff; explore strategies for learning, time management and overcoming challenges; and build plans for succeeding in college and in their career.

UNST 150 Global Learning and Living: Madison International**Credits** 1-3 **Grading Basis** Graded

This course provides an opportunity for a diverse cohort of international and U.S. students to learn from and about each other through stimulating discussions, intercultural residence hall programs and the opportunity to participate in service learning. Repeatable up to a maximum of six credit hours.

Prerequisites: Participation in the Madison International Learning Community or permission of the instructor.

UNST 151 Making Sense of Beliefs and Values: A Guided Tour for Global Citizens**Credits** 1-3 **Grading Basis** Graded

This course explores the origin and nature of beliefs and values and how they are linked to actions, policies and practices around the world. These processes are examined through a range of big picture issues (e.g., religious, political, environmental, gender-based, cultural) that are relevant to all global citizens. Through dynamic speakers, discussions, readings, activities and lectures, this course helps students develop a deeper understanding of self, others and the larger world. Repeatable up to a maximum of six credit hours.

Prerequisites: Participation in the Madison International Learning Community or permission of the instructor.

UNST 190 Special Topics in University Studies**Credits** 1-3 **Grading Basis** Graded

Special topics course with topics approached from a variety of academic perspectives. Designed for lower-level elective credit.

UNST 250 Alternative Break Leadership Training**Credits** 2 **Grading Basis** Graded

A leadership training curriculum that covers a broad range of topics essential to leading an alternative break trip. This course incorporates service learning pedagogy. This will be accomplished through the use of affective and cognitive approaches. Personal growth as it relates to core concepts of leadership will be prominent. Emphasis is also placed on peer interaction, active participation, extensive reading, case work, and a range of guest speakers. Attendance is required.

UNST 251 Alternative Break Leadership Practicum**Credits** 1 **Grading Basis** Graded

Hands-on practicum of leadership strategies and techniques. Collaborative learning is enhanced when students apply what they learn in class by describing relevant lessons learned through experiences outside the classroom. The focus of this course is to provide students with the opportunity to lead in an observed setting and receive constant feedback and mentoring on their demonstrated leadership skills, critical reflection, inquiry, dialogue, and group interaction.

Prerequisites: UNST 250.

UNST 300 Integrative General Education [C1CT]**Credits** 3 **Grading Basis** Graded

This innovative topics course is designed to meet the General Education program's learning outcomes for critical and integrative thinking by providing opportunities for students to synthesize prior learning and apply it to solve contemporary real-world problems. Students complete a portion of the class assignments in teams to develop essential real-world skills in collaboration and communication. Topics vary by instructor. May be repeated when course content changes.

UNST 307 Peer Education and Facilitation Studies**Credits** 2 **Grading Basis** Graded

This is a two-credit course designed to provide students with skills and knowledge to be effective peer educators. Students will receive health and identity content knowledge while increasing their group facilitation, presentation and processing skills.

UNST 389 Campus Health Communication**Credits** 2 **Grading Basis** Graded

This course provides students an opportunity to learn about, create and communicate messages about health and wellness topics. Students develop the content produced in the campus publication Potty Mouth. Students develop skills in media literacy, self-directed learning and systems thinking, and they create media products. Through group discussion and interviews, students learn how to effectively communicate health behaviors to their peers.

UNST 390 Special Studies in University Studies**Credits** 1-3 **Grading Basis** Graded

Designed to give students an opportunity to complete independent study and/or research under faculty supervision in university studies.

UNST 398 Practicum in University Studies**Credits** 1-3 **Grading Basis** Graded

Selected practicum experiences, which provide students with supervised practicum experiences.

UNST 475 Dollars and Sense**Credits** 3 **Grading Basis** Graded

This practical course will review the affect a personal philosophy on money, and management of personal finances, has on all aspects of life when it comes to securing the American Dream. Students will learn real life skills in the areas of eliminating debt, creating a budget, understanding investments and insurance, saving money, planning for retirement, shopping for a house and other topics dealing with financial issues faced in daily life.

UNST 480 Internship Preparation, Participation and Professionalism**Credits** 1-3 **Grading Basis** Graded

This course provides preparation and support during and reflection after internship experiences in a variety of disciplines. It will cover the internship experience, professionalism, networking and reflection. Open to all majors for elective credit.

UNST 490 Advanced Special Studies in University Studies**Credits** 1-3 **Grading Basis** Graded

Exploration of a significant topic in depth.

UNST 498 Internship in University Studies**Credits** 1-6 **Grading Basis** Graded

The course allows students to receive academic credit for work experienced in an agency or organization related to university studies.

Vocational Education (VOED)

VOED 383 Curriculum and Instructional Procedures in Vocational Education**Credits** 3 **Grading Basis** Graded

The study and development of techniques and methods to provide vocational education instruction to secondary school students. Competencies to be developed will include planning for instruction, applying different methodologies, and assessing student performance and progress. The relationship of vocational education to other curricular areas will also be addressed.

Women's, Gender and Sexuality Studies (WGSS)

WGSS 200 Introduction to Women's, Gender and Sexuality Studies [C5SD]

Credits 3 Grading Basis Graded

Interdisciplinary introduction to theories and scholarship in Women's, Gender and Sexuality Studies. Examines the social construction of gender, how gender affects access to opportunity, and the experiences and contributions of women throughout history. Provides a foundation for subsequent work in the women's, gender, and sexuality studies minor. May be used for general education credit.

WGSS 230 Biology, Gender and Society

Cross-listed with: BIO 230

Credits 3 Grading Basis Graded

Students in this course will explore the biology of sex and gender and consider their interactions and interrelationships through a lens that considers intersectional identities and societal expectations. Topics include reproductive biology, sexual differentiation, gender development and expression, and the interactions between science and society that inform gendered bodies and practices.

Prerequisites: Either BIO 103, BIO 150, CHEM 120, CHEM 131, GEOL 110, GEOL 210, or ISAT 112.

WGSS 300 Special Topics in Women's, Gender and Sexuality Studies

Credits 3 Grading Basis Graded

Examination of selected topics of importance to the field of women's, gender and sexuality studies.

WGSS 301 Feminist Blogging: Writing for Shout Out! JMU

Cross-listed with: SCOM 301

Credits 3 Grading Basis Graded

The goal of this course is to work in a feminist collective to write daily blog posts for the Shout Out! JMU blog. Students will organize the blog, research and write blog posts, publicize the blog, and create dialogue by commenting on one another's blog posts as a means of consciousness raising.

WGSS 302 Third Wave Ecofeminism

Cross-listed with: SCOM 302

Credits 3 Grading Basis Graded

This course explores the historically strong association between women and nature, in which the image of Mother Earth is central, and critiques the power-as-domination assumption of our culture shown in the exploitation of women and of the earth itself. Religious, psychological, social, historical and scientific manifestations of this assumption will be examined, along with alternative models of power and responsibility. Students will explore the considerable research on ecofeminism.

WGSS 305 Theories and Methods in Women's, Gender and Sexuality Studies

Credits 3 Grading Basis Graded

This course introduces students to interdisciplinary theories and methods informing scholarship in women's, gender and sexuality studies.

WGSS 315 Transnational Feminisms

Credits 3 Grading Basis Graded

This multidisciplinary course explores the complex issues that women and LGBTQ communities encounter in non-Western regions of the world. May be focused regionally or topically drawing on multiple disciplinary perspectives.

WGSS 325 Gender and Violence

Credits 3 Grading Basis Graded

This course explores the public nature of private violence, specifically violence committed against women in U.S. culture. Students will investigate the social, political and personal meaning of violence within a gendered context. Throughout the course students will analyze the ways in which demographic, social, cultural, economic and political factors teach us to think about women in violent terms as well as help perpetuate violence against women. Students will consider violence not only in its physical dimension, but also in its symbolic and structural manifestations. Students will also examine the ways in which ideas about race, ethnicity, class and sexuality affect the degree and types of violence committed against women.

WGSS 337 Sociology of Gender

Cross-listed with: SOCI 337

Credits 3 Grading Basis Graded

This course covers foundational sociological theory, research and examination of the lived realities of sex and gender in society. Using a sociological lens, the course explores the interactions, identities and institutions that are implicated in the social construction of gender and gender inequality. A core emphasis is on how individuals learn and "do" gender, how various institutions (e.g., school, work, media and family) produce and perpetuate gender inequality, how gender inequality intersects with other systems of inequality in social life, and movements for gender liberation and justice.

WGSS 341 Gender and Justice

Cross-listed with: JUST 341

Credits 3 Grading Basis Graded

This course is an interdisciplinary examination of the causes, structure and consequences of gender oppression. Consistent with the social justice track of the major, notions of fairness, justice, and equality with respect to gendered social, political and economic relations will be examined.

WGSS 347 Queer Theories of Gender and Inequality

Cross-listed with: SOCI 347

Credits 3 Grading Basis Graded

This course looks at contemporary approaches to the culture and social structure of gender and gender binaries, as they intersect with power, knowledge, science/biology and inequality. Method involves examining and challenging the cultural classification systems that inform knowledge, human action and social institutions. Requires college level reading, writing and intellectual engagement.

WGSS 348 Communication and Gender

Cross-listed with: SCOM 348

Credits 3 Grading Basis Graded

This course explores the role of communication in the social construction of gender across a variety of contexts. Students will critically examine the interconnectedness of sex, gender, sexuality, race and class toward understanding systems of privilege and oppression in communication practices and social institutions. Focus is on communication as a tool for personal and social change.

Prerequisites: Any 100-level SCOM course.

WGSS 350 The Philosophy of Feminism

Cross-listed with: PHIL 350

Credits 3 Grading Basis Graded

An intermediate-level examination of philosophical problems in feminist theory and feminist contributions to philosophy.

WGSS 355 American Women at War**Credits 3 Grading Basis Graded**

This course invites students to engage a series of issues about the role of women in the US military. This course will examine the contributions and experiences of women who served during the American Revolution, the U.S. Civil War, WW I & II, Korea, Vietnam and the Persian Gulf War(s). Also included in this course is an examination of how women in military service both past and present are an instrument for societal change in America specifically in promoting the cause of women's rights.

WGSS 367 Sociology of Sexuality**Cross-listed with: SOCI 367****Credits 3 Grading Basis Graded**

This course examines sociological theory and research on sexual behaviors, identities, cultures and social movements, investigating how sexuality is shaped by society and its social institutions. In addition, the course examines how sociological research on sexuality is conducted, how society shapes the sociological study of sexuality, the unique ethical concerns and methodological challenges in researching sexuality and the place of sociology in shaping discourse and social policy on relevant social issues.

WGSS 368 Women's Literature**Cross-listed with: ENG 368****Credits 3 Grading Basis Graded**

A study of literature by women.

WGSS 369 Feminist Literary Theory**Cross-listed with: ENG 369****Credits 3 Grading Basis Graded**

An intensive study of a variety of feminist critical approaches and their applications to literature.

WGSS 370 Queer Literature**Cross-listed with: ENG 370****Credits 3 Grading Basis Graded**

An exploration of texts and issues in literature written by and about lesbian, gay, bisexual, transgender and queer writers, including critical and theoretical issues as well as questions of canon. Text studied may include fiction, poetry, drama, essays and memoirs written primarily, but not exclusively, in the 20th century.

WGSS 376 Anthropology of Reproduction**Cross-listed with: ANTH 376****Credits 3 Grading Basis Graded**

This course provides students with a critical and cross-cultural perspective on human reproduction. Examining how individuals draw on social and symbolic resources to sort out complicated private decisions, we will discuss how reproductive experiences are embedded in local, national and transnational politics. Topics covered may include: cross-cultural perspectives on childbearing and childlessness, kinship, and the globalization of new reproductive technologies such as in vitro fertilization.

Prerequisites: ANTH 195 or permission of the instructor.

WGSS 383 Women and Politics in Comparative Perspective**Cross-listed with: POSC 383****Credits 3 Grading Basis Graded**

A study of the causes and consequences of women's political marginalization in the United States and abroad. The course examines socioeconomic and political dimensions of gender inequality, exploring how women have worked through social movements, electoral politics, and public policy initiatives to overcome obstacles to their political empowerment.

WGSS 386 Sexual Orientation, Law and Justice**Cross-listed with: JUST 386****Credits 3 Grading Basis Graded**

This course explores considerations of justice in shaping law and policy related to sexual orientation. The concept of sexual orientation is a relatively recent development in academic, legal, medical and other policy contexts in the United States and globally. Topics addressed in the course include the dehumanization, criminalization, medicalization and oppression of sexual orientation minorities, and legal and regulatory interventions and movements for social justice, criminal justice reform and human rights.

WGSS 387 Gender Identity, Law and Justice**Cross-listed with: JUST 387****Credits 3 Grading Basis Graded**

This course explores considerations of justice in shaping law and policy related to gender identity. The concept of gender identity is a relatively recent development in academic, legal, medical, and other policy contexts in the United States and globally. Topics addressed in the course include the dehumanization, criminalization, medicalization, and oppression of transgender, non-binary, and gender-expansive populations and legal and regulatory interventions and movements for gender self-determination, social justice, criminal justice reform, and human rights.

Prerequisites: JUST major prereqs: JUST 200 & one other 200-level JUST course, excluding JUST 225. SJS minor: JUST 221 or 223. WGSS minor: WGSS 200 & permission of minor coordinator. QUST minor: permission of instructor.

WGSS 400 Issues and Research in Women's, Gender and Sexuality Studies**Credits 3 Grading Basis Graded**

The capstone seminar for the Women's, Gender and Sexuality Studies minor. Focuses on readings in feminist philosophy, history and literature. Students will engage in research in critical issues affecting women's lives.

Prerequisites: WGSS 200 and nine additional credits approved by the Women's, Gender and Sexuality Studies Coordinator.

WGSS 417 Women's Global Health and Human Rights**Cross-listed with: HTH 417, NSG 417****Credits 3 Grading Basis Graded**

An international and human rights approach providing an overview of health issues within the context of a woman's life cycle. Attention will be given to critical issues of women's health such as access to health care and gender-based violence. Such issues as sexuality, nutrition, diseases affecting women, violence, harmful traditional practices, and sex trafficking will be discussed.

WGSS 420 Feminist Rhetorics**Cross-listed with: SCOM 420, WRTC 420****Credits 3 Grading Basis Graded**

Surveys key women figures in classical and contemporary rhetorical traditions and challenges the strategies used to historicize this tradition from feminist perspectives. Explores diverse feminist rhetorical discourses informed by race, sexual orientation, ethnicity and social class.

Prerequisites: WRTC 103 or equivalent and junior or senior standing; or permission of the instructor.

WGSS 423 Advanced Studies in Gender and Sexuality in Literature**Cross-listed with:** ENG 423**Credits 3 Grading Basis** Graded

Advanced study of a topic using a gender and sexuality studies approach to literary texts. This course will explore how gender and sexuality and their representation in literature are shaped by social, cultural, historical, and political contexts. Course may be repeated as topic changes. Course may be repeated as topic changes.

WGSS 466 Advanced Studies in Women's Literature**Cross-listed with:** ENG 466**Credits 3 Grading Basis** Graded

Advanced study of women's literary achievements in several cultural and historical contexts. May be focused by theme. May be repeated as course content varies.

Prerequisites: ENG 368/WGSS 368 or ENG 369/WGSS 369.

WGSS 485 Gender Issues in Science**Cross-listed with:** ISAT 485**Credits 3 Grading Basis** Graded

An interdisciplinary course that looks at the scientific process, science practitioners, and science students through the lens of gender analysis. Students read literature, lead discussions, perform experiments, and analyze both data and processes to address the effects of educational systems on the preparation and careers of scientists, the influence of politics and culture on scientific inquiry, and the effects of critiques grounded in gender analyses on understanding the scientific process.

WGSS 490 Independent Studies in Women's, Gender and Sexuality Studies**Credits 3 Grading Basis** Graded

Designed to give capable students in women's, gender and sexuality studies an opportunity to complete independent study under faculty supervision.

WGSS 492 Internship in Women's, Gender and Sexuality Studies**Credits 1-3 Grading Basis** Graded

Provides the student with practical experience in employing and refining women's, gender and sexuality studies concepts in a public or private agency, under faculty supervision.

WGSS 495 Special Topics in Women's, Gender and Sexuality Studies**Credits 3 Grading Basis** Graded

In-depth examination of selected topics of current importance to the field of women's, gender and sexuality studies. Offered only with approval of the program coordinator and dean of the College of Arts and Letters. May be repeated for credit when course content changes.

Prerequisites: WGSS 200 or consent of the instructor.

World Languages and Cultures (WLC)

WLC 330 Food and Culture of France**Credits 3 Grading Basis** Graded

This class explores the history and impact of French food on contemporary culture. Through a wide range of experiences and on-site lectures, students will be exposed to the history of French cuisine and its importance in the modern dining experience. Instruction is in English.

WLC 361 Modern Japanese Culture**Credits 3 Grading Basis** Graded

This course is an introduction to modern Japan and its culture. Students will explore some of the key concepts surrounding cultural values, attitudes, behavior patterns and communication styles seen in modern Japan. Students examine the roots of these concepts by learning about Japan's rich history, traditions, arts and more. Topics such as the concept of harmony, Japanese aesthetics, relationships and pop culture will be covered. Through the class activities and assignments, students develop a deeper appreciation for diverse cultures and become equipped for intercultural communications in the future global scenes. Instruction is in English.

WLC 391 Murder in Hispanic Literature**Credits 3 Grading Basis** Graded

This course offers an exploration of Spanish literature focused on the theme of murder. Students will engage with masterpieces of Spanish literature which include the theme of murder and its contexts of abusive power, social discrimination, war confrontation and political controversy.

WLC 421 Chinese Culture through Film**Credits 3 Grading Basis** Graded

This course offers an immersive exploration of Chinese culture, history and society through the lens of its vibrant and diverse cinematic traditions. Students will engage with a curated selection of films spanning different eras and genres. By analyzing narrative structures, visual aesthetics, thematic concerns and historical contexts, students will gain a deeper understanding of key cultural values, societal transformations and political dynamics that have shaped China. The course will cover significant film movements, influential directors and the evolving relationship between cinema and national identity, providing insights into topics such as family structures, gender roles, traditional philosophies and the impact of globalization on Chinese society.

WLC 422 Chinese Traditions: Kung Fu, Tea and Calligraphy**Credits 3 Grading Basis** Graded

This course offers a broad survey of traditional Chinese culture through three interrelated art forms that embody China's aesthetic, philosophical and social values. The course combines lectures, readings, multimedia materials and in-class experiential activities to introduce students to the cultural significance of martial arts, tea culture and calligraphy. A guest instructor will provide demonstrations and guided practice sessions to deepen students' appreciation of these traditions. Emphasis is placed on understanding how these practices reflect Chinese thought, history and daily life, as well as their influence in a global context.

WLC 440 Viking Culture and Myth**Credits 3 Grading Basis** Graded

This course requires students to develop a broad understanding of the Viking period by examining a wide variety of cultural artifacts (archaeological, literary and historical) and the perspectives of contemporaneous and subsequent historians writing about the Vikings. Students analyze historical accounts of the Viking period by members of the vanquished cultures, the myths of the Vikings themselves, the literary texts about these myths compiled 500 years later, as well as modern renditions of the Viking Age in film, literature and pop culture today. They learn to question the veracity and accuracy of these sources as well as the political and cultural motivations behind depictions of the Vikings. They then use the information from these sources to develop their own assessment of the Viking period and its treatment in primary and secondary sources.

WLC 446 Special Topics: World Literatures in Translation**Credits 3 Grading Basis** Graded

This course is a study of a particular topic in world literature in translation. The course may be repeated if content changes.

WLC 447 Special Topics: World Cultures in Translation**Credits 3 Grading Basis** Graded

This course studies a particular world culture in translation. The course may be repeated if content changes.

WLC 448 Special Topics: Linguistics in World Languages**Credits 3 Grading Basis** Graded

Students will study linguistic structures of a world language in translation. Topics could include an introduction to sociolinguistics and psycholinguistics. This course may be repeated for credit if the content changes.

WLC 451 Dante: from Hell to Paradise**Credits 3 Grading Basis** Graded

This course offers an in-depth exploration of Dante Alighieri's *Divine Comedy*. Students will engage with this foundational text of Italian and world literature, focusing on its themes of redemption and the human condition as Dante journeys through Hell, Purgatory and Paradise.

WLC 452 The Woman Question in Italian Culture**Credits 3 Grading Basis** Graded

This course examines the evolution of the "woman question" and the "querelle des femmes" in Italian and European literature, exploring how these discussions have shaped and reflected societal views on gender and women's roles throughout history. Students will investigate a range of texts from the Middle Ages to the modern era, focusing on key literary works that address the complexities of gender, identity and empowerment.

WLC 456 Translating Italian Literature**Credits 3 Grading Basis** Graded

This course offers an in-depth exploration of Italian literature through the lens of translation, aimed at students pursuing a minor or major in Italian. Focusing on the practical challenges and creative opportunities involved in translating works from Italian to English, the course provides both theoretical foundations and hands-on experience. Instruction is in English.

Prerequisites: ITAL 212 or ITAL 232, or by permission of the instructor.

WLC 490 Special Studies: World Languages and Cultures in Translation**Credits 1-4 Grading Basis** Graded

This course allows students an opportunity to complete independent studies under faculty supervision. Work may be done in all languages offered in the department but may not replace standard course offerings. This course may be repeated for credit if the content changes.

Writing, Rhetoric, and Technical Communication (WRTC)

WRTC 100 Reading and Writing Workshop**Credits 3 Grading Basis** Graded

An introduction to writing process and structure in a workshop setting. This is an elective course for students who want to select an introductory-level college writing course before taking WRTC 103.

WRTC 103 Rhetorical Reading and Writing [C1W]**Credits 3 Grading Basis** Graded

Rhetorical Reading and Writing teaches students how to write effectively in situations they have not yet encountered. By practicing reading and writing as rhetorical activities, students learn how to explore and research topics; design writing for diverse audiences, purposes and occasions; and use feedback to improve their writing. WRTC 103 fulfills the General Education Madison Foundations writing requirement and is a prerequisite for all WRTC courses numbered 200 or above. May be used for general education credit. May not be used for major credit.

WRTC 200 Introduction to Studies in Writing, Rhetoric and Technical Communication**Credits 3 Grading Basis** Graded

This is the initial core course and portal of entry into the School of Writing, Rhetoric and Technical Communication. Students will read and discuss foundational articles, undertake course projects and explore the roles that writers, rhetoricians and technical communicators are called upon to fill in their internships and jobs.

Prerequisites: WRTC 103 or equivalent.

WRTC 300 Professional Editing**Credits 3 Grading Basis** Graded

Introduction to the conceptual and technical editing of a wide range of documents for diverse audiences and different purposes. Consideration of genre, tone, style and syntax. Students will learn to interact with authors and clients and will practice both hard copy and electronic editing. Course topics allow students to encounter a wide range of editorial experiences to prepare them for the workplace.

Corequisites: WRTC 200 or permission of instructor.

WRTC 301 Language, Law and Ethics**Credits 3 Grading Basis** Graded

Broad examination of the nexus of law, ethics and language. Exploration of a variety of genres in legal writing, providing a background in the role of law and ethics in society and helping students understand how language operates in the construction of legal and moral codes. Primary topics include intellectual property, privacy, legal issues in communication, the history of legal language and the ownership and ethical use of information.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 314 Writing in the Public Sphere**Credits 3 Grading Basis** Graded

Introduction to the concept of the public sphere and examination of a variety of texts and media that illustrate the function of the public sphere. Students apply theoretical knowledge to the analysis of public discourse and present their analyses in oral and written formats. Students gain important insights into their own roles and responsibilities as citizens within the public sphere and learn to use language effectively in multiple rhetorical situations.

Prerequisites: WRTC 103 or equivalent.

WRTC 316 Research Methodologies in WRTC**Credits 3 Grading Basis** Graded

This course is an introduction to the process of conducting research grounded in inquiry. Students use a variety of research methodologies to gather information from secondary and primary sources. Students evaluate information for accuracy and usability and interpret information for the audience and rhetorical context they have defined. Students in this course design a research study, carry it out and write a subject appropriate report.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 318 Intercultural Professional Communication**Credits 3 Grading Basis Graded**

Focus on the importance of culture to professional communication, both in print and online, by using an intercultural perspective to examine audience, purpose, persona, context, language, page and screen design, graphics and color. Includes a consideration of basic models of culture developed in professional environments, incorporating management, teamwork and translation issues, as well as how American culture differs from other cultures worldwide.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 326 Environmental Communication and Advocacy**Cross-listed with: SCOM 354****Credits 3 Grading Basis Graded**

This course is an exploration of the multifaceted aspects of environmental controversies including the rhetoric, advocacy campaigns and decision-making processes that produce and attempt to manage environmental conflict. Emphasis on persuasive efforts by, interest groups, corporations, resource managers, government agencies, scientific experts, politicians and citizens to influence public understanding of environmental issues, adoption of sustainable behavior and lifestyles, and public policy outcomes.

Prerequisites: WRTC 103 or equivalent.

WRTC 328 Practicum**Credits 1-3 Grading Basis Graded**

Allows students to engage in practical experience opportunities in the field of Writing, Rhetoric, and Technical Communication. Students may apply no more than 3 practicum credit hours toward completion of WRTC major or minor requirements.

Prerequisites: WRTC 103 or equivalent.

WRTC 330 Rhetorical Analysis and Criticism**Credits 3 Grading Basis Graded**

This course is a survey and application of a range of rhetorical approaches to analyze print, oral, visual and multimodal forms of everyday communicative practices. Primary topics include the origins of rhetoric, the manifestations of contemporary forms of communication, the rhetorical theories used to explain those forms, and the criticisms developed to respond to them.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 332 Creative Approaches to Digital Storytelling**Credits 3 Grading Basis Graded**

This course introduces students to the history and methods of digital storytelling and provides them with the opportunity to critically evaluate a range of digital stories and to develop their own digital storytelling projects. Digital stories are narratives that are composed in digital environments using sound, audio, still images, video, and interactive elements like maps, charts and timelines. By the end of the course, students will be able to adapt digital storytelling techniques to suit a variety of genres and audiences. Throughout the course, students learn to assess digital storytelling artifacts, such as short video documentaries and mixed media narratives, and their usefulness for particular audiences and contexts. In addition to these digital composing techniques, students learn to assess and use a variety of online platforms where digital stories can be published, critique and improve their own and others' work, collaborate with others and create digital storytelling artifacts for clients.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 334 Introduction to Popular Writing**Credits 3 Grading Basis Graded**

A theoretical and practical overview of the growing field of popular writing. Students will analyze a broad range of genres – including reviews, commentaries, profiles, blogs and ads – from a broad range of publications, including newspapers, magazines and the Web, with the goal of acquiring a critical understanding of the rhetorical aims and practices of popular writing.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 336 Writing Consultation**Credits 3 Grading Basis Graded**

Integrates the theory and practice of writing consultation in academic and professional settings and is suited for developing tutors, teachers, consultants and editors. The course includes mentorship and practice in the University Writing Center, and it provides students opportunities to develop as writers, scholars and professionals. Upon completion of the course, students will be eligible for, but not guaranteed, employment in the University Writing Center.

Prerequisites: WRTC 103 and permission of the instructor.

WRTC 338 Sound Writing**Credits 3 Grading Basis Graded**

This course will focus on composing about and with sound. The course combines analytical and reflective projects with practical exercises in audio recording and editing, and rhetorically discovering and using online audio sources such as music, sounds and interviews. The course may be used to explore topics such as podcasts, audio tours and professional profiles.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 340 Writing as Leading**Credits 3 Grading Basis Graded**

Investigation of contemporary leadership theories as they apply to writing; students will apply these principles and techniques to their own writing. The course will explore how writers lead readers and how leaders employ writing and use writers to influence their audiences. Students will gain experience writing in typical leadership genres, such as the position paper and the op-ed piece.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 342 Writing Place**Credits 3 Grading Basis Graded**

Examines the relationship between language and location by analyzing print and virtual rhetorics of the social and natural environment. Students will learn about the rhetorical tradition of place-centered expression and the importance of place in society.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 350 Foundations of Technical Communication**Credits 3 Grading Basis Graded**

This course provides an introduction to the major theories, issues and contributors in the field of technical and scientific communication. Students explore global, print and electronic communication, ethical and legal issues, and the project cycle. They start with rhetorical analyses of audiences and progress through deliverables. Students learn to problem solve through a variety of technical communication projects and develop their career interests in chosen technical communication fields.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 351 Visual Rhetoric**Cross-listed with:** SCOM 351, WRTC 351L**Credits 3 Grading Basis Graded**

A study of the rhetorical foundations of visual and verbal arguments in academic disciplines and popular culture. Students will analyze and produce visual and verbal arguments in a variety of rhetorical contexts. Prerequisites: WRTC 103 and any 100-level communication studies course or permission of the instructor.

WRTC 351L Visual Rhetoric**Cross-listed with:** SCOM 351, WRTC 351**Credits 3 Grading Basis Graded**

A study of the rhetorical foundations of visual and verbal arguments in academic disciplines and popular culture. Students will analyze and produce visual and verbal arguments in a variety of rhetorical contexts. Prerequisites: WRTC 103 and any 100-level communication studies course or permission of the instructor.

WRTC 352 Online Design**Credits 3 Grading Basis Graded**

Students learn to design user-centered web and app content with an emphasis on inclusive design. Students increase their familiarity with industry-accepted tools and programs for developing and polishing online content while also developing the ability to recognize and resolve basic HTML/CSS issues that affect content presentation. Students shape portfolio pieces through a user-centered design process, with an emphasis on building skills at the design stage. Students improve their ability to assess integrated AI and LLM resources for web and app developers.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 354 Document Design**Credits 3 Grading Basis Graded**

This course is an examination of the principles of design and the importance of the project cycle in designing documents. Students use layout and graphics programs to create professional brochures, flyers, posters, newsletters and manuals. Students work individually and collaboratively on their projects, producing excellent portfolio pieces. This course gives students flexibility for a wide variety of career opportunities in business and industry, the non-profit sector, and government.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 358 Writing About Science and Technology**Credits 3 Grading Basis Graded**

This course focuses on the development and application of rhetorical strategies and tools used in writing about science and technology for a contemporary, general audience. As citizen-interpreters, students analyze and produce writing designed to make even 'hard' science and technology accessible, with particular attention to narrative, to the framing of policy issues, and to establishing relevance.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 360 British Media and Society**Cross-listed with:** HUM 251L, SMAD 360, SCOM 360**Credits 3 Grading Basis Graded**

Study the history, nature and impact of mass media in the United Kingdom. Emphasis on the impact modern media has on society, and society has on media. Consideration of similarities and differences in mass media in the United States and Great Britain. Consideration of the relationship between mass media and the arts. Focus on 20th century mass media in London, one of the world's preeminent and most influential centers. London term only.

WRTC 400 Advanced Editing**Credits 3 Grading Basis Graded**

Advanced Editing initiates students into the profession of editing. Content focuses on work with longer texts and real clients to introduce advanced editing skills and considerations such as document design, universal design, project management, production and freelancing. Students will learn how to initiate professional communication with clients, edit for users with diverse needs and work as part of an editing team.

Prerequisites: WRTC 300 or permission of the instructor.

WRTC 412 Content Strategy**Credits 3 Grading Basis Graded**

Focused study of the theory and practice of building a content strategy applicable for a variety of organizations (business, nonprofit, government). Students will learn content strategy theories, perform case study analyses and apply their findings to the design of content strategies with specific organizations in mind.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 416 Rhetoric of Environmental Science and Technology**Cross-listed with:** SCOM 465**Credits 3 Grading Basis Graded**

This course offers an advanced study of the way the public receives, makes sense of, and influences scientific and technical information about environmental issues. Implications of these processes on environmental policy will be analyzed. Readings and assignments will concentrate on the interactions between technical and public spheres of communication, with an in-depth examination of the way the media facilitates the transfer of information between scientific communities and public audiences.

Prerequisites: WRTC 103 or equivalent and junior or senior standing; or permission of the instructor.

WRTC 420 Feminist Rhetorics**Cross-listed with:** SCOM 420, WGSS 420**Credits 3 Grading Basis Graded**

Surveys key women figures in classical and contemporary rhetorical traditions and challenges the strategies used to historicize this tradition from feminist perspectives. Explores diverse feminist rhetorical discourses informed by race, sexual orientation, ethnicity and social class.

Prerequisites: WRTC 103 or equivalent and junior or senior standing; or permission of the instructor.

WRTC 426 Special Topics in Writing, Rhetoric and Technical Communication**Credits 3 Grading Basis Graded**

Focused, in-depth study of specific areas or subjects in writing, rhetoric, and technical communication. Topics may pertain to issues relevant to the discipline, to the study of particular theories and practices, or to the study of significant figures in the field. Seminars may be repeated for credit when course content changes.

Prerequisites: WRTC 103 or equivalent and junior or senior standing, or permission of the instructor.

WRTC 432 Rhetoric of the Personal Narrative**Credits 3 Grading Basis Graded**

Examination of the rhetorical elements of personal narrative. Students will read examples of personal narratives ranging from essays to longer memoirs and autobiographies, in order to examine questions related to purpose, audience, voice, and style. Discussion will include what makes a piece of writing personal, what makes it a narrative, and what makes it effective.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 433 Animal Rhetorics**Credits 3 Grading Basis Graded**

Students will explore the ways various peoples have defined the human as opposed to an animal "other." What it is to be a human, a person and an animal is not static; these definitions change over time and across cultures. They are the results of ongoing acts of persuasion. What is universal, though, is the idea of a boundary between human and animal, and the ways we define either category helps define the other. Animal Rhetorics will feature readings from the history of rhetoric, biology, biosemiotics and philosophy, tracing our definitions of what it means to be a human and what it means to be rhetorical, from pre-history to the present. Assignments will include creative work, presentation and a research component.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 434 Advanced Popular Writing**Credits 3 Grading Basis Graded**

Advanced focus on a particular genre in popular writing, such as reviews, commentaries, opinion pieces, profiles, blogs, or ads. Students will engage with a specific genre to acquire both a critical understanding of its rhetorical aims and practices as well as the skills to practice writing within that genre. Students in this course have the opportunity to develop several professional portfolio pieces.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 446 Rhetorics of Design and Culture**Credits 3 Grading Basis Graded**

This course examines the central role cultures and communities play in the design process of visual and digital campaigns. Students will develop an awareness of cultural markers in images and branding campaigns; discuss and apply theories of multimodality, visual rhetoric, and design; and apply their learning in the creation of an original campaign.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 447 Interaction Design**Cross-listed with: ISAT 447****Credits 3 Grading Basis Graded**

Study of and practice with processes, principles, tools, models, and techniques for designing interactions between humans and digital products and systems. Topics include physiological and psychological factors affecting interaction design, interaction design process, interaction models, styles, and paradigms, design notations and representations, prototyping, and interaction design evaluation.

Prerequisites: Junior or senior standing only.

WRTC 450 Digital Rhetoric**Credits 3 Grading Basis Graded**

Introduces the rhetoric of digital design in a variety of contexts. Students learn what makes for effective static and interactive digital designs and practice analyzing and creating digital designs. Student will compose technical documents within diverse traditions, which include digital rhetoric, mixed media and visual rhetoric.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 454 Publication Management**Credits 3 Grading Basis Graded**

Exploration of the publication production process that addresses the theory and practice of project management for professional print and electronic documents. Students work collaboratively to examine managerial and editorial responsibilities. Topics include defining editorial policy, defining management roles, working with project teams, creating document publication schedules, reviewing and editing submissions for publication and collaborating with authors.

Prerequisites: WRTC 300 or permission of the instructor.

WRTC 456 Usability and User Experience**Credits 3 Grading Basis Graded**

This course provides a theoretical and practical study of the principles, techniques and tools of user experience (UX) in technical communication, including user and task analysis, iterative and human-centered design, and usability testing. Students will research audience needs, define content needs, consider issues of inclusion and accessibility, design and plan usability and UX evaluations, conduct tests, analyze test data, draft prototypes, and prepare reports.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 458 Scientific and Medical Communication**Credits 3 Grading Basis Graded**

Introduction to the context and use of language in scientific and medical disciplines. Emphasis is placed on understanding the rhetorical nature of scientific discourse. Primary topics include examining different forms of scientific and medical writing in traditional and digital contexts, the nature of communication within professional communities and composing texts for general readers.

Prerequisites: WRTC 200 or permission of the instructor.

WRTC 460 Legal Writing**Credits 3 Grading Basis Graded**

Study of the central components of legal writing and the language of law with an emphasis on the genres of the legal professions. Primary focus is on legal analysis, intellectual property and fair use, and the creation of genre-relevant documents, including memoranda, briefs and legal summaries.

Prerequisites: WRTC 300 and WRTC 301, or permission of the instructor.

WRTC 462 Government Writing**Credits 3 Grading Basis Graded**

This course provides intensive practice in government genres or genres that occur at the intersection between governmental and political writing at the local and national levels. Democratic institutions create writing exigencies that require rhetorical intervention and technical knowledge, and this course offers writing practice in relevant genres, including news releases, consultant reports (i.e., feasibility studies), lobbyist communication and staff reports (i.e. monthly updates and annual reports). The authors of such genres include elected officials, appointed officials, citizen groups and individual citizens. Relevant issues include taxation, zoning, recycling, water treatment, sewer and wastewater treatment, environmental issues, transportation and any other timely infrastructure or policy question.

Prerequisites: WRTC 300 and WRTC 301, or permission of the instructor.

WRTC 464 Grant and Proposal Writing**Credits 3 Grading Basis Graded**

This course introduces students to the principles and practices of proposal and grant writing for research, education and community projects. Students will learn how funding organizations operate at local, federal and private levels and practice strategies for producing competitive proposals. Emphasis is placed on audience awareness, ethical considerations and key genres such as needs statements, project descriptions, budgets and timelines. By the end of the course, students will be able to identify funding opportunities and create professional proposals that demonstrate clarity, purpose and persuasion.

Prerequisites: WRTC 200 and WRTC 300, or permission of the instructor.

WRTC 470 Disability Rhetoric**Credits 3 Grading Basis** Graded

This course presents the political and social positioning of disability in multiple contexts, including the rhetorical traditions of ancient Greece, the Commonwealth of Virginia's eugenics laws and the Americans with Disabilities Act. Students will examine how the concept of disability is a rhetorical construct and will build knowledge about how language is utilized to shape current and prior models of disability, including rhetorical, political, historical, educational, legal and medical frameworks. The course culminates in the creation of an accessible online project that highlights individual students' scholarly interests, created following universal design principles.

Prerequisites: WRTC 103 or equivalent.

WRTC 480 Writing for Business and Industry**Credits 3 Grading Basis** Graded

This course offers an introduction to the communication and discourse practices of the business community through a community-based learning model. Emphasis is placed on working directly with a business organization. Primary topics include language and ethics in business, the understanding of audience for business communication and the creation of business documents including proposals and business plans.

Prerequisites: WRTC 300 or permission of the instructor.

WRTC 484 Writing for Nonprofits**Credits 3 Grading Basis** Graded

This course provides an introduction to the nonprofit sector through a community-based learning model. Emphasis is placed on working directly with a nonprofit agency in the local community. Primary topics include the role of the nonprofit in society, especially as an organization for change; creation of internal and public documents, including proposals, grants and publicity materials; and the legal requirements for nonprofit status.

Prerequisites: WRTC 300 or permission of the instructor.

WRTC 486 Writing in the Community**Credits 3 Grading Basis** Graded

This course offers an introduction to political and social engagement at the community level using multiple texts and a community-based learning model. Emphasis is placed on writing, reflection and hands-on service projects with community agencies. Primary topics include an examination of the central role of rhetoric in citizenship, leadership, social justice and social change.

Prerequisites: WRTC 300 or permission of the instructor.

WRTC 488 Writing in the Health Sciences**Credits 3 Grading Basis** Graded

This course offers an introduction to the medical field through a community-based learning model. Emphasis is placed on communication within the medical field and the translation of medical language for lay audiences. Primary topics include the roles of the practitioner and audience in medical communication, power relationships among clinicians and patients, and the creation of medical documents, including reports, proposals and technical articles.

Prerequisites: WRTC 300 or permission of the instructor.

WRTC 490 Independent Study in Writing, Rhetoric and Technical Communication**Credits 3 Grading Basis** Graded

Individualized projects in Writing, Rhetoric and Technical Communication. Available only to junior or senior majors, though exceptions may be made at the director's discretion. May be repeated with the director's approval when course content changes.

Prerequisites: WRTC 103 or equivalent and junior or senior standing, or permission of the instructor.

WRTC 495 Internship in Writing, Rhetoric and Technical Communication**Credits 3 Grading Basis** Graded

Allows students to incorporate field experience with WRTC course work through internships in government, business, industry, or education where they can observe communication processes and apply effective written, interpersonal and public communication skills. Students must complete an application process and be approved before receiving a permission number to enroll in the course (see Internship page on WRTC website for requirements and forms).

Prerequisites: WRTC 300, WRTC 301, junior or senior standing, and permission of the instructor.

WRTC 496 Capstone in Writing, Rhetoric and Technical Communication**Credits 3 Grading Basis** Graded

This course provides a capstone experience in which students reflect on their course work in the major with the aim of articulating how the major has broadened and deepened their understanding of the theories and practices of written communication and rhetoric as they apply to academic and professional contexts; demonstrating how WRTC student learning objectives tie together to inform their academic and professional work; and articulating short-term and long-term career goals. Students participate in seminar-style discussions, compose a personal philosophy of writing/editing/design through analysis of the work they have created in the major, and create a professional digital portfolio containing their strongest work from WRTC courses, internships and professional positions that are most representative of their career goals.

Prerequisites: WRTC 200, WRTC 300, WRTC 301, WRTC 316, WRTC 350, WRTC 495 and senior standing, and permission of the instructor.

WRTC 499 Honors**Credits 1-6 Grading Basis** Graded

Offered fall and spring.

Youth Justice (YTHJ)

YTHJ 300 Introduction to Youth Justice**Credits 3 Grading Basis** Graded

The course introduces students to the structure and functions of the American juvenile justice system (JJS) and situates the JJS within the web of other youth-serving systems, most importantly, the child welfare system (CWS). The course also covers the characteristics of adolescents who come into contact with the JJS and CWS. A developmental perspective is used to understand the diverse pathways of juvenile delinquency, emphasizing the pivotal role that family, trauma, mental health difficulties and school play in youths' involvement in crime. This is a required course for the youth justice minor.

YTHJ 400 Transforming Youth Justice Integrative Seminar**Credits 3 Grading Basis** Graded

This course is designed to integrate and apply knowledge from the student's major and the youth justice minor. The course emphasizes the application of knowledge through the development and implementation of a substantial, theory-based, in-depth individualized project around issues and/or challenges in the juvenile justice system with an emphasis on the population of dual status youth. Students may adapt the project to fit their career objectives and/or plans for graduate study.

Prerequisites: YTHJ 300; two additional courses approved as electives for the youth justice minor and junior or senior standing.

YTHJ 405 Internship in Transformative Youth Justice Studies**Credits 3 Grading Basis** Graded

This is an independent study course designed to allow students to receive academic credit for work experience in a youth justice setting related to the youth justice minor. Students enrolled in YTHJ 405 will arrange their work schedule with their internship provider and will consult with the faculty member of record for the course to obtain approval for their final research paper topic. Course may be taken twice.

Prerequisites: YTHJ 300; two additional courses in the YTHJ minor; and approval of the youth justice minor coordinator.

NON-DISCRIMINATION STATEMENT

Notice of Non-Discrimination and Equal Opportunity

James Madison University does not discriminate and prohibits discrimination in its employment, programs, activities, and admissions on the basis of age, color, disability, gender expression, gender identity, genetic information (including family medical history), marital status, military status (including veteran status), national origin (including ethnicity), parental status, political affiliation, pregnancy (including childbirth or related medical conditions), race, religion, sex, sexual orientation, or on any basis protected by law, unless otherwise permitted or required by law. JMU complies with all applicable federal and state laws regarding discrimination, harassment and retaliation. The responsibility for overall coordination, monitoring, and information dissemination about JMU's program of equal opportunity and non-discrimination is assigned to the Office of Equal Opportunity as described within Policies 1302 and 1324. Inquiries or complaints may be directed to the Office of Equal Opportunity via the OEO website (<https://www.jmu.edu/access-and-enrollment/o eo/index.shtml>), email (oeo@jmu.edu), or phone (540-568-6991).

JMU's prohibition of discrimination on the basis of sex includes dating and domestic violence, sexual assault, sexual exploitation, sexual harassment, stalking, and non-consensual relationships as defined in Policy 1340 and Policy 1346. The responsibility for overall coordination, monitoring, and information dissemination about JMU's Title IX program is assigned to the Title IX Coordinator. Inquiries or complaints may be directed to the Title IX Coordinator, Laura Sider Jost, via the Title IX Office website (<https://www.jmu.edu/access-and-enrollment/title-ix/index.shtml>), email (titleix@jmu.edu) or phone (540-568-5219).

YOUR RIGHT TO KNOW

Your personal safety and the security of the campus community are of vital concern to James Madison University. Information regarding campus security and personal safety, including topics such as crime prevention, sexual assault, available resources, the law enforcement authority of the James Madison University Police Department, emergency notification protocols, protocols for resident students who are reported missing, fire safety and residence hall fire statistics, and crime reporting policies can be found in the James Madison University Annual Security Report and Annual Fire Safety Report. The publication also contains reported crime statistics for the most recent three-year period that occurred on campus, in certain off-campus buildings or property owned or controlled by James Madison University or its affiliates, and on public property within, or immediately adjacent to and accessible from, the university's Harrisonburg campus and three international campuses.

The reports are available from the JMU Police Department website (<https://www.jmu.edu/police/clerycompliance/Annual-Security-and-Fire-Safety-Report.shtml>).

You can obtain a copy of these reports by contacting:

Police Department
James Madison University
821 South Main Street
MSC 6810
Harrisonburg, VA 22807

You can request that a copy be mailed to you by calling (540) 568-6769/6913.

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